

Compositional Monitoring of Energy-Mix Drift on the Simplex

Premier Data Output — CNT v3.1.0 + CNQ v2.0.0

CoDaWork 2026

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Coimbra, Portugal · 1-5 June 2026

P. Higgins, Rogue Wave Audio

This document contains the full premier data output of the CNT and CNQ engines applied to the 9-country EMBER electricity-generation corpus (2000–2025).

Each country section includes:

- Stage 1 plates · three orthogonal projections (XY scatter, XZ bearings, YZ CLR) per year + course plot (26–27 plates)
- Stage 2/3 plates · group barycenter distance matrix, helmsman frequency, pairwise divergence ranking, triadic area ranking, carrier interaction matrix, navigation summary table (7 plates)
- CNQ plates · quaternion-view dashboard (Higgins scale, angular velocity, bearing tensor magnitude, helmsman channel, κ^{HS} metric energy, CHSH + twin-quaternion + attractor diagnostics) + bearing trajectories at D=2 / D=3 / D=4 twin-factor (2 plates)

Generated 2026-05-13. Conforms to HUF Publication Standards (HUF-STD-001).

The instrument reads. The expert decides. The hashes carry the receipts. The vocabulary holds the line.

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§ AUS Australia

EMBER electricity-generation series · 2000-2025 · N = 26 · D = 9

INPUT

source: ember_AUS*_generation_TWh.csv
N records: 26
D carriers: 9
Bioenergy
Coal
Gas
Hydro
Nuclear
Other Fossil
Other Renewables
Solar
Wind
source_sha256: 4c872aa6bb2f4ac1c9897ff1...

HEADLINE METRICS

Helmsman flips: ?
Helmsman stability S_σ : {'global': 0.6666666666666667, 'rolling': [0.5714285714285714, 0.5714285714285714]}
IR class: ?
Attractor amplitude: ?

CHSH S: ?
Tsirelson bound: $2\sqrt{2} \approx 2.828$

Bearing pairs $(D(D-1)/2)$: 36

CNT v3.1.0 content_sha: ?...
CNQ v2.0.0 content_sha: ?...

Following pages: Stage 1 (XY/XZ/YZ × N years + course) → Stage 2/3 (7 pages) → CNQ (2 pages).

SECTION PLATE t=0 | Year 2000 | D=9 N=26 pairs=36

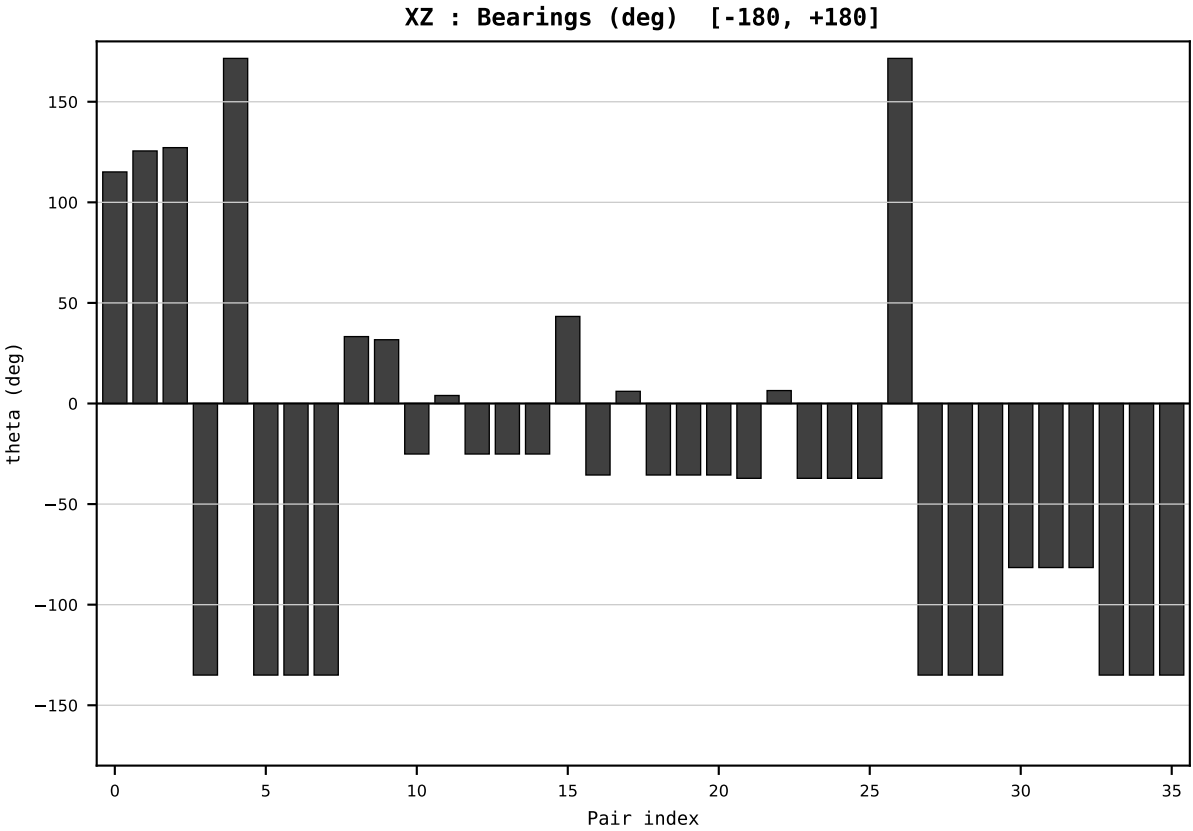
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2000
t : 0 / 25
D : 9 carriers
Pairs : 36 channels

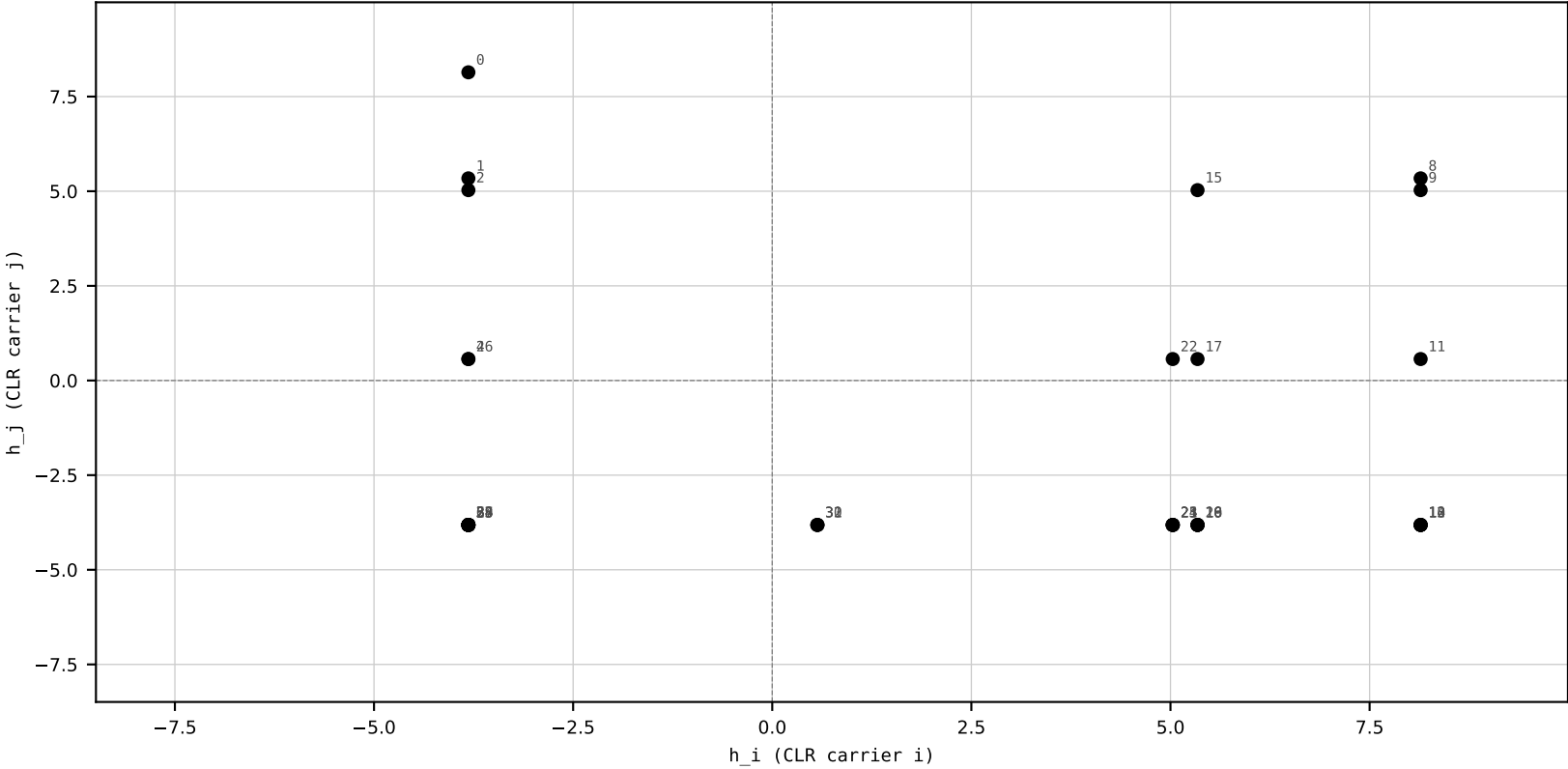
Hs = 0.825462
Ring = Hs-5
E_metric= 193.1704
kappa_HS= 155760.00
omega = 0.0000 deg
d_A = 0.000000
Helm = ---
Helm d = ---
DR = 11.956 HLR
DR ratio= 155760.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

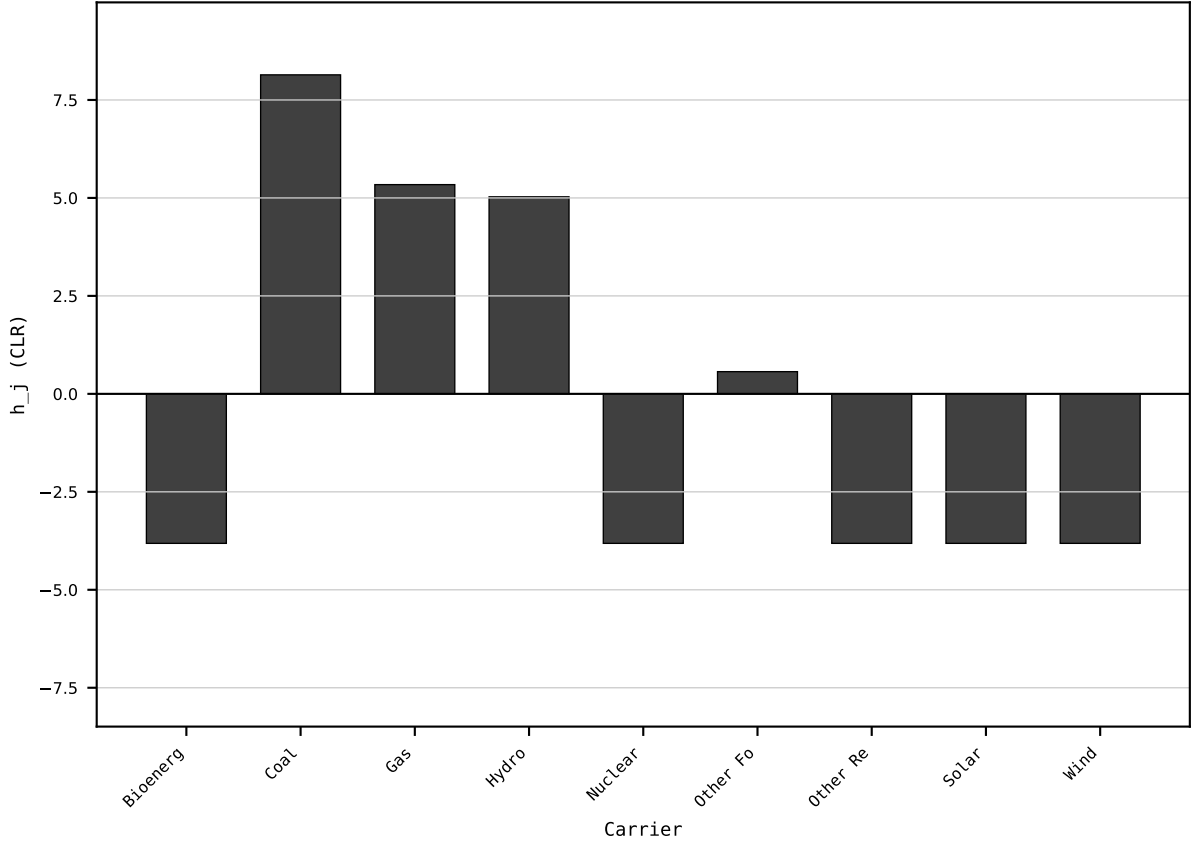
PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



XY : Plan View [-8.5, 10.0]



YZ : CLR [-8.5, 10.0]



SECTION PLATE t=1 | Year 2001 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

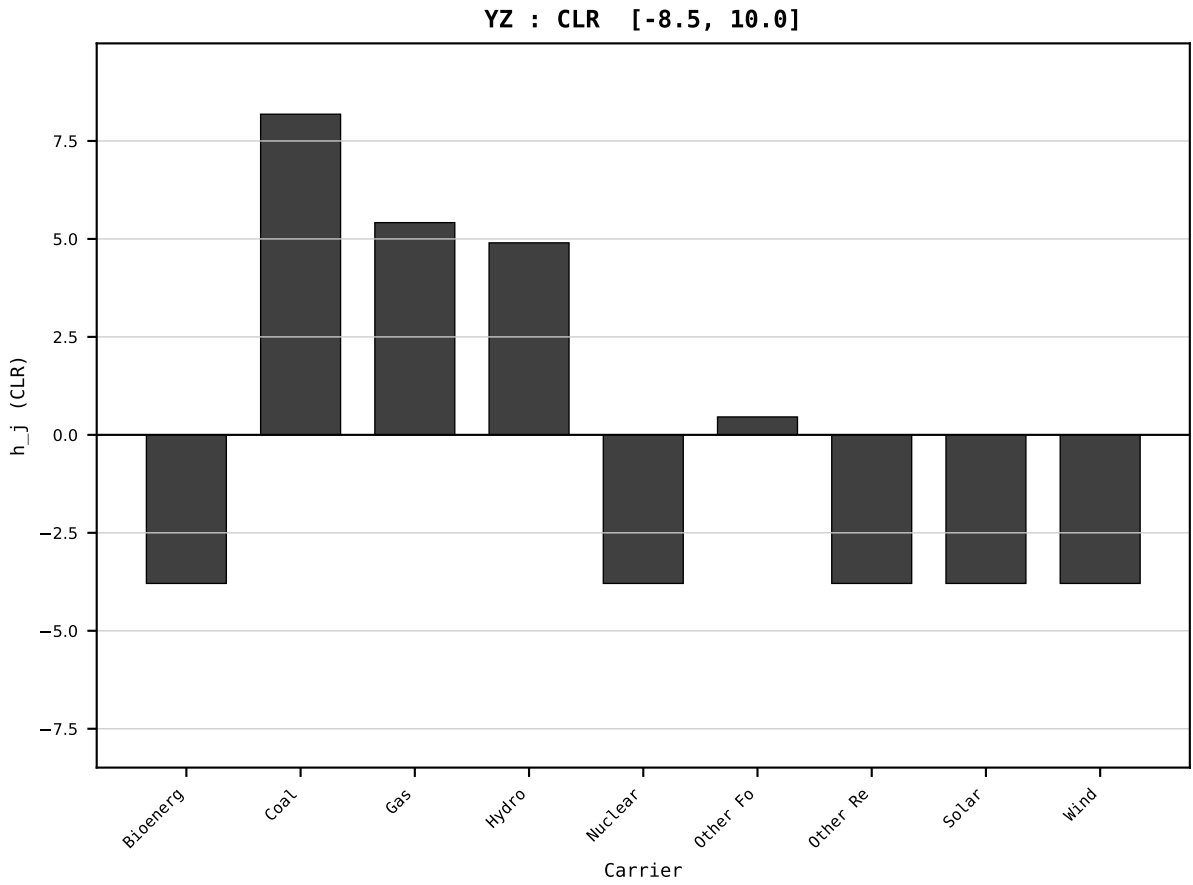
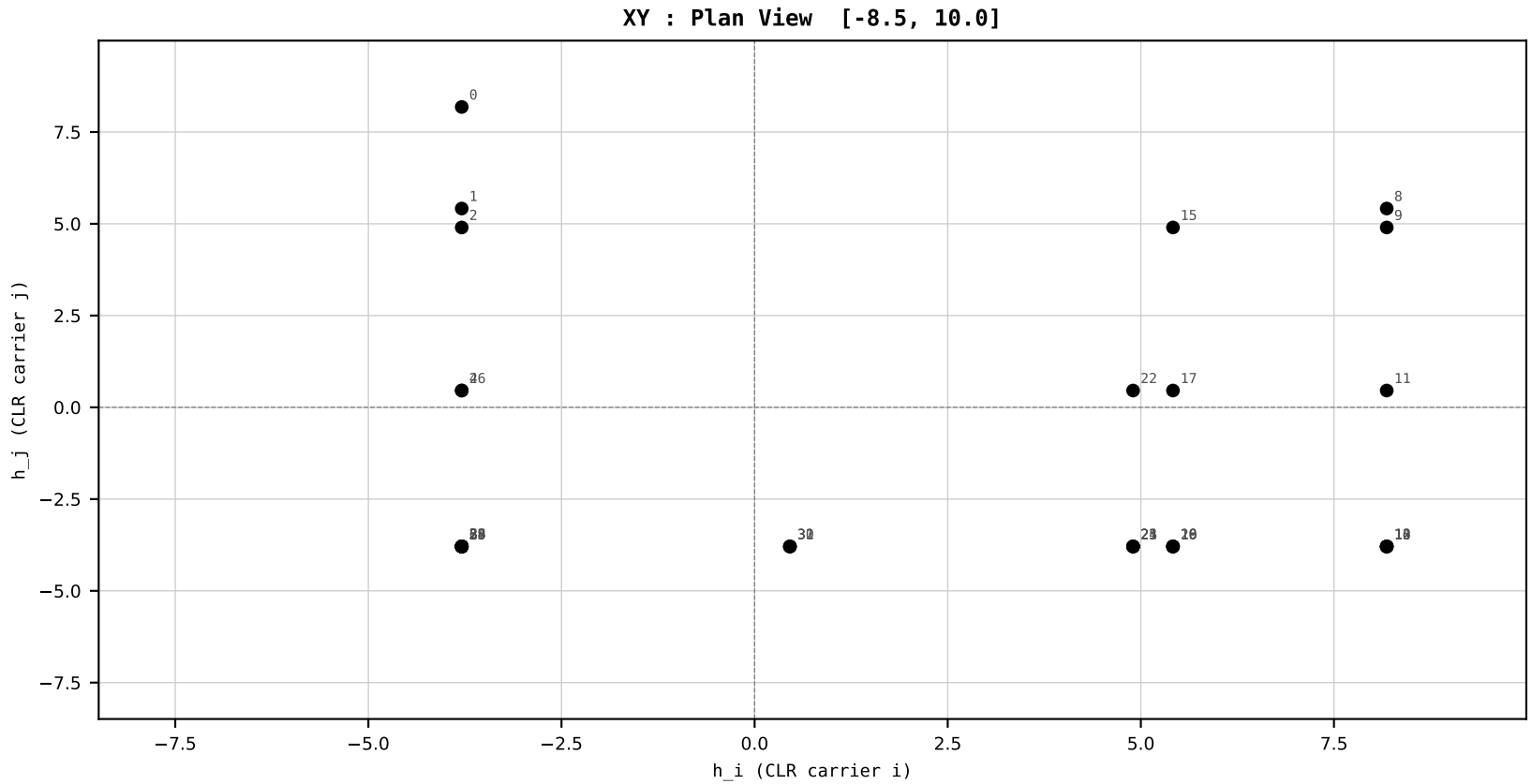
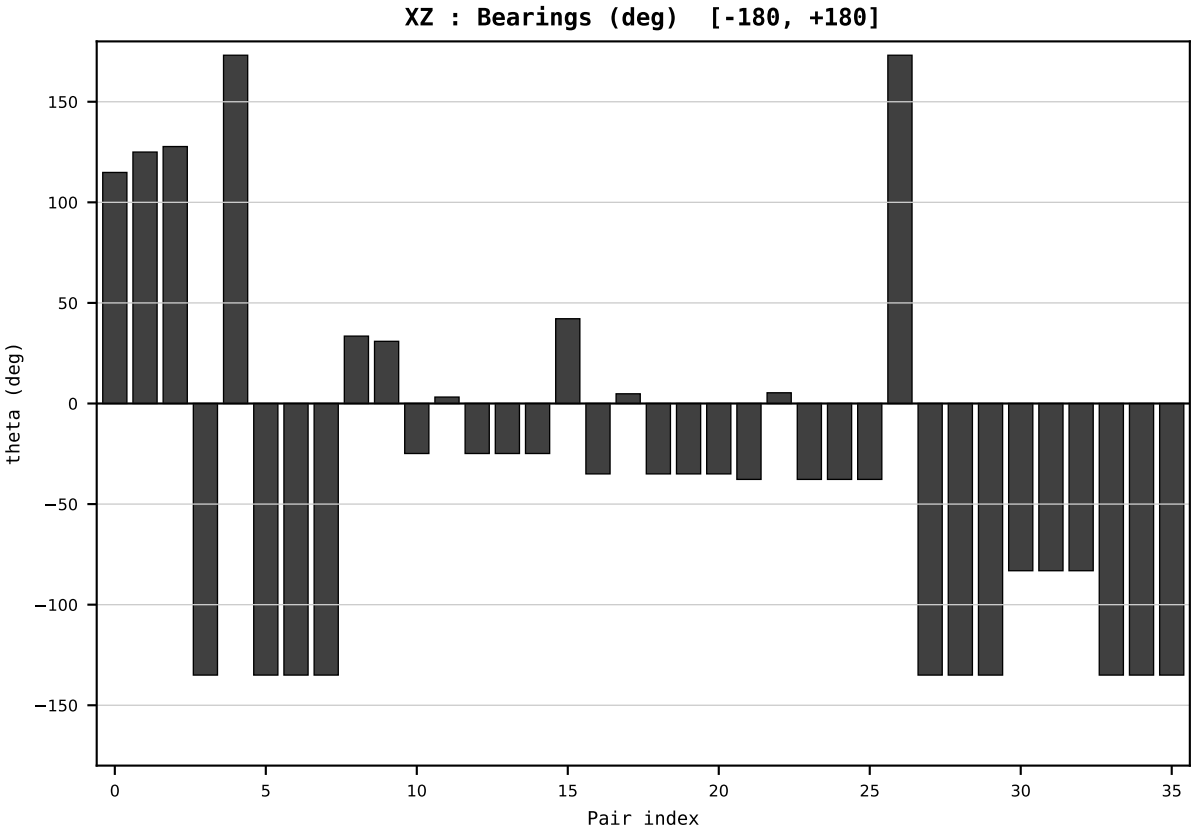
Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2001
t : 1 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.831929
Ring = Hs-5
E_metric= 192.3868
kappa_HS= 158670.00
omega = 0.8070 deg
d_A = 0.197590
Helm = Hydro
Helm d = -0.128582
DR = 11.975 HLR
DR ratio= 158670.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=2 | Year 2002 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

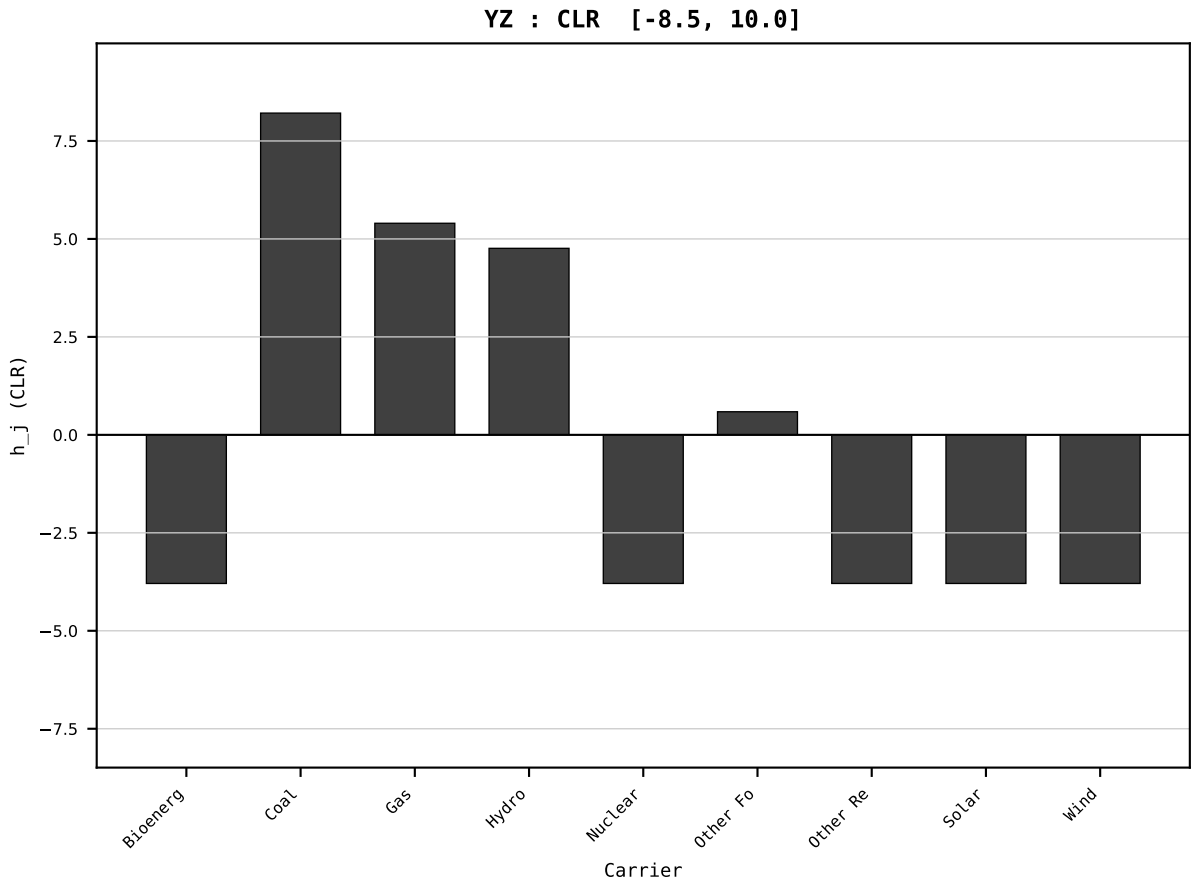
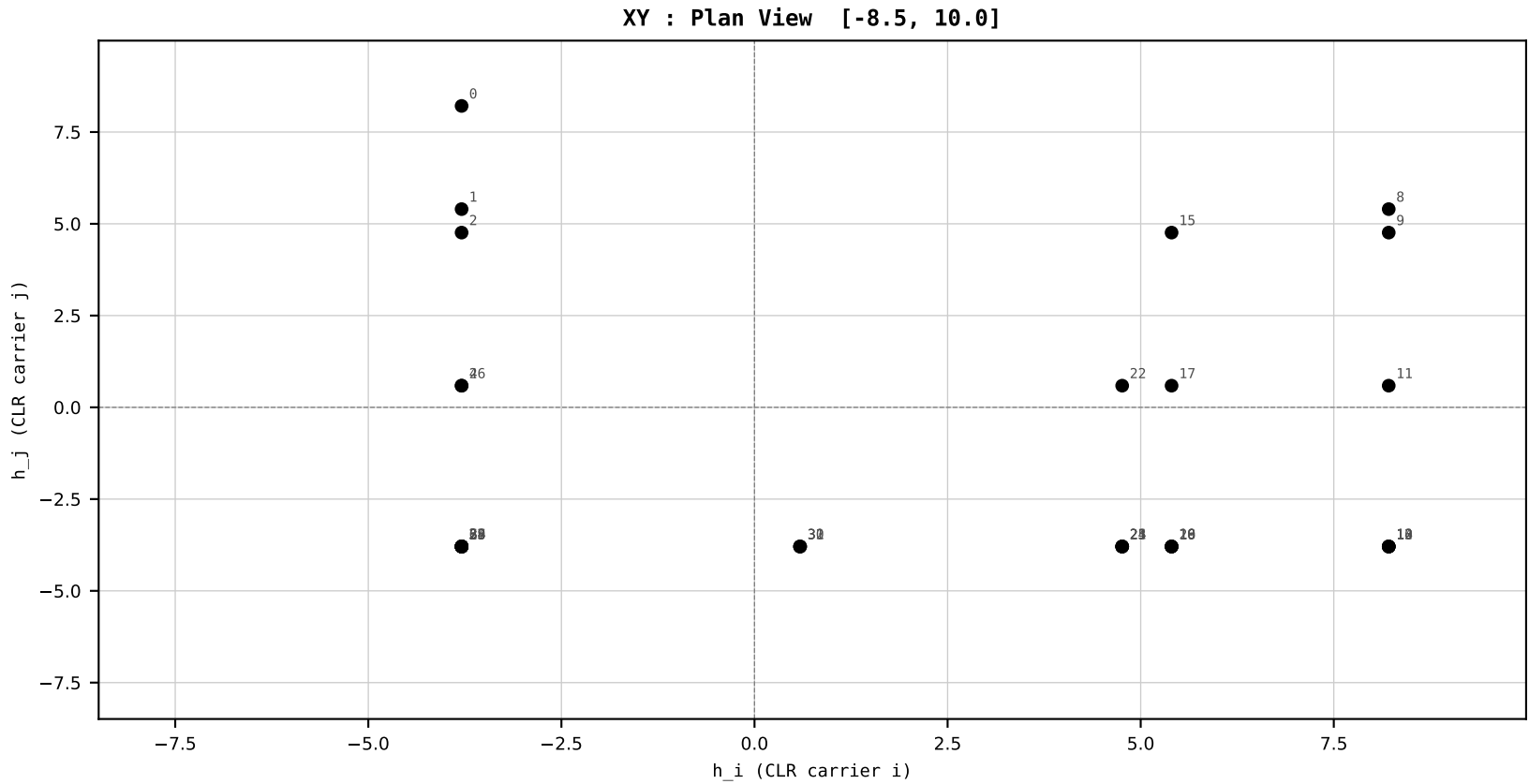
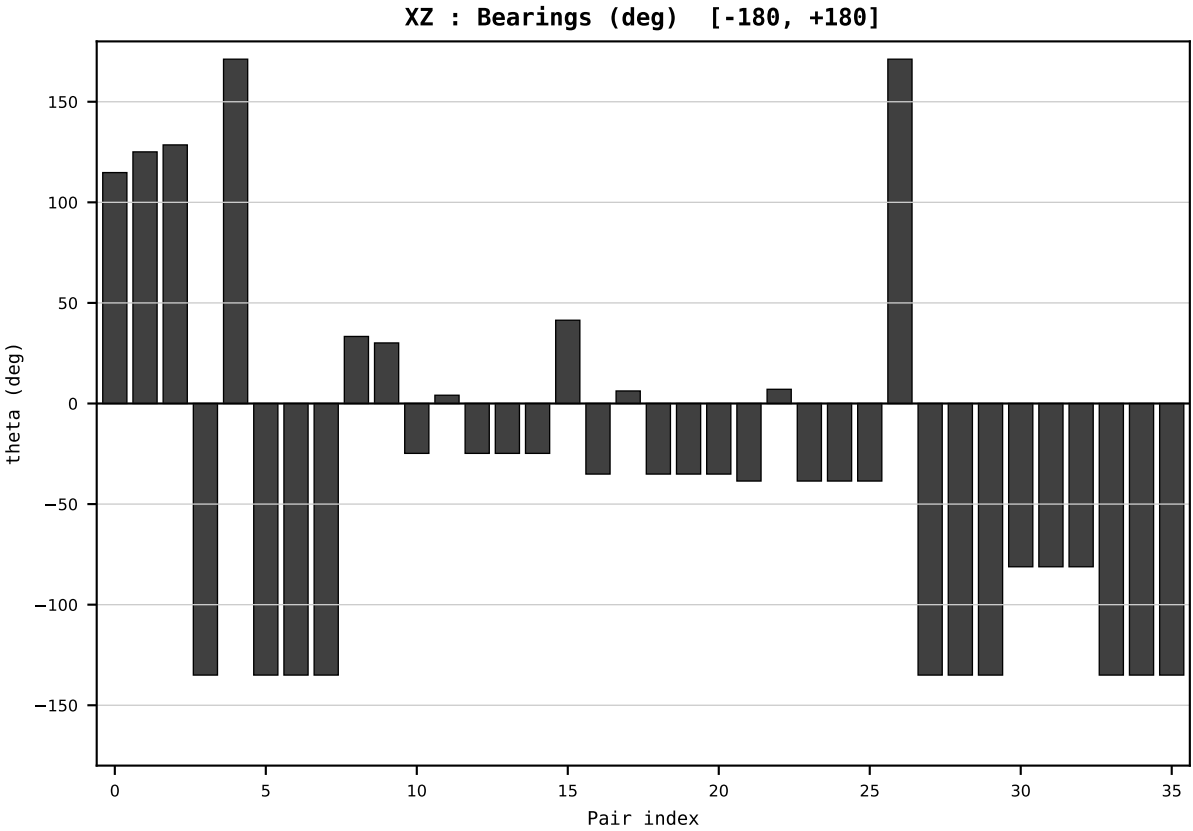
Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2002
t : 2 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.842056
Ring = Hs-5
E_metric= 191.4975
kappa_HS= 163320.00
omega = 0.7962 deg
d_A = 0.195173
Helm = Hydro
Helm d = -0.139550
DR = 12.003 HLR
DR ratio= 163320.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=3 | Year 2003 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

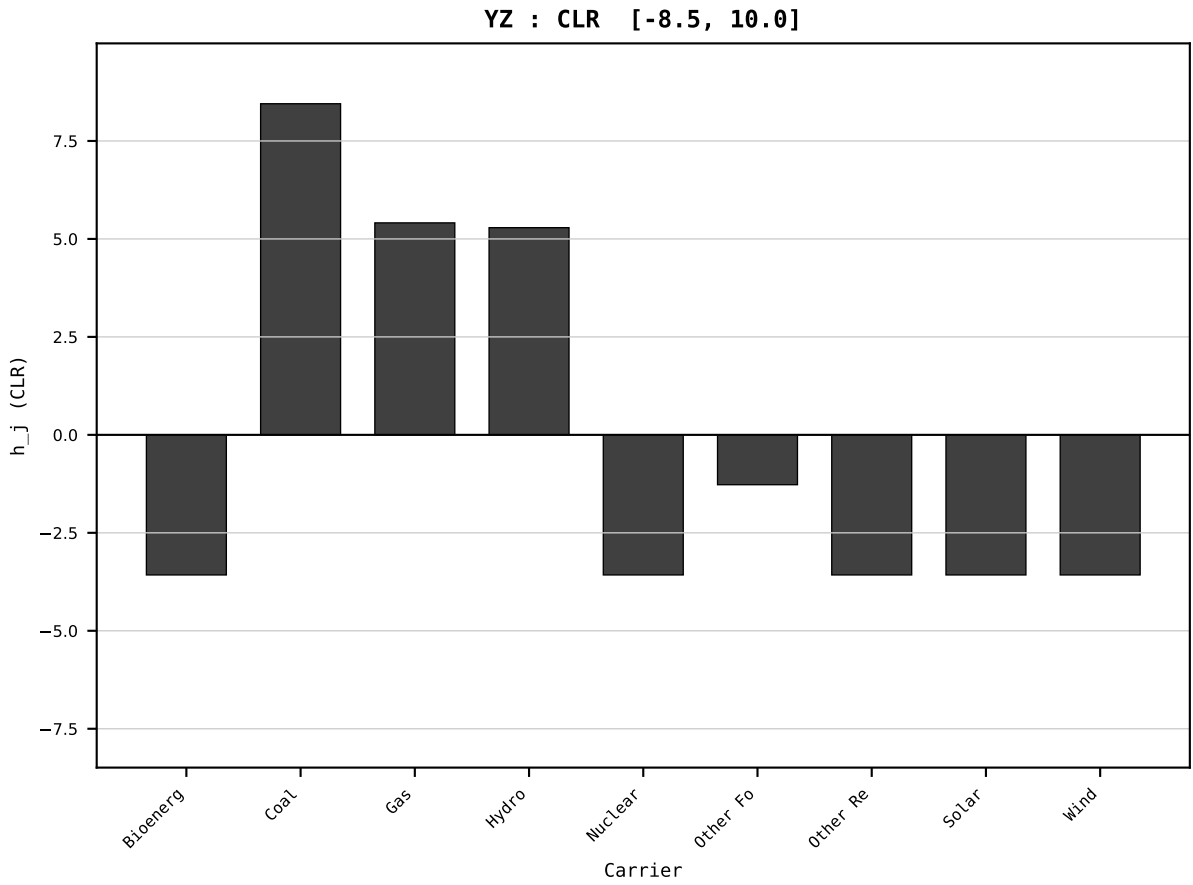
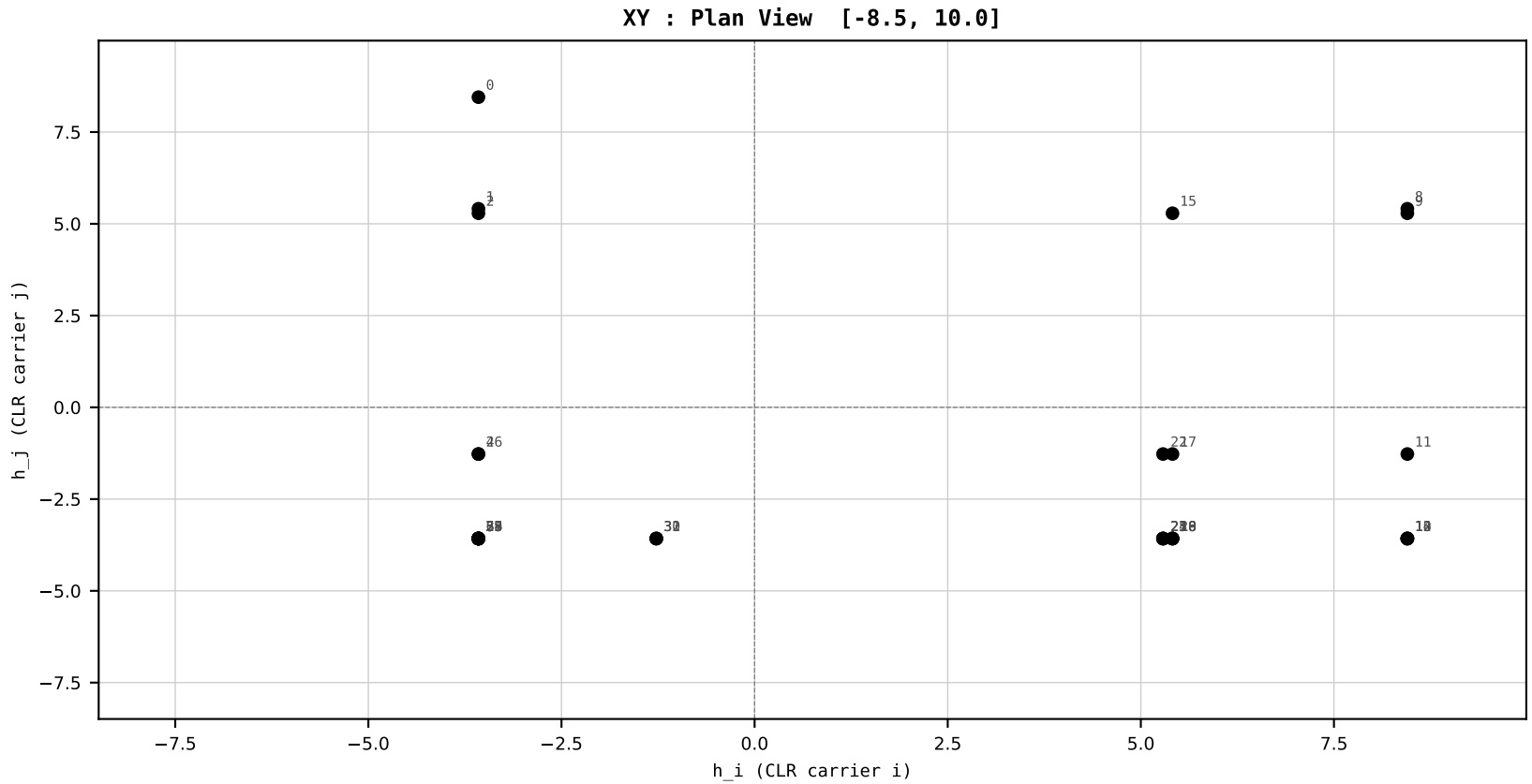
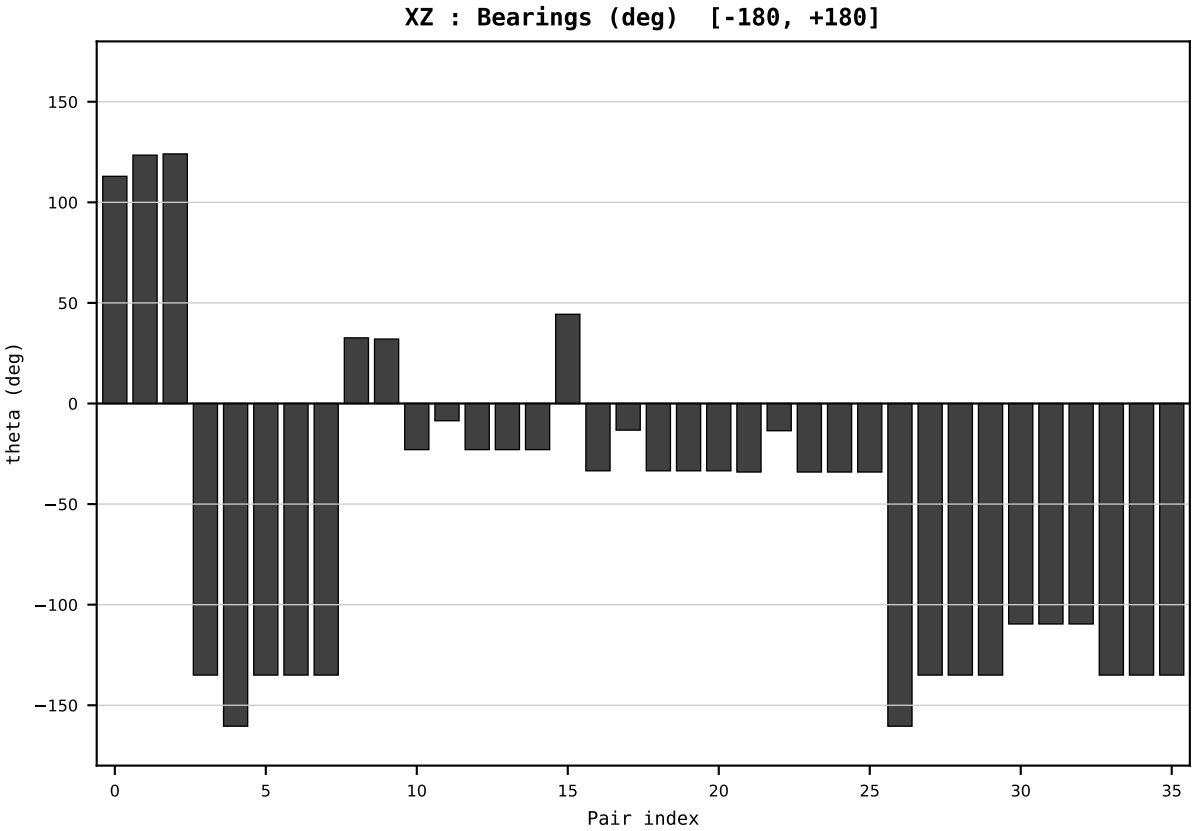
Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2003
t : 3 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.843732
Ring = Hs-5
E_metric= 194.1424
kappa_HS= 166860.00
omega = 8.2899 deg
d_A = 2.009584
Helm = Other Fossil
Helm d = -1.862126
DR = 12.025 HLR
DR ratio= 166860.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=4 | Year 2004 | D=9 N=26 pairs=36

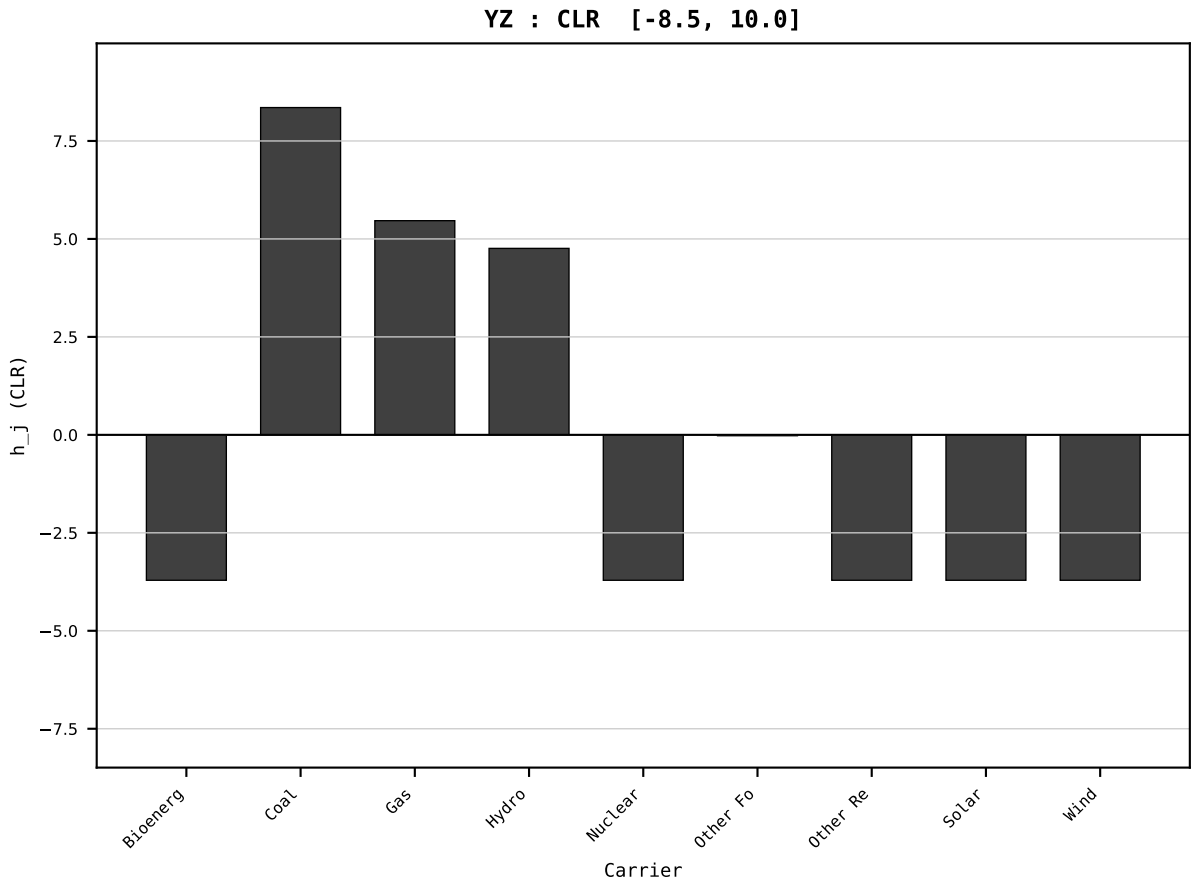
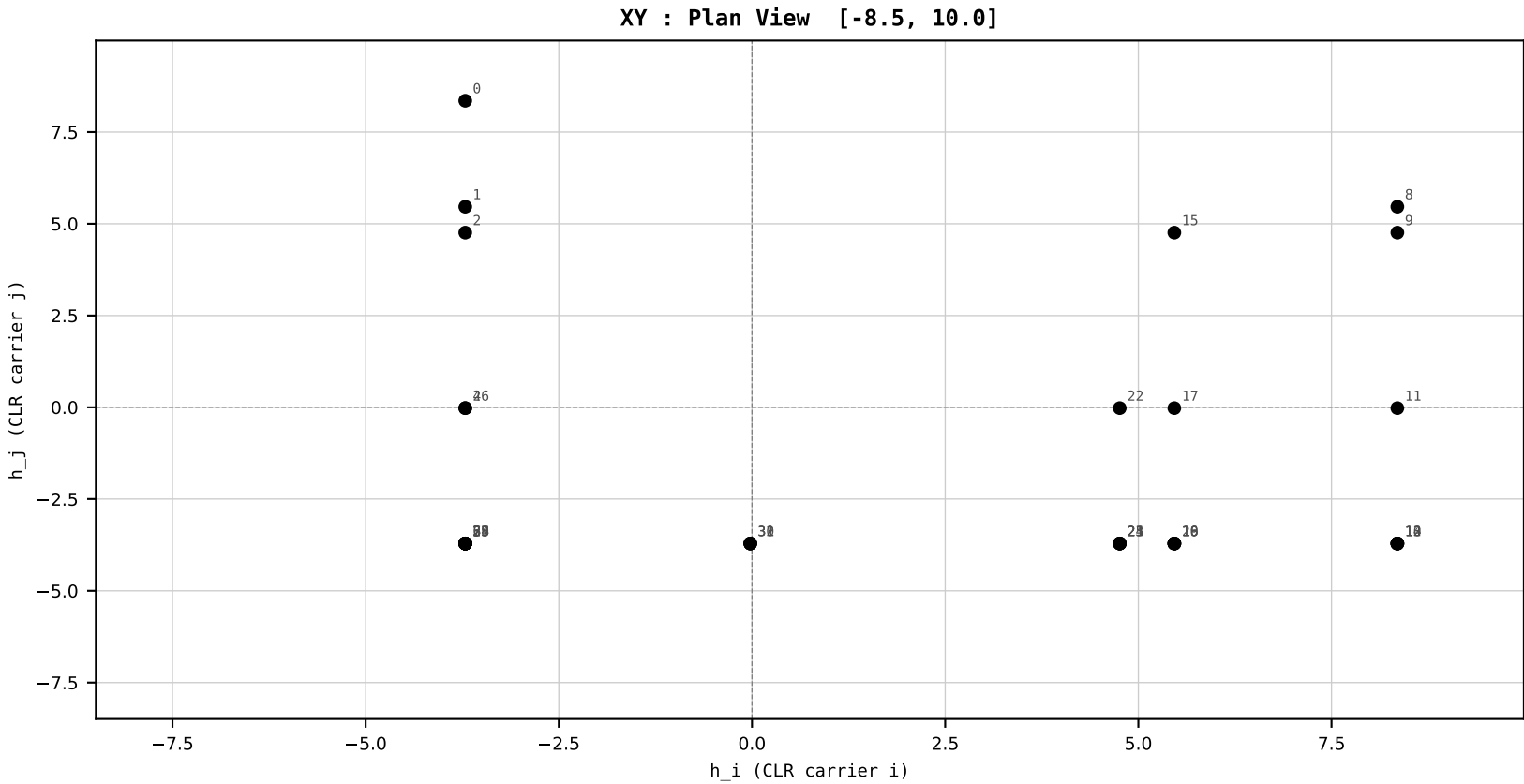
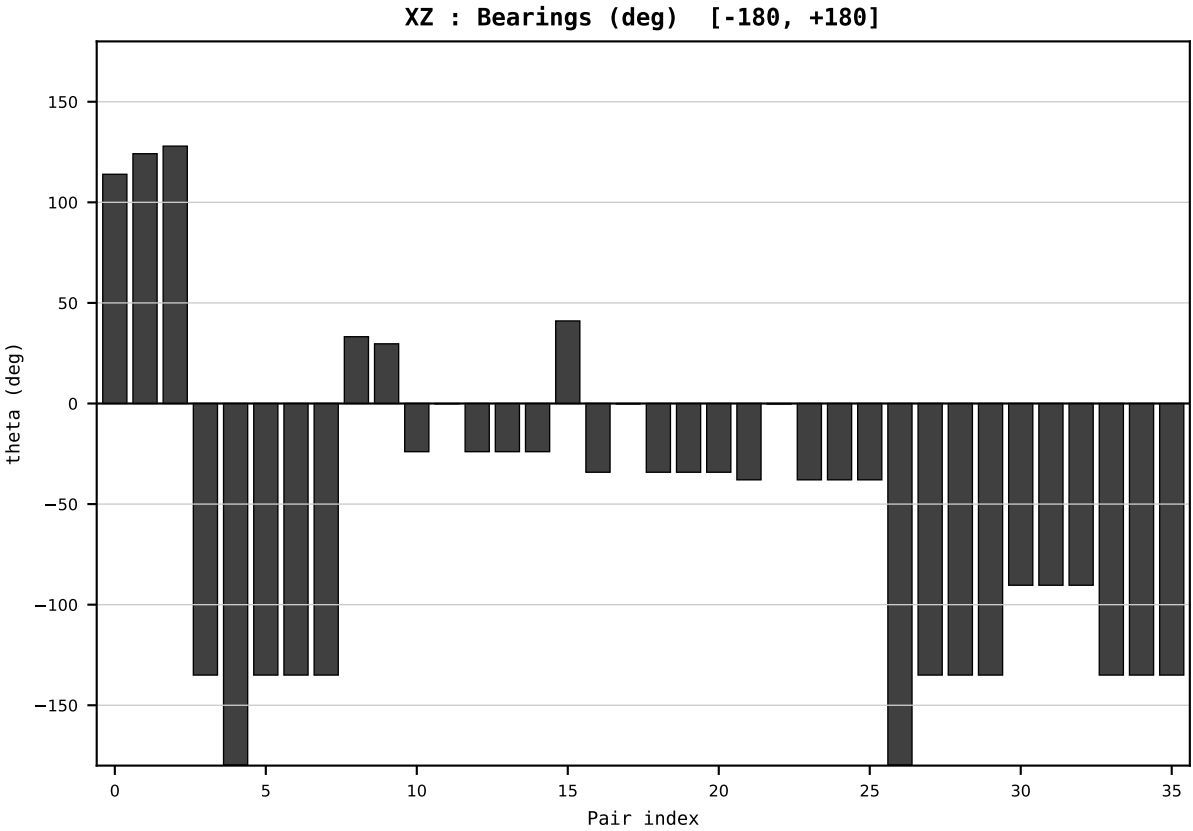
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2004
t : 4 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.853390
Ring = Hs-5
E_metric= 191.1428
kappa_HS= 173390.00
omega = 5.7457 deg
d_A = 1.395449
Helm = Other Fossil
Helm d = +1.250220
DR = 12.063 HLR
DR ratio= 173390.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=5 | Year 2005 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

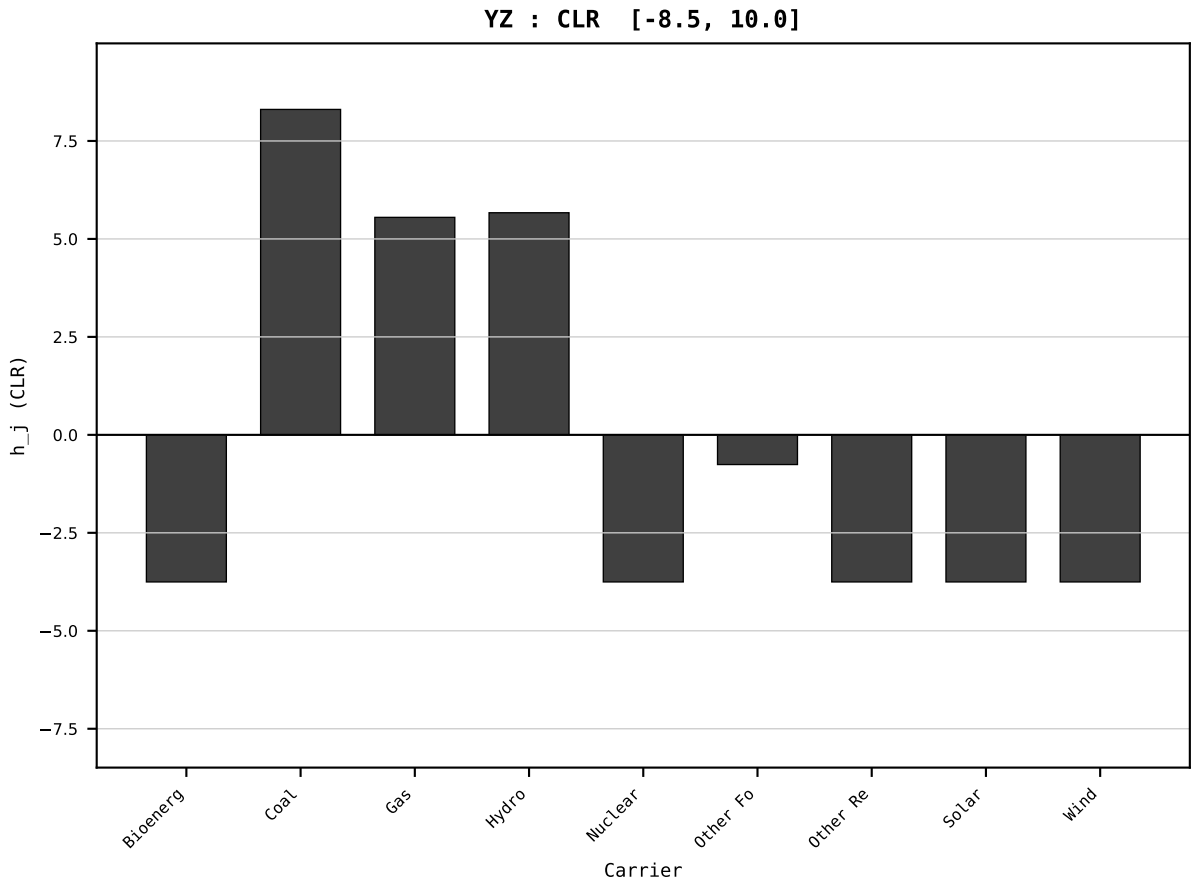
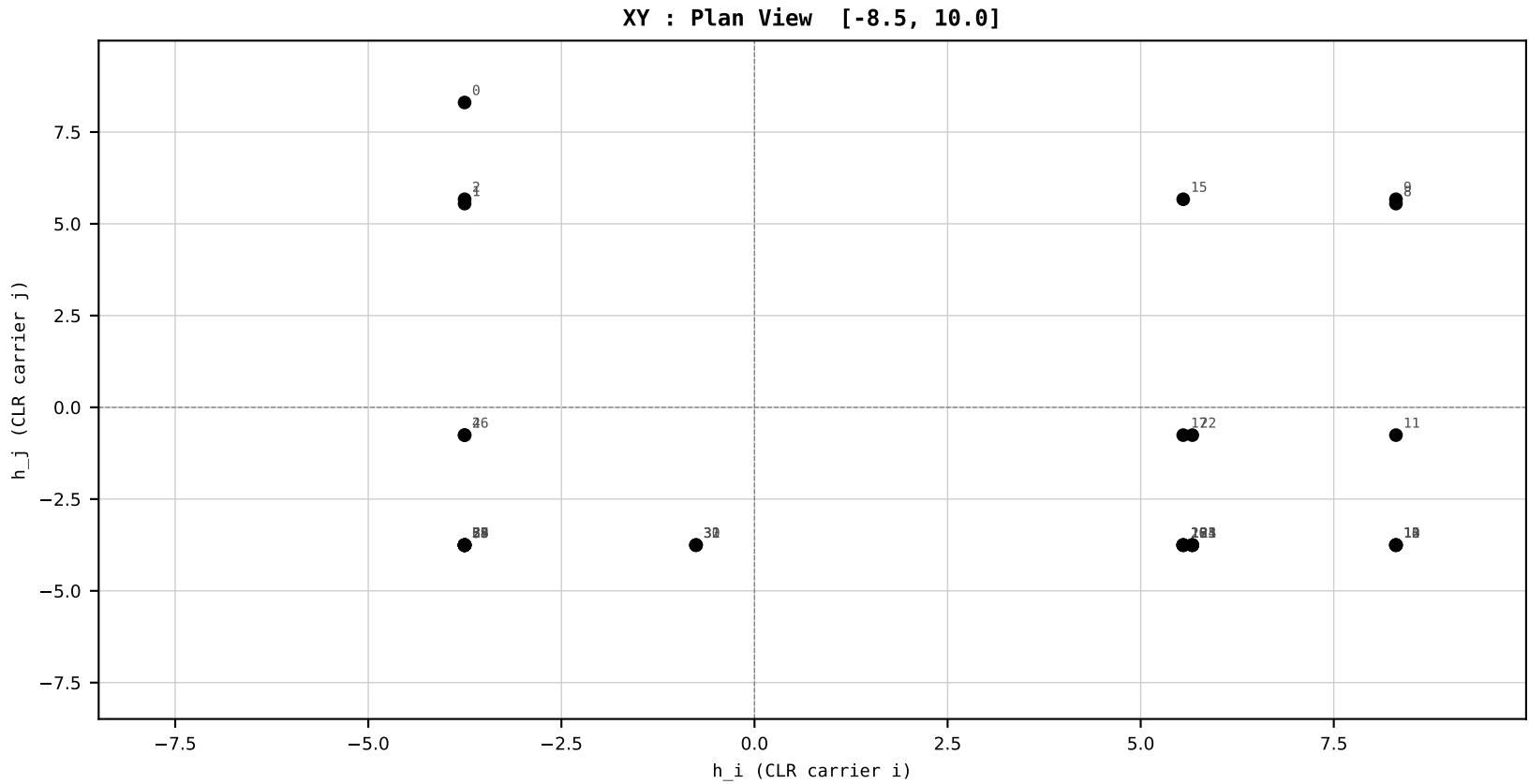
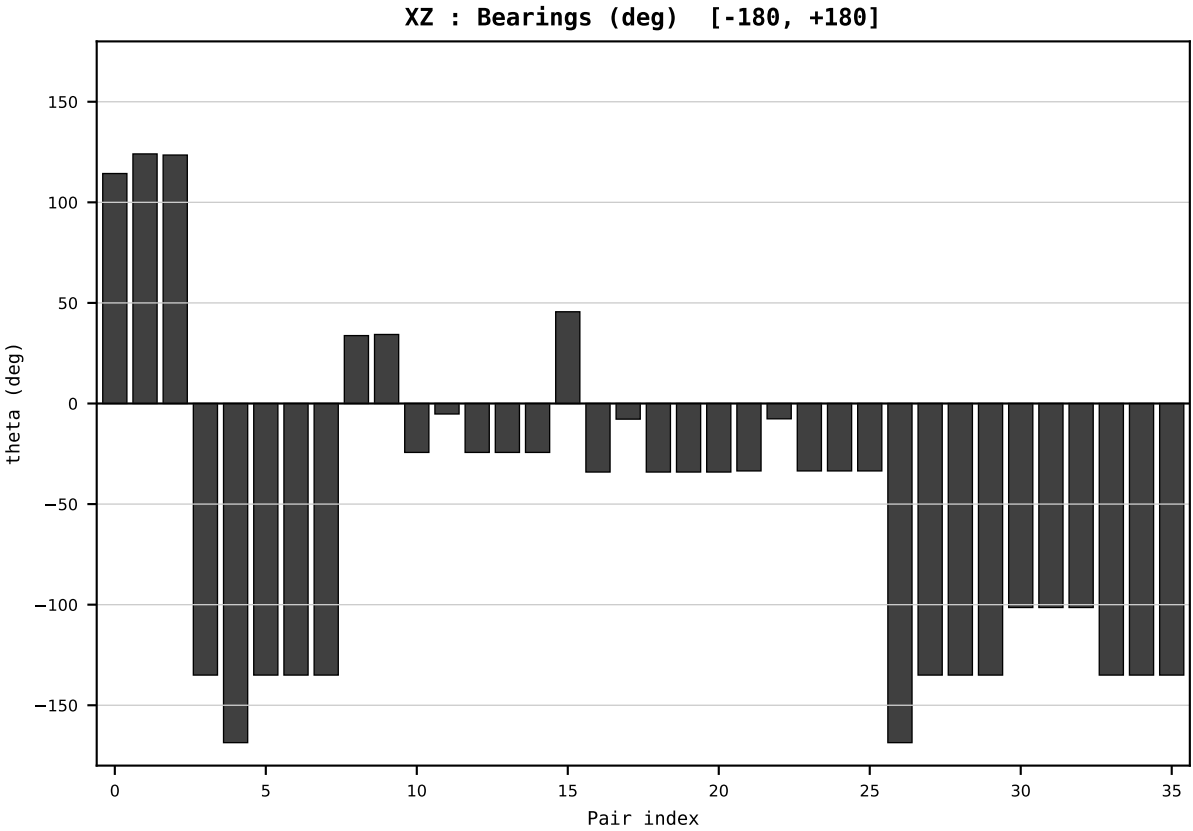
Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2005
t : 5 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.795717
Ring = Hs-5
E_metric= 202.9186
kappa_HS= 172520.00
omega = 4.4914 deg
d_A = 1.177094
Helm = Hydro
Helm d = +0.909068
DR = 12.058 HLR
DR ratio= 172520.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=6 | Year 2006 | D=9 N=26 pairs=36

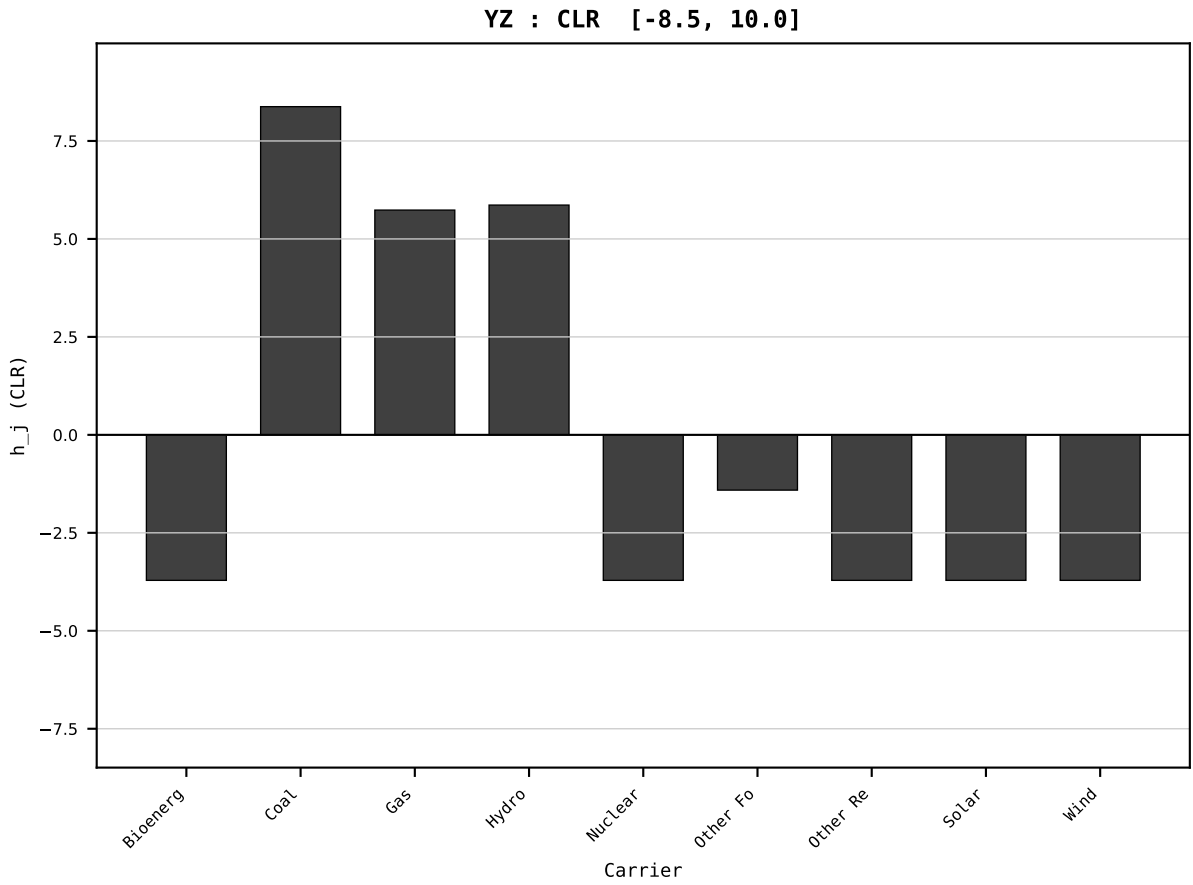
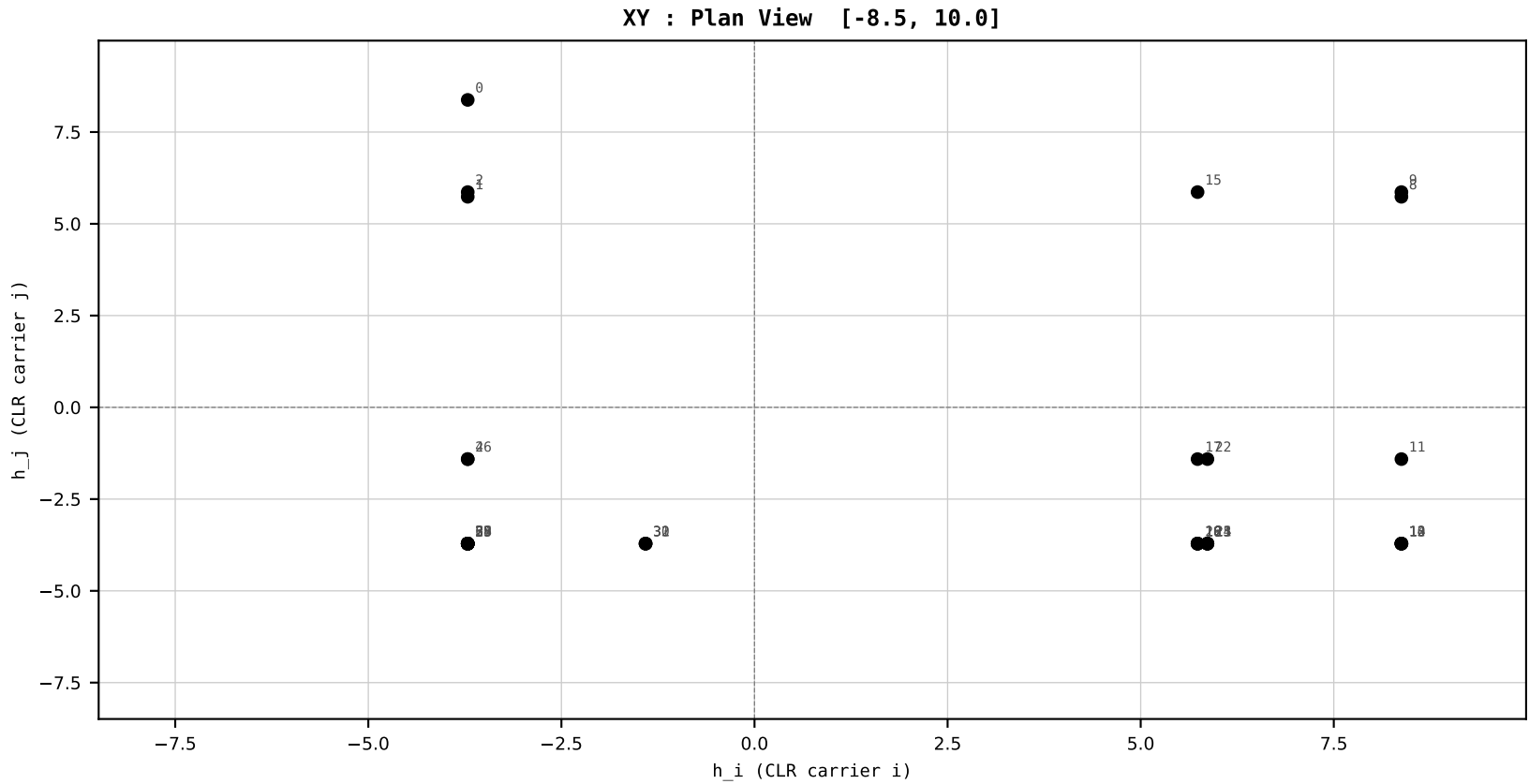
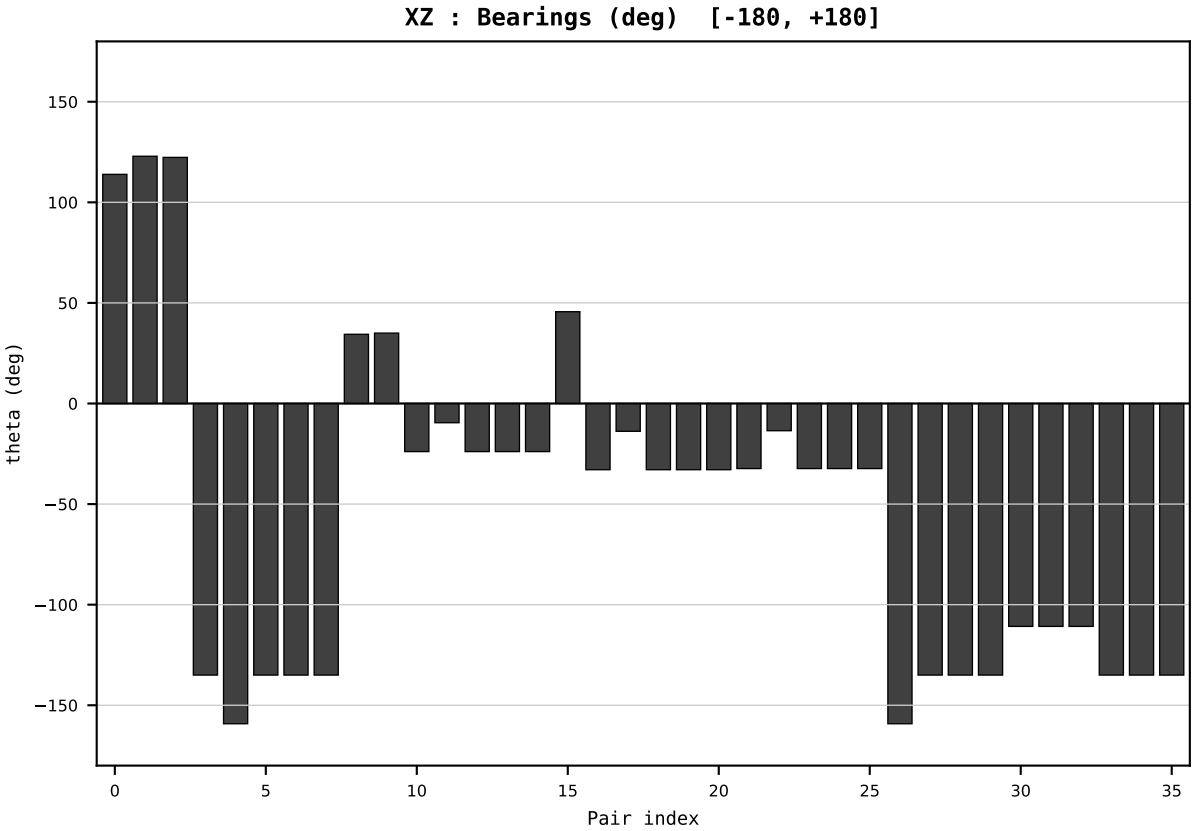
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2006
t : 6 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.780110
Ring = Hs-5
E_metric= 208.3655
kappa_HS= 177840.00
omega = 2.7575 deg
d_A = 0.715719
Helm = Other Fossil
Helm d = -0.652970
DR = 12.089 HLR
DR ratio= 177840.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=7 | Year 2007 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

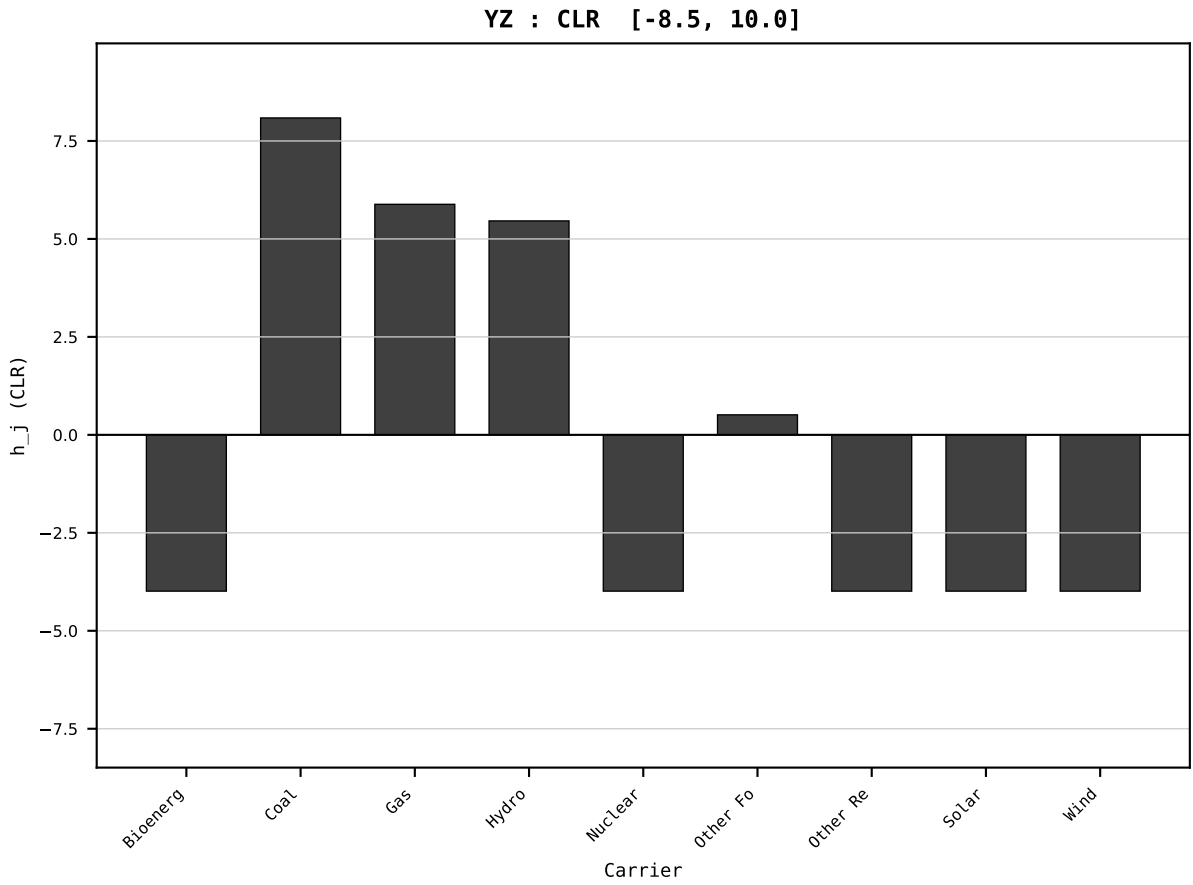
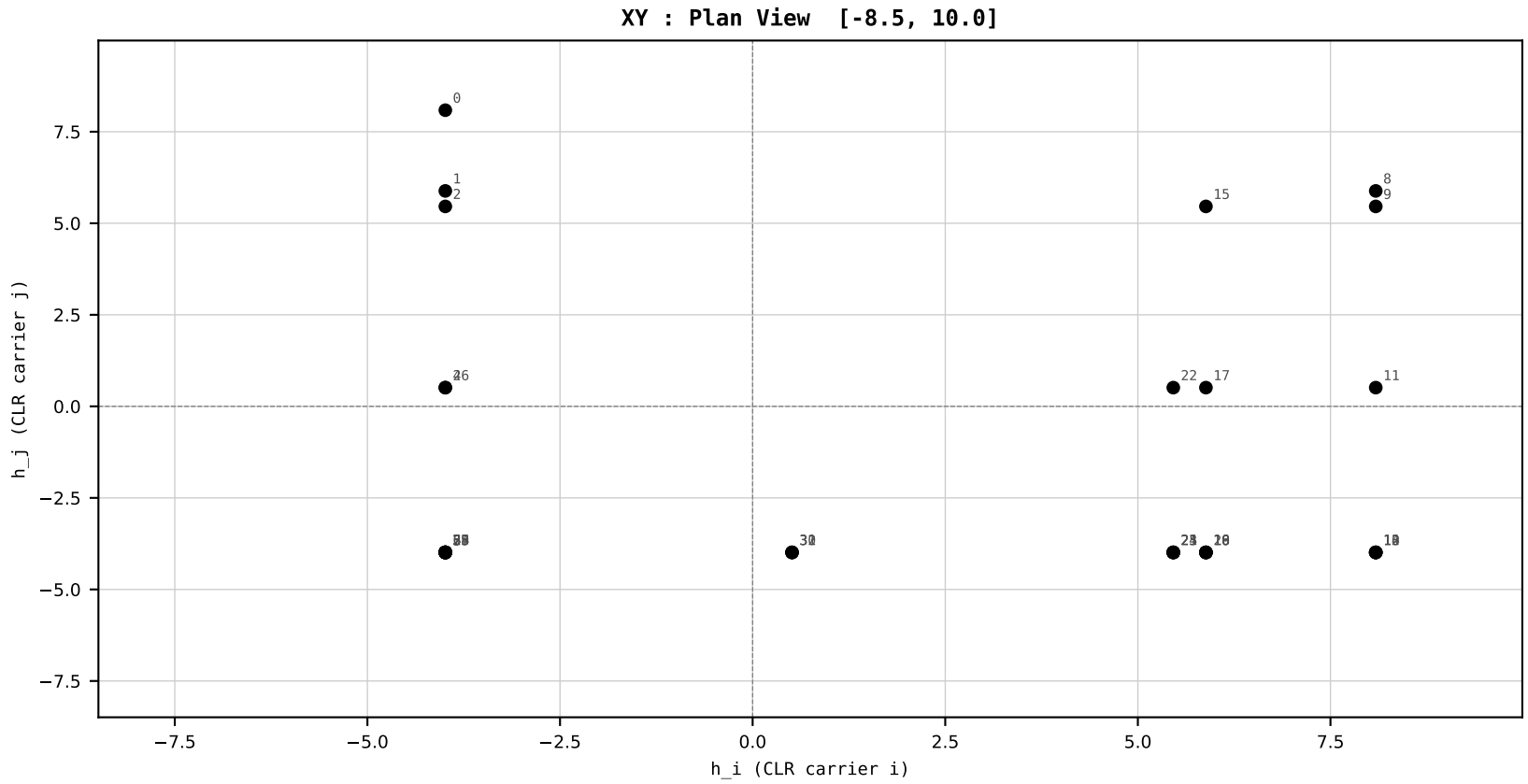
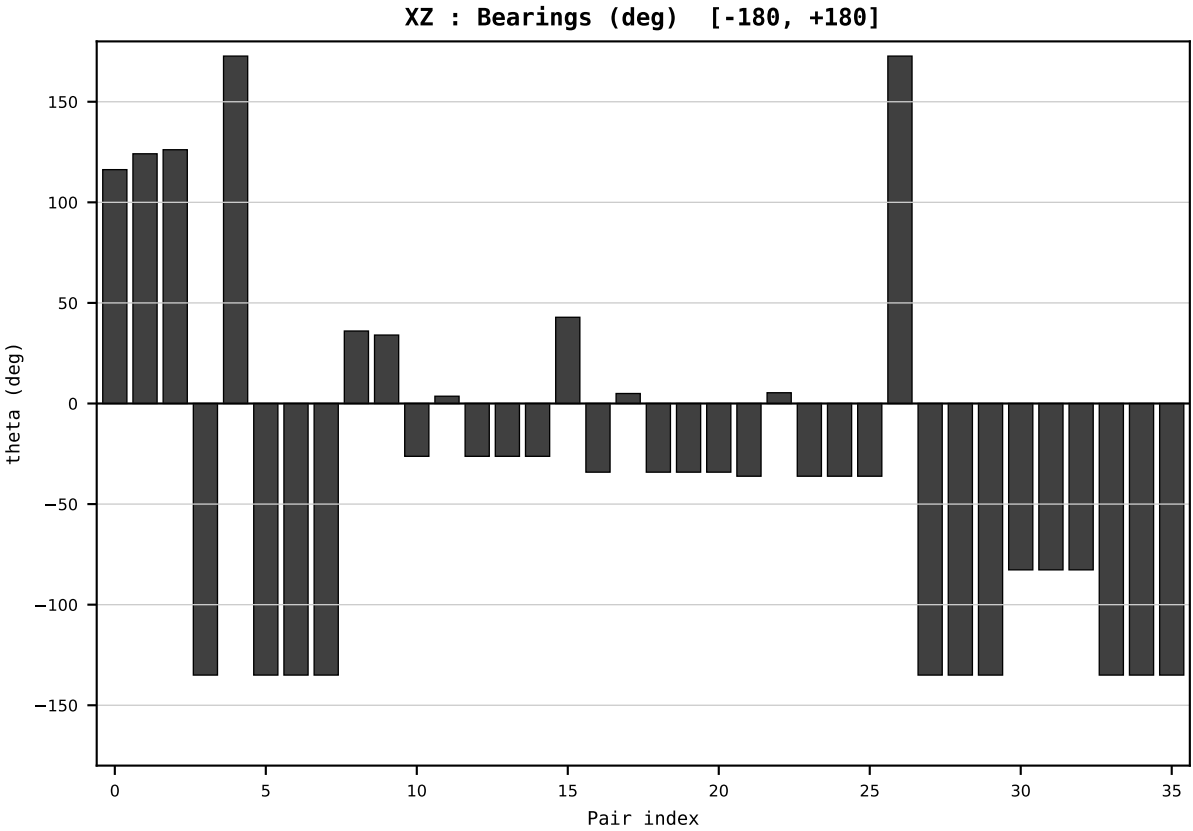
Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2007
t : 7 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.755279
Ring = Hs-5
E_metric= 209.6435
kappa_HS= 175670.00
omega = 8.2619 deg
d_A = 2.083316
Helm = Other Fossil
Helm d = +1.921770
DR = 12.076 HLR
DR ratio= 175670.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=8 | Year 2008 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

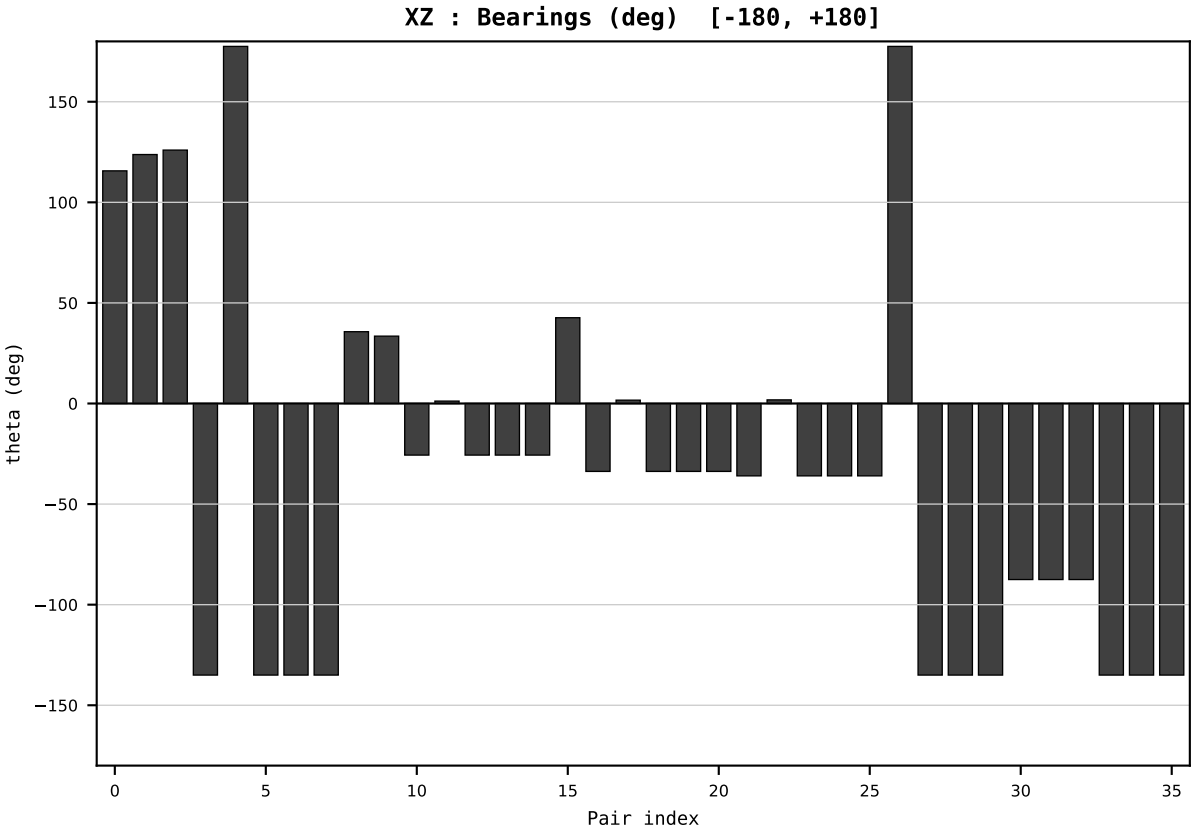
Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2008
t : 8 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.771736
Ring = Hs-5
E_metric= 207.4798
kappa_HS= 179120.00
omega = 1.4930 deg
d_A = 0.383687
Helm = Other Fossil
Helm d = -0.340748
DR = 12.096 HLR
DR ratio= 179120.0x

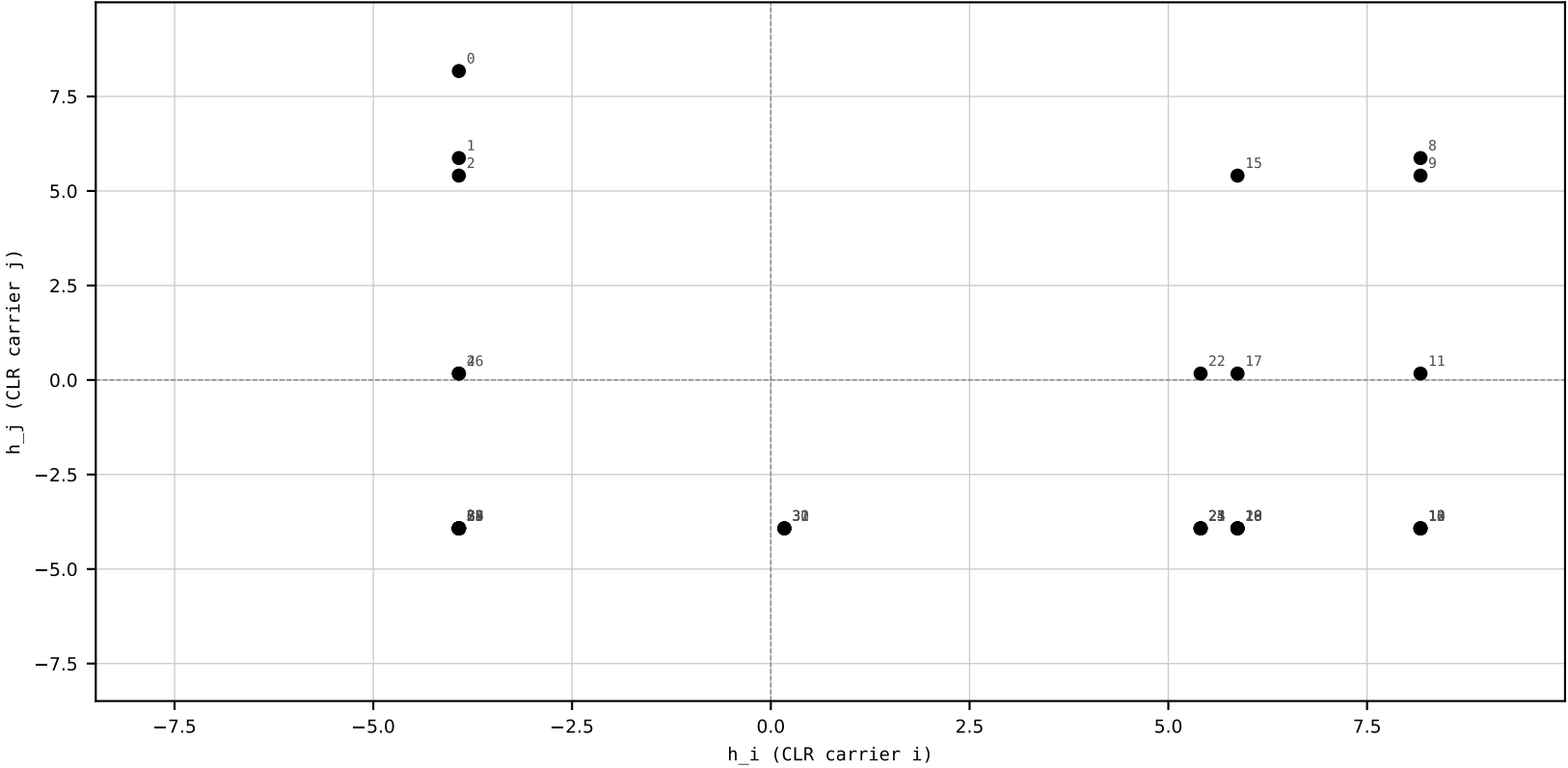
XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:

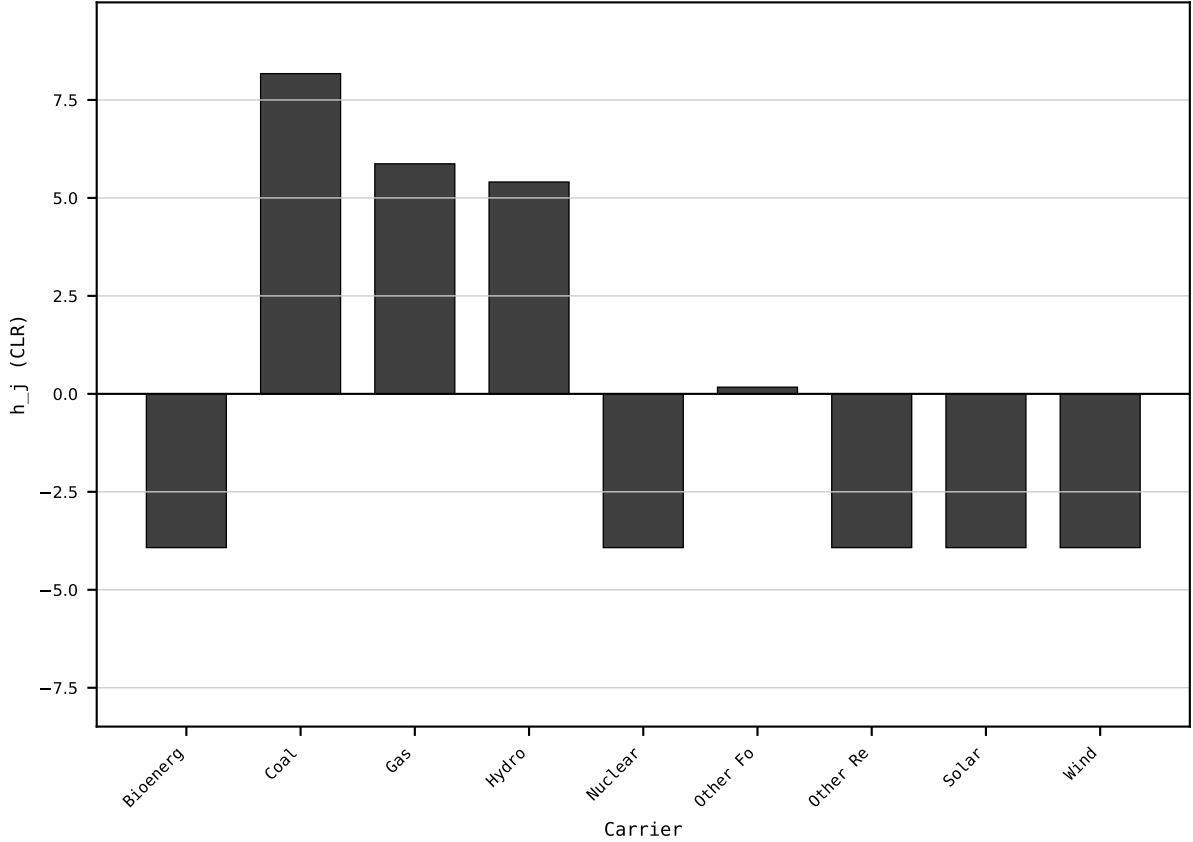
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



XY : Plan View [-8.5, 10.0]



YZ : CLR [-8.5, 10.0]



HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

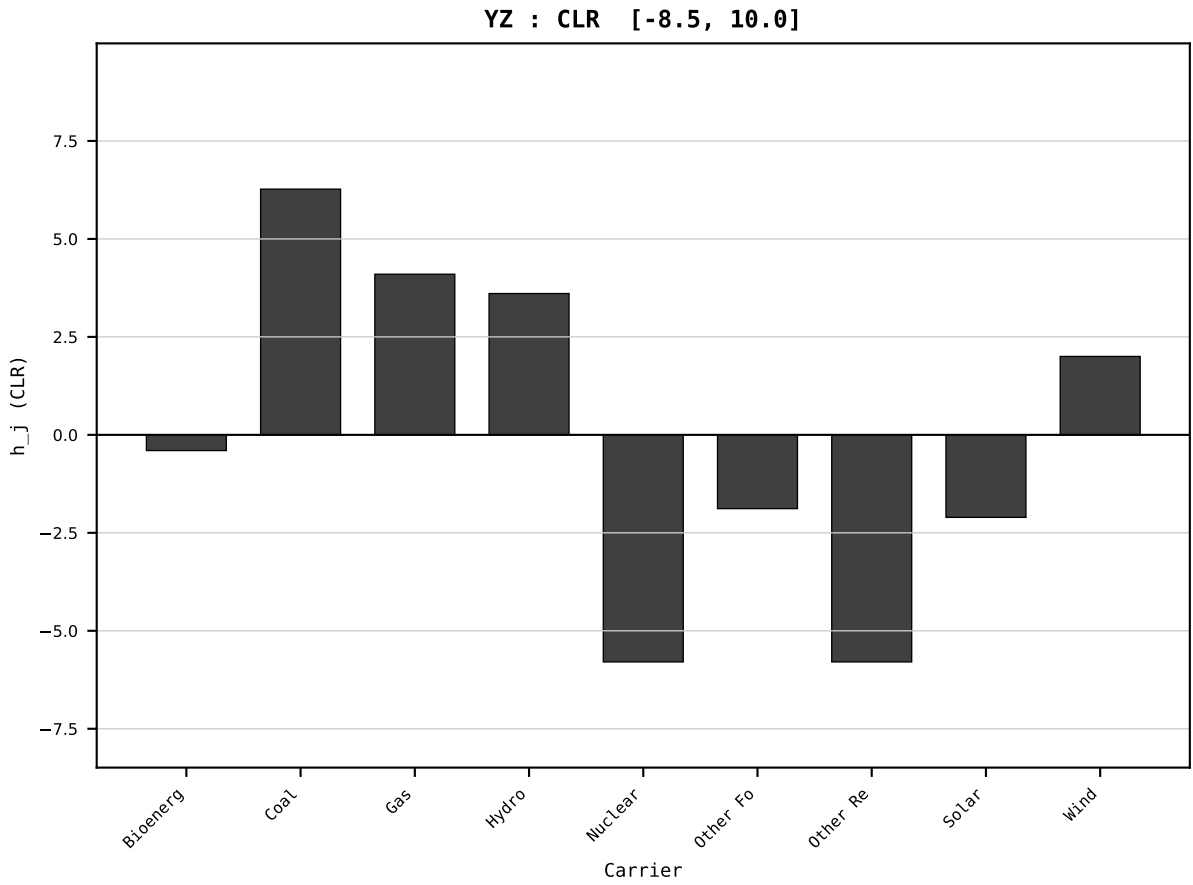
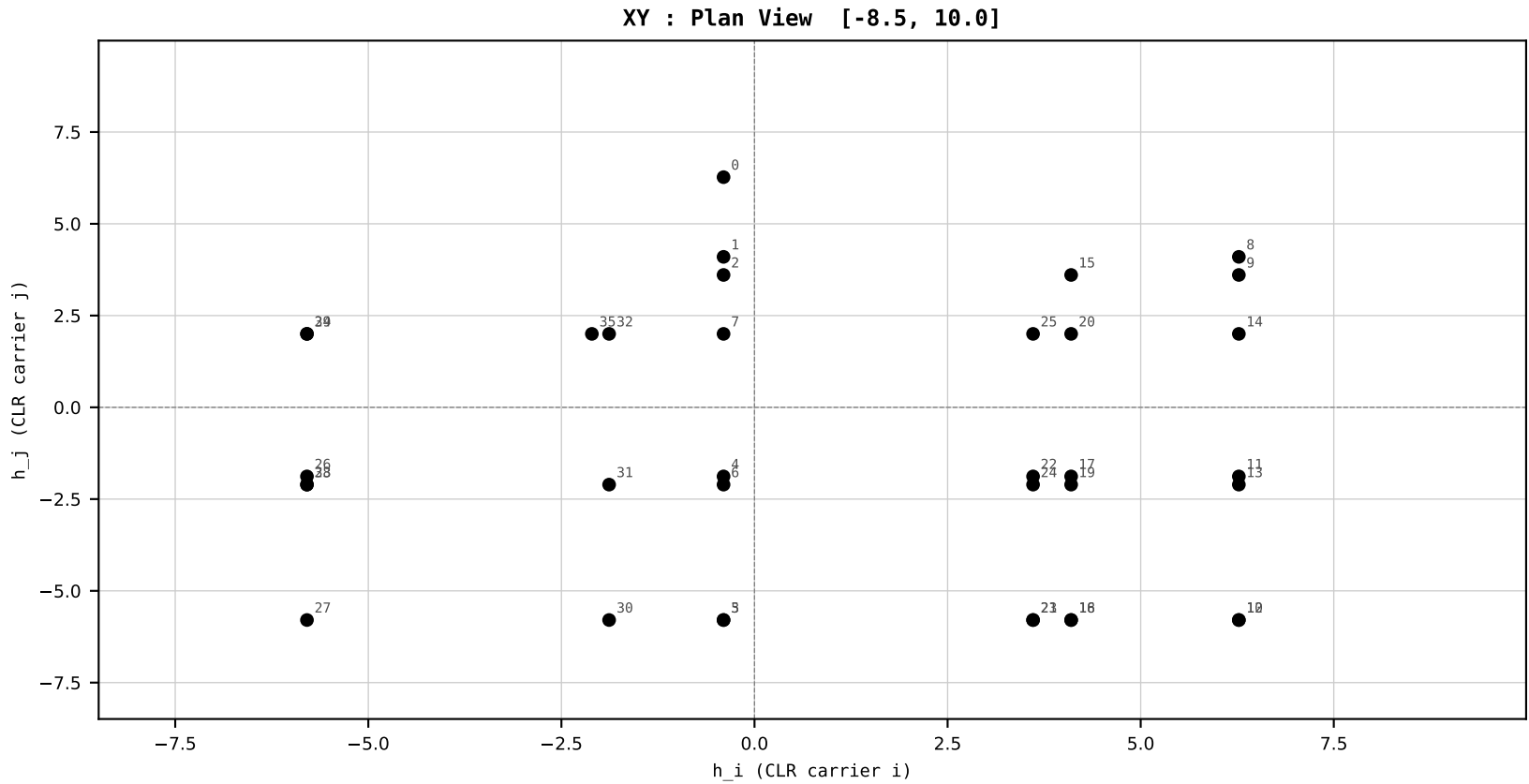
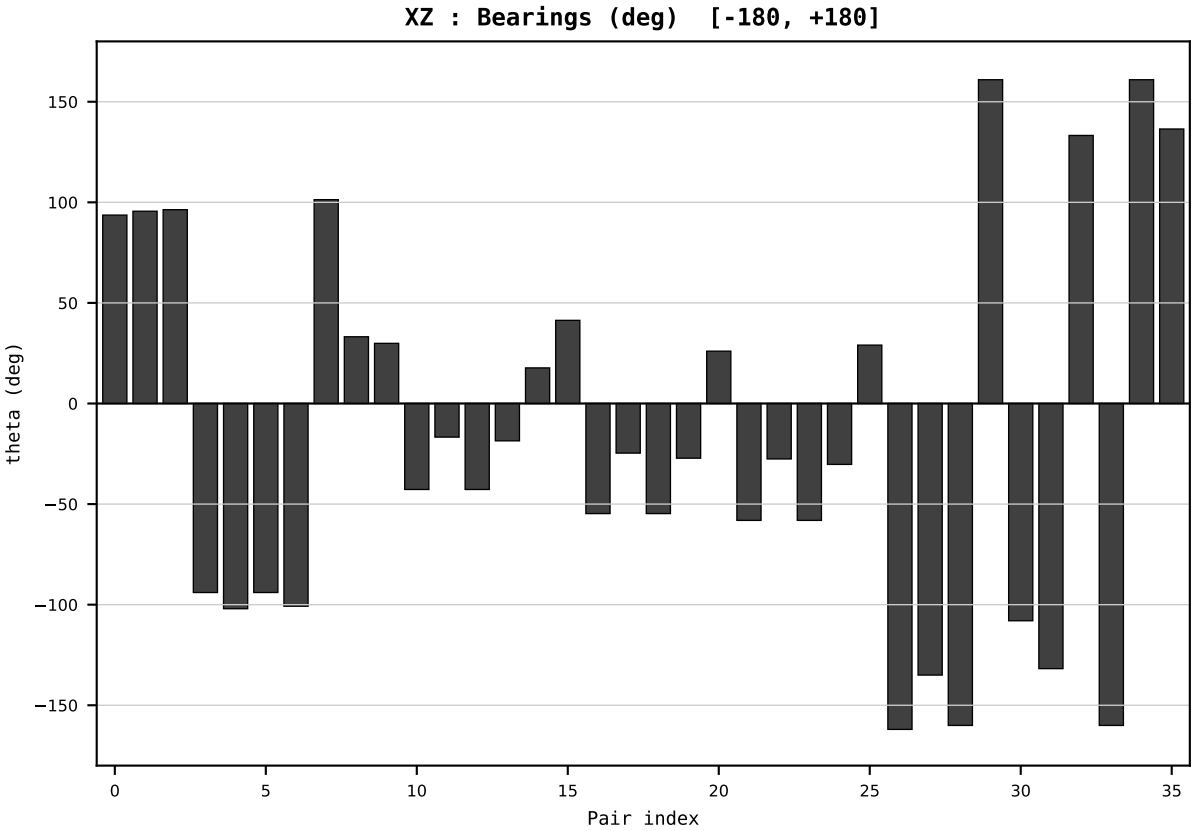
Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2009
t : 9 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.725332
Ring = Hs-4
E_metric= 148.4295
kappa_HS= 173730.00
omega = 36.0173 deg
d_A = 8.486789
Helm = Wind
Helm d = +5.924974
DR = 12.065 HLR
DR ratio= 173730.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=10 | Year 2010 | D=9 N=26 pairs=36

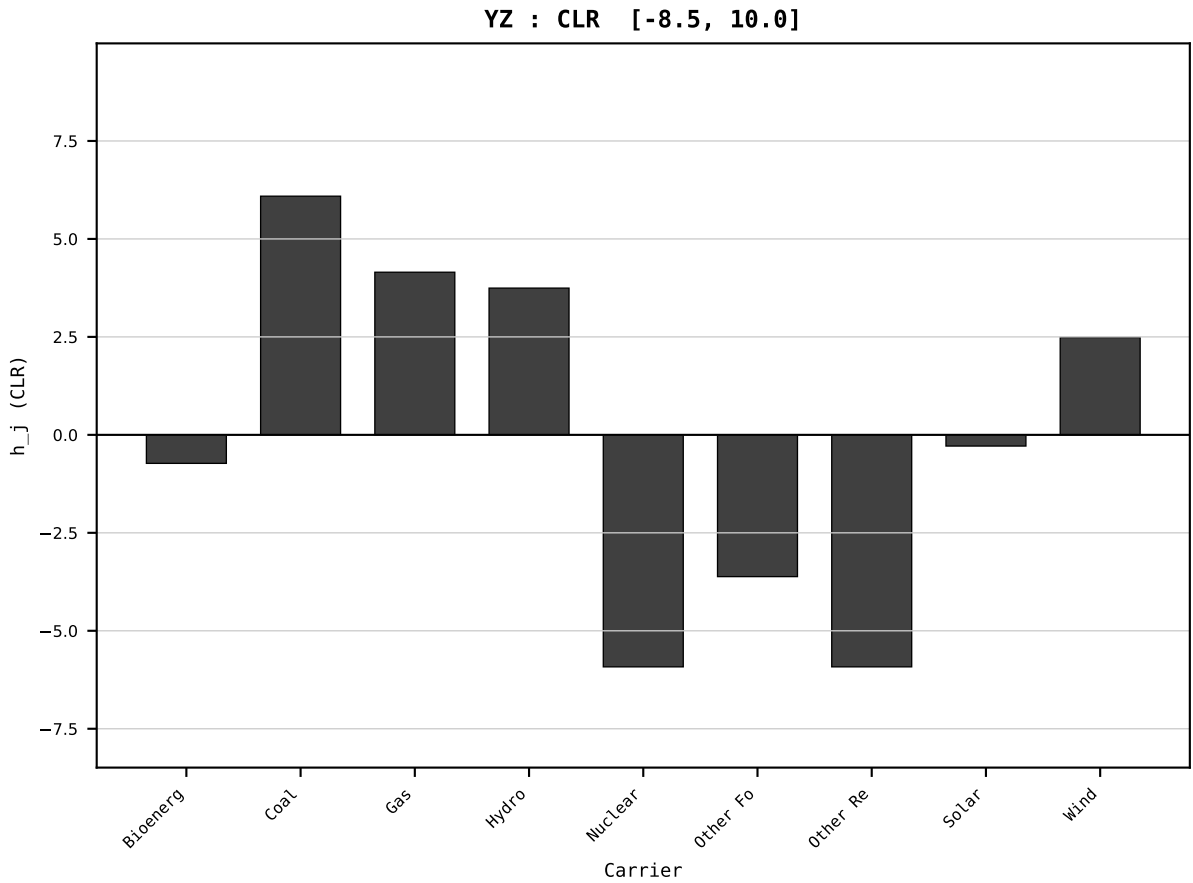
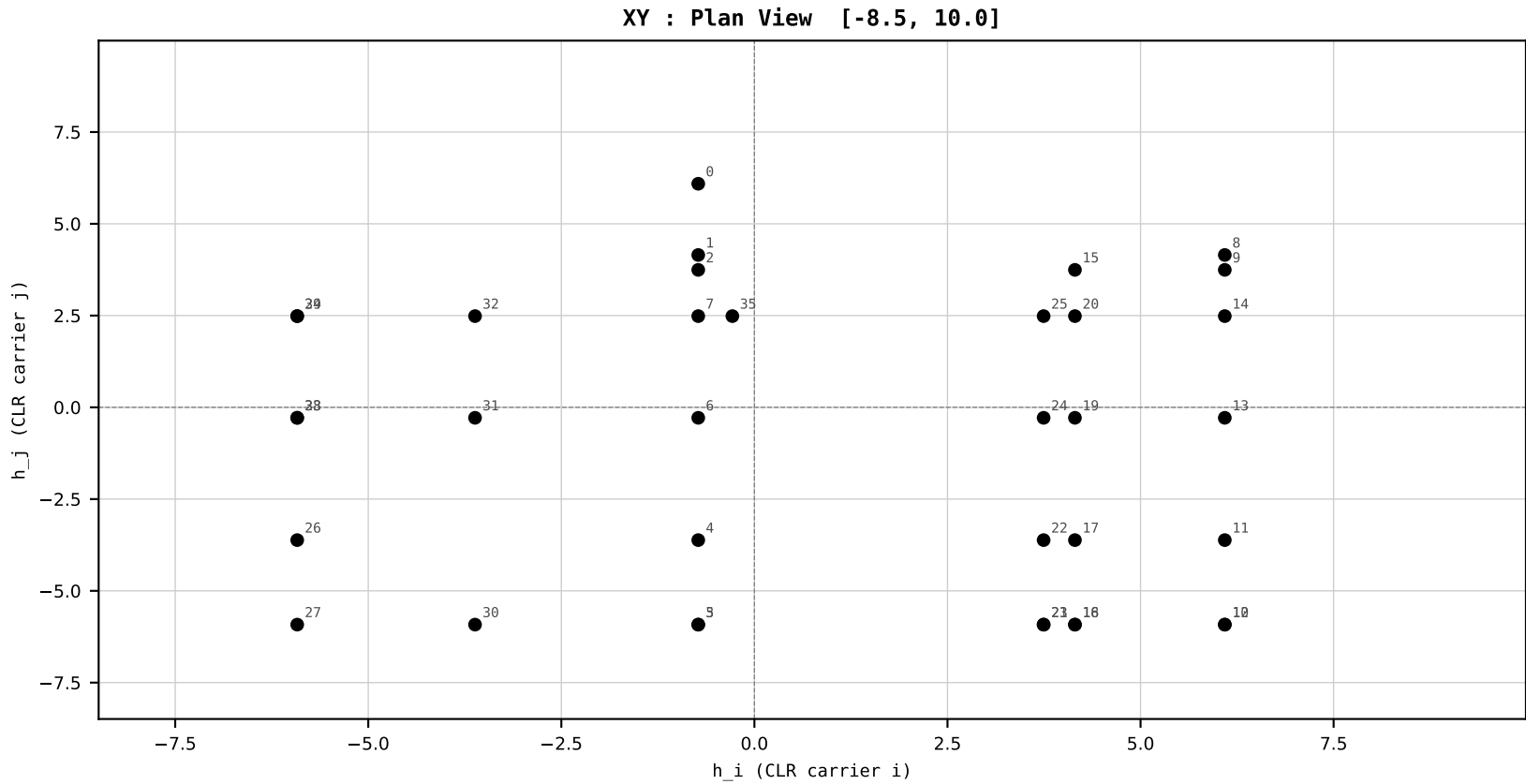
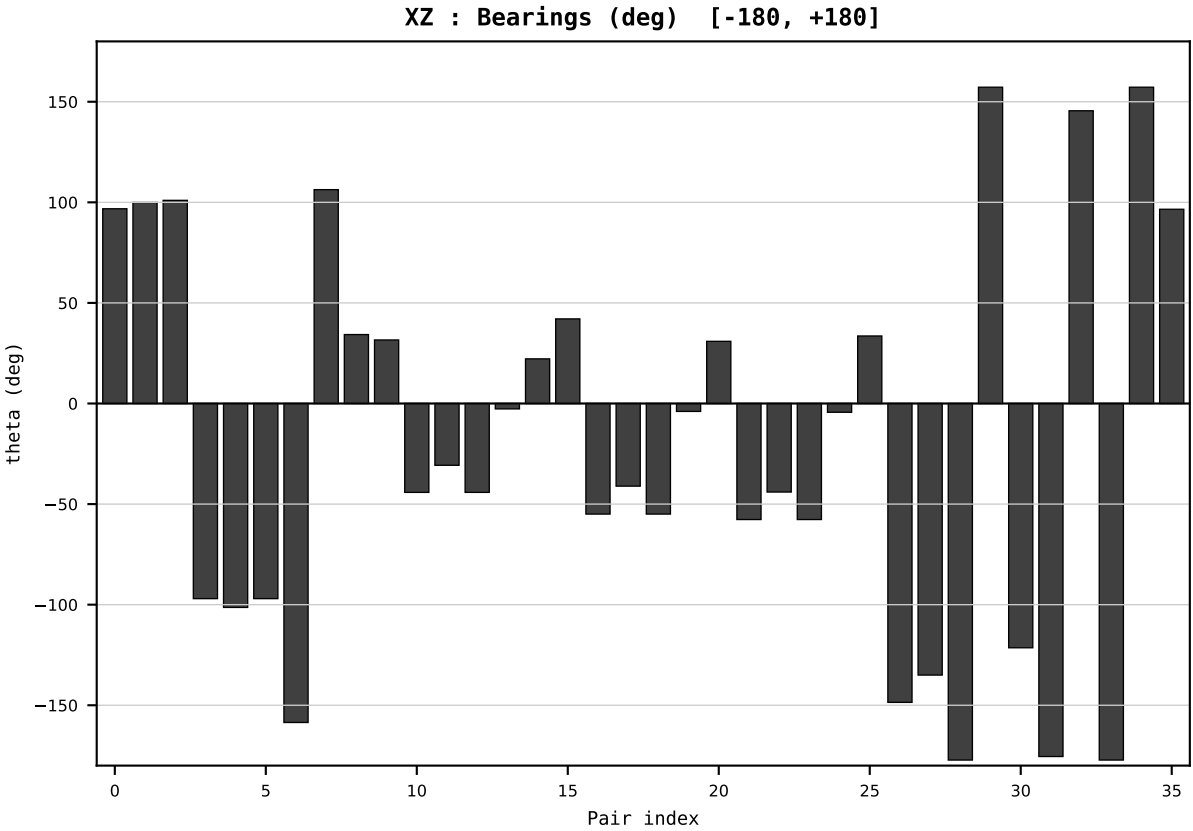
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2010
t : 10 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.669015
Ring = Hs-4
E_metric= 158.3215
kappa_HS= 164580.00
omega = 11.9015 deg
d_A = 2.598112
Helm = Solar
Helm d = +1.820108
DR = 12.011 HLR
DR ratio= 164580.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=11 | Year 2011 | D=9 N=26 pairs=36

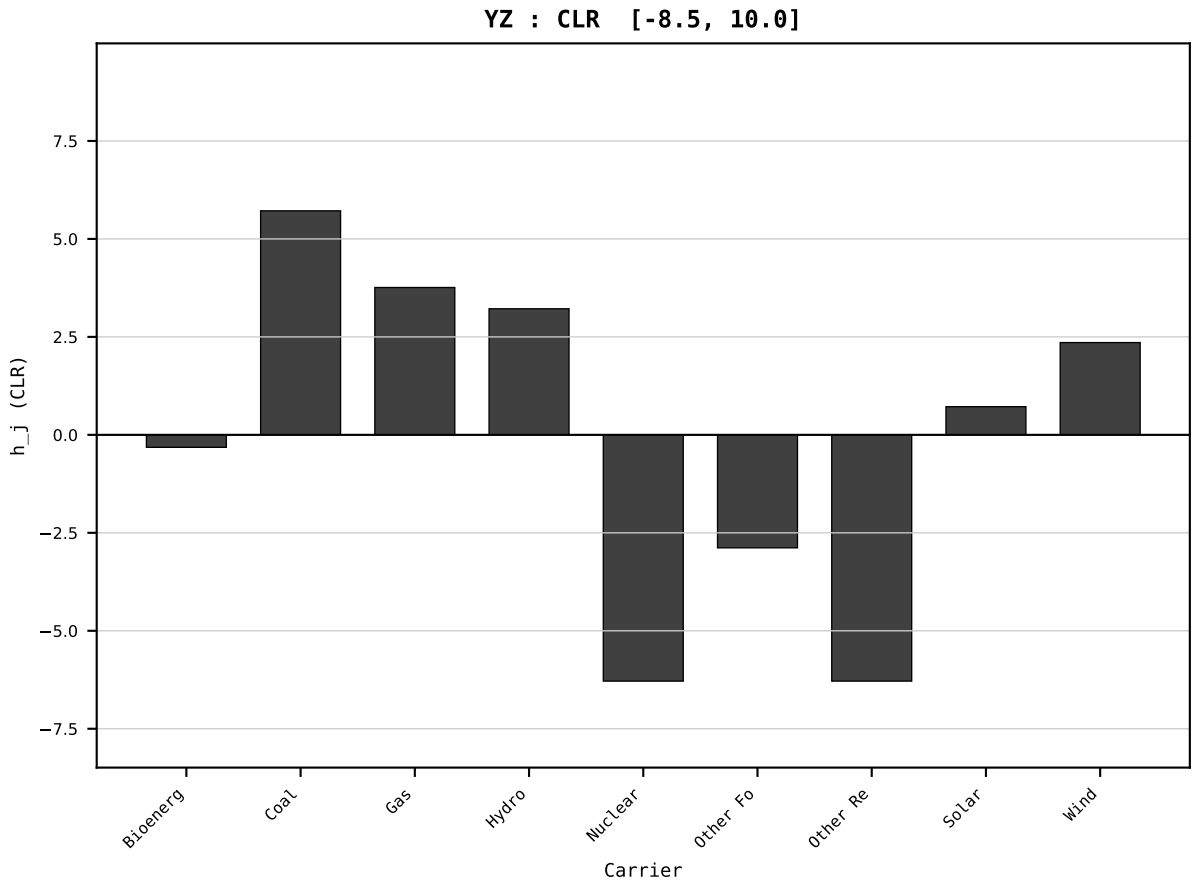
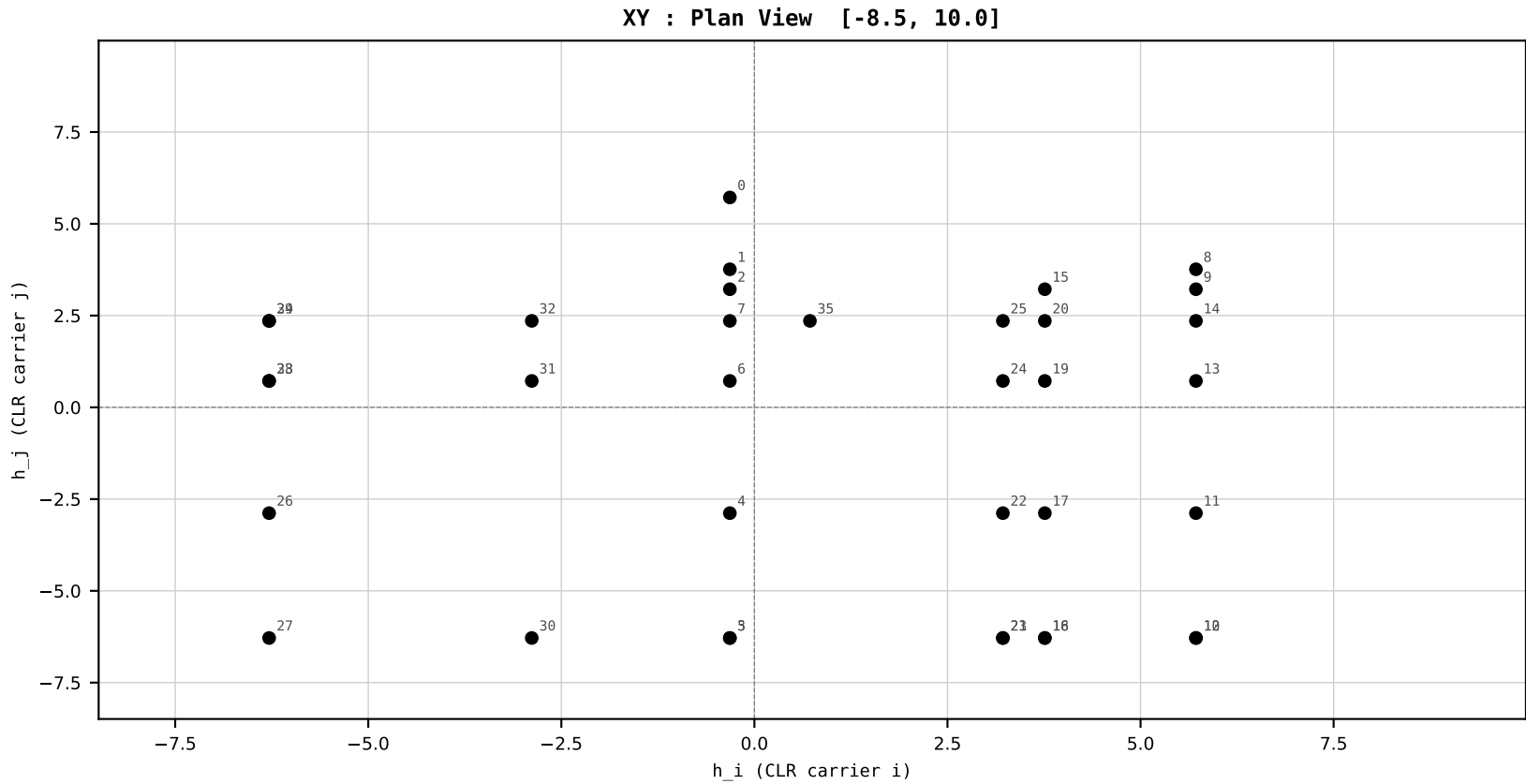
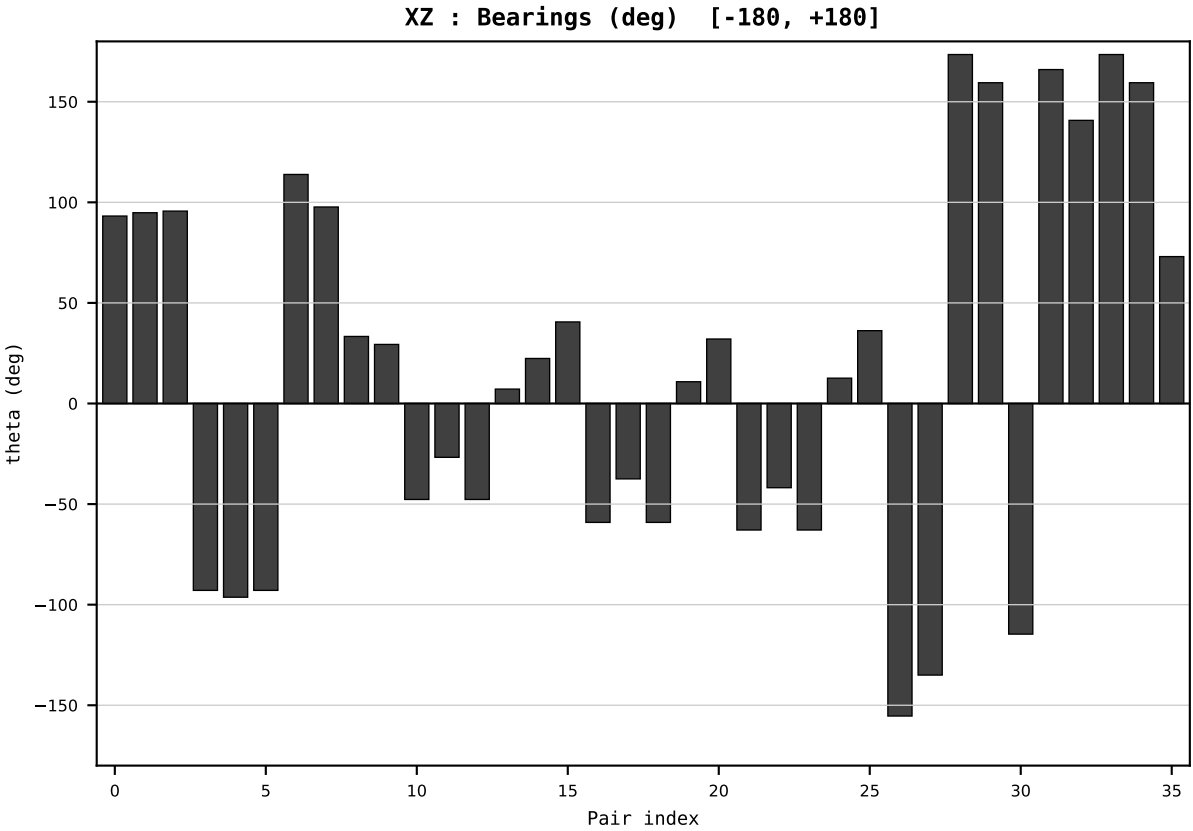
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2011
t : 11 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.659385
Ring = Hs-4
E_metric= 150.6365
kappa_HS= 162980.00
omega = 7.2543 deg
d_A = 1.602461
Helm = Solar
Helm d = +1.004418
DR = 12.001 HLR
DR ratio= 162980.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=12 | Year 2012 | D=9 N=26 pairs=36

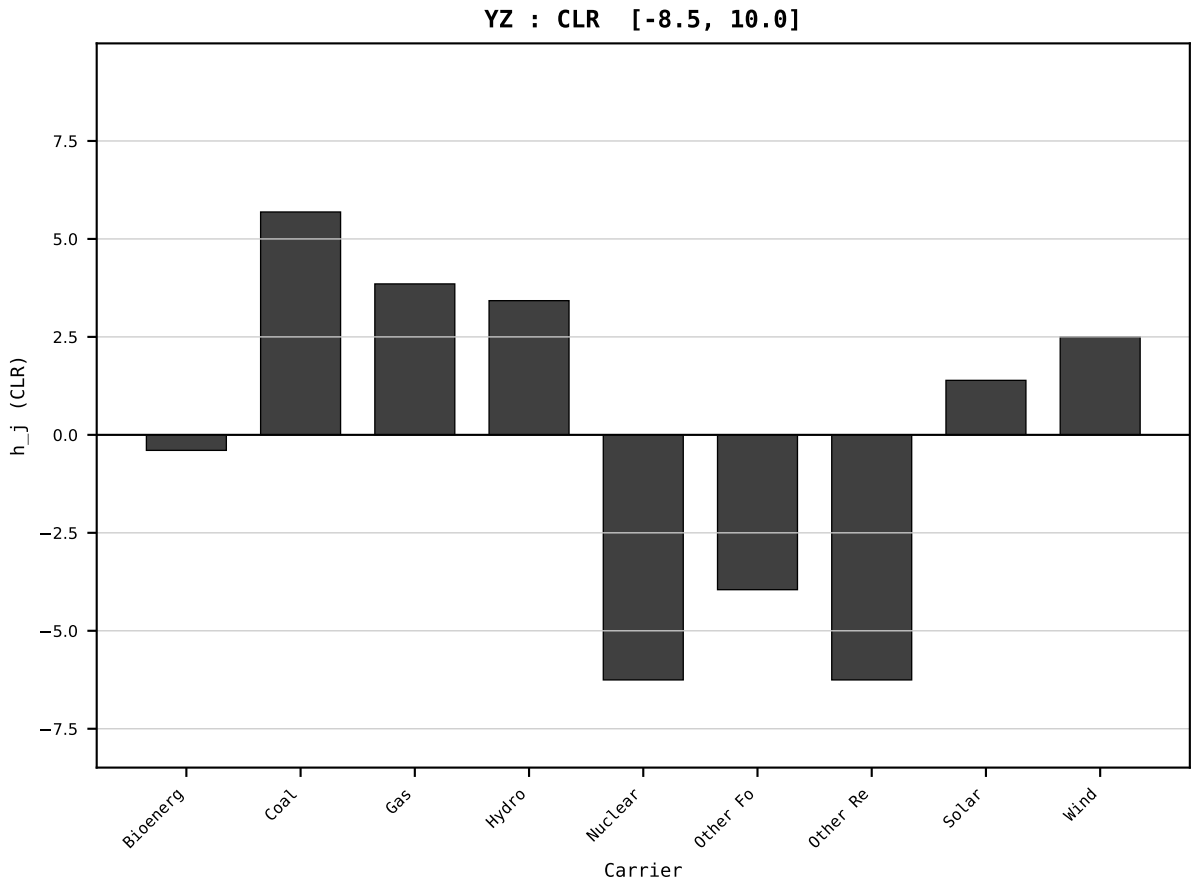
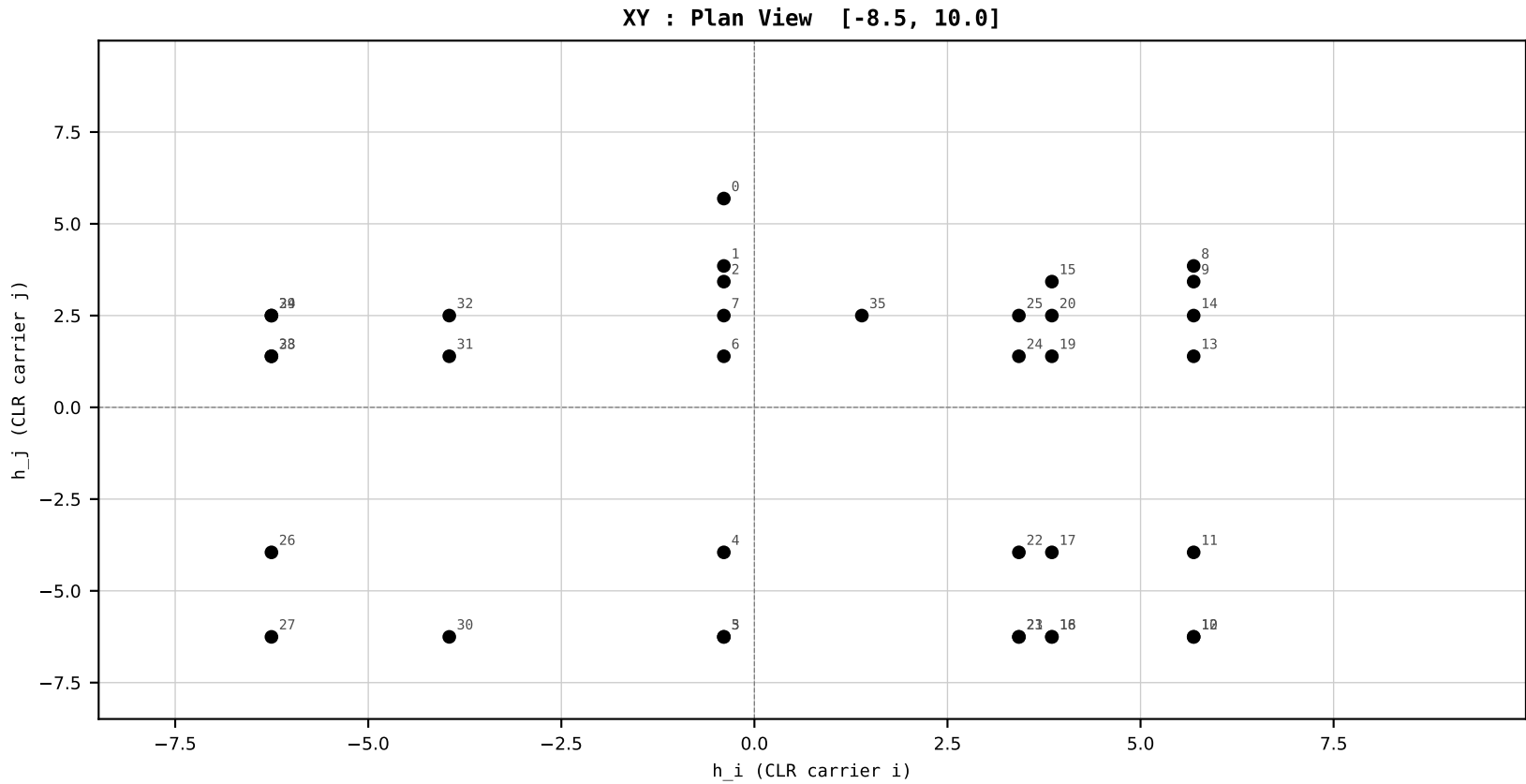
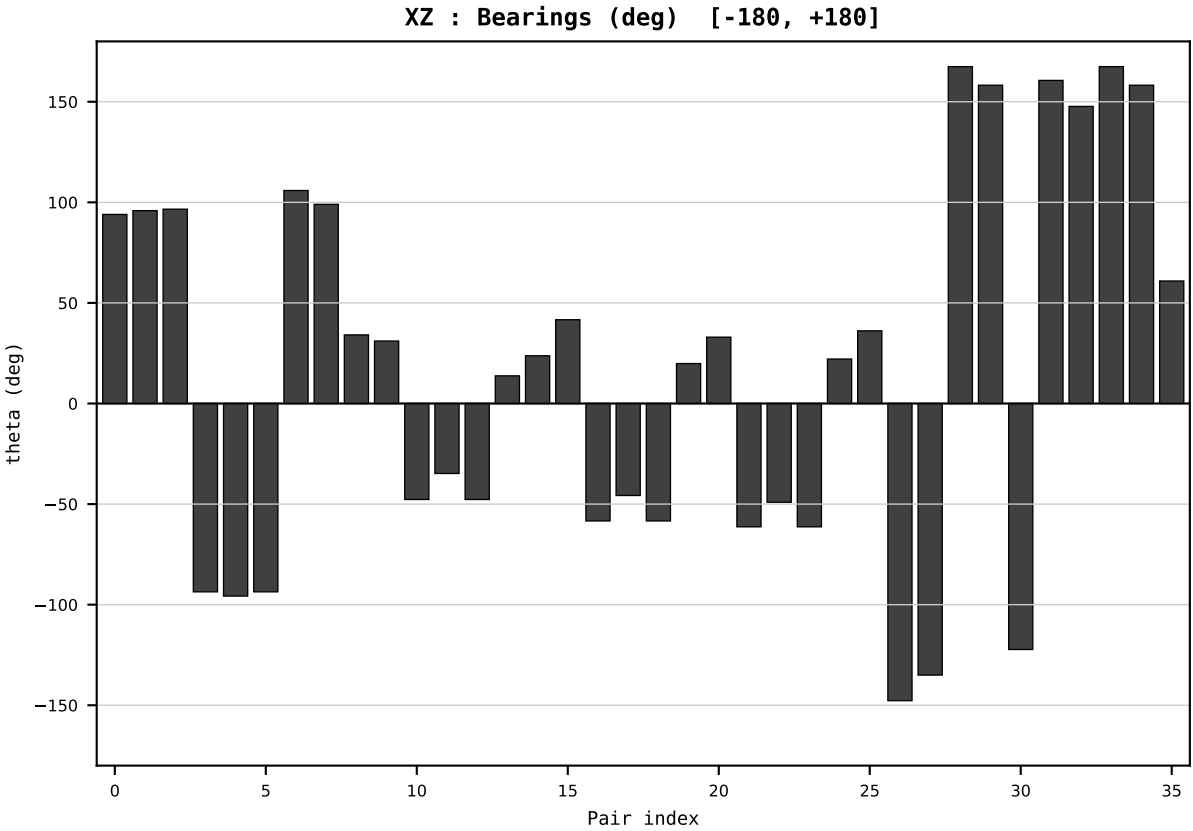
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2012
t : 12 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.620841
Ring = Hs-4
E_metric= 161.0838
kappa_HS= 153490.00
omega = 5.6224 deg
d_A = 1.293789
Helm = Other Fossil
Helm d = -1.068245
DR = 11.941 HLR
DR ratio= 153490.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=13 | Year 2013 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

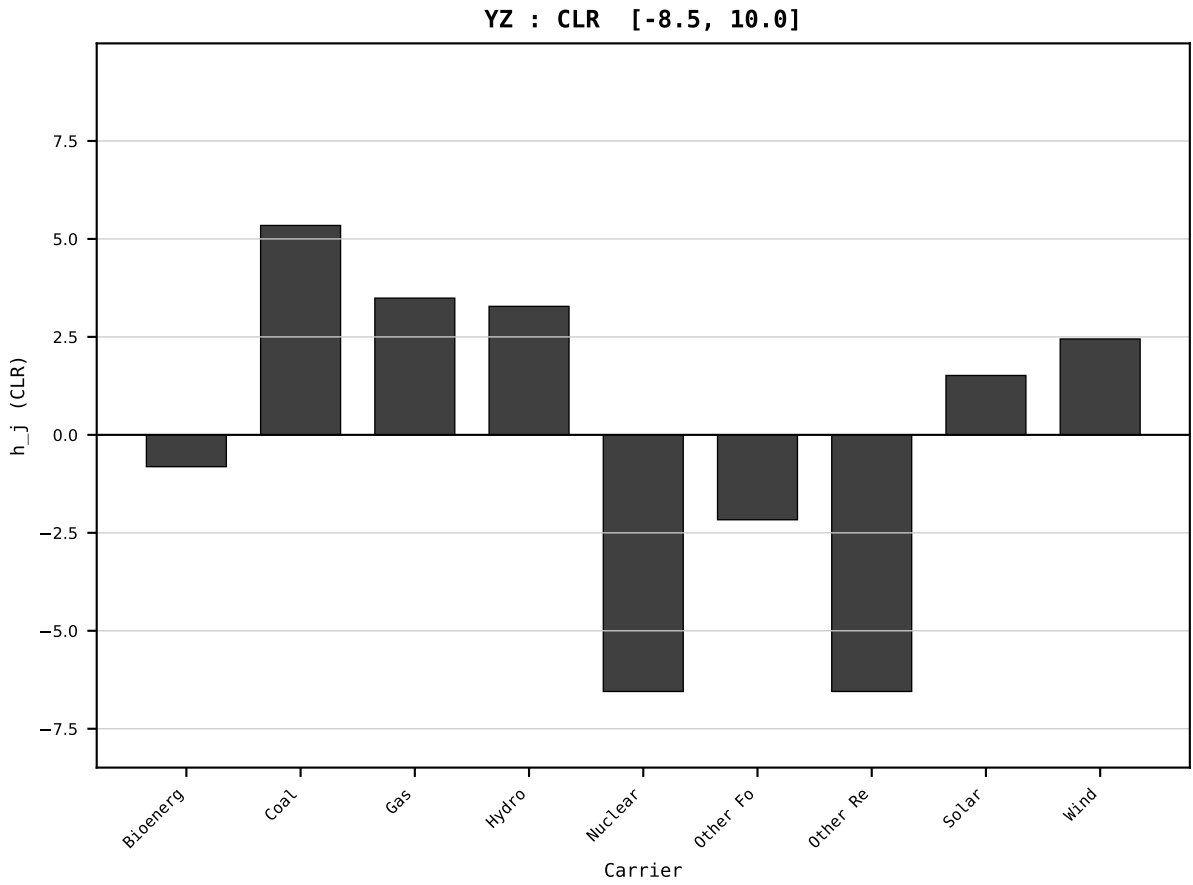
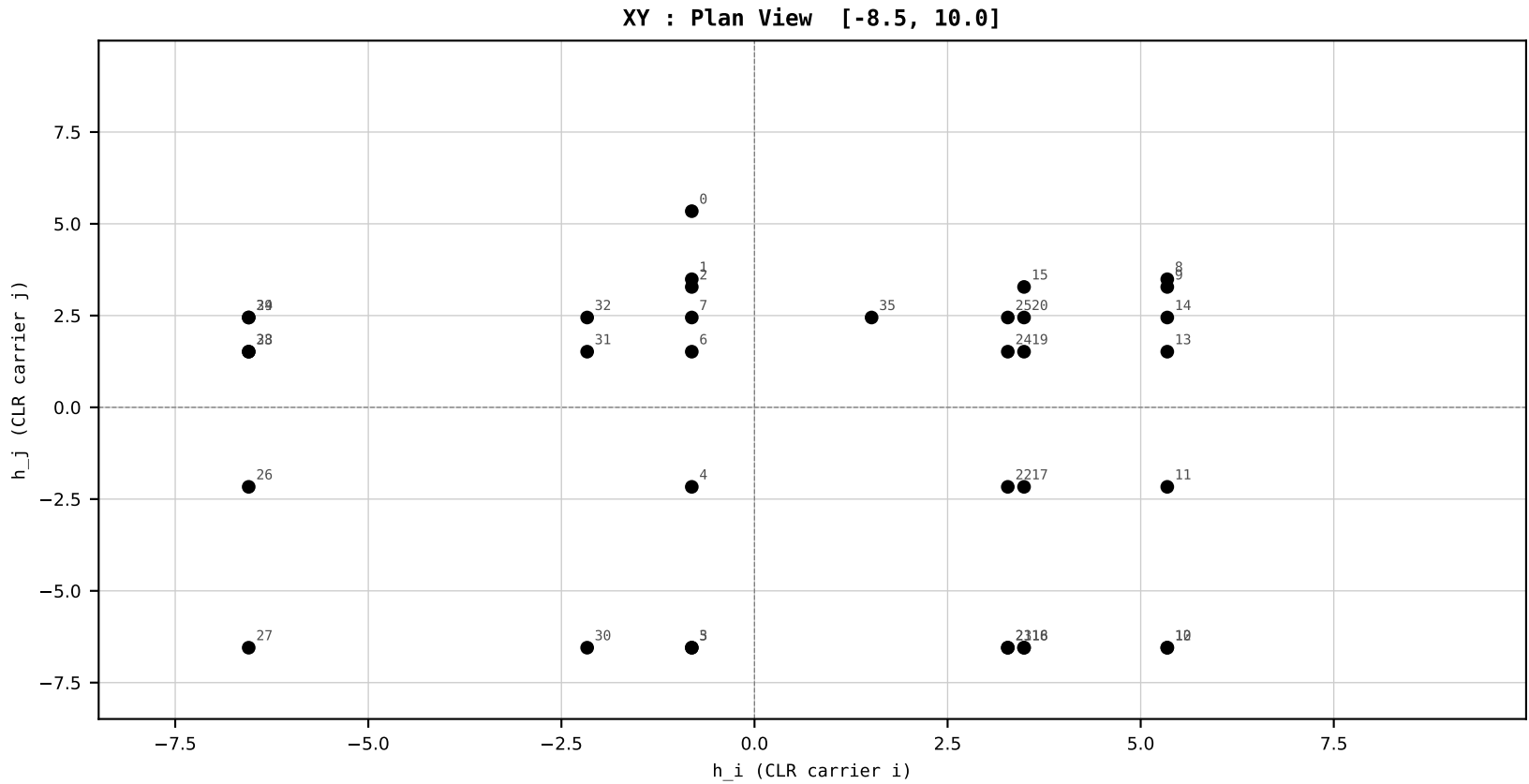
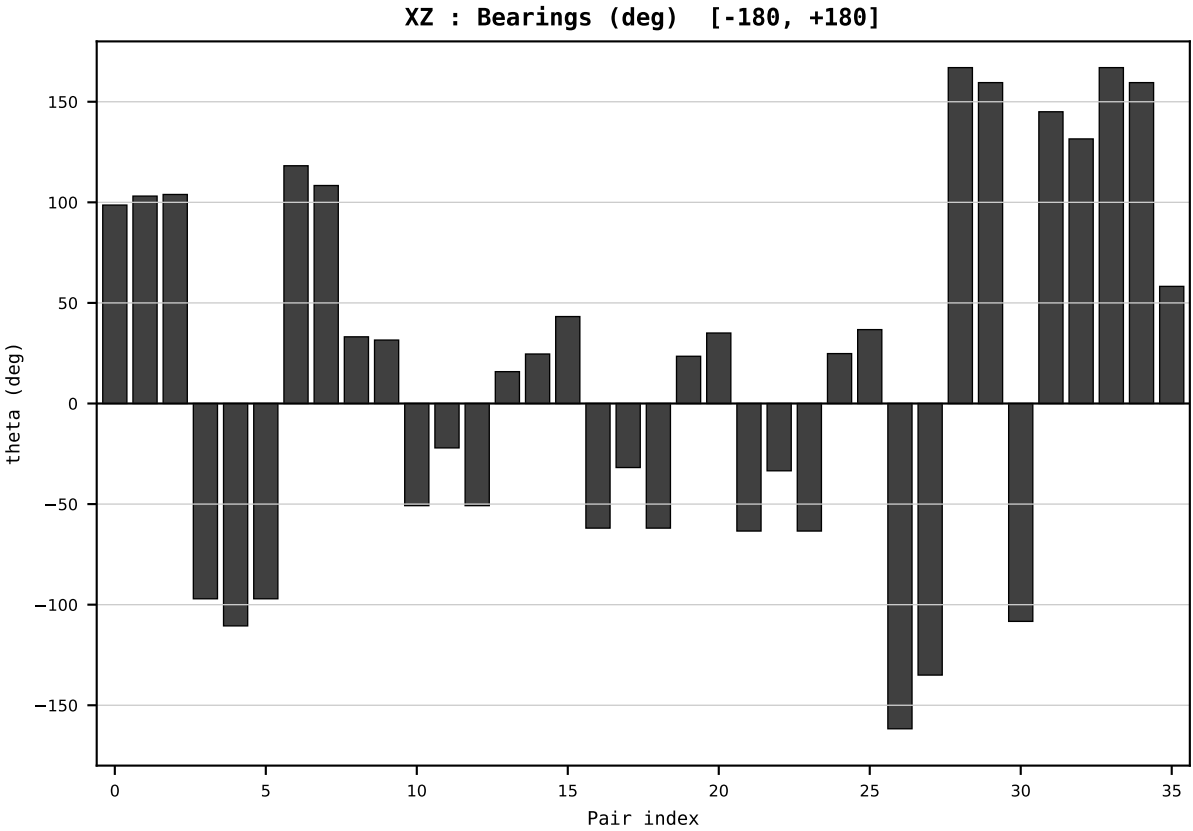
Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2013
t : 13 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.587671
Ring = Hs-4
E_metric= 150.9049
kappa_HS= 146210.00
omega = 8.7801 deg
d_A = 1.954538
Helm = Other Fossil
Helm d = +1.784479
DR = 11.893 HLR
DR ratio= 146210.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=14 | Year 2014 | D=9 N=26 pairs=36

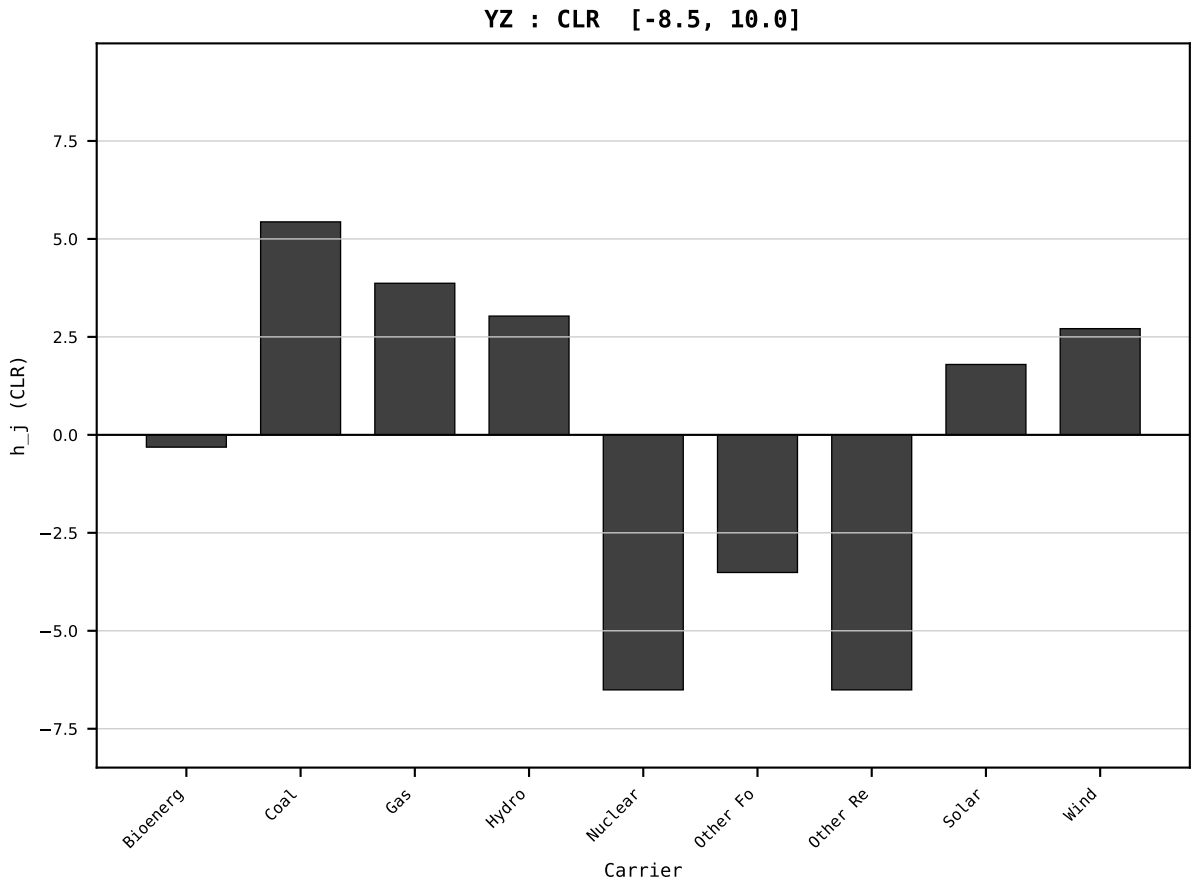
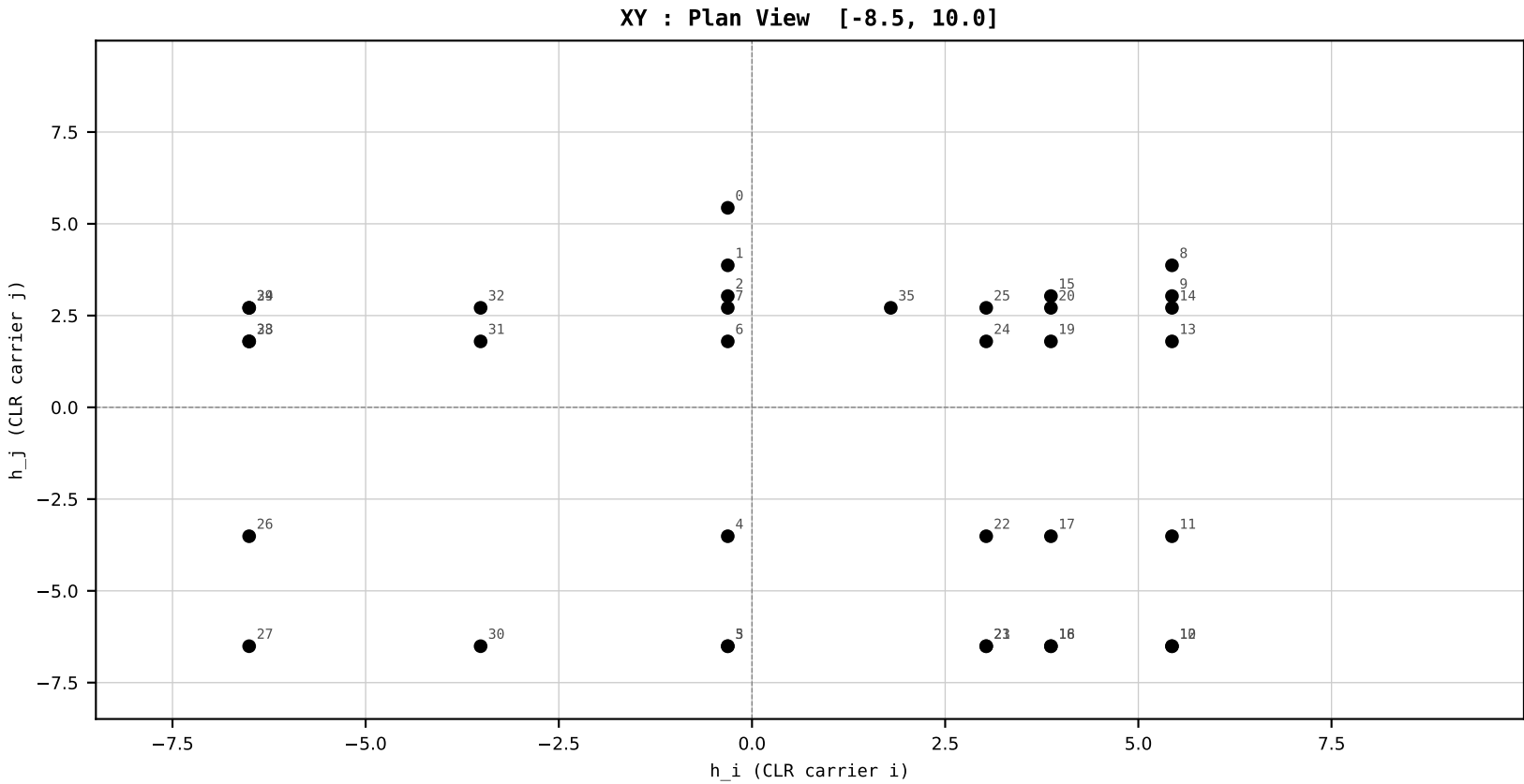
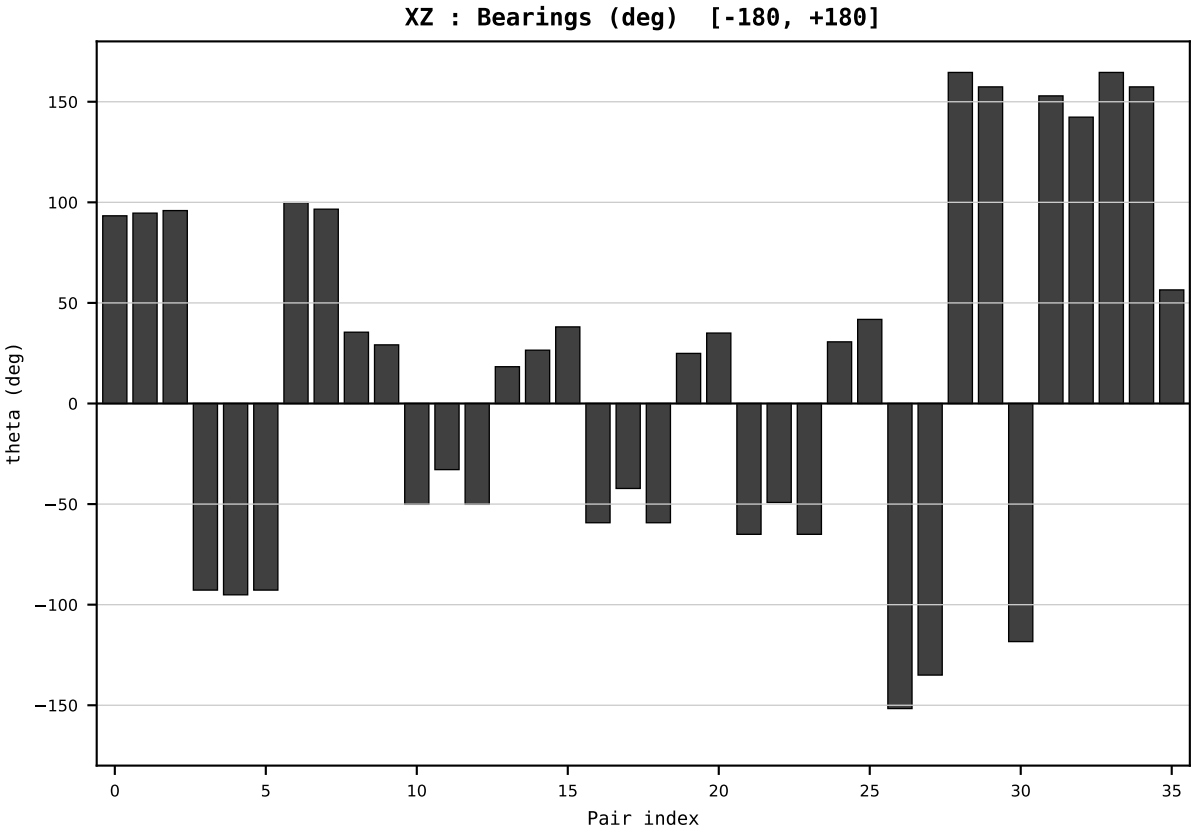
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2014
t : 14 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.575153
Ring = Hs-4
E_metric= 161.4074
kappa_HS= 153760.00
omega = 6.8792 deg
d_A = 1.556841
Helm = Other Fossil
Helm d = -1.345462
DR = 11.943 HLR
DR ratio= 153760.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=15 | Year 2015 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

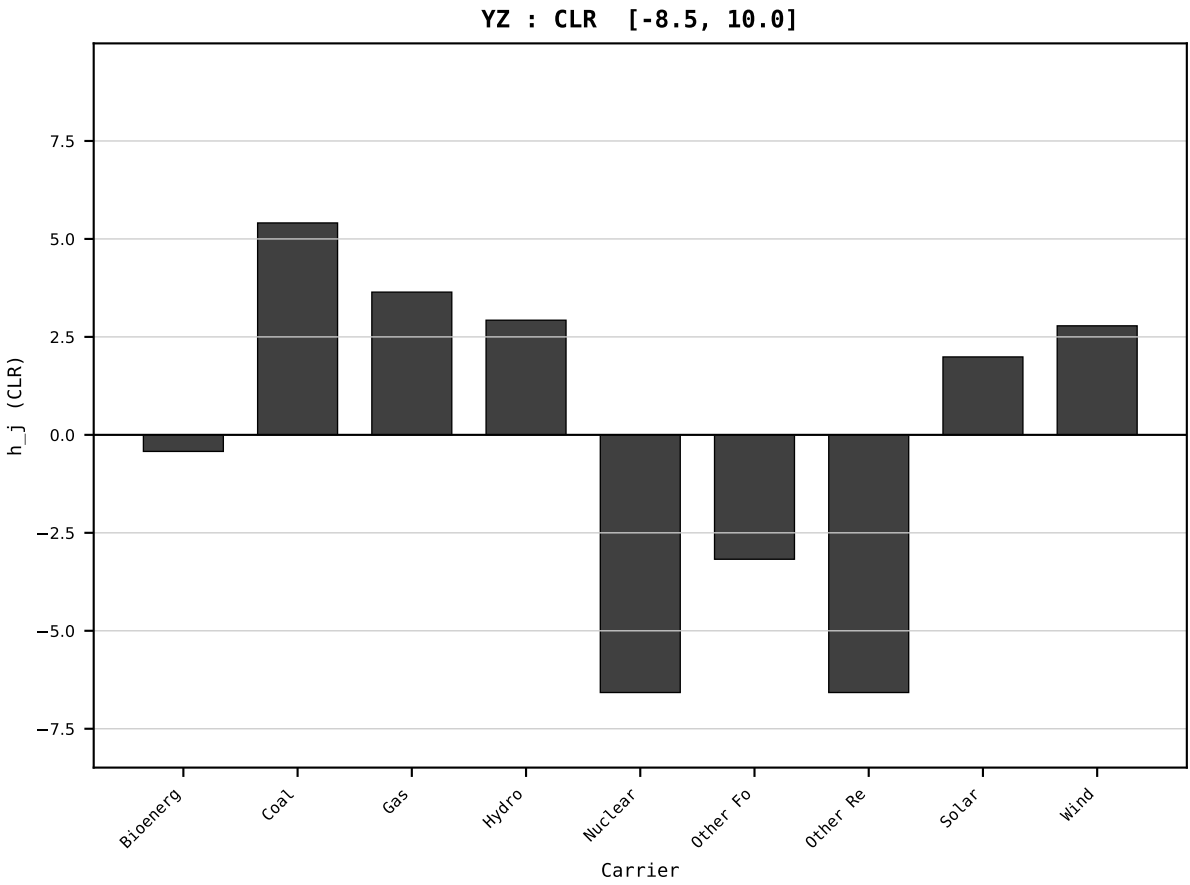
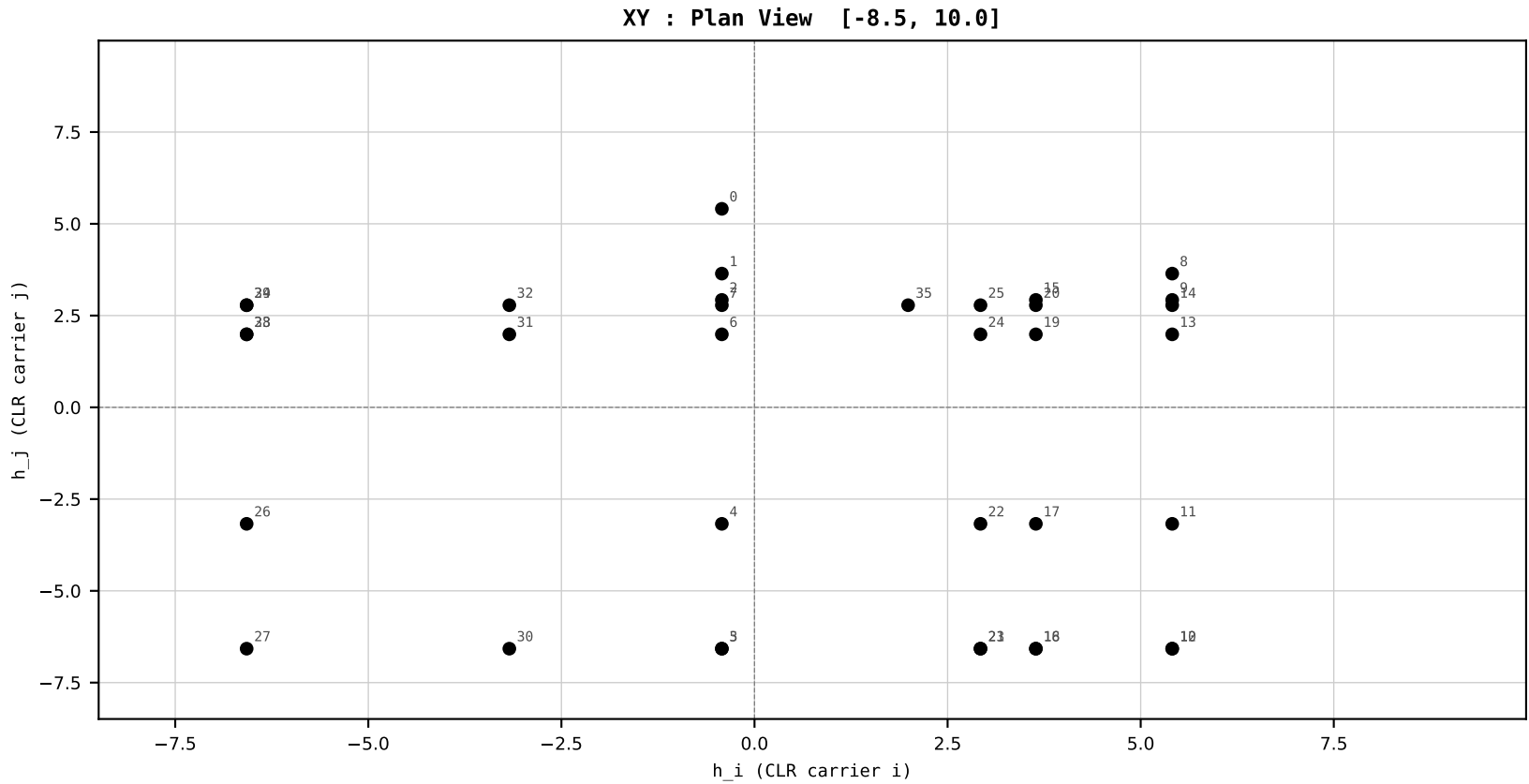
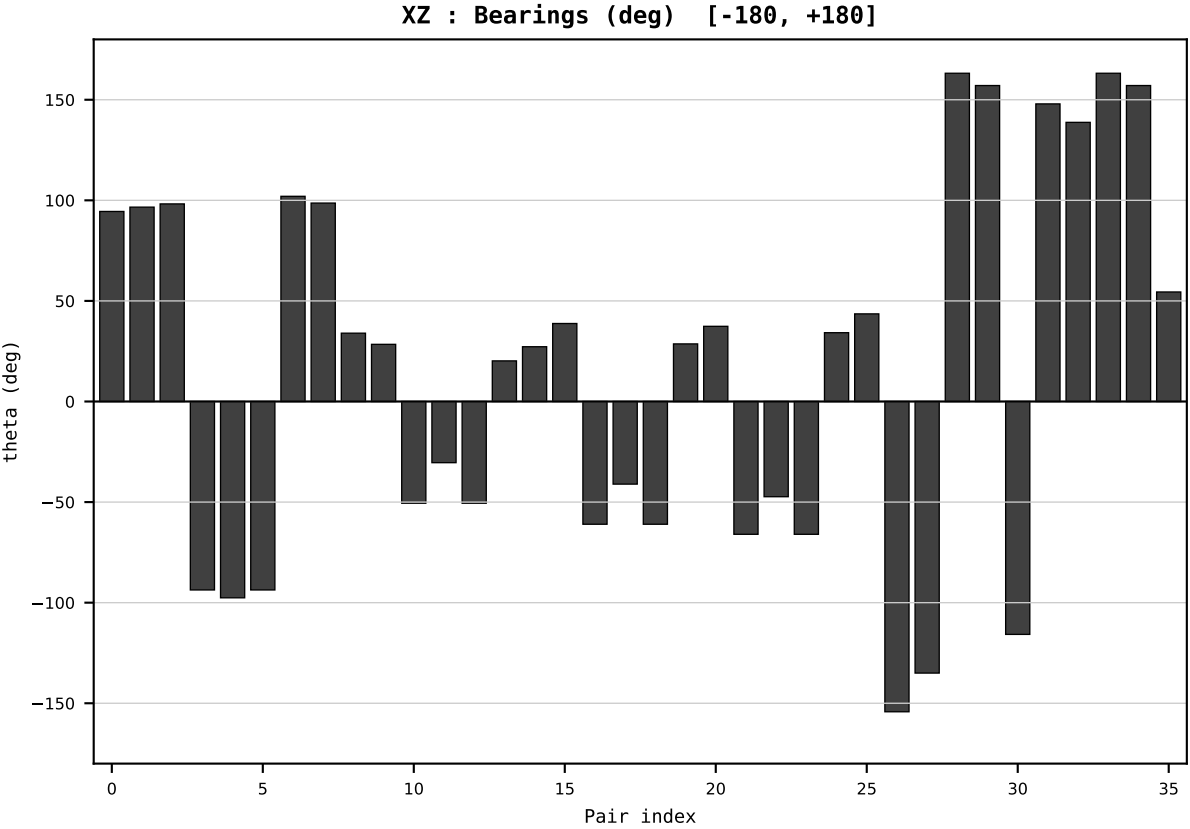
Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2015
t : 15 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.581737
Ring = Hs-4
E_metric= 159.5021
kappa_HS= 160070.00
omega = 2.1911 deg
d_A = 0.490179
Helm = Other Fossil
Helm d = +0.337914
DR = 11.983 HLR
DR ratio= 160070.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=16 | Year 2016 | D=9 N=26 pairs=36

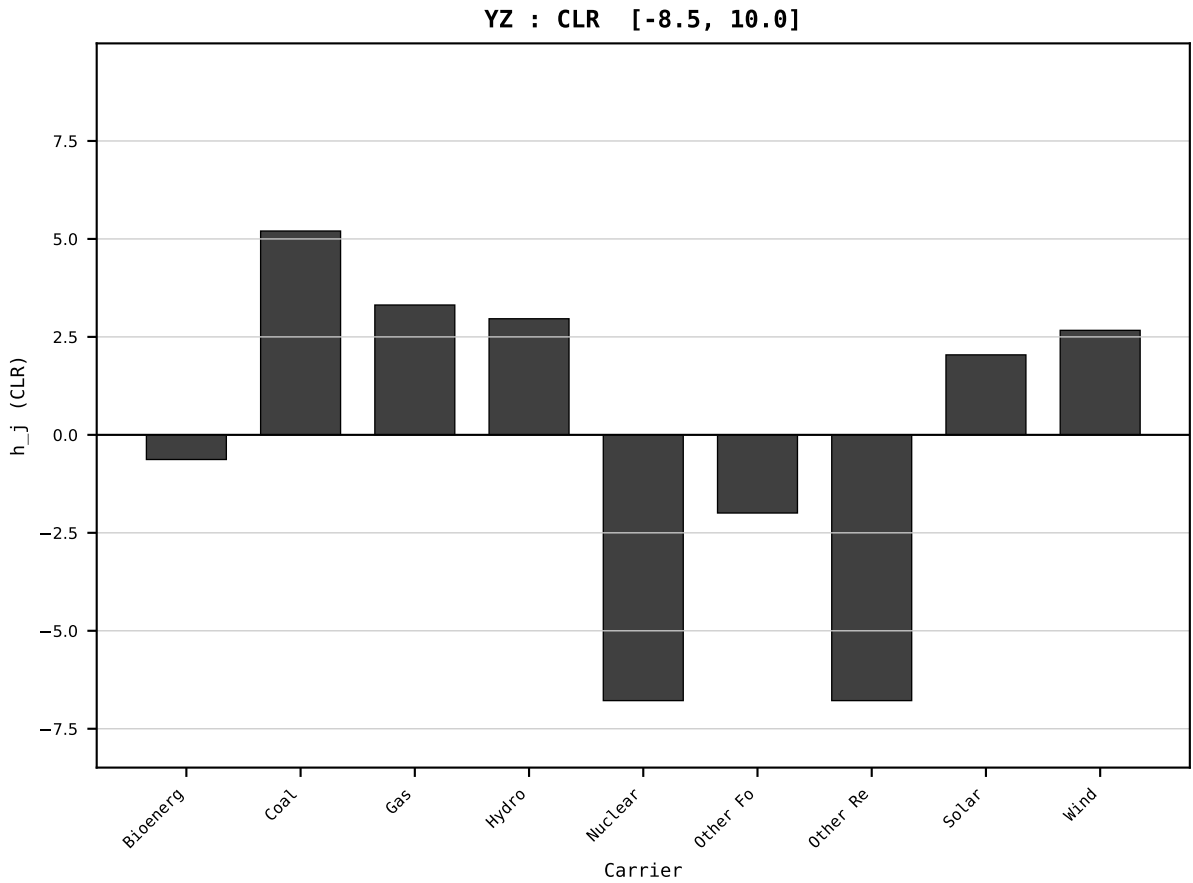
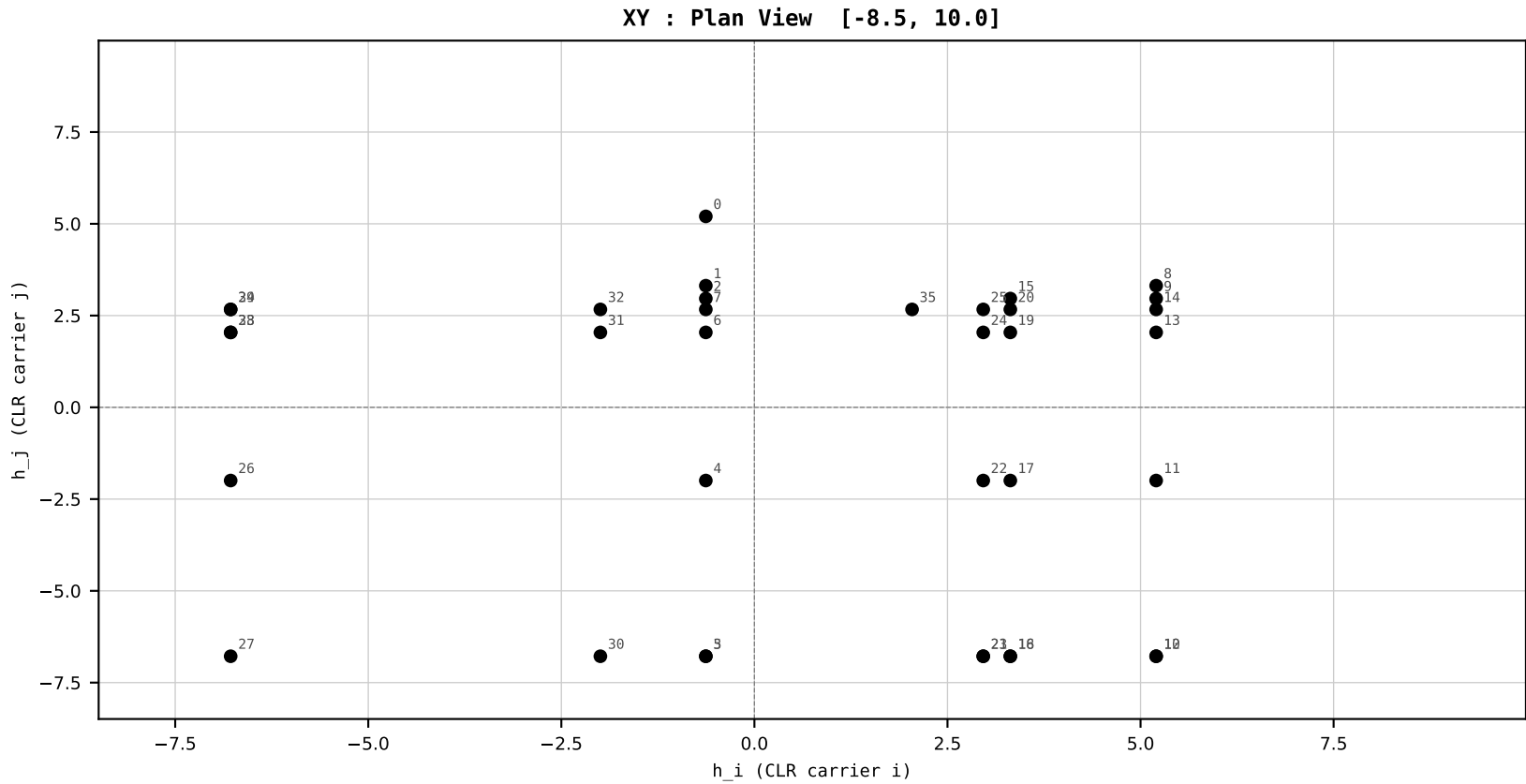
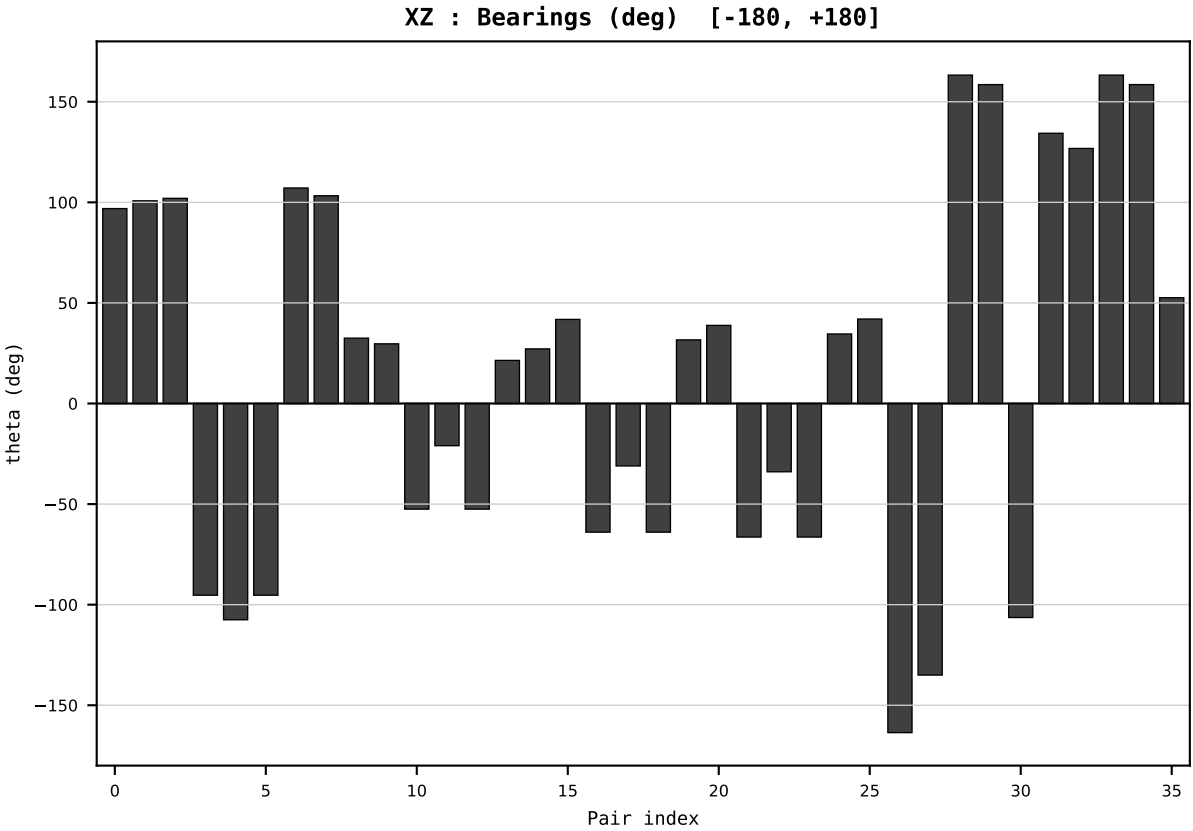
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2016
t : 16 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.562143
Ring = Hs-4
E_metric= 154.4679
kappa_HS= 160170.00
omega = 5.8725 deg
d_A = 1.299181
Helm = Other Fossil
Helm d = +1.179791
DR = 11.984 HLR
DR ratio= 160170.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=17 | Year 2017 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

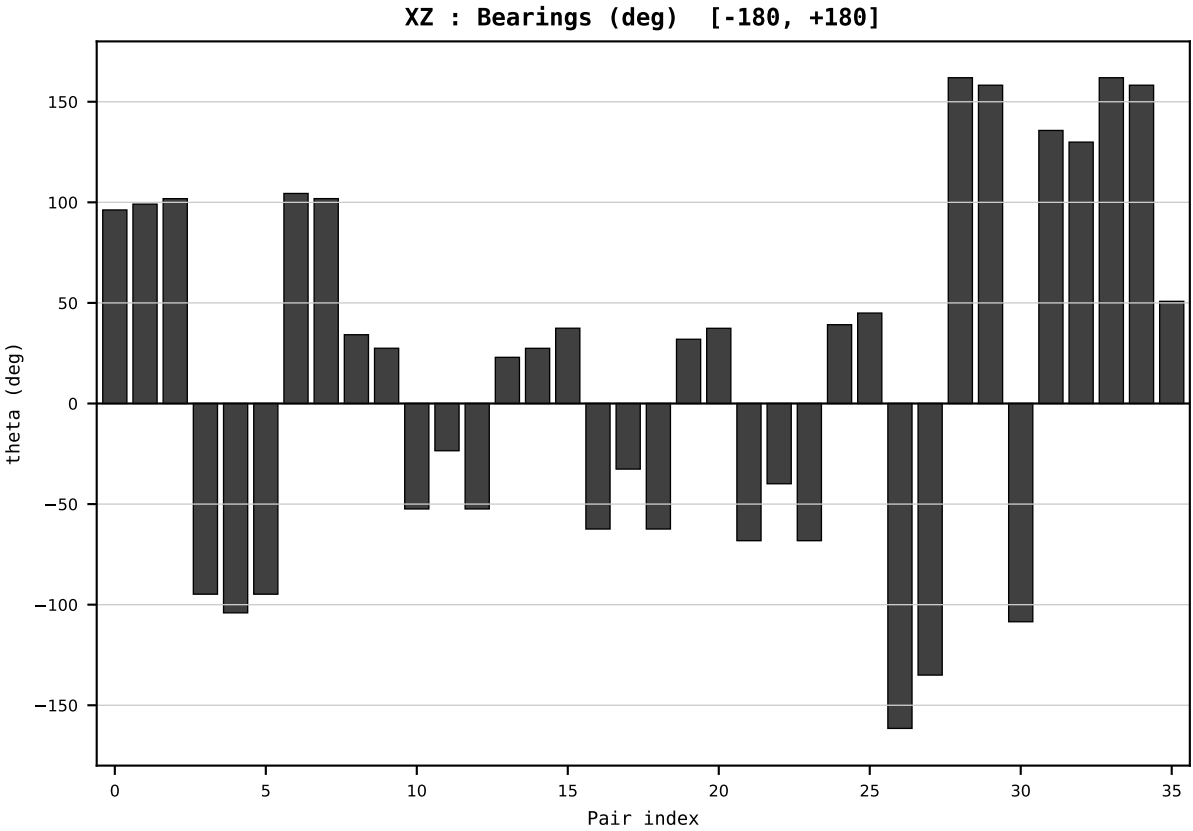
Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2017
t : 17 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.554731
Ring = Hs-4
E_metric= 155.8956
kappa_HS= 157110.00
omega = 2.1363 deg
d_A = 0.467967
Helm = Other Fossil
Helm d = -0.266487
DR = 11.965 HLR
DR ratio= 157110.0x

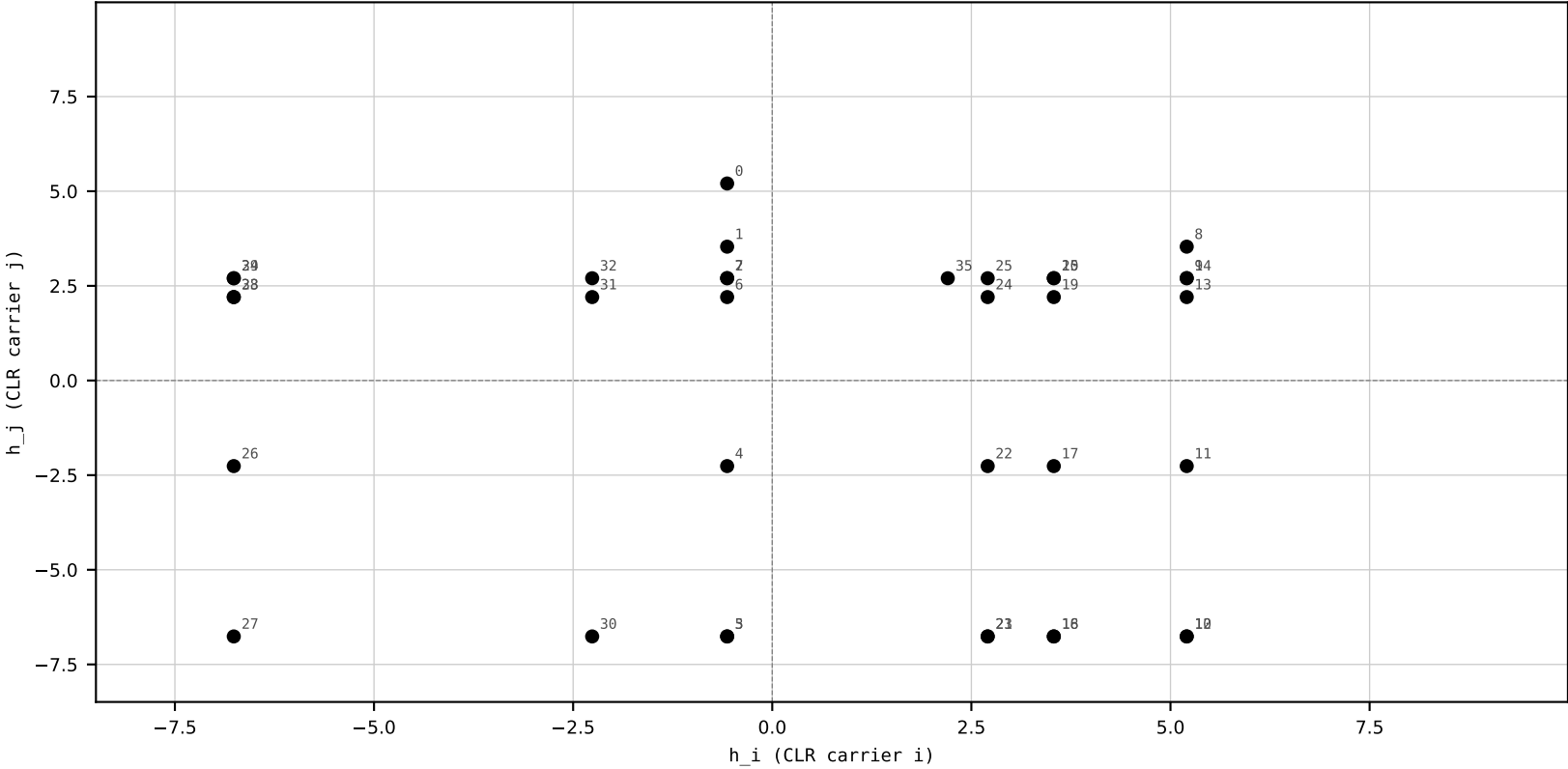
XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:

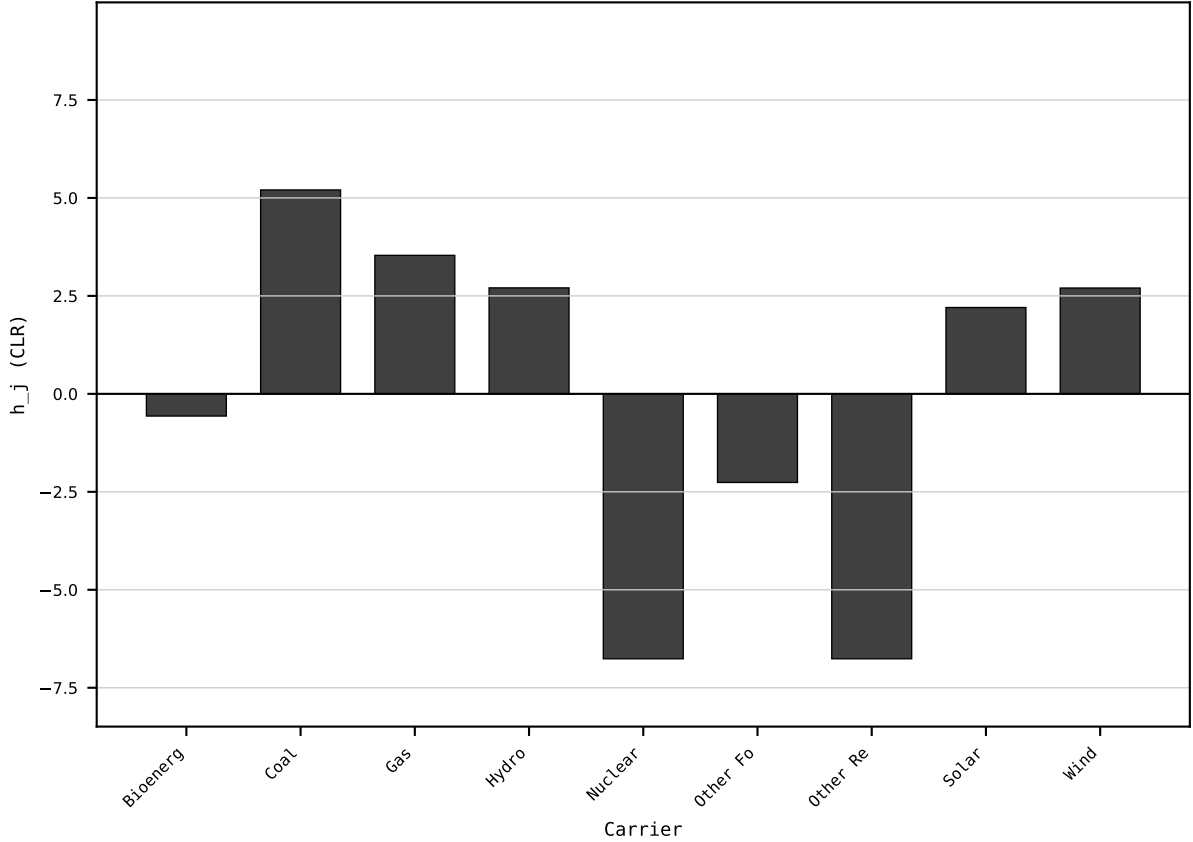
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



XY : Plan View [-8.5, 10.0]



YZ : CLR [-8.5, 10.0]



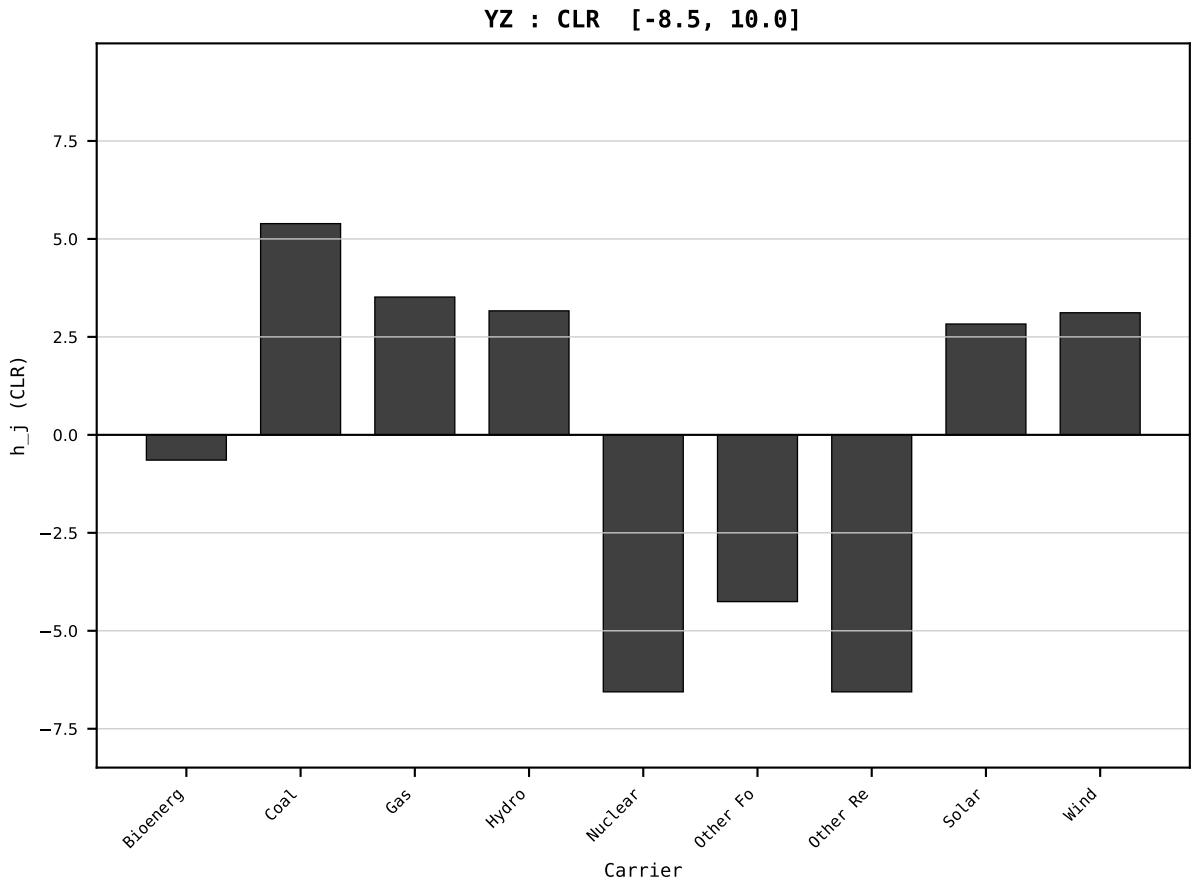
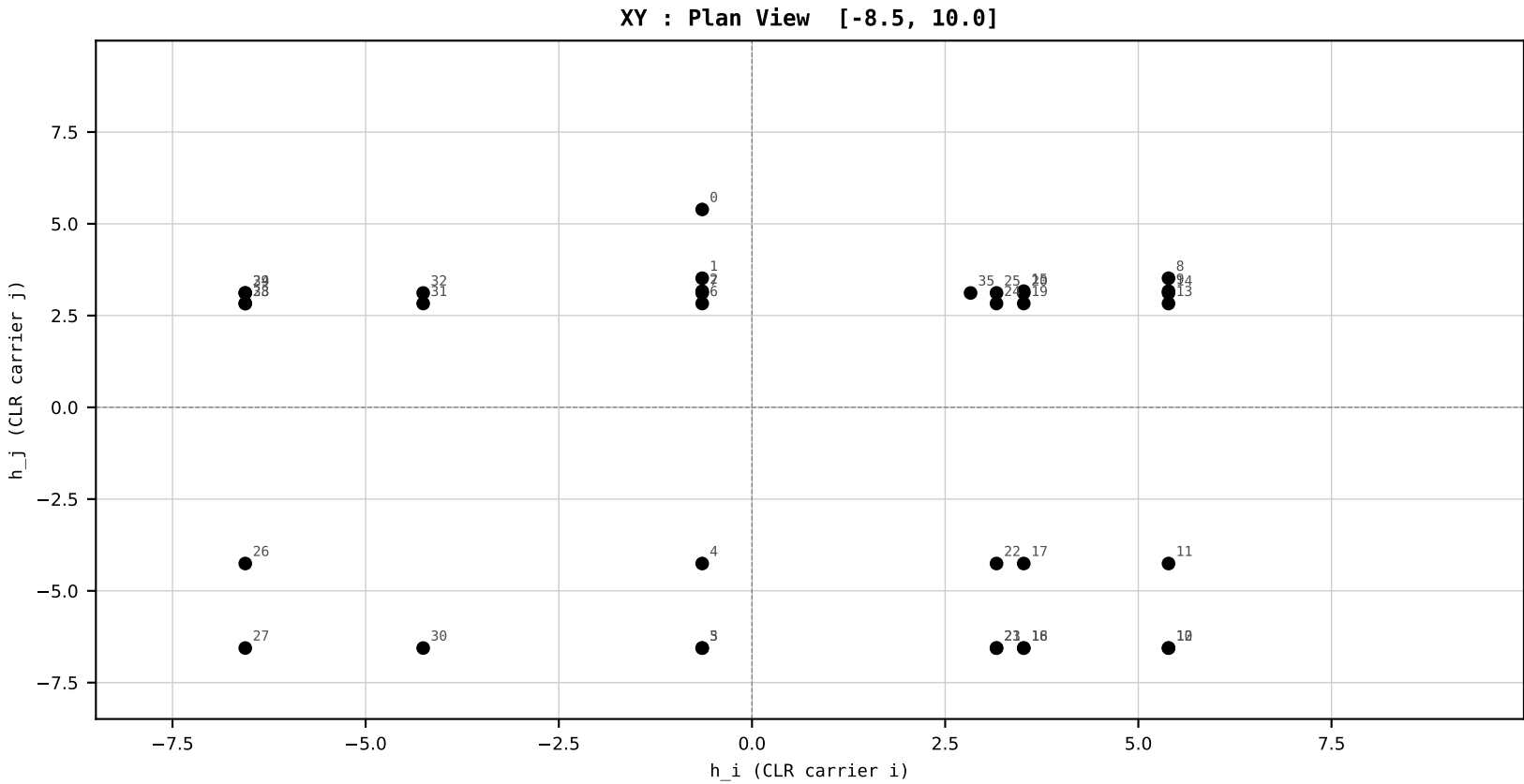
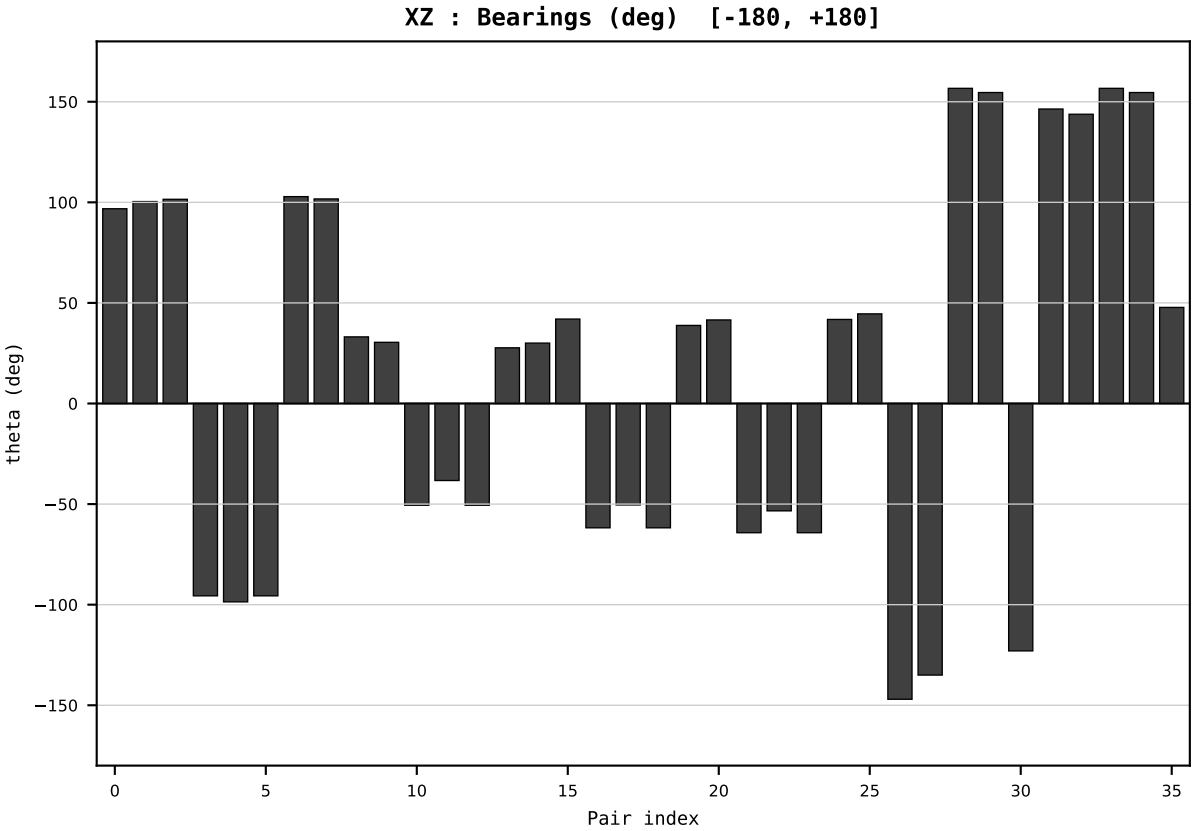
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2018
t : 18 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.525552
Ring = Hs-4
E_metric= 173.6939
kappa_HS= 154620.00
omega = 9.3738 deg
d_A = 2.208080
Helm = Other Fossil
Helm d = -1.994594
DR = 11.949 HLR
DR ratio= 154620.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



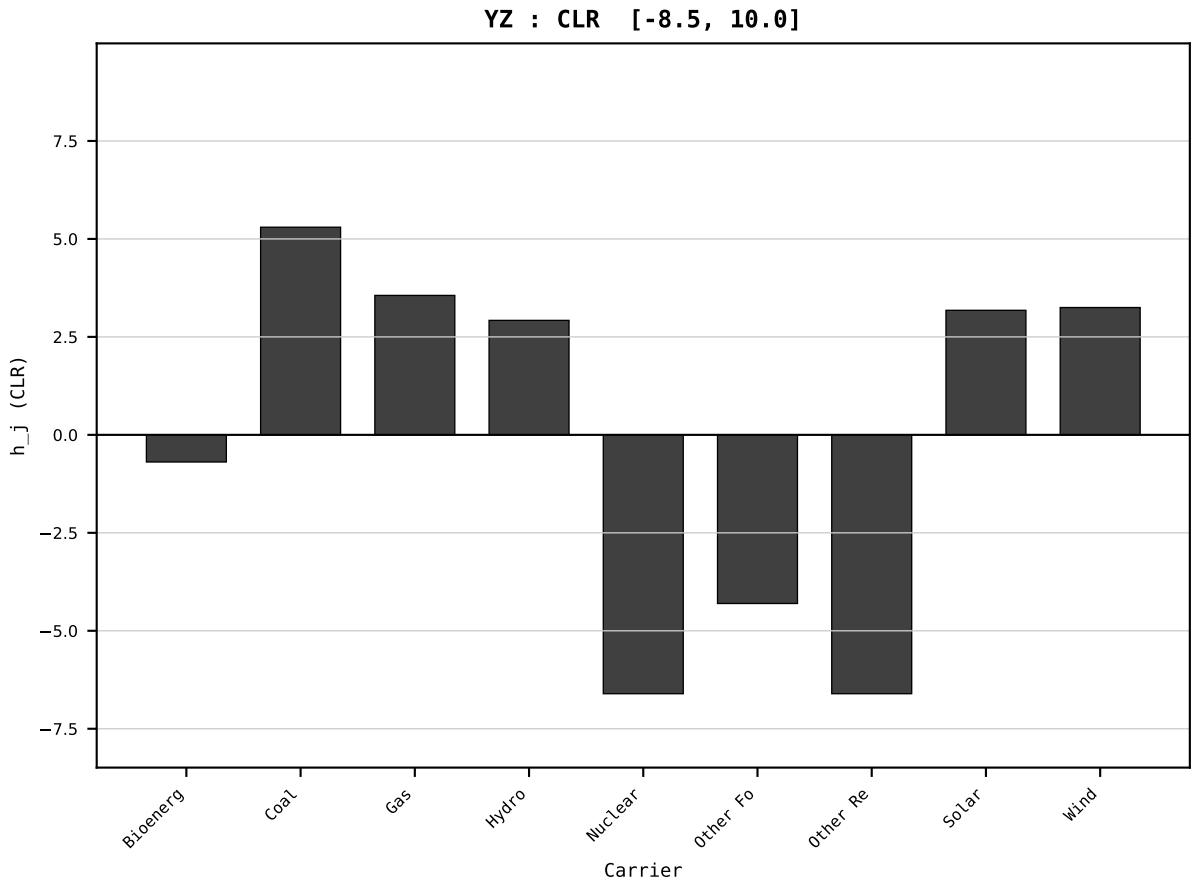
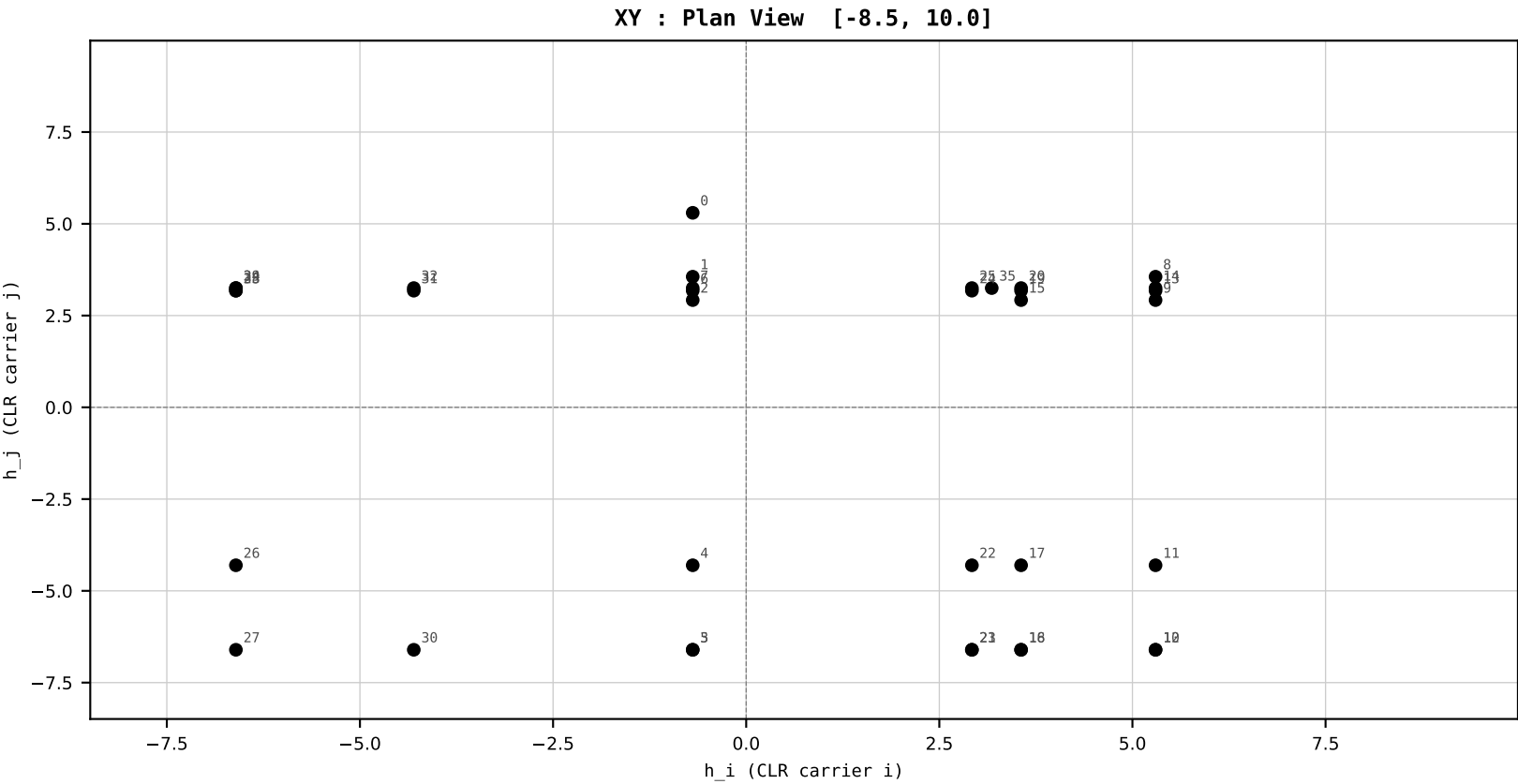
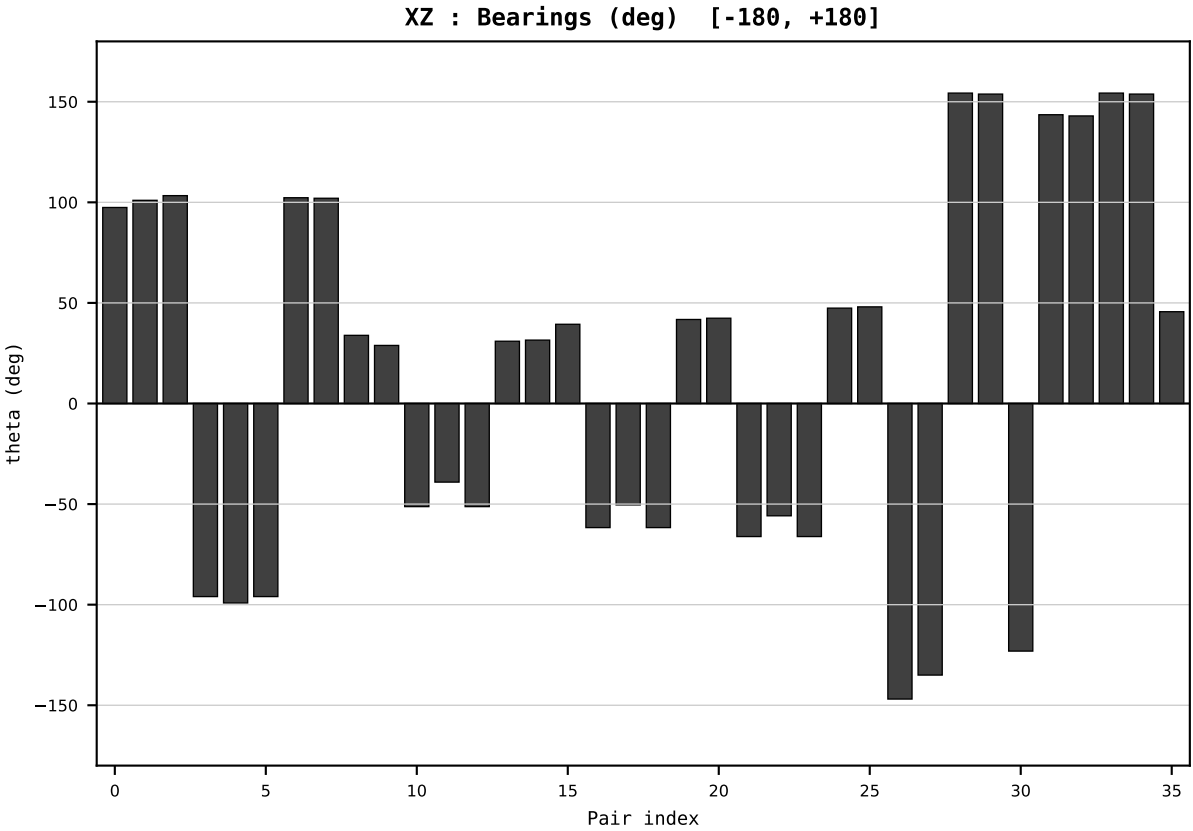
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2019
t : 19 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.492113
Ring = Hs-3
E_metric= 176.2070
kappa_HS= 147990.00
omega = 1.9812 deg
d_A = 0.467089
Helm = Solar
Helm d = +0.349727
DR = 11.905 HLR
DR ratio= 147990.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=20 | Year 2020 | D=9 N=26 pairs=36

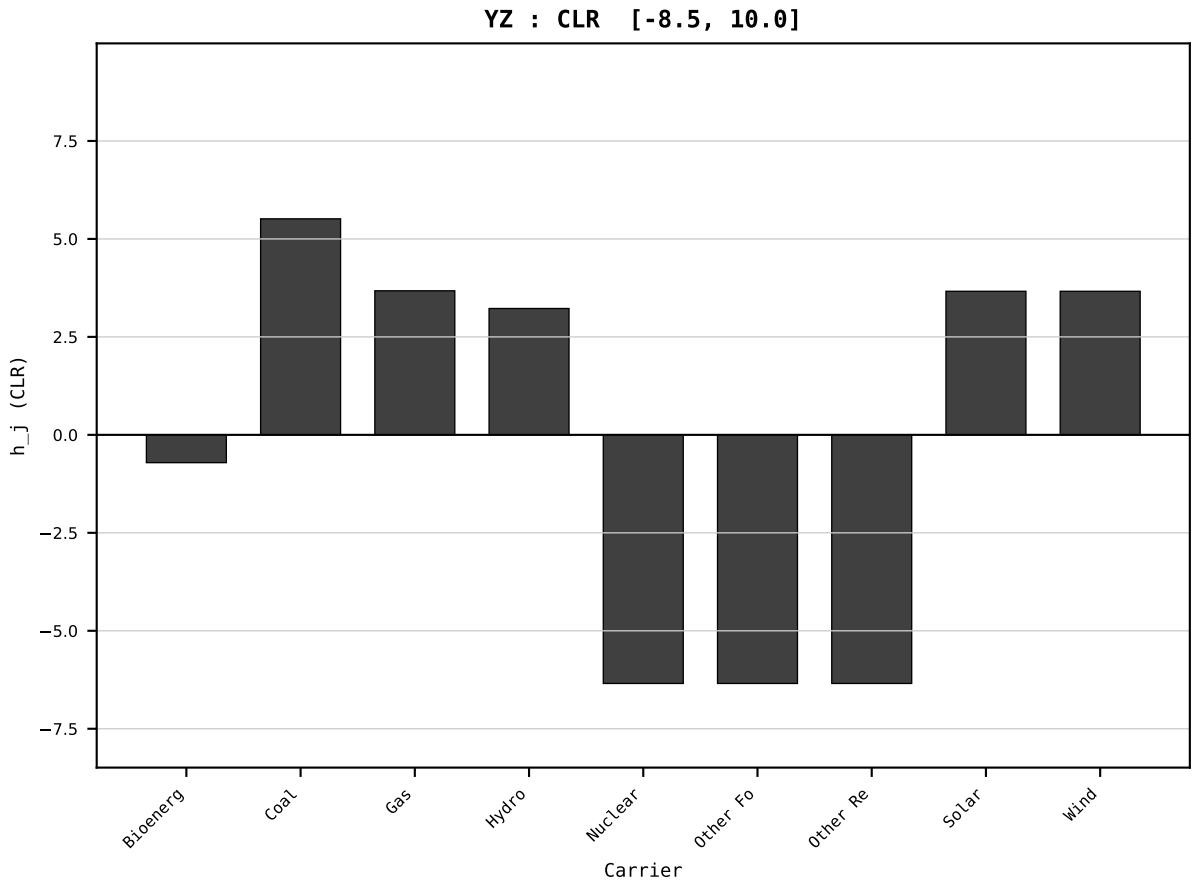
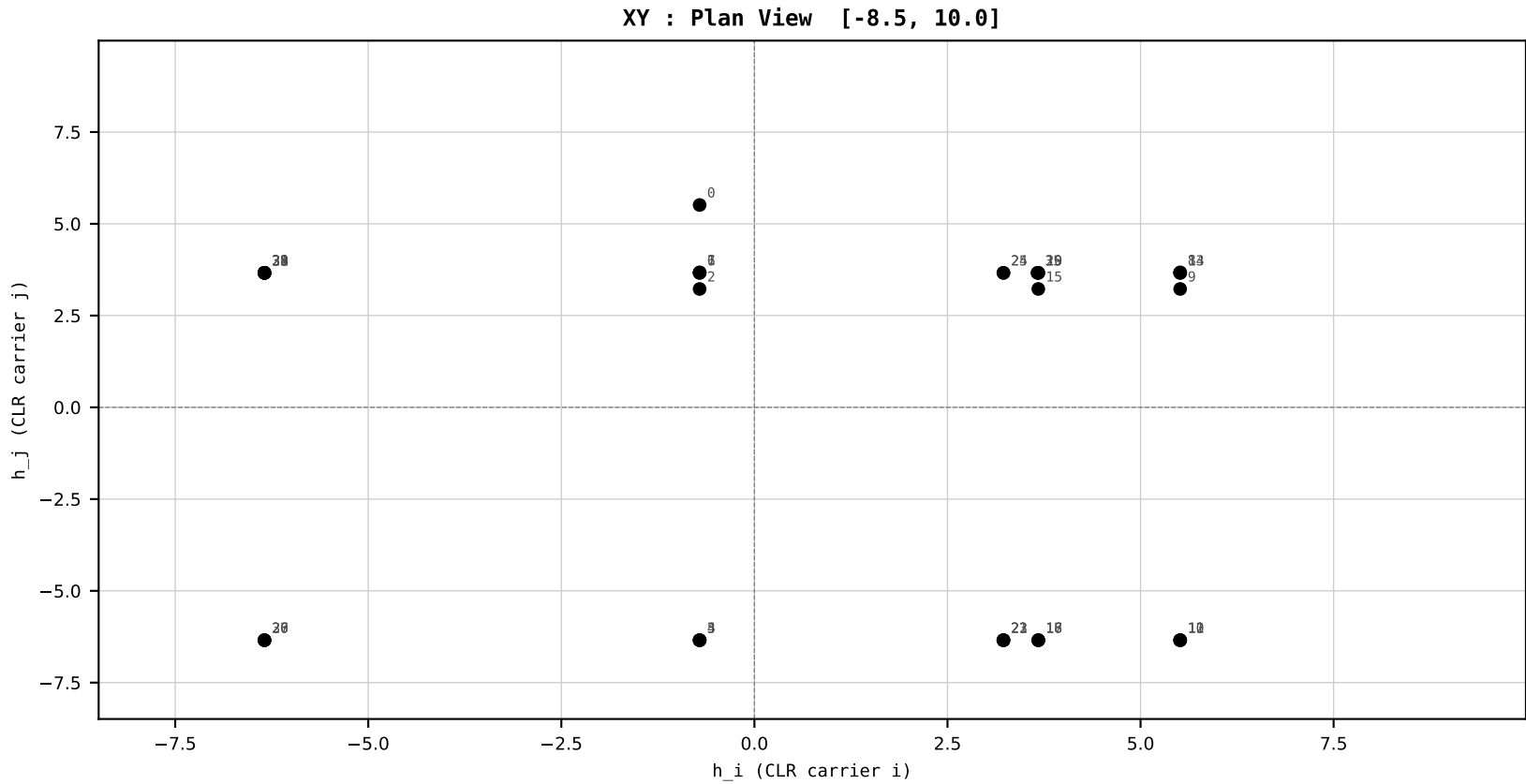
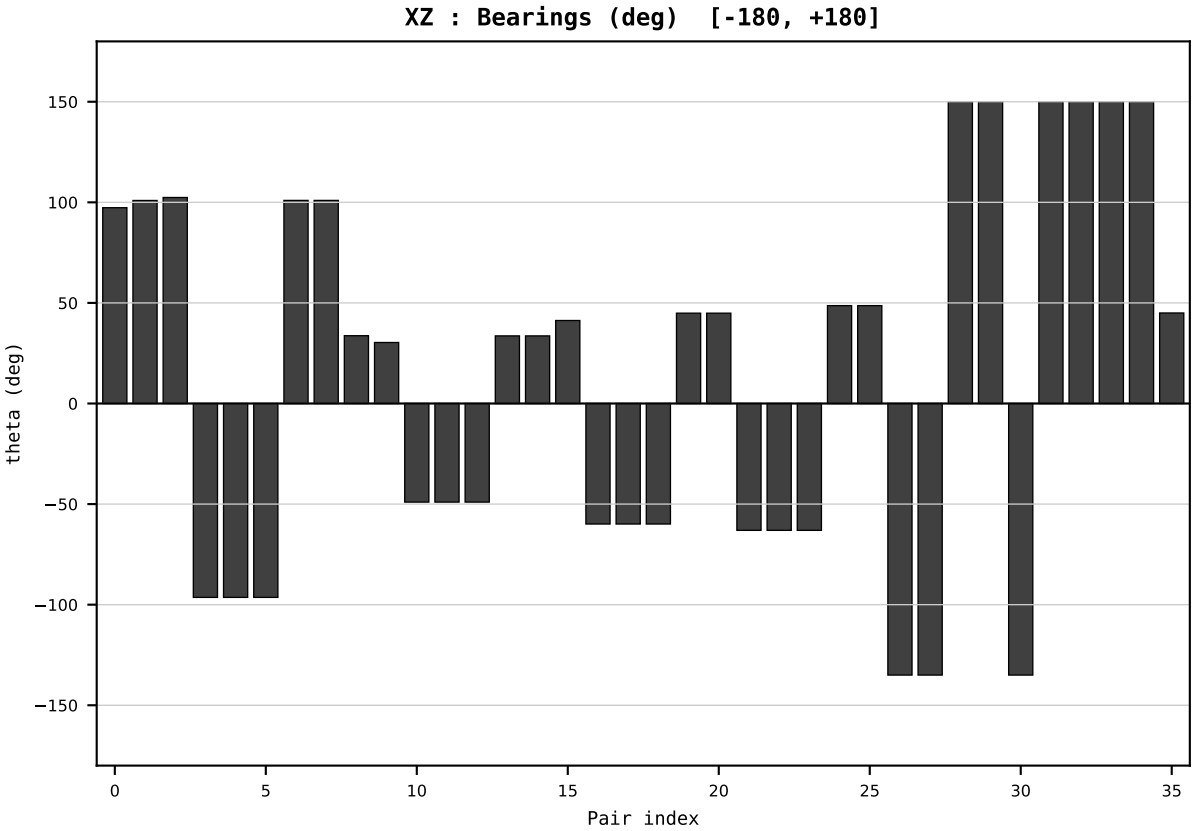
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2020
t : 20 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.469557
Ring = Hs-3
E_metric= 202.4007
kappa_HS= 141070.00
omega = 8.2986 deg
d_A = 2.204987
Helm = Other Fossil
Helm d = -2.040967
DR = 11.857 HLR
DR ratio= 141070.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=21 | Year 2021 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

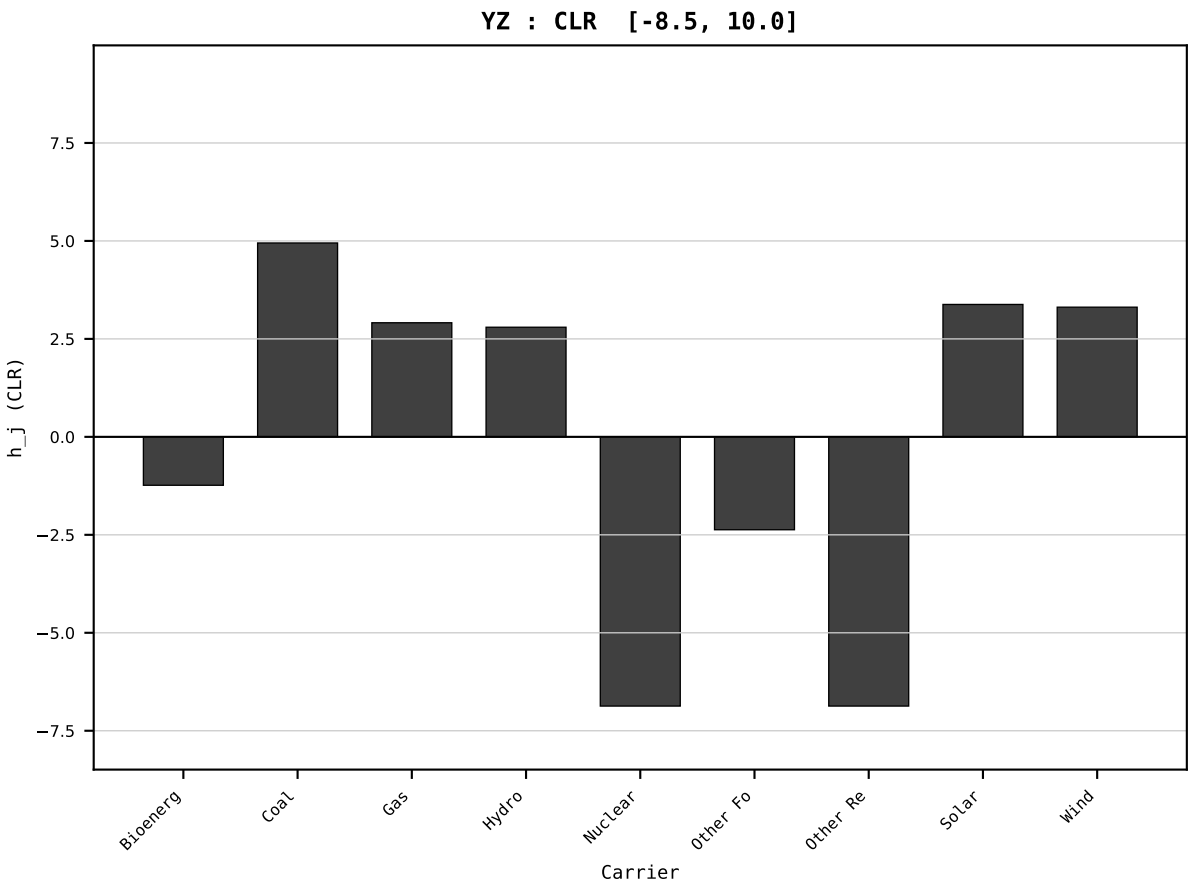
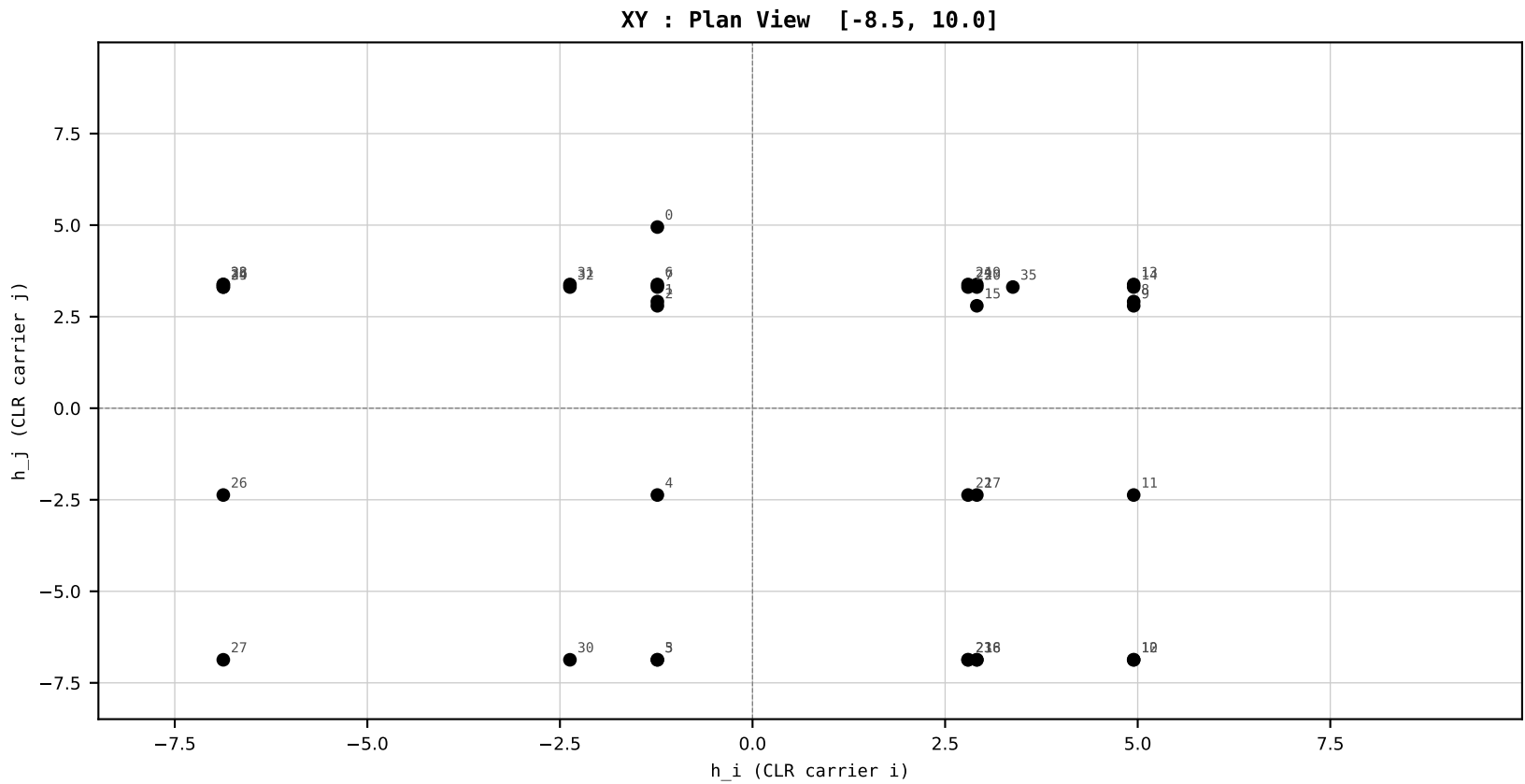
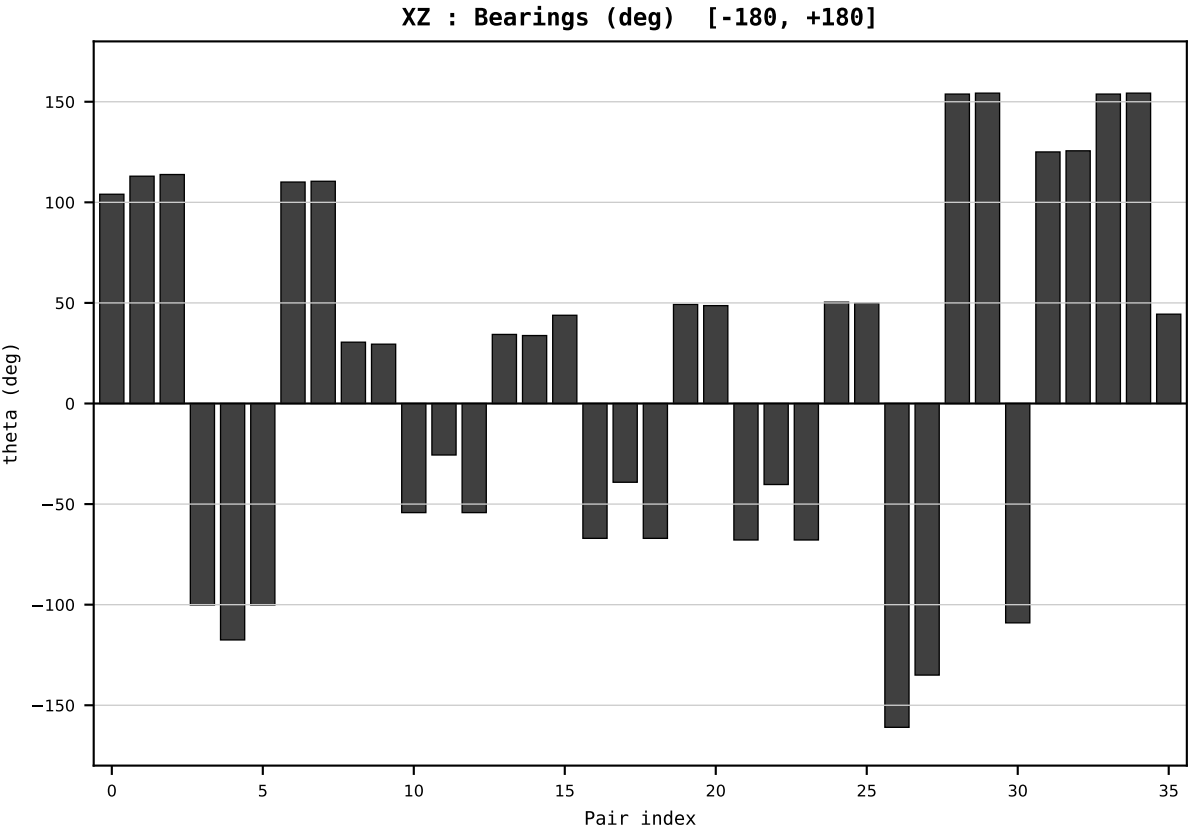
Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2021
t : 21 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.446631
Ring = Hs-3
E_metric= 164.6967
kappa_HS= 135640.00
omega = 17.0092 deg
d_A = 4.232518
Helm = Other Fossil
Helm d = +3.973835
DR = 11.818 HLR
DR ratio= 135640.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



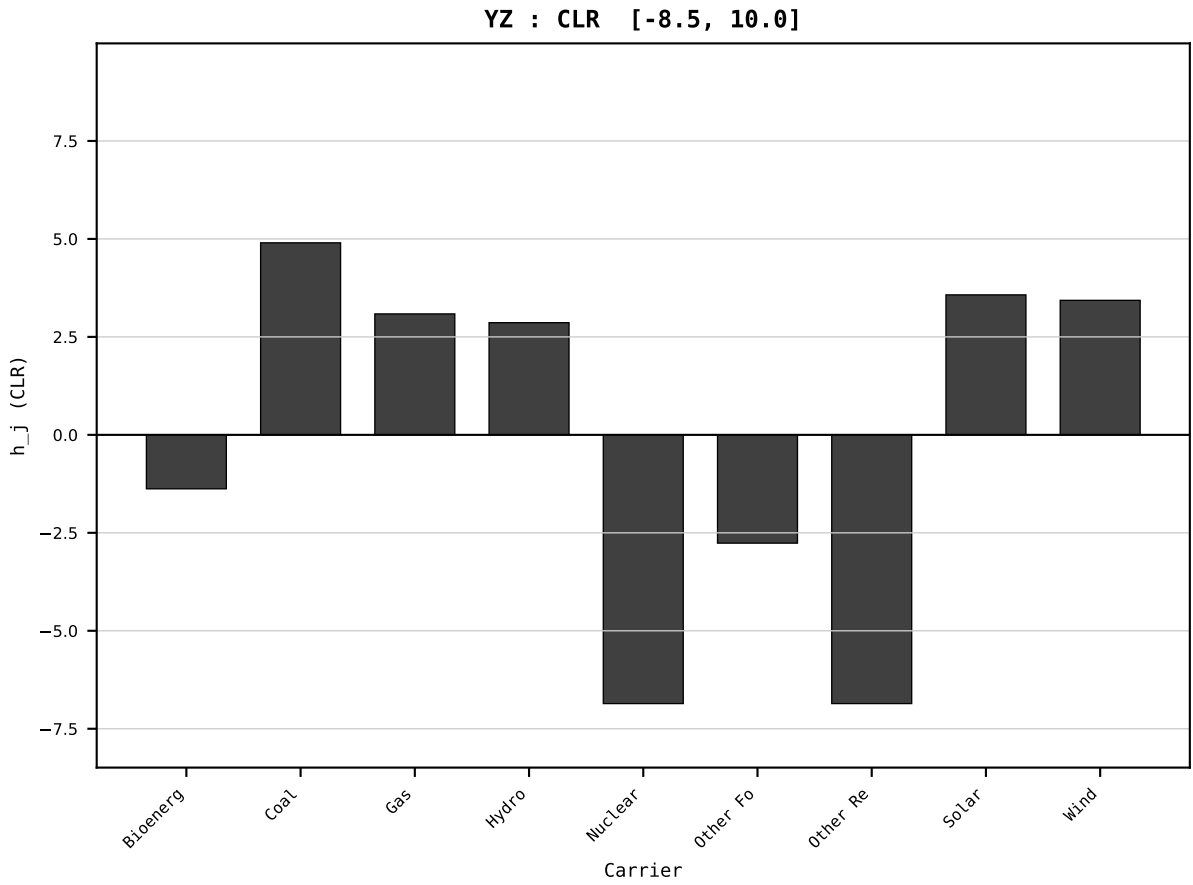
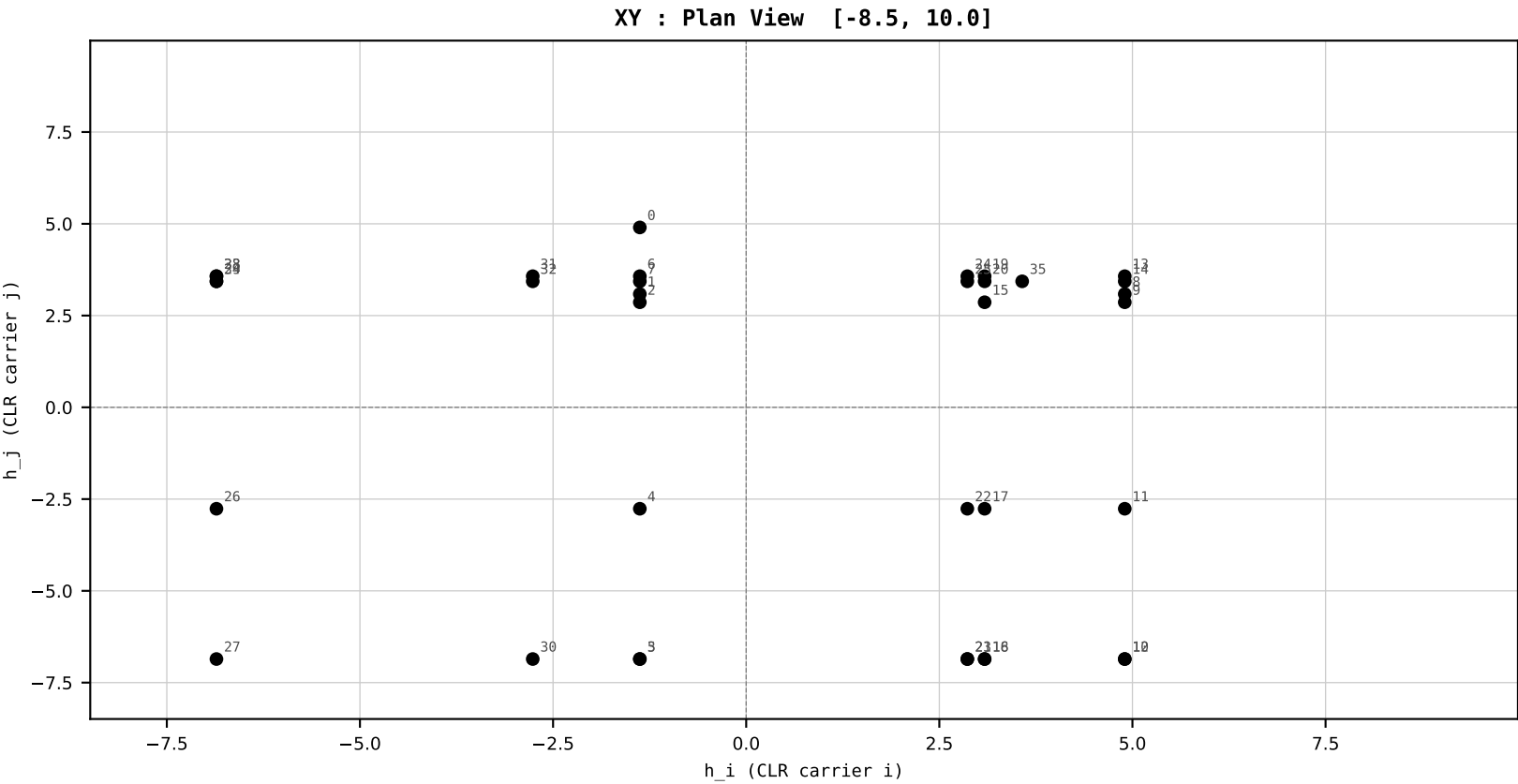
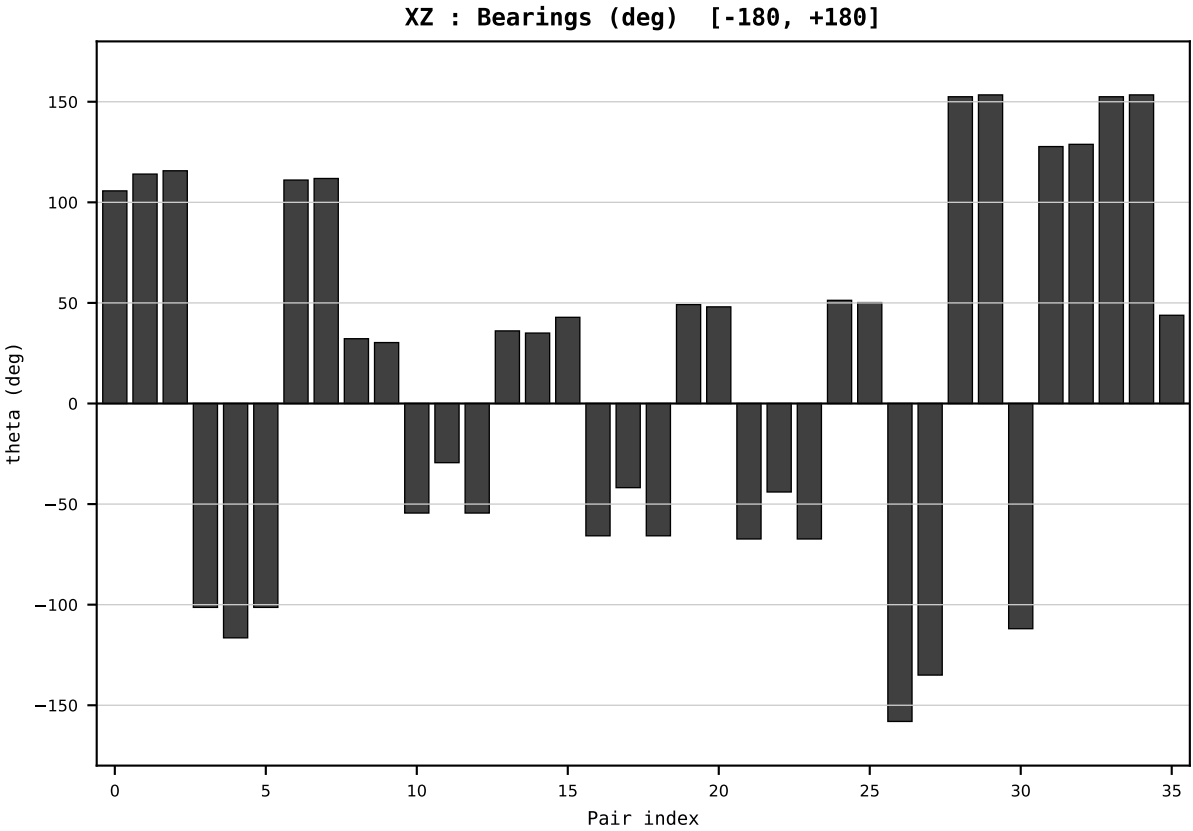
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2022
t : 22 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.412813
Ring = Hs-3
E_metric= 169.8678
kappa_HS= 127790.00
omega = 2.0987 deg
d_A = 0.514148
Helm = Other Fossil
Helm d = -0.392917
DR = 11.758 HLR
DR ratio= 127790.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



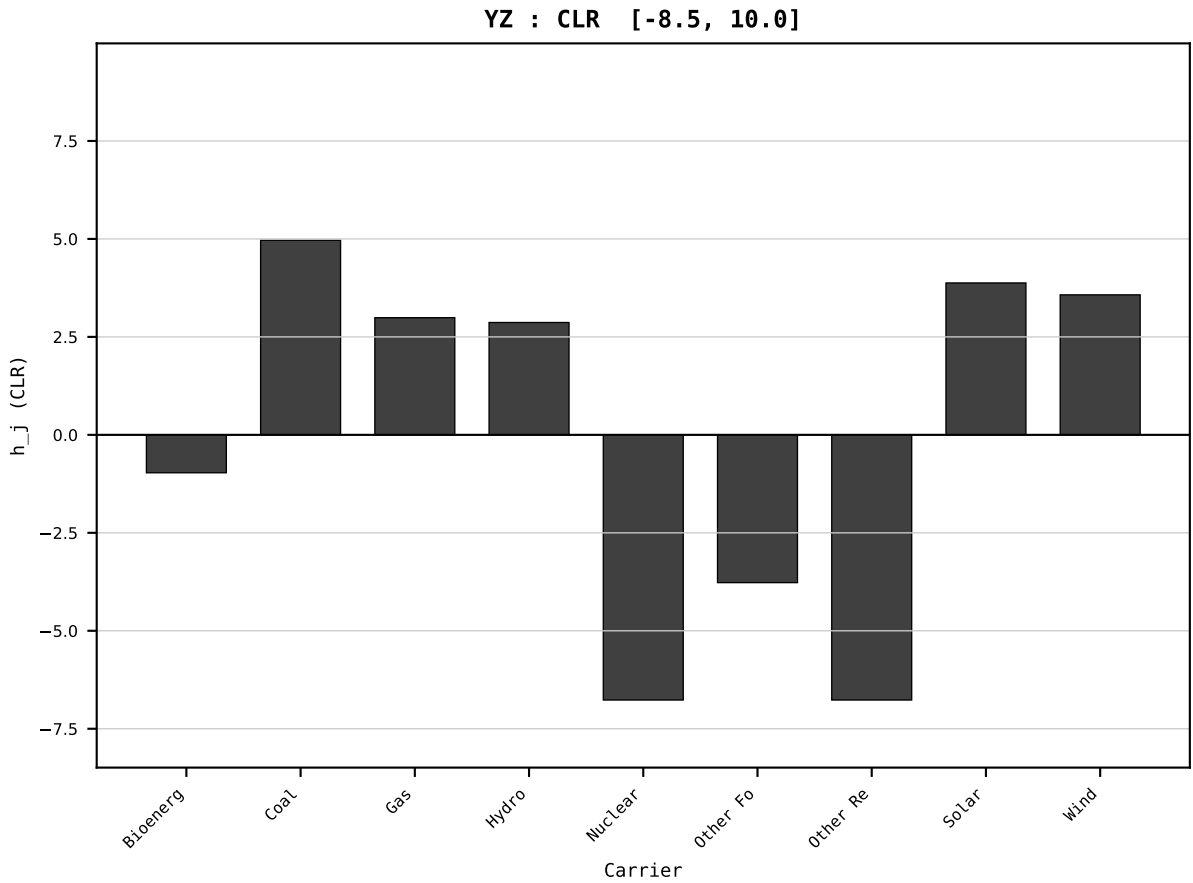
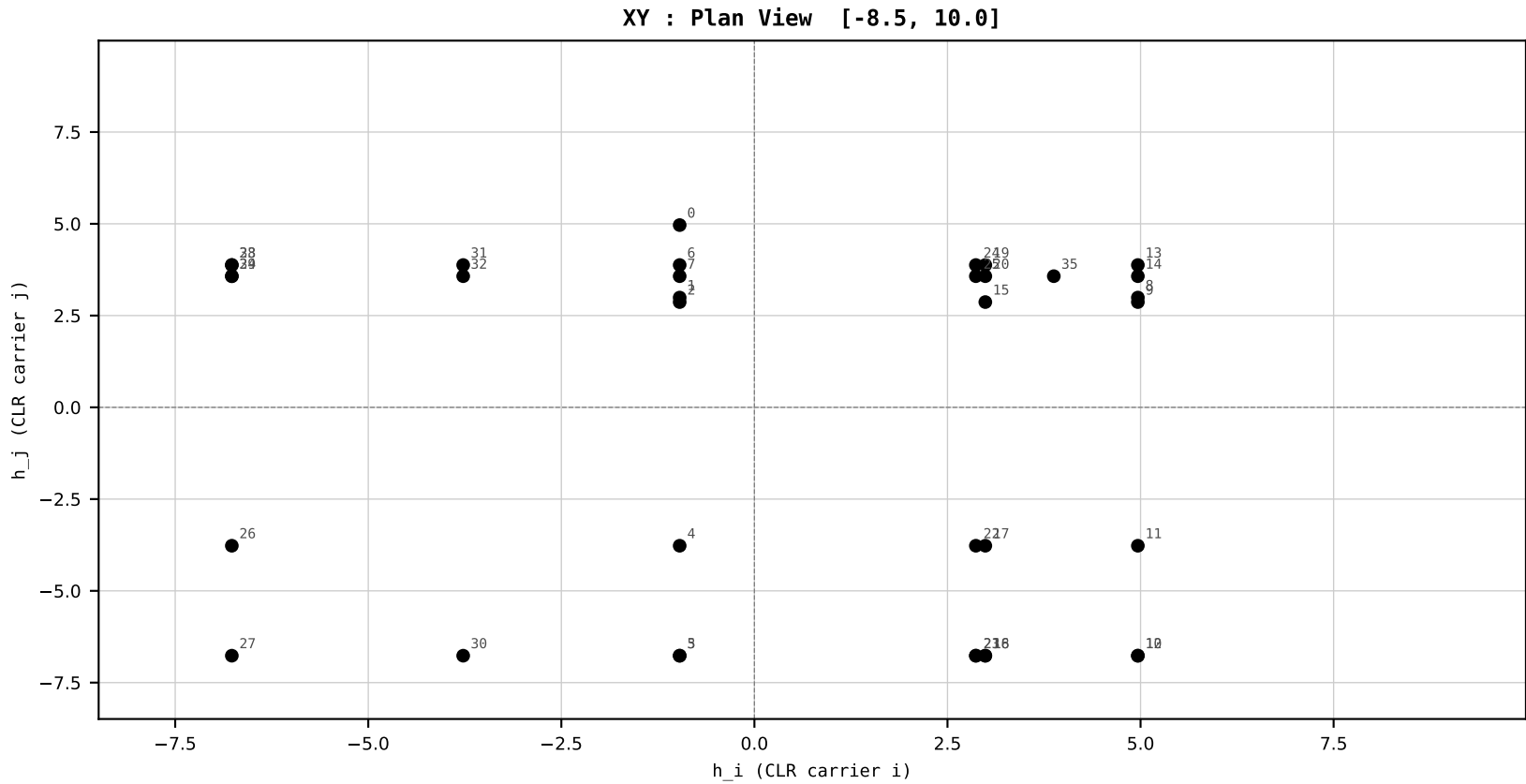
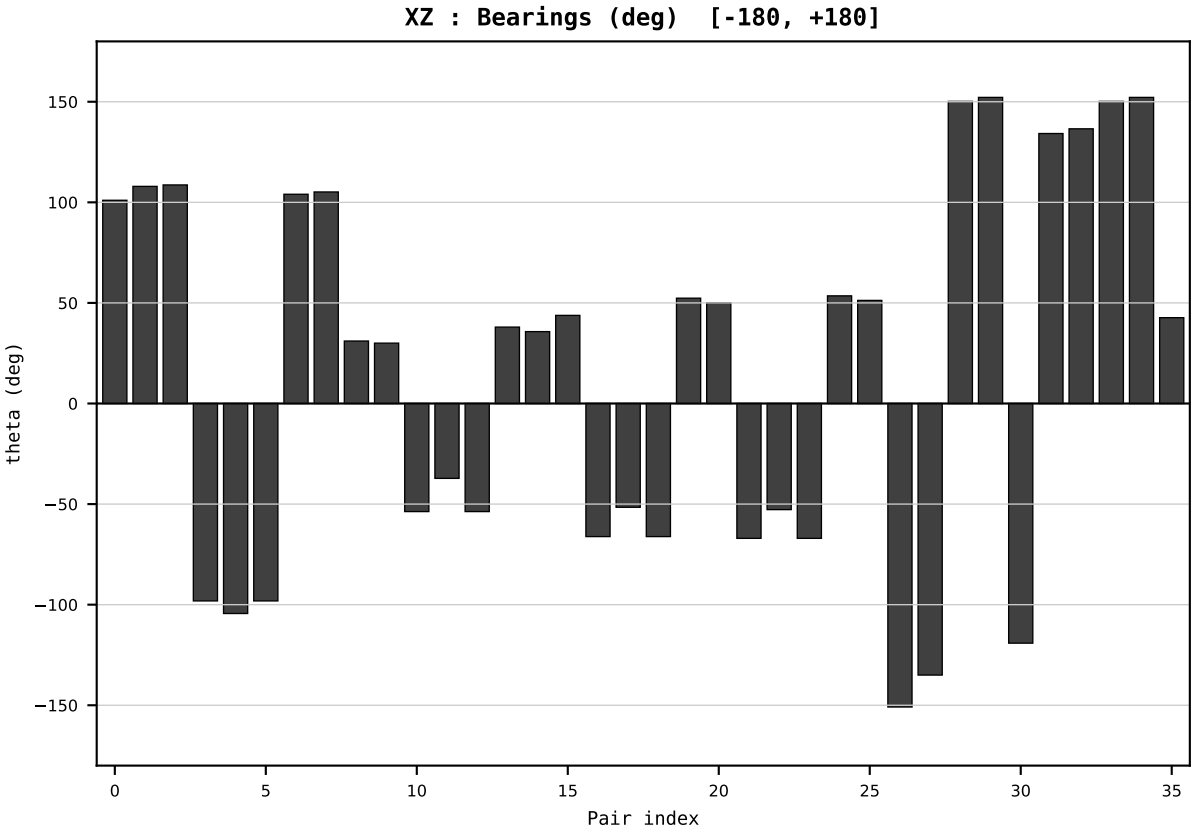
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2023
t : 23 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.409750
Ring = Hs-3
E_metric= 176.3471
kappa_HS= 124490.00
omega = 4.8995 deg
d_A = 1.151287
Helm = Other Fossil
Helm d = -1.007865
DR = 11.732 HLR
DR ratio= 124490.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



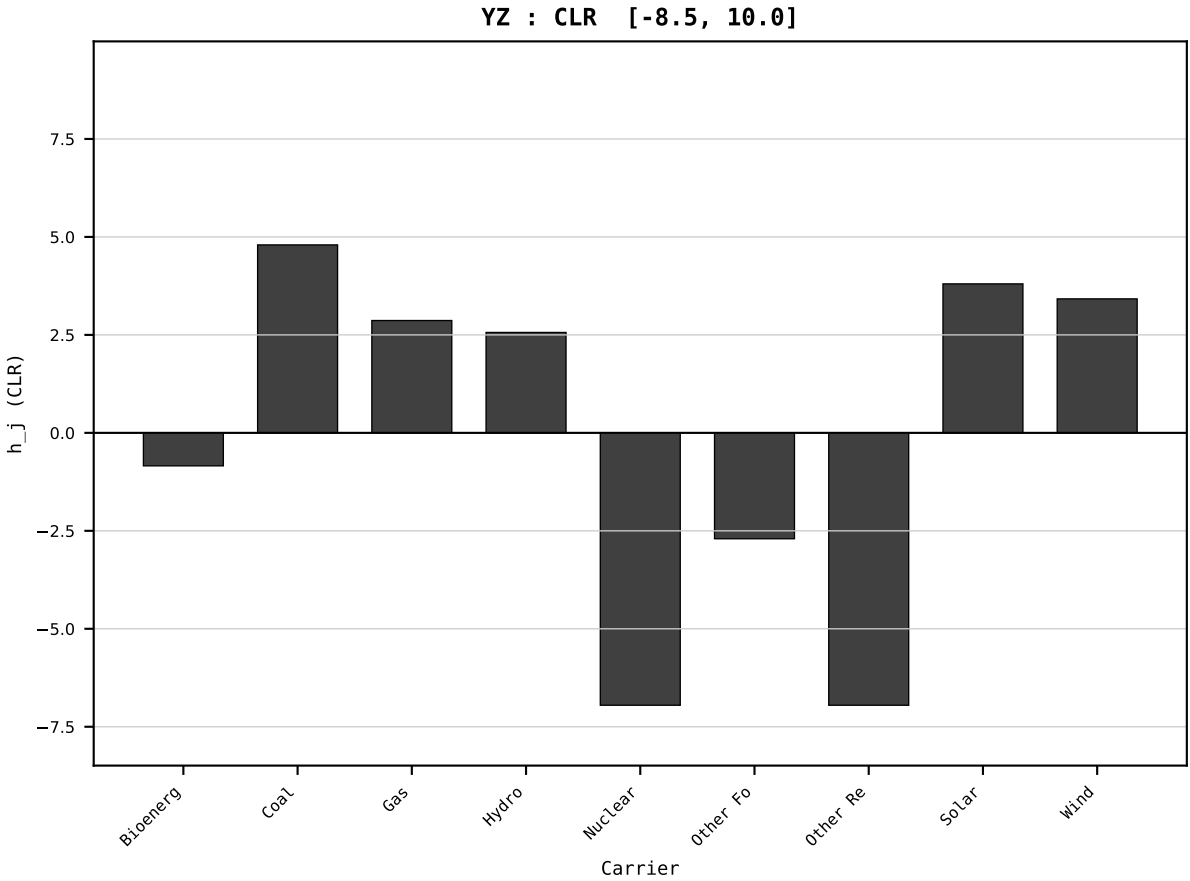
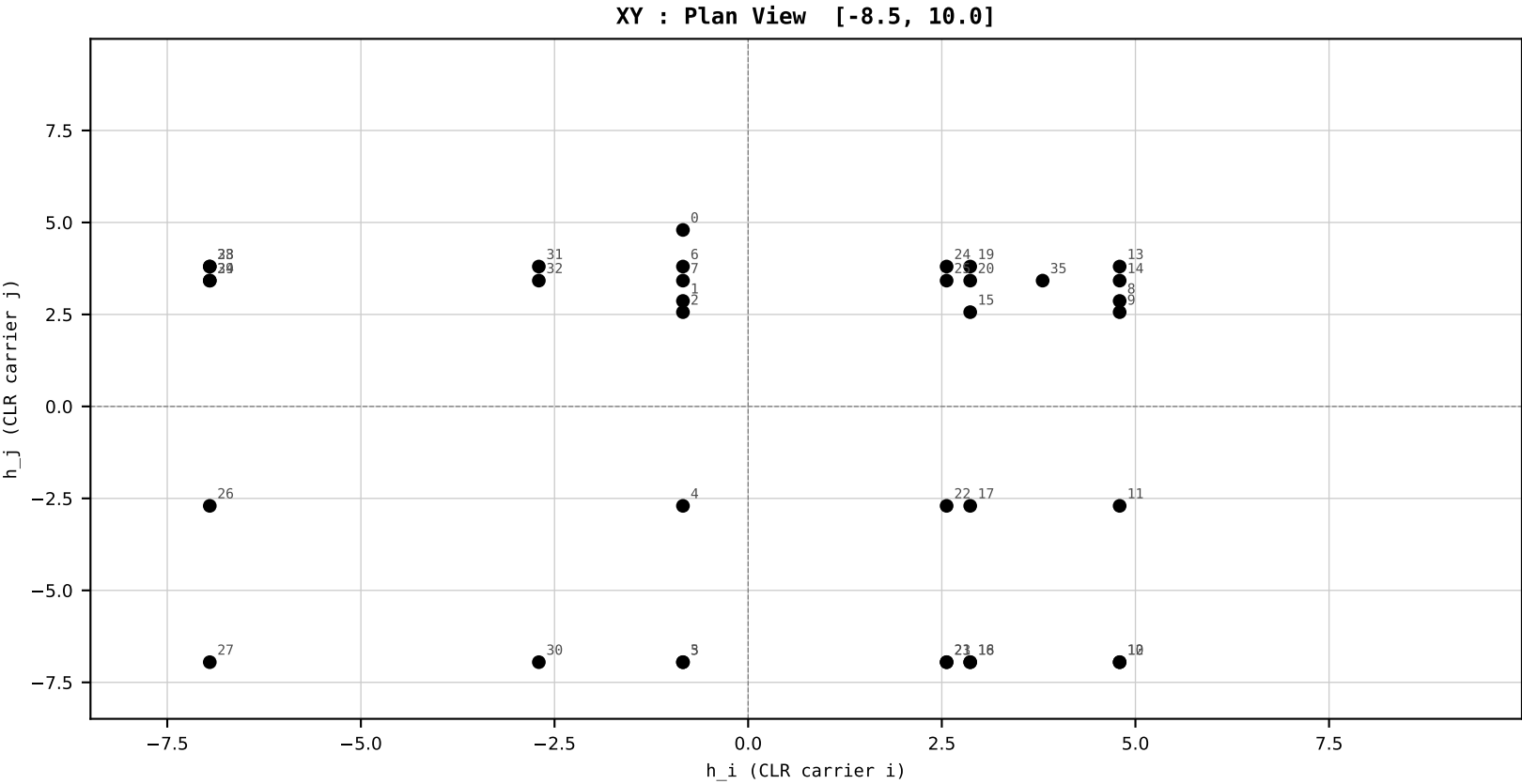
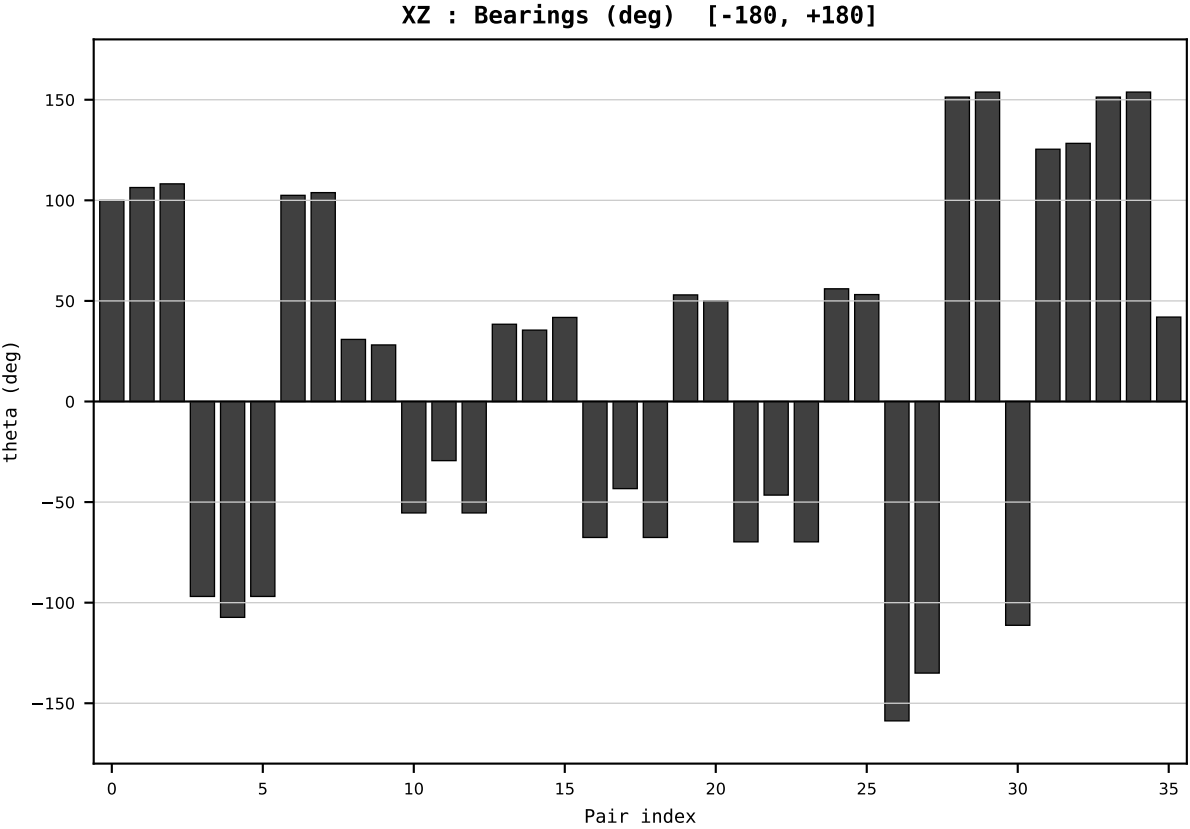
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2024
t : 24 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.407970
Ring = Hs-3
E_metric= 168.5832
kappa_HS= 126430.00
omega = 4.9844 deg
d_A = 1.179587
Helm = Other Fossil
Helm d = +1.068460
DR = 11.747 HLR
DR ratio= 126430.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=25 | Year 2025 | D=9 N=26 pairs=36

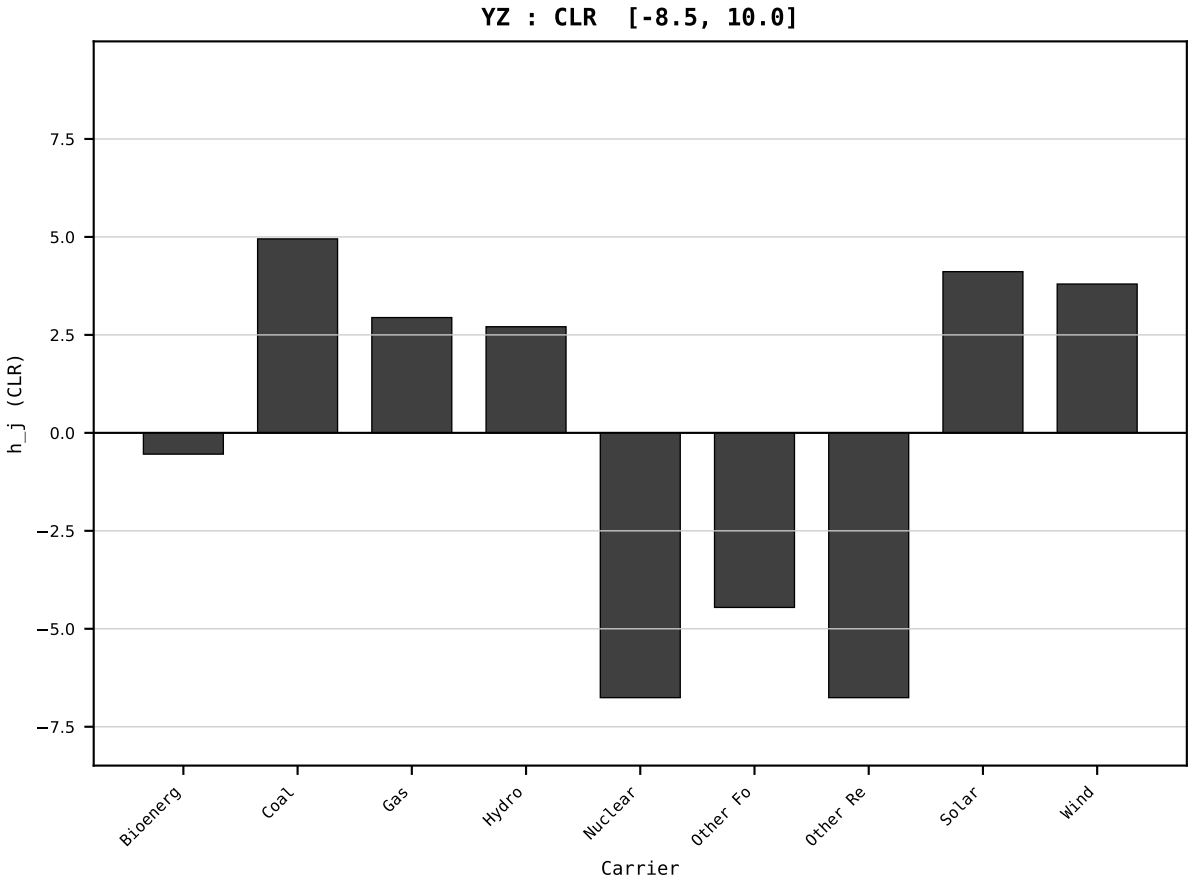
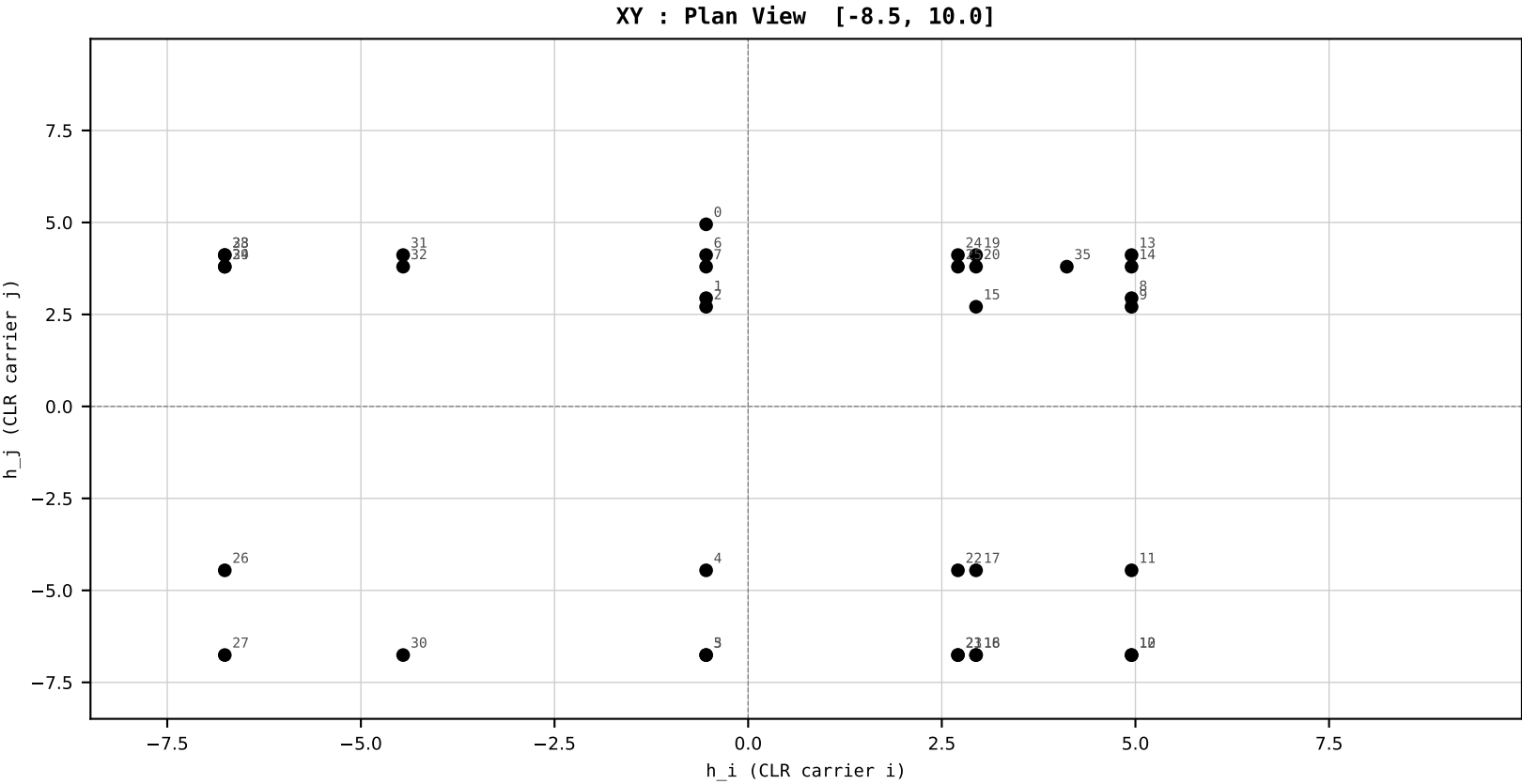
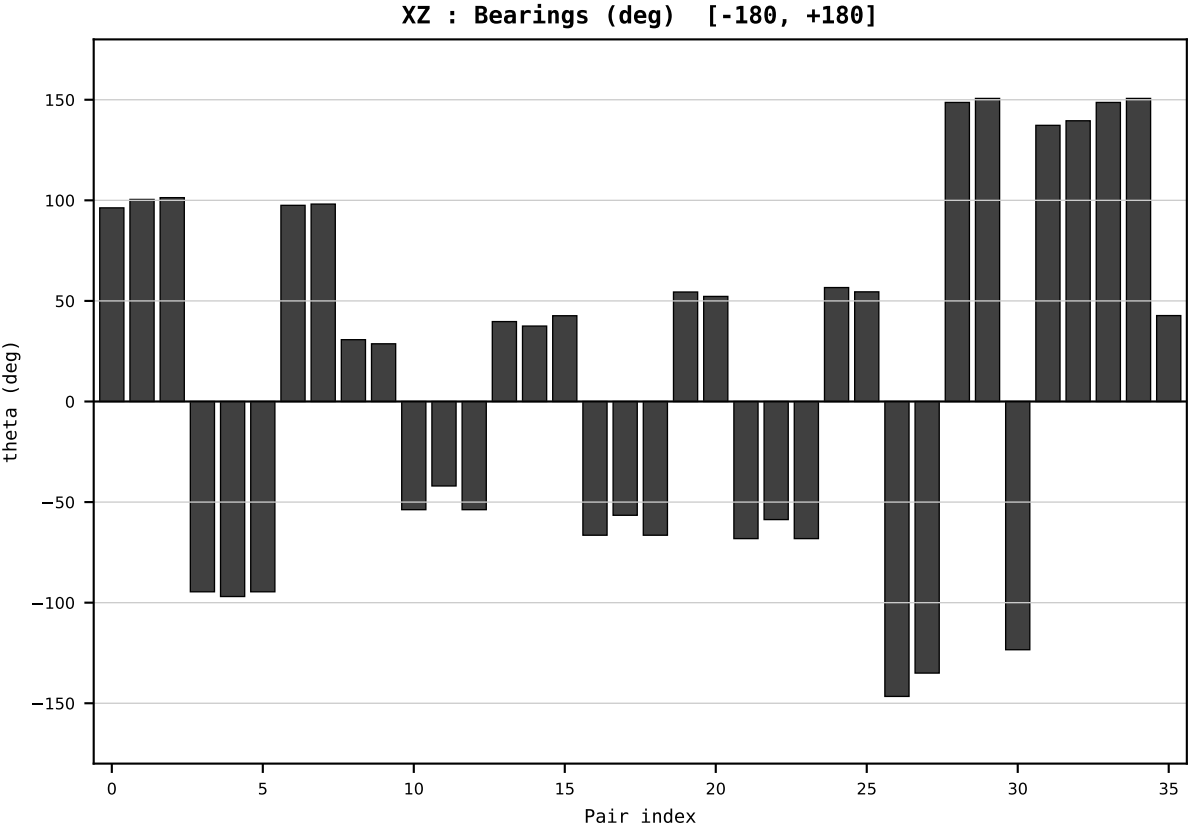
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

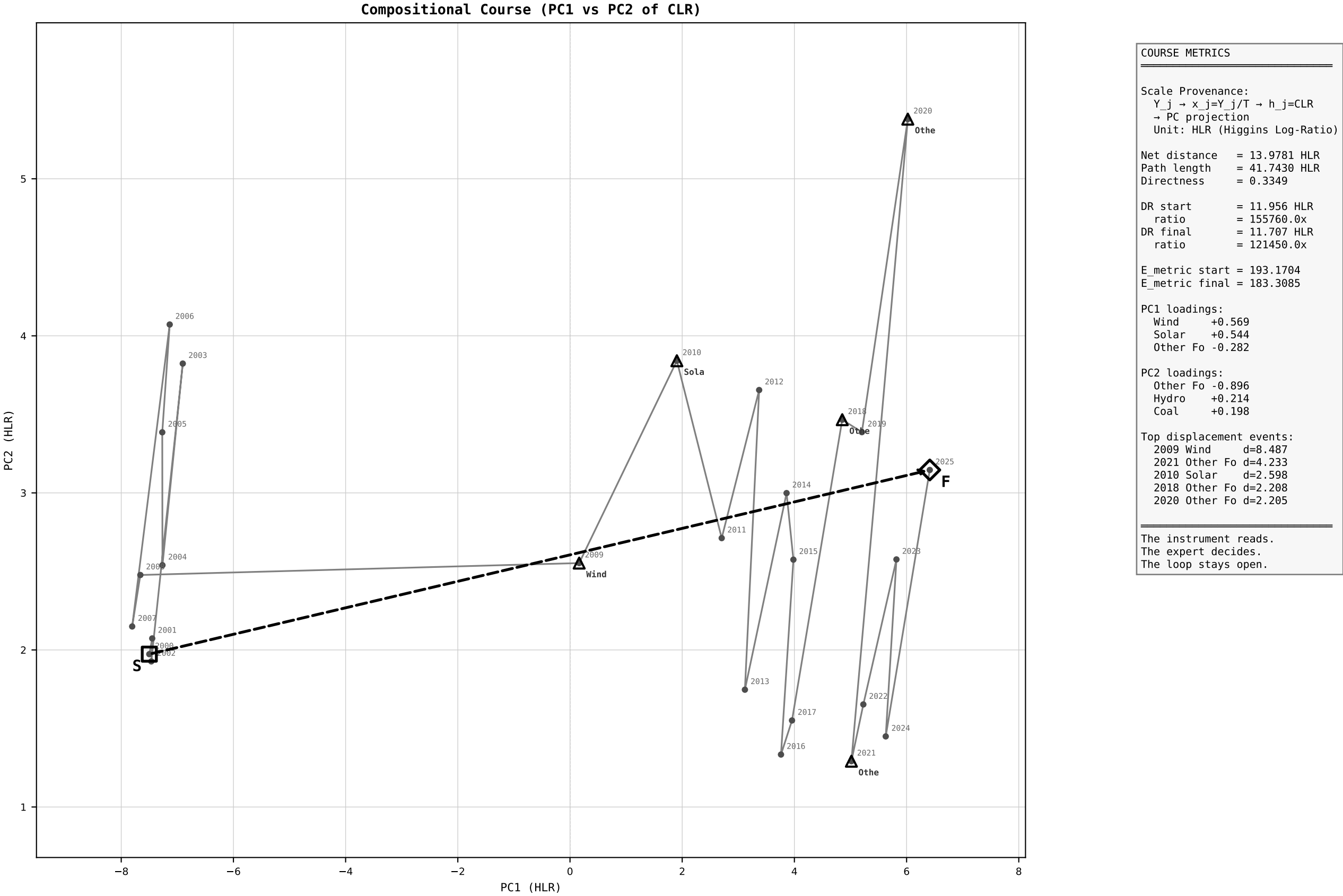
Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2025
t : 25 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.398765
Ring = Hs-3
E_metric= 183.3085
kappa_HS= 121450.00
omega = 7.7580 deg
d_A = 1.877835
Helm = Other Fossil
Helm d = -1.752225
DR = 11.707 HLR
DR ratio= 121450.0x

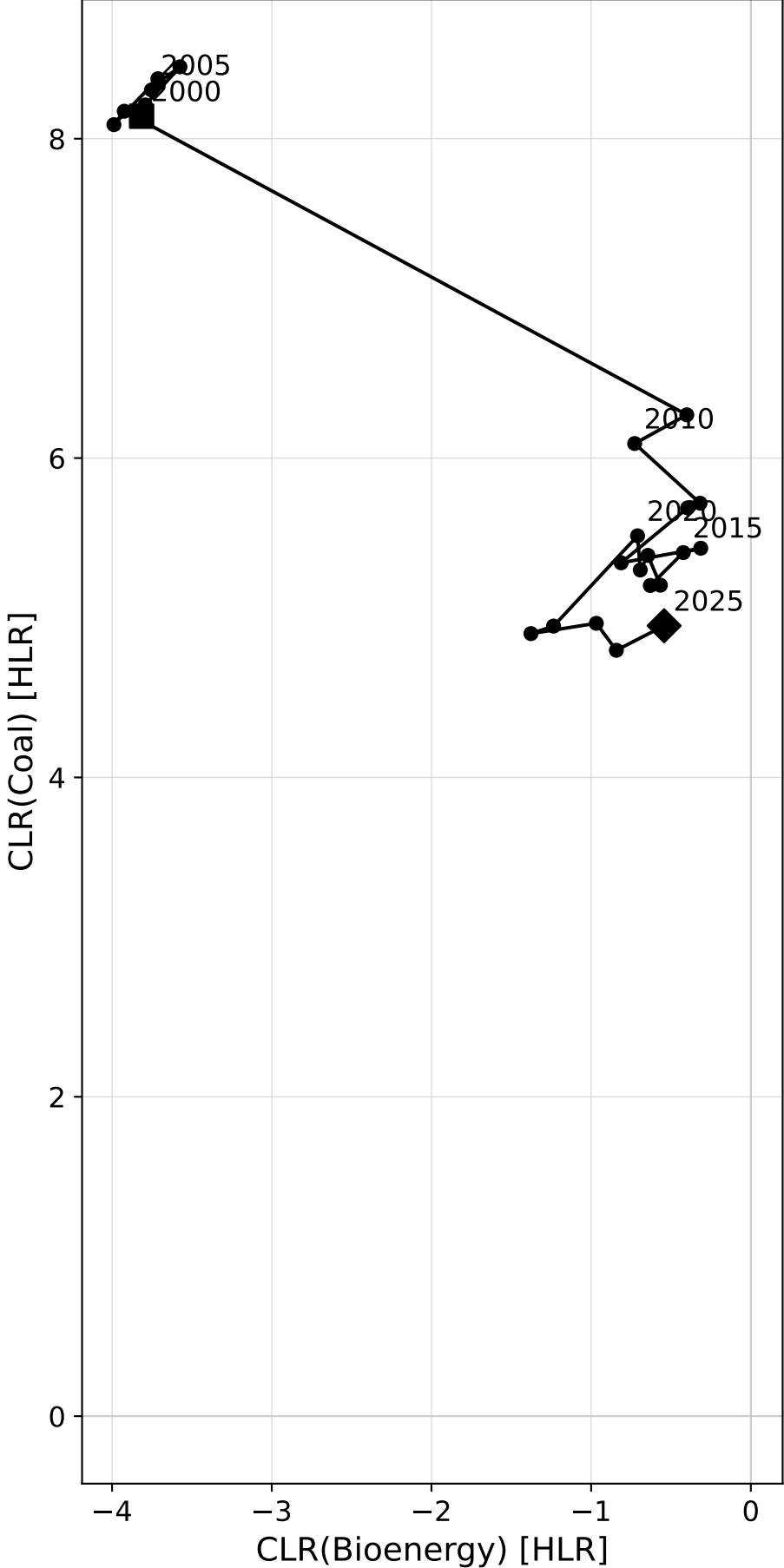
XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind

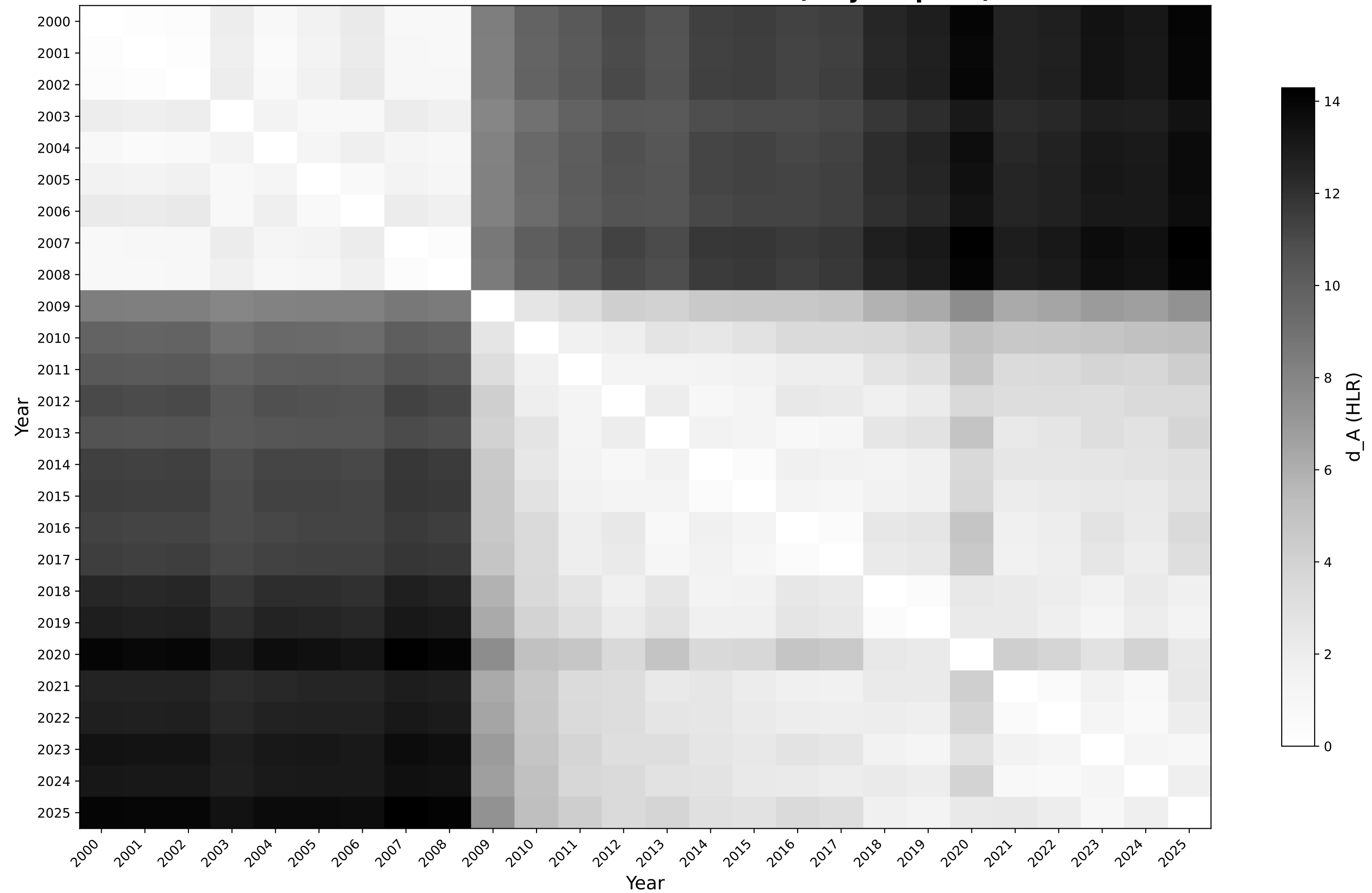




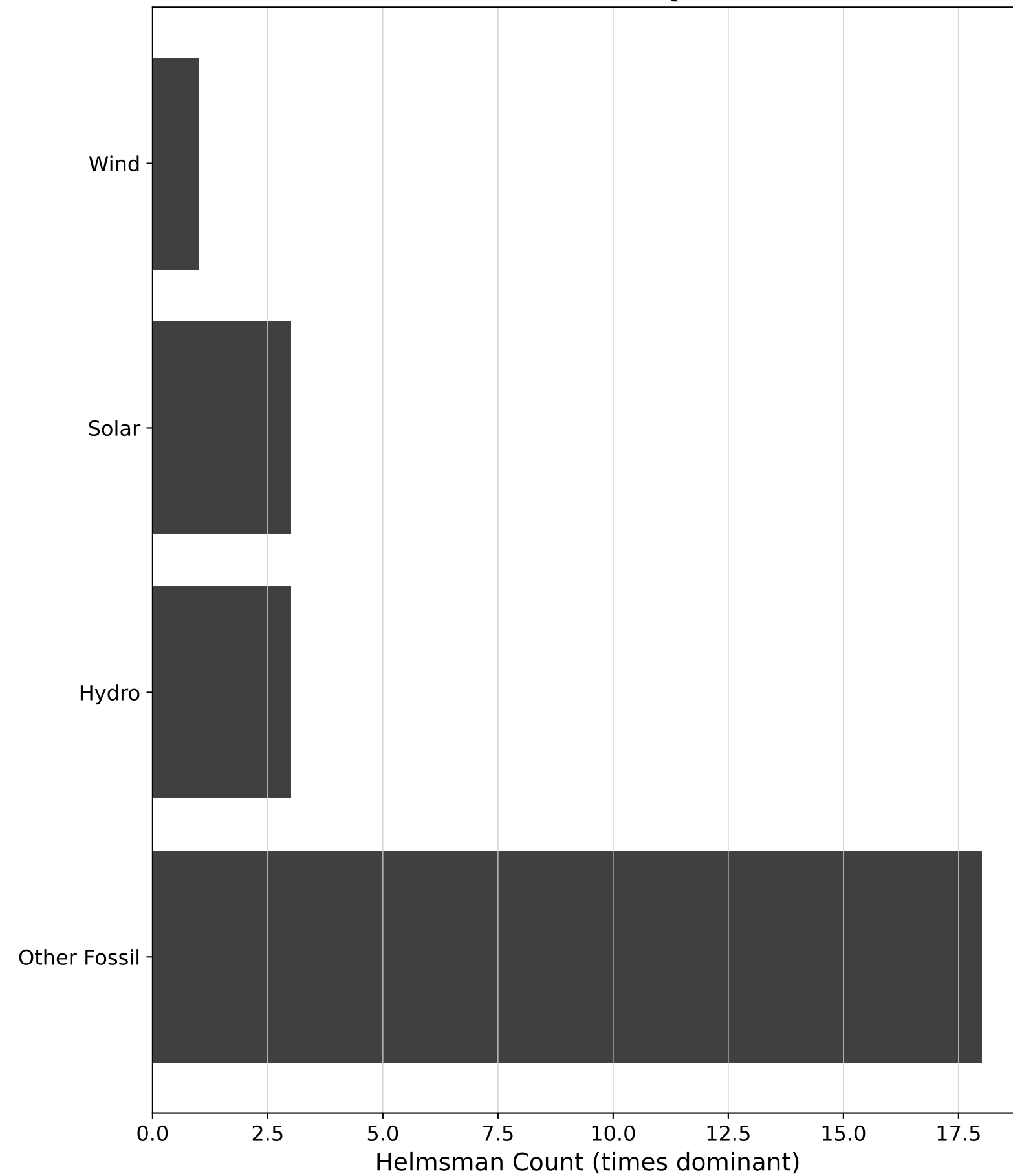
SYSTEM COURSE PLOT — XY Plan View (CLR Space)



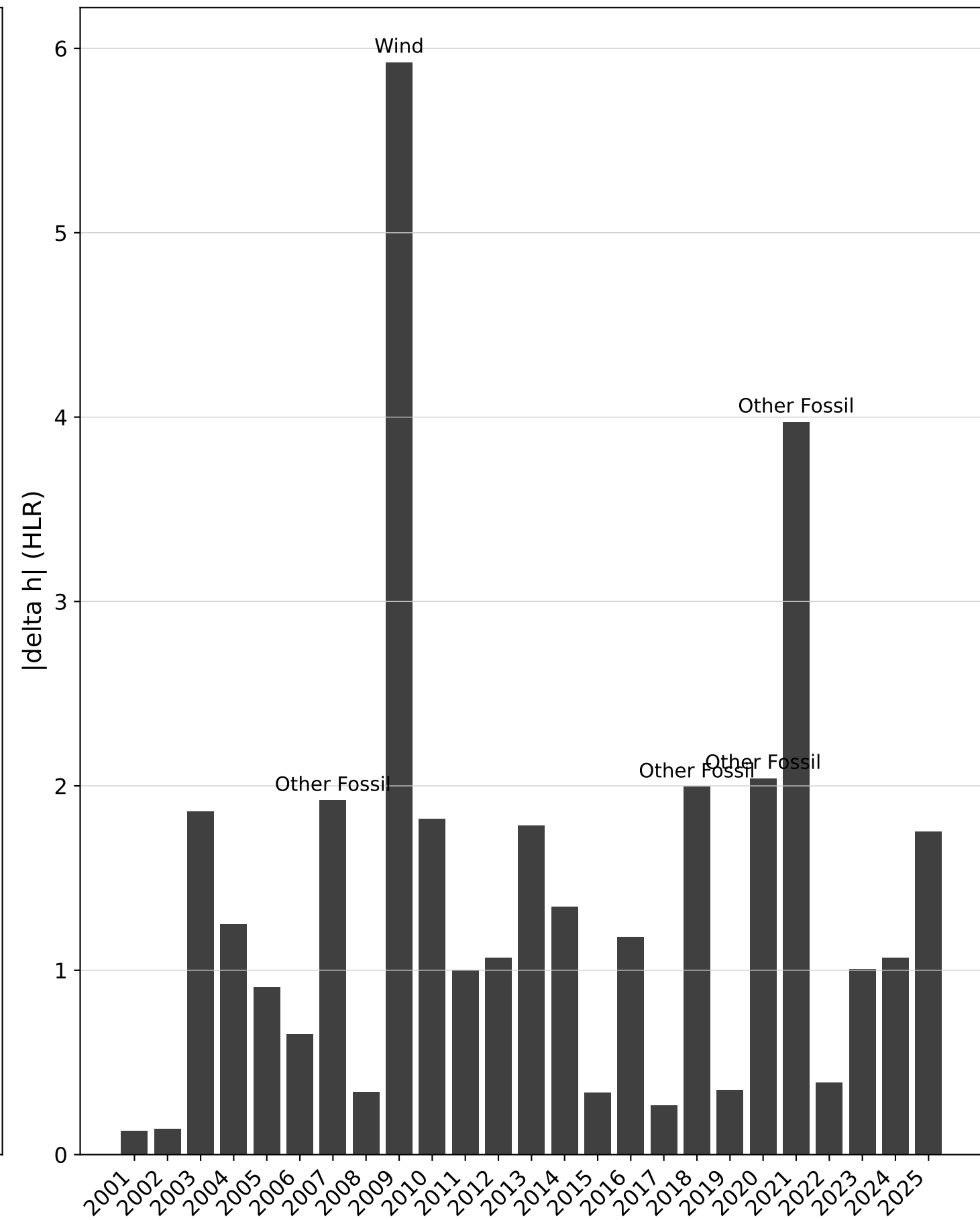
STAGE 2 — Aitchison Distance Matrix (all year pairs)



HELMSMAN FREQUENCY



HELMSMAN DISPLACEMENT PER YEAR



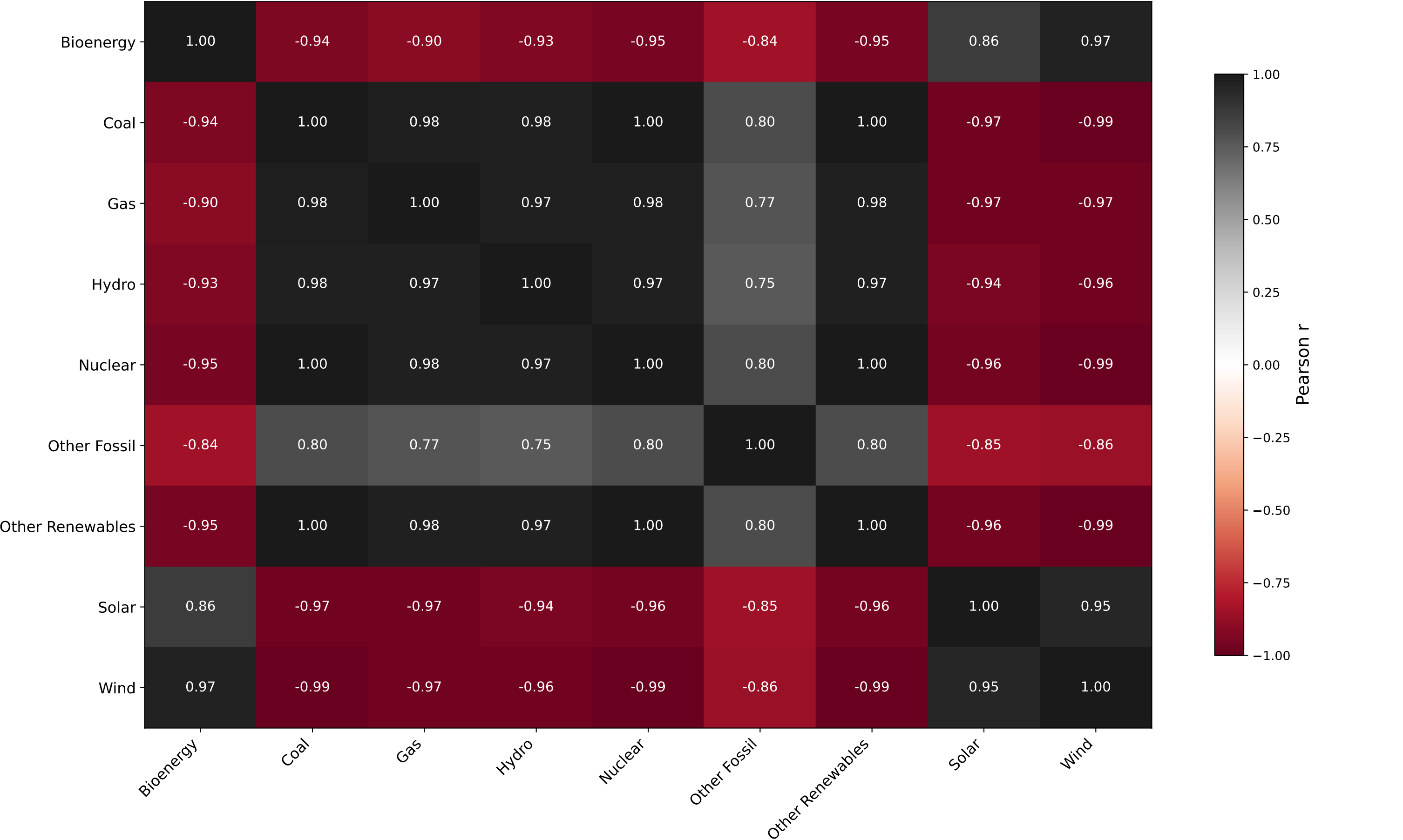
STAGE 2 — Pairwise Divergence Ranking (Top 15)

Rank	Year A	Year B	d_A (HLR)	Angle (deg)	Dominant Carrier
1	2007	2025	14.2906	61.2	Solar
2	2007	2020	14.2291	59.4	Solar
3	2008	2025	14.1193	60.6	Solar
4	2008	2020	14.0101	58.6	Solar
5	2000	2025	13.9781	61.2	Solar
6	2000	2020	13.9643	59.5	Solar
7	2002	2025	13.9500	61.3	Solar
8	2002	2020	13.9430	59.6	Solar
9	2001	2025	13.9193	61.0	Solar
10	2001	2020	13.8894	59.2	Solar
11	2004	2025	13.7119	60.1	Solar
12	2005	2025	13.7028	59.0	Solar
13	2007	2023	13.6669	58.8	Solar
14	2004	2020	13.6104	58.0	Solar
15	2006	2025	13.6028	58.1	Solar

STAGE 3 — Triadic Metric Triangle Area Ranking (Top 10)

Rank	Year A	Year B	Year C	Area (HLR^2)	Imbalance
1	2006	2009	2020	28.4334	1.75
2	2005	2009	2020	27.6942	1.77
3	2007	2009	2020	27.5941	1.88
4	2008	2009	2020	27.5393	1.85
5	2003	2009	2020	27.4006	1.72
6	2007	2020	2021	26.8451	3.36
7	2008	2020	2021	26.7023	3.31
8	2007	2020	2024	26.6745	3.59
9	2004	2009	2020	26.6131	1.79
10	2000	2009	2020	26.5906	1.84

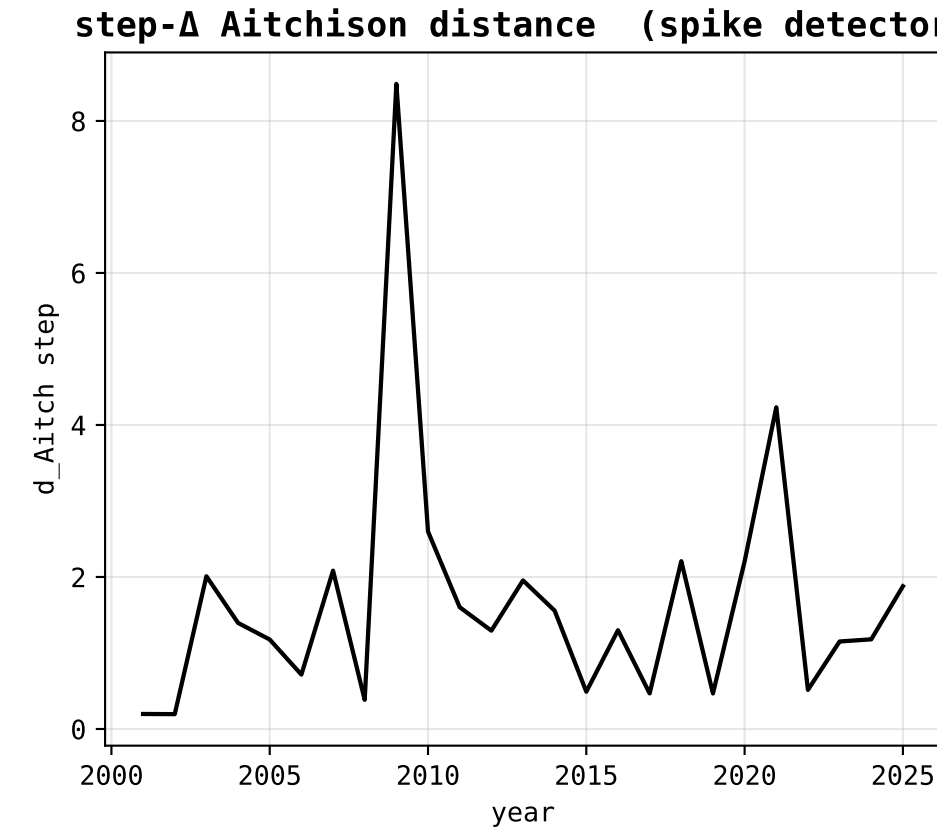
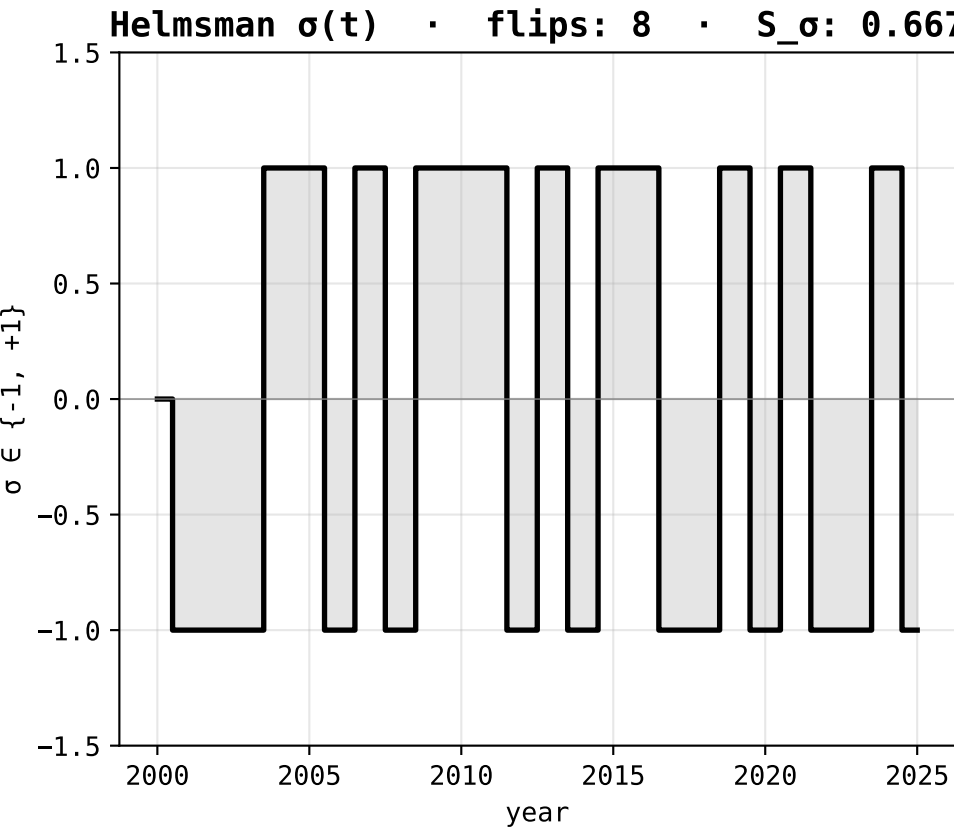
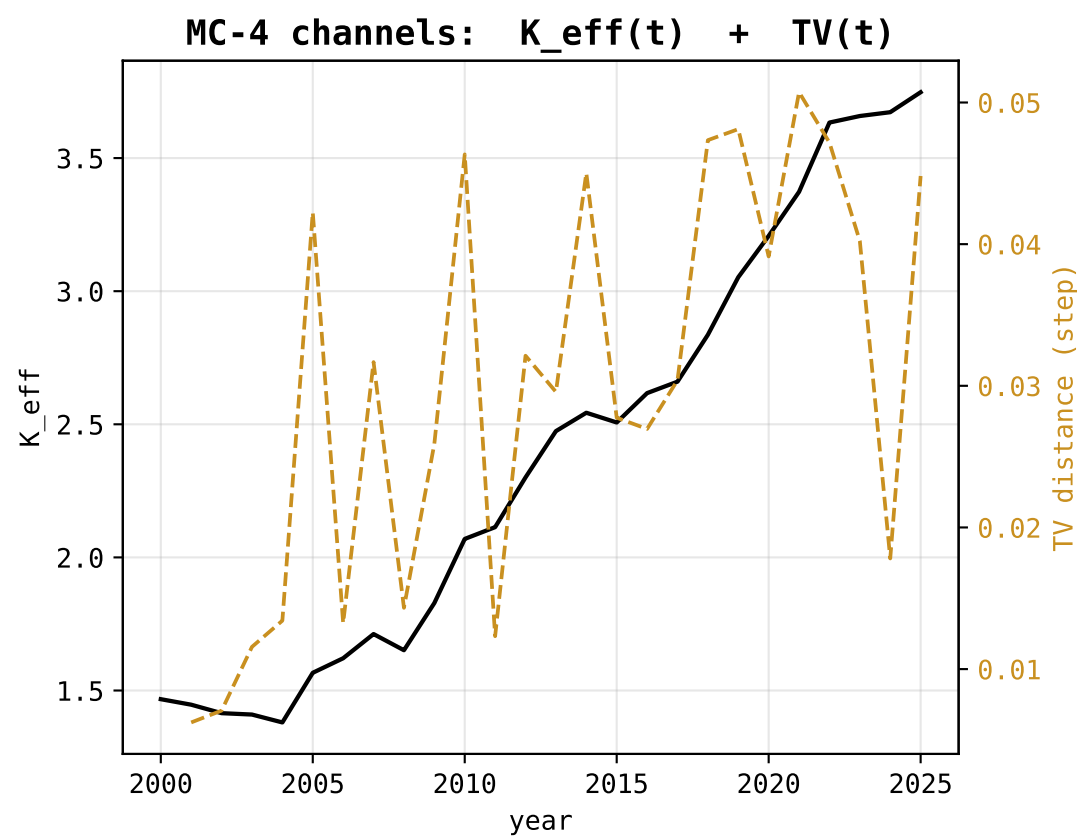
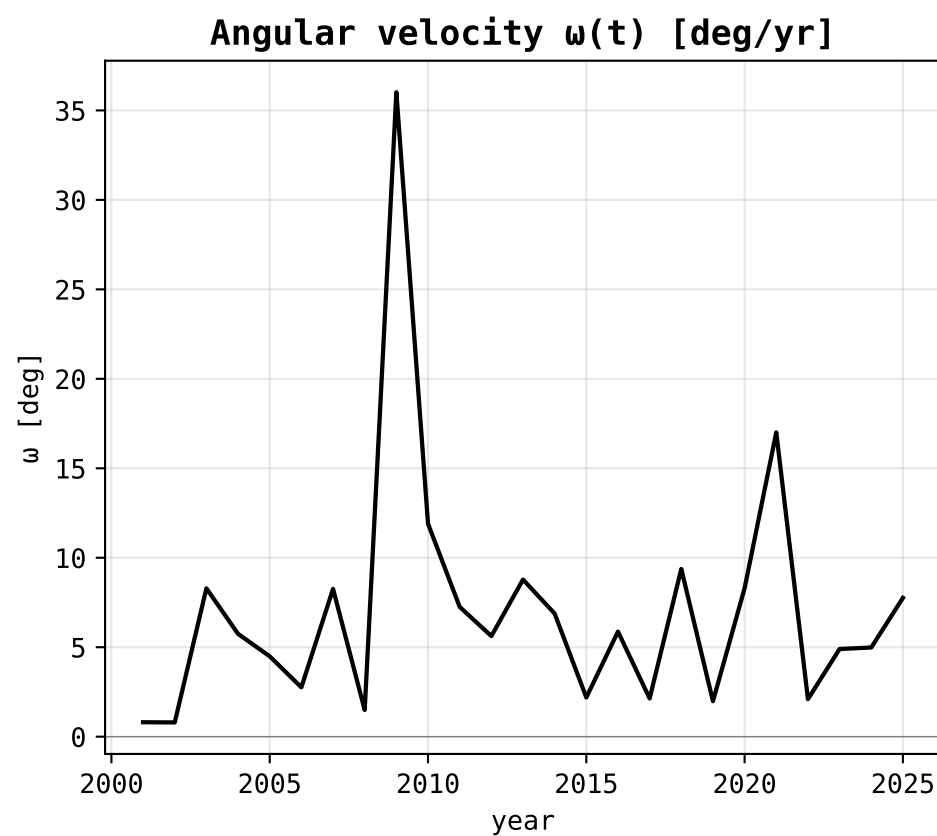
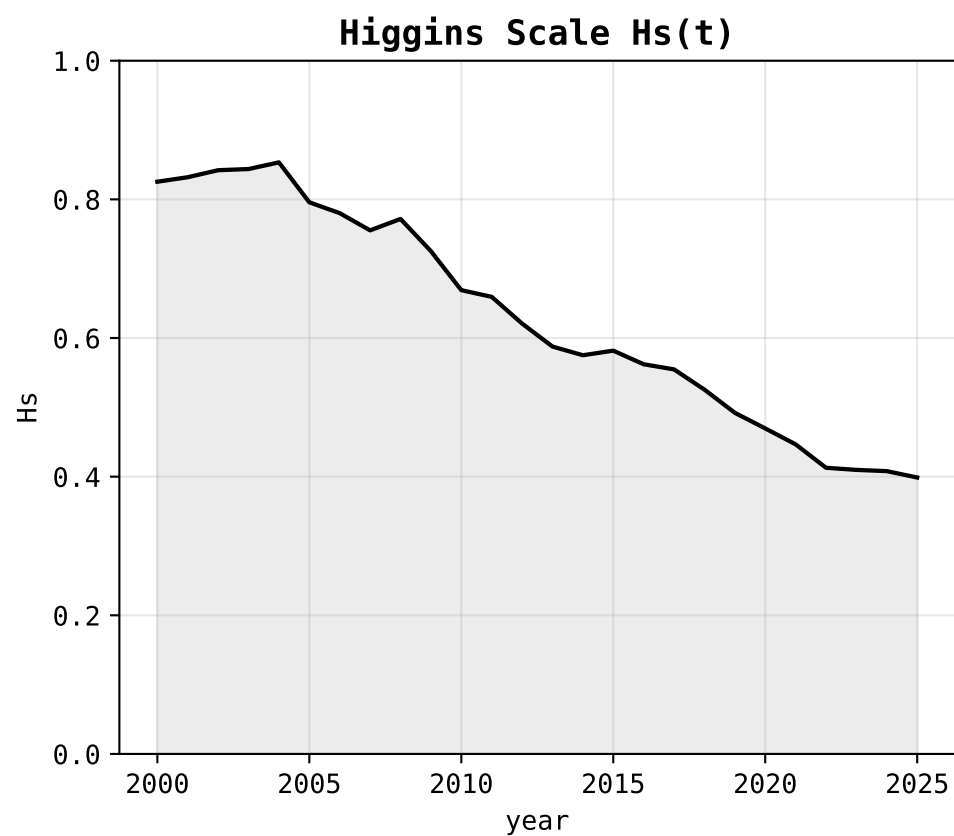
STAGE 3 — Carrier CLR Trajectory Correlation



NAVIGATION SUMMARY — All Stage 1 Metrics by Year

Year	Hs	Ring	E_metric	omega(deg)	d_A	Helmsman	kappa_max
2000	0.8255	Hs-5	193.2	0.00	0.0000	---	172235
2001	0.8319	Hs-5	192.4	0.81	0.1976	Hydro	174665
2002	0.8421	Hs-5	191.5	0.80	0.1952	Hydro	178405
2003	0.8437	Hs-5	194.1	8.29	2.0096	Other Fossil	181915
2004	0.8534	Hs-5	191.1	5.75	1.3954	Other Fossil	187875
2005	0.7957	Hs-5	202.9	4.49	1.1771	Hydro	195875
2006	0.7801	Hs-5	208.4	2.76	0.7157	Other Fossil	204985
2007	0.7553	Hs-5	209.6	8.26	2.0833	Other Fossil	207835
2008	0.7717	Hs-5	207.5	1.49	0.3837	Other Fossil	208385
2009	0.7253	Hs-4	148.4	36.02	8.4868	Wind	208382
2010	0.6690	Hs-4	158.3	11.90	2.5981	Solar	208922
2011	0.6594	Hs-4	150.6	7.25	1.6025	Solar	206562
2012	0.6208	Hs-4	161.1	5.62	1.2938	Other Fossil	202722
2013	0.5877	Hs-4	150.9	8.78	1.9545	Other Fossil	199292
2014	0.5752	Hs-4	161.4	6.88	1.5568	Other Fossil	214382
2015	0.5817	Hs-4	159.5	2.19	0.4902	Other Fossil	218182
2016	0.5621	Hs-4	154.5	5.87	1.2992	Other Fossil	221562
2017	0.5547	Hs-4	155.9	2.14	0.4680	Other Fossil	220892
2018	0.5256	Hs-4	173.7	9.37	2.2081	Other Fossil	223262
2019	0.4921	Hs-3	176.2	1.98	0.4671	Solar	224862
2020	0.4696	Hs-3	202.4	8.30	2.2050	Other Fossil	222553
2021	0.4466	Hs-3	164.7	17.01	4.2325	Other Fossil	224142
2022	0.4128	Hs-3	169.9	2.10	0.5141	Other Fossil	228882
2023	0.4098	Hs-3	176.3	4.90	1.1513	Other Fossil	230252
2024	0.4080	Hs-3	168.6	4.98	1.1796	Other Fossil	237522
2025	0.3988	Hs-3	183.3	7.76	1.8778	Other Fossil	242222

CNQ v2.0.0 – Quaternion View – AUS Australia · D=9 · N=26



CNQ DIAGNOSTICS · AUS (D = 9)

CHSH joint-coherence:
not computed at D = 9
(CHSH requires D=8 twin factoring;
D = 9 does not factor as 4x4)

Twin-quaternion factor:
not computed at D = 9
(twin requires D=8 = 4+4 partition)

Attractor fit:
amplitude A: 1.438
contraction λ : -0.0211
period: 1
fitted: False

Helmsman family:
total flips: 8
stability S_σ : 0.667

§ CHN China

EMBER electricity-generation series · 2000-2025 · N = 26 · D = 8

INPUT

```
source: ember_CHN_*_generation_TWh.csv
N records: 26
D carriers: 8
Bioenergy
  Coal
  Gas
  Hydro
  Nuclear
  Other Fossil
  Solar
  Wind
source_sha256: f47e4af61ed44a1760162a13...
```

HEADLINE METRICS

```

Helmsman flips: ?
Helmsman stability S_σ: {'global': 0.5, 'rolling': [0.5714285714285714, 0.4285714285714286, 0.5714285714285714]}
IR class: ?
Attractor amplitude: ?

CHSH S: ?
Tsirelson bound:  $2\sqrt{2} \approx 2.828$ 

Bearing pairs  $(D(D-1)/2)$ : 28

CNT v3.1.0 content_sha: ?...
CNQ v2.0.0 content_sha: ?...

```

Following pages: Stage 1 (XY/XZ/YZ $\times$ N years + course) $\rightarrow$ Stage 2/3 (7 pages) $\rightarrow$ CNQ (2 pages).

SECTION PLATE t=0 | Year 2000 | D=8 N=26 pairs=28

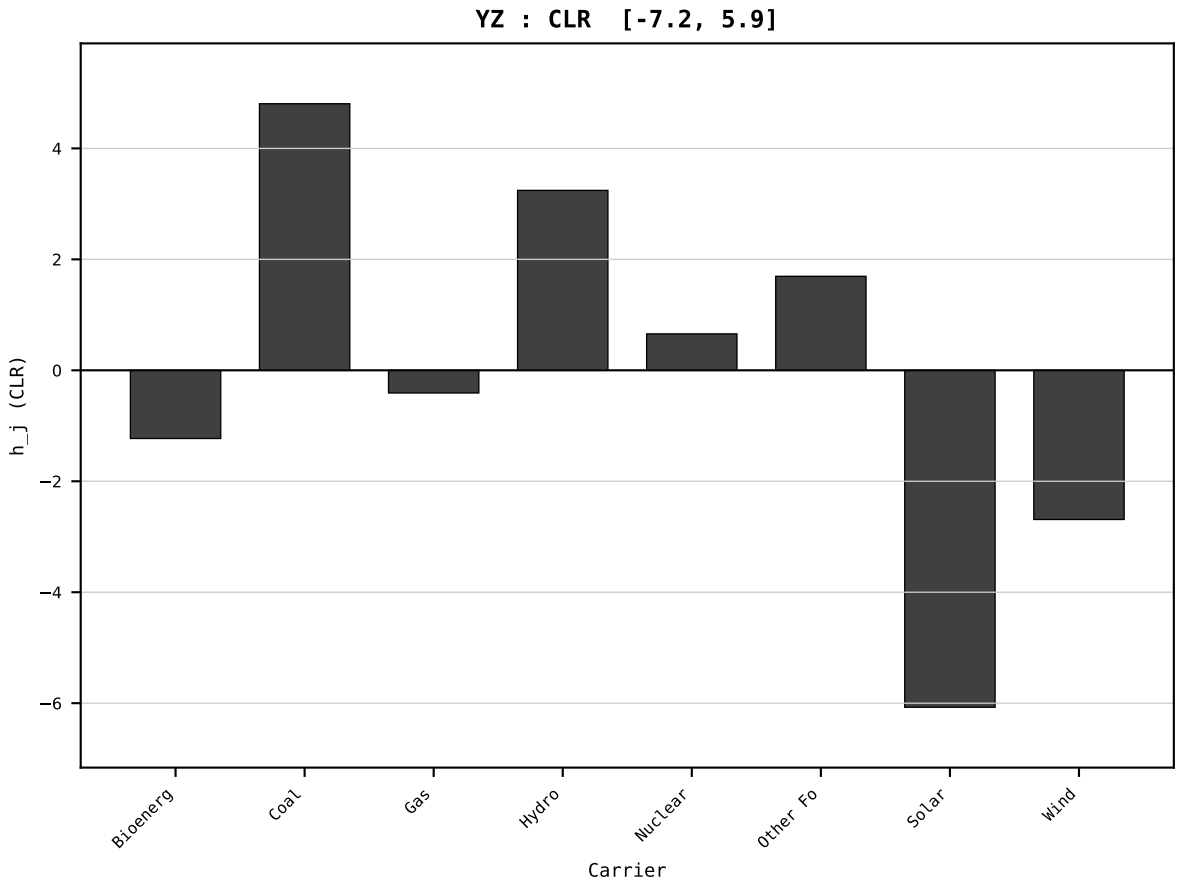
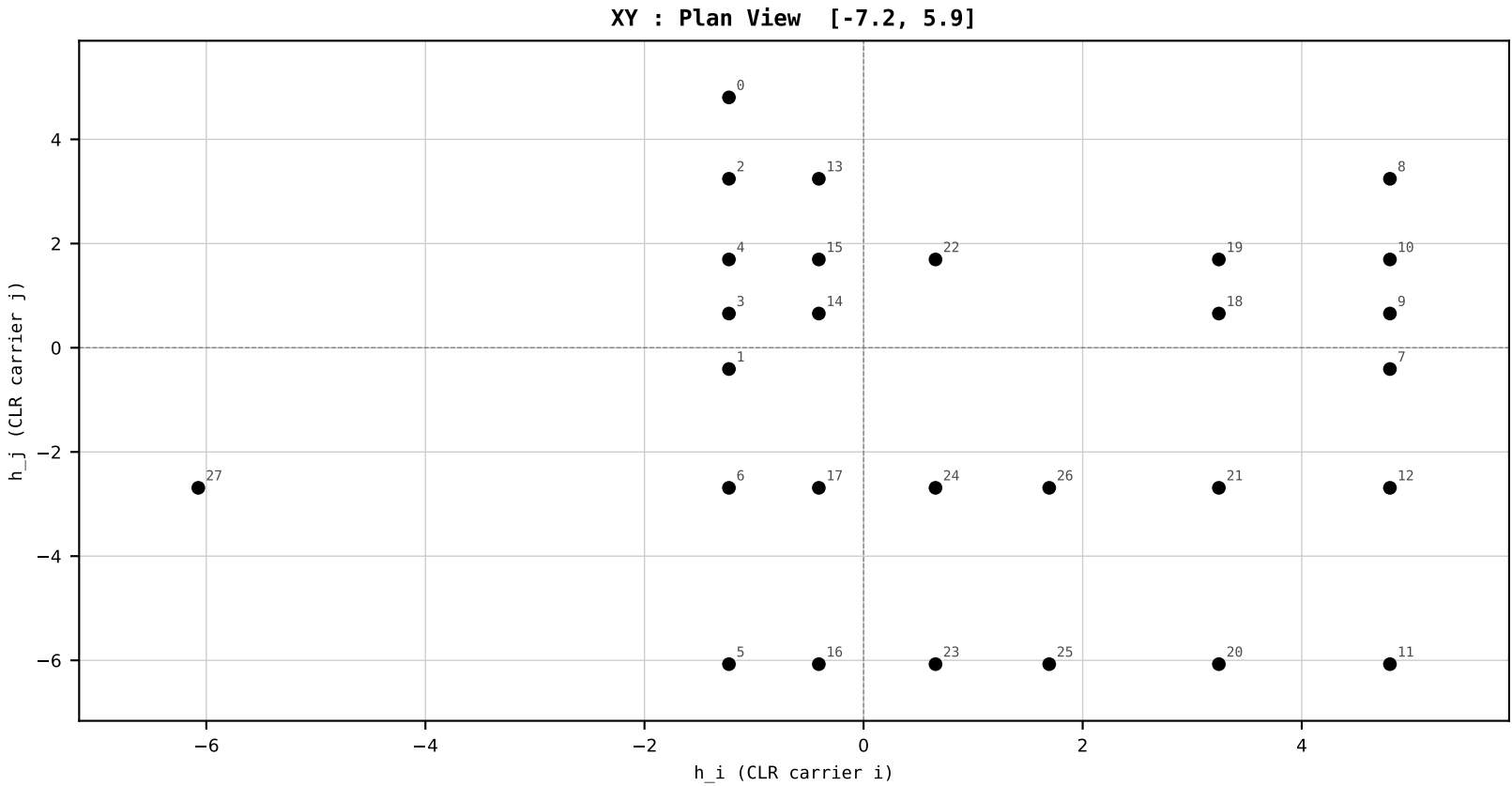
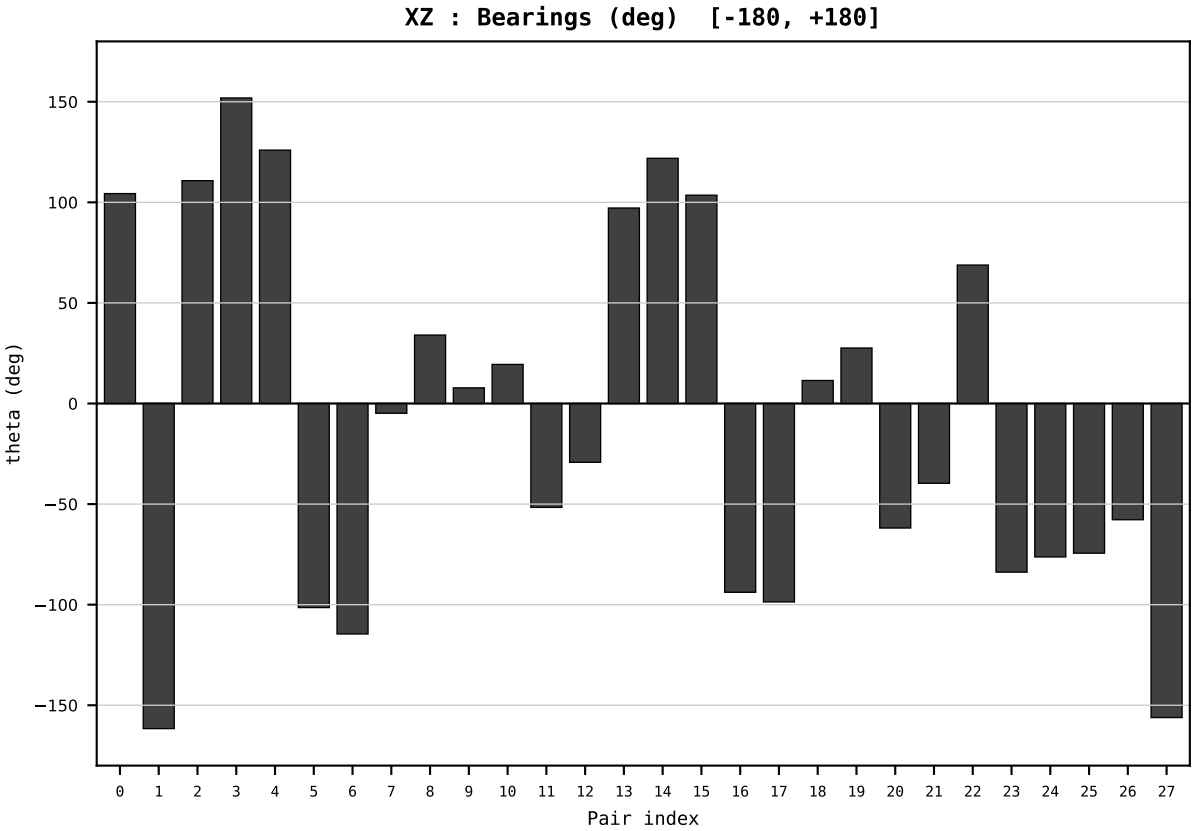
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2000
t : 0 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.664059
Ring = Hs-4
E_metric= 82.7022
kappa_HS= 53013.00
omega = 0.0000 deg
d_A = 0.000000
Helm = ---
Helm d = ---
DR = 10.878 HLR
DR ratio= 53013.0x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=1 | Year 2001 | D=8 N=26 pairs=28

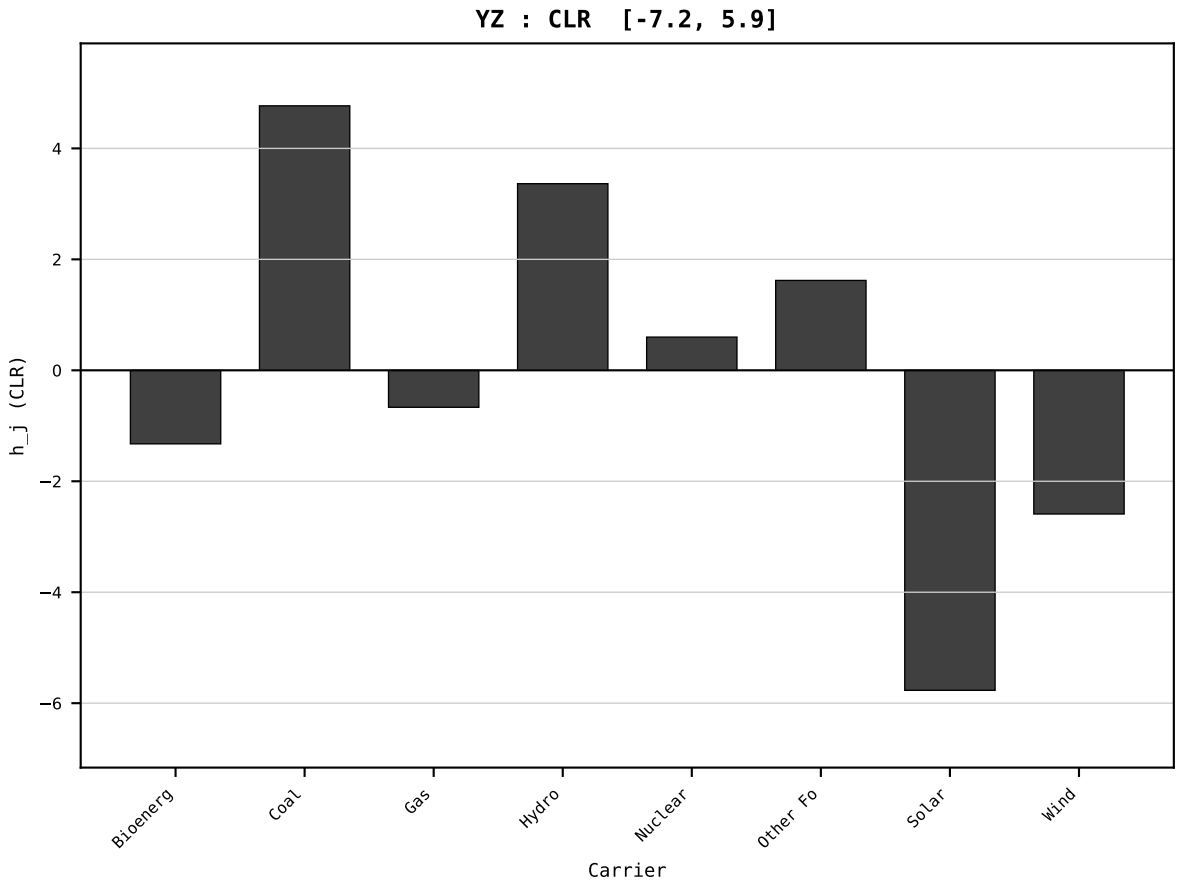
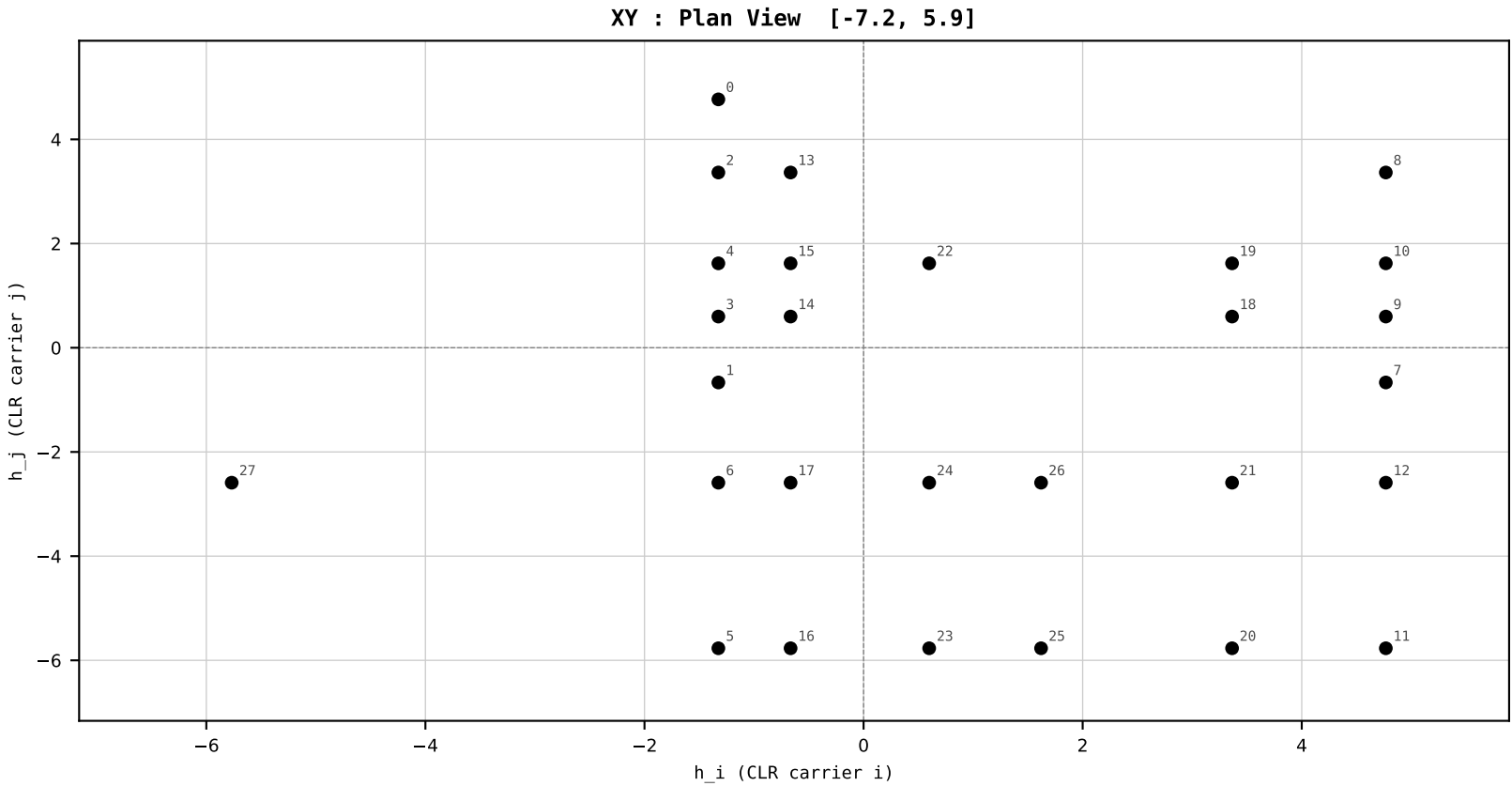
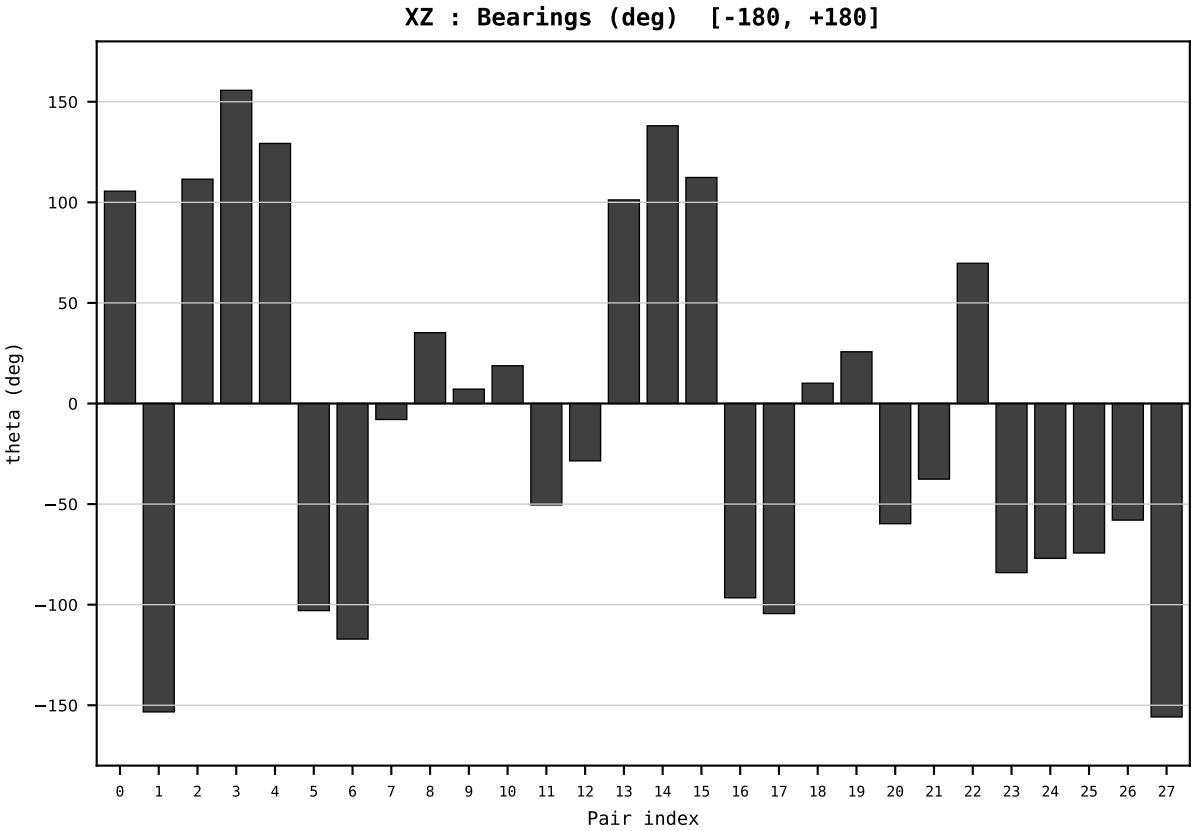
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2001
t : 1 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.654342
Ring = Hs-4
E_metric= 79.2118
kappa_HS= 37638.00
omega = 2.5929 deg
d_A = 0.450949
Helm = Solar
Helm d = +0.304970
DR = 10.536 HLR
DR ratio= 37638.0x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=2 | Year 2002 | D=8 N=26 pairs=28

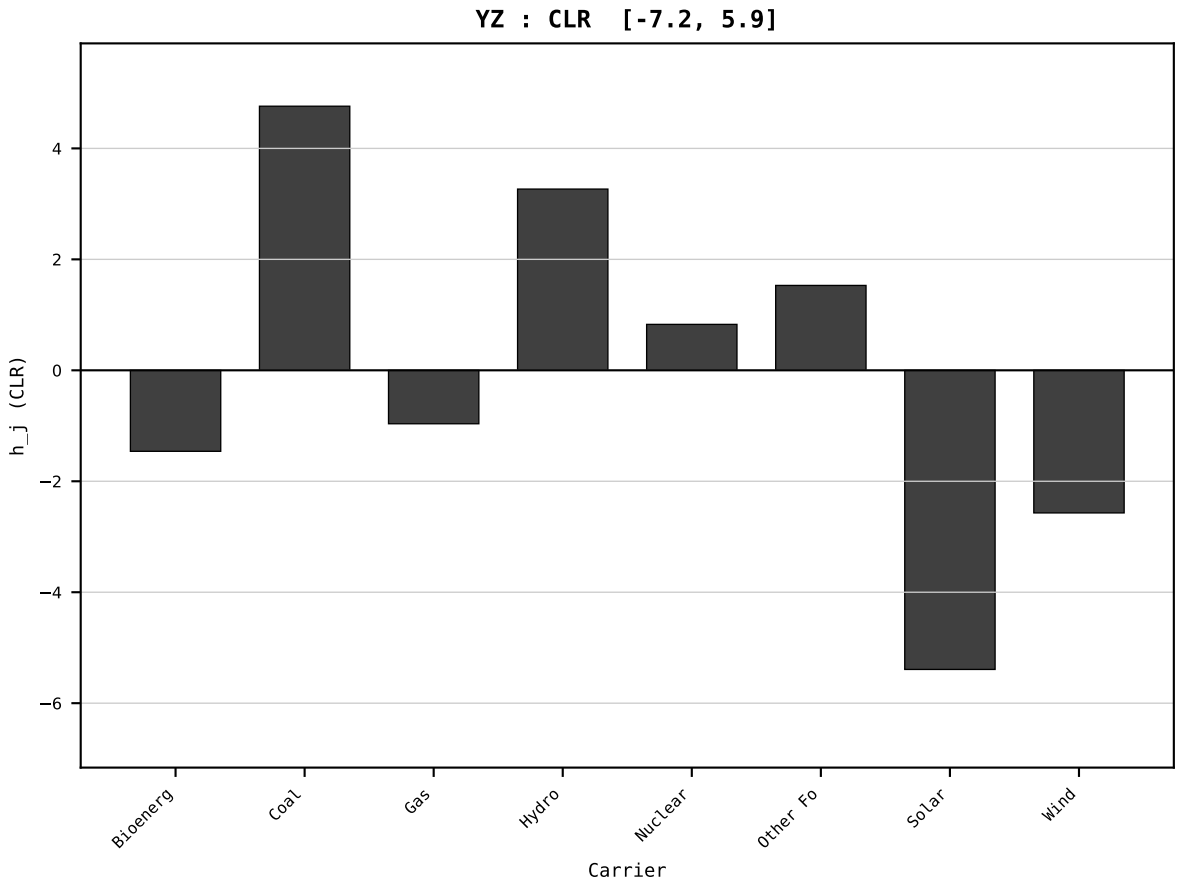
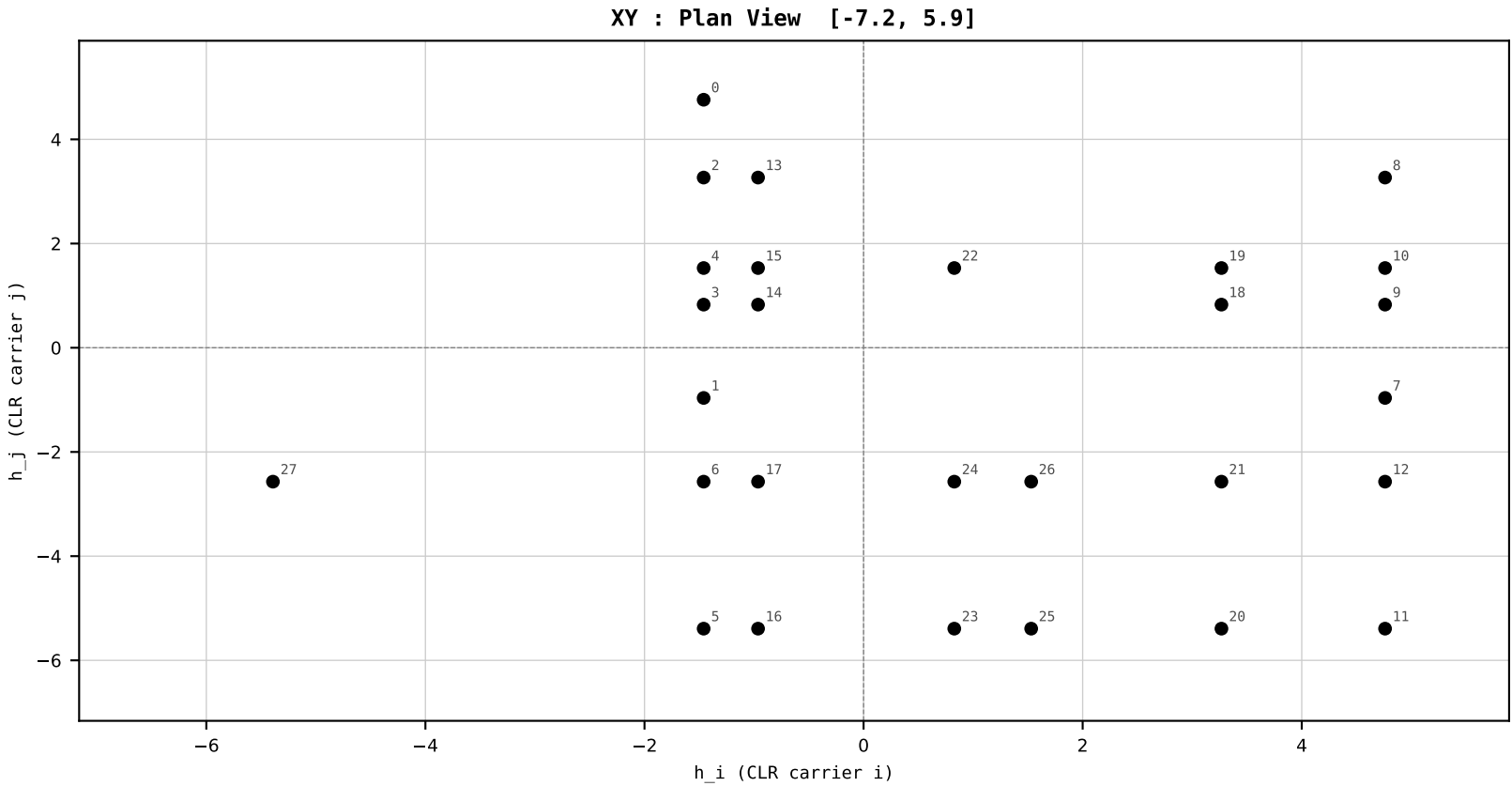
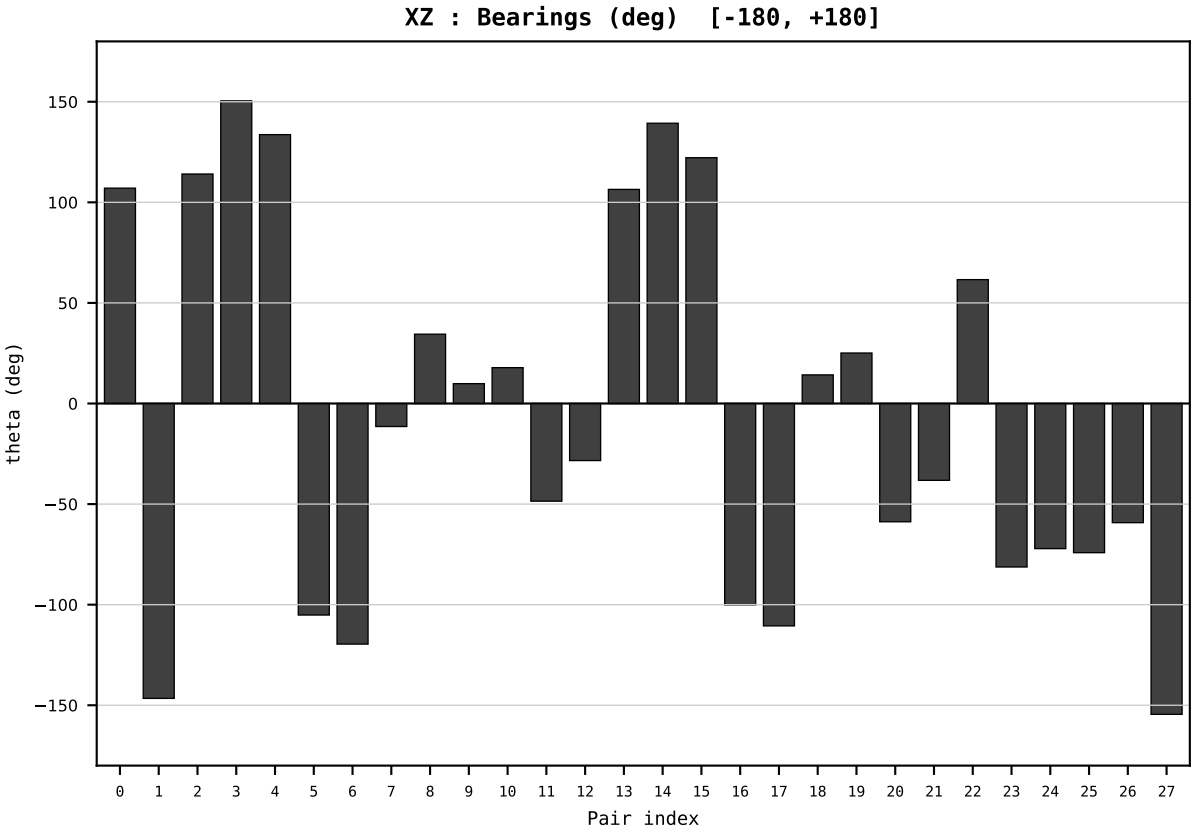
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2002
t : 2 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.662736
Ring = Hs-4
E_metric= 75.0996
kappa_HS= 25651.40
omega = 3.3508 deg
d_A = 0.564384
Helm = Solar
Helm d = +0.376481
DR = 10.152 HLR
DR ratio= 25651.4x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=3 | Year 2003 | D=8 N=26 pairs=28

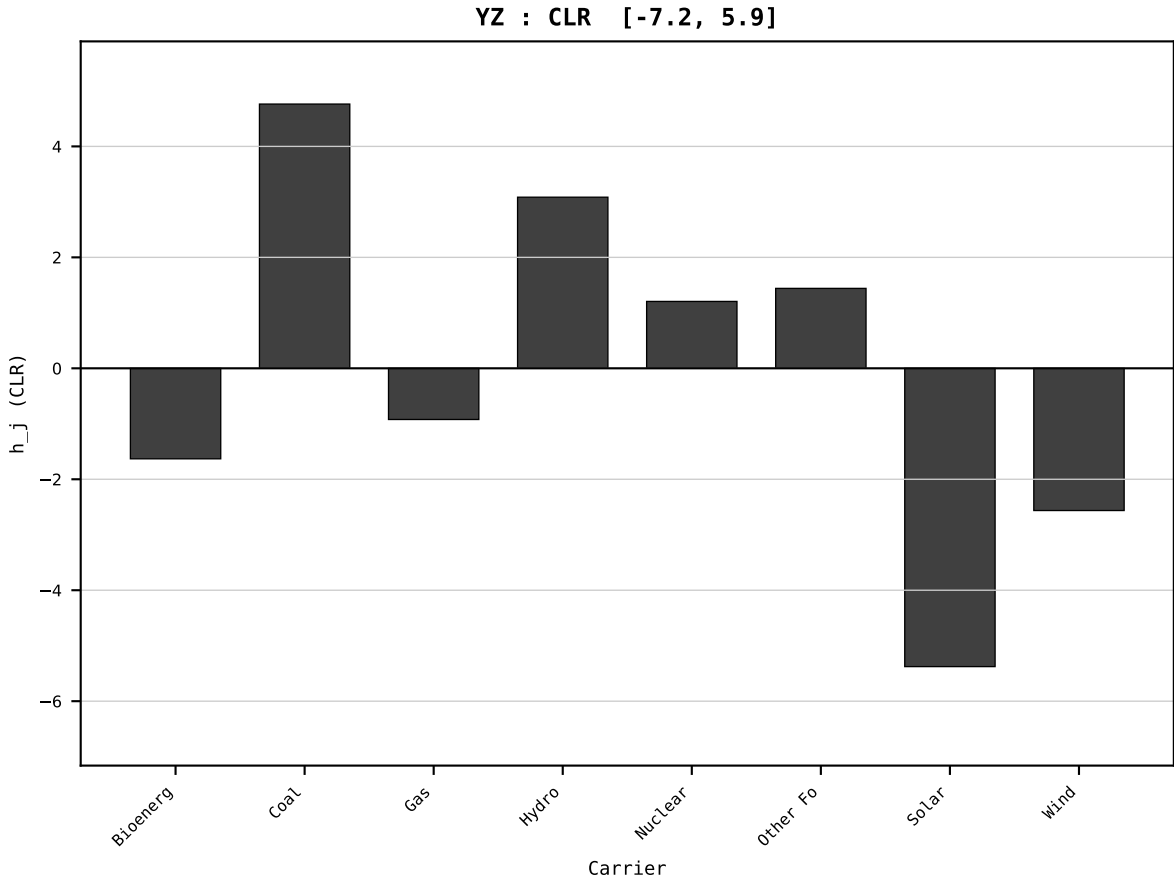
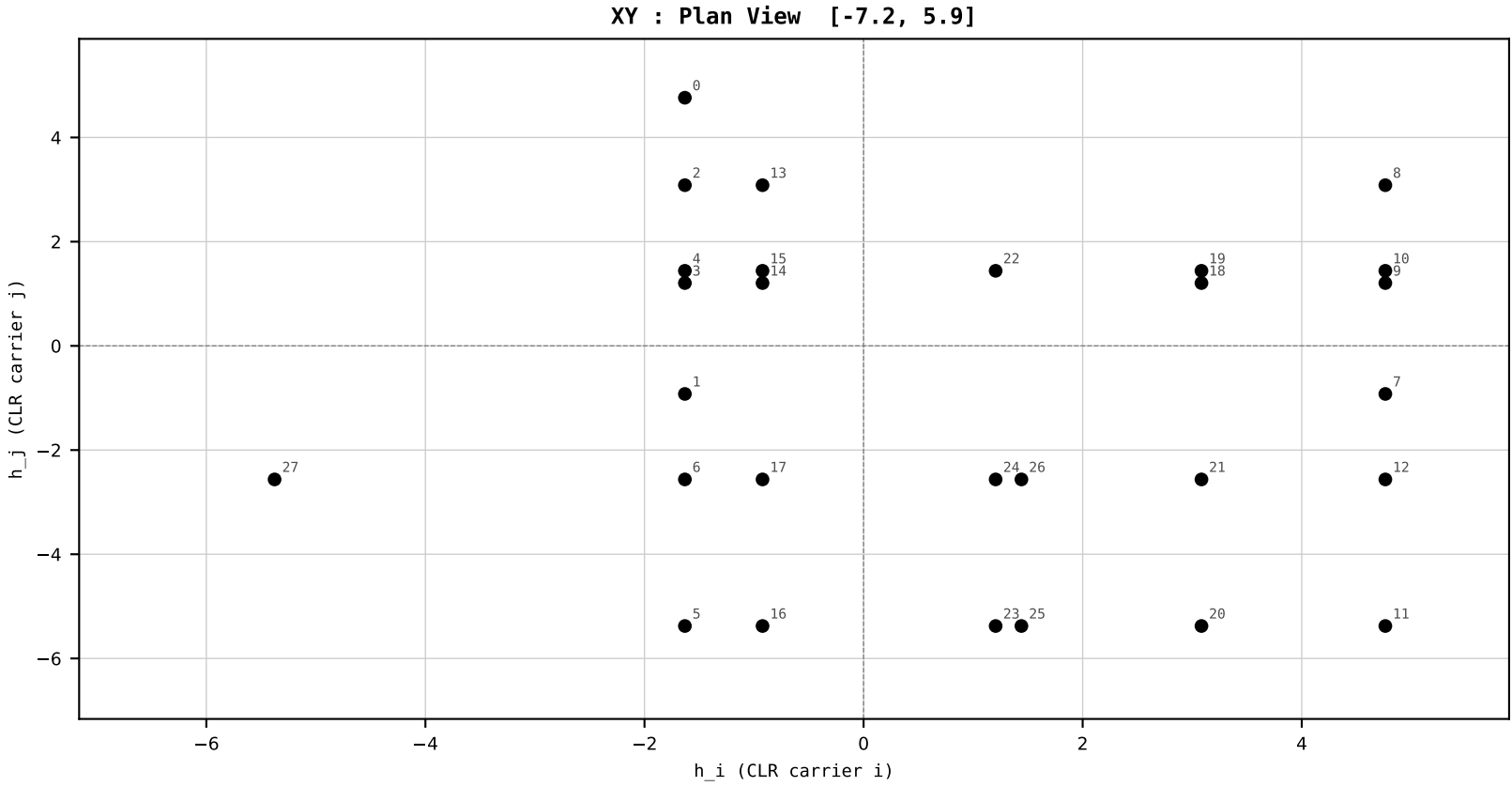
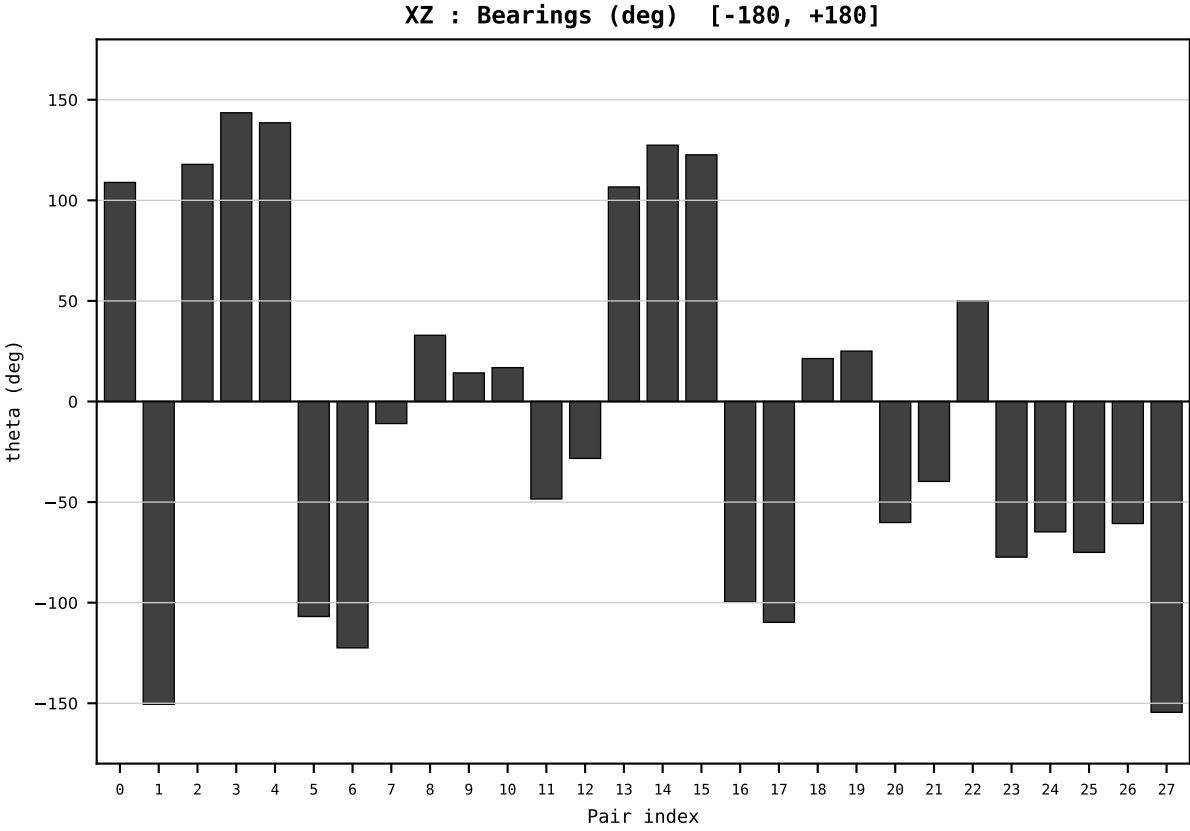
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2003
t : 3 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.672028
Ring = Hs-4
E_metric= 74.7231
kappa_HS= 25332.83
omega = 3.0675 deg
d_A = 0.463827
Helm = Nuclear
Helm d = +0.377648
DR = 10.140 HLR
DR ratio= 25332.8x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=4 | Year 2004 | D=8 N=26 pairs=28

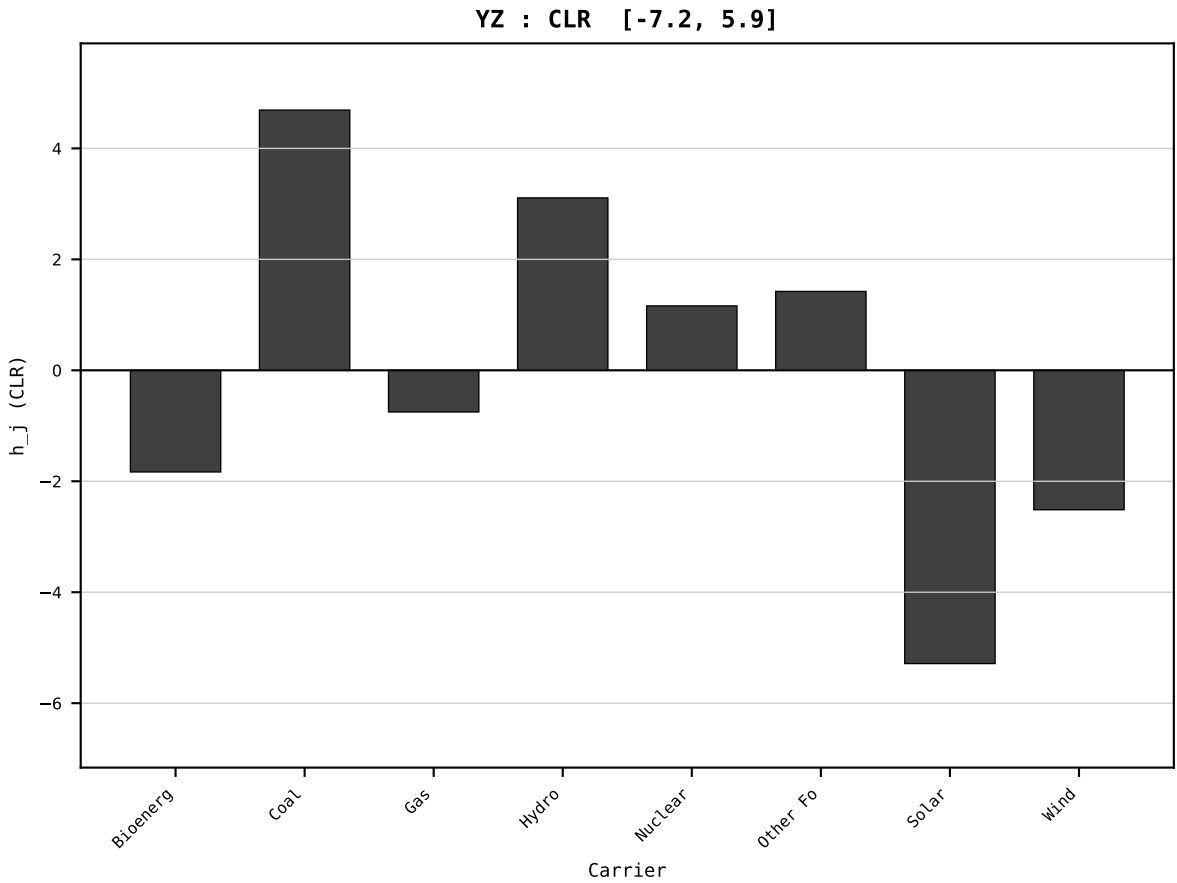
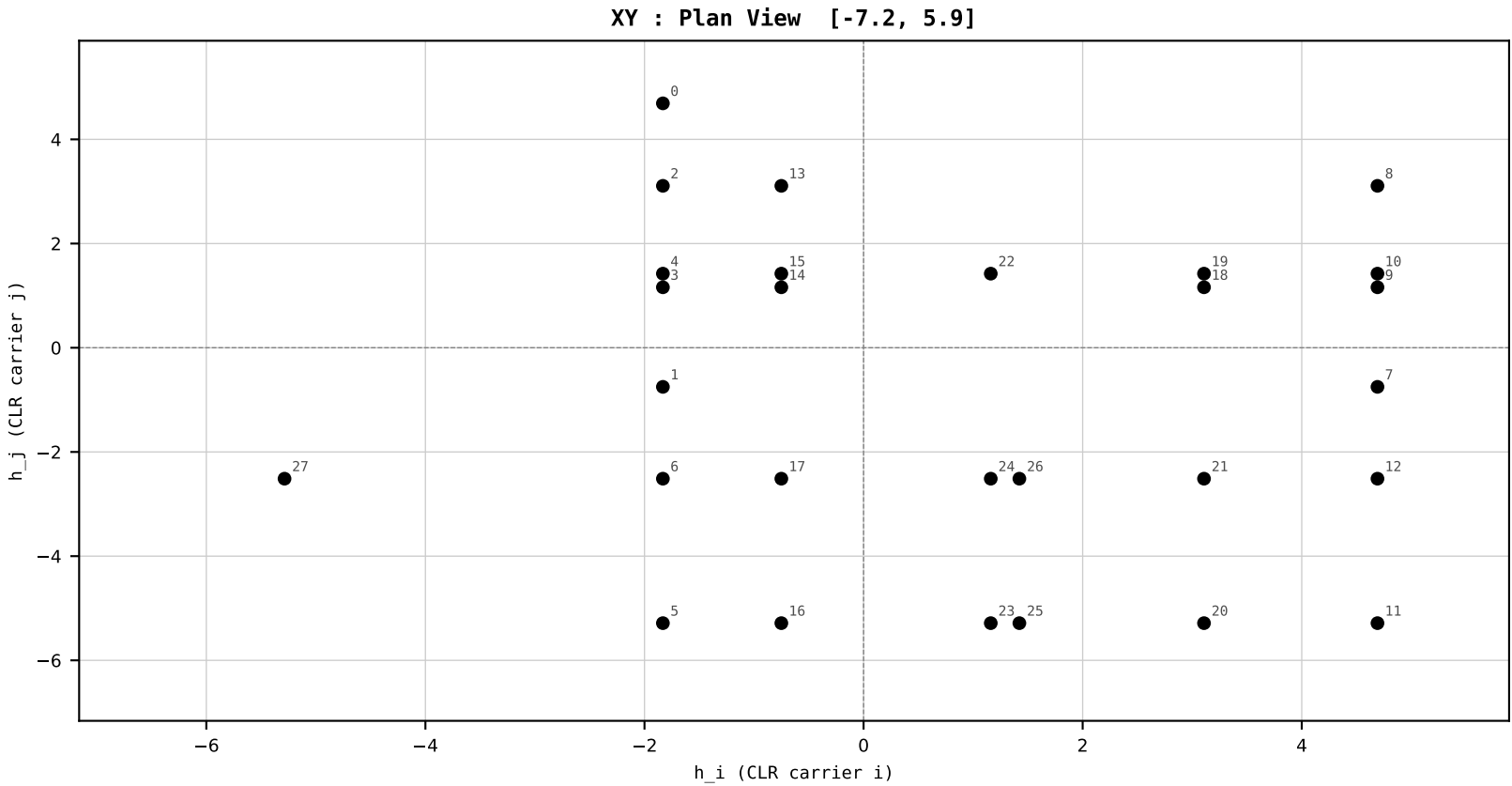
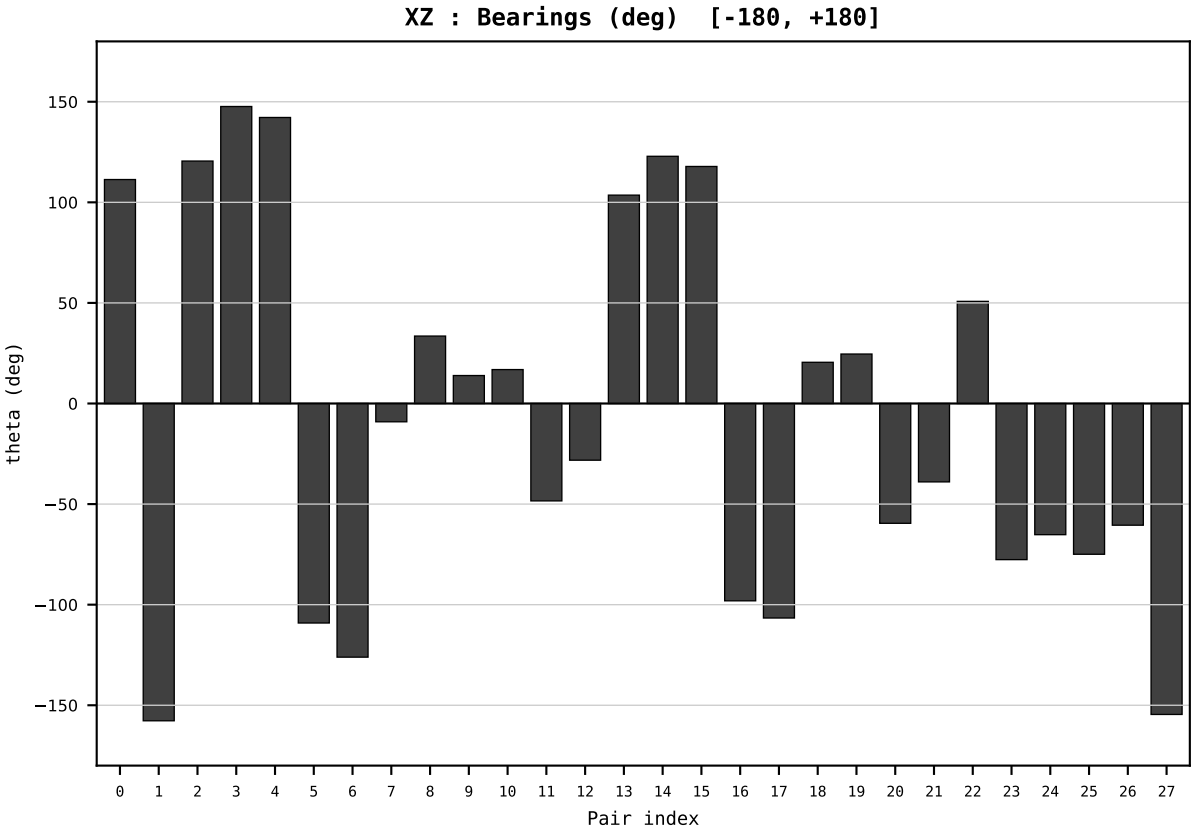
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2004
t : 4 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.659108
Ring = Hs-4
E_metric= 73.2154
kappa_HS= 21530.25
omega = 1.8952 deg
d_A = 0.297664
Helm = Bioenergy
Helm d = -0.200817
DR = 9.977 HLR
DR ratio= 21530.2x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=5 | Year 2005 | D=8 N=26 pairs=28

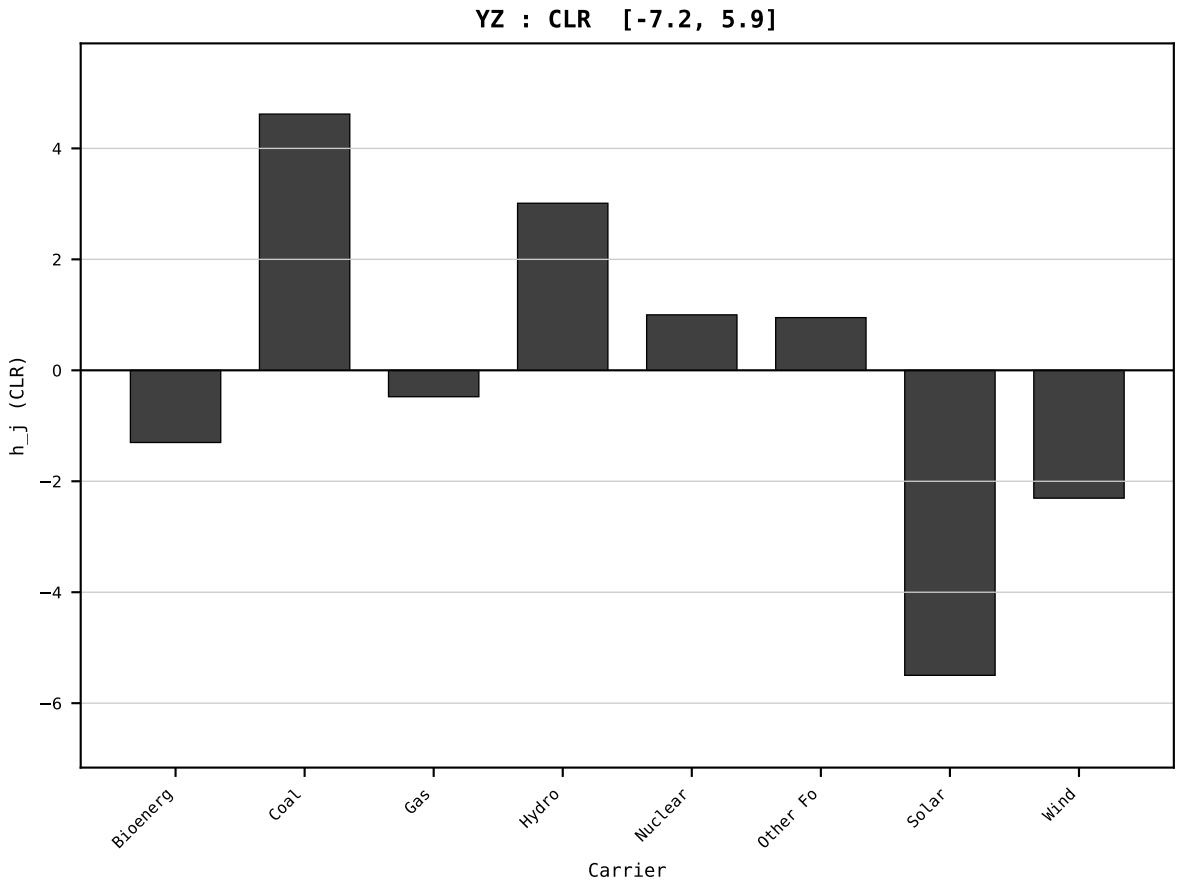
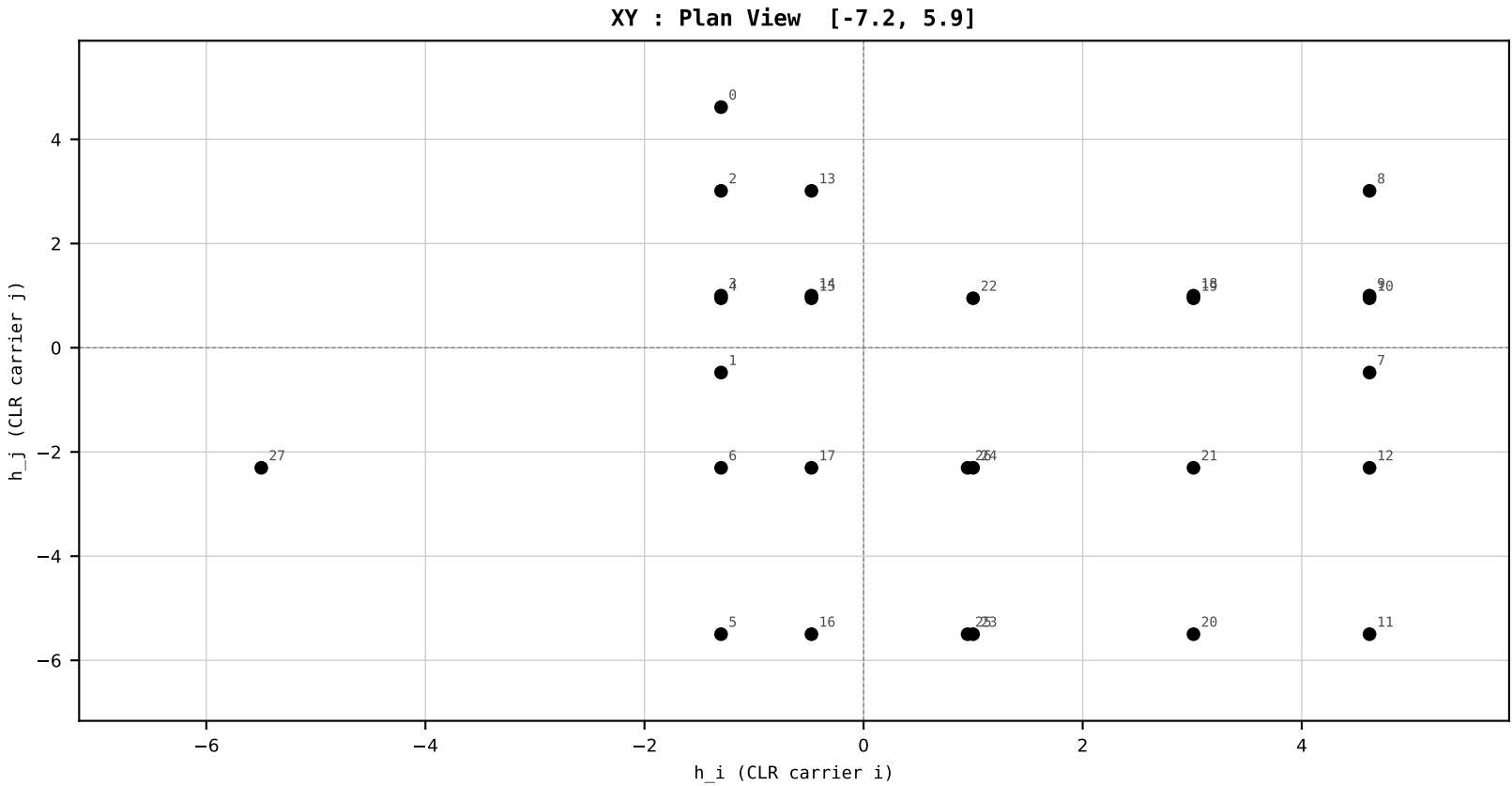
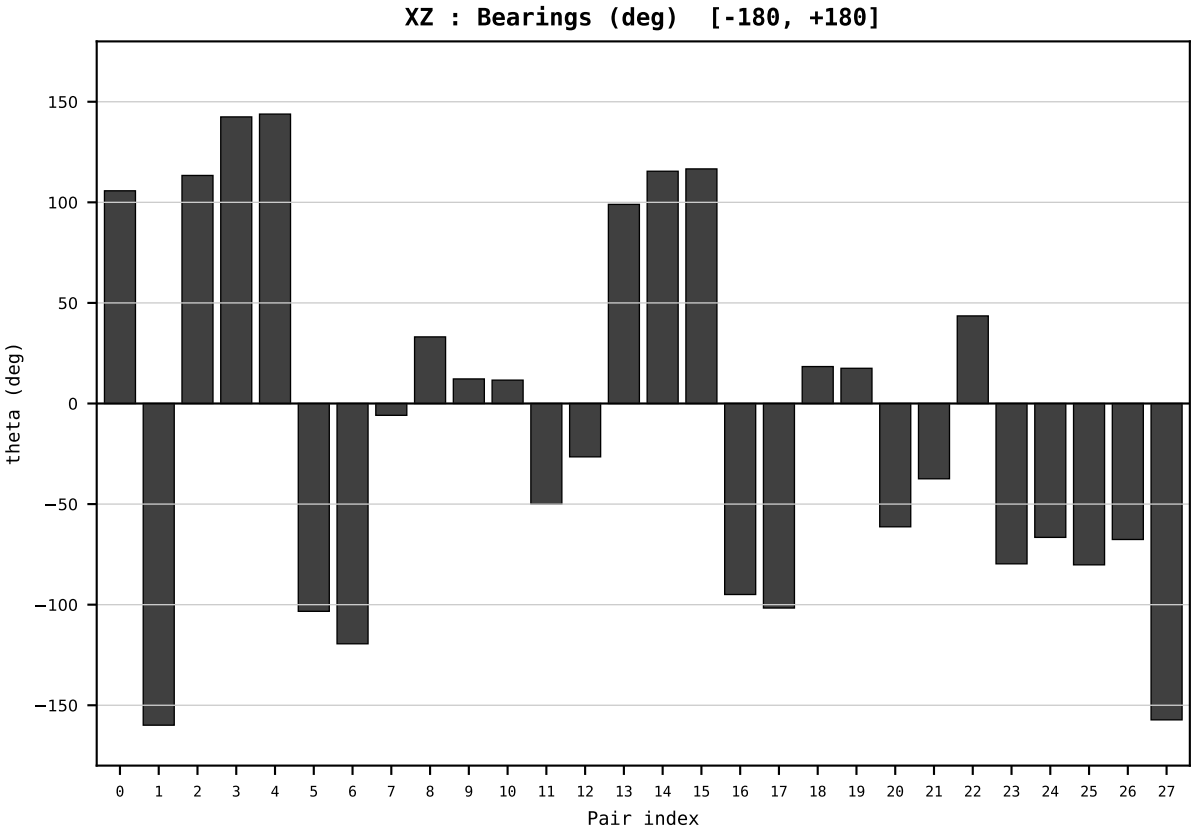
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2005
t : 5 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.671833
Ring = Hs-4
E_metric= 69.7610
kappa_HS= 24751.88
omega = 5.5429 deg
d_A = 0.842658
Helm = Bioenergy
Helm d = +0.531144
DR = 10.117 HLR
DR ratio= 24751.9x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=6 | Year 2006 | D=8 N=26 pairs=28

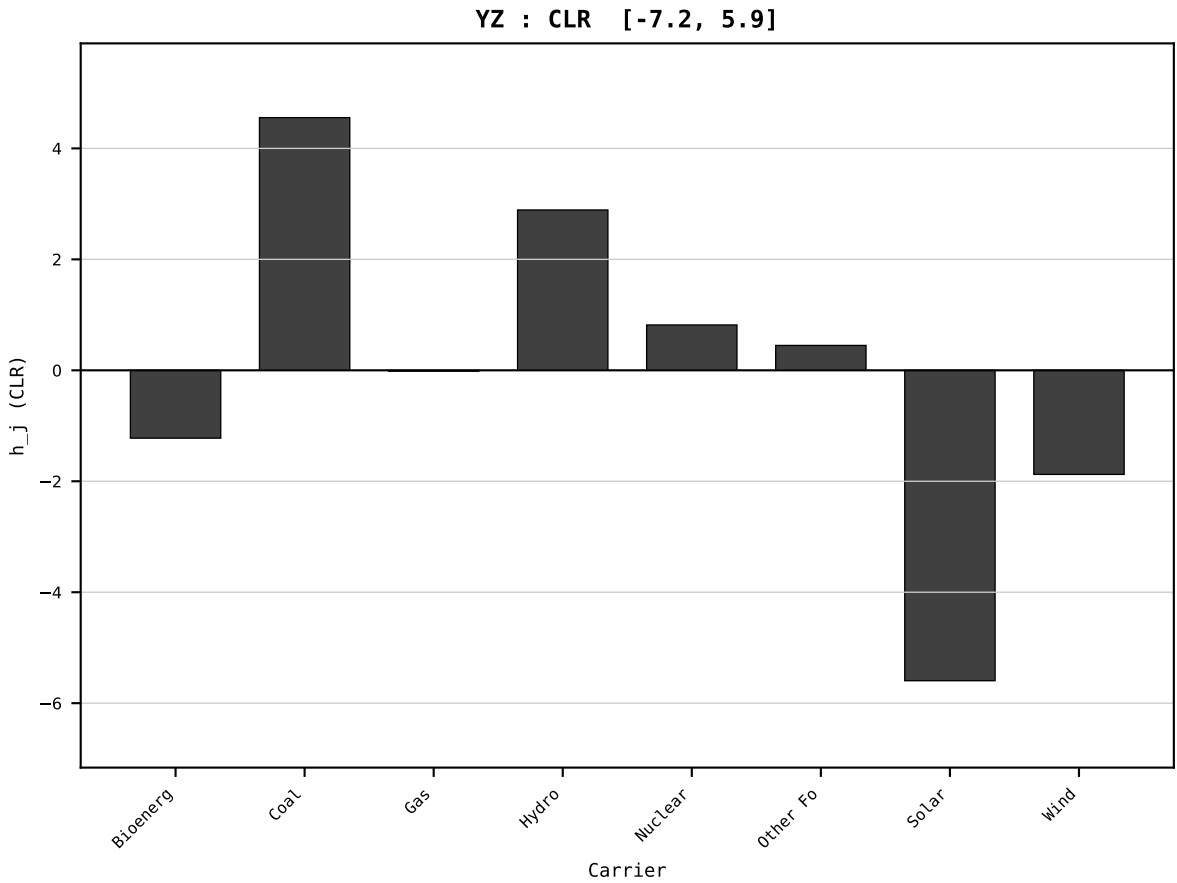
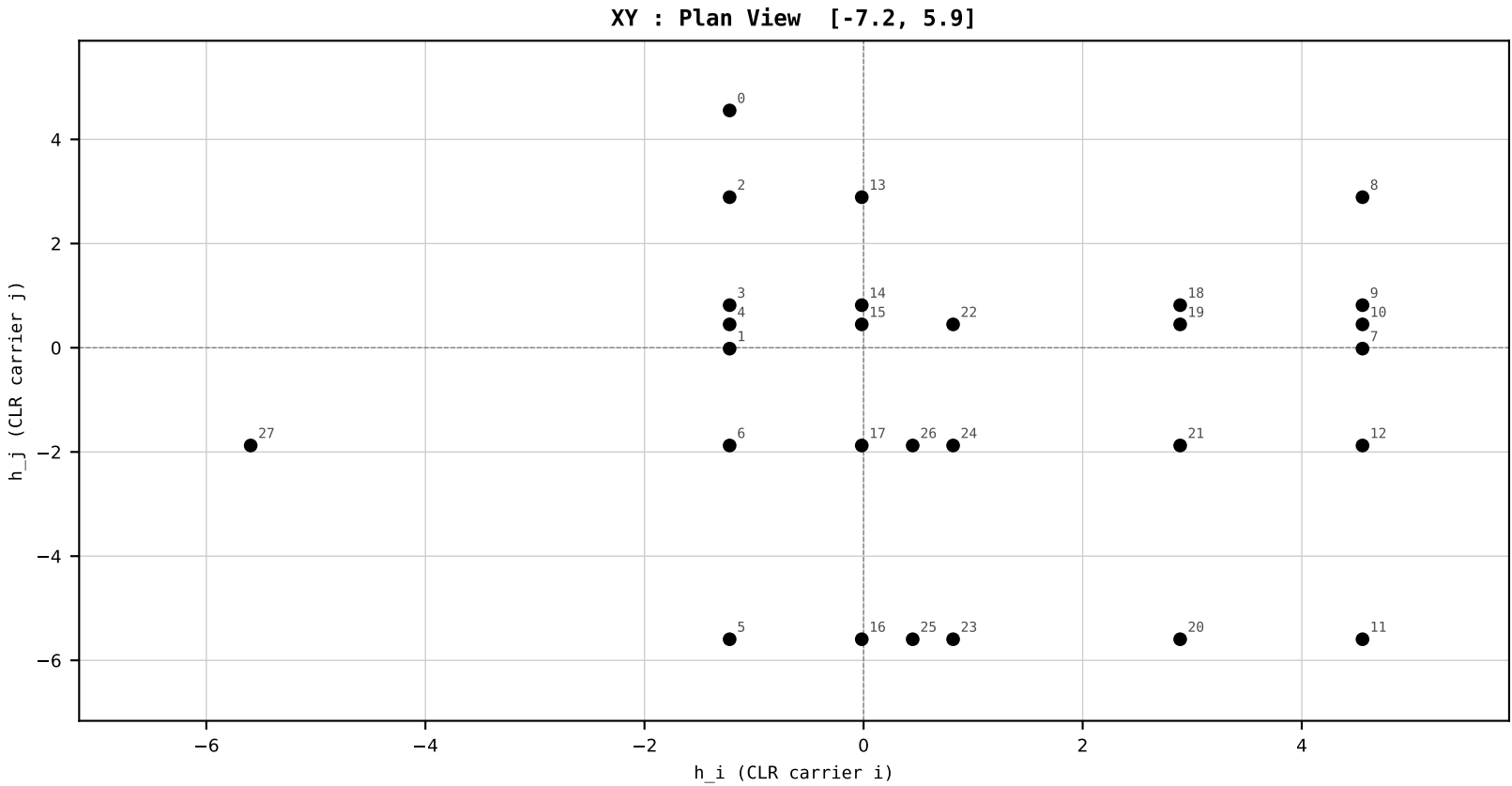
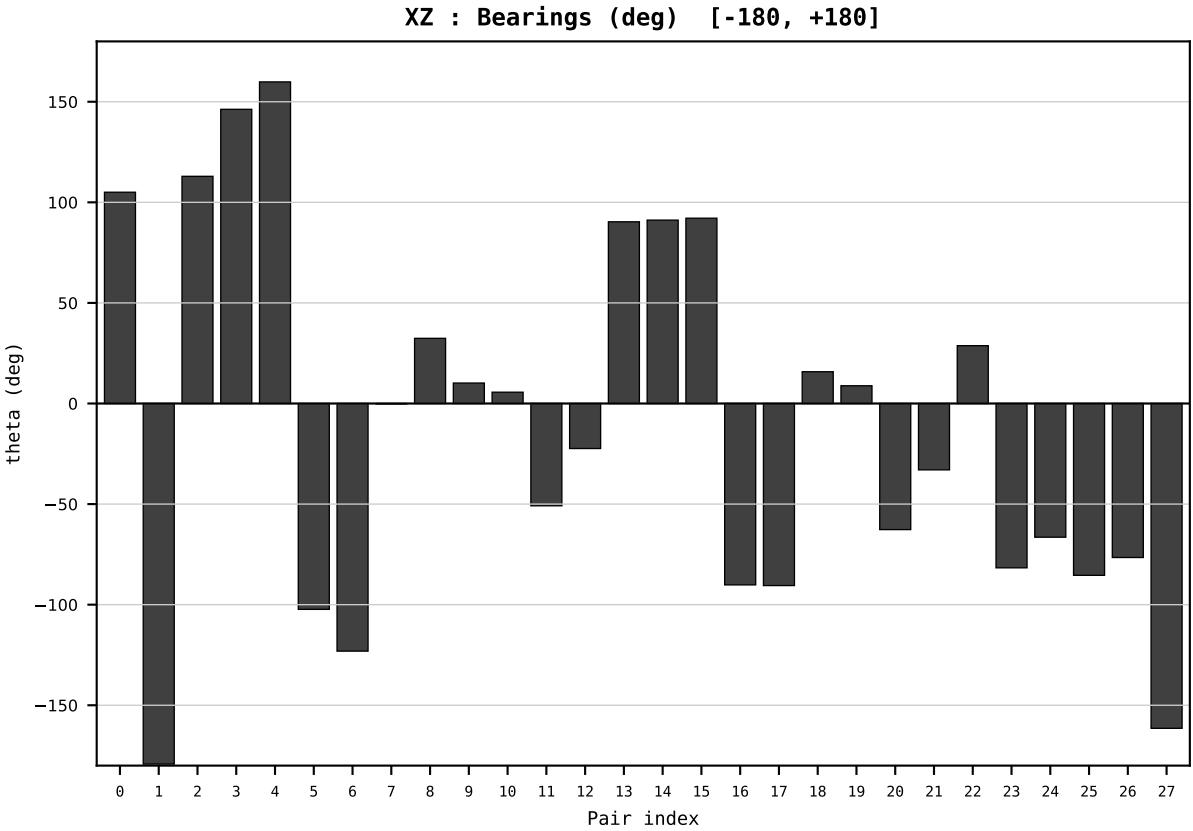
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2006
t : 6 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.683157
Ring = Hs-4
E_metric= 66.2857
kappa_HS= 25582.44
omega = 5.6860 deg
d_A = 0.844723
Helm = Other Fossil
Helm d = -0.501386
DR = 10.150 HLR
DR ratio= 25582.4x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=7 | Year 2007 | D=8 N=26 pairs=28

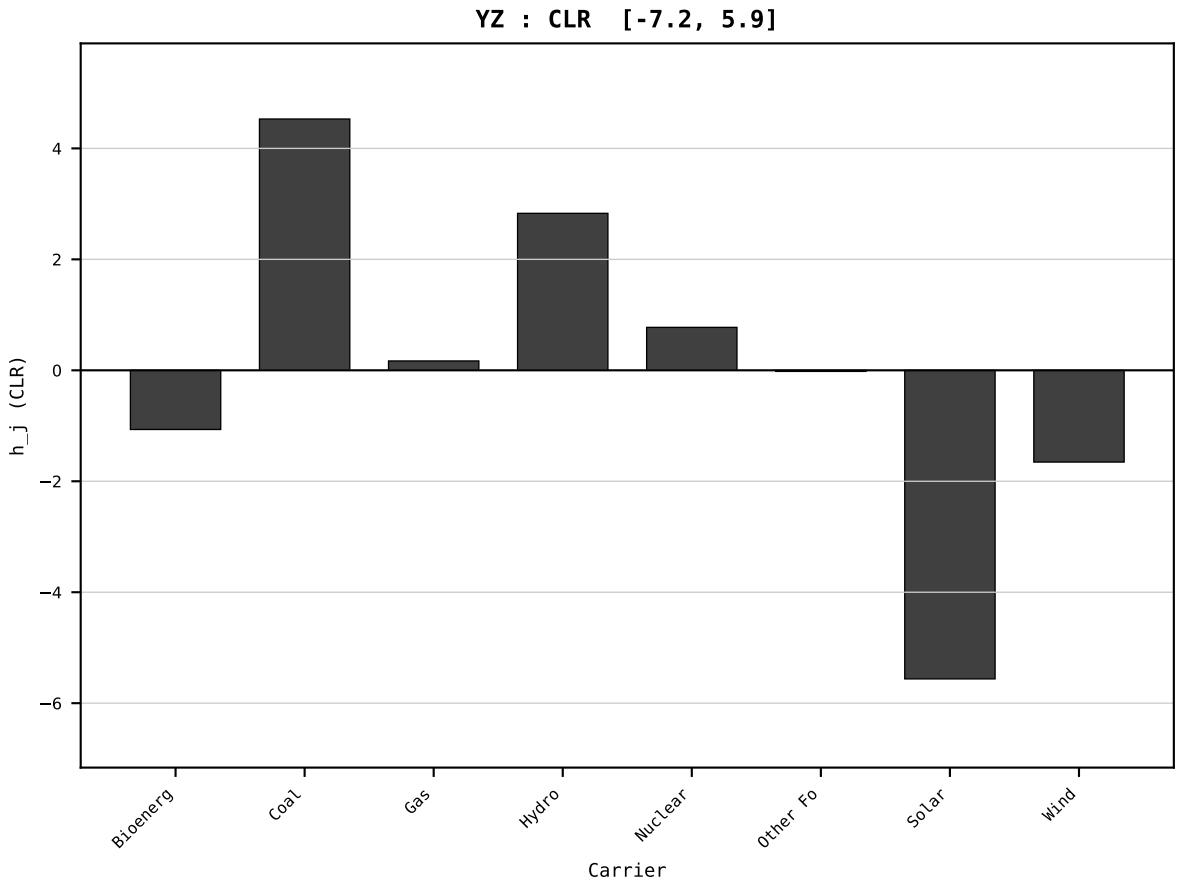
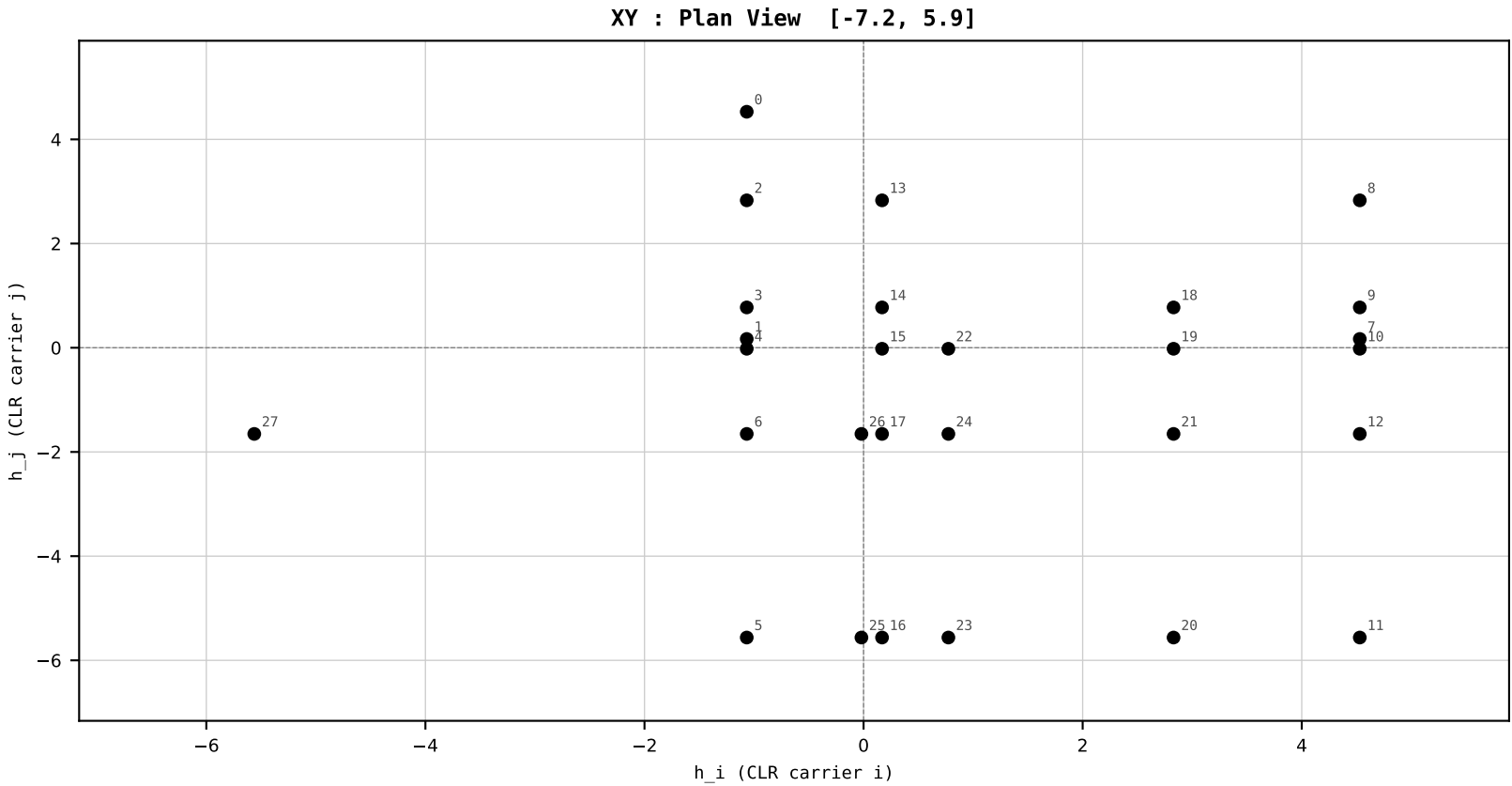
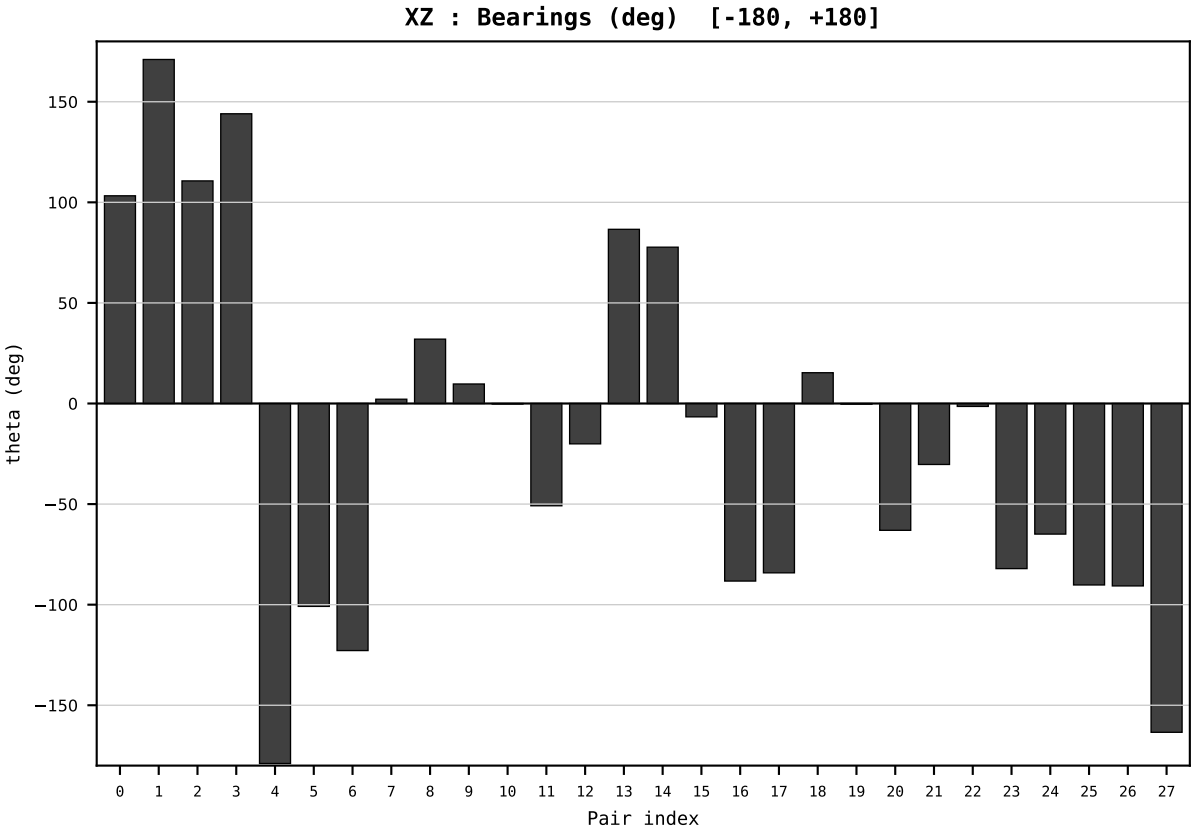
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2007
t : 7 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.689708
Ring = Hs-4
E_metric= 63.9666
kappa_HS= 24151.82
omega = 3.9773 deg
d_A = 0.578182
Helm = Other Fossil
Helm d = -0.468002
DR = 10.092 HLR
DR ratio= 24151.8x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=8 | Year 2008 | D=8 N=26 pairs=28

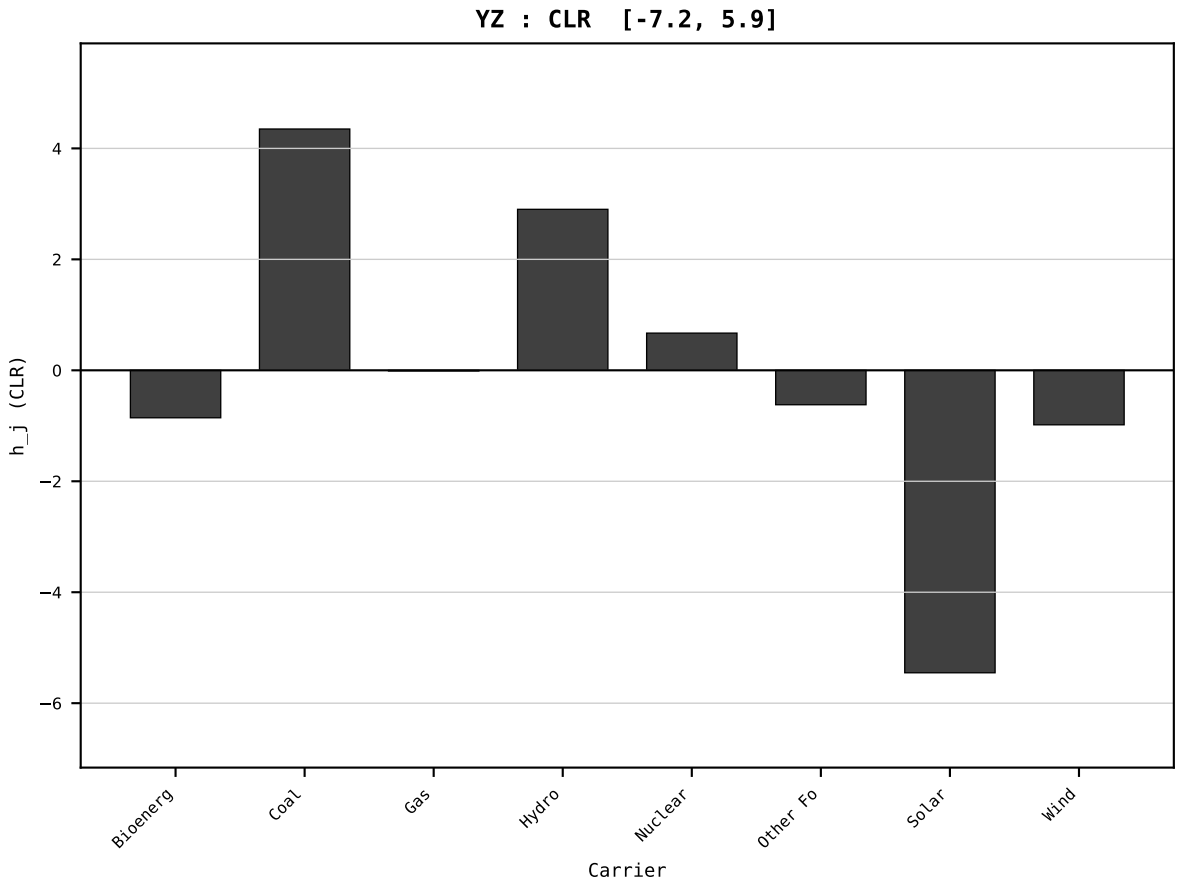
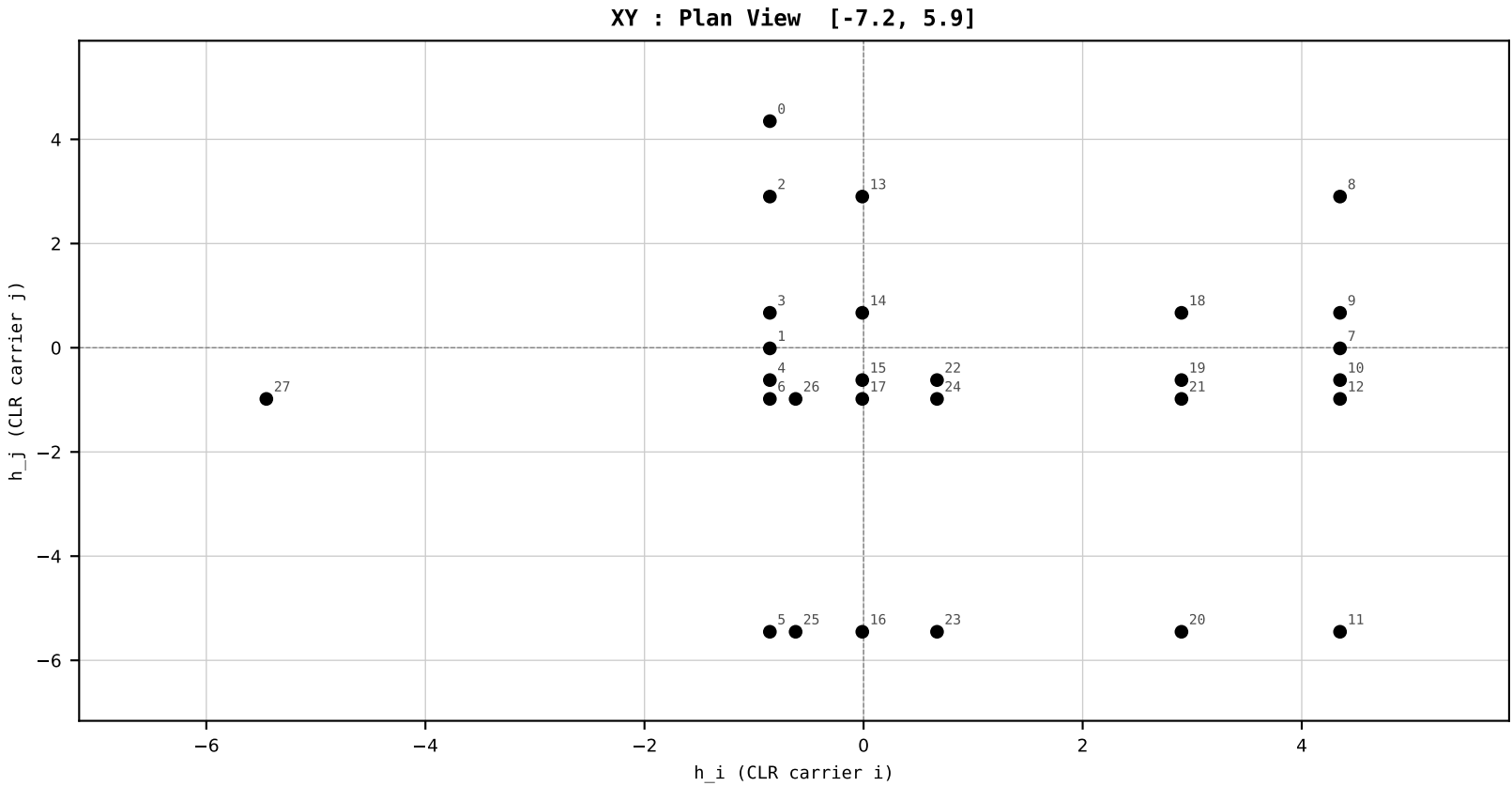
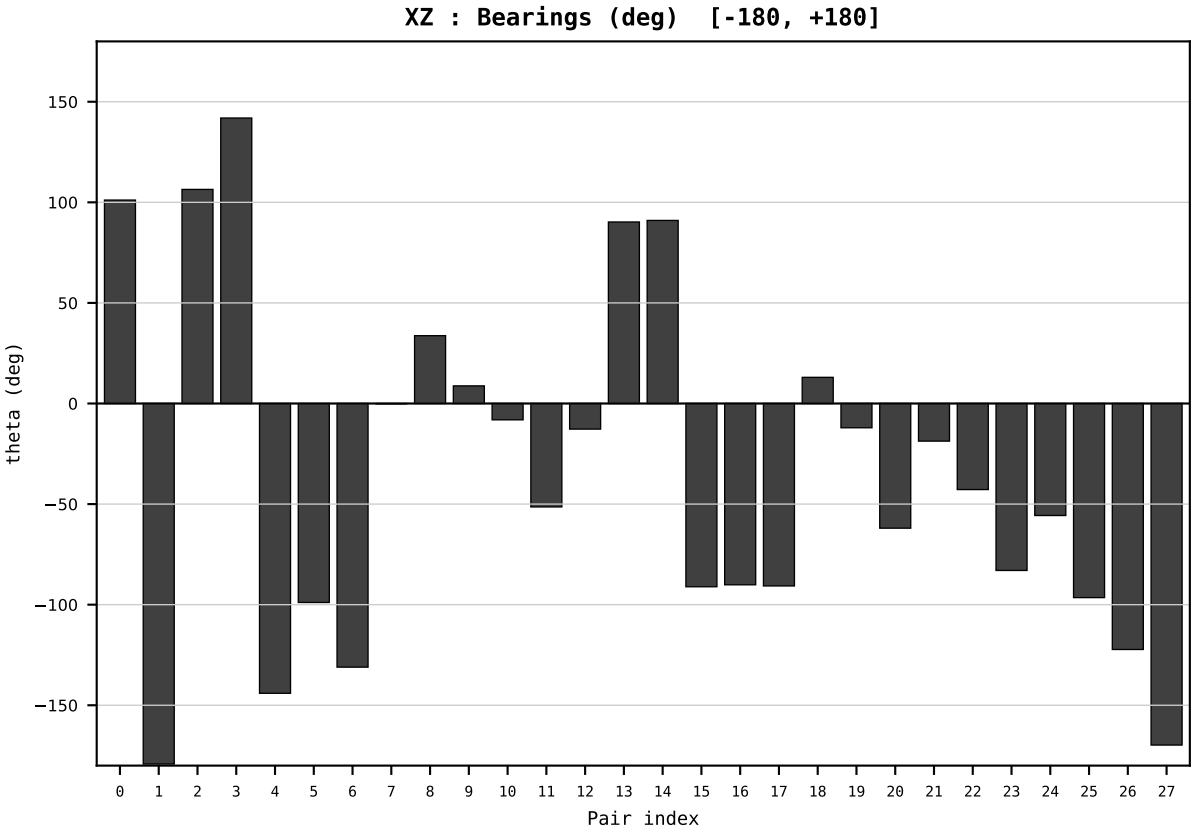
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2008
t : 8 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.661852
Ring = Hs-4
E_metric= 59.5937
kappa_HS= 18059.40
omega = 6.8155 deg
d_A = 0.974683
Helm = Wind
Helm d = +0.671725
DR = 9.801 HLR
DR ratio= 18059.4x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



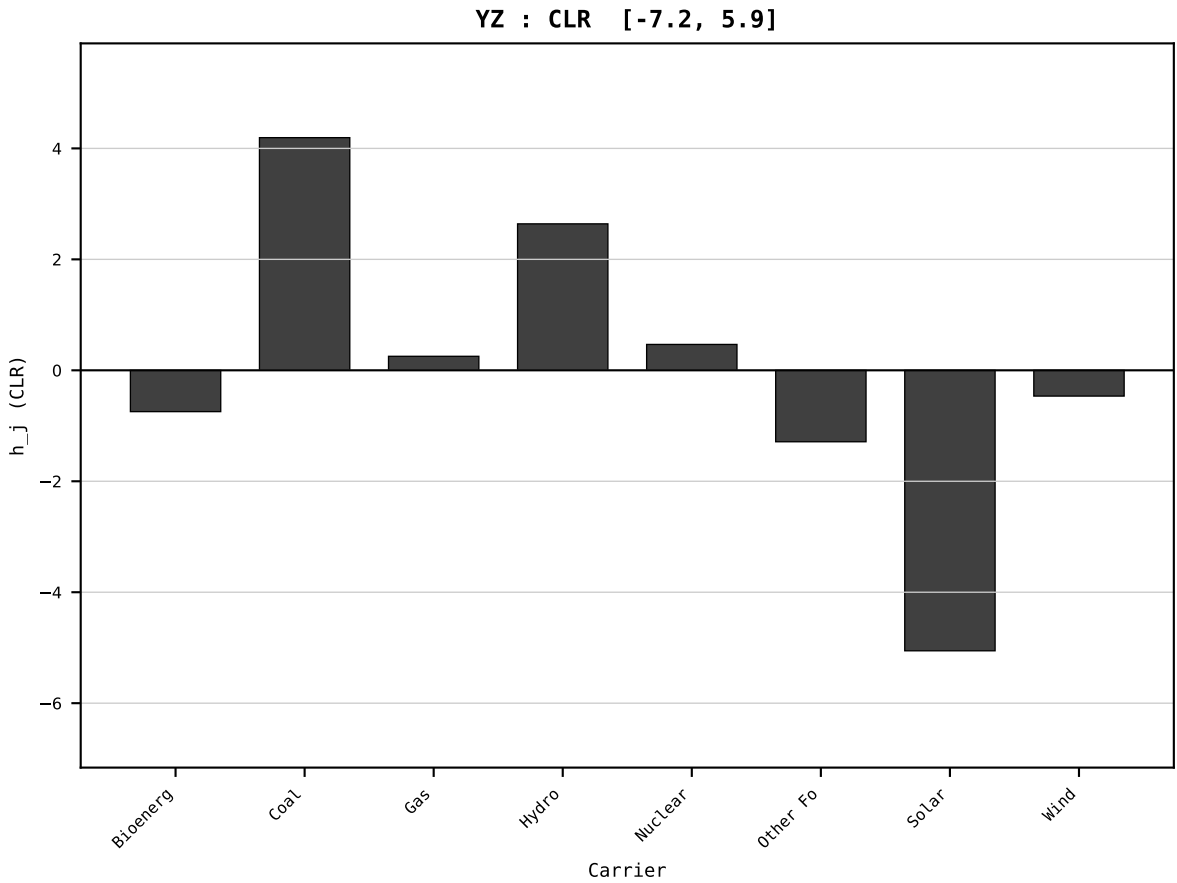
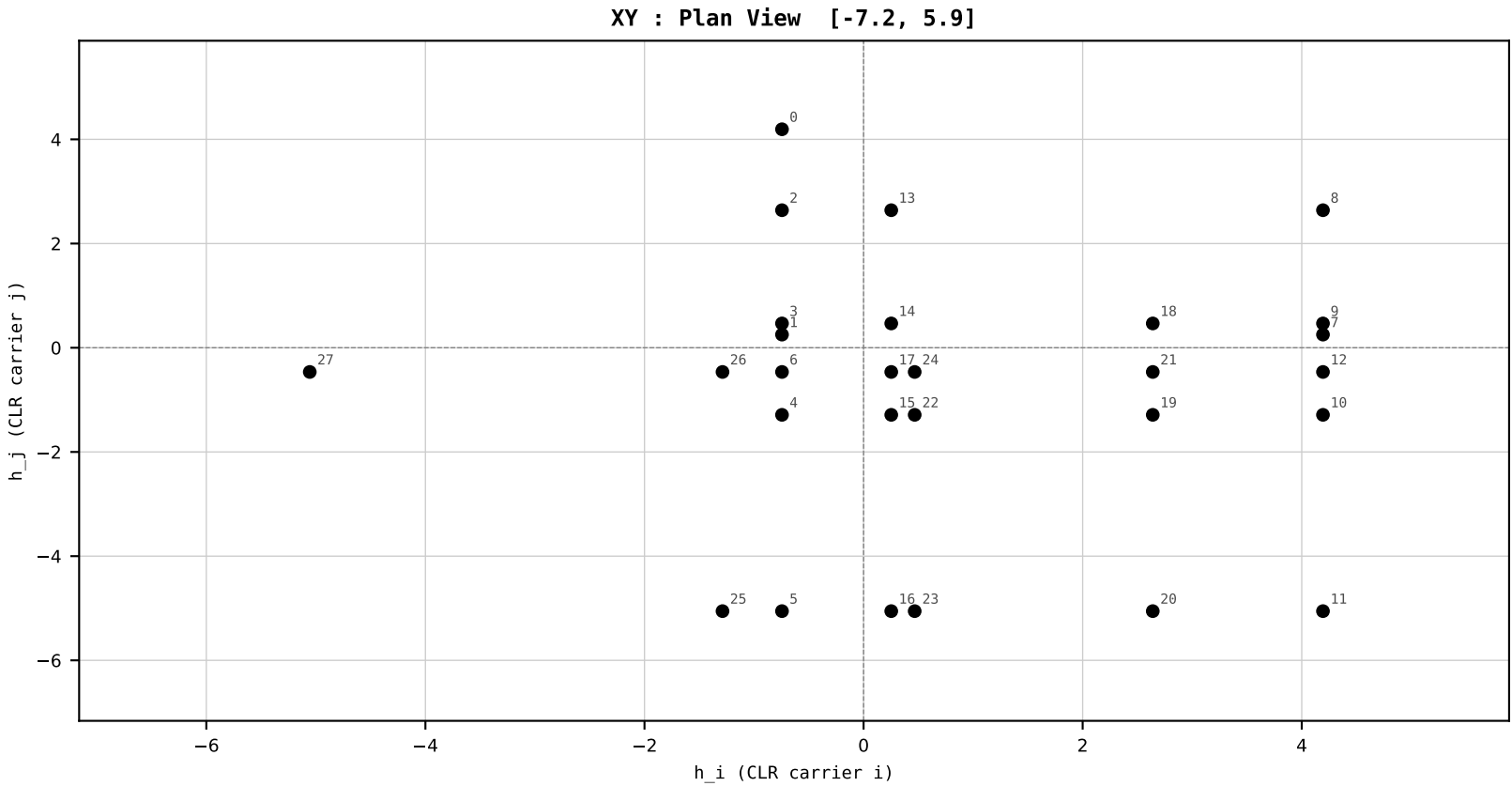
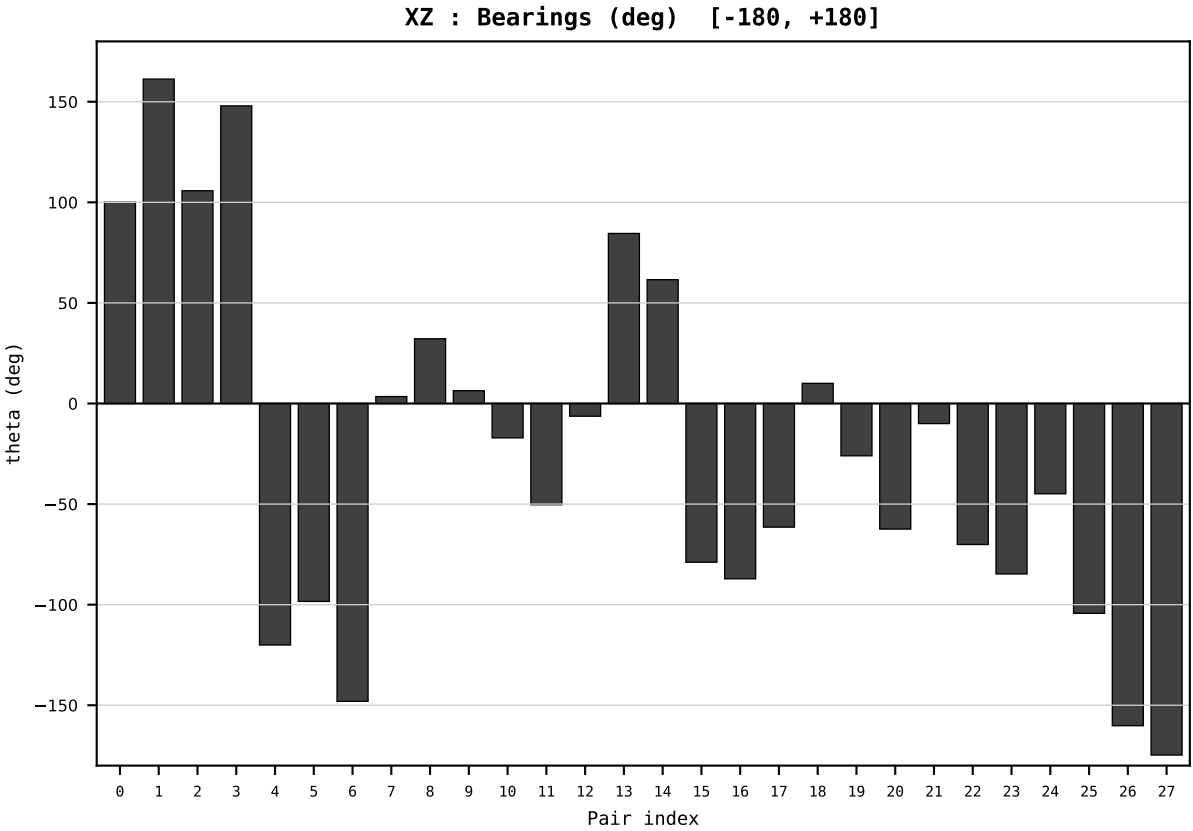
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2009
t : 9 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.657431
Ring = Hs-4
E_metric= 52.8271
kappa_HS= 10398.29
omega = 7.1934 deg
d_A = 1.042625
Helm = Other Fossil
Helm d = -0.667414
DR = 9.249 HLR
DR ratio= 10398.3x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=10 | Year 2010 | D=8 N=26 pairs=28

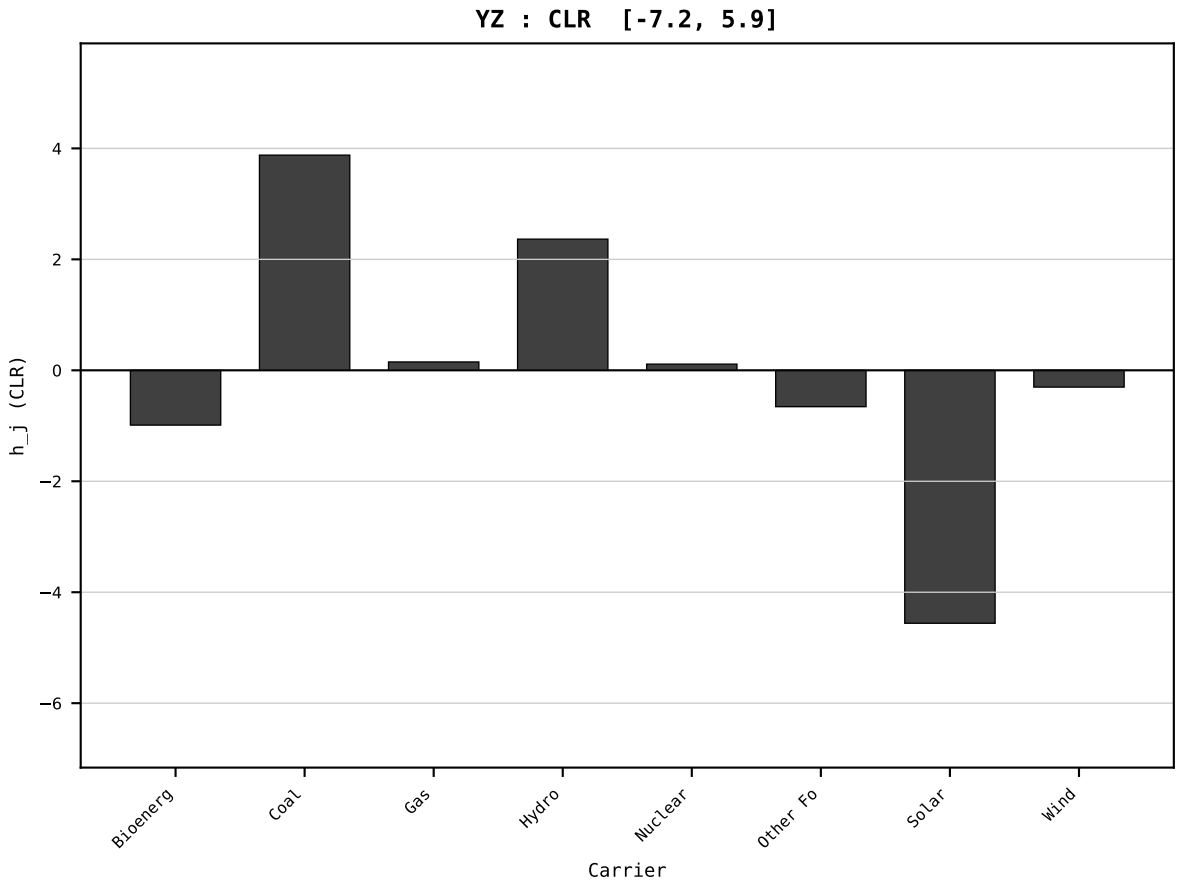
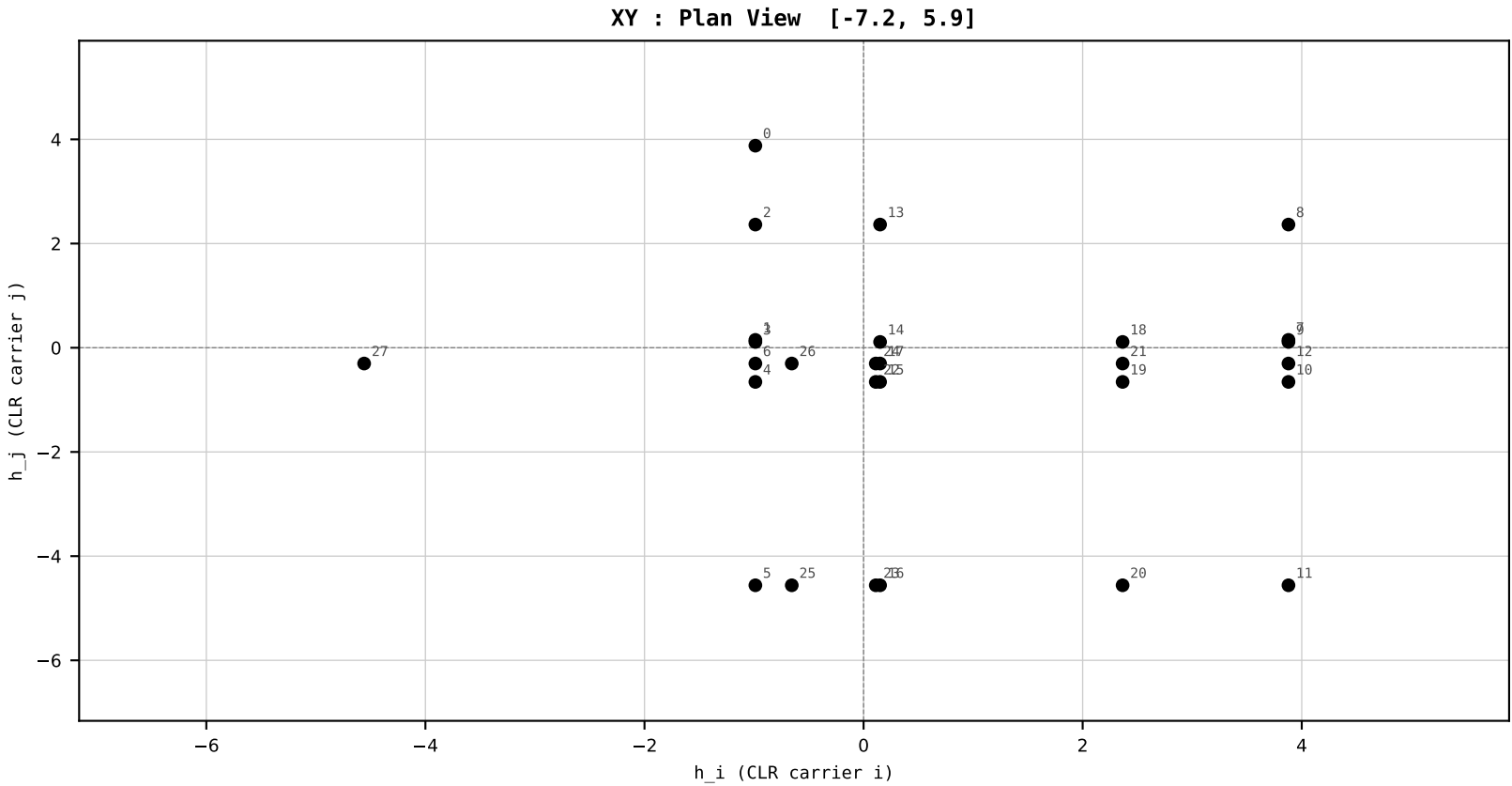
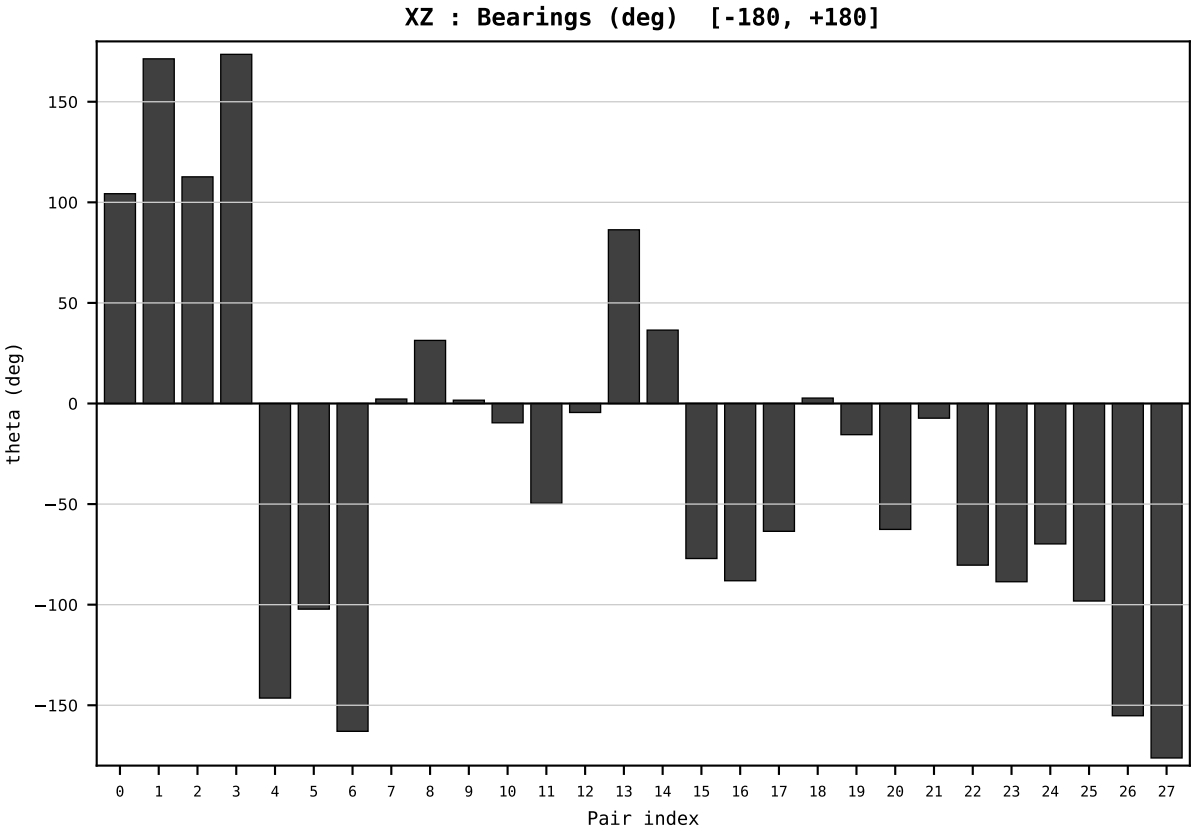
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2010
t : 10 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.628874
Ring = Hs-4
E_metric= 42.9552
kappa_HS= 4619.43
omega = 6.0622 deg
d_A = 1.021213
Helm = Other Fossil
Helm d = +0.632236
DR = 8.438 HLR
DR ratio= 4619.4x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=11 | Year 2011 | D=8 N=26 pairs=28

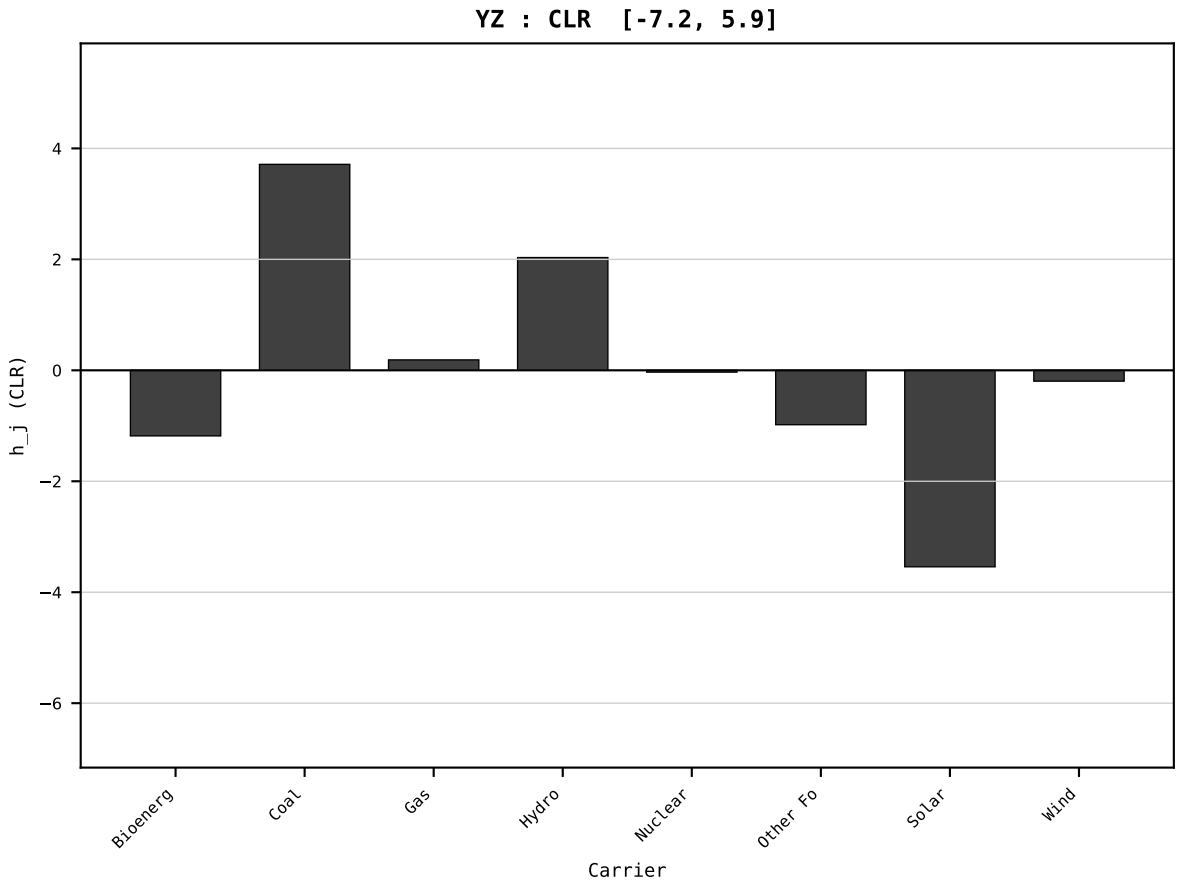
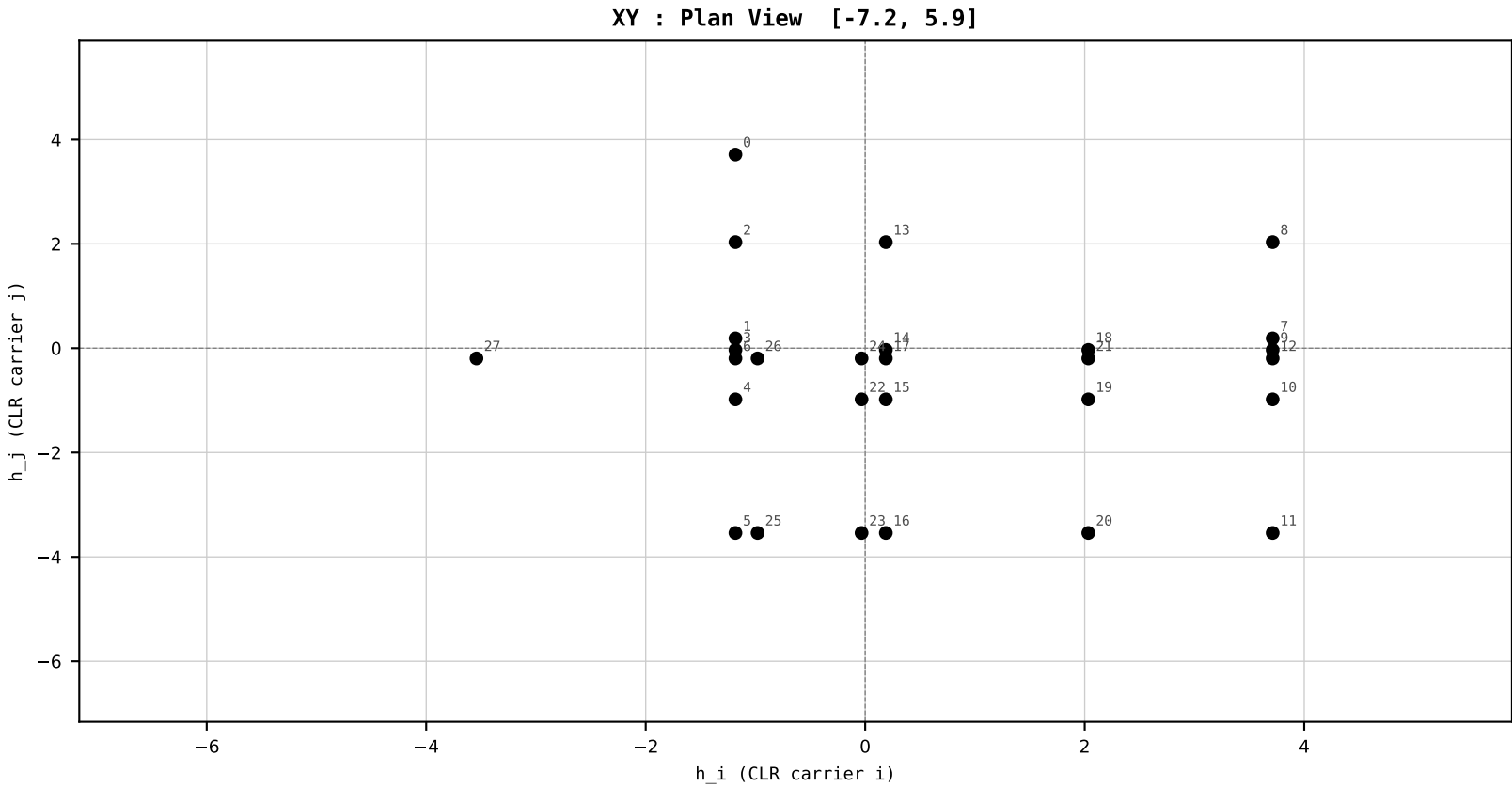
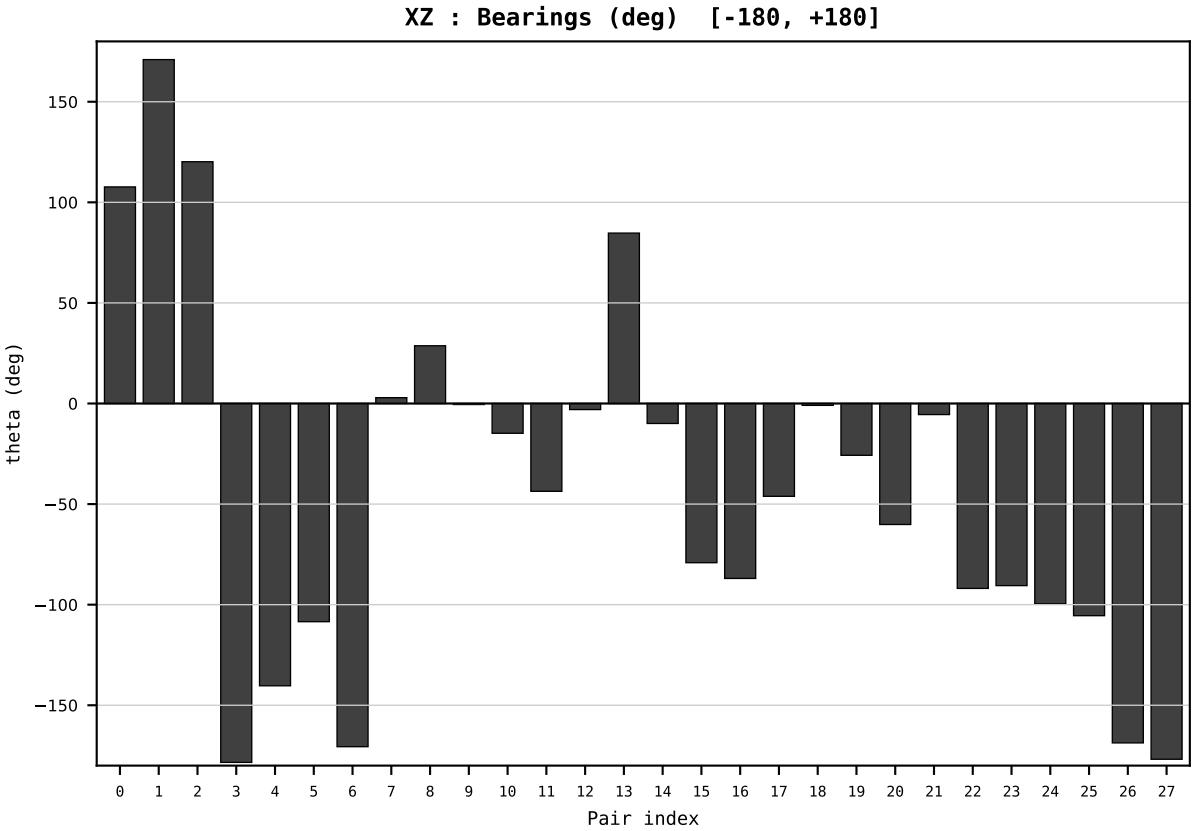
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2011
t : 11 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.630601
Ring = Hs-4
E_metric= 32.8929
kappa_HS= 1414.13
omega = 7.7114 deg
d_A = 1.162016
Helm = Solar
Helm d = +1.017519
DR = 7.254 HLR
DR ratio= 1414.1x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=12 | Year 2012 | D=8 N=26 pairs=28

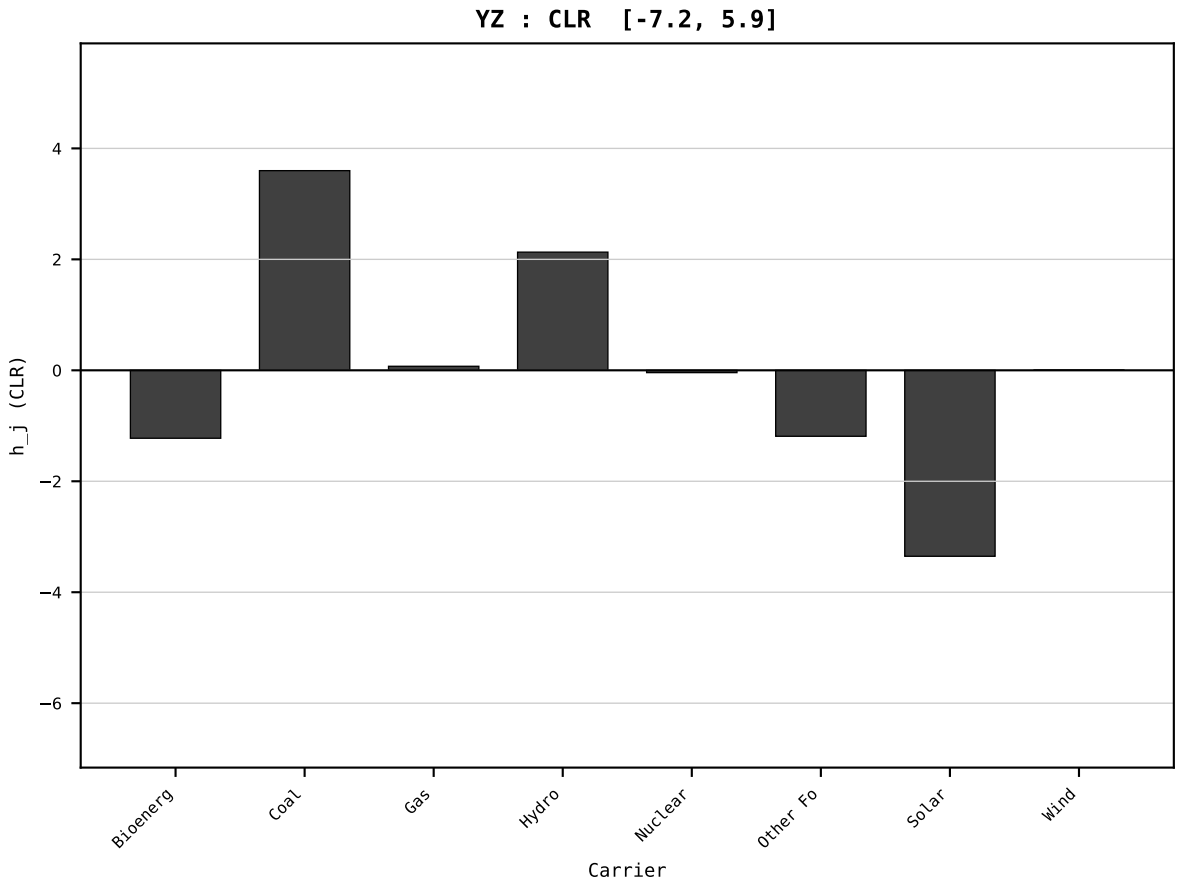
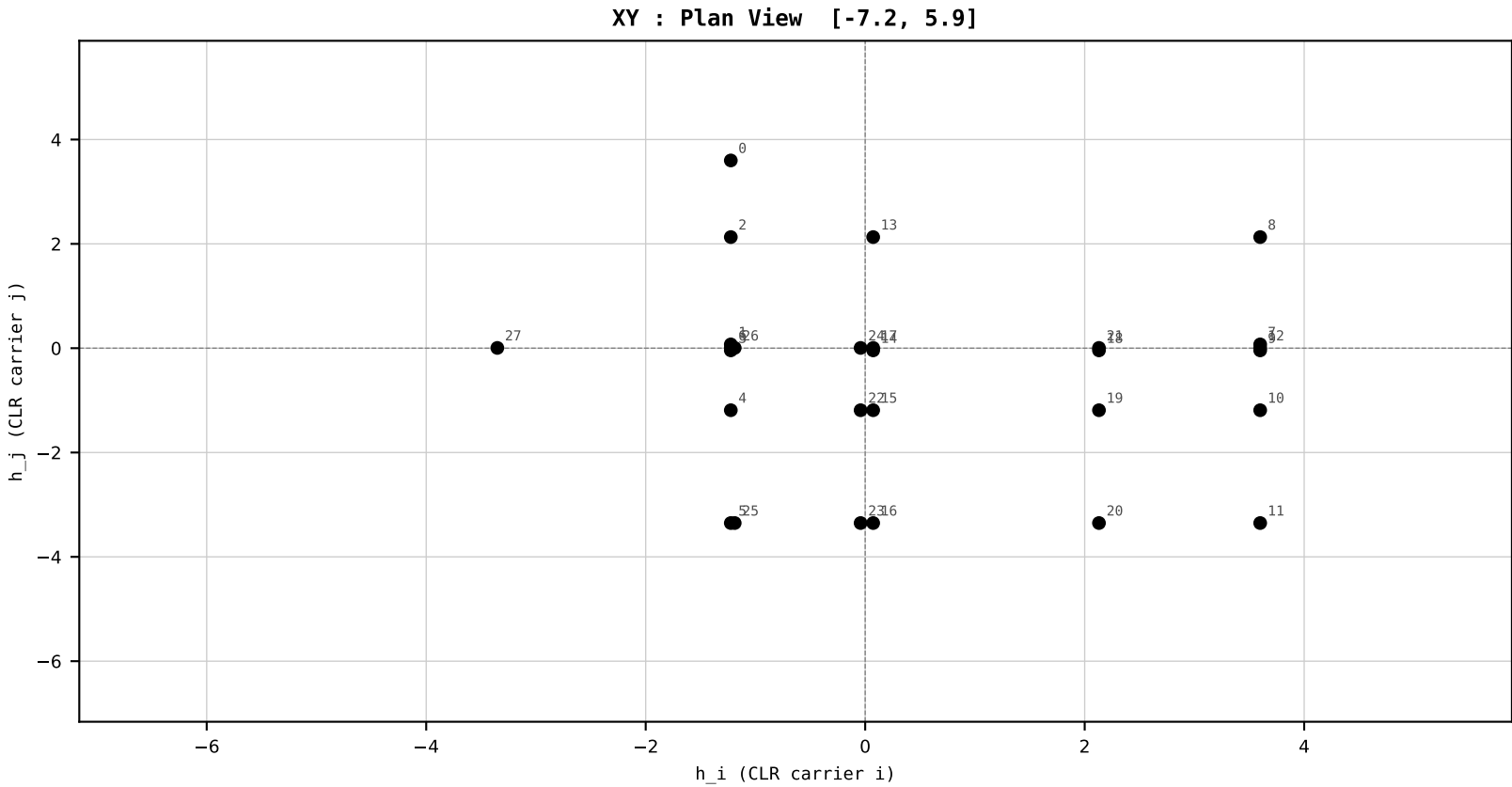
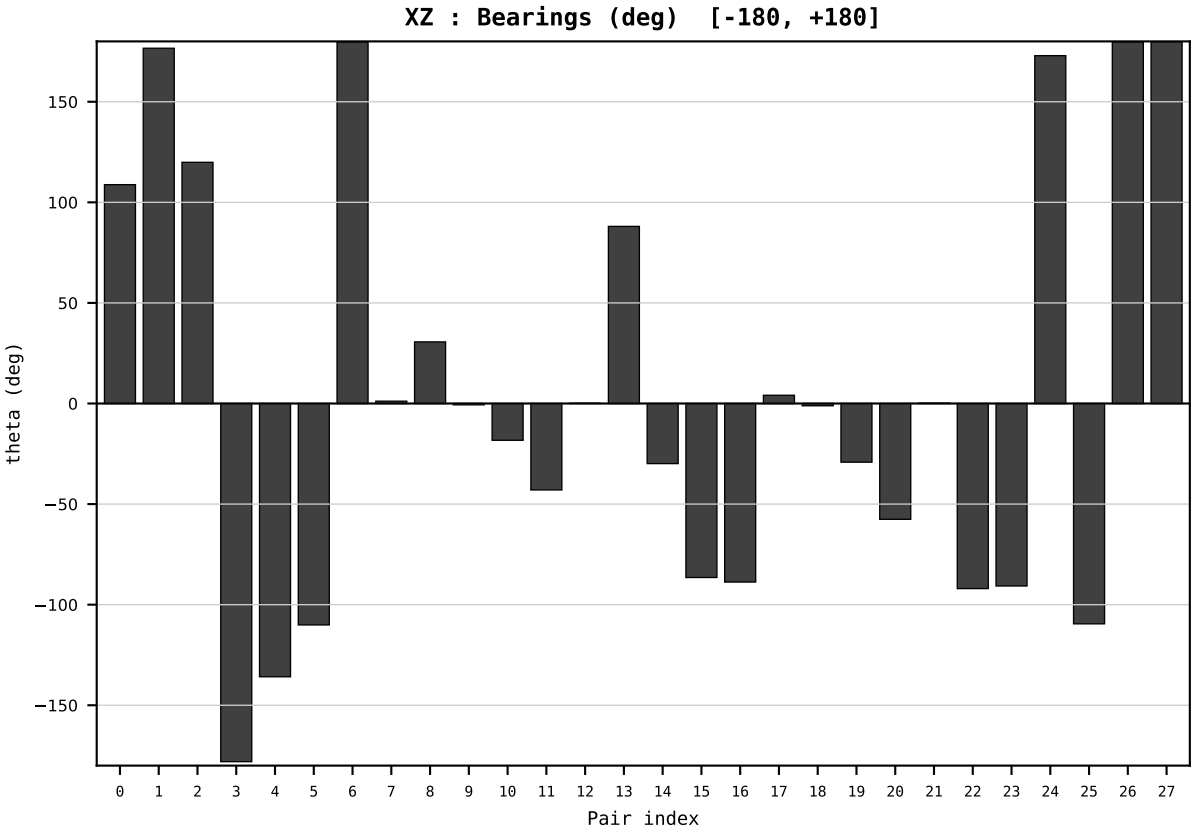
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2012
t : 12 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.601887
Ring = Hs-4
E_metric= 31.6455
kappa_HS= 1044.06
omega = 3.8459 deg
d_A = 0.396692
Helm = Other Fossil
Helm d = -0.208288
DR = 6.951 HLR
DR ratio= 1044.1x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=13 | Year 2013 | D=8 N=26 pairs=28

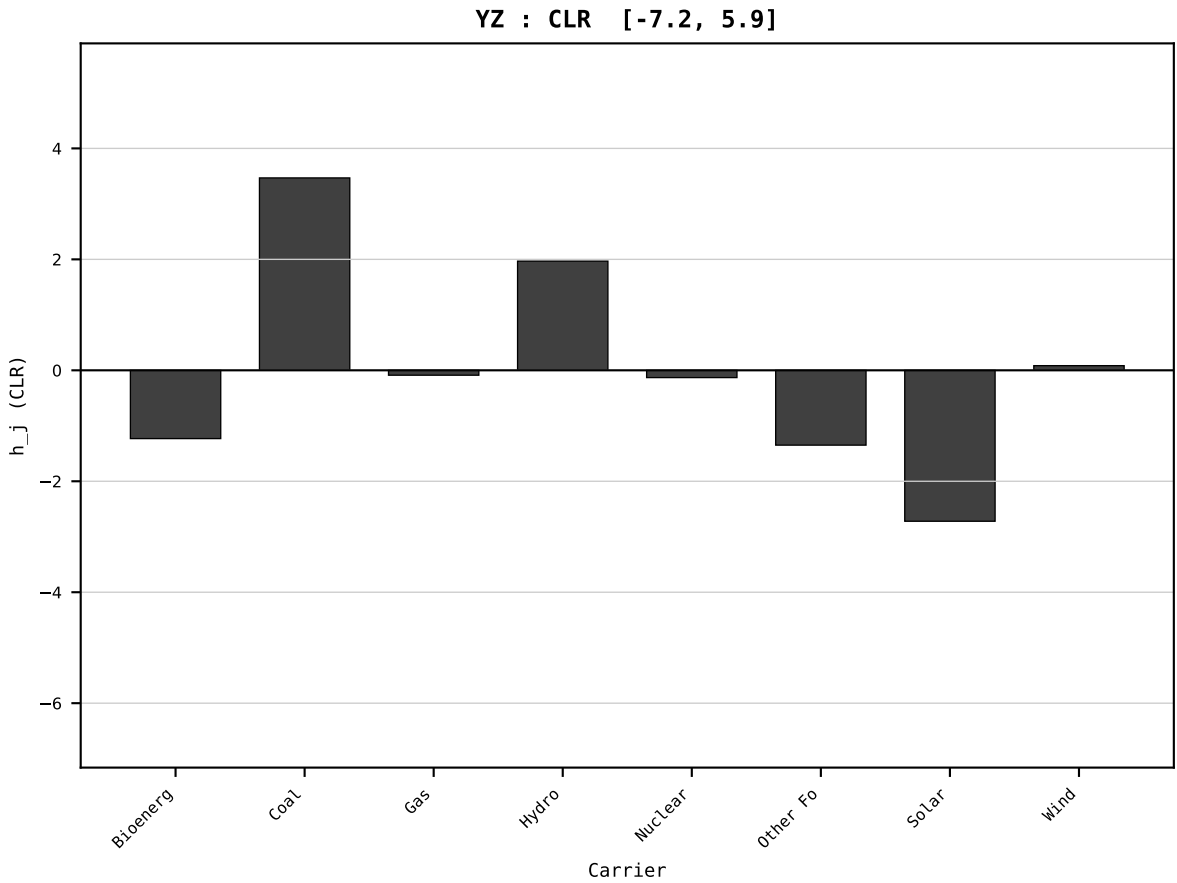
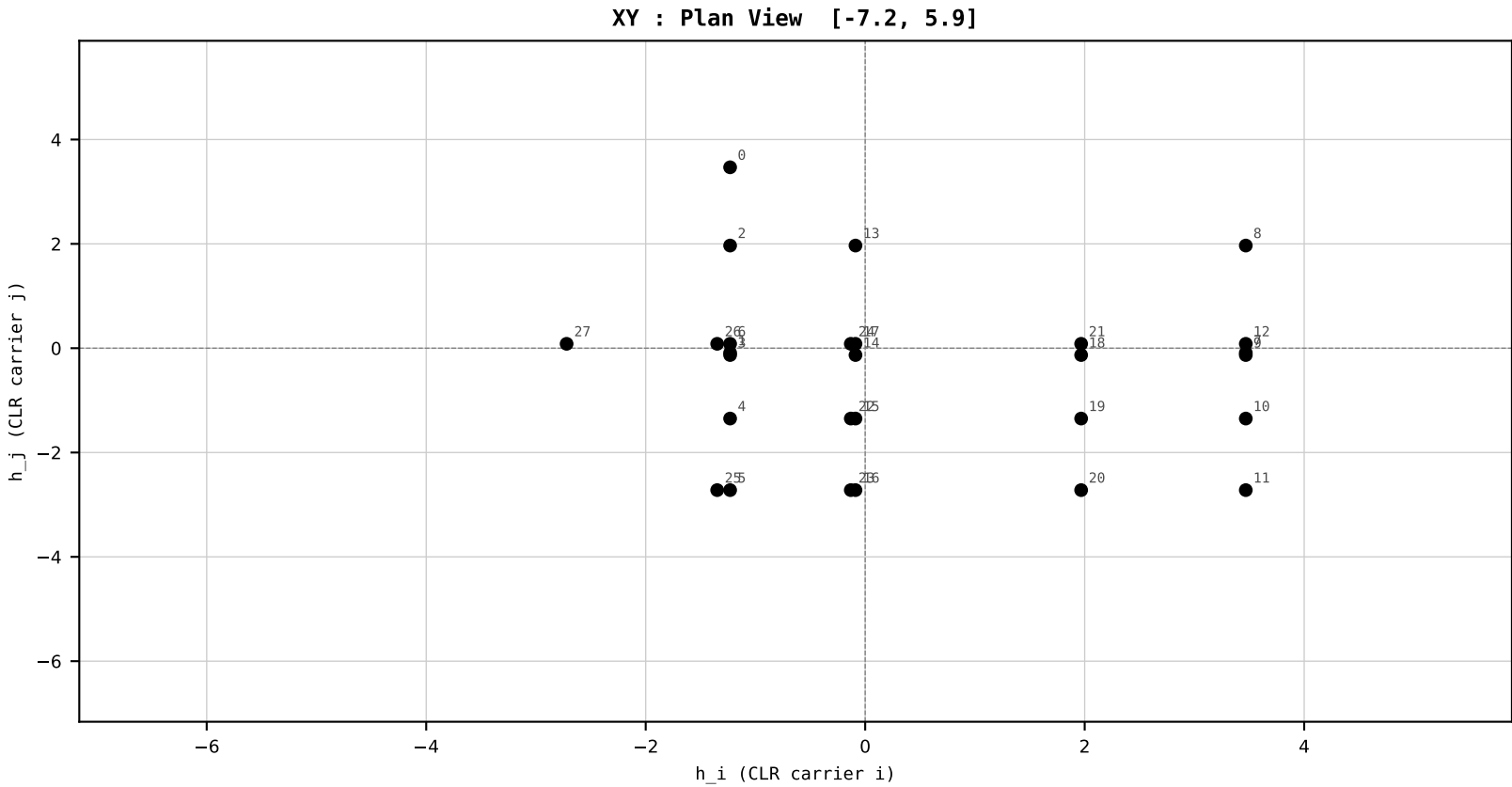
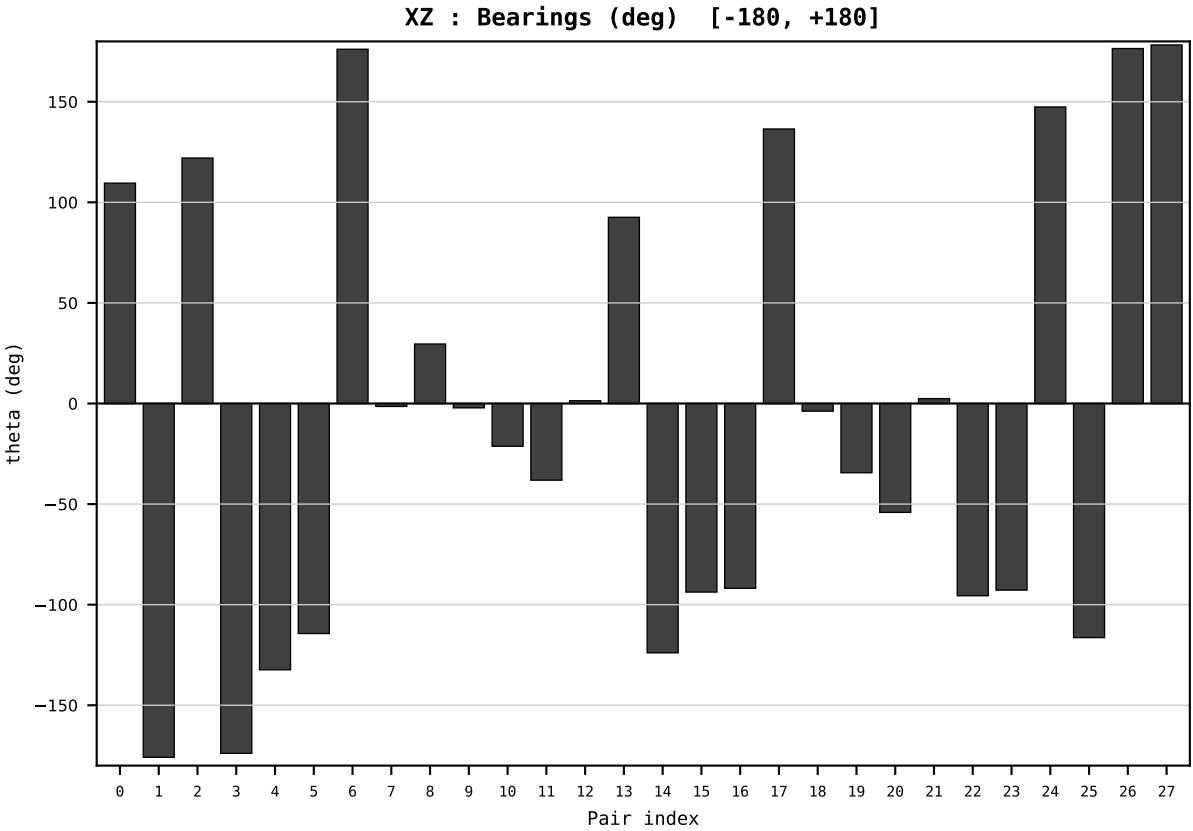
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2013
t : 13 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.593561
Ring = Hs-4
E_metric= 26.6666
kappa_HS= 487.14
omega = 5.7742 deg
d_A = 0.712559
Helm = Solar
Helm d = +0.631278
DR = 6.189 HLR
DR ratio= 487.1x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=14 | Year 2014 | D=8 N=26 pairs=28

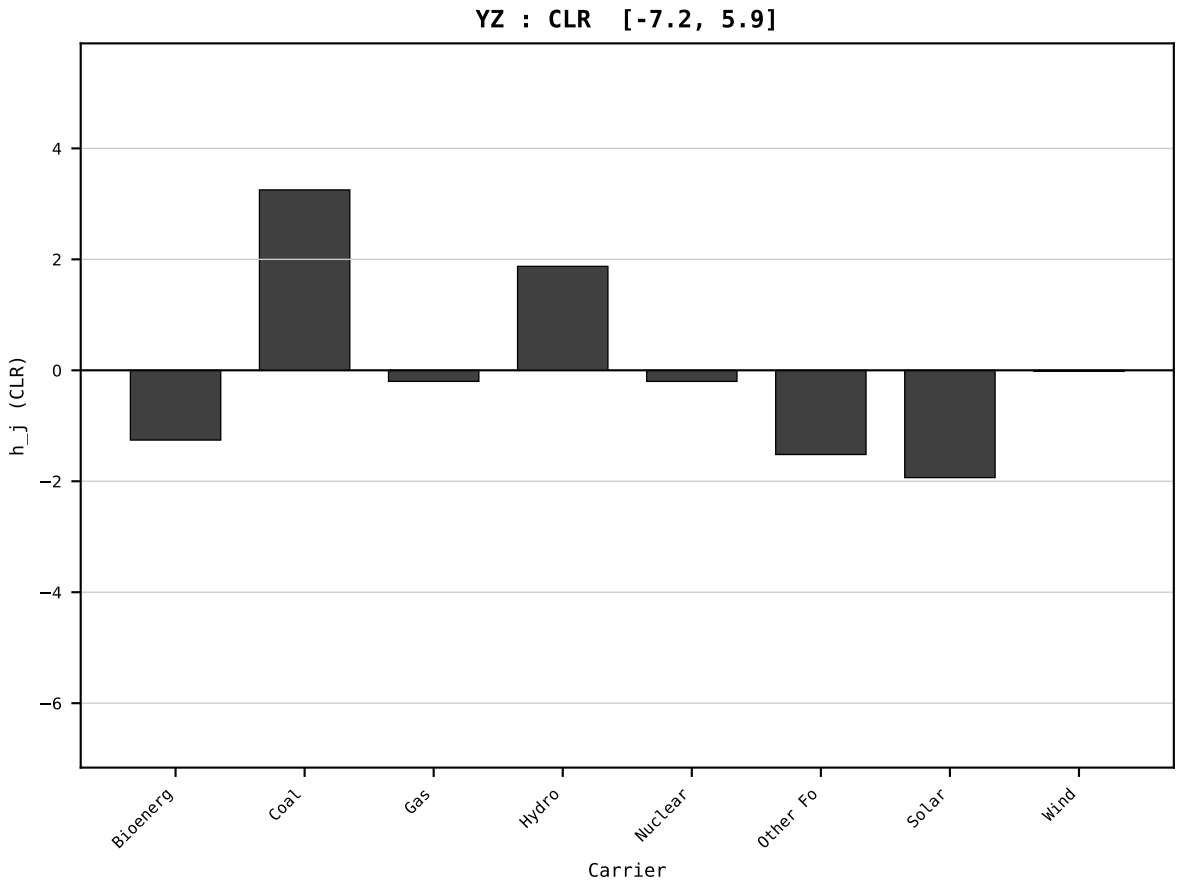
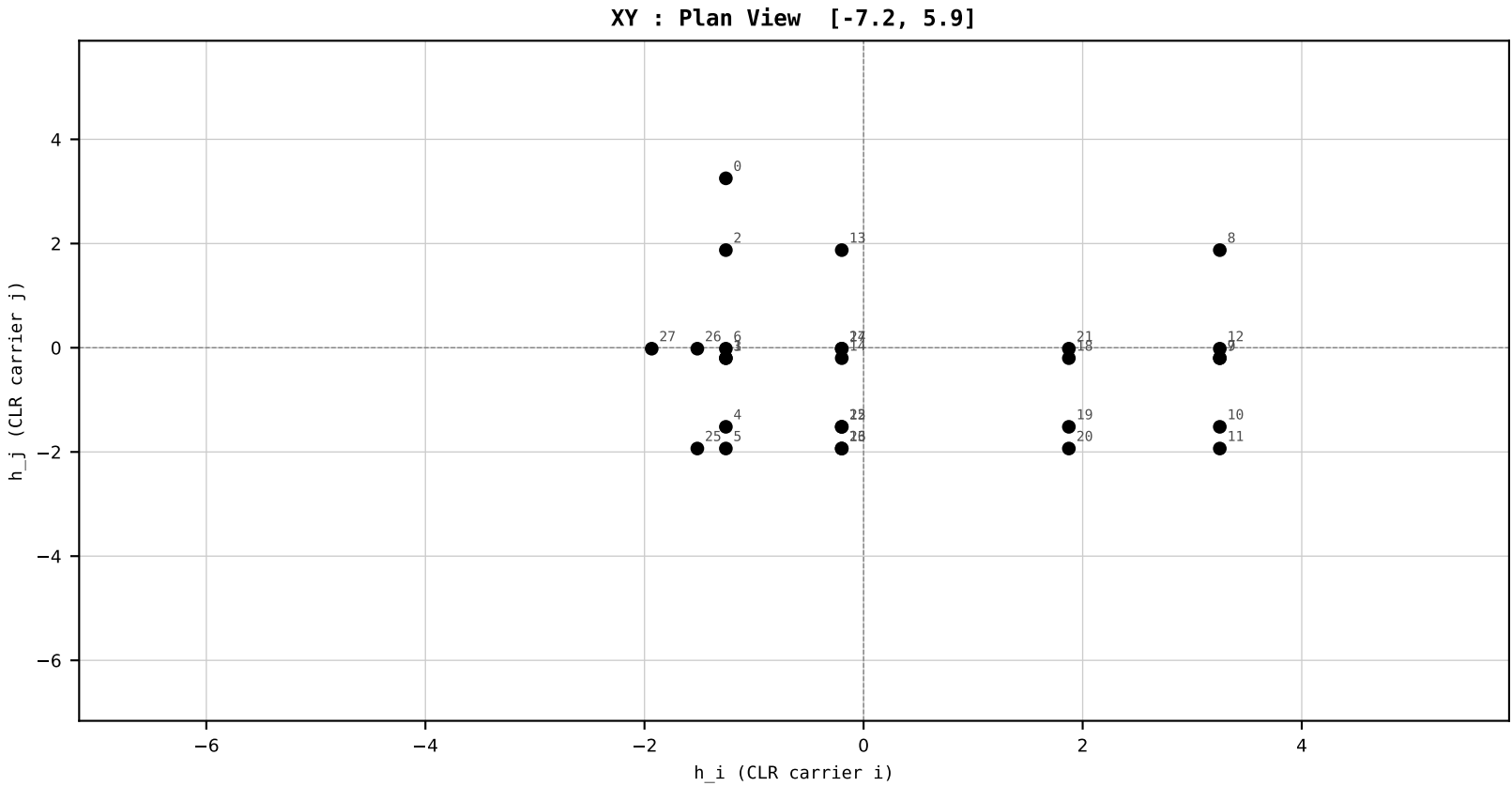
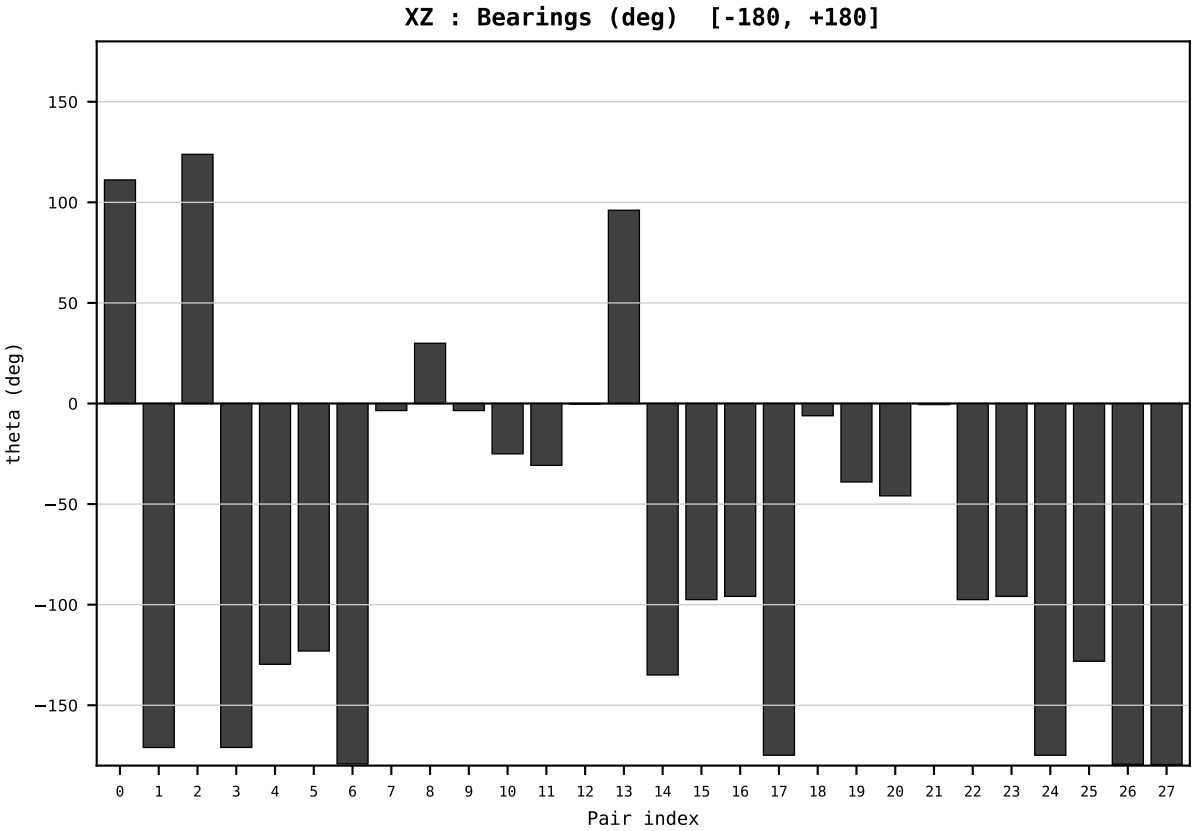
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2014
t : 14 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.563173
Ring = Hs-4
E_metric= 21.7919
kappa_HS= 178.78
omega = 8.1303 deg
d_A = 0.854634
Helm = Solar
Helm d = +0.786340
DR = 5.186 HLR
DR ratio= 178.8x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=15 | Year 2015 | D=8 N=26 pairs=28

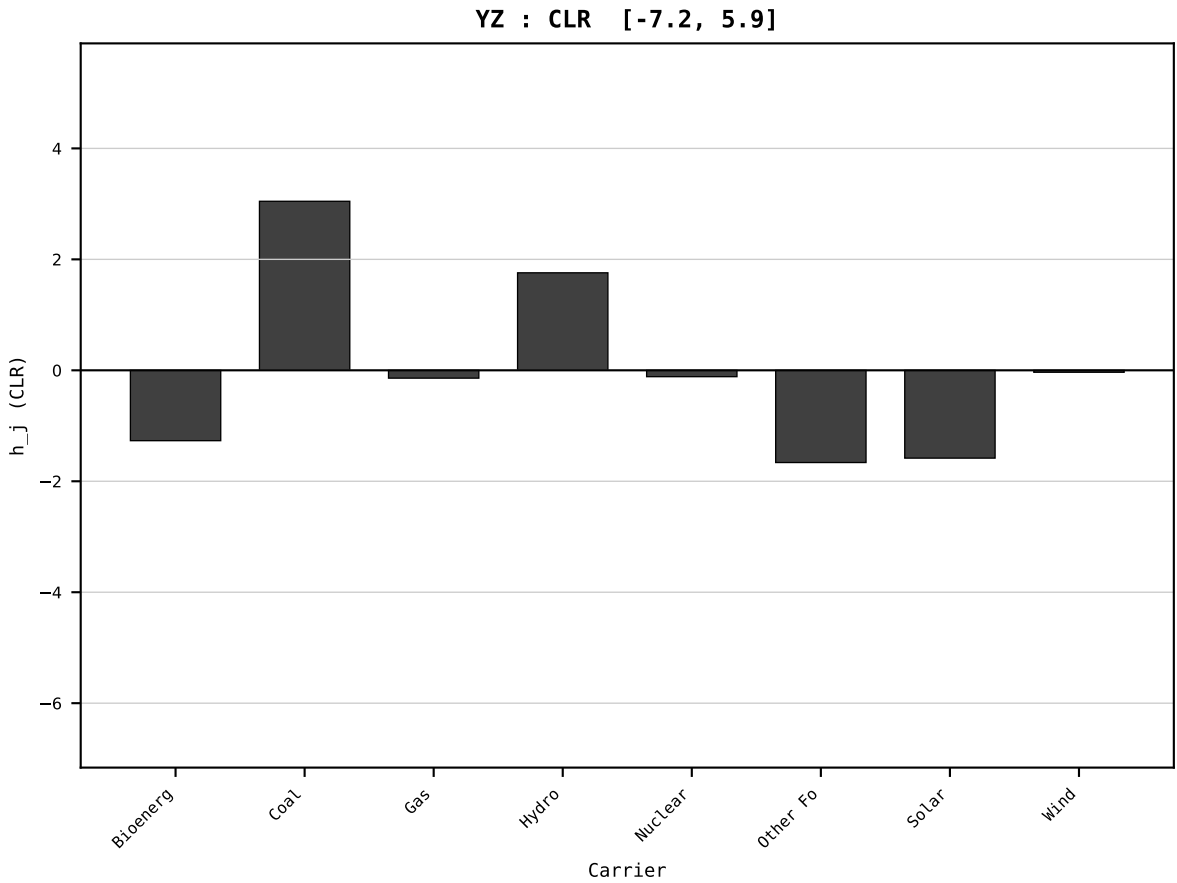
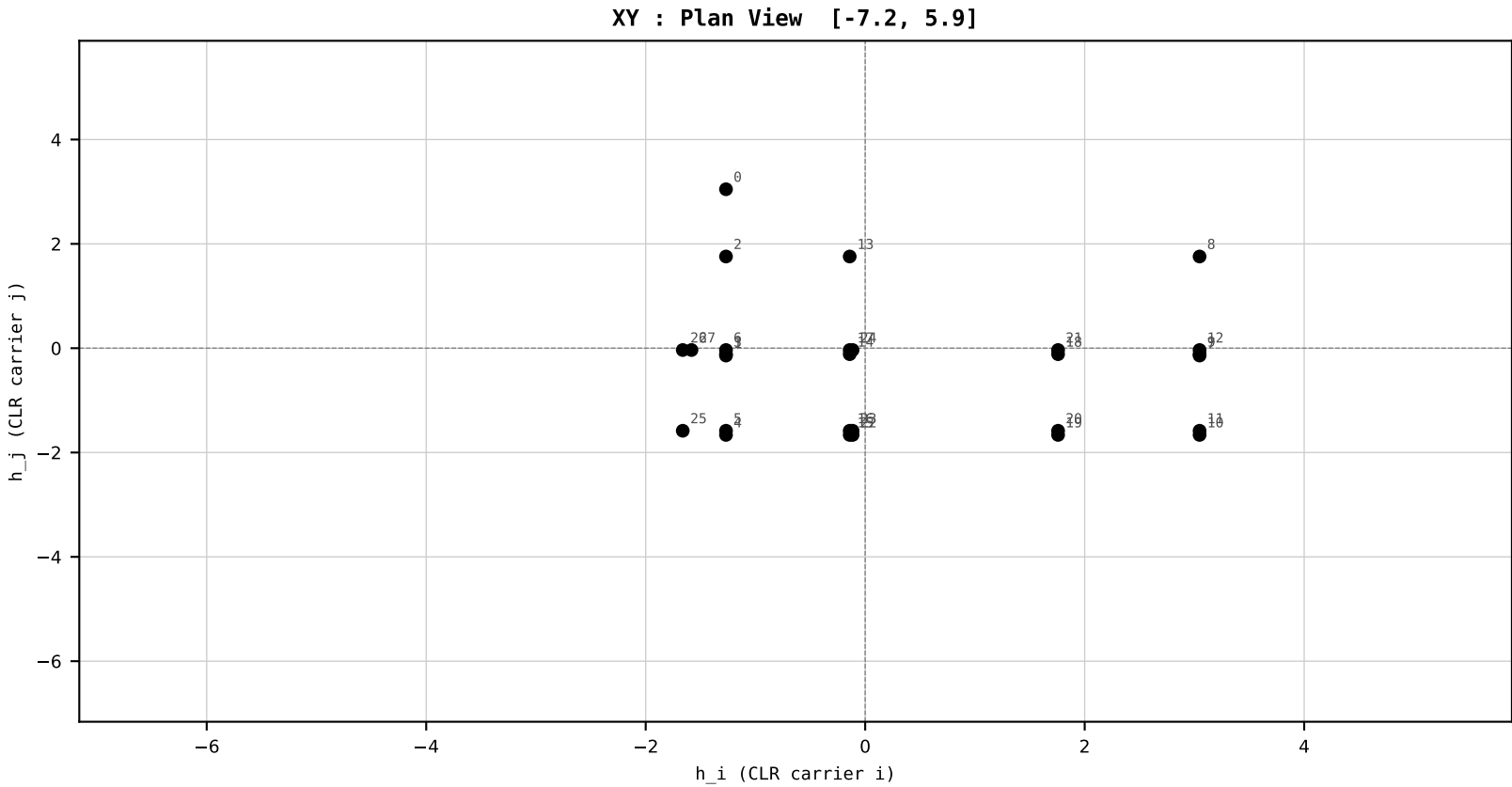
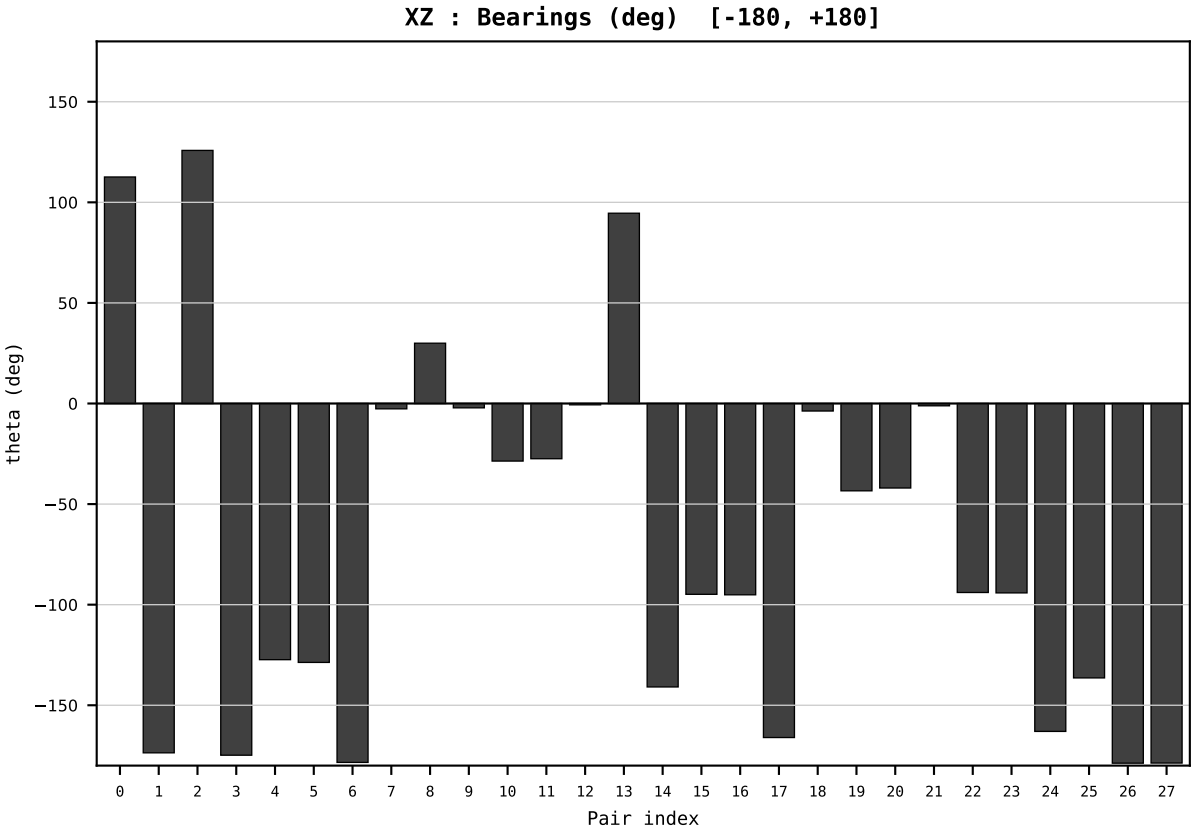
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2015
t : 15 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.522035
Ring = Hs-4
E_metric= 19.2846
kappa_HS= 110.98
omega = 4.6414 deg
d_A = 0.459397
Helm = Solar
Helm d = +0.351431
DR = 4.709 HLR
DR ratio= 111.0x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=16 | Year 2016 | D=8 N=26 pairs=28

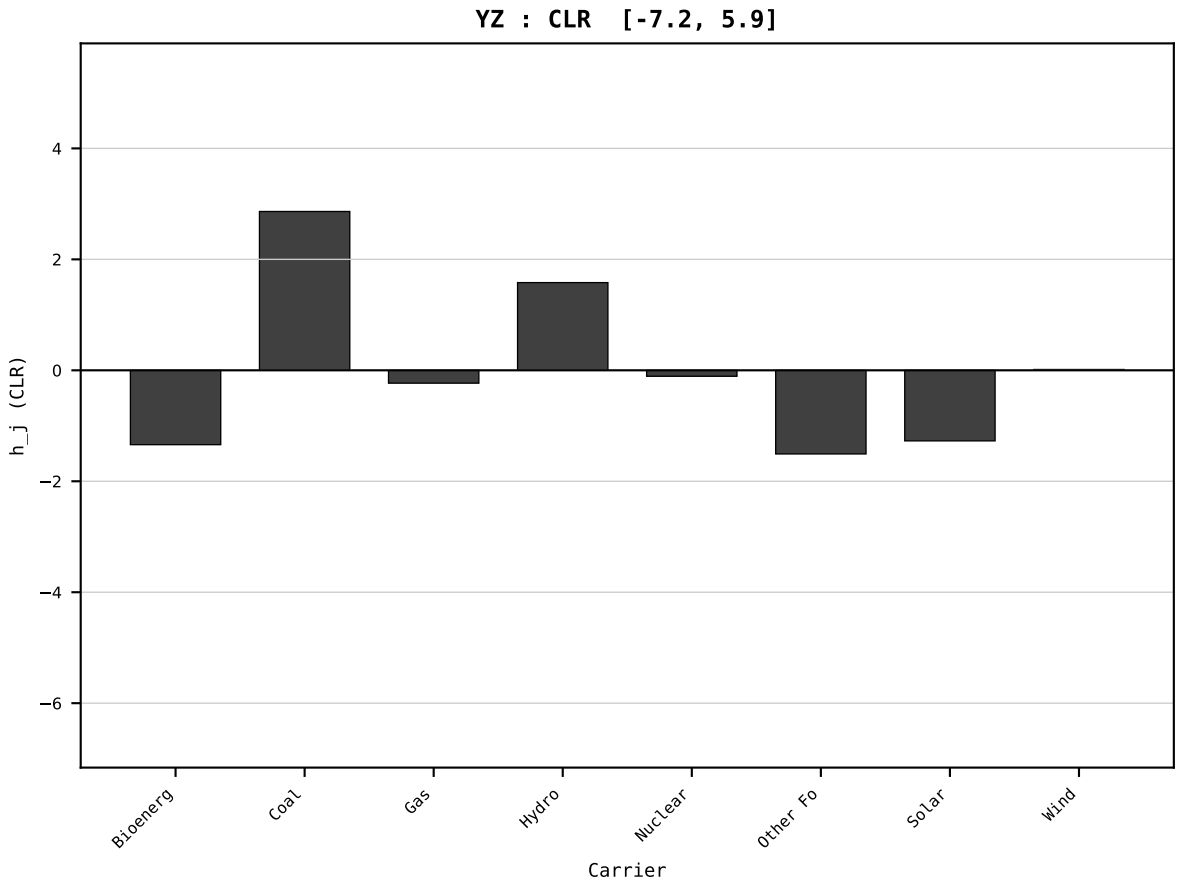
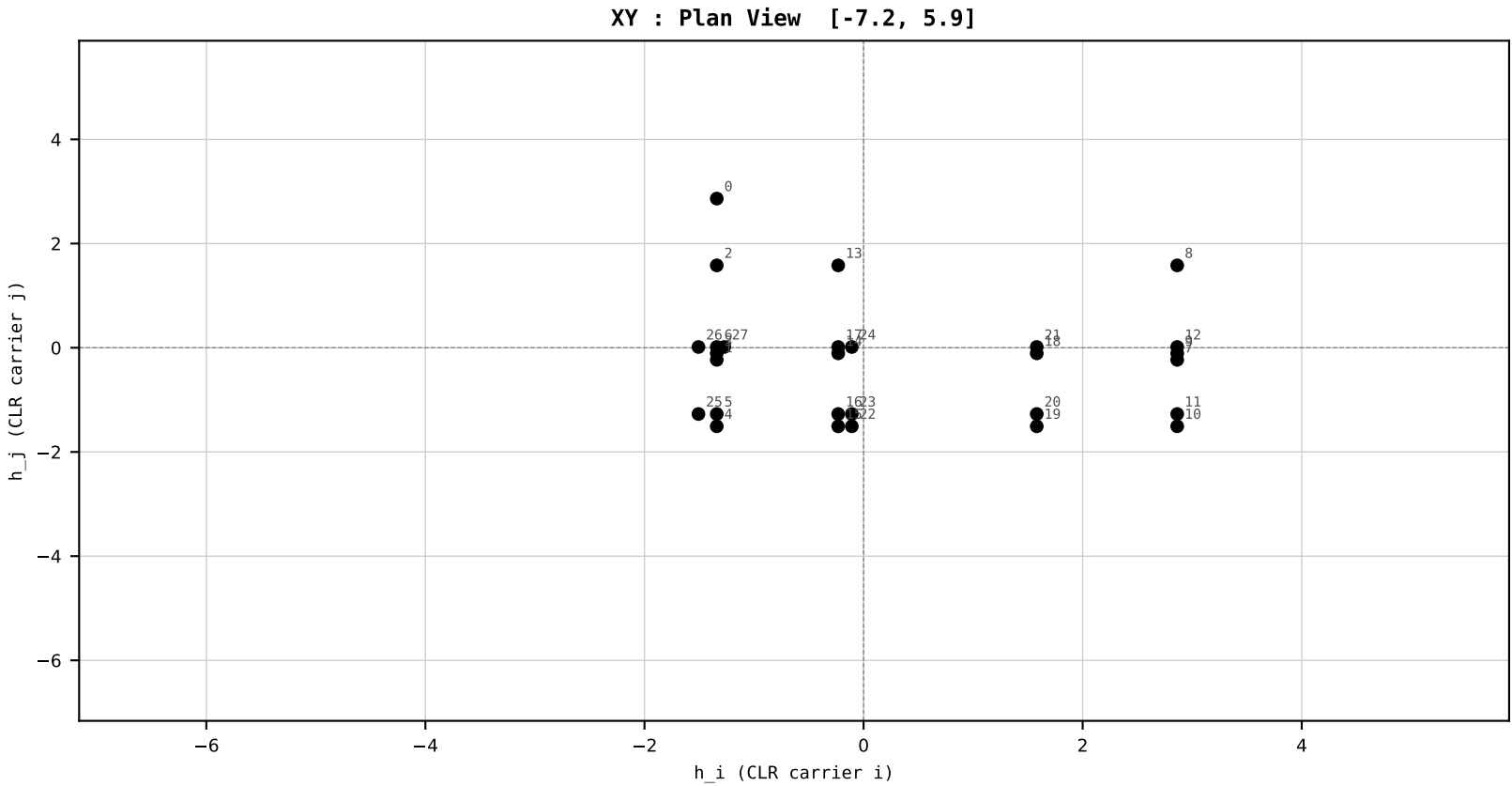
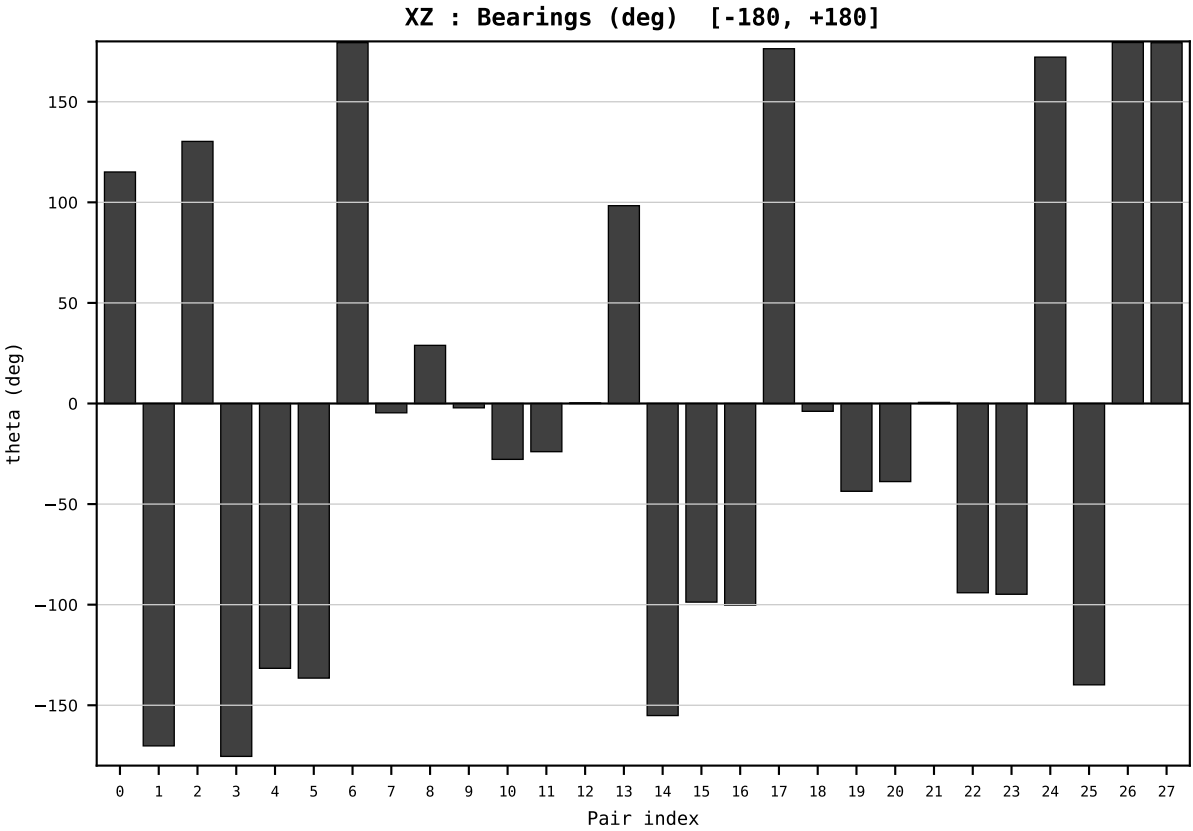
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2016
t : 16 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.487785
Ring = Hs-3
E_metric= 16.4472
kappa_HS= 79.08
omega = 4.0472 deg
d_A = 0.449060
Helm = Solar
Helm d = +0.310984
DR = 4.370 HLR
DR ratio= 79.1x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=17 | Year 2017 | D=8 N=26 pairs=28

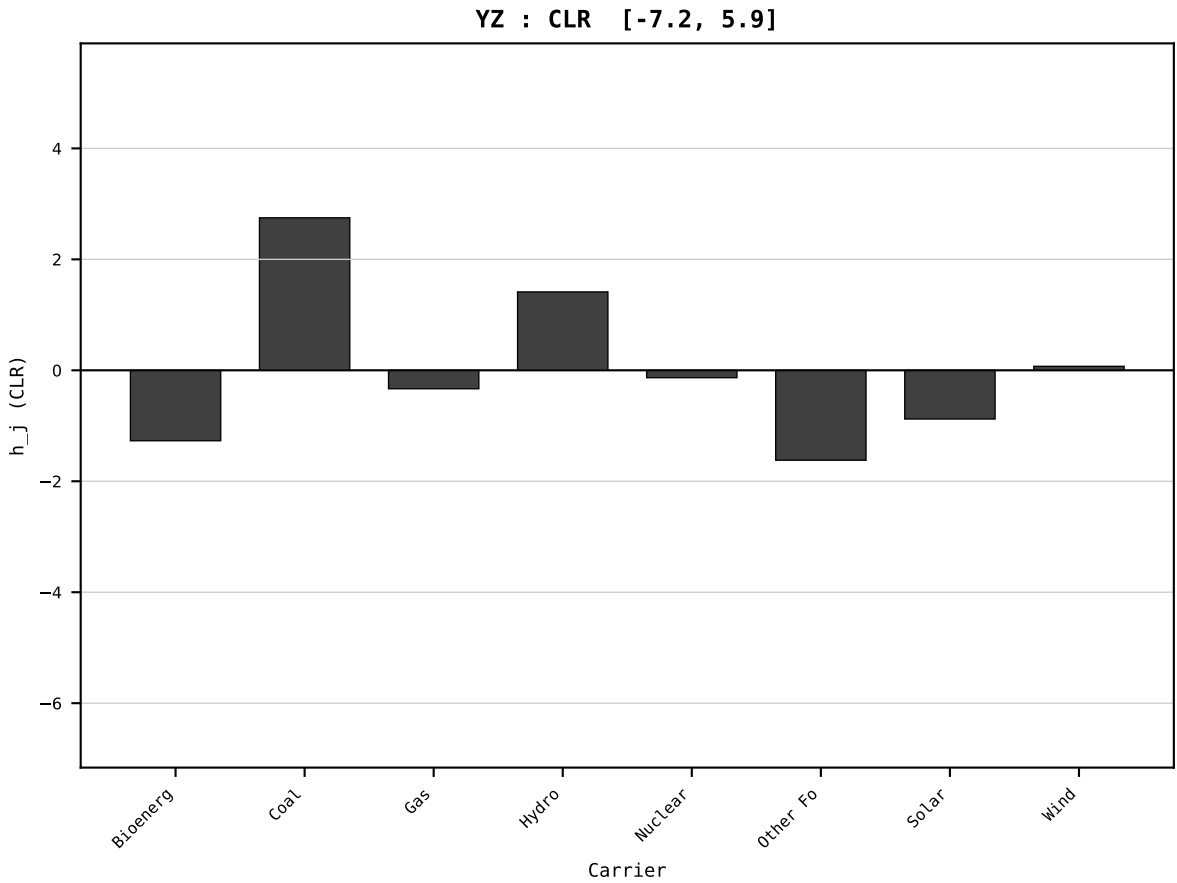
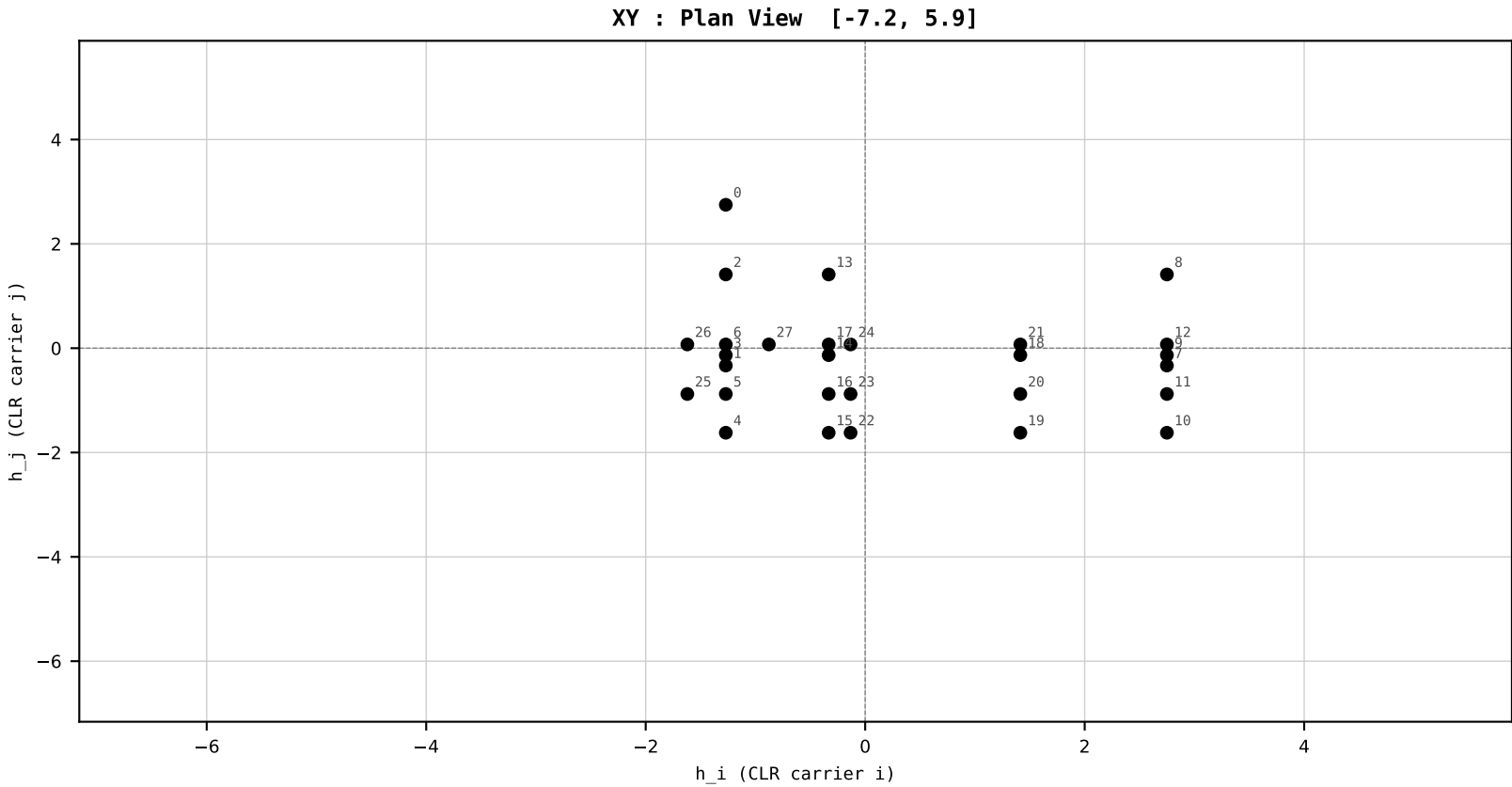
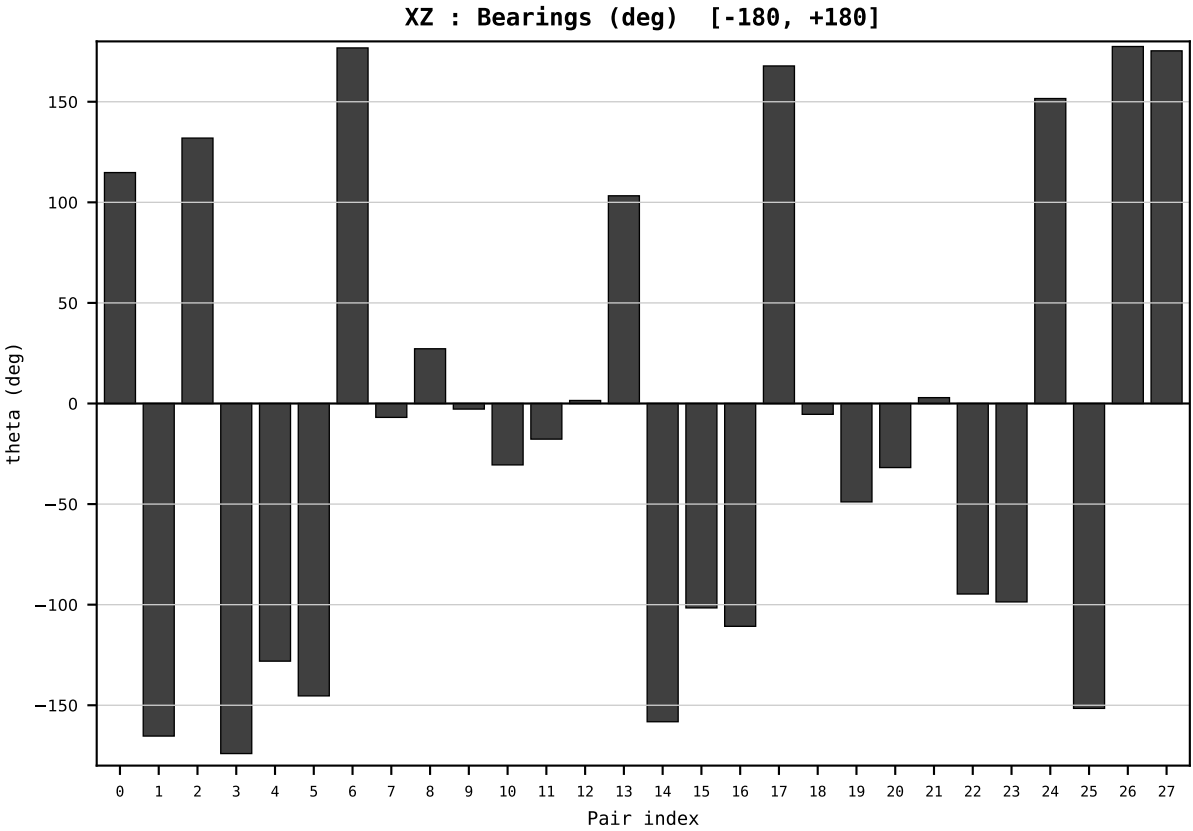
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2017
t : 17 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.465364
Ring = Hs-3
E_metric= 14.7002
kappa_HS= 79.04
omega = 6.1498 deg
d_A = 0.477487
Helm = Solar
Helm d = +0.393824
DR = 4.370 HLR
DR ratio= 79.0x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=18 | Year 2018 | D=8 N=26 pairs=28

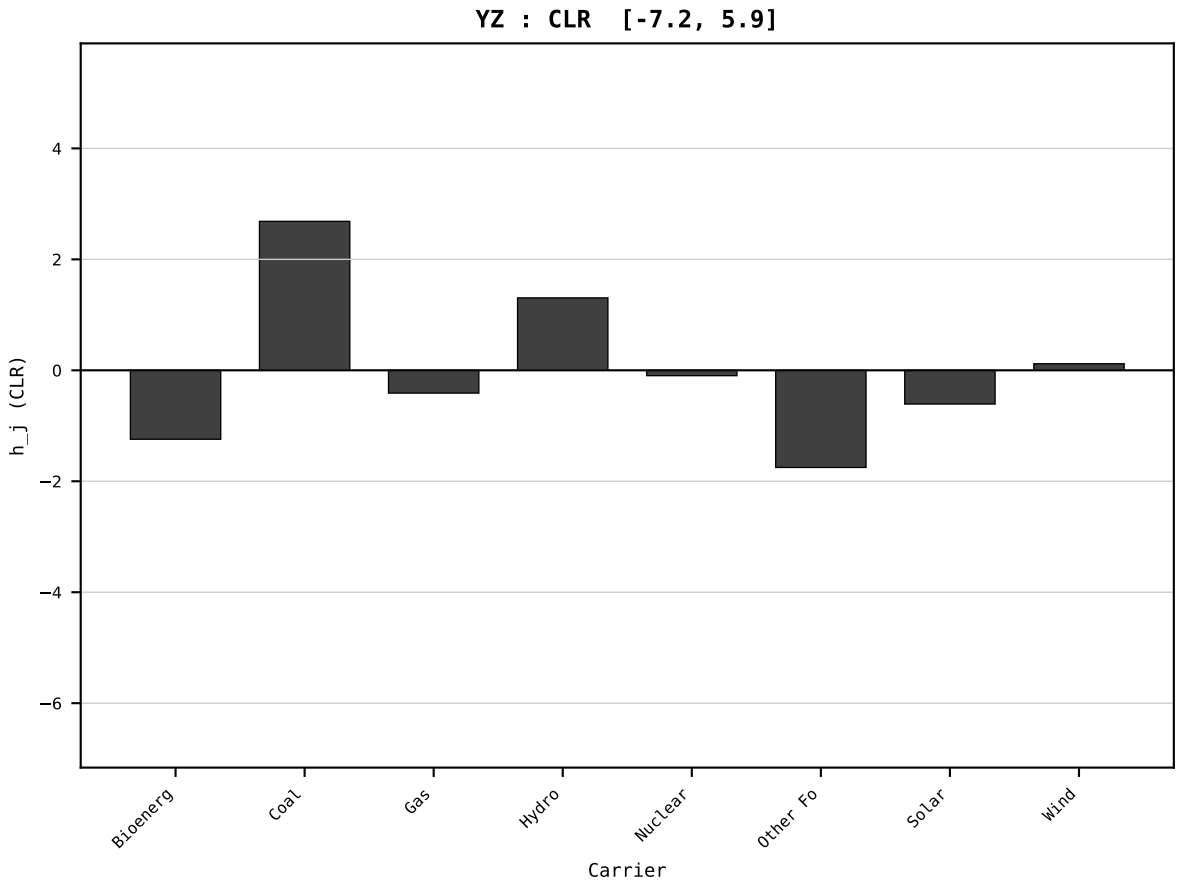
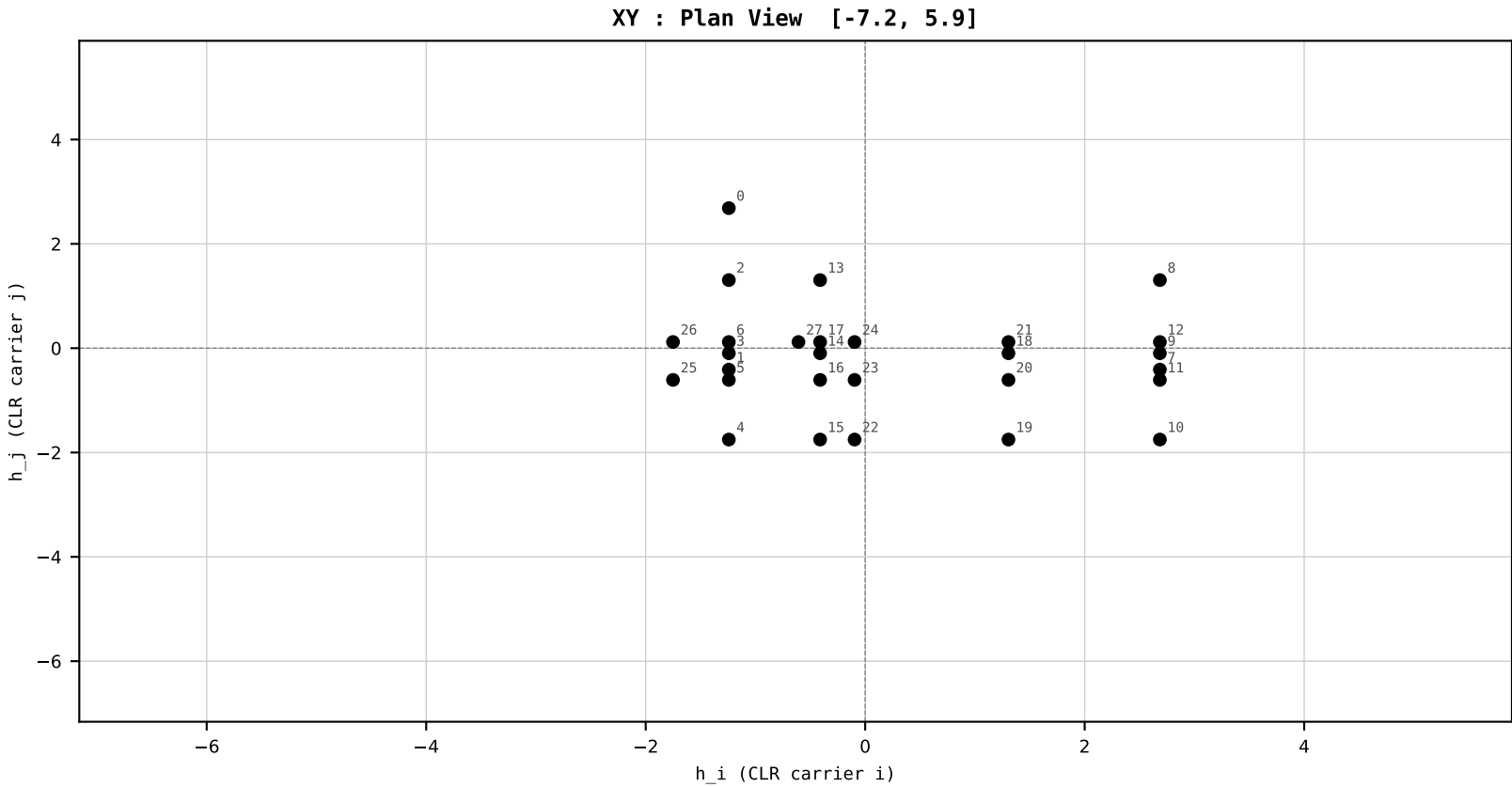
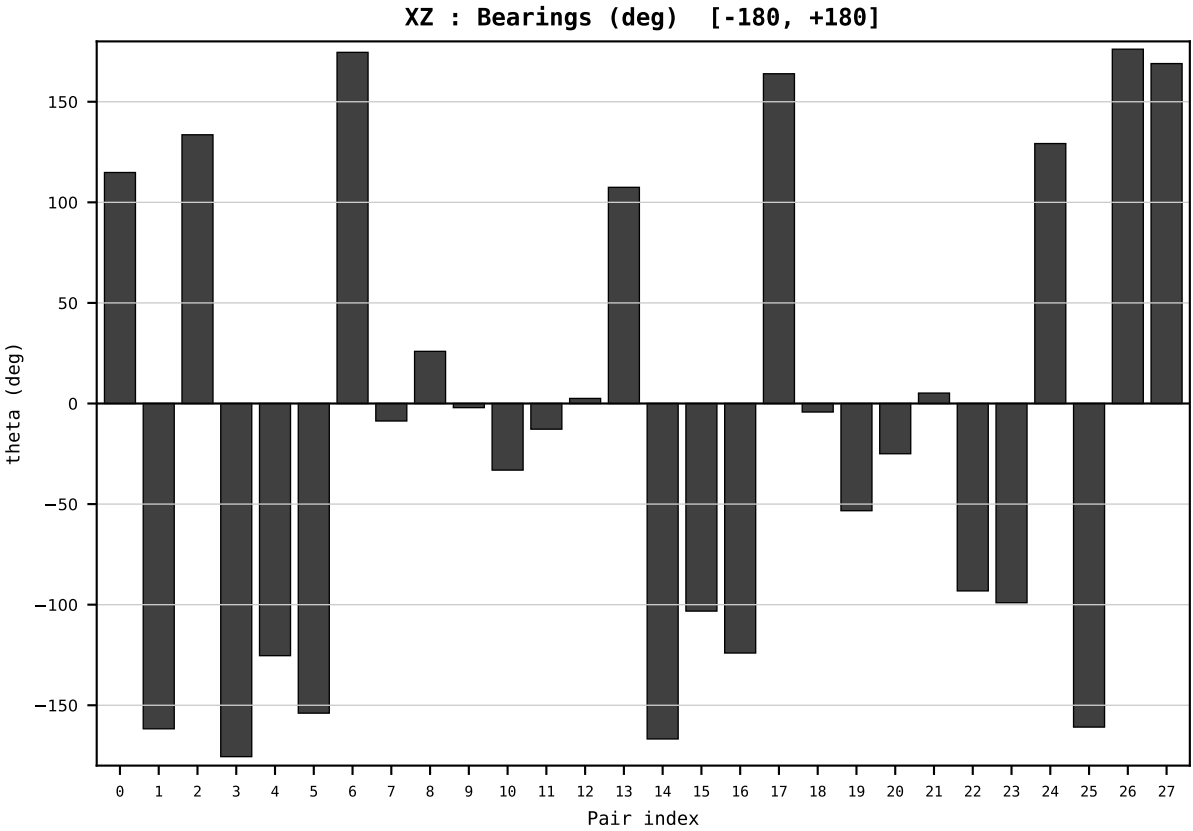
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2018
t : 18 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.449187
Ring = Hs-3
E_metric= 14.0866
kappa_HS= 84.47
omega = 4.9977 deg
d_A = 0.340525
Helm = Solar
Helm d = +0.269941
DR = 4.436 HLR
DR ratio= 84.5x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=19 | Year 2019 | D=8 N=26 pairs=28

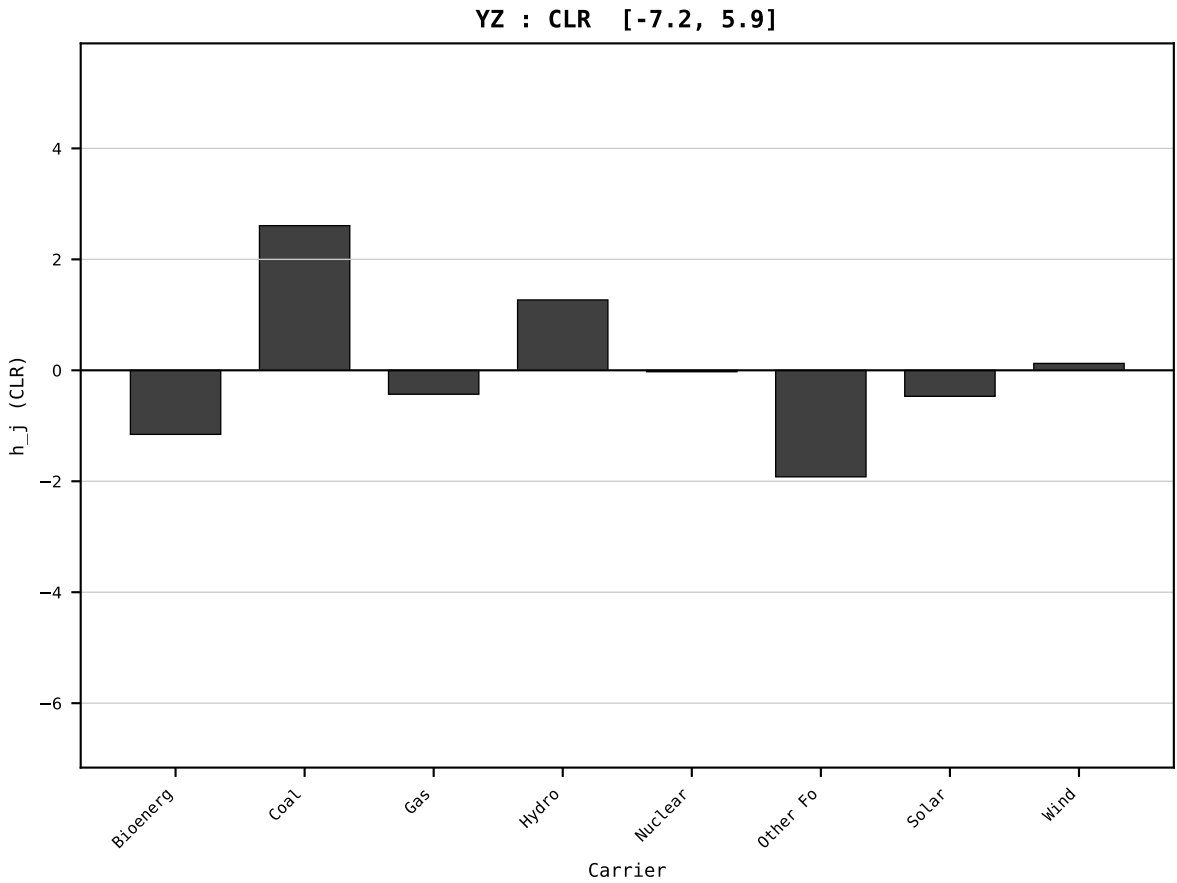
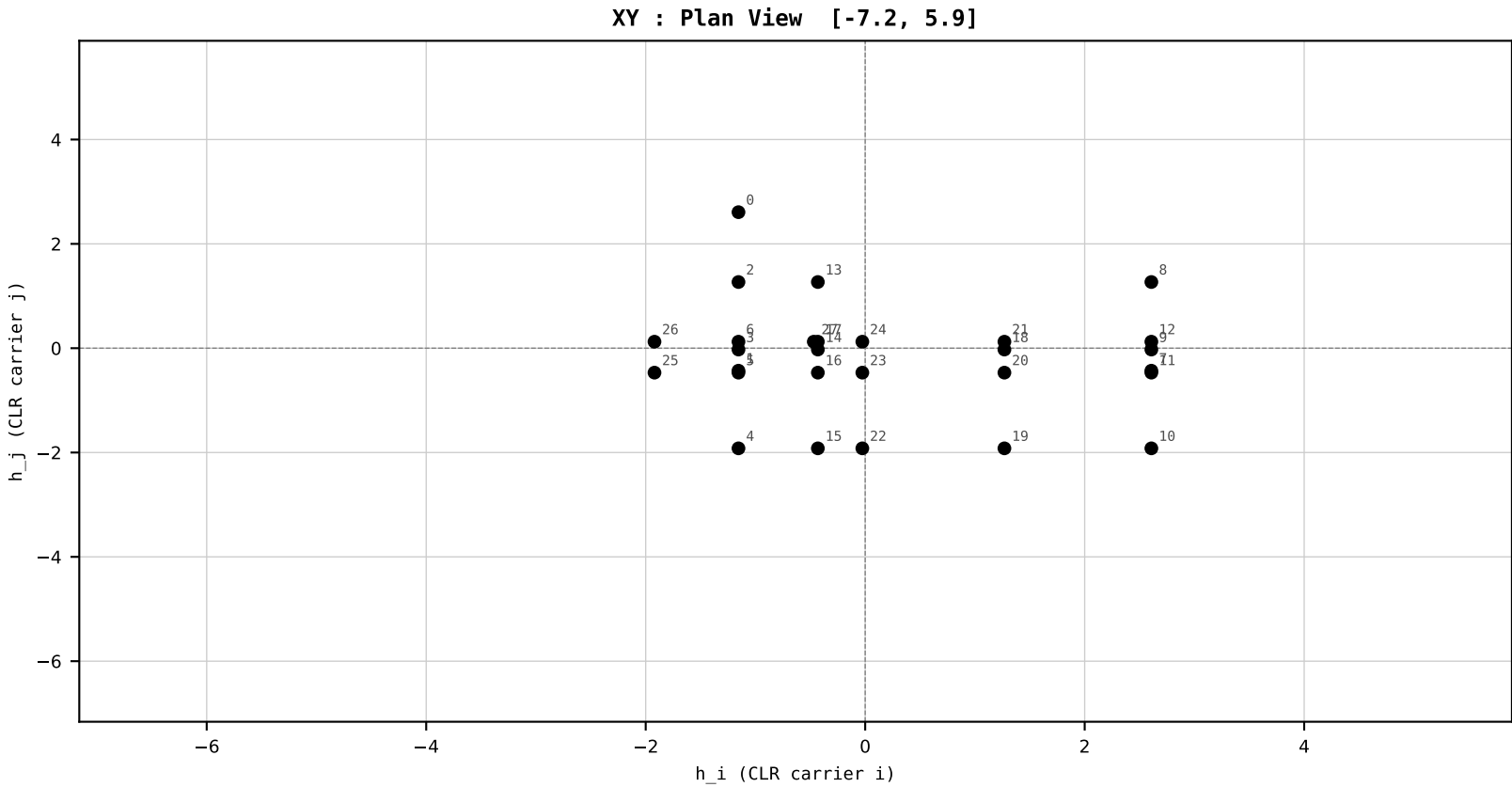
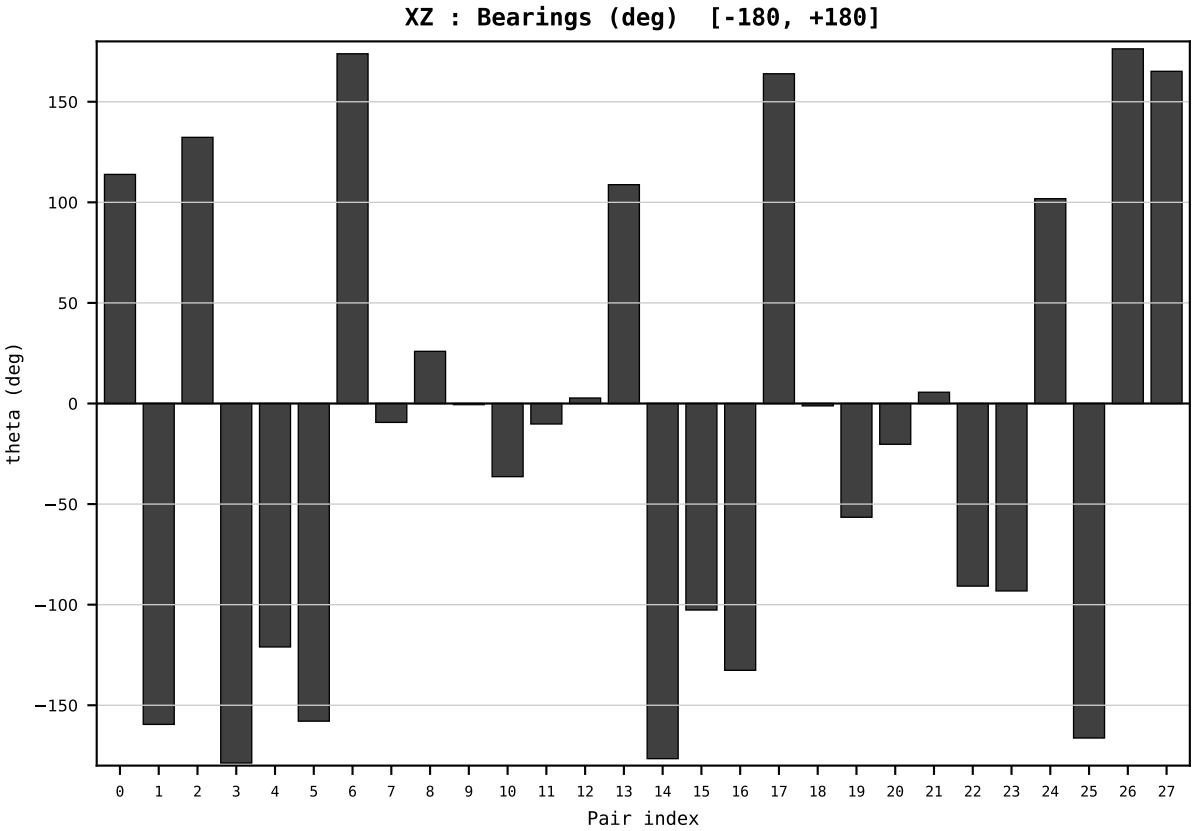
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2019
t : 19 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.426218
Ring = Hs-3
E_metric= 13.8498
kappa_HS= 92.53
omega = 3.9842 deg
d_A = 0.261754
Helm = Other Fossil
Helm d = -0.168801
DR = 4.528 HLR
DR ratio= 92.5x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=20 | Year 2020 | D=8 N=26 pairs=28

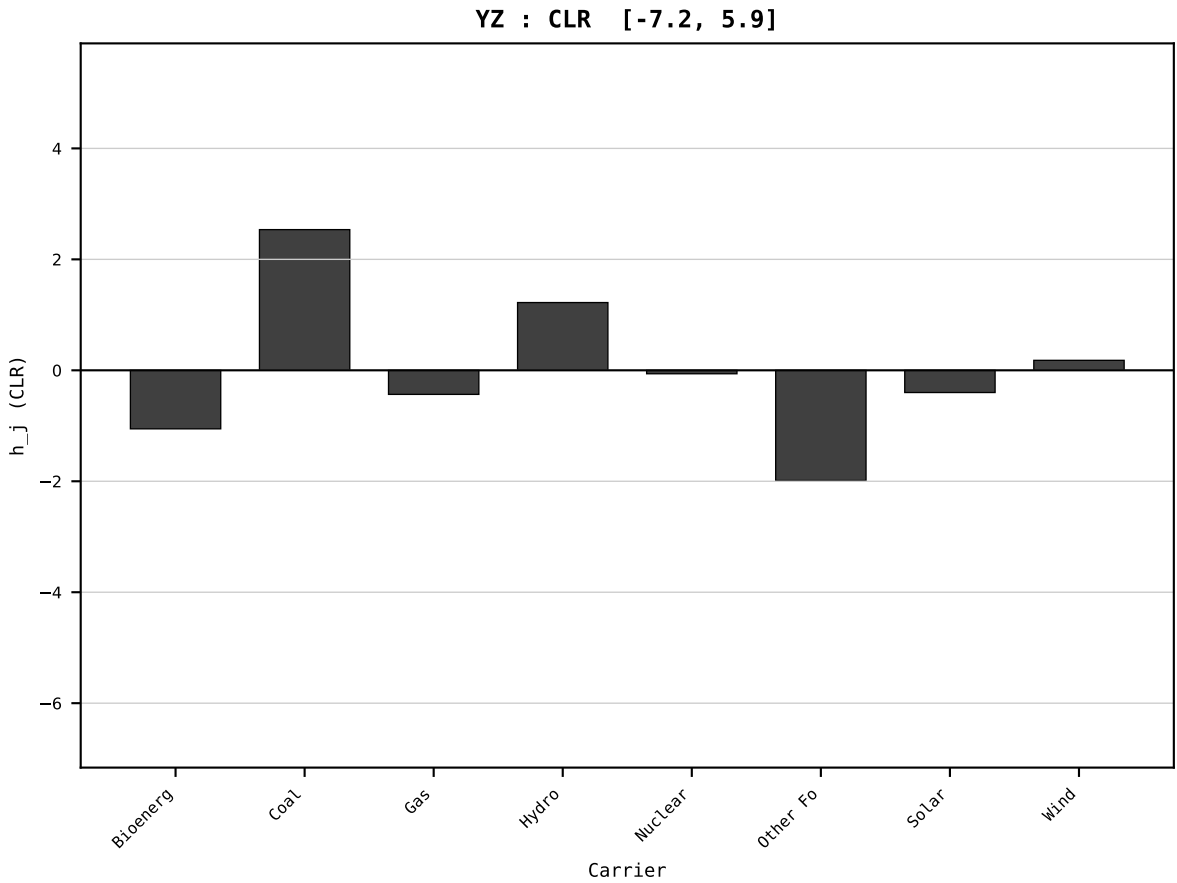
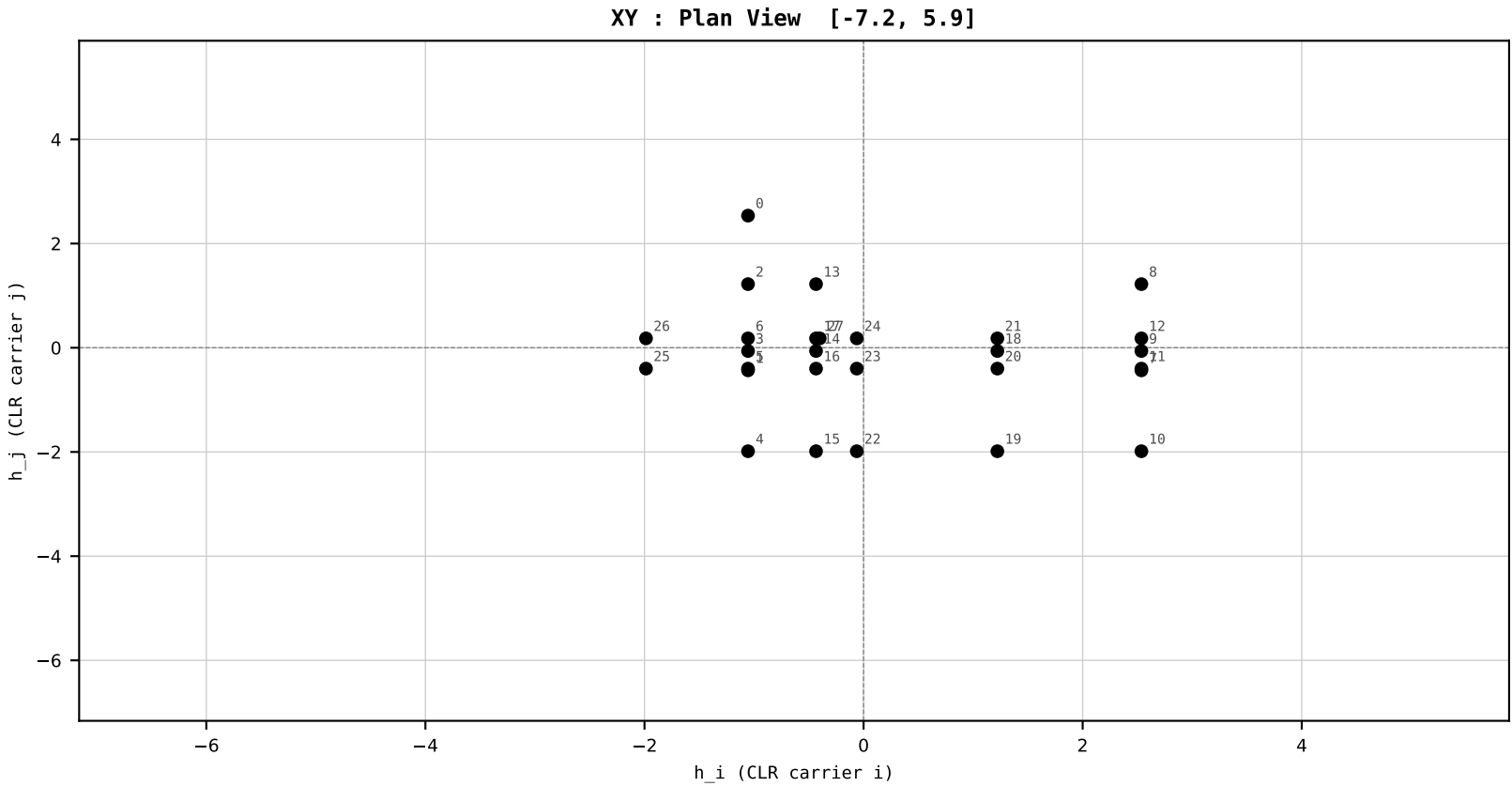
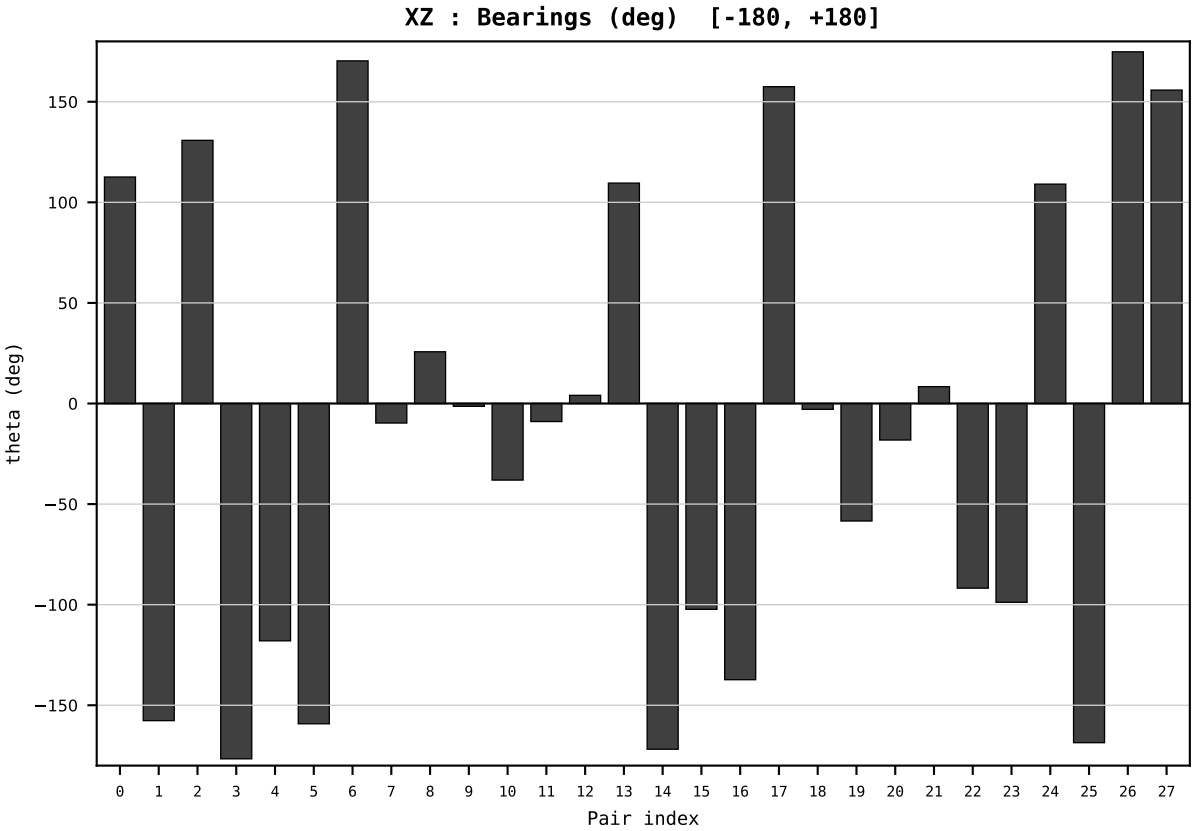
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2020
t : 20 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.406843
Ring = Hs-3
E_metric= 13.3663
kappa_HS= 92.05
omega = 2.5244 deg
d_A = 0.175223
Helm = Bioenergy
Helm d = +0.099748
DR = 4.522 HLR
DR ratio= 92.0x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=21 | Year 2021 | D=8 N=26 pairs=28

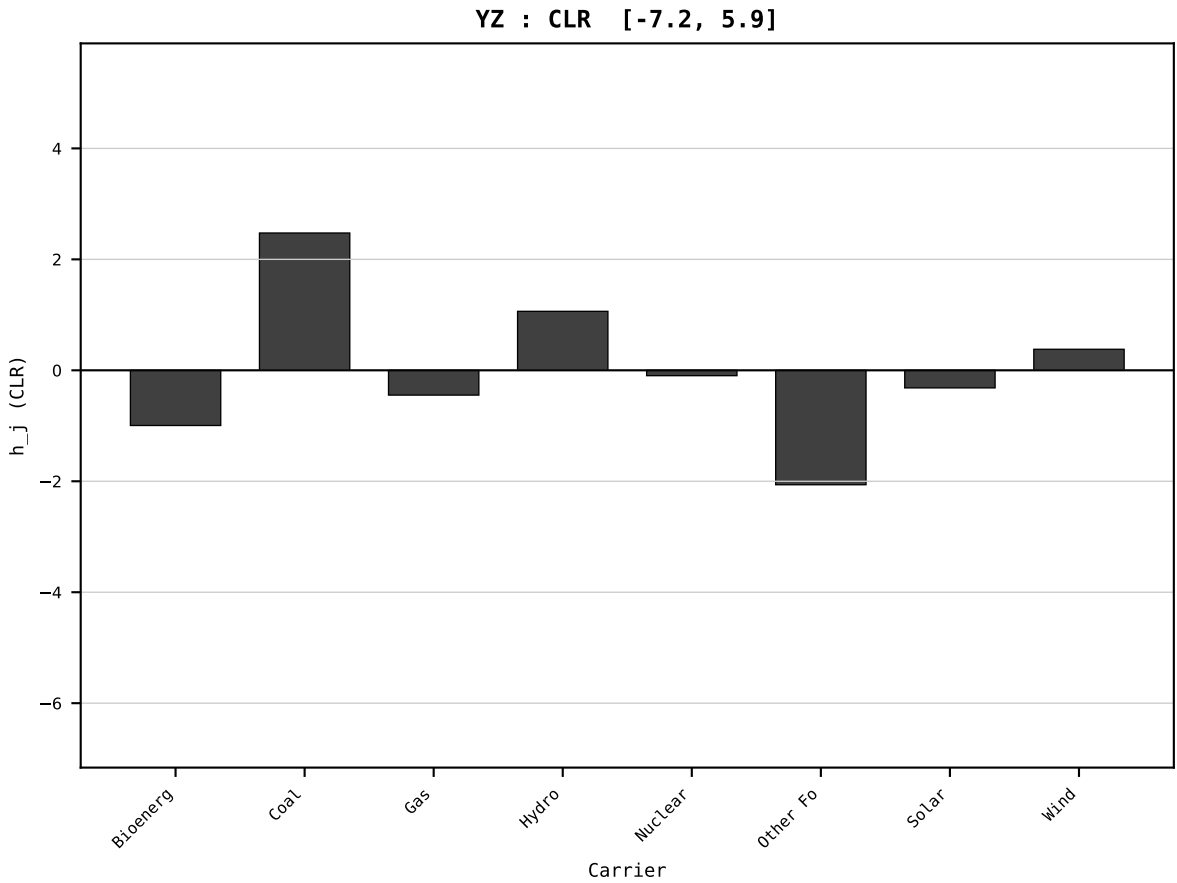
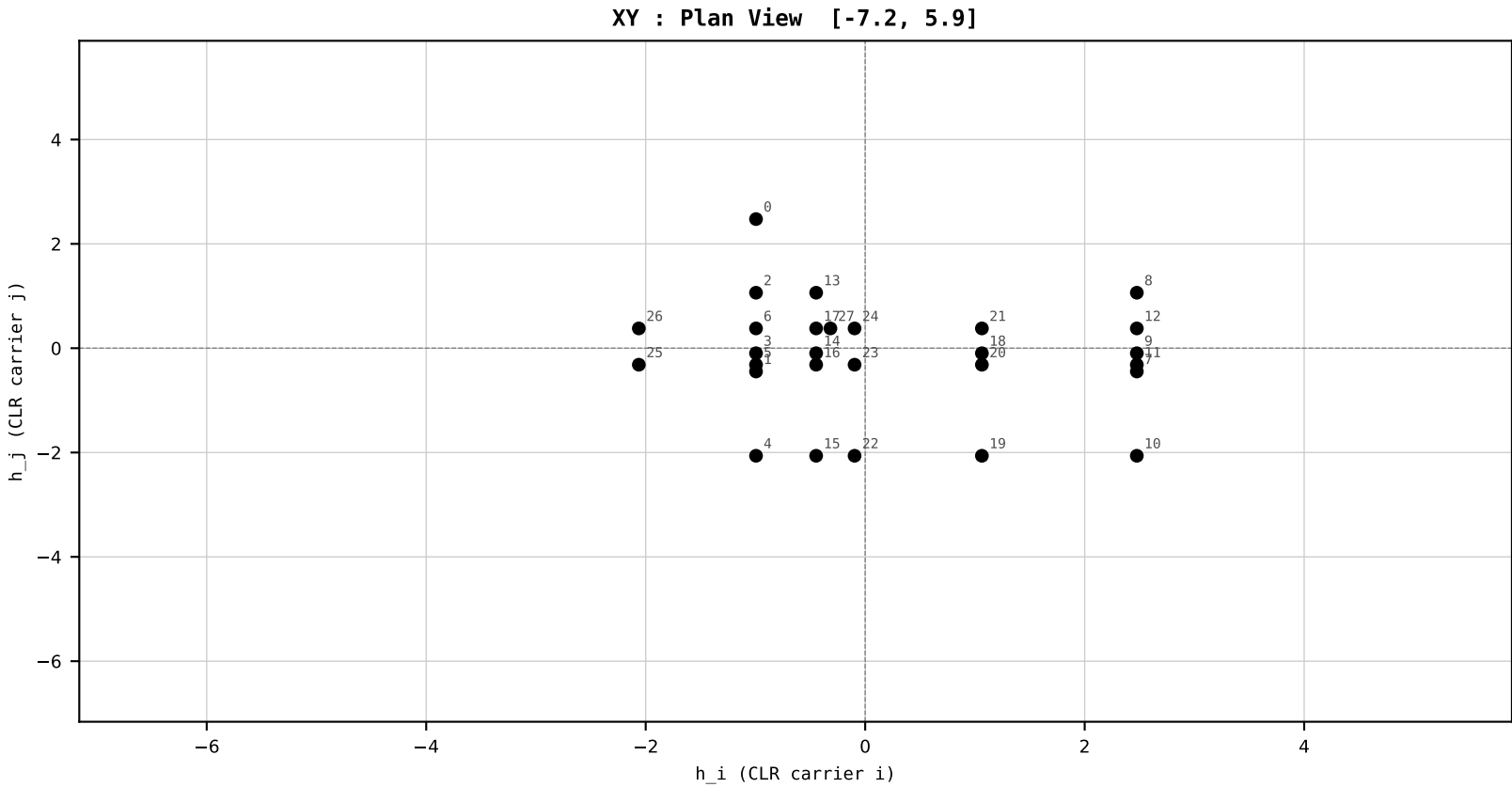
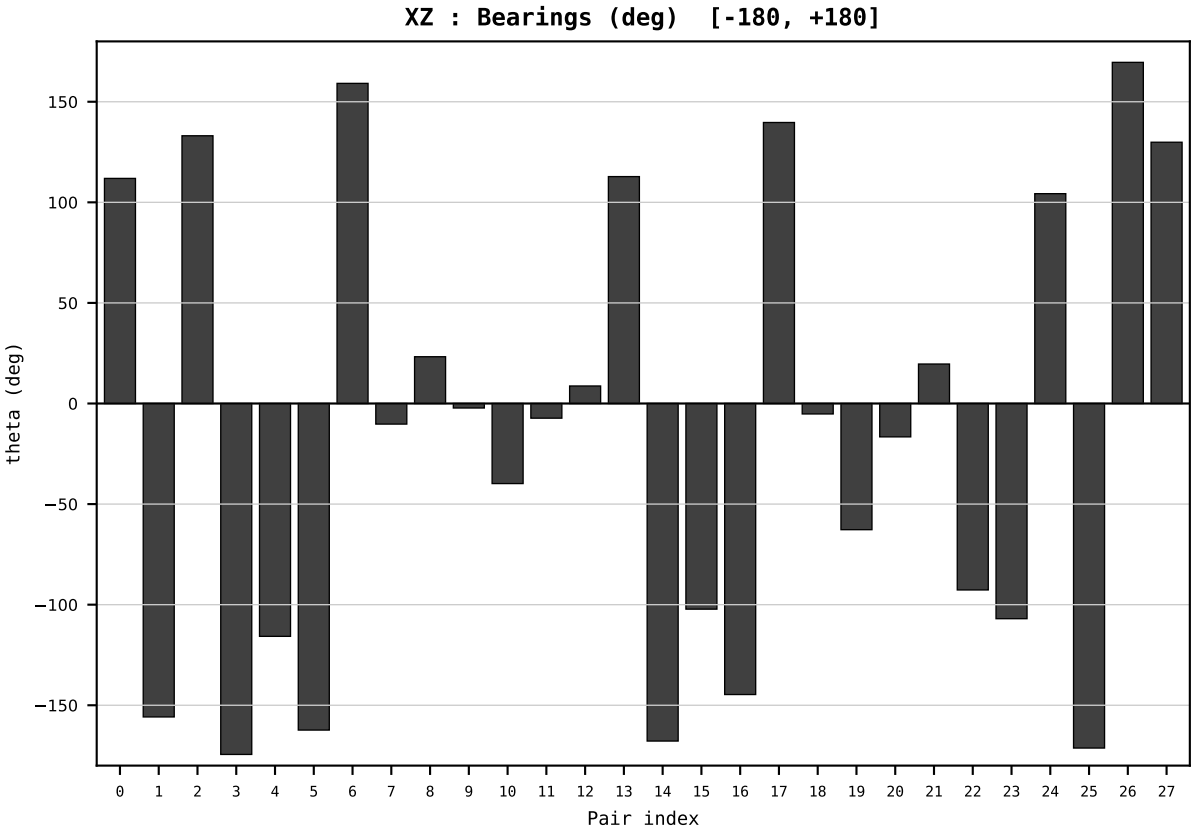
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2021
t : 21 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.388284
Ring = Hs-3
E_metric= 12.9540
kappa_HS= 93.48
omega = 4.5501 deg
d_A = 0.293552
Helm = Wind
Helm d = +0.199189
DR = 4.538 HLR
DR ratio= 93.5x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=22 | Year 2022 | D=8 N=26 pairs=28

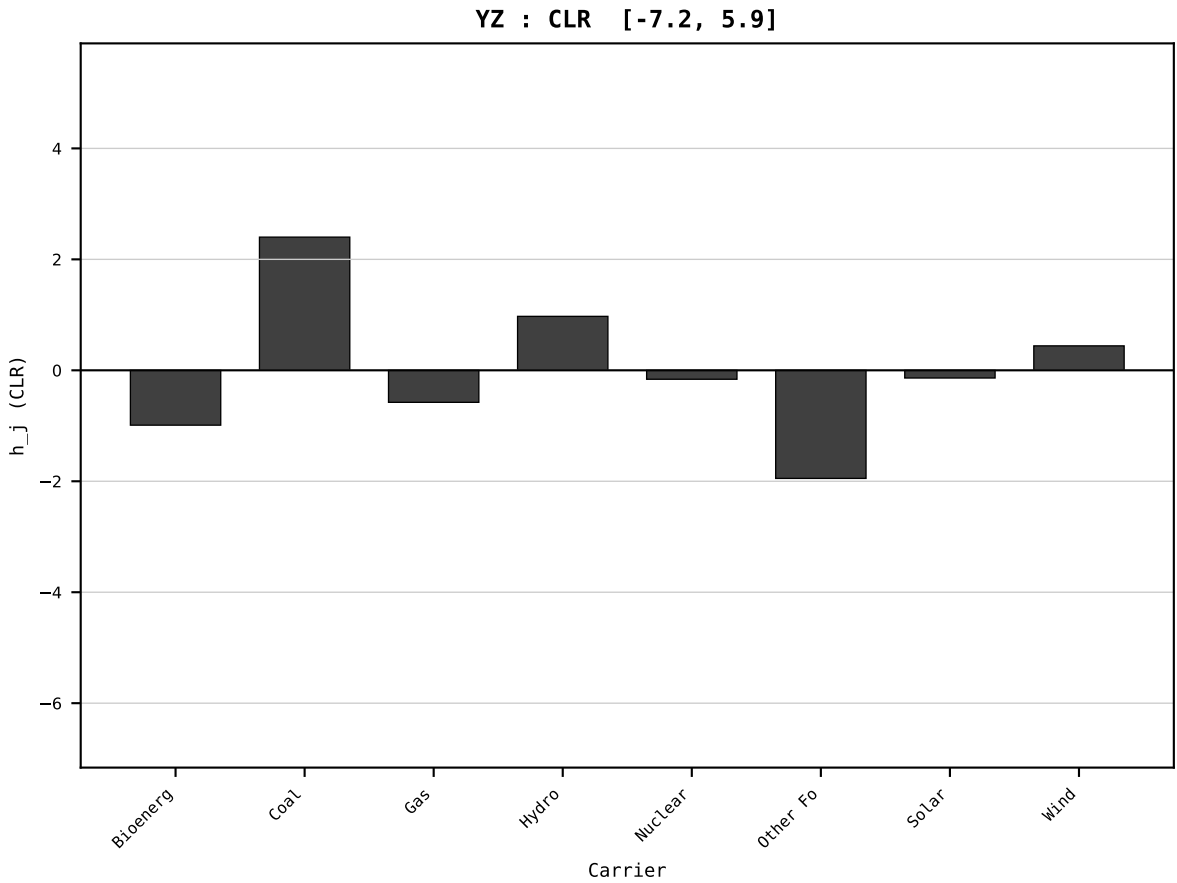
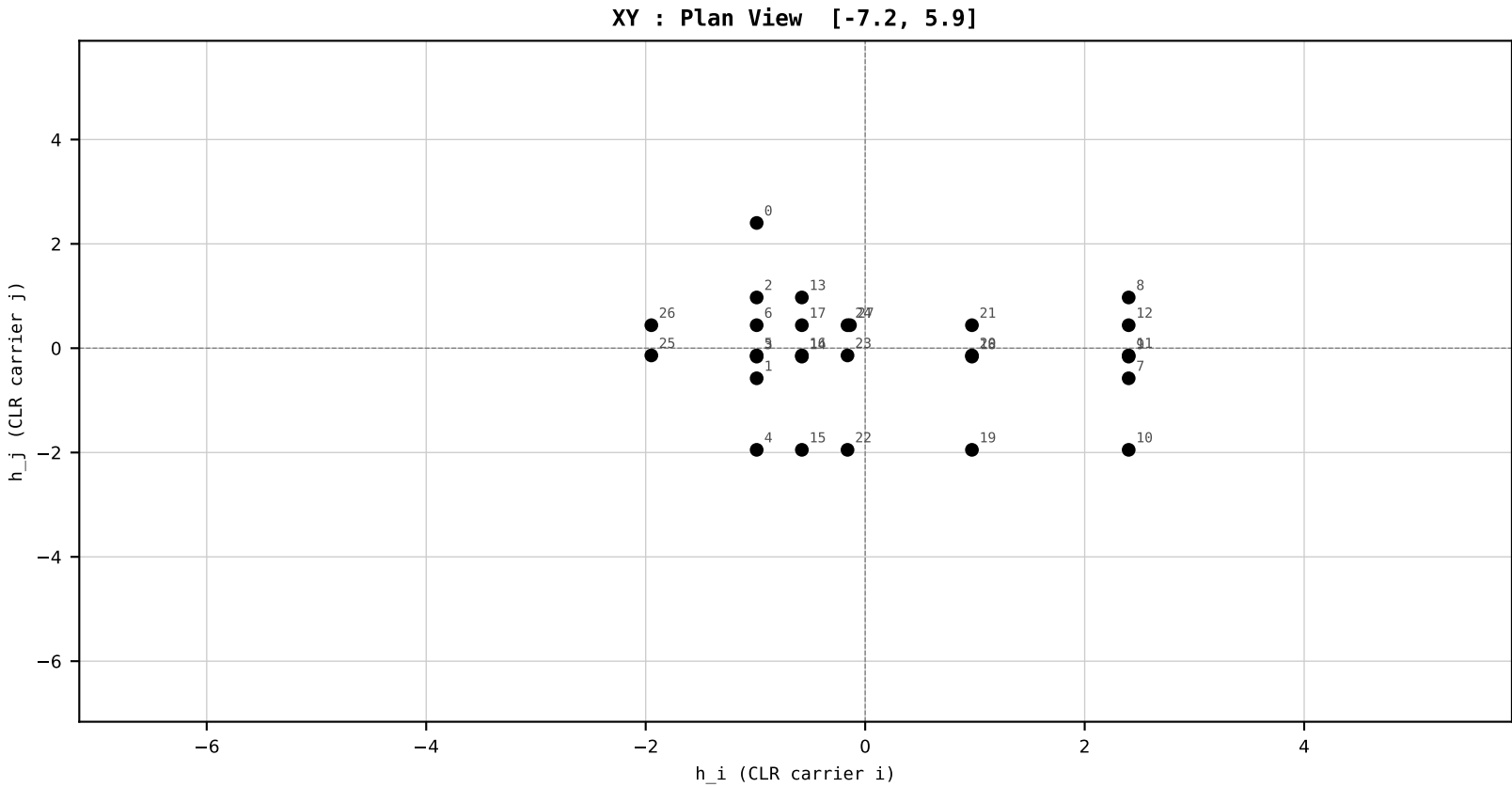
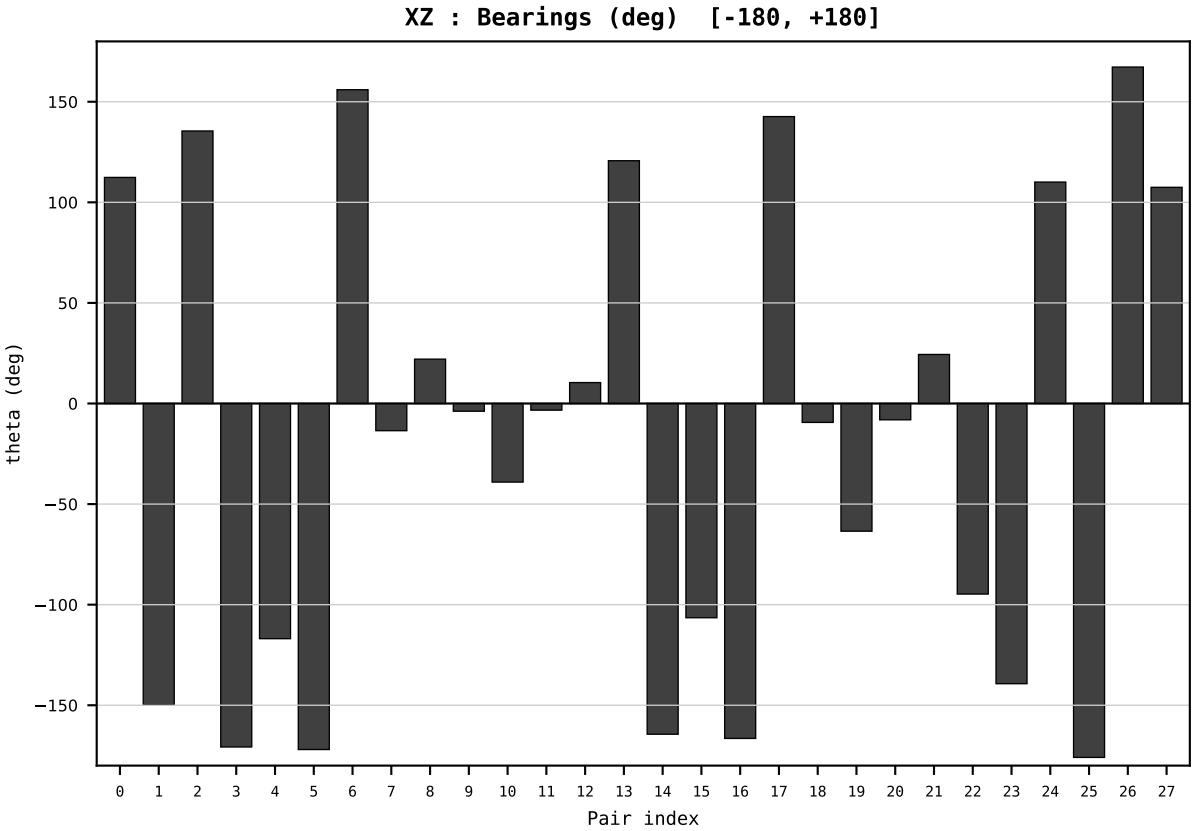
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2022
t : 22 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.369899
Ring = Hs-3
E_metric= 12.0569
kappa_HS= 77.43
omega = 4.2062 deg
d_A = 0.288819
Helm = Solar
Helm d = +0.178100
DR = 4.349 HLR
DR ratio= 77.4x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=23 | Year 2023 | D=8 N=26 pairs=28

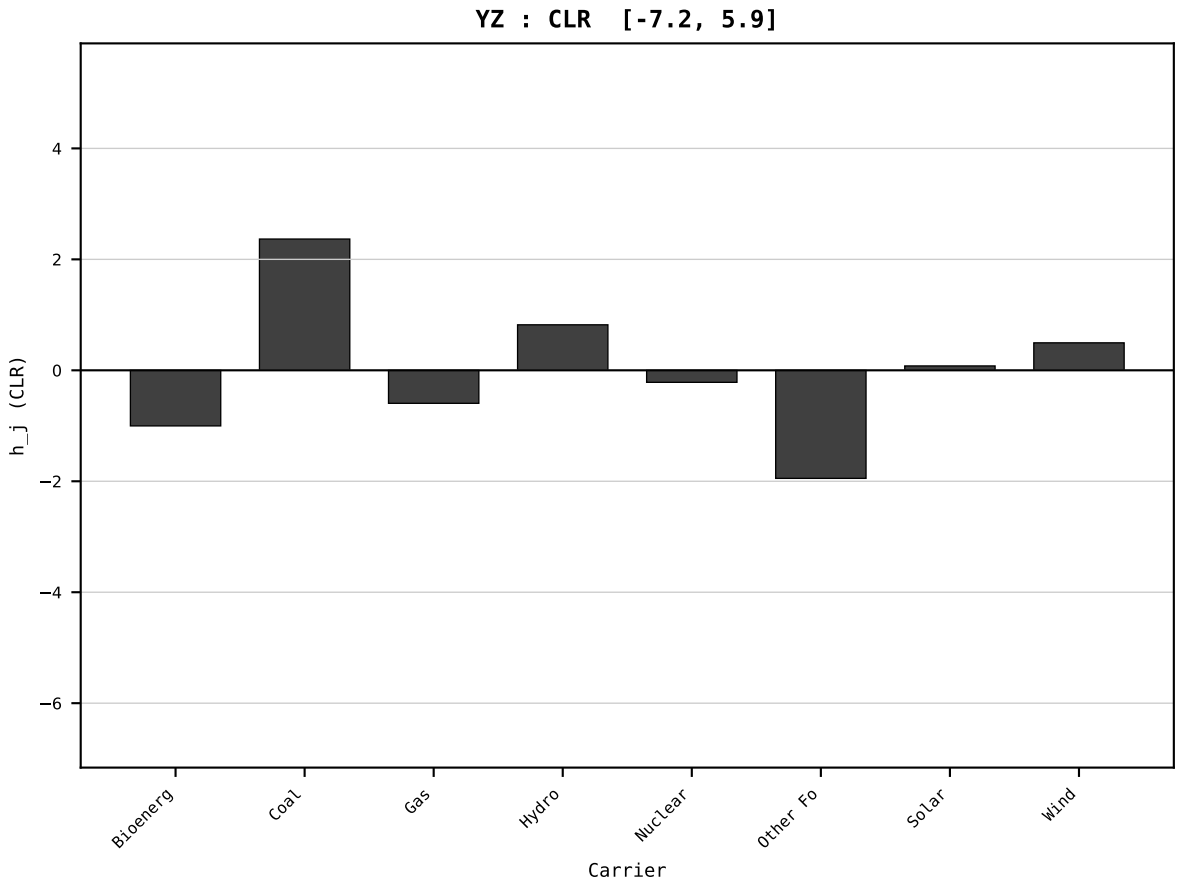
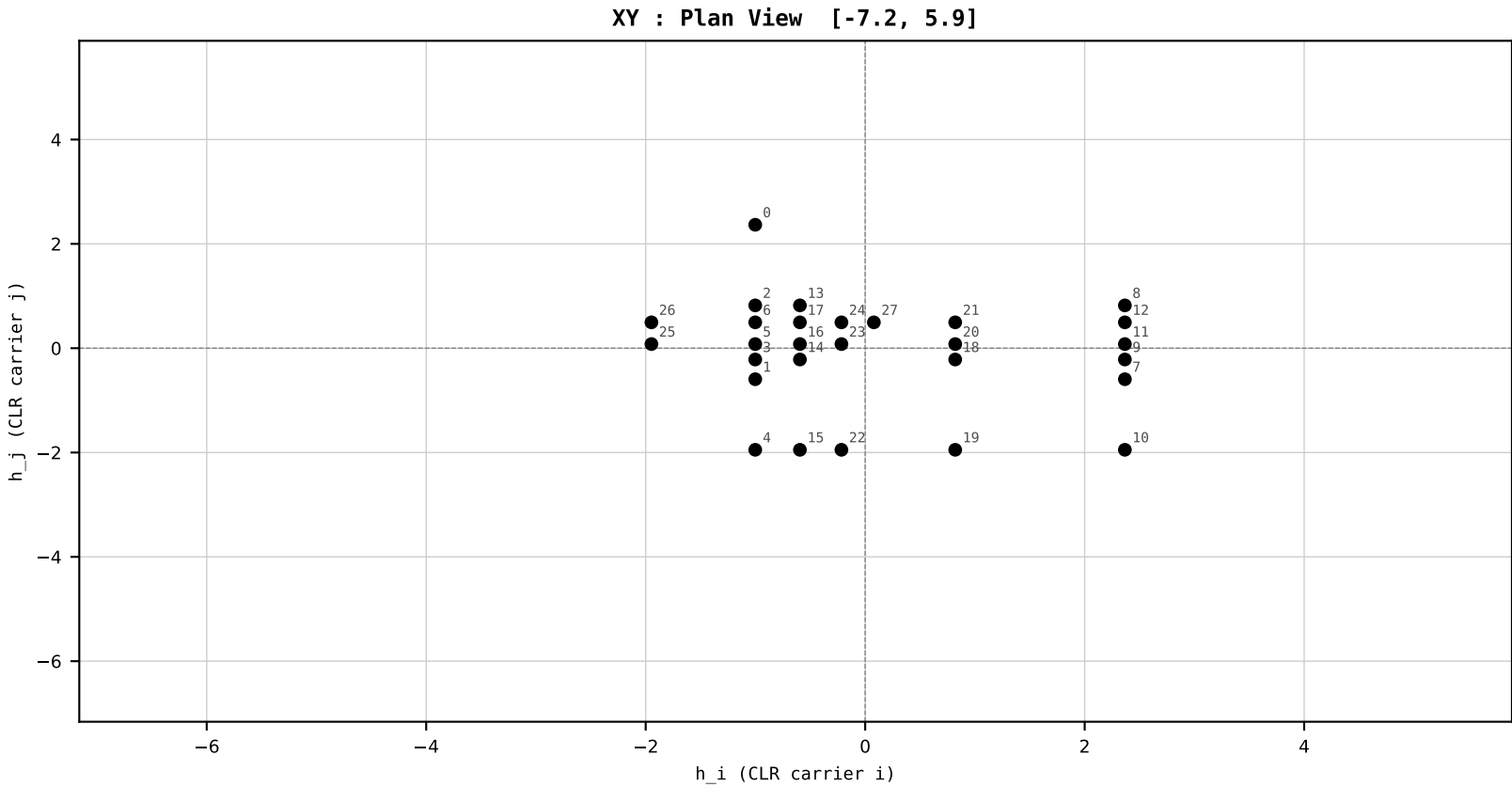
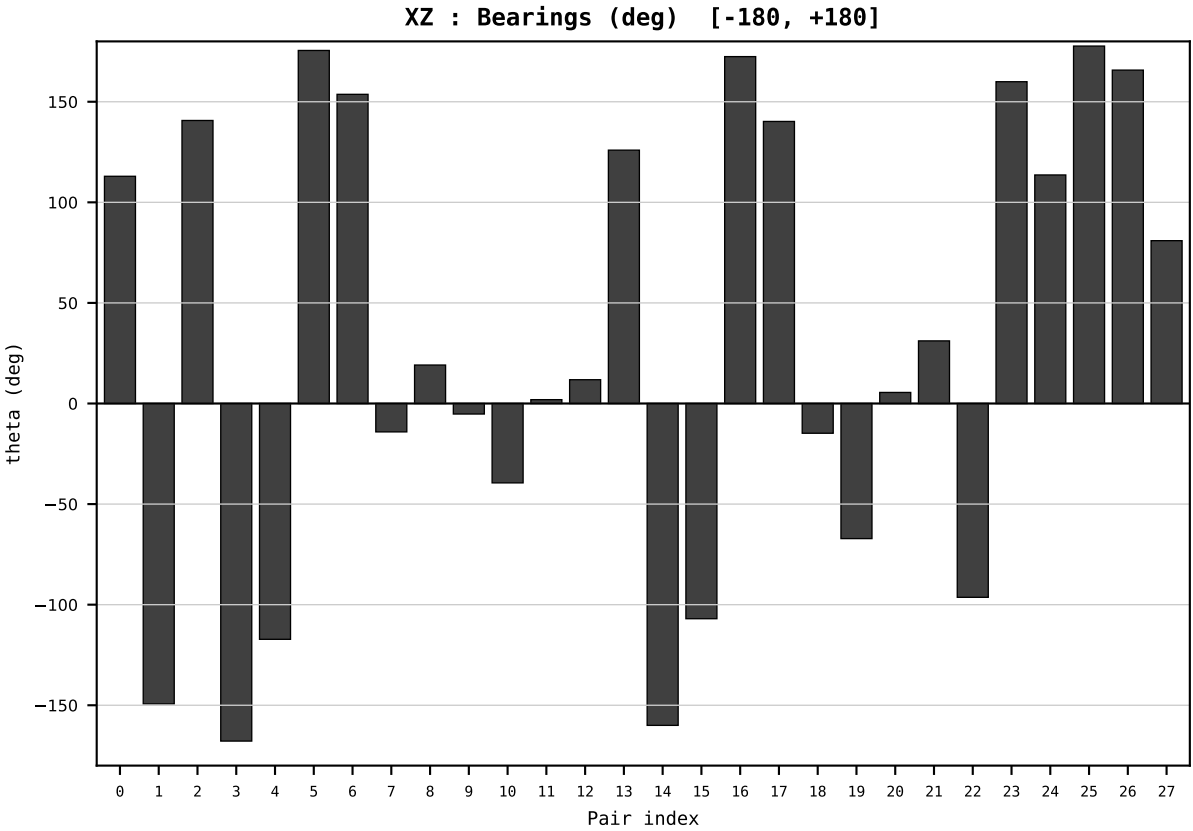
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2023
t : 23 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.359597
Ring = Hs-3
E_metric= 11.7219
kappa_HS= 74.74
omega = 4.5800 deg
d_A = 0.279787
Helm = Solar
Helm d = +0.217626
DR = 4.314 HLR
DR ratio= 74.7x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=24 | Year 2024 | D=8 N=26 pairs=28

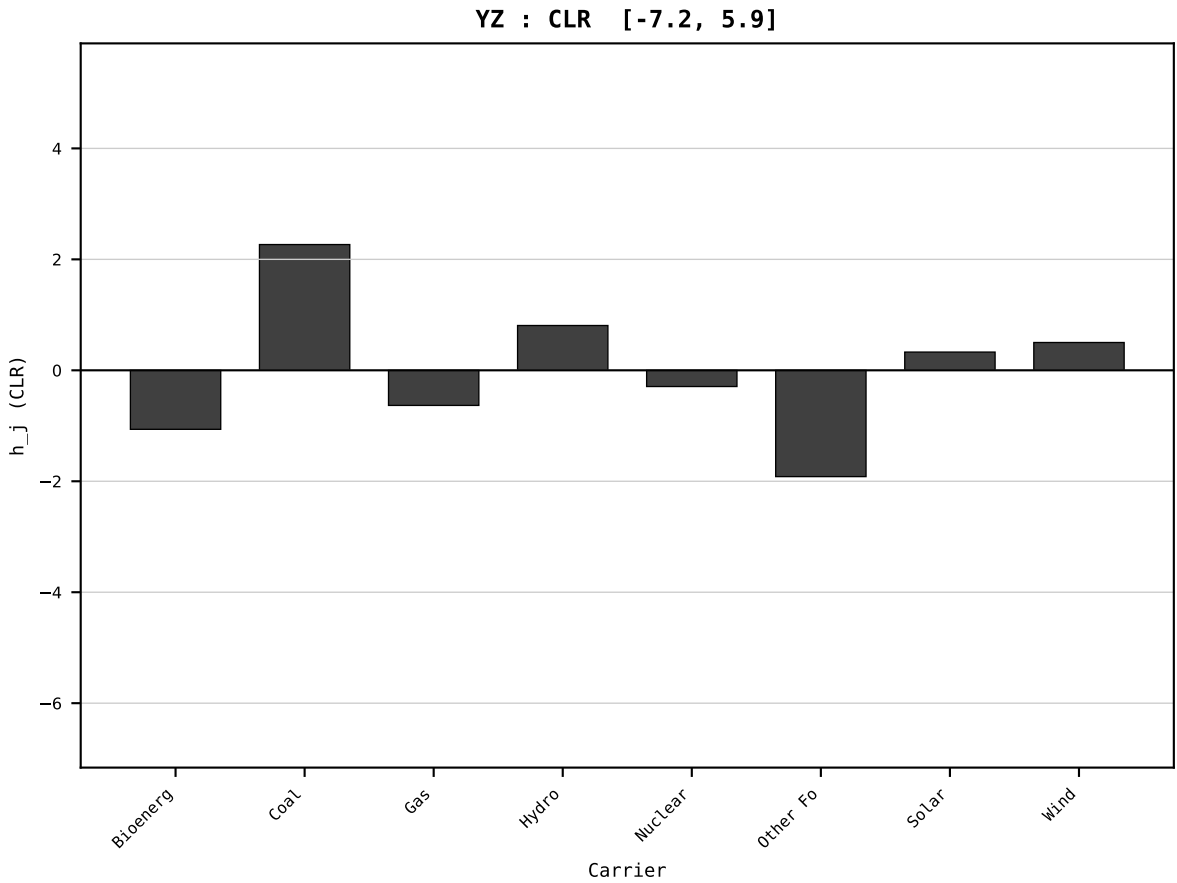
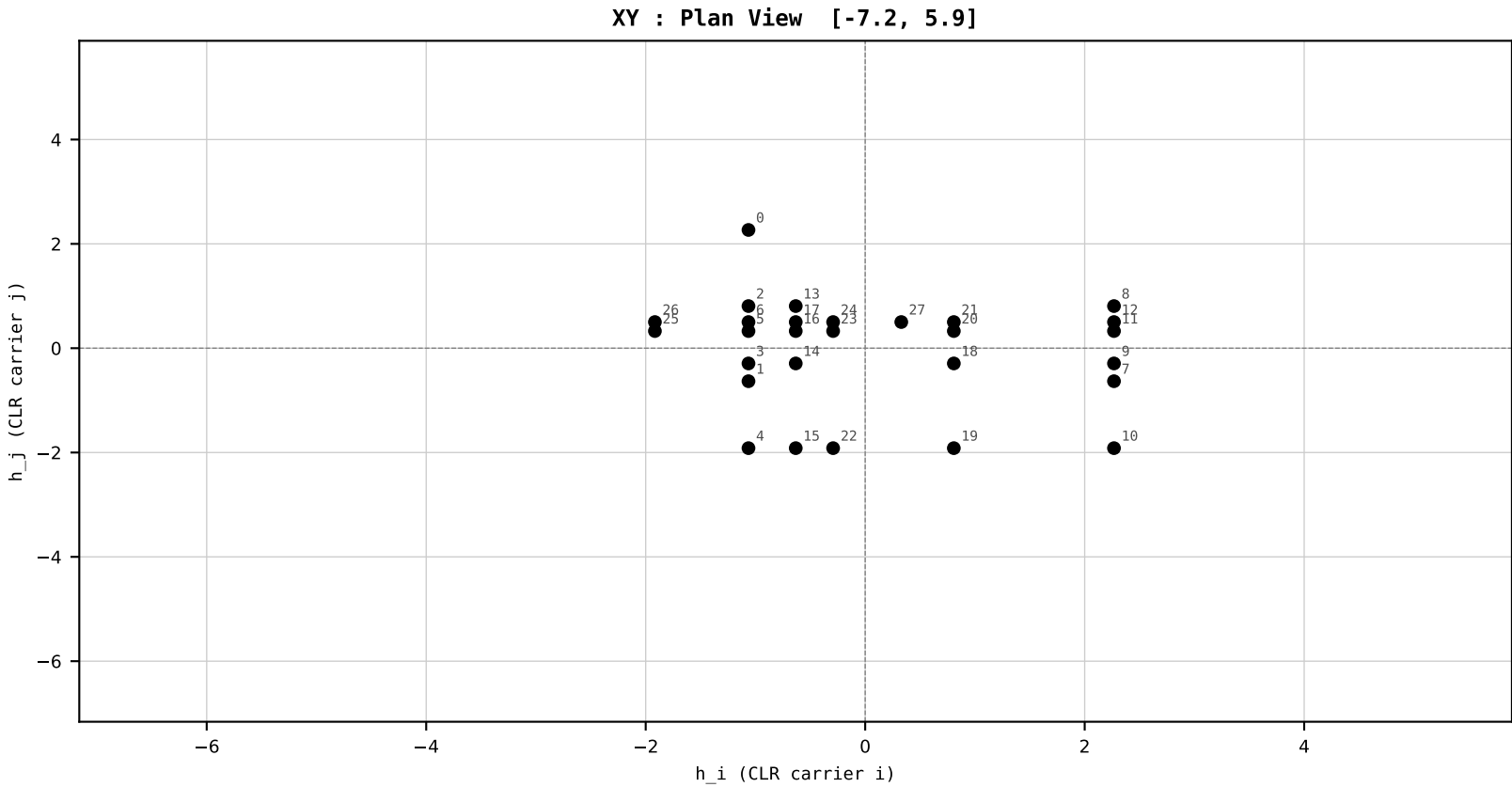
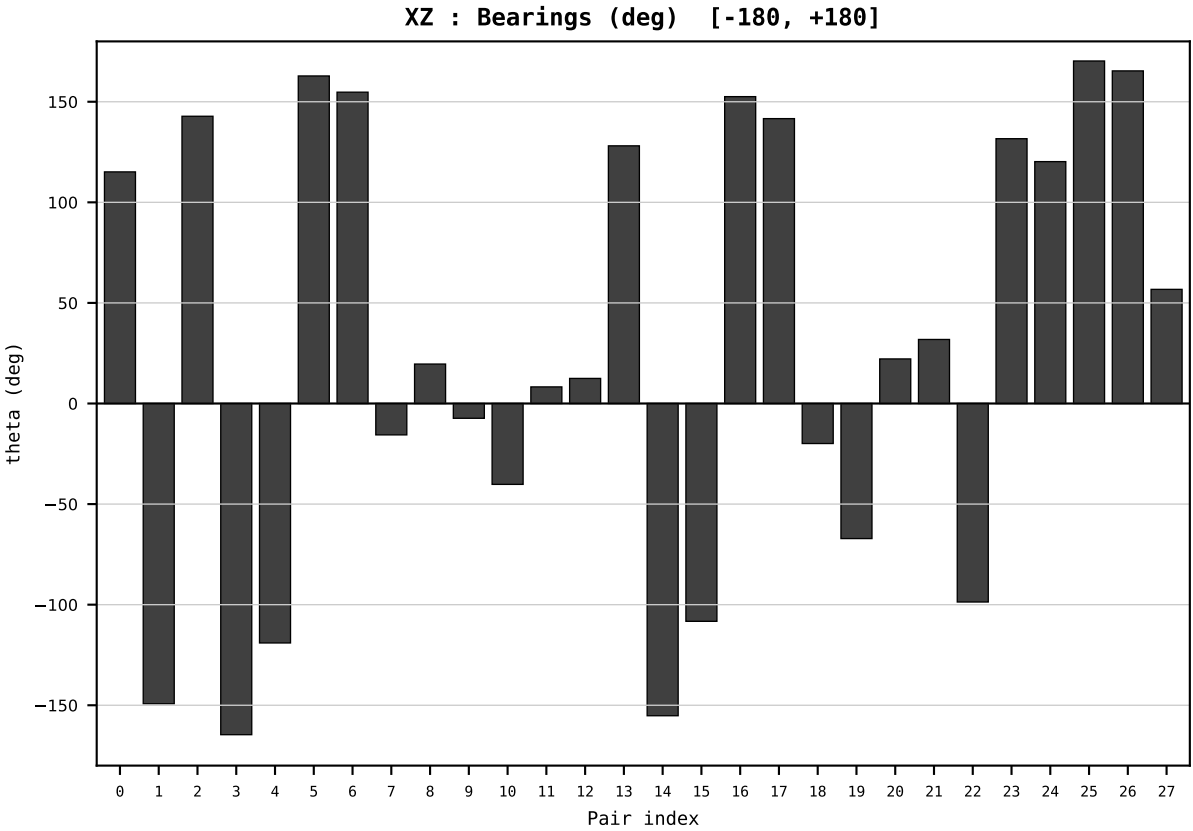
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2024
t : 24 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.330309
Ring = Hs-3
E_metric= 11.4392
kappa_HS= 65.57
omega = 4.8451 deg
d_A = 0.290657
Helm = Solar
Helm d = +0.249869
DR = 4.183 HLR
DR ratio= 65.6x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=25 | Year 2025 | D=8 N=26 pairs=28

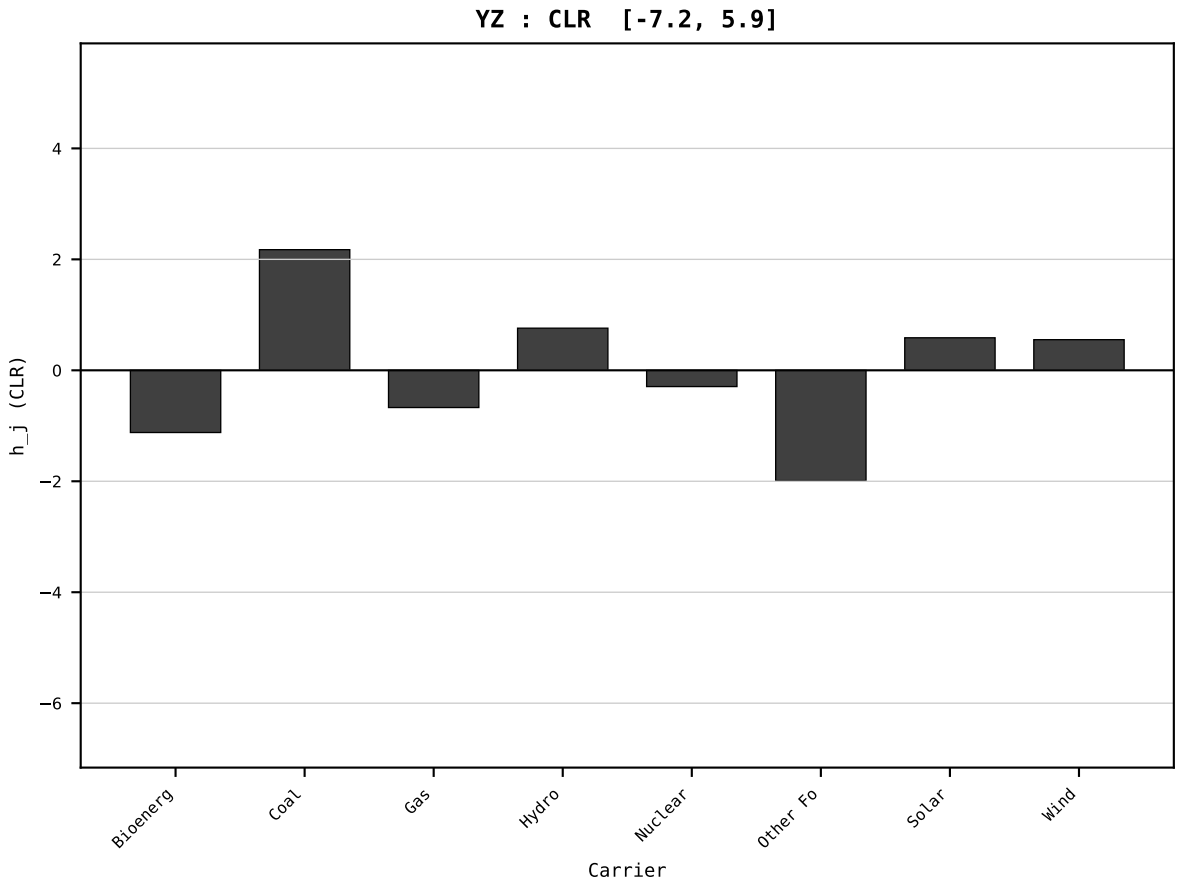
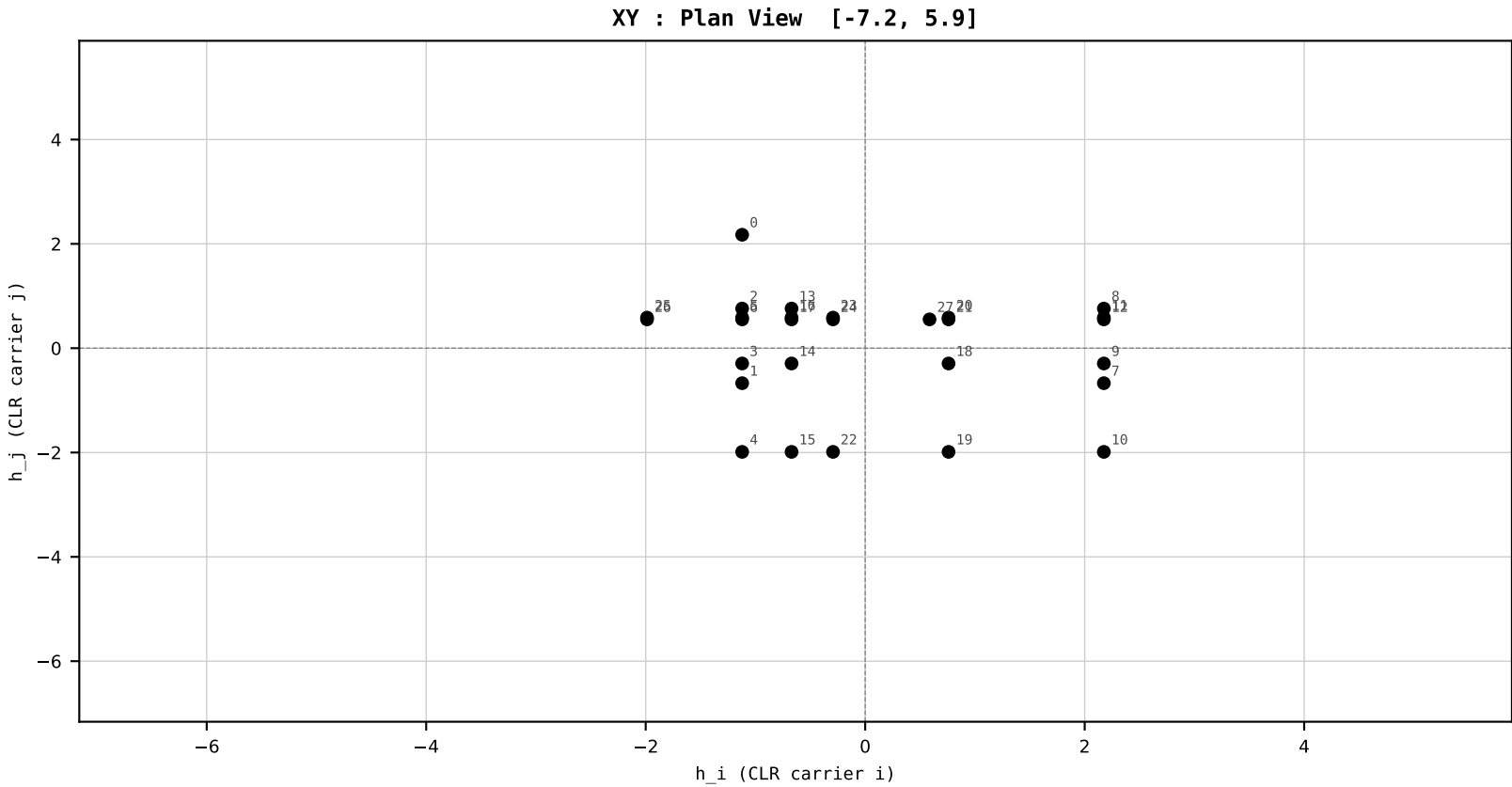
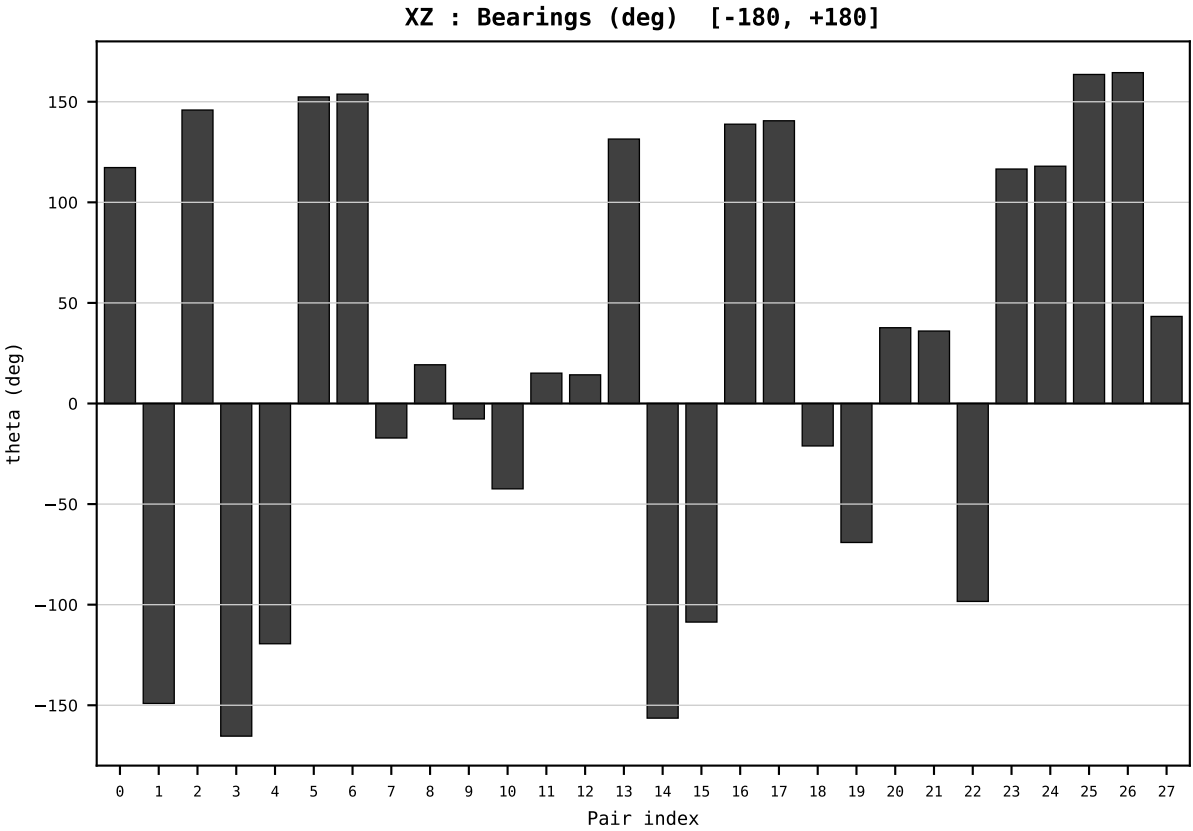
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

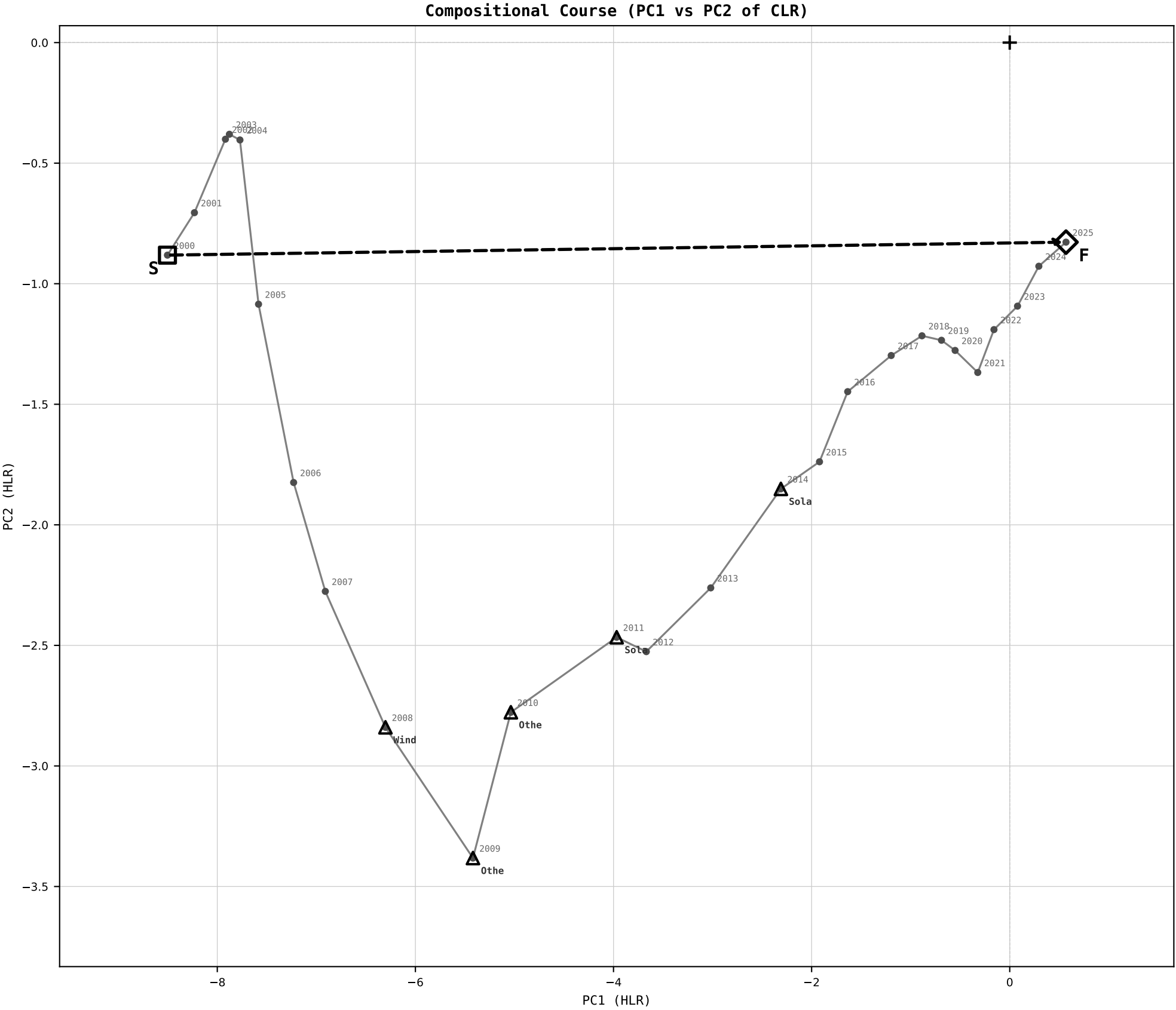
Dataset : ember_CHN_China_generation_TWh.csv
Year : 2025
t : 25 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.301686
Ring = Hs-3
E_metric= 11.6993
kappa_HS= 64.23
omega = 5.0047 deg
d_A = 0.299453
Helm = Solar
Helm d = +0.257596
DR = 4.162 HLR
DR ratio= 64.2x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind





COURSE METRICS

Scale Provenance:
 $Y_j \rightarrow x_j = Y_j / T \rightarrow h_j = \text{CLR}$
 $\rightarrow$ PC projection
Unit: HLR (Higgins Log-Ratio)

Net distance = 9.0822 HLR
Path length = 13.8225 HLR
Directness = 0.6571

DR start = 10.878 HLR
ratio = 53013.0x
DR final = 4.162 HLR
ratio = 64.2x

E_metric start = 82.7022
E_metric final = 11.6993

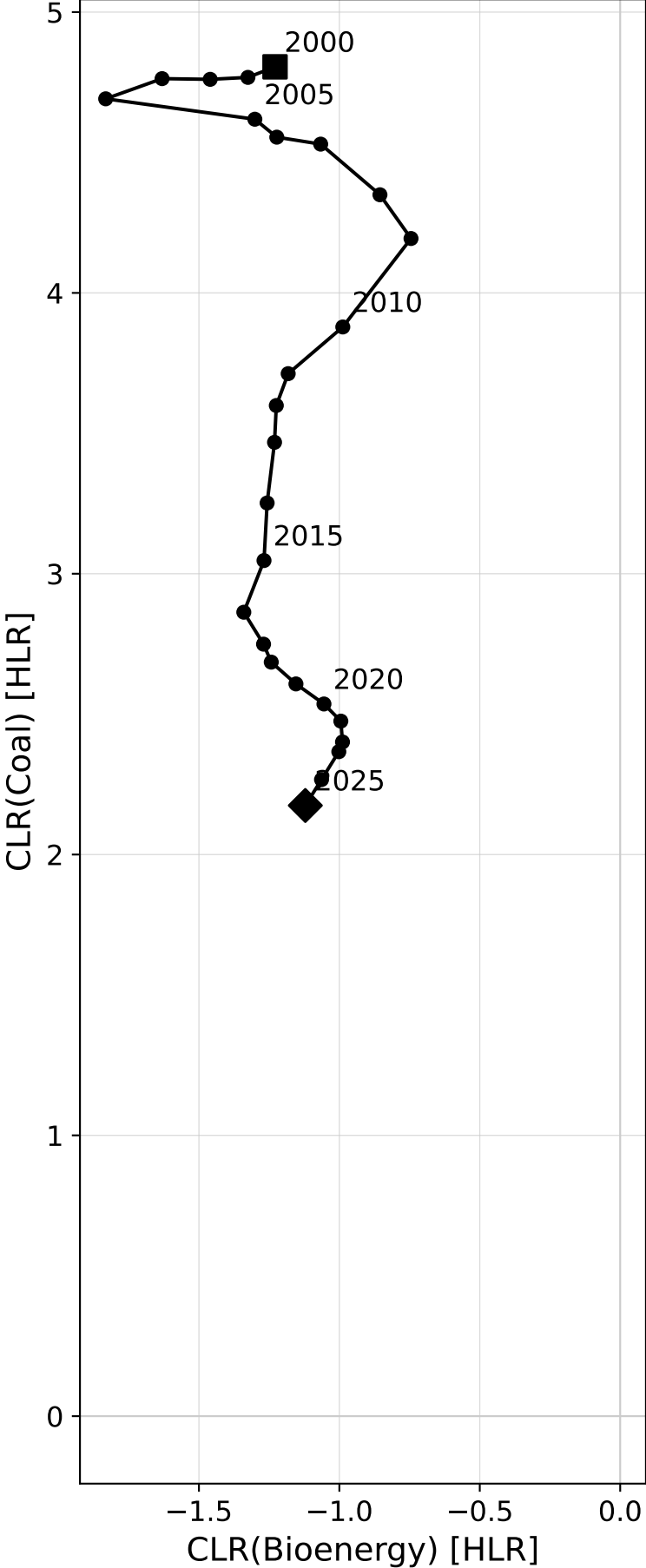
PC1 loadings:
Solar +0.735
Other Fo -0.396
Wind +0.347

PC2 loadings:
Other Fo +0.557
Solar +0.502
Wind -0.472

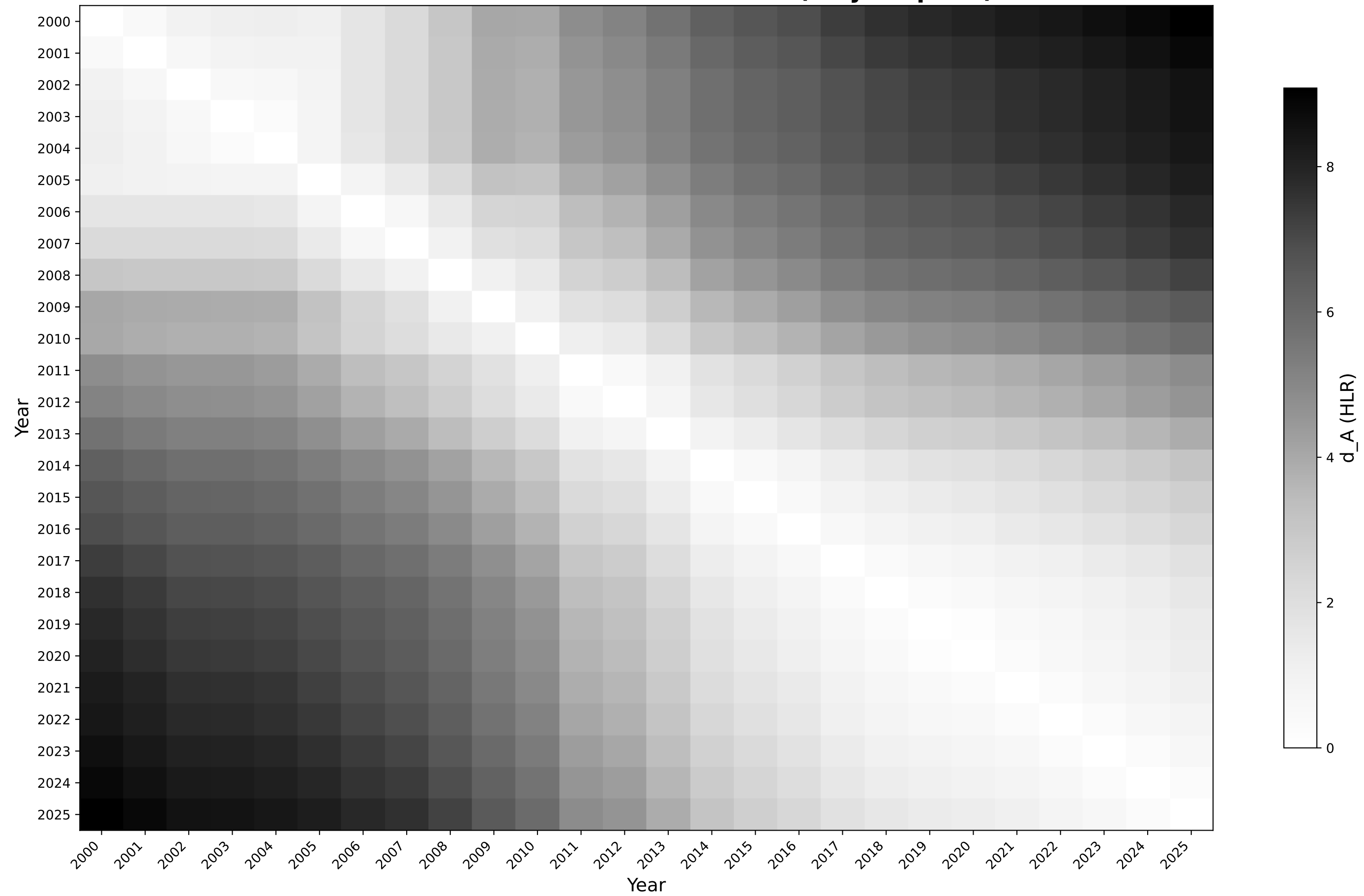
Top displacement events:
2011 Solar d=1.162
2009 Other Fo d=1.043
2010 Other Fo d=1.021
2008 Wind d=0.975
2014 Solar d=0.855

The instrument reads.
The expert decides.
The loop stays open.

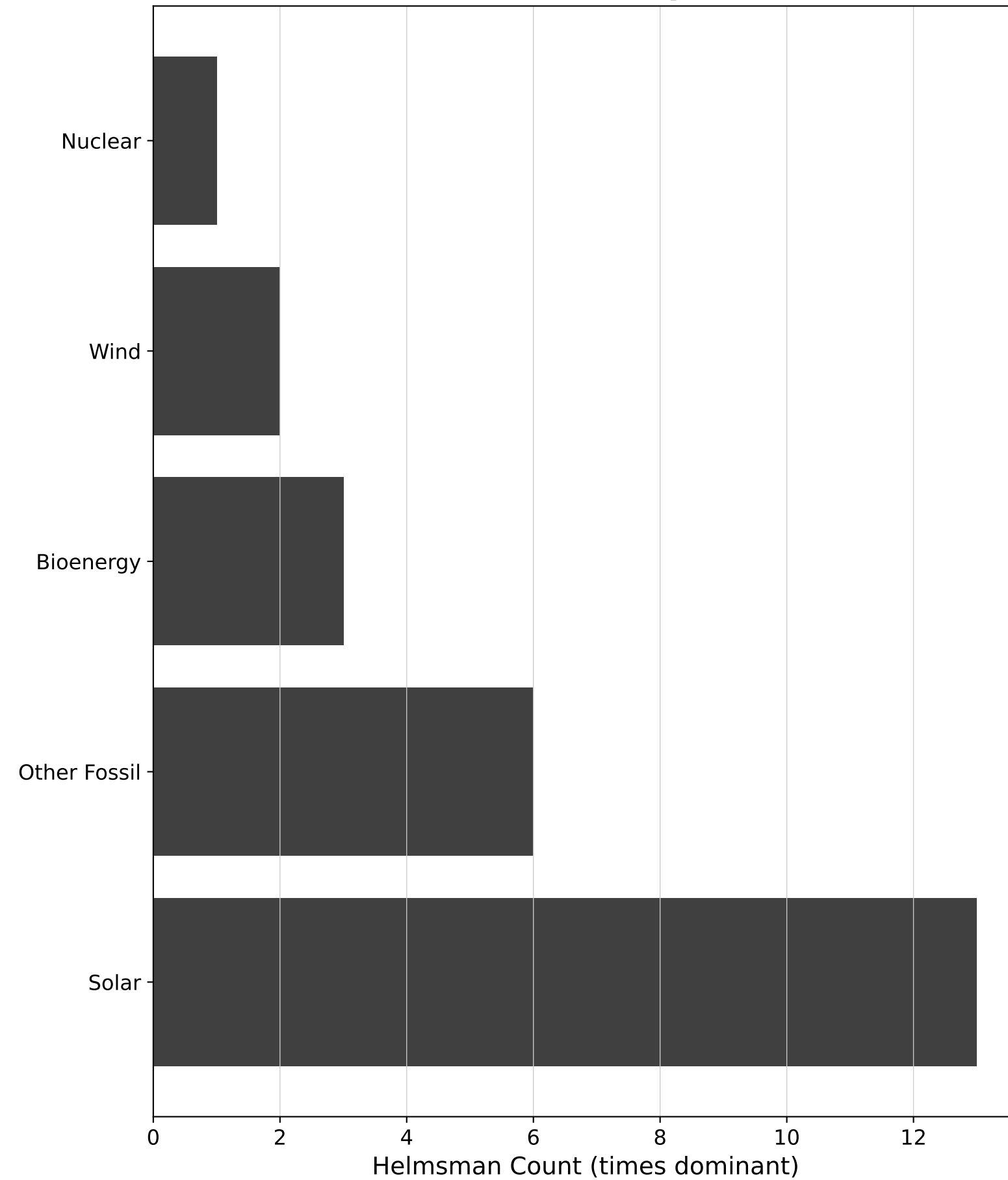
SYSTEM COURSE PLOT — XY Plan View (CLR Space)



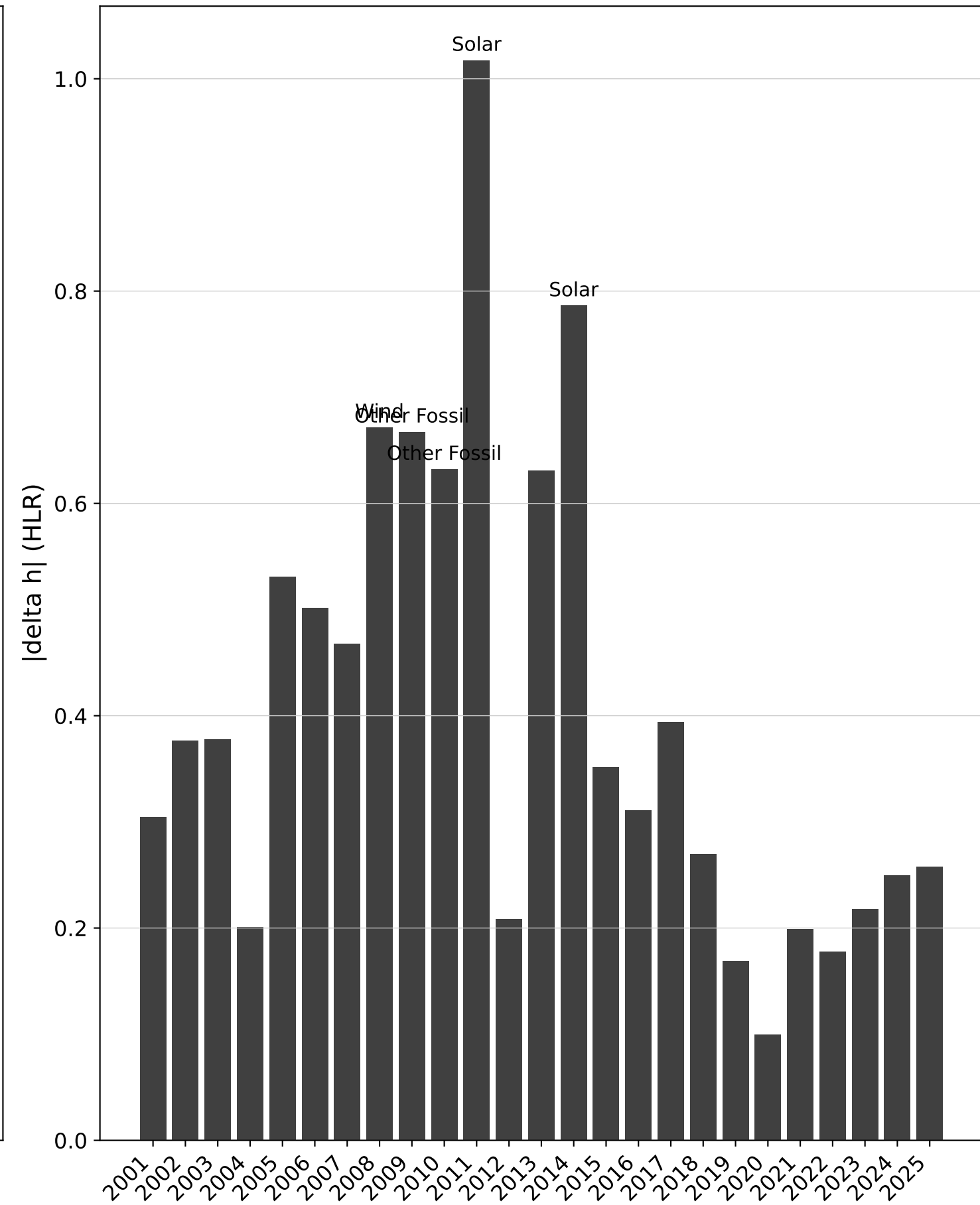
STAGE 2 — Aitchison Distance Matrix (all year pairs)



HELMSMAN FREQUENCY



HELMSMAN DISPLACEMENT PER YEAR



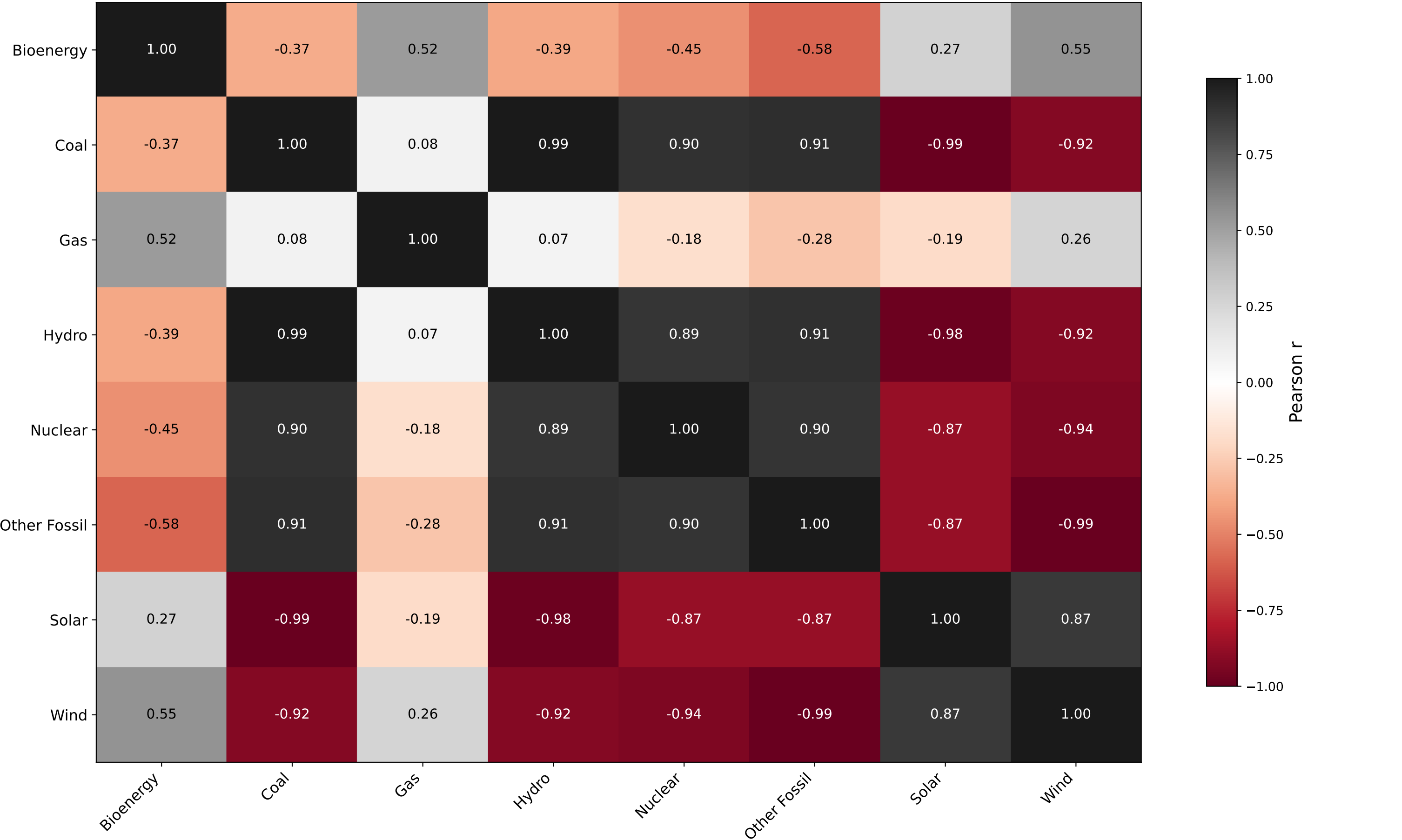
STAGE 2 — Pairwise Divergence Ranking (Top 15)

Rank	Year A	Year B	d_A (HLR)	Angle (deg)	Dominant Carrier
1	2000	2025	9.0822	79.0	Solar
2	2001	2025	8.8101	77.4	Solar
3	2000	2024	8.8065	74.4	Solar
4	2000	2023	8.5979	70.8	Solar
5	2001	2024	8.5376	72.8	Solar
6	2002	2025	8.5027	75.8	Solar
7	2003	2025	8.4674	75.6	Solar
8	2004	2025	8.3657	75.2	Solar
9	2000	2022	8.3653	66.9	Solar
10	2001	2023	8.3338	69.4	Solar
11	2002	2024	8.2356	71.4	Solar
12	2000	2021	8.2141	64.5	Solar
13	2003	2024	8.2025	71.2	Solar
14	2005	2025	8.1607	74.9	Solar
15	2004	2024	8.1021	70.8	Solar

STAGE 3 — Triadic Metric Triangle Area Ranking (Top 10)

Rank	Year A	Year B	Year C	Area (HLR^2)	Imbalance
1	2002	2009	2025	12.2454	2.17
2	2003	2009	2025	12.2205	2.17
3	2004	2009	2025	12.1074	2.18
4	2001	2009	2025	11.8897	2.24
5	2000	2009	2025	11.7785	2.25
6	2002	2009	2024	11.7128	2.10
7	2003	2009	2024	11.6863	2.10
8	2004	2009	2024	11.5763	2.11
9	2001	2009	2024	11.3801	2.17
10	2000	2009	2024	11.2806	2.18

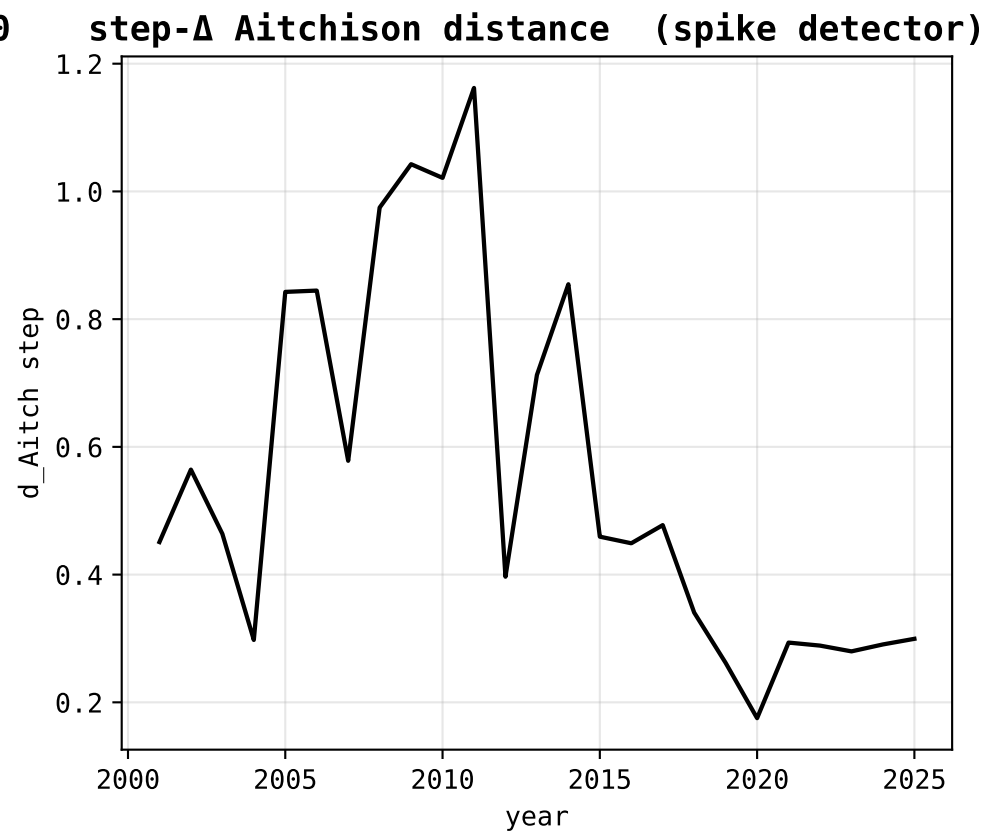
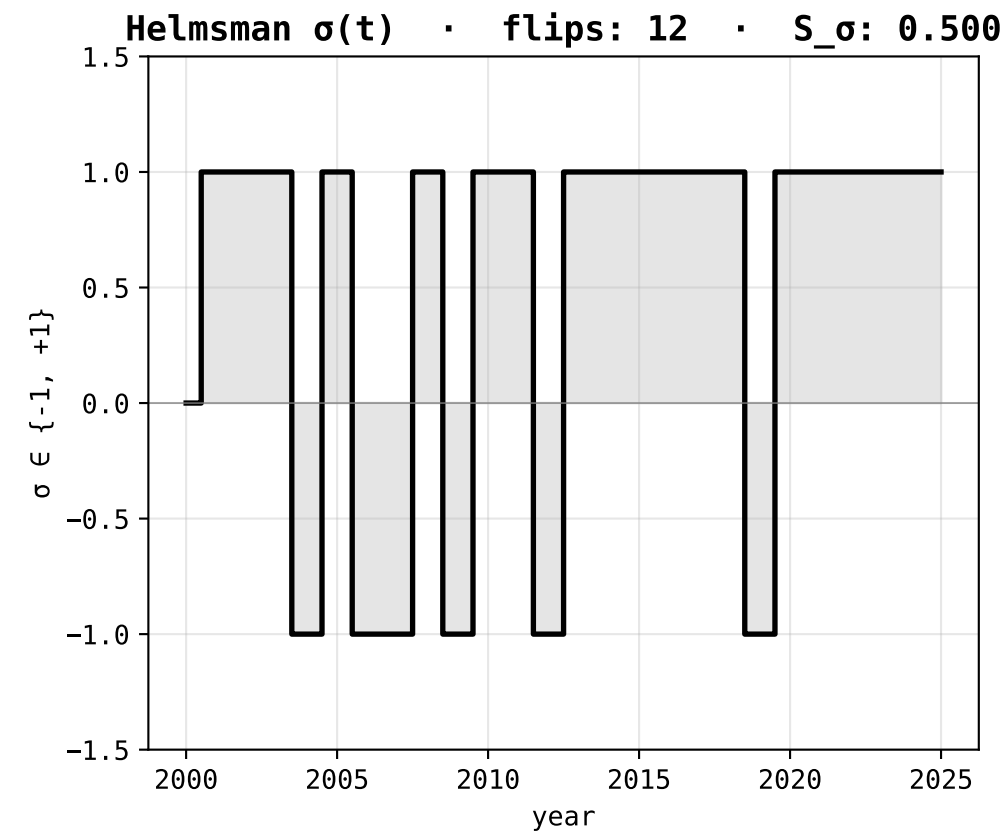
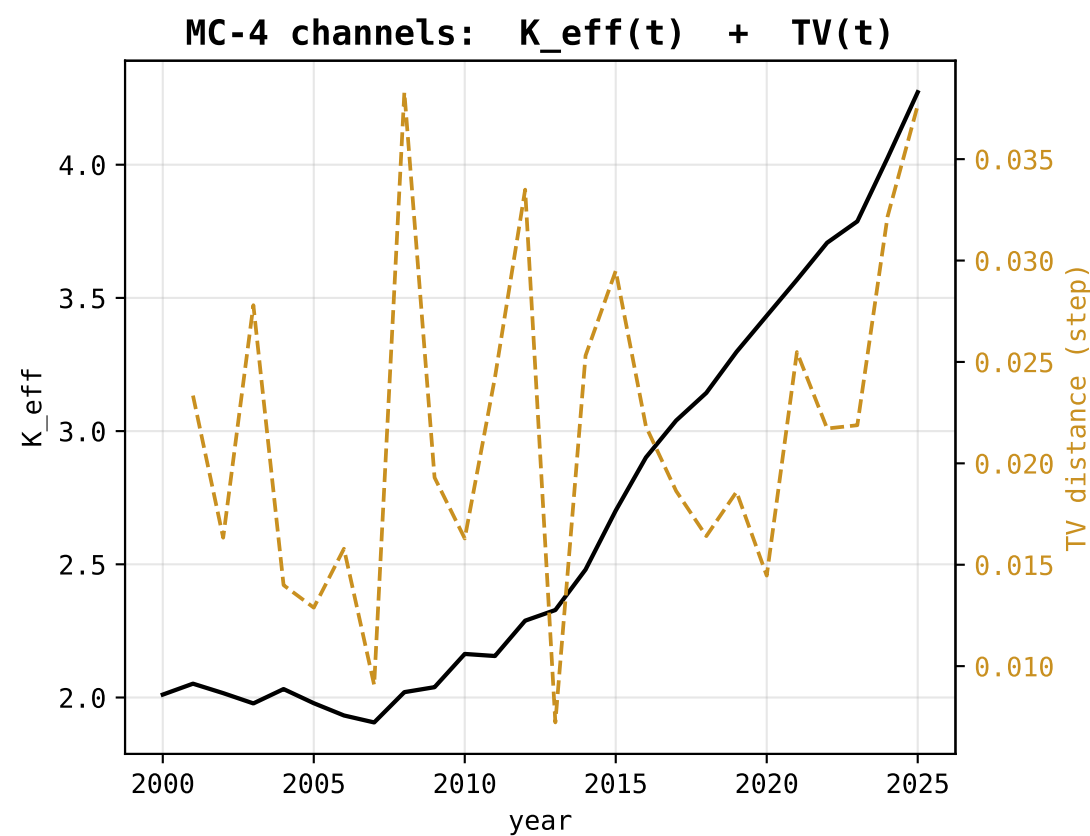
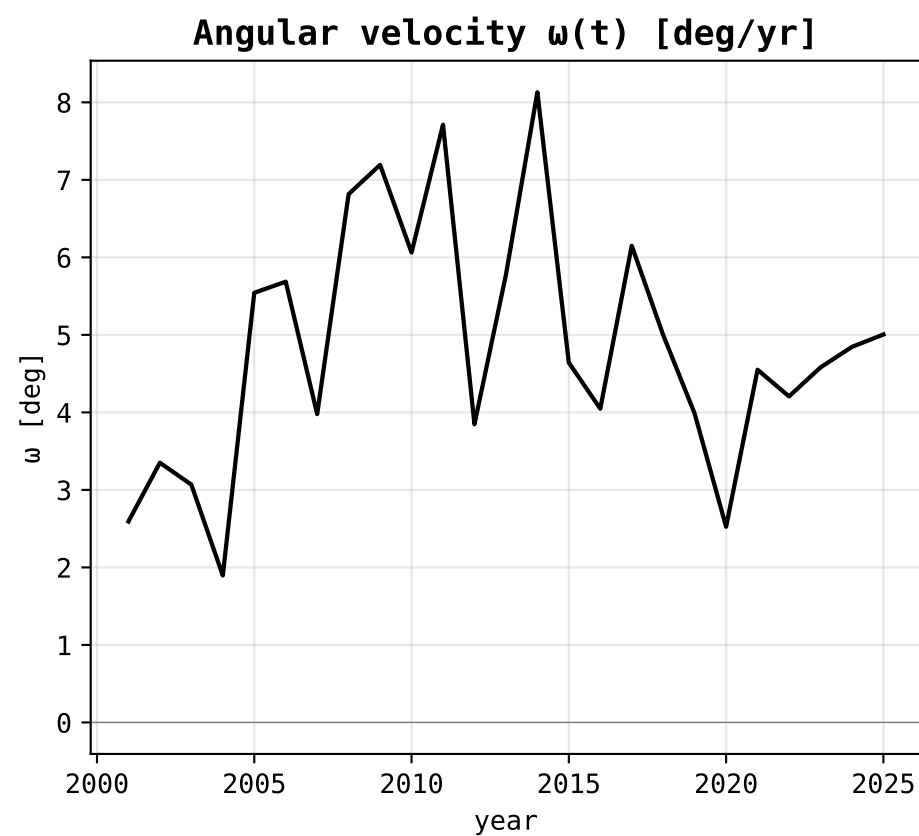
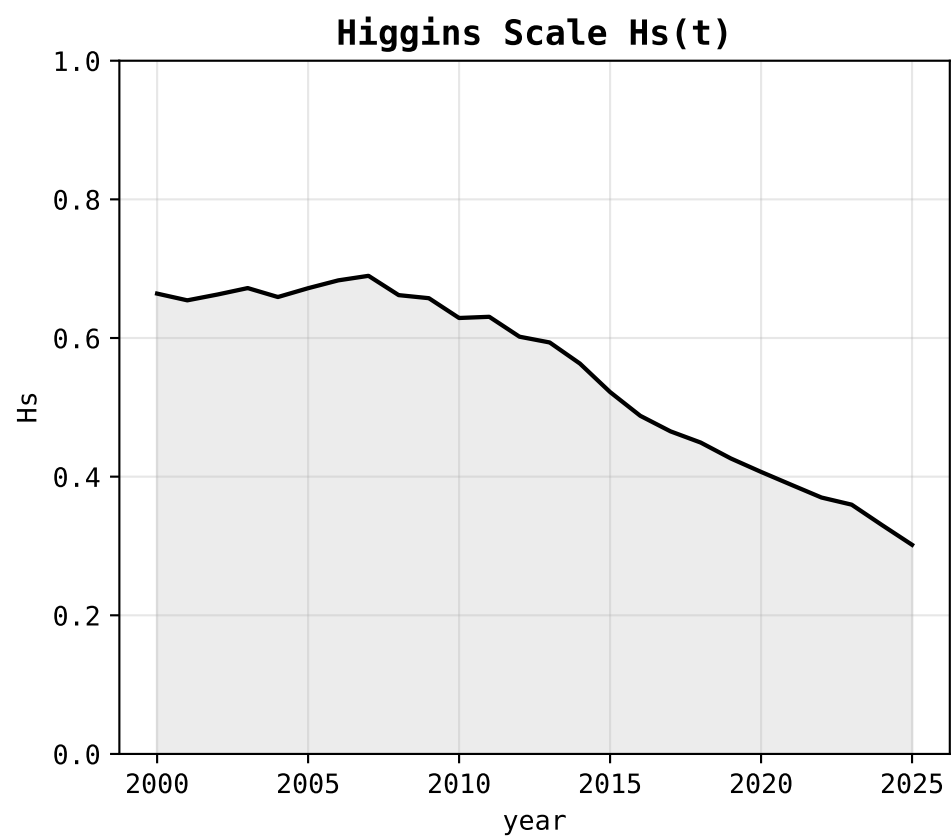
STAGE 3 — Carrier CLR Trajectory Correlation



NAVIGATION SUMMARY — All Stage 1 Metrics by Year

Year	Hs	Ring	E_metric	omega(deg)	d_A	Helmsman	kappa_max
2000	0.6641	Hs-4	82.7	0.00	0.0000	---	67780
2001	0.6543	Hs-4	79.2	2.59	0.4509	Solar	49360
2002	0.6627	Hs-4	75.1	3.35	0.5644	Solar	33080
2003	0.6720	Hs-4	74.7	3.07	0.4638	Nuclear	31843
2004	0.6591	Hs-4	73.2	1.90	0.2977	Bioenergy	27541
2005	0.6718	Hs-4	69.8	5.54	0.8427	Bioenergy	31253
2006	0.6832	Hs-4	66.3	5.69	0.8447	Other Fossil	31841
2007	0.6897	Hs-4	64.0	3.98	0.5782	Other Fossil	29832
2008	0.6619	Hs-4	59.6	6.82	0.9747	Wind	23305
2009	0.6574	Hs-4	52.8	7.19	1.0426	Other Fossil	13267
2010	0.6289	Hs-4	43.0	6.06	1.0212	Other Fossil	6010
2011	0.6306	Hs-4	32.9	7.71	1.1620	Solar	1806
2012	0.6019	Hs-4	31.6	3.85	0.3967	Other Fossil	1389
2013	0.5936	Hs-4	26.7	5.77	0.7126	Solar	649
2014	0.5632	Hs-4	21.8	8.13	0.8546	Solar	246
2015	0.5220	Hs-4	19.3	4.64	0.4594	Solar	159
2016	0.4878	Hs-3	16.4	4.05	0.4491	Solar	117
2017	0.4654	Hs-3	14.7	6.15	0.4775	Solar	118
2018	0.4492	Hs-3	14.1	5.00	0.3405	Solar	127
2019	0.4262	Hs-3	13.8	3.98	0.2618	Other Fossil	143
2020	0.4068	Hs-3	13.4	2.52	0.1752	Bioenergy	145
2021	0.3883	Hs-3	13.0	4.55	0.2936	Wind	150
2022	0.3699	Hs-3	12.1	4.21	0.2888	Solar	127
2023	0.3596	Hs-3	11.7	4.58	0.2798	Solar	123
2024	0.3303	Hs-3	11.4	4.85	0.2907	Solar	113
2025	0.3017	Hs-3	11.7	5.00	0.2995	Solar	118

CNQ v2.0.0 – Quaternion View – CHN China · D=8 · N=26



CNQ DIAGNOSTICS · CHN (D = 8)

CHSH joint-coherence:
S value: 0.8800
classical bound: 2.0
Tsirelson bound: 2.8284271247461903
verdict: independent

Twin-quaternion factor:
enabled: True
partition axis: 6
max |residual|: 2.8745e+00

Attractor fit:
amplitude A: 0.942
contraction λ : -0.0324
period: 1
fitted: False

Helmsman family:
total flips: 12
stability S_σ : 0.500

§ DEU Germany

EMBER electricity-generation series · 2000-2025 · N = 26 · D = 9

INPUT

source: ember_DEU*_generation_TWh.csv
N records: 26
D carriers: 9
Bioenergy
Coal
Gas
Hydro
Nuclear
Other Fossil
Other Renewables
Solar
Wind
source_sha256: 8d52a1c20cd36340bef83792...

HEADLINE METRICS

Helmsman flips: ?
Helmsman stability S_σ: {'global': 0.4583333333333337, 'rolling': [0.5714285714285714, 0.4285714285714286]}
IR class: ?
Attractor amplitude: ?

CHSH S: ?
Tsirelson bound: $2\sqrt{2} \approx 2.828$

Bearing pairs (D(D-1)/2): 36

CNT v3.1.0 content_sha: ?...
CNQ v2.0.0 content_sha: ?...

Following pages: Stage 1 (XY/XZ/YZ × N years + course) → Stage 2/3 (7 pages) → CNQ (2 pages).

SECTION PLATE t=0 | Year 2000 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

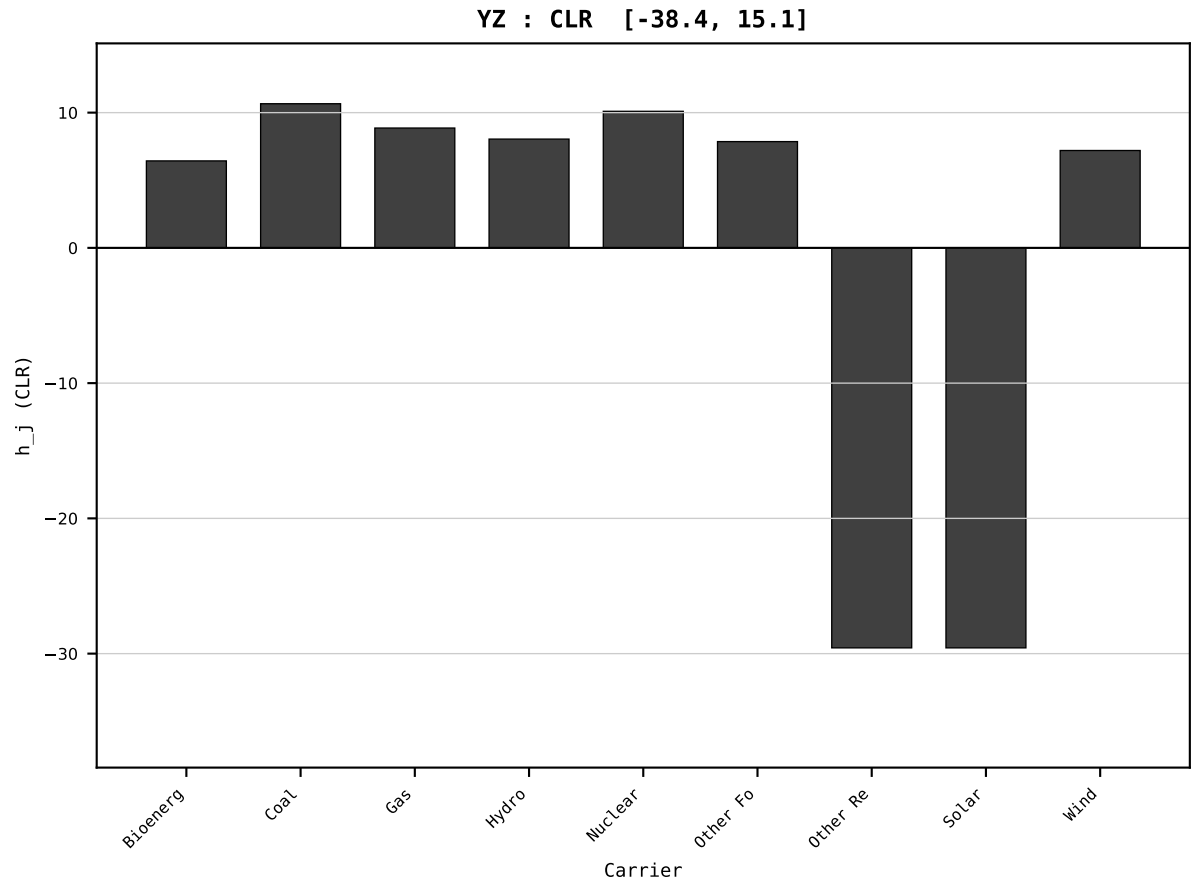
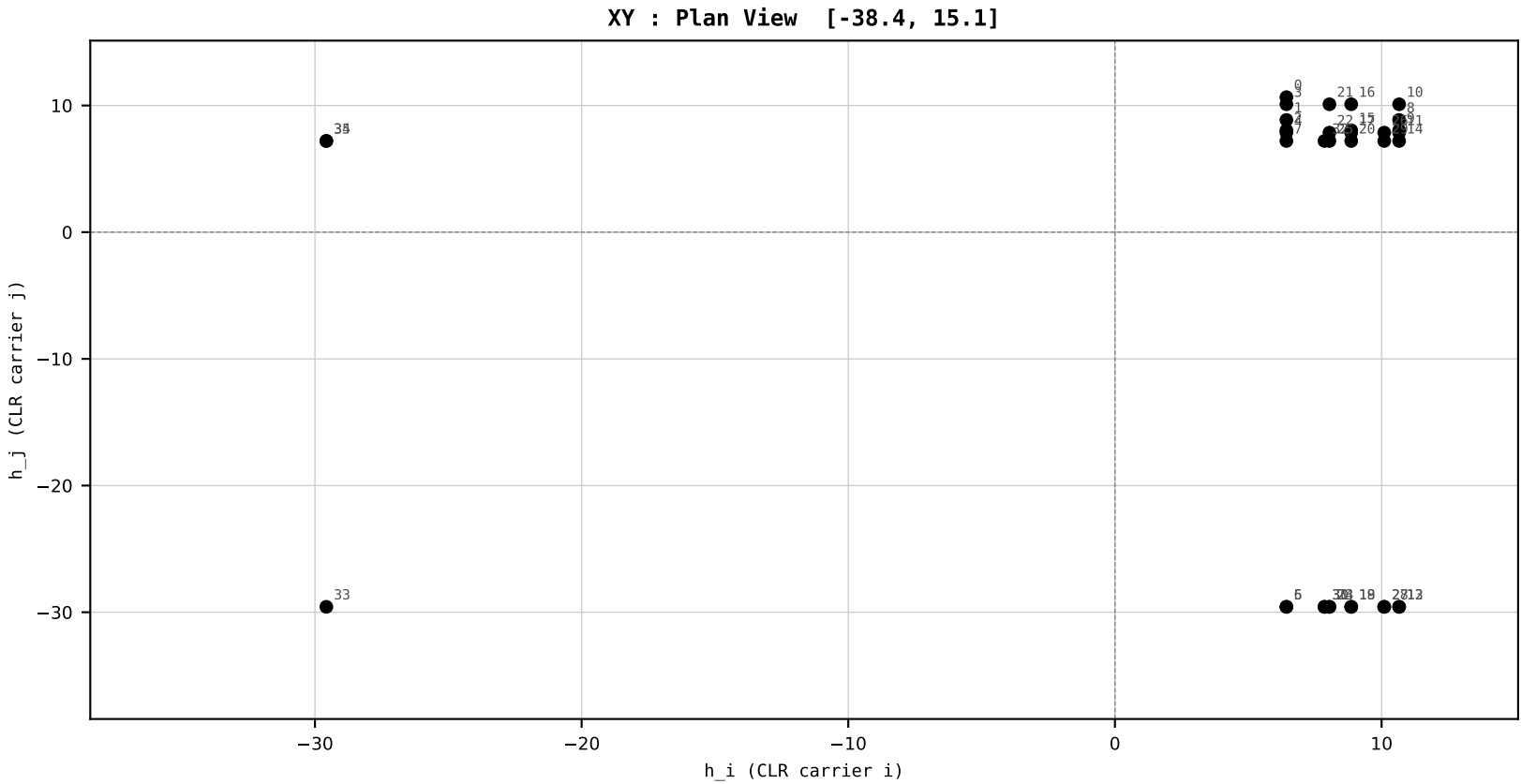
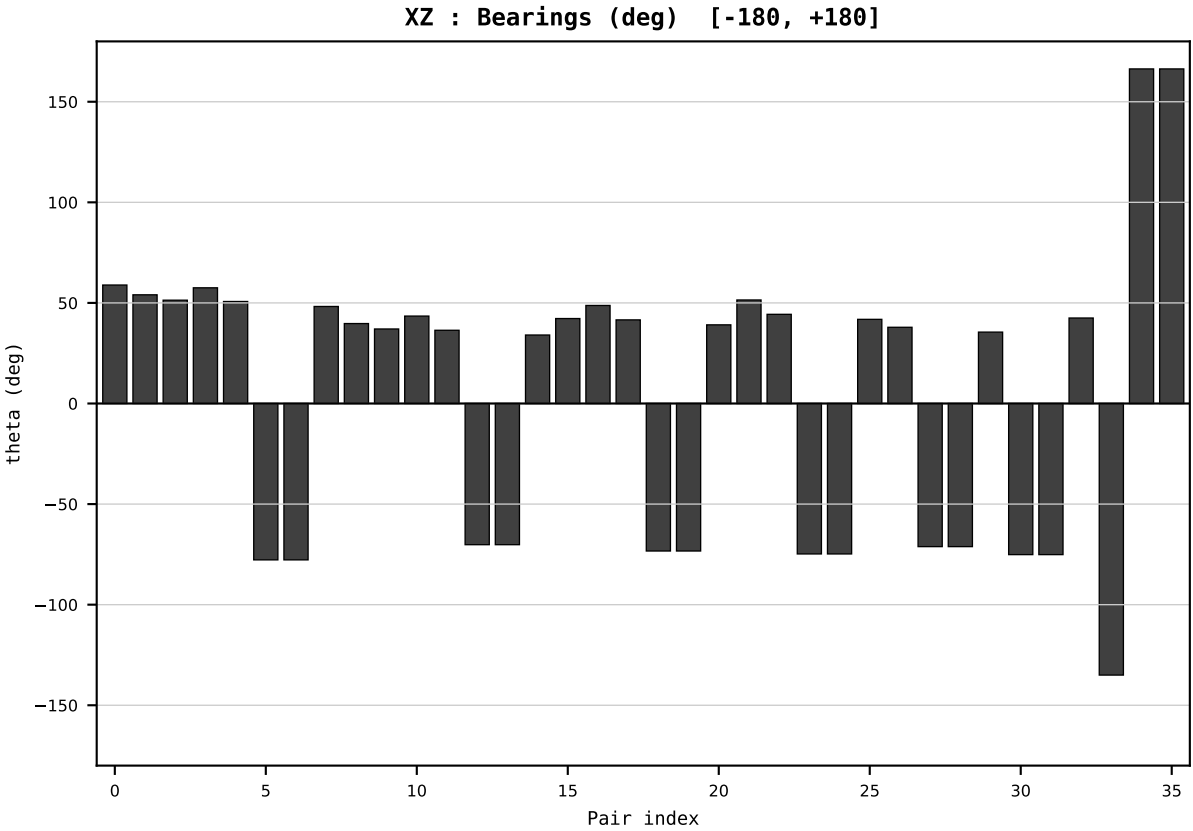
Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2000
t : 0 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.430719
Ring = Hs-3
E_metric= 1575.1016
kappa_HS= 296680000000000000.00
omega = 0.0000 deg
d_A = 0.000000
Helm = ---
Helm d = ---
DR = 40.231 HLR
DR ratio= 296679999472192320.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=1 | Year 2001 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

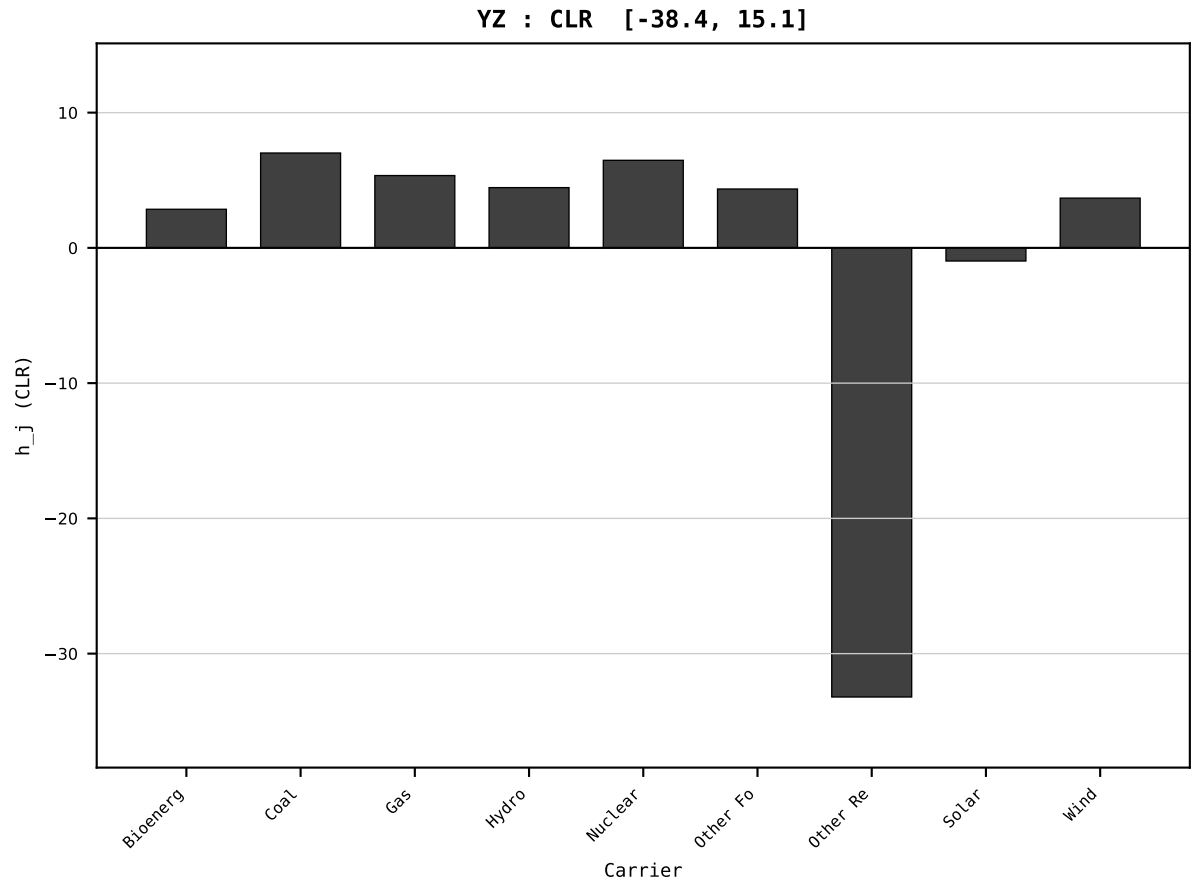
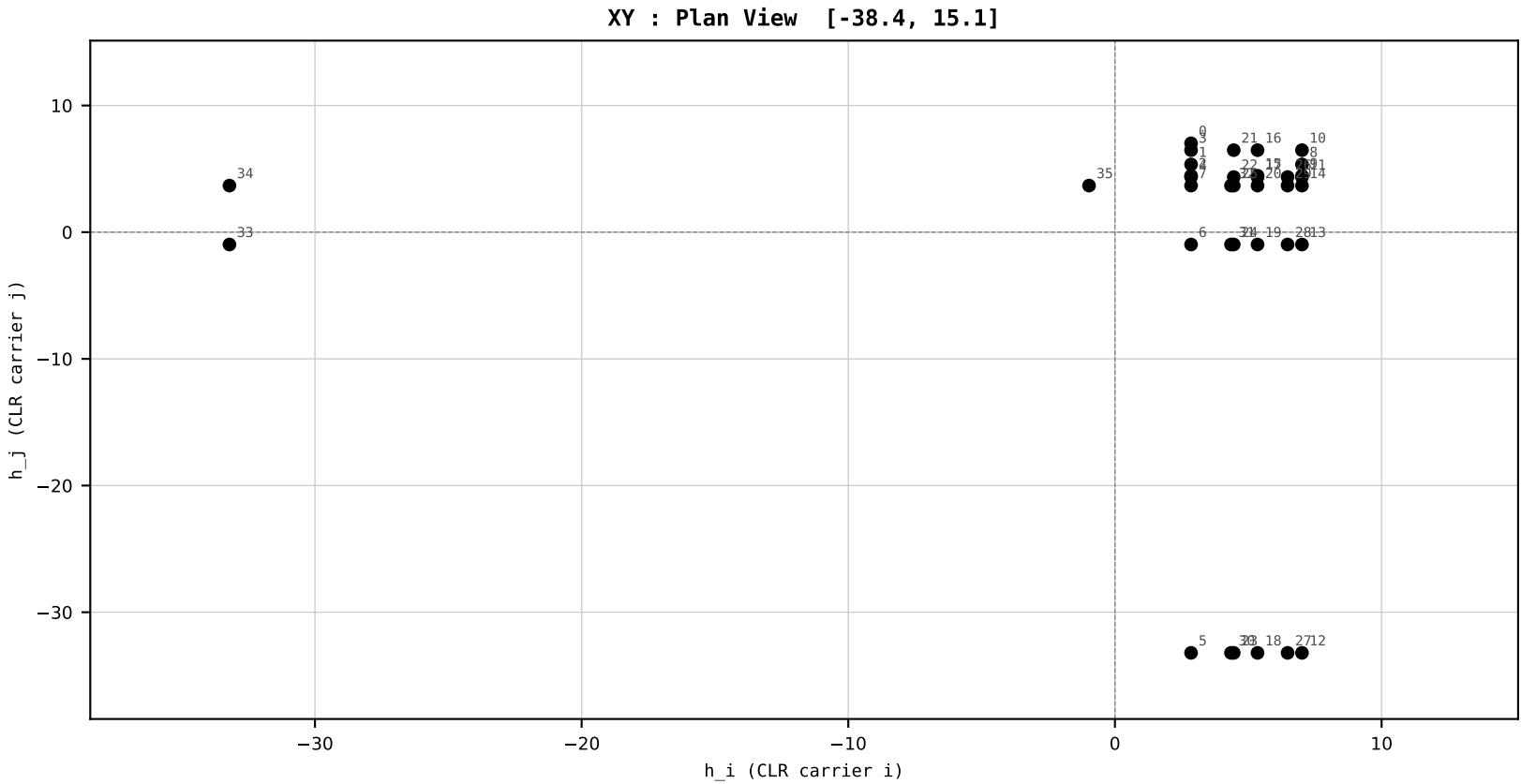
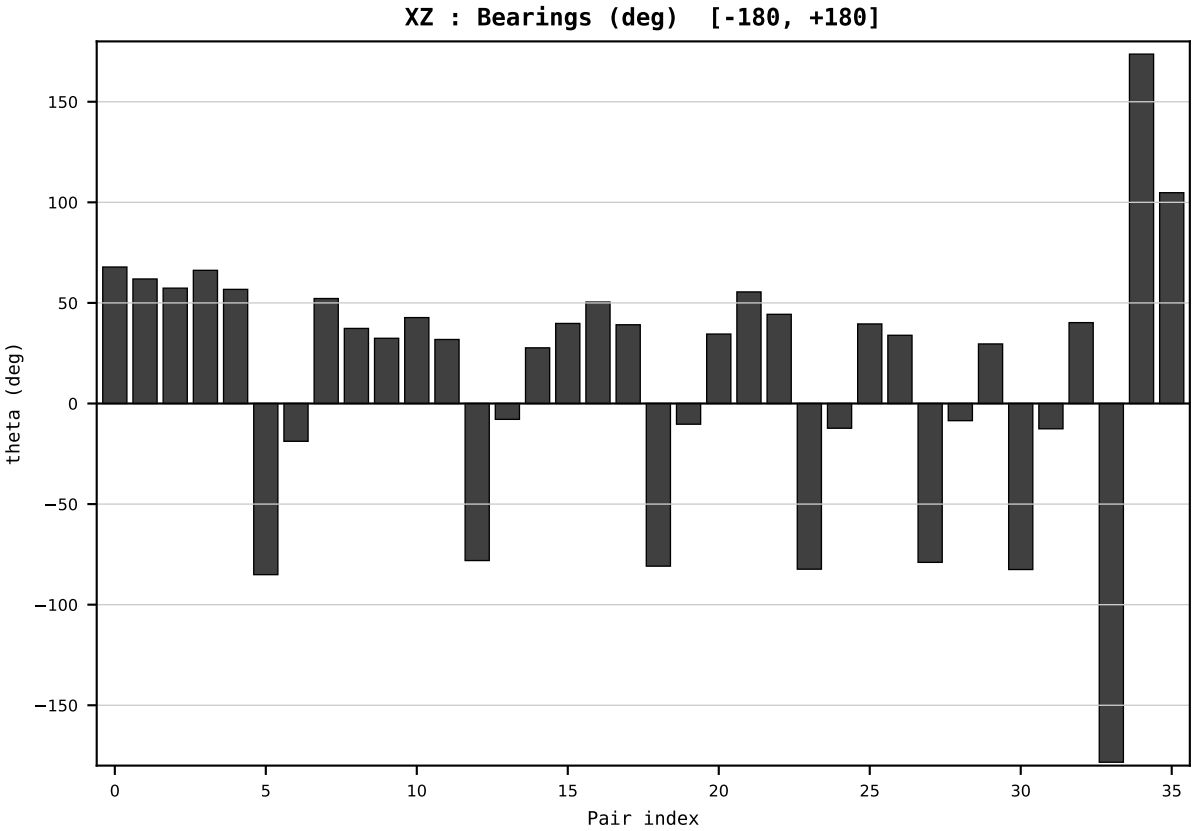
Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2001
t : 1 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.414163
Ring = Hs-3
E_metric= 897.3656
kappa_HS= 29374000000000000.00
omega = 39.6073 deg
d_A = 30.338234
Helm = Solar
Helm d = +28.602816
DR = 40.221 HLR
DR ratio= 293740000811779456.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=2 | Year 2002 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

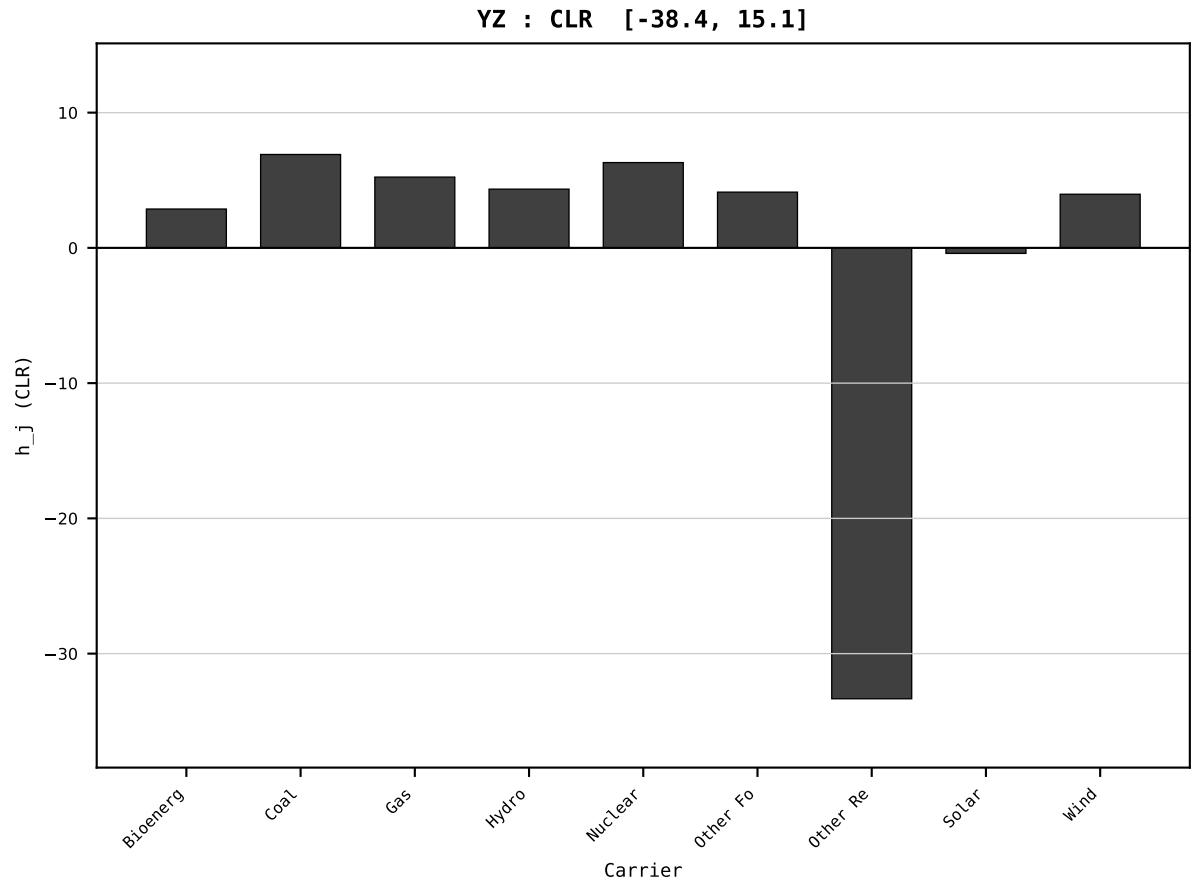
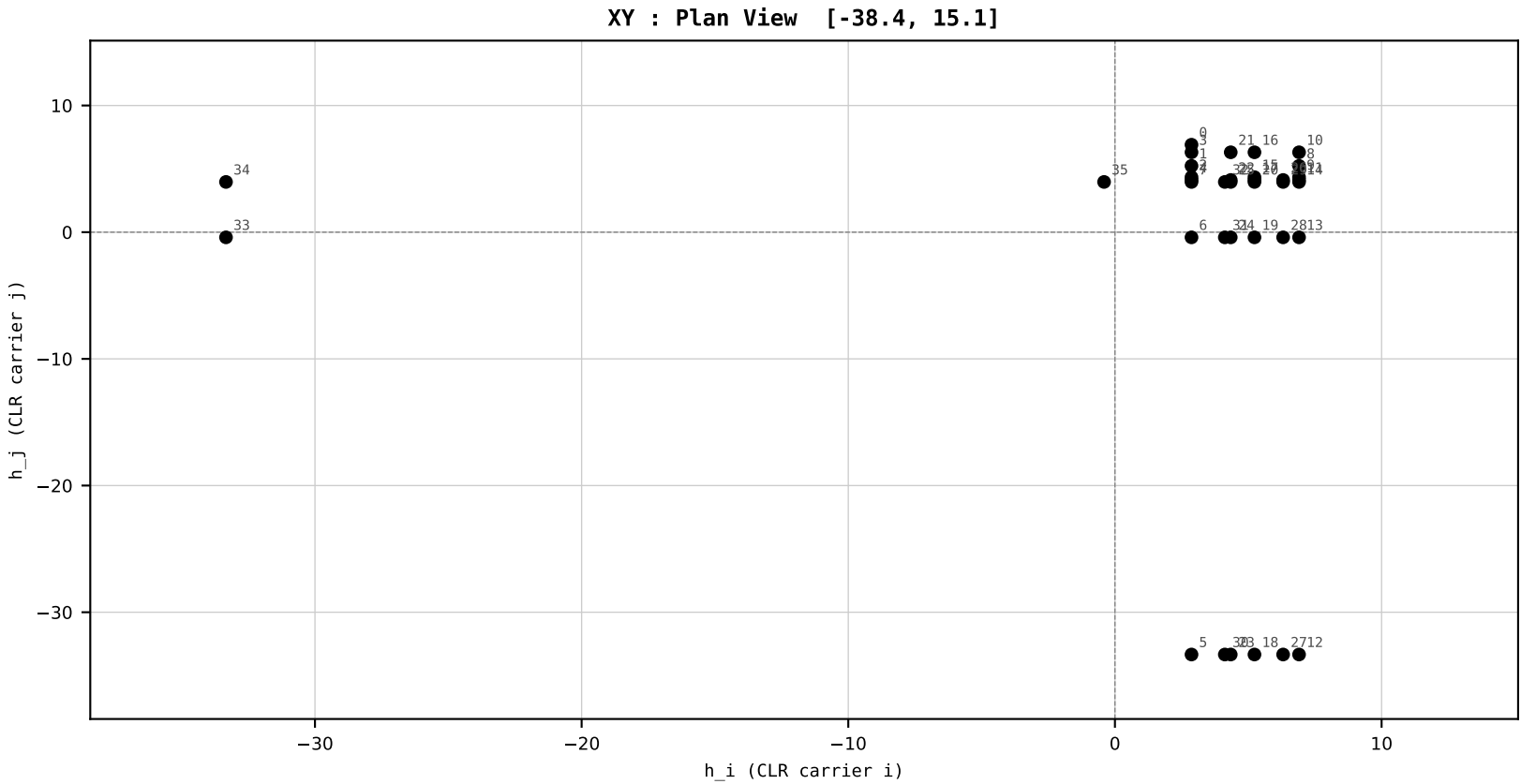
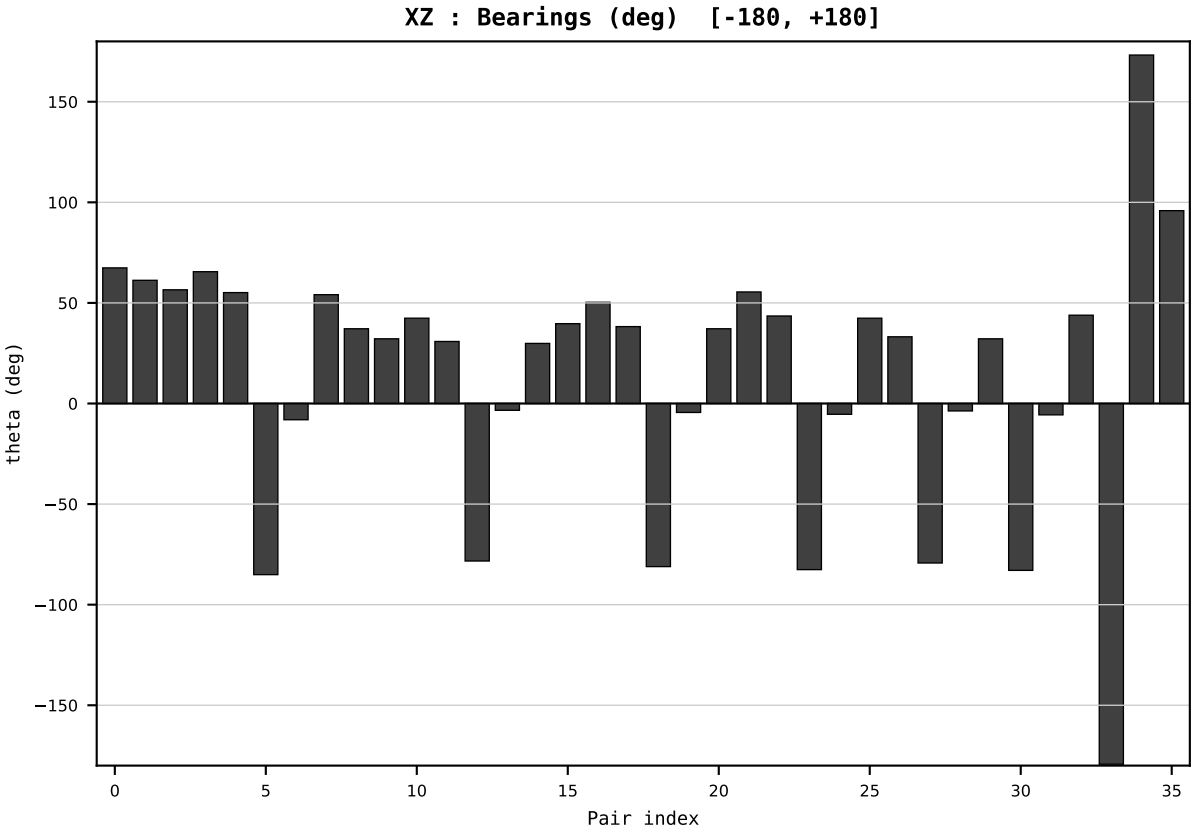
Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2002
t : 2 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.405699
Ring = Hs-3
E_metric= 897.5397
kappa_HS= 299600000000000000.00
omega = 1.1699 deg
d_A = 0.732611
Helm = Solar
Helm d = +0.563622
DR = 40.241 HLR
DR ratio= 299600001032496448.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=3 | Year 2003 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

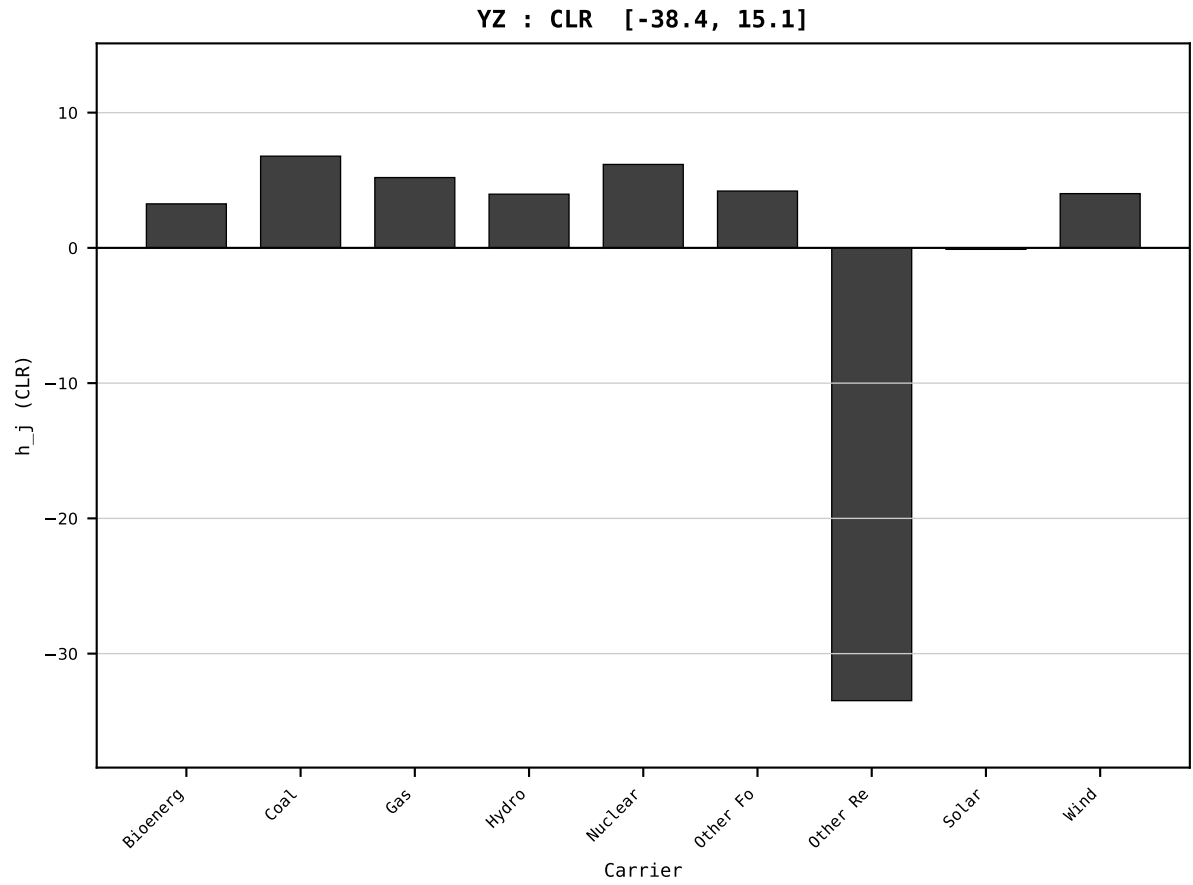
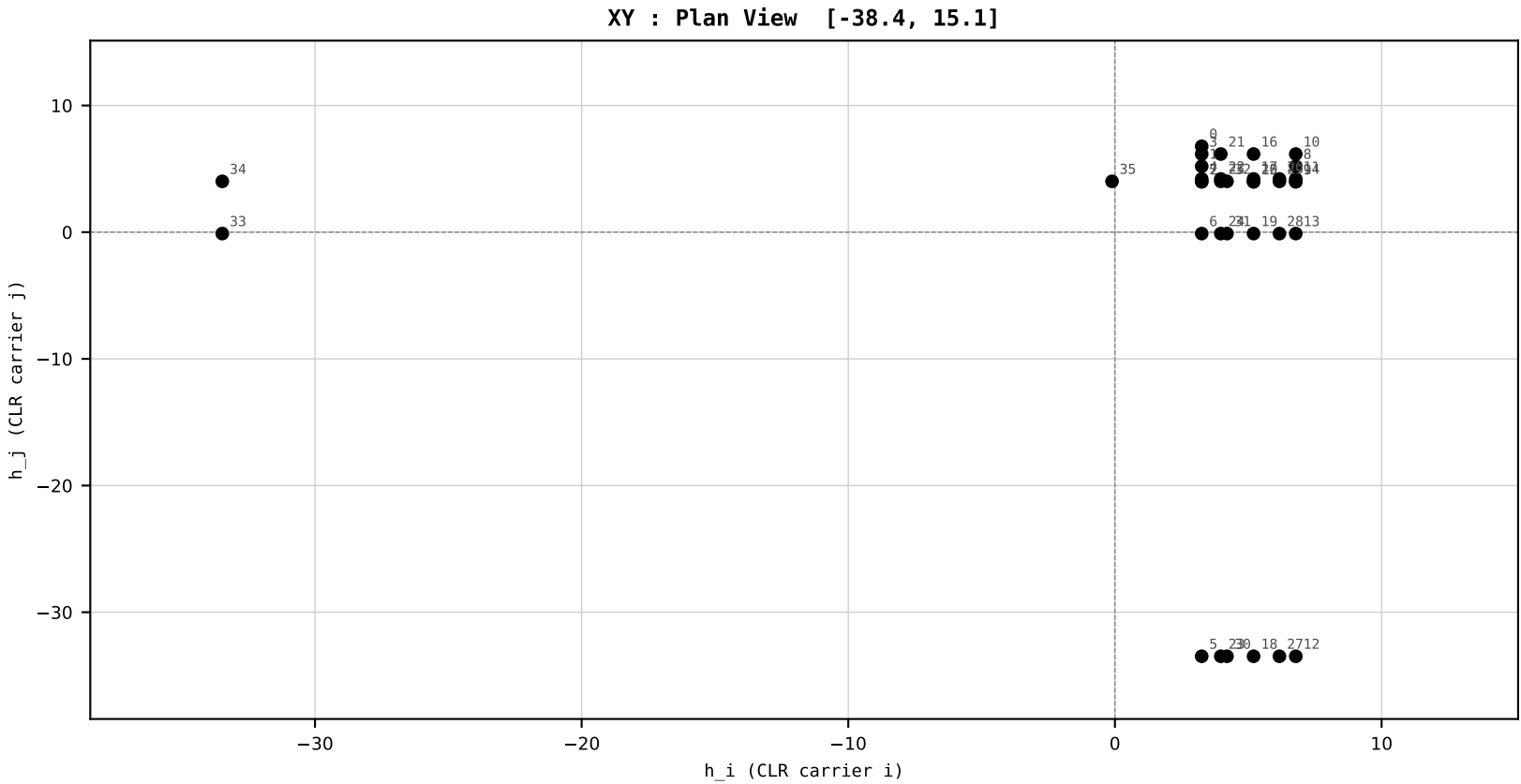
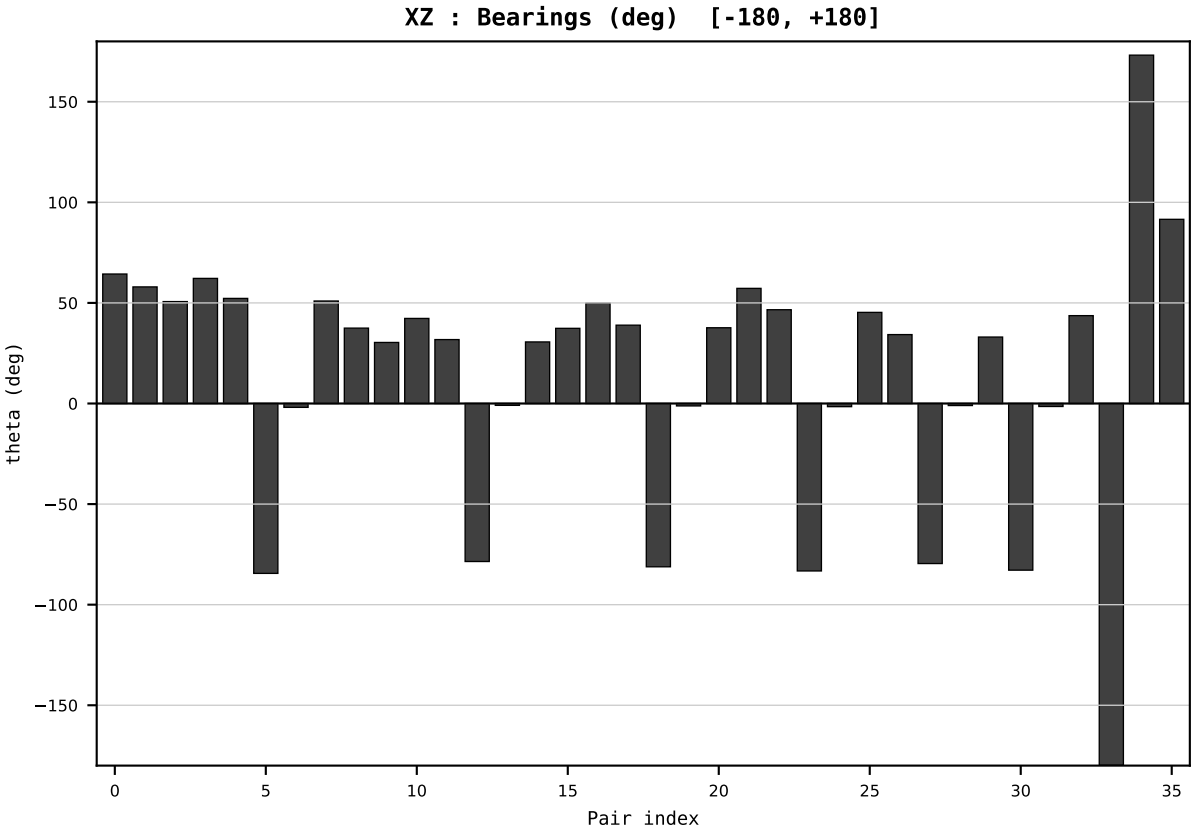
Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2003
t : 3 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.389220
Ring = Hs-3
E_metric= 899.7401
kappa_HS= 304629999999999936.00
omega = 1.0479 deg
d_A = 0.661385
Helm = Bioenergy
Helm d = +0.382457
DR = 40.258 HLR
DR ratio= 304629999895005696.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=4 | Year 2004 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

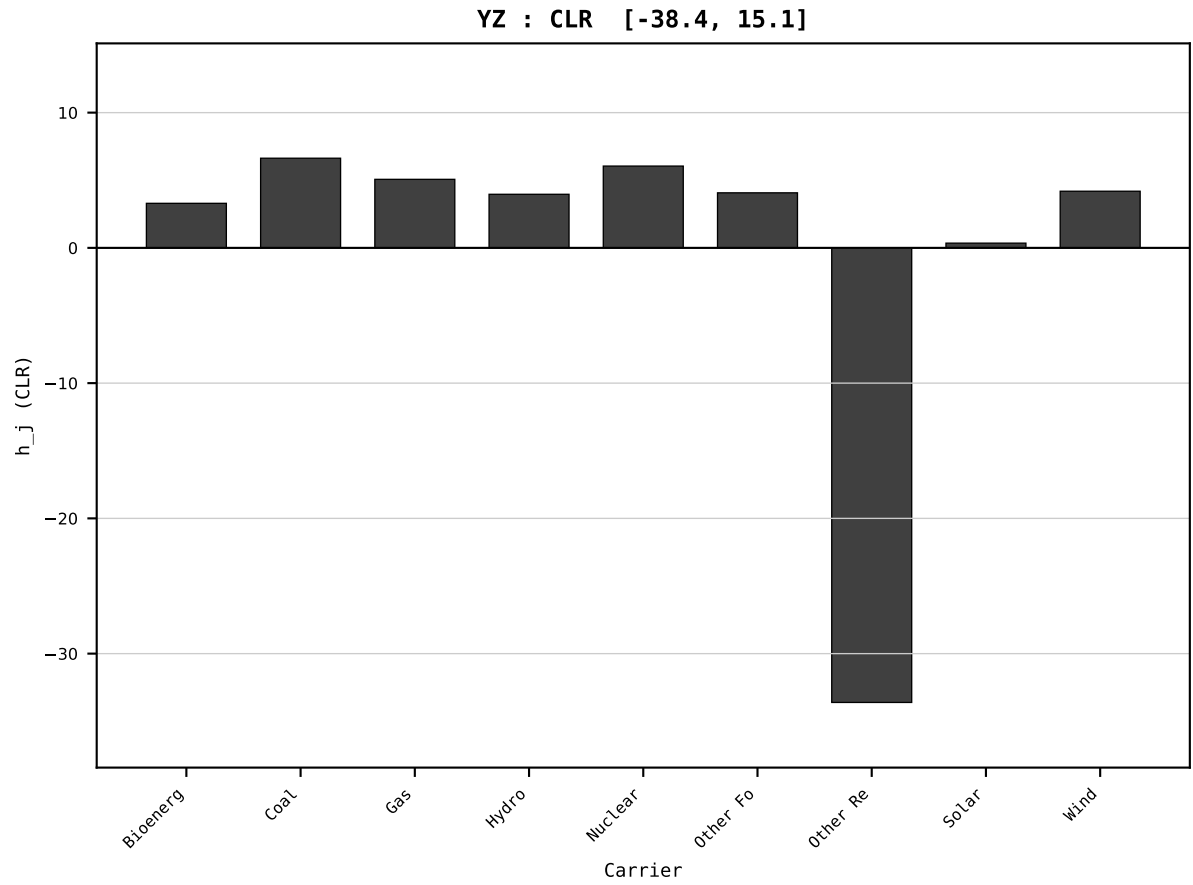
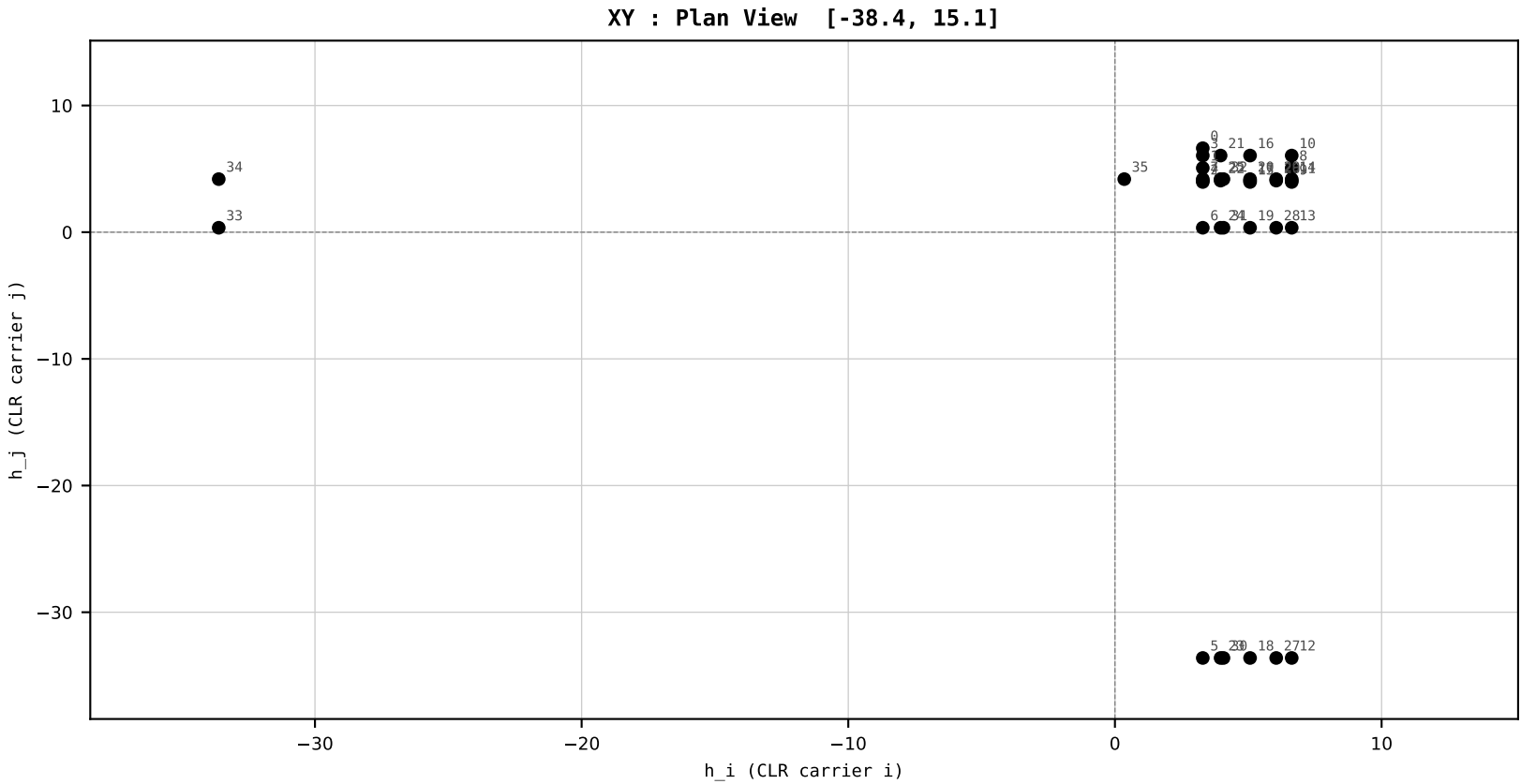
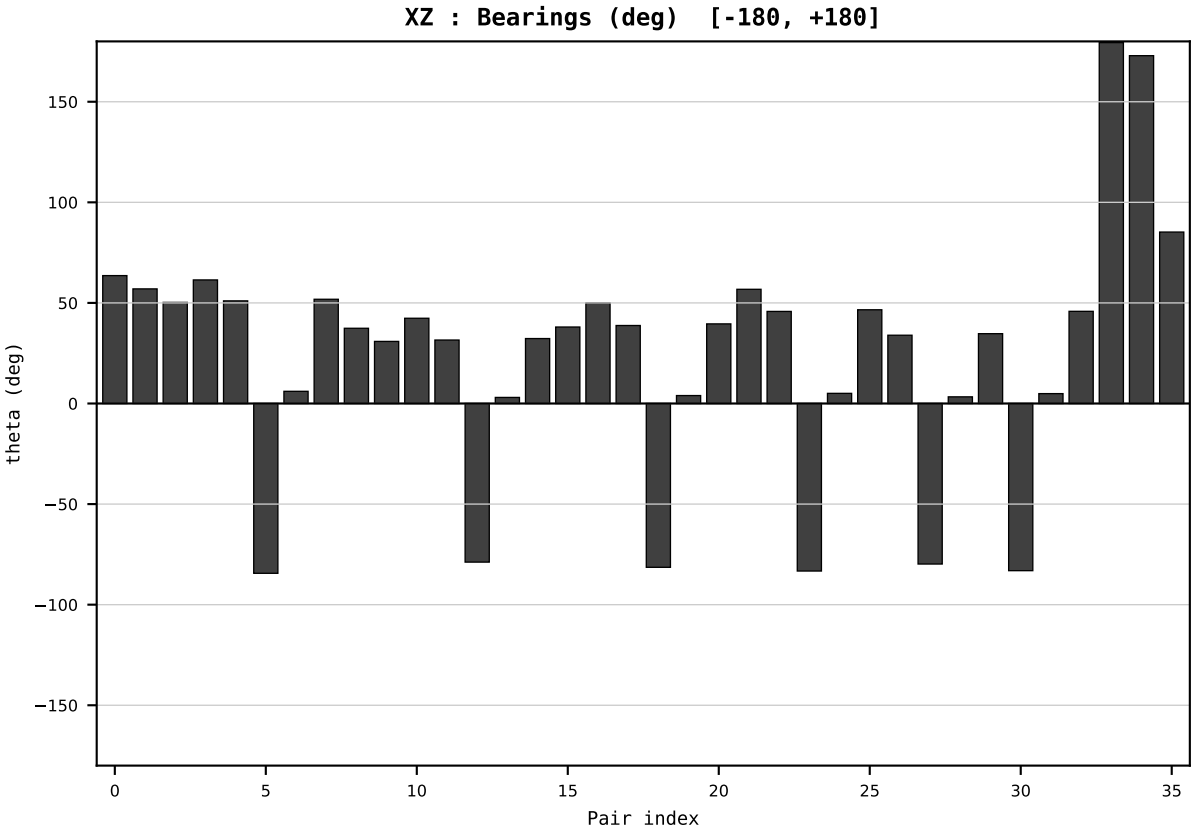
Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2004
t : 4 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.367842
Ring = Hs-3
E_metric= 902.0612
kappa_HS= 298769999999999936.00
omega = 0.9133 deg
d_A = 0.577156
Helm = Solar
Helm d = +0.458667
DR = 40.238 HLR
DR ratio= 298769999545169920.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=5 | Year 2005 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

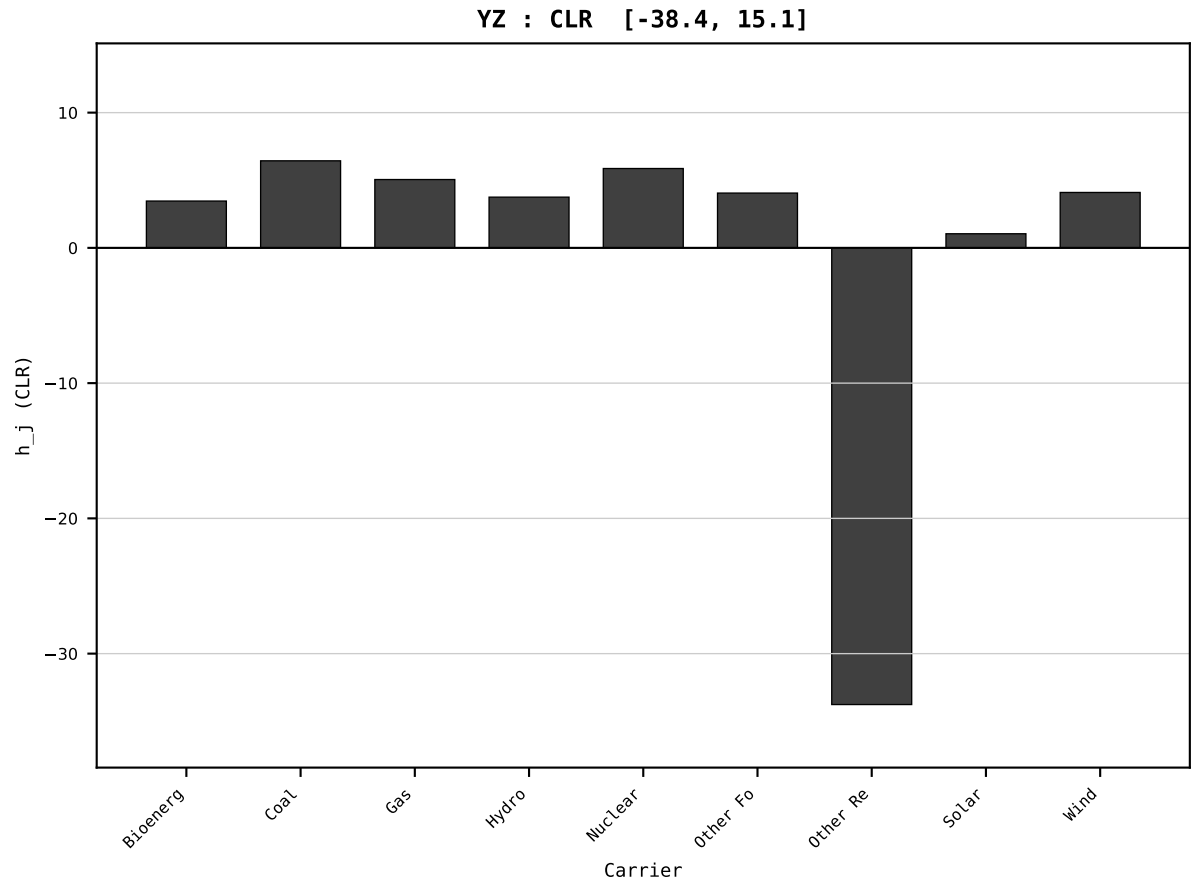
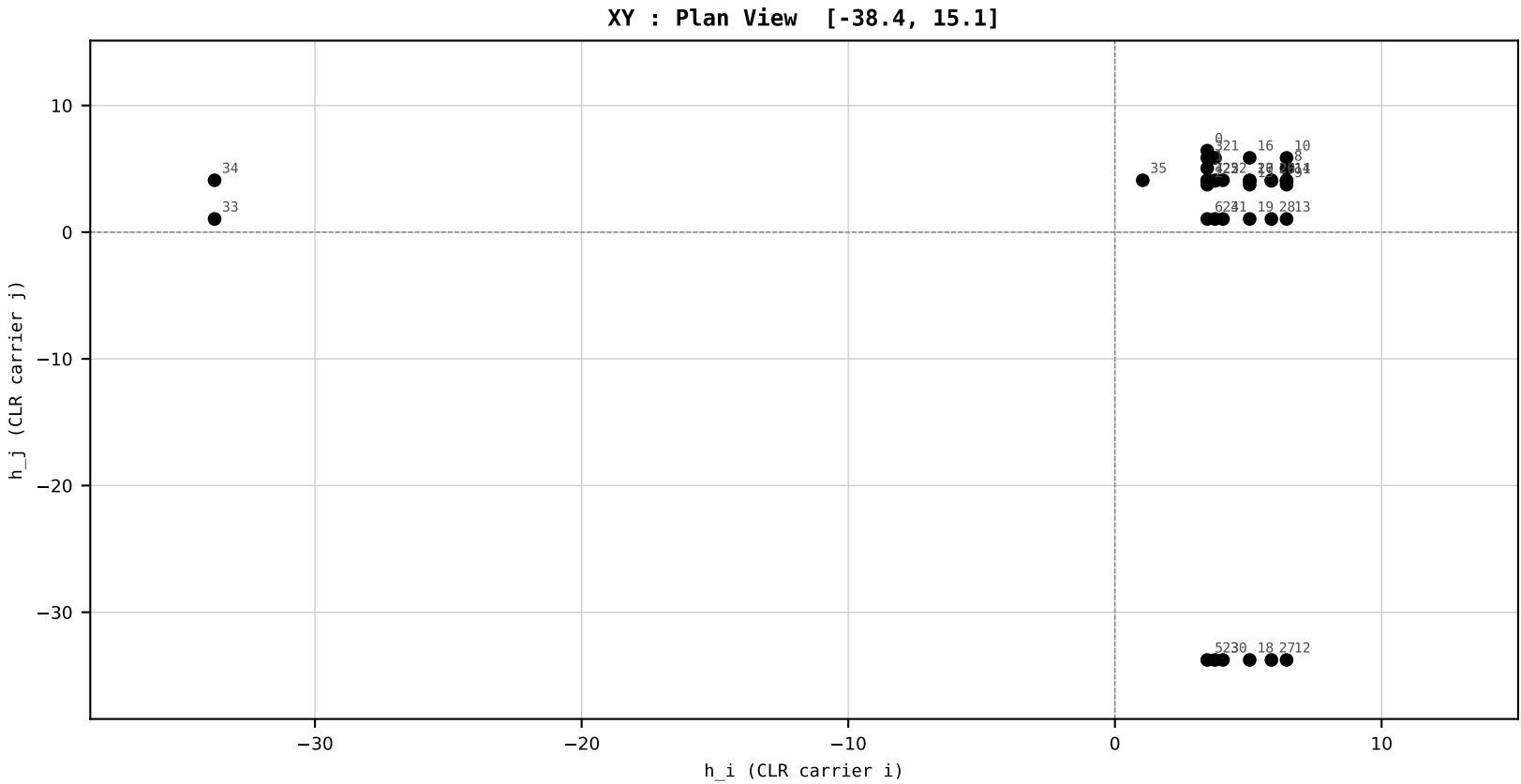
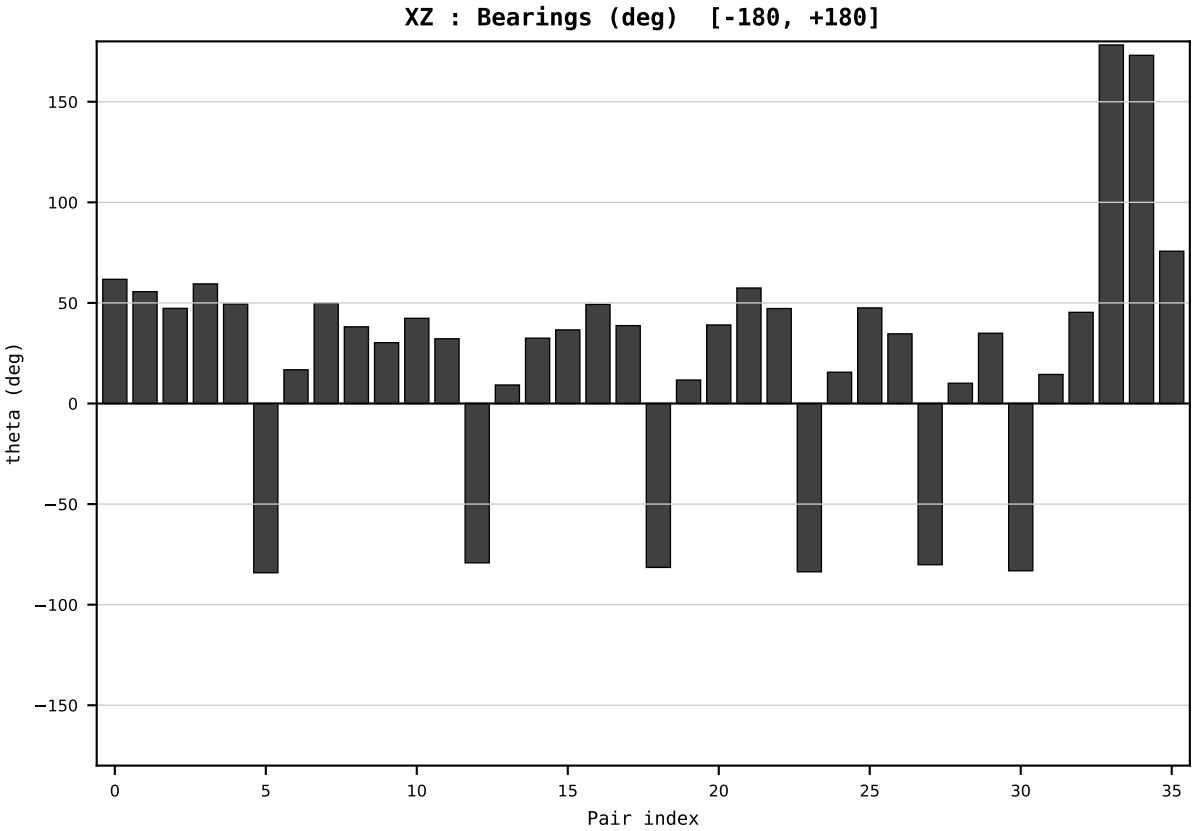
Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2005
t : 5 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.341114
Ring = Hs-3
E_metric= 904.9926
kappa_HS= 28813999999999968.00
omega = 1.2830 deg
d_A = 0.810406
Helm = Solar
Helm d = +0.694037
DR = 40.202 HLR
DR ratio= 288140000563598304.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=6 | Year 2006 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

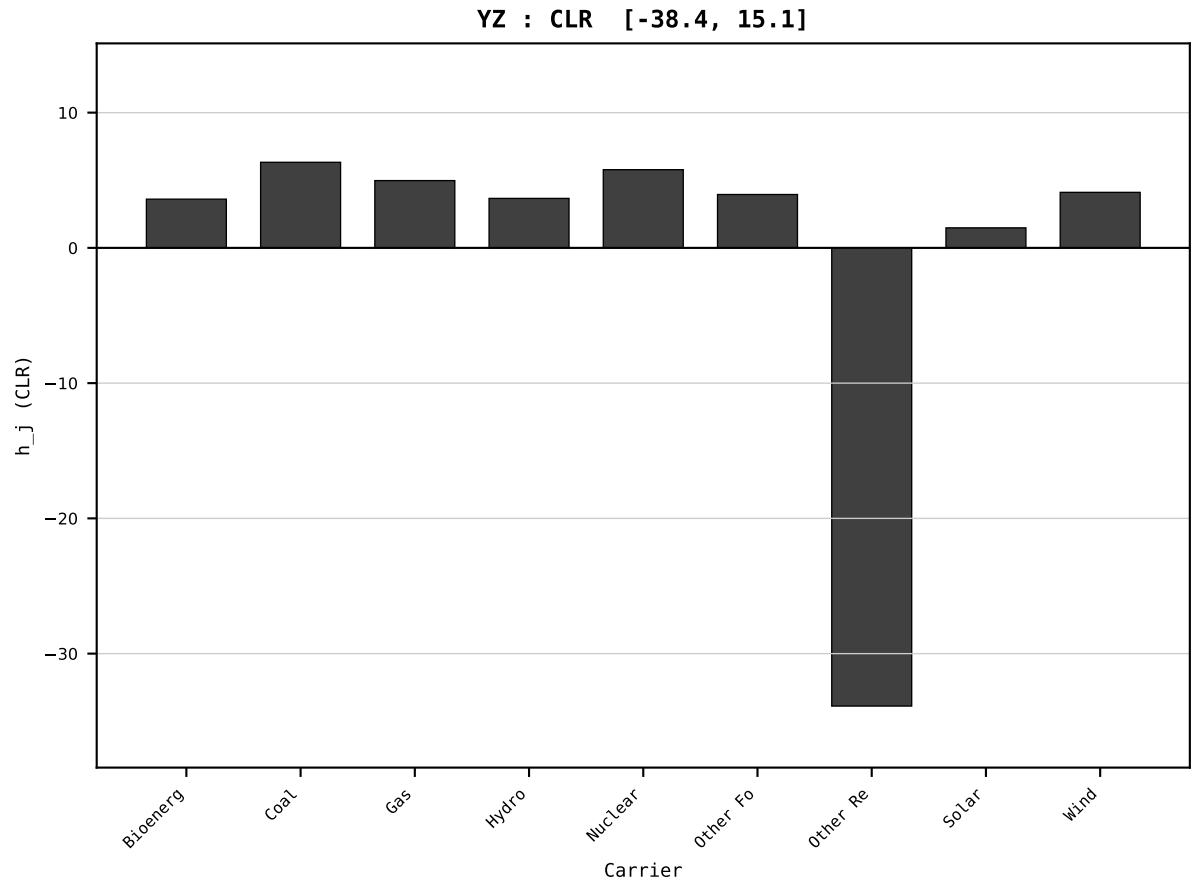
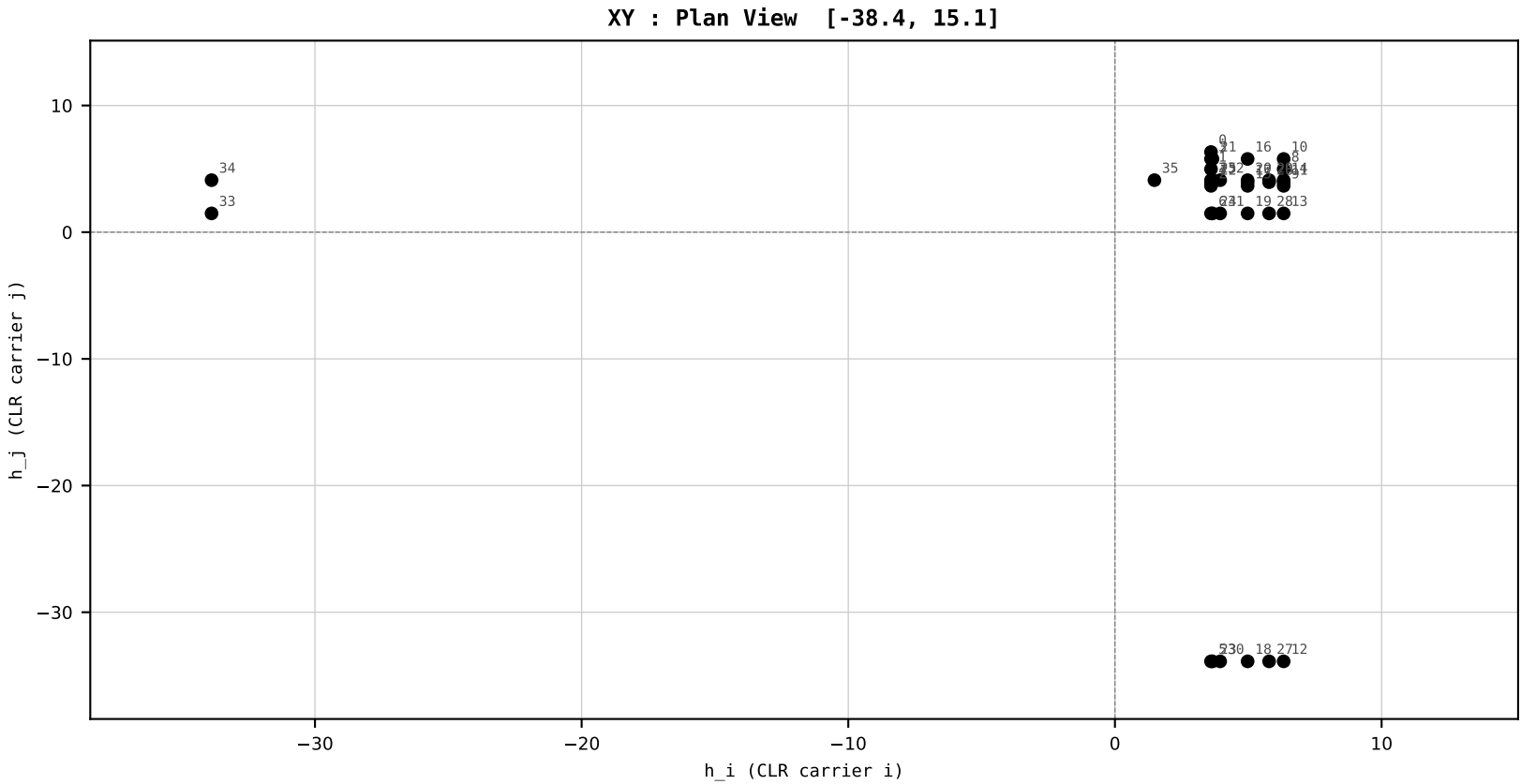
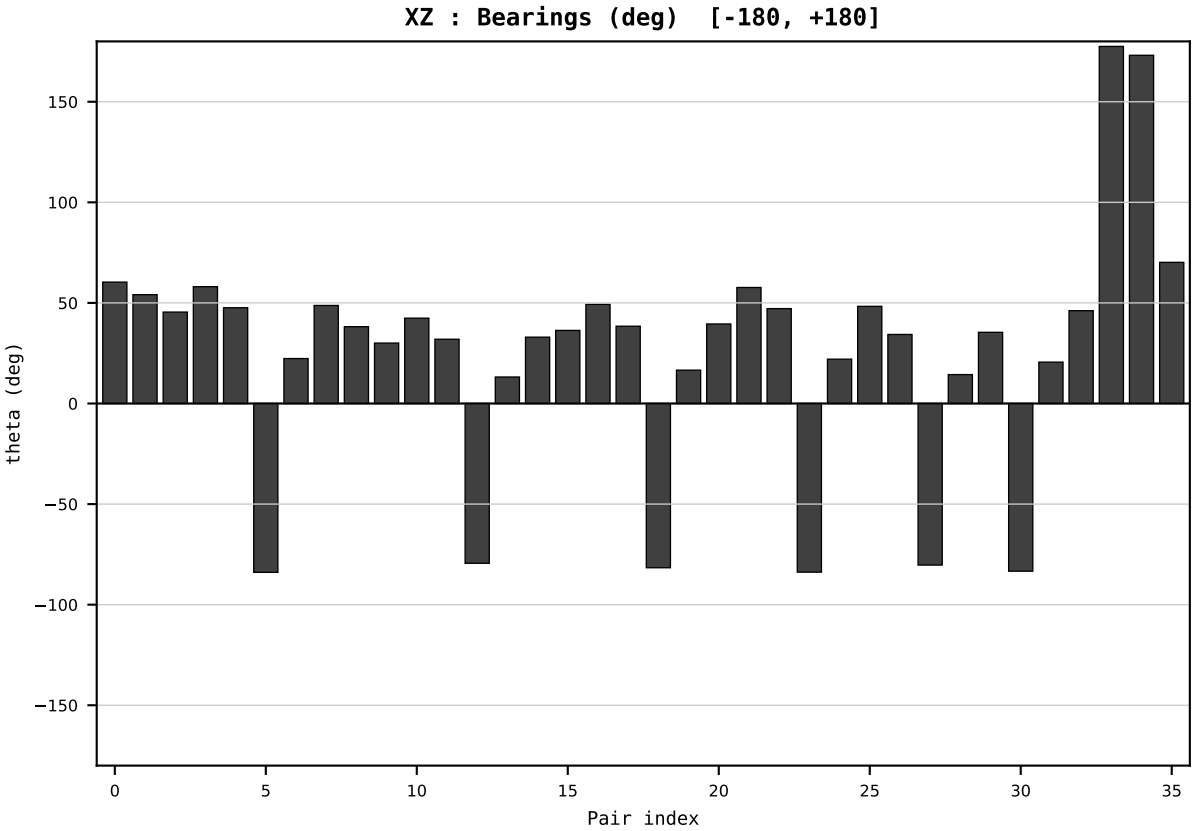
Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2006
t : 6 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.326053
Ring = Hs-3
E_metric= 907.0614
kappa_HS= 288930000000000000.00
omega = 0.8154 deg
d_A = 0.518705
Helm = Solar
Helm d = +0.437297
DR = 40.205 HLR
DR ratio= 288930000245136960.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=7 | Year 2007 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

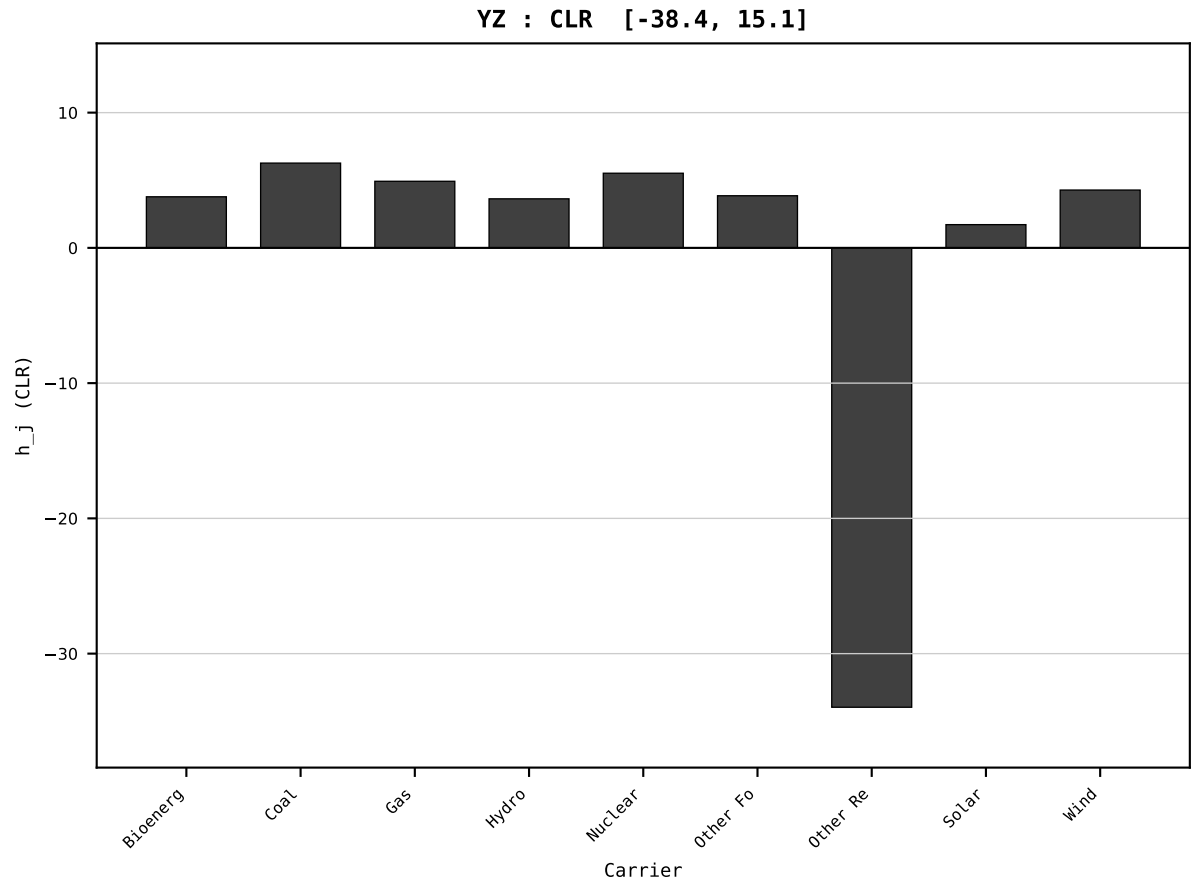
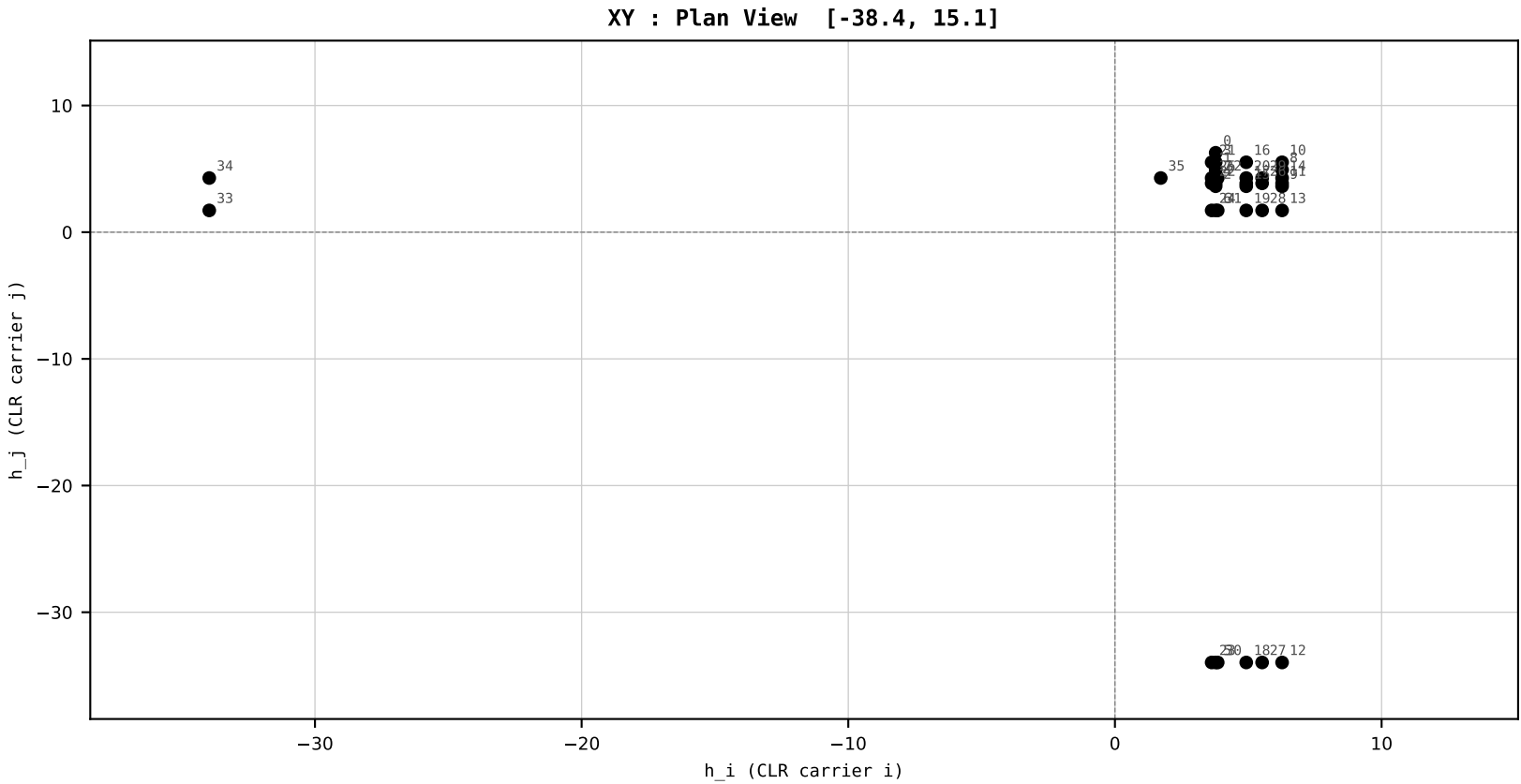
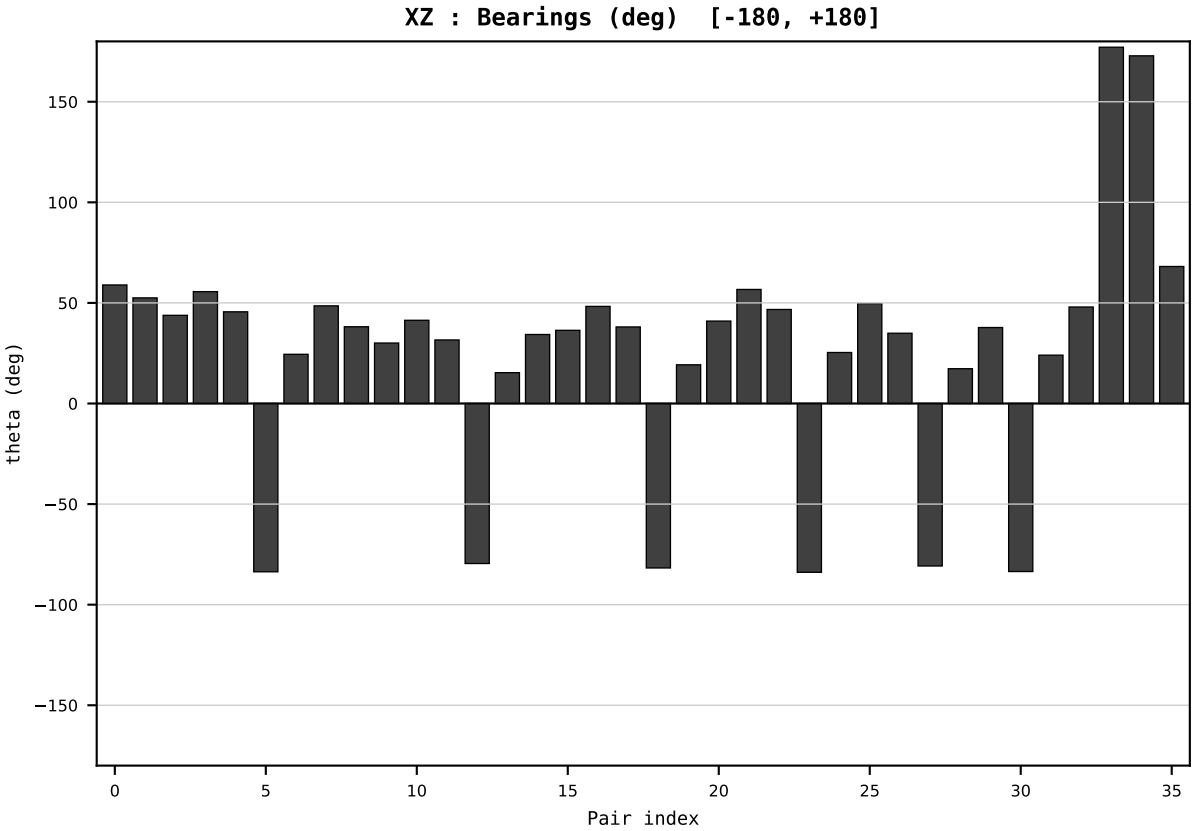
Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2007
t : 7 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.306825
Ring = Hs-3
E_metric= 910.0899
kappa_HS= 297100000000000000.00
omega = 0.7153 deg
d_A = 0.455473
Helm = Nuclear
Helm d = -0.261293
DR = 40.233 HLR
DR ratio= 297099997807627008.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=8 | Year 2008 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

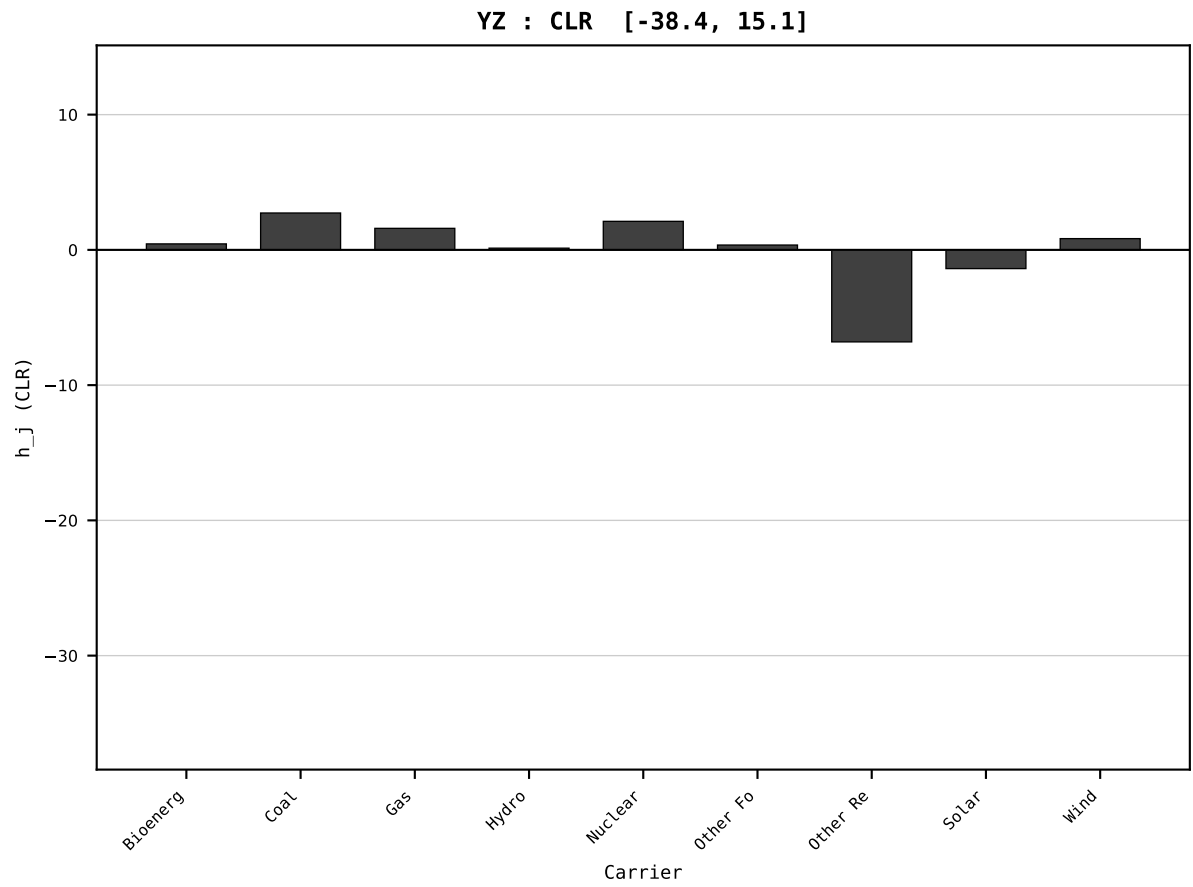
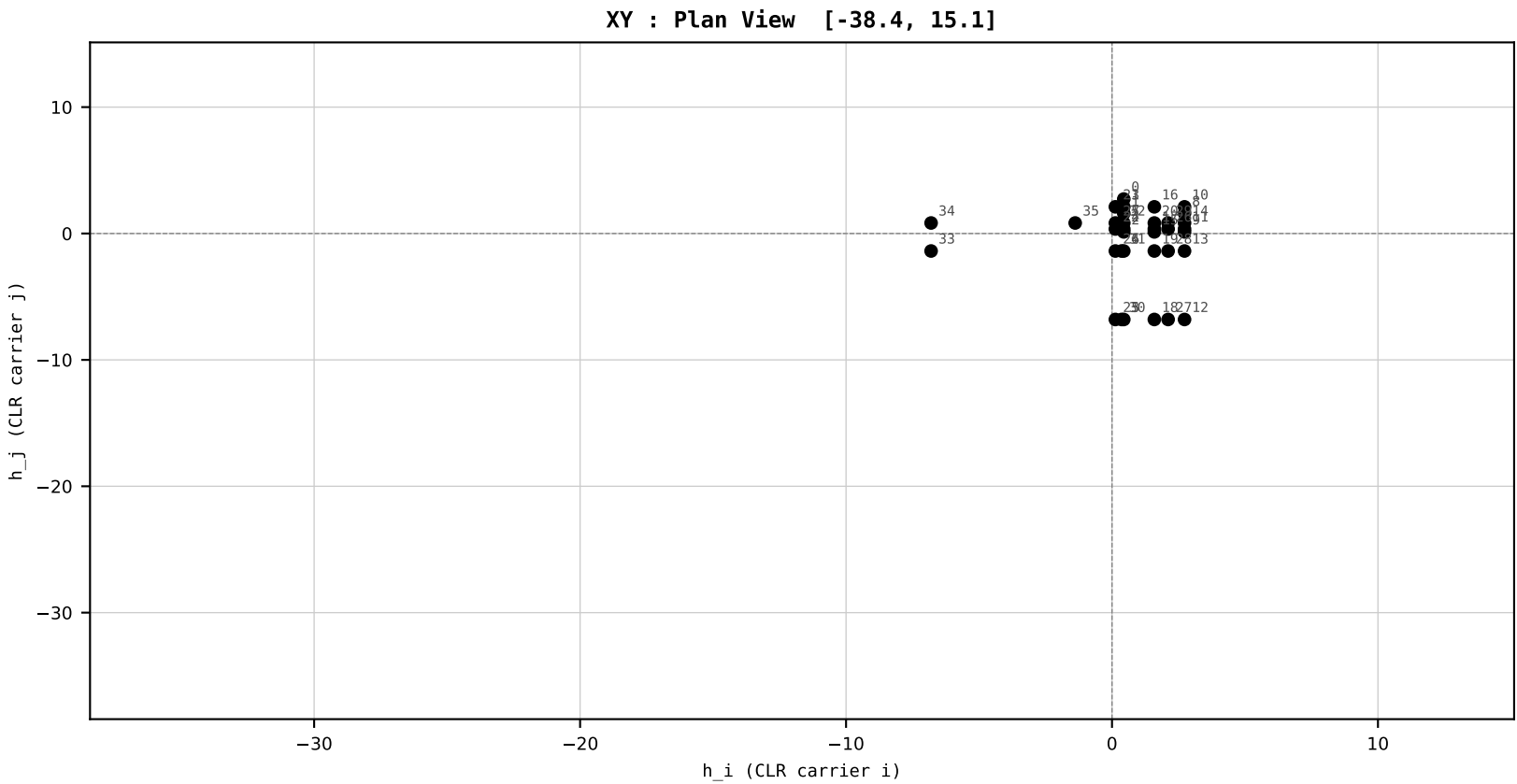
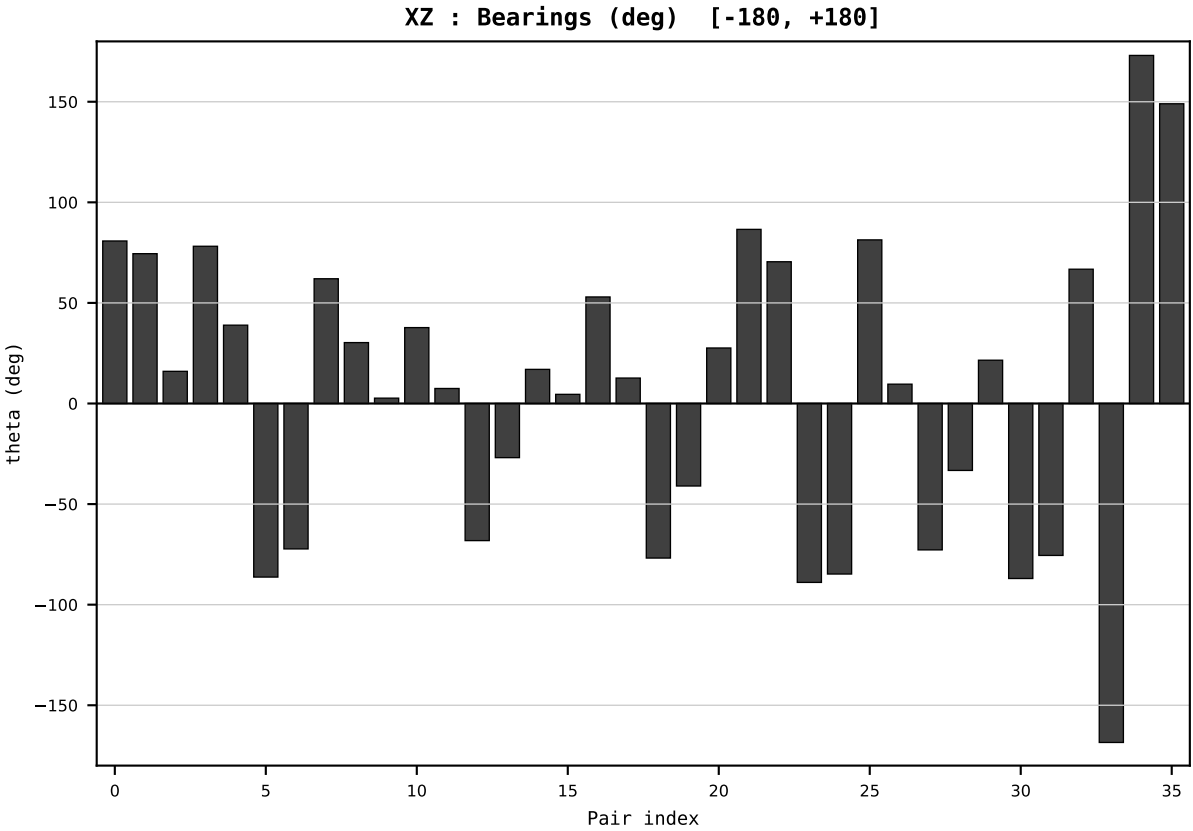
Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2008
t : 8 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.285057
Ring = Hs-2
E_metric= 63.6470
kappa_HS= 13760.50
omega = 19.4937 deg
d_A = 28.811916
Helm = Other Renewables
Helm d = +27.161871
DR = 9.530 HLR
DR ratio= 13760.5x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

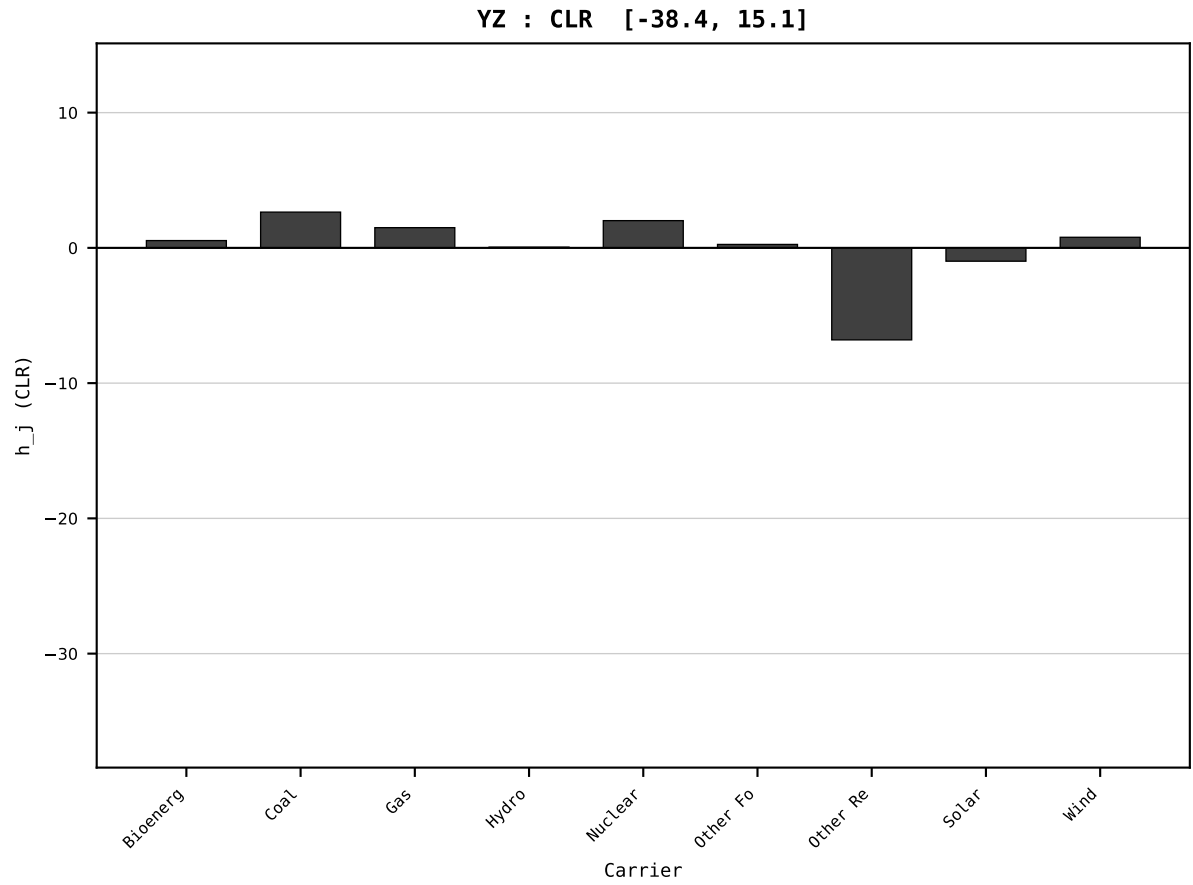
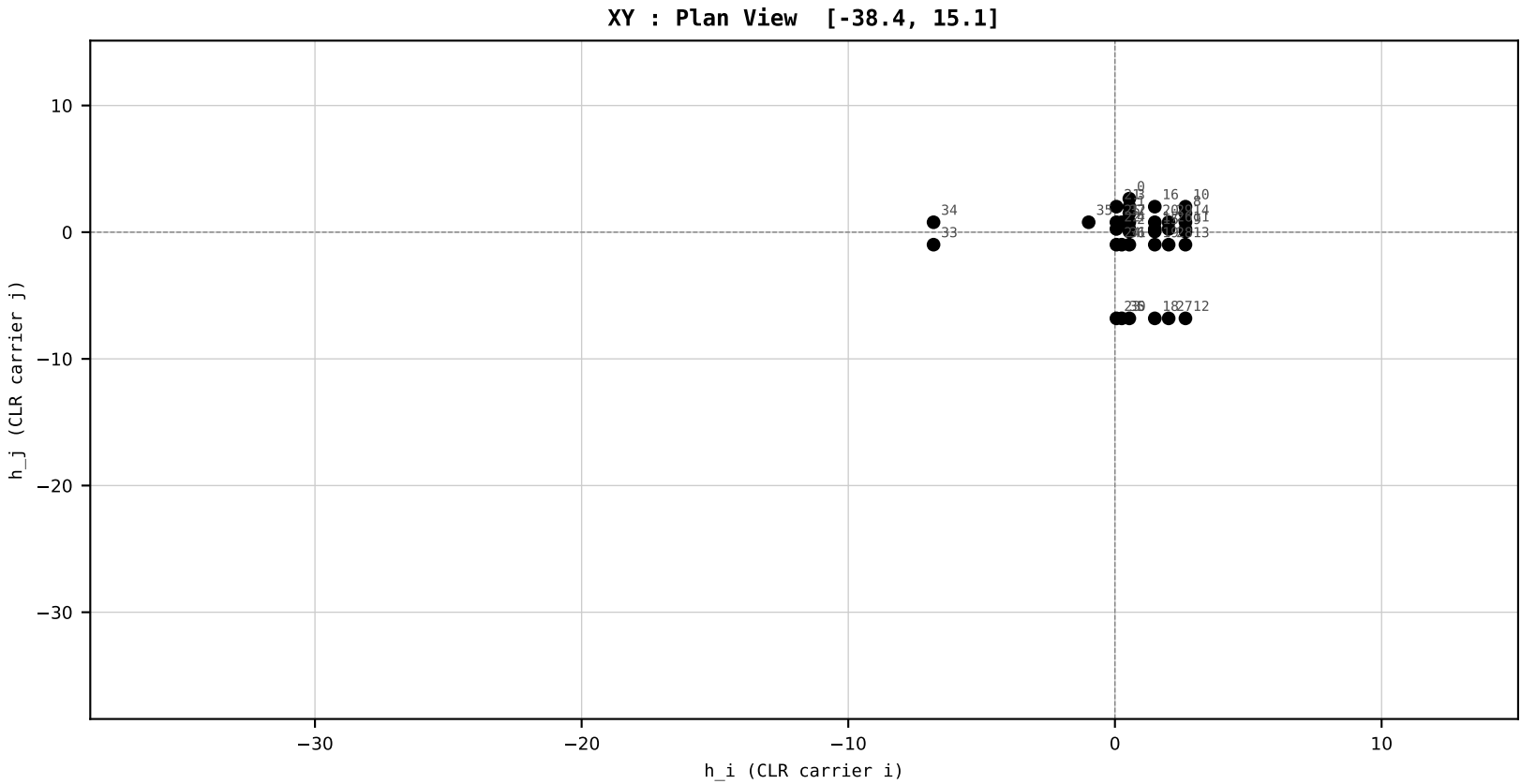
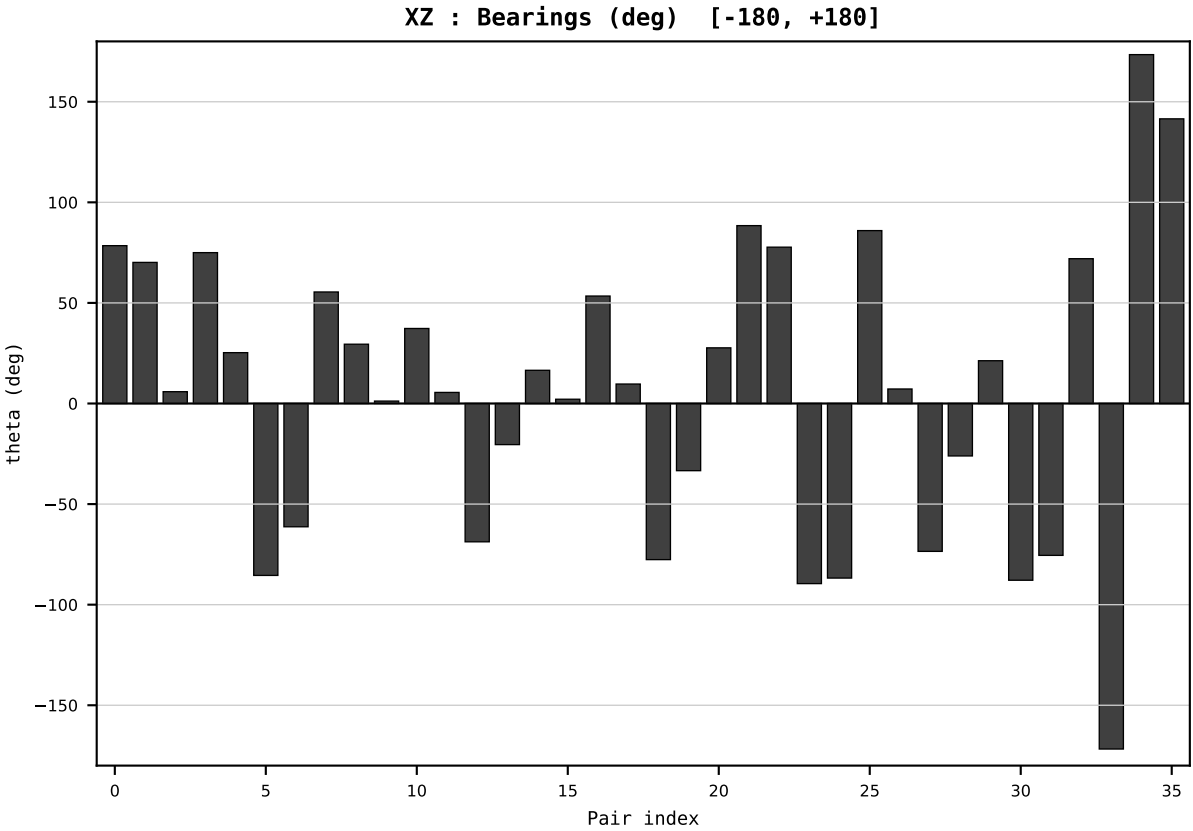
Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2009
t : 9 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.272489
Ring = Hs-2
E_metric= 61.5055
kappa_HS= 12672.50
omega = 3.1880 deg
d_A = 0.460403
Helm = Solar
Helm d = +0.398992
DR = 9.447 HLR
DR ratio= 12672.5x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=10 | Year 2010 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

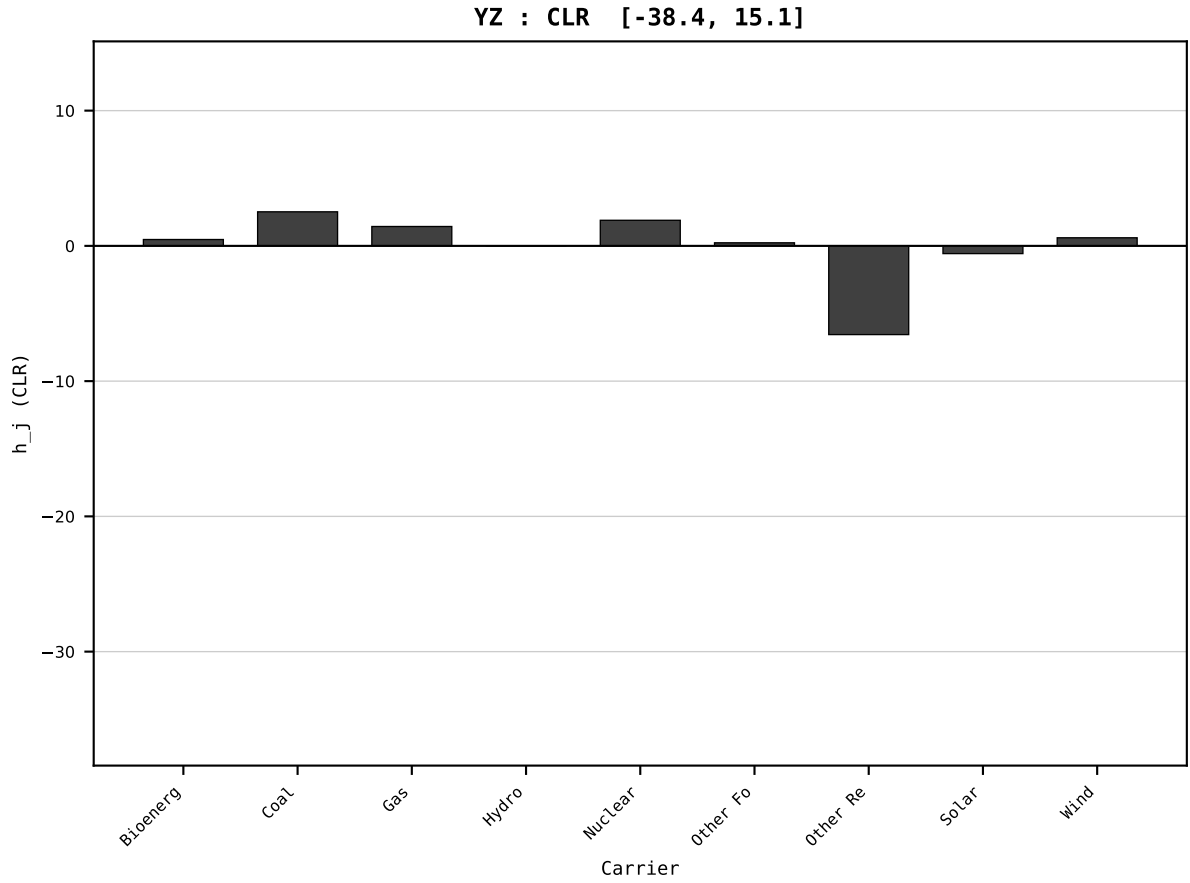
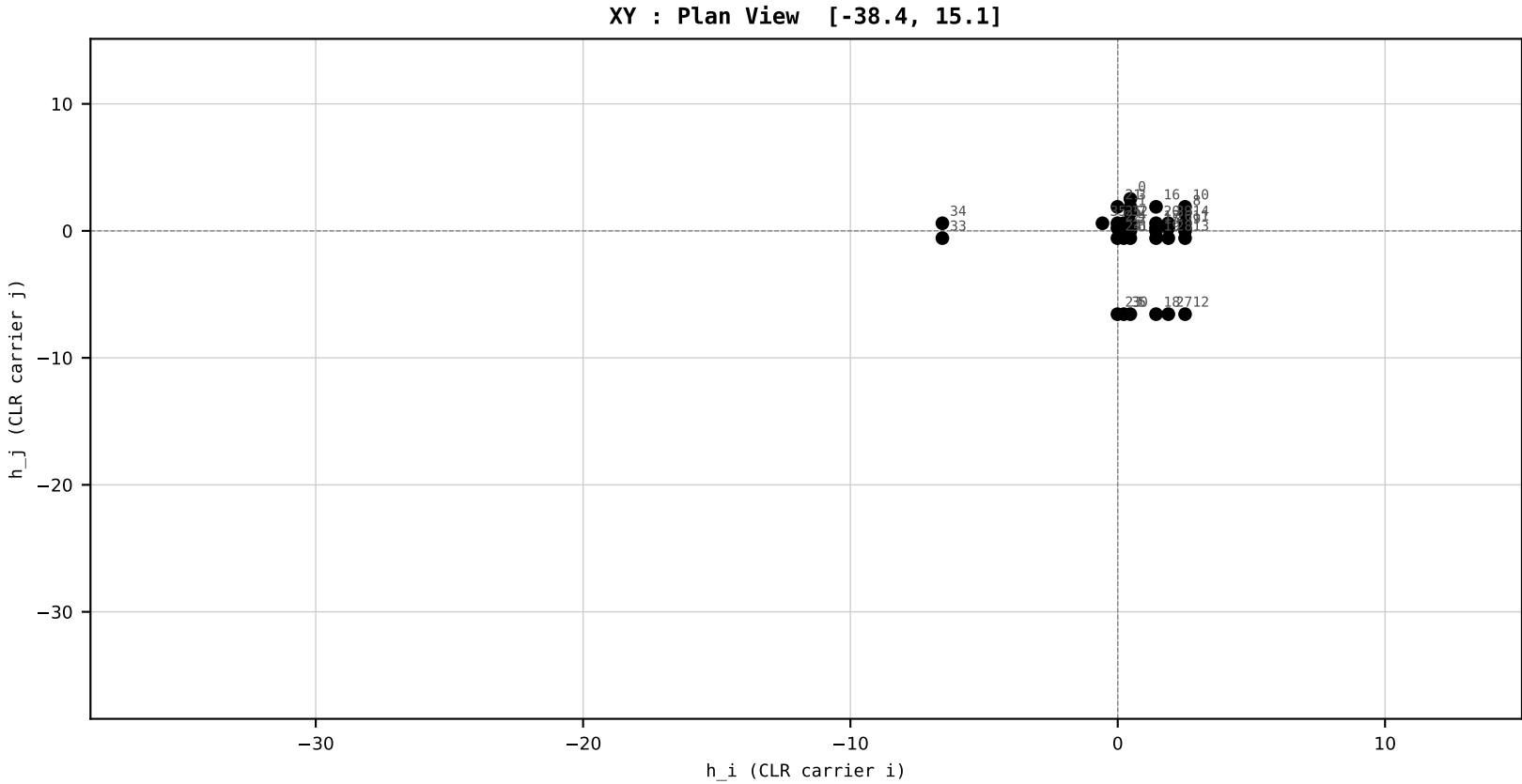
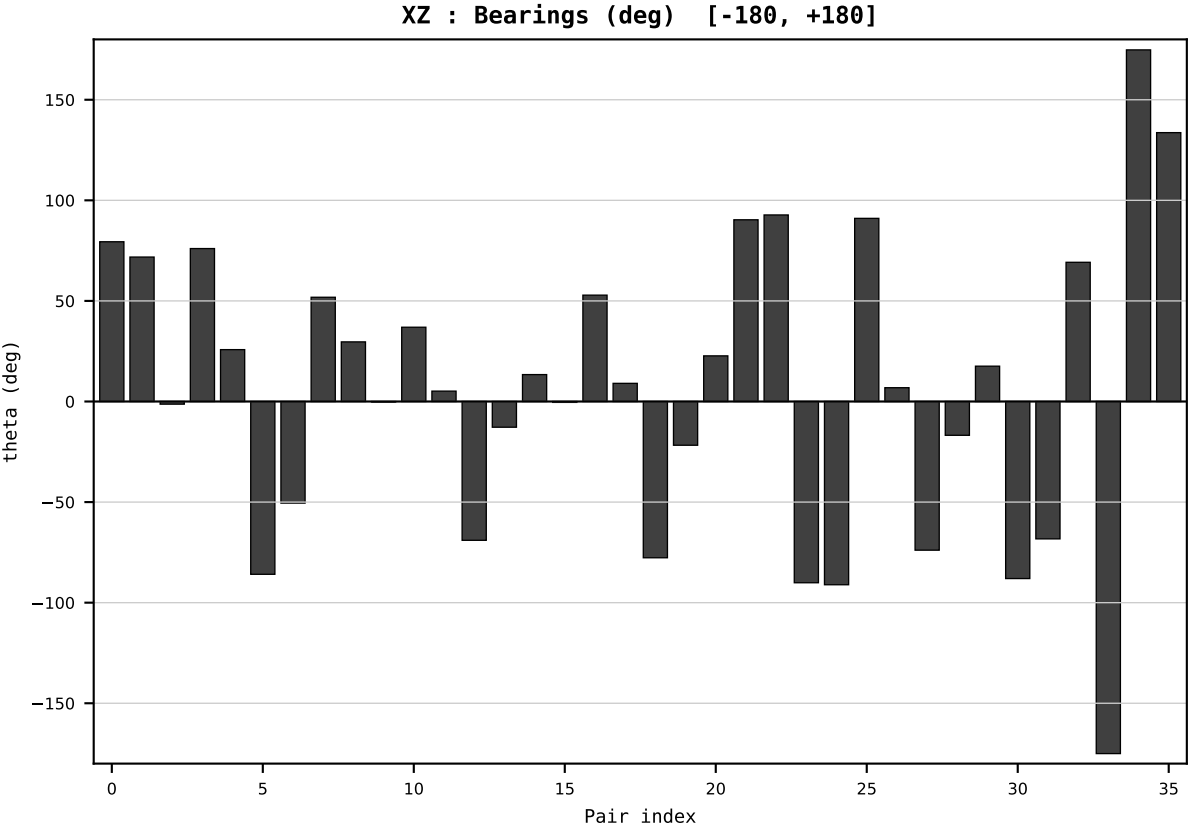
Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2010
t : 10 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.257193
Ring = Hs-2
E_metric= 55.9662
kappa_HS= 8763.00
omega = 3.1589 deg
d_A = 0.555847
Helm = Solar
Helm d = +0.413991
DR = 9.078 HLR
DR ratio= 8763.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=11 | Year 2011 | D=9 N=26 pairs=36

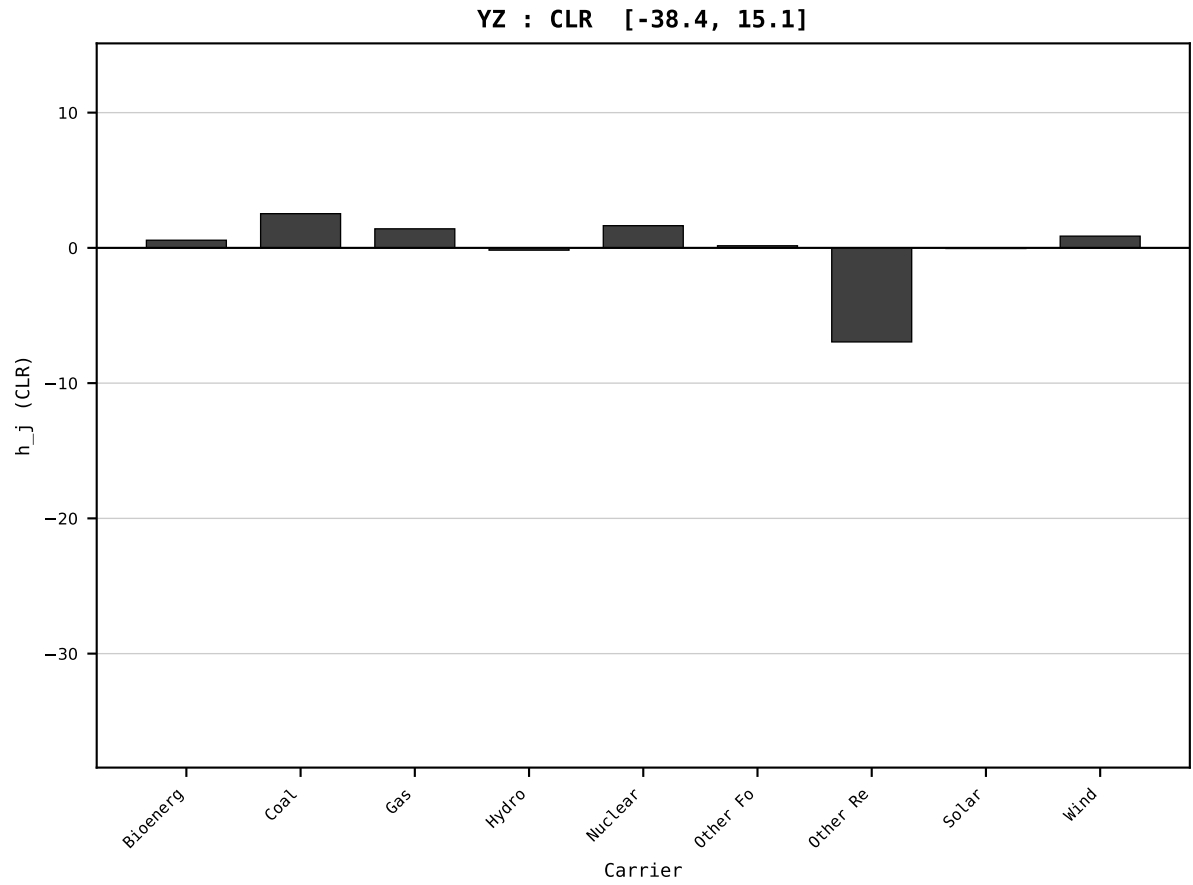
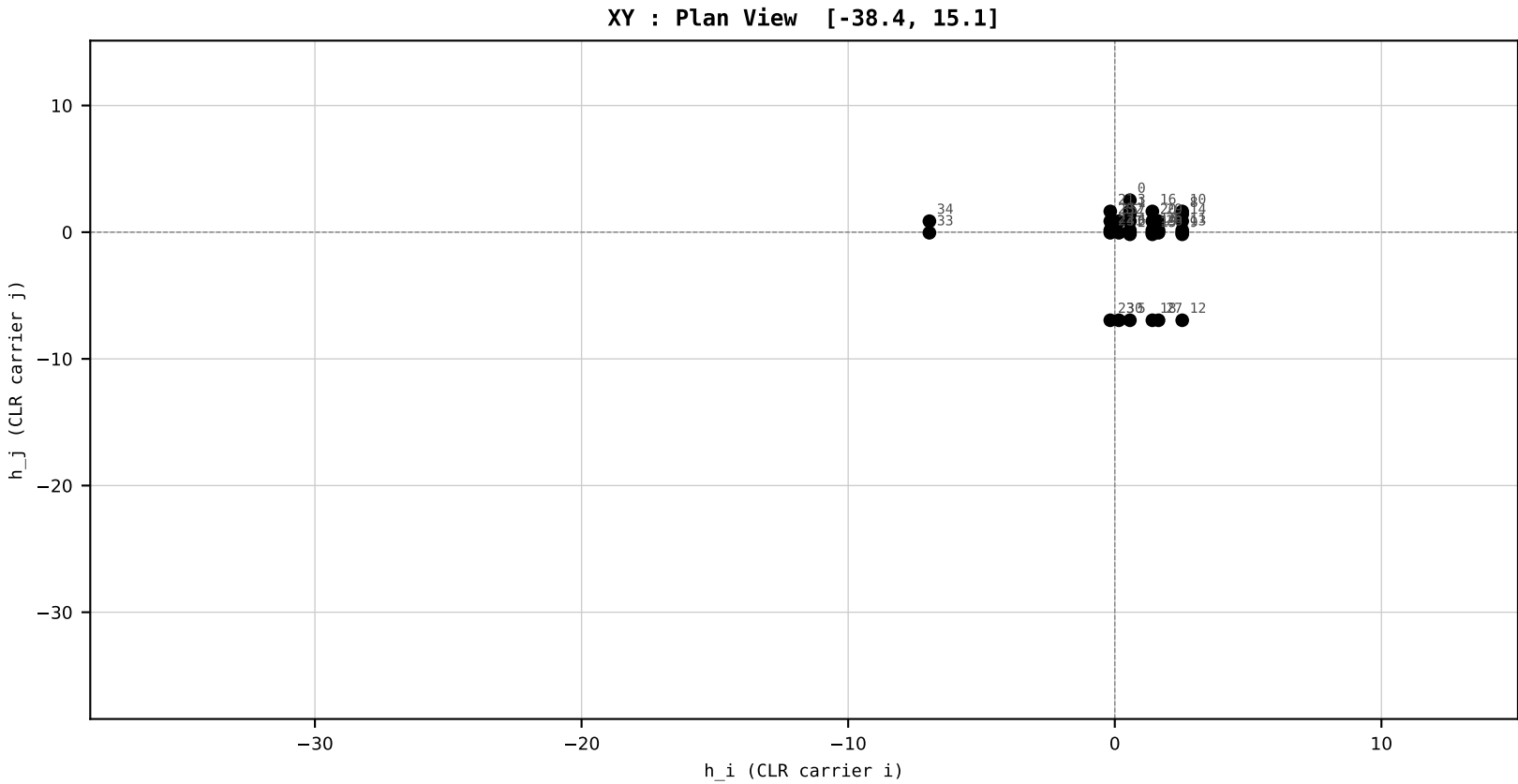
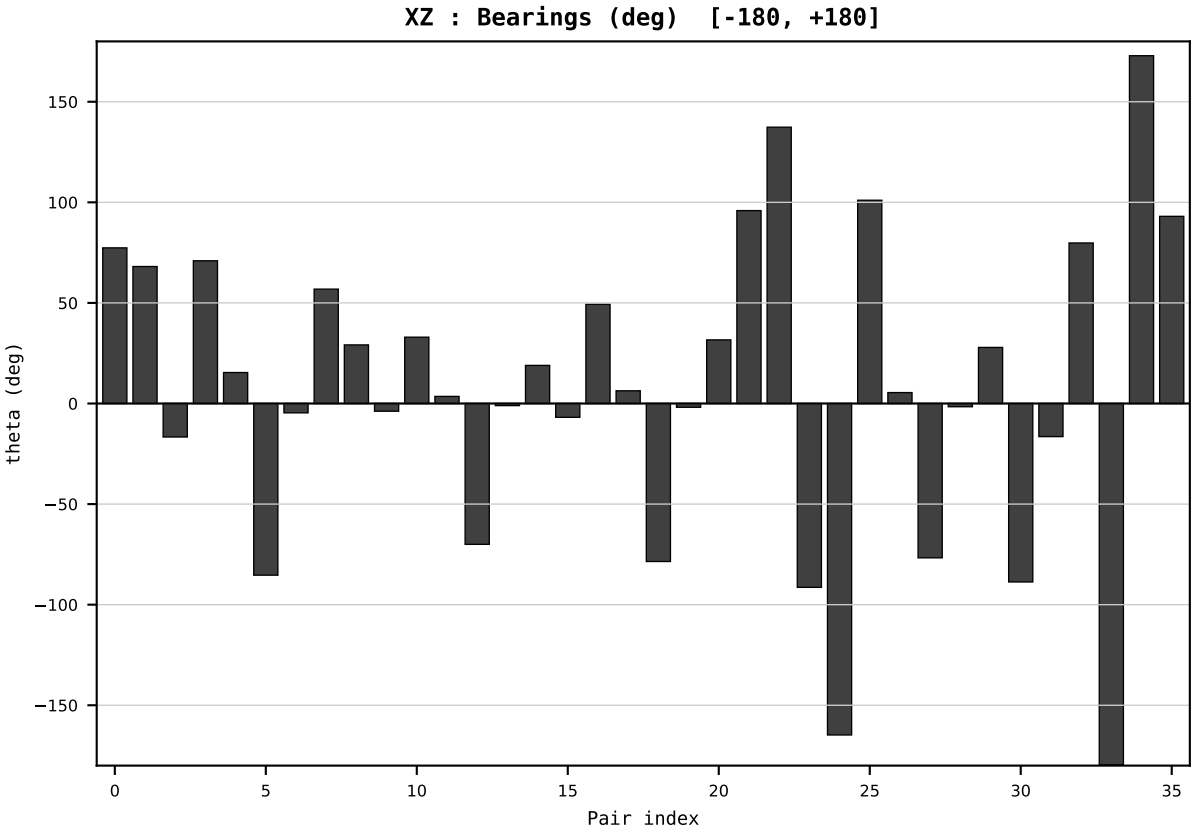
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2011
t : 11 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.240465
Ring = Hs-2
E_metric= 60.5504
kappa_HS= 13123.00
omega = 5.4004 deg
d_A = 0.779100
Helm = Solar
Helm d = +0.525310
DR = 9.482 HLR
DR ratio= 13123.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=12 | Year 2012 | D=9 N=26 pairs=36

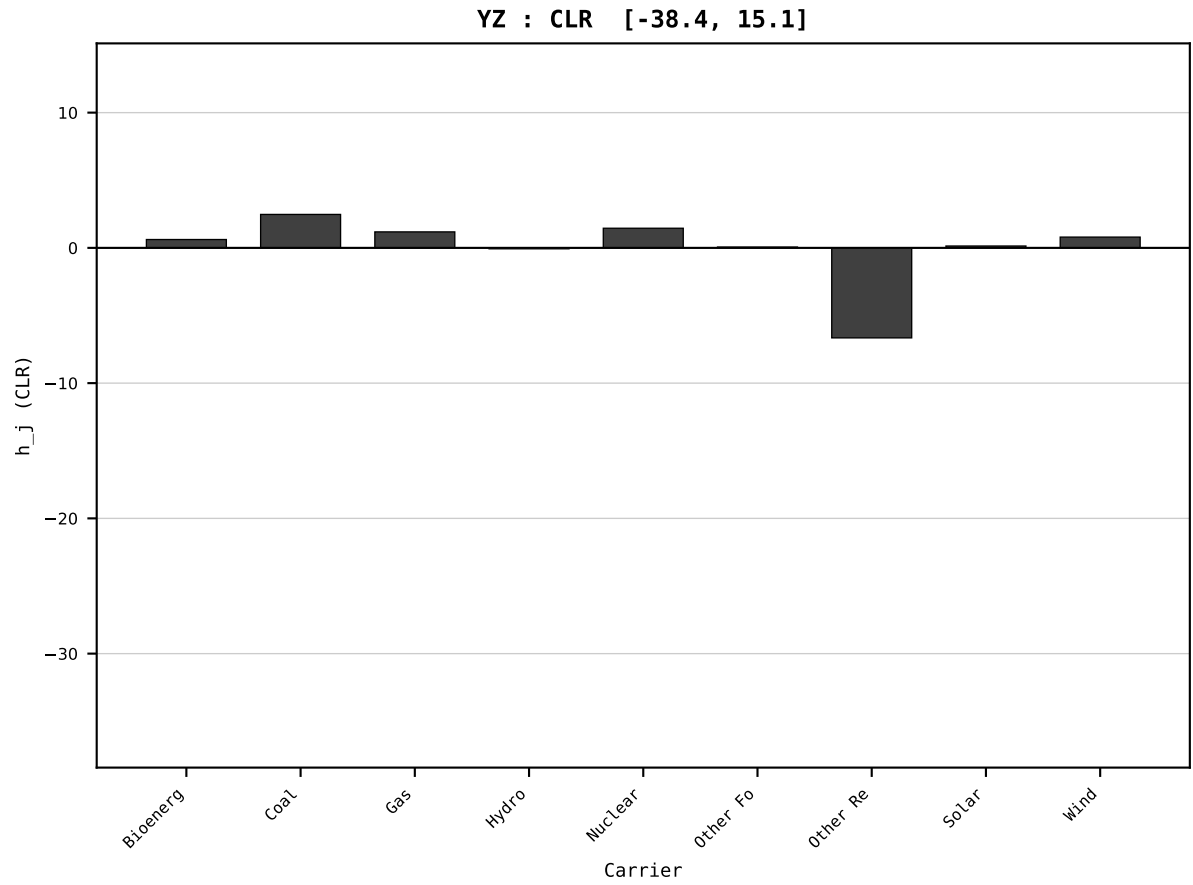
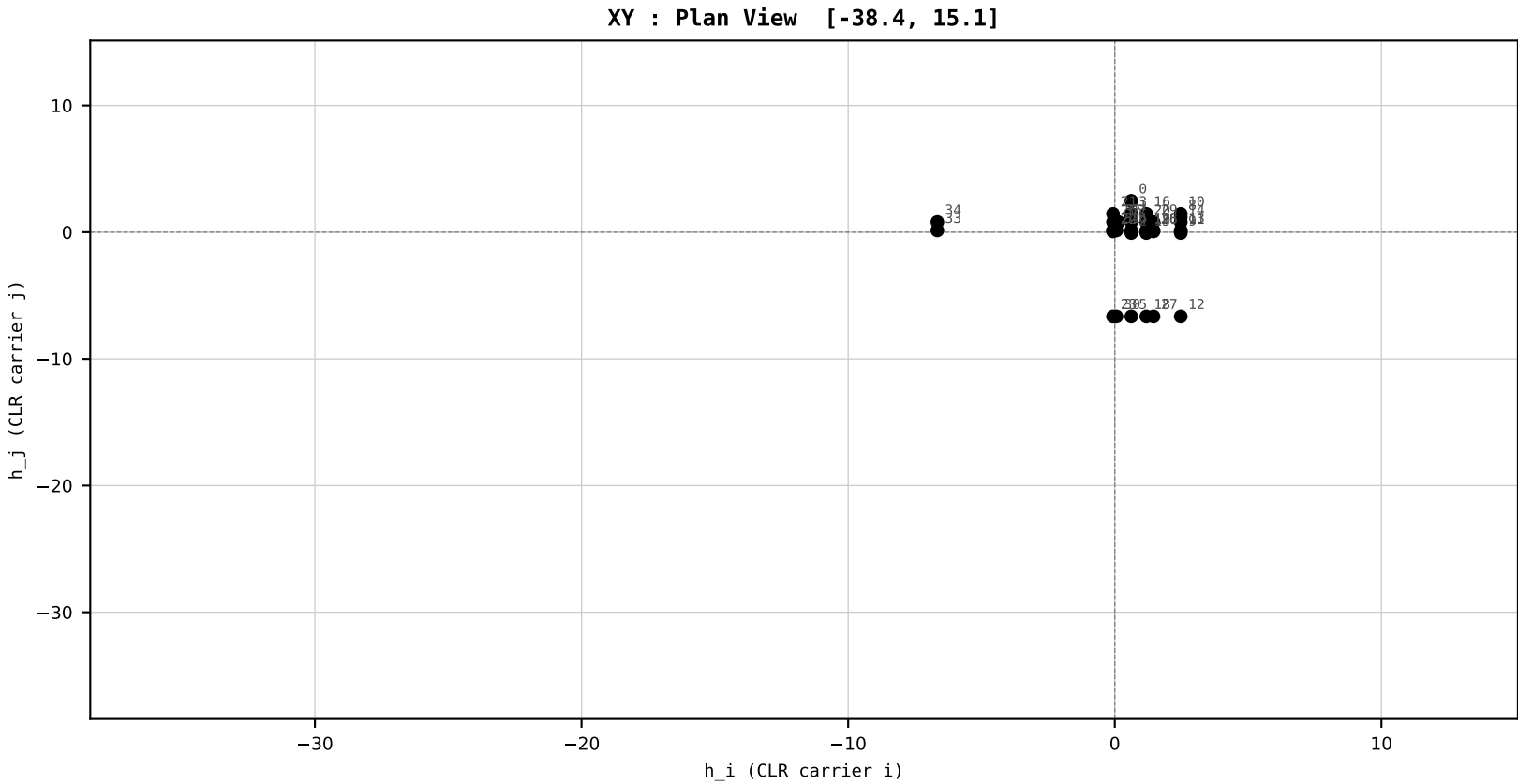
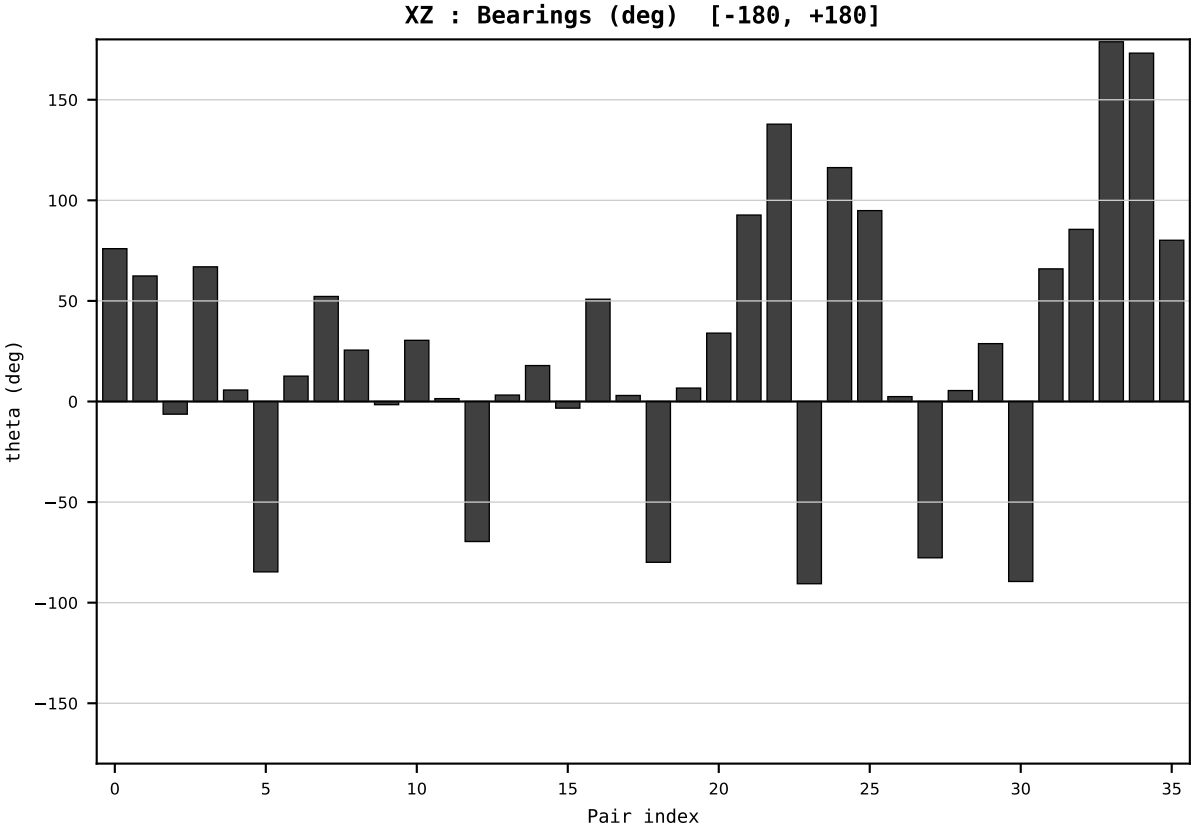
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2012
t : 12 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.232607
Ring = Hs-2
E_metric= 54.9495
kappa_HS= 9207.67
omega = 2.4380 deg
d_A = 0.490208
Helm = Other Renewables
Helm d = +0.298944
DR = 9.128 HLR
DR ratio= 9207.7x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=13 | Year 2013 | D=9 N=26 pairs=36

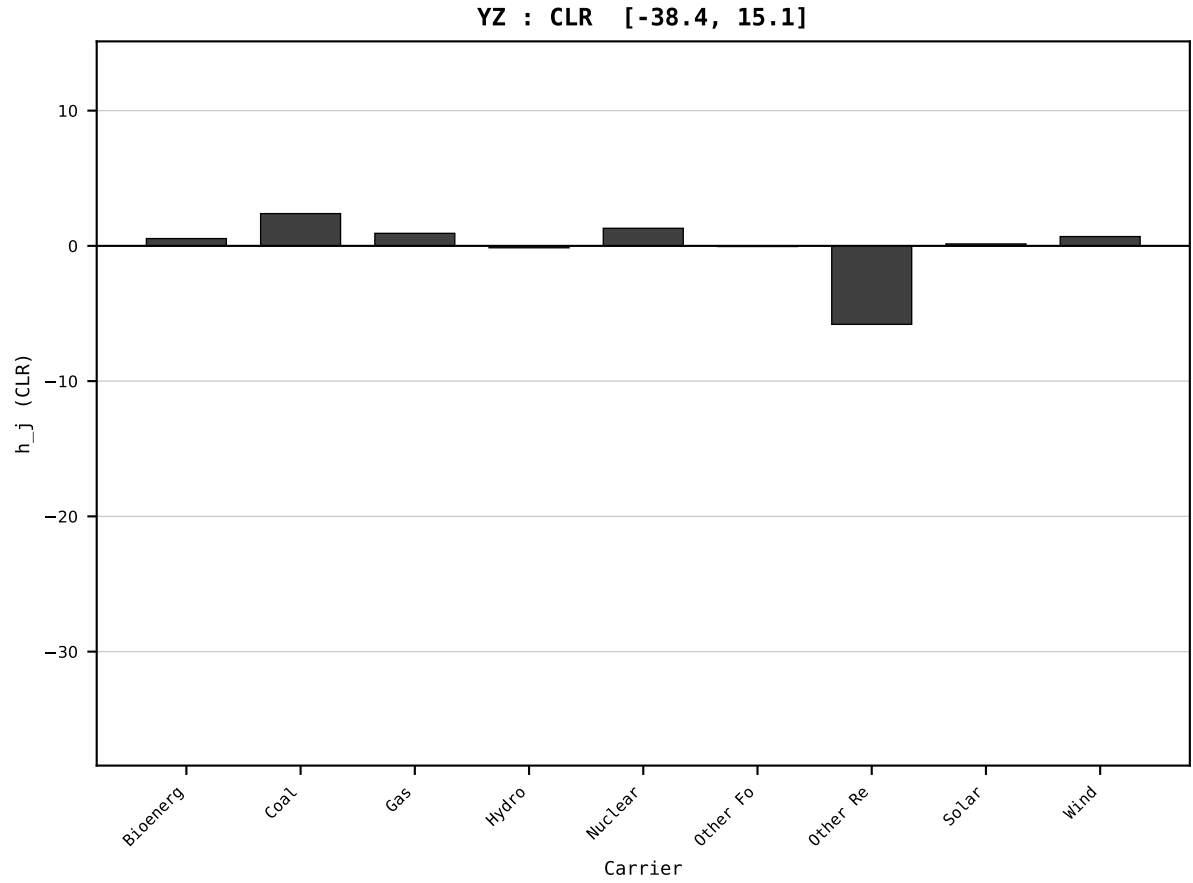
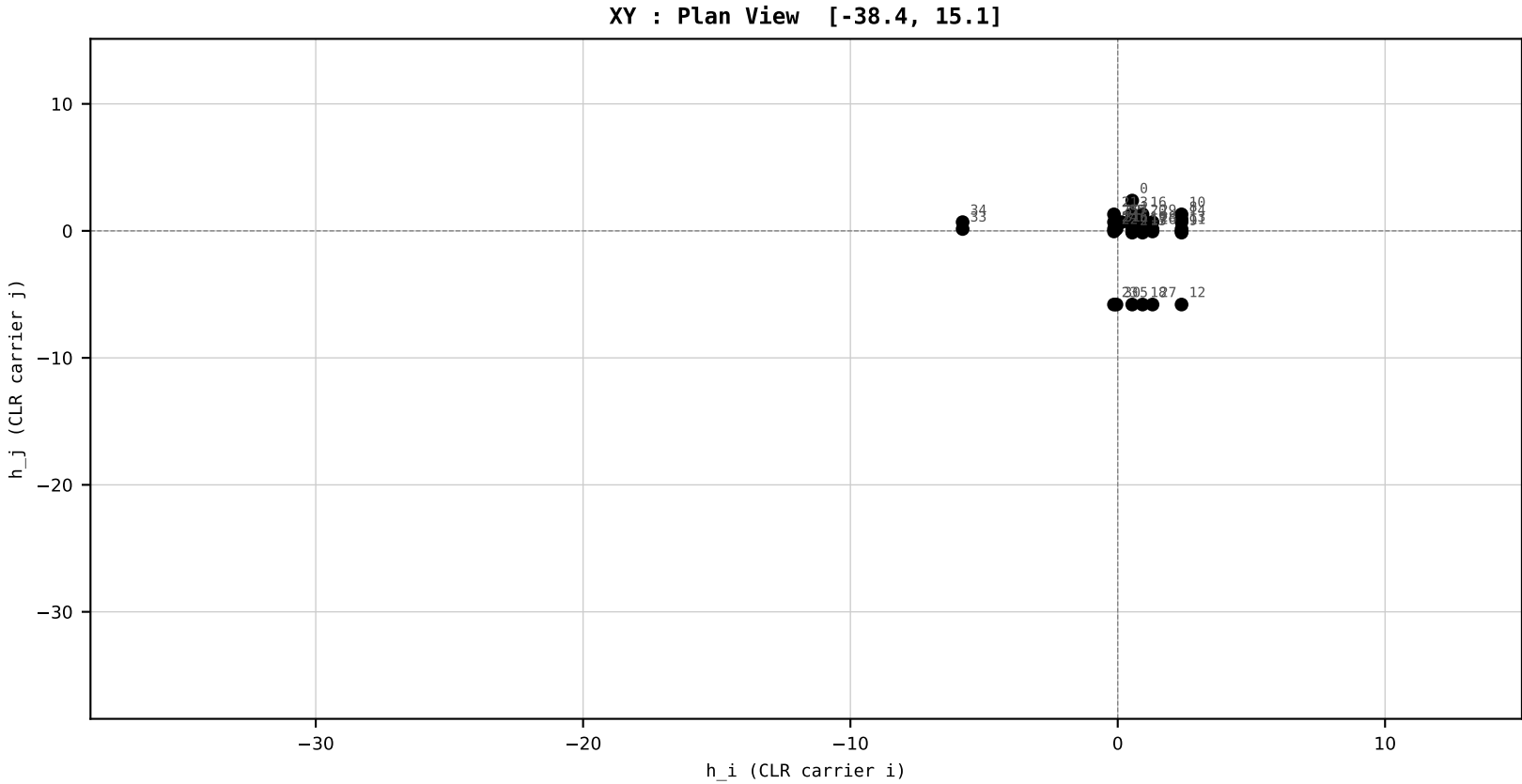
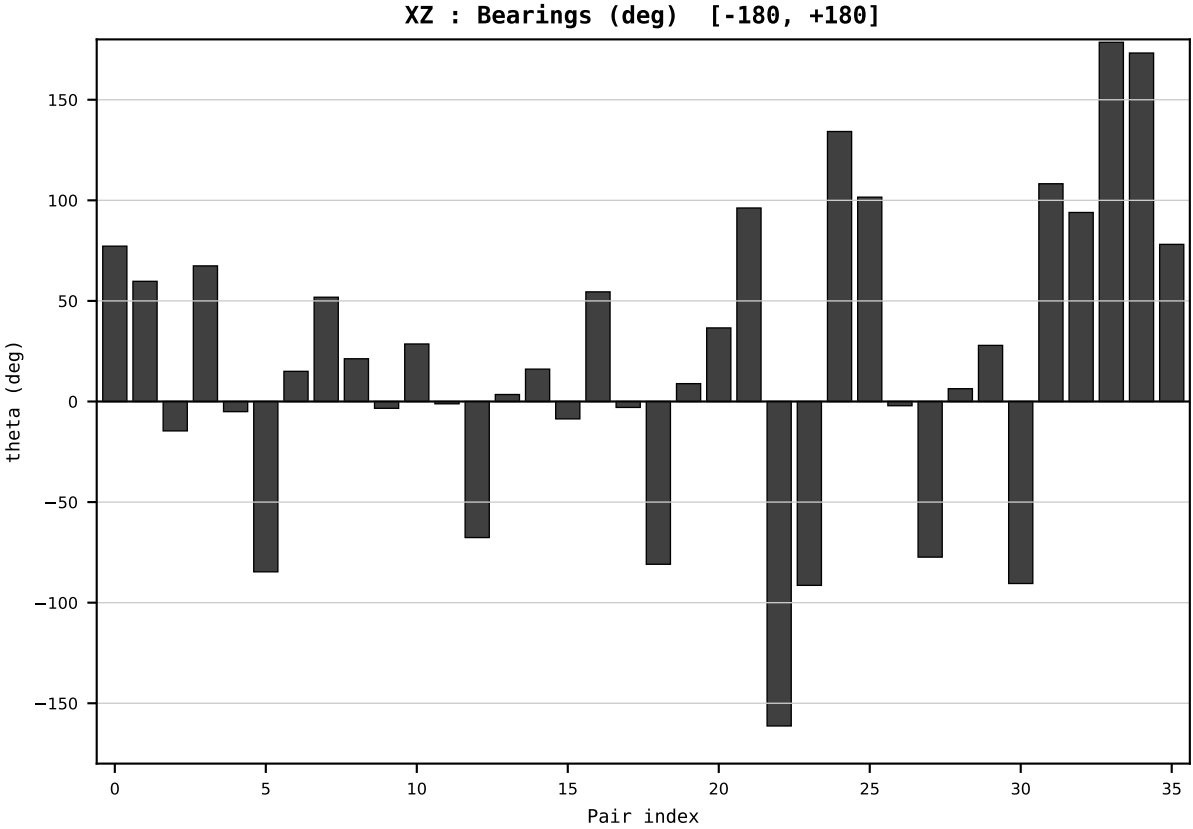
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2013
t : 13 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.235275
Ring = Hs-2
E_metric= 42.7299
kappa_HS= 3602.50
omega = 2.4484 deg
d_A = 0.925102
Helm = Other Renewables
Helm d = +0.852077
DR = 8.189 HLR
DR ratio= 3602.5x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=14 | Year 2014 | D=9 N=26 pairs=36

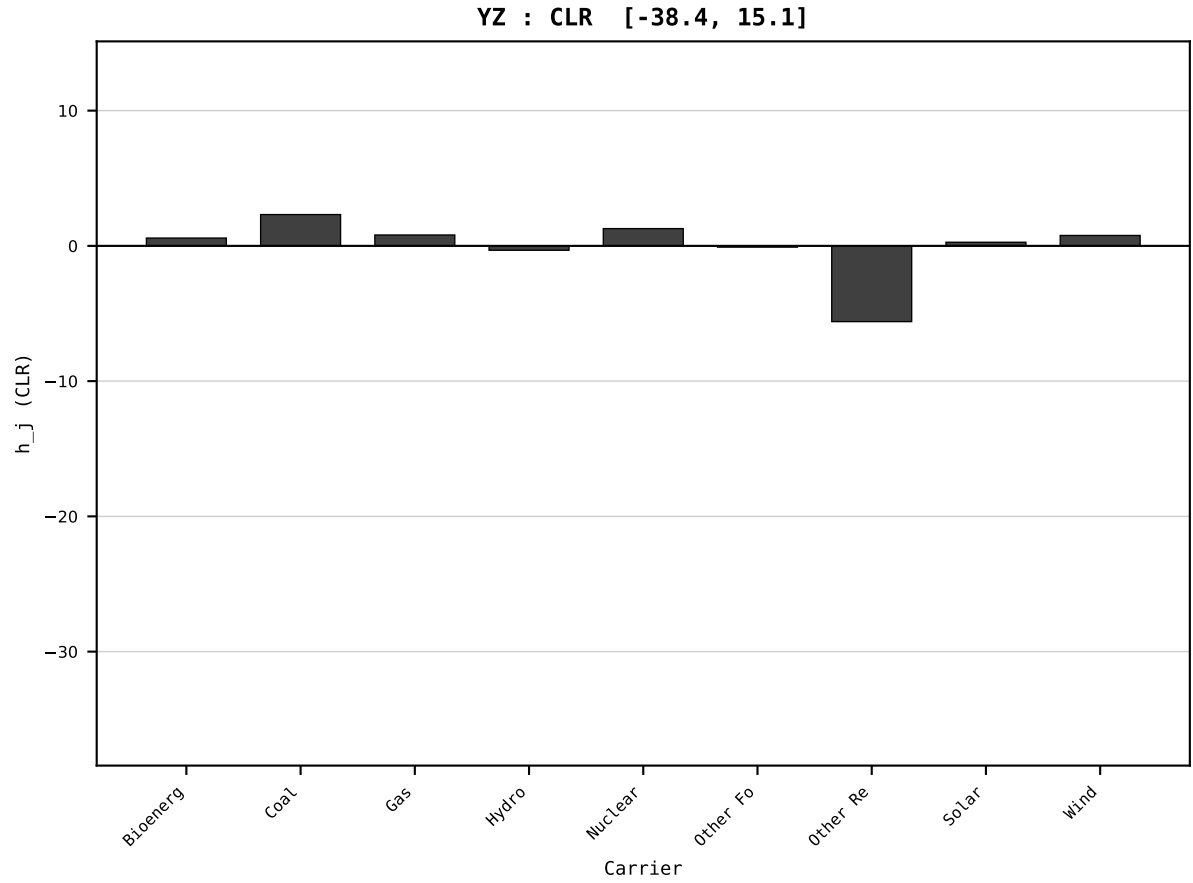
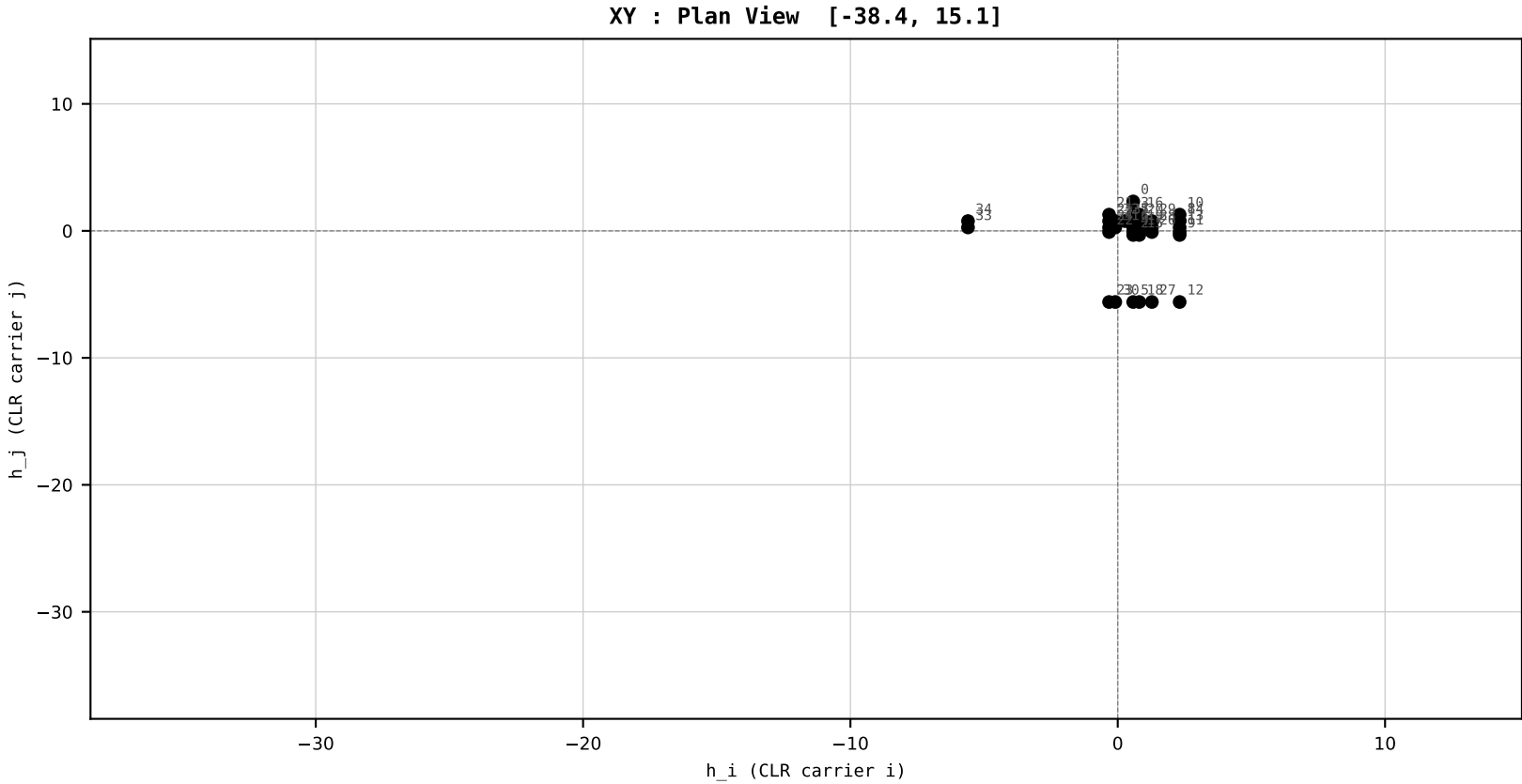
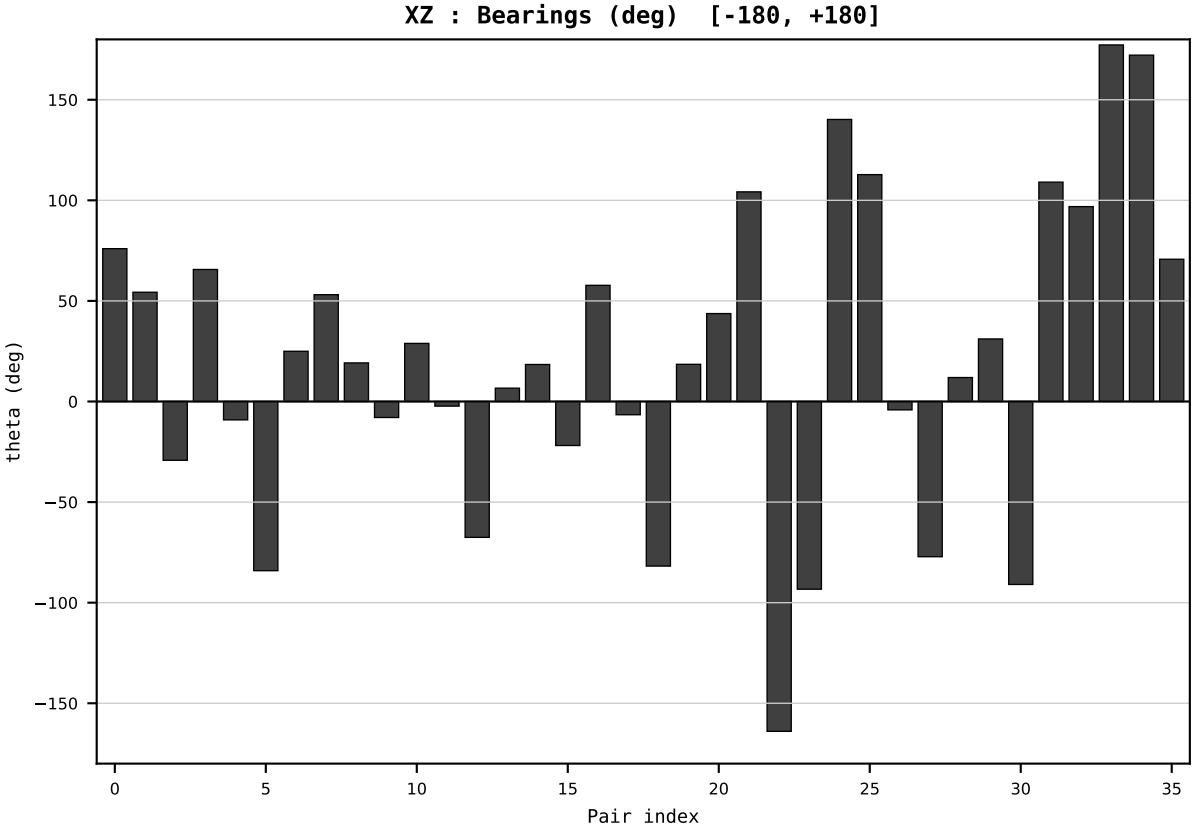
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2014
t : 14 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.224515
Ring = Hs-2
E_metric= 40.1329
kappa_HS= 2744.10
omega = 2.5057 deg
d_A = 0.346263
Helm = Other Renewables
Helm d = +0.201151
DR = 7.917 HLR
DR ratio= 2744.1x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=15 | Year 2015 | D=9 N=26 pairs=36

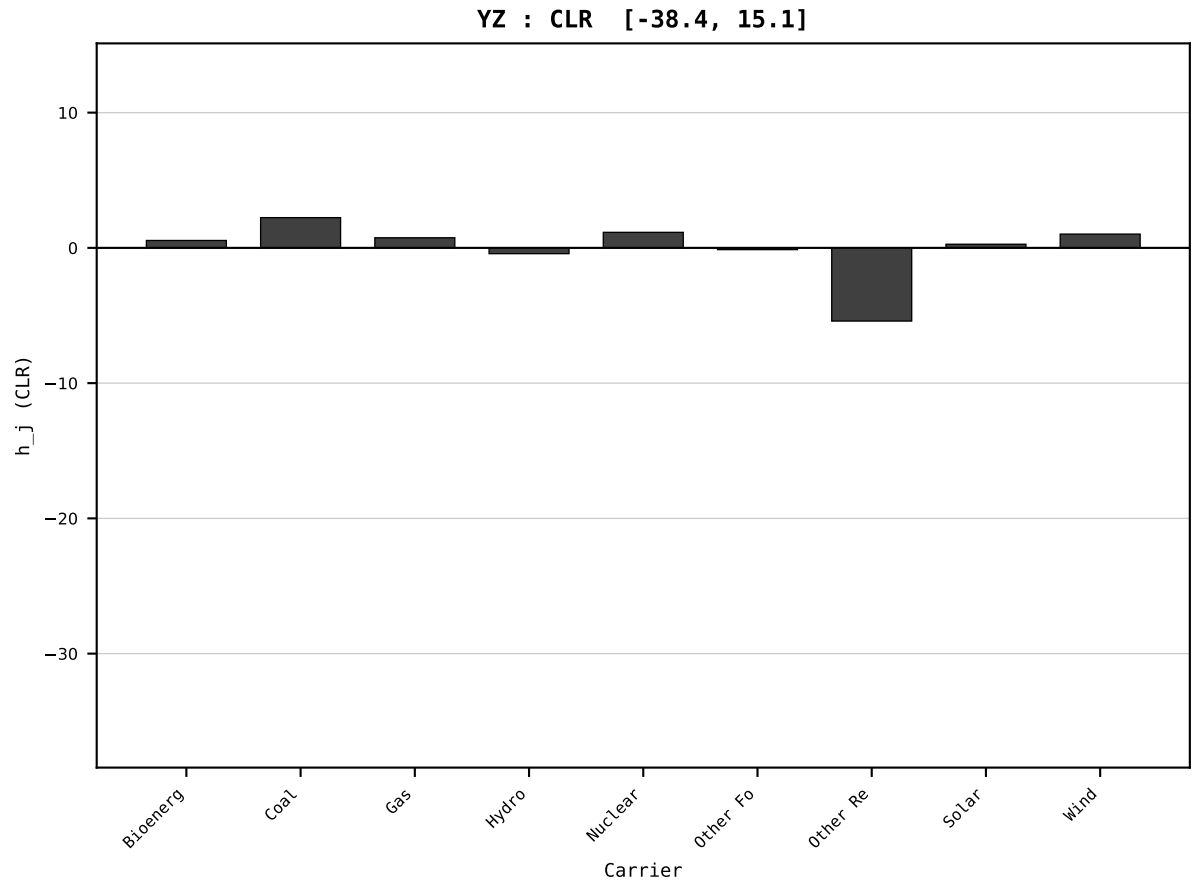
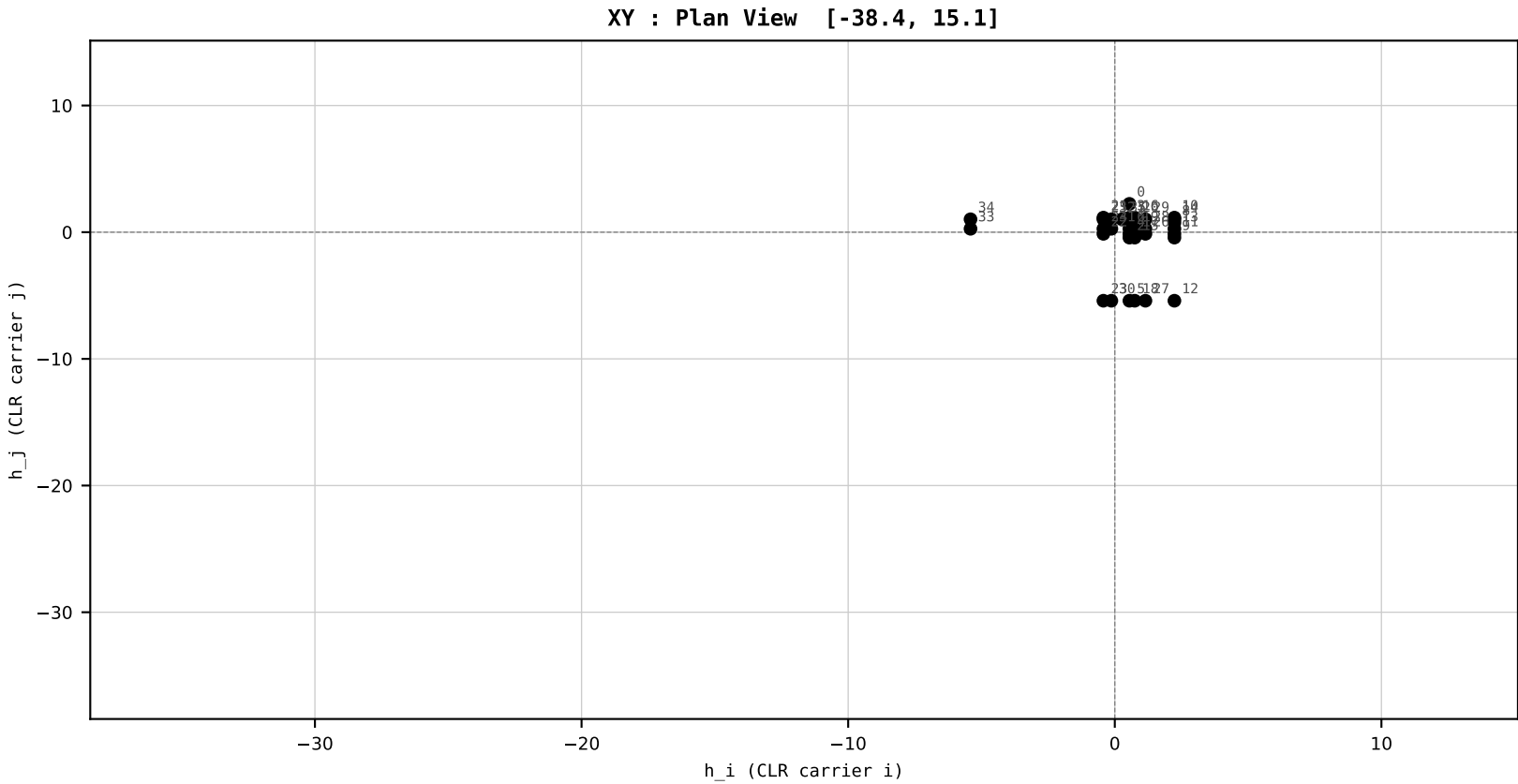
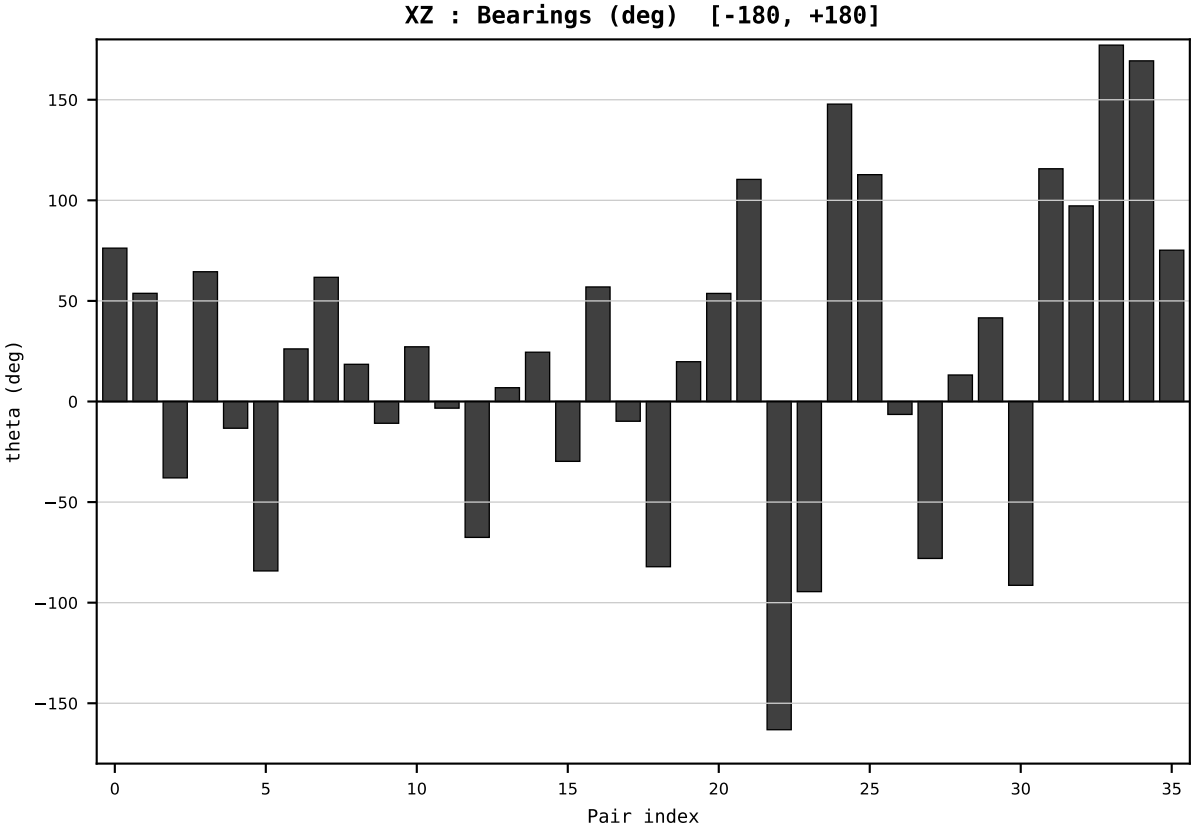
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2015
t : 15 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.211885
Ring = Hs-2
E_metric= 37.7664
kappa_HS= 2093.85
omega = 2.9247 deg
d_A = 0.370645
Helm = Wind
Helm d = +0.248533
DR = 7.647 HLR
DR ratio= 2093.8x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=16 | Year 2016 | D=9 N=26 pairs=36

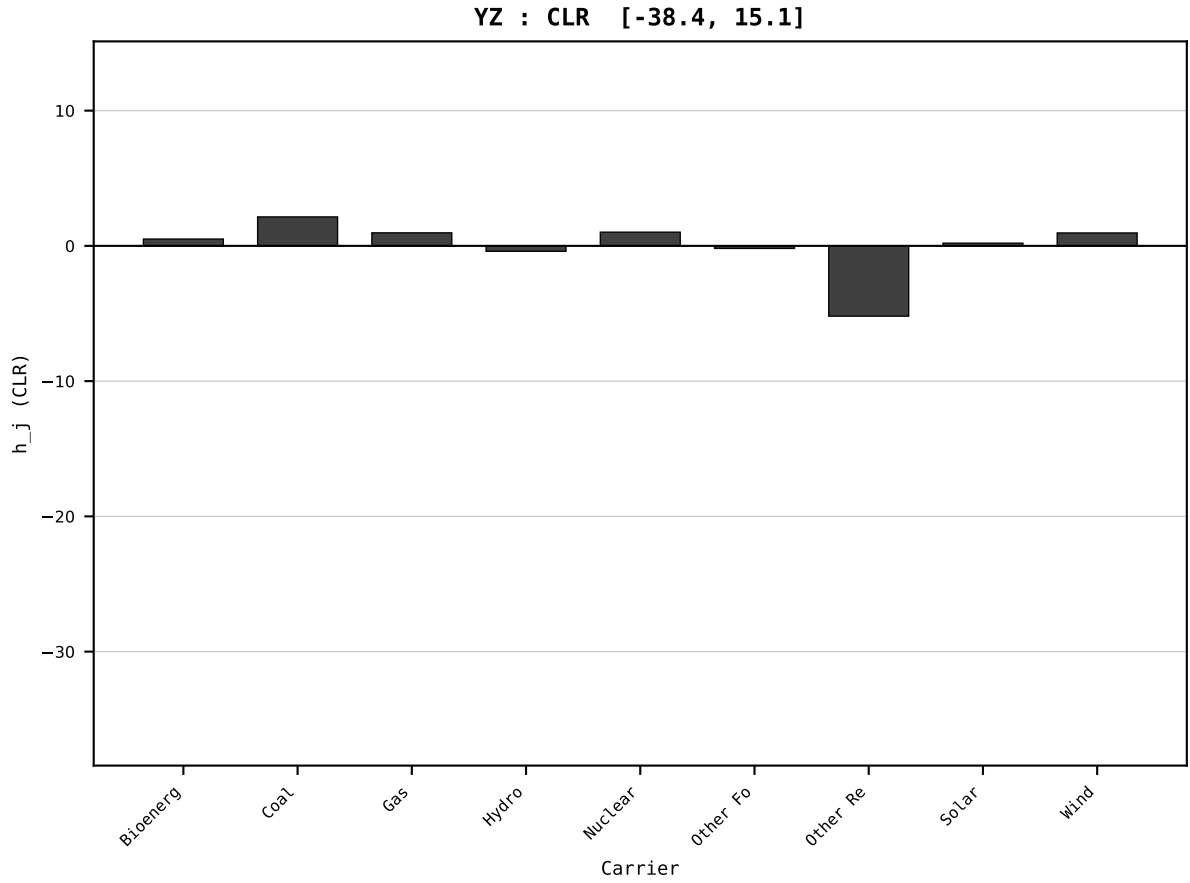
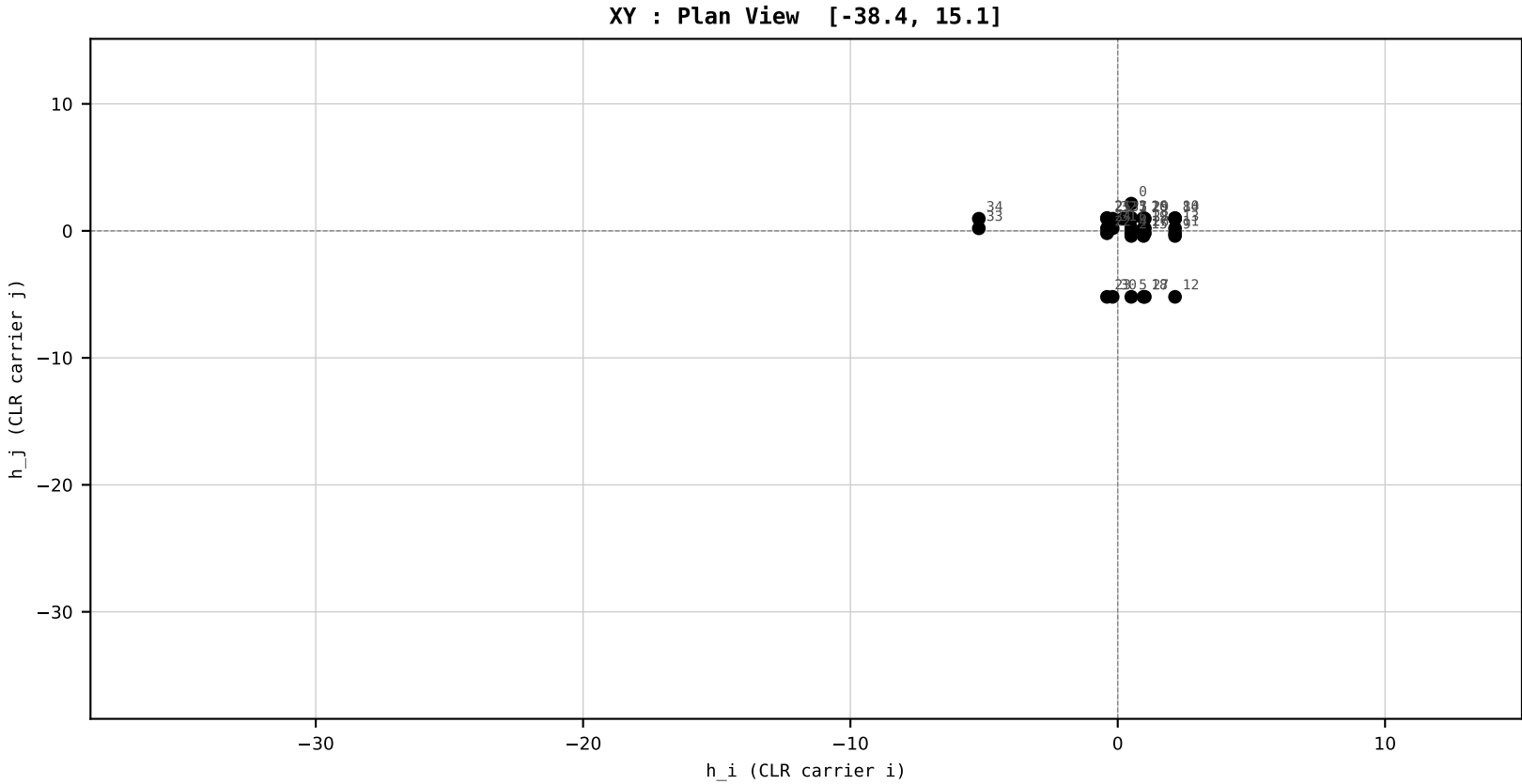
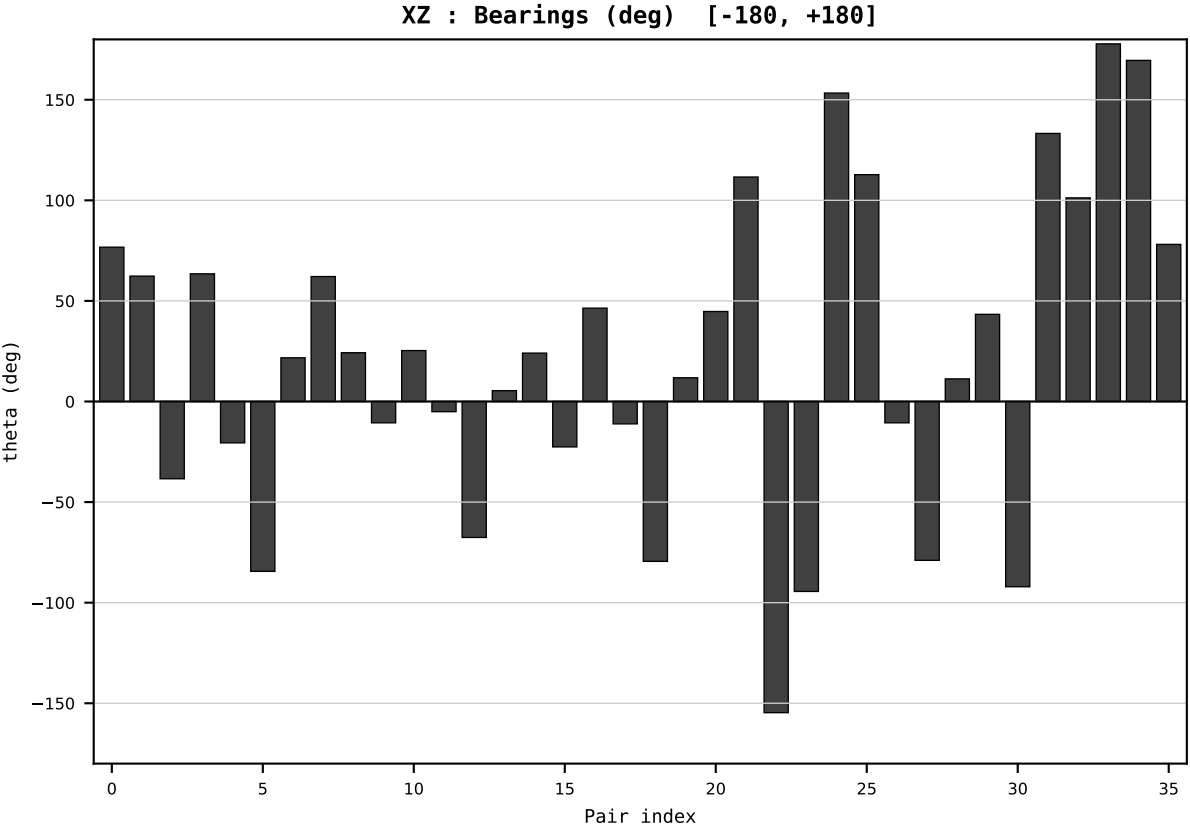
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2016
t : 16 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.198712
Ring = Hs-2
E_metric= 34.9636
kappa_HS= 1539.71
omega = 2.7002 deg
d_A = 0.367040
Helm = Gas
Helm d = +0.217765
DR = 7.339 HLR
DR ratio= 1539.7x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=17 | Year 2017 | D=9 N=26 pairs=36

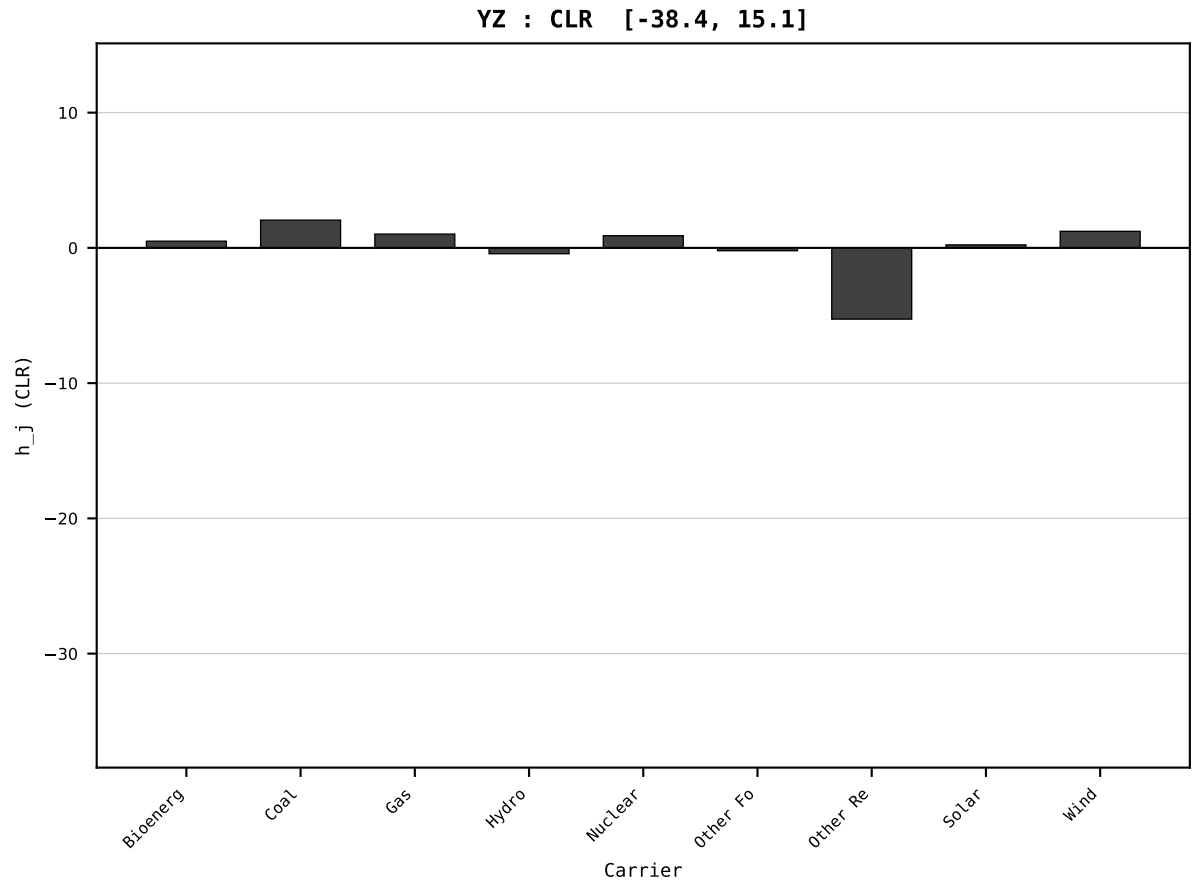
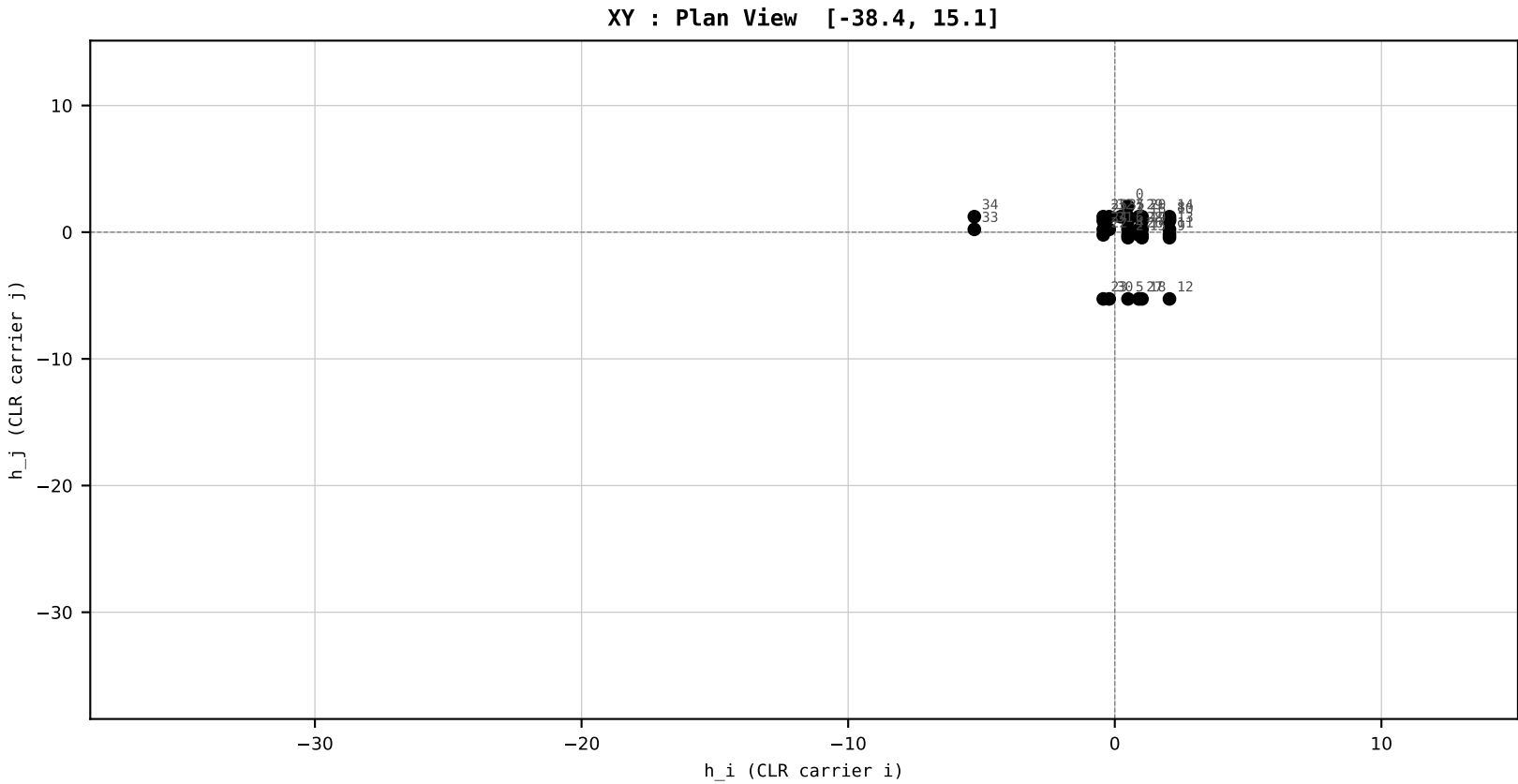
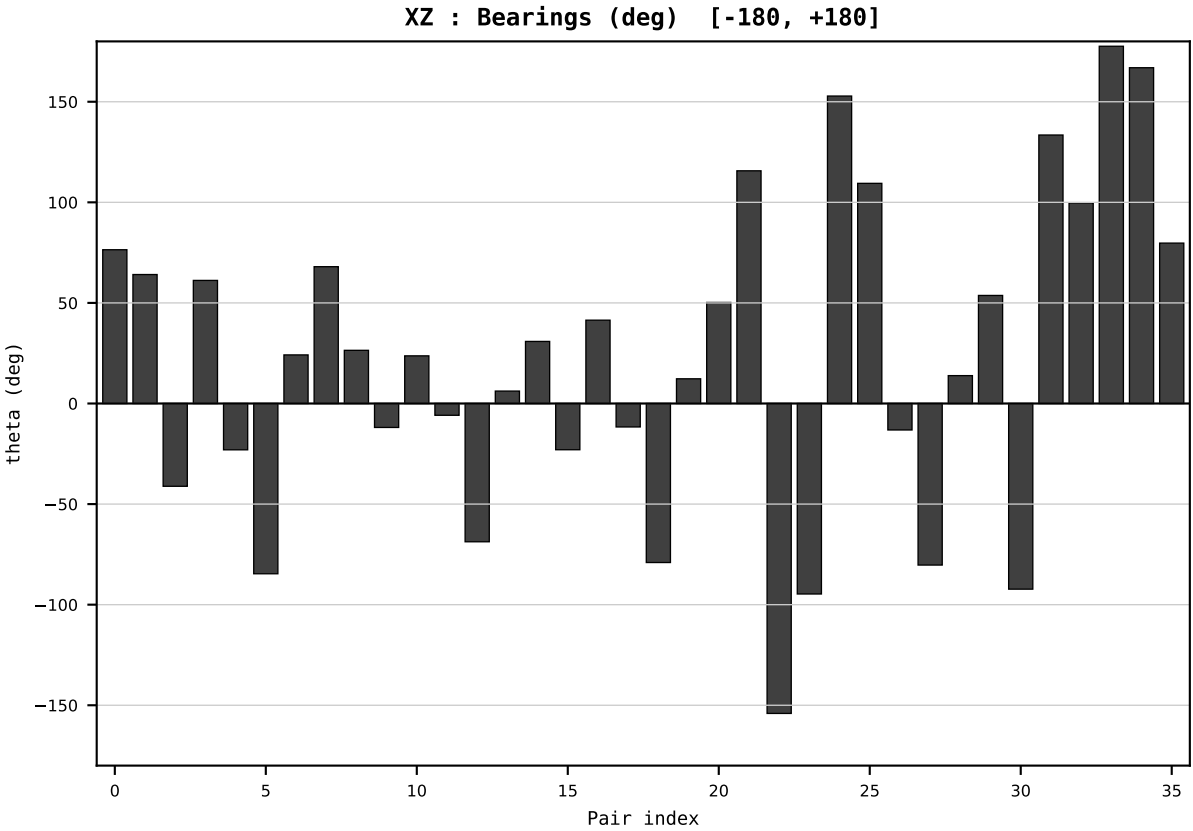
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2017
t : 17 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.184230
Ring = Hs-2
E_metric= 35.8305
kappa_HS= 1507.88
omega = 3.0206 deg
d_A = 0.321962
Helm = Wind
Helm d = +0.268201
DR = 7.318 HLR
DR ratio= 1507.9x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=18 | Year 2018 | D=9 N=26 pairs=36

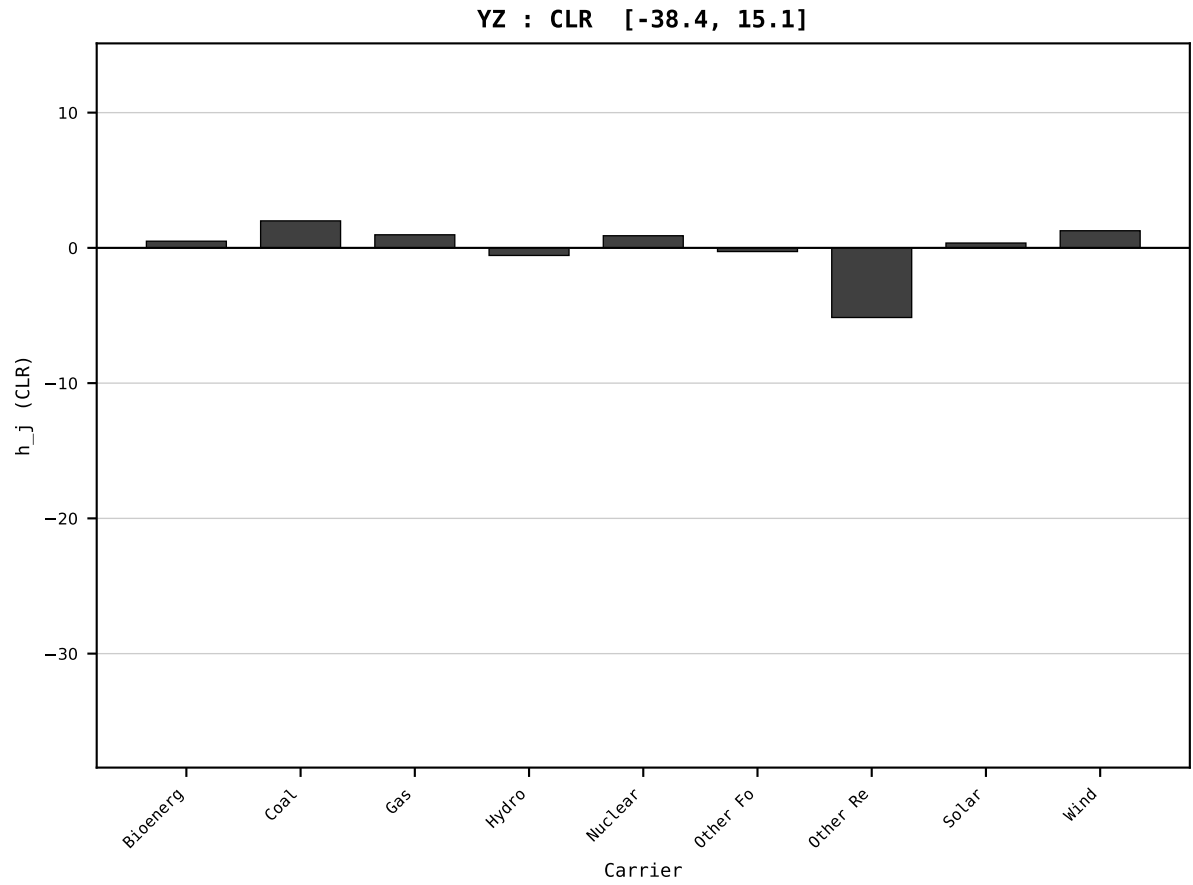
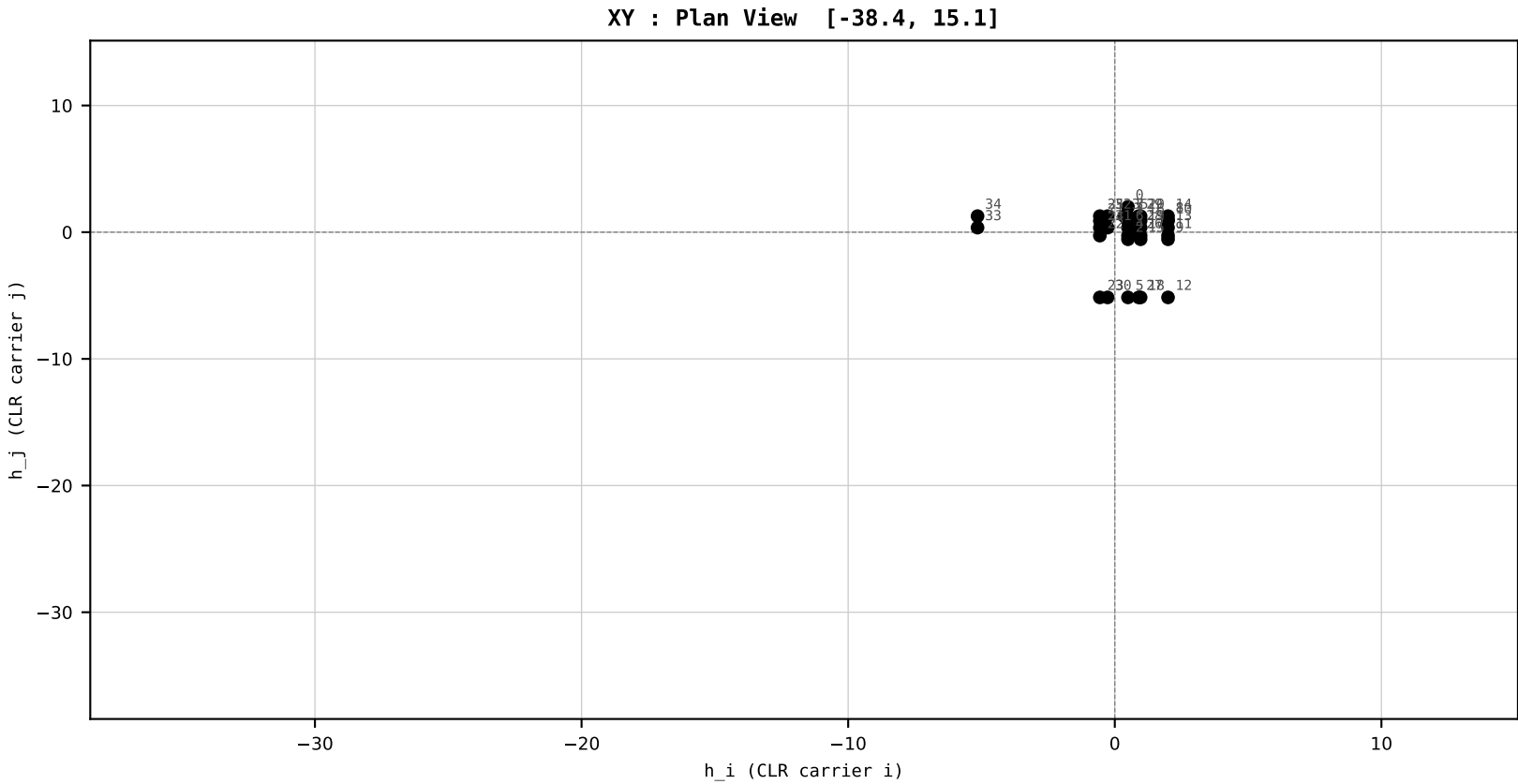
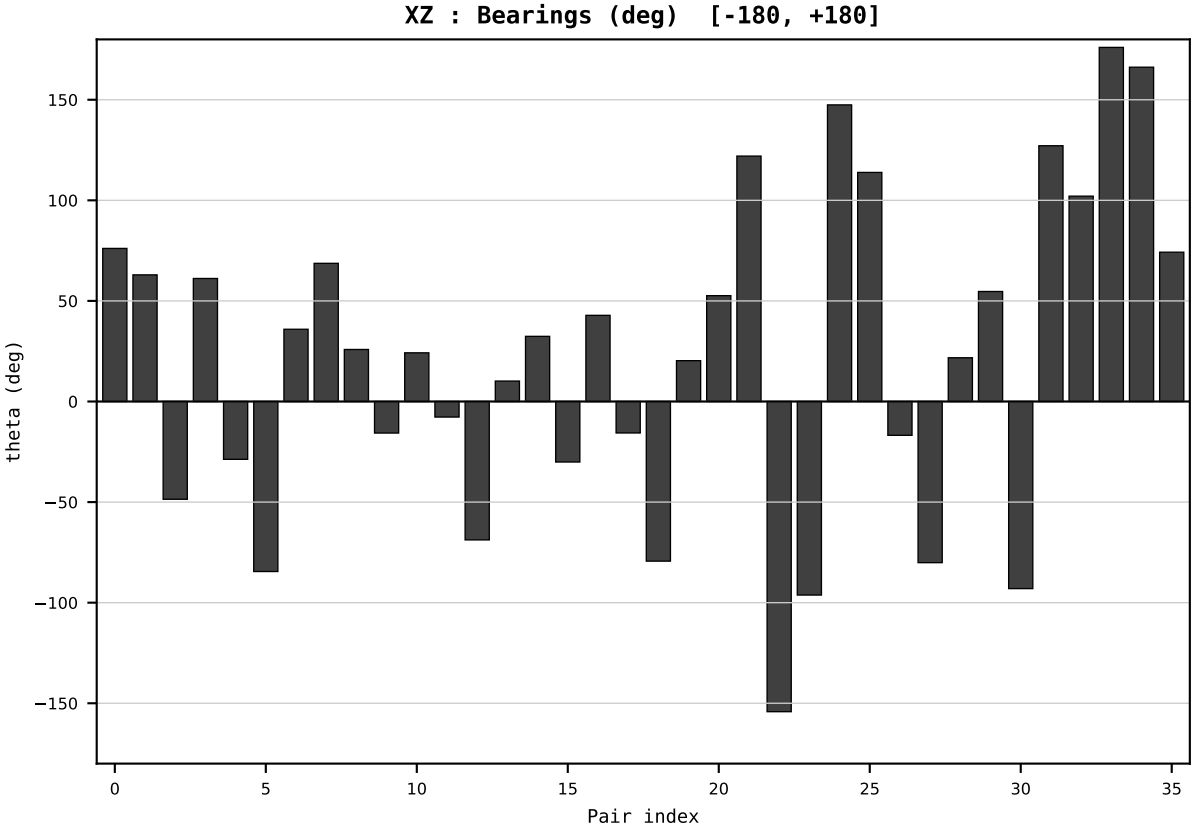
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2018
t : 18 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.178543
Ring = Hs-2
E_metric= 34.5969
kappa_HS= 1267.56
omega = 2.1475 deg
d_A = 0.245475
Helm = Solar
Helm d = +0.136009
DR = 7.145 HLR
DR ratio= 1267.6x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



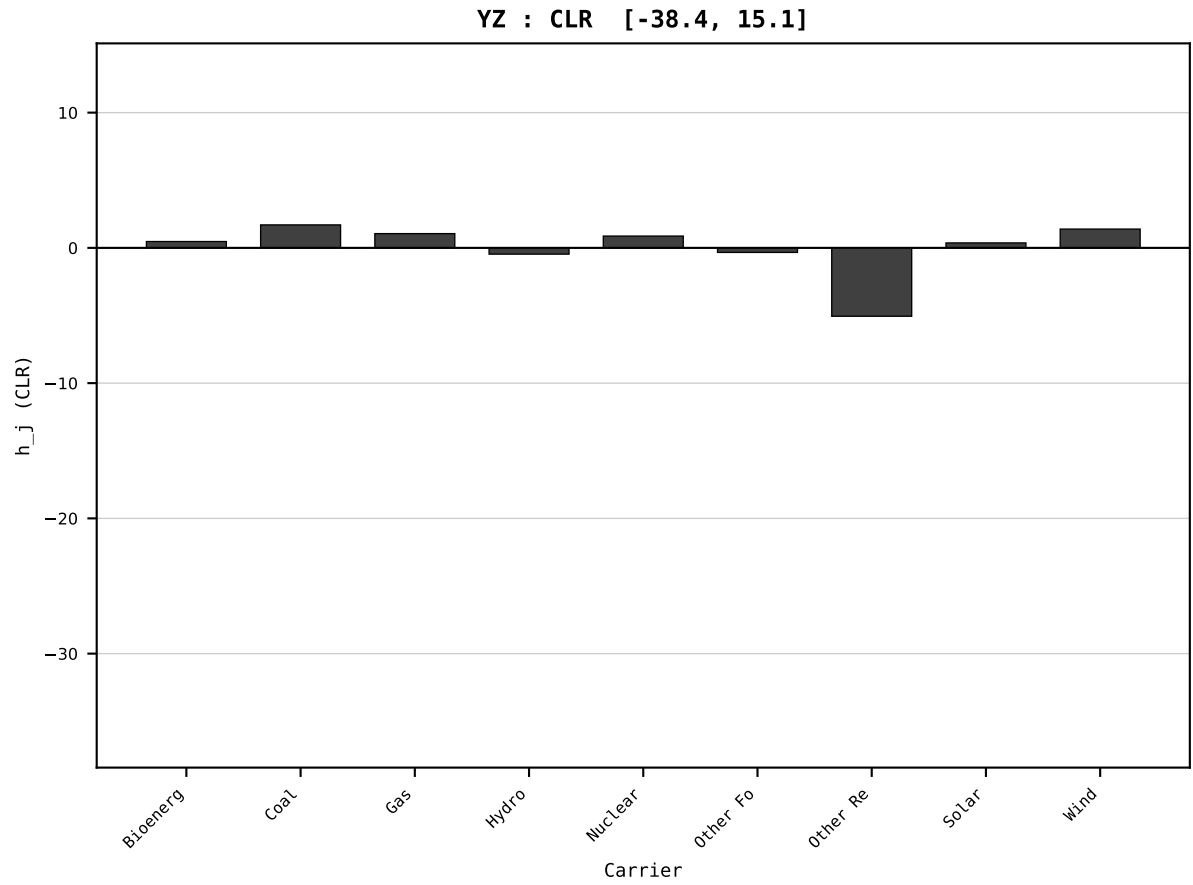
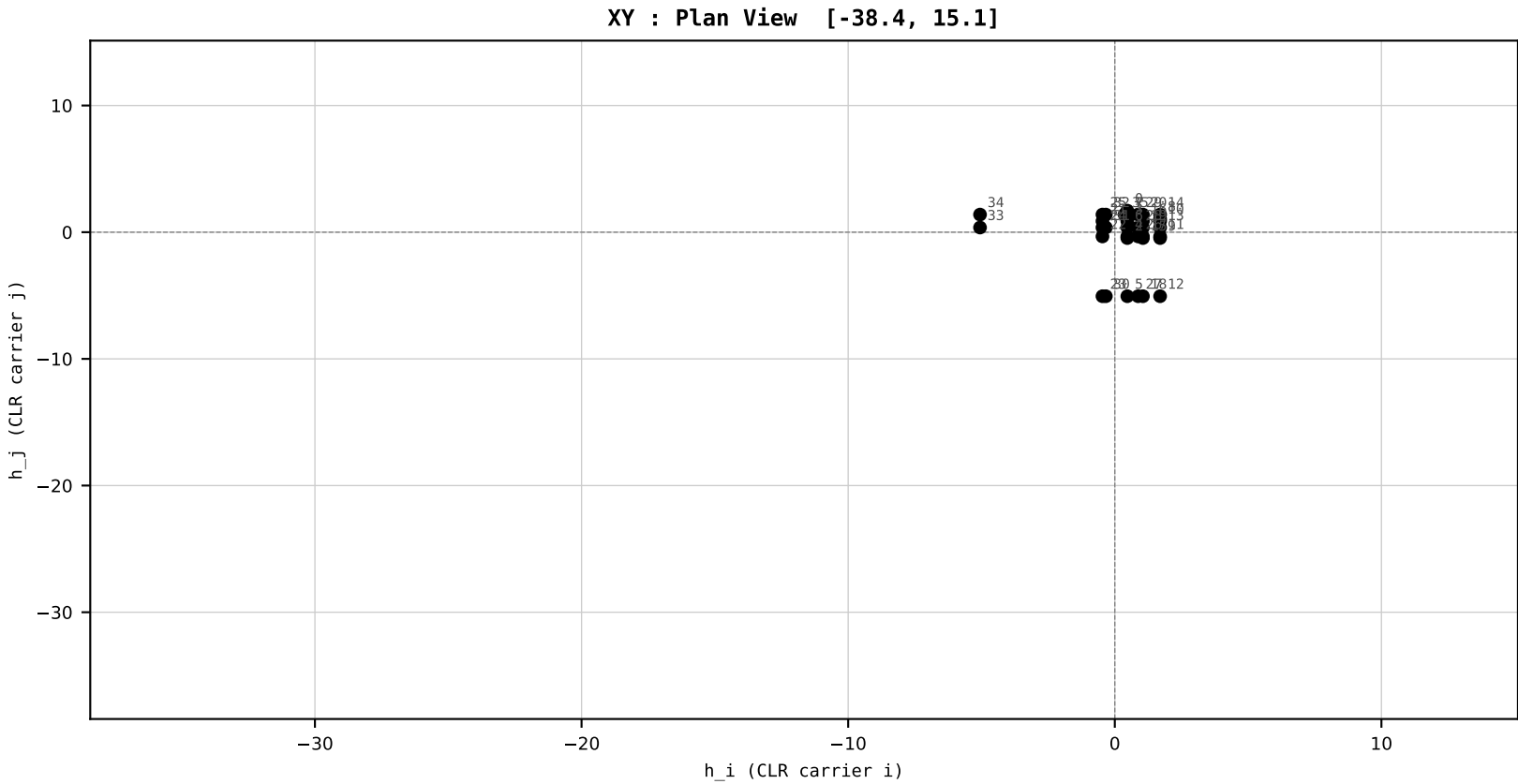
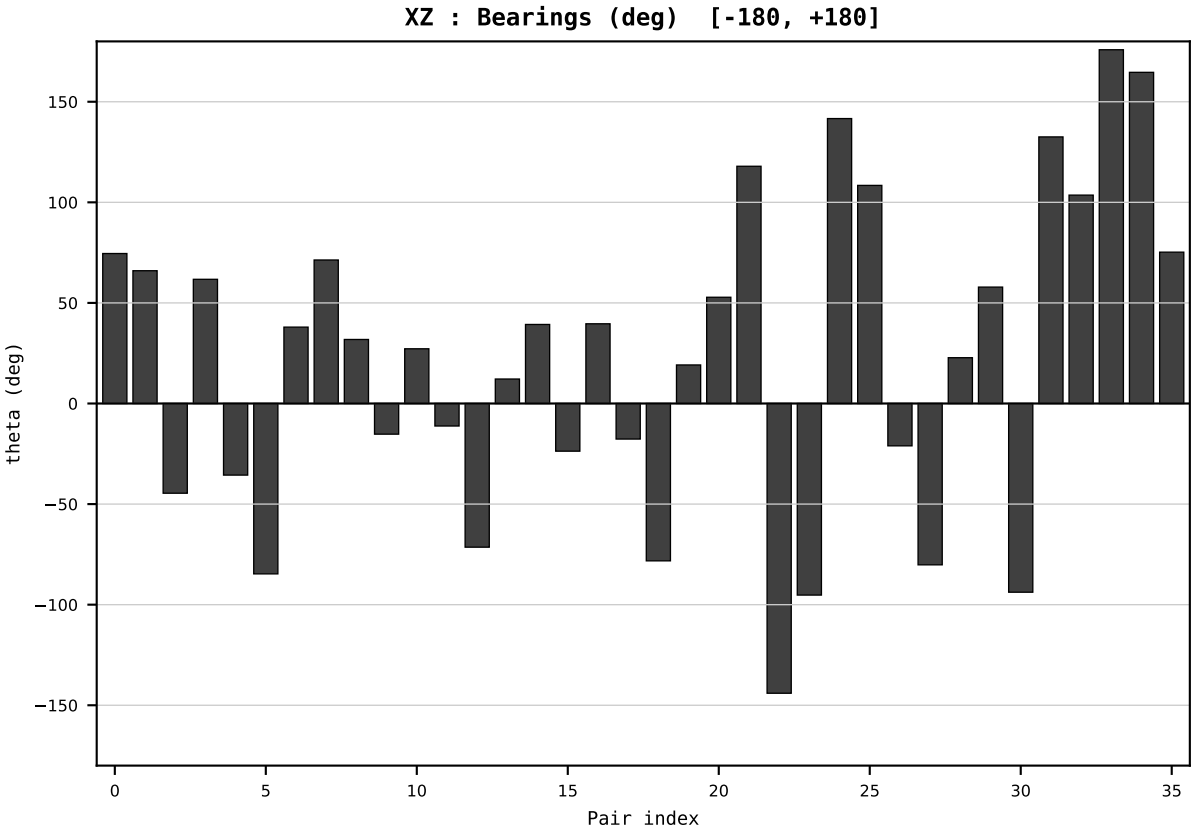
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2019
t : 19 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.148916
Ring = Hs-1
E_metric= 32.9256
kappa_HS= 857.25
omega = 3.3361 deg
d_A = 0.367532
Helm = Coal
Helm d = -0.297295
DR = 6.754 HLR
DR ratio= 857.2x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=20 | Year 2020 | D=9 N=26 pairs=36

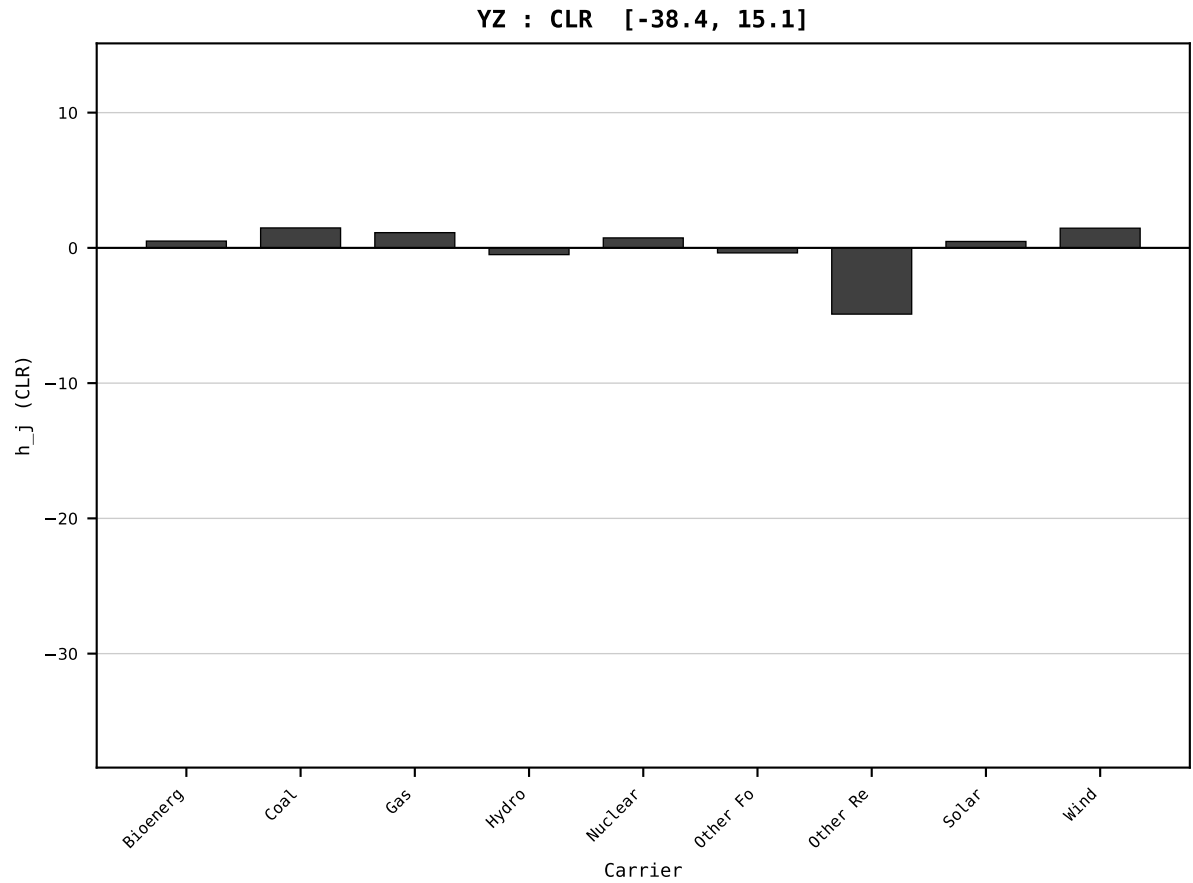
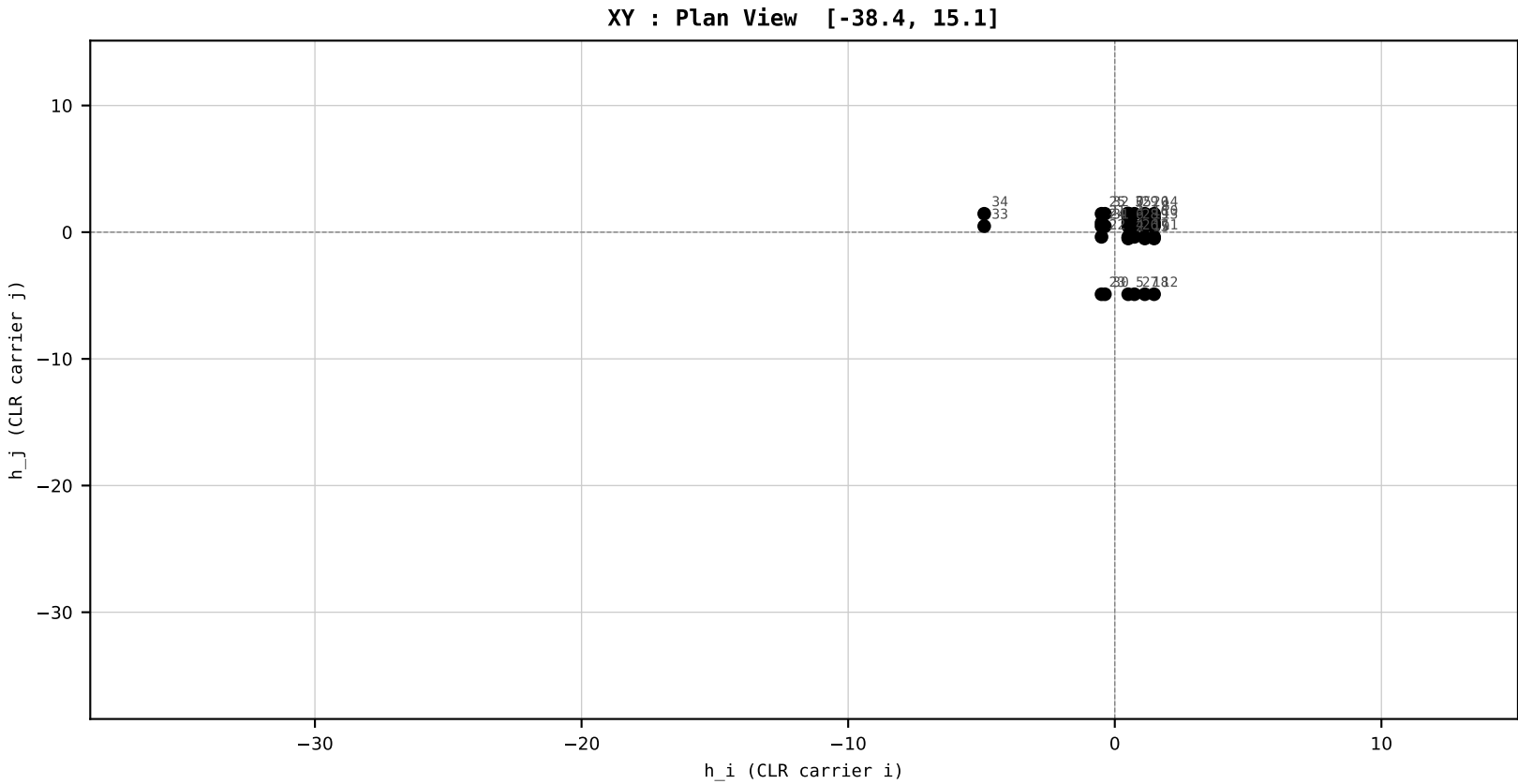
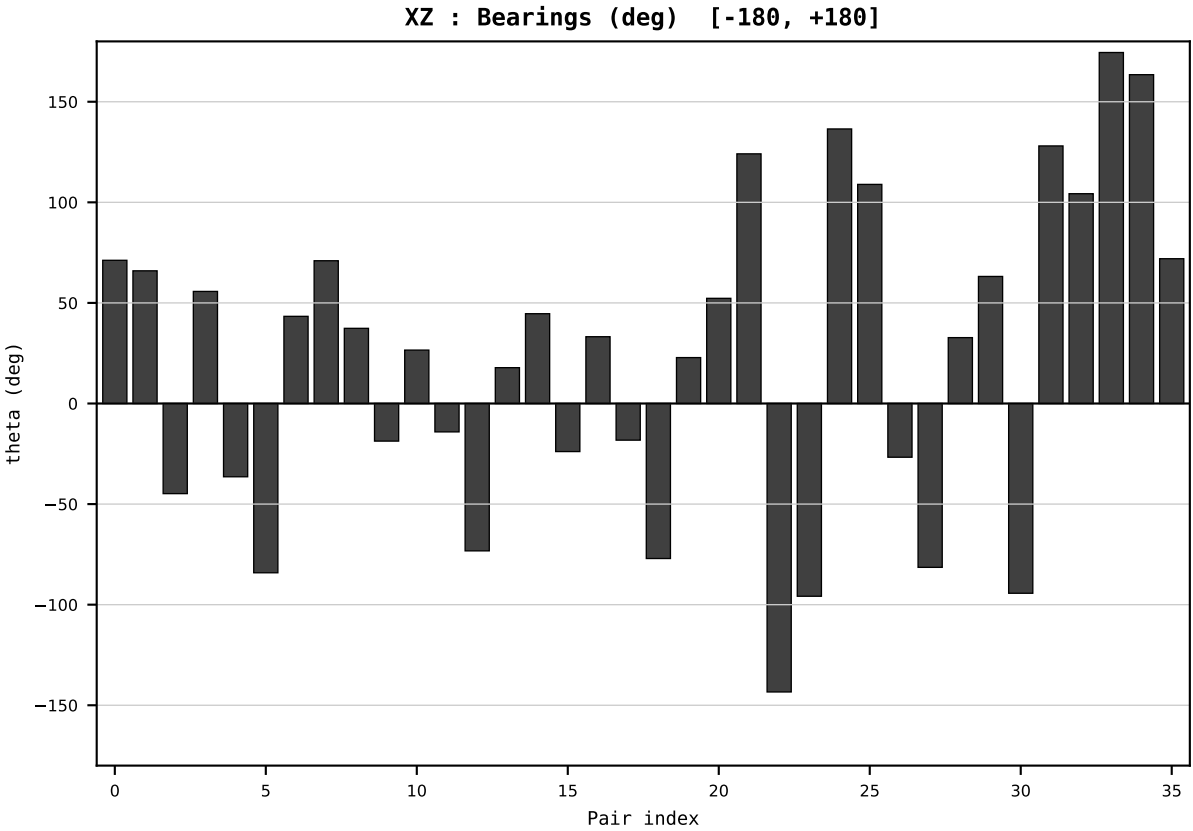
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2020
t : 20 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.137161
Ring = Hs-1
E_metric= 30.9528
kappa_HS= 585.22
omega = 3.0042 deg
d_A = 0.343833
Helm = Coal
Helm d = -0.225193
DR = 6.372 HLR
DR ratio= 585.2x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=21 | Year 2021 | D=9 N=26 pairs=36

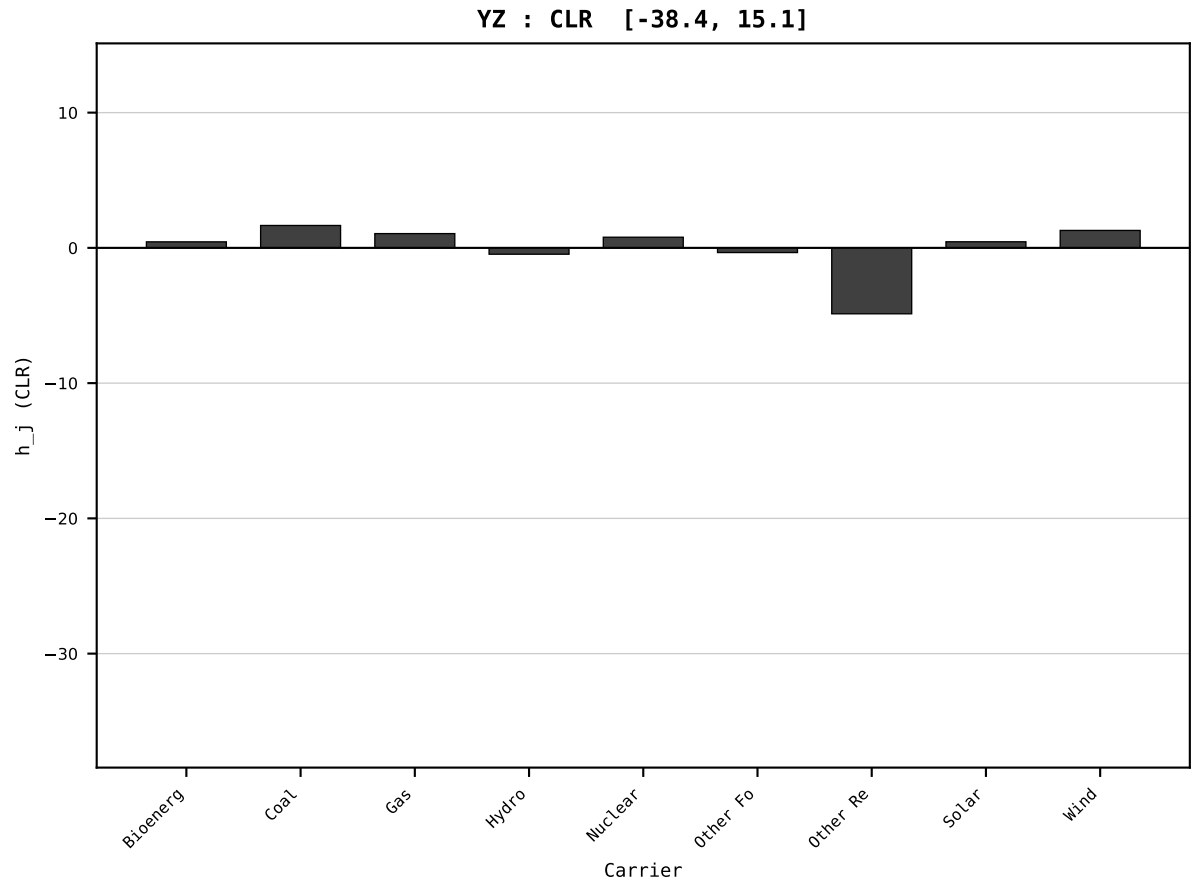
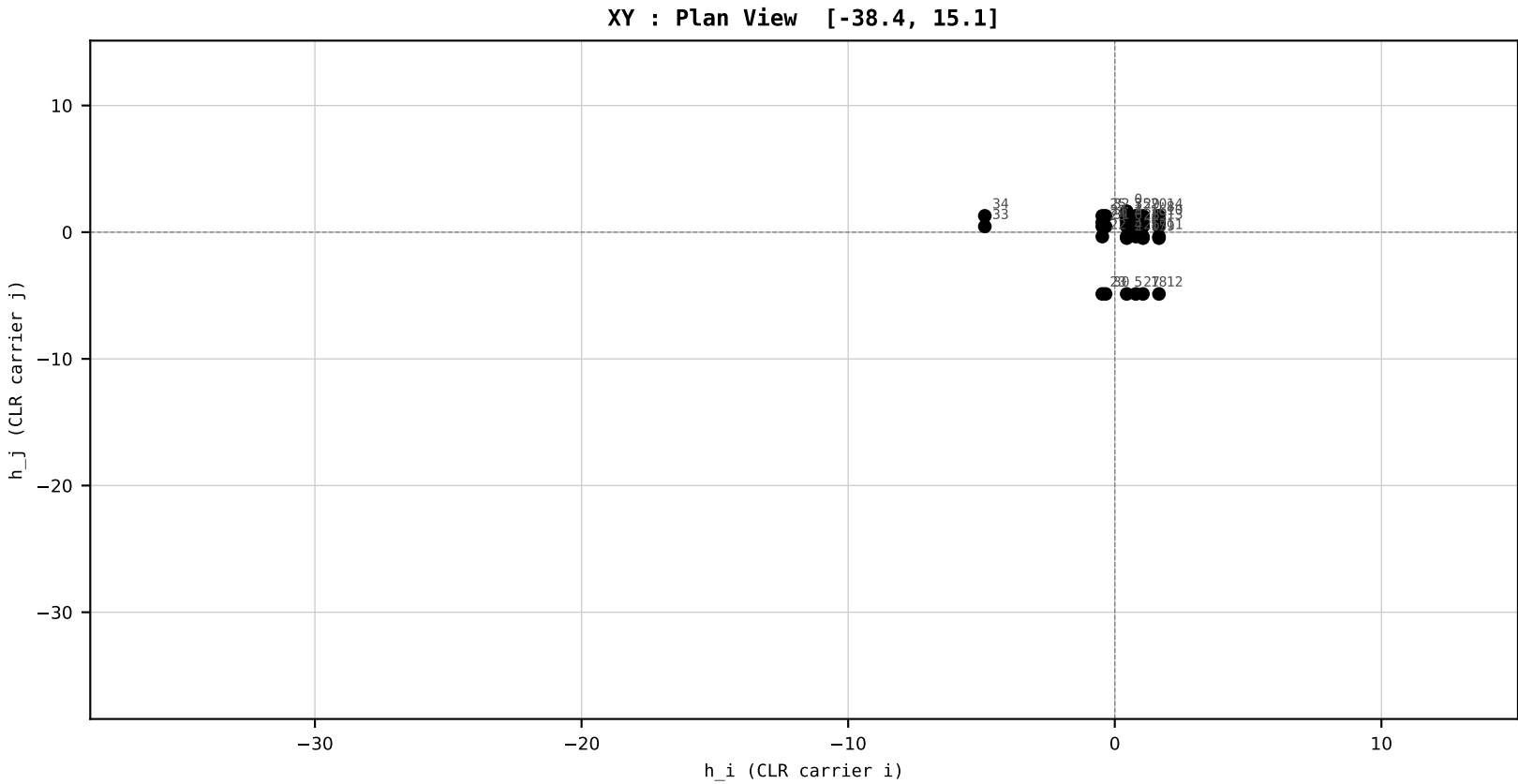
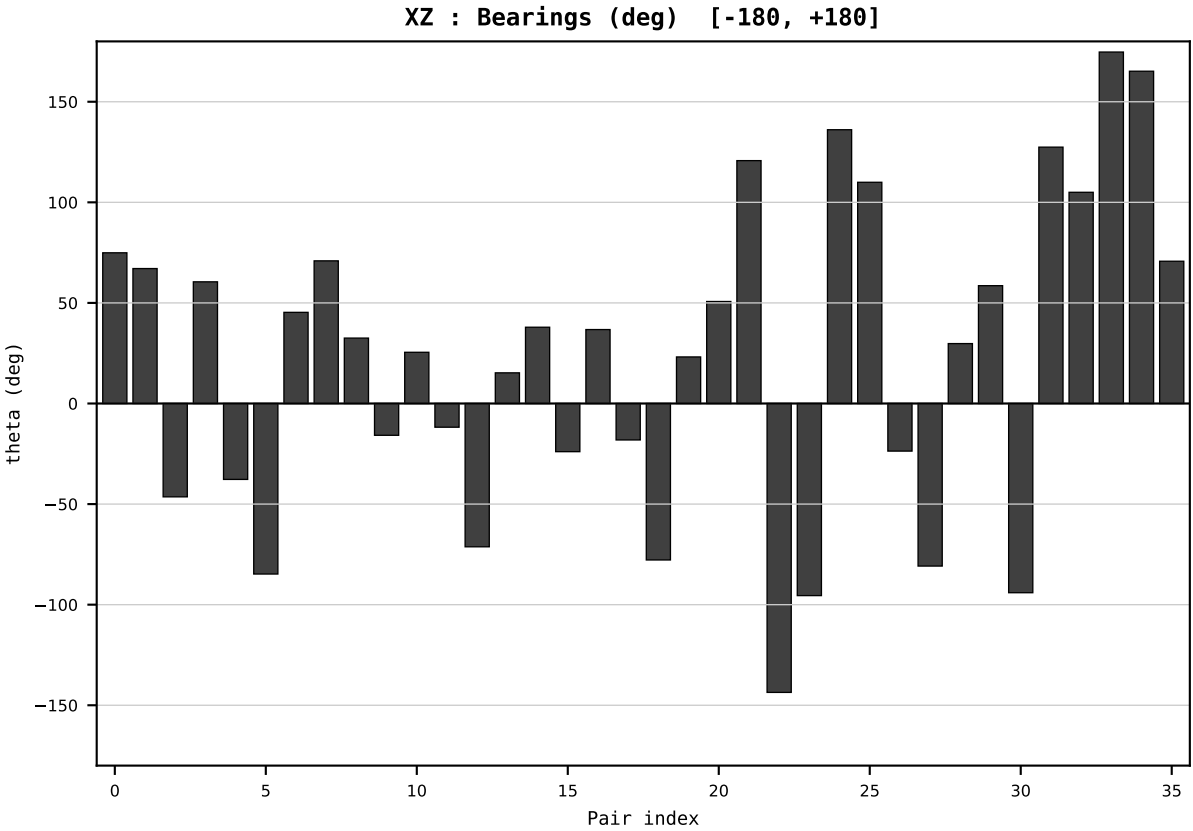
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2021
t : 21 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.142585
Ring = Hs-1
E_metric= 30.6481
kappa_HS= 686.04
omega = 2.7904 deg
d_A = 0.271644
Helm = Coal
Helm d = +0.182416
DR = 6.531 HLR
DR ratio= 686.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



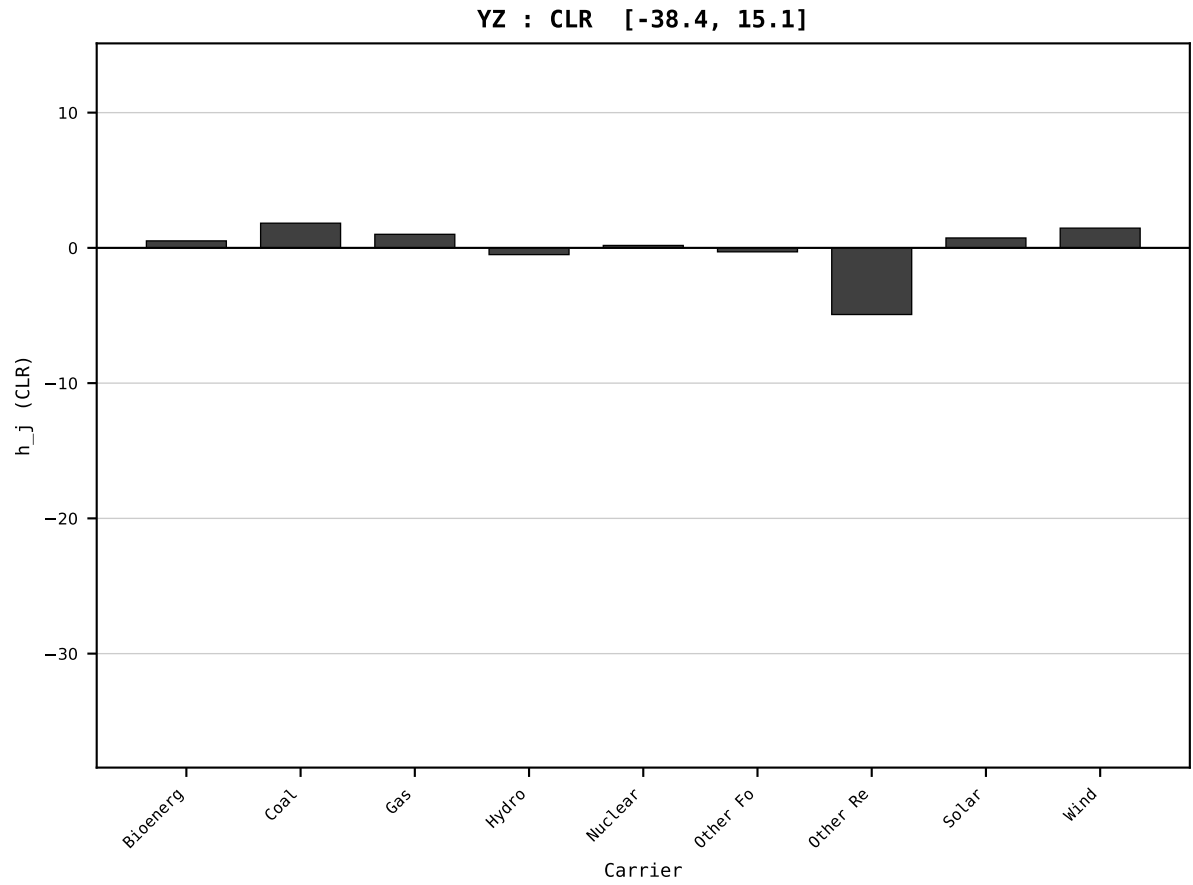
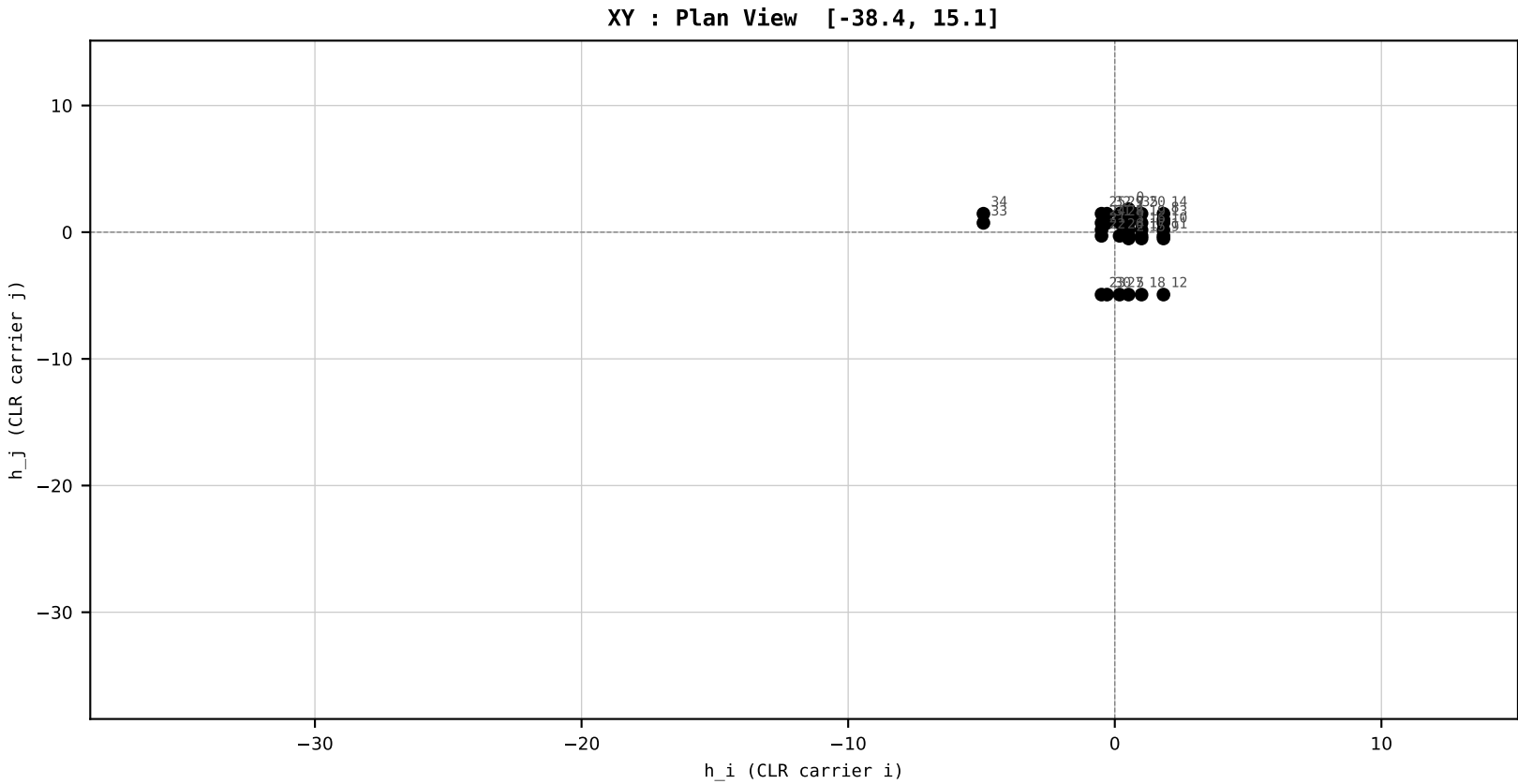
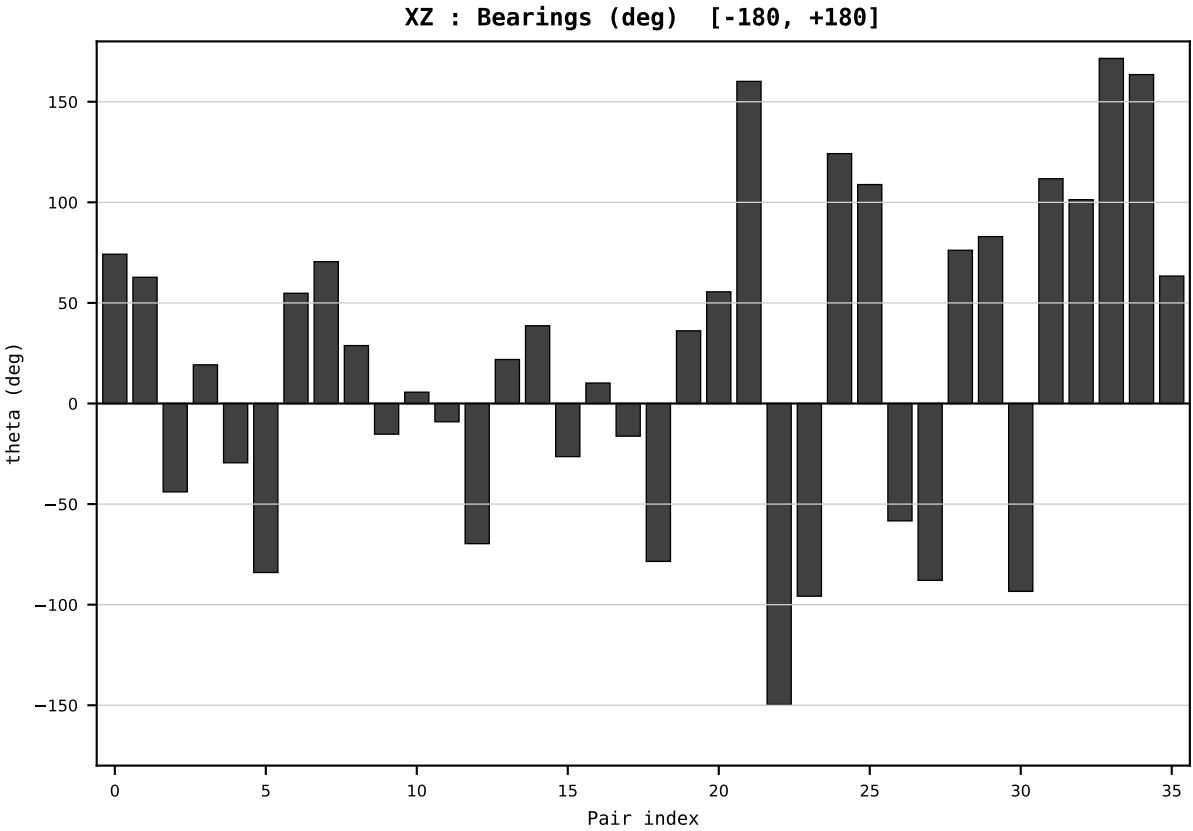
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2022
t : 22 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.168193
Ring = Hs-2
E_metric= 31.9204
kappa_HS= 856.81
omega = 7.3067 deg
d_A = 0.721749
Helm = Nuclear
Helm d = -0.608539
DR = 6.753 HLR
DR ratio= 856.8x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=23 | Year 2023 | D=9 N=26 pairs=36

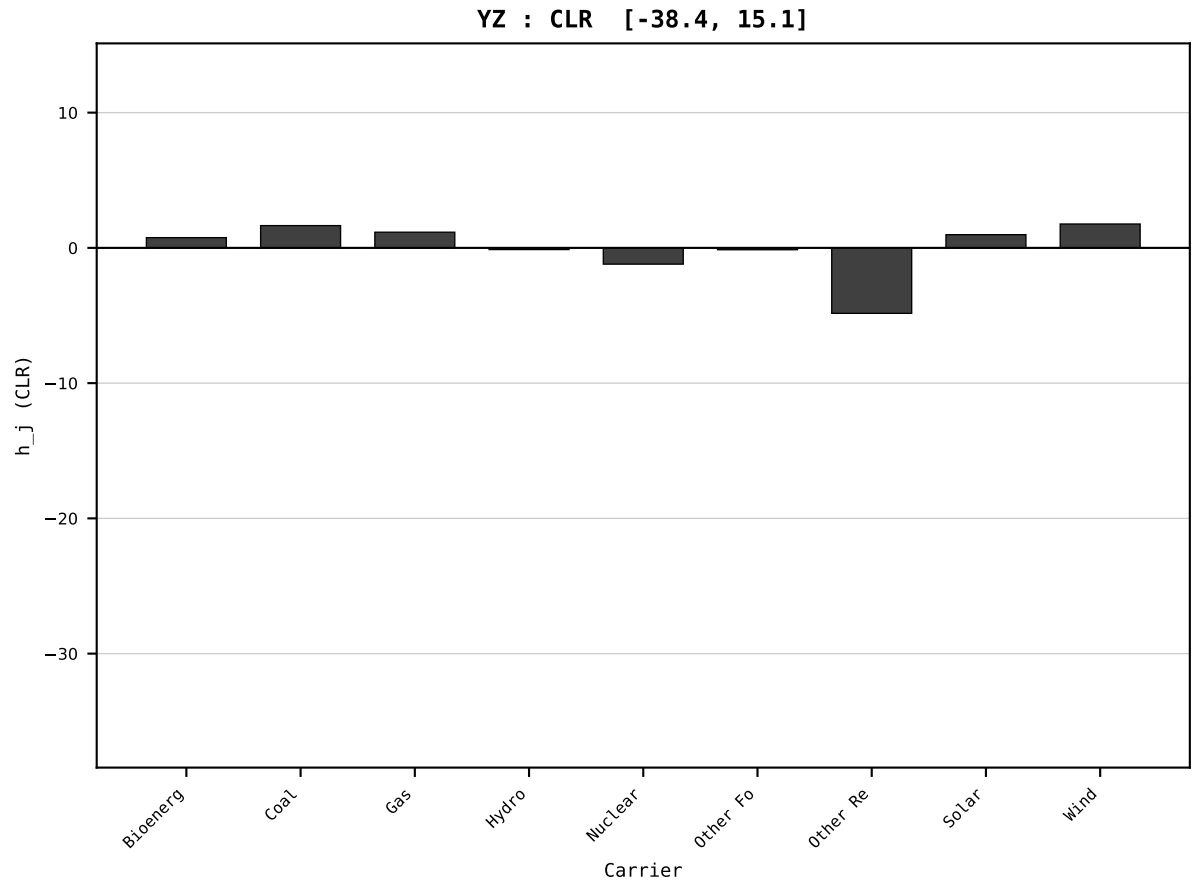
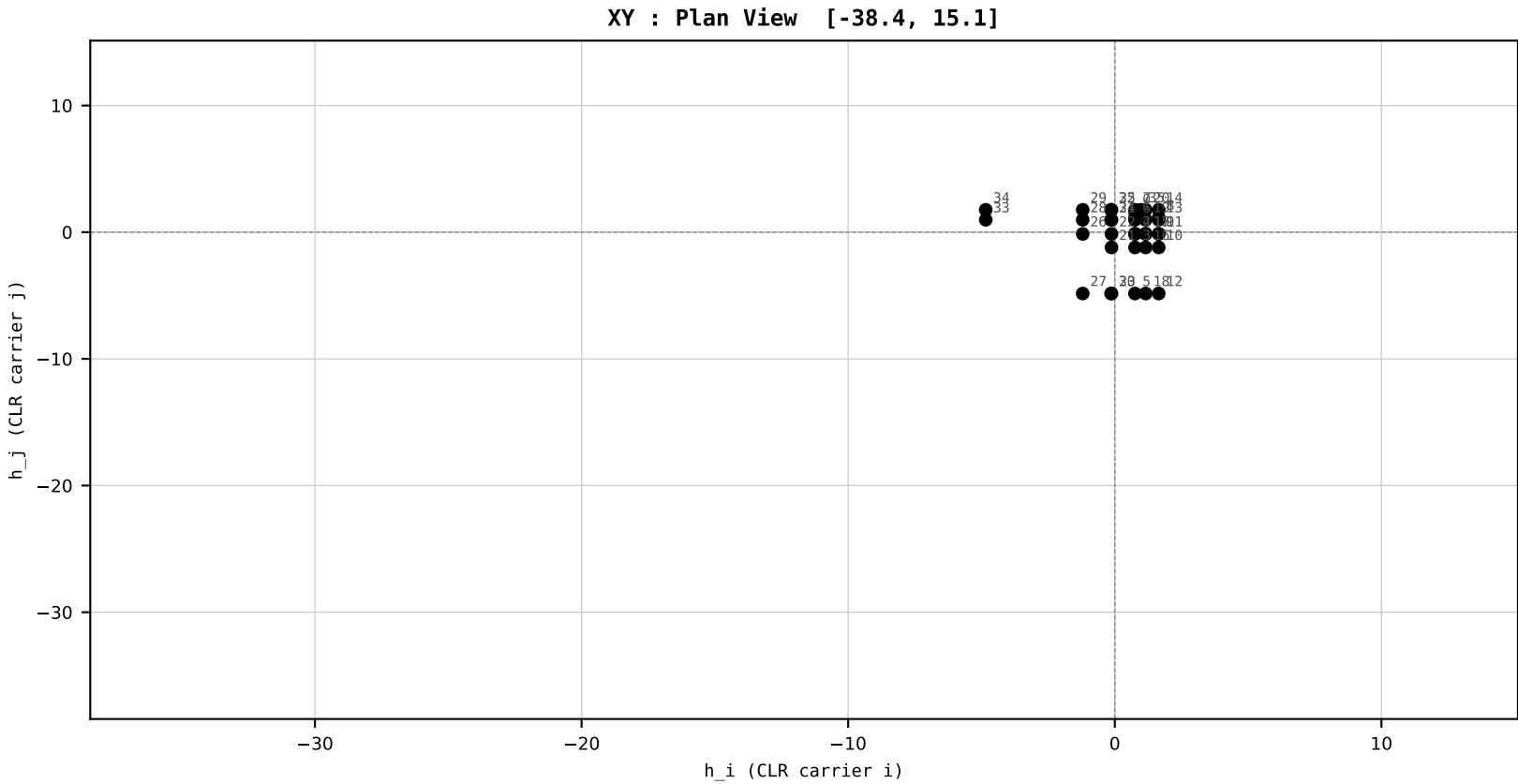
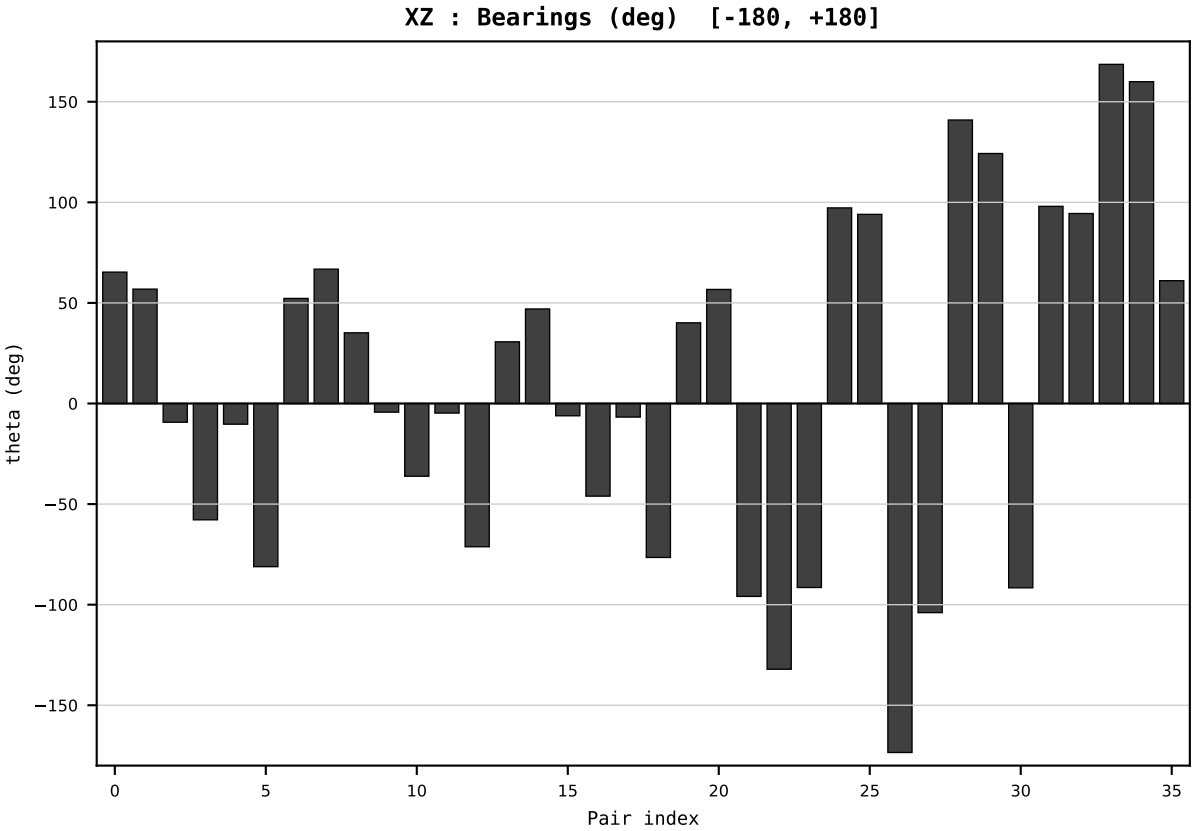
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2023
t : 23 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.177336
Ring = Hs-2
E_metric= 33.6139
kappa_HS= 739.68
omega = 15.3266 deg
d_A = 1.533573
Helm = Nuclear
Helm d = -1.383172
DR = 6.606 HLR
DR ratio= 739.7x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=24 | Year 2024 | D=9 N=26 pairs=36

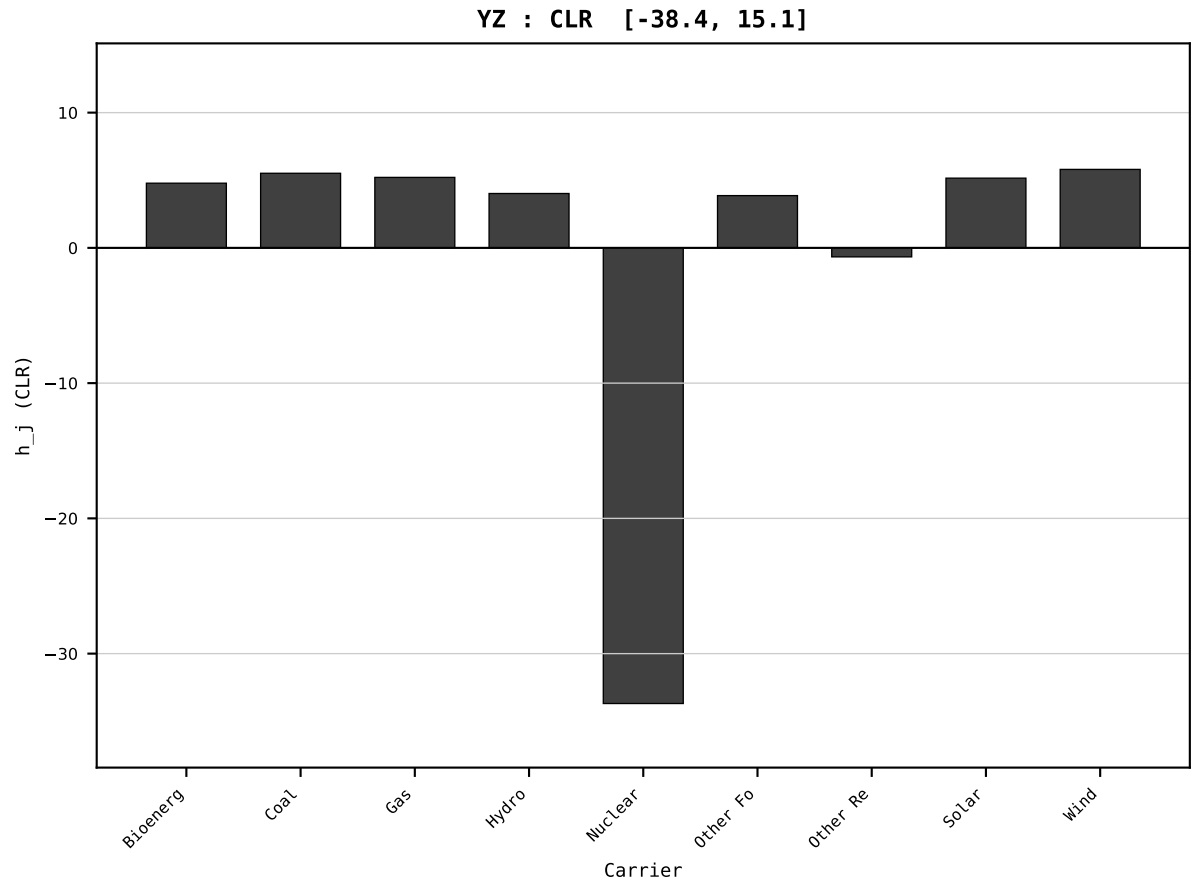
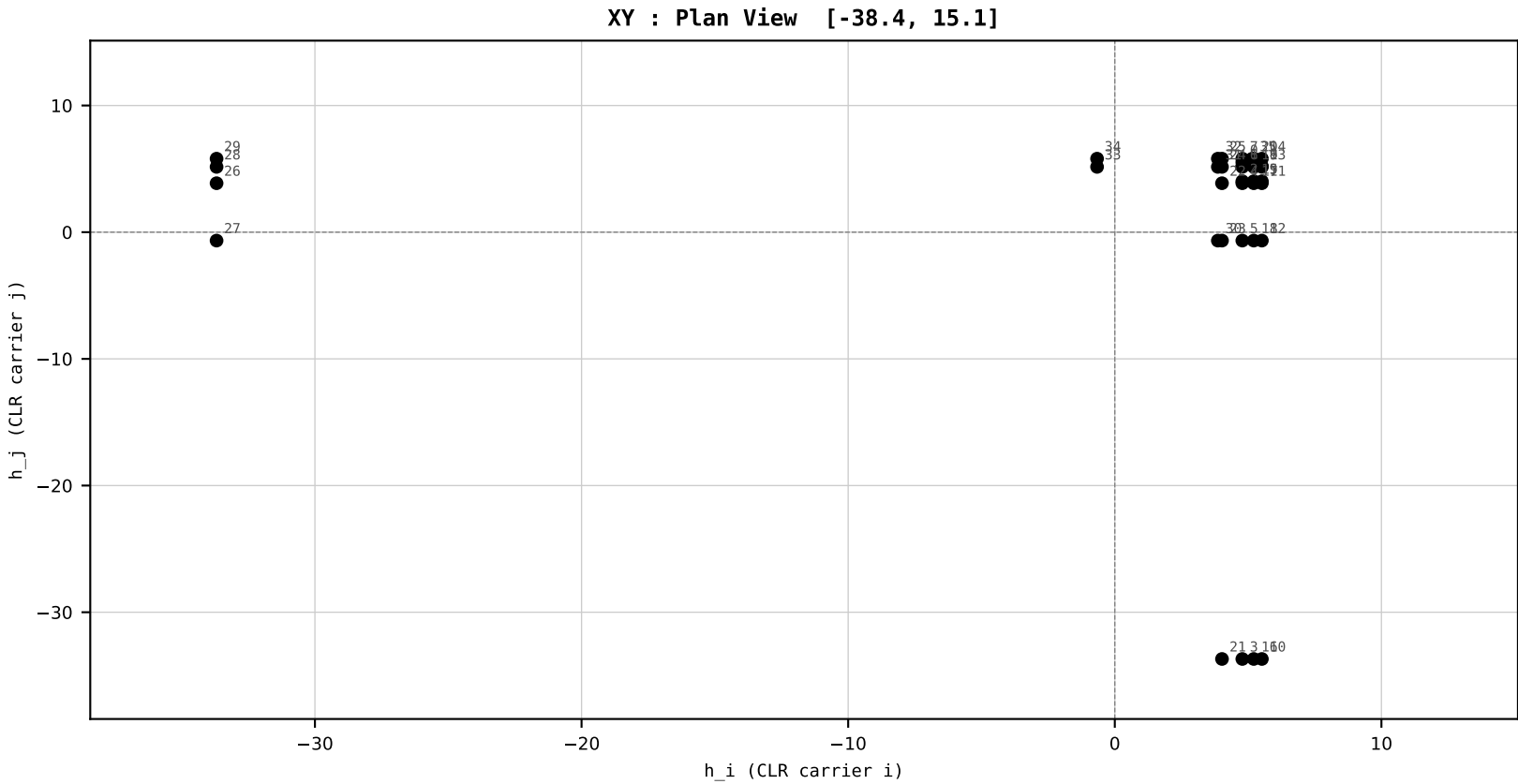
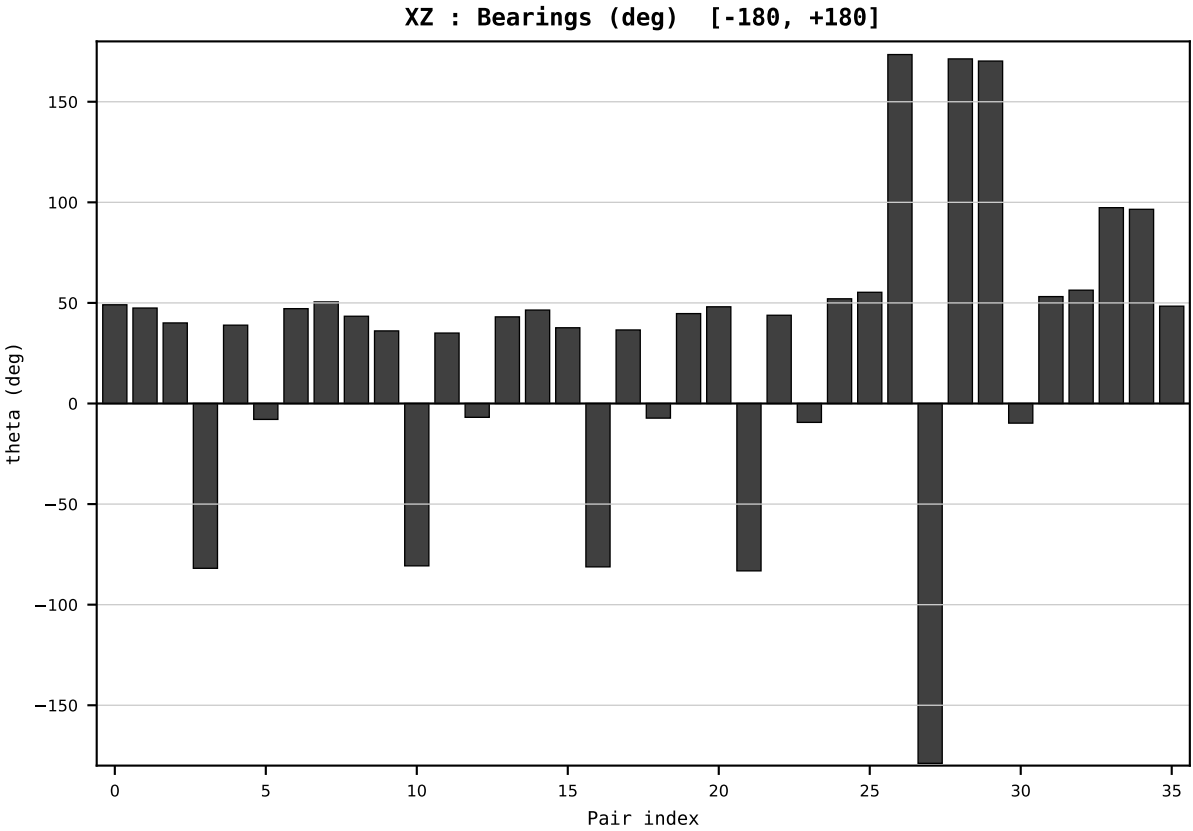
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2024
t : 24 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.190666
Ring = Hs-2
E_metric= 923.2696
kappa_HS= 14159999999999984.00
omega = 68.5237 deg
d_A = 34.457790
Helm = Nuclear
Helm d = -32.486075
DR = 39.492 HLR
DR ratio= 141599999126284592.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=25 | Year 2025 | D=9 N=26 pairs=36

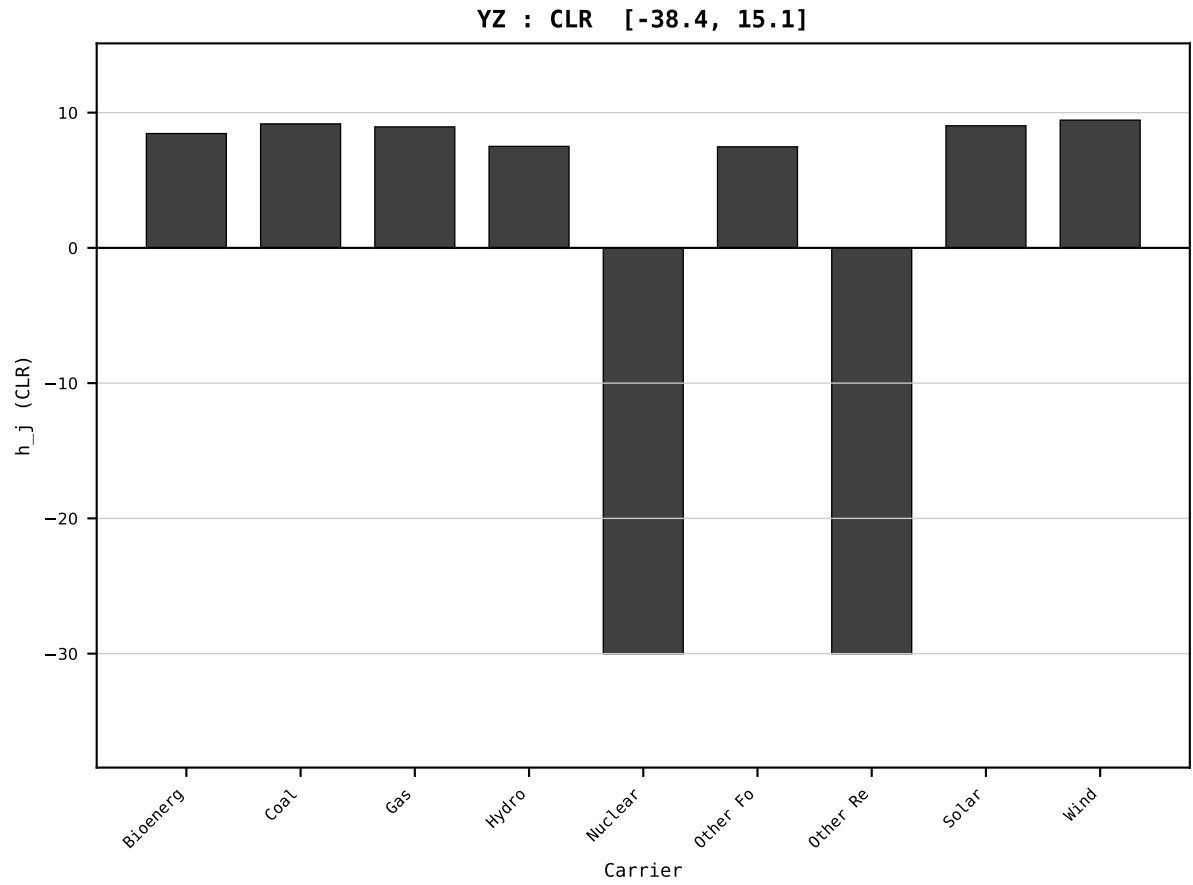
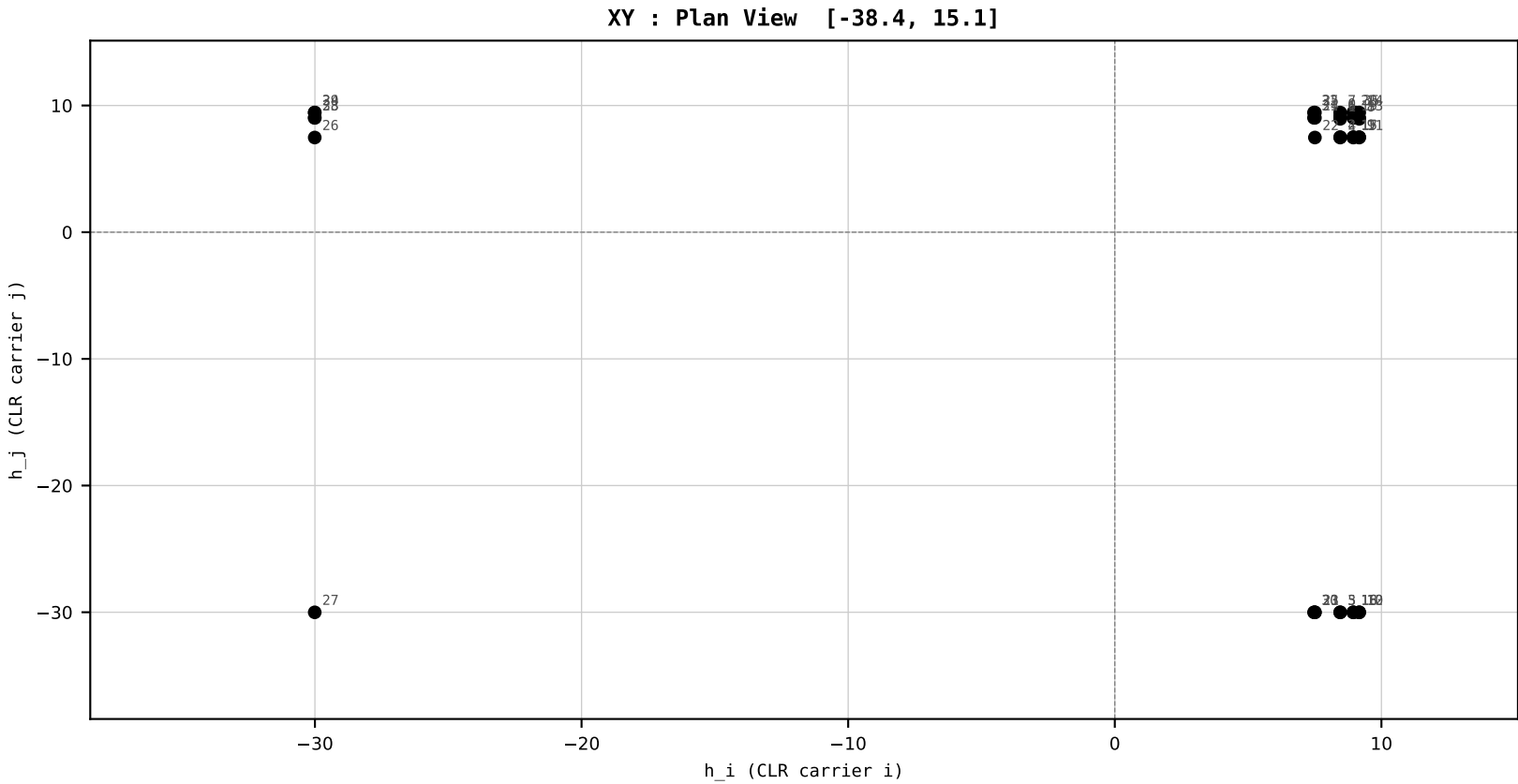
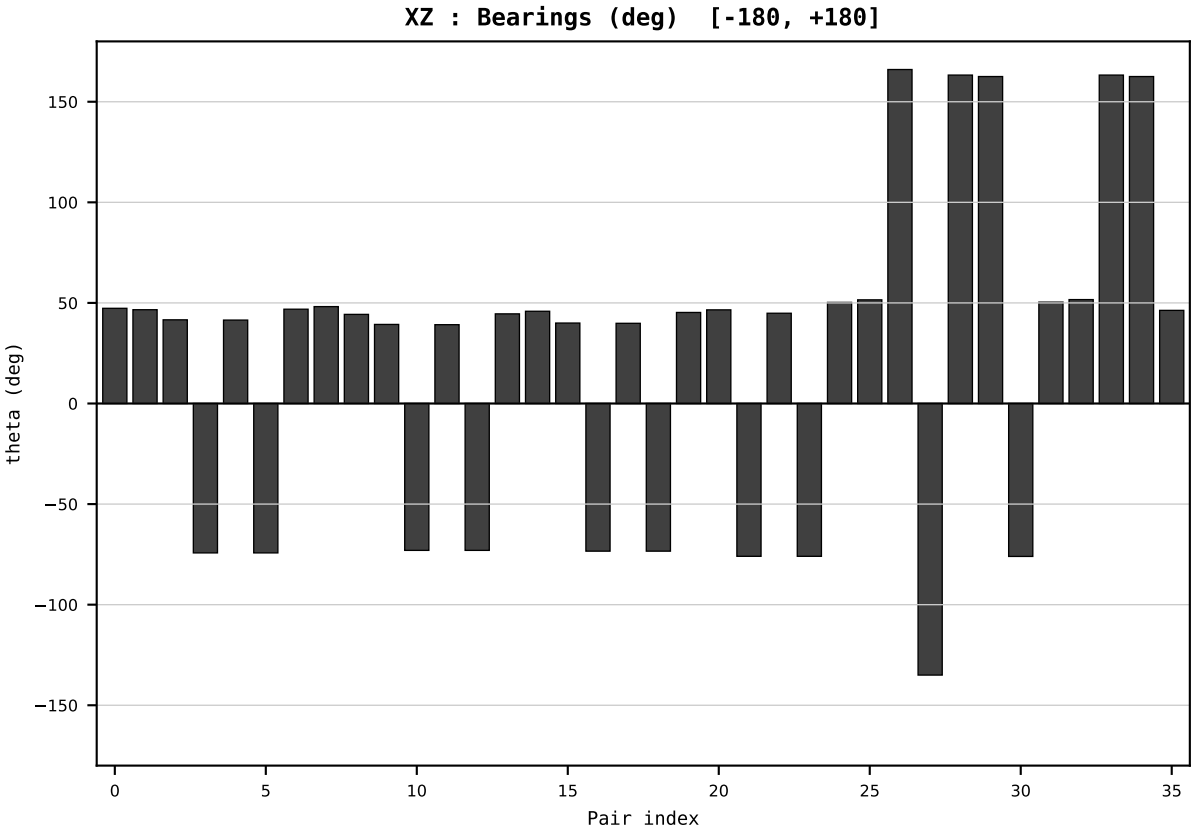
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

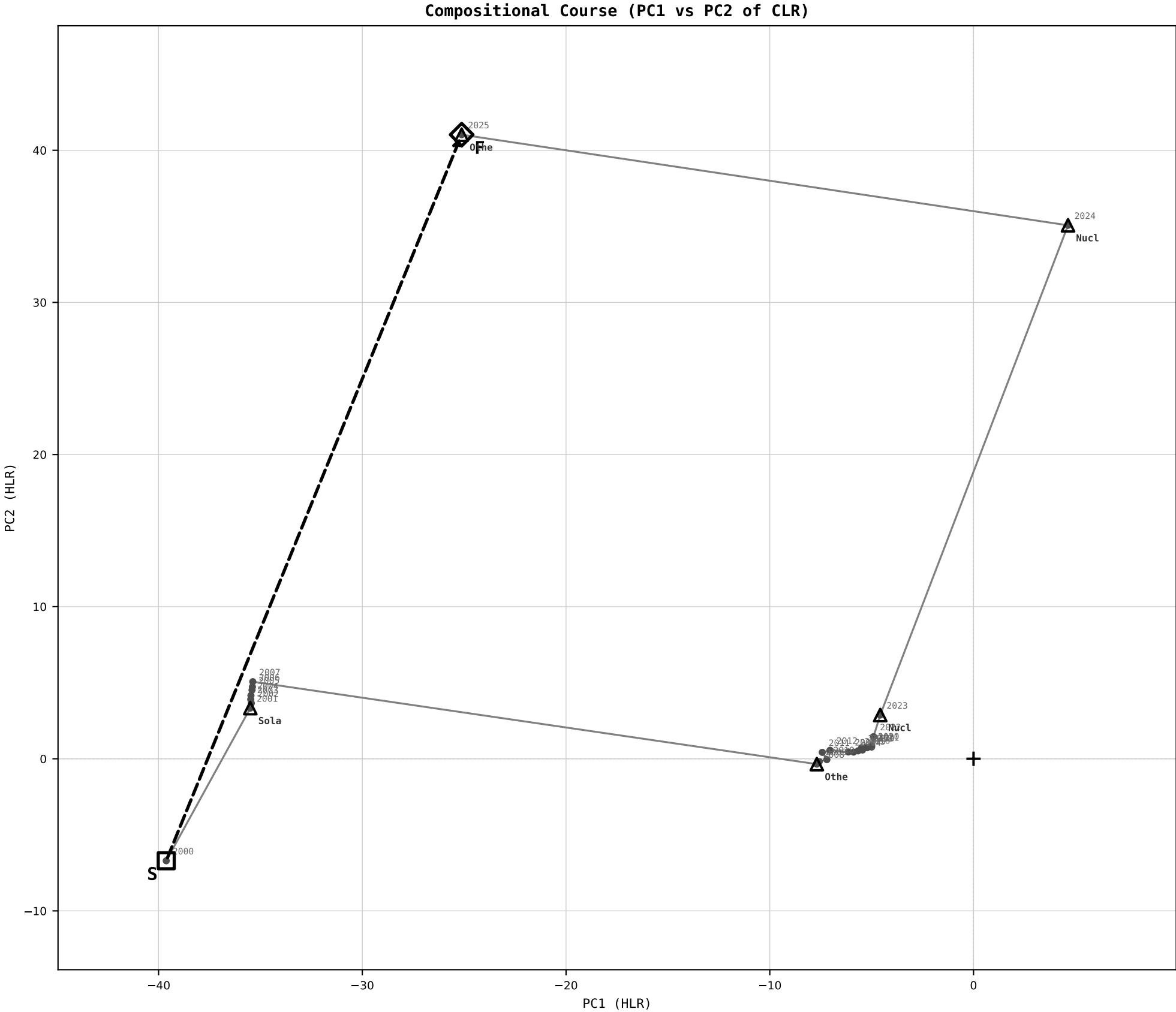
Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2025
t : 25 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.195789
Ring = Hs-2
E_metric= 1633.3021
kappa_HS= 13602999999999984.00
omega = 40.2530 deg
d_A = 31.124159
Helm = Other Renewables
Helm d = -29.342858
DR = 39.452 HLR
DR ratio= 136030000740543728.0x

XZ [-180,180]
CLR[-38.4,15.1]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind





COURSE METRICS

Scale Provenance:
 $Y_j \rightarrow x_j = Y_j / T \rightarrow h_j = \text{CLR}$
 $\rightarrow$ PC projection
Unit: HLR (Higgins Log-Ratio)

Net distance = 55.7717 HLR
Path length = 136.5882 HLR
Directness = 0.4083

DR start = 40.231 HLR
ratio = 296679999472192320.0x
DR final = 39.452 HLR
ratio = 136030000740543728.0x

E_metric start = 2262.9463
E_metric final = 2319.2562

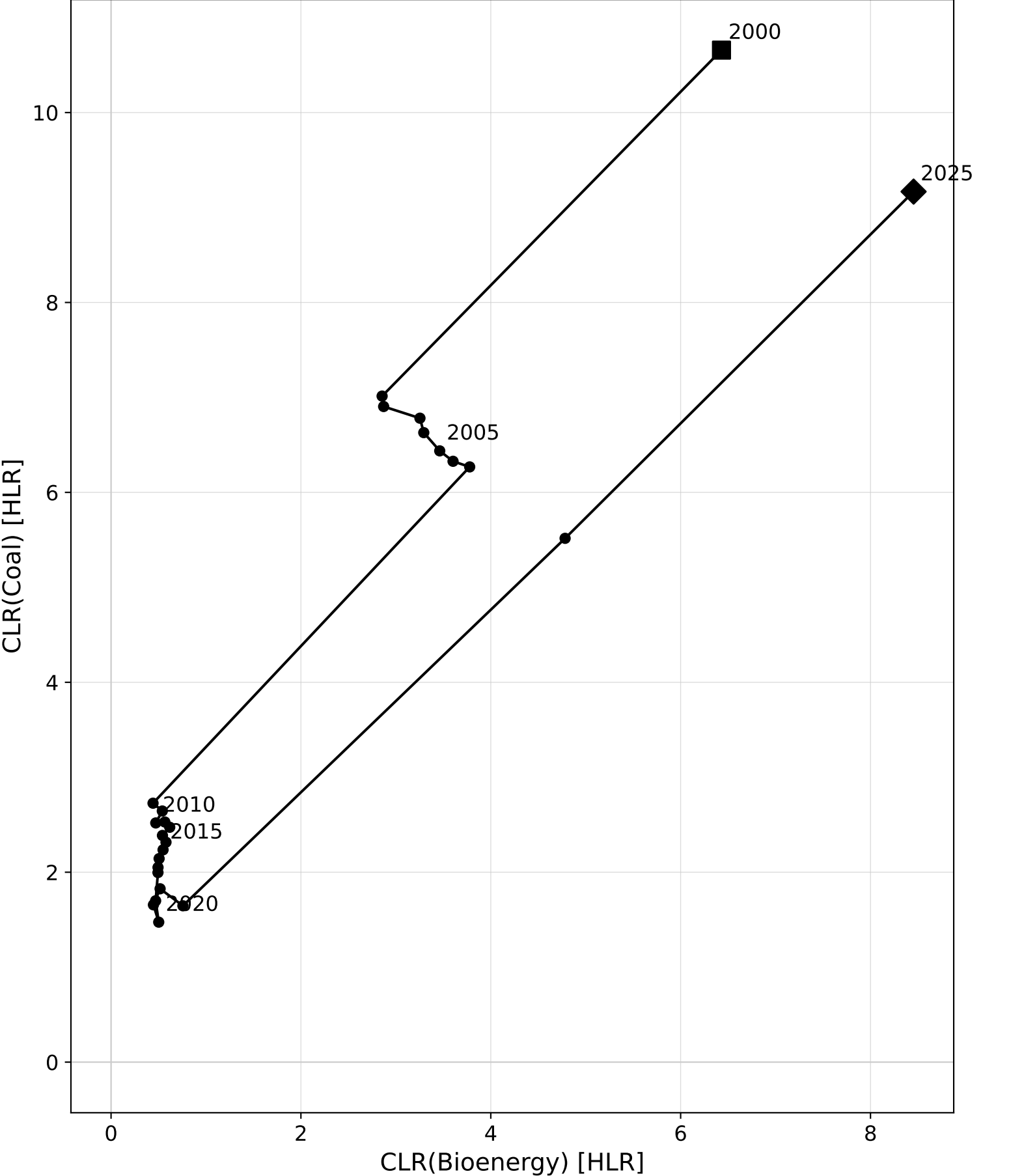
PC1 loadings:
Other Re +0.903
Nuclear -0.248
Coal -0.157

PC2 loadings:
Nuclear -0.881
Solar +0.310
Other Re -0.180

Top displacement events:
2024 Nuclear d=34.458
2025 Other Re d=31.124
2001 Solar d=30.338
2008 Other Re d=28.812
2023 Nuclear d=1.534

The instrument reads.
The expert decides.
The loop stays open.

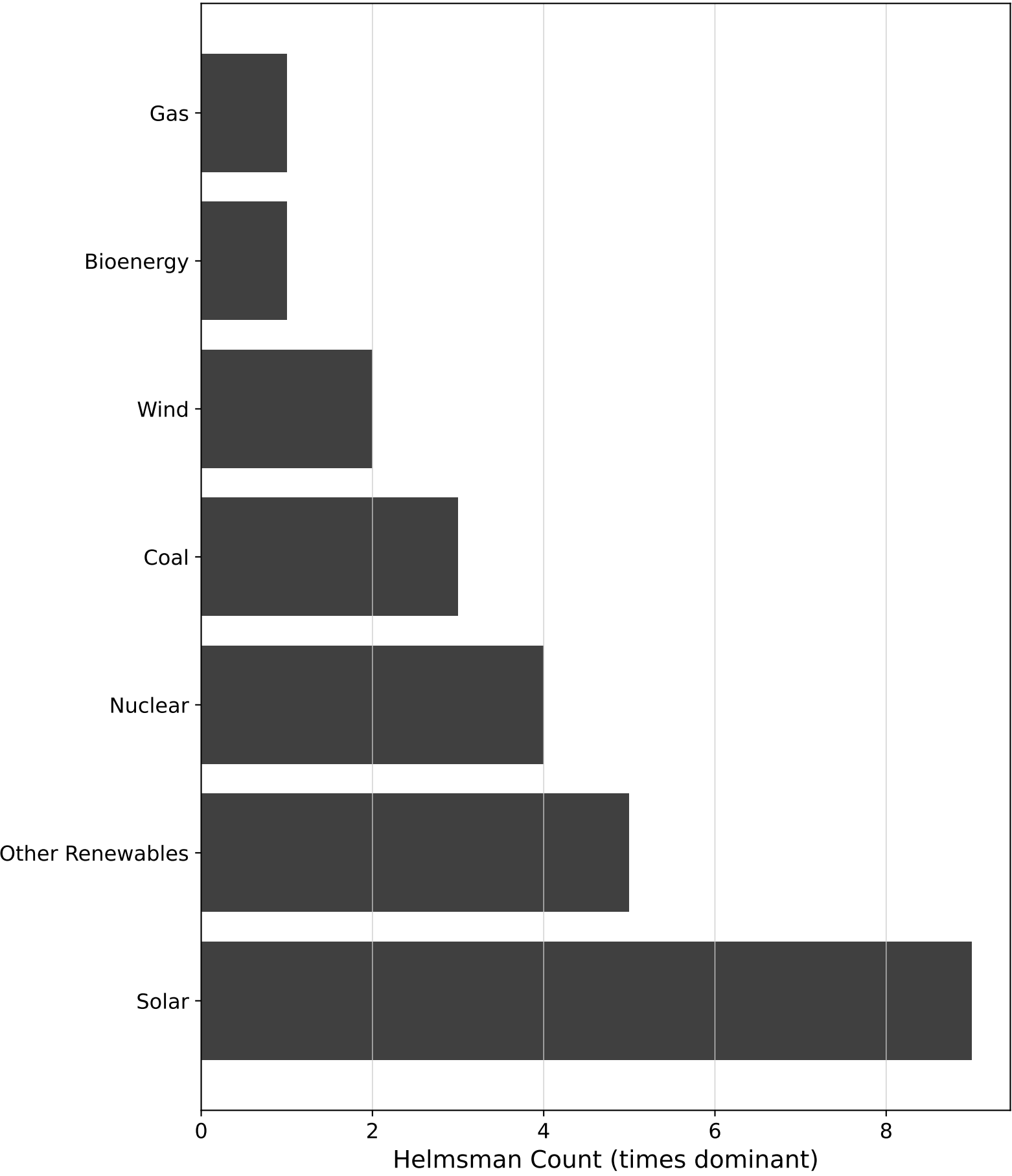
SYSTEM COURSE PLOT — XY Plan View (CLR Space)



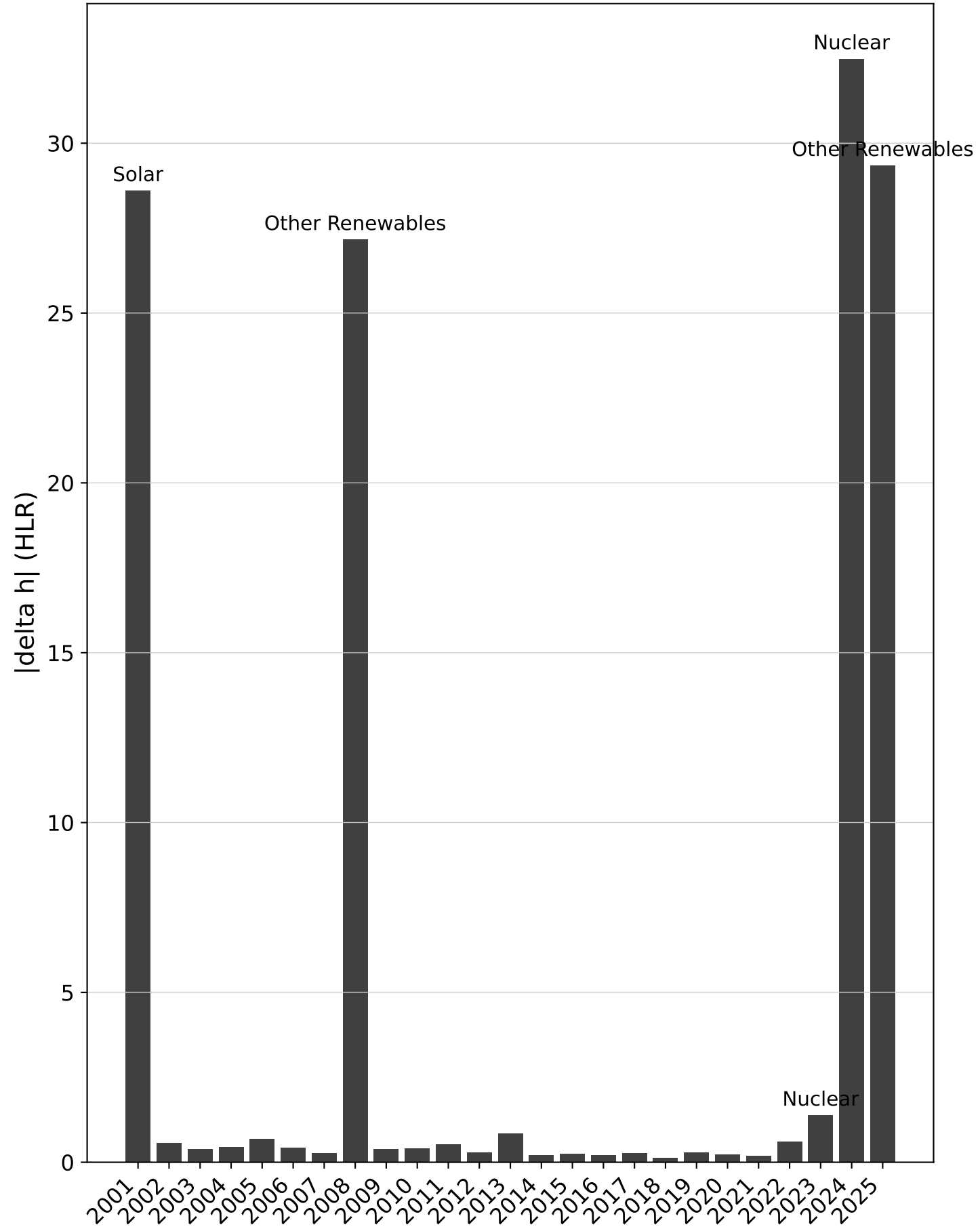
STAGE 2 — Aitchison Distance Matrix (all year pairs)


$$d_A(\text{HLR})$$

HELMSMAN FREQUENCY



HELMSMAN DISPLACEMENT PER YEAR



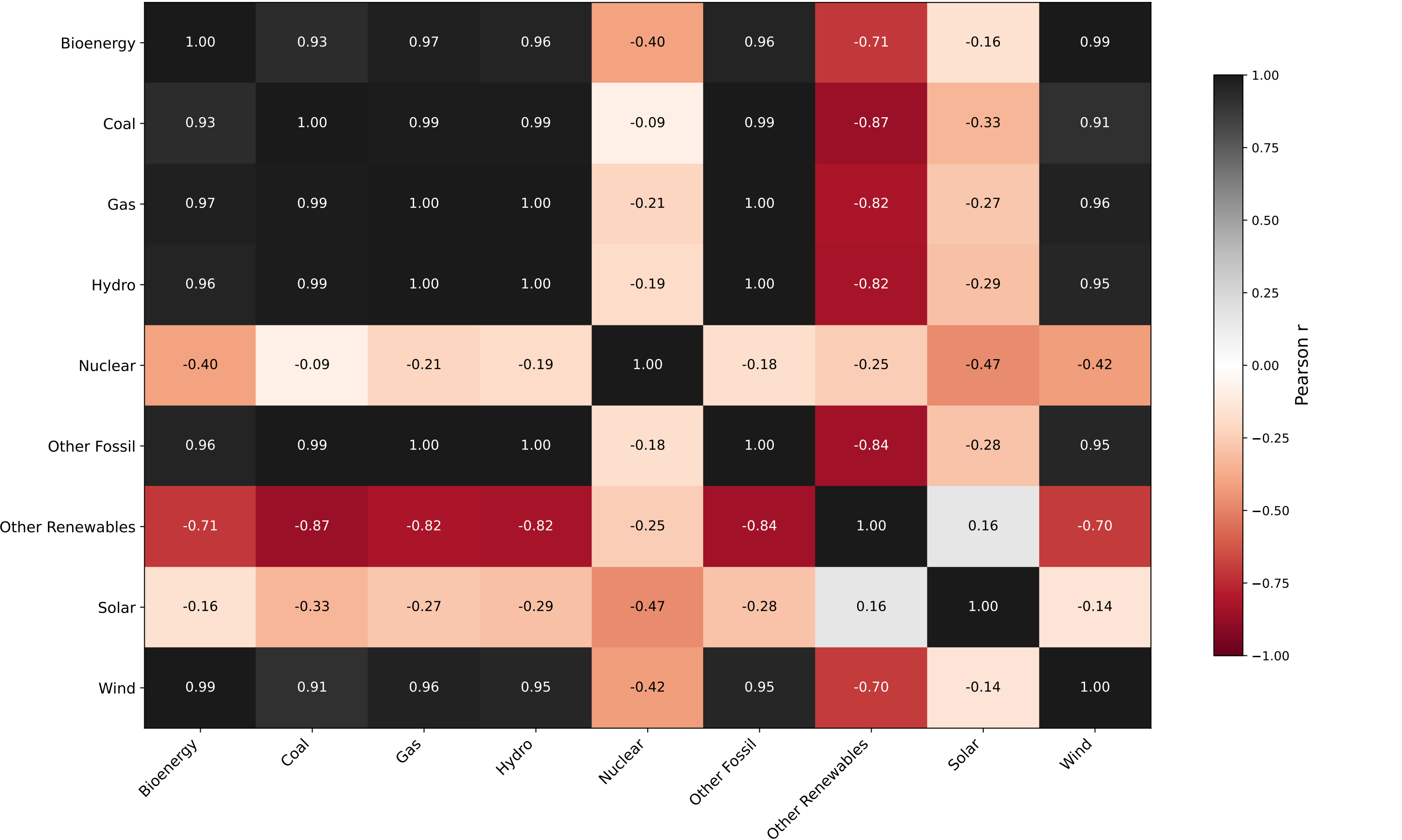
STAGE 2 — Pairwise Divergence Ranking (Top 15)

Rank	Year A	Year B	d_A (HLR)	Angle (deg)	Dominant Carrier
1	2000	2024	63.5278	97.8	Nuclear
2	2000	2025	55.7717	71.3	Nuclear
3	2001	2024	52.1597	92.9	Nuclear
4	2002	2024	52.0311	92.5	Nuclear
5	2003	2024	51.9639	92.2	Nuclear
6	2004	2024	51.9001	92.0	Nuclear
7	2005	2024	51.7971	91.6	Nuclear
8	2006	2024	51.7650	91.4	Nuclear
9	2007	2024	51.5963	91.0	Nuclear
10	2021	2025	45.0284	52.8	Nuclear
11	2016	2025	45.0028	54.7	Nuclear
12	2019	2025	44.9708	53.4	Nuclear
13	2015	2025	44.9648	55.5	Nuclear
14	2020	2025	44.9643	52.2	Nuclear
15	2014	2025	44.9451	56.2	Nuclear

STAGE 3 — Triadic Metric Triangle Area Ranking (Top 10)

Rank	Year A	Year B	Year C	Area (HLR^2)	Imbalance
1	2000	2021	2025	970.4208	1.26
2	2000	2020	2025	969.6442	1.26
3	2000	2019	2025	965.7014	1.27
4	2000	2022	2025	964.6295	1.26
5	2000	2018	2025	963.3611	1.27
6	2000	2016	2025	961.1902	1.28
7	2000	2015	2025	958.6905	1.28
8	2000	2017	2025	958.0769	1.28
9	2000	2014	2025	955.5446	1.28
10	2000	2023	2025	951.2340	1.29

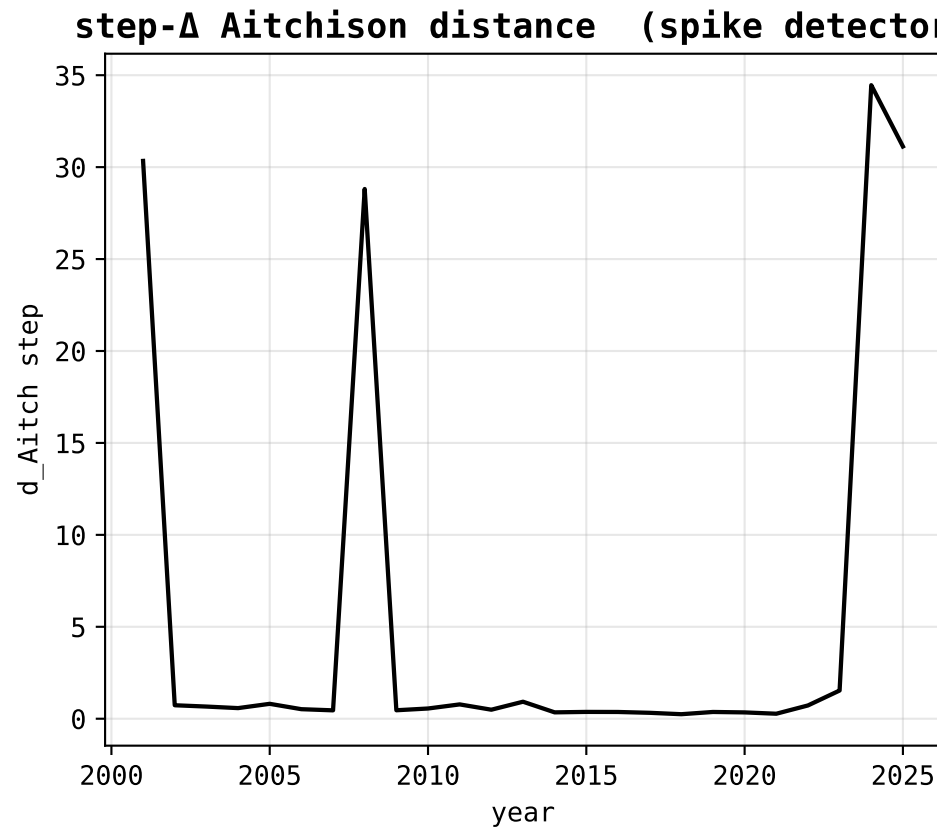
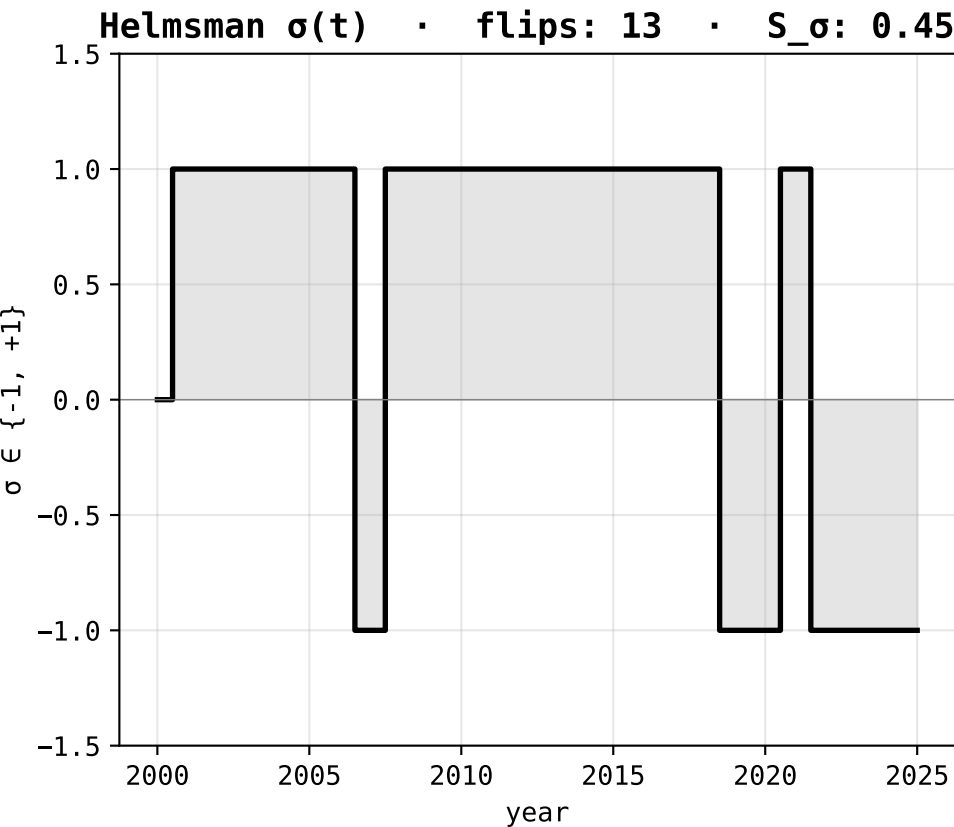
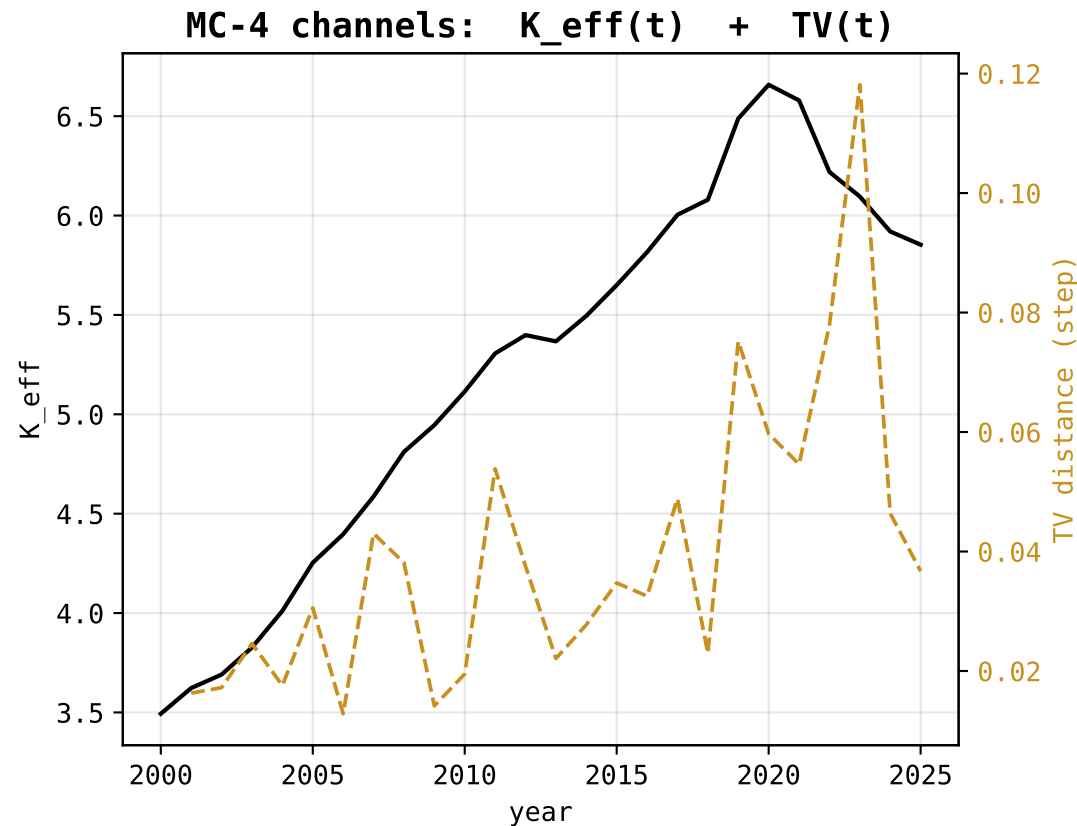
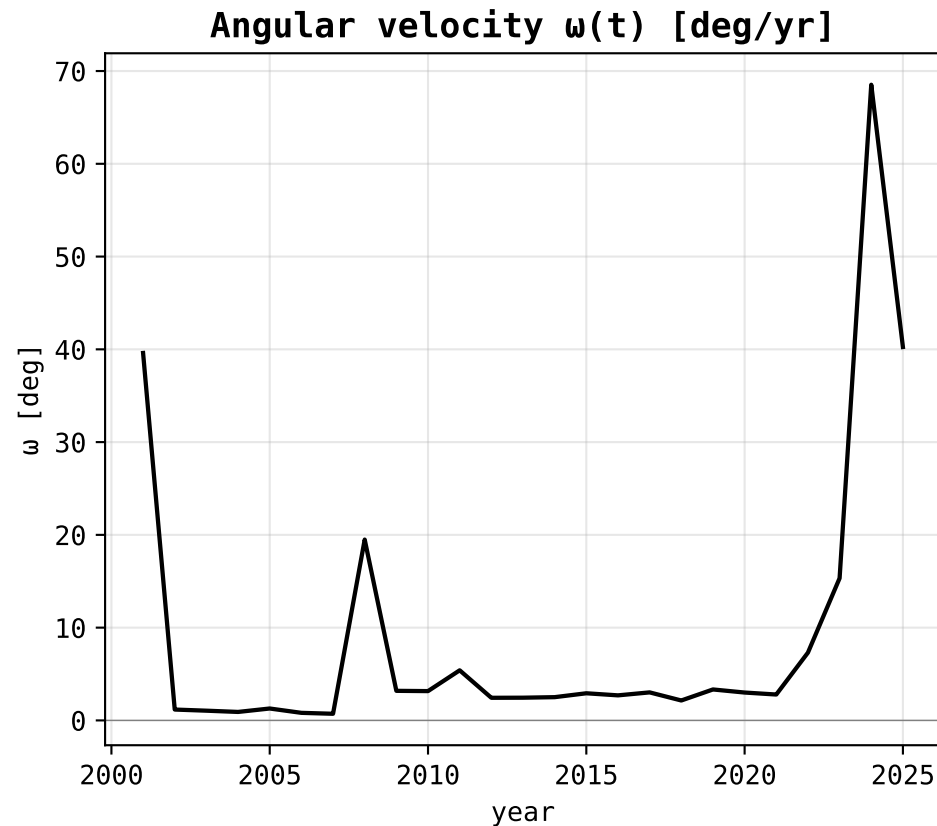
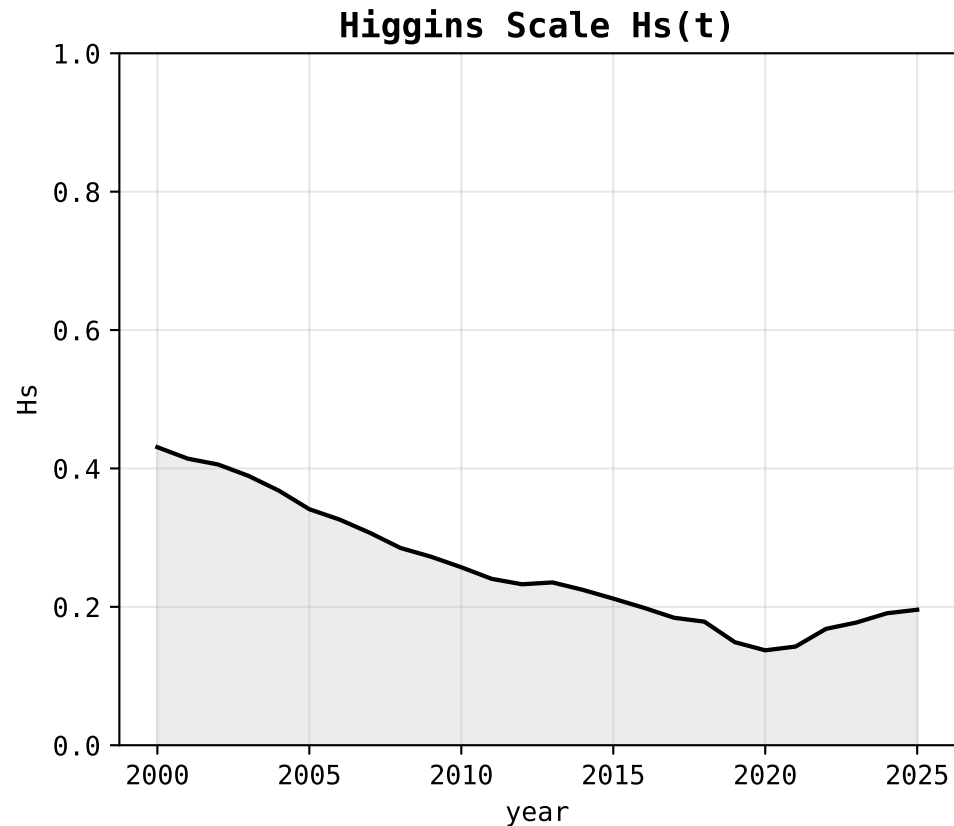
STAGE 3 — Carrier CLR Trajectory Correlation



NAVIGATION SUMMARY — All Stage 1 Metrics by Year

Year	Hs	Ring	E_metric	omega(deg)	d_A	Helmsman	kappa_max
2000	0.4307	Hs-3	1575.1	0.00	0.0000	---	1000000000000000
2001	0.4142	Hs-3	897.4	39.61	30.3382	Solar	1000000000000000
2002	0.4057	Hs-3	897.5	1.17	0.7326	Solar	1000000000000000
2003	0.3892	Hs-3	899.7	1.05	0.6614	Bioenergy	1000000000000000
2004	0.3678	Hs-3	902.1	0.91	0.5772	Solar	1000000000000000
2005	0.3411	Hs-3	905.0	1.28	0.8104	Solar	1000000000000000
2006	0.3261	Hs-3	907.1	0.82	0.5187	Solar	1000000000000000
2007	0.3068	Hs-3	910.1	0.72	0.4555	Nuclear	1000000000000000
2008	0.2851	Hs-2	63.6	19.49	28.8119	Other Renewables	31614
2009	0.2725	Hs-2	61.5	3.19	0.4604	Solar	29398
2010	0.2572	Hs-2	56.0	3.16	0.5558	Solar	20807
2011	0.2405	Hs-2	60.6	5.40	0.7791	Solar	30250
2012	0.2326	Hs-2	54.9	2.44	0.4902	Other Renewables	20660
2013	0.2353	Hs-2	42.7	2.45	0.9251	Other Renewables	7871
2014	0.2245	Hs-2	40.1	2.51	0.3463	Other Renewables	6188
2015	0.2119	Hs-2	37.8	2.92	0.3706	Wind	4917
2016	0.1987	Hs-2	35.0	2.70	0.3670	Gas	3774
2017	0.1842	Hs-2	35.8	3.02	0.3220	Wind	4028
2018	0.1785	Hs-2	34.6	2.15	0.2455	Solar	3513
2019	0.1489	Hs-1	32.9	3.34	0.3675	Coal	3000
2020	0.1372	Hs-1	31.0	3.00	0.3438	Coal	2464
2021	0.1426	Hs-1	30.6	2.79	0.2716	Coal	2412
2022	0.1682	Hs-2	31.9	7.31	0.7217	Nuclear	2699
2023	0.1773	Hs-2	33.6	15.33	1.5336	Nuclear	2667
2024	0.1907	Hs-2	923.3	68.52	34.4578	Nuclear	1000000000000000
2025	0.1958	Hs-2	1633.3	40.25	31.1242	Other Renewables	1000000000000000

CNQ v2.0.0 – Quaternion View – DEU Germany · D=9 · N=26



CNQ DIAGNOSTICS · DEU (D = 9)

CHSH joint-coherence:
not computed at D = 9
(CHSH requires D=8 twin factoring;
D = 9 does not factor as 4x4)

Twin-quaternion factor:
not computed at D = 9
(twin requires D=8 = 4+4 partition)

Attractor fit:
amplitude A: 6.824
contraction λ : 0.0122
period: 1
fitted: False

Helmsman family:
total flips: 13
stability S_σ : 0.458

§ FRA France

EMBER electricity-generation series · 2000-2025 · N = 26 · D = 9

INPUT

```
source: ember_FRA_*_generation_TWh.csv
N records: 26
D carriers: 9
Bioenergy
  Coal
  Gas
  Hydro
  Nuclear
  Other Fossil
  Other Renewables
  Solar
  Wind
source_sha256: 7d929d9c4c7d1b47754582e5...
```

HEADLINE METRICS

```

Helmsman flips: ?
Helmsman stability S_σ: {'global': 0.5, 'rolling': [0.8571428571428572, 0.8571428571428572, 0.8571428571428572]}
IR class: ?
Attractor amplitude: ?

CHSH S: ?
Tsirelson bound:  $2\sqrt{2} \approx 2.828$ 

Bearing pairs  $(D(D-1)/2)$ : 36

CNT v3.1.0 content_sha: ?...
CNQ v2.0.0 content_sha: ?...

```

Following pages: Stage 1 (XY/XZ/YZ $\times$ N years + course) $\rightarrow$ Stage 2/3 (7 pages) $\rightarrow$ CNQ (2 pages).

SECTION PLATE t=0 | Year 2000 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

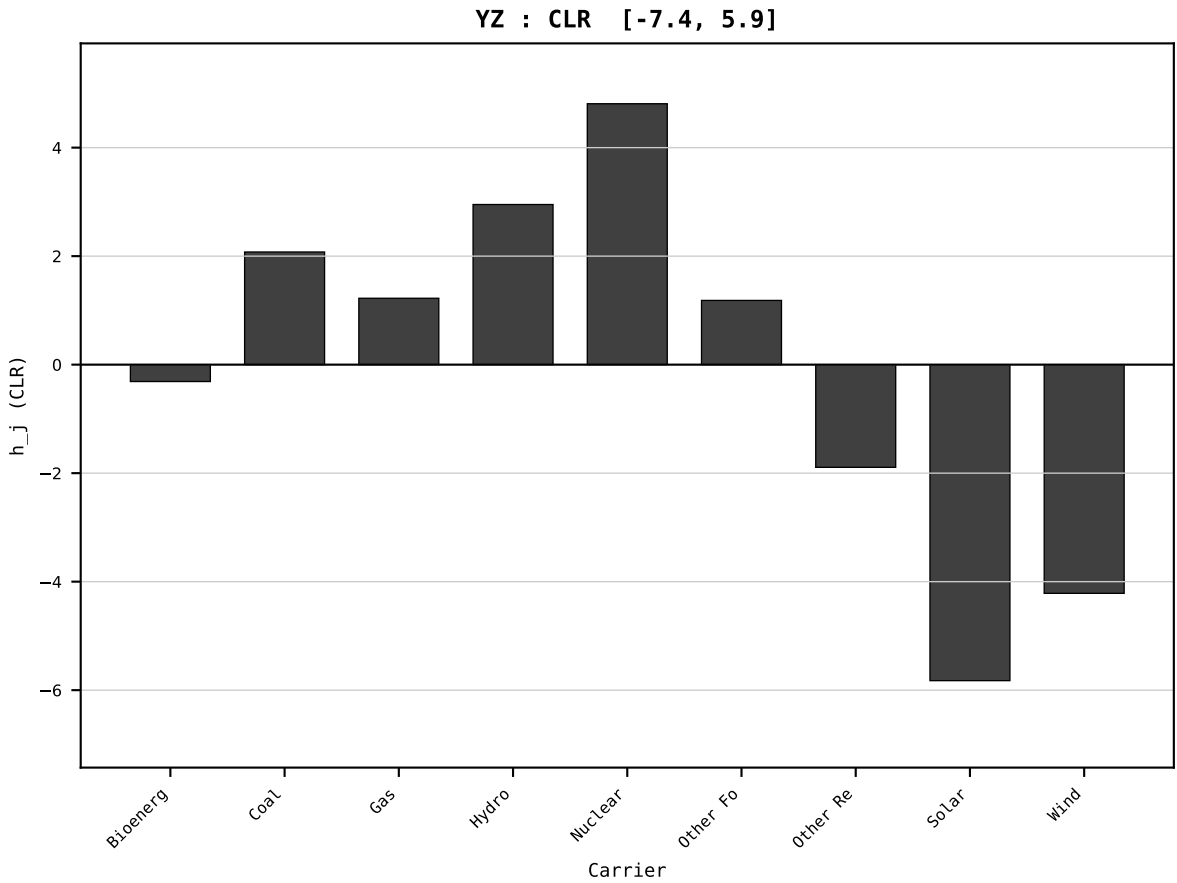
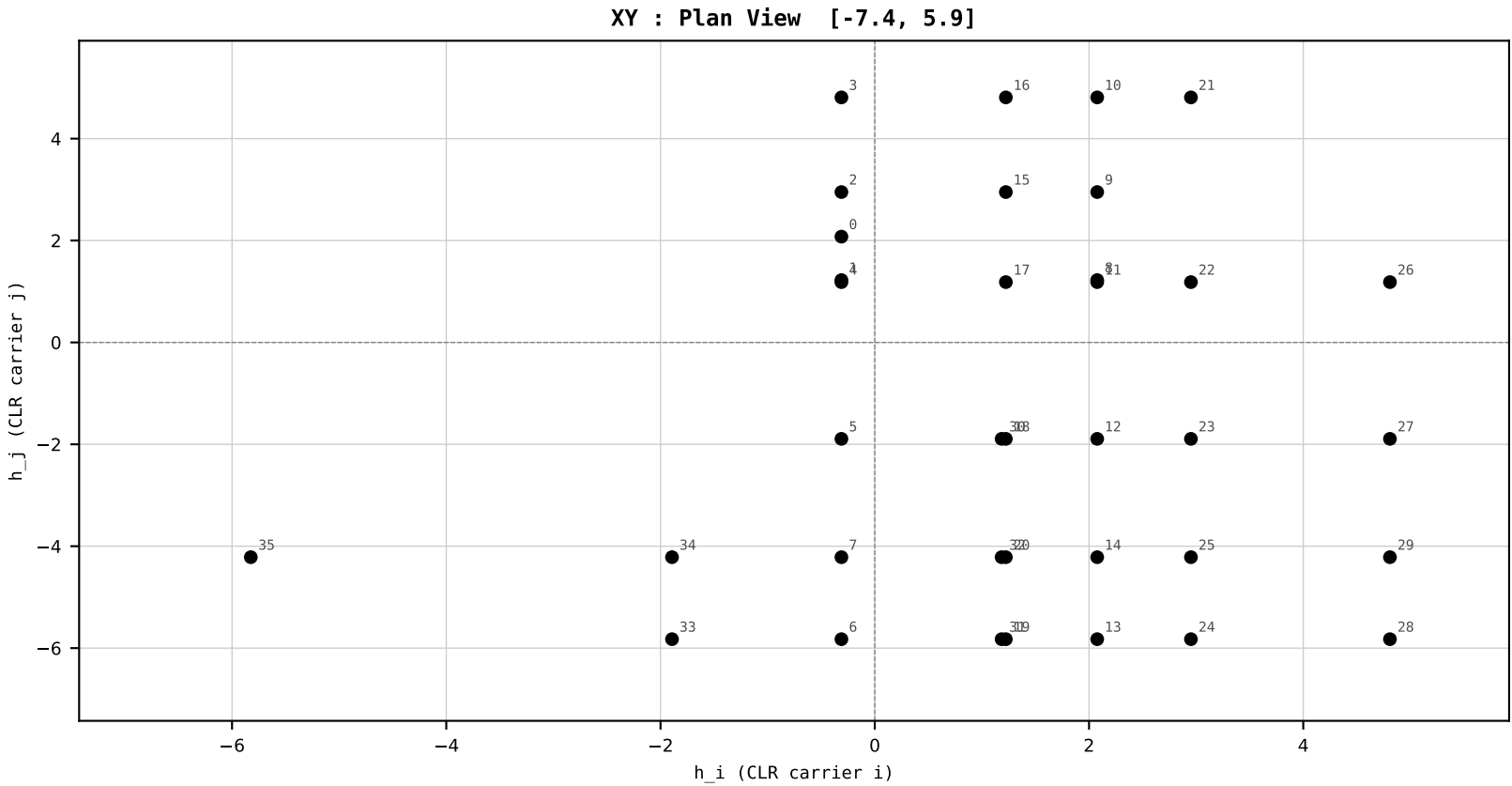
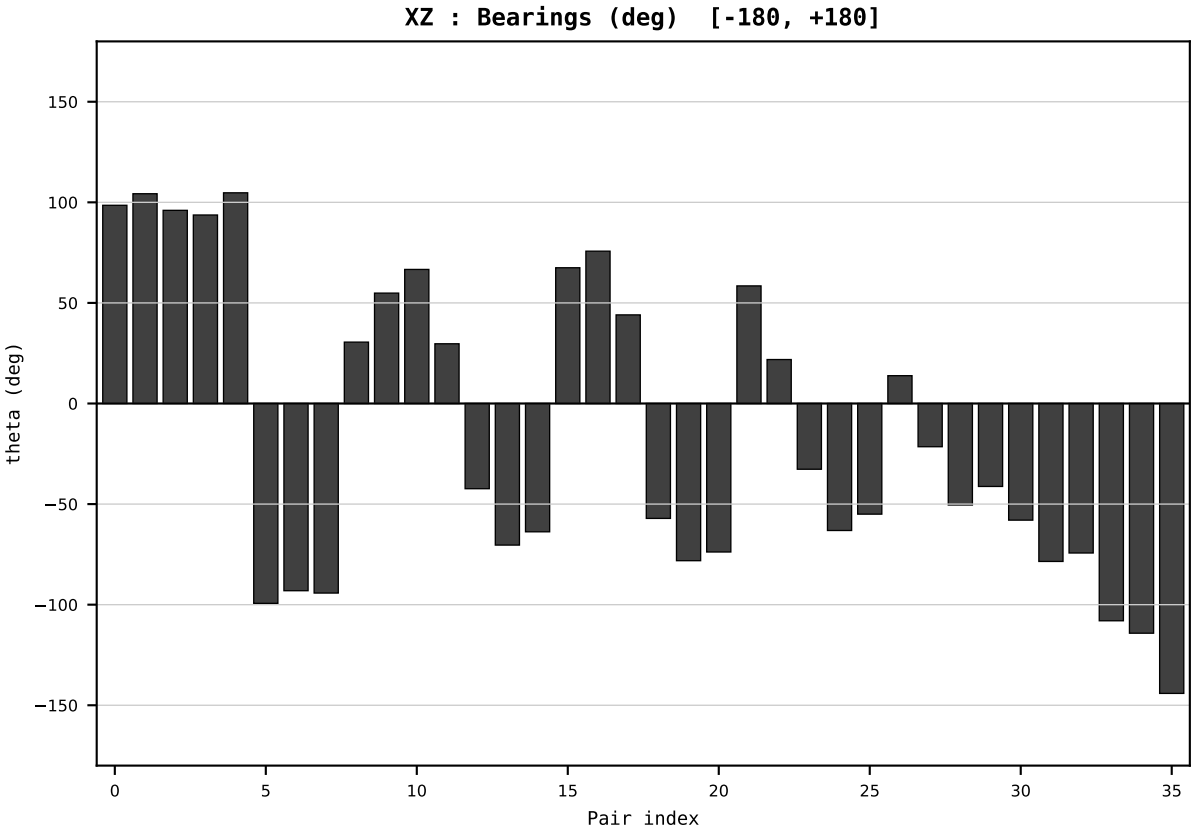
Dataset : ember_FRA_France_generation_TWh.csv
Year : 2000
t : 0 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.636980
Ring = Hs-4
E_metric= 94.4242
kappa_HS= 41516.00
omega = 0.0000 deg
d_A = 0.000000
Helm = ---
Helm d = ---
DR = 10.634 HLR
DR ratio= 41516.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=1 | Year 2001 | D=9 N=26 pairs=36

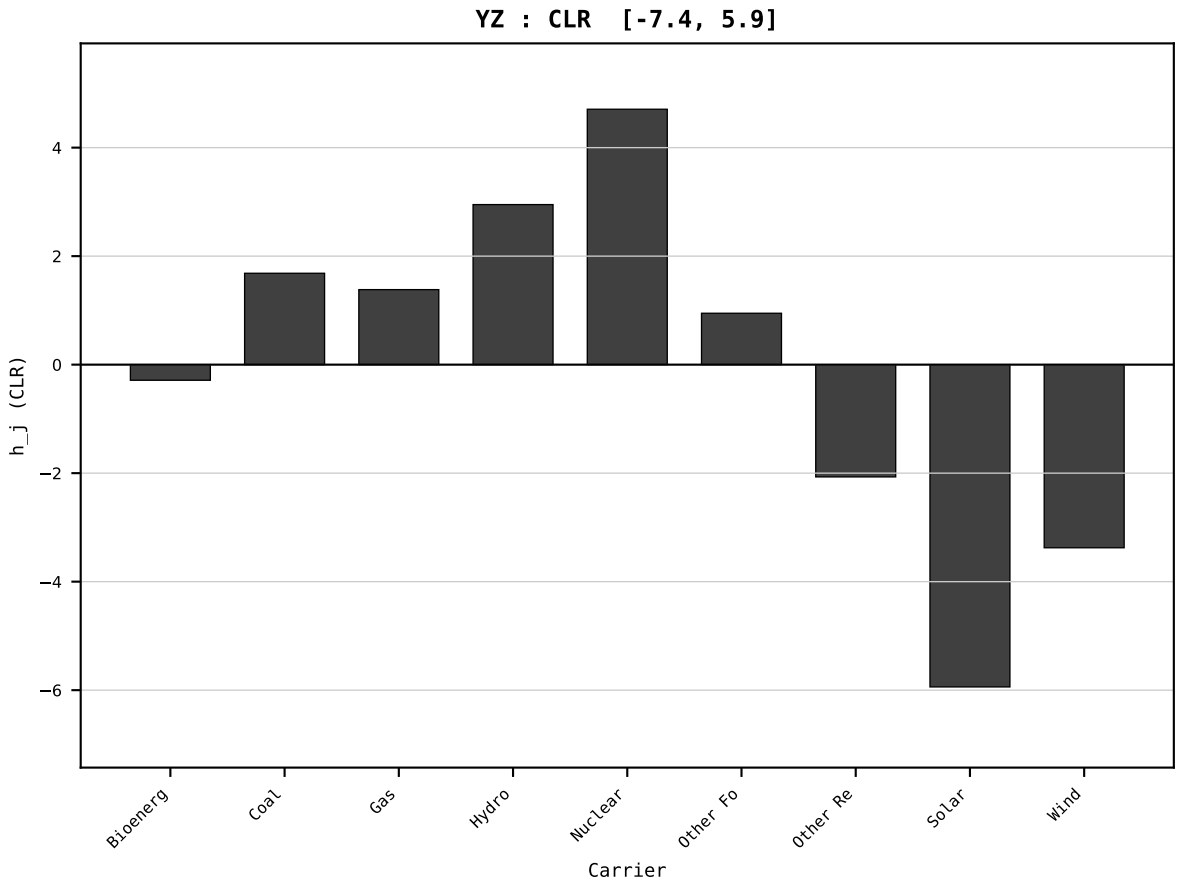
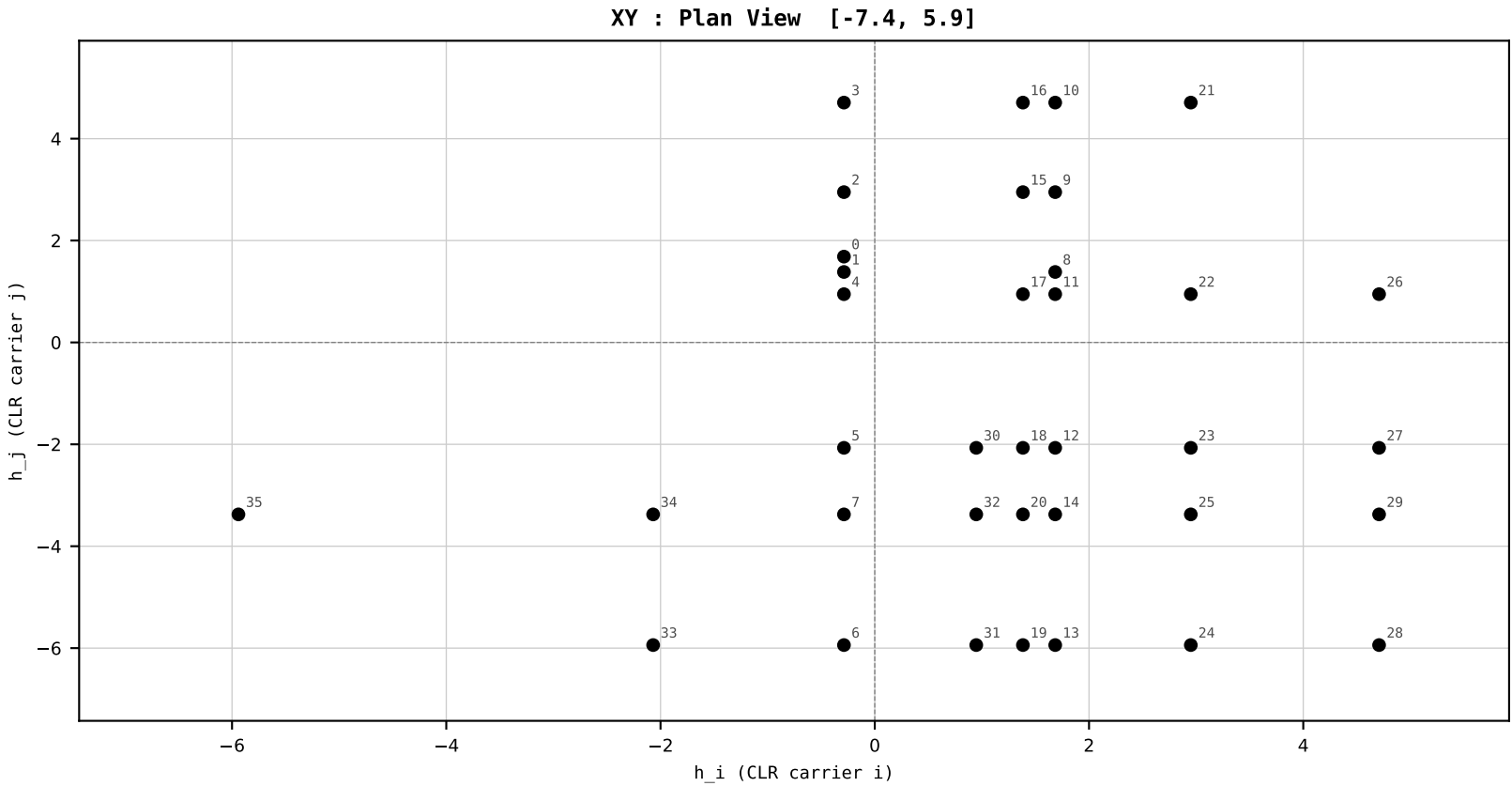
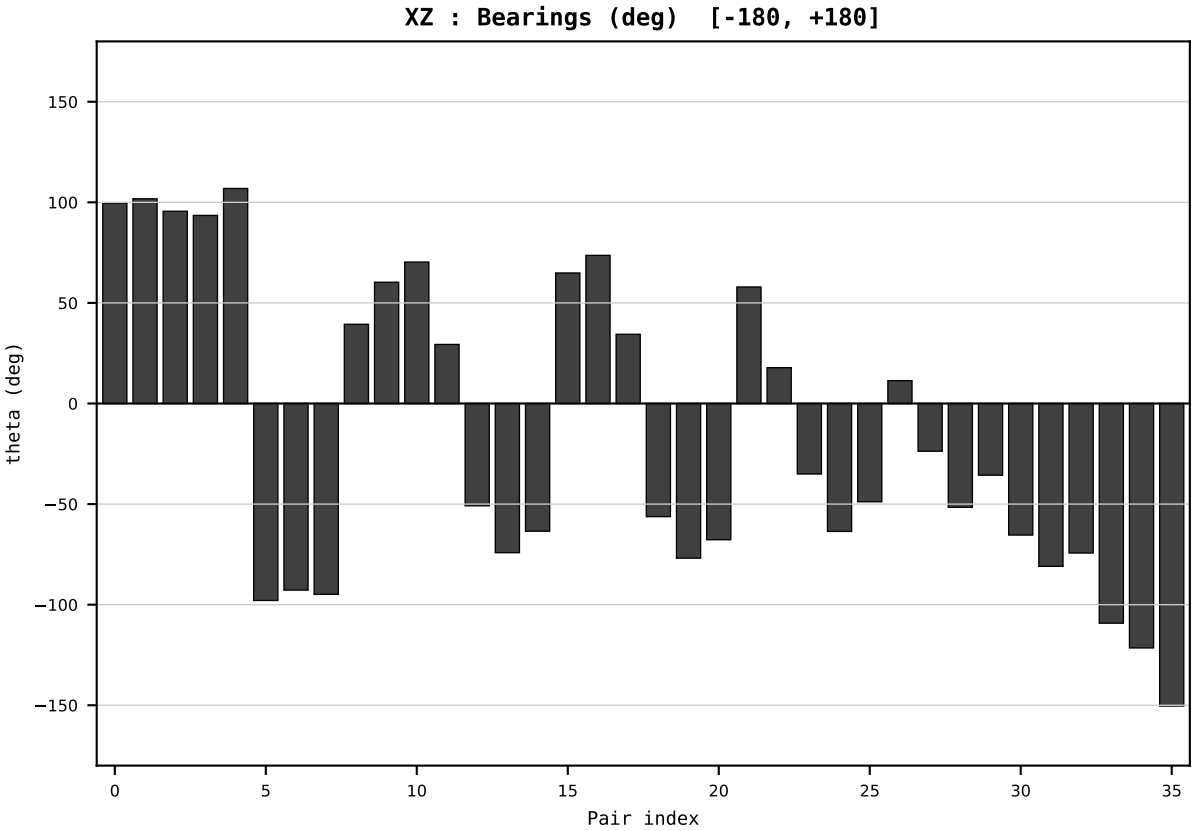
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2001
t : 1 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.636809
Ring = Hs-4
E_metric= 87.5562
kappa_HS= 42108.00
omega = 5.5939 deg
d_A = 0.997829
Helm = Wind
Helm d = +0.839979
DR = 10.648 HLR
DR ratio= 42108.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=2 | Year 2002 | D=9 N=26 pairs=36

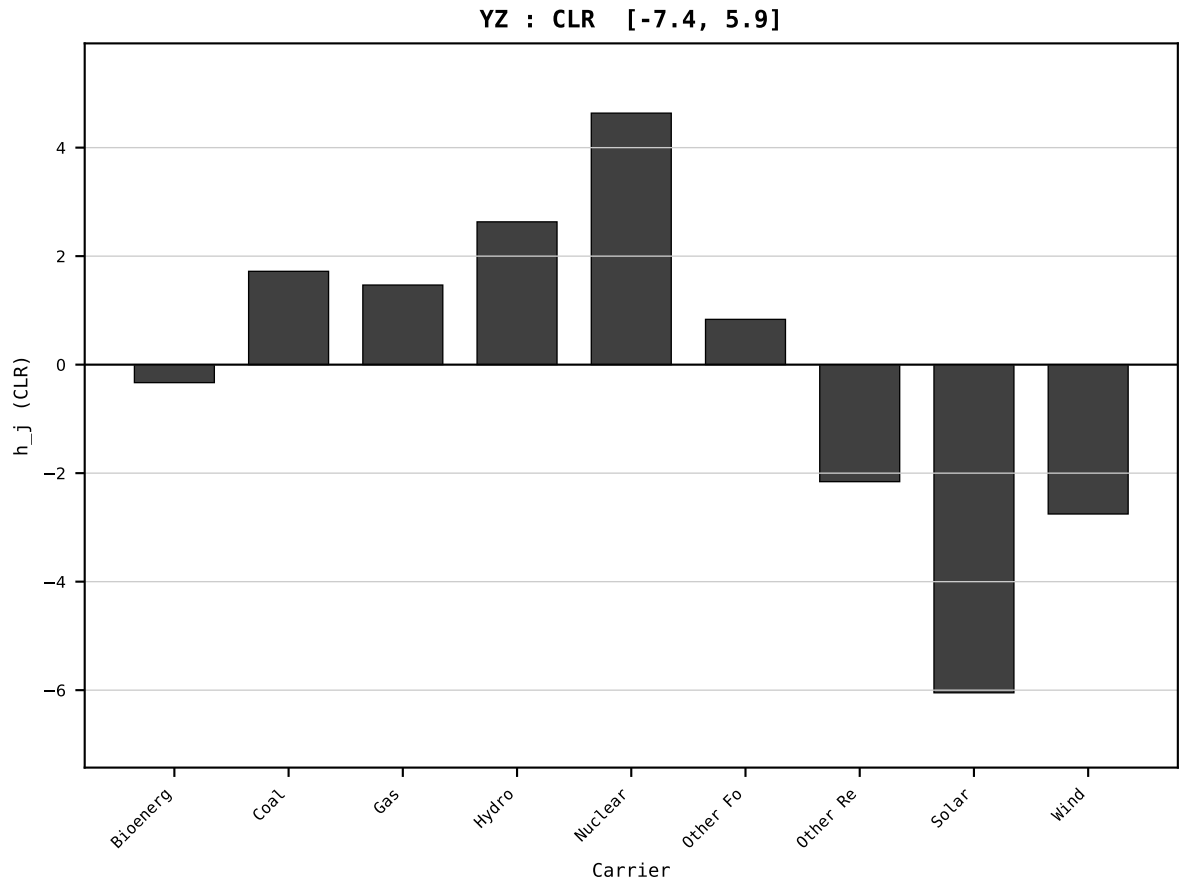
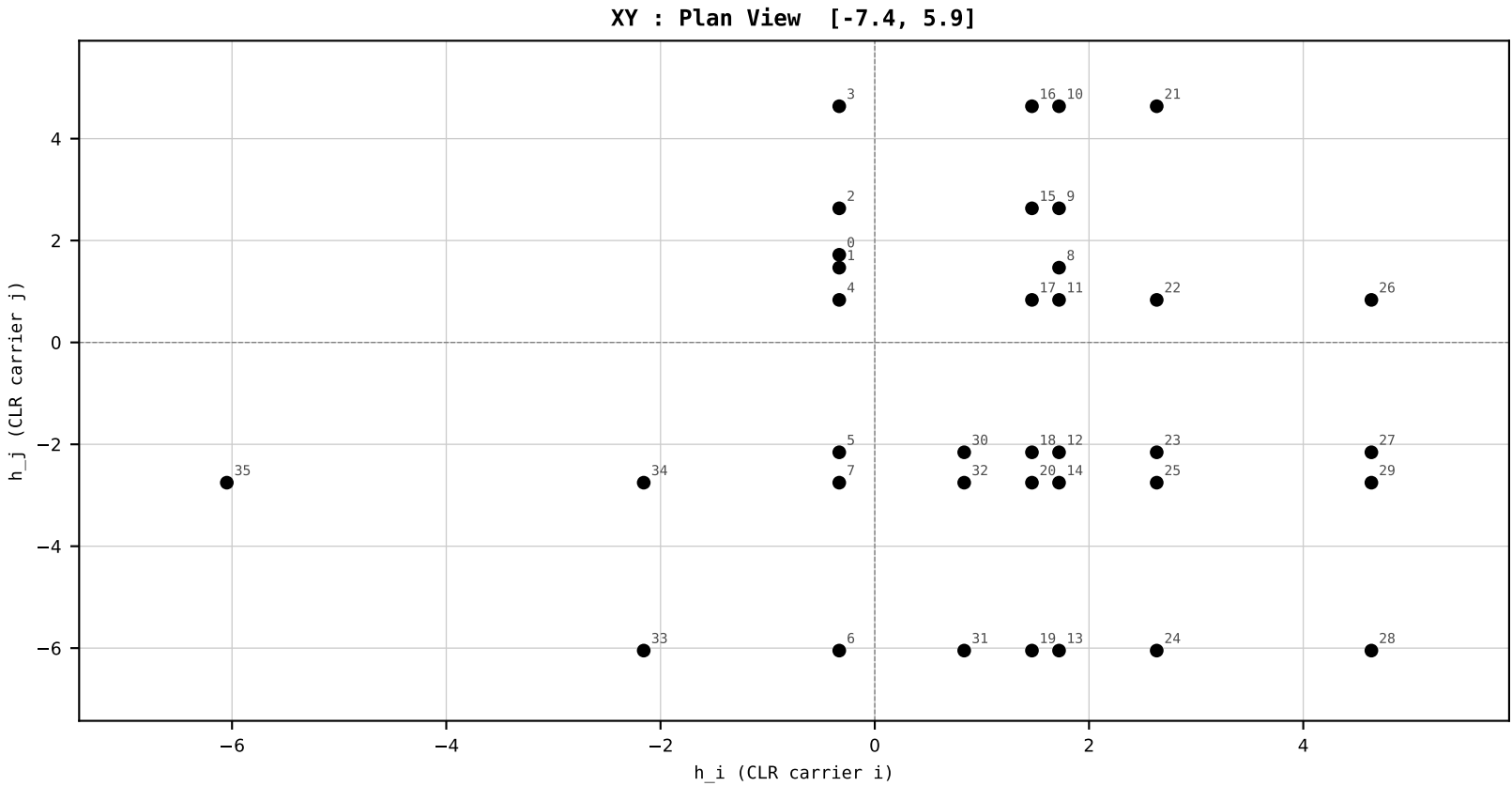
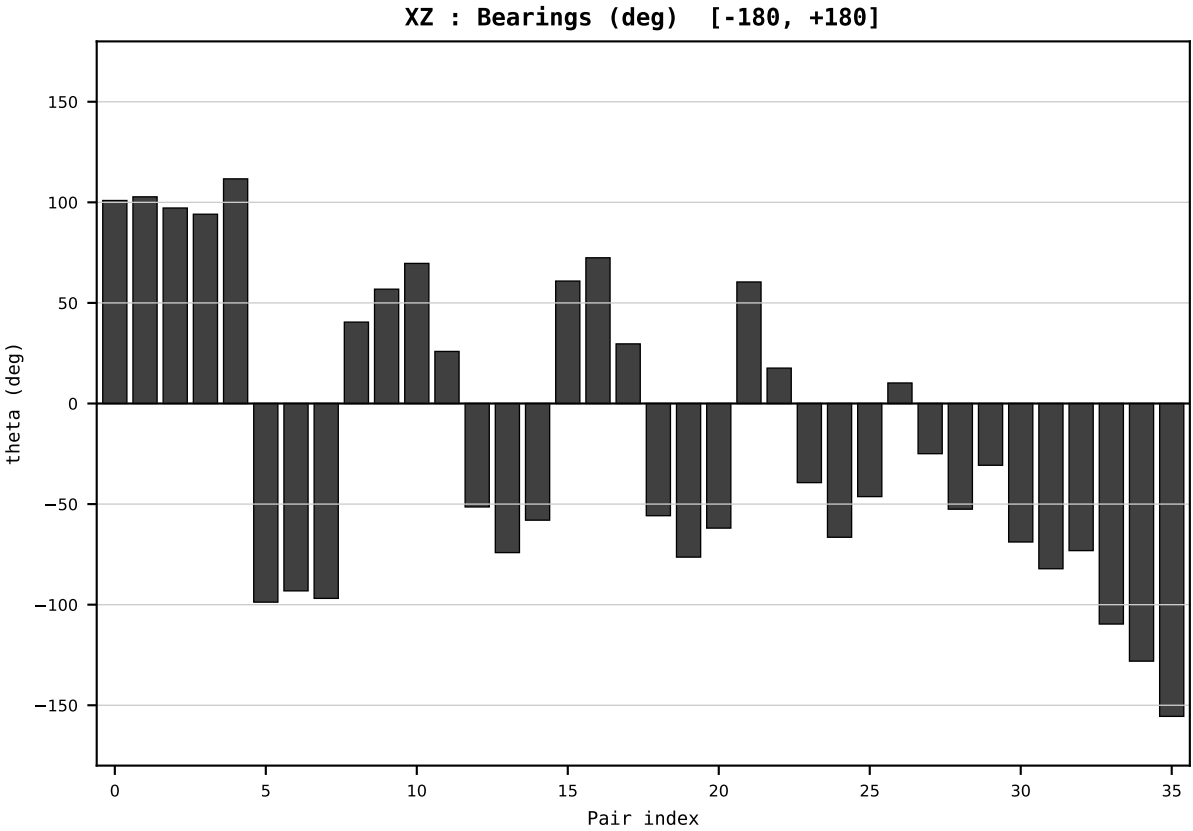
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2002
t : 2 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.644046
Ring = Hs-4
E_metric= 83.1543
kappa_HS= 43676.00
omega = 4.2963 deg
d_A = 0.732324
Helm = Wind
Helm d = +0.622462
DR = 10.685 HLR
DR ratio= 43676.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=3 | Year 2003 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

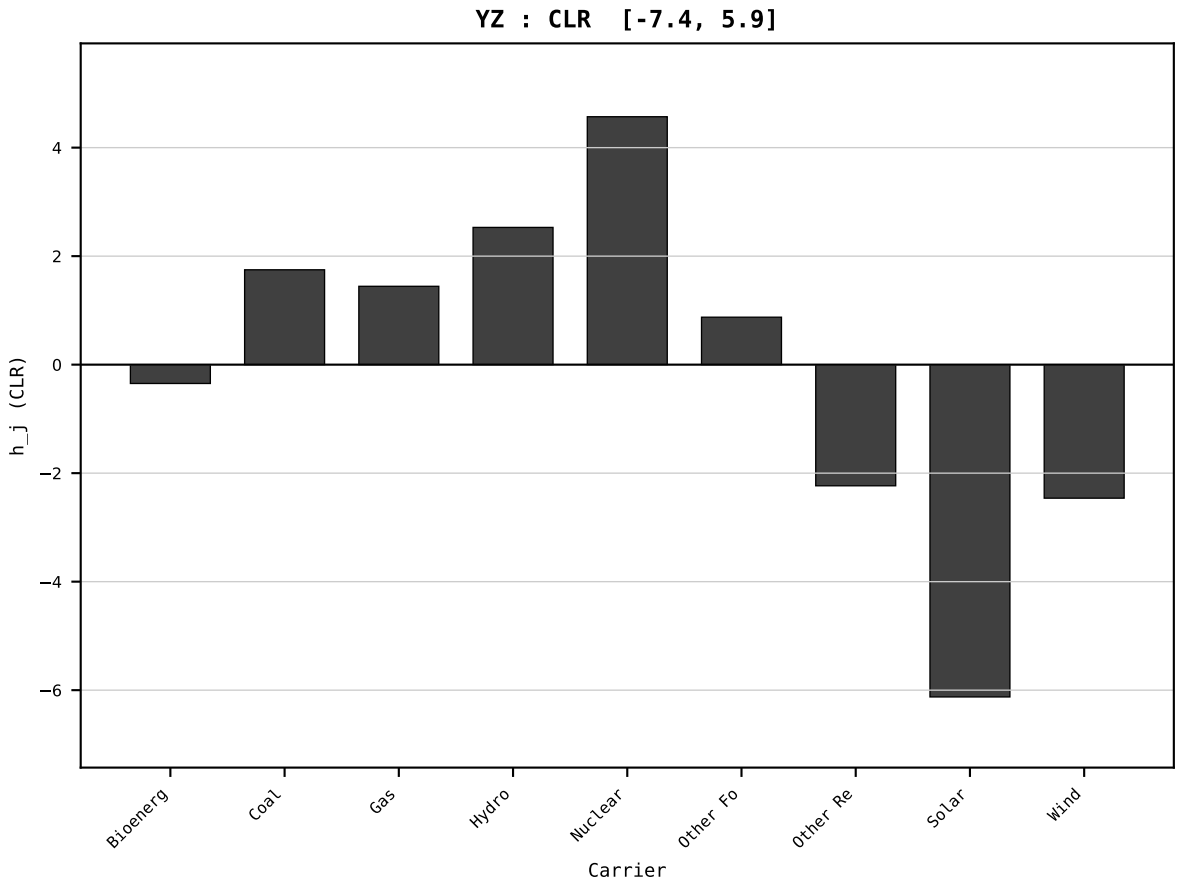
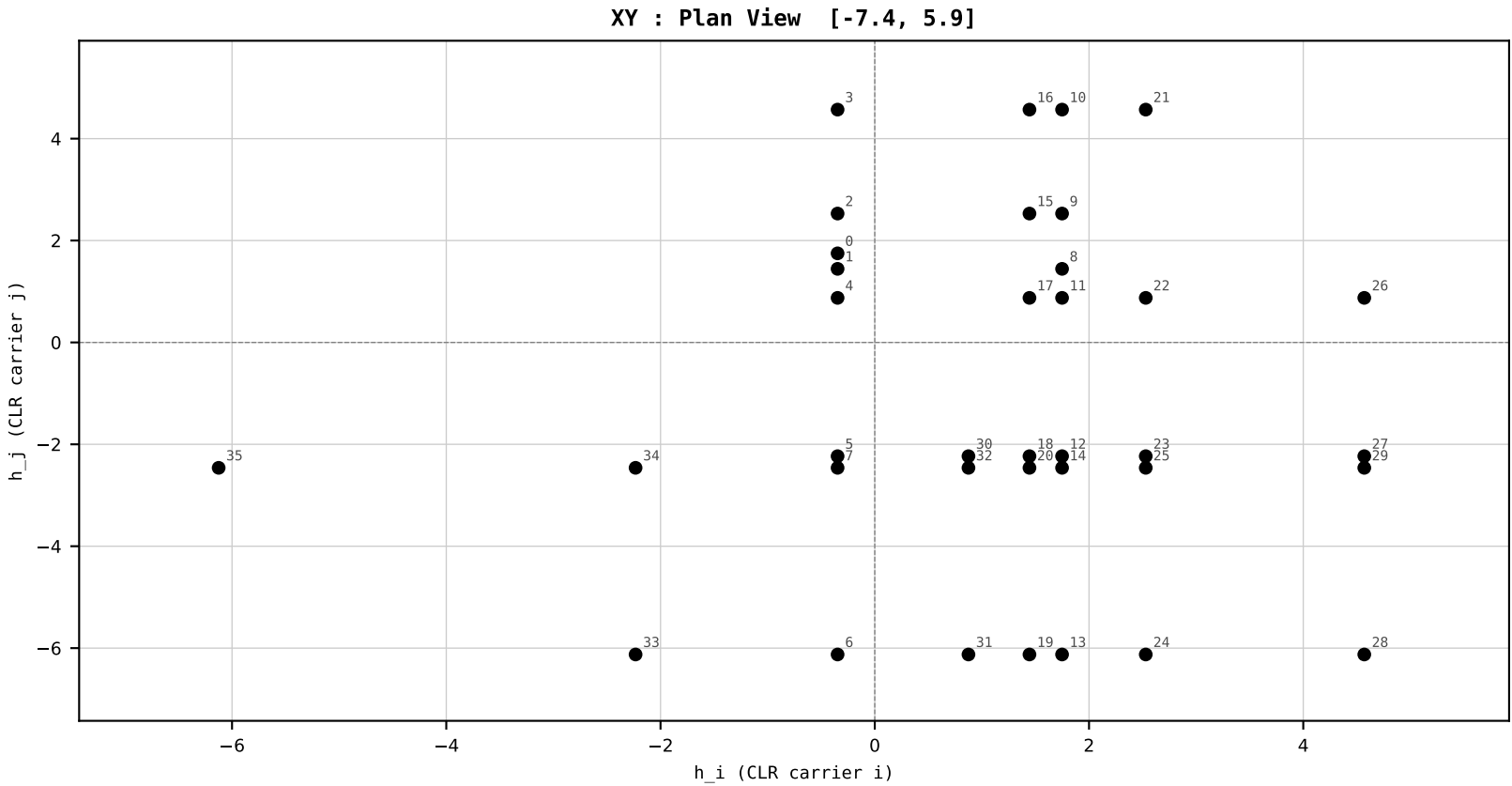
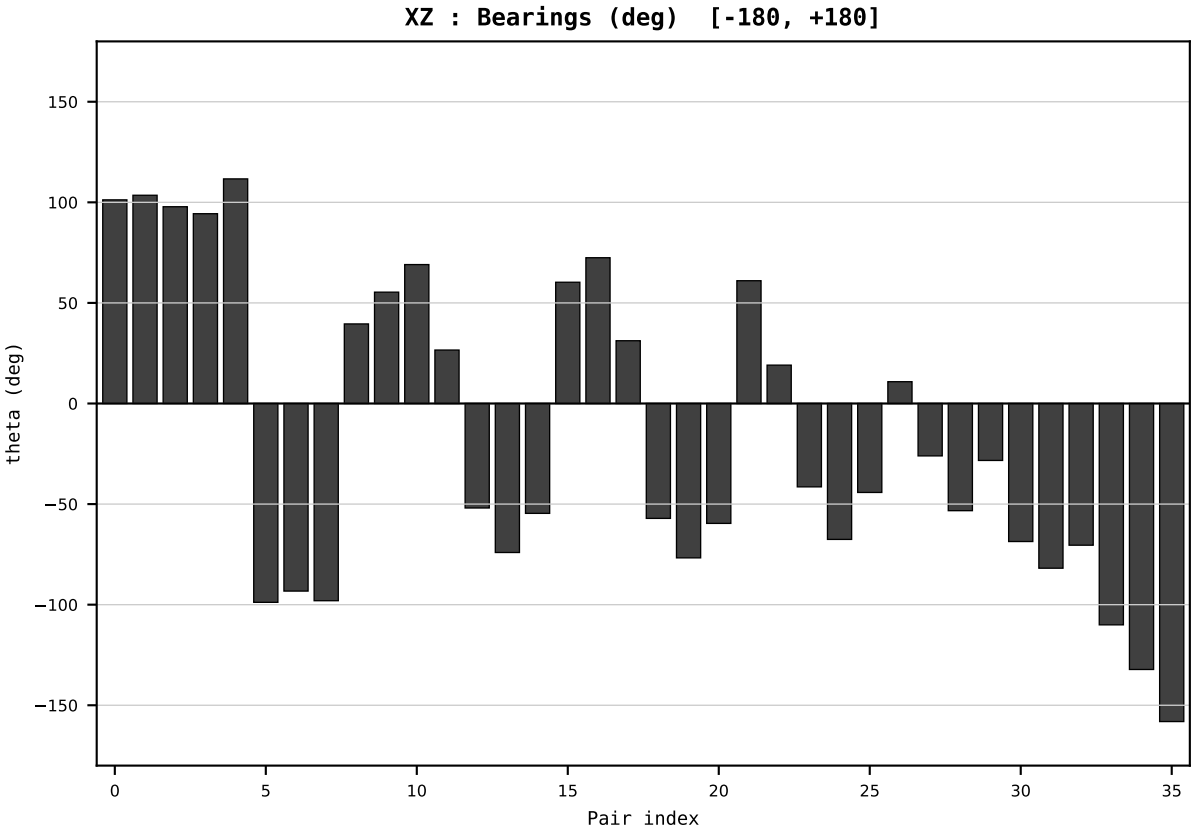
Dataset : ember_FRA_France_generation_TWh.csv
Year : 2003
t : 3 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.636320
Ring = Hs-4
E_metric= 81.8645
kappa_HS= 44107.00
omega = 2.0886 deg
d_A = 0.338625
Helm = Wind
Helm d = +0.291606
DR = 10.694 HLR
DR ratio= 44107.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=4 | Year 2004 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

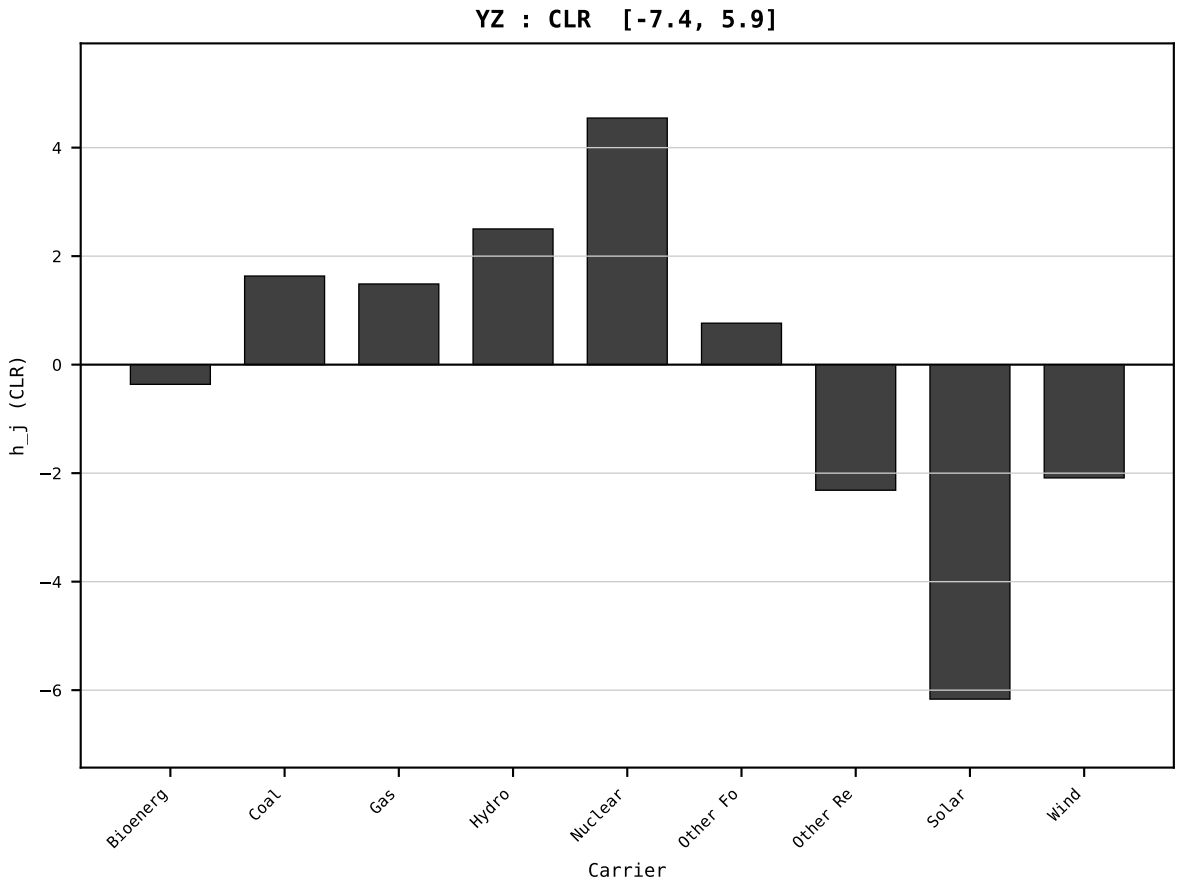
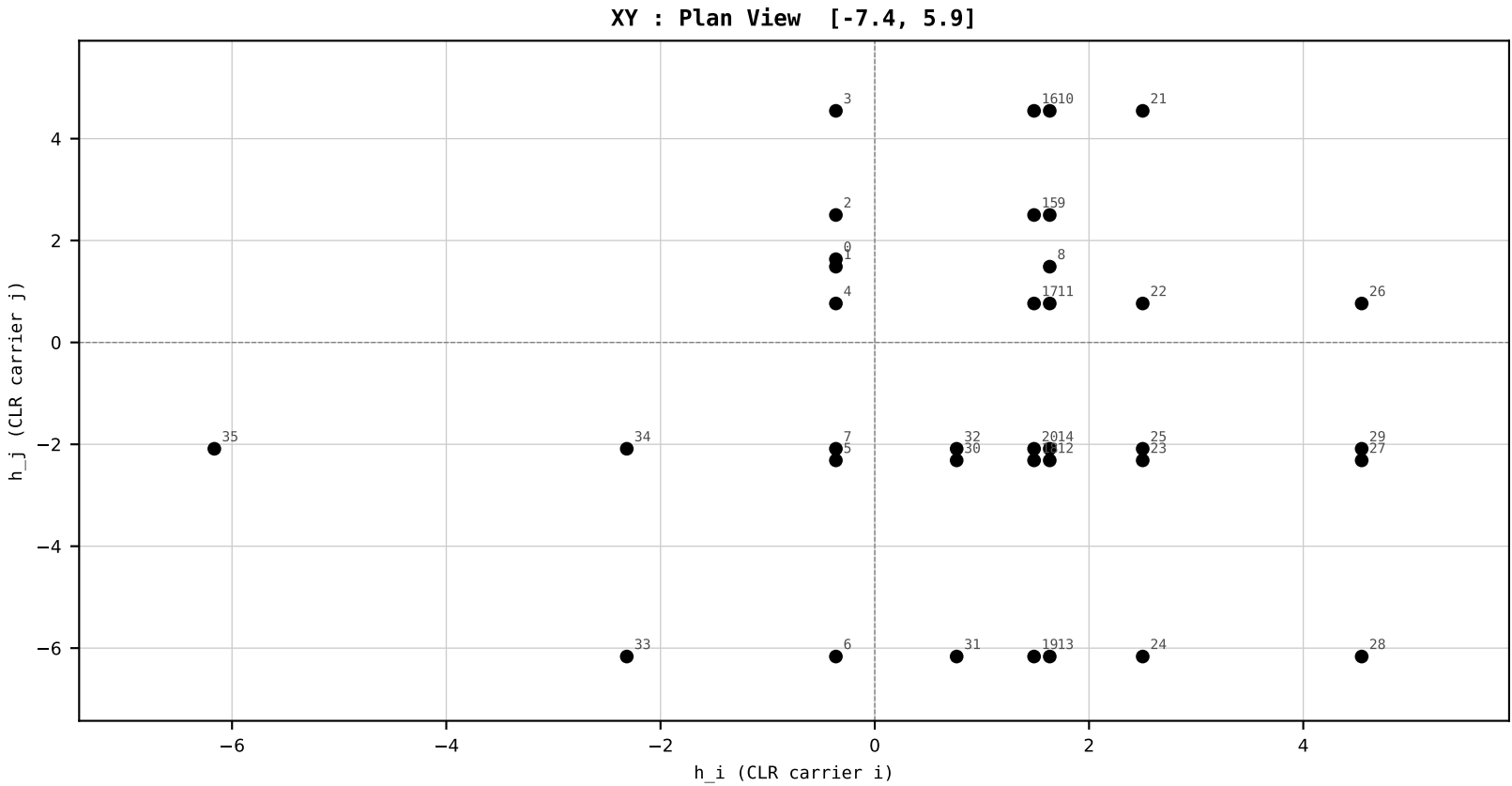
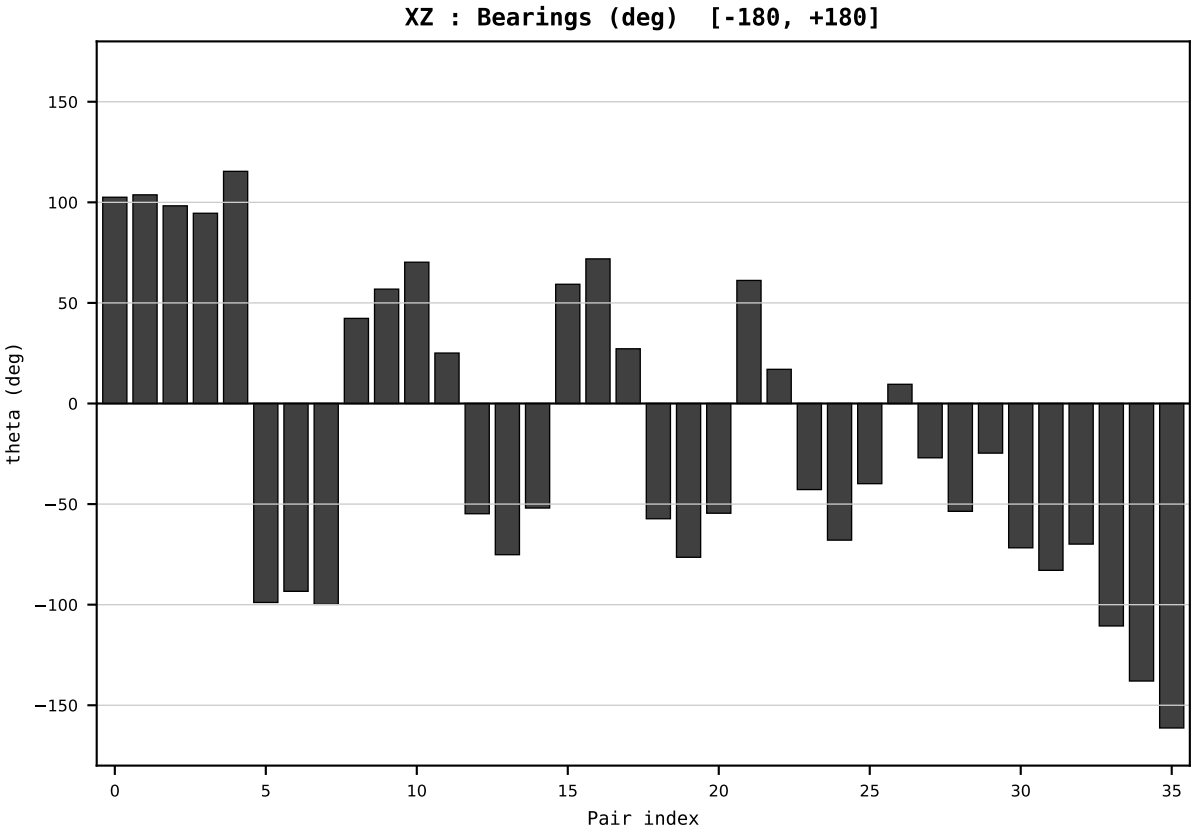
Dataset : ember_FRA_France_generation_TWh.csv
Year : 2004
t : 4 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.639536
Ring = Hs-4
E_metric= 80.2373
kappa_HS= 44824.00
omega = 2.6139 deg
d_A = 0.420503
Helm = Wind
Helm d = +0.373595
DR = 10.710 HLR
DR ratio= 44824.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

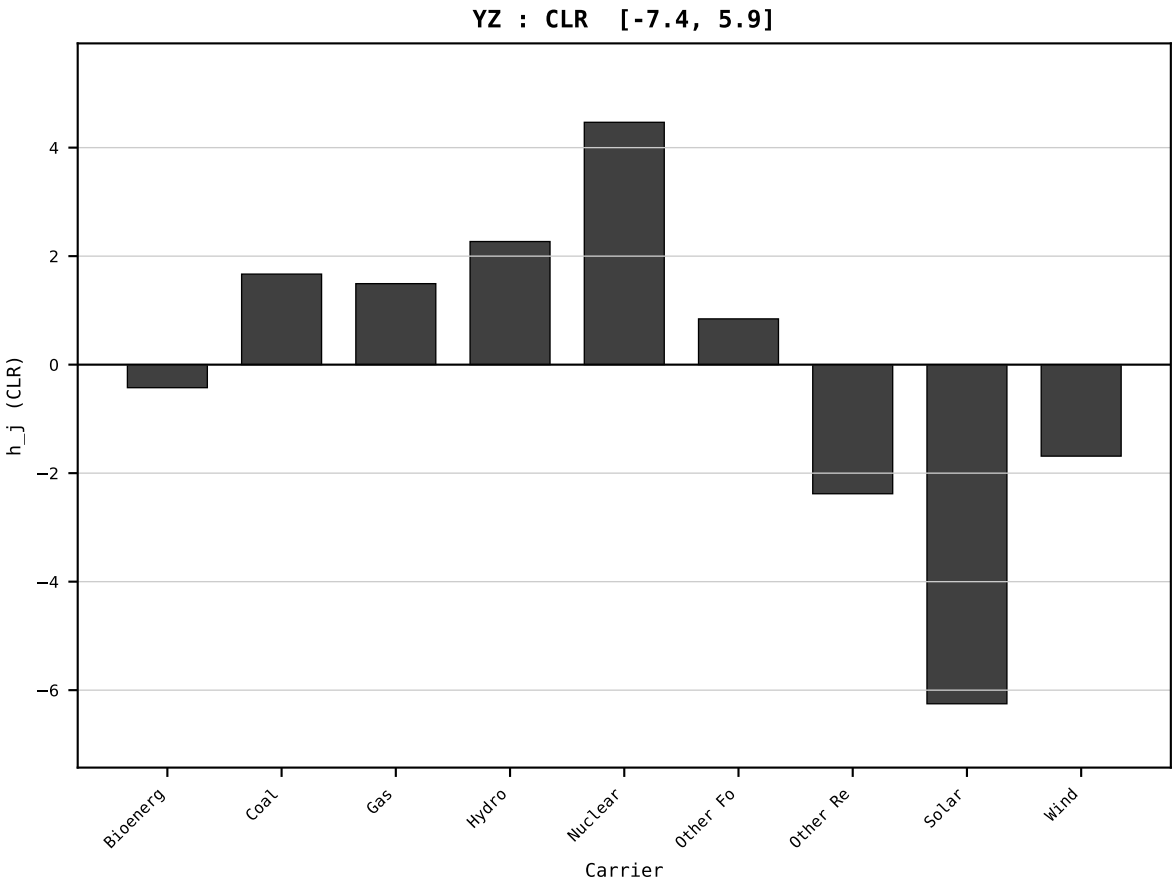
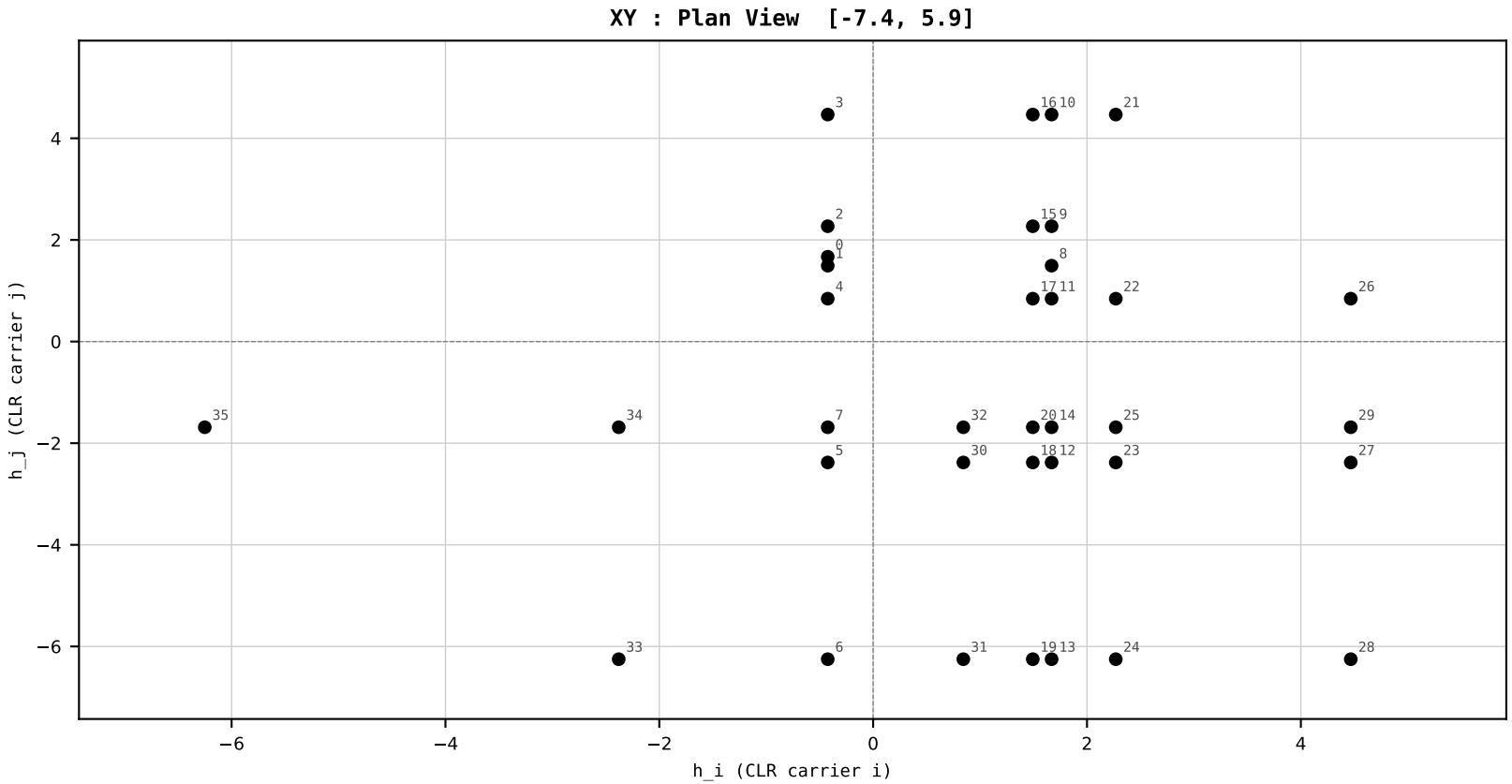
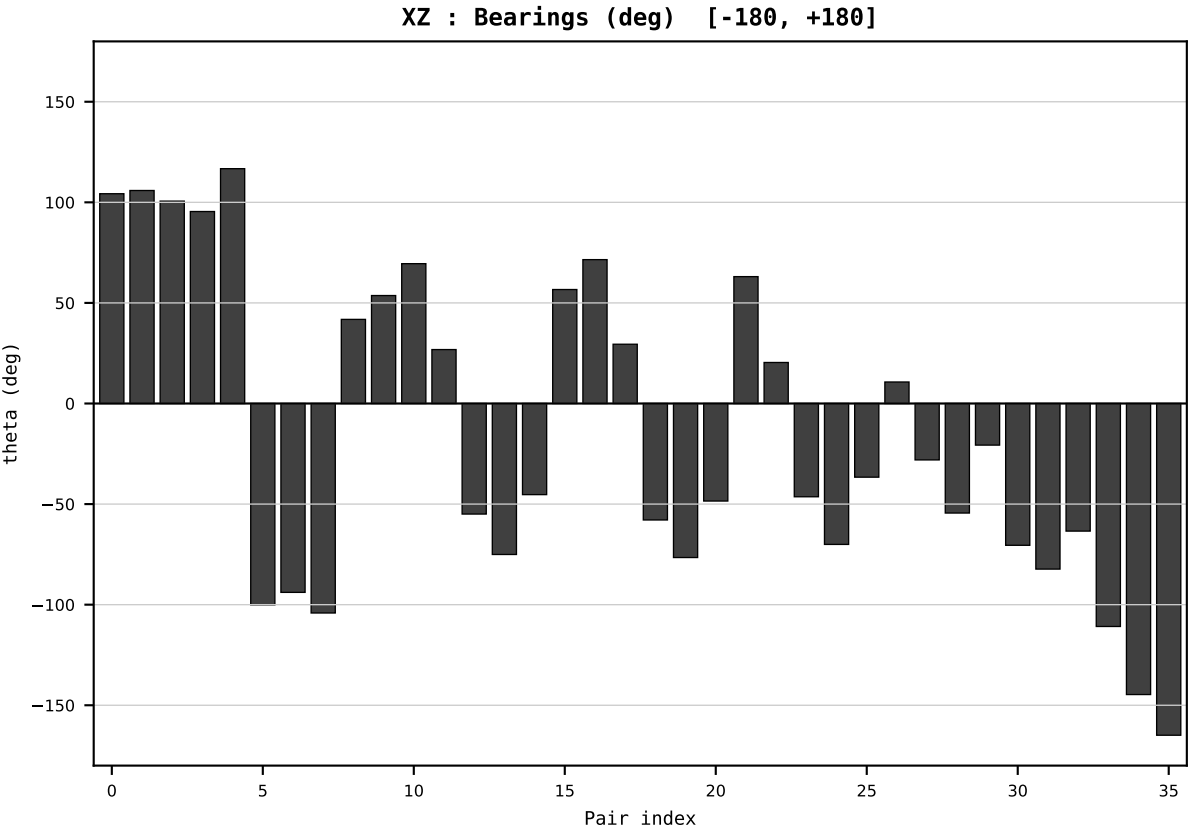
Dataset : ember_FRA_France_generation_TWh.csv
Year : 2005
t : 5 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.634489
Ring = Hs-4
E_metric= 78.5884
kappa_HS= 45153.00
omega = 3.1185 deg
d_A = 0.493709
Helm = Wind
Helm d = +0.401223
DR = 10.718 HLR
DR ratio= 45153.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=6 | Year 2006 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

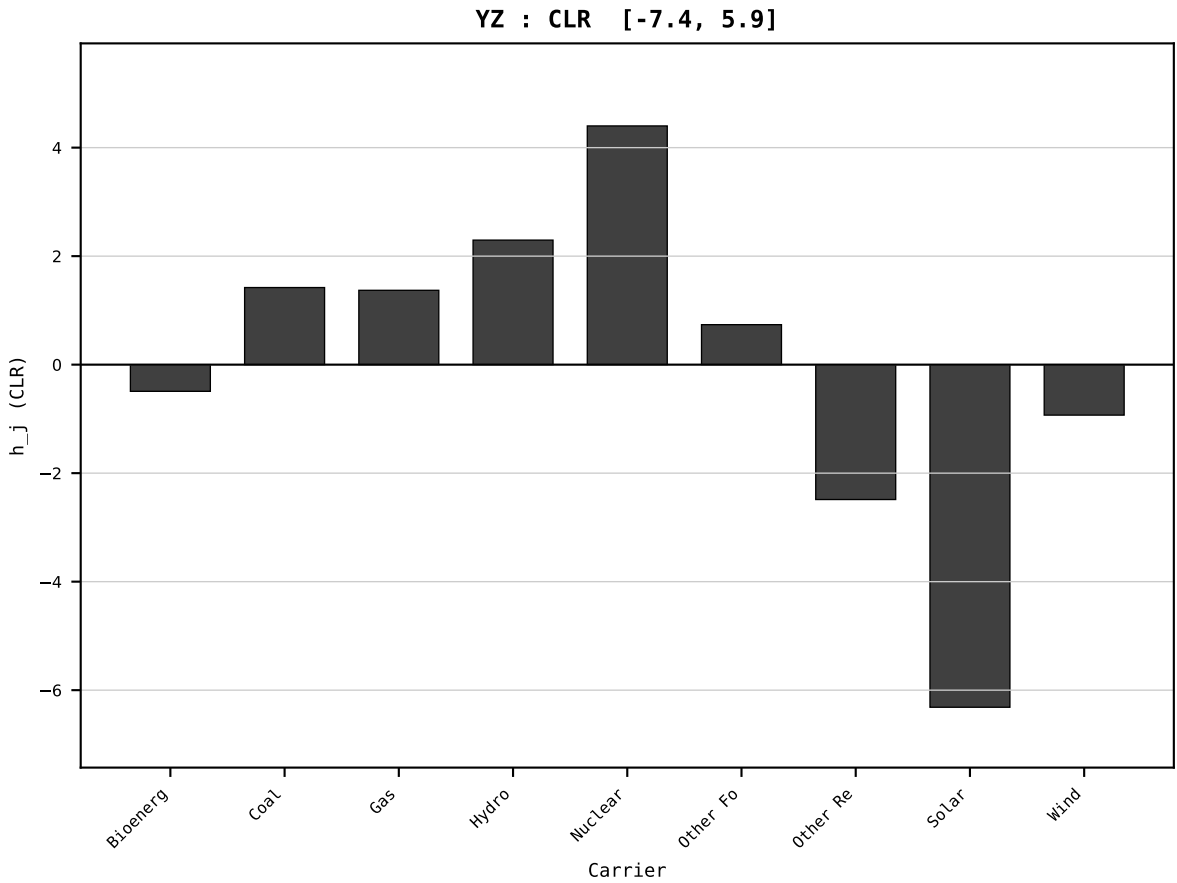
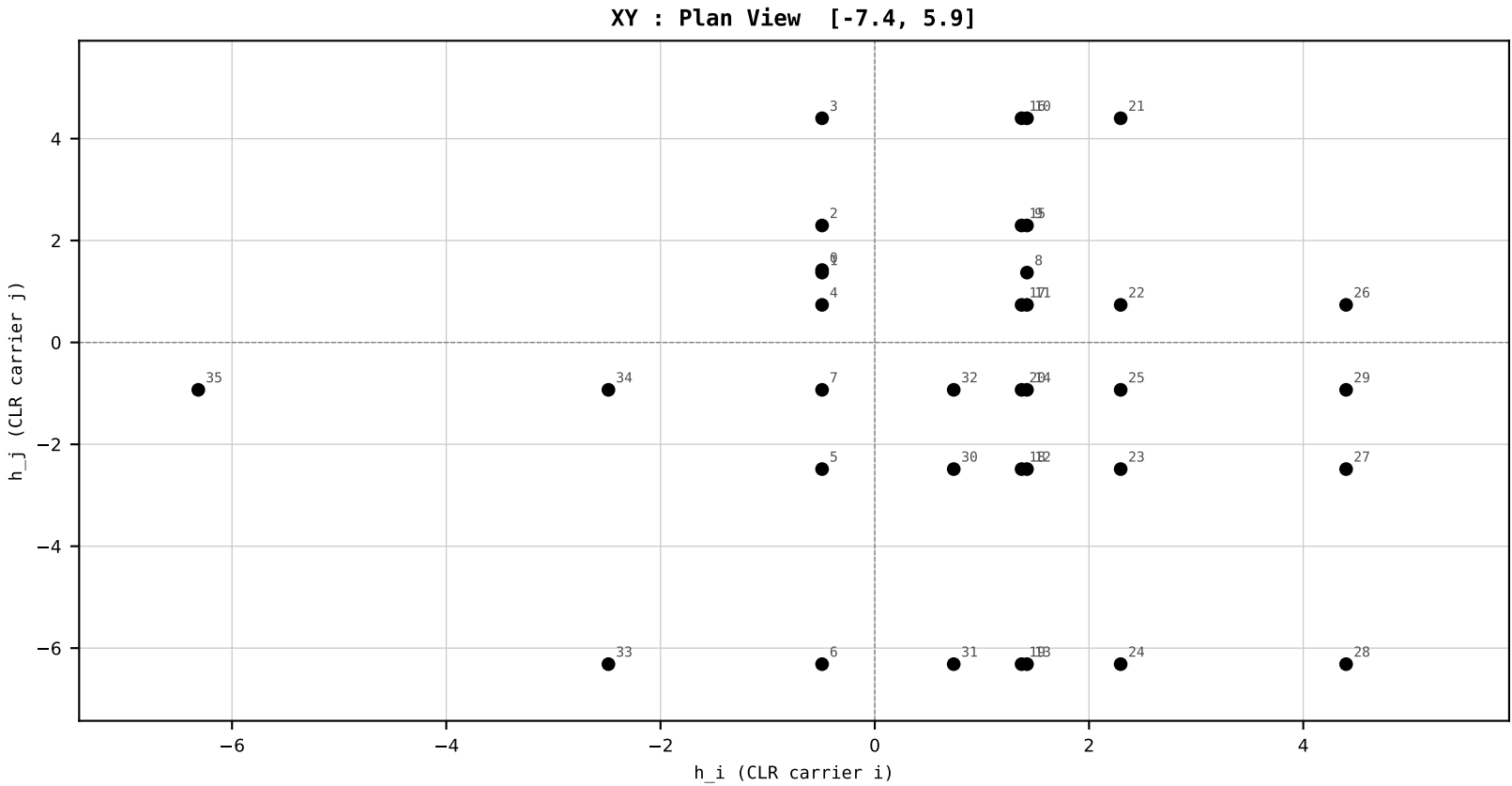
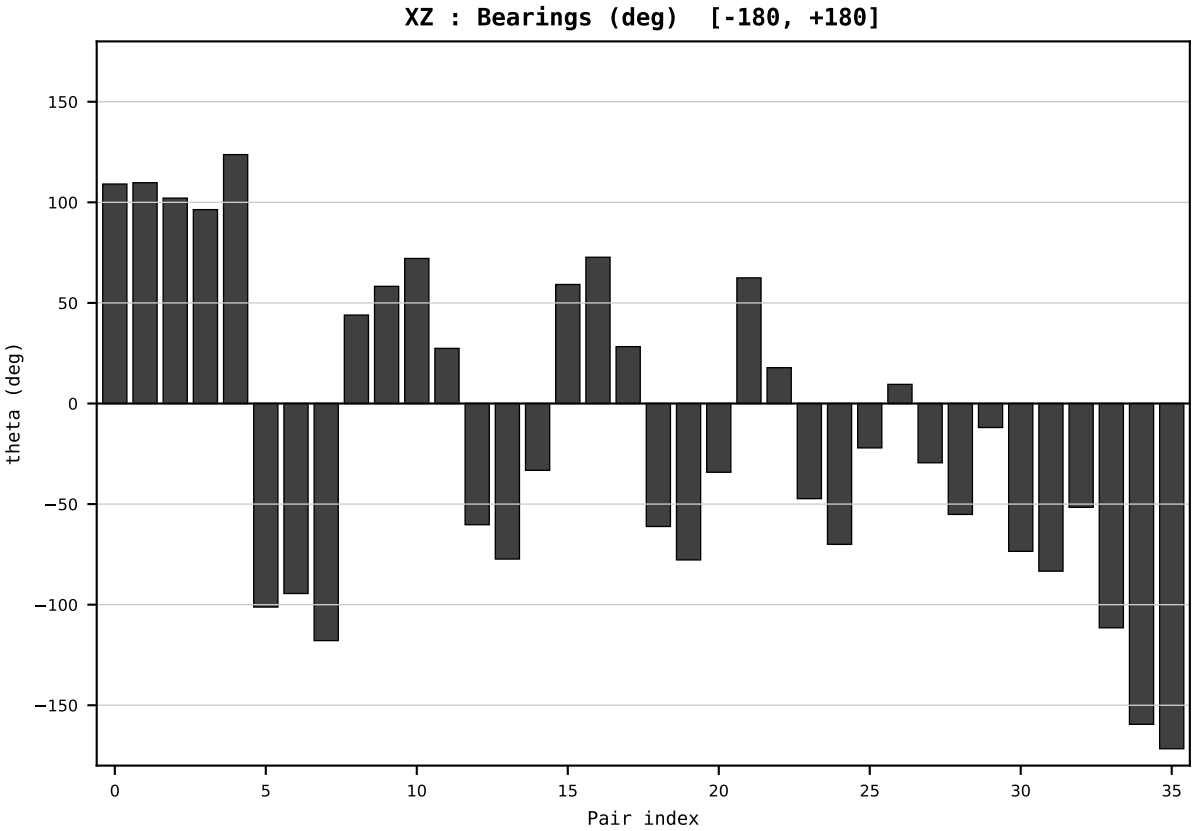
Dataset : ember_FRA_France_generation_TWh.csv
Year : 2006
t : 6 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.635362
Ring = Hs-4
E_metric= 76.2355
kappa_HS= 45019.00
omega = 5.3196 deg
d_A = 0.827413
Helm = Wind
Helm d = +0.756002
DR = 10.715 HLR
DR ratio= 45019.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=7 | Year 2007 | D=9 N=26 pairs=36

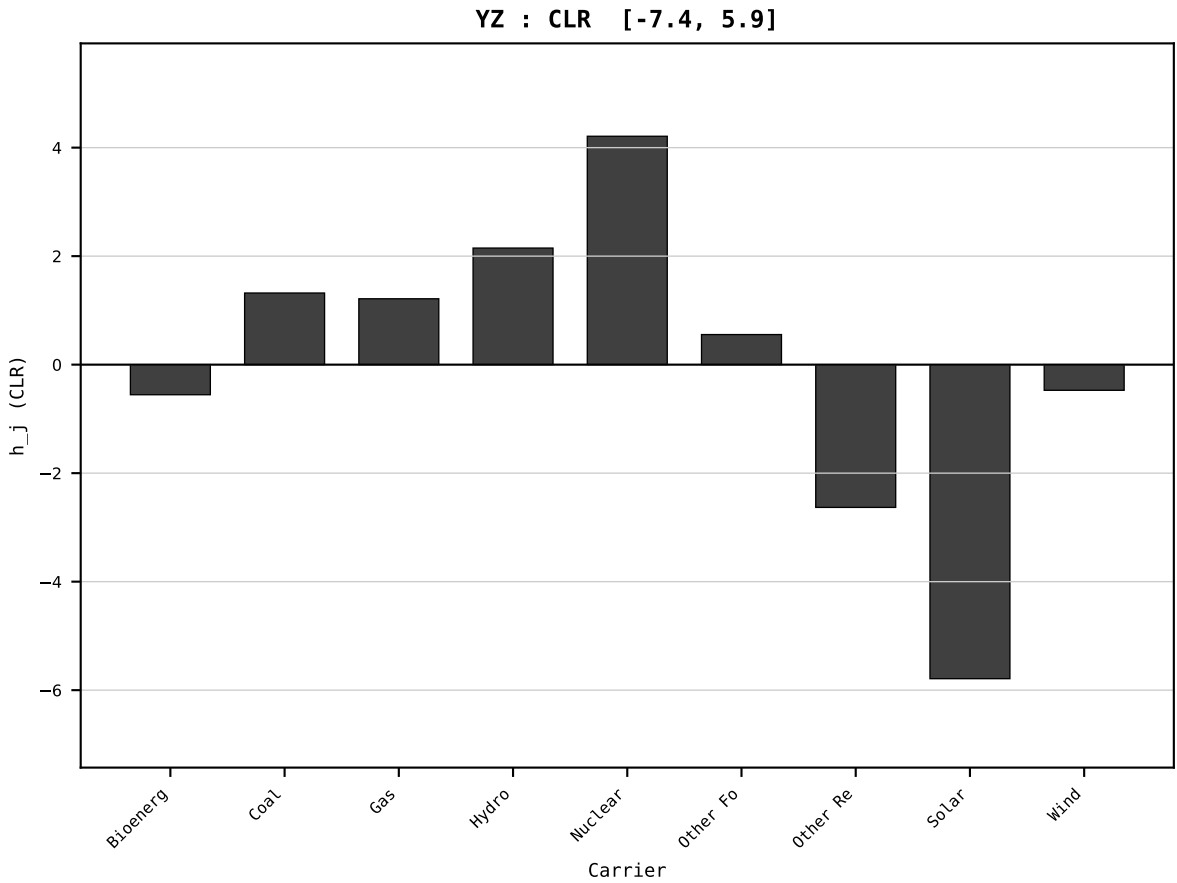
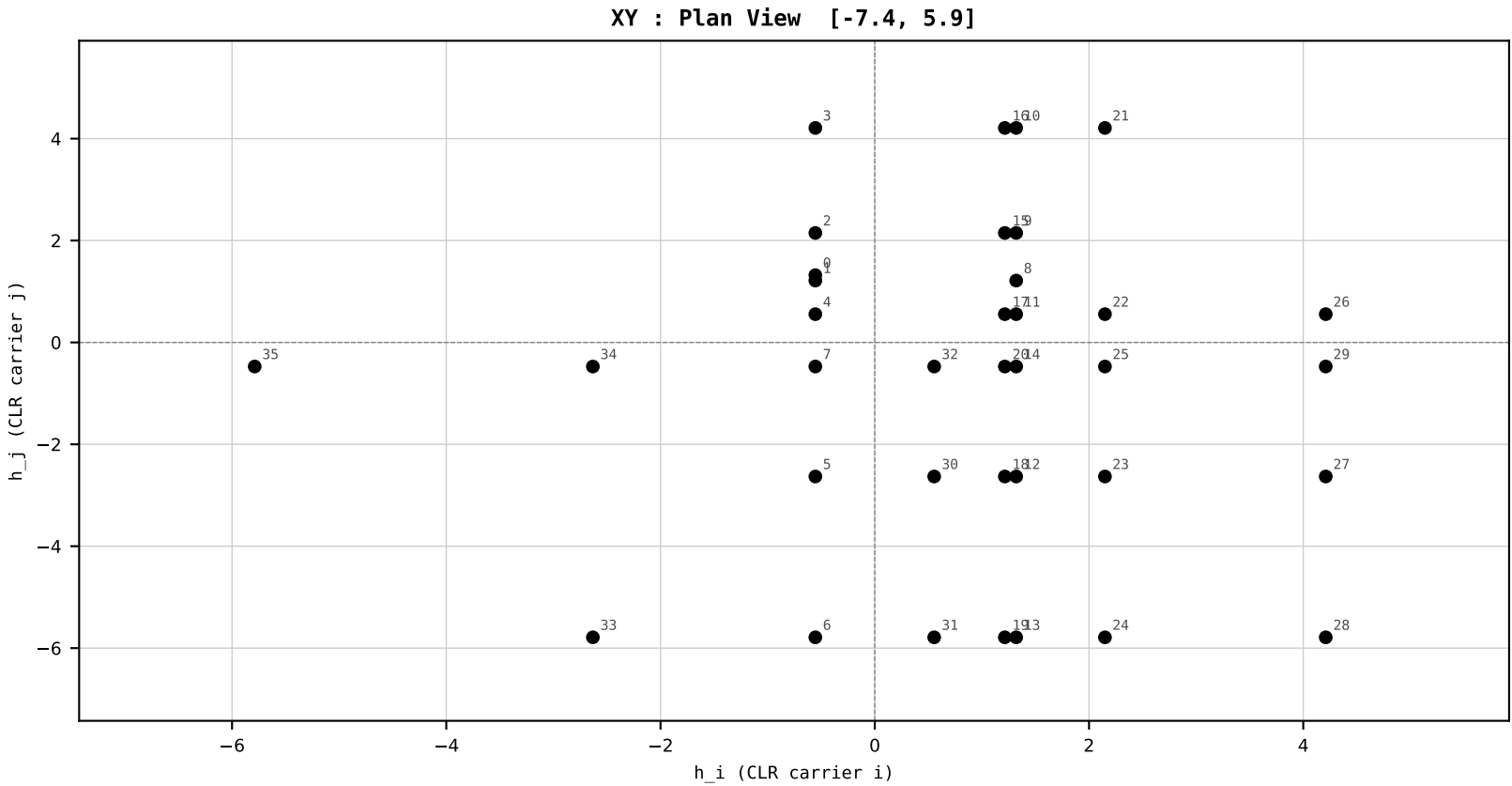
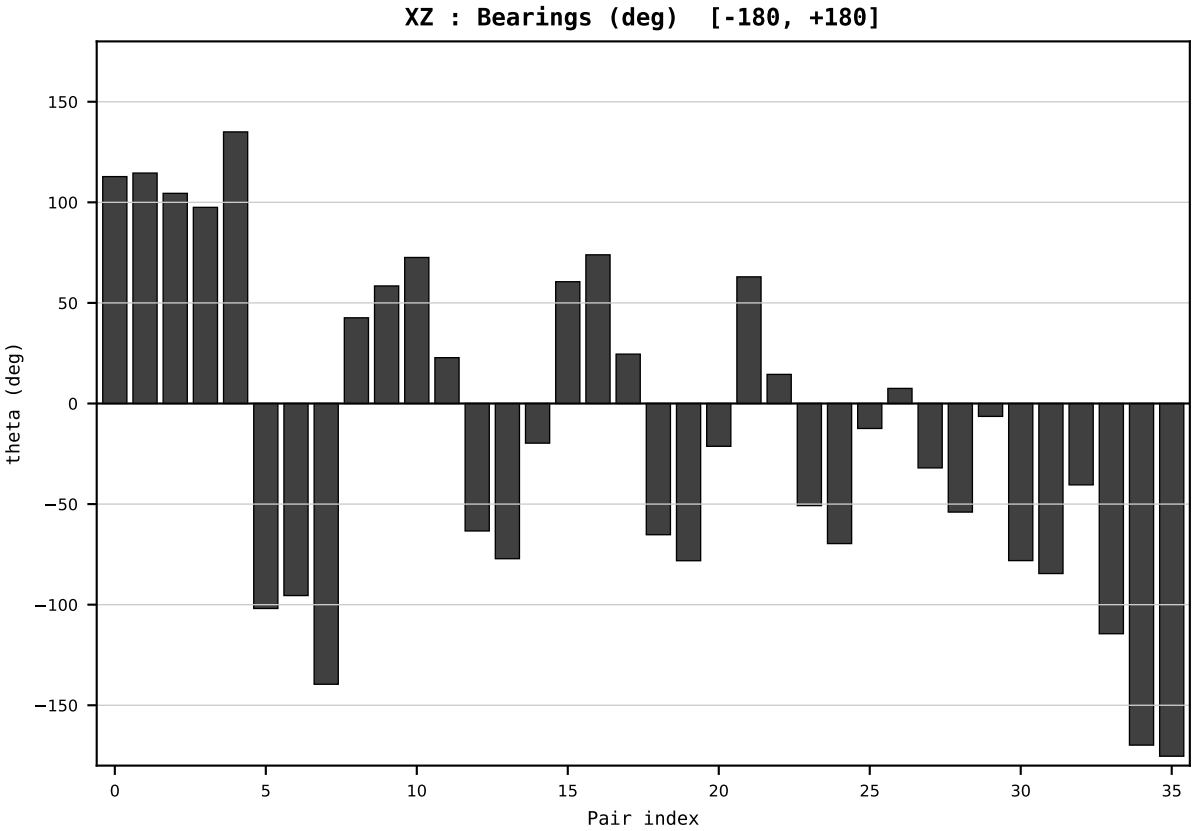
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2007
t : 7 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.618062
Ring = Hs-4
E_metric= 66.8255
kappa_HS= 21986.50
omega = 3.8806 deg
d_A = 0.798187
Helm = Solar
Helm d = +0.526469
DR = 9.998 HLR
DR ratio= 21986.5x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=8 | Year 2008 | D=9 N=26 pairs=36

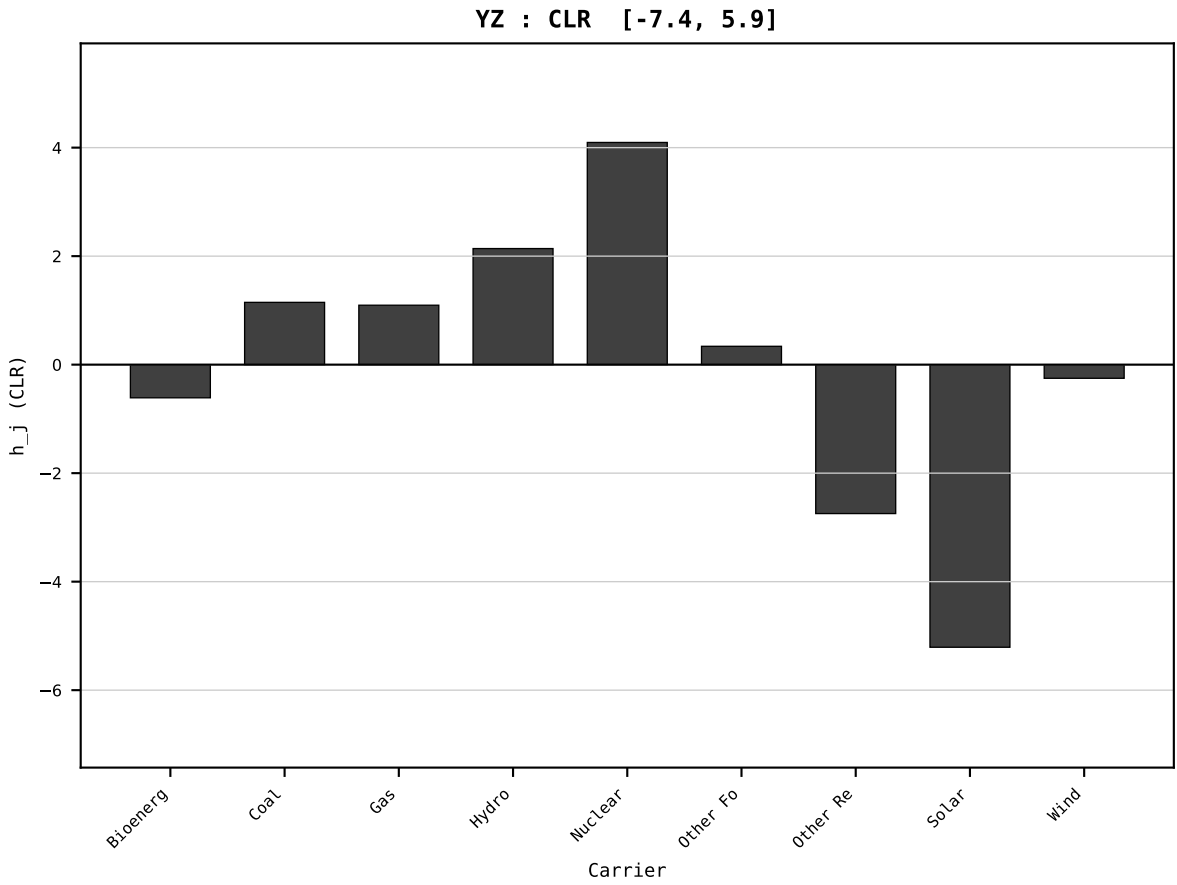
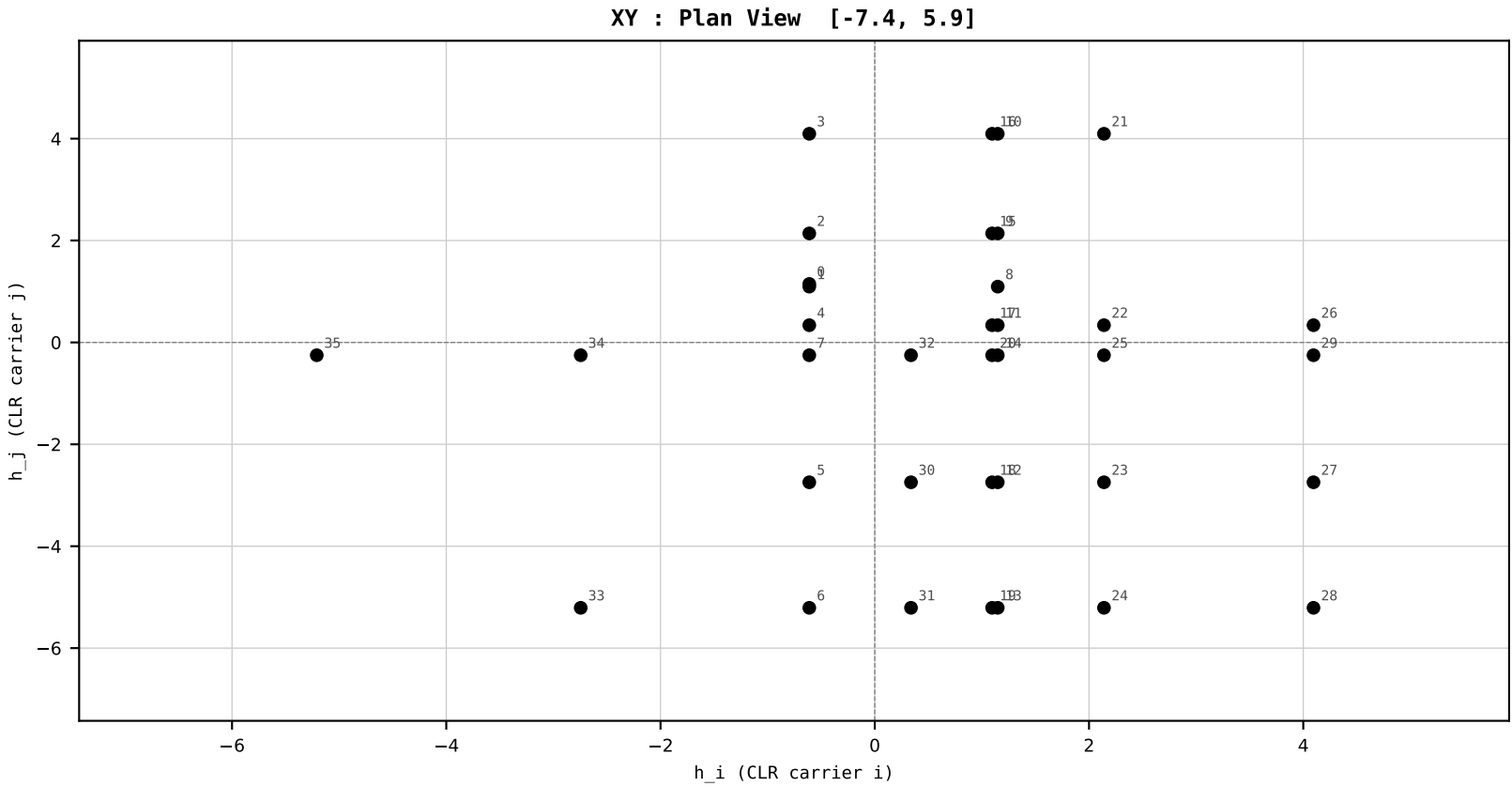
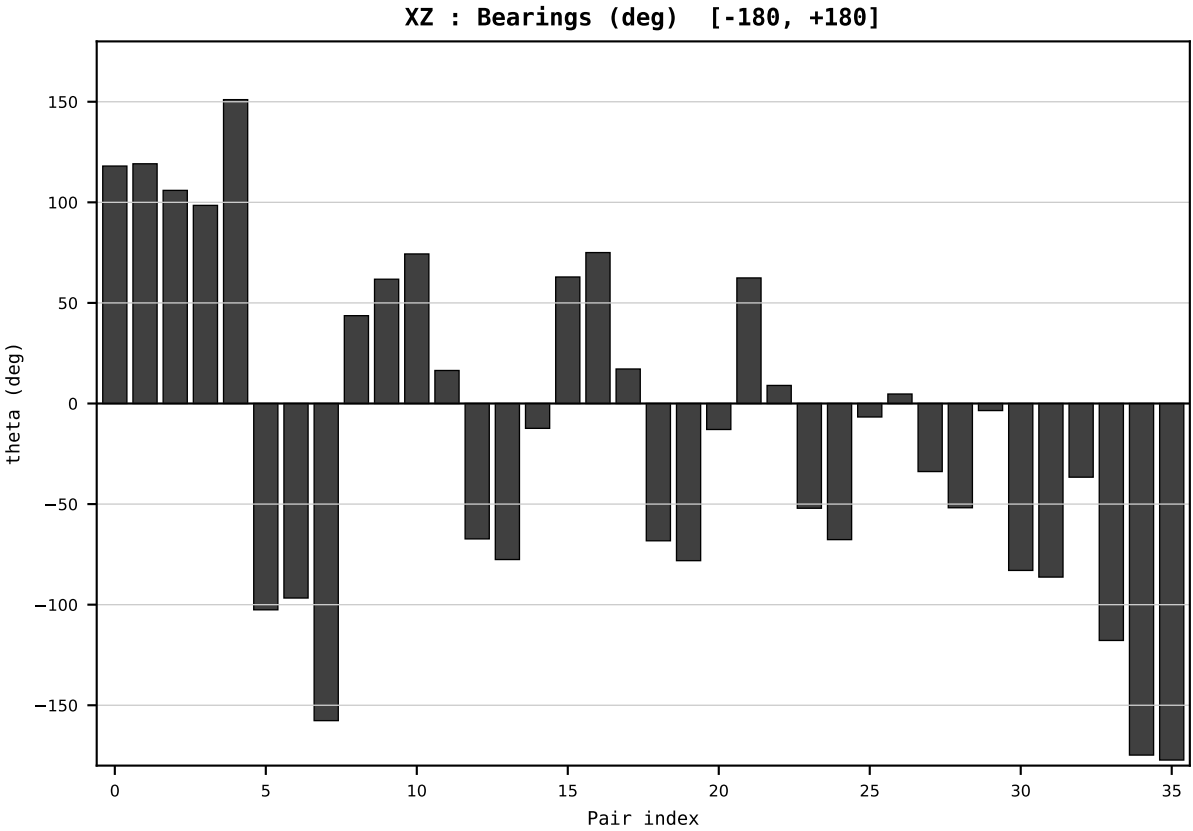
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2008
t : 8 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.610524
Ring = Hs-4
E_metric= 59.0888
kappa_HS= 10986.25
omega = 3.7337 deg
d_A = 0.710392
Helm = Solar
Helm d = +0.579497
DR = 9.304 HLR
DR ratio= 10986.3x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



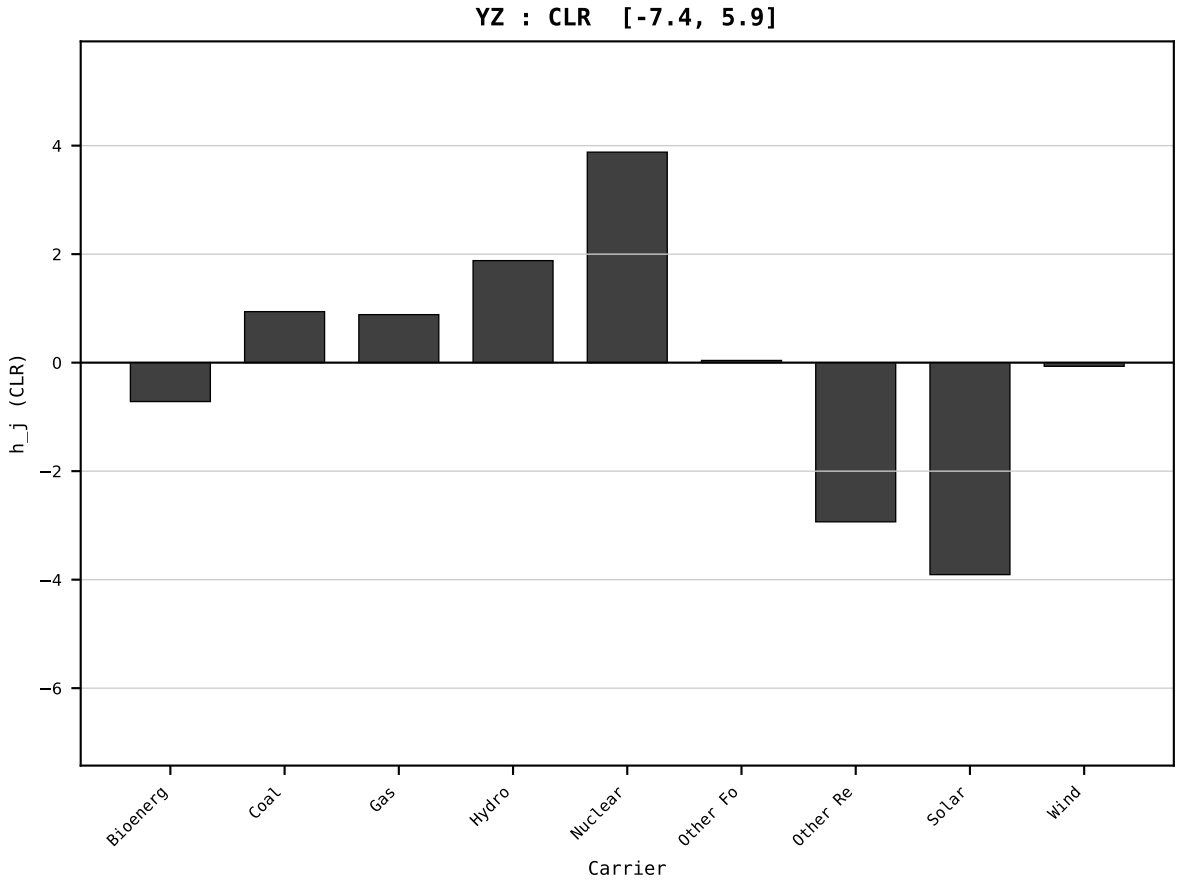
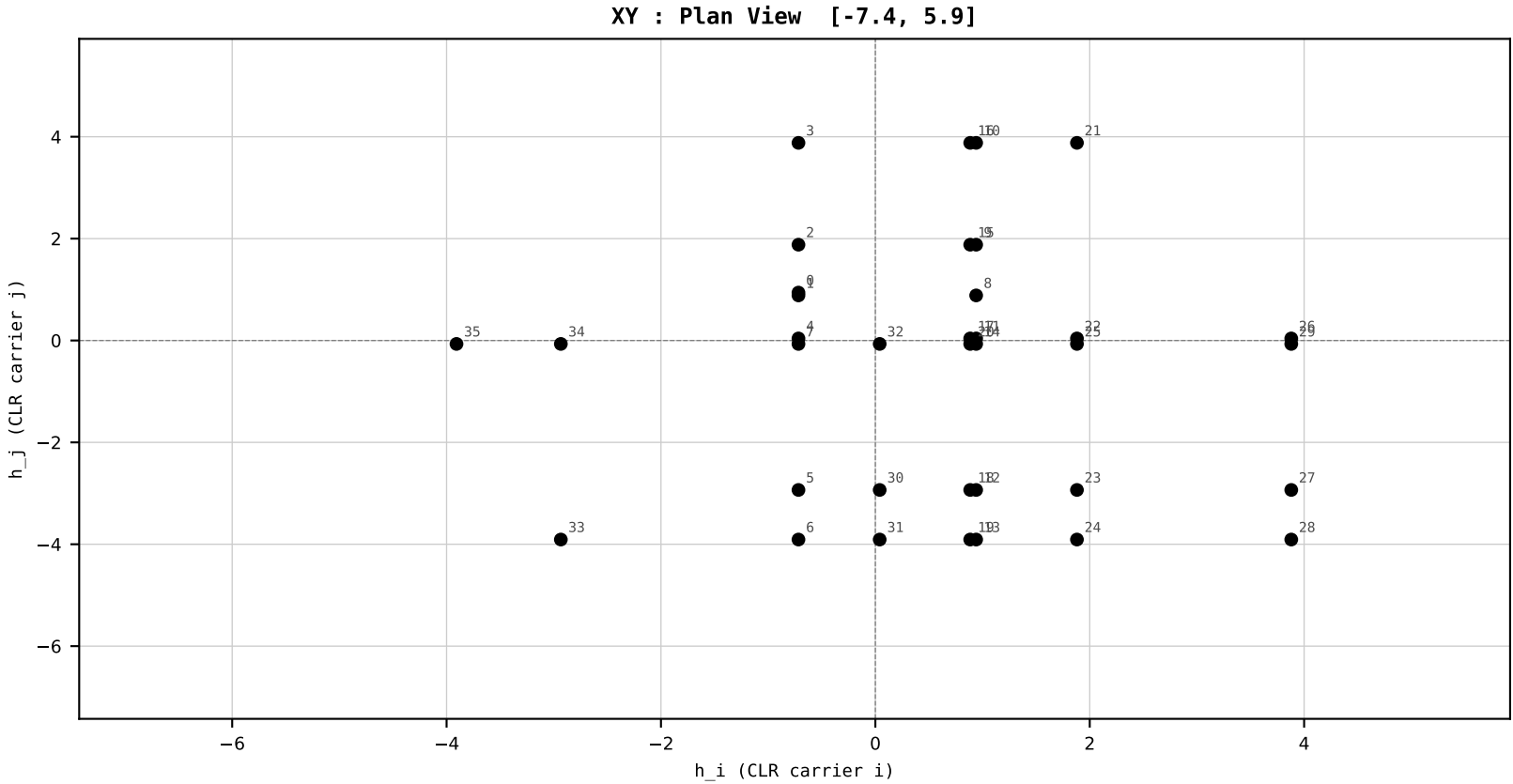
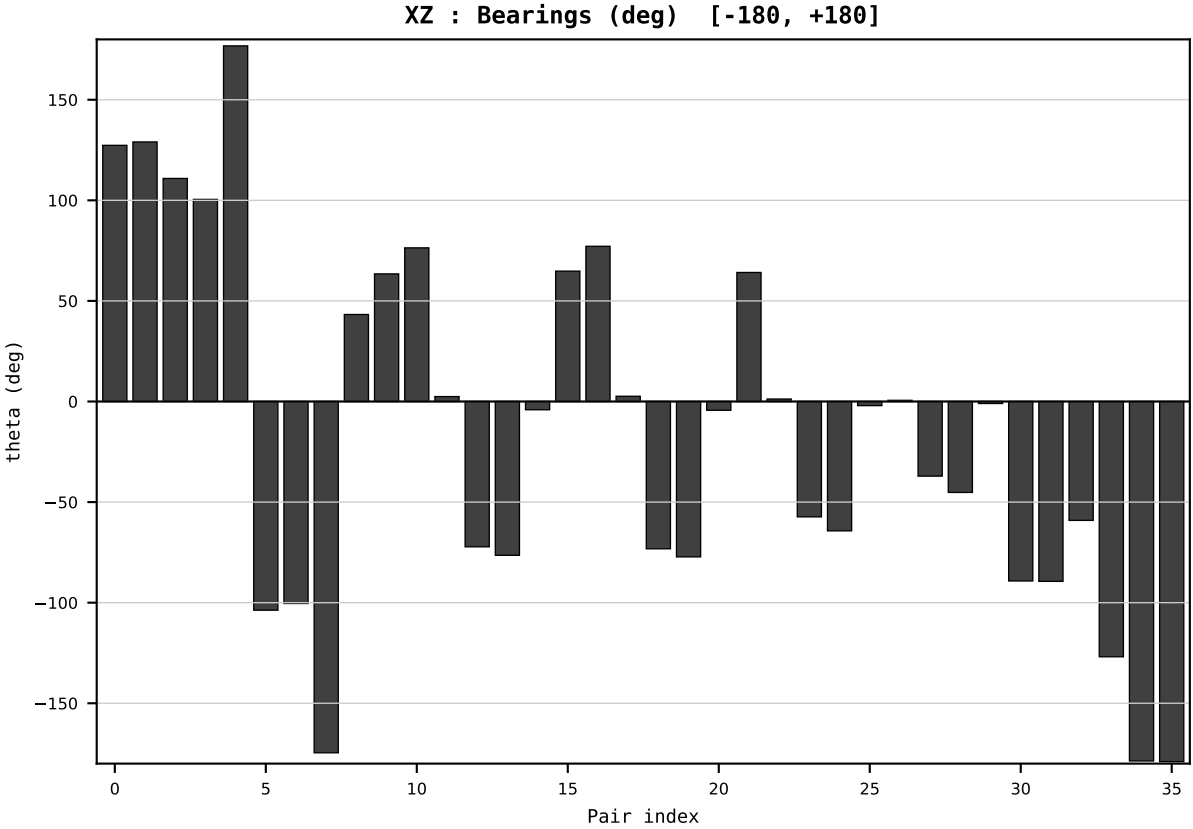
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2009
t : 9 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.604621
Ring = Hs-4
E_metric= 44.6563
kappa_HS= 2410.24
omega = 8.2198 deg
d_A = 1.436738
Helm = Solar
Helm d = +1.301456
DR = 7.787 HLR
DR ratio= 2410.2x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=10 | Year 2010 | D=9 N=26 pairs=36

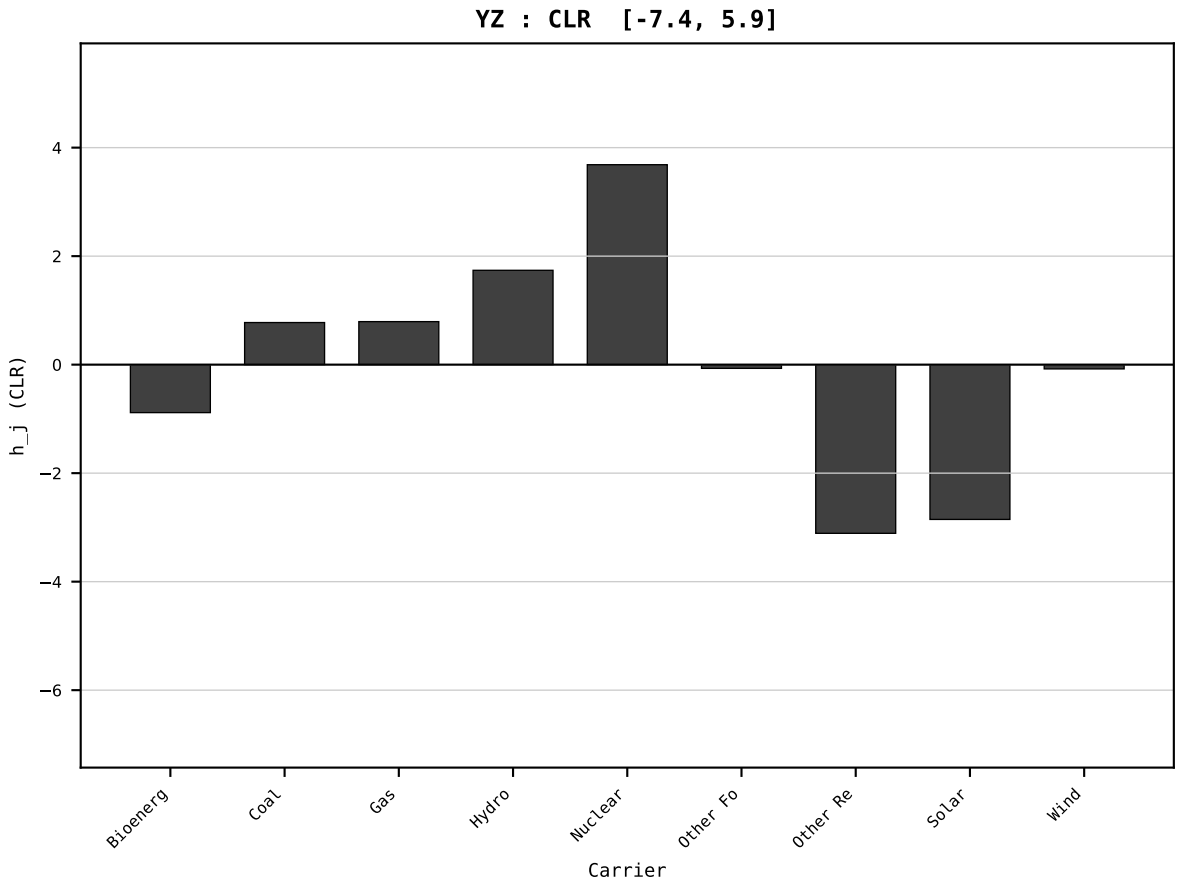
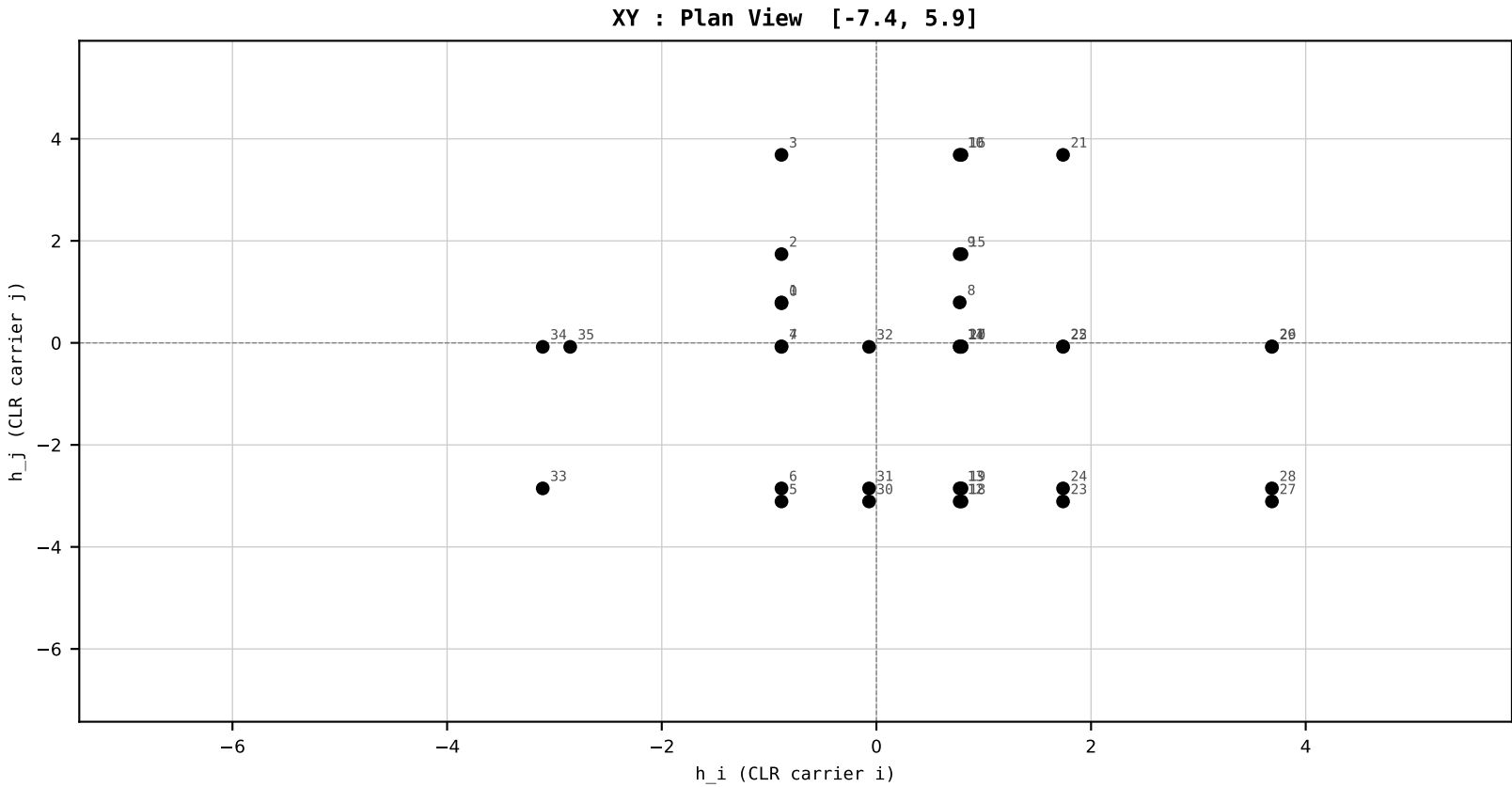
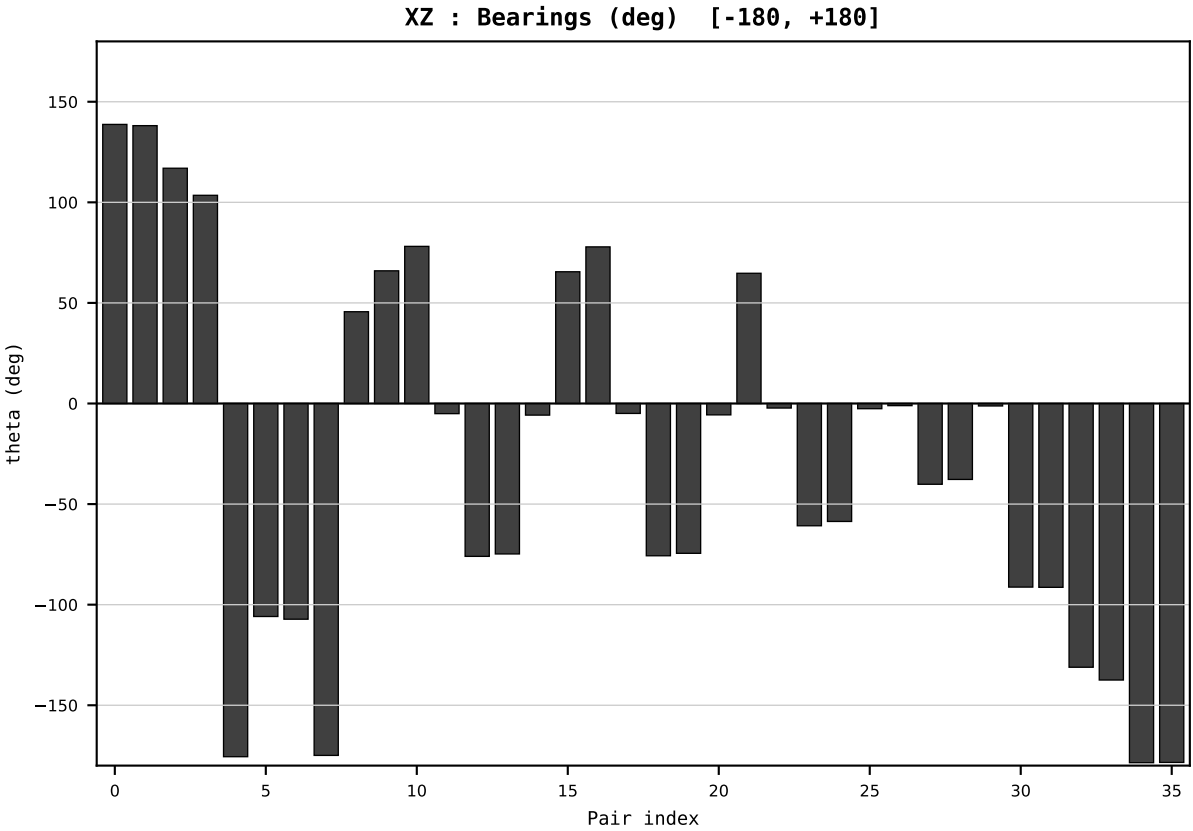
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2010
t : 10 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.586166
Ring = Hs-4
E_metric= 36.4358
kappa_HS= 892.75
omega = 8.3636 deg
d_A = 1.129480
Helm = Solar
Helm d = +1.054432
DR = 6.794 HLR
DR ratio= 892.7x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=11 | Year 2011 | D=9 N=26 pairs=36

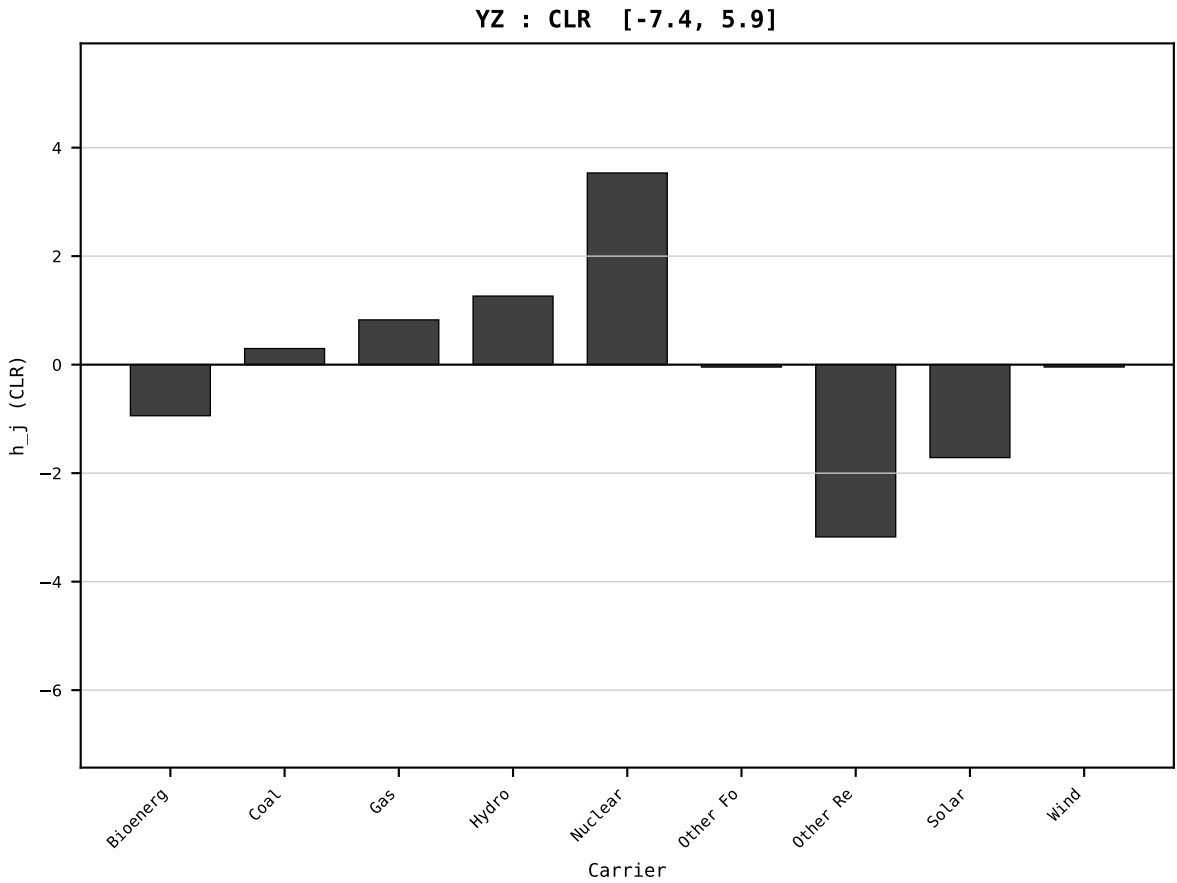
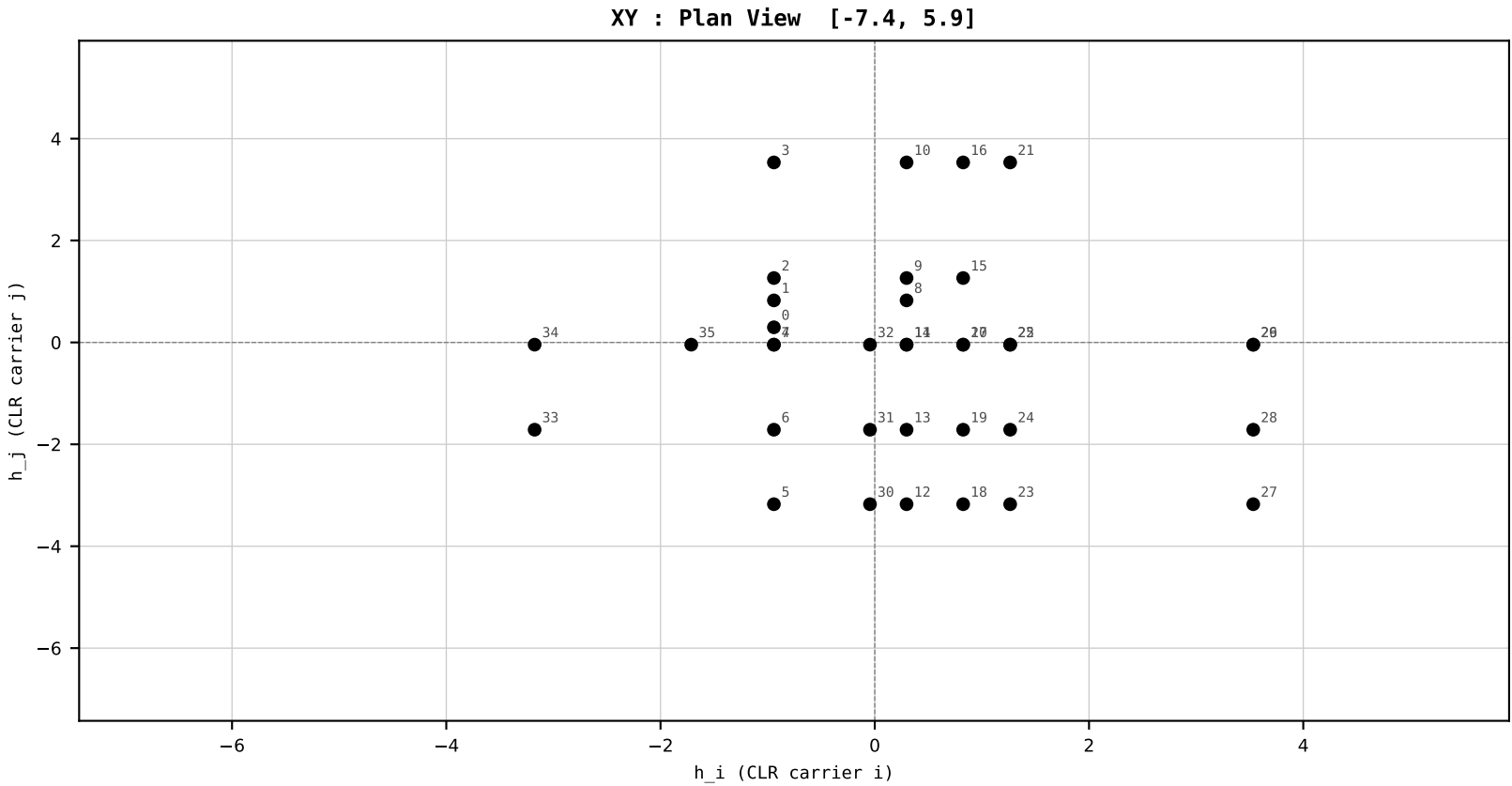
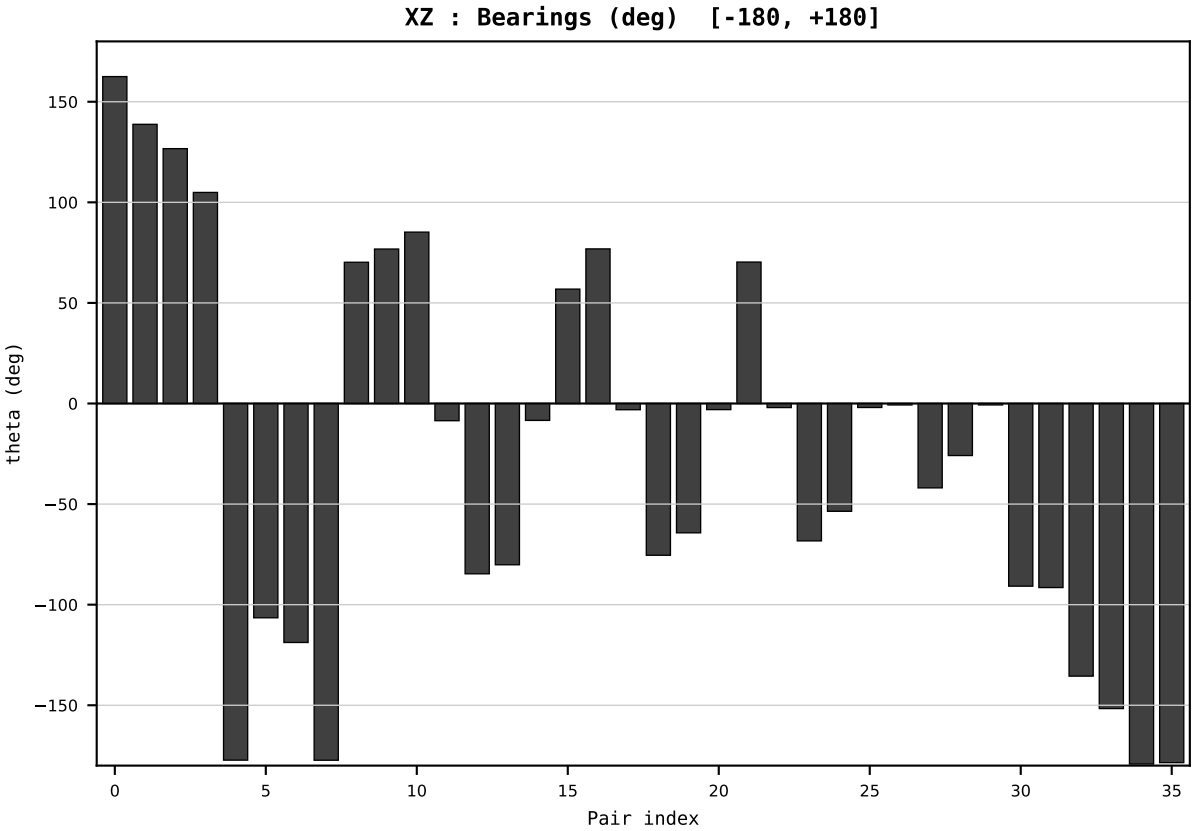
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2011
t : 11 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.592444
Ring = Hs-4
E_metric= 28.7571
kappa_HS= 819.24
omega = 11.6539 deg
d_A = 1.337292
Helm = Solar
Helm d = +1.139834
DR = 6.708 HLR
DR ratio= 819.2x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



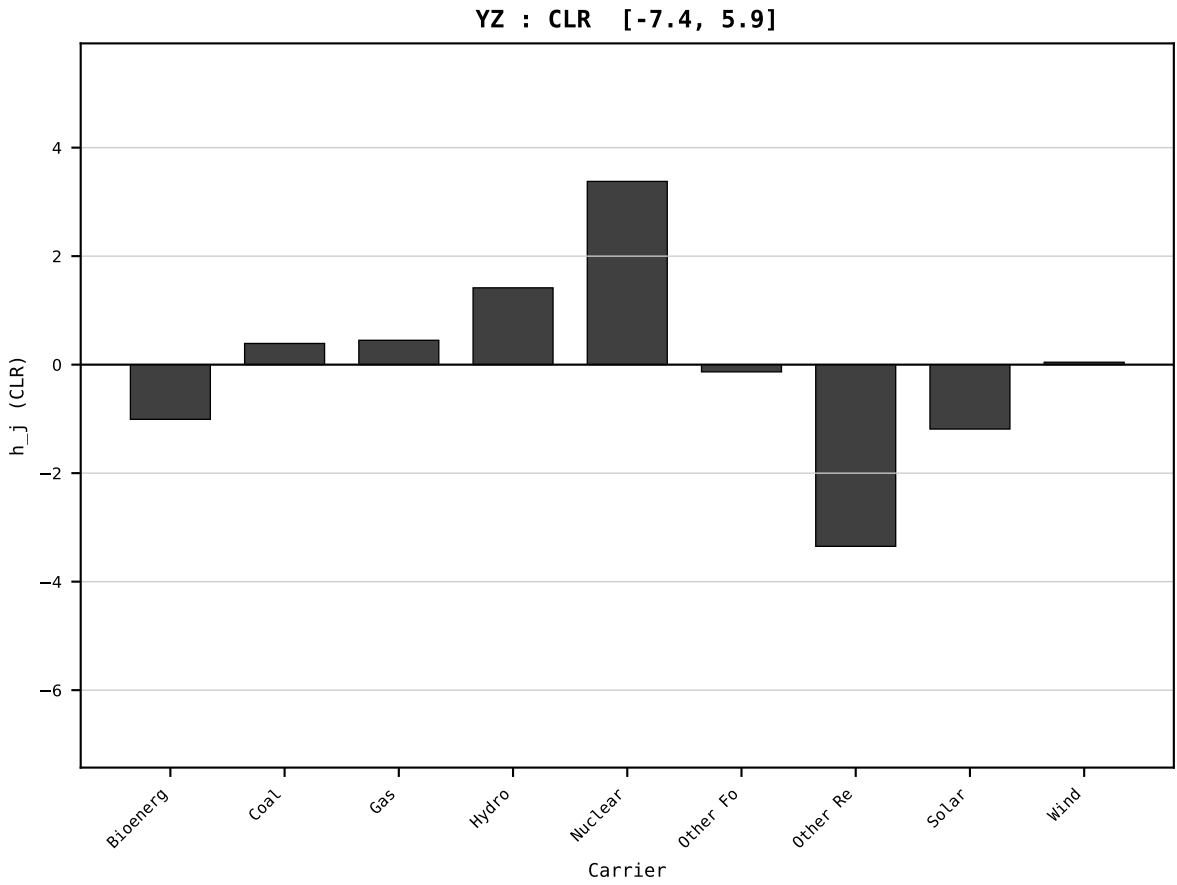
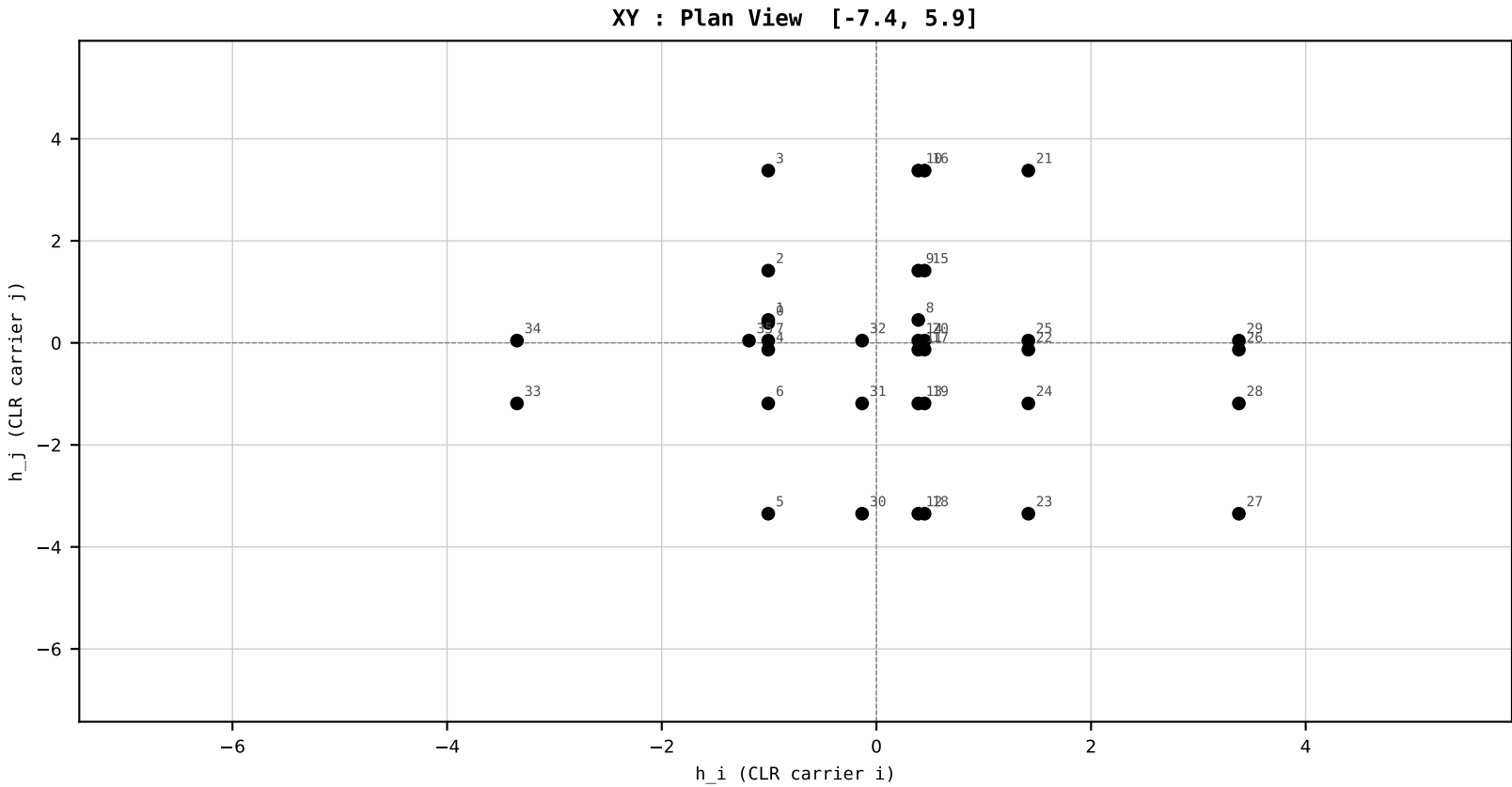
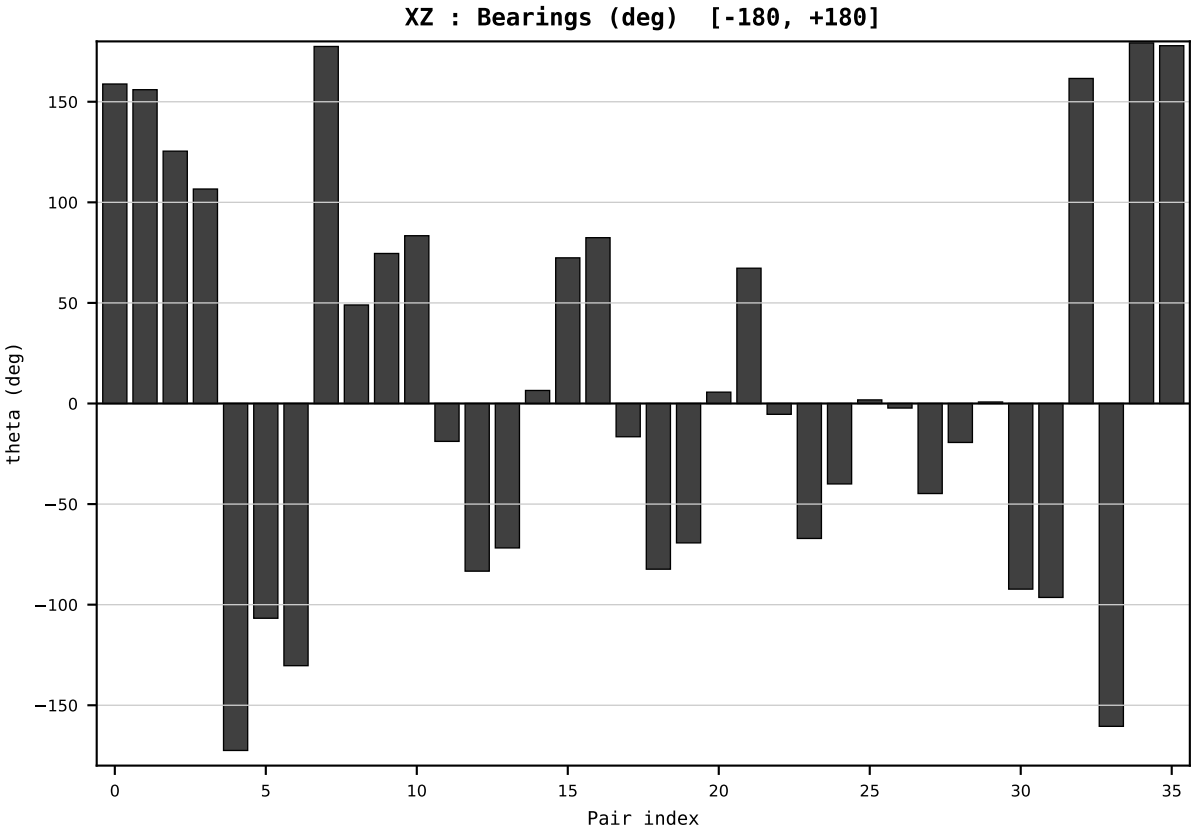
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2012
t : 12 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.555897
Ring = Hs-4
E_metric= 27.4266
kappa_HS= 834.14
omega = 7.7165 deg
d_A = 0.724138
Helm = Solar
Helm d = +0.526031
DR = 6.726 HLR
DR ratio= 834.1x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=13 | Year 2013 | D=9 N=26 pairs=36

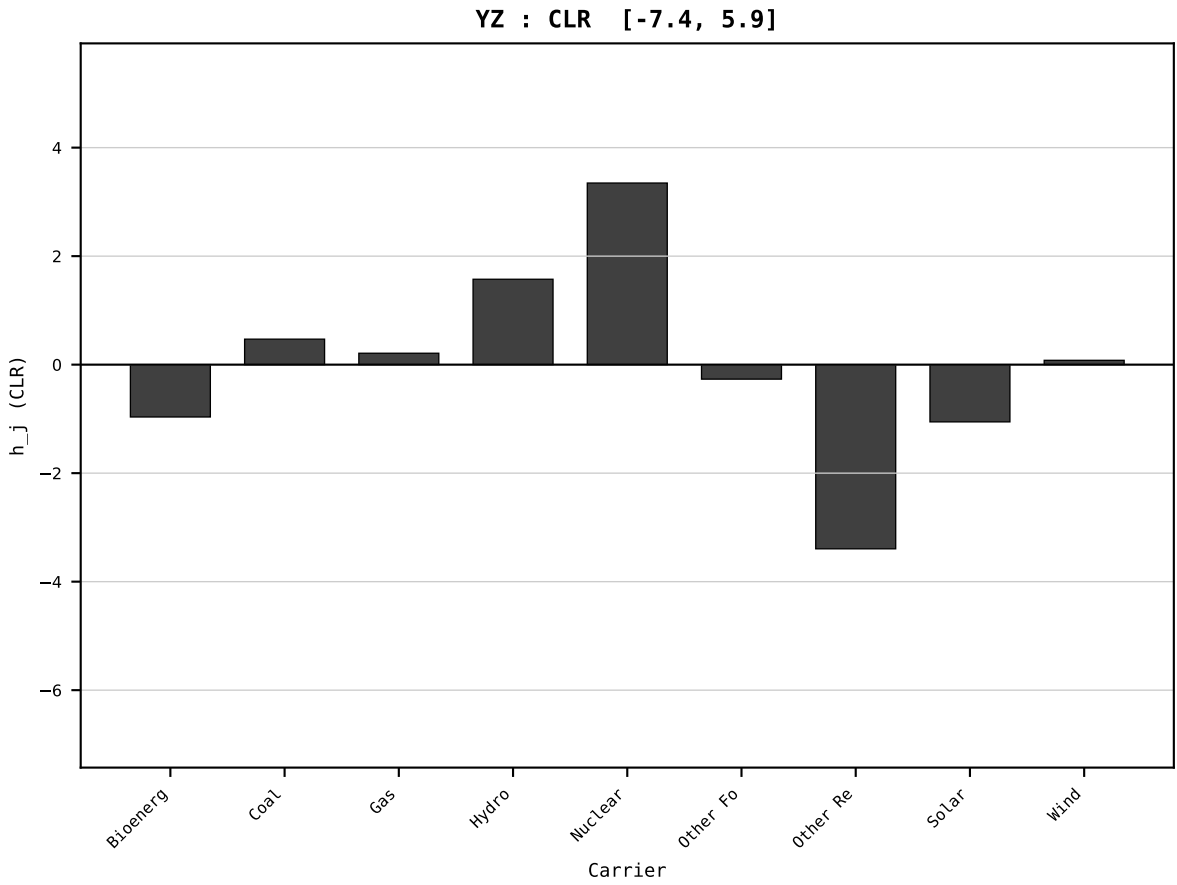
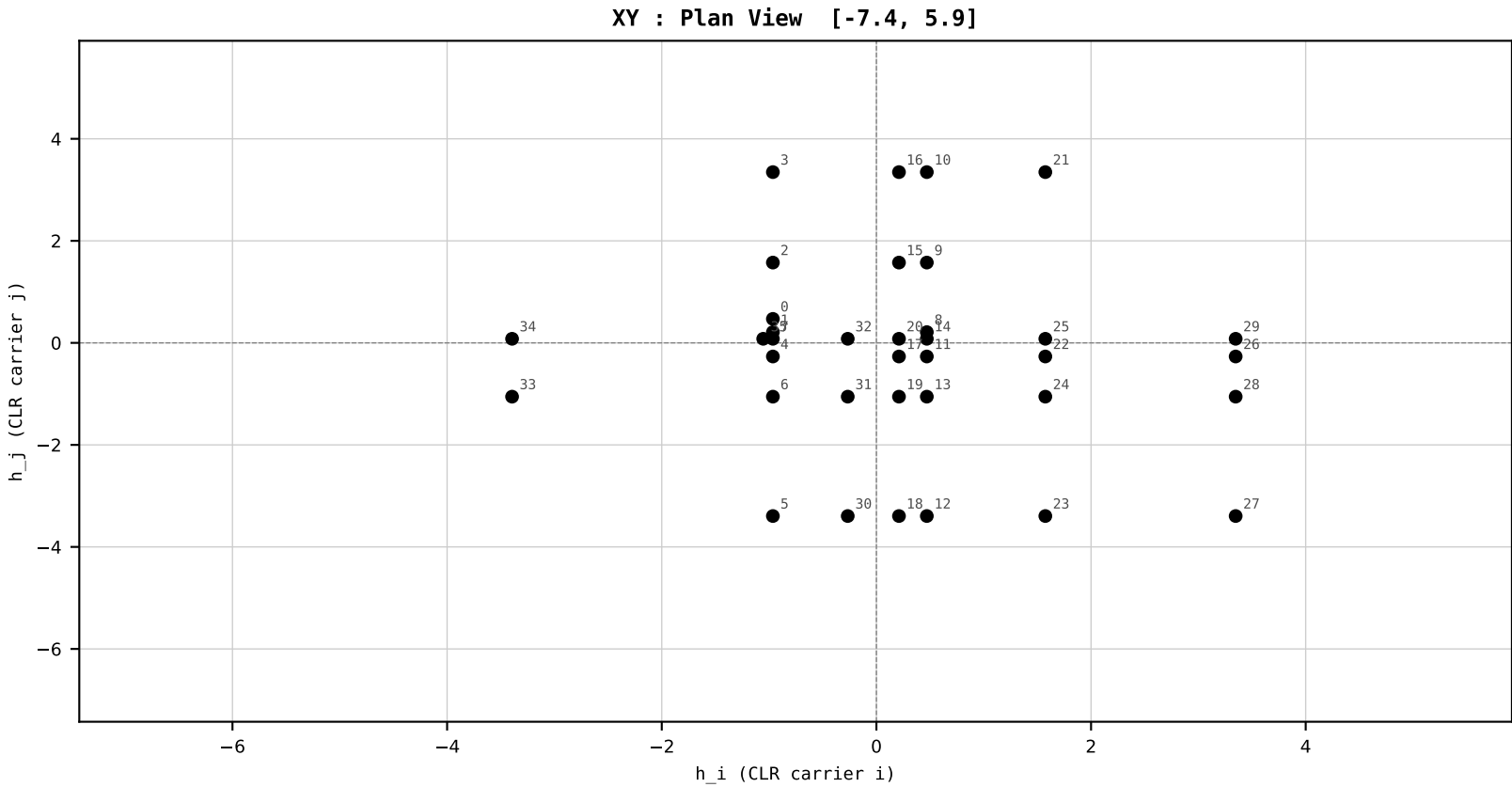
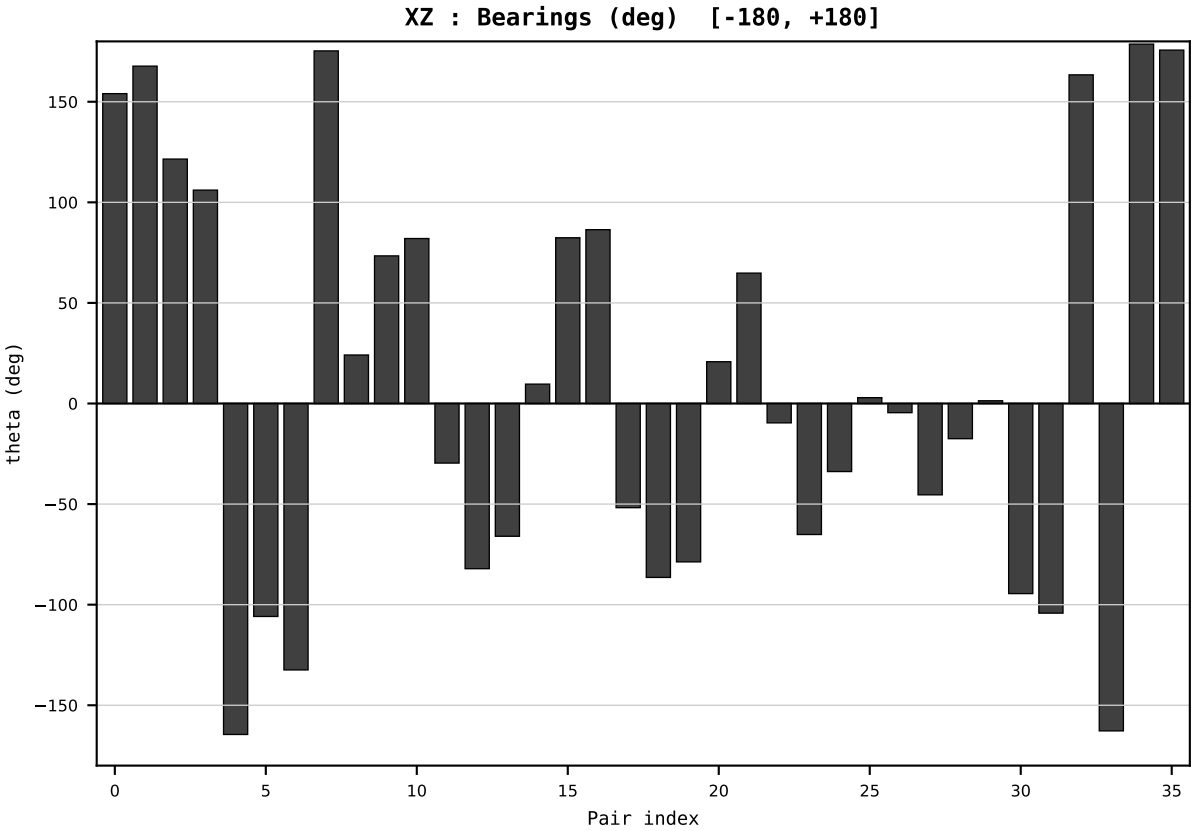
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2013
t : 13 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.545141
Ring = Hs-4
E_metric= 27.5924
kappa_HS= 847.36
omega = 3.9329 deg
d_A = 0.360297
Helm = Gas
Helm d = -0.238712
DR = 6.742 HLR
DR ratio= 847.4x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=14 | Year 2014 | D=9 N=26 pairs=36

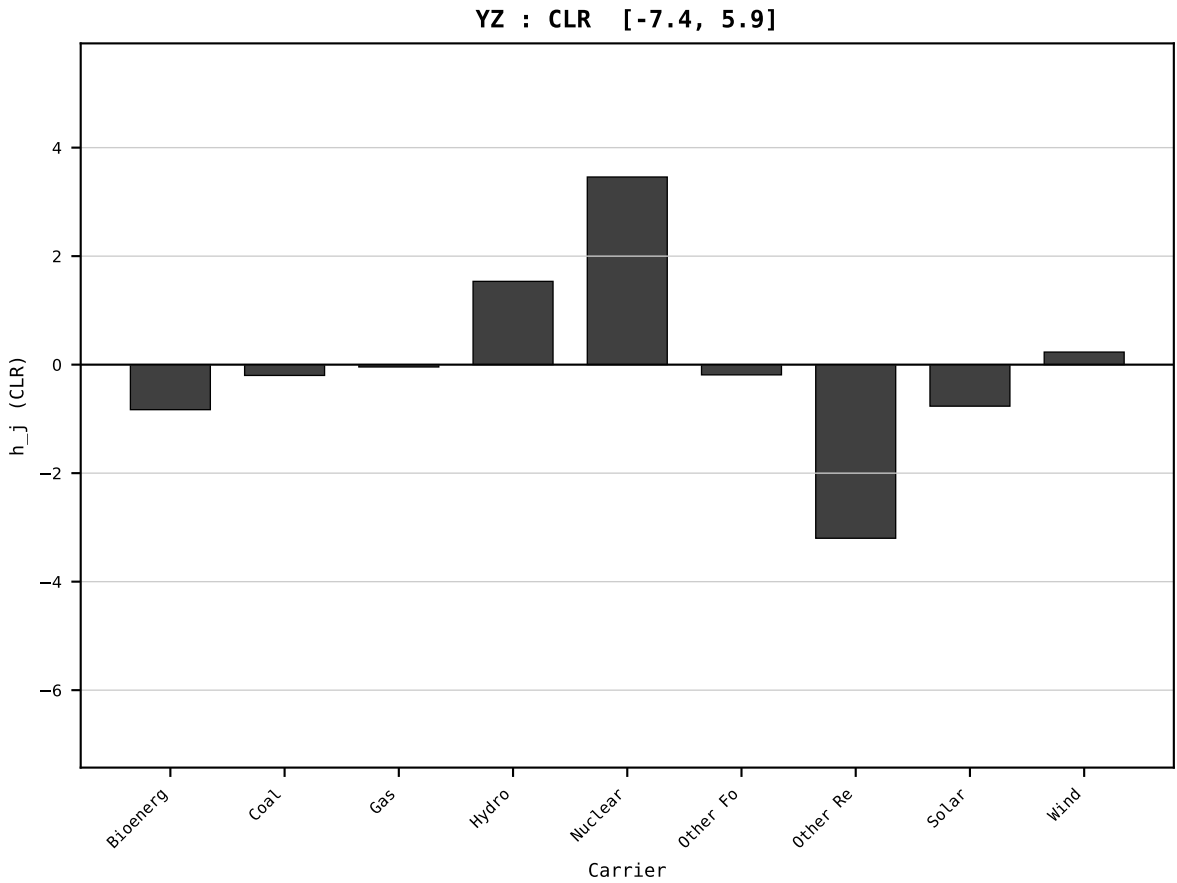
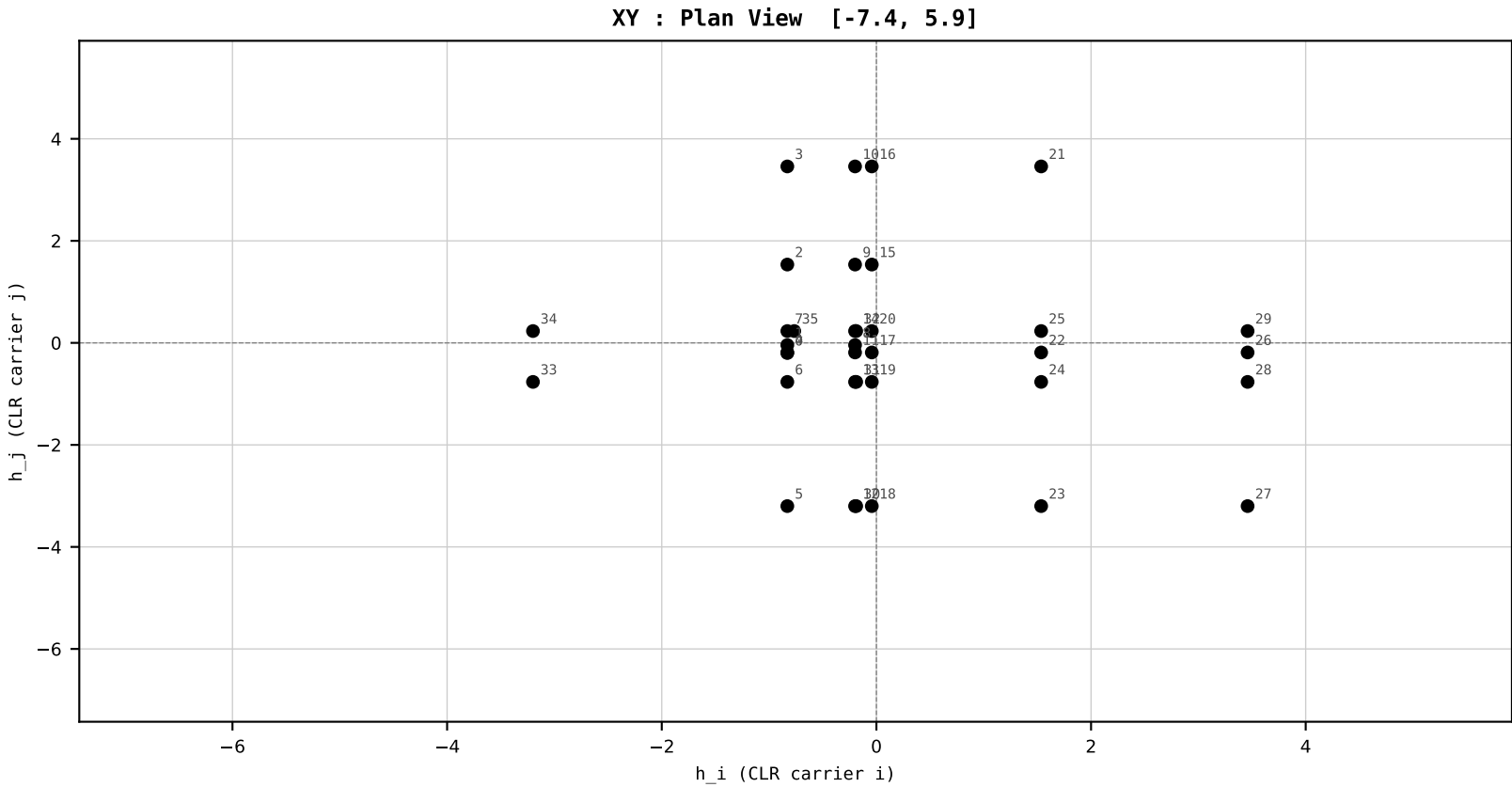
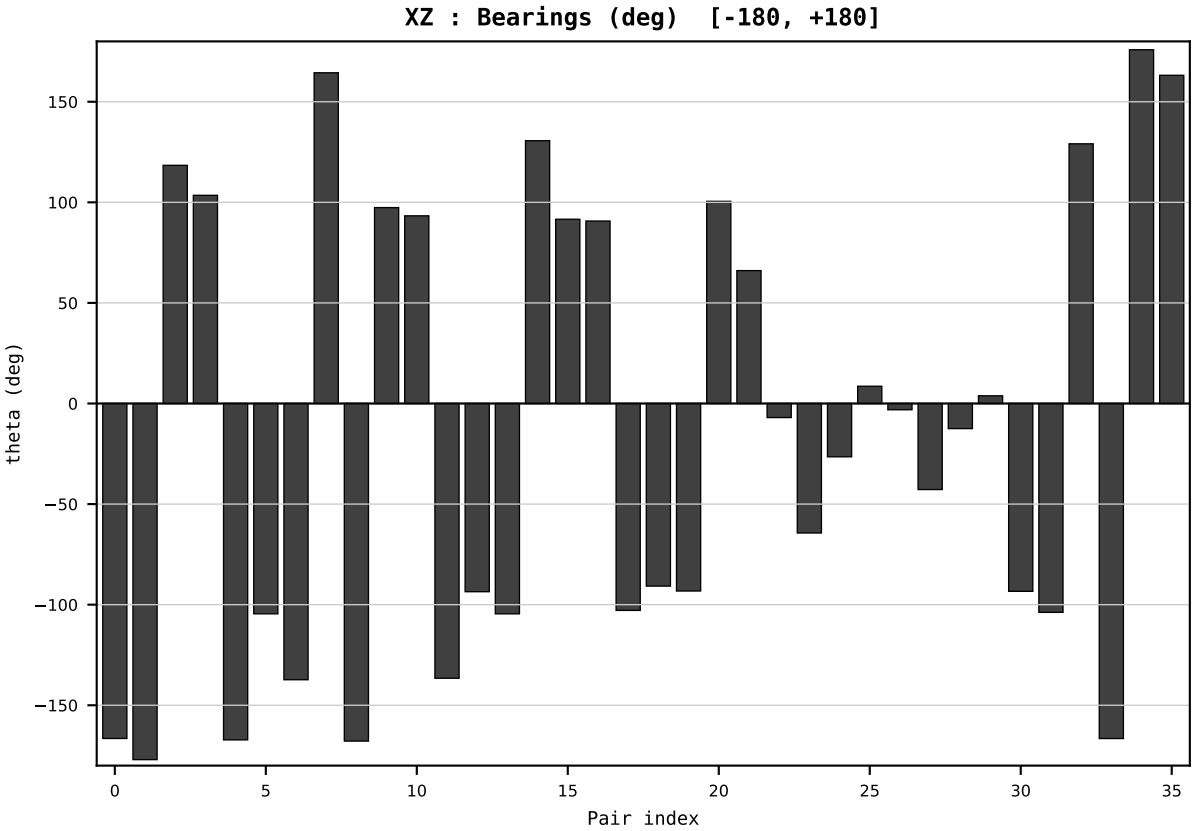
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2014
t : 14 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.589178
Ring = Hs-4
E_metric= 25.9637
kappa_HS= 779.43
omega = 9.0709 deg
d_A = 0.833206
Helm = Coal
Helm d = -0.668681
DR = 6.659 HLR
DR ratio= 779.4x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=15 | Year 2015 | D=9 N=26 pairs=36

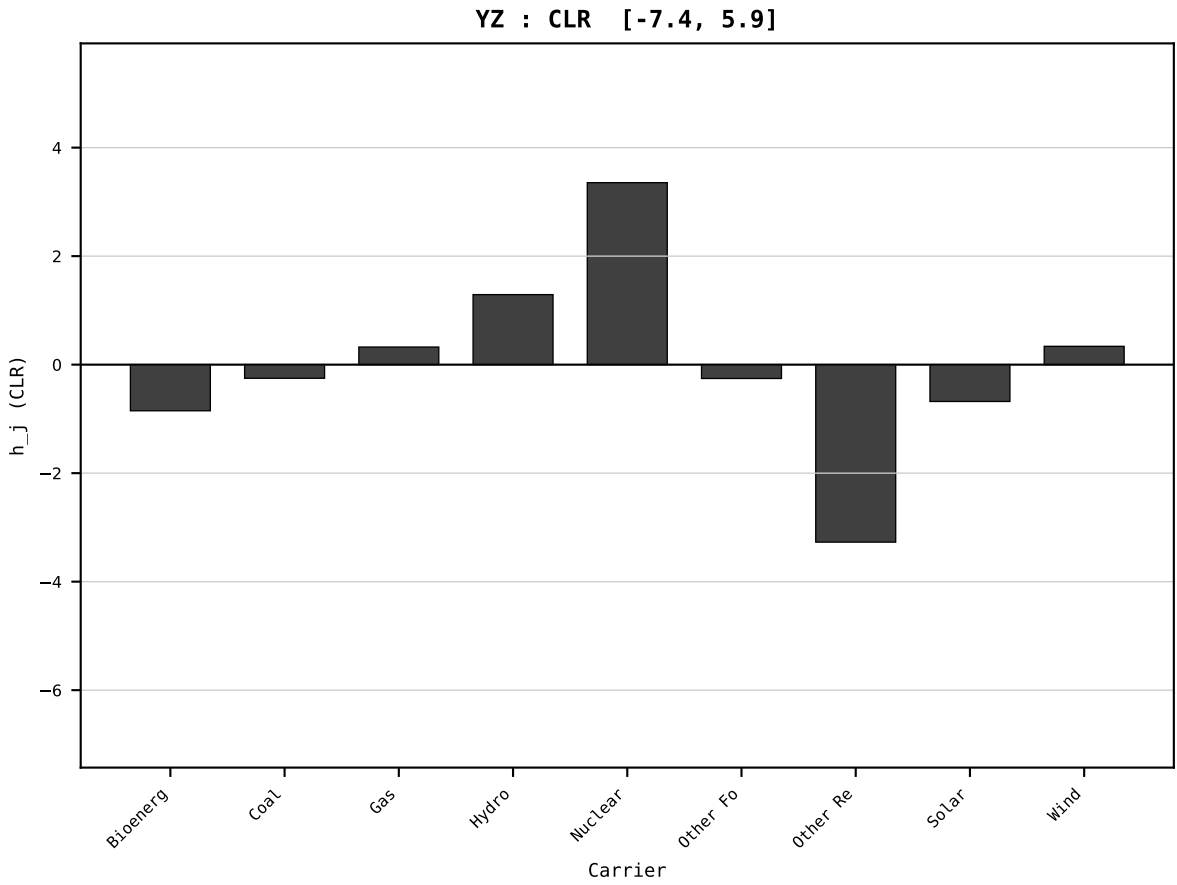
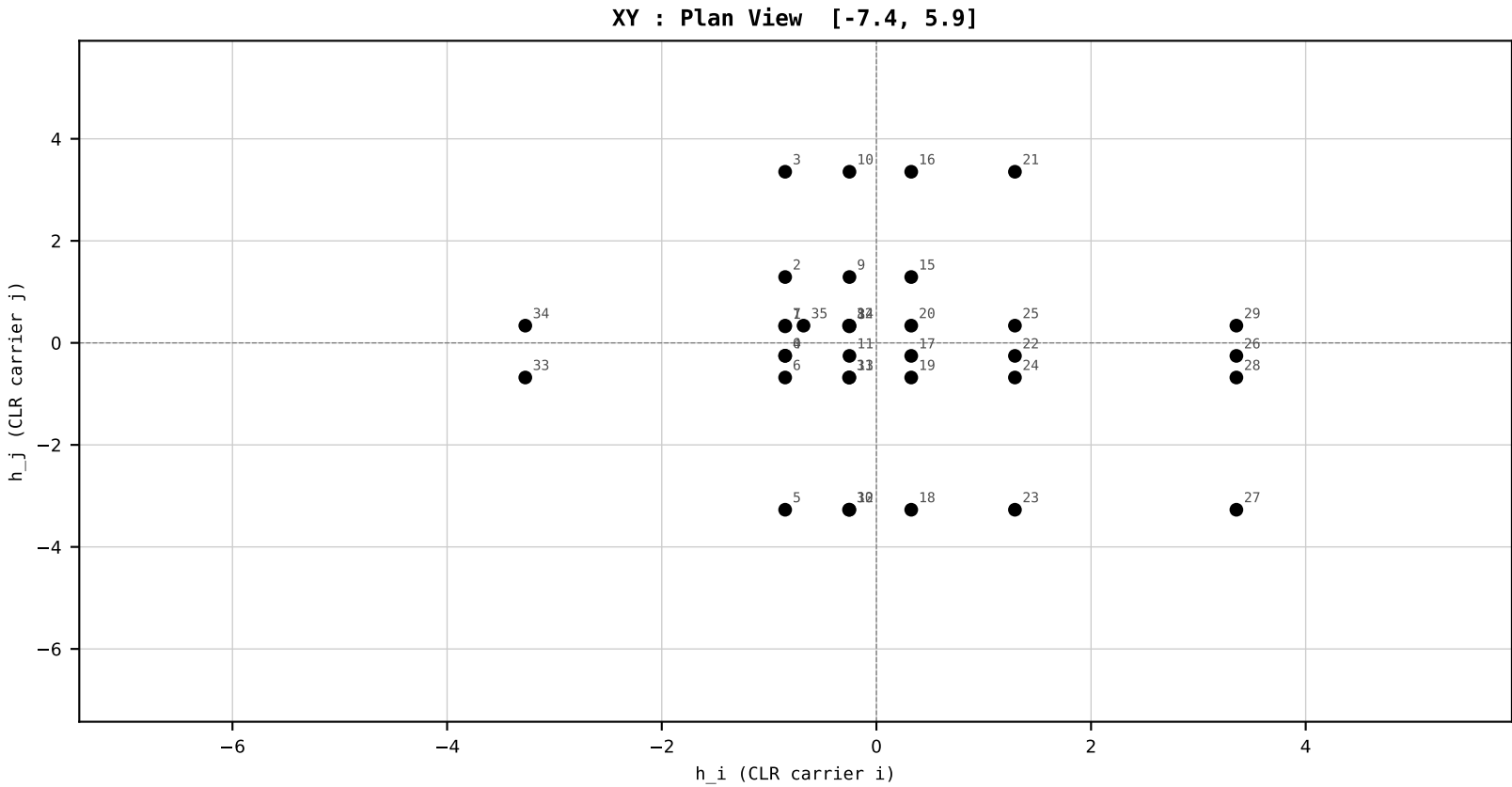
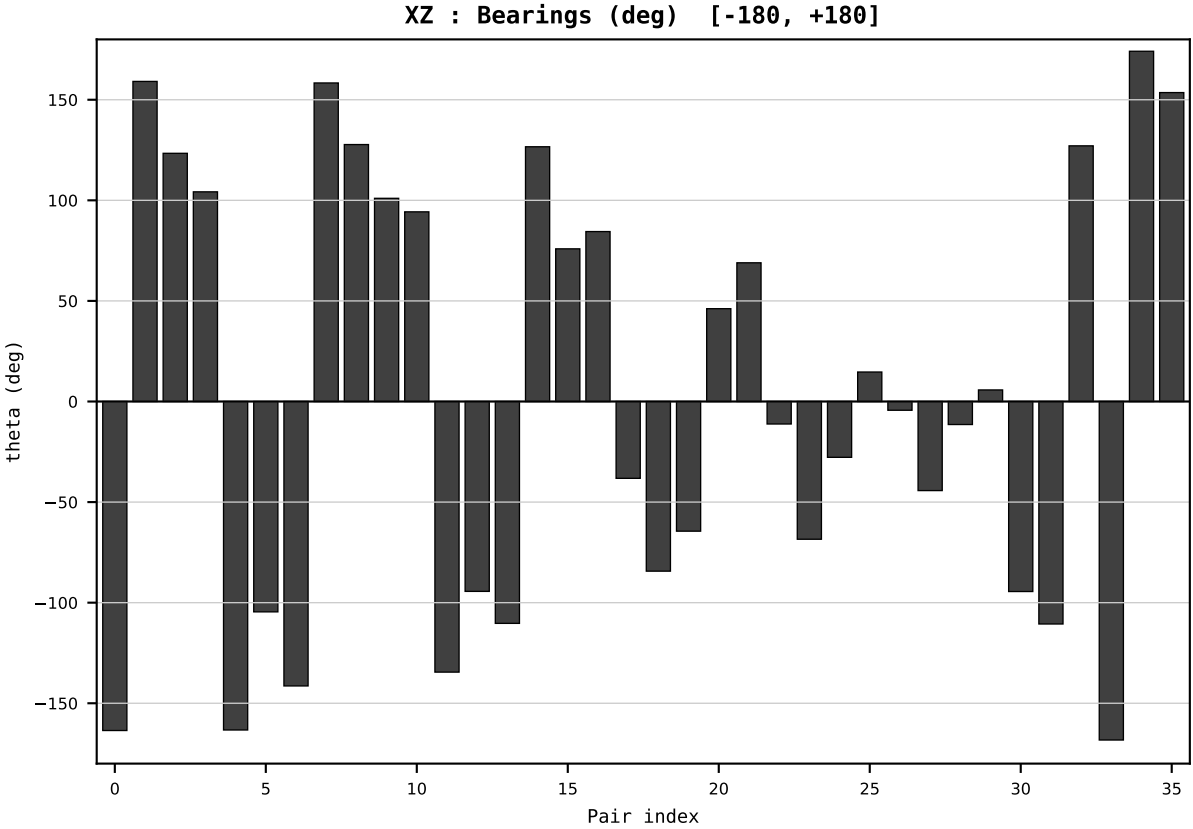
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2015
t : 15 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.565912
Ring = Hs-4
E_metric= 25.1505
kappa_HS= 754.19
omega = 5.4442 deg
d_A = 0.486840
Helm = Gas
Helm d = +0.367403
DR = 6.626 HLR
DR ratio= 754.2x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



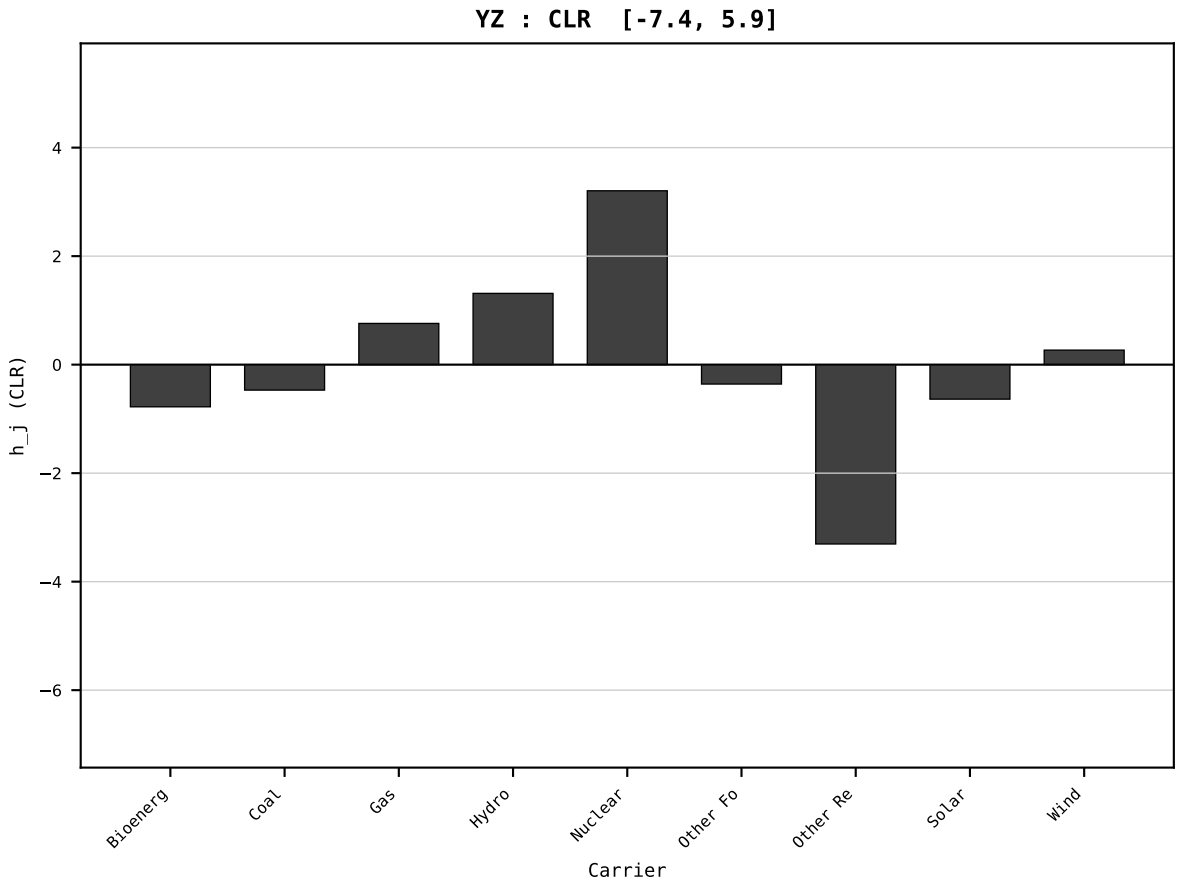
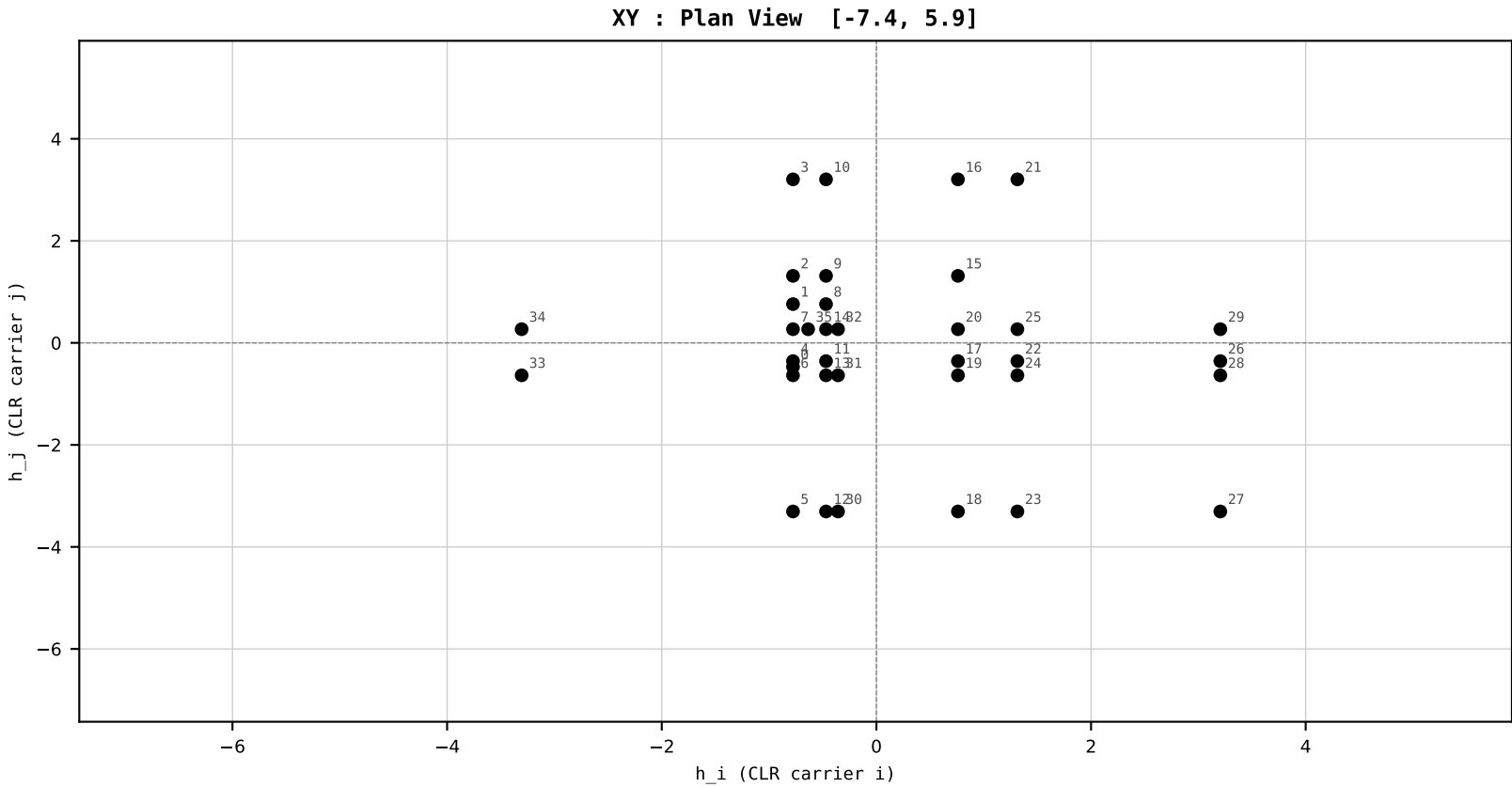
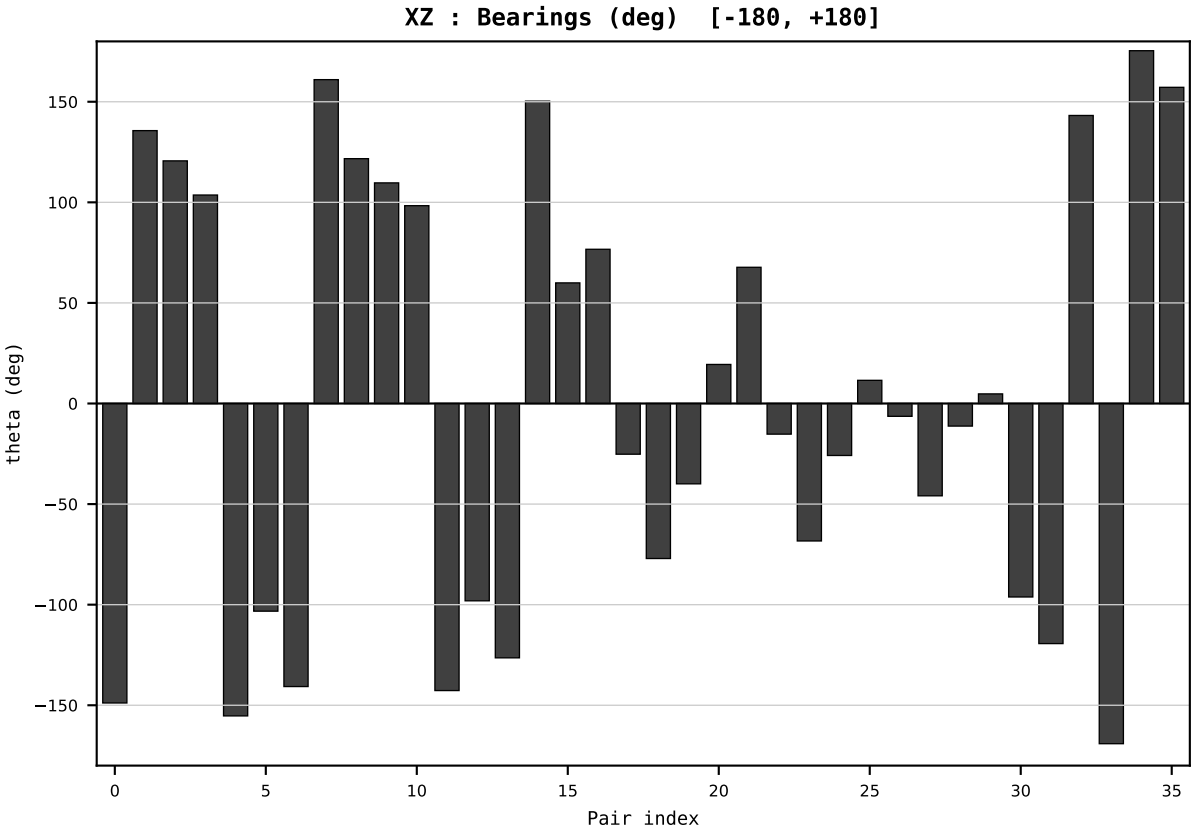
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2016
t : 16 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.518507
Ring = Hs-4
E_metric= 24.9278
kappa_HS= 672.00
omega = 6.0973 deg
d_A = 0.532714
Helm = Gas
Helm d = +0.435313
DR = 6.510 HLR
DR ratio= 672.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=17 | Year 2017 | D=9 N=26 pairs=36

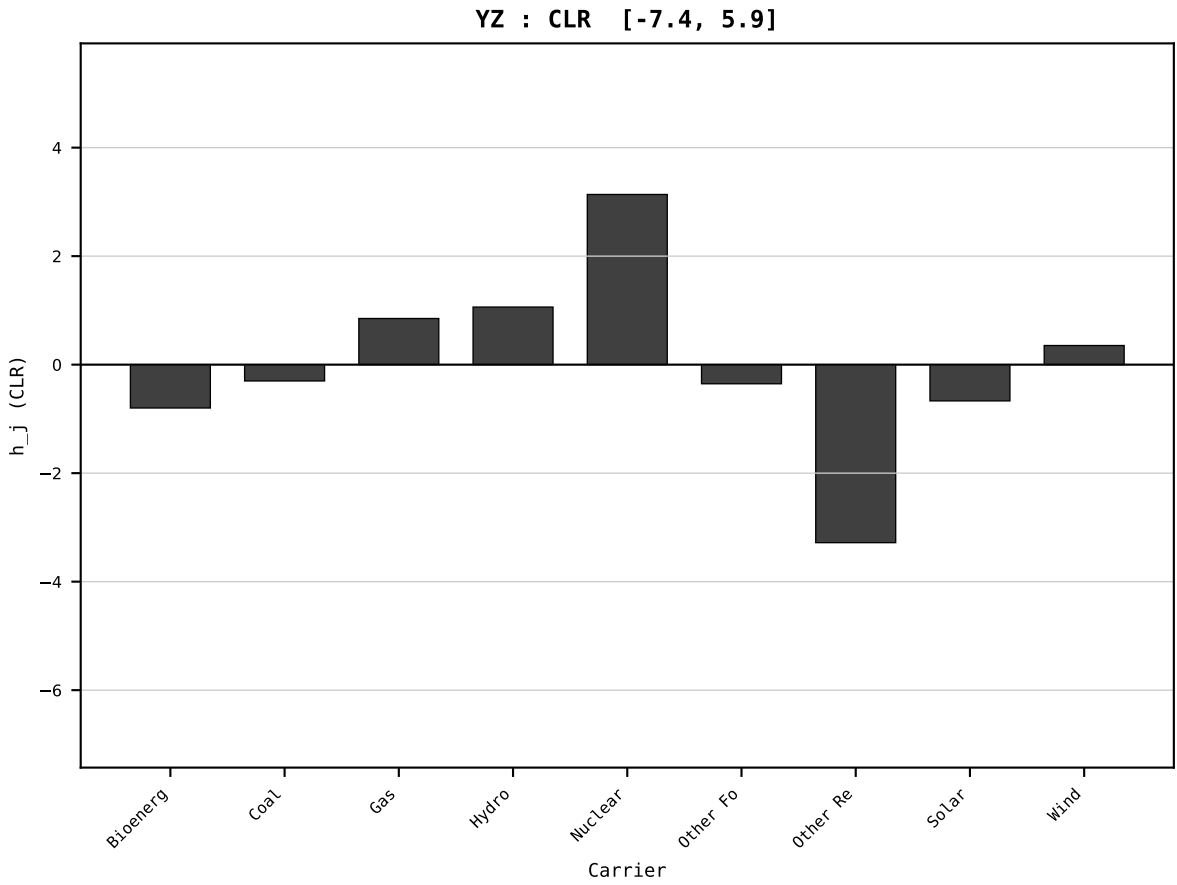
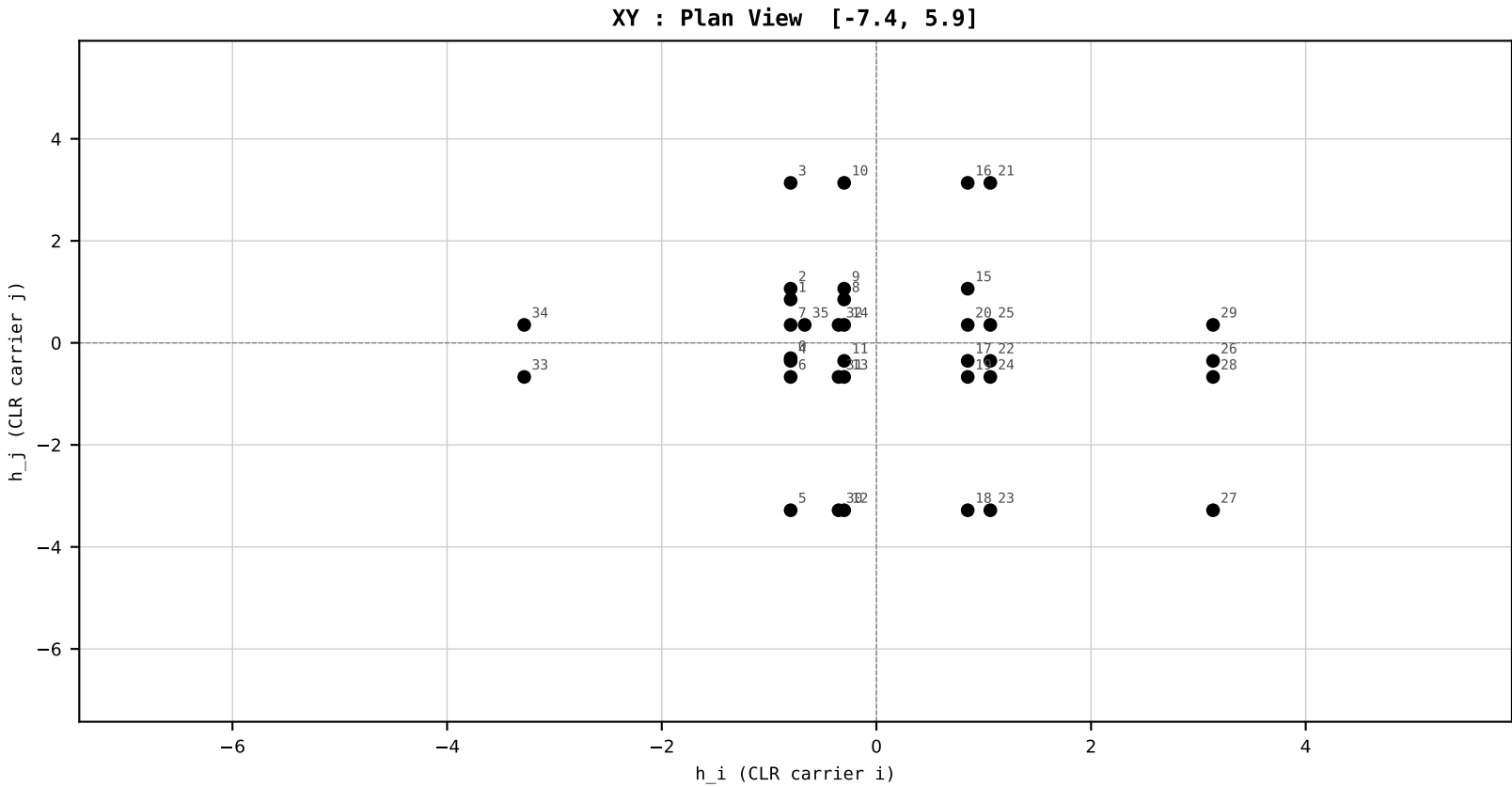
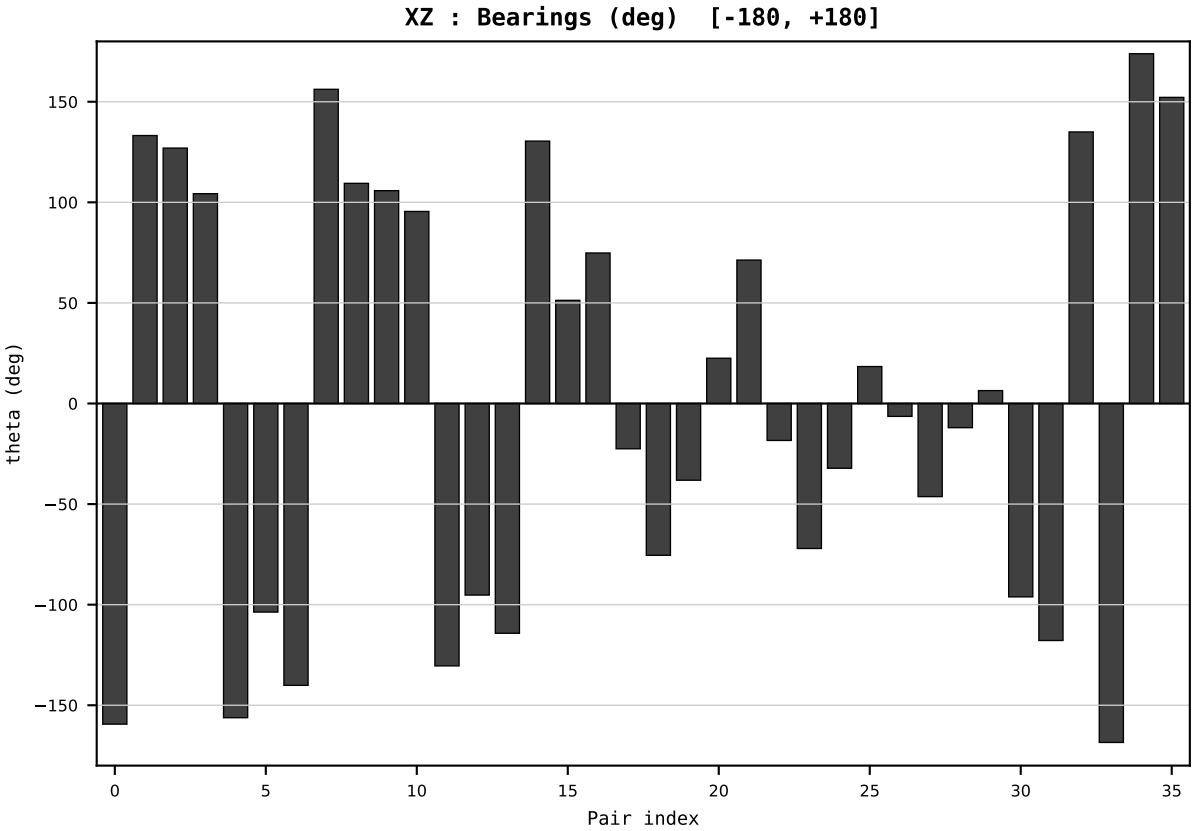
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2017
t : 17 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.504621
Ring = Hs-4
E_metric= 23.8804
kappa_HS= 612.86
omega = 3.7247 deg
d_A = 0.338103
Helm = Hydro
Helm d = -0.252129
DR = 6.418 HLR
DR ratio= 612.9x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



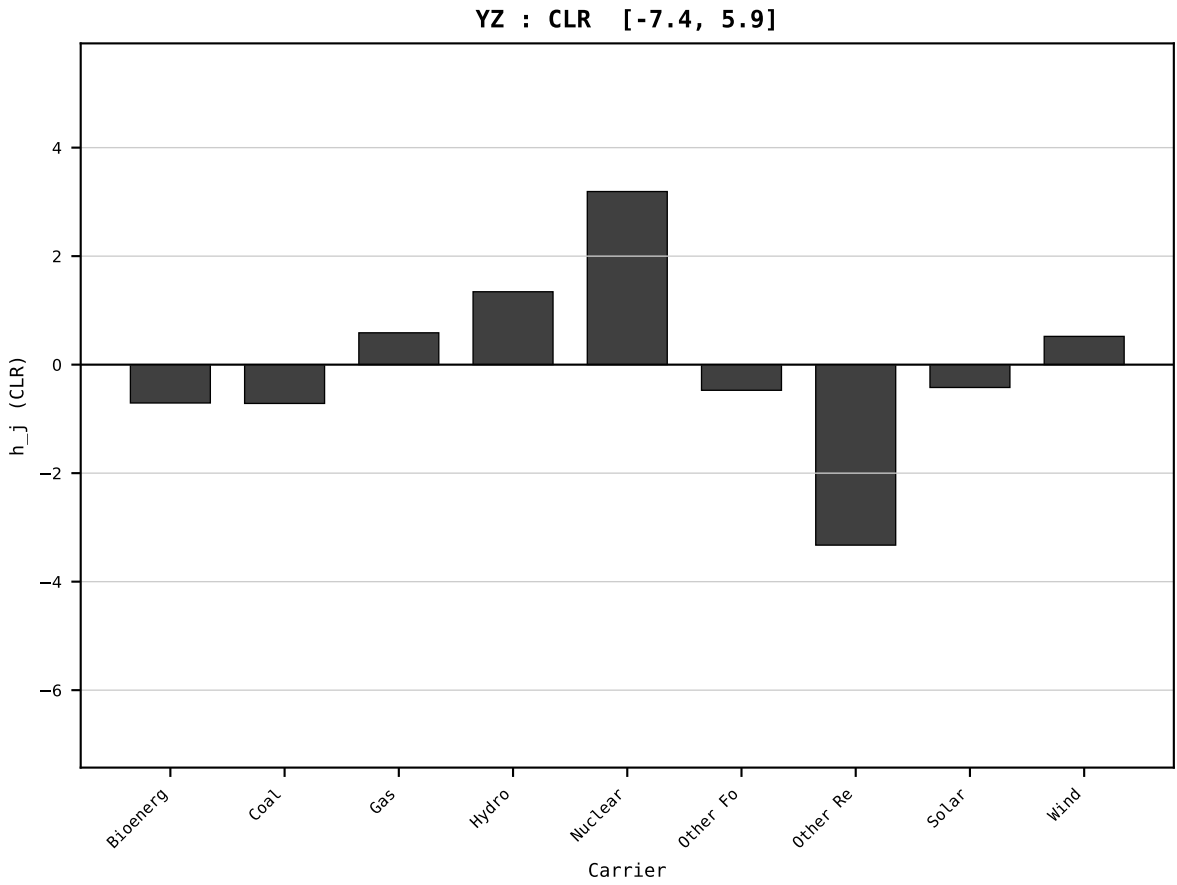
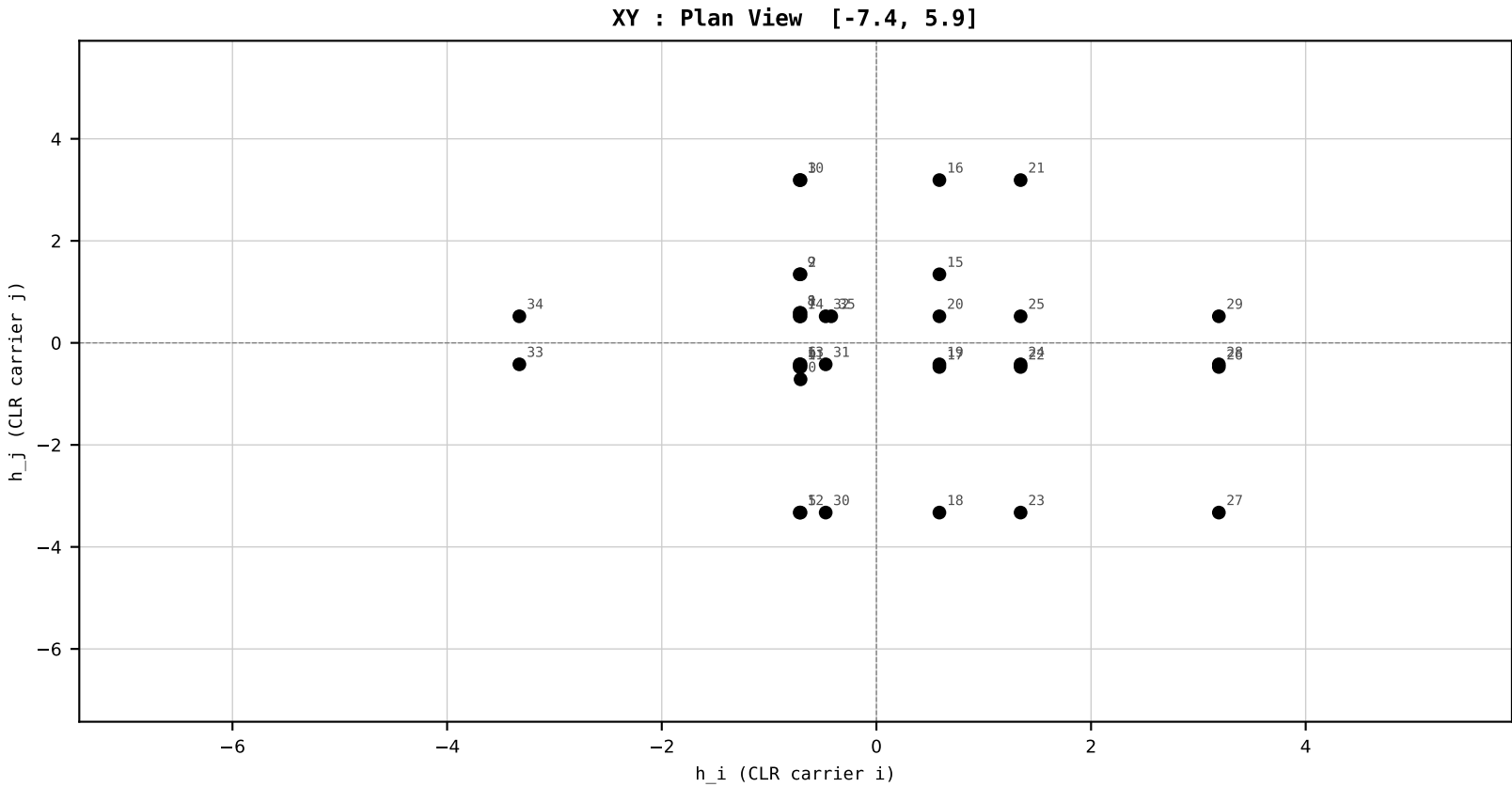
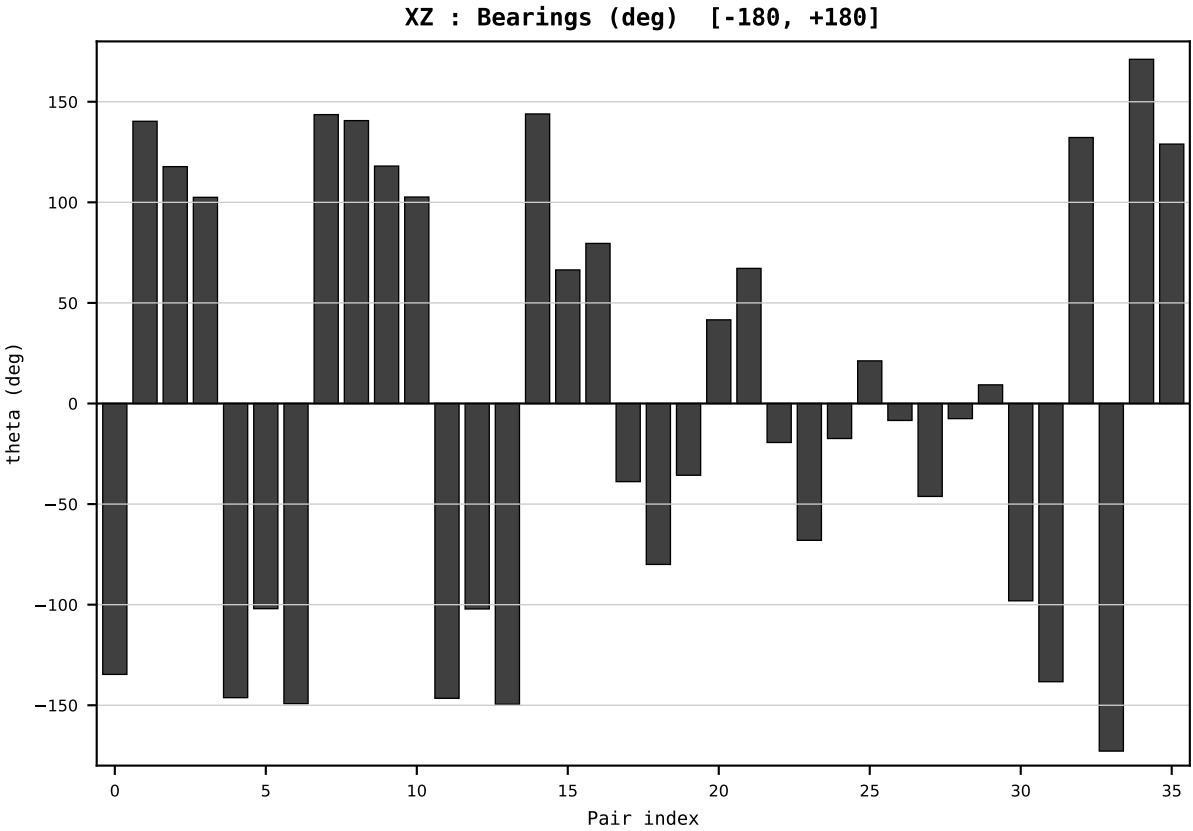
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2018
t : 18 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.513194
Ring = Hs-4
E_metric= 25.0810
kappa_HS= 676.95
omega = 7.5442 deg
d_A = 0.662124
Helm = Coal
Helm d = -0.414639
DR = 6.518 HLR
DR ratio= 677.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=19 | Year 2019 | D=9 N=26 pairs=36

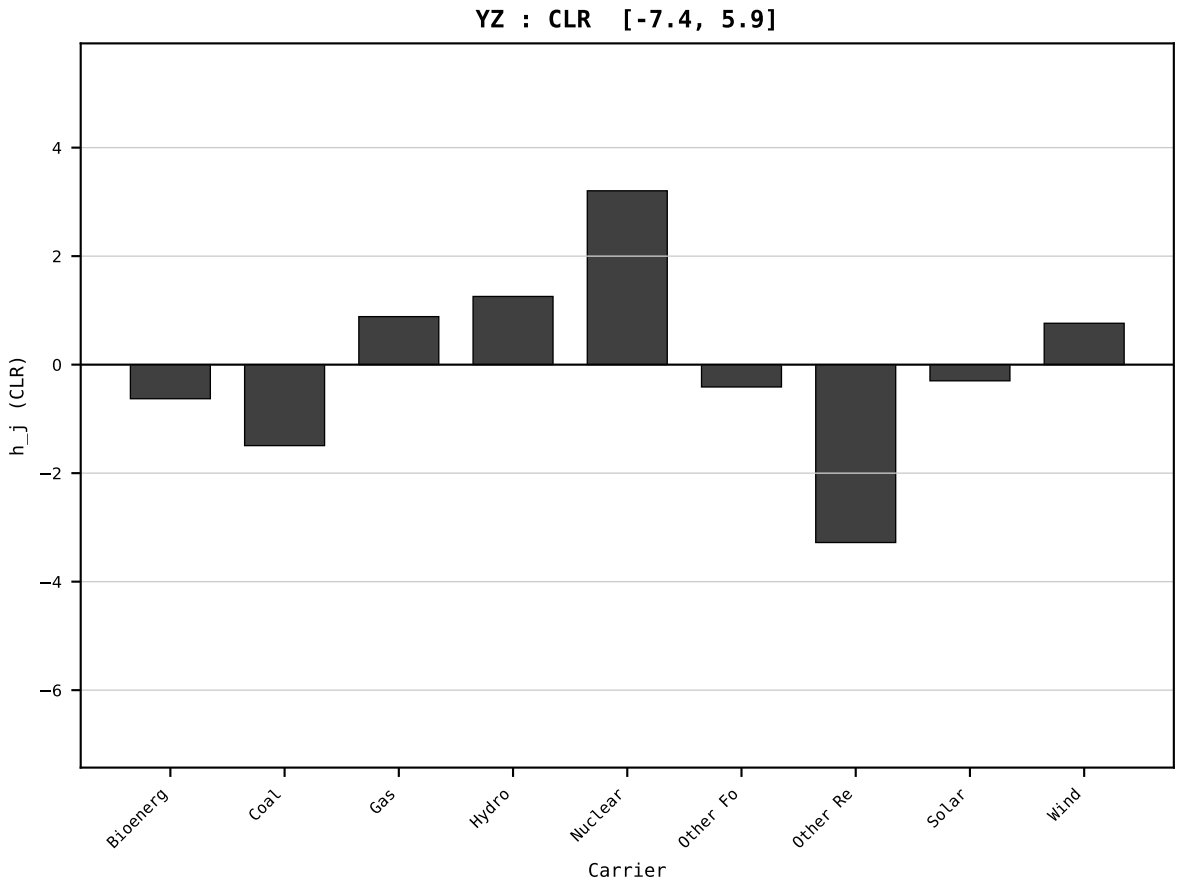
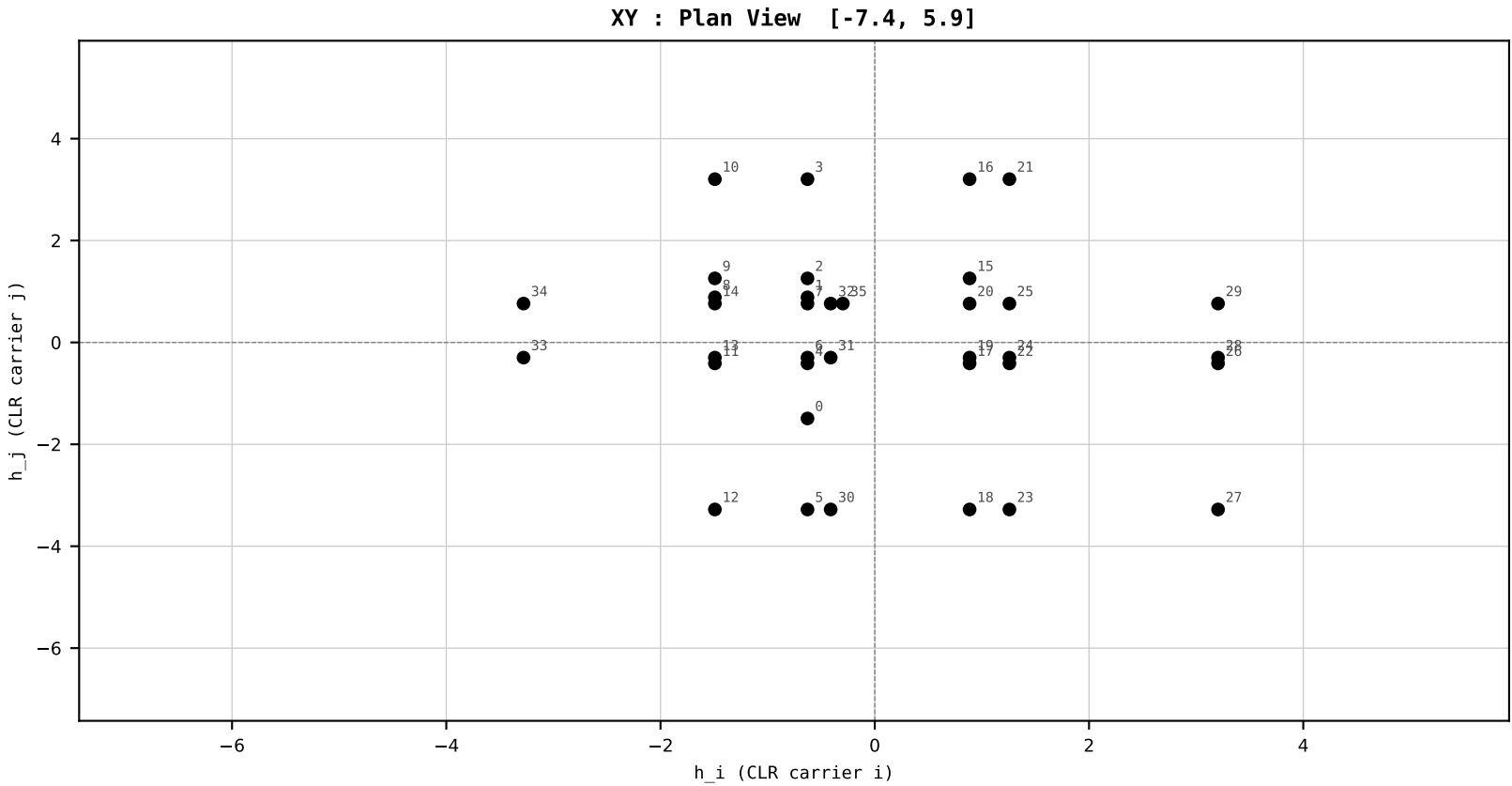
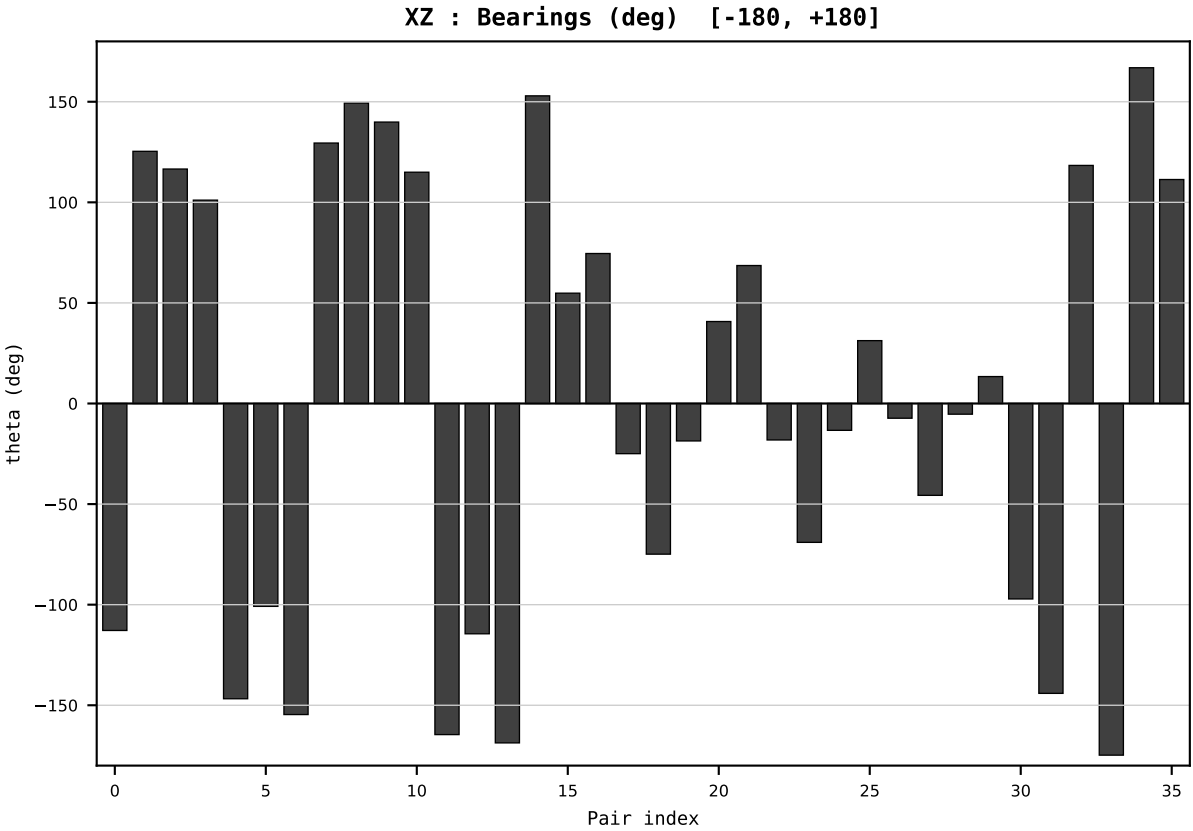
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2019
t : 19 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.501913
Ring = Hs-4
E_metric= 26.8448
kappa_HS= 654.11
omega = 9.7988 deg
d_A = 0.887160
Helm = Coal
Helm d = -0.777550
DR = 6.483 HLR
DR ratio= 654.1x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=20 | Year 2020 | D=9 N=26 pairs=36

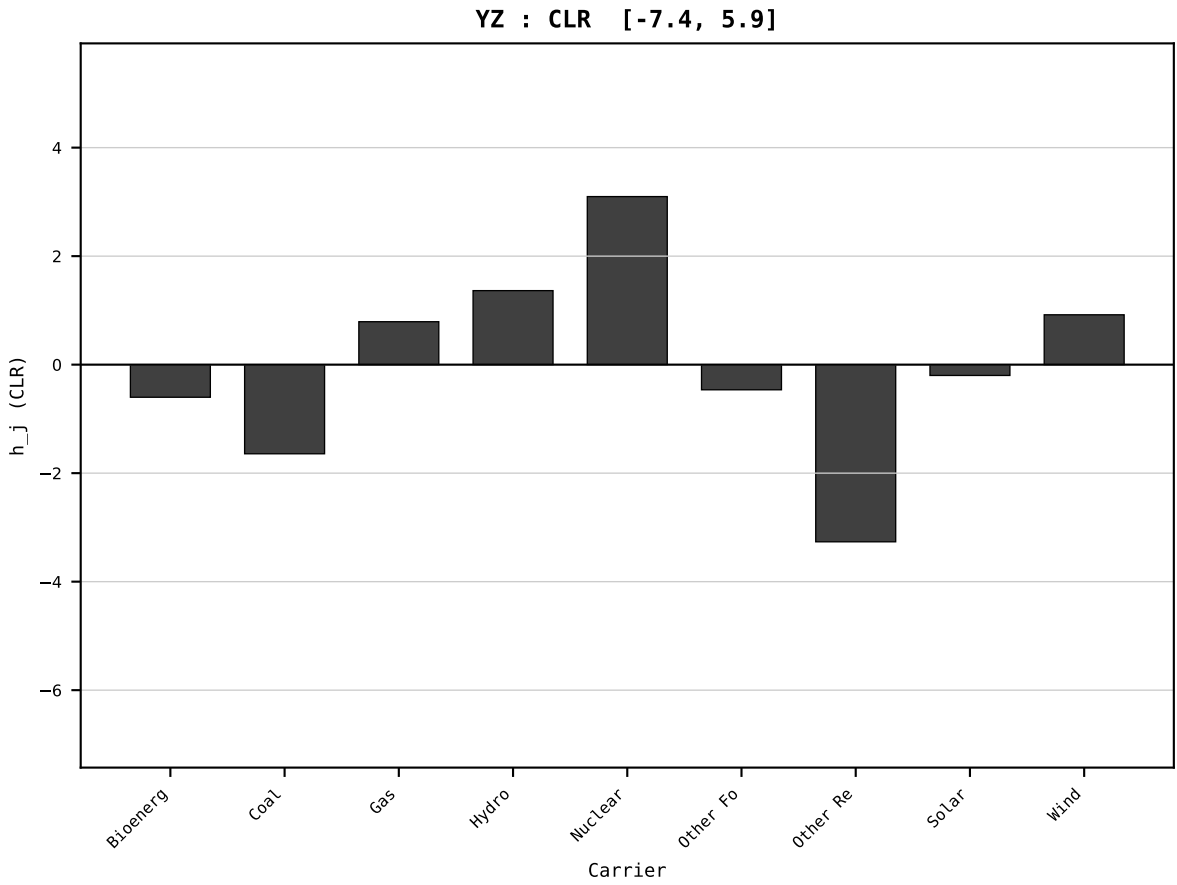
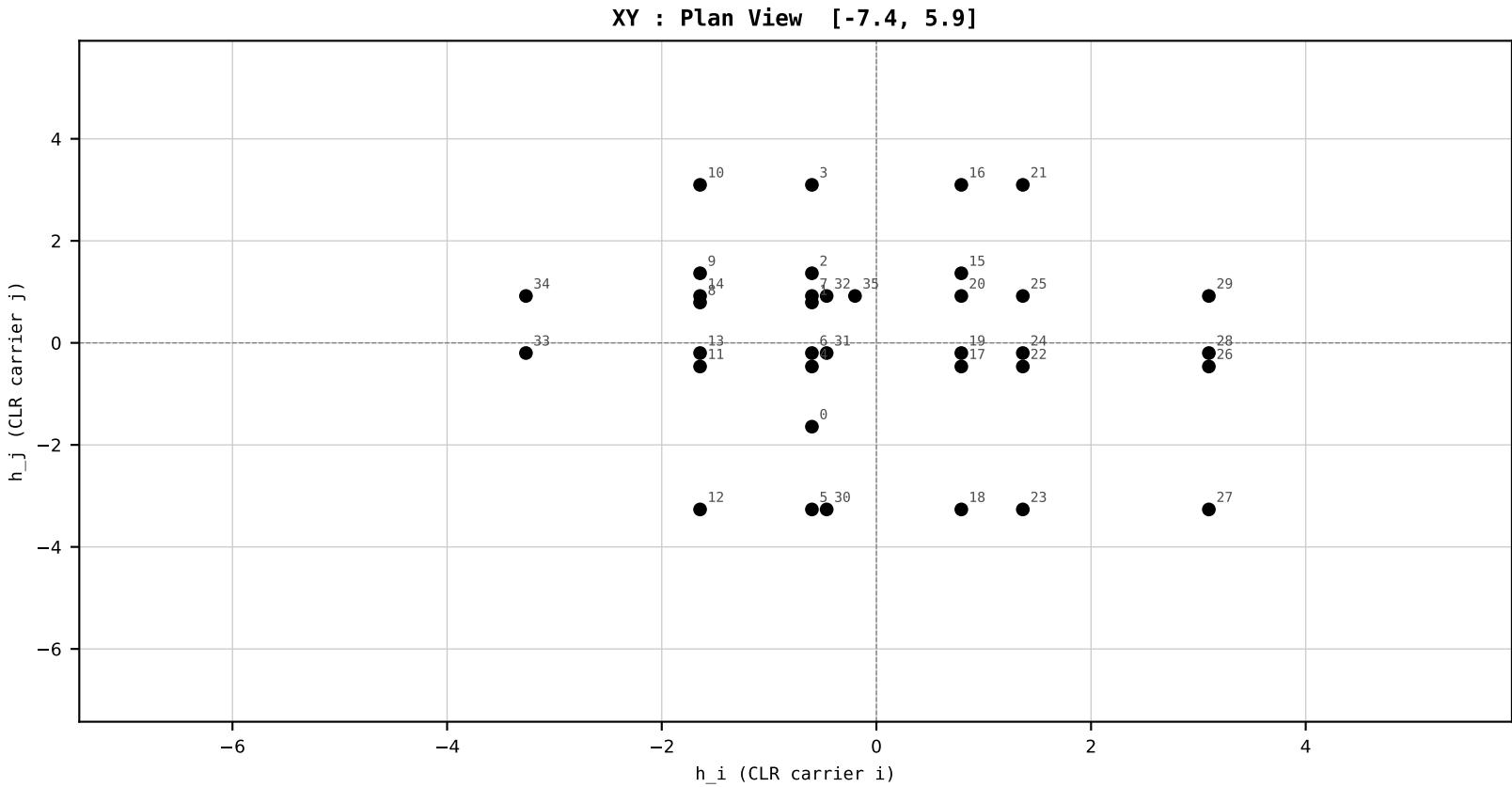
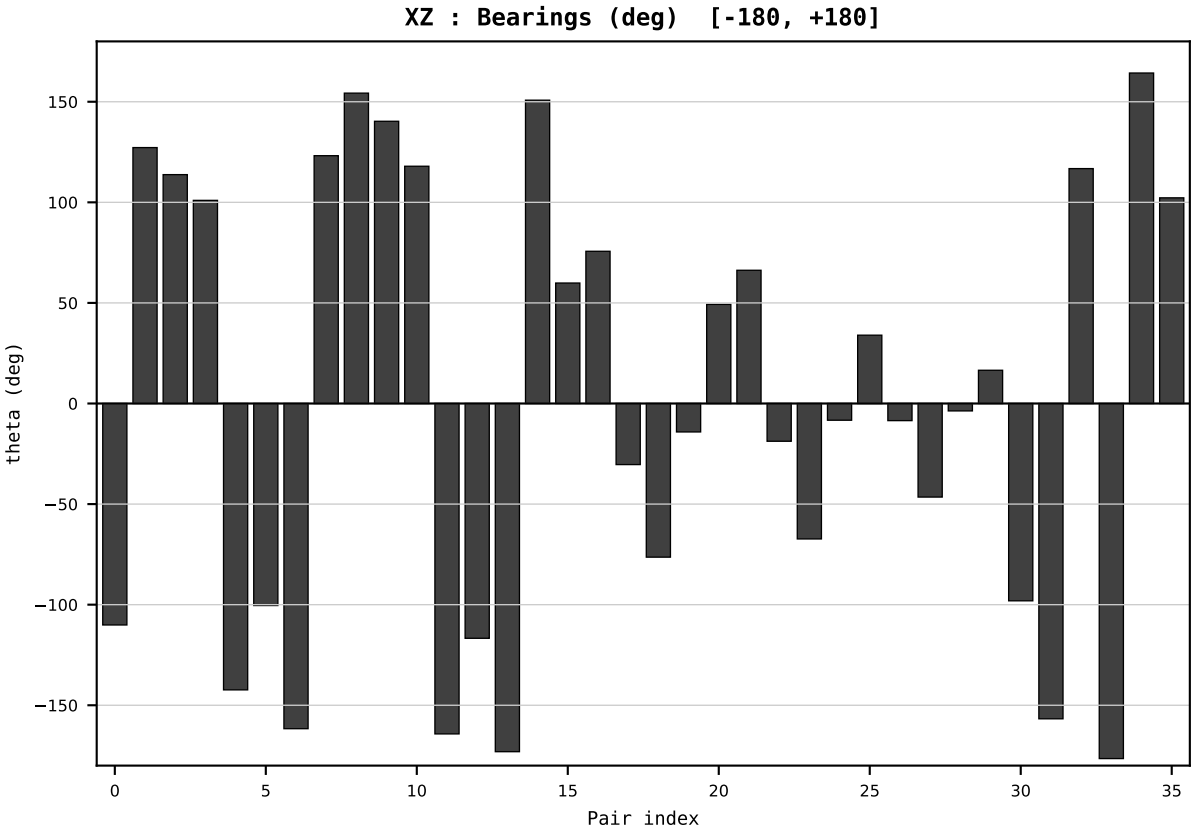
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2020
t : 20 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.467323
Ring = Hs-3
E_metric= 26.9059
kappa_HS= 580.05
omega = 3.3546 deg
d_A = 0.303539
Helm = Wind
Helm d = +0.156233
DR = 6.363 HLR
DR ratio= 580.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=21 | Year 2021 | D=9 N=26 pairs=36

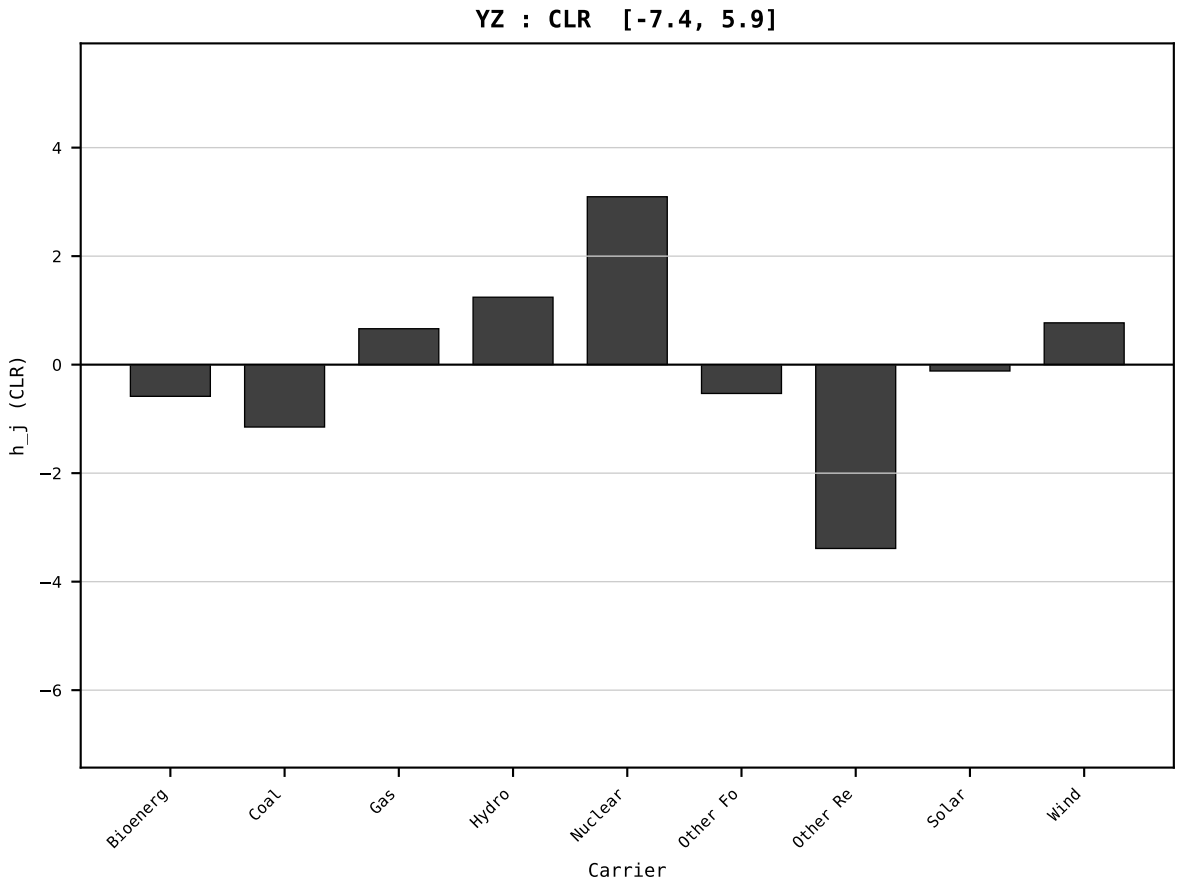
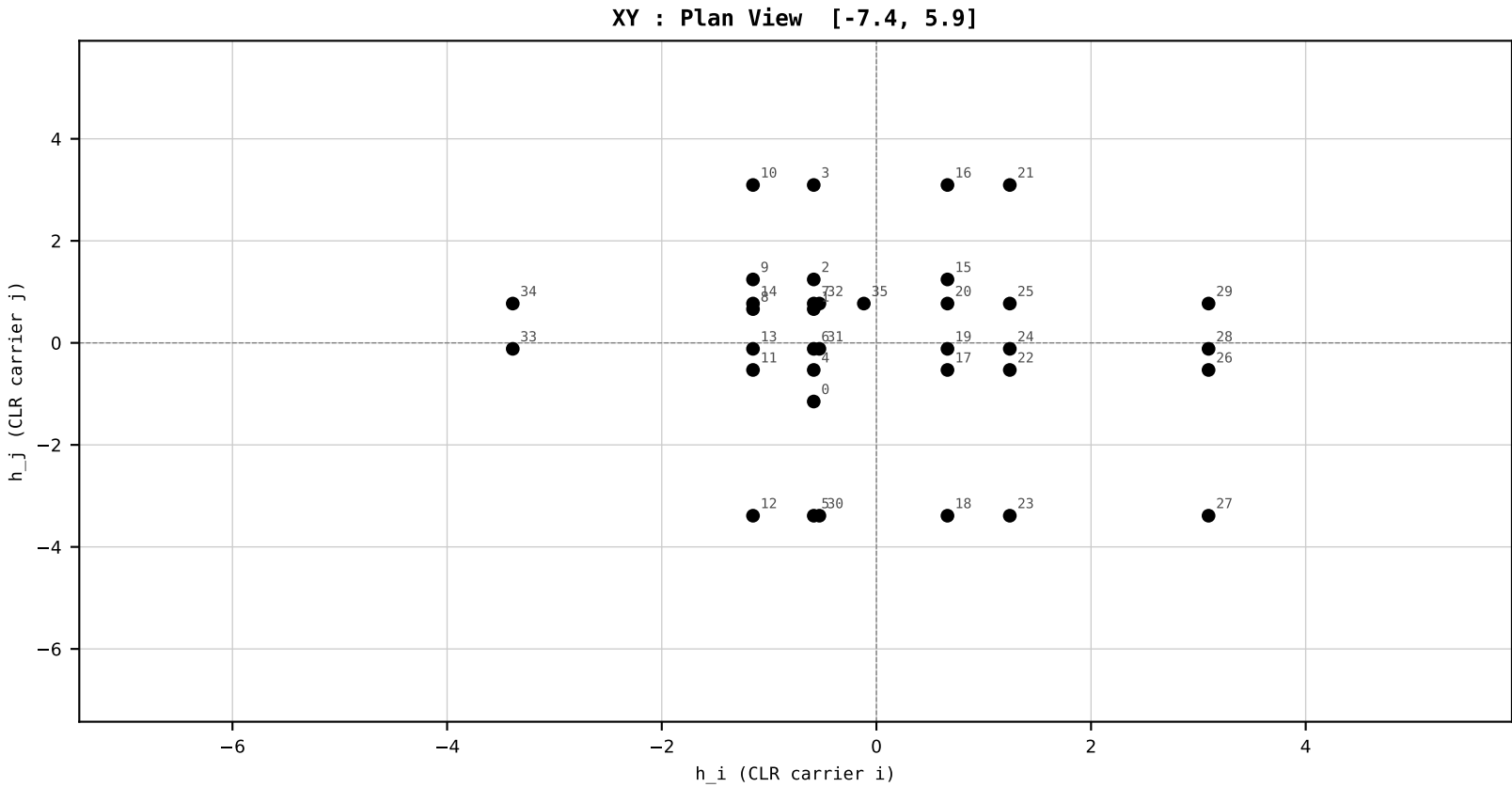
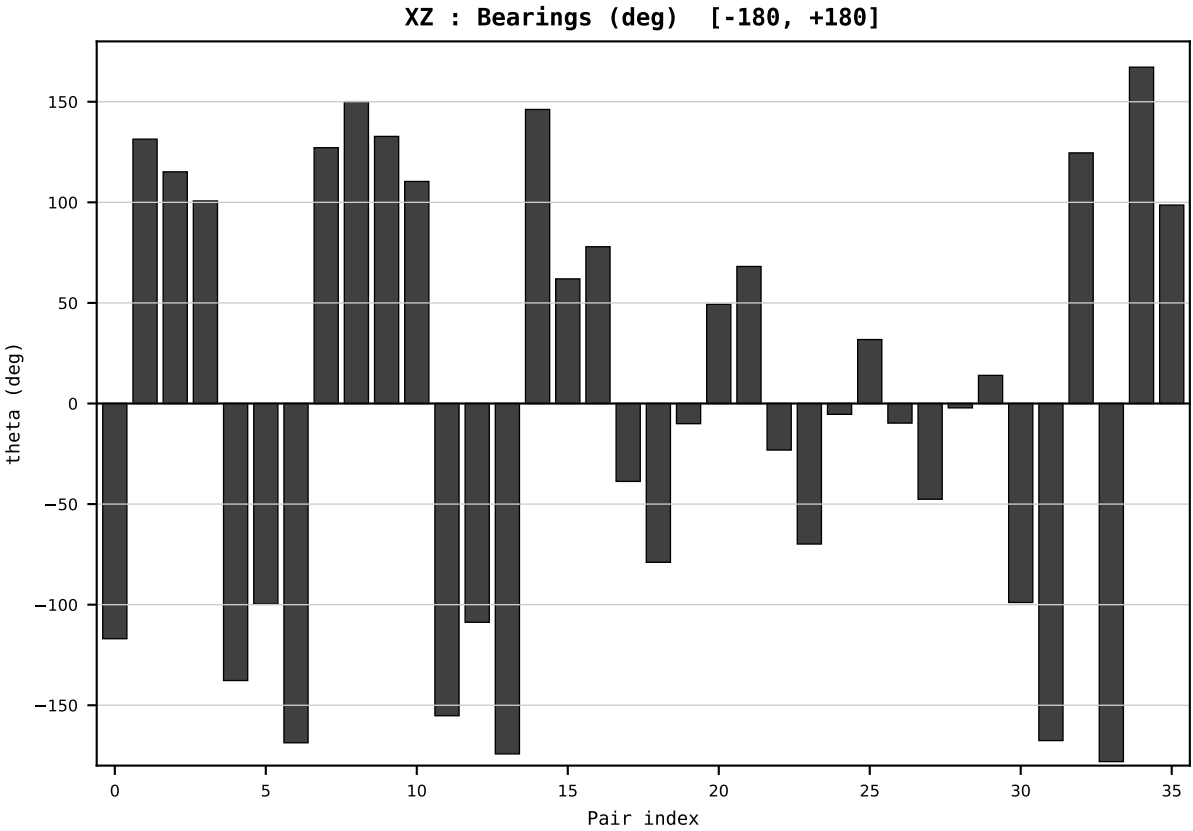
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2021
t : 21 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.478912
Ring = Hs-3
E_metric= 25.5954
kappa_HS= 654.07
omega = 6.1997 deg
d_A = 0.568610
Helm = Coal
Helm d = +0.493100
DR = 6.483 HLR
DR ratio= 654.1x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



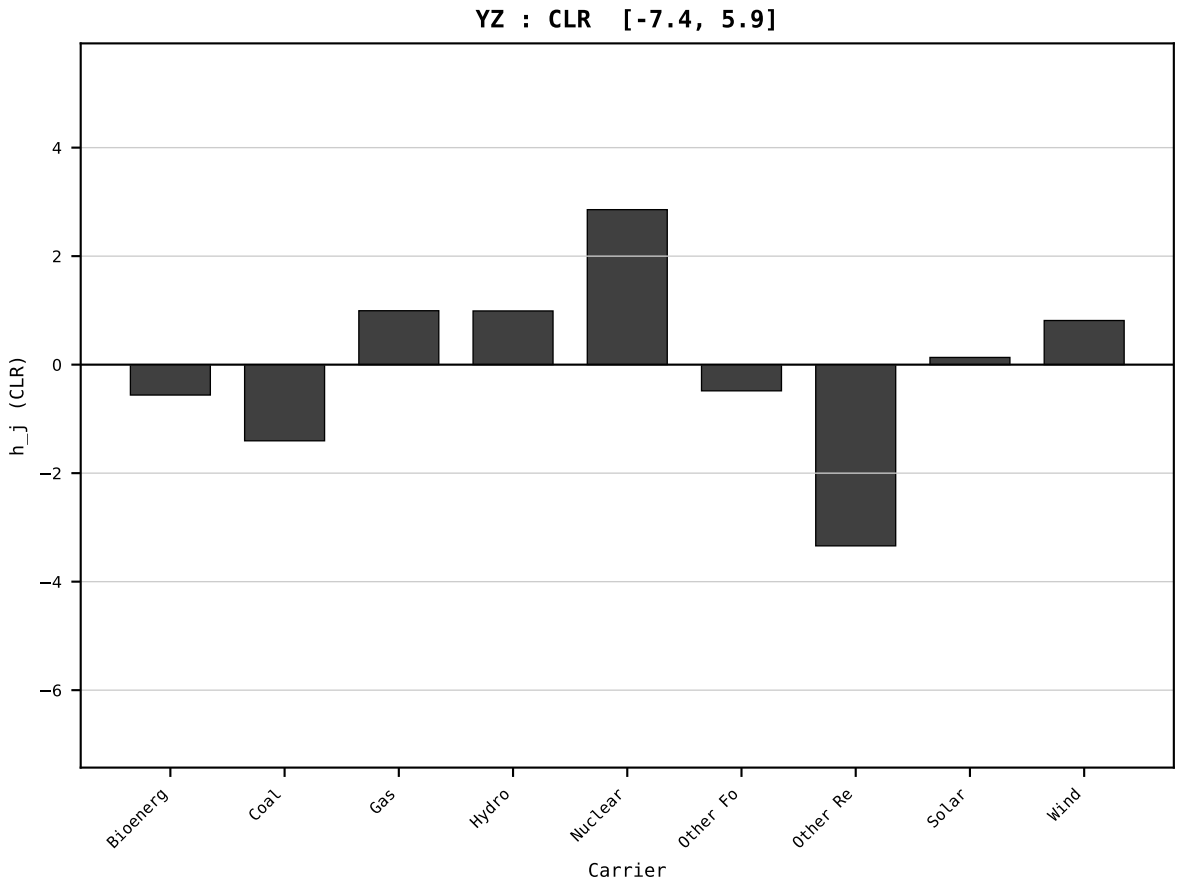
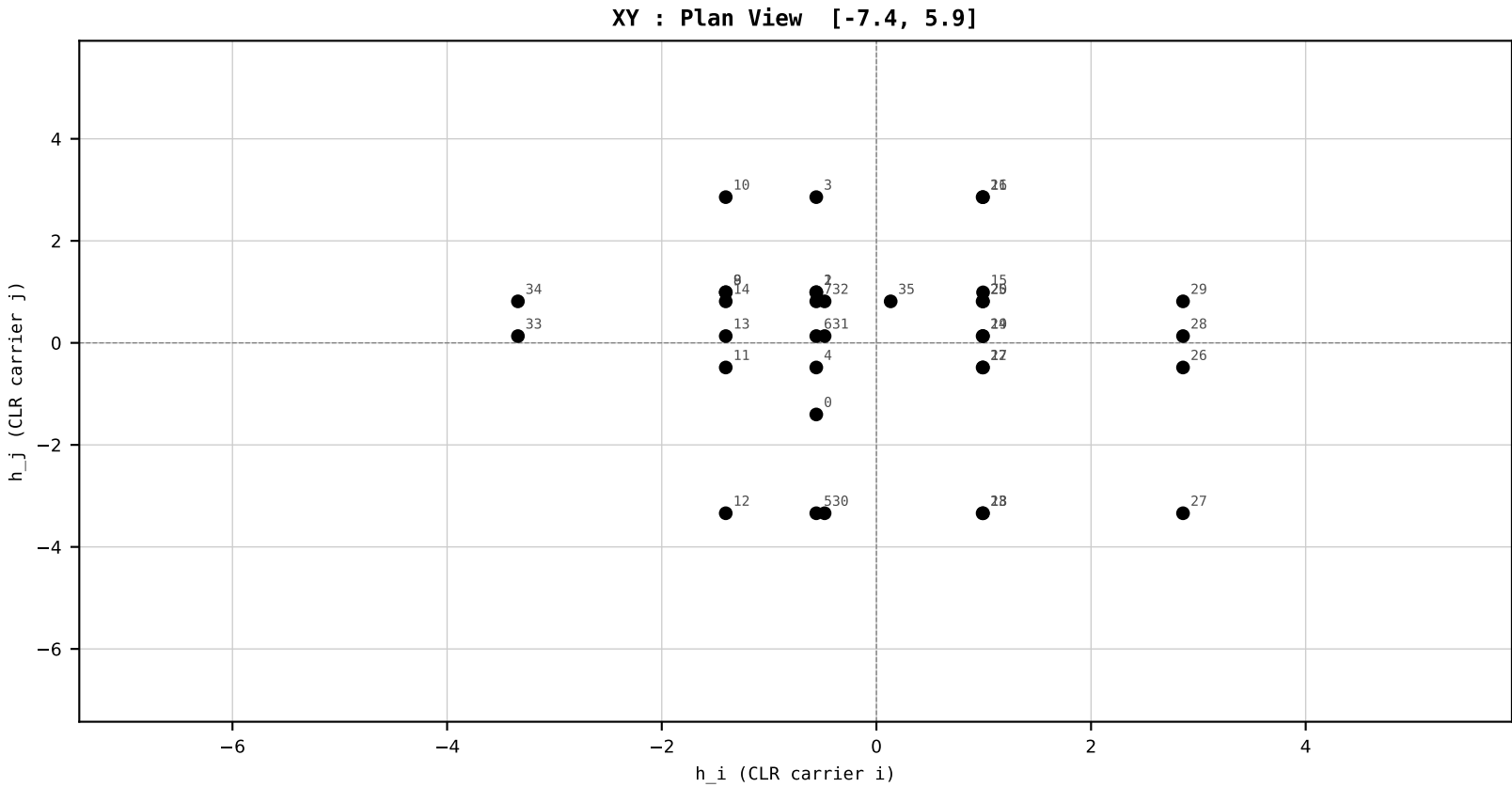
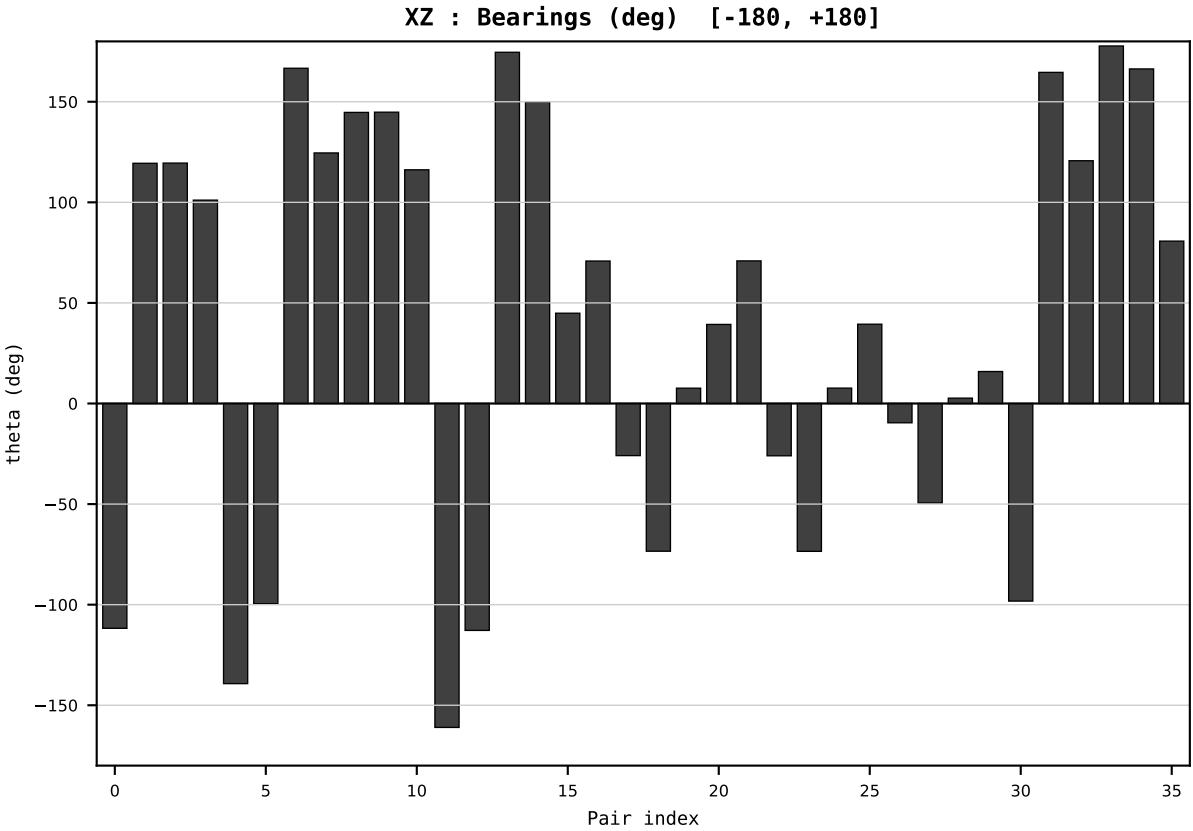
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2022
t : 22 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.409828
Ring = Hs-3
E_metric= 24.4815
kappa_HS= 491.22
omega = 6.8017 deg
d_A = 0.603939
Helm = Gas
Helm d = +0.331511
DR = 6.197 HLR
DR ratio= 491.2x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



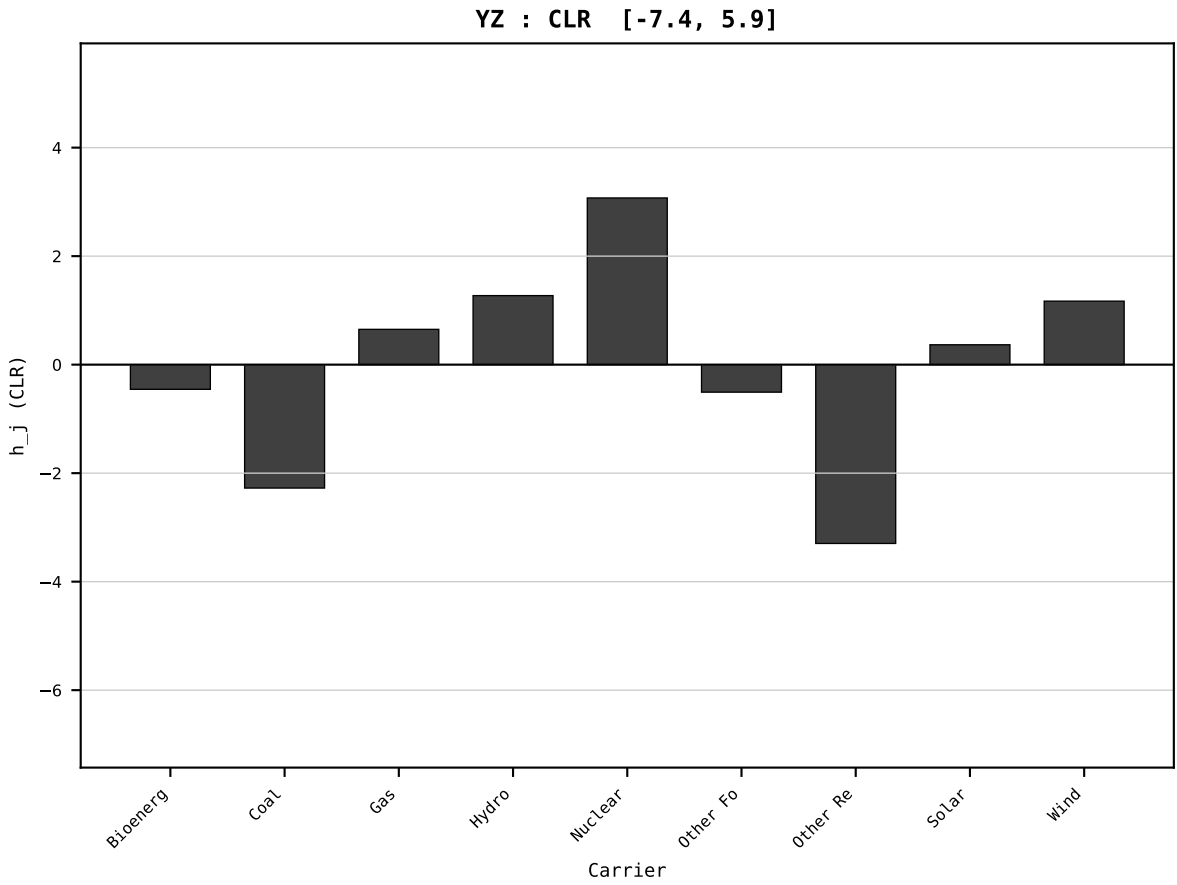
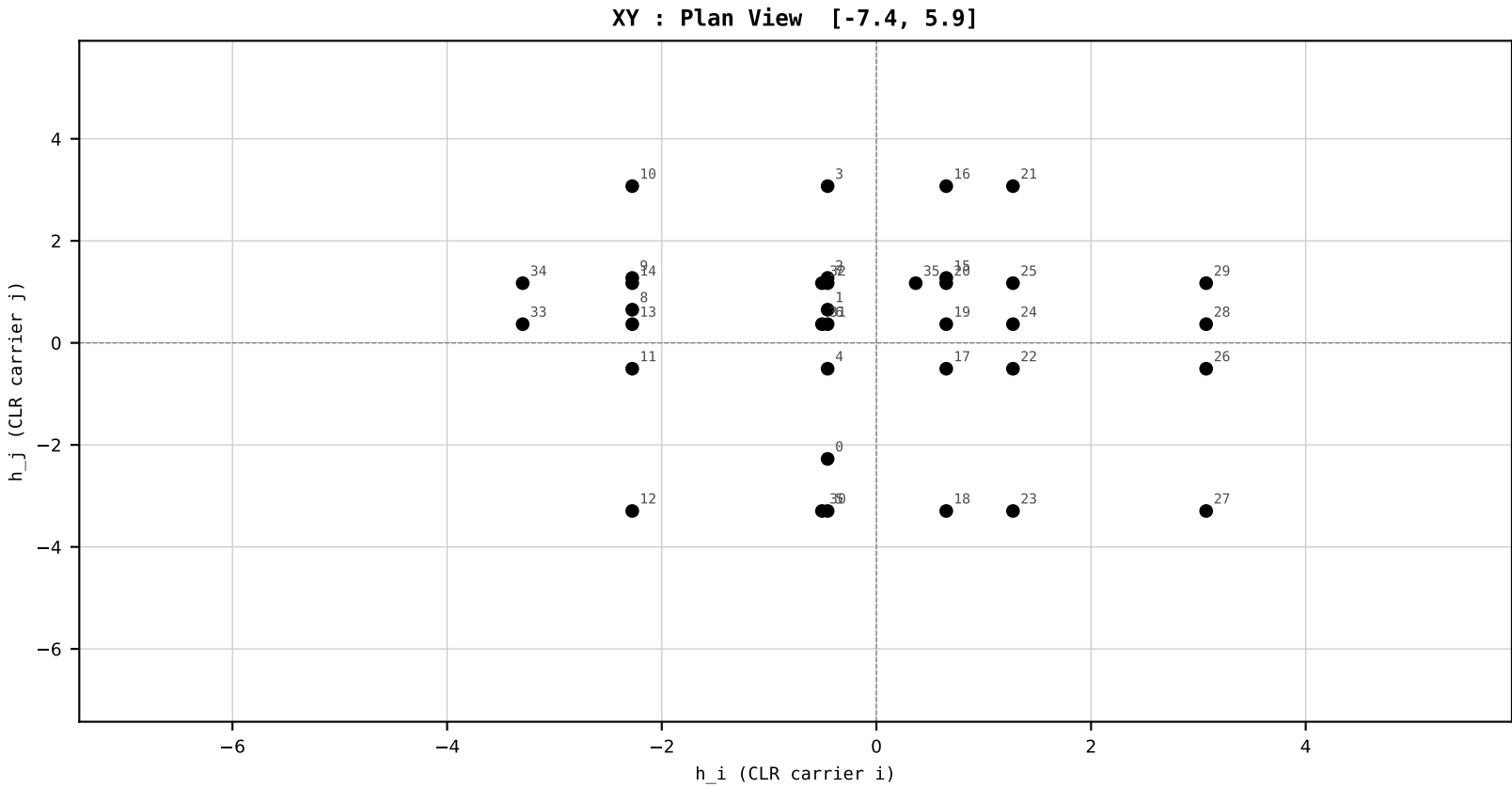
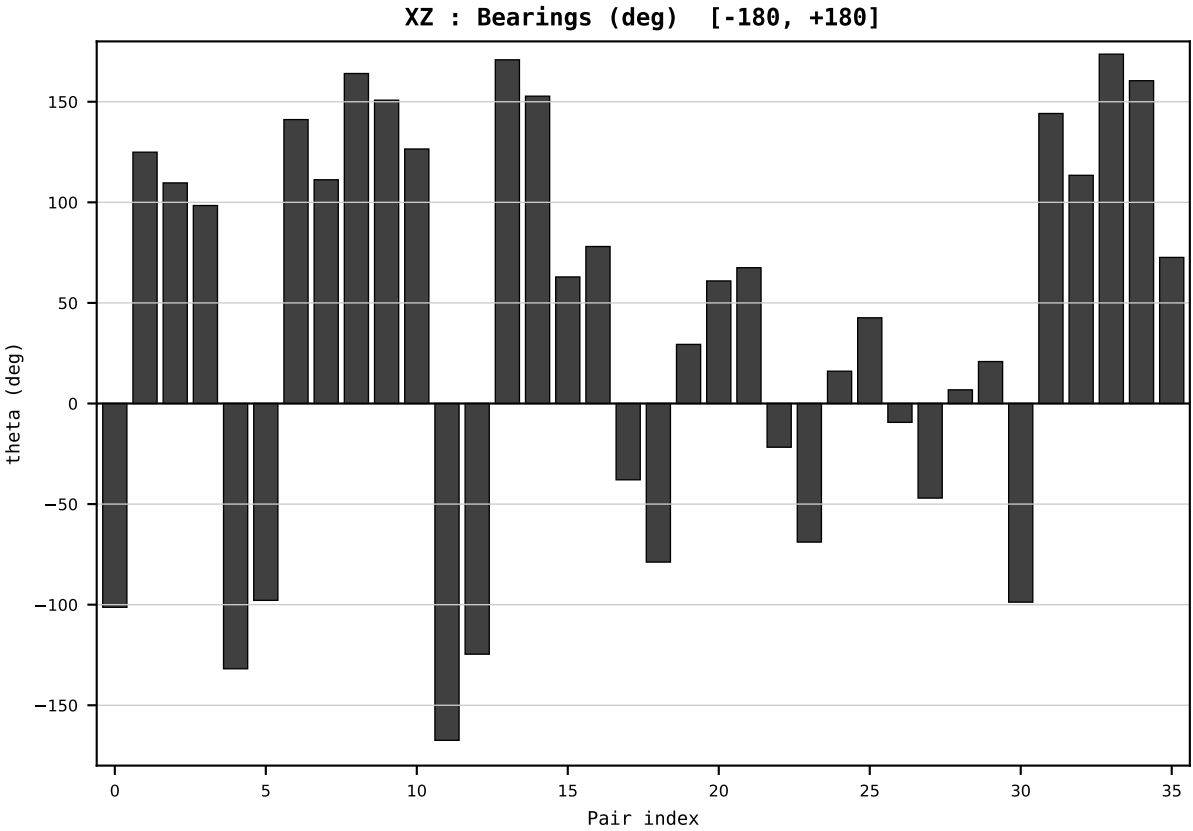
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2023
t : 23 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.444201
Ring = Hs-3
E_metric= 29.4888
kappa_HS= 583.10
omega = 10.8790 deg
d_A = 1.094786
Helm = Coal
Helm d = -0.871139
DR = 6.368 HLR
DR ratio= 583.1x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=24 | Year 2024 | D=9 N=26 pairs=36

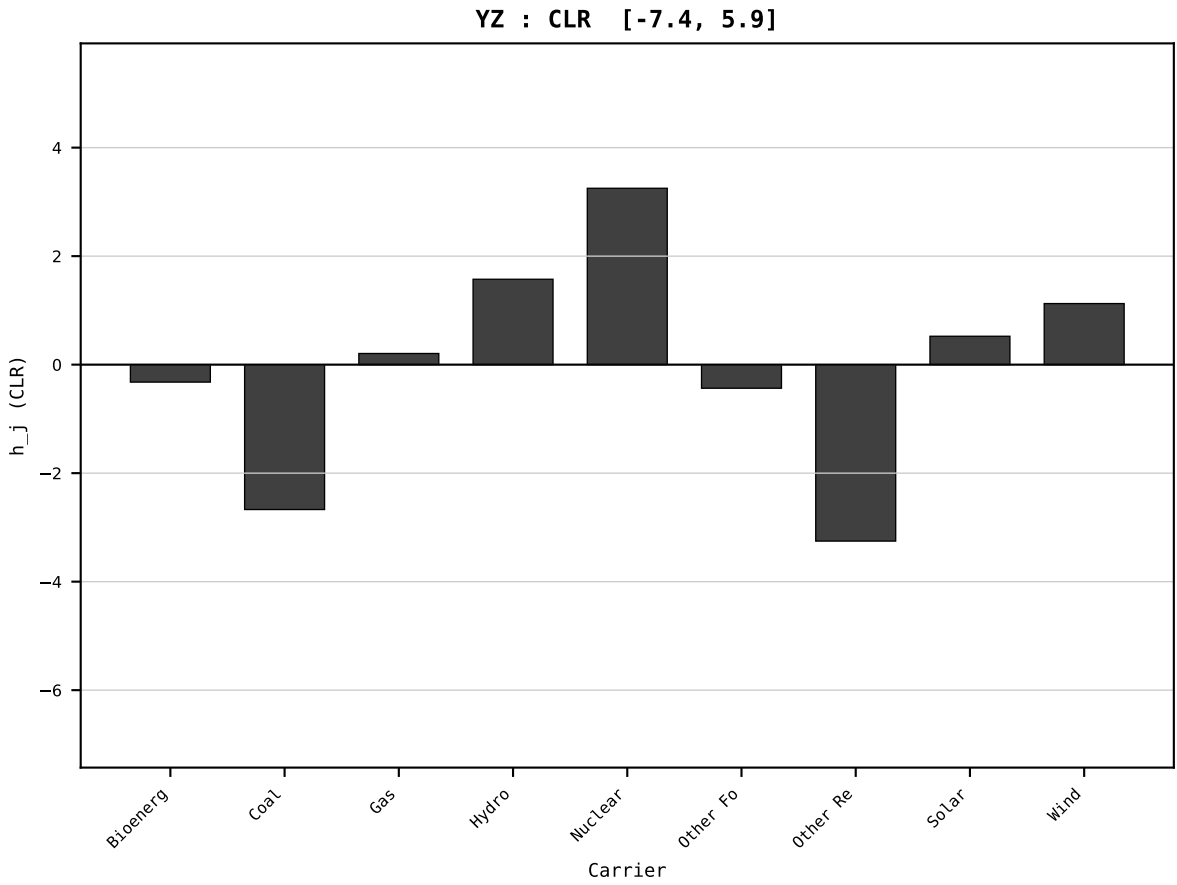
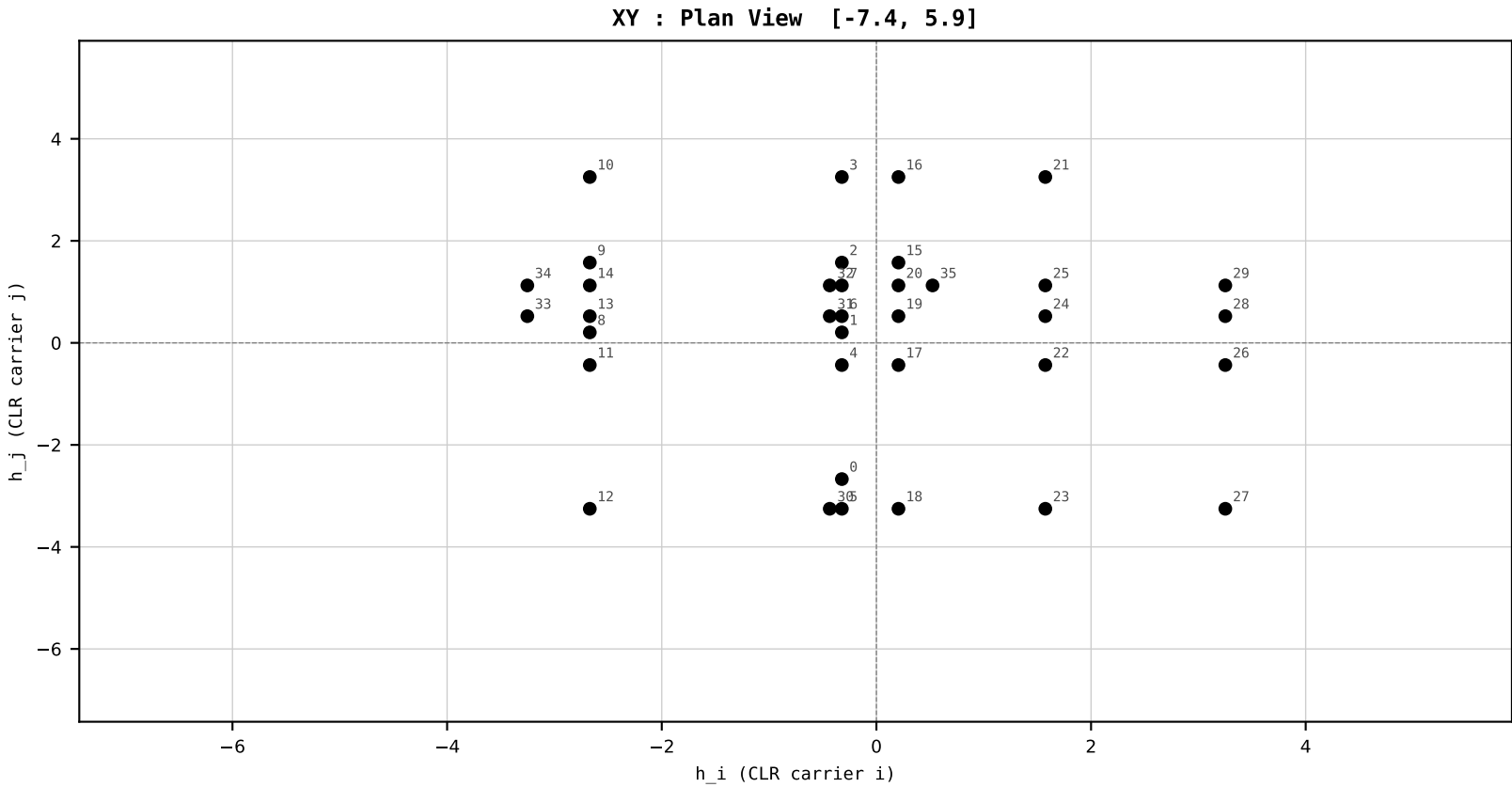
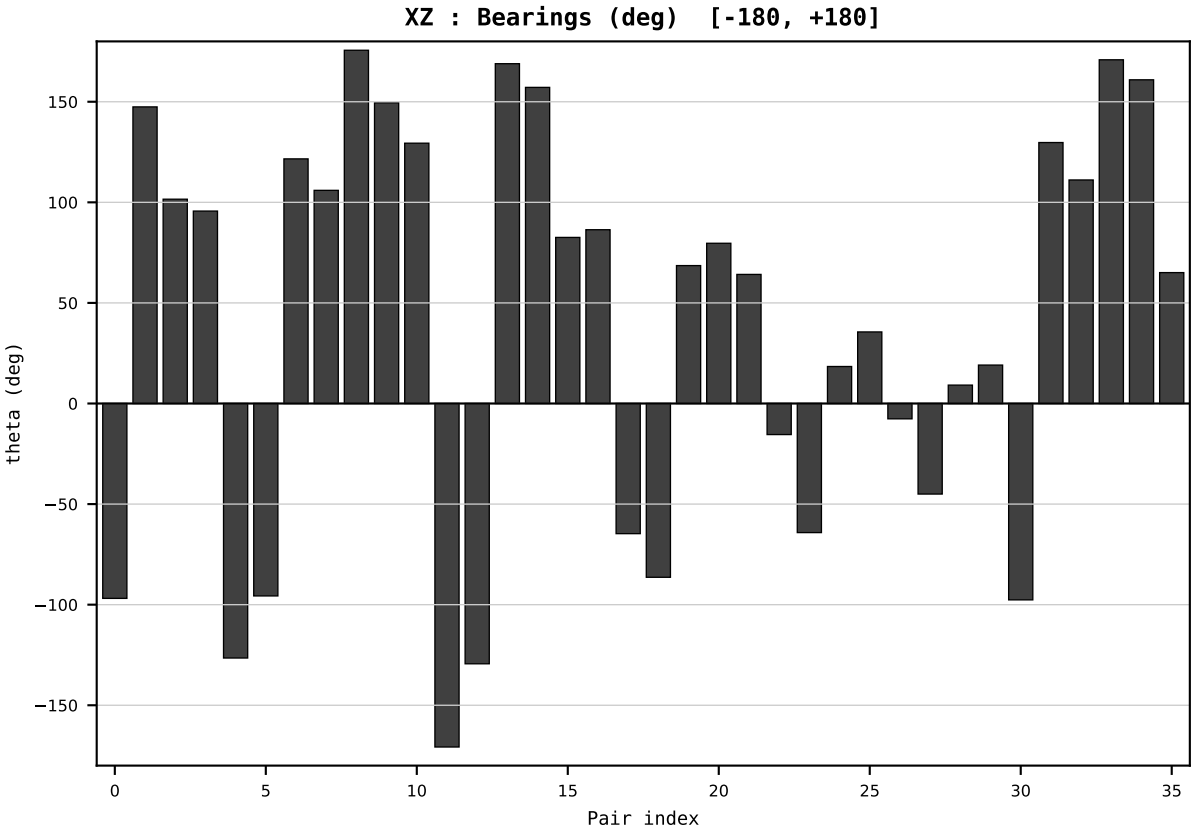
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year : 2024
t : 24 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.480863
Ring = Hs-3
E_metric= 32.6333
kappa_HS= 667.46
omega = 6.8989 deg
d_A = 0.727222
Helm = Gas
Helm d = -0.445163
DR = 6.503 HLR
DR ratio= 667.5x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=25 | Year 2025 | D=9 N=26 pairs=36

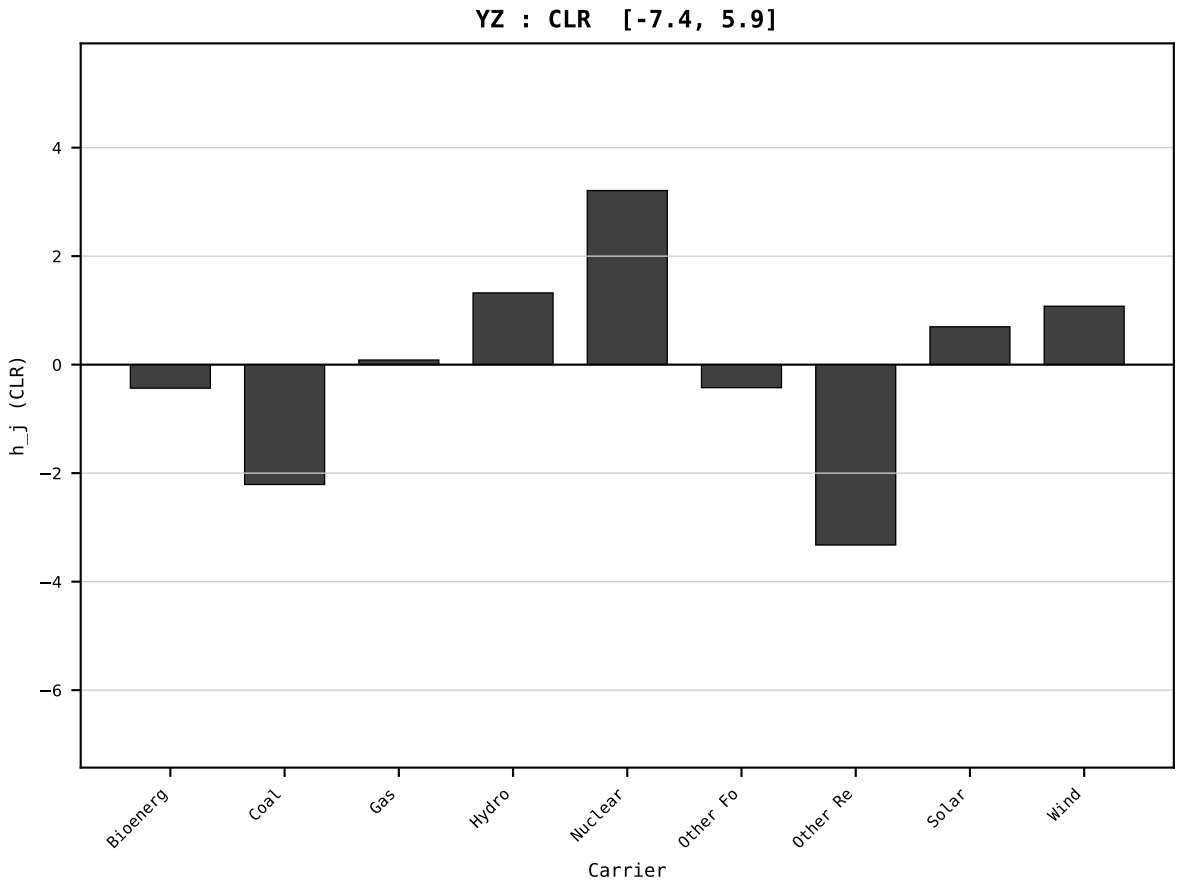
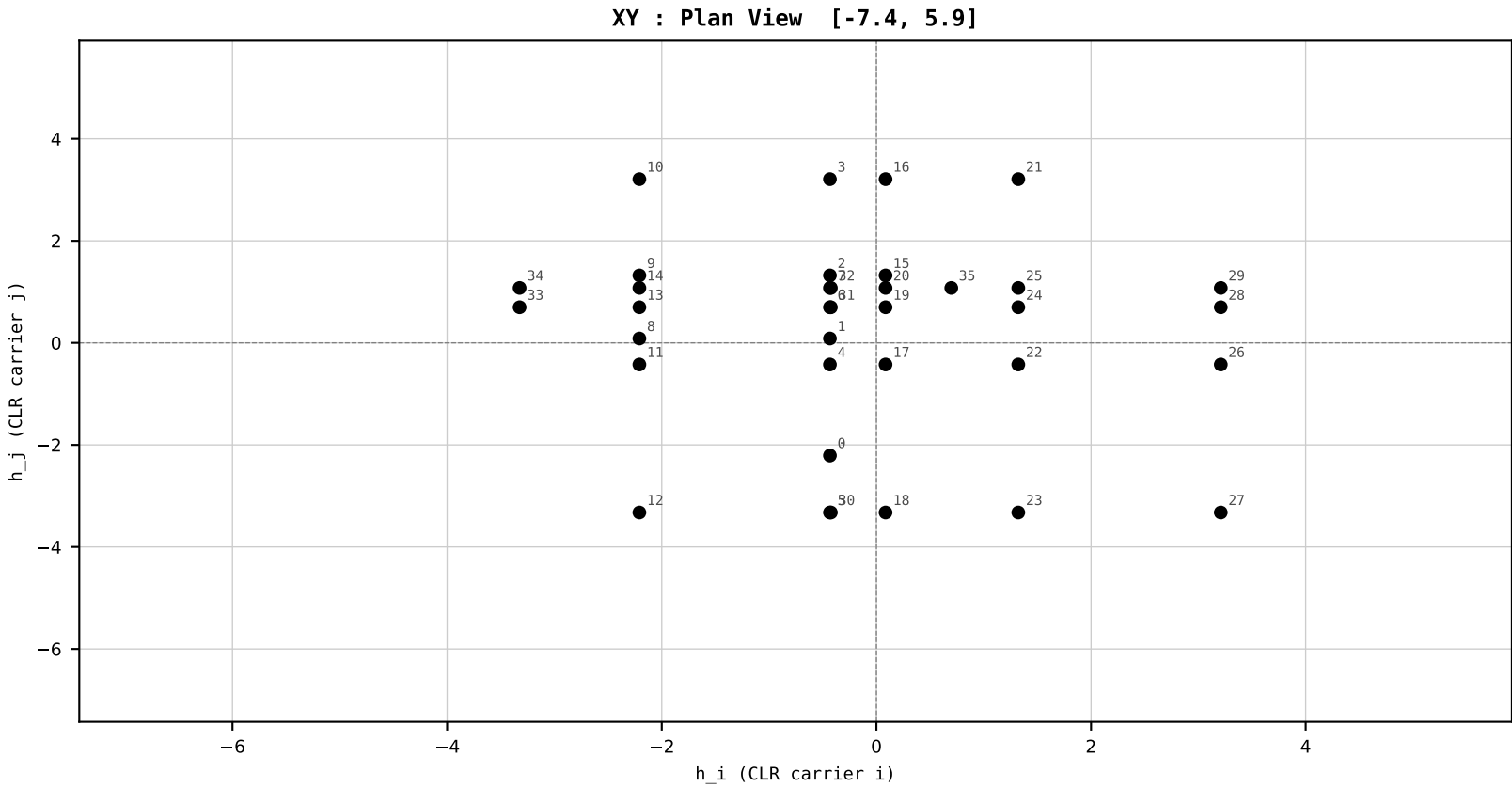
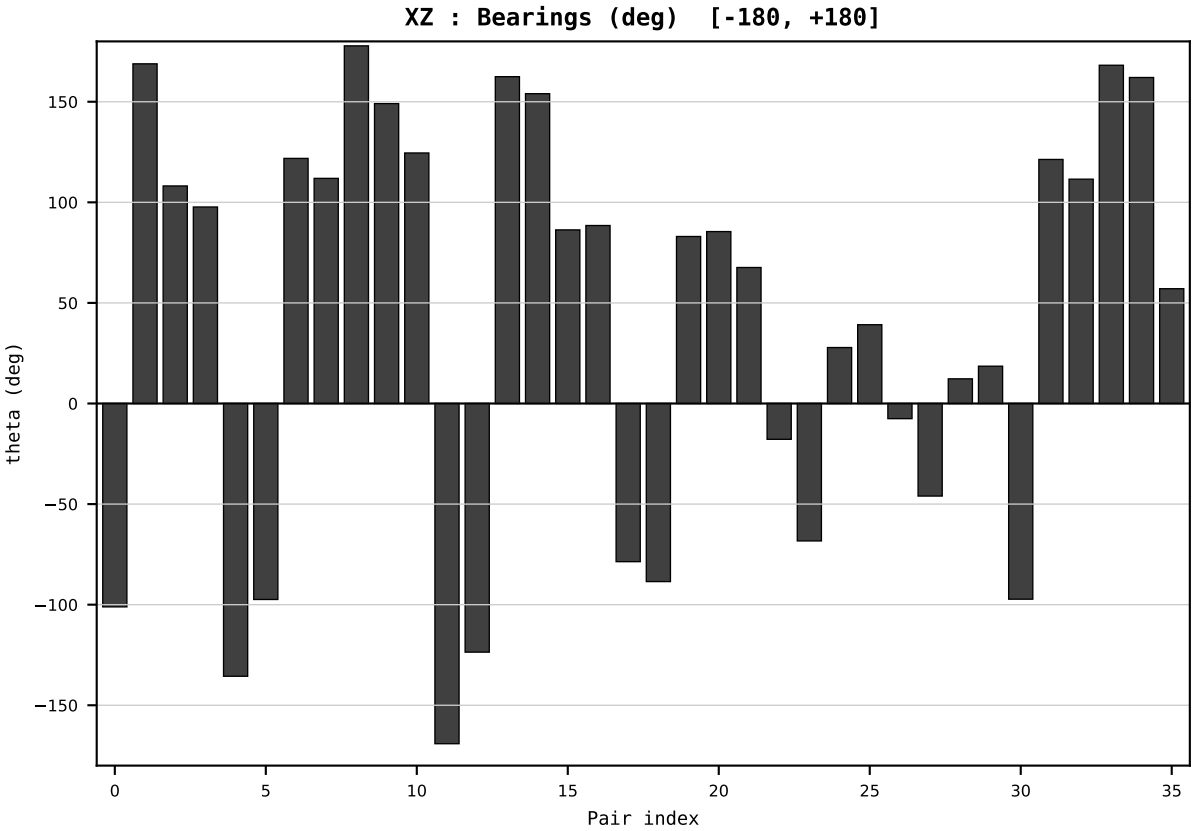
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

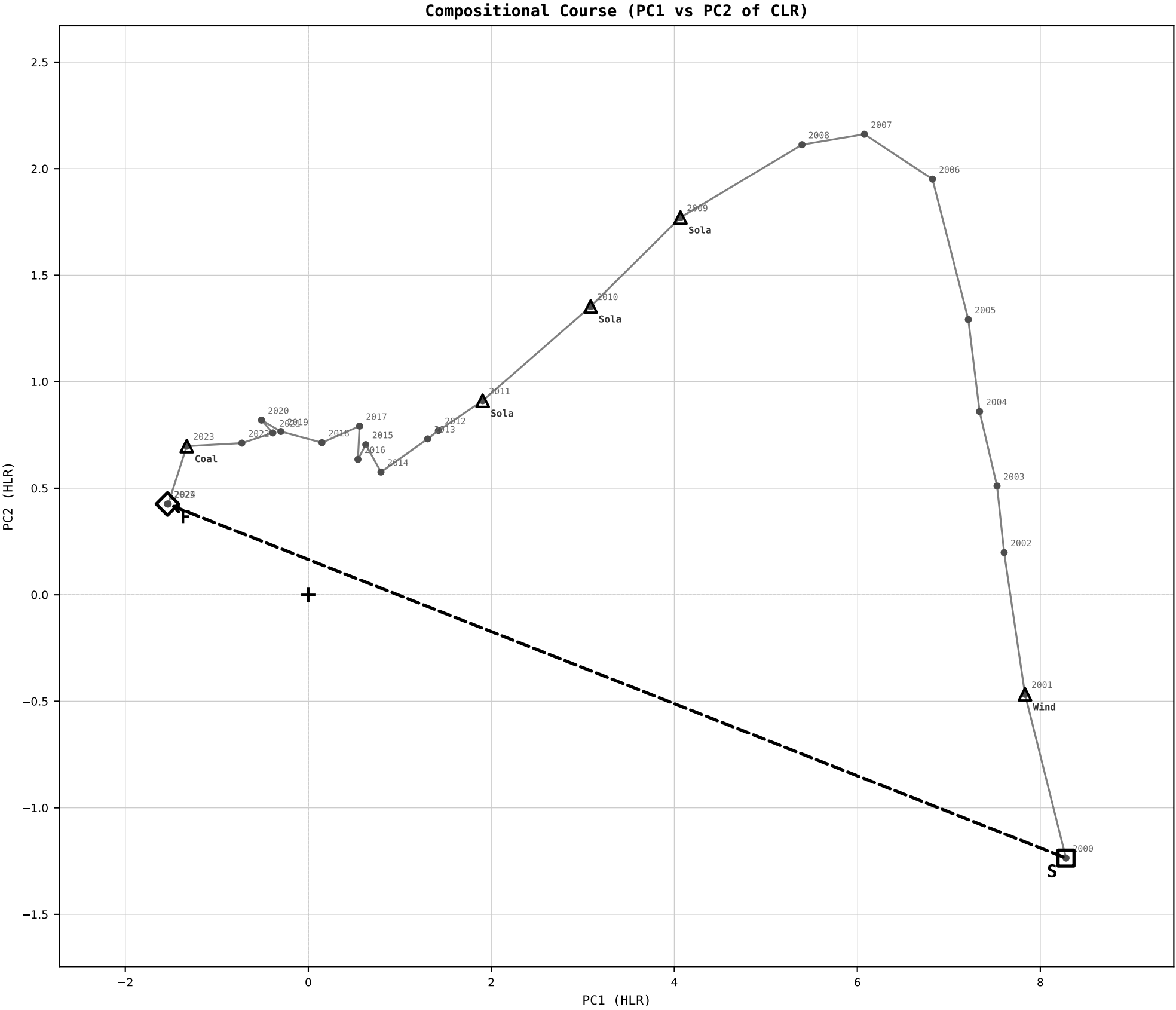
Dataset : ember_FRA_France_generation_TWh.csv
Year : 2025
t : 25 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.483790
Ring = Hs-3
E_metric= 29.9972
kappa_HS= 687.84
omega = 5.5004 deg
d_A = 0.586196
Helm = Coal
Helm d = +0.461974
DR = 6.534 HLR
DR ratio= 687.8x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind





COURSE METRICS

Scale Provenance:
Y_j → x_j=Y_j/T → h_j=CLR
→ PC projection
Unit: HLR (Higgins Log-Ratio)

Net distance = 10.0035 HLR
Path length = 17.9314 HLR
Directness = 0.5579

DR start = 10.634 HLR
ratio = 41516.0x
DR final = 6.534 HLR
ratio = 687.8x

E_{metric} start = 94.4242
E_{metric} final = 29.9972

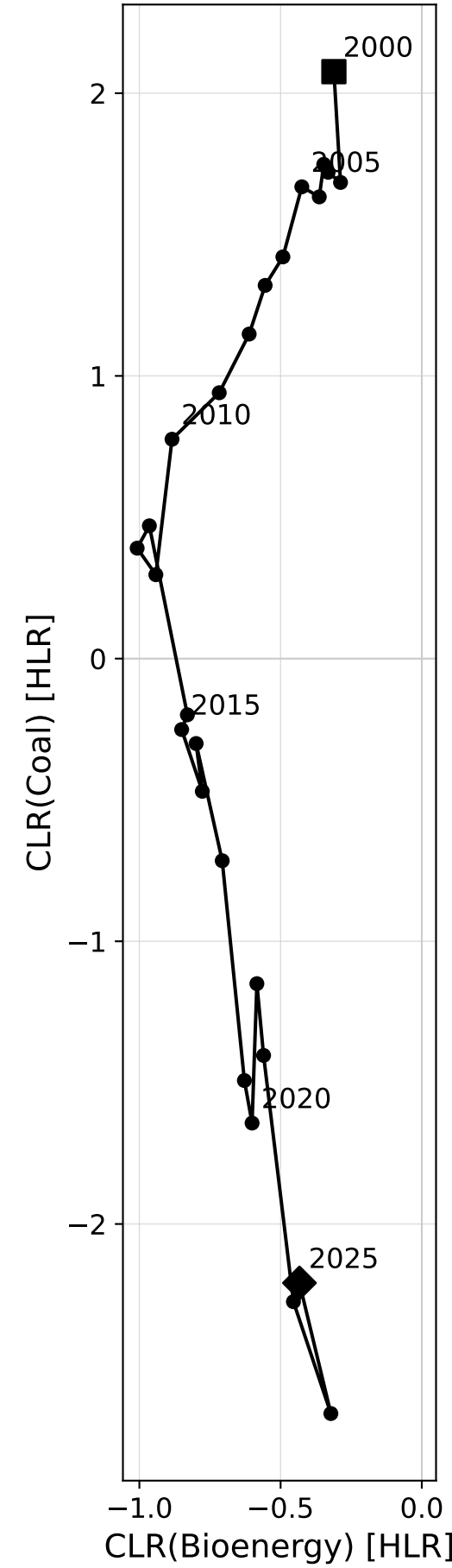
PC1 loadings:
Solar -0.768
Coal +0.387
Wind -0.385

PC2 loadings:
Wind +0.841
Solar -0.448
Hydro -0.160

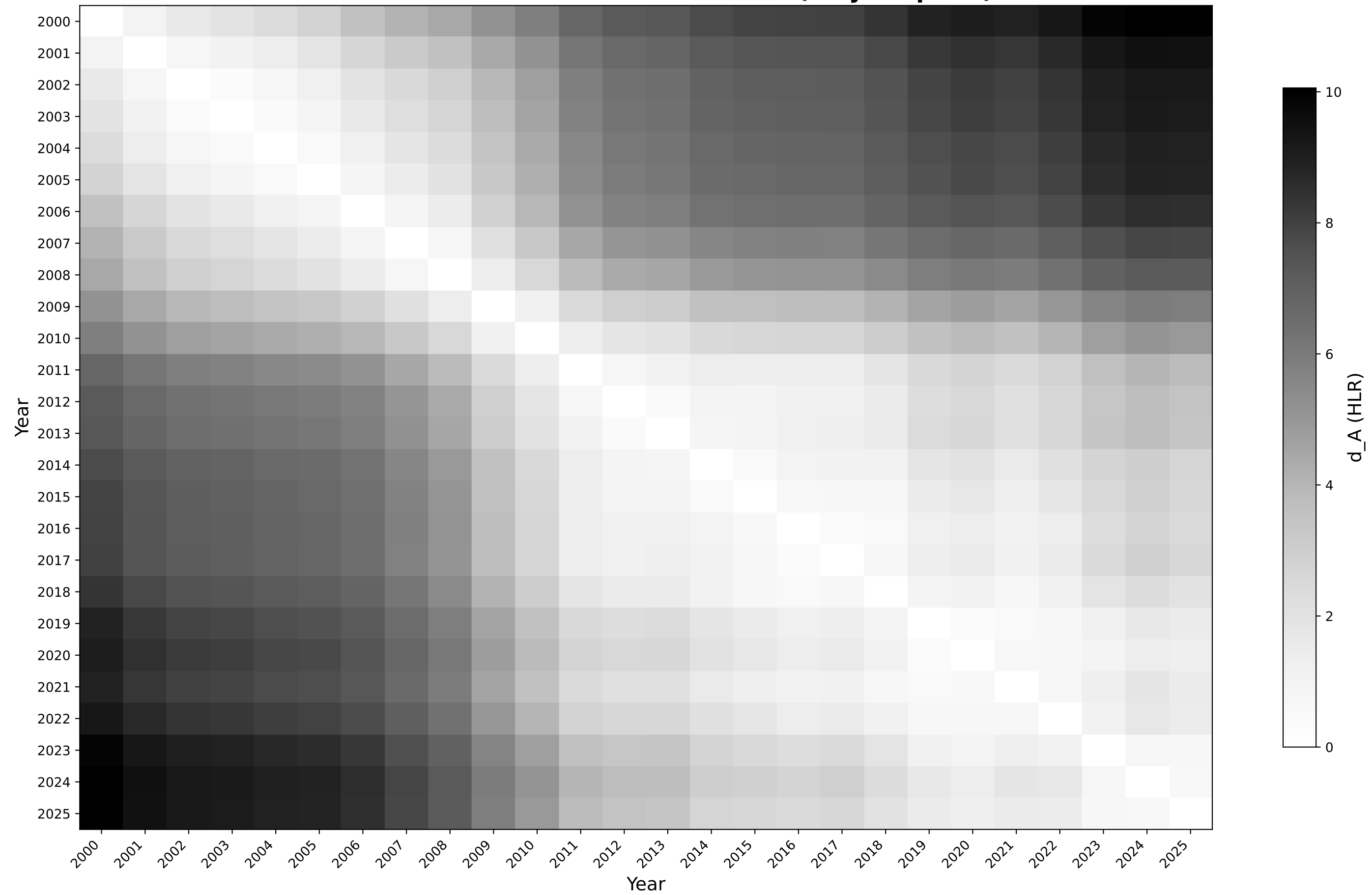
Top displacement events:
2009 Solar d=1.437
2011 Solar d=1.337
2010 Solar d=1.129
2023 Coal d=1.095
2001 Wind d=0.998

The instrument reads.
The expert decides.
The loop stays open.

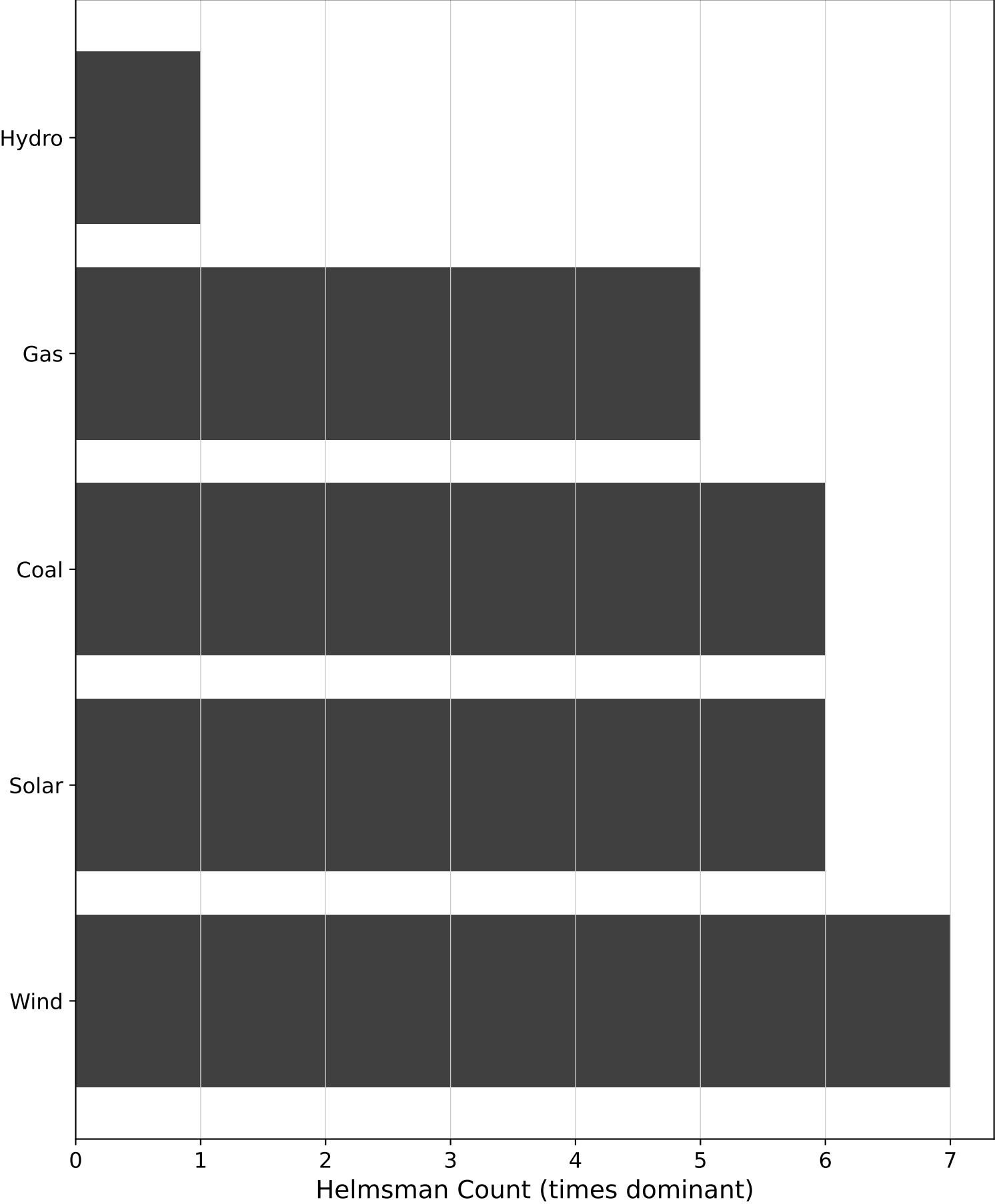
SYSTEM COURSE PLOT — XY Plan View (CLR Space)



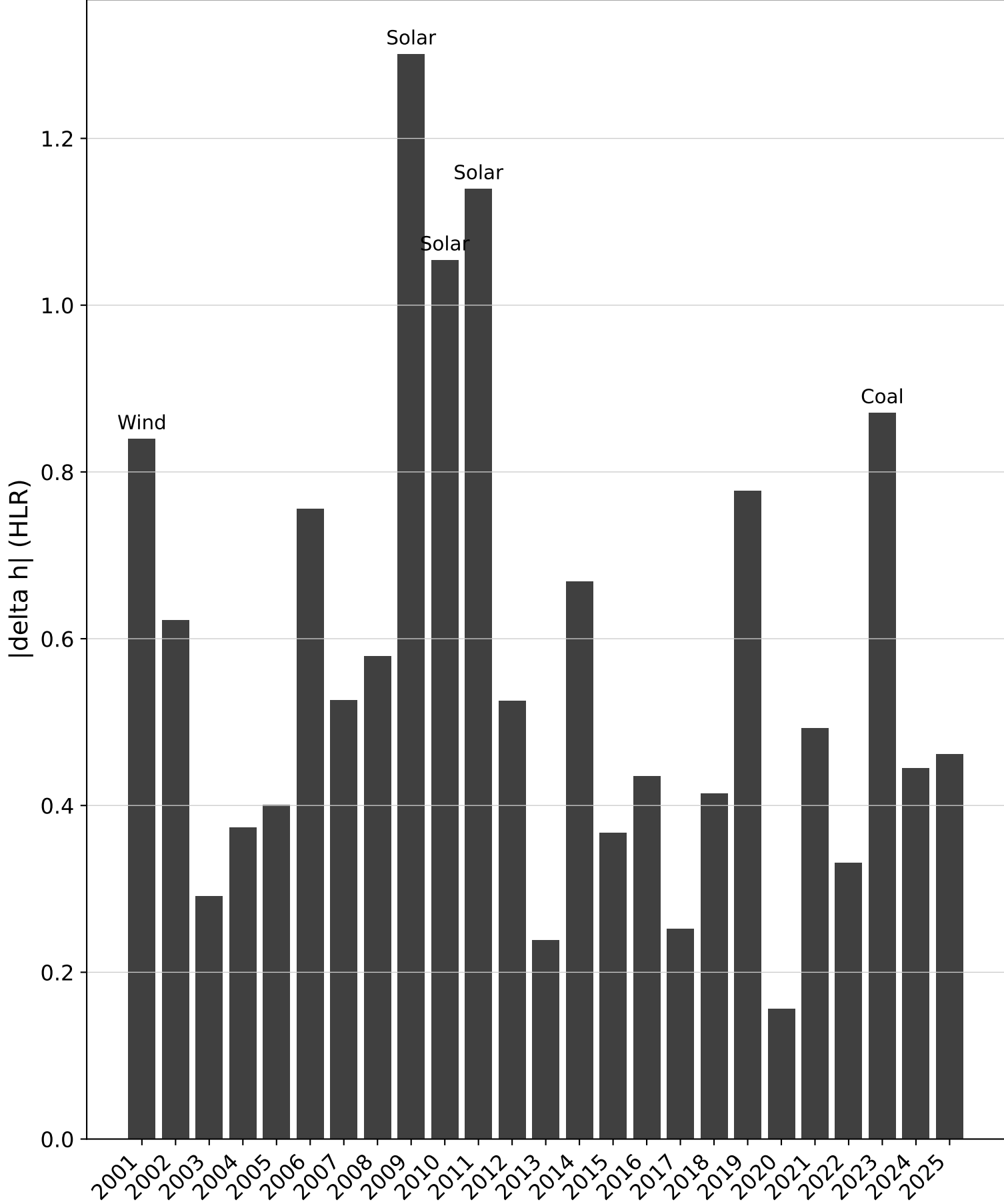
STAGE 2 — Aitchison Distance Matrix (all year pairs)



HELMSMAN FREQUENCY



HELMSMAN DISPLACEMENT PER YEAR



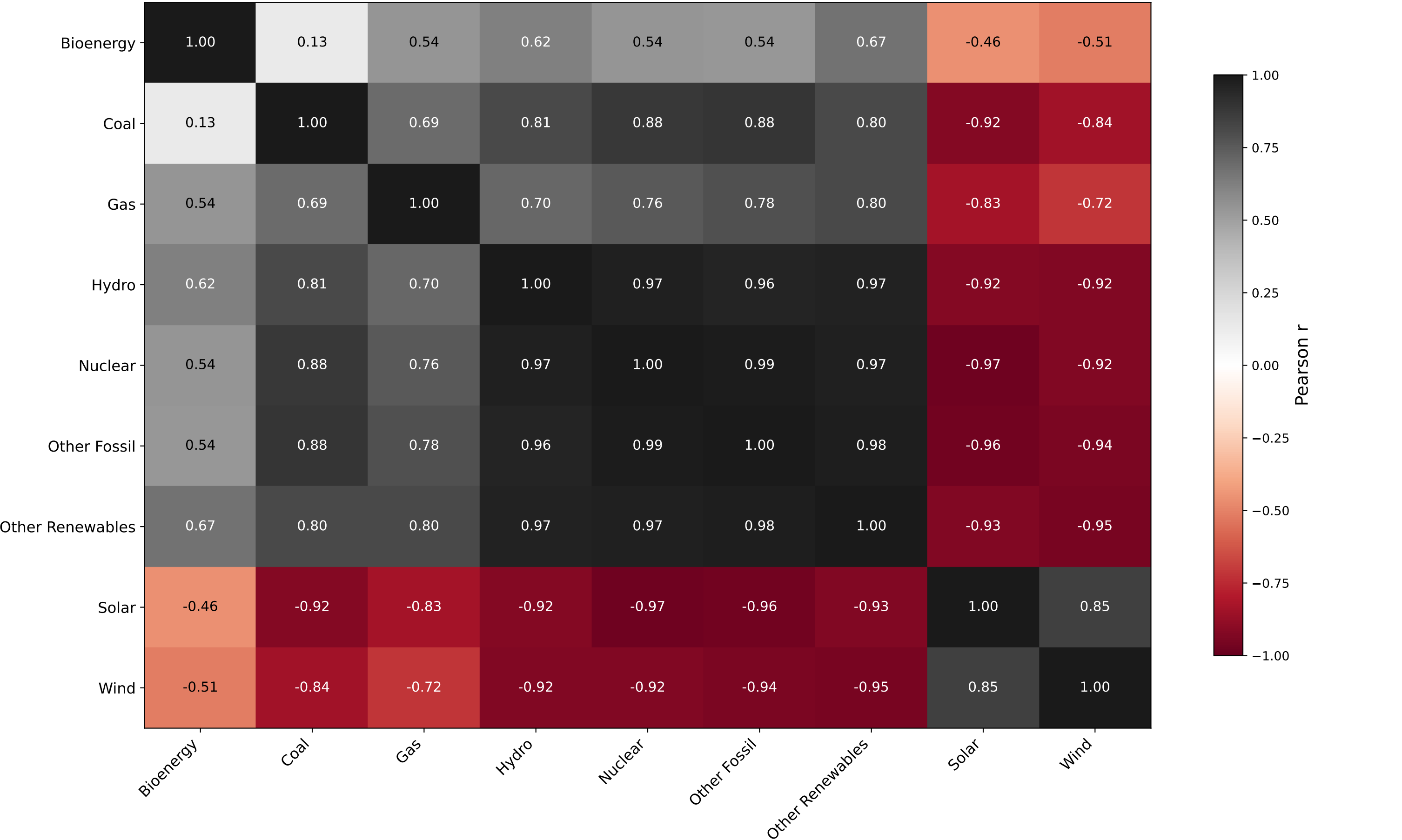
STAGE 2 — Pairwise Divergence Ranking (Top 15)

Rank	Year A	Year B	d_A (HLR)	Angle (deg)	Dominant Carrier
1	2000	2024	10.0592	76.5	Solar
2	2000	2025	10.0035	76.8	Solar
3	2000	2023	9.8631	75.4	Solar
4	2001	2024	9.4719	73.4	Solar
5	2001	2025	9.4392	73.9	Solar
6	2001	2023	9.2621	72.1	Solar
7	2000	2022	9.2488	69.7	Solar
8	2002	2024	9.2171	72.8	Solar
9	2002	2025	9.1780	73.2	Solar
10	2003	2024	9.1396	72.6	Solar
11	2003	2025	9.0974	72.9	Solar
12	2000	2020	9.0640	67.1	Solar
13	2002	2023	8.9766	71.1	Solar
14	2004	2024	8.9488	71.3	Solar
15	2004	2025	8.9135	71.7	Solar

STAGE 3 — Triadic Metric Triangle Area Ranking (Top 10)

Rank	Year A	Year B	Year C	Area (HLR^2)	Imbalance
1	2000	2007	2024	15.0050	2.47
2	2000	2007	2025	14.9541	2.46
3	2000	2006	2024	14.6841	2.83
4	2000	2006	2025	14.6763	2.81
5	2000	2007	2023	14.3006	2.42
6	2000	2008	2024	14.2167	2.27
7	2000	2008	2025	14.1315	2.26
8	2000	2006	2023	14.1174	2.77
9	2000	2008	2023	13.4178	2.22
10	2000	2007	2022	13.3358	2.27

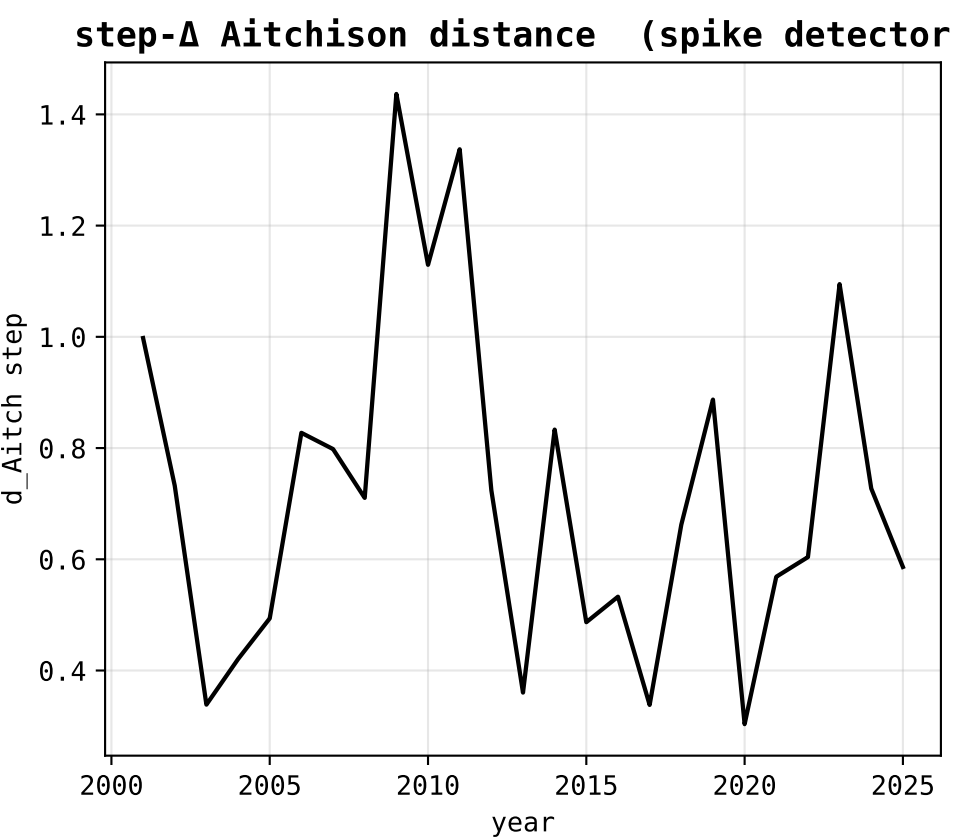
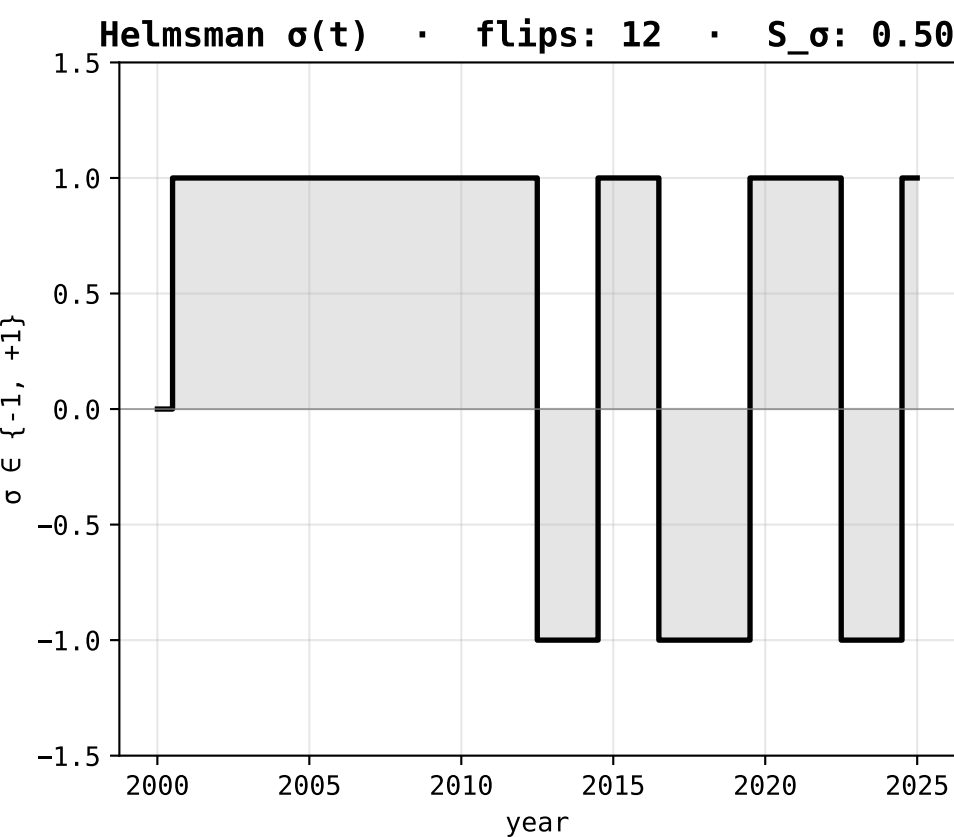
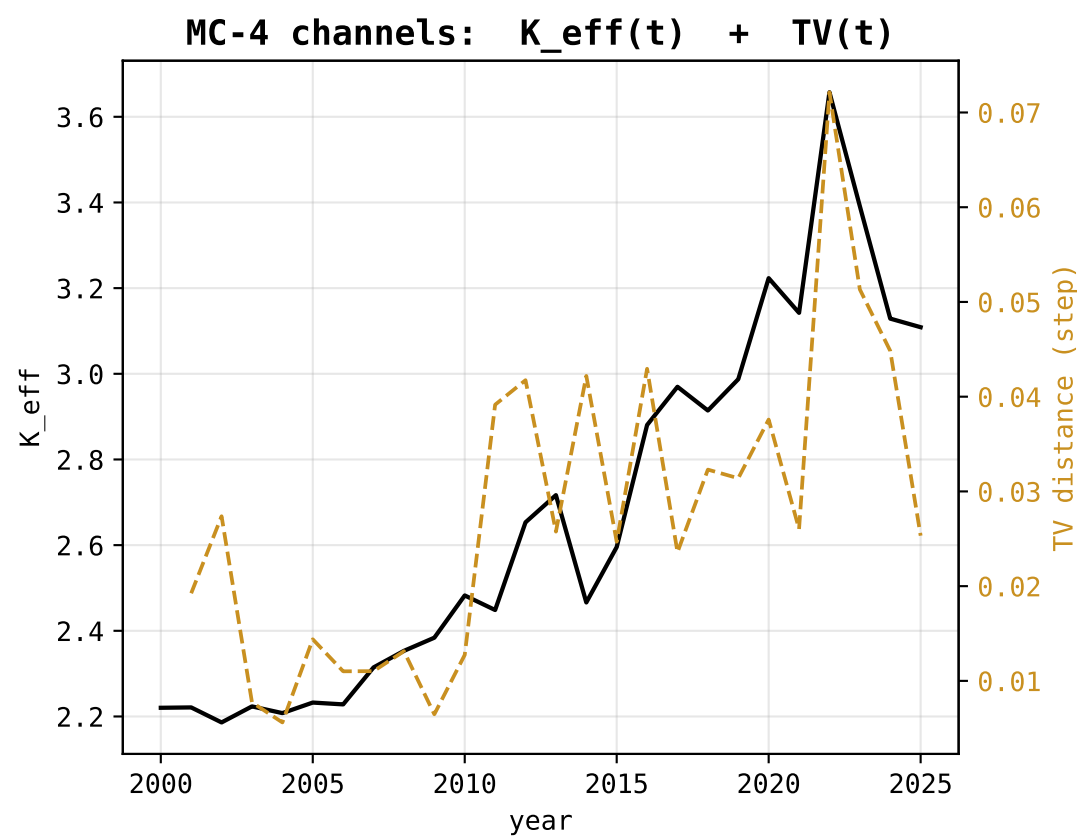
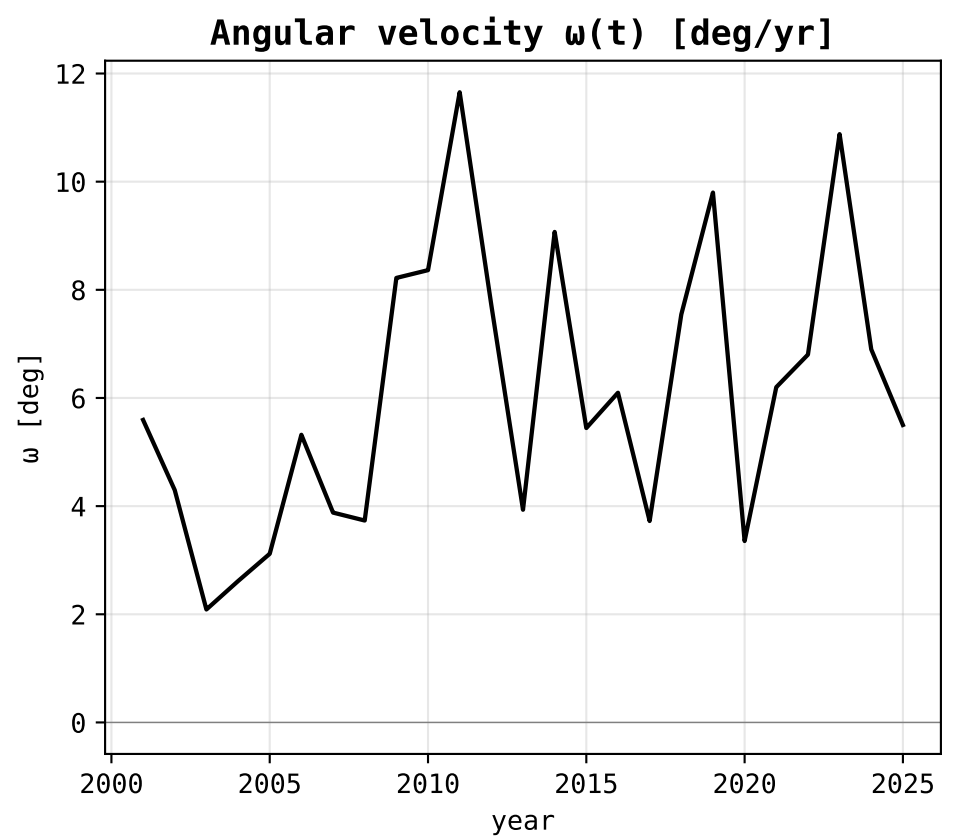
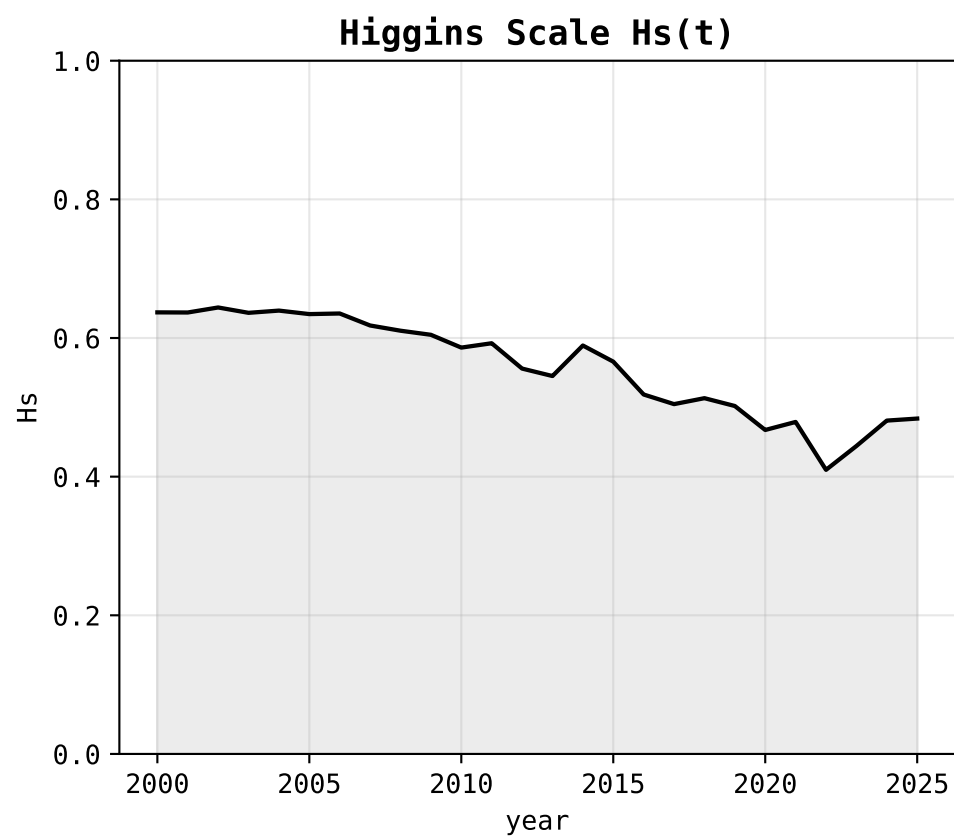
STAGE 3 — Carrier CLR Trajectory Correlation



NAVIGATION SUMMARY — All Stage 1 Metrics by Year

Year	Hs	Ring	E_metric	omega(deg)	d_A	Helmsman	kappa_max
2000	0.6370	Hs-4	94.4	0.00	0.0000	---	53256
2001	0.6368	Hs-4	87.6	5.59	0.9978	Wind	54260
2002	0.6440	Hs-4	83.2	4.30	0.7323	Wind	55124
2003	0.6363	Hs-4	81.9	2.09	0.3386	Wind	55913
2004	0.6395	Hs-4	80.2	2.61	0.4205	Wind	56630
2005	0.6345	Hs-4	78.6	3.12	0.4937	Wind	56914
2006	0.6354	Hs-4	76.2	5.32	0.8274	Wind	56730
2007	0.6181	Hs-4	66.8	3.88	0.7982	Solar	28092
2008	0.6105	Hs-4	59.1	3.73	0.7104	Solar	14174
2009	0.6046	Hs-4	44.7	8.22	1.4367	Solar	3111
2010	0.5862	Hs-4	36.4	8.36	1.1295	Solar	1172
2011	0.5924	Hs-4	28.8	11.65	1.3373	Solar	1051
2012	0.5559	Hs-4	27.4	7.72	0.7241	Solar	1113
2013	0.5451	Hs-4	27.6	3.93	0.3603	Gas	1154
2014	0.5892	Hs-4	26.0	9.07	0.8332	Coal	1011
2015	0.5659	Hs-4	25.2	5.44	0.4868	Gas	990
2016	0.5185	Hs-4	24.9	6.10	0.5327	Gas	931
2017	0.5046	Hs-4	23.9	3.72	0.3381	Hydro	855
2018	0.5132	Hs-4	25.1	7.54	0.6621	Coal	945
2019	0.5019	Hs-4	26.8	9.80	0.8872	Coal	927
2020	0.4673	Hs-3	26.9	3.35	0.3035	Wind	864
2021	0.4789	Hs-3	25.6	6.20	0.5686	Coal	949
2022	0.4098	Hs-3	24.5	6.80	0.6039	Gas	781
2023	0.4442	Hs-3	29.5	10.88	1.0948	Coal	894
2024	0.4809	Hs-3	32.6	6.90	0.7272	Gas	986
2025	0.4838	Hs-3	30.0	5.50	0.5862	Coal	1000

CNQ v2.0.0 – Quaternion View – FRA France · D=9 · N=26



CNQ DIAGNOSTICS · FRA (D = 9)

CHSH joint-coherence:
not computed at D = 9
(CHSH requires D=8 twin factoring;
D = 9 does not factor as 4x4)

Twin-quaternion factor:
not computed at D = 9
(twin requires D=8 = 4+4 partition)

Attractor fit:
amplitude A: 0.177
contraction λ : 0.0428
period: 1
fitted: False

Helmsman family:
total flips: 12
stability S_σ : 0.500

§ GBR United Kingdom

EMBER electricity-generation series · 2000-2025 · N = 26 · D = 9

INPUT

source: ember_GBR*_generation_TWh.csv
N records: 26
D carriers: 9
Bioenergy
Coal
Gas
Hydro
Nuclear
Other Fossil
Other Renewables
Solar
Wind
source_sha256: b1df2a0ae6f21f77c0d6fb5b...

HEADLINE METRICS

Helmsman flips: ?
Helmsman stability S_σ : {'global': 0.375, 'rolling': [0.2857142857142857, 0.14285714285714285]}
IR class: ?
Attractor amplitude: ?

CHSH S: ?
Tsirelson bound: $2\sqrt{2} \approx 2.828$

Bearing pairs $(D(D-1)/2)$: 36

CNT v3.1.0 content_sha: ?...
CNQ v2.0.0 content_sha: ?...

Following pages: Stage 1 (XY/XZ/YZ × N years + course) → Stage 2/3 (7 pages) → CNQ (2 pages).

SECTION PLATE t=0 | Year 2000 | D=9 N=26 pairs=36

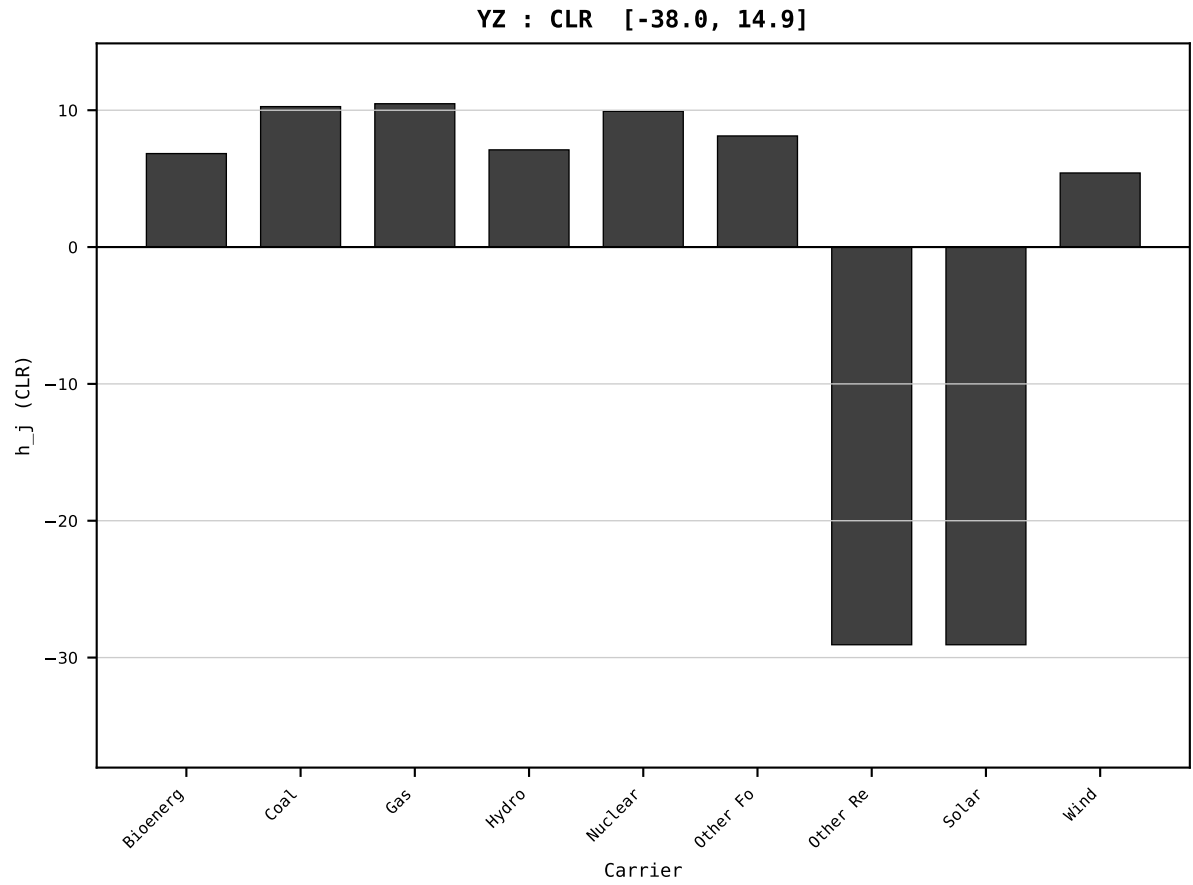
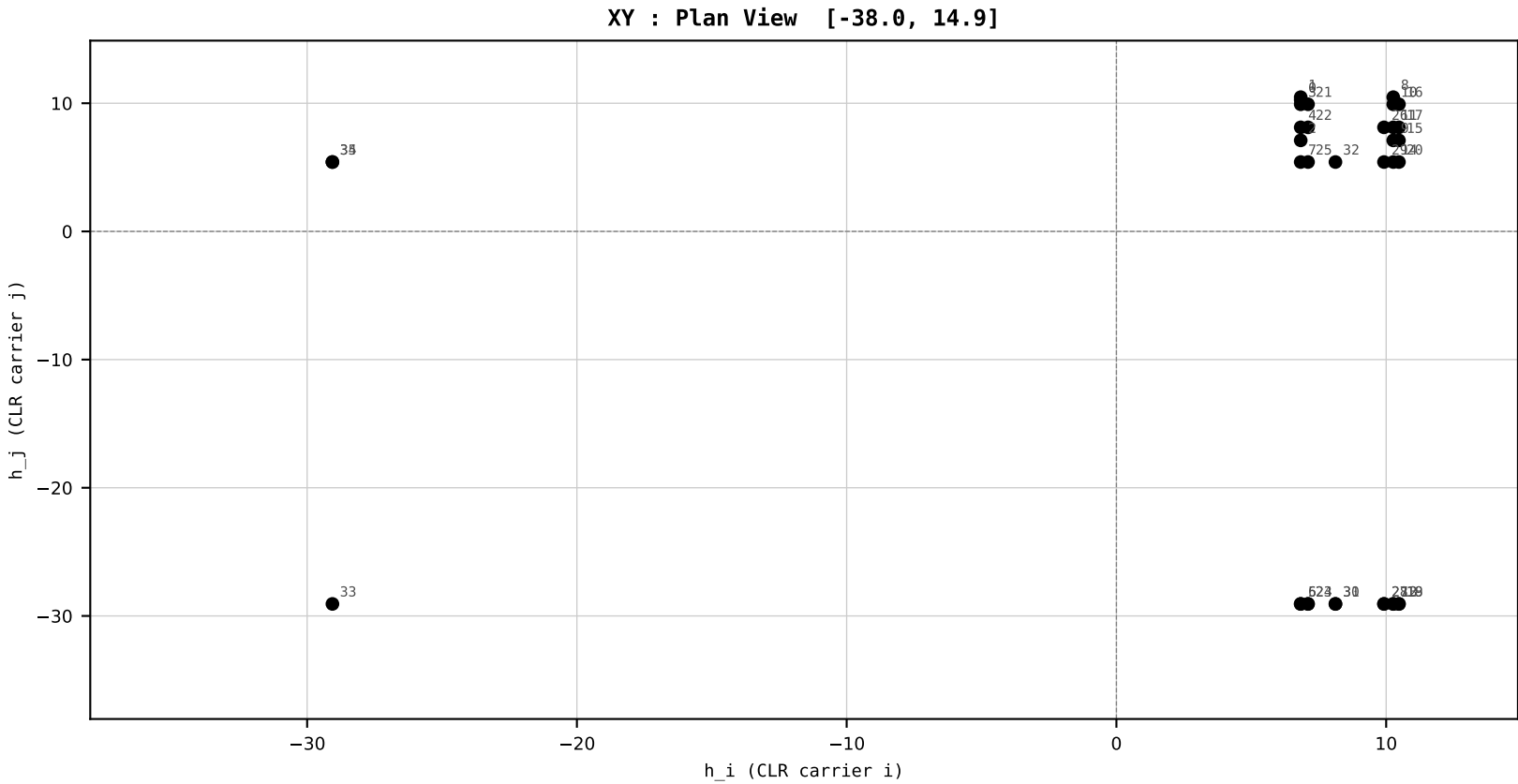
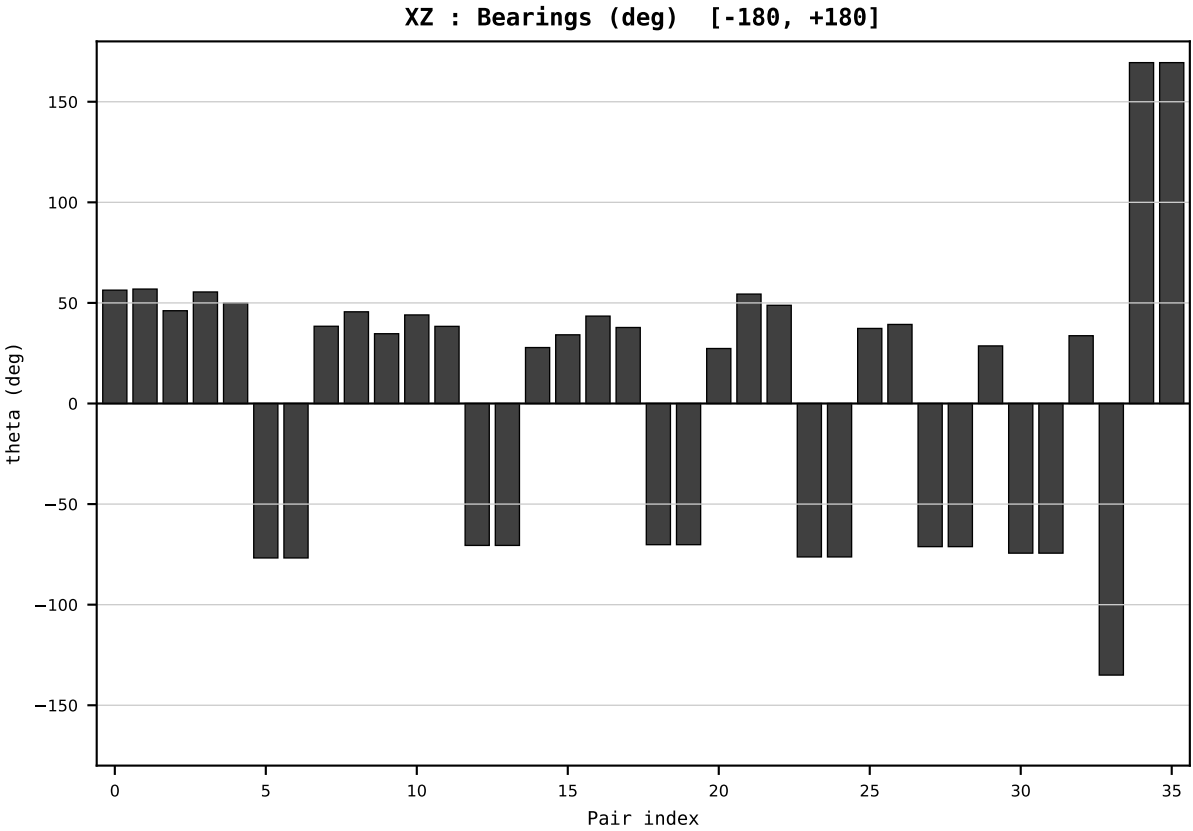
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2000
t : 0 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.403740
Ring = Hs-3
E_metric= 1560.1795
kappa_HS= 148080000000000032.00
omega = 0.0000 deg
d_A = 0.000000
Helm = ---
Helm d = ---
DR = 39.537 HLR
DR ratio= 148079999529708608.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=1 | Year 2001 | D=9 N=26 pairs=36

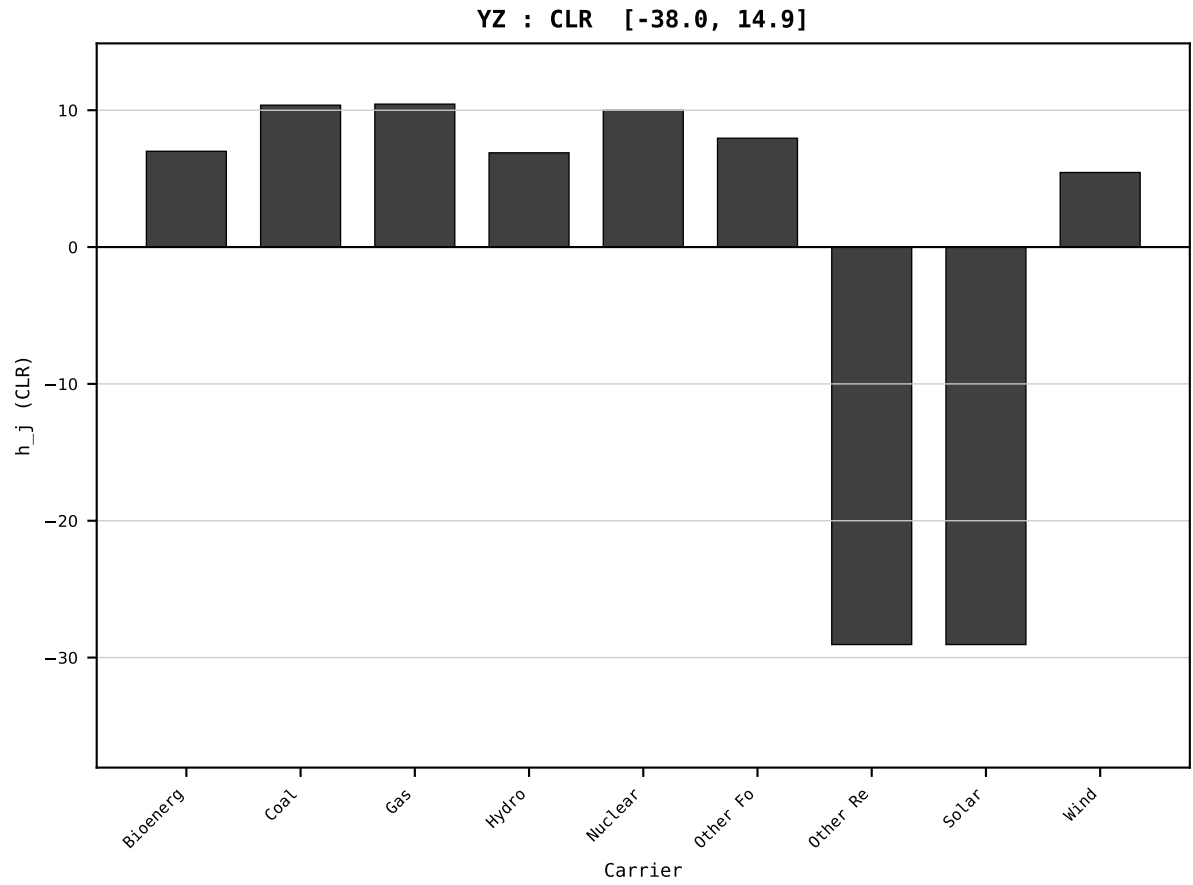
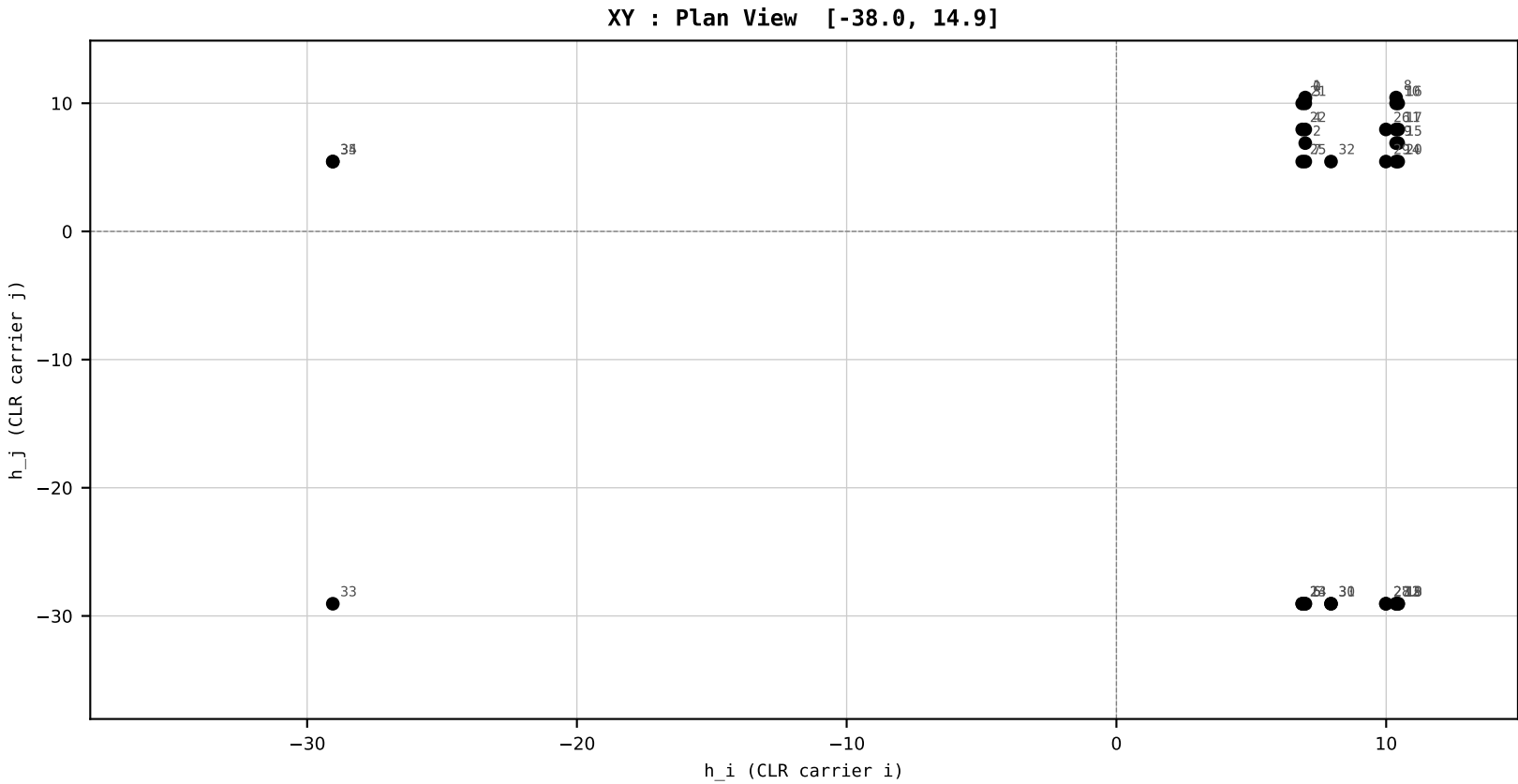
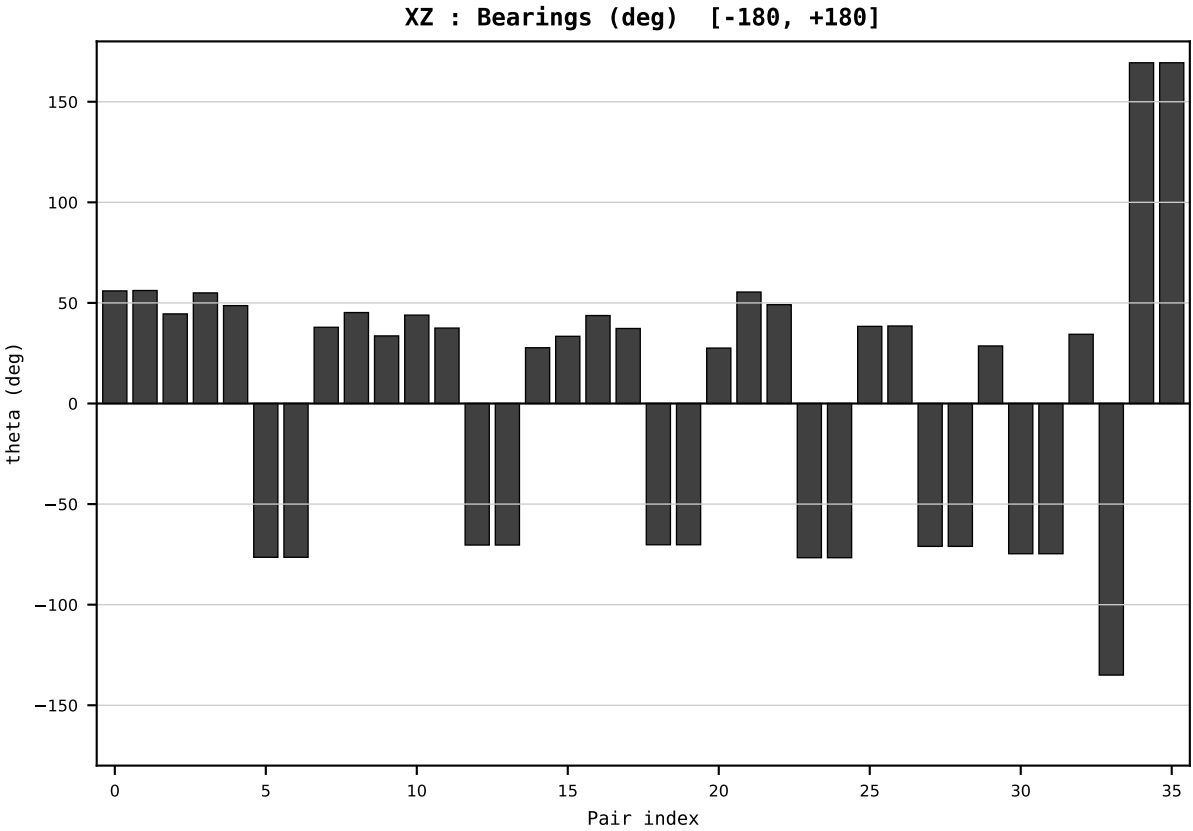
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2001
t : 1 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.409899
Ring = Hs-3
E_metric= 1556.9878
kappa_HS= 14190999999999984.00
omega = 0.4234 deg
d_A = 0.346528
Helm = Hydro
Helm d = -0.214796
DR = 39.494 HLR
DR ratio= 141910000176375712.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=2 | Year 2002 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

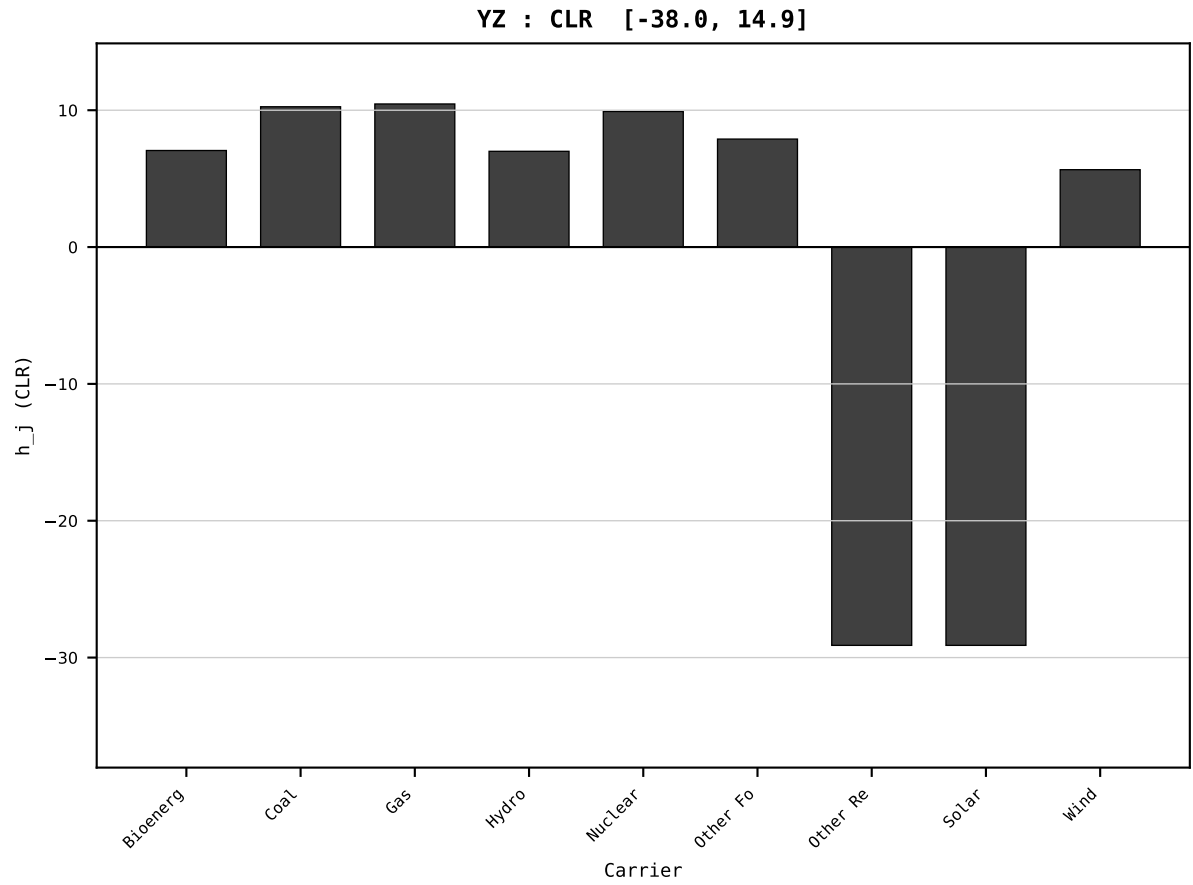
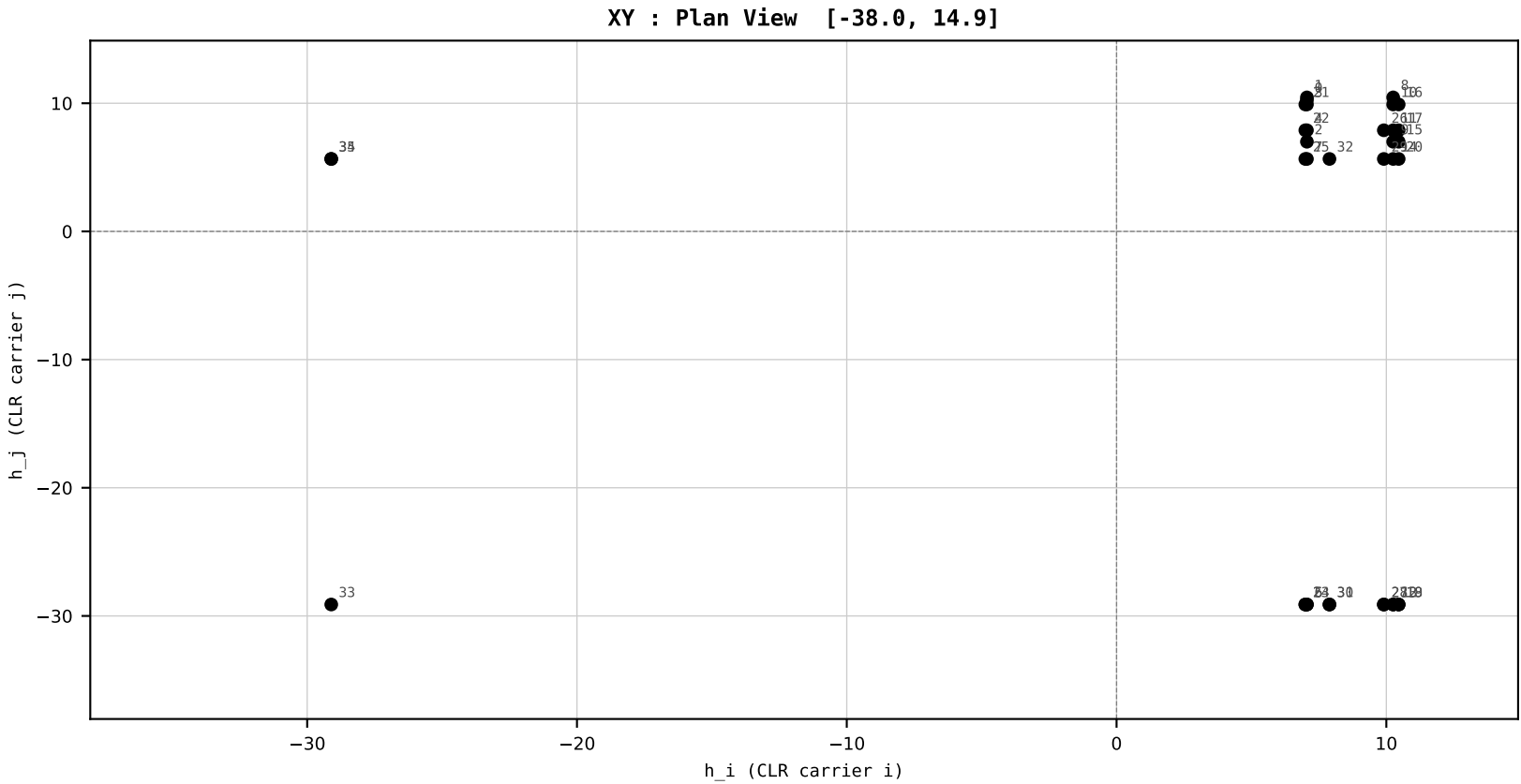
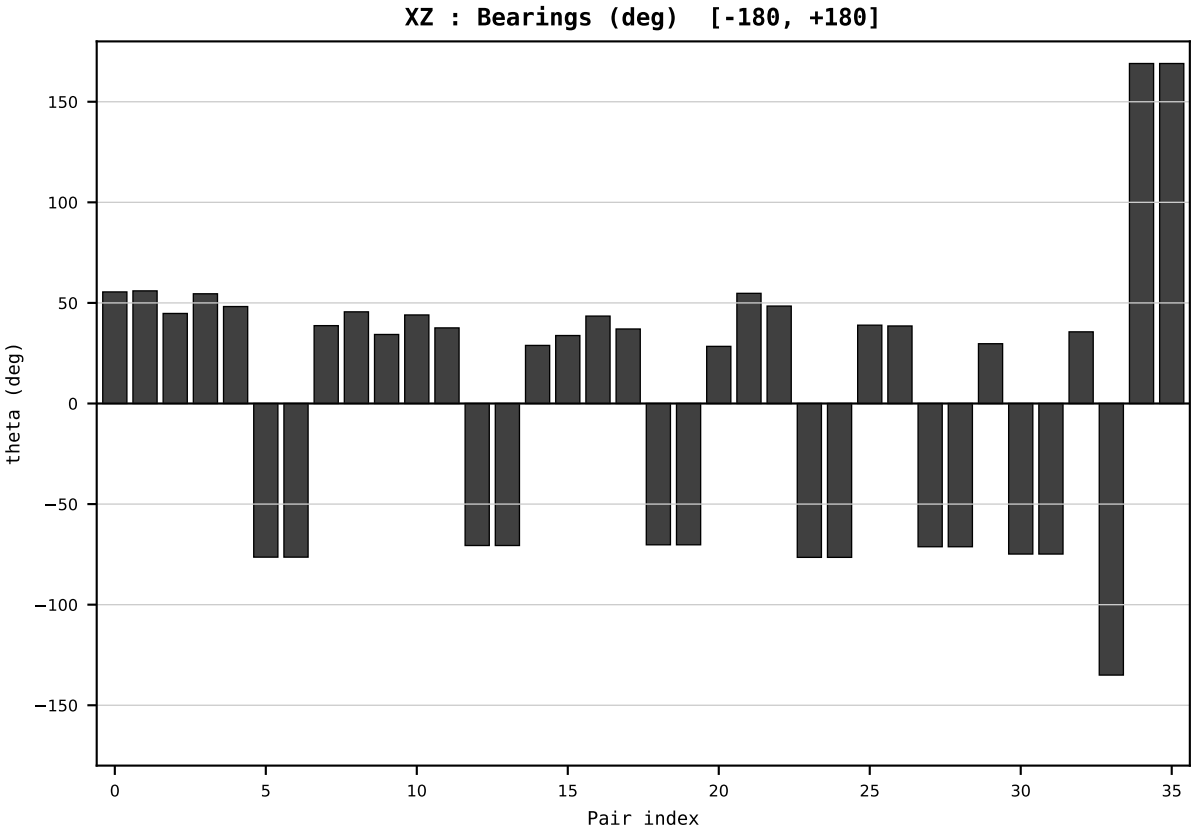
Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2002
t : 2 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.406615
Ring = Hs-3
E_metric= 1561.5638
kappa_HS= 15228000000000000.00
omega = 0.3533 deg
d_A = 0.297482
Helm = Wind
Helm d = +0.205028
DR = 39.564 HLR
DR ratio= 152279999021603904.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=3 | Year 2003 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

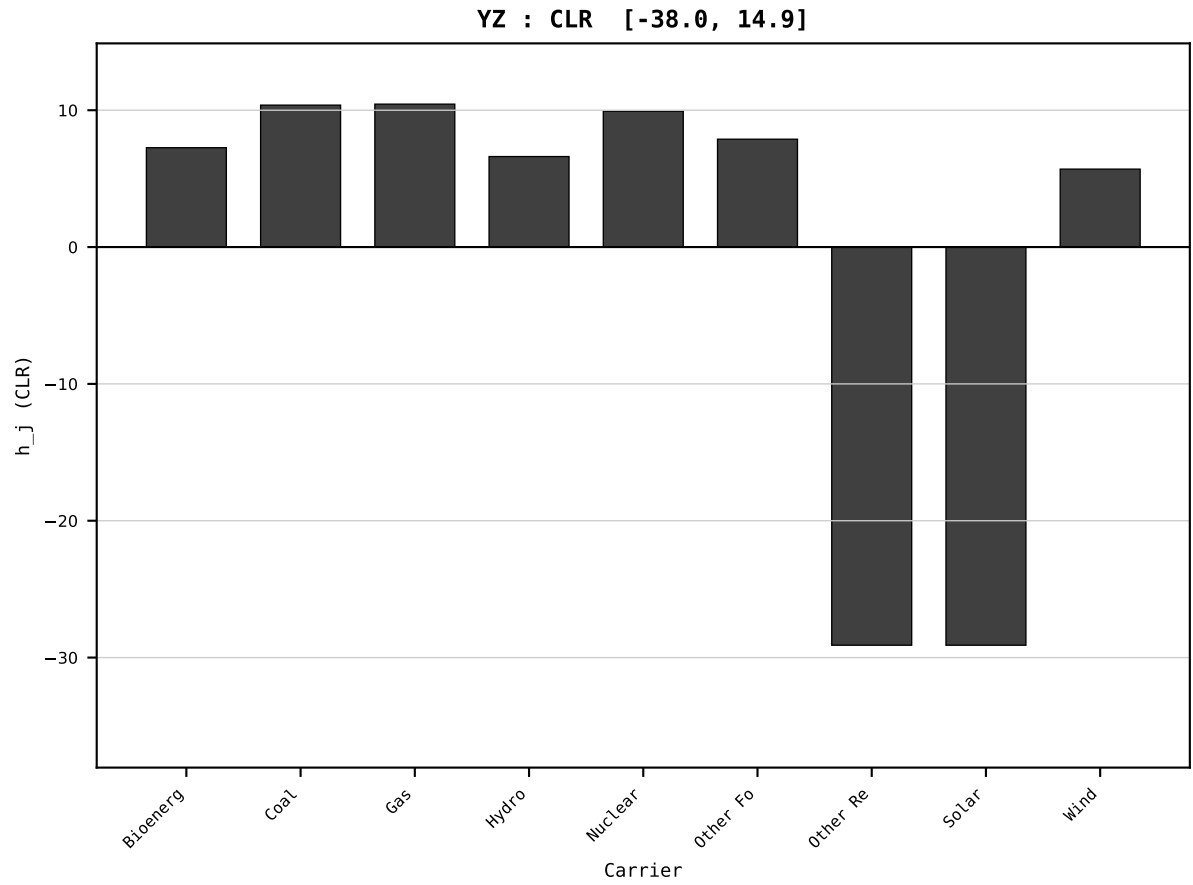
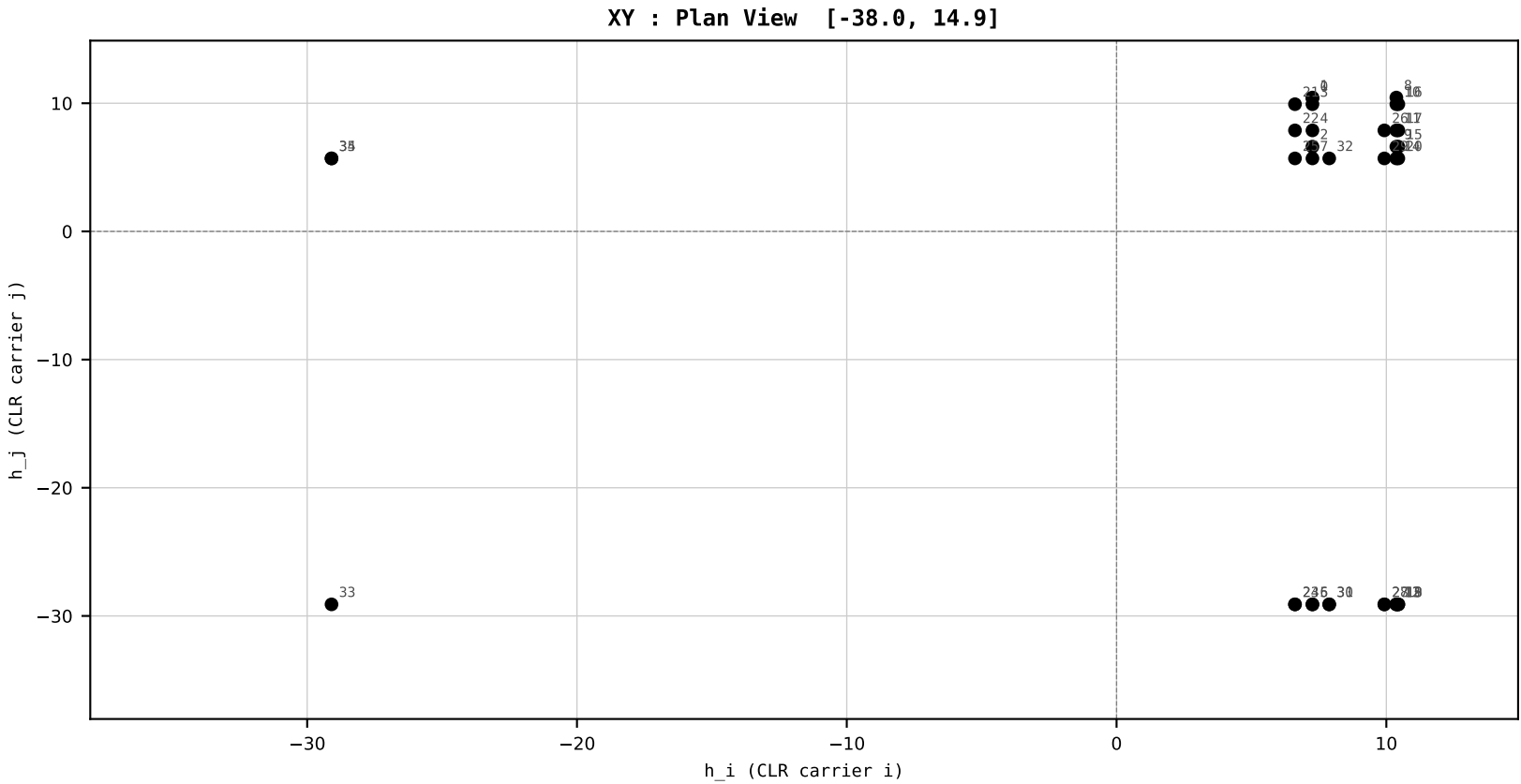
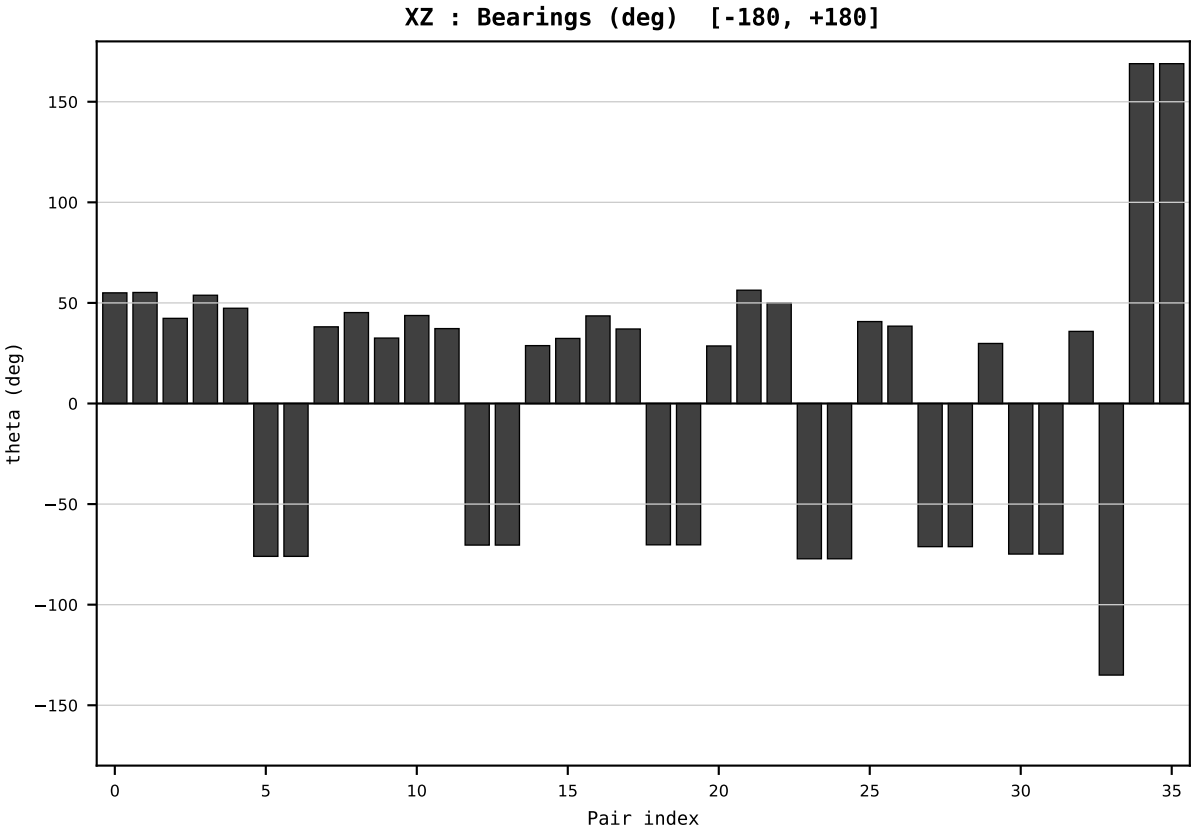
Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2003
t : 3 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.411016
Ring = Hs-3
E_metric= 1558.4988
kappa_HS= 148880000000000000.00
omega = 0.5541 deg
d_A = 0.453589
Helm = Hydro
Helm d = -0.383810
DR = 39.542 HLR
DR ratio= 148880000410211776.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=4 | Year 2004 | D=9 N=26 pairs=36

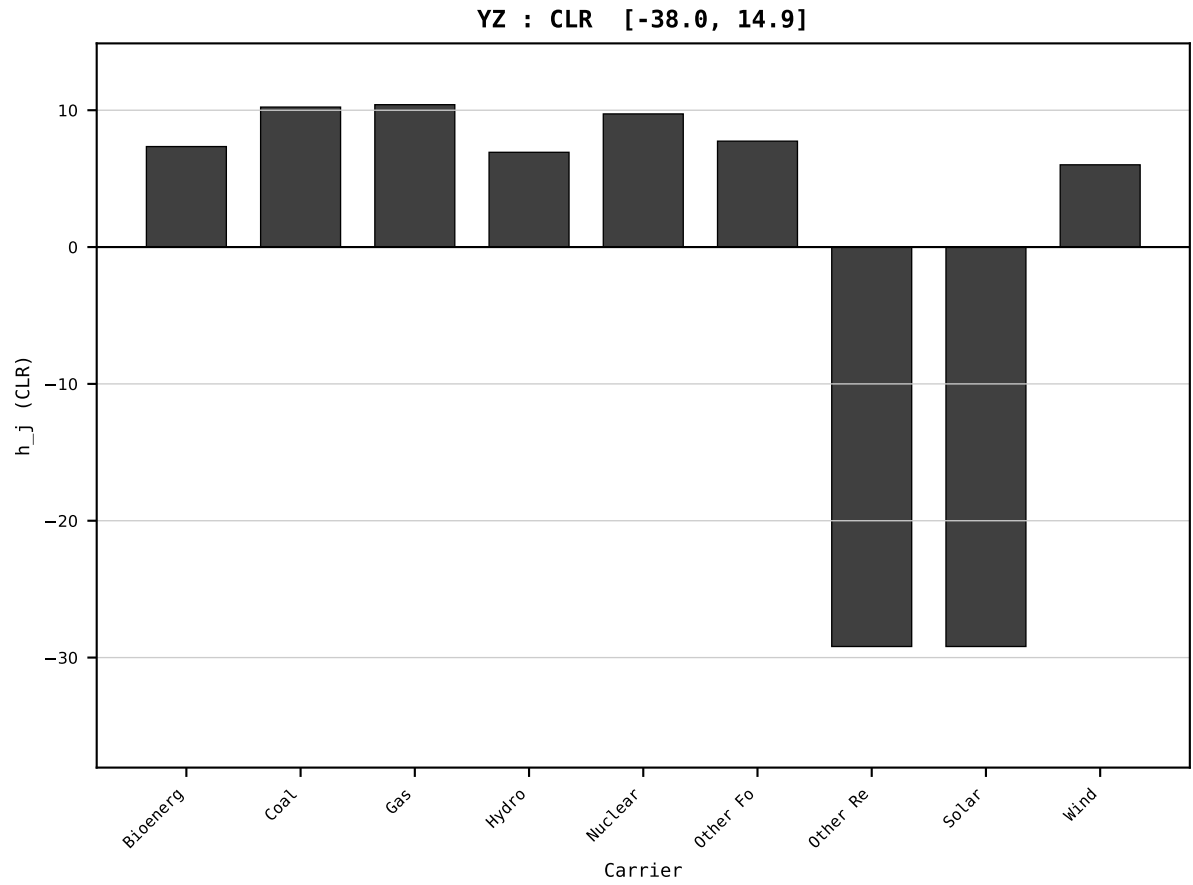
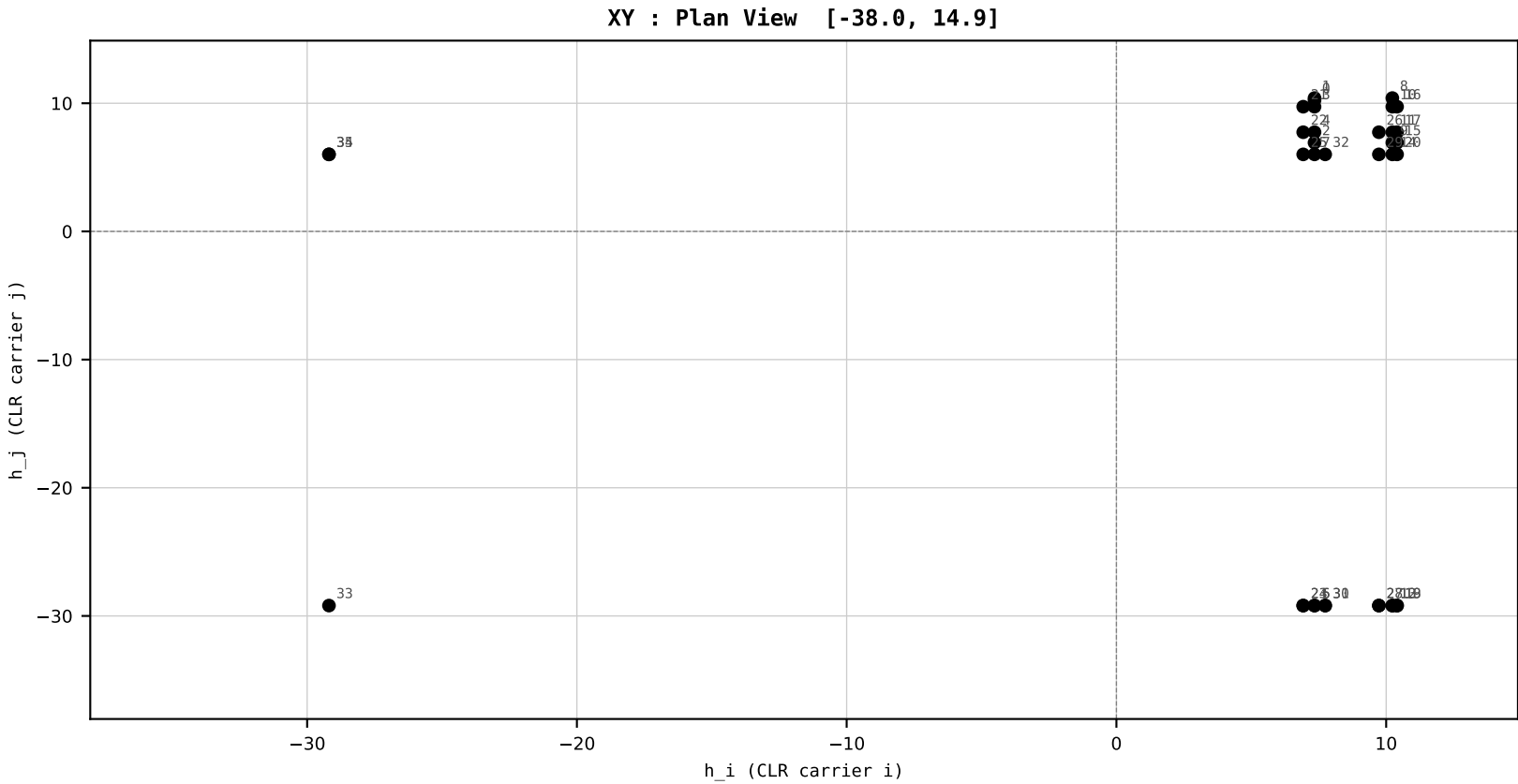
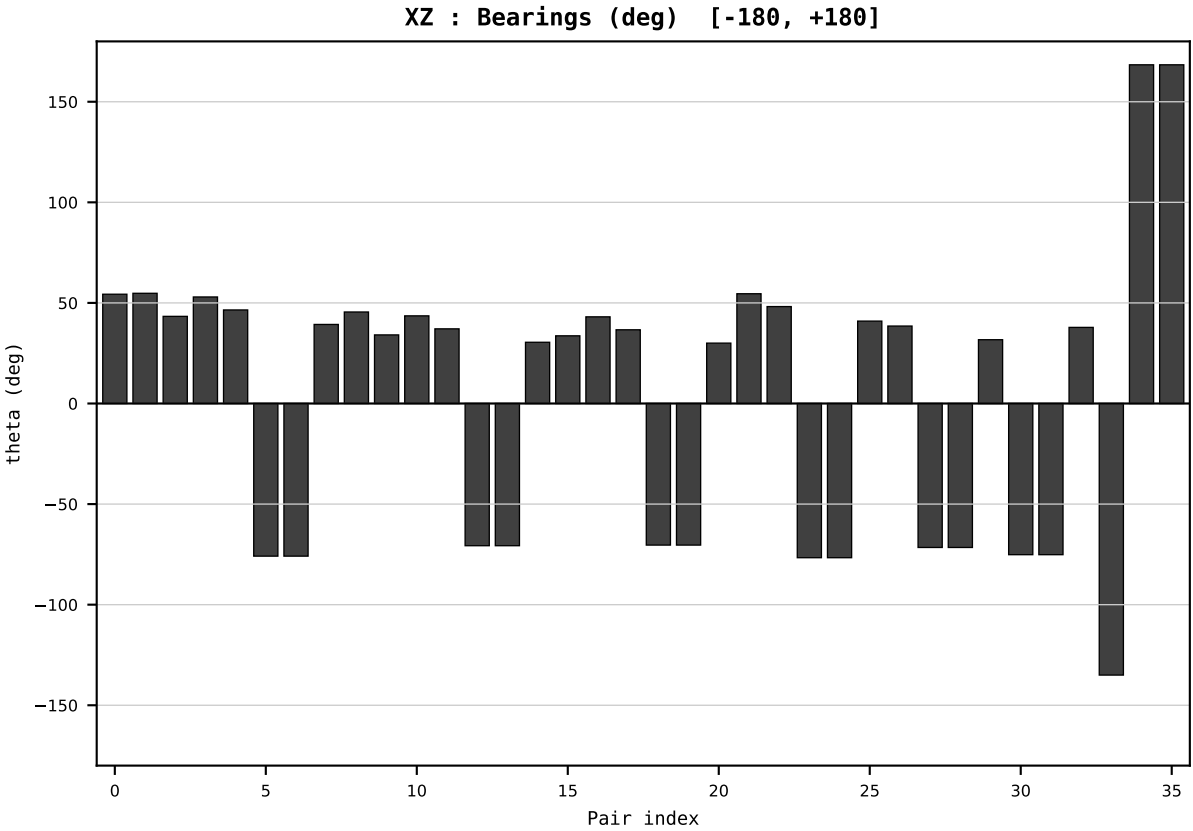
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2004
t : 4 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.403458
Ring = Hs-3
E_metric= 1567.4455
kappa_HS= 15706000000000000.00
omega = 0.6567 deg
d_A = 0.548838
Helm = Wind
Helm d = +0.314727
DR = 39.595 HLR
DR ratio= 157059999551214208.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=5 | Year 2005 | D=9 N=26 pairs=36

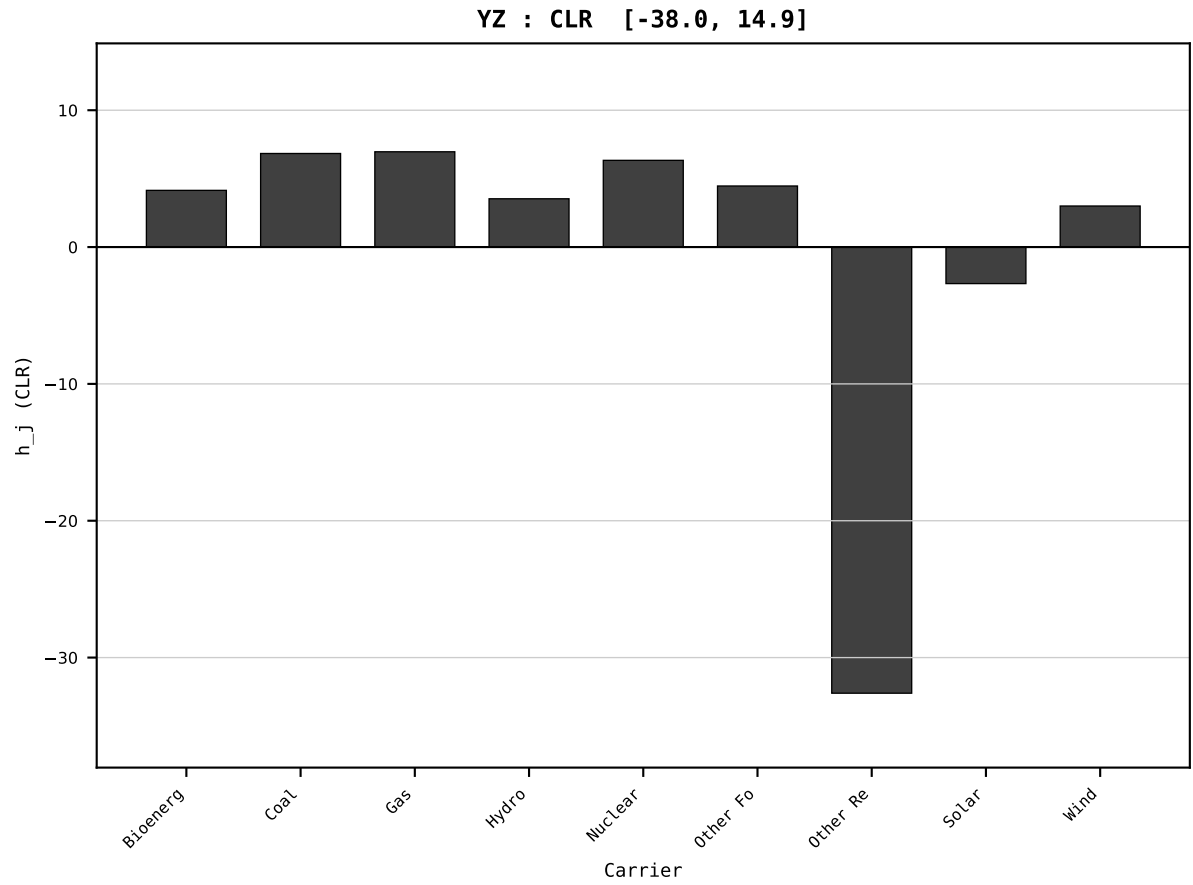
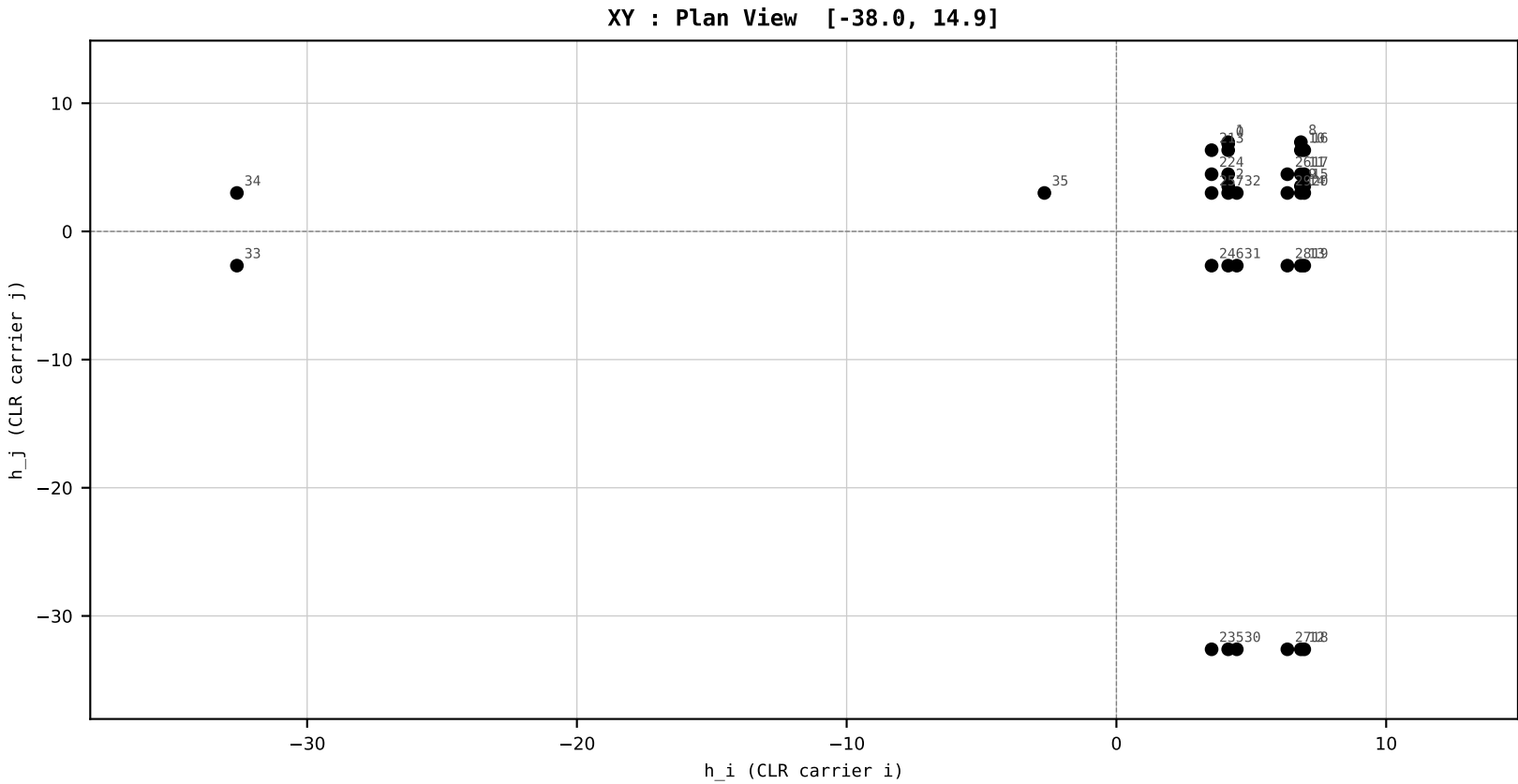
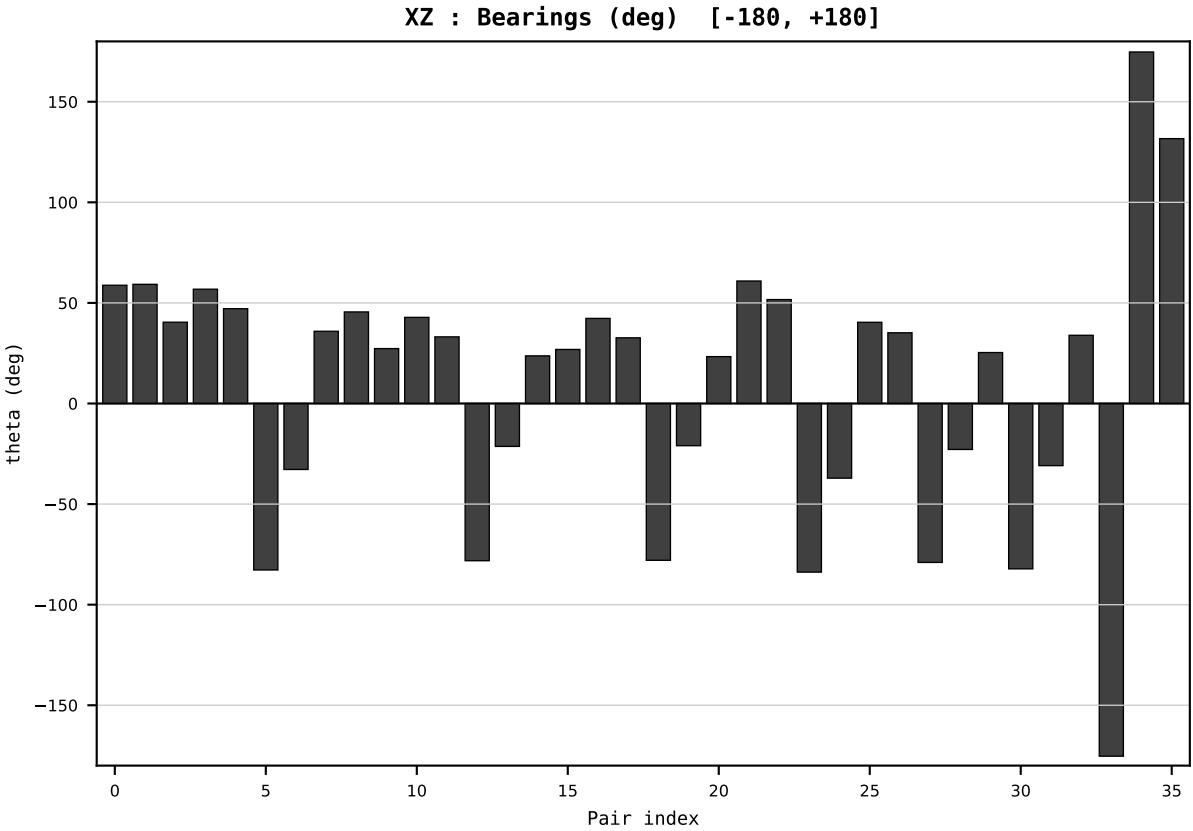
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2005
t : 5 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.388111
Ring = Hs-3
E_metric= 905.5074
kappa_HS= 15264000000000000.00
omega = 36.6332 deg
d_A = 28.132514
Helm = Solar
Helm d = +26.521056
DR = 39.567 HLR
DR ratio= 152640001127908160.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=6 | Year 2006 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

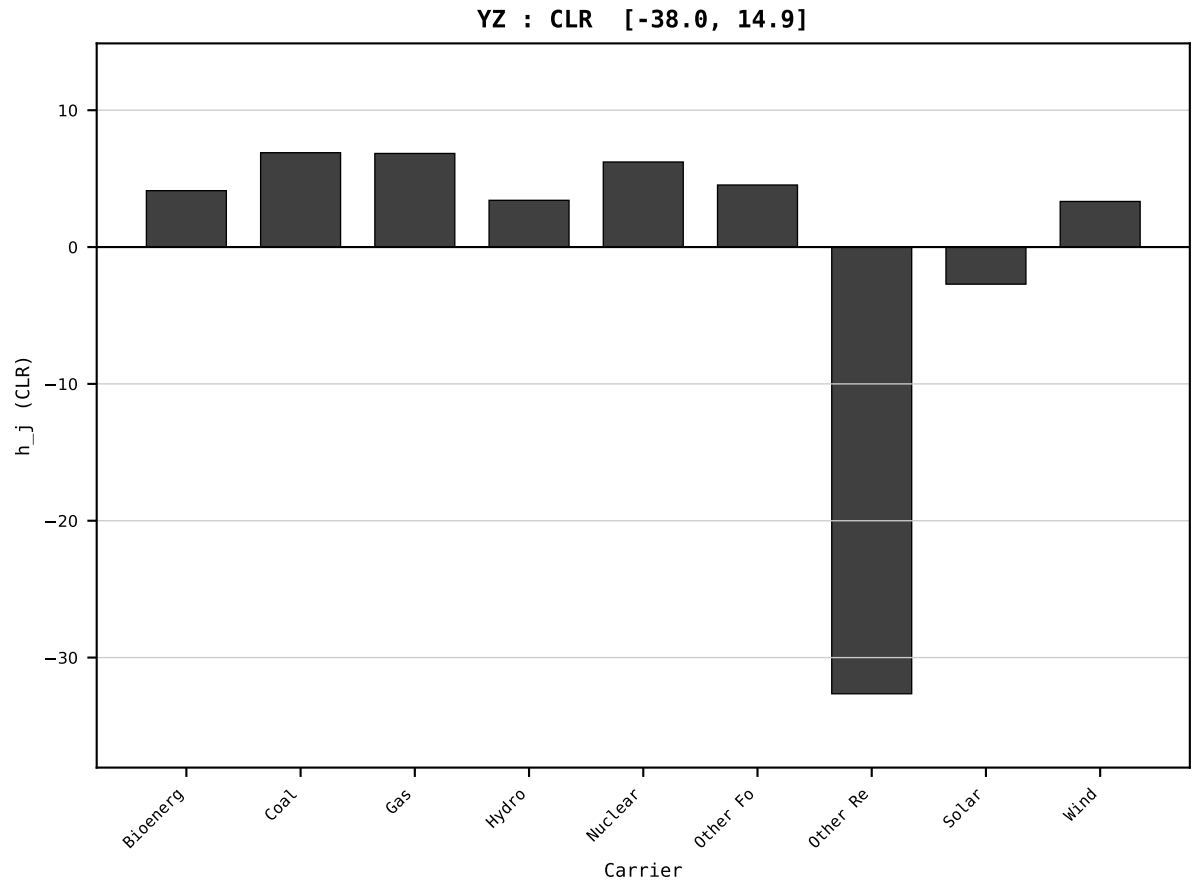
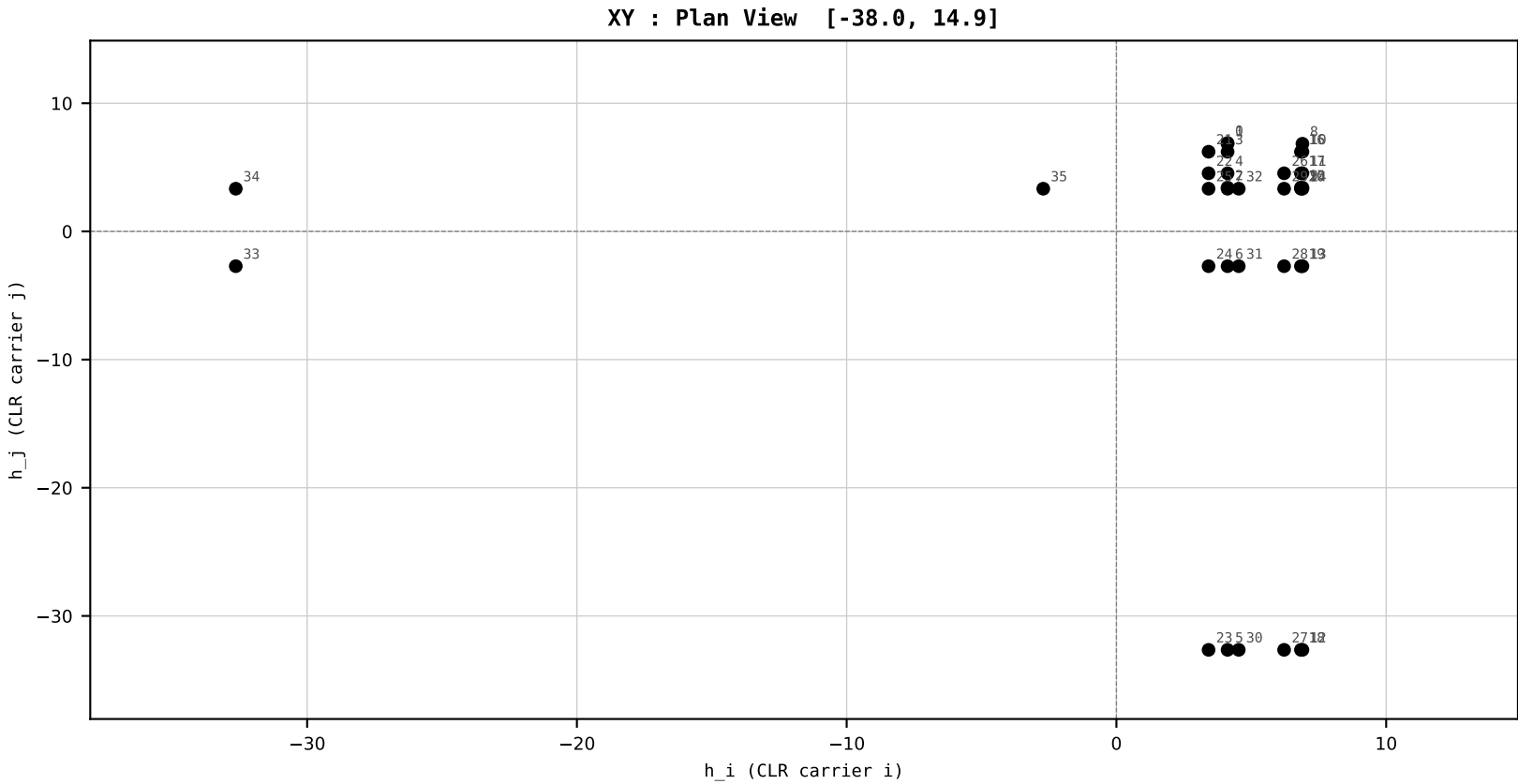
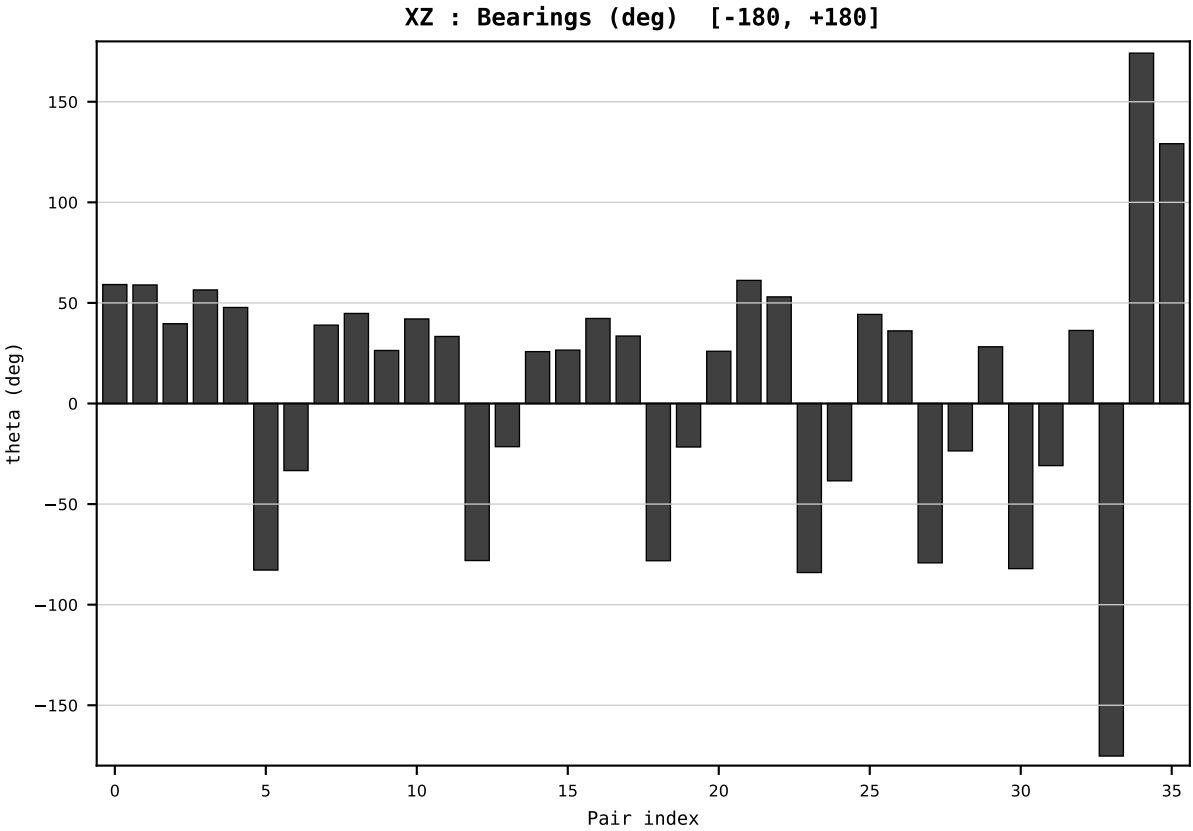
Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2006
t : 6 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.382444
Ring = Hs-3
E_metric= 907.5146
kappa_HS= 14885000000000000.00
omega = 0.6534 deg
d_A = 0.406998
Helm = Wind
Helm d = +0.332784
DR = 39.542 HLR
DR ratio= 148849999646857824.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=7 | Year 2007 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

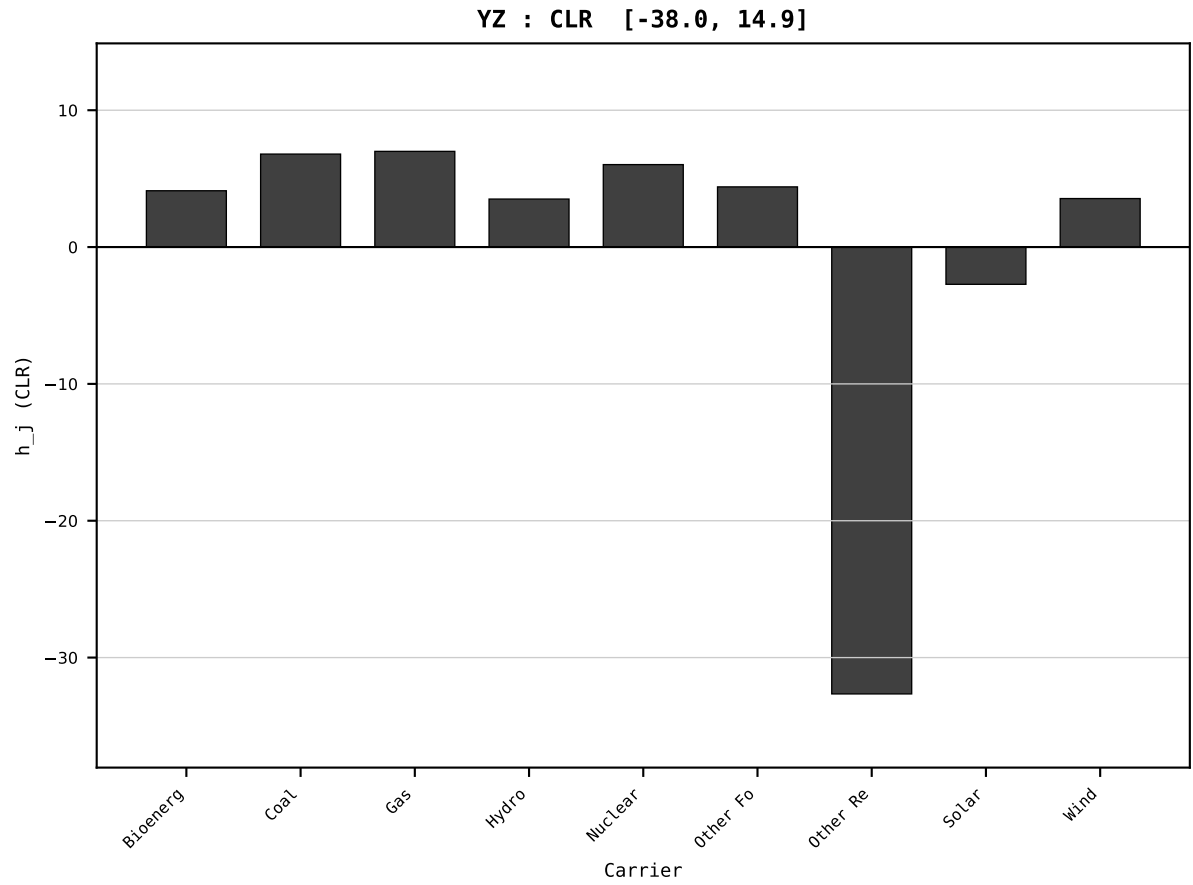
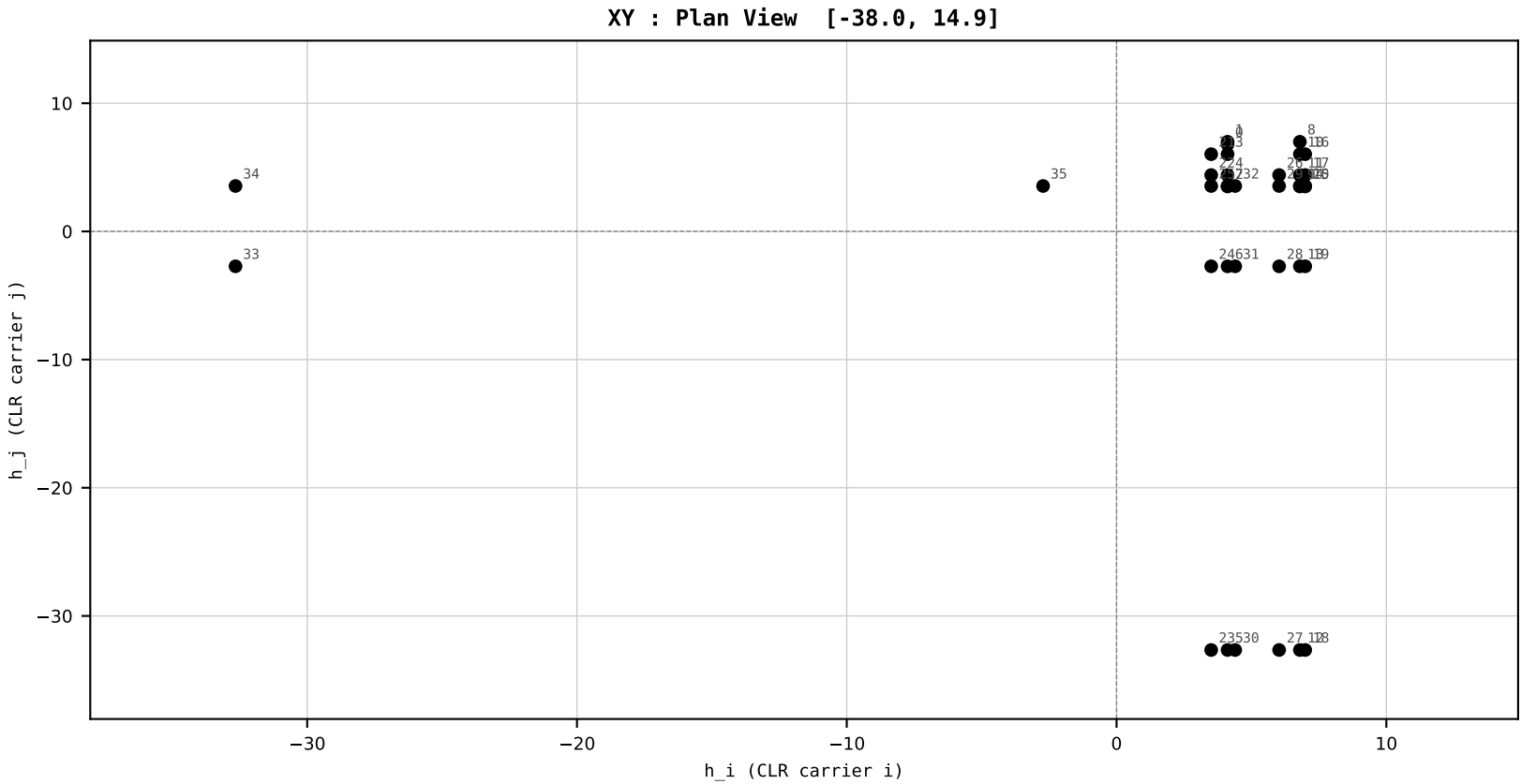
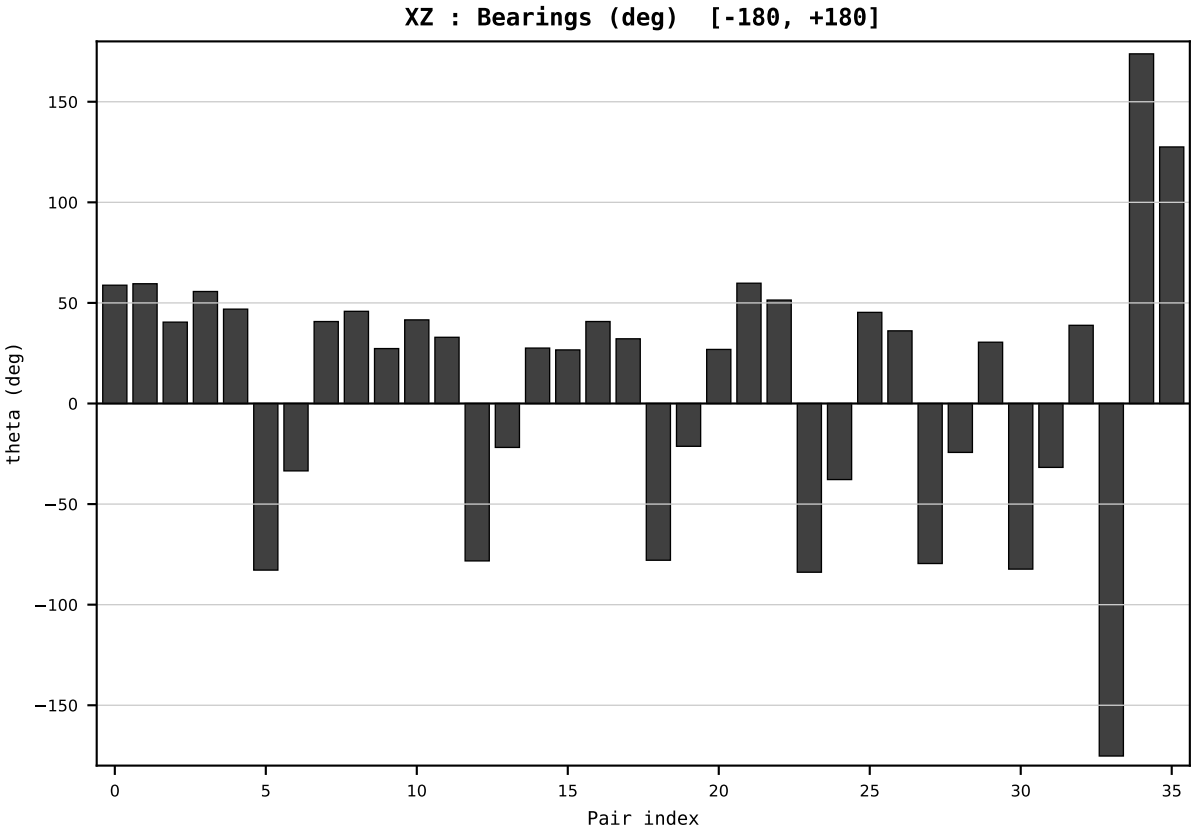
Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2007
t : 7 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.393078
Ring = Hs-3
E_metric= 907.4010
kappa_HS= 16578999999999968.00
omega = 0.6066 deg
d_A = 0.376749
Helm = Wind
Helm d = +0.211765
DR = 39.649 HLR
DR ratio= 165789999641552096.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=8 | Year 2008 | D=9 N=26 pairs=36

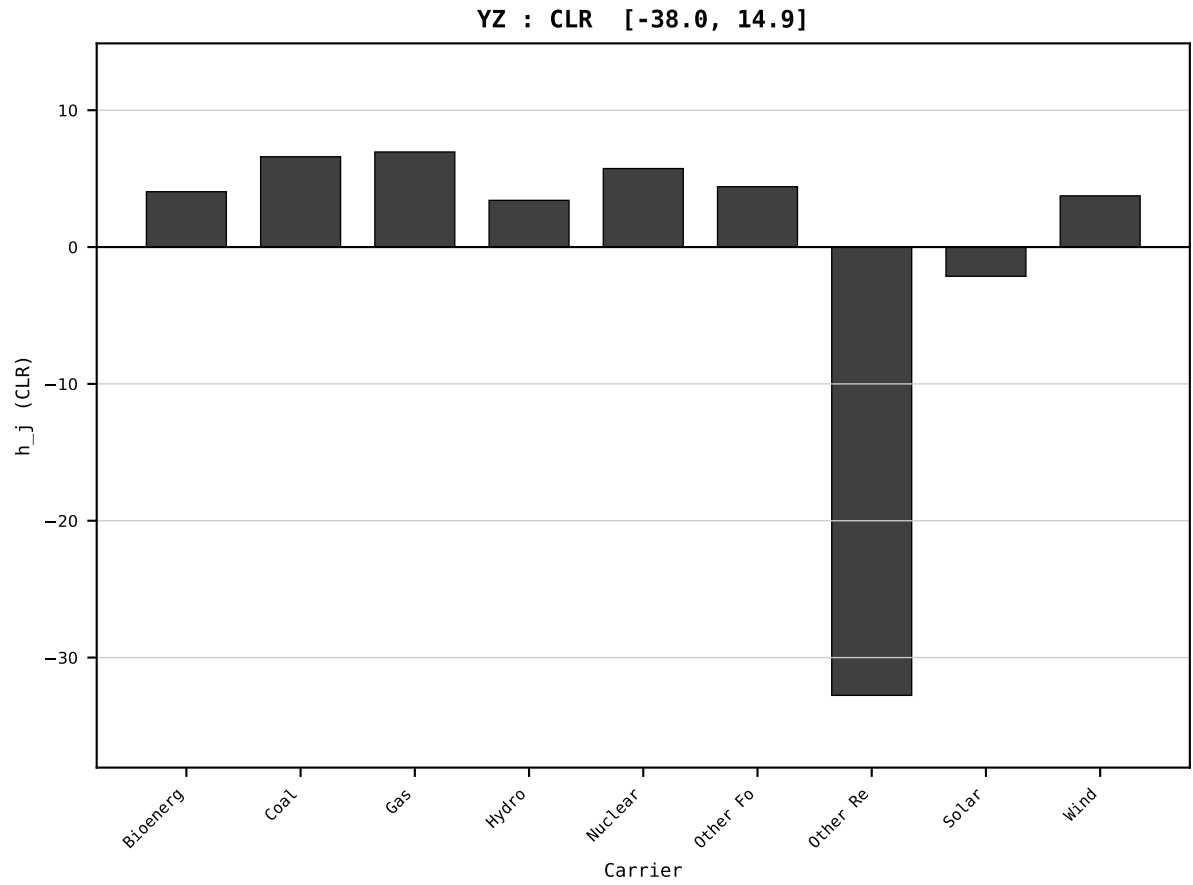
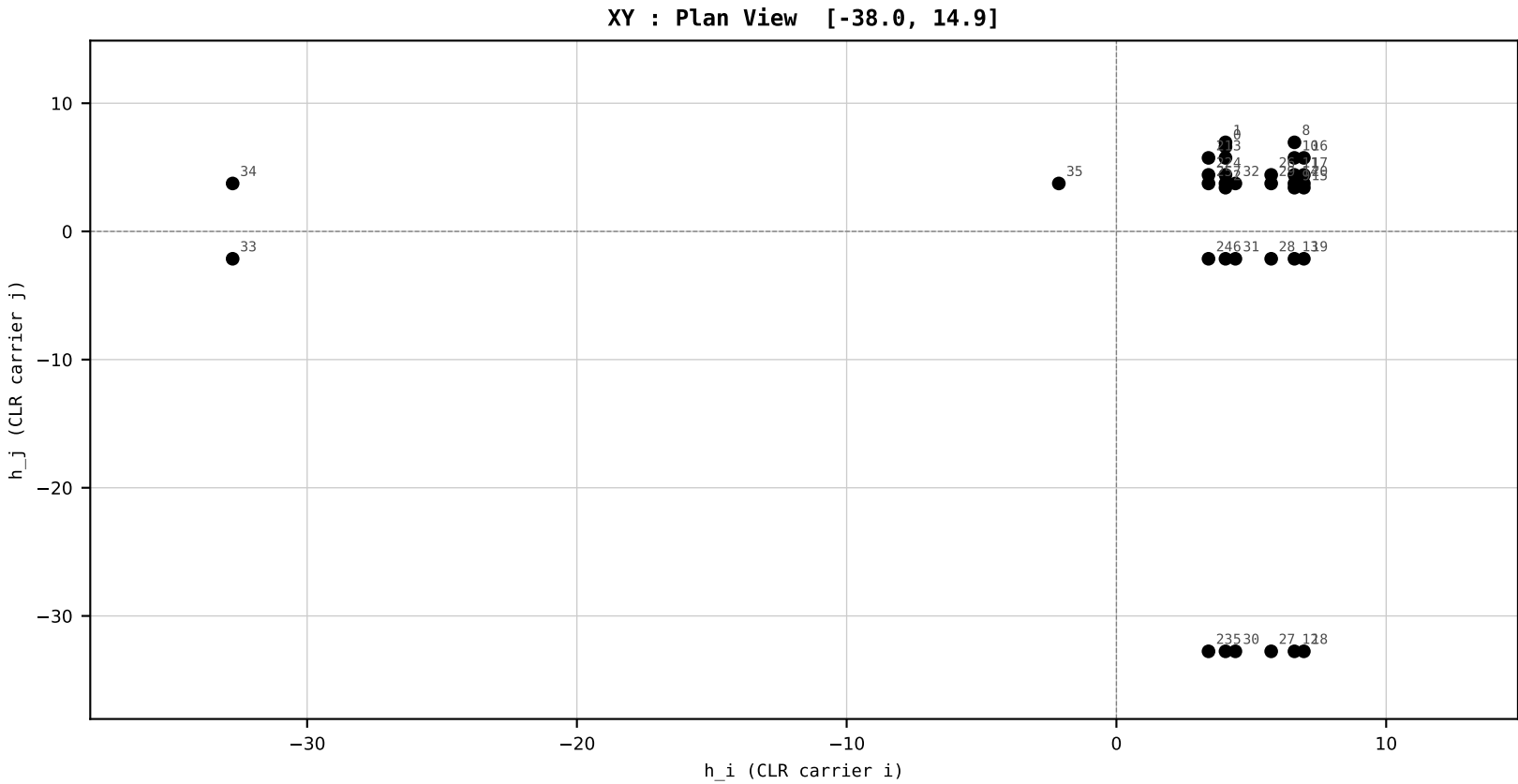
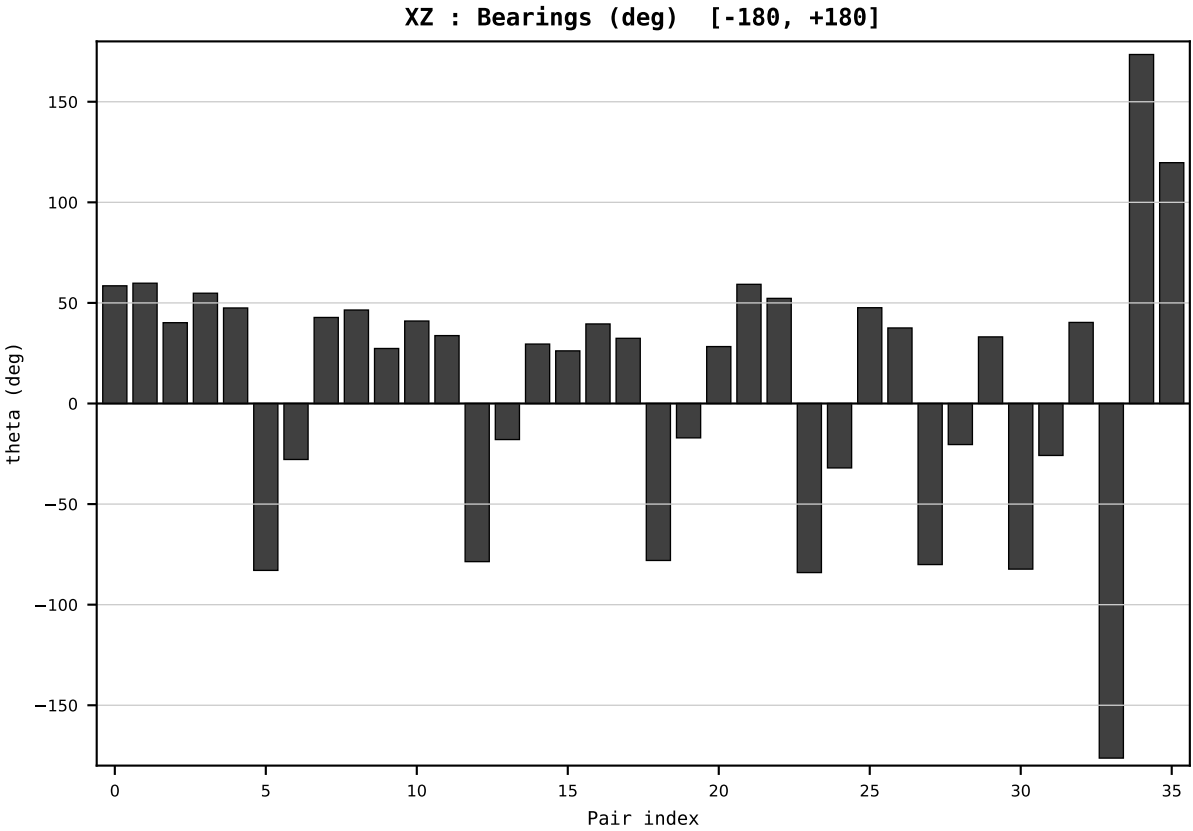
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2008
t : 8 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.392183
Ring = Hs-3
E_metric= 904.9892
kappa_HS= 17622000000000000.00
omega = 1.1739 deg
d_A = 0.729475
Helm = Solar
Helm d = +0.587604
DR = 39.711 HLR
DR ratio= 176220000114672288.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=9 | Year 2009 | D=9 N=26 pairs=36

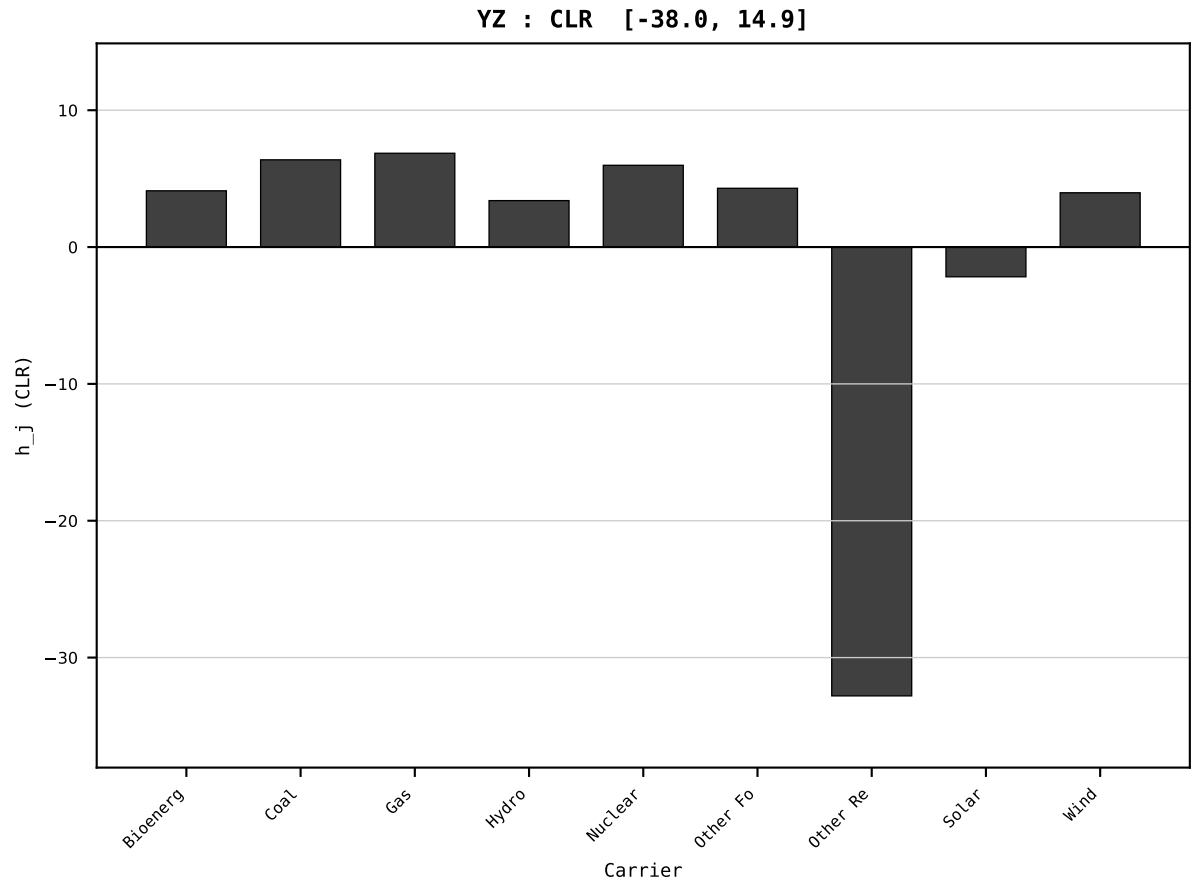
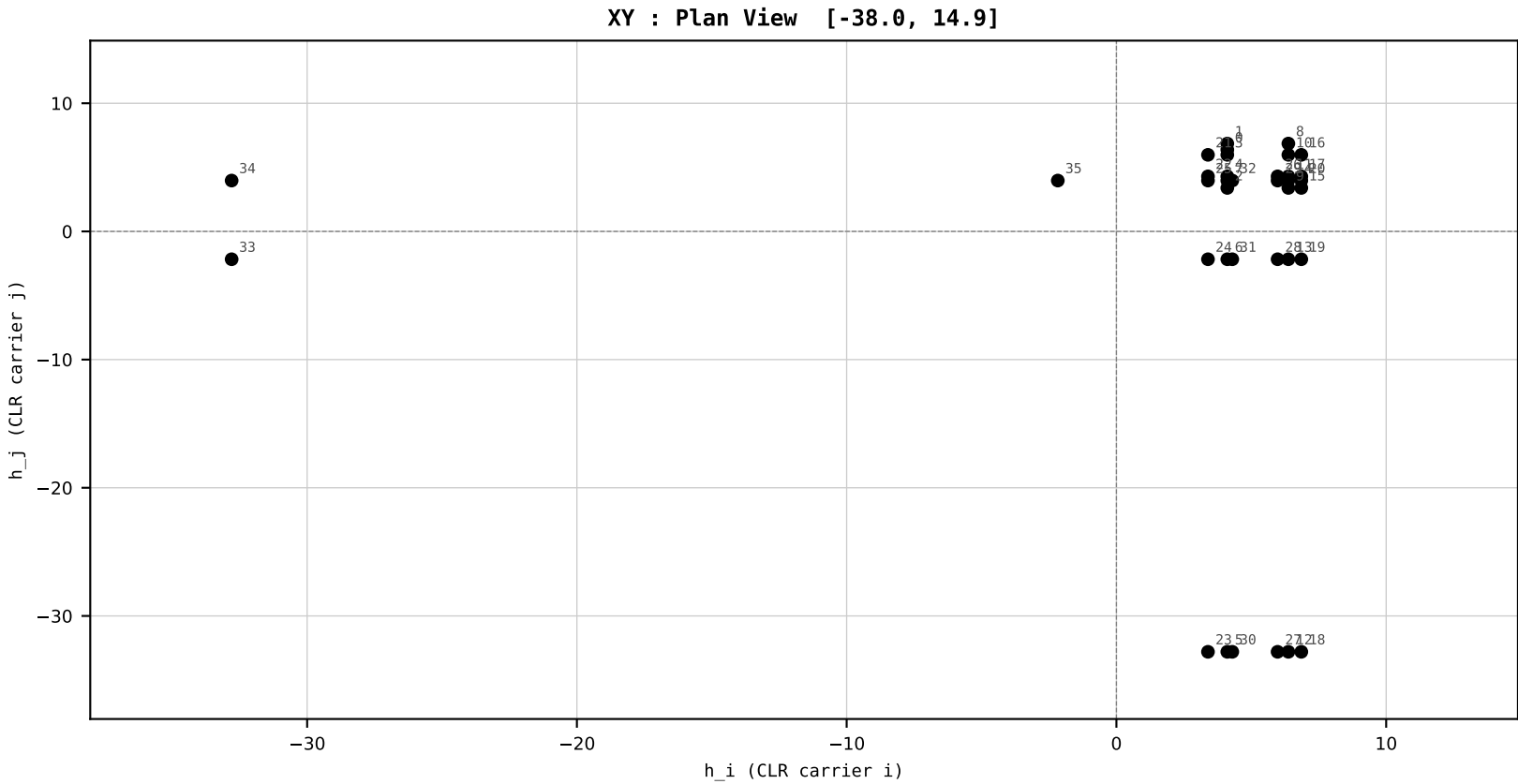
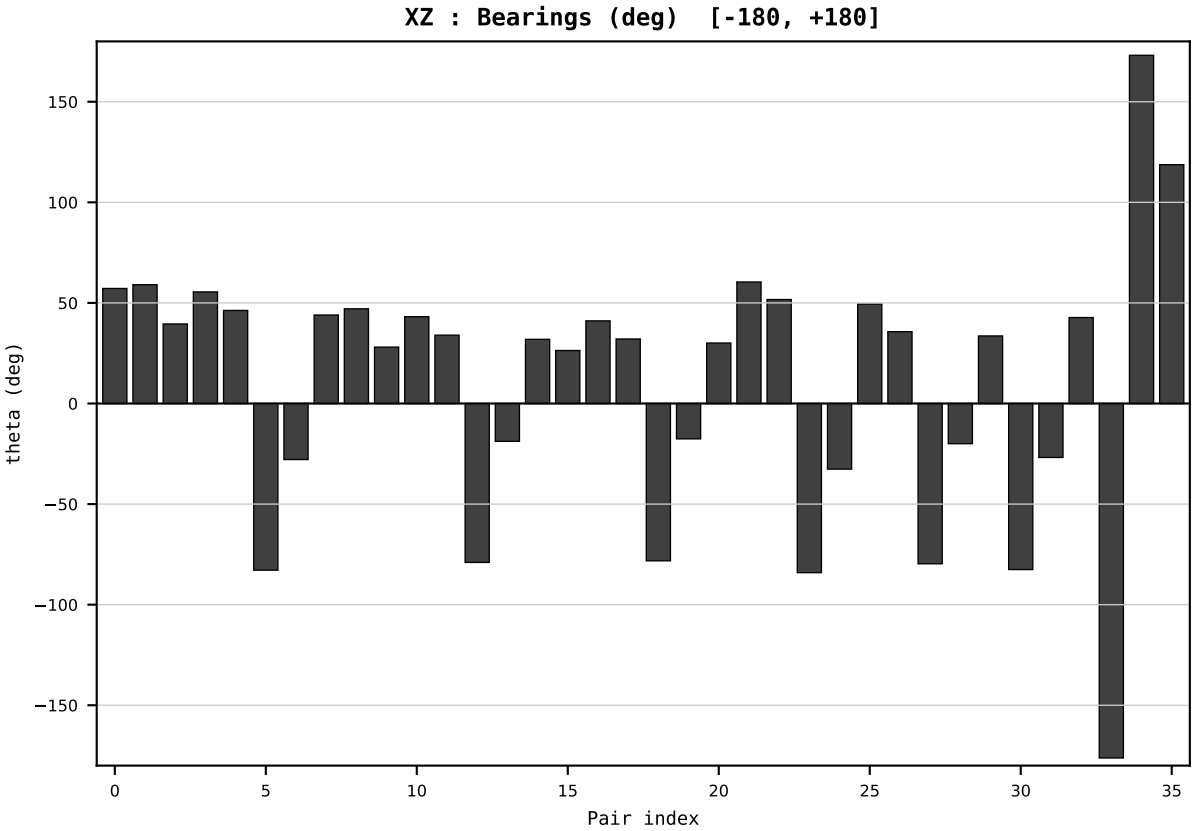
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2009
t : 9 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.365440
Ring = Hs-3
E_metric= 908.6087
kappa_HS= 16650000000000000.00
omega = 0.6972 deg
d_A = 0.434051
Helm = Nuclear
Helm d = +0.237518
DR = 39.654 HLR
DR ratio= 166499999278857920.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=10 | Year 2010 | D=9 N=26 pairs=36

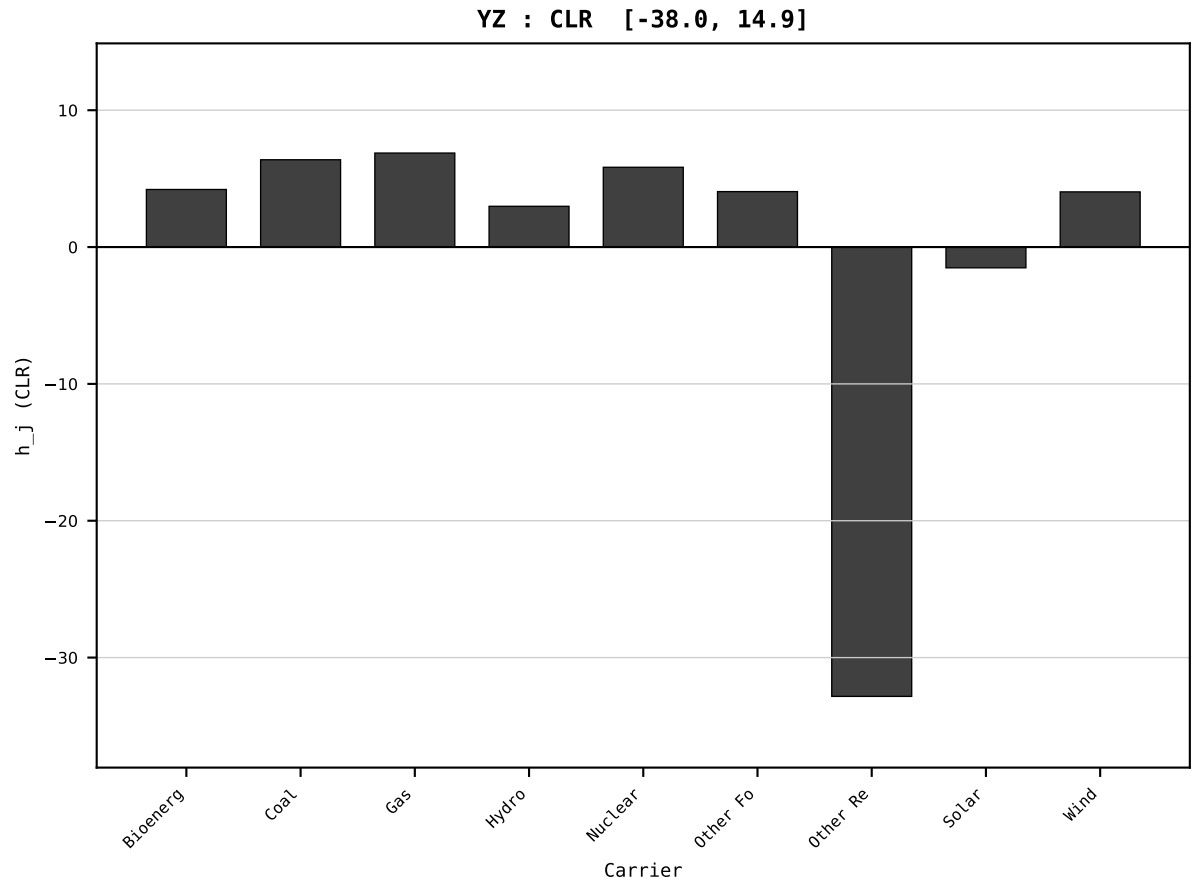
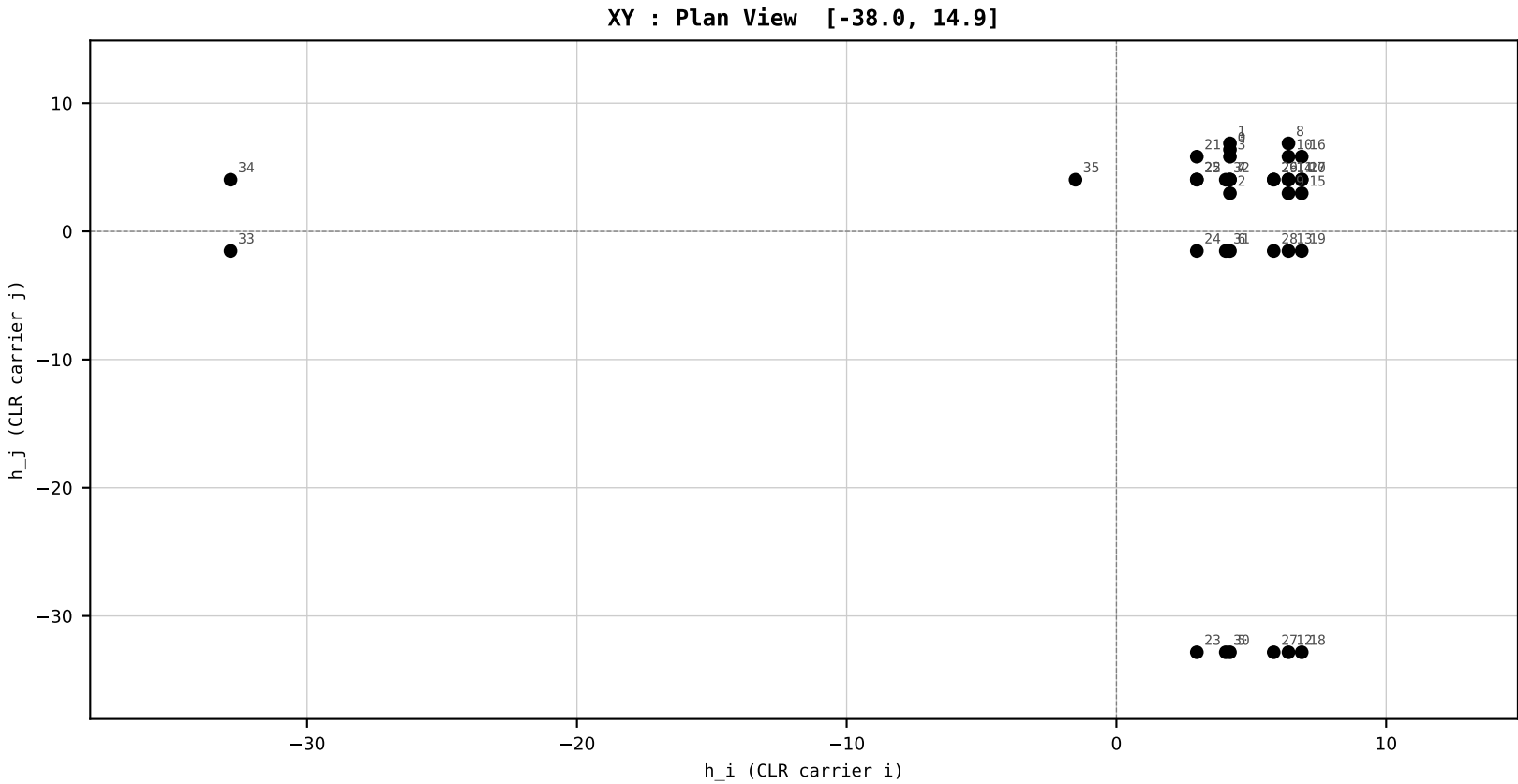
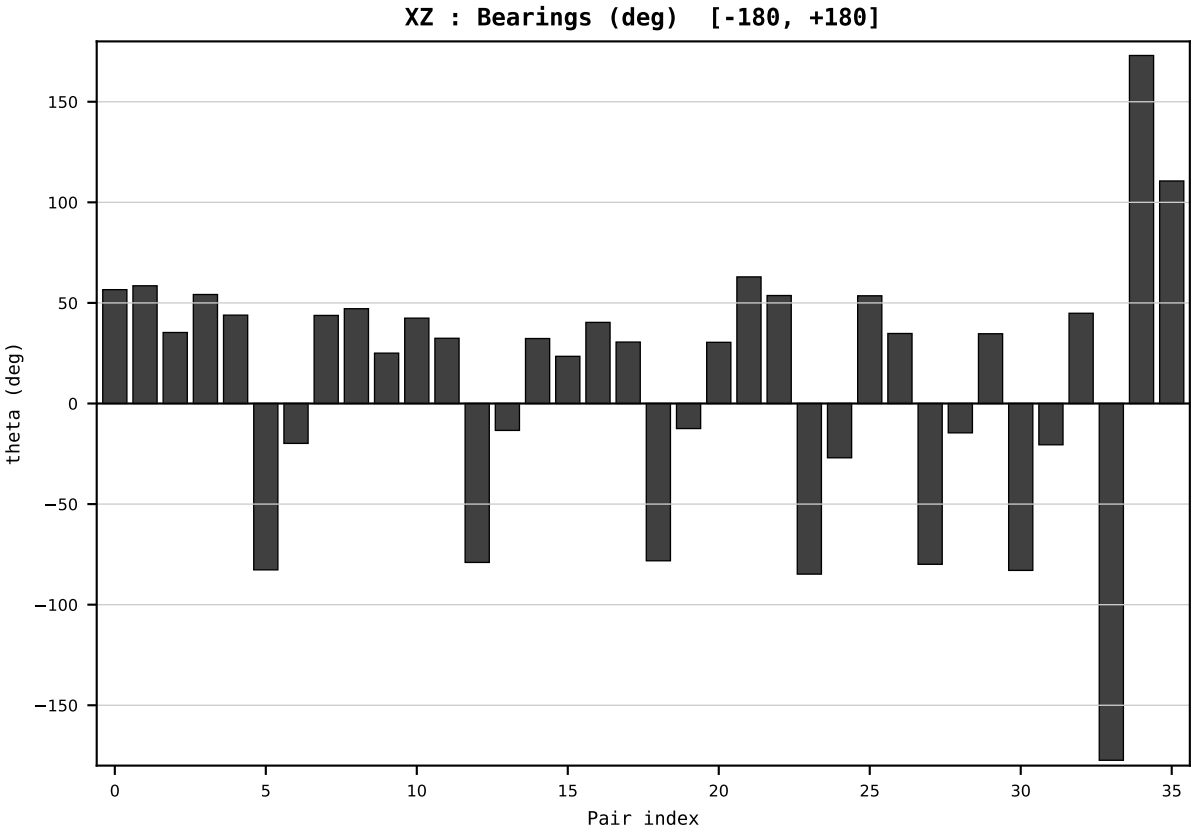
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2010
t : 10 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.380664
Ring = Hs-3
E_metric= 902.7101
kappa_HS= 17565000000000000.00
omega = 1.3400 deg
d_A = 0.834110
Helm = Solar
Helm d = +0.655191
DR = 39.707 HLR
DR ratio= 175649999363704736.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=11 | Year 2011 | D=9 N=26 pairs=36

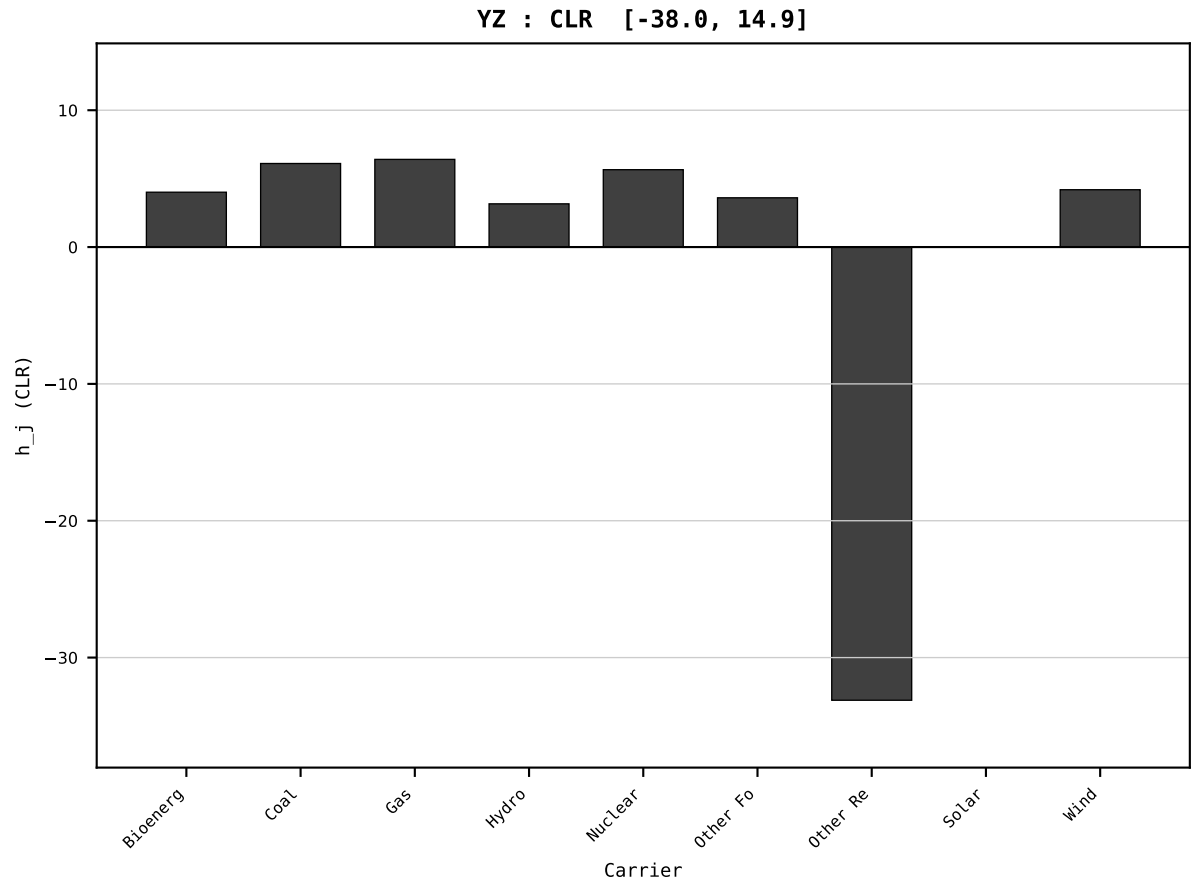
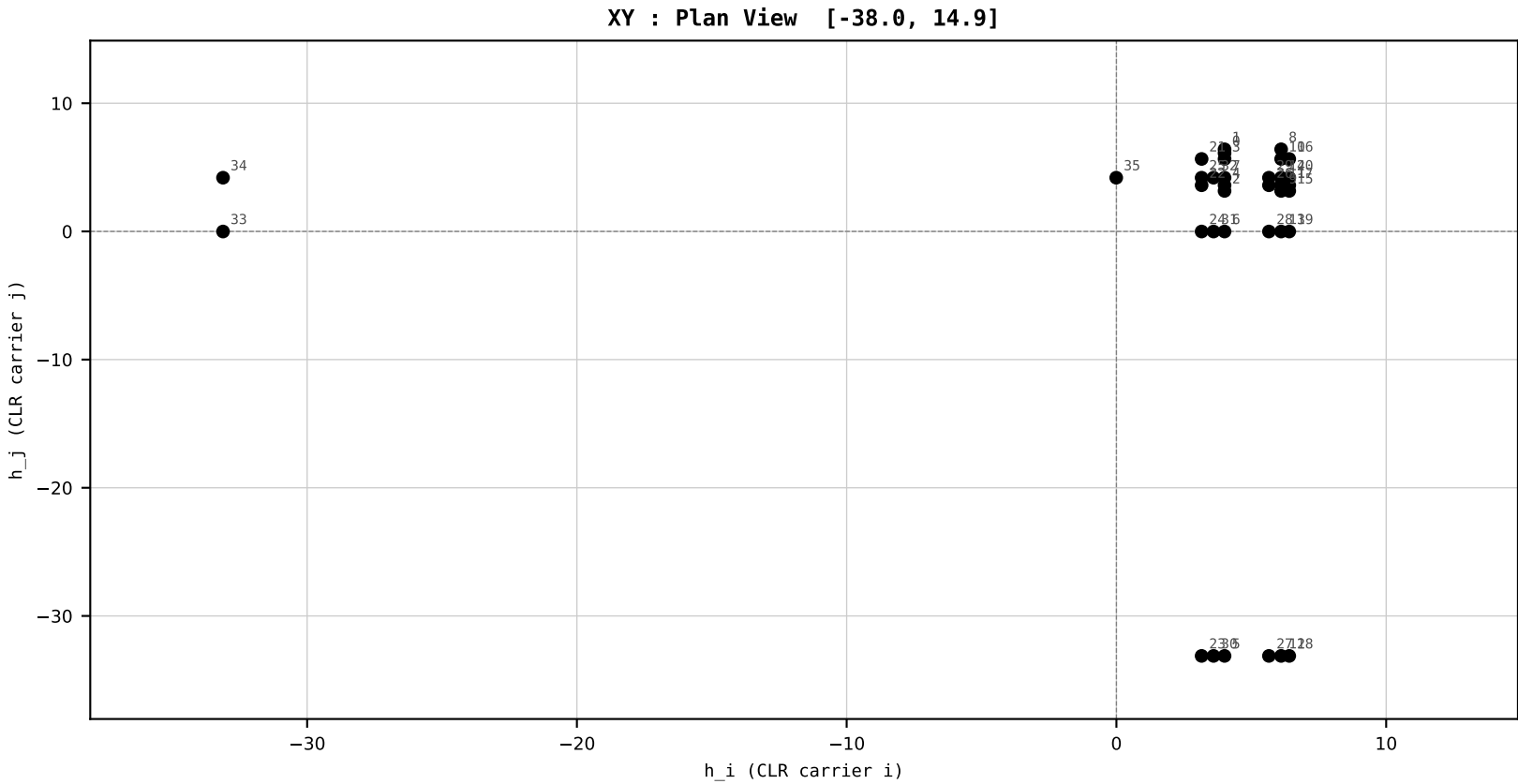
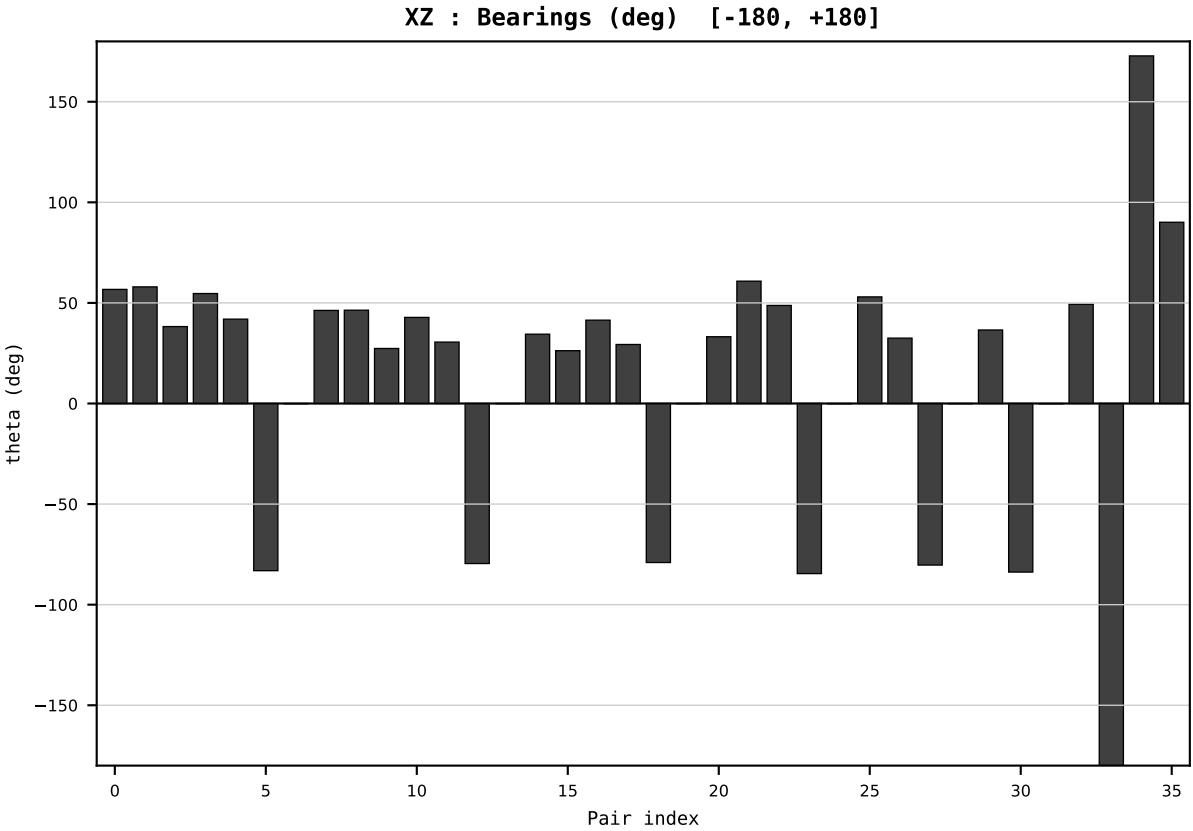
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2011
t : 11 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.337503
Ring = Hs-3
E_metric= 903.4163
kappa_HS= 14649999999999968.00
omega = 2.7842 deg
d_A = 1.726842
Helm = Solar
Helm d = +1.510285
DR = 39.526 HLR
DR ratio= 146499999506615968.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

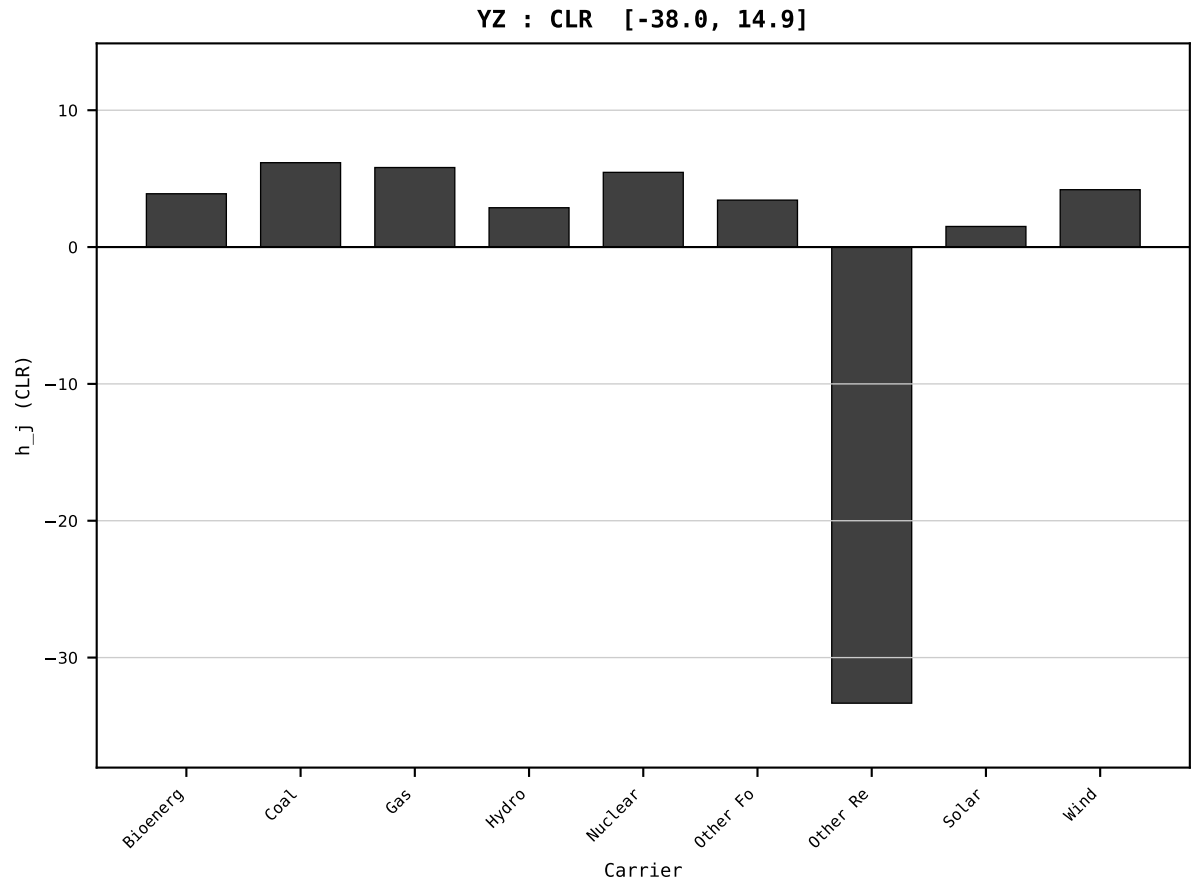
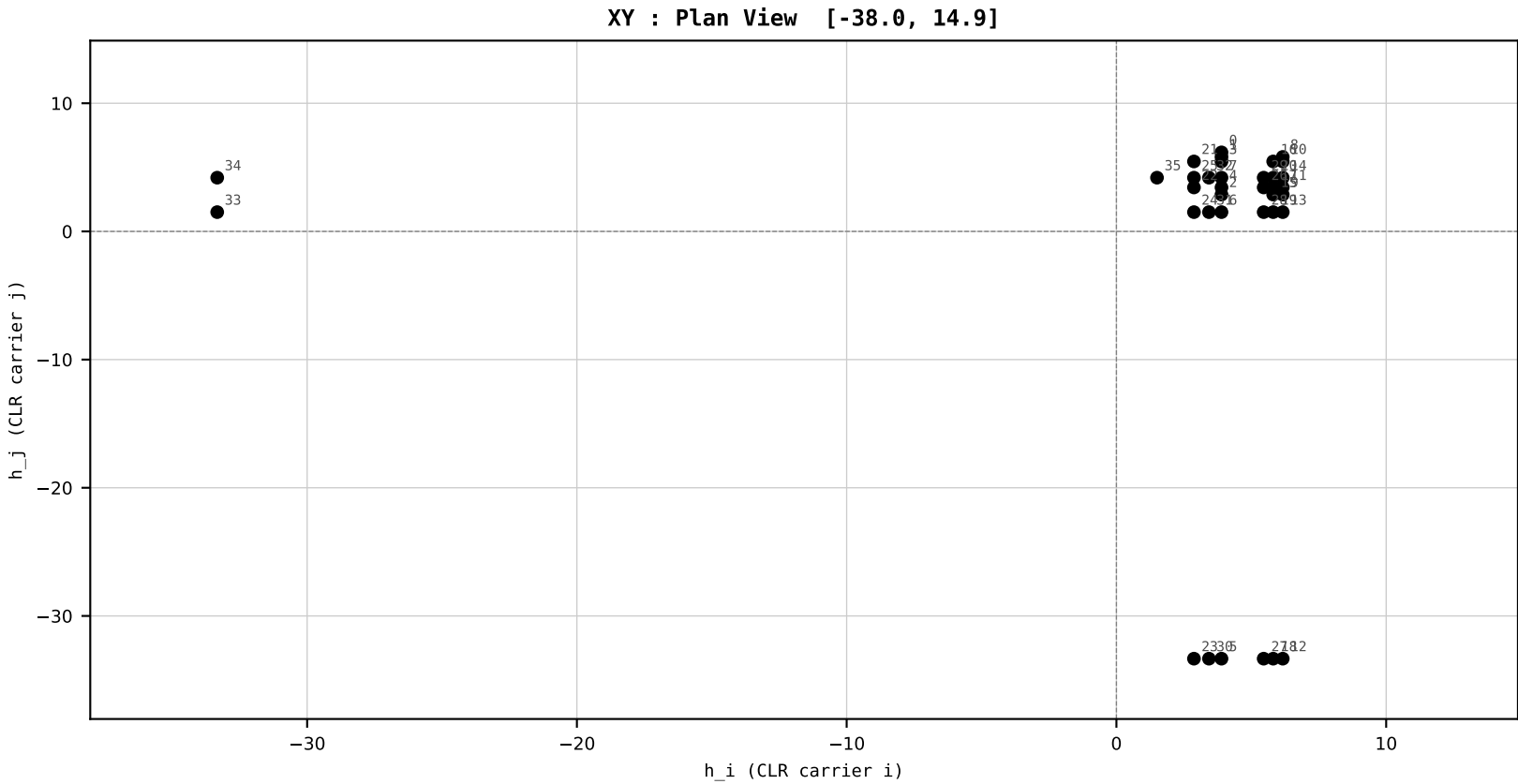
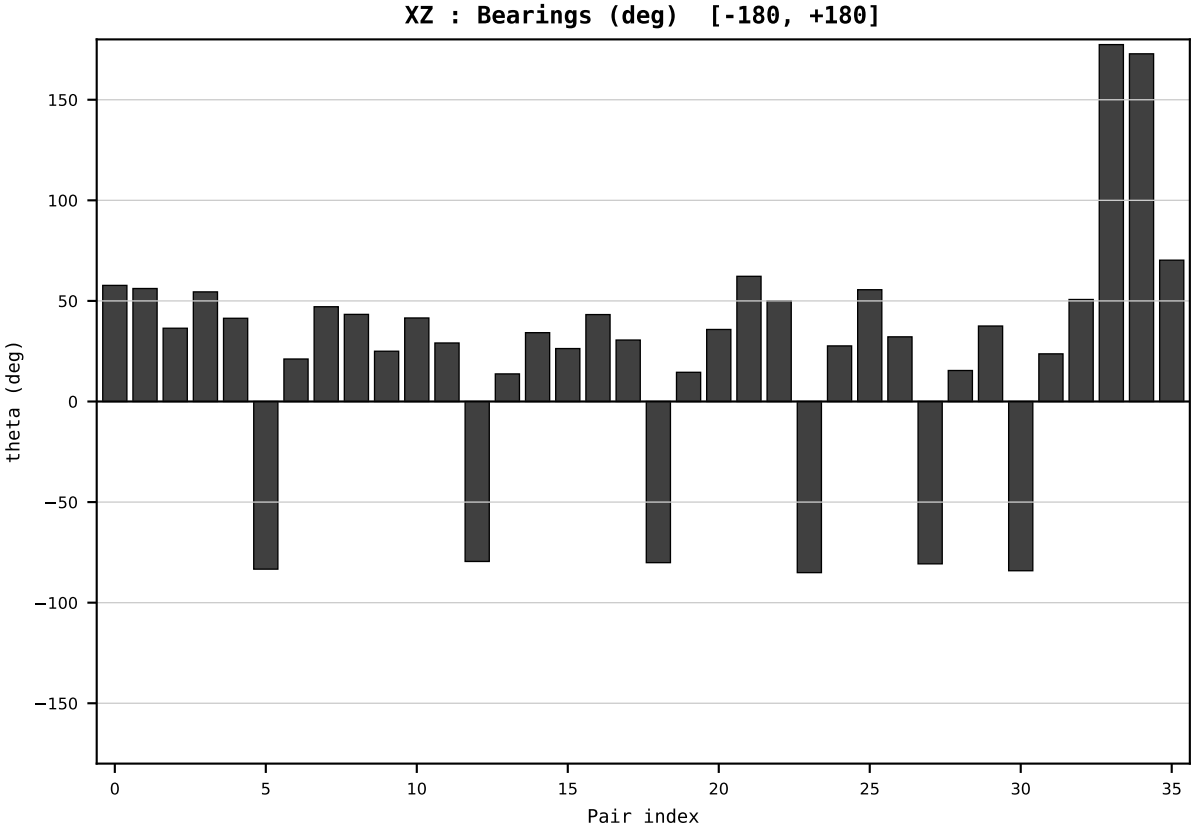
Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2012
t : 12 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.315326
Ring = Hs-3
E_metric= 905.5799
kappa_HS= 142789999999999984.00
omega = 2.7179 deg
d_A = 1.688485
Helm = Solar
Helm d = +1.511822
DR = 39.500 HLR
DR ratio= 142789999378451088.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=13 | Year 2013 | D=9 N=26 pairs=36

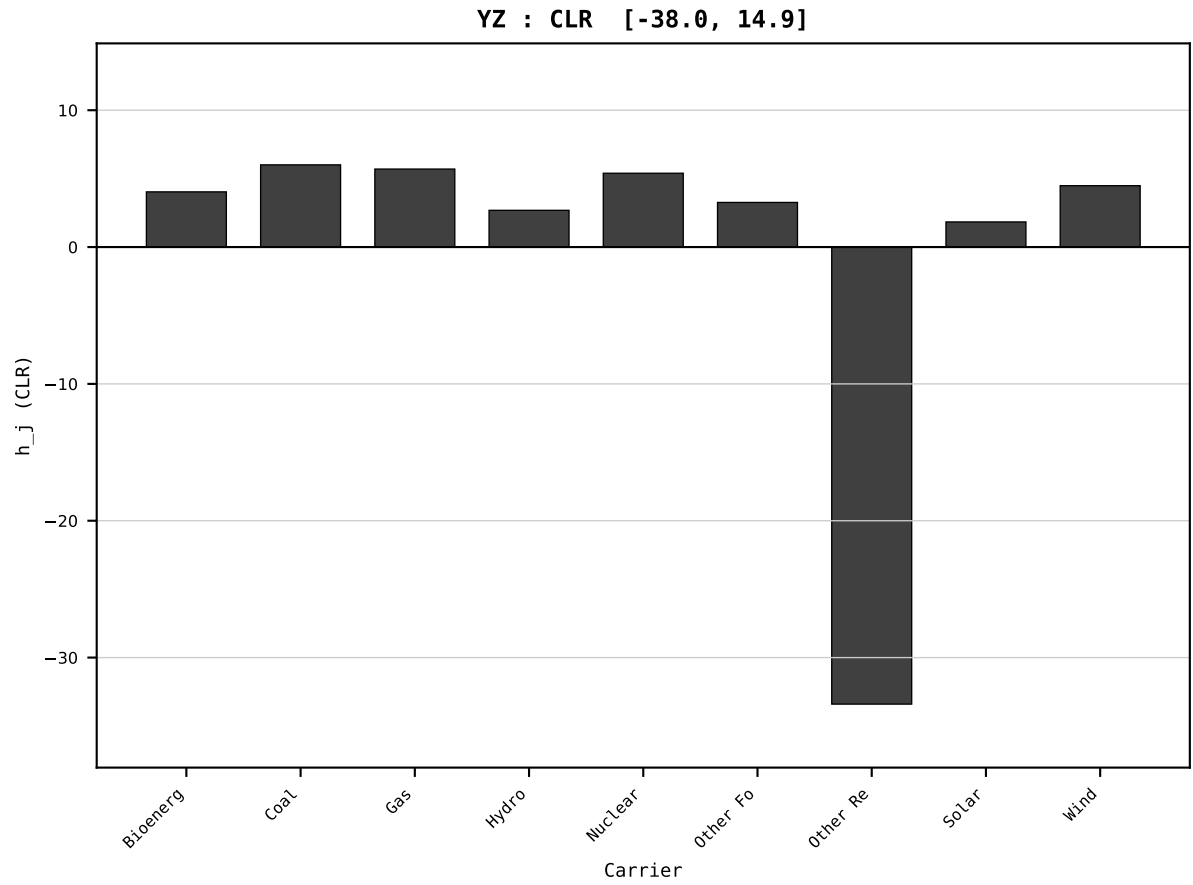
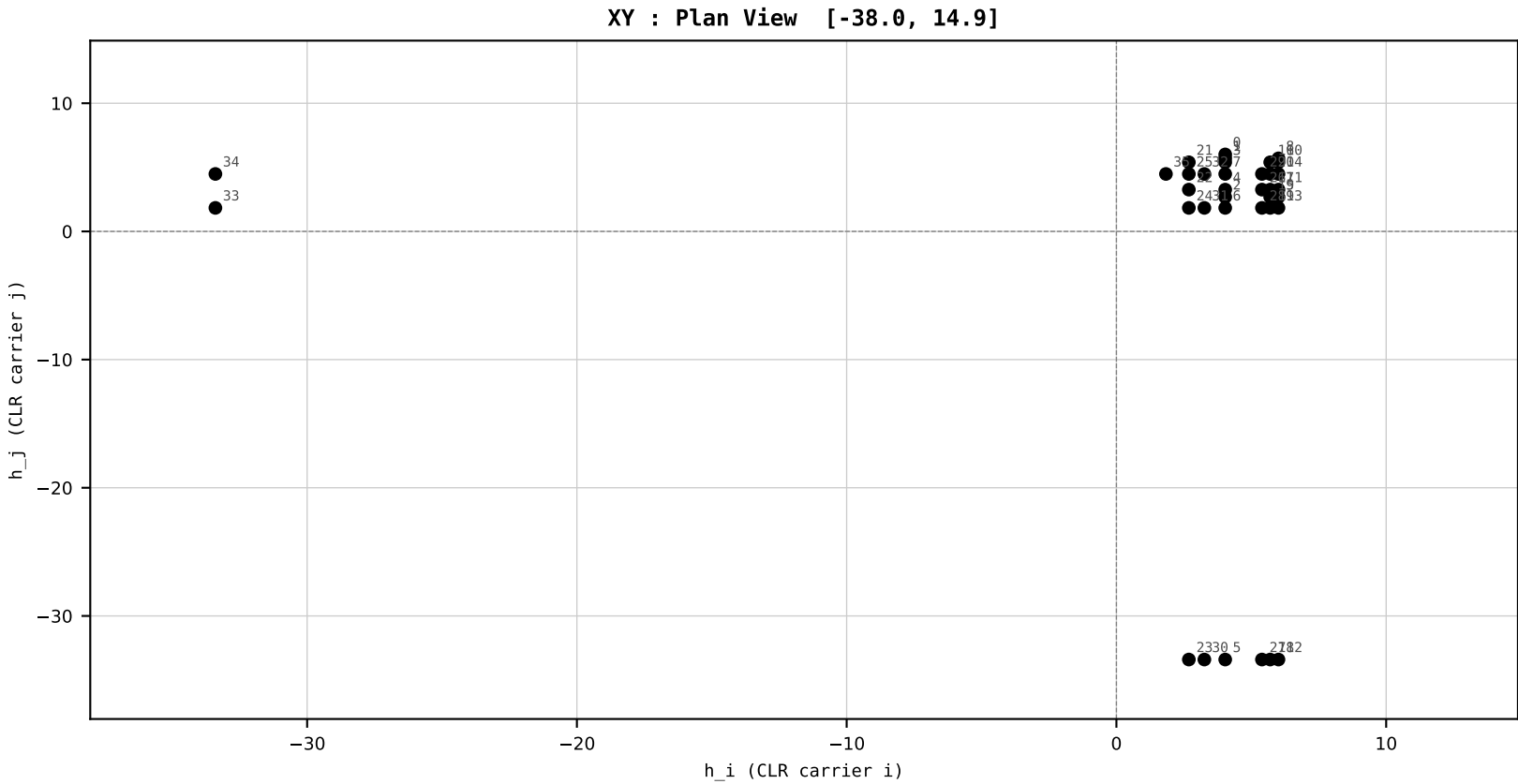
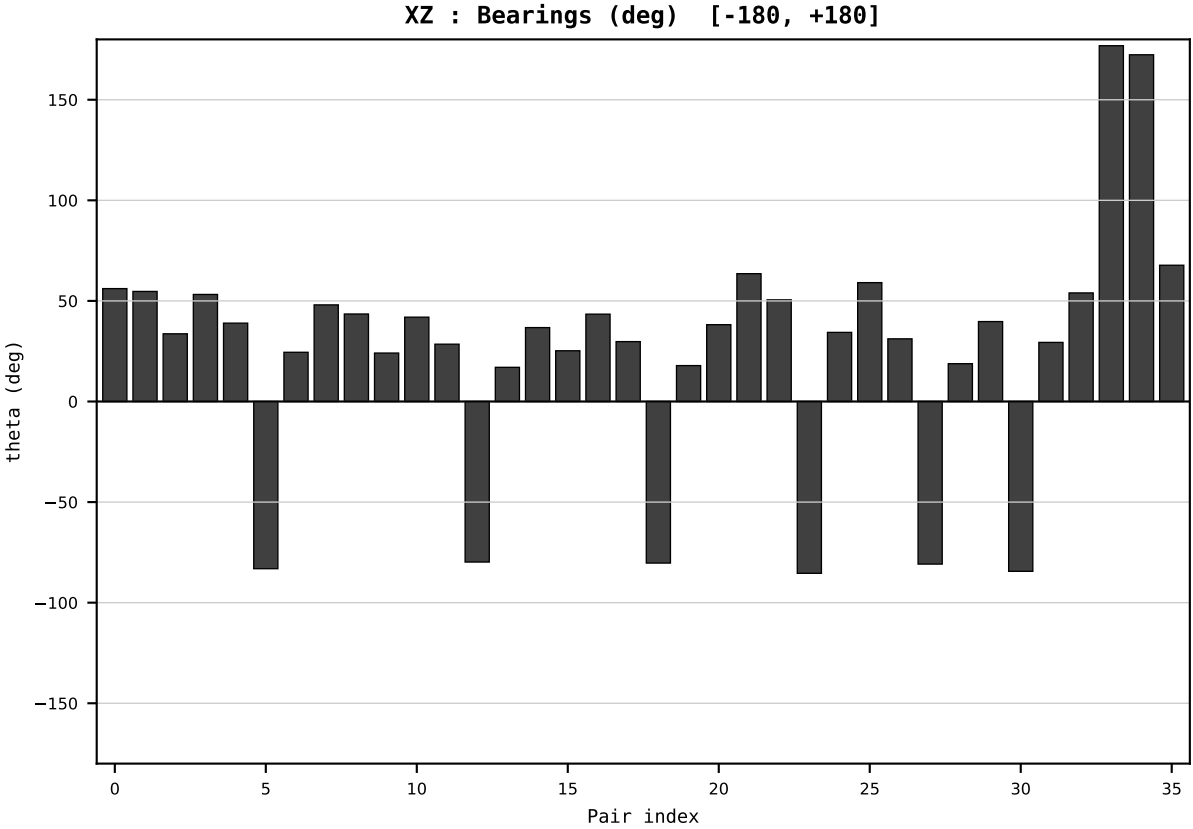
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2013
t : 13 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.287261
Ring = Hs-2
E_metric= 908.7712
kappa_HS= 130259999999999984.00
omega = 0.9132 deg
d_A = 0.569543
Helm = Solar
Helm d = +0.330853
DR = 39.408 HLR
DR ratio= 130260000256749840.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=14 | Year 2014 | D=9 N=26 pairs=36

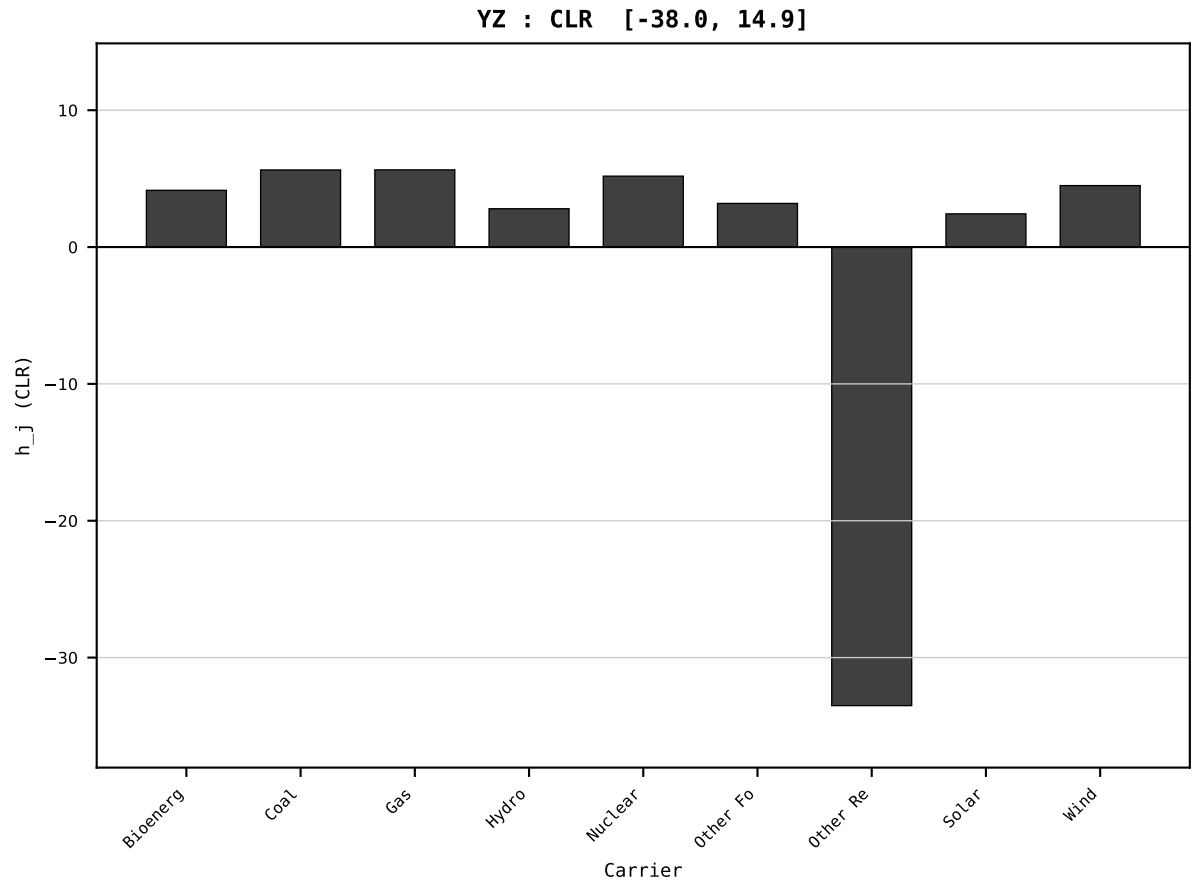
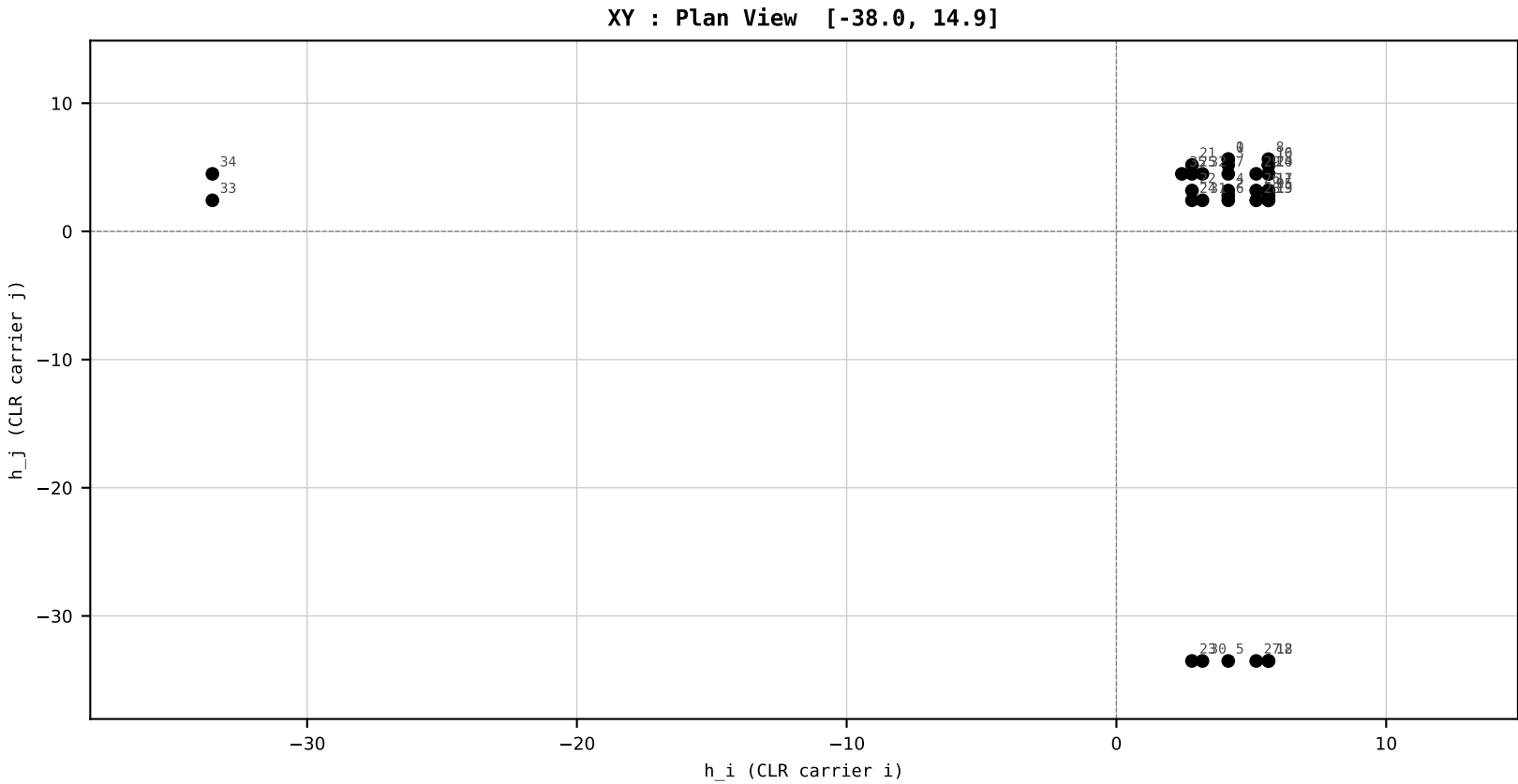
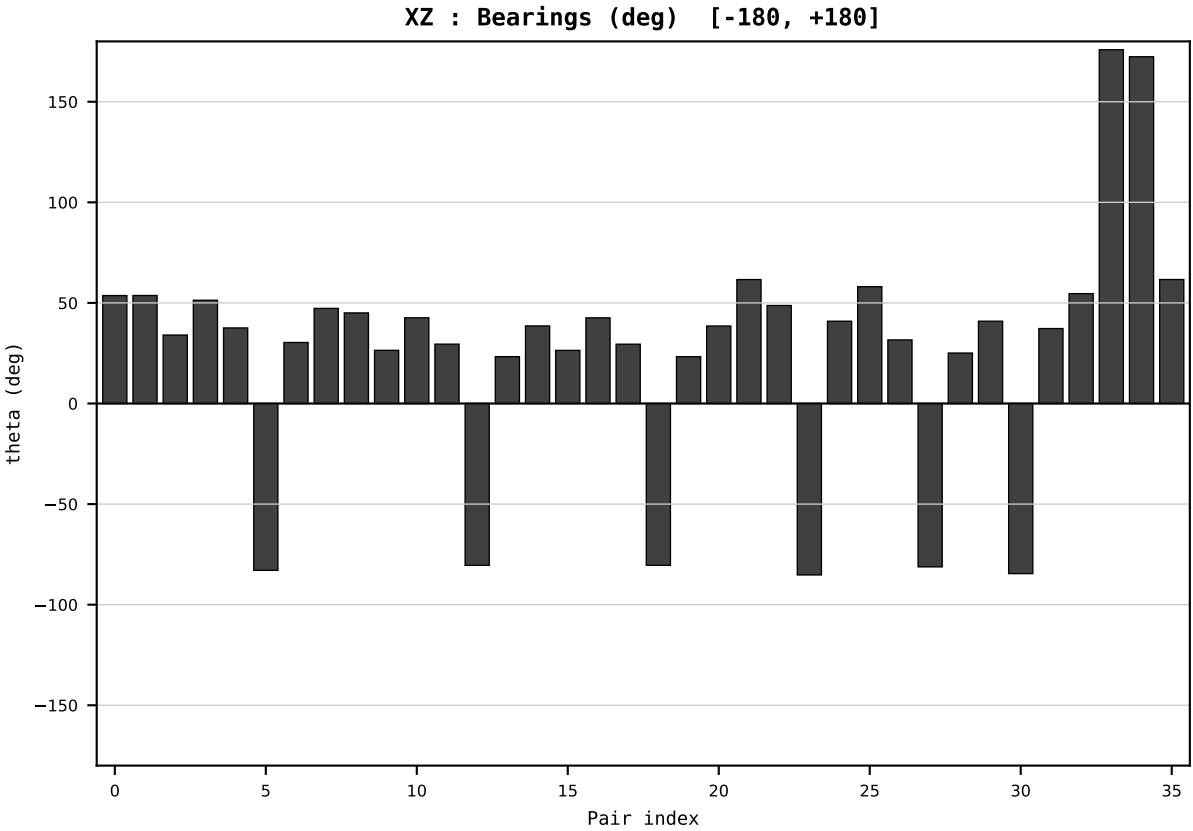
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2014
t : 14 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.245653
Ring = Hs-2
E_metric= 914.5648
kappa_HS= 100889999999999984.00
omega = 1.2190 deg
d_A = 0.760916
Helm = Solar
Helm d = +0.590262
DR = 39.153 HLR
DR ratio= 1008900000067498000.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=15 | Year 2015 | D=9 N=26 pairs=36

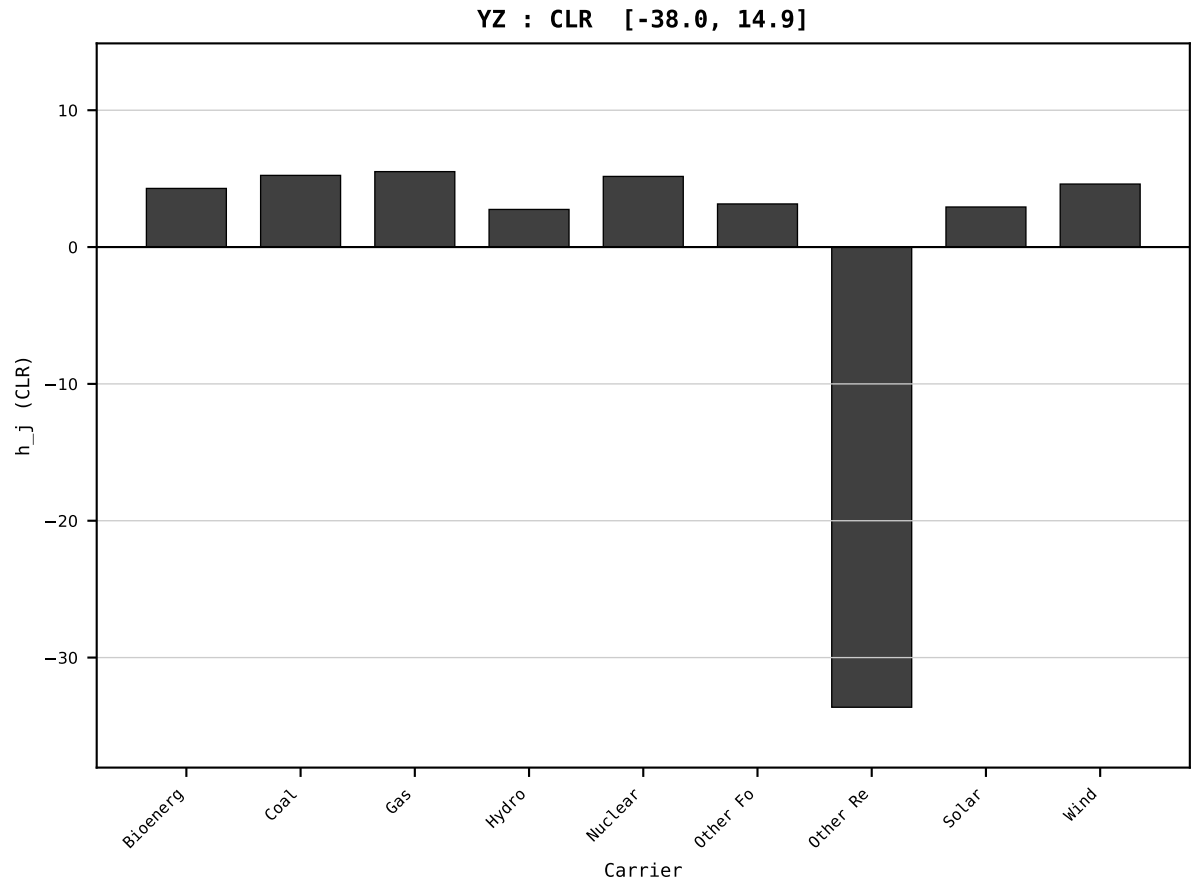
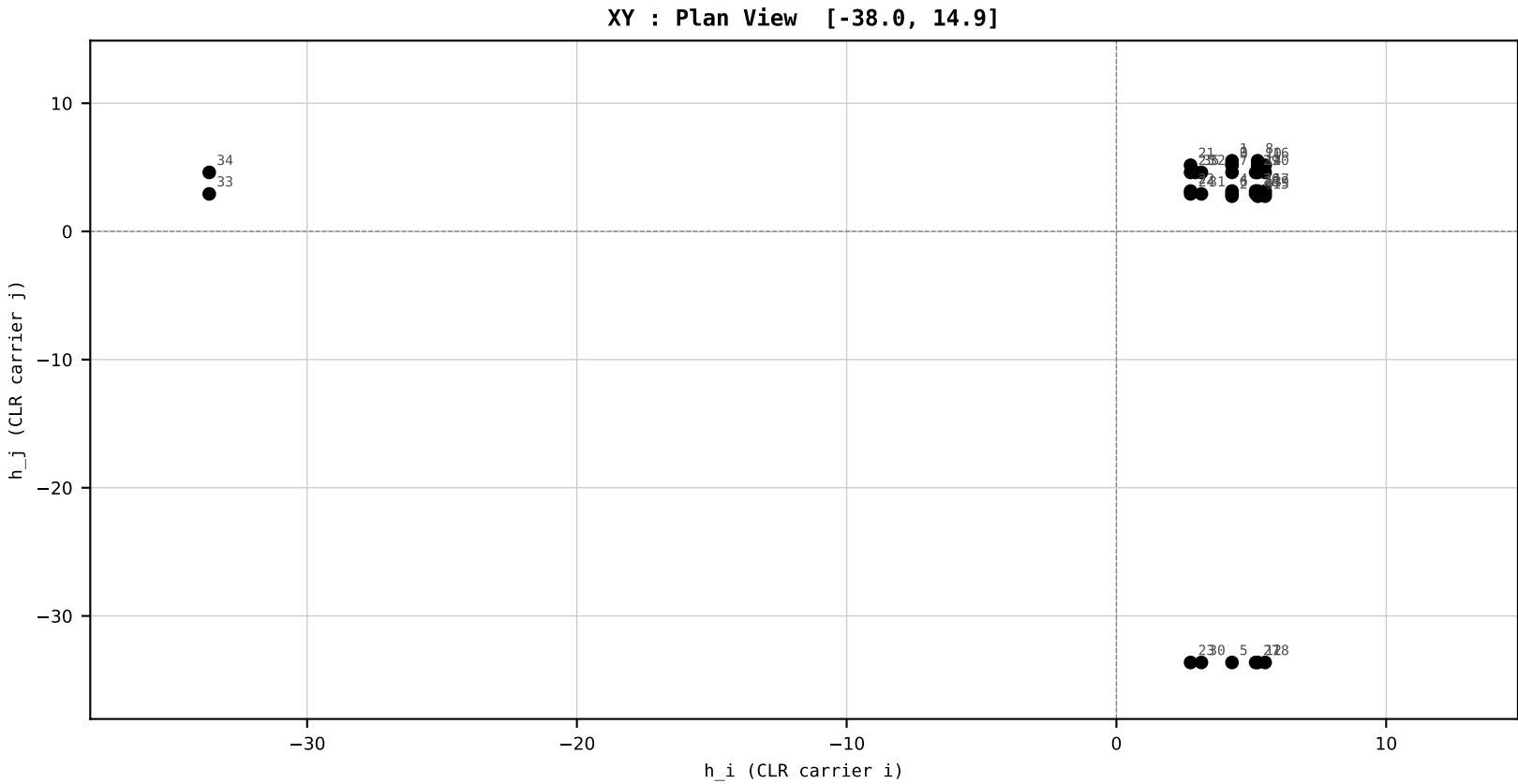
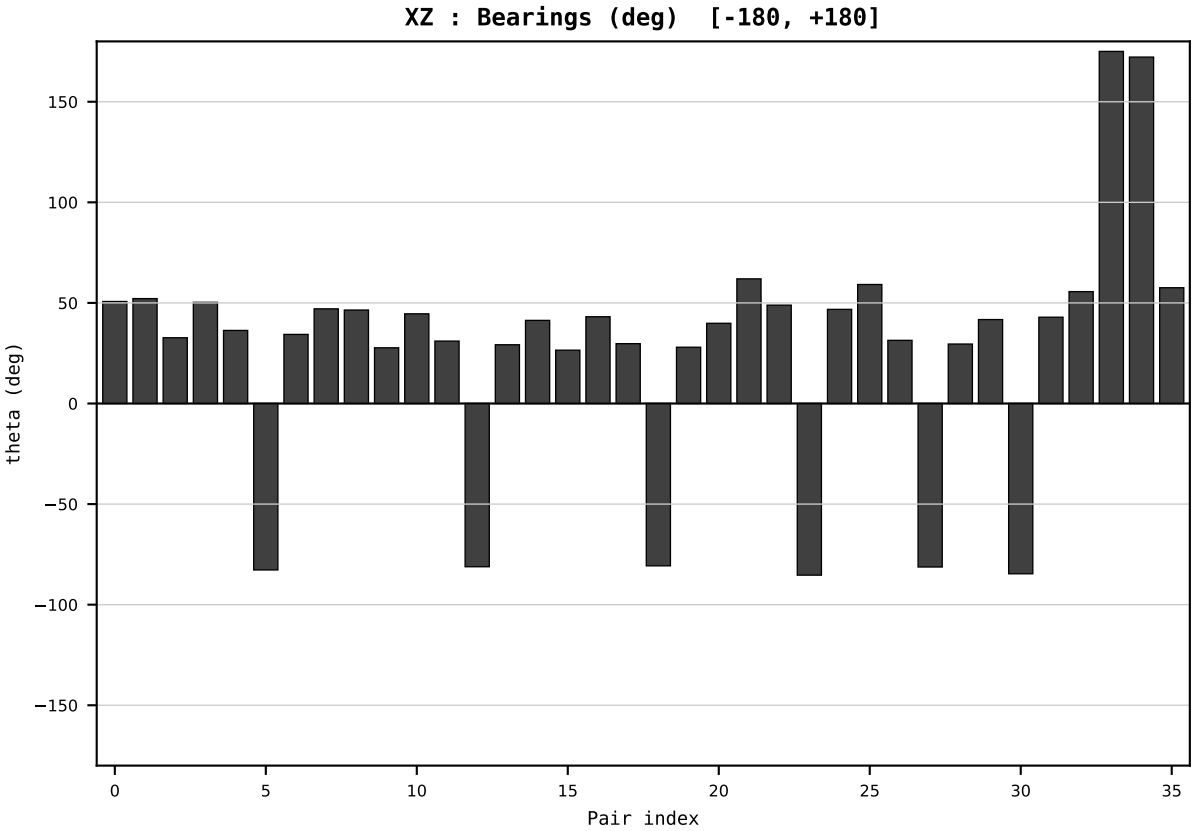
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2015
t : 15 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.206119
Ring = Hs-2
E_metric= 919.4045
kappa_HS= 9987999999999984.00
omega = 1.0971 deg
d_A = 0.690200
Helm = Solar
Helm d = +0.501757
DR = 39.143 HLR
DR ratio= 99879999967813408.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=16 | Year 2016 | D=9 N=26 pairs=36

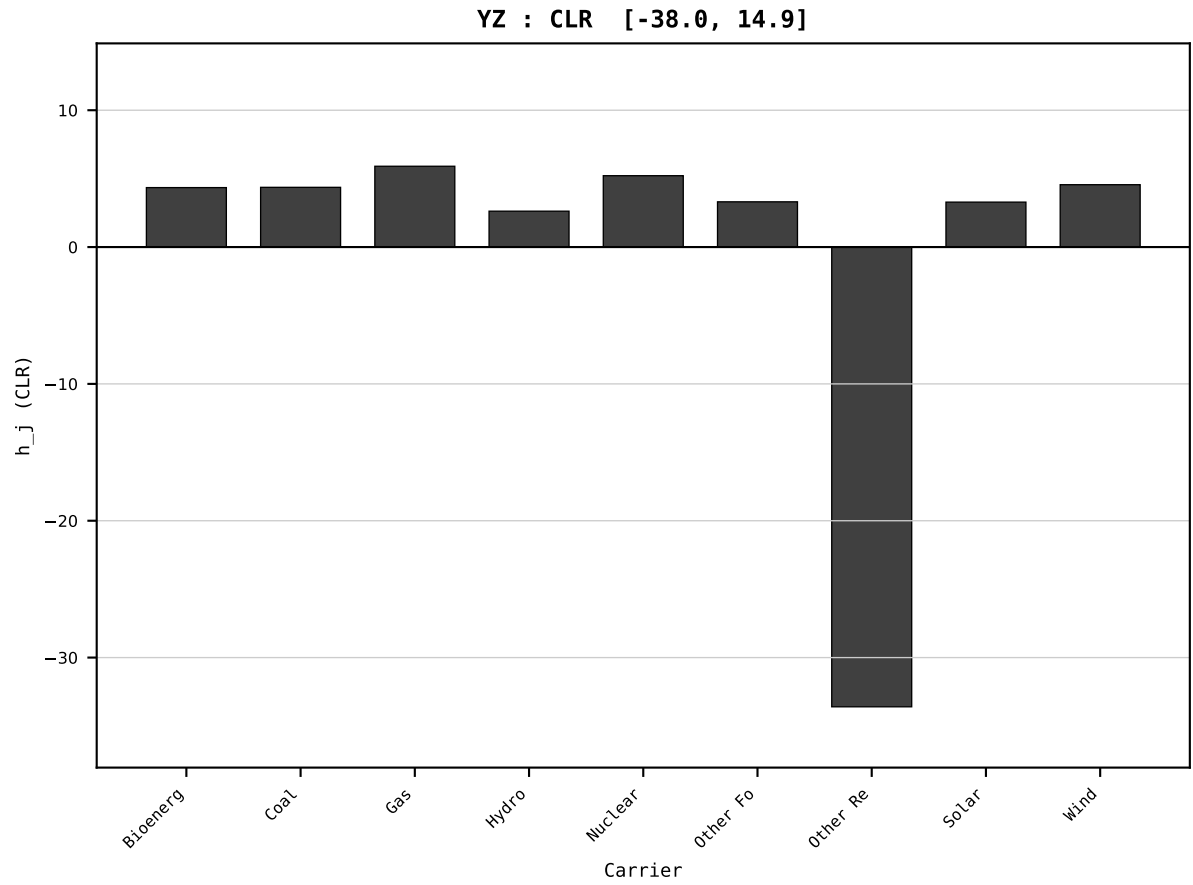
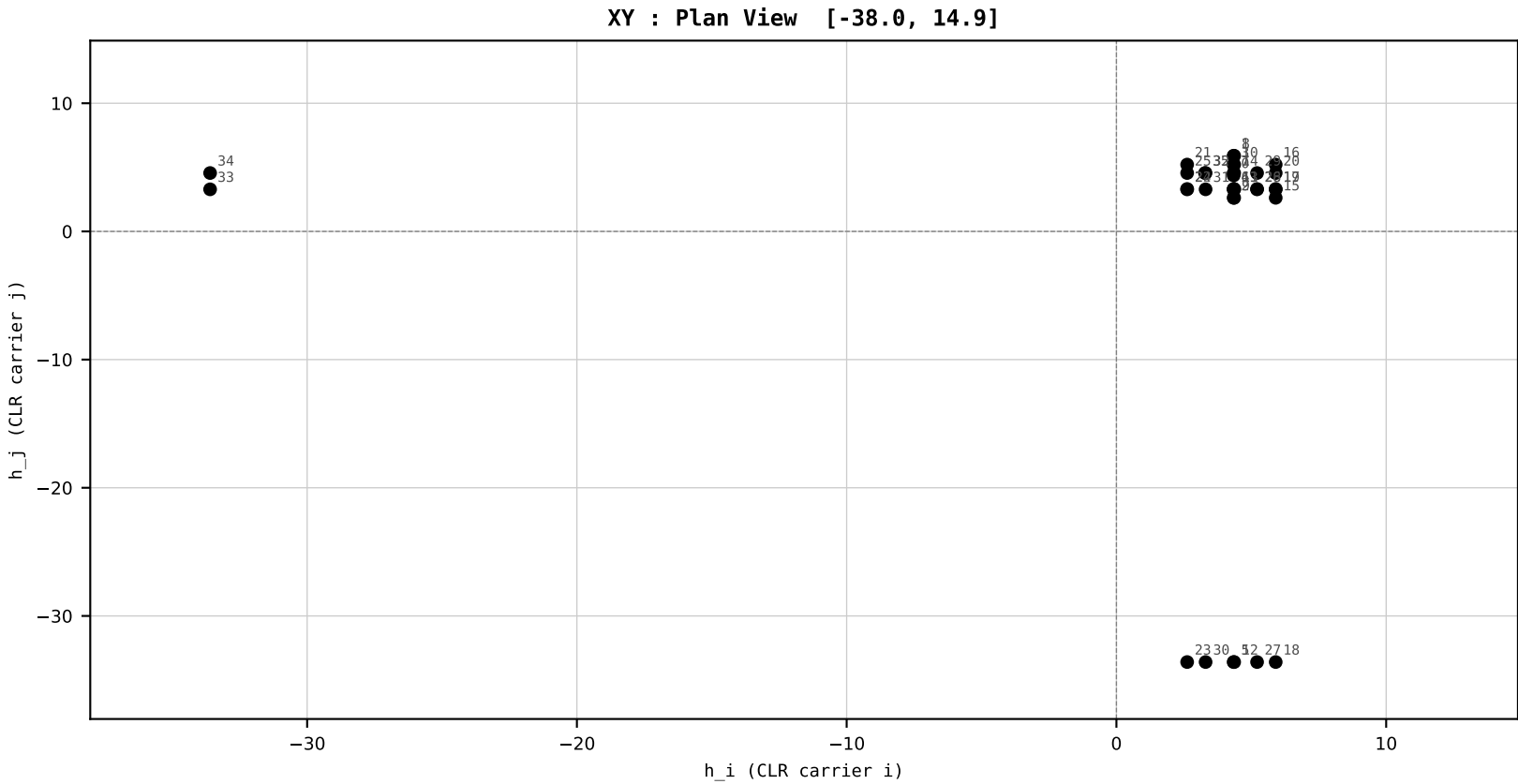
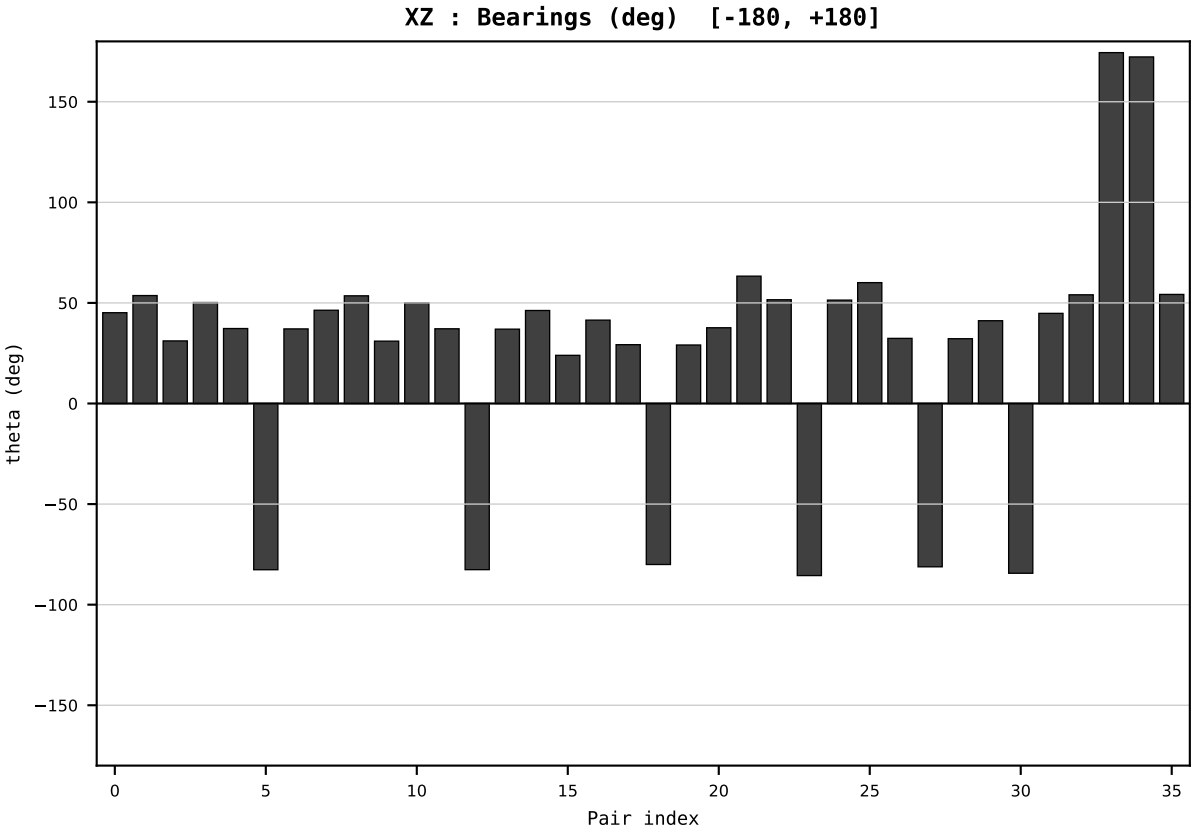
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2016
t : 16 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.249877
Ring = Hs-2
E_metric= 916.7901
kappa_HS= 143129999999999984.00
omega = 1.6738 deg
d_A = 1.045760
Helm = Coal
Helm d = -0.873359
DR = 39.503 HLR
DR ratio= 143129999555321712.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=17 | Year 2017 | D=9 N=26 pairs=36

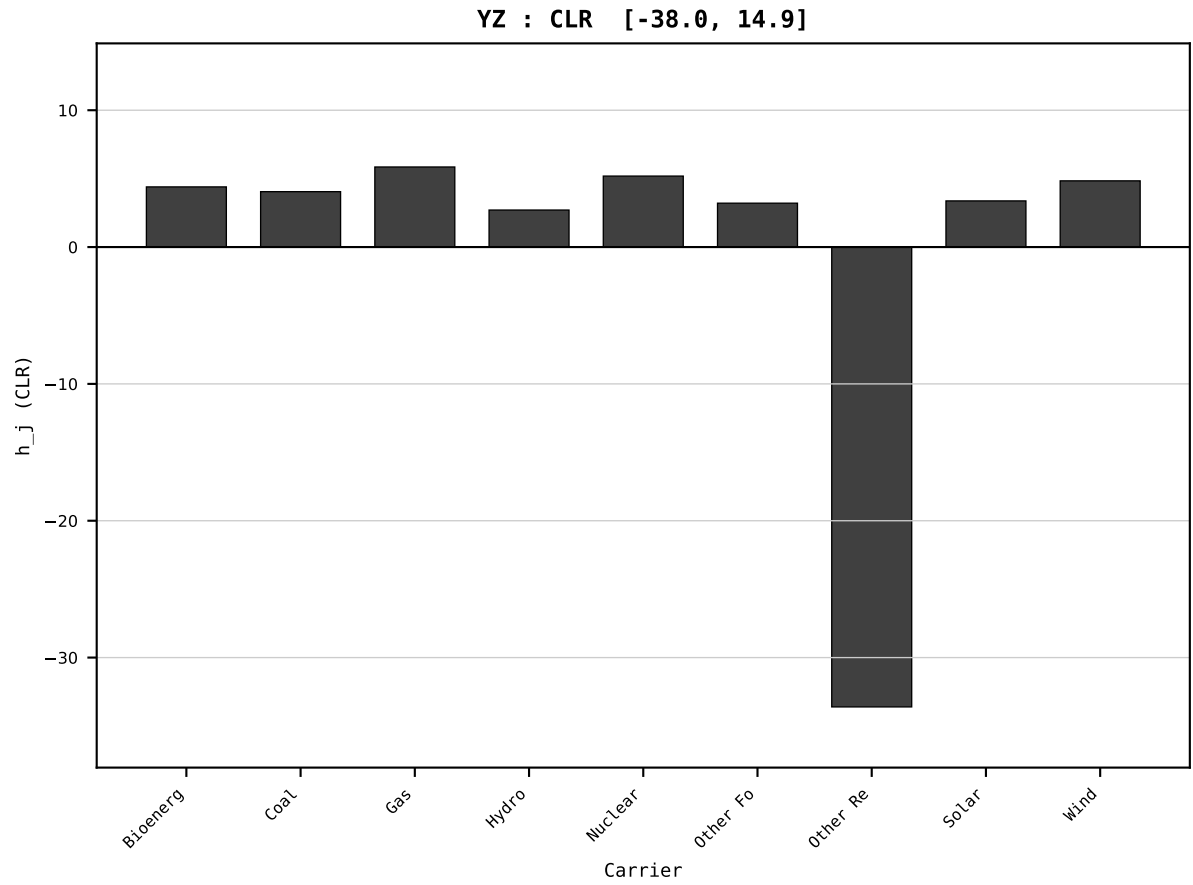
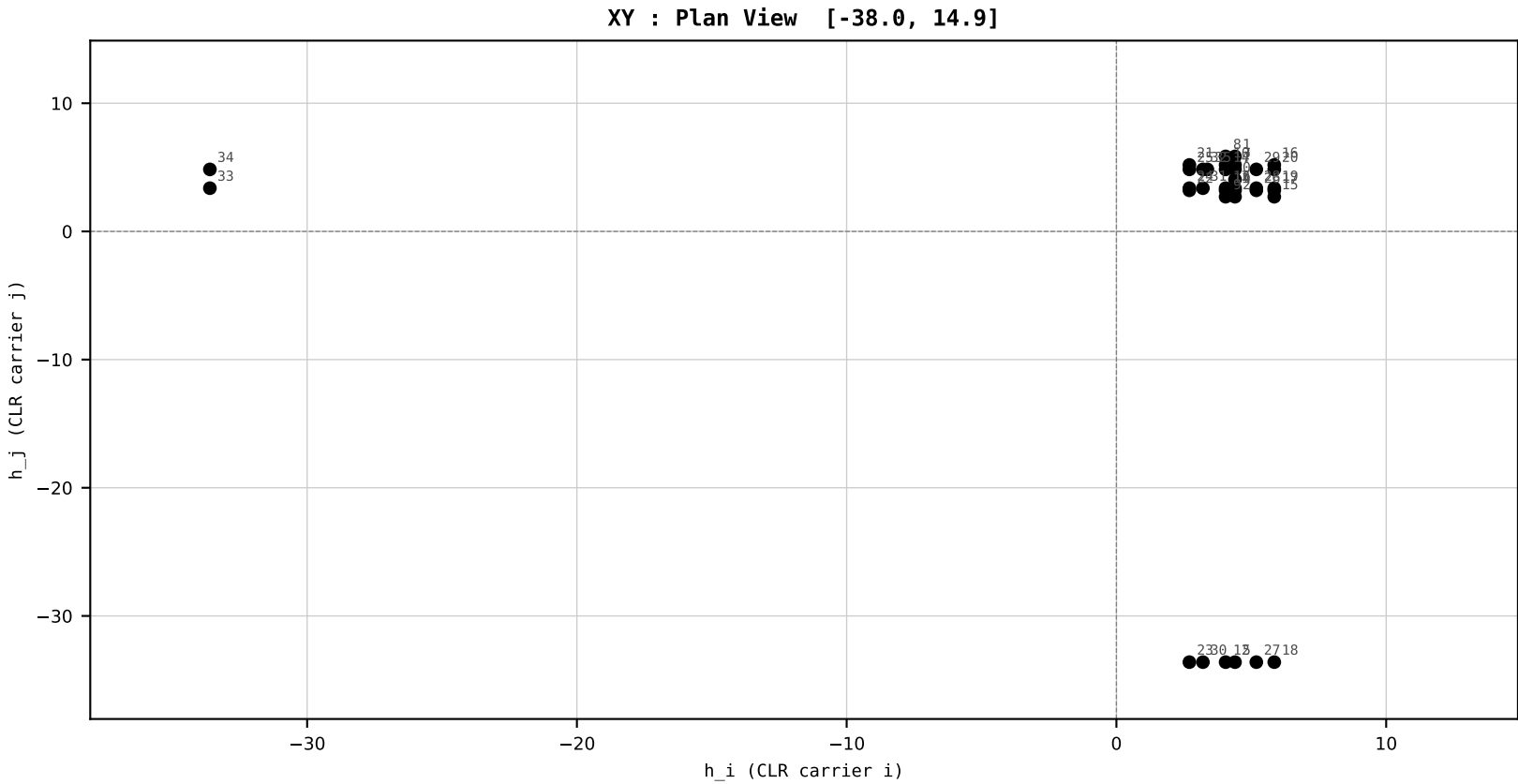
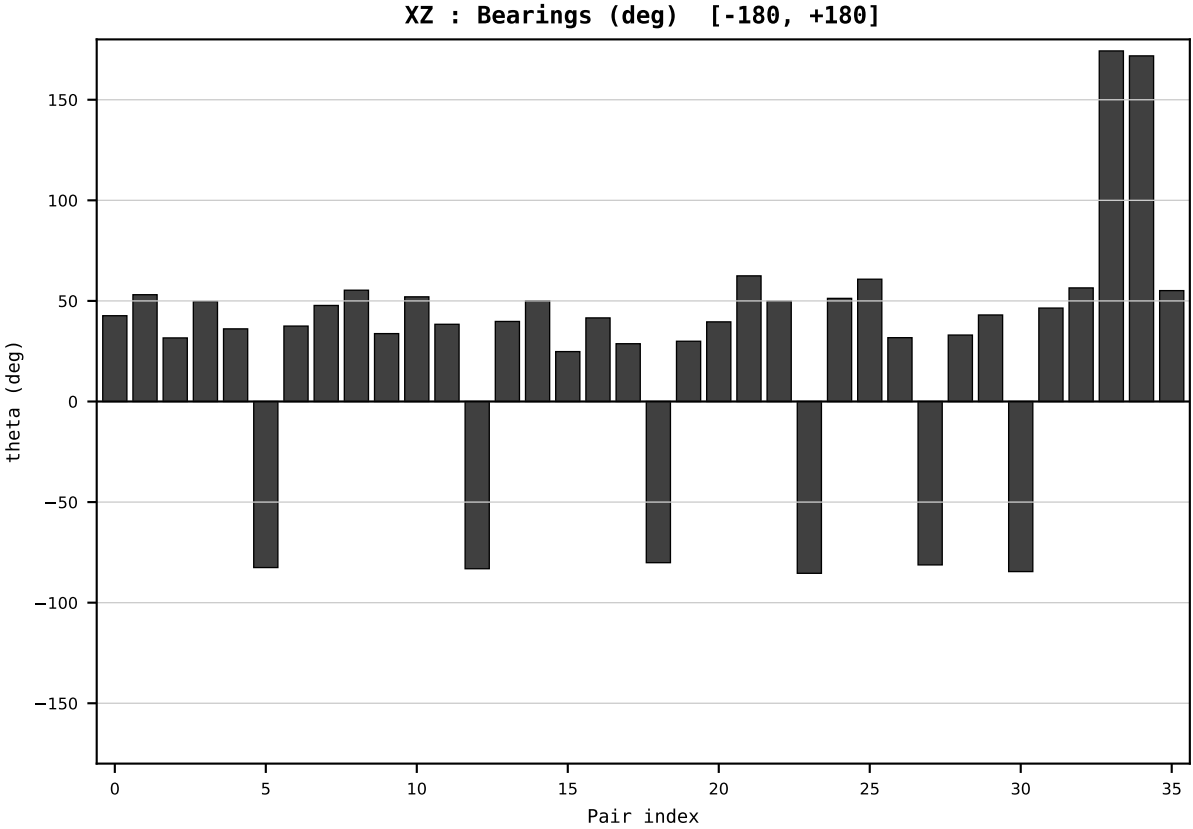
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2017
t : 17 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.242414
Ring = Hs-2
E_metric= 917.2914
kappa_HS= 13674999999999984.00
omega = 0.7349 deg
d_A = 0.458615
Helm = Coal
Helm d = -0.316282
DR = 39.457 HLR
DR ratio= 136749999150403536.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=18 | Year 2018 | D=9 N=26 pairs=36

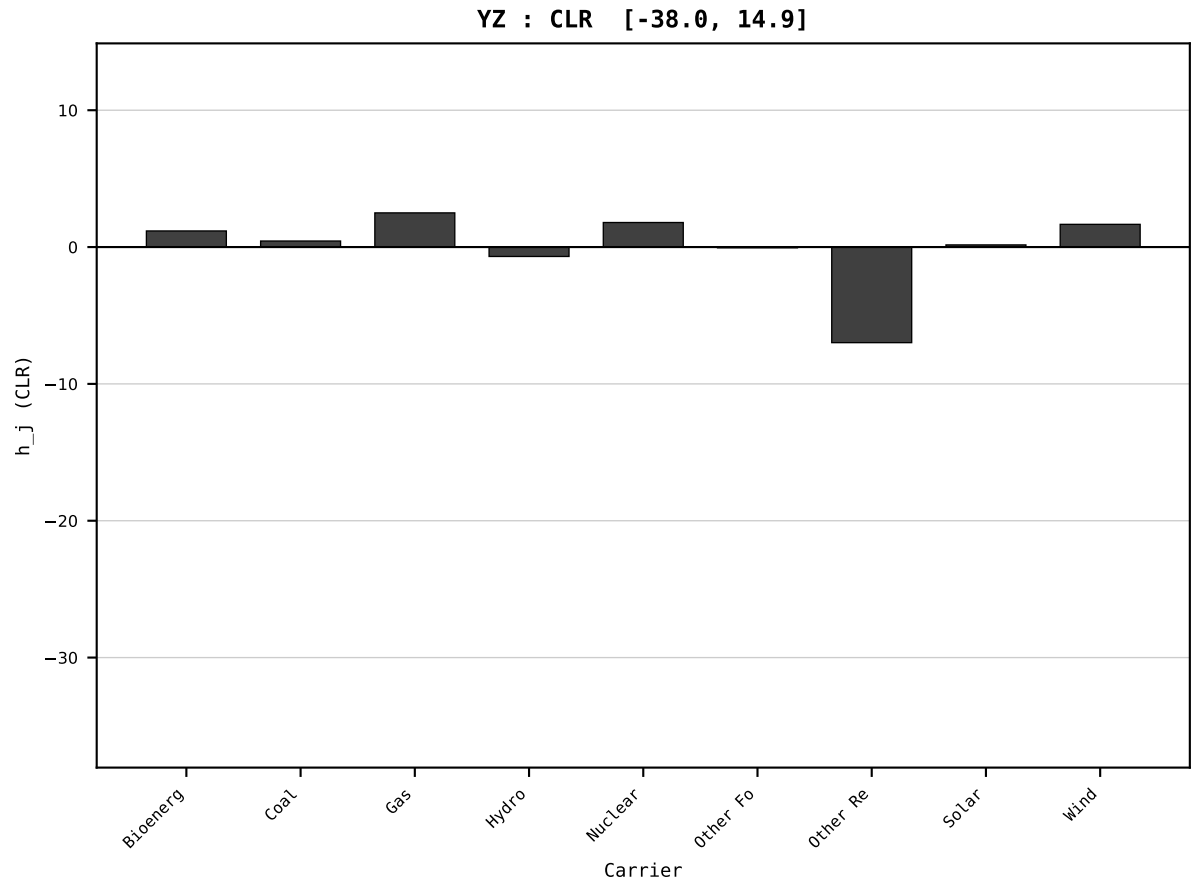
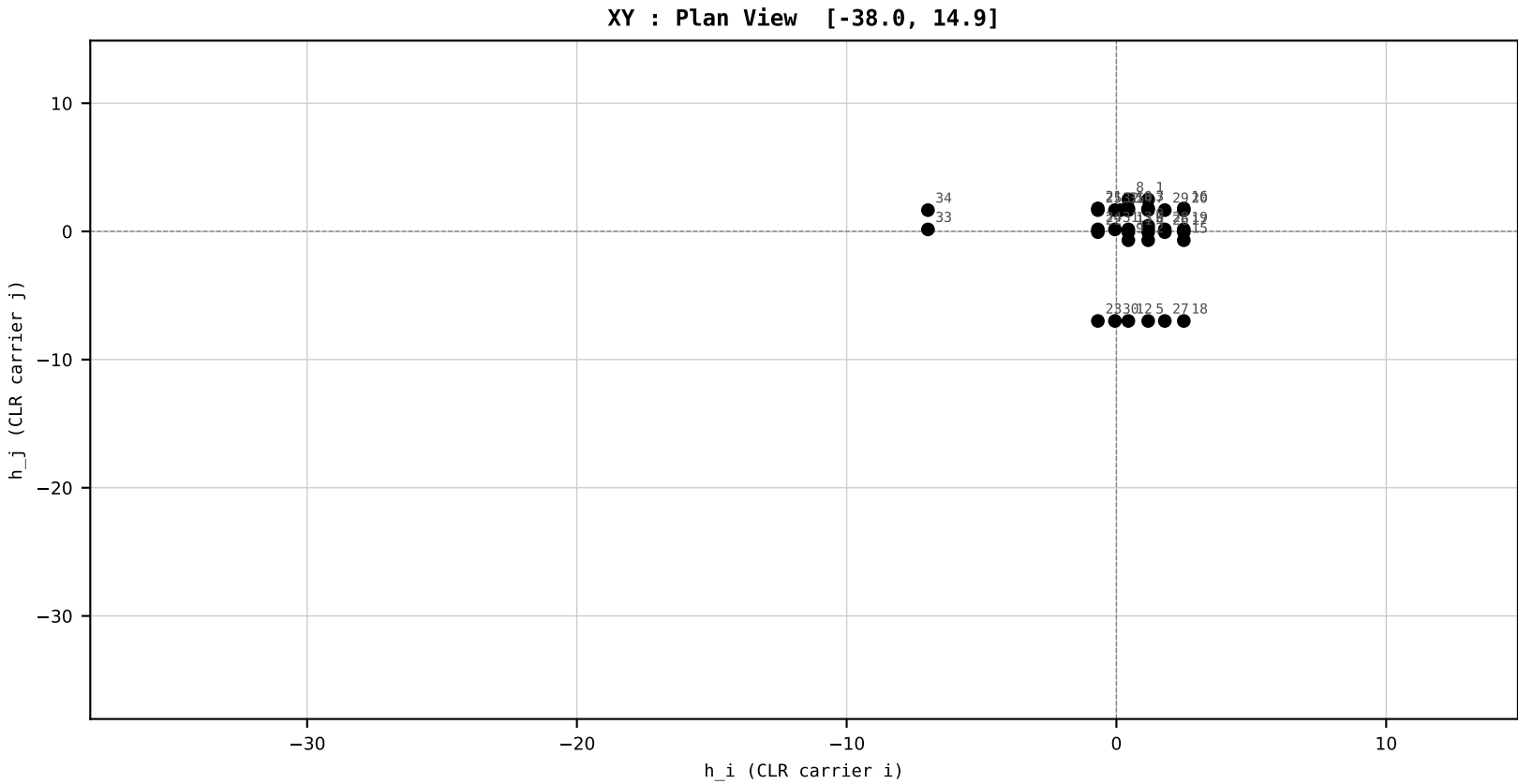
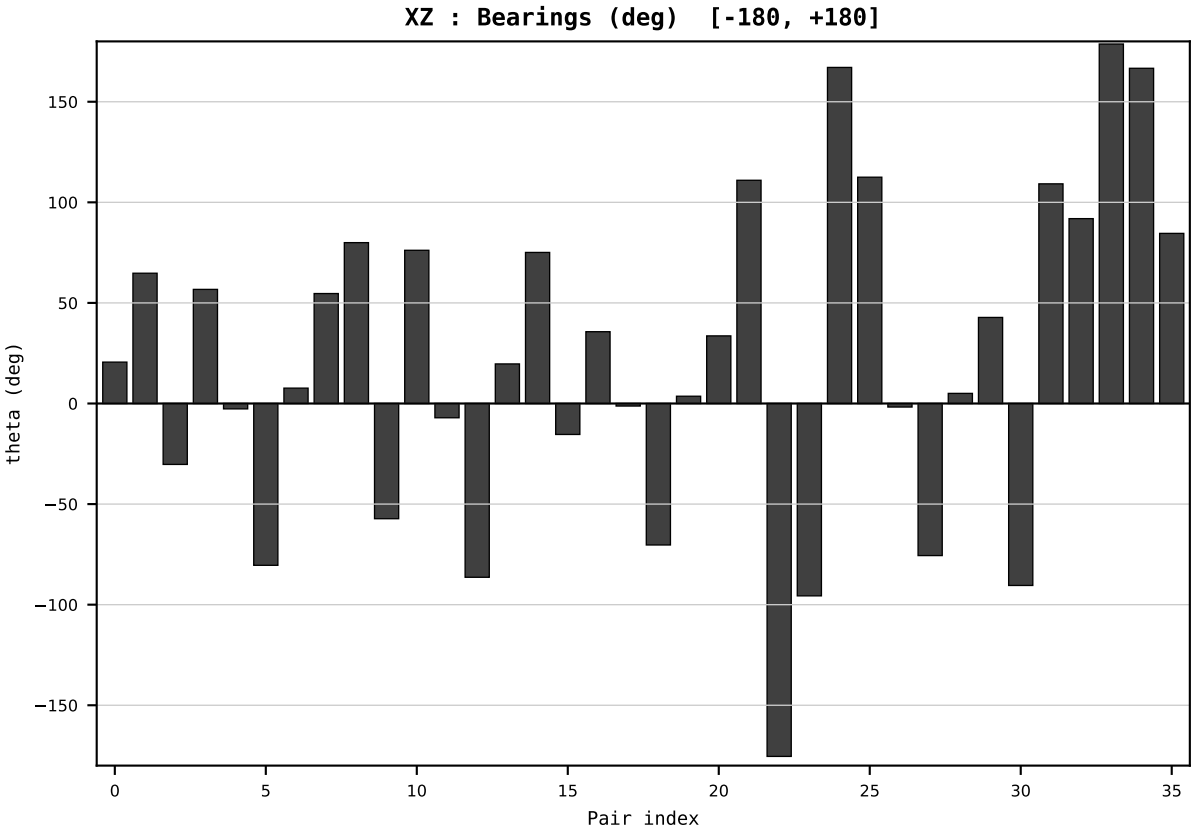
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2018
t : 18 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.238427
Ring = Hs-2
E_metric= 63.1052
kappa_HS= 13149.00
omega = 16.6103 deg
d_A = 28.236007
Helm = Other Renewables
Helm d = +26.618855
DR = 9.484 HLR
DR ratio= 13149.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=19 | Year 2019 | D=9 N=26 pairs=36

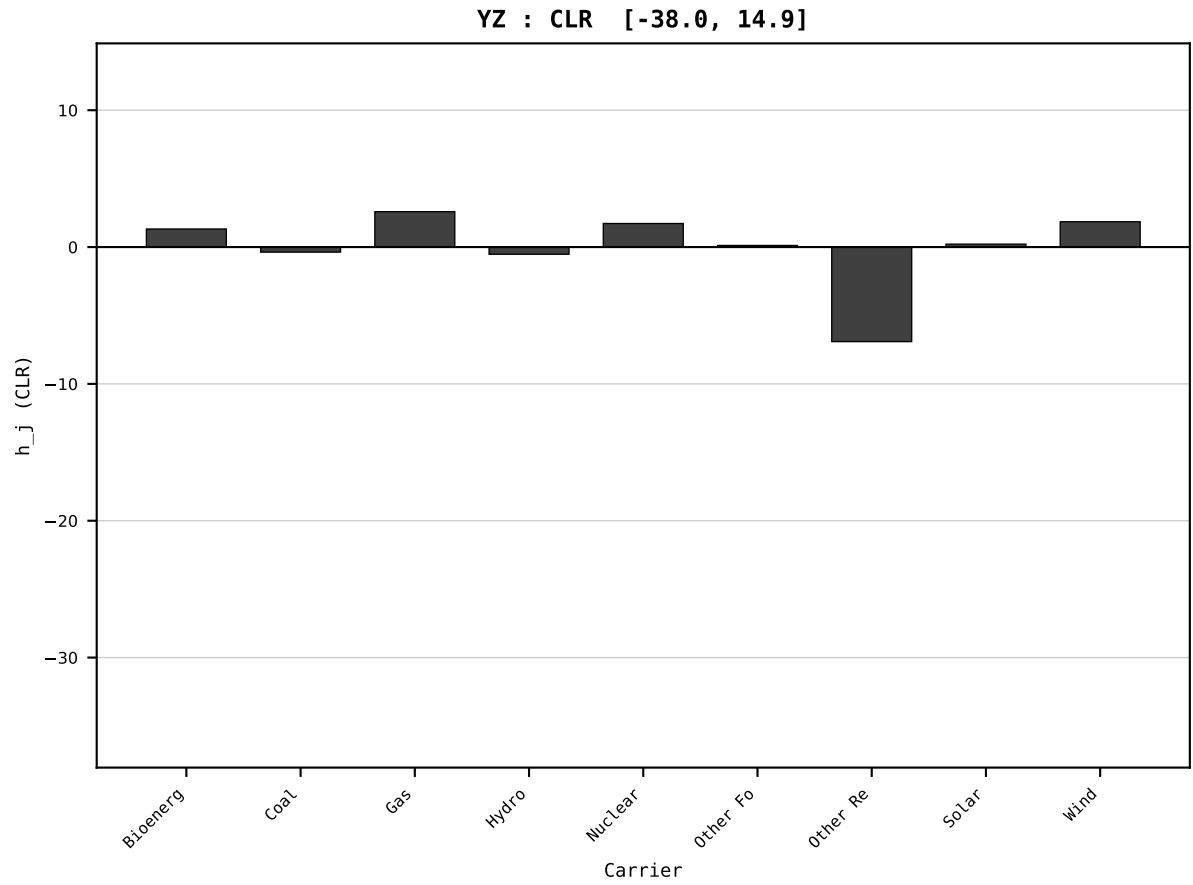
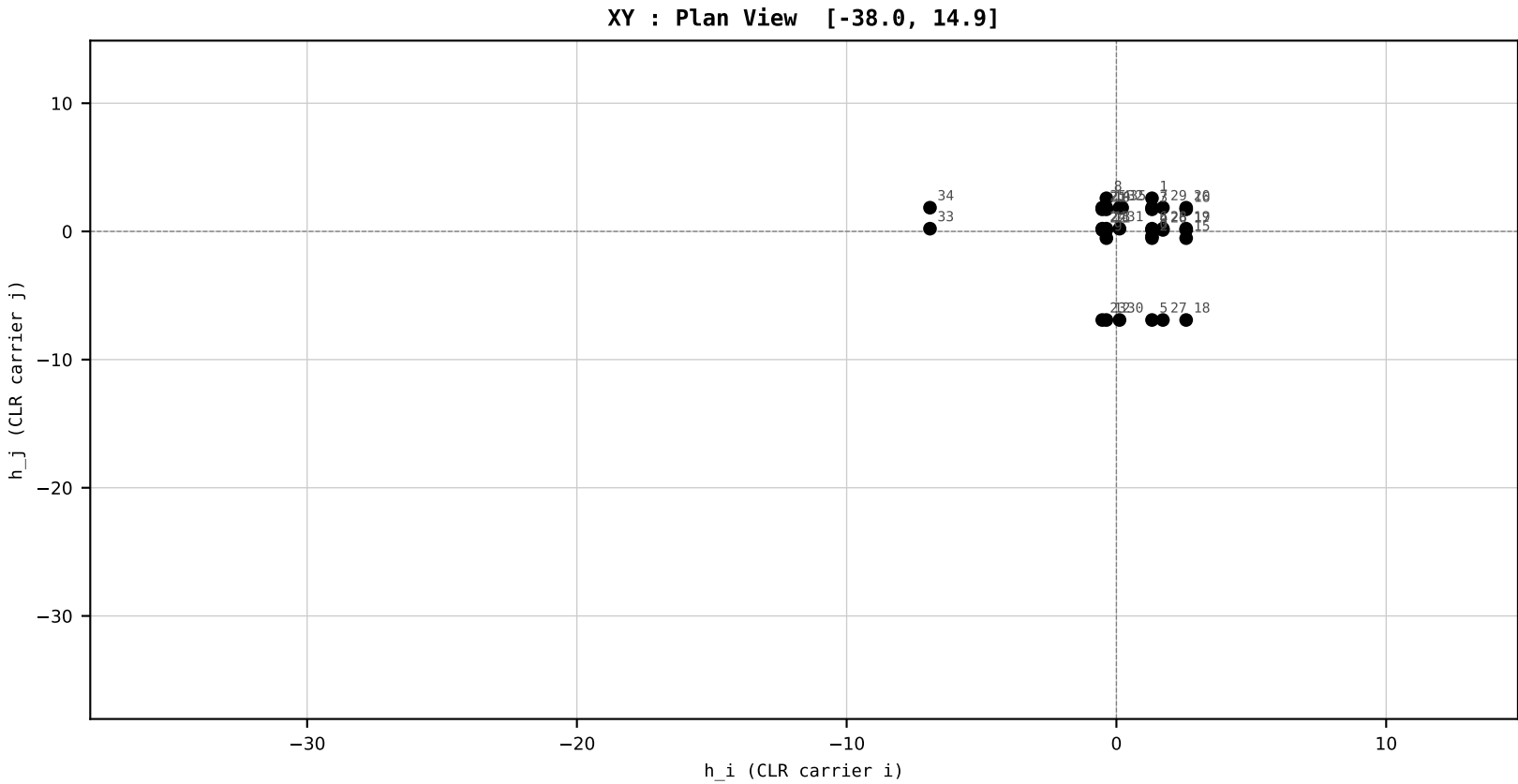
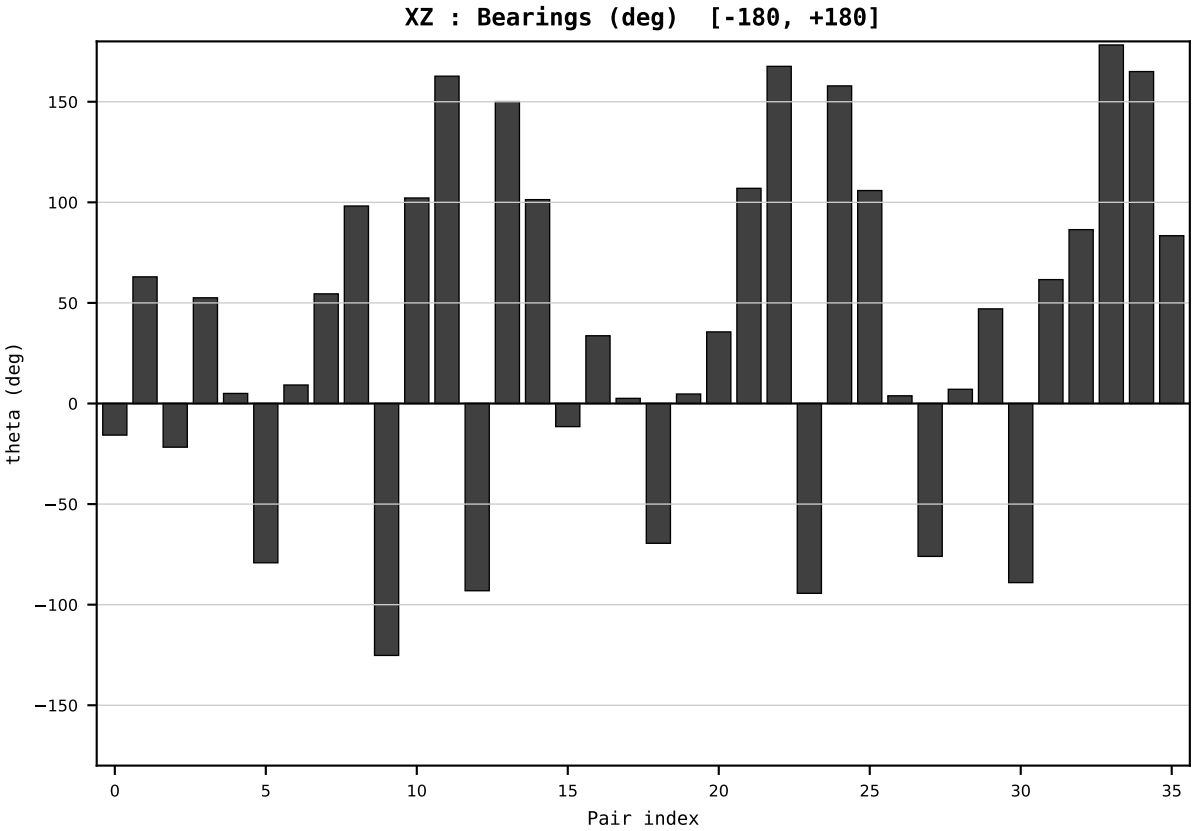
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2019
t : 19 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.257979
Ring = Hs-2
E_metric= 63.0521
kappa_HS= 13309.00
omega = 6.4343 deg
d_A = 0.891439
Helm = Coal
Helm d = -0.813158
DR = 9.496 HLR
DR ratio= 13309.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=20 | Year 2020 | D=9 N=26 pairs=36

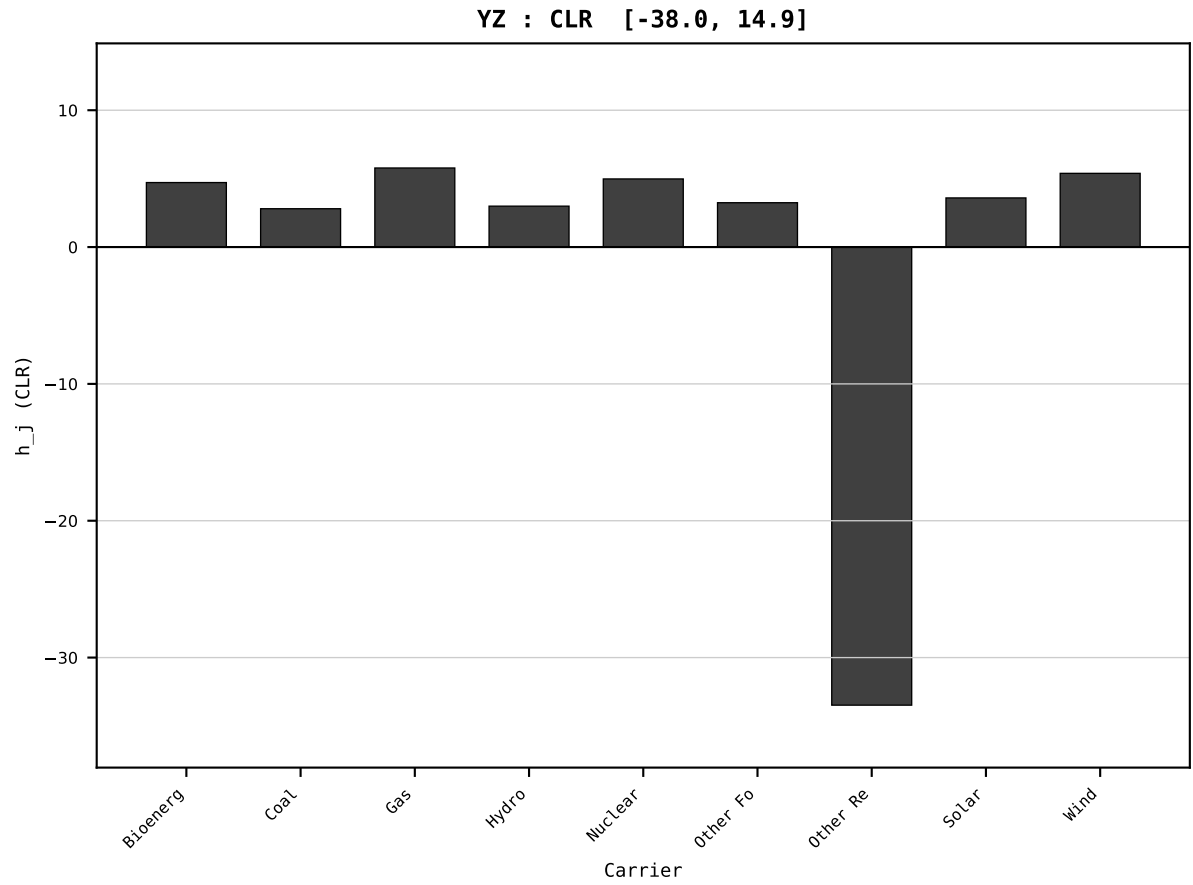
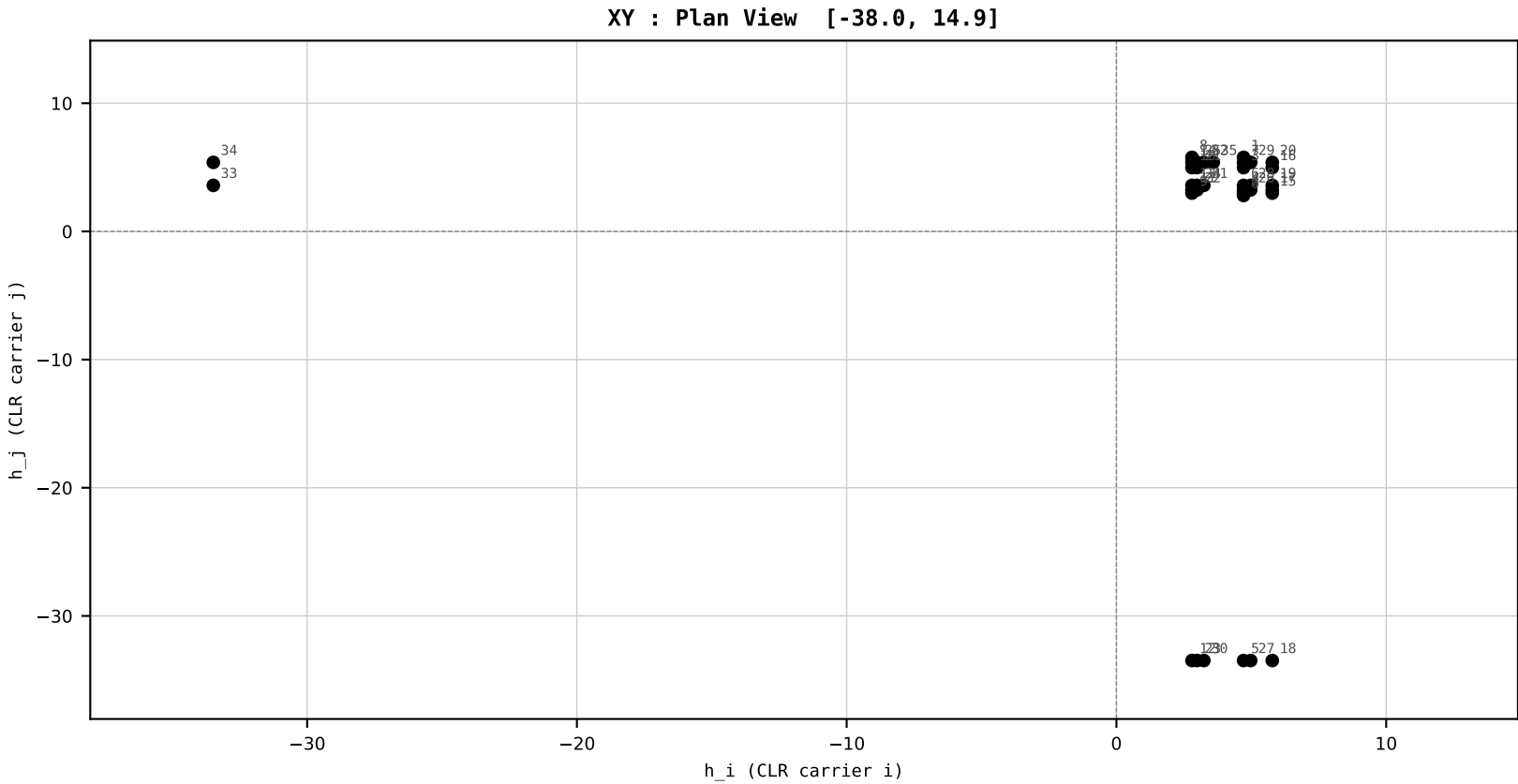
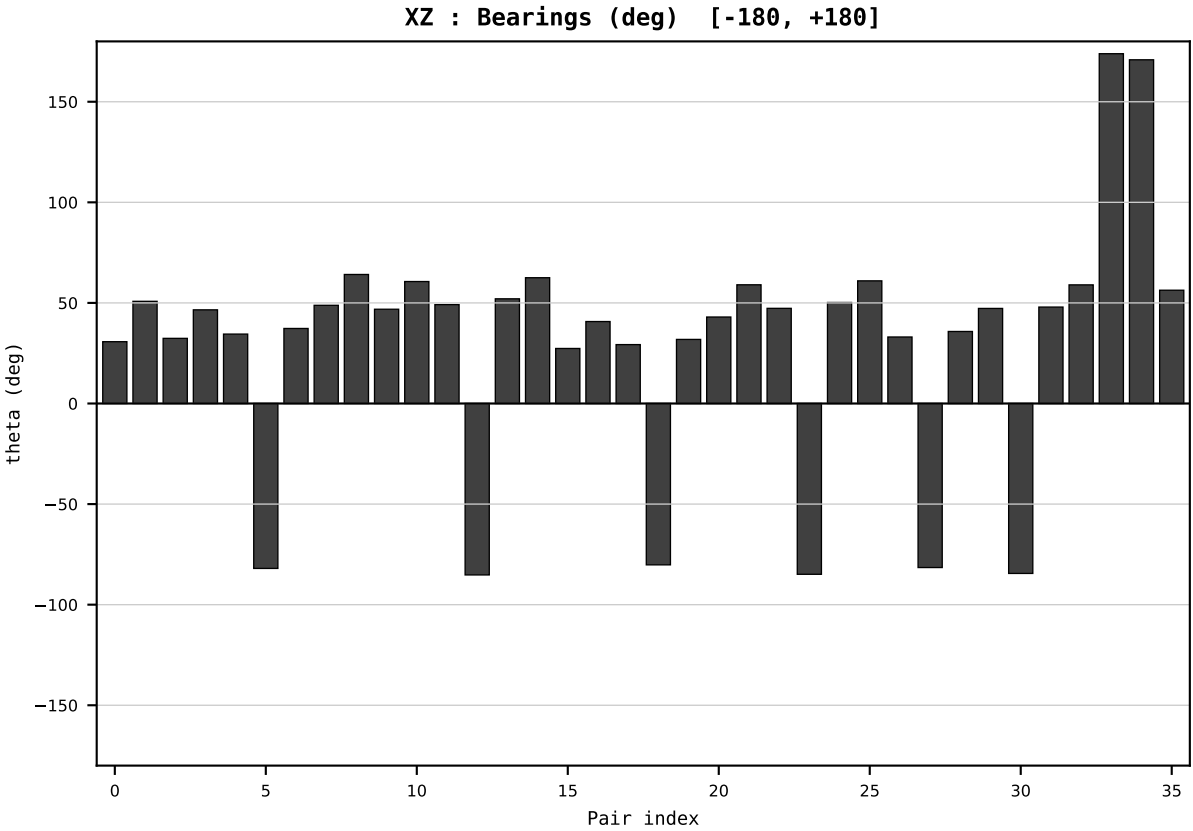
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2020
t : 20 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.246847
Ring = Hs-2
E_metric= 915.4463
kappa_HS= 11188999999999984.00
omega = 17.7152 deg
d_A = 28.181681
Helm = Other Renewables
Helm d = -26.567038
DR = 39.256 HLR
DR ratio= 111889999918284832.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=21 | Year 2021 | D=9 N=26 pairs=36

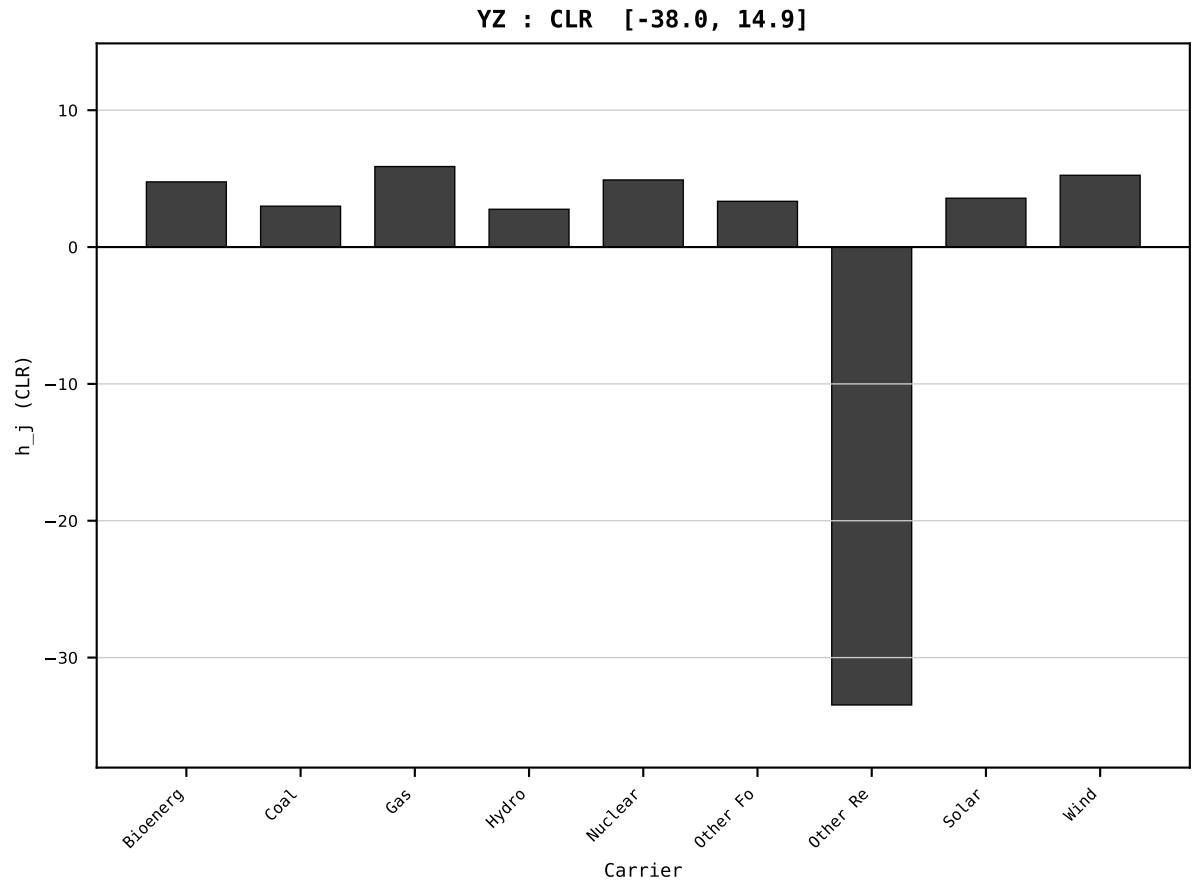
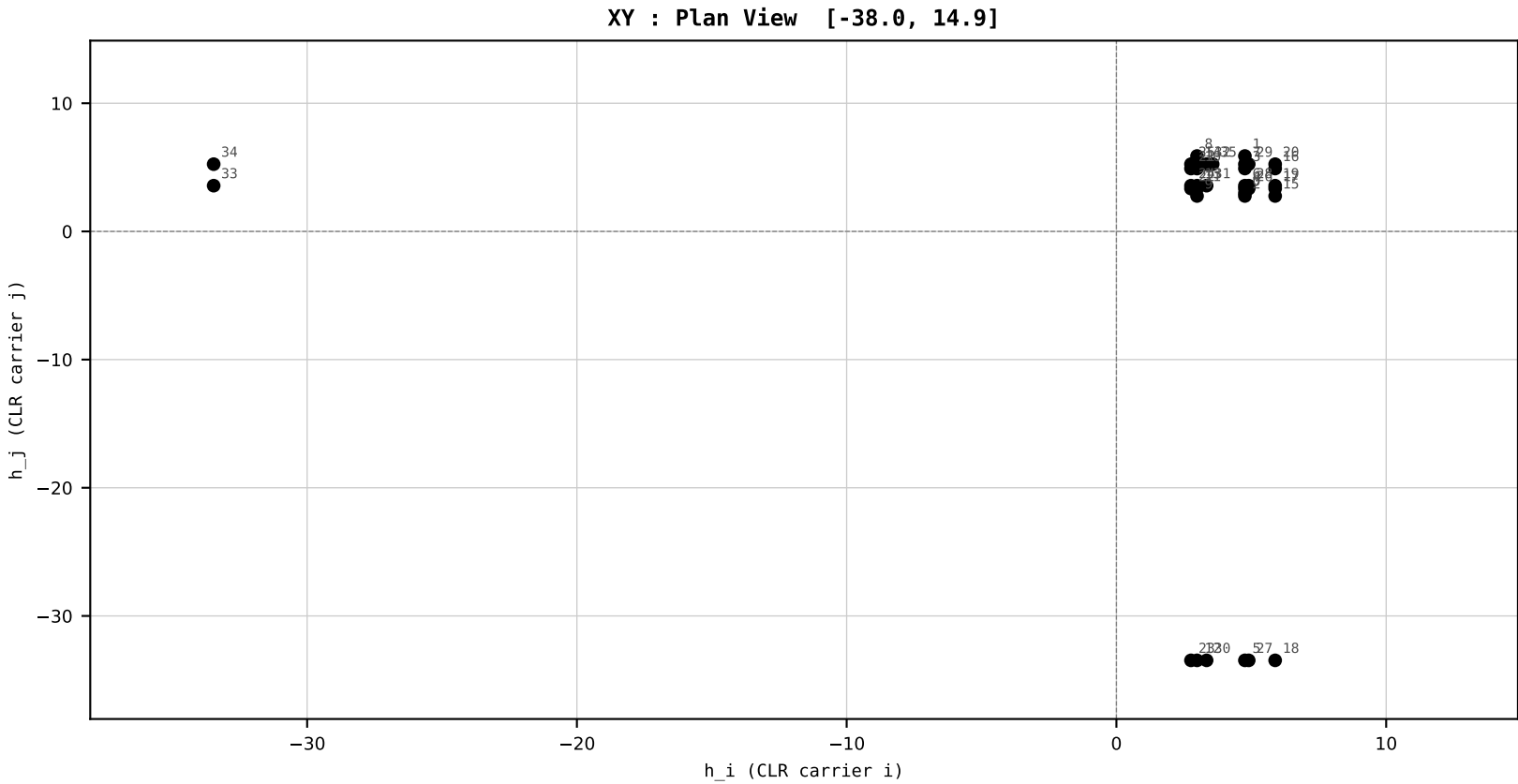
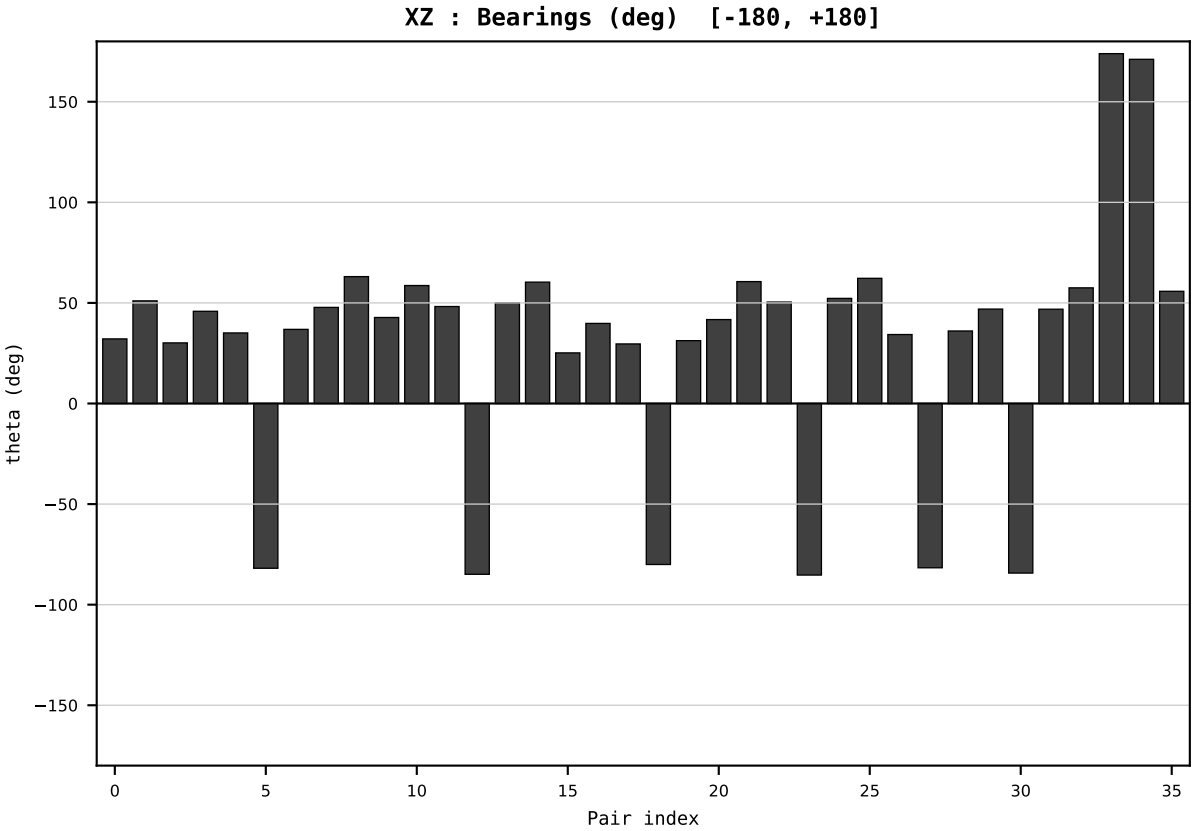
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2021
t : 21 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.255549
Ring = Hs-2
E_metric= 914.9922
kappa_HS= 12284000000000000.00
omega = 0.5927 deg
d_A = 0.368914
Helm = Hydro
Helm d = -0.225695
DR = 39.350 HLR
DR ratio= 122839999940630944.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=22 | Year 2022 | D=9 N=26 pairs=36

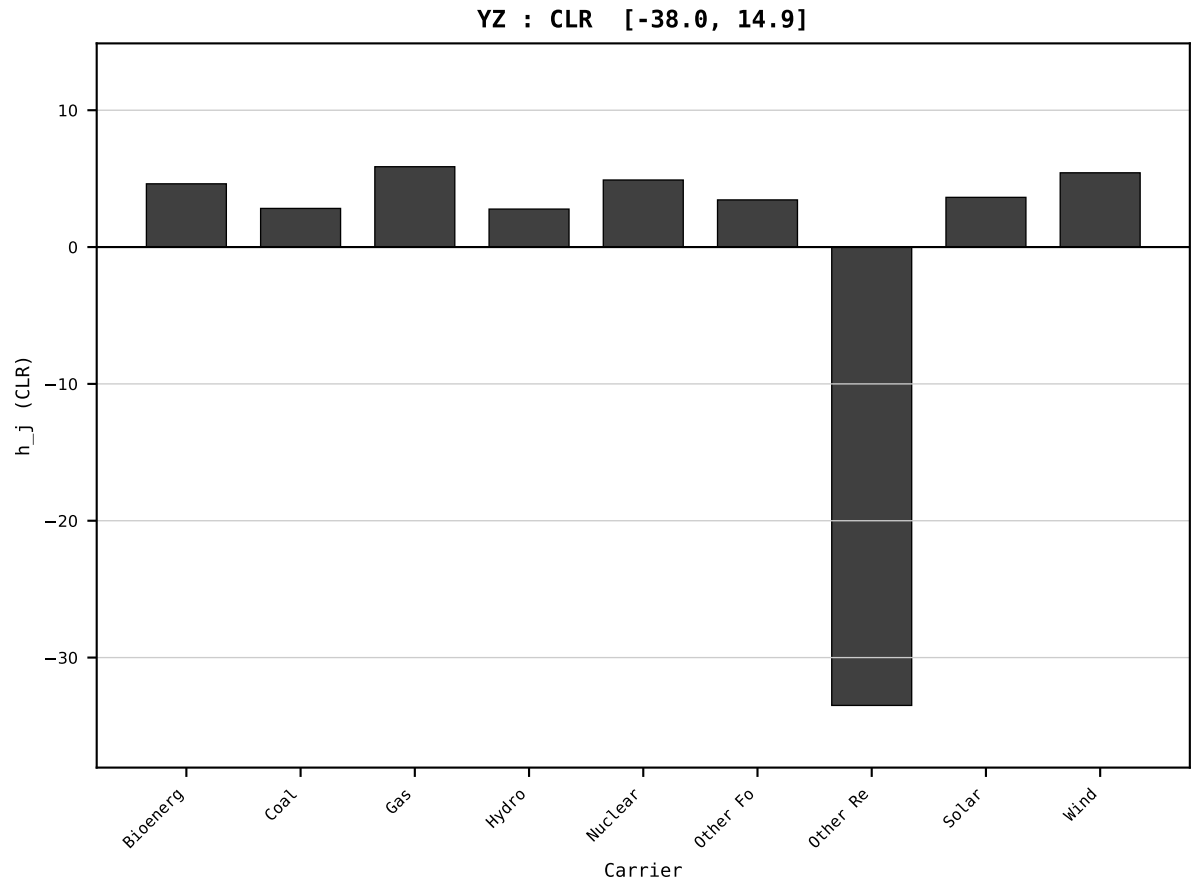
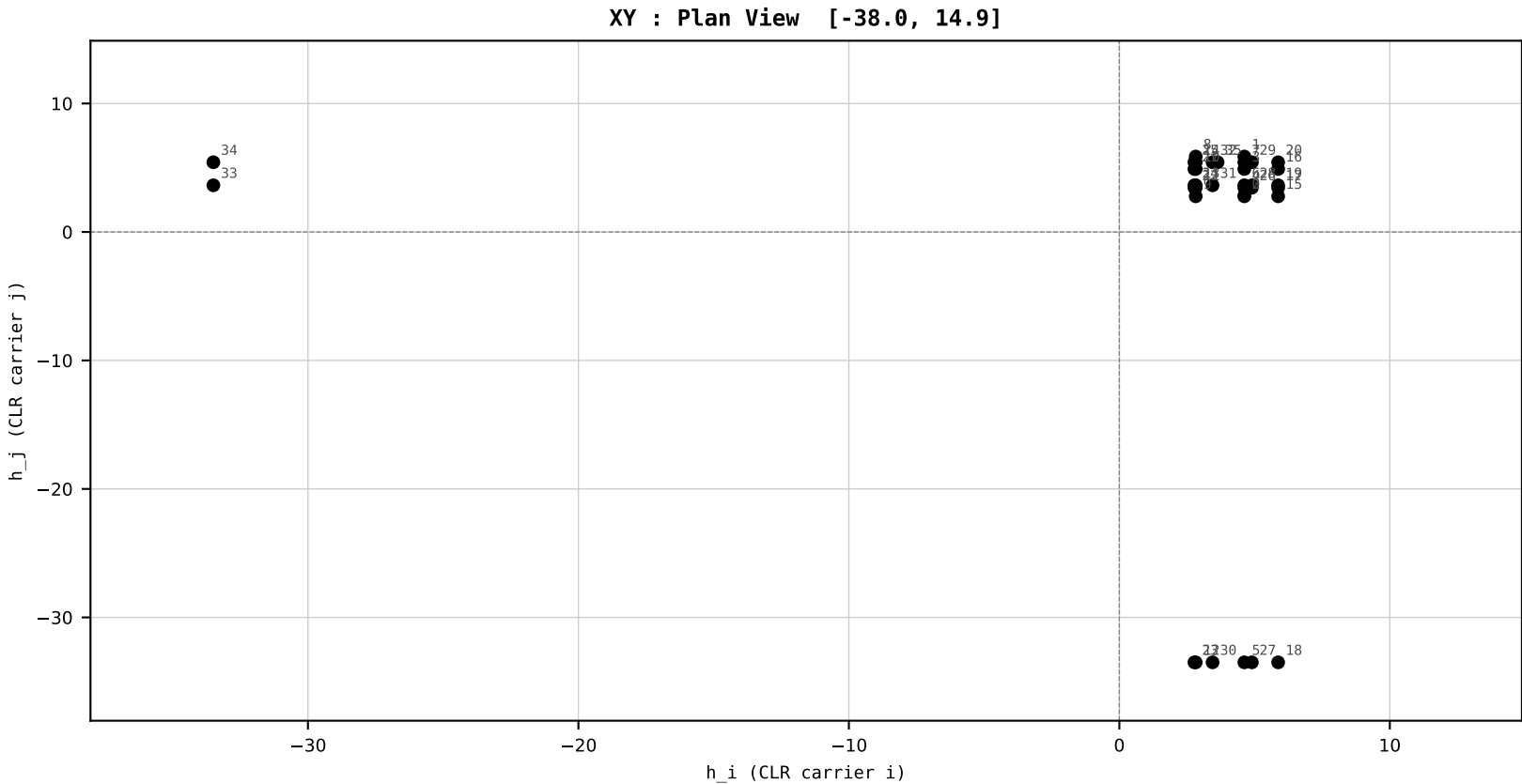
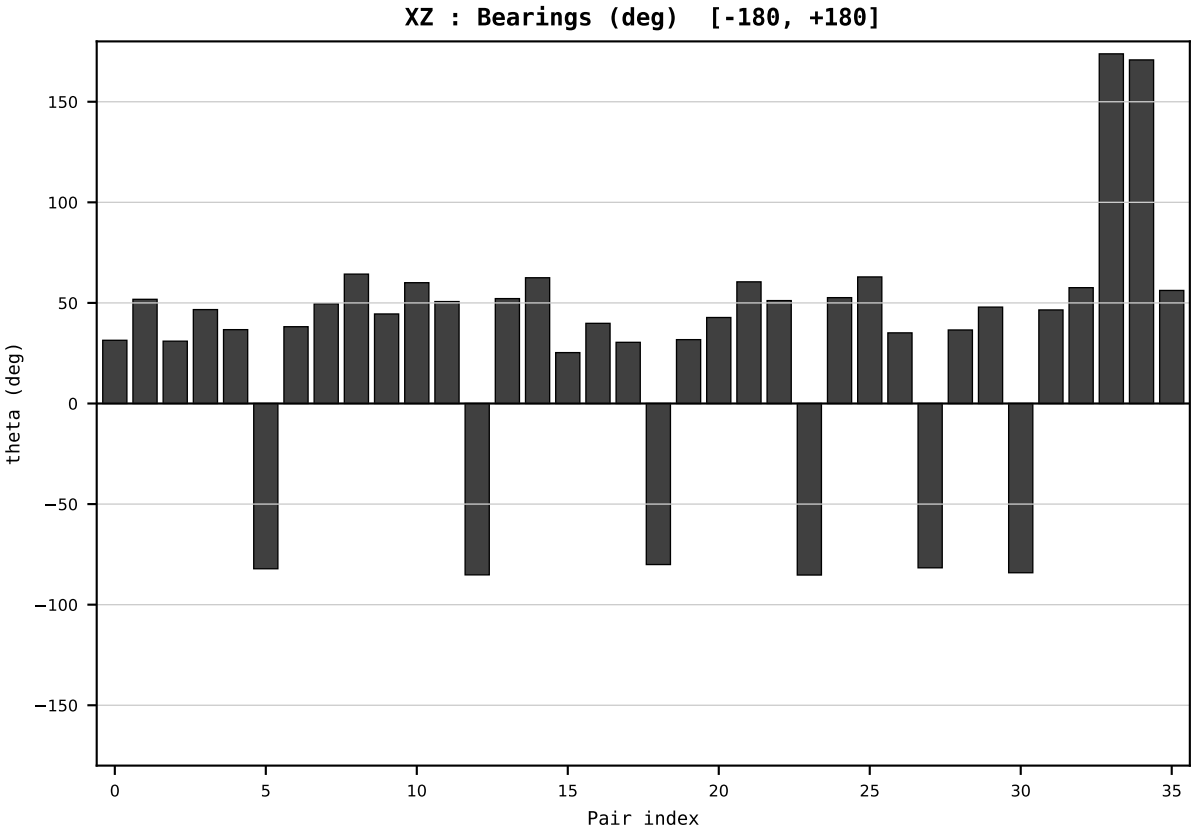
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2022
t : 22 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.259322
Ring = Hs-2
E_metric= 914.4047
kappa_HS= 125109999999999984.00
omega = 0.4928 deg
d_A = 0.309051
Helm = Wind
Helm d = +0.179503
DR = 39.368 HLR
DR ratio= 125110000595524400.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



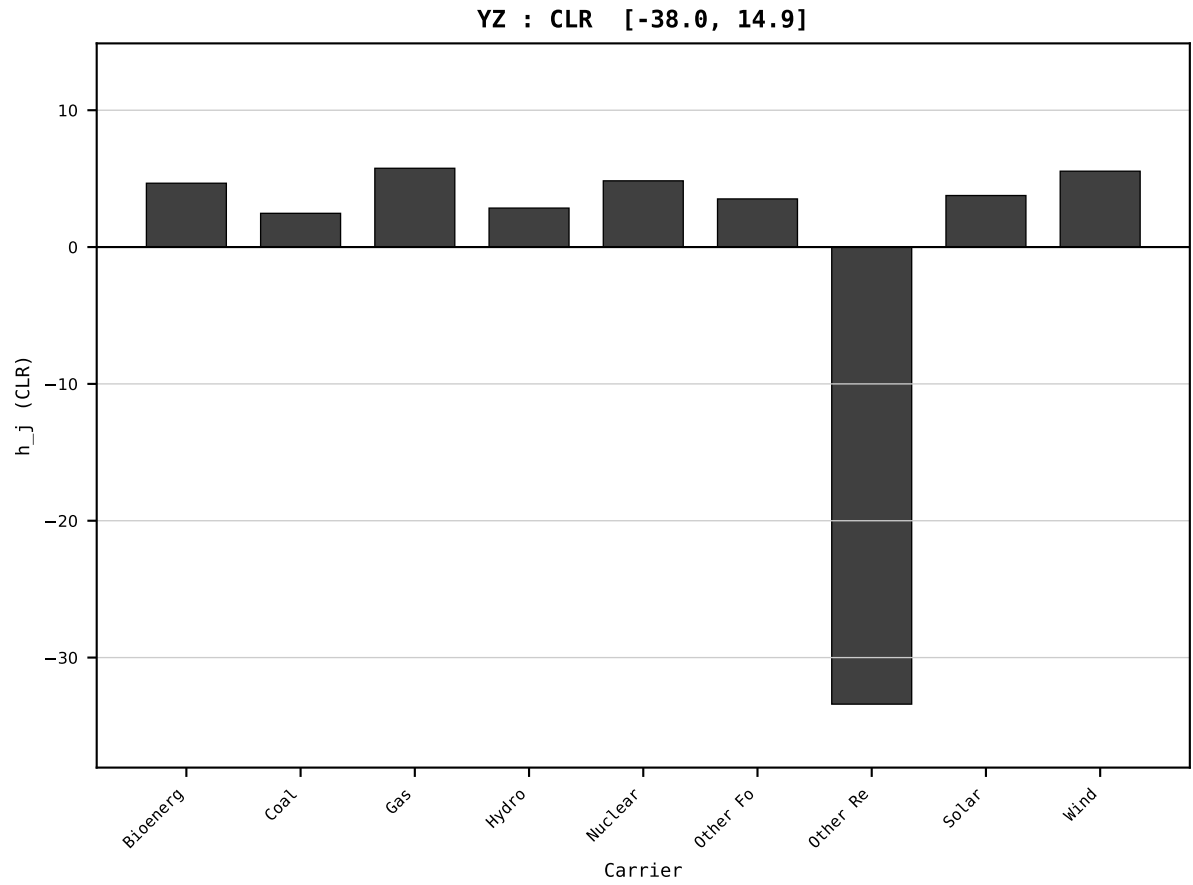
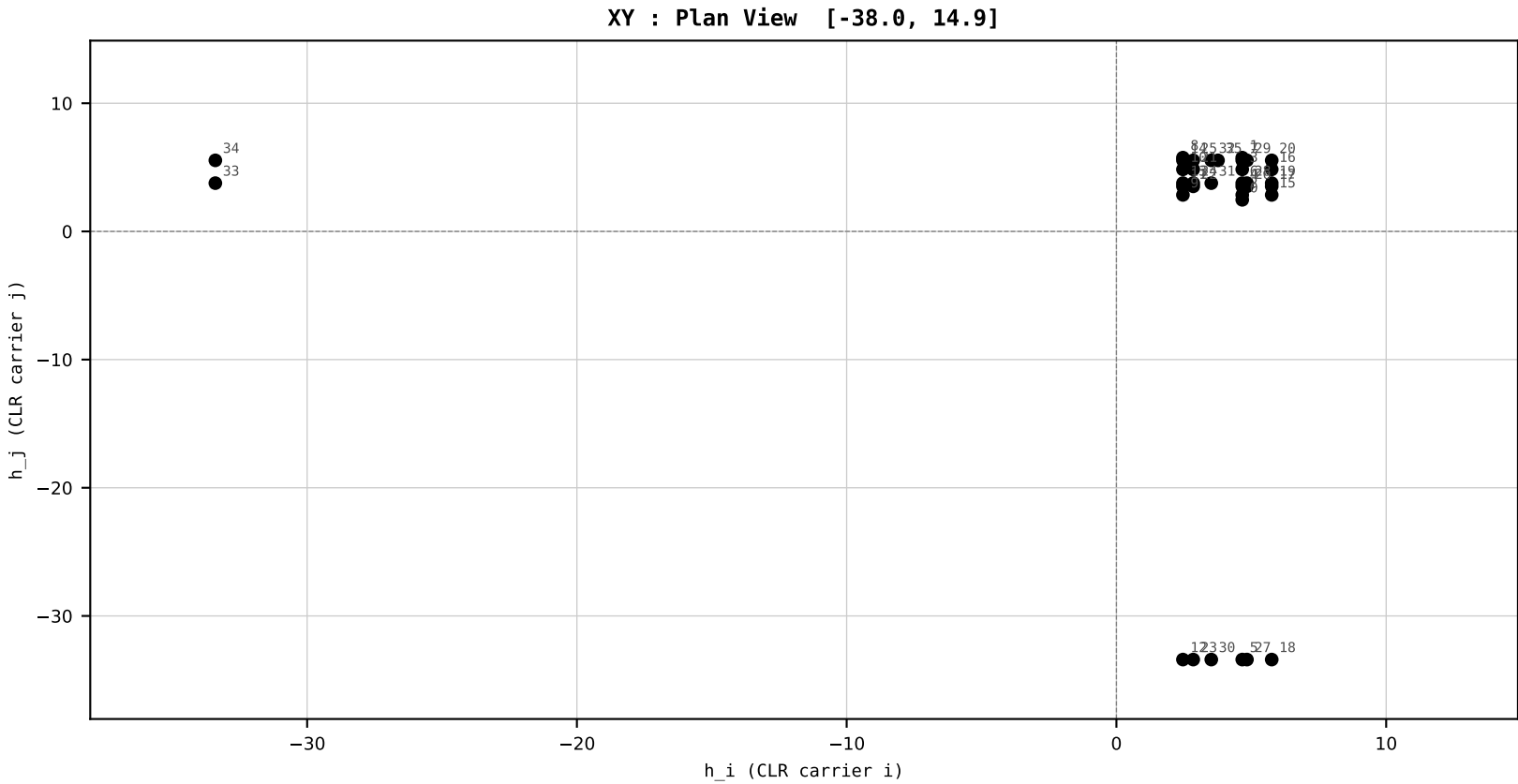
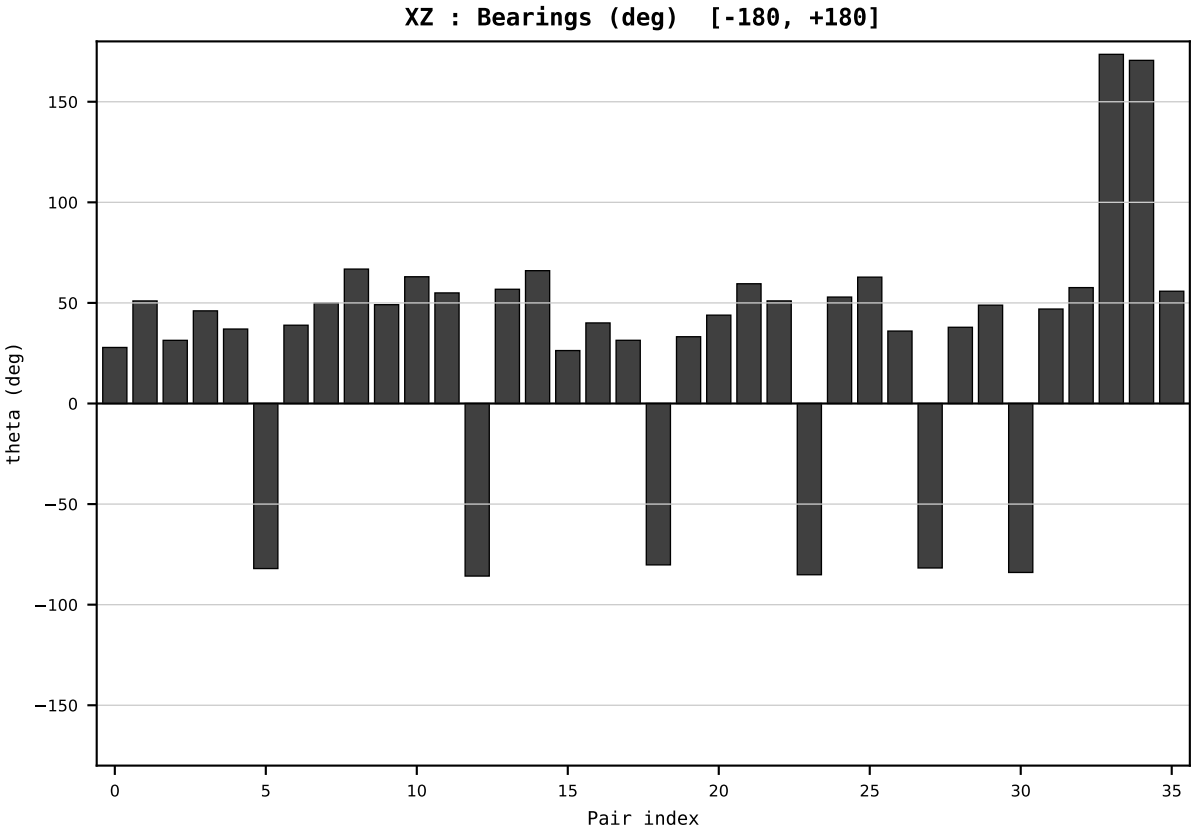
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2023
t : 23 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.250656
Ring = Hs-2
E_metric= 914.8761
kappa_HS= 10165999999999984.00
omega = 0.6998 deg
d_A = 0.444546
Helm = Coal
Helm d = -0.358081
DR = 39.160 HLR
DR ratio= 101660000312153280.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=24 | Year 2024 | D=9 N=26 pairs=36

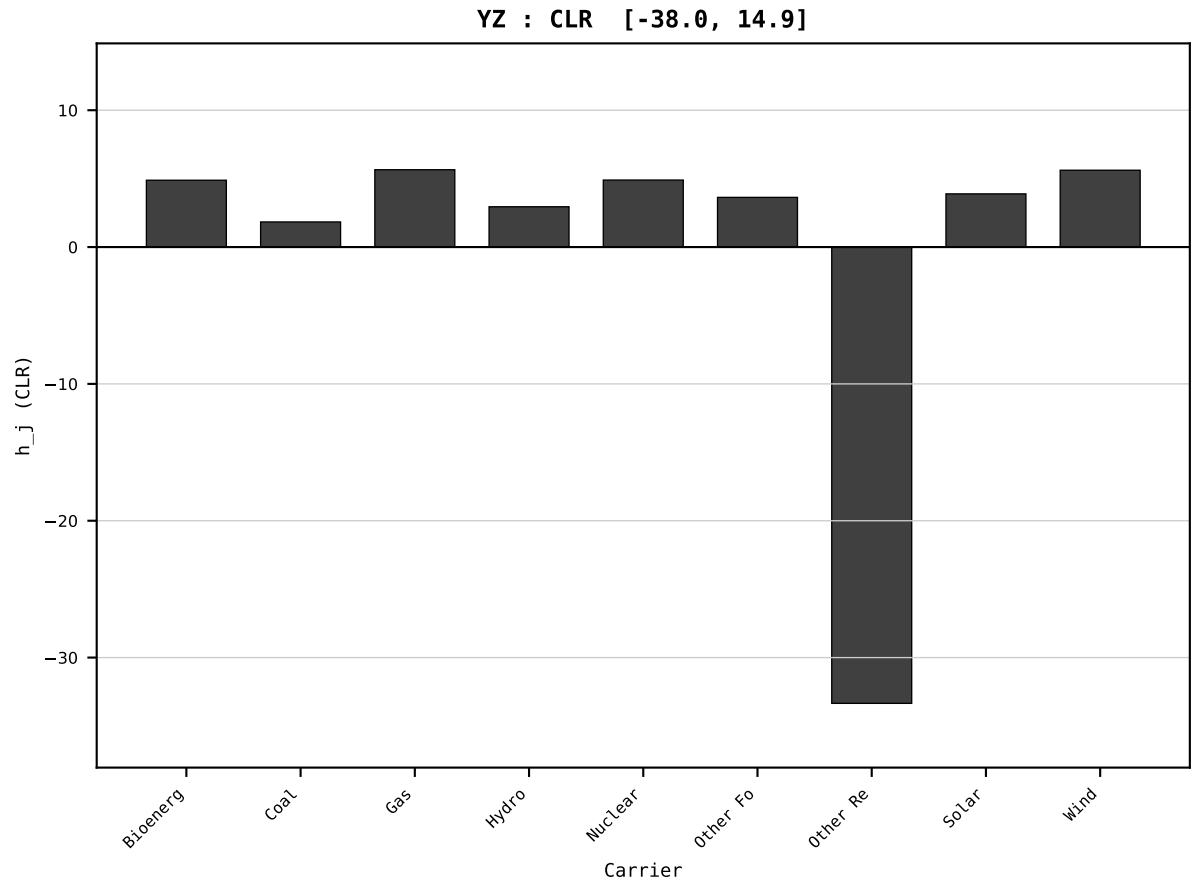
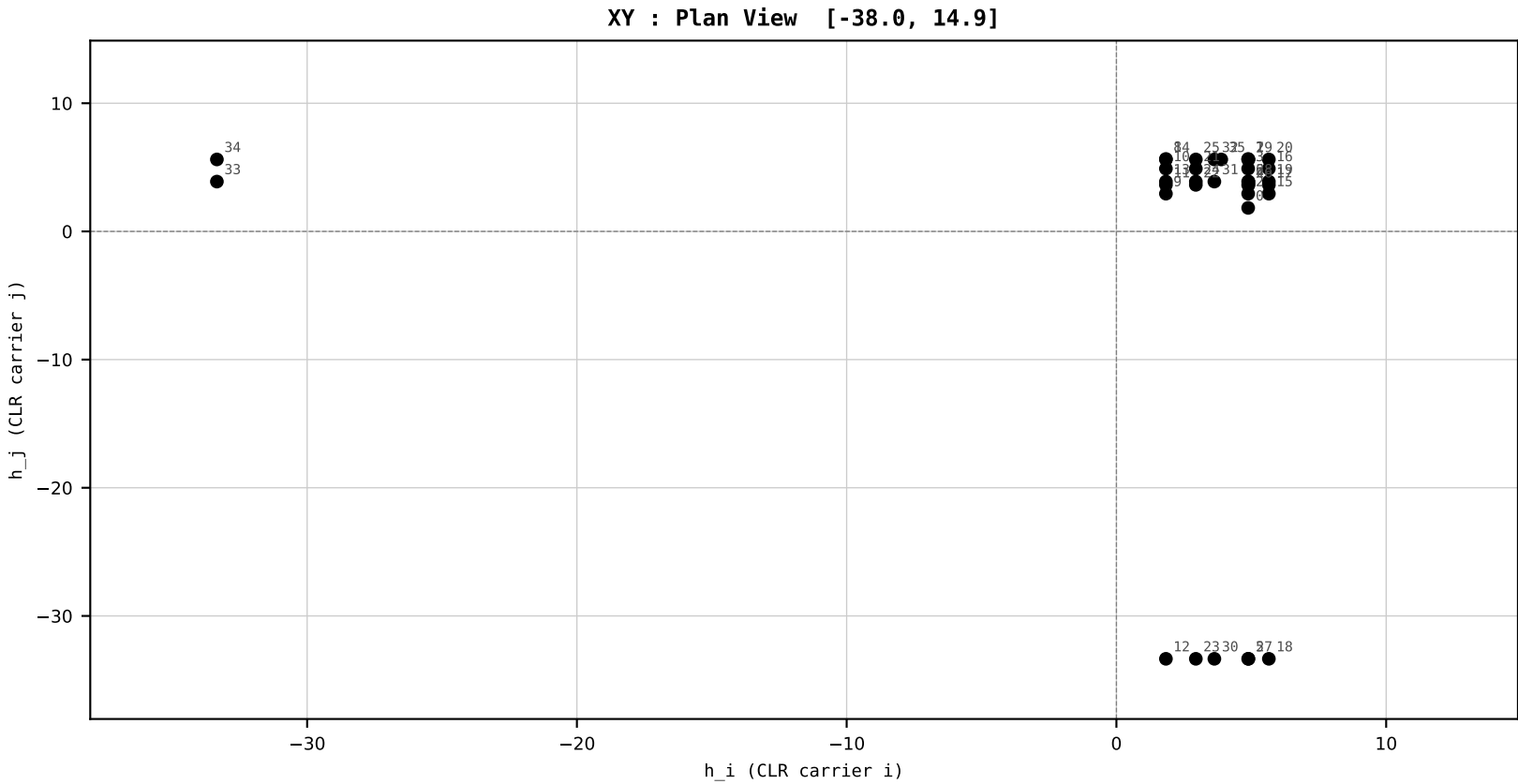
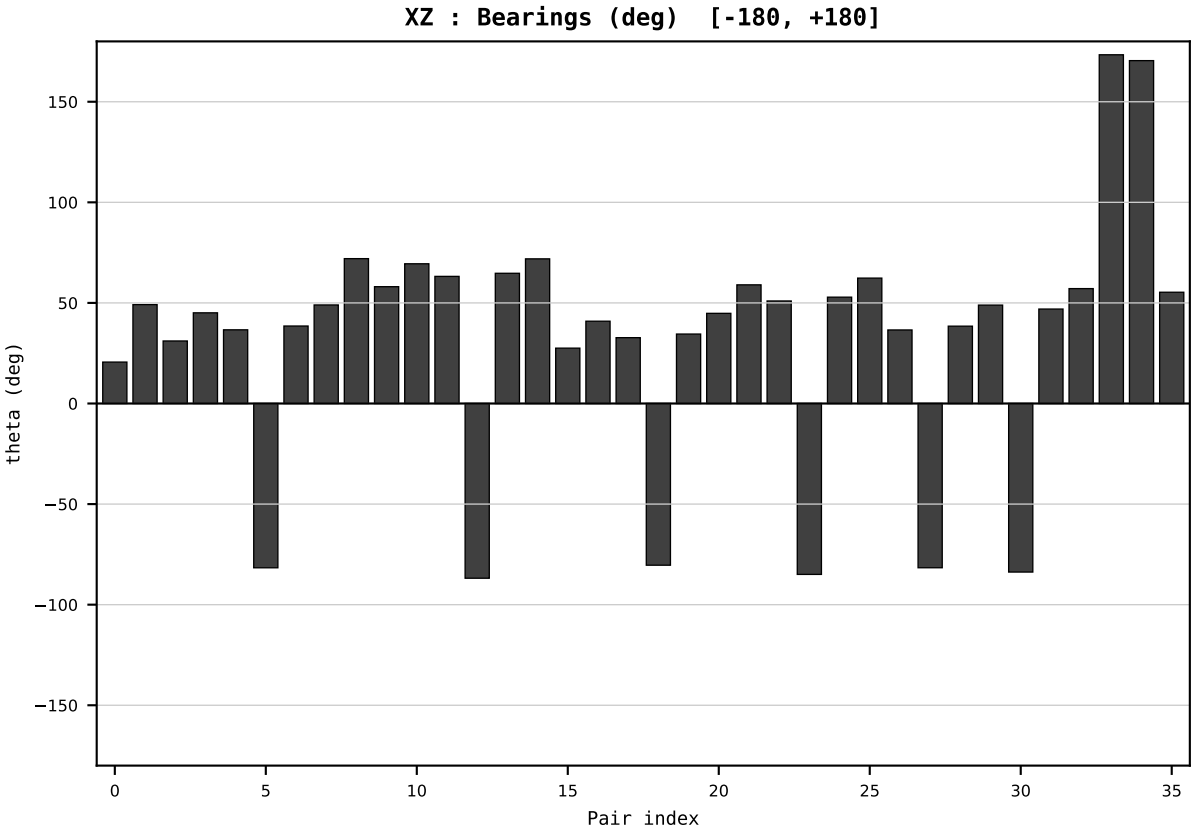
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2024
t : 24 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.239203
Ring = Hs-2
E_metric= 915.1572
kappa_HS= 8630000000000000.00
omega = 1.1445 deg
d_A = 0.710901
Helm = Coal
Helm d = -0.630382
DR = 38.997 HLR
DR ratio= 86300000604094160.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=25 | Year 2025 | D=9 N=26 pairs=36

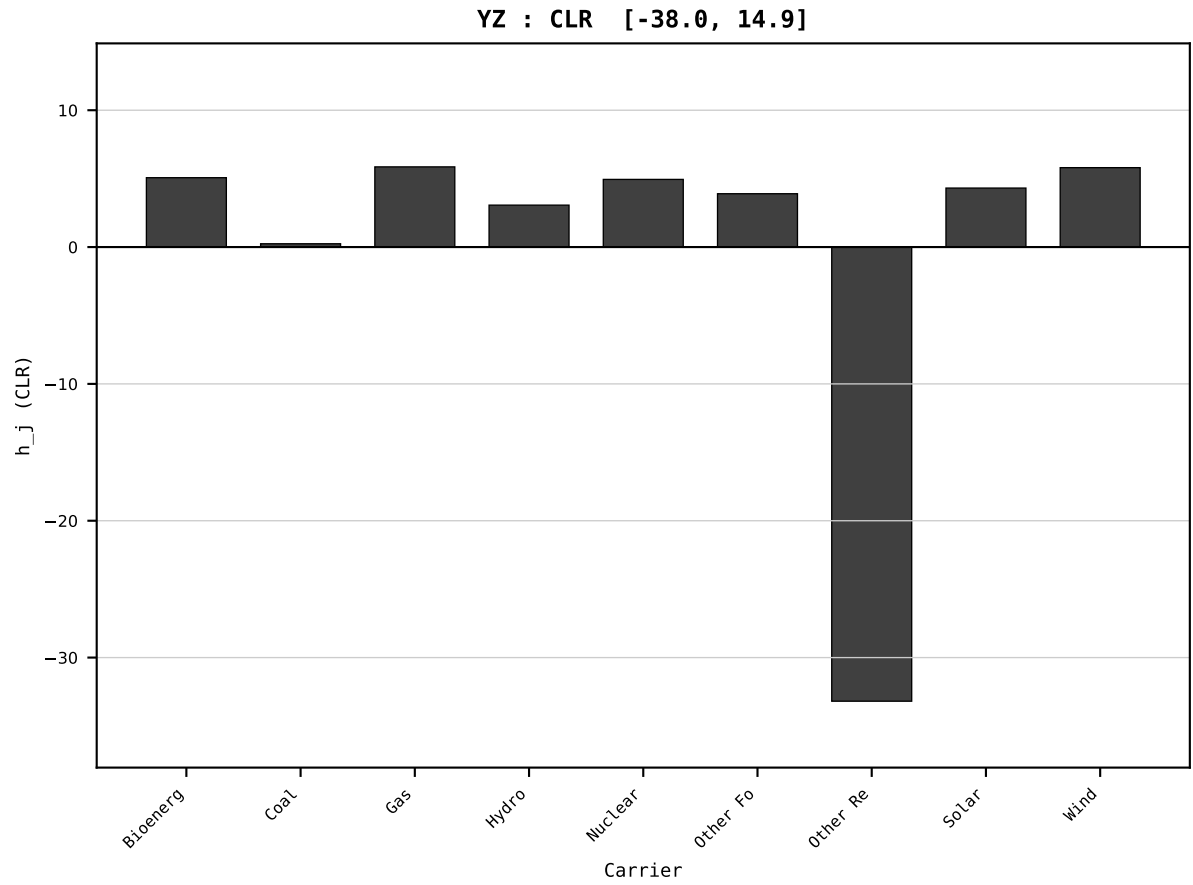
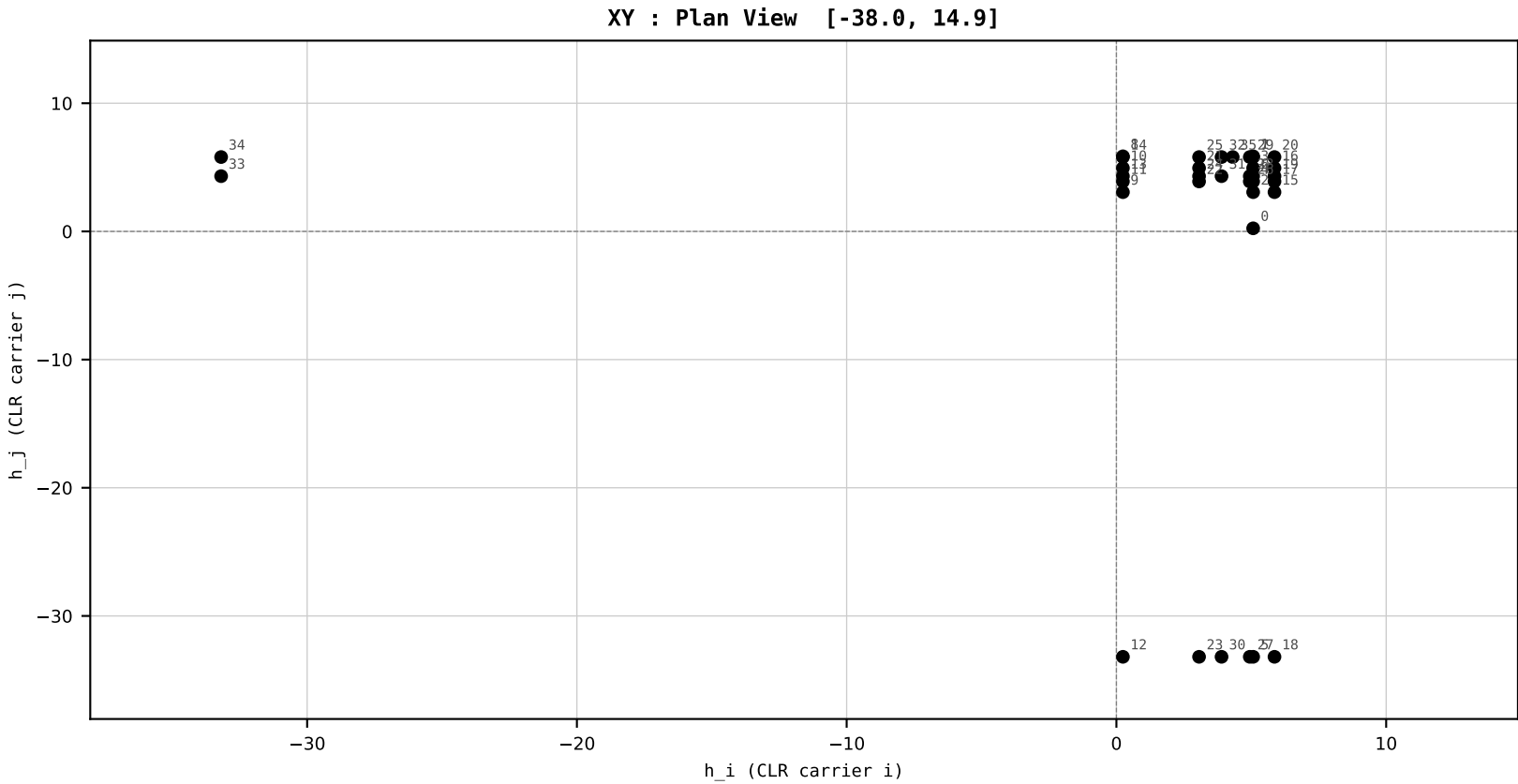
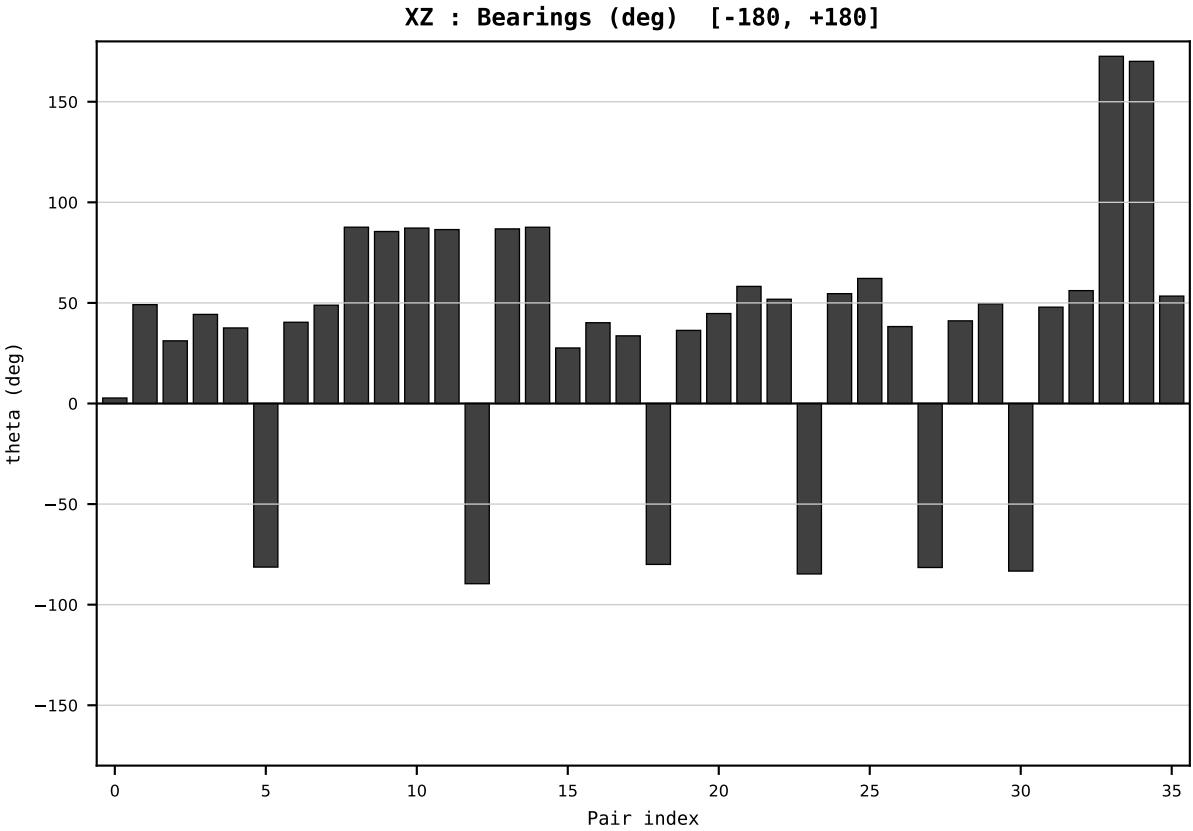
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

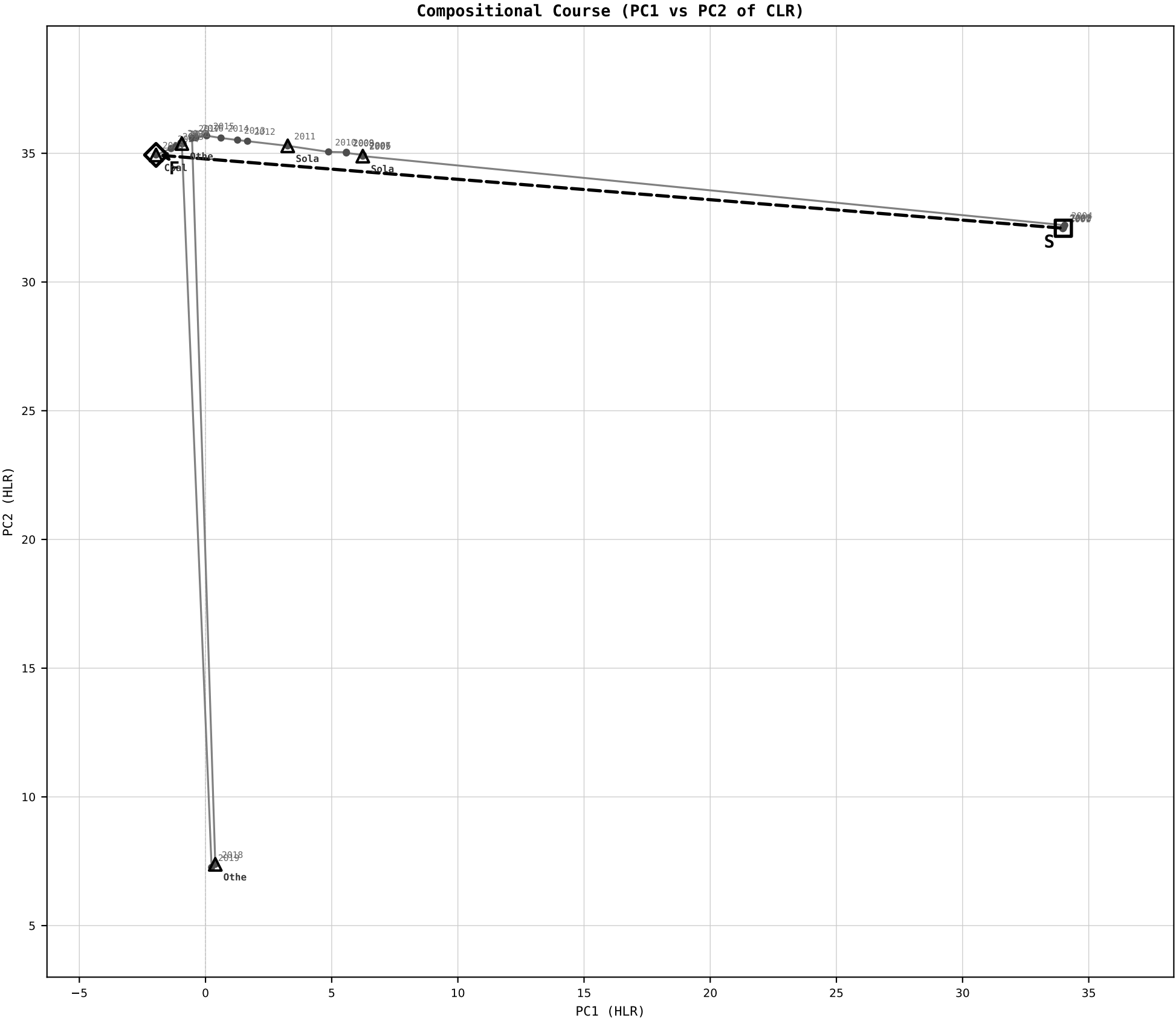
Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2025
t : 25 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.245474
Ring = Hs-2
E_metric= 914.6317
kappa_HS= 9090999999999984.00
omega = 2.7663 deg
d_A = 1.715852
Helm = Coal
Helm d = -1.593740
DR = 39.049 HLR
DR ratio= 909099999905049104.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind





COURSE METRICS

Scale Provenance:

$Y_j \rightarrow x_j = Y_j / T \rightarrow h_j = \text{CLR}$

$\rightarrow$ PC projection

Unit: HLR (Higgins Log-Ratio)

Net distance = 36.2591 HLR

Path length = 100.3591 HLR

Directness = 0.3613

DR start = 39.537 HLR

ratio = 148079999529708608.0x

DR final = 39.049 HLR

ratio = 90909999905049104.0x

E_metric start = 2195.0806

E_metric final = 1262.8551

PC1 loadings:

Solar -0.931

Coal +0.202

Nuclear +0.151

PC2 loadings:

Other Re -0.941

Coal +0.160

Other Fo +0.123

Top displacement events:

2018 Other Re d=28.236

2020 Other Re d=28.182

2005 Solar d=28.133

2011 Solar d=1.727

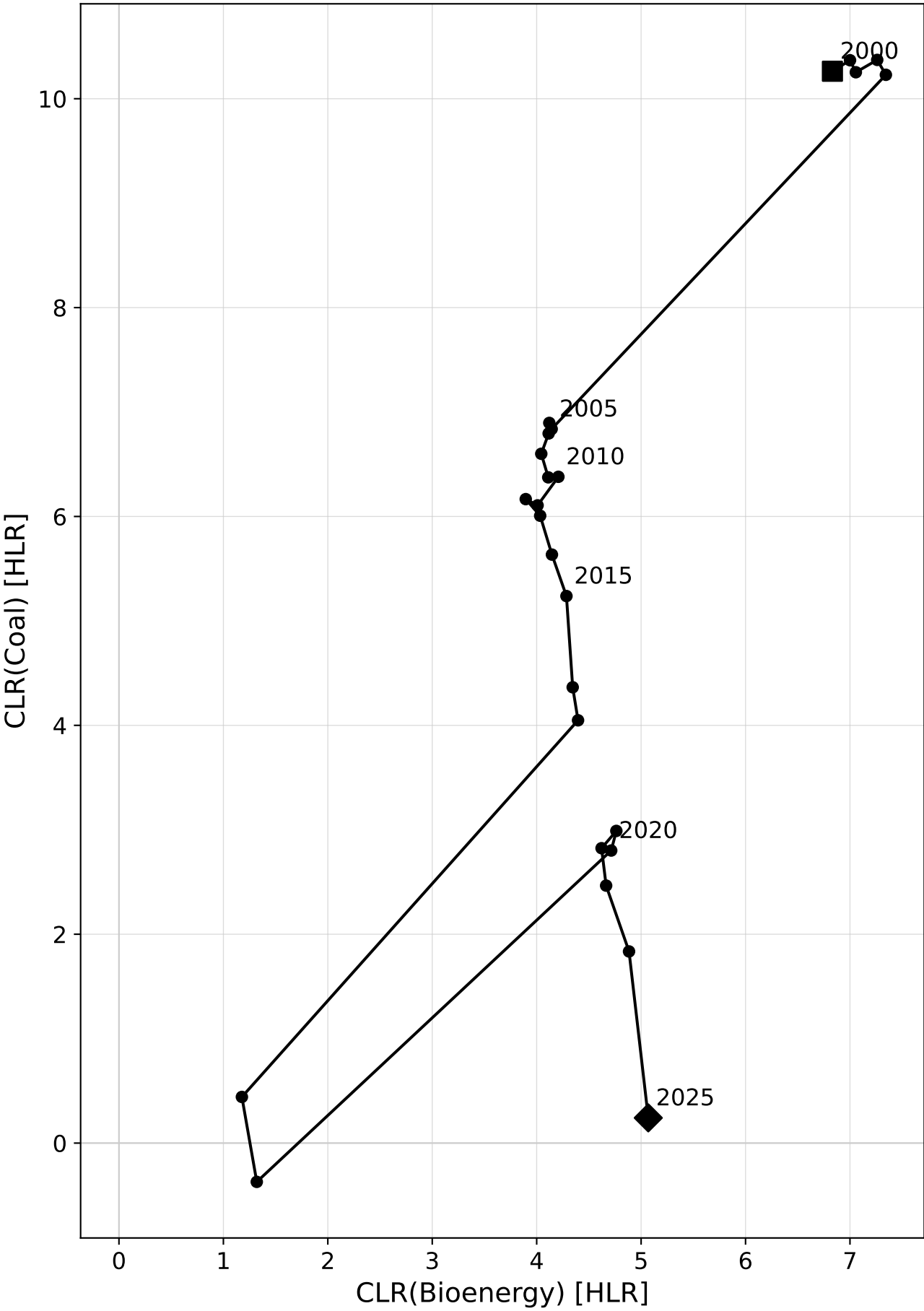
2025 Coal d=1.716

The instrument reads.

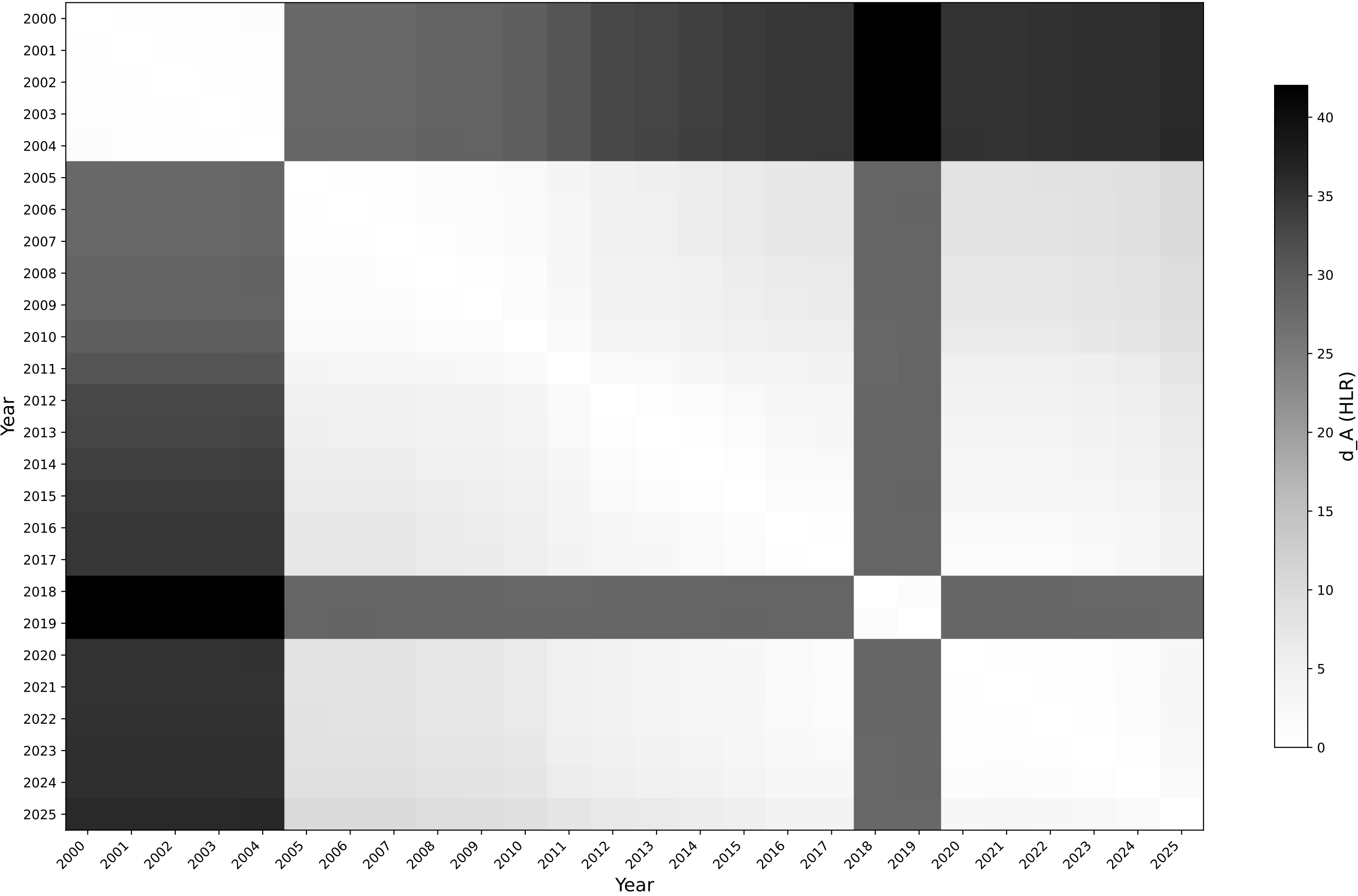
The expert decides.

The loop stays open.

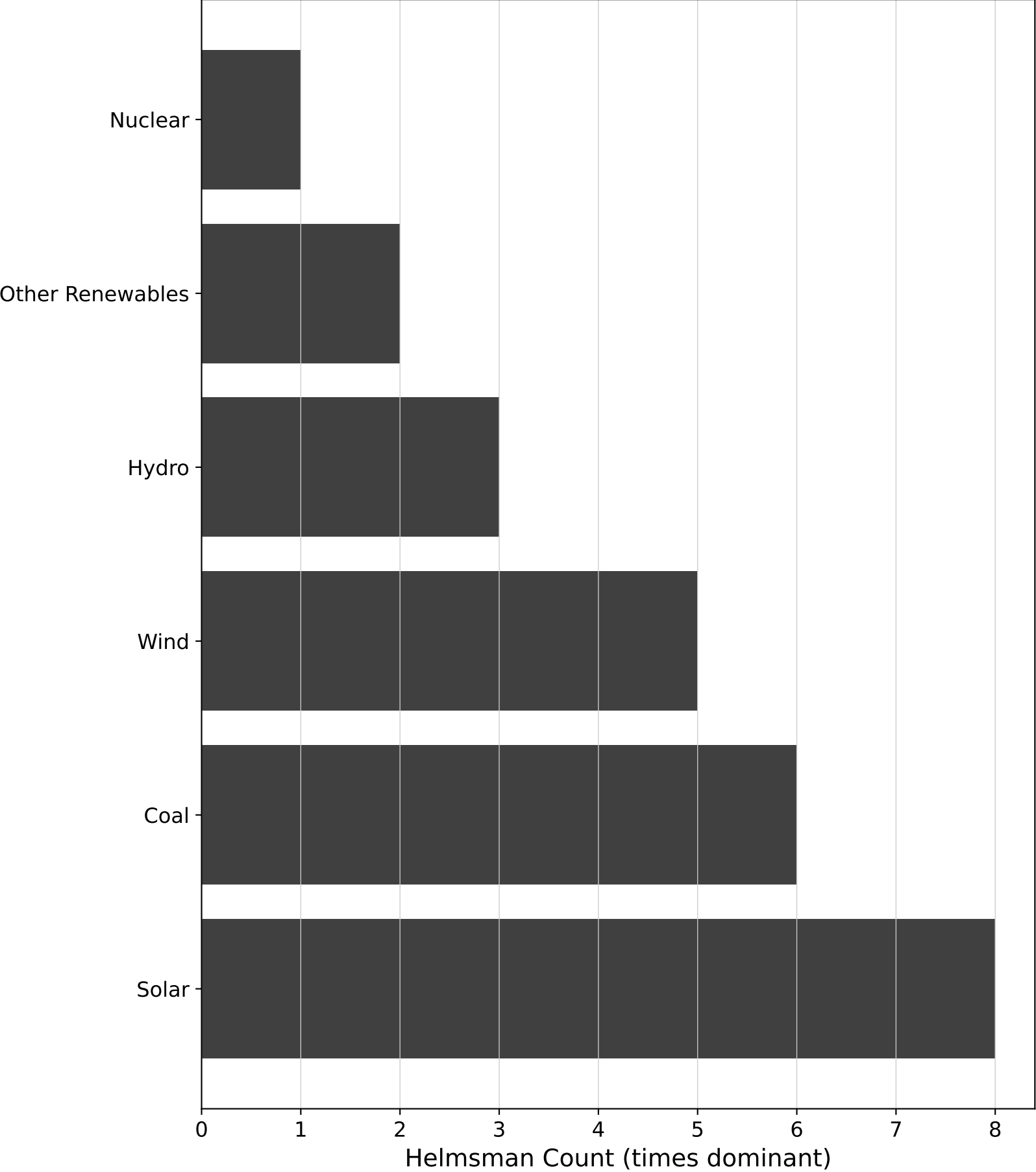
SYSTEM COURSE PLOT — XY Plan View (CLR Space)



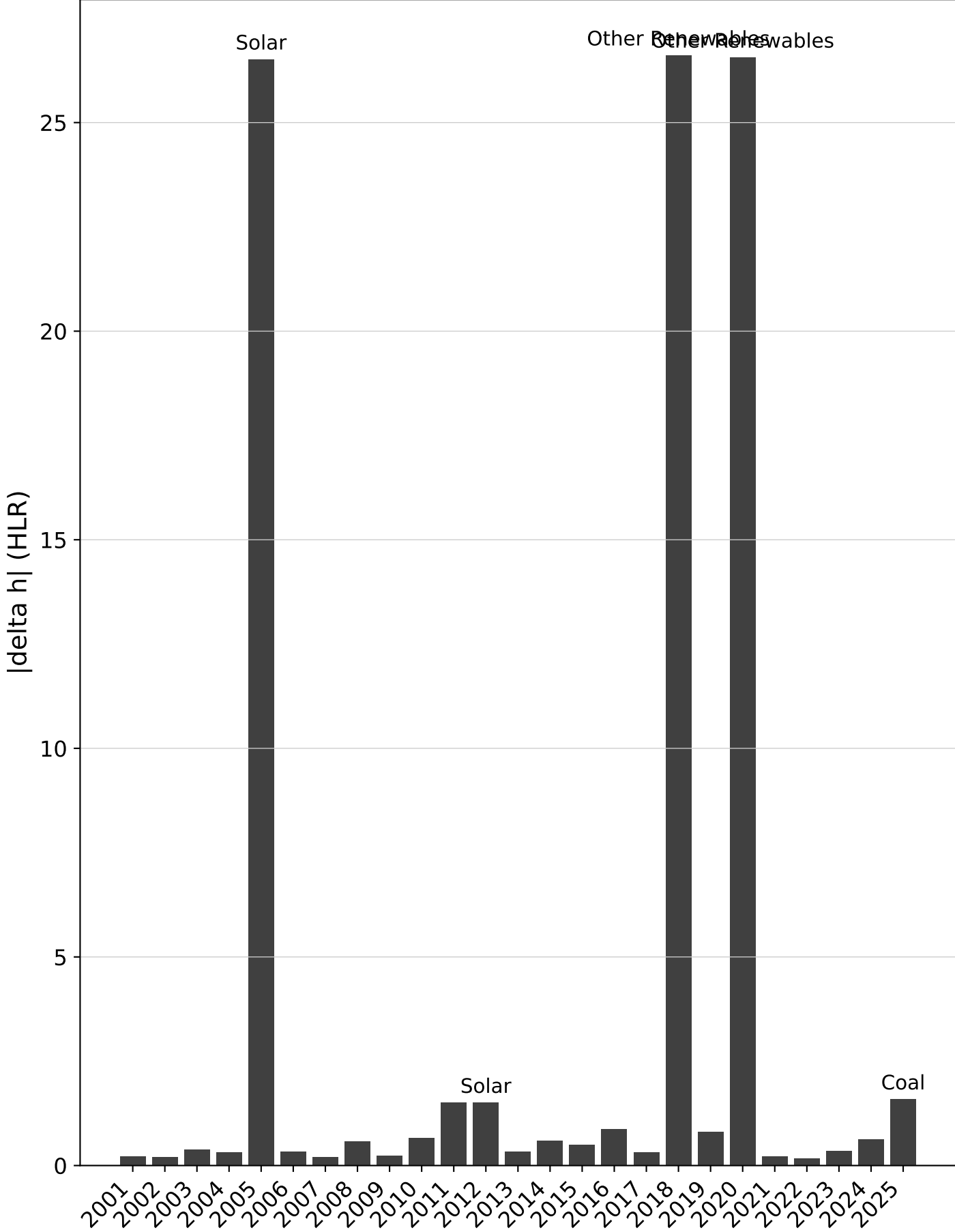
STAGE 2 — Aitchison Distance Matrix (all year pairs)



HELMSMAN FREQUENCY



HELMSMAN DISPLACEMENT PER YEAR



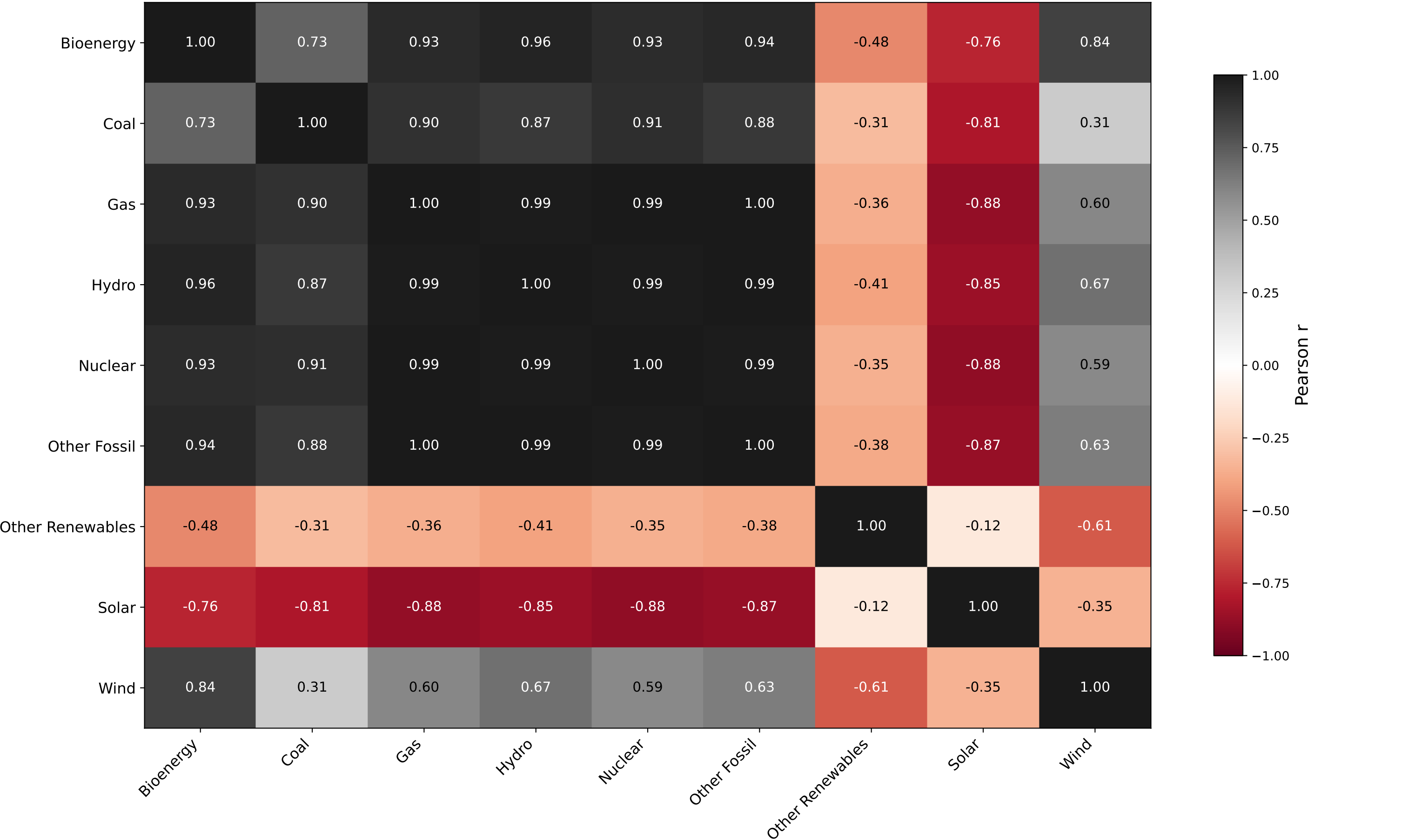
STAGE 2 — Pairwise Divergence Ranking (Top 15)

Rank	Year A	Year B	d_A (HLR)	Angle (deg)	Dominant Carrier
1	2004	2019	42.0223	47.2	Solar
2	2002	2019	41.9369	47.4	Solar
3	2003	2019	41.9196	47.3	Solar
4	2000	2019	41.9006	47.5	Solar
5	2001	2019	41.8750	47.4	Solar
6	2004	2018	41.8489	45.7	Solar
7	2002	2018	41.7611	45.8	Solar
8	2003	2018	41.7407	45.7	Solar
9	2000	2018	41.7240	46.0	Solar
10	2001	2018	41.6954	45.9	Solar
11	2004	2025	36.2846	49.8	Solar
12	2002	2025	36.2613	49.9	Solar
13	2000	2025	36.2591	49.9	Solar
14	2003	2025	36.2581	49.9	Solar
15	2001	2025	36.2497	49.9	Solar

STAGE 3 — Triadic Metric Triangle Area Ranking (Top 10)

Rank	Year A	Year B	Year C	Area (HLR^2)	Imbalance
1	2004	2019	2025	501.5179	1.50
2	2004	2018	2025	501.2999	1.49
3	2002	2019	2025	500.9550	1.50
4	2003	2019	2025	500.8574	1.50
5	2000	2019	2025	500.8000	1.49
6	2002	2018	2025	500.7174	1.49
7	2003	2018	2025	500.6060	1.49
8	2001	2019	2025	500.6035	1.49
9	2000	2018	2025	500.5544	1.49
10	2001	2018	2025	500.3433	1.49

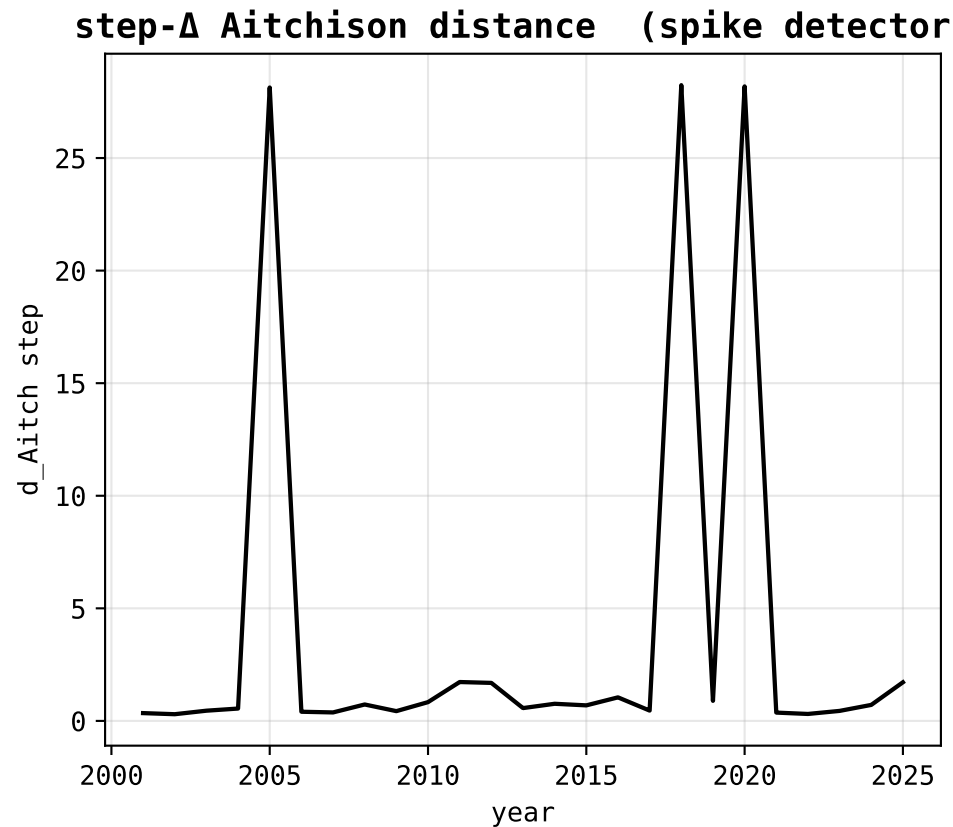
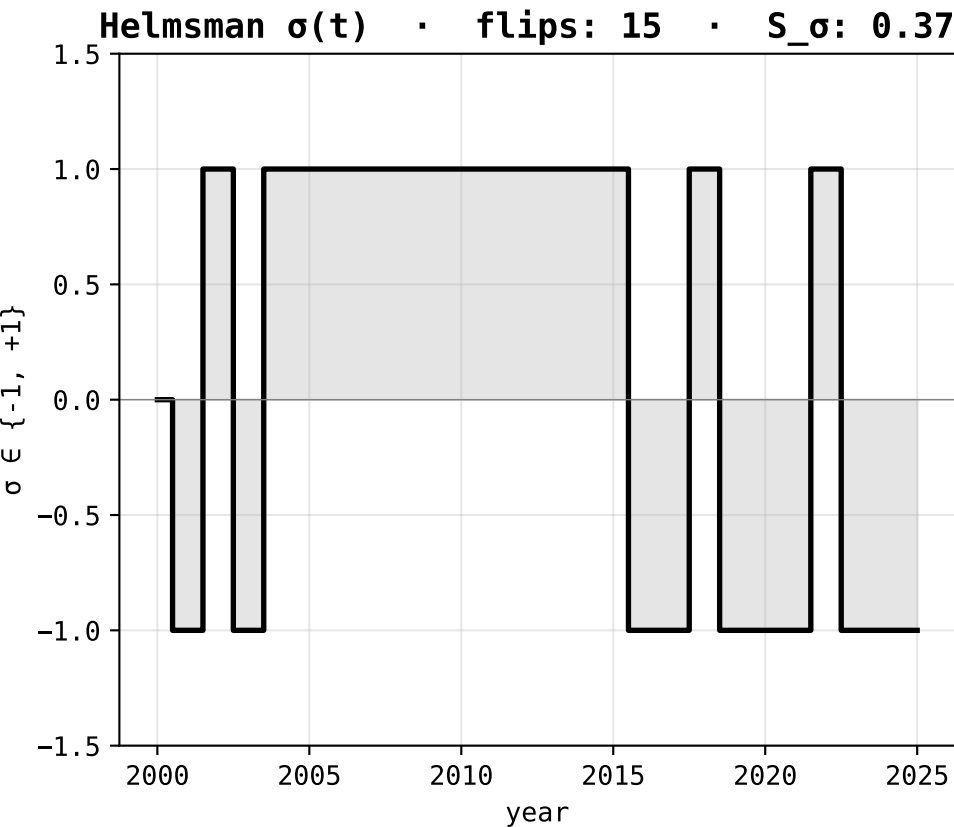
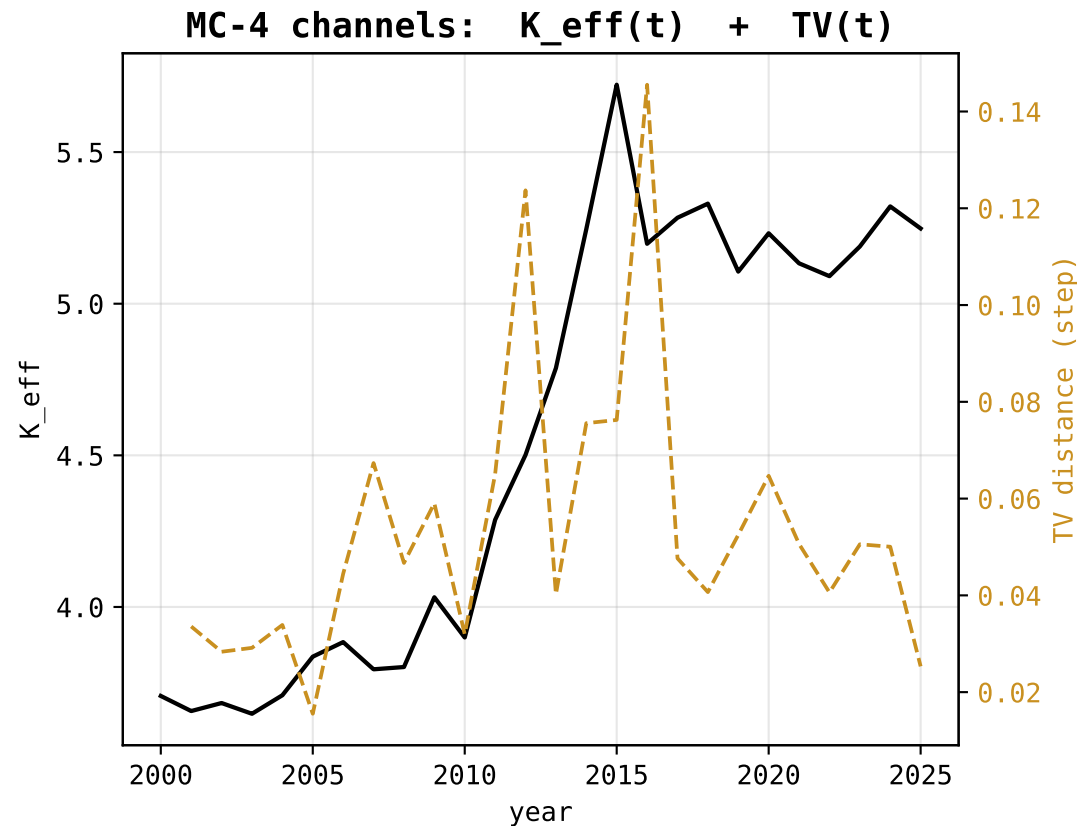
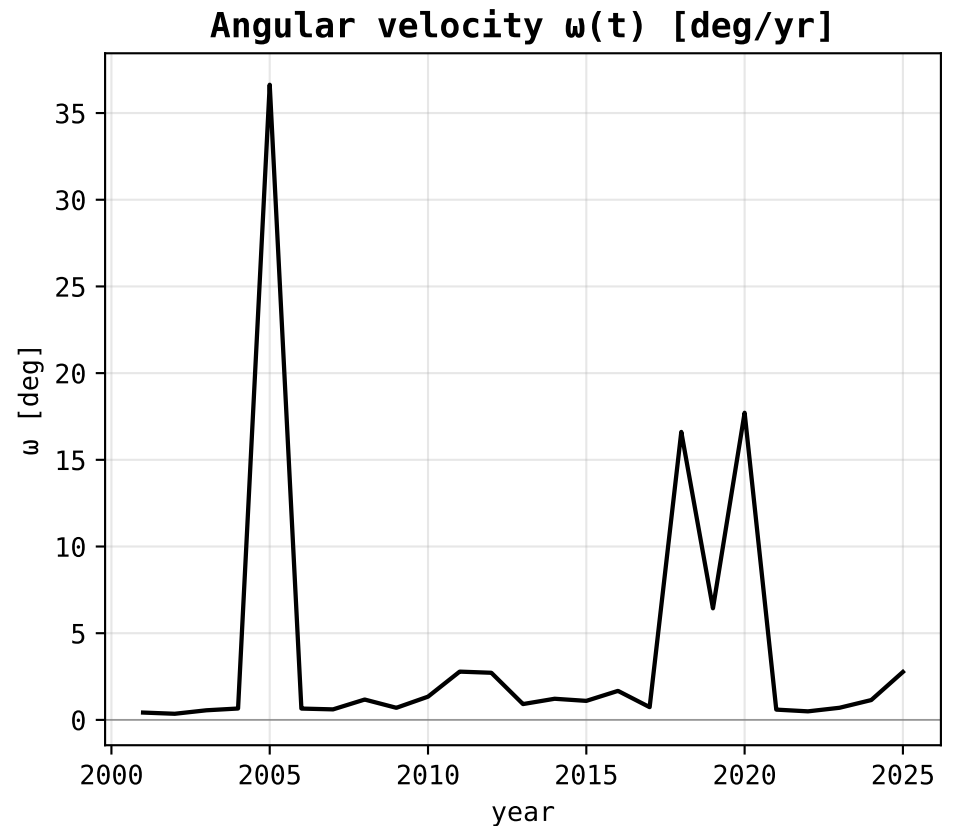
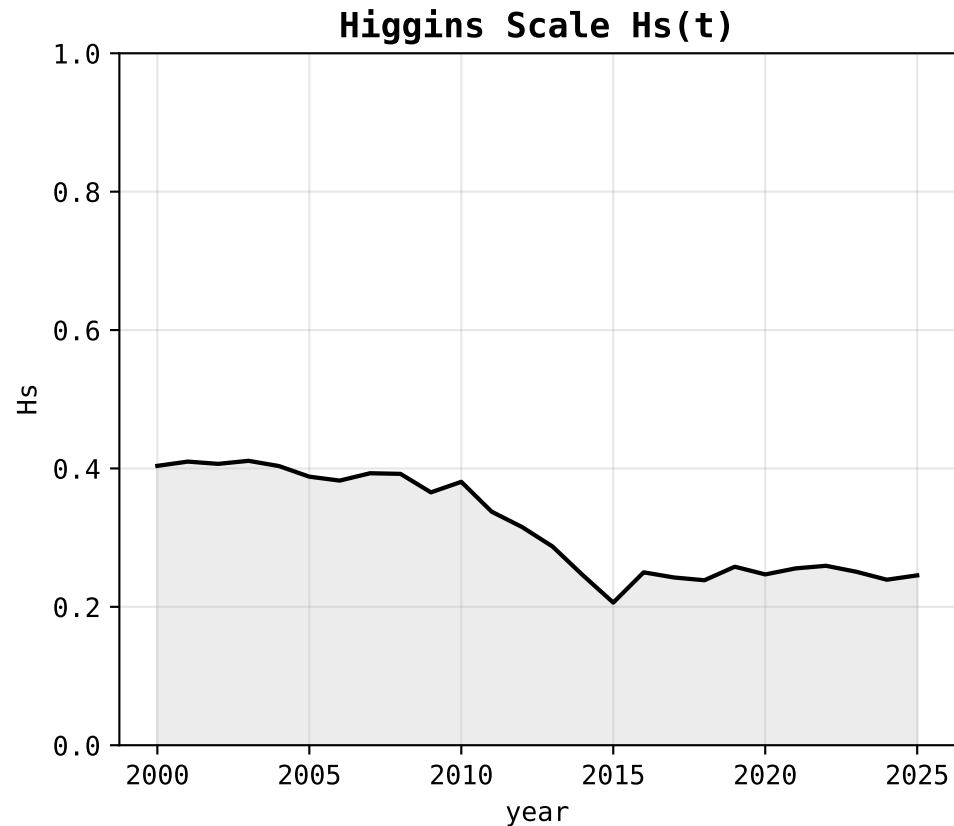
STAGE 3 — Carrier CLR Trajectory Correlation



NAVIGATION SUMMARY — All Stage 1 Metrics by Year

Year	Hs	Ring	E_metric	omega(deg)	d_A	Helmsman	kappa_max
2000	0.4037	Hs-3	1560.2	0.00	0.0000	---	1000000000000000
2001	0.4099	Hs-3	1557.0	0.42	0.3465	Hydro	1000000000000000
2002	0.4066	Hs-3	1561.6	0.35	0.2975	Wind	1000000000000000
2003	0.4110	Hs-3	1558.5	0.55	0.4536	Hydro	1000000000000000
2004	0.4035	Hs-3	1567.4	0.66	0.5488	Wind	1000000000000000
2005	0.3881	Hs-3	905.5	36.63	28.1325	Solar	1000000000000000
2006	0.3824	Hs-3	907.5	0.65	0.4070	Wind	1000000000000000
2007	0.3931	Hs-3	907.4	0.61	0.3767	Wind	1000000000000000
2008	0.3922	Hs-3	905.0	1.17	0.7295	Solar	1000000000000000
2009	0.3654	Hs-3	908.6	0.70	0.4341	Nuclear	1000000000000000
2010	0.3807	Hs-3	902.7	1.34	0.8341	Solar	1000000000000000
2011	0.3375	Hs-3	903.4	2.78	1.7268	Solar	1000000000000000
2012	0.3153	Hs-3	905.6	2.72	1.6885	Solar	1000000000000000
2013	0.2873	Hs-2	908.8	0.91	0.5695	Solar	1000000000000000
2014	0.2457	Hs-2	914.6	1.22	0.7609	Solar	1000000000000000
2015	0.2061	Hs-2	919.4	1.10	0.6902	Solar	1000000000000000
2016	0.2499	Hs-2	916.8	1.67	1.0458	Coal	1000000000000000
2017	0.2424	Hs-2	917.3	0.73	0.4586	Coal	1000000000000000
2018	0.2384	Hs-2	63.1	16.61	28.2360	Other Renewables	33376
2019	0.2580	Hs-2	63.1	6.43	0.8914	Coal	32719
2020	0.2468	Hs-2	915.4	17.72	28.1817	Other Renewables	1000000000000000
2021	0.2555	Hs-2	915.0	0.59	0.3689	Hydro	1000000000000000
2022	0.2593	Hs-2	914.4	0.49	0.3091	Wind	1000000000000000
2023	0.2507	Hs-2	914.9	0.70	0.4445	Coal	1000000000000000
2024	0.2392	Hs-2	915.2	1.14	0.7109	Coal	1000000000000000
2025	0.2455	Hs-2	914.6	2.77	1.7159	Coal	1000000000000000

CNQ v2.0.0 – Quaternion View – GBR United Kingdom · D=9 · N=26



CNQ DIAGNOSTICS · GBR (D = 9)

CHSH joint-coherence:
not computed at D = 9
(CHSH requires D=8 twin factoring;
D = 9 does not factor as 4x4)

Twin-quaternion factor:
not computed at D = 9
(twin requires D=8 = 4+4 partition)

Attractor fit:
amplitude A: 7.505
contraction λ : 0.0078
period: 1
fitted: False

Helmsman family:
total flips: 15
stability S_σ : 0.375

§ IND India

EMBER electricity-generation series · 2000–2025 · N = 26 · D = 8

INPUT

source: ember_IND*_generation_TWh.csv
N records: 26
D carriers: 8
Bioenergy
Coal
Gas
Hydro
Nuclear
Other Fossil
Solar
Wind
source_sha256: 05cde94fa8f37b98ae695073...

HEADLINE METRICS

Helmsman flips: ?
Helmsman stability S_σ : {'global': 0.375, 'rolling': [0.2857142857142857, 0.14285714285714285]}
IR class: ?
Attractor amplitude: ?

CHSH S: ?
Tsirelson bound: $2\sqrt{2} \approx 2.828$

Bearing pairs $(D(D-1)/2)$: 28

CNT v3.1.0 content_sha: ?...
CNQ v2.0.0 content_sha: ?...

Following pages: Stage 1 (XY/XZ/YZ × N years + course) → Stage 2/3 (7 pages) → CNQ (2 pages).

SECTION PLATE t=0 | Year 2000 | D=8 N=26 pairs=28

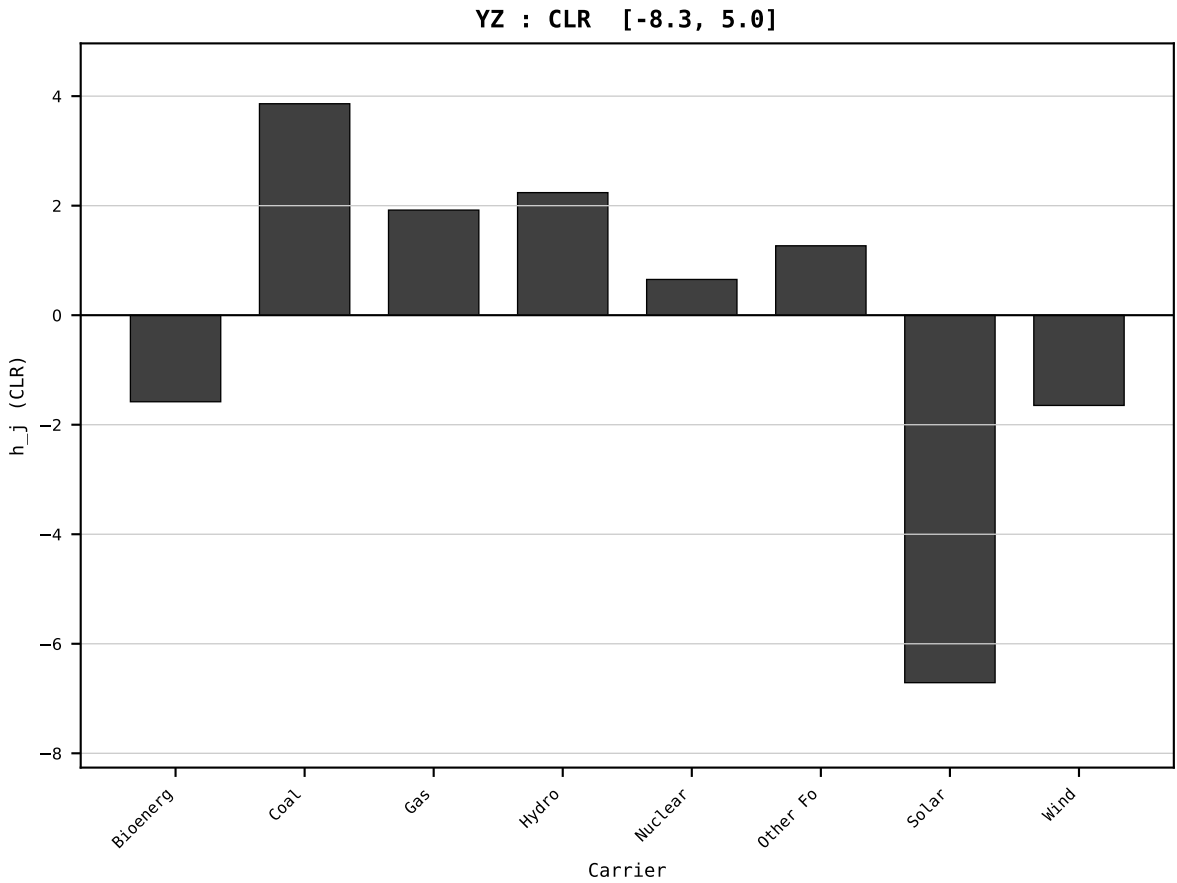
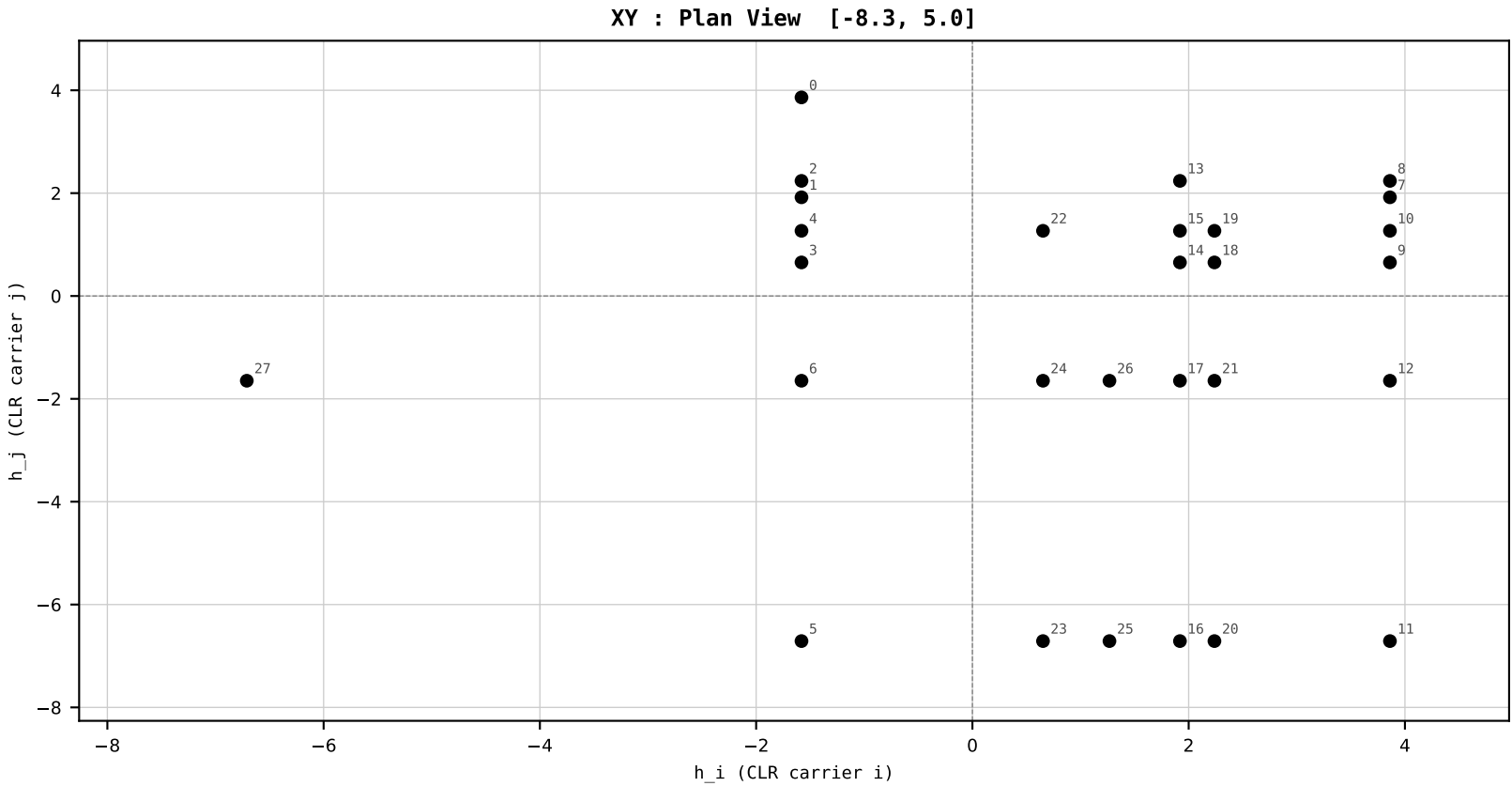
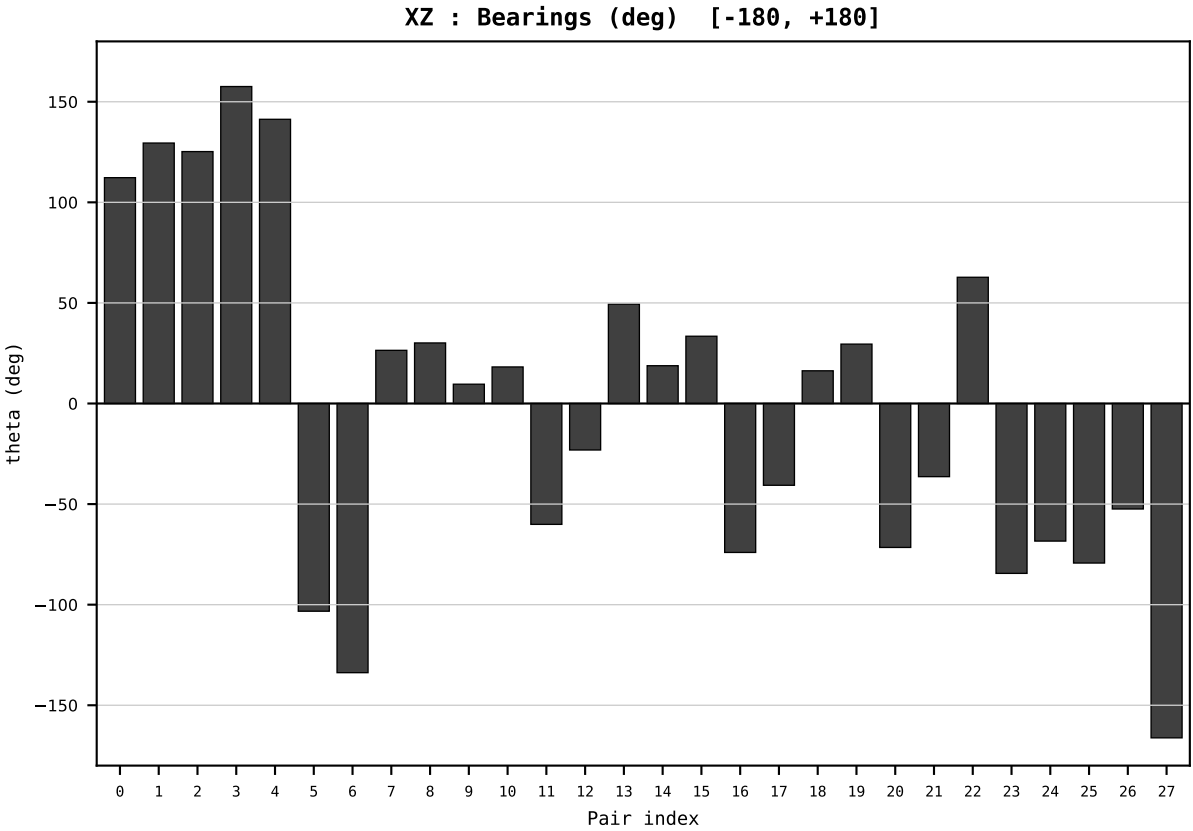
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2000
t : 0 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.498576
Ring = Hs-3
E_metric= 75.8817
kappa_HS= 39023.00
omega = 0.0000 deg
d_A = 0.000000
Helm = ---
Helm d = ---
DR = 10.572 HLR
DR ratio= 39023.0x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=1 | Year 2001 | D=8 N=26 pairs=28

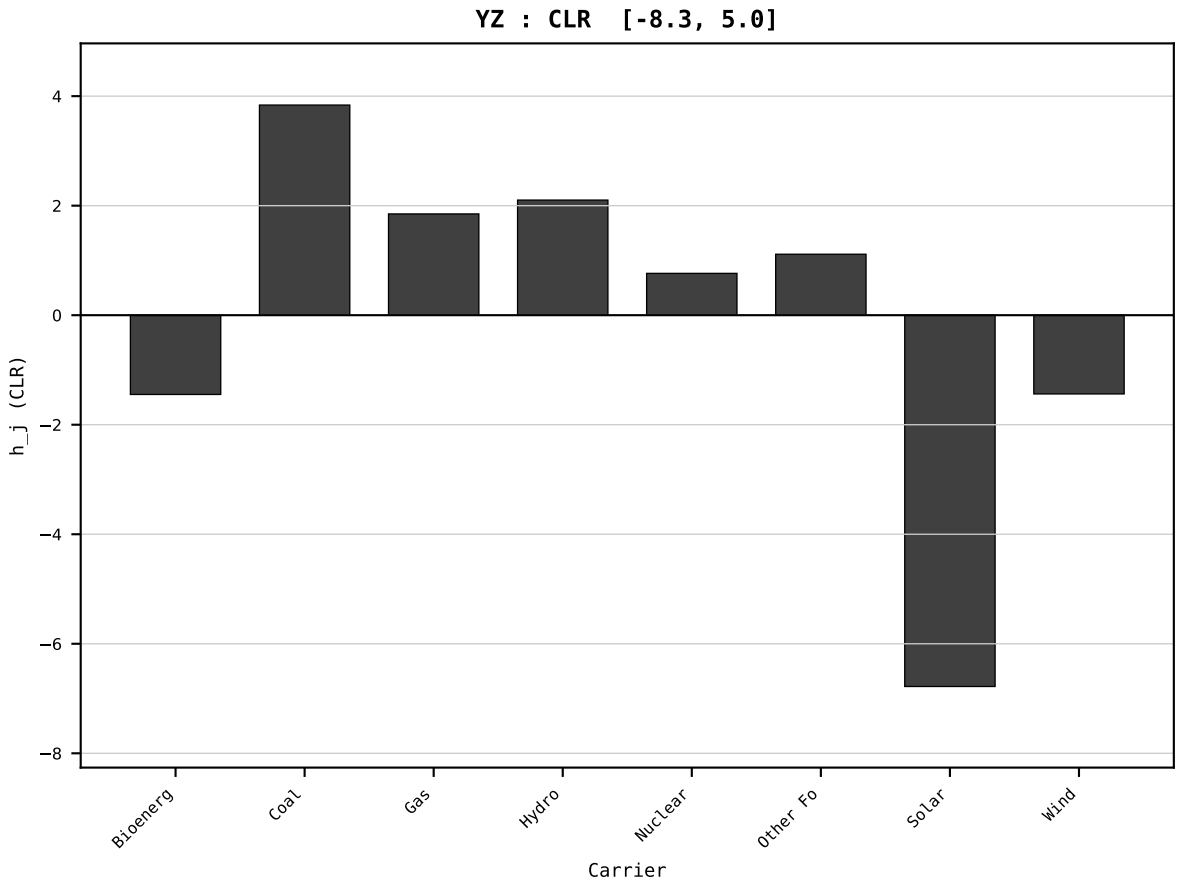
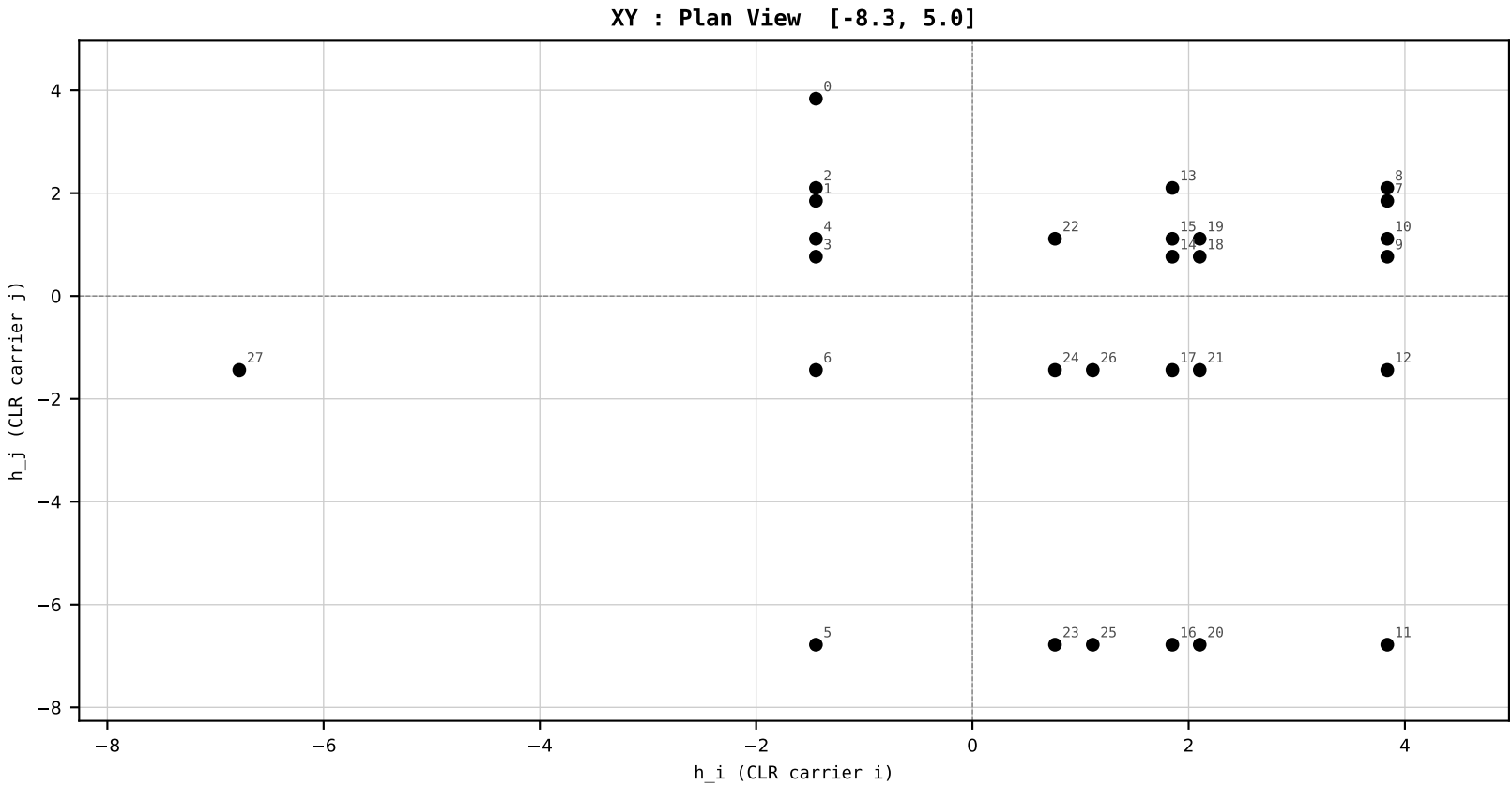
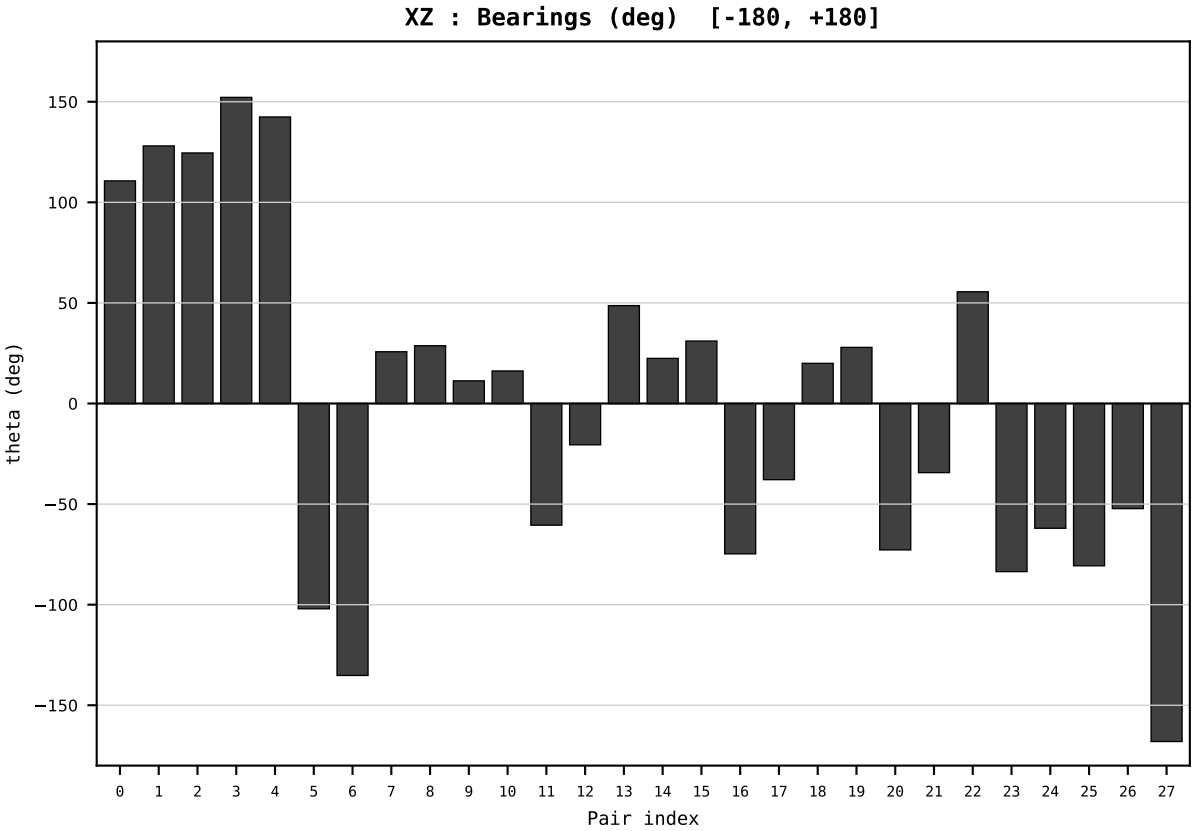
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2001
t : 1 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.506780
Ring = Hs-4
E_metric= 74.5149
kappa_HS= 40829.00
omega = 2.2964 deg
d_A = 0.356346
Helm = Wind
Helm d = +0.210203
DR = 10.617 HLR
DR ratio= 40829.0x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=2 | Year 2002 | D=8 N=26 pairs=28

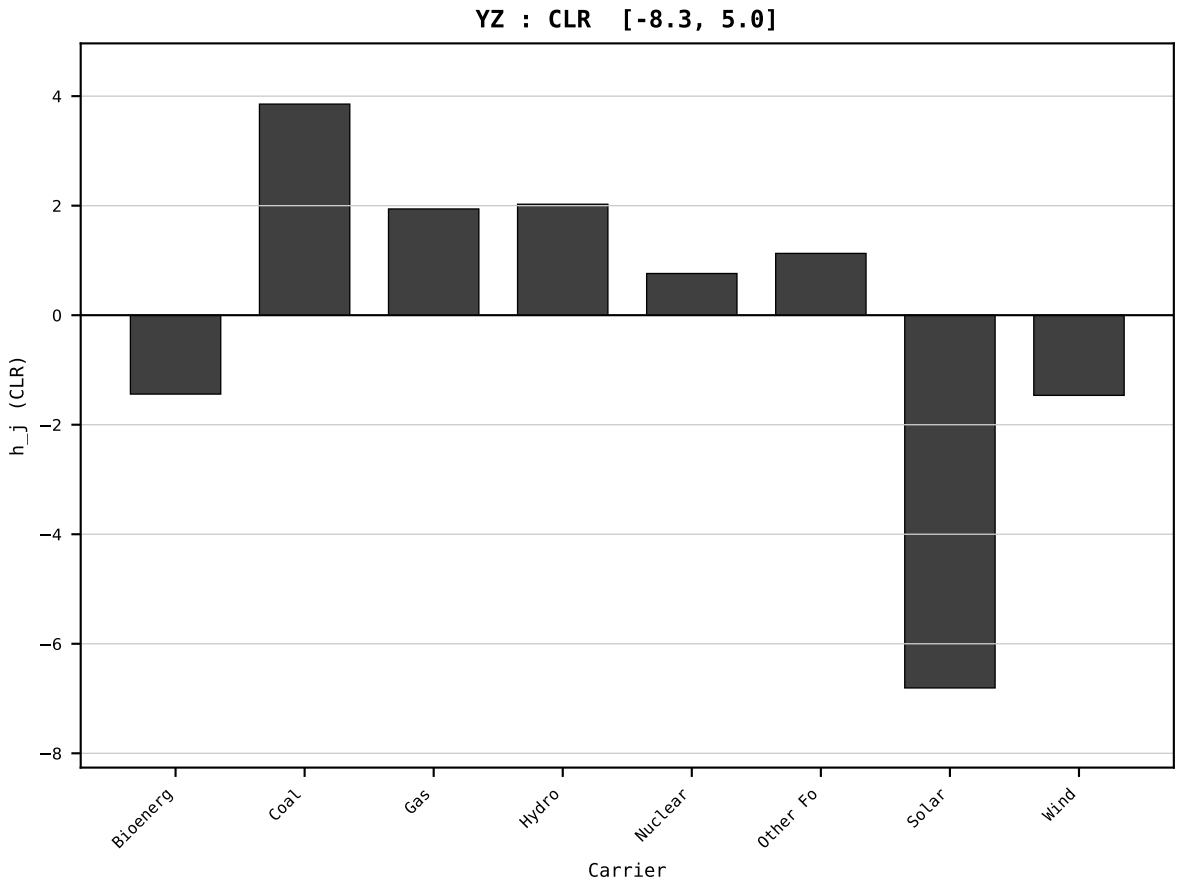
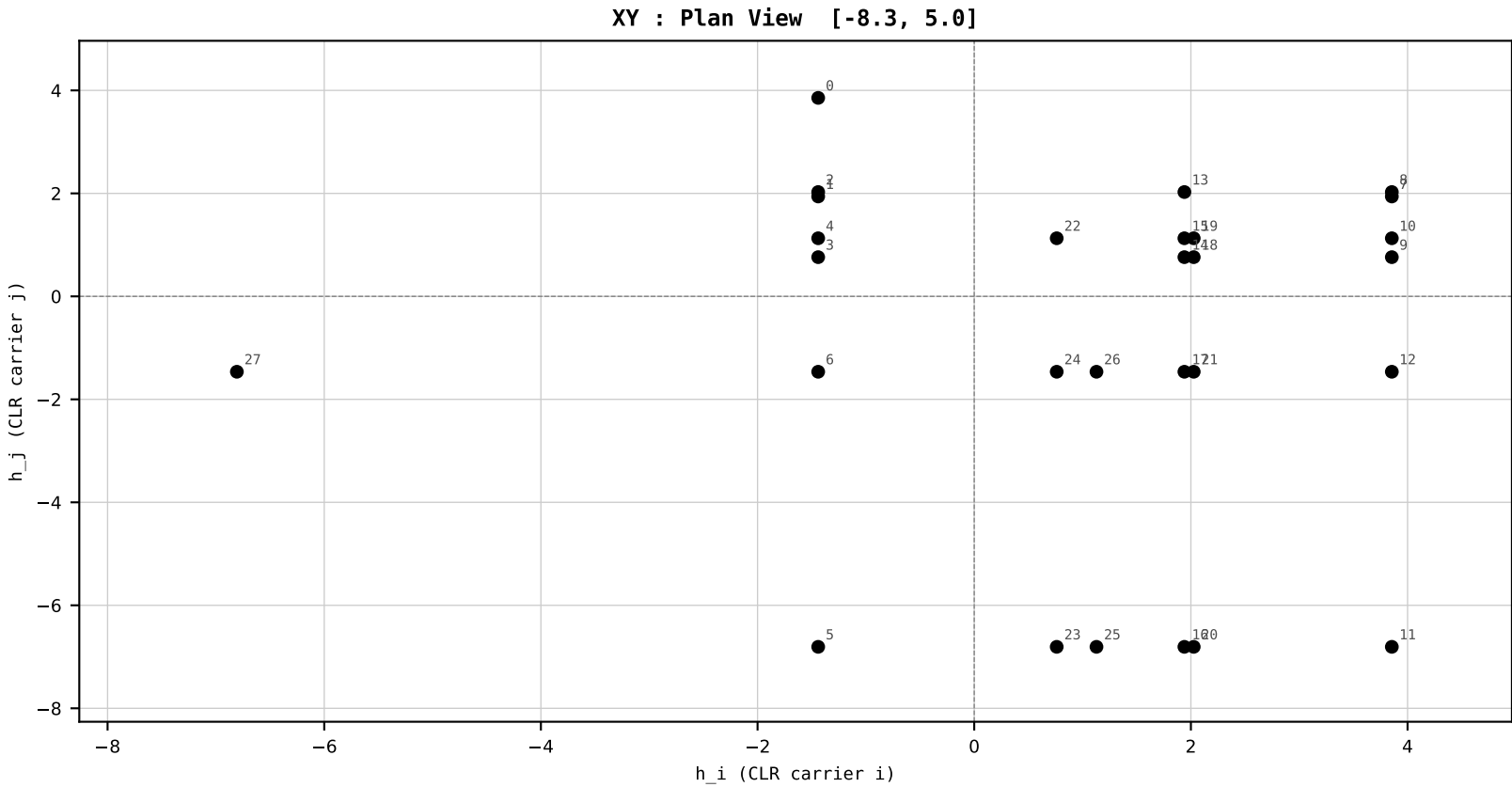
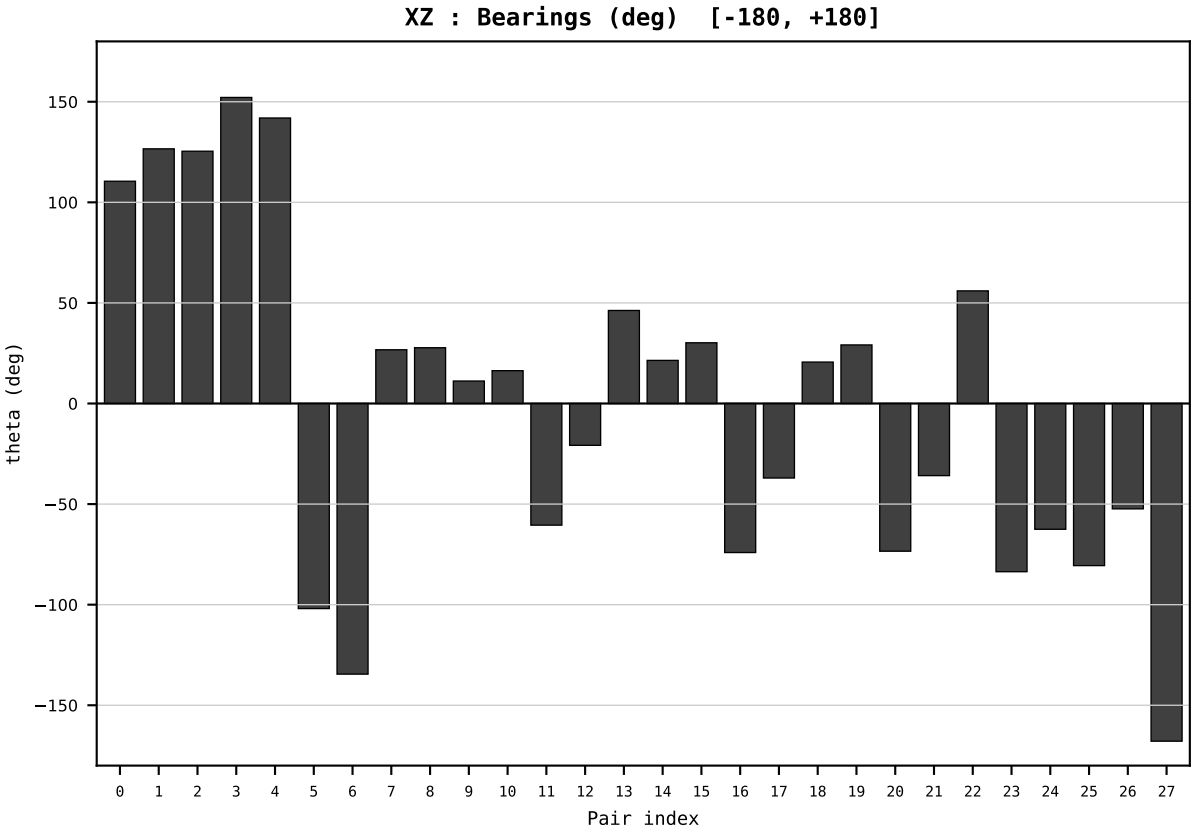
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2002
t : 2 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.509498
Ring = Hs-4
E_metric= 75.1276
kappa_HS= 42670.00
omega = 0.8041 deg
d_A = 0.126449
Helm = Gas
Helm d = +0.090790
DR = 10.661 HLR
DR ratio= 42670.0x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=3 | Year 2003 | D=8 N=26 pairs=28

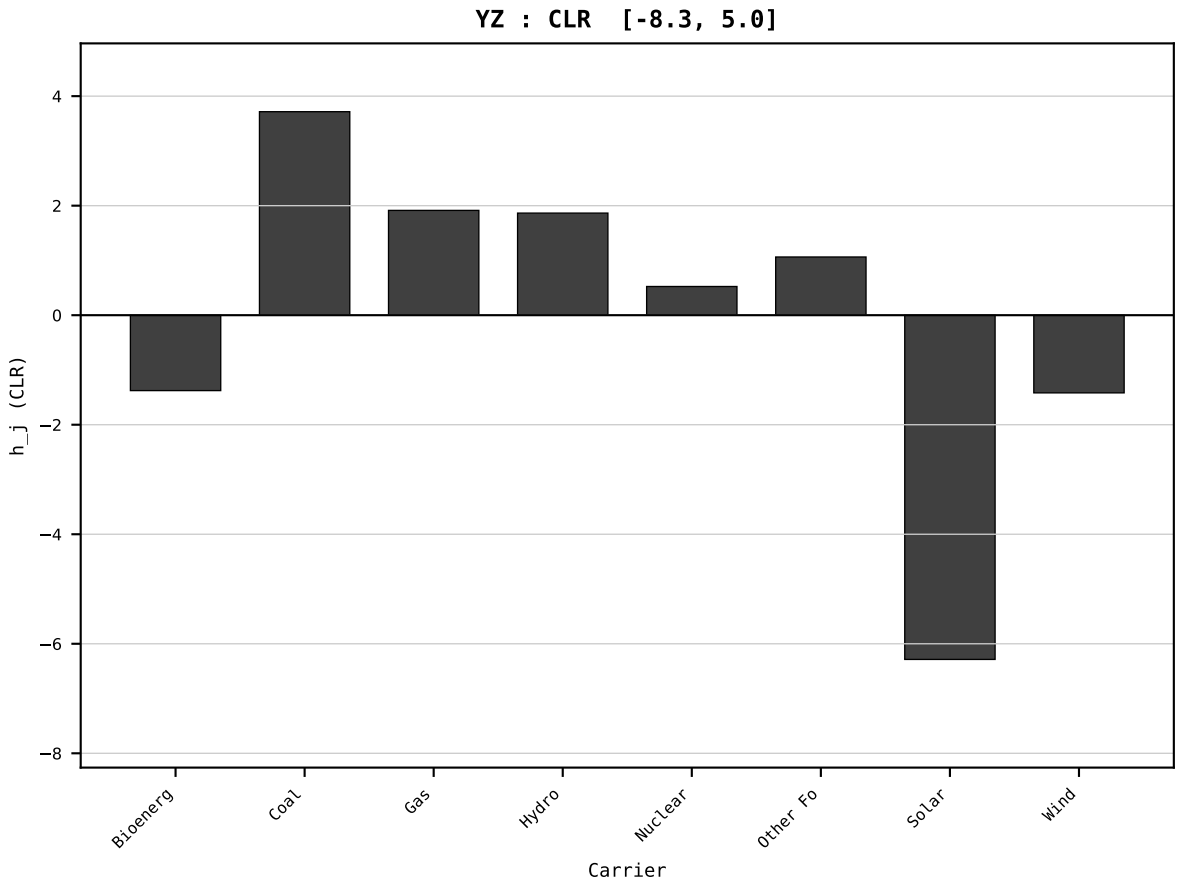
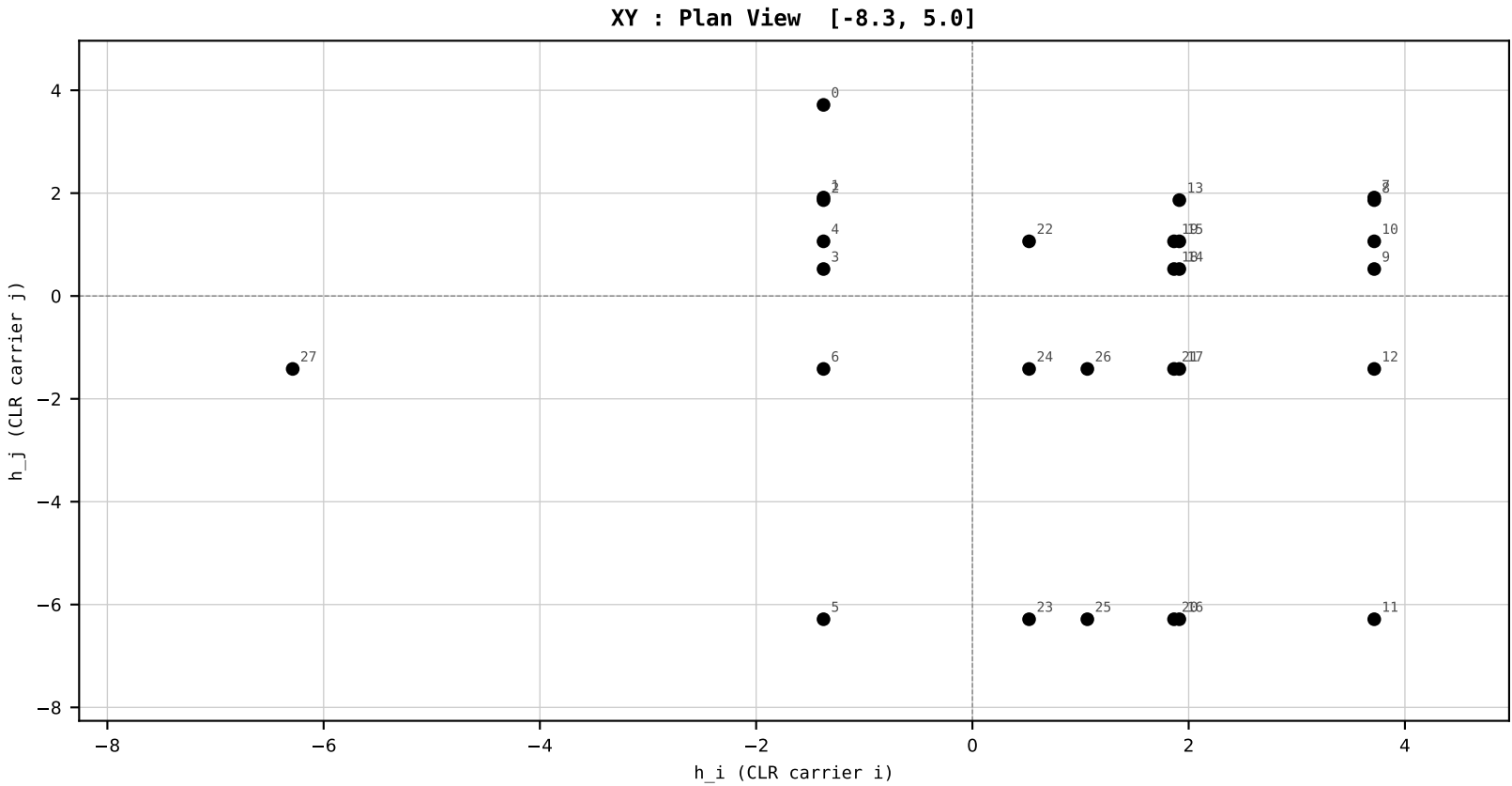
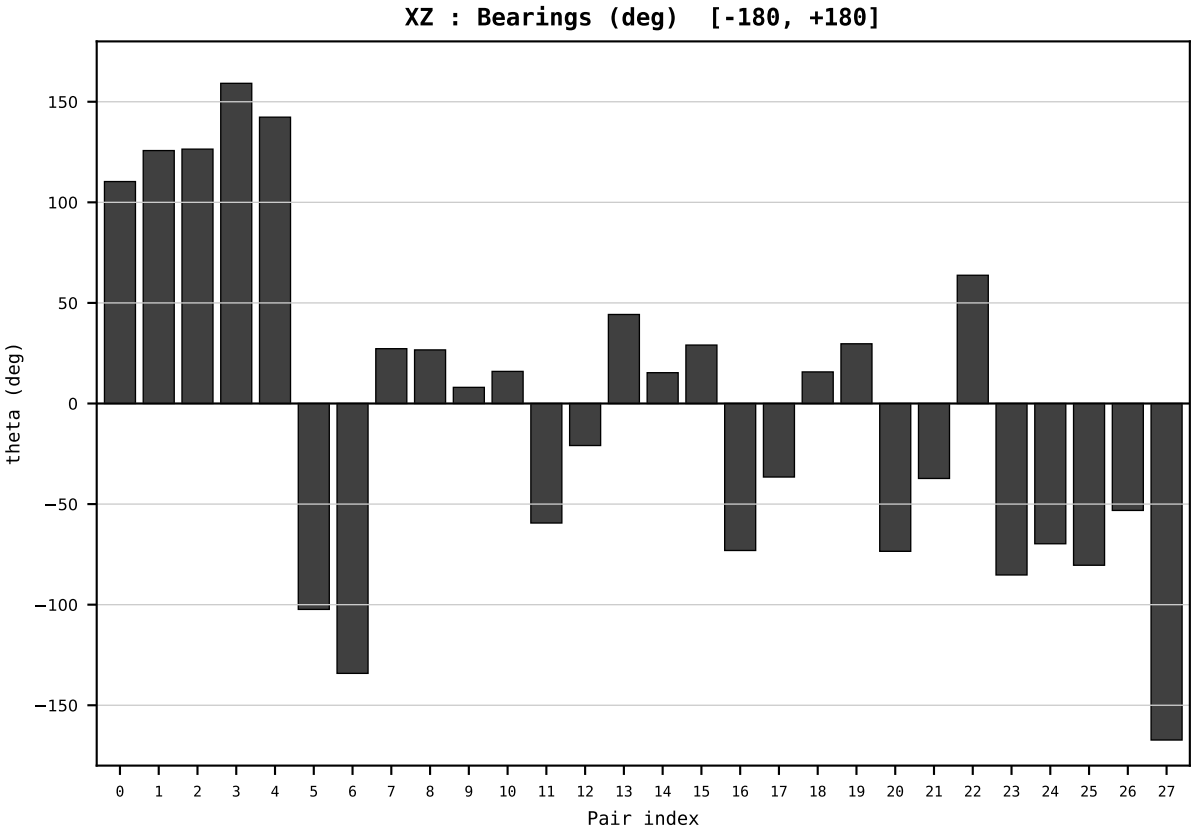
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2003
t : 3 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.500822
Ring = Hs-4
E_metric= 65.7748
kappa_HS= 22073.50
omega = 1.8448 deg
d_A = 0.619370
Helm = Solar
Helm d = +0.520423
DR = 10.002 HLR
DR ratio= 22073.5x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=4 | Year 2004 | D=8 N=26 pairs=28

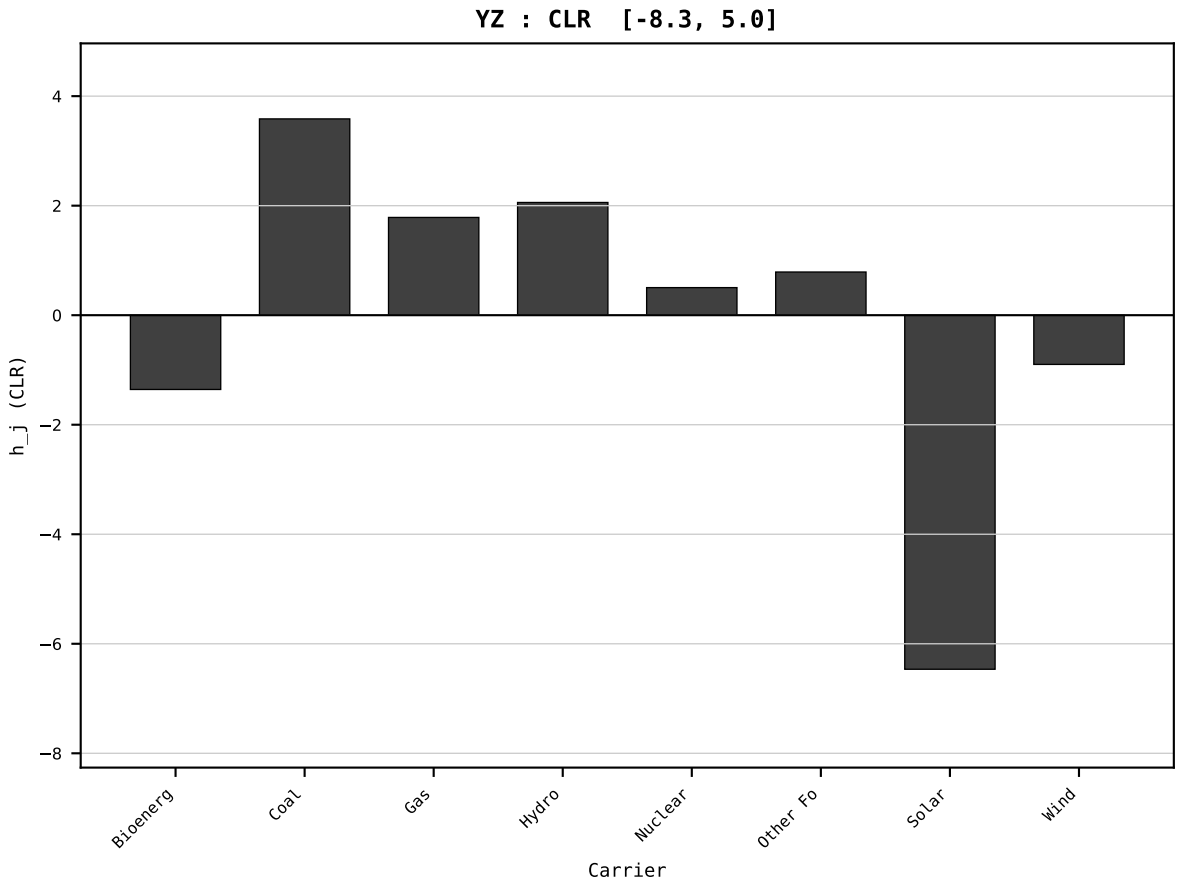
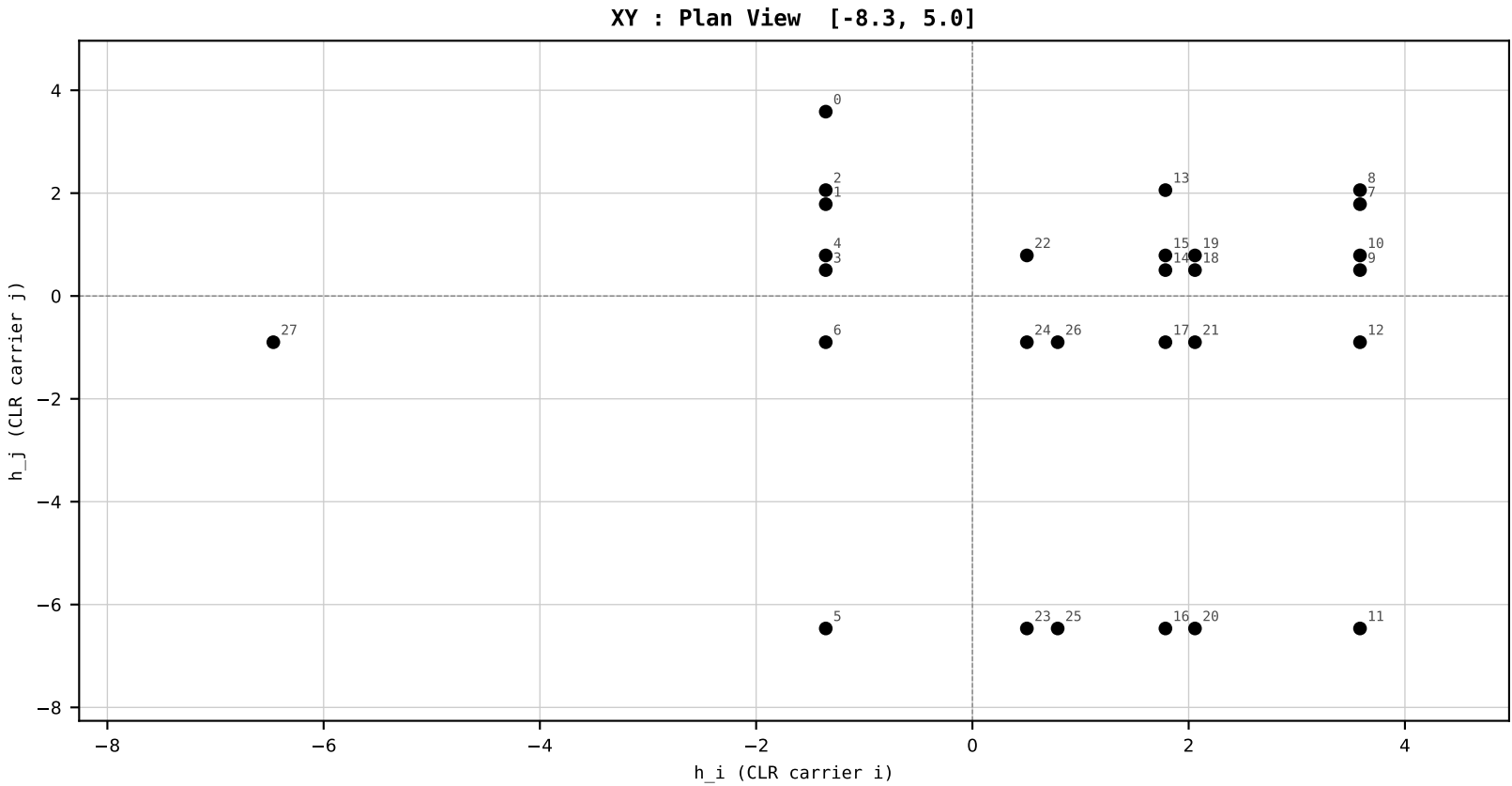
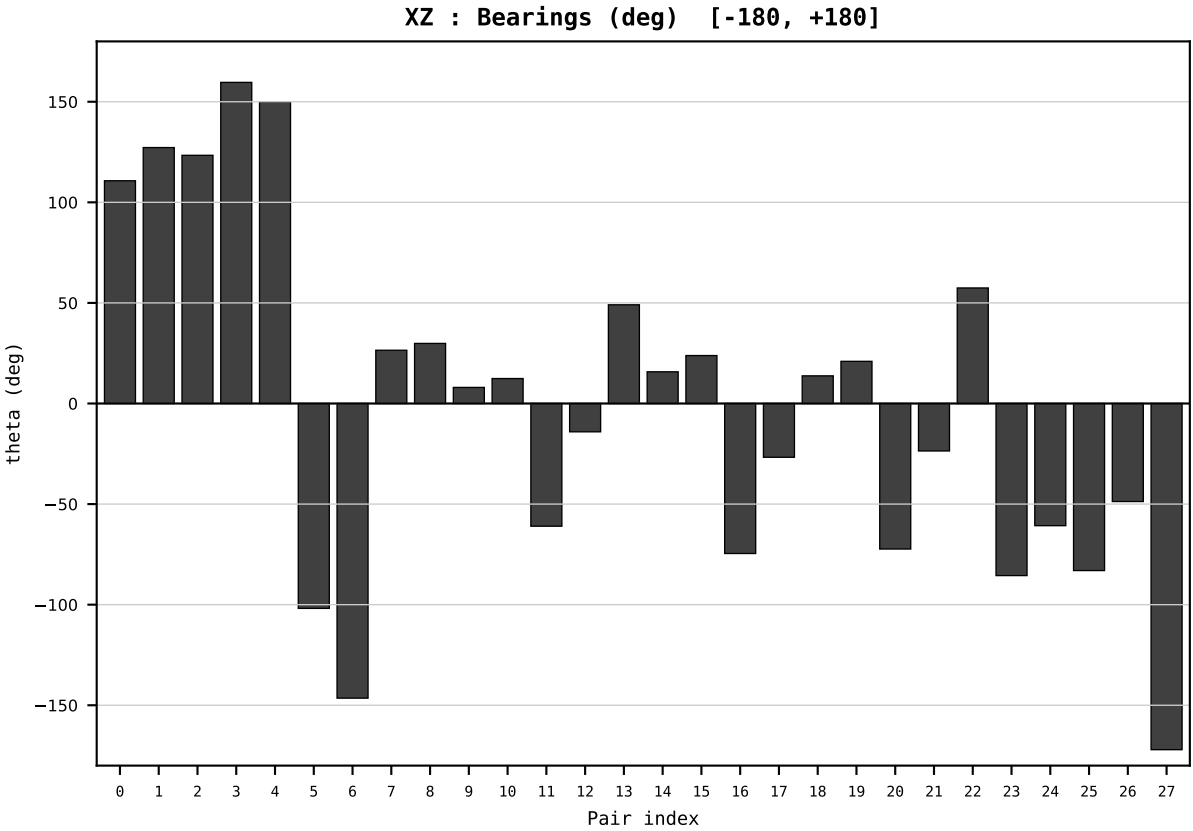
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2004
t : 4 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.474628
Ring = Hs-3
E_metric= 65.5944
kappa_HS= 23150.50
omega = 4.7416 deg
d_A = 0.670616
Helm = Wind
Helm d = +0.519664
DR = 10.050 HLR
DR ratio= 23150.5x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=5 | Year 2005 | D=8 N=26 pairs=28

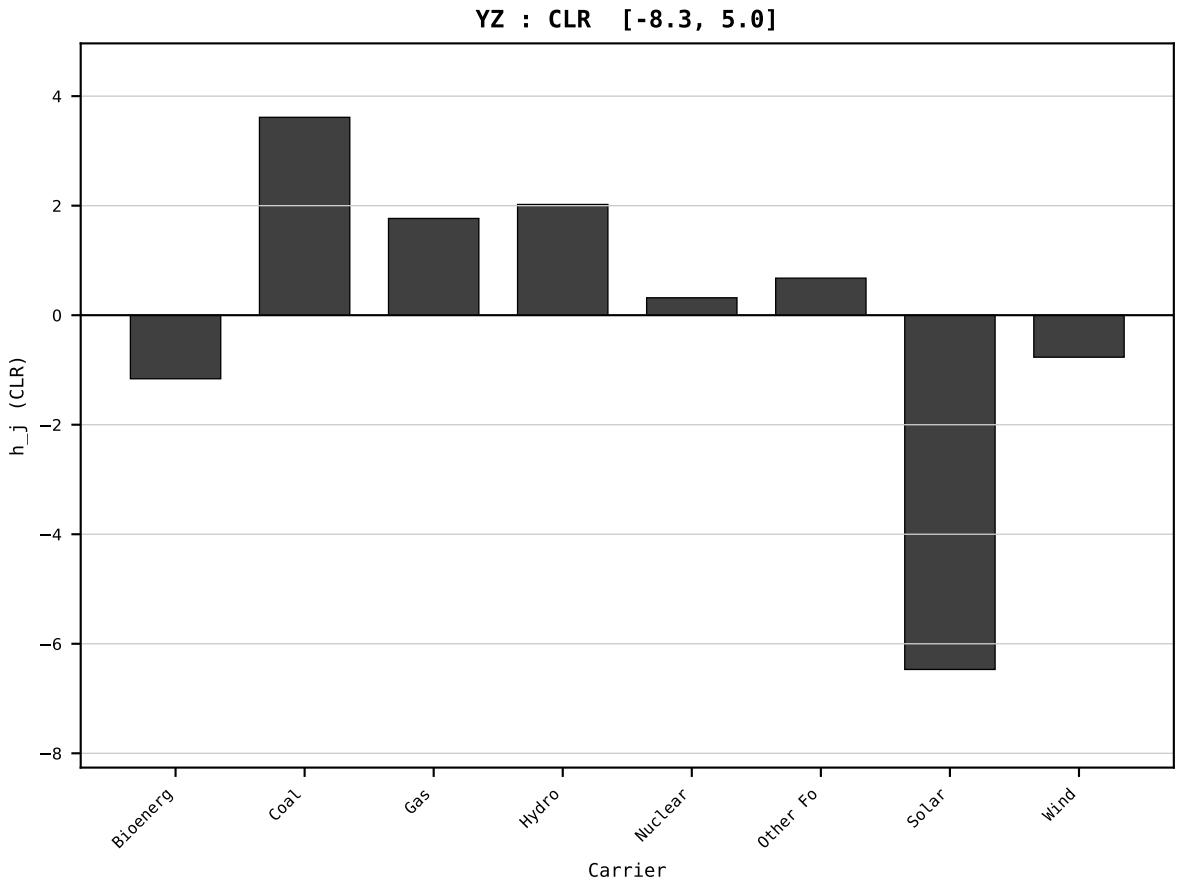
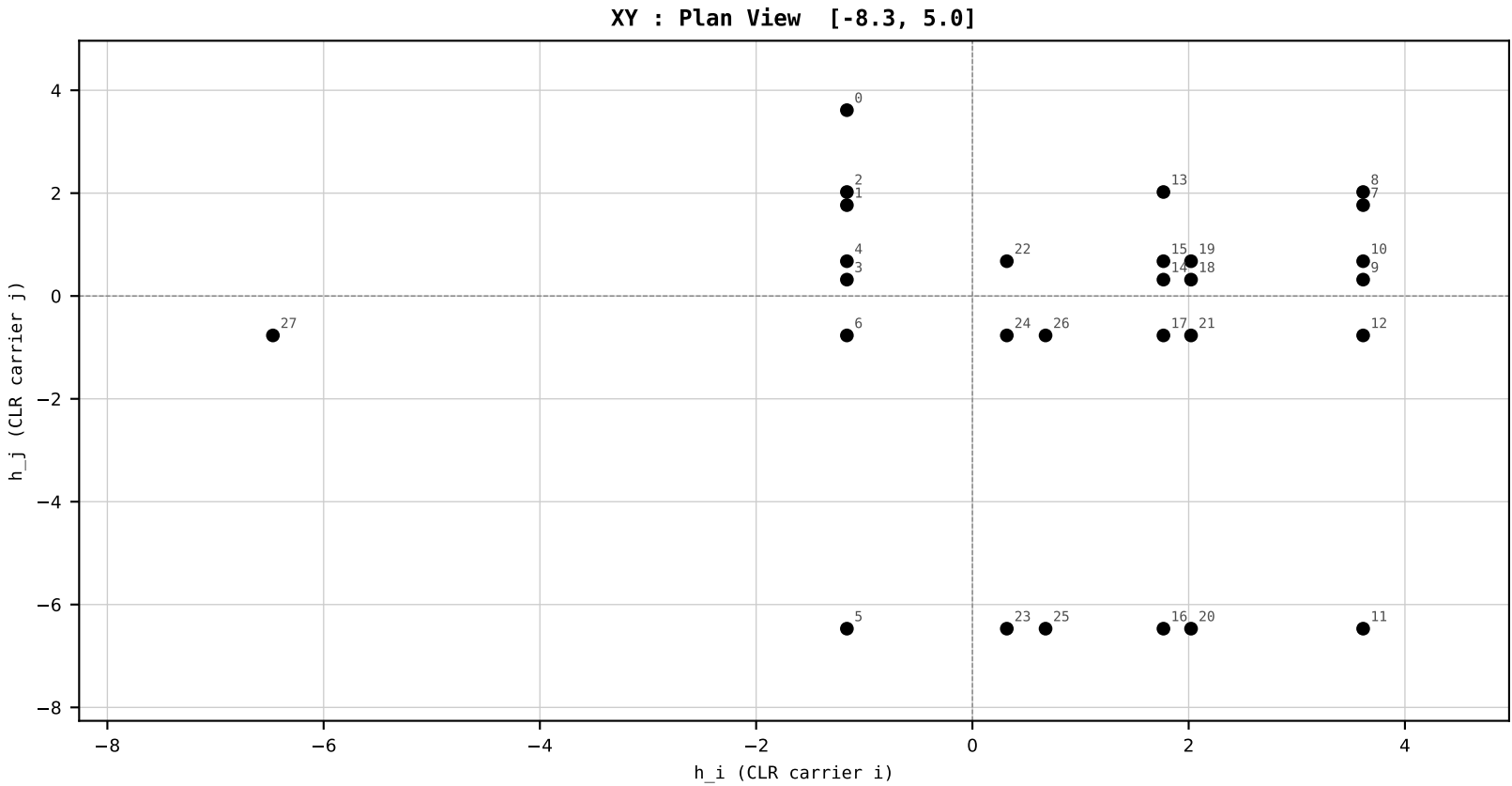
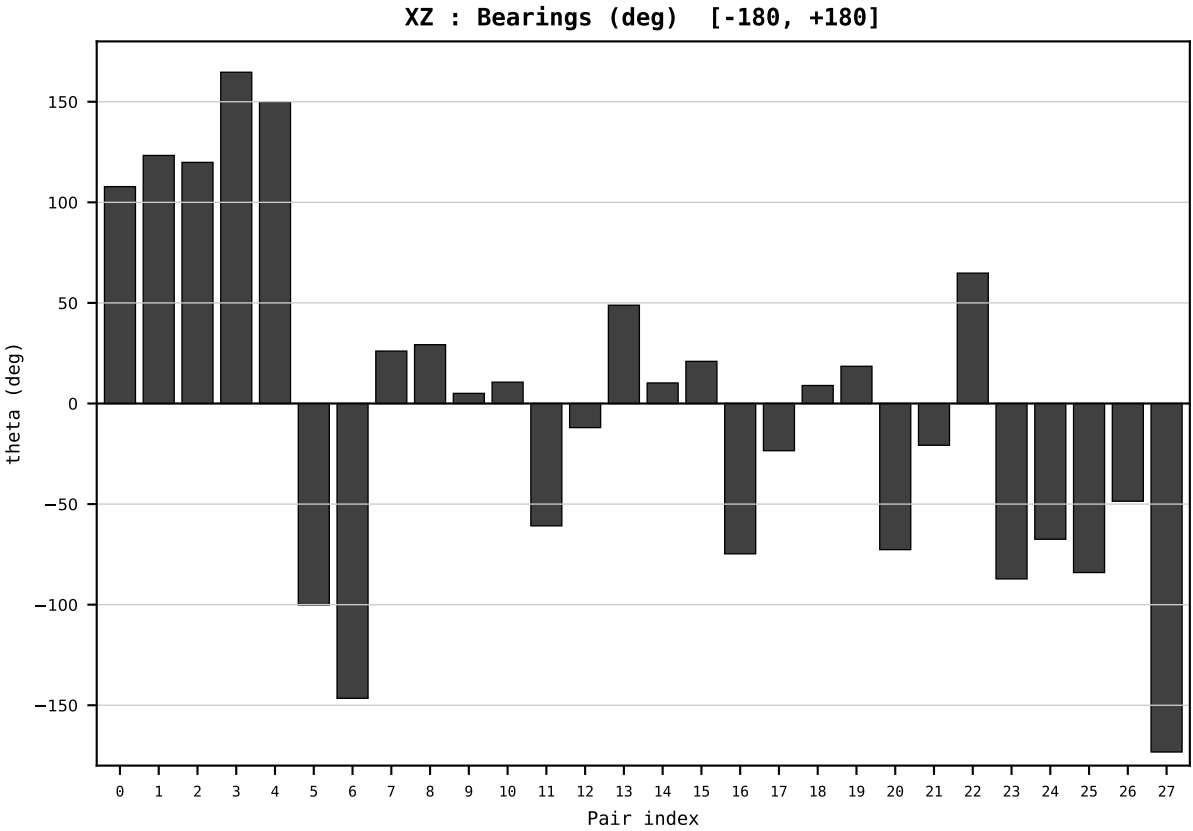
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2005
t : 5 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.491003
Ring = Hs-3
E_metric= 64.6110
kappa_HS= 23924.00
omega = 2.2606 deg
d_A = 0.324101
Helm = Bioenergy
Helm d = +0.195431
DR = 10.083 HLR
DR ratio= 23924.0x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=6 | Year 2006 | D=8 N=26 pairs=28

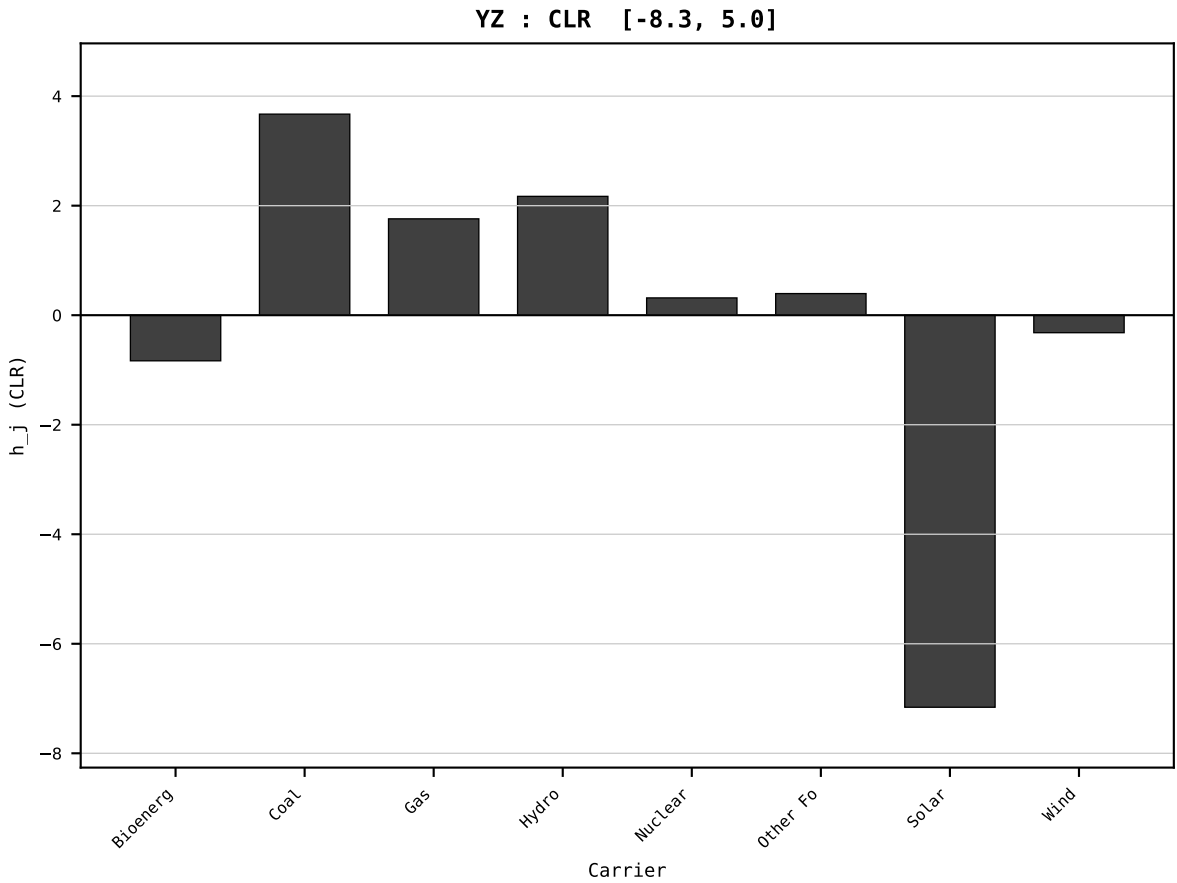
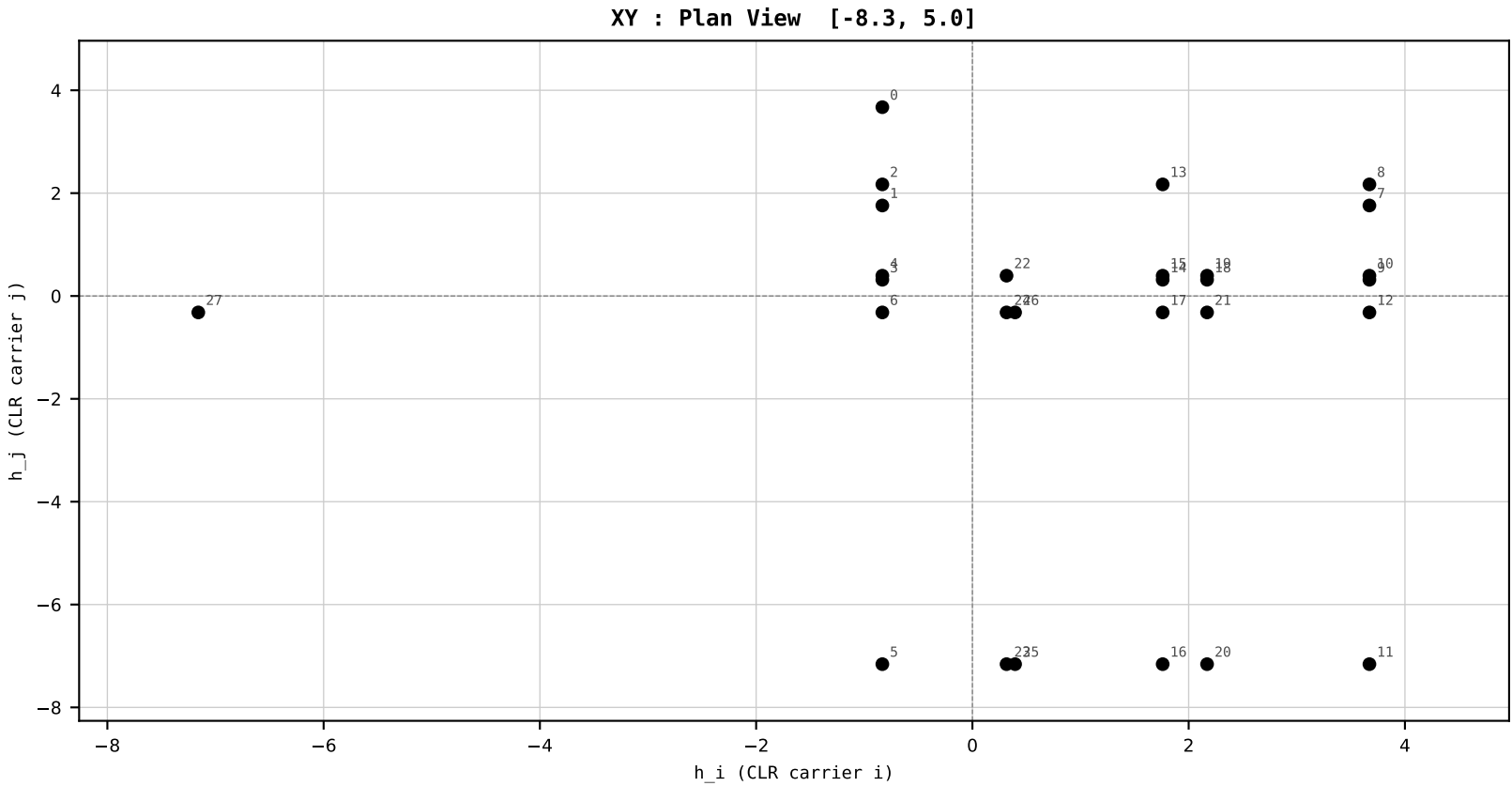
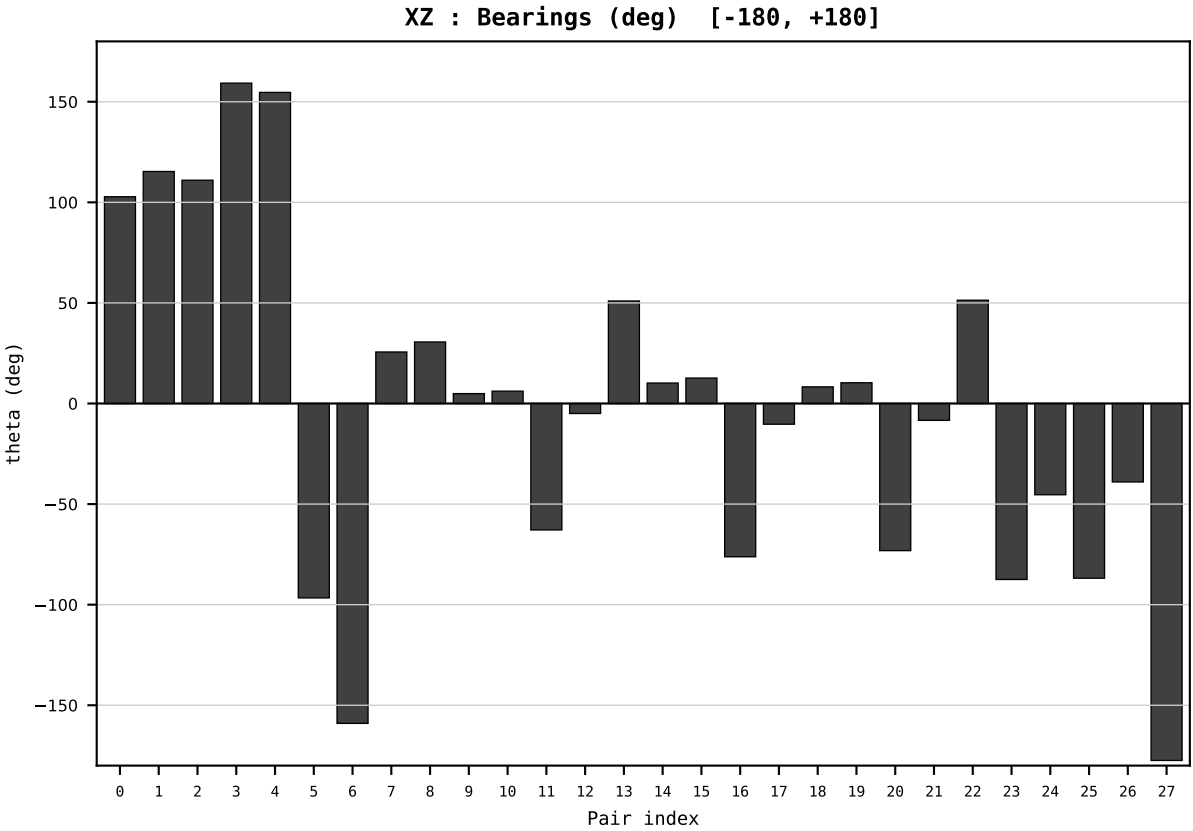
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2006
t : 6 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.493296
Ring = Hs-3
E_metric= 73.5896
kappa_HS= 50546.00
omega = 5.3311 deg
d_A = 0.942605
Helm = Solar
Helm d = -0.689998
DR = 10.831 HLR
DR ratio= 50546.0x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=7 | Year 2007 | D=8 N=26 pairs=28

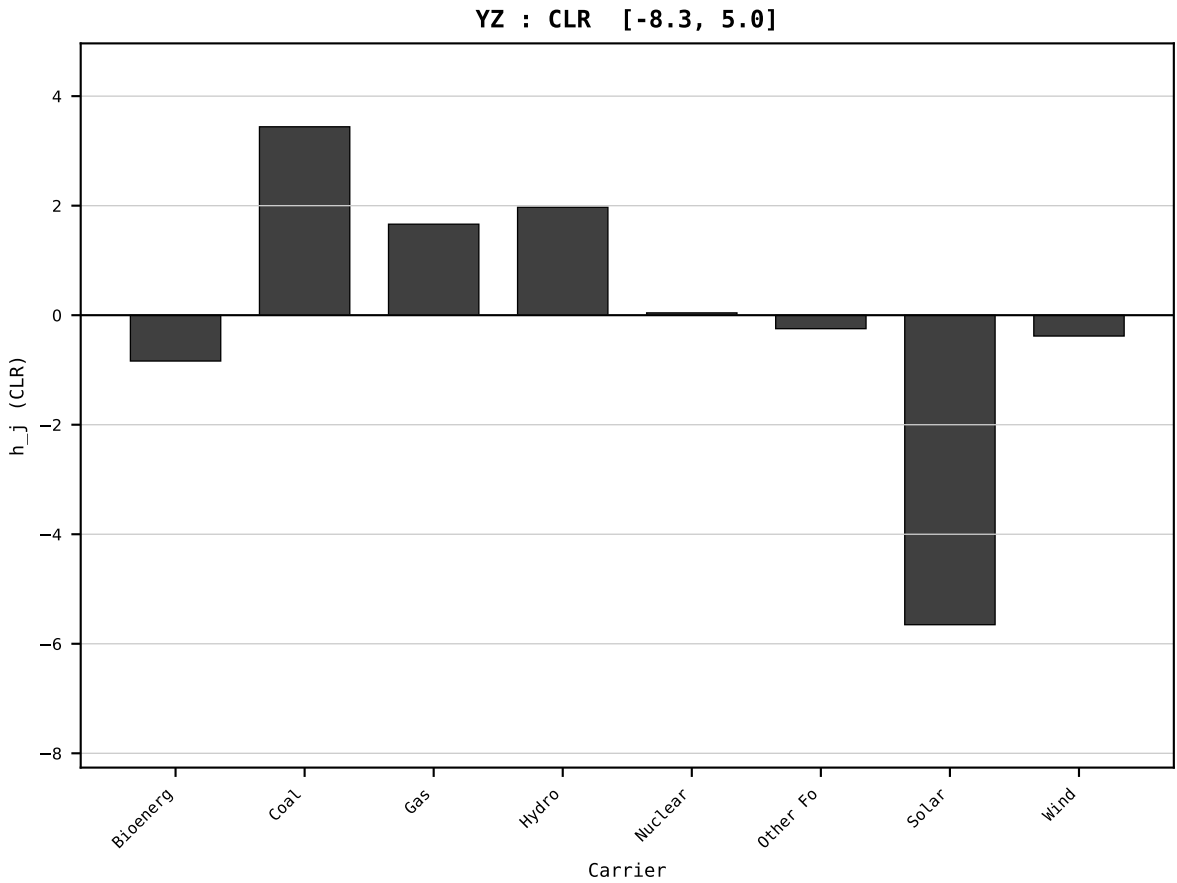
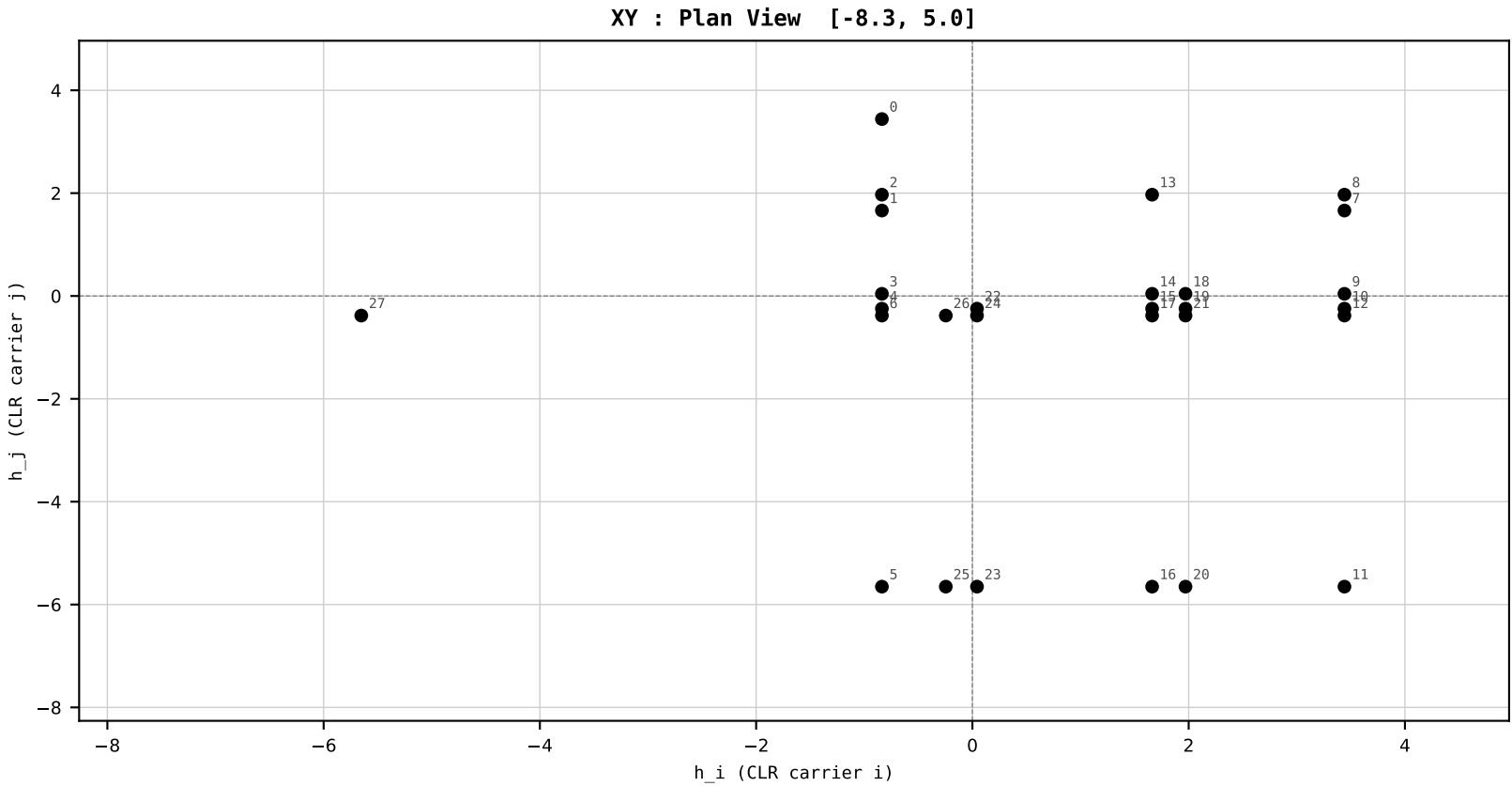
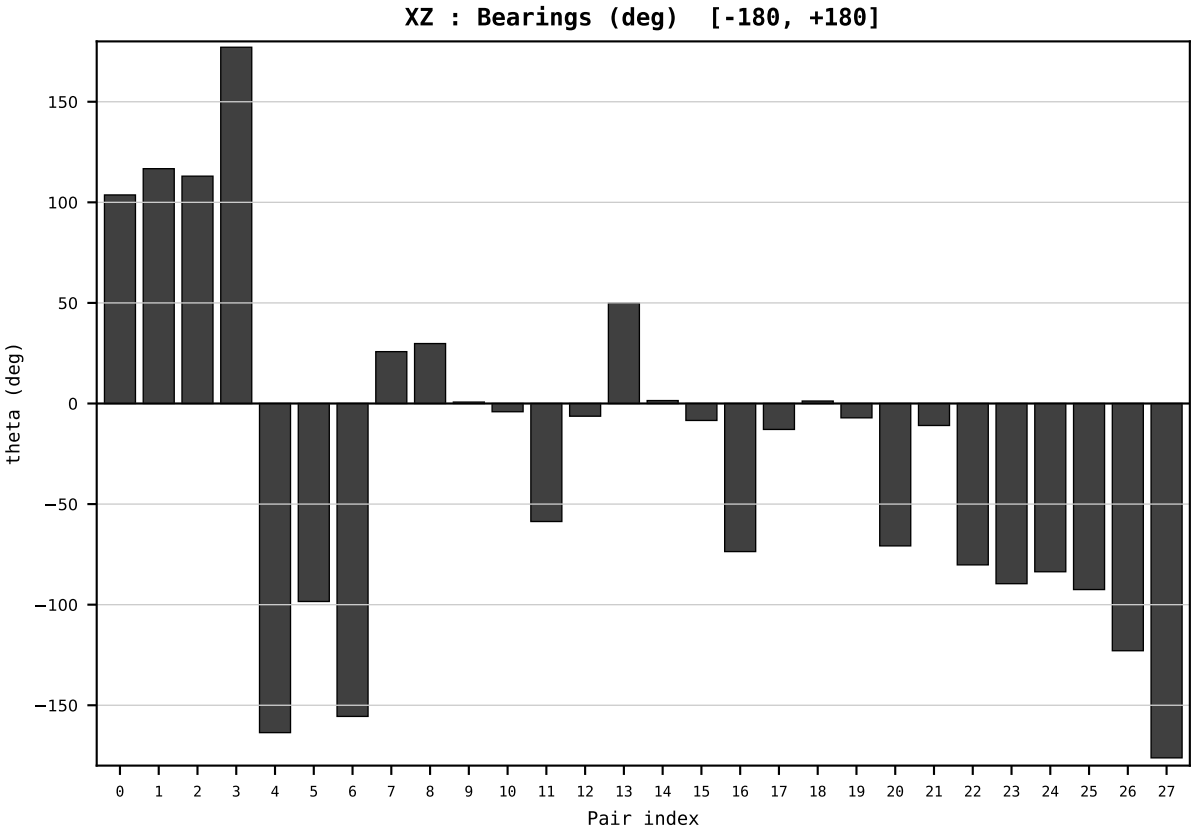
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2007
t : 7 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.488879
Ring = Hs-3
E_metric= 51.3339
kappa_HS= 8887.83
omega = 6.8016 deg
d_A = 1.692216
Helm = Solar
Helm d = +1.507238
DR = 9.092 HLR
DR ratio= 8887.8x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=8 | Year 2008 | D=8 N=26 pairs=28

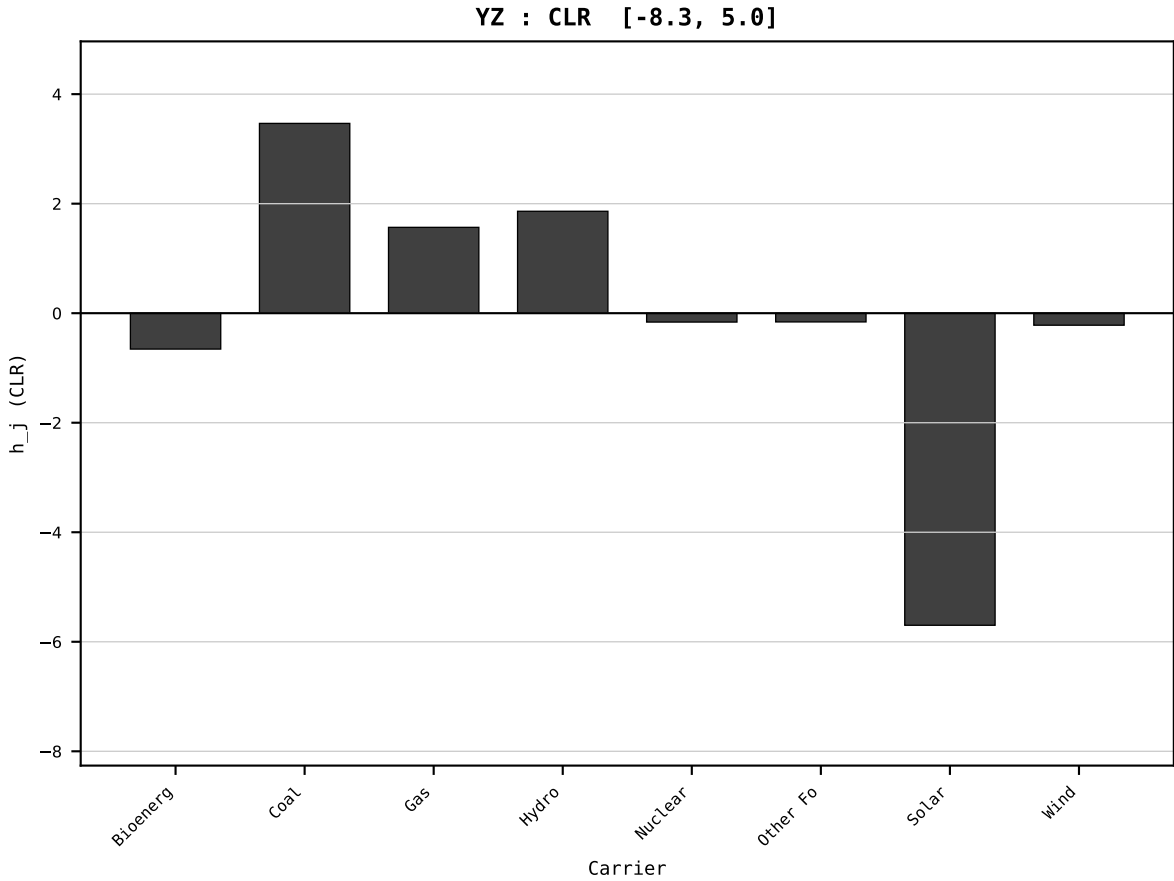
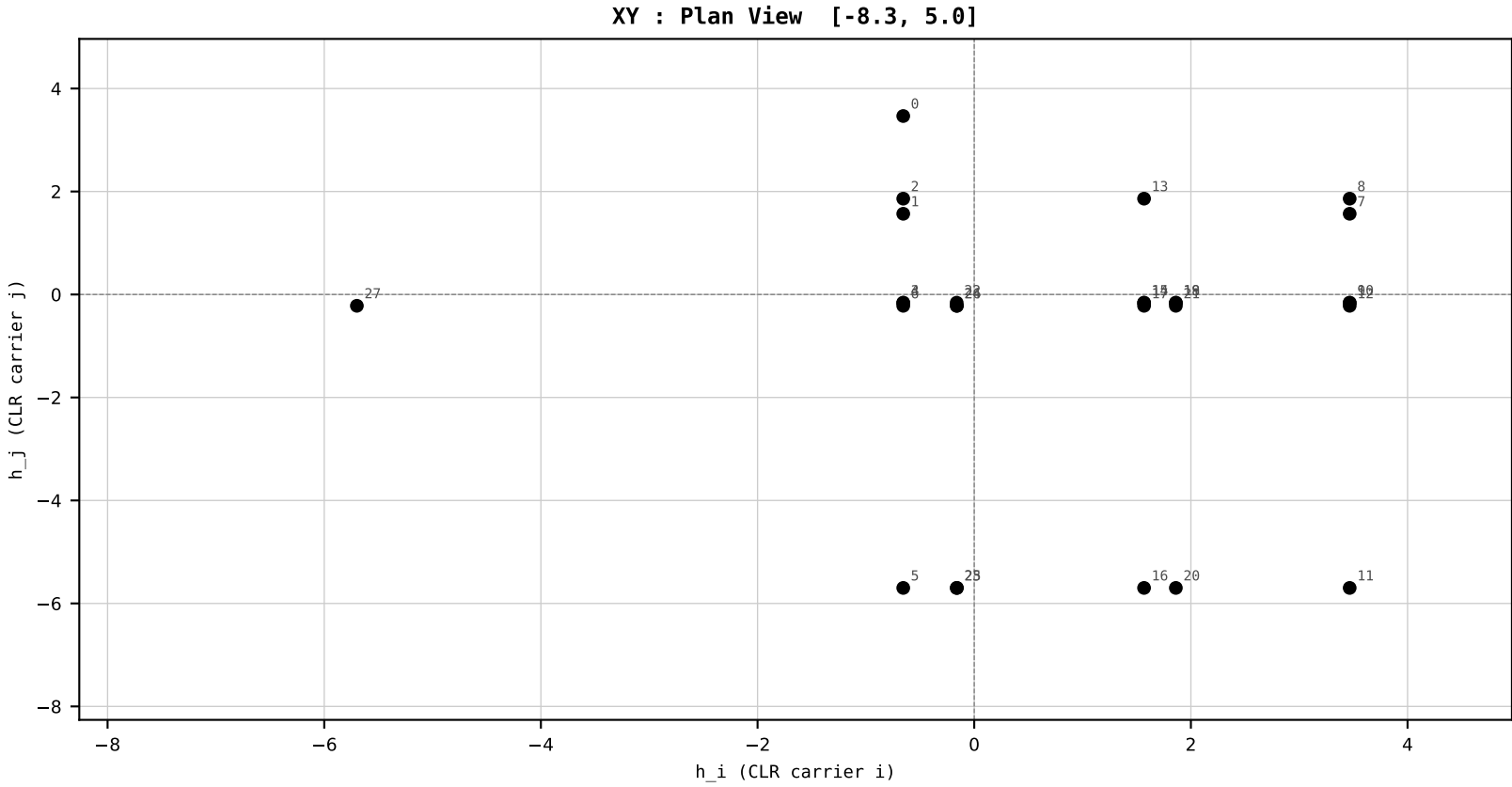
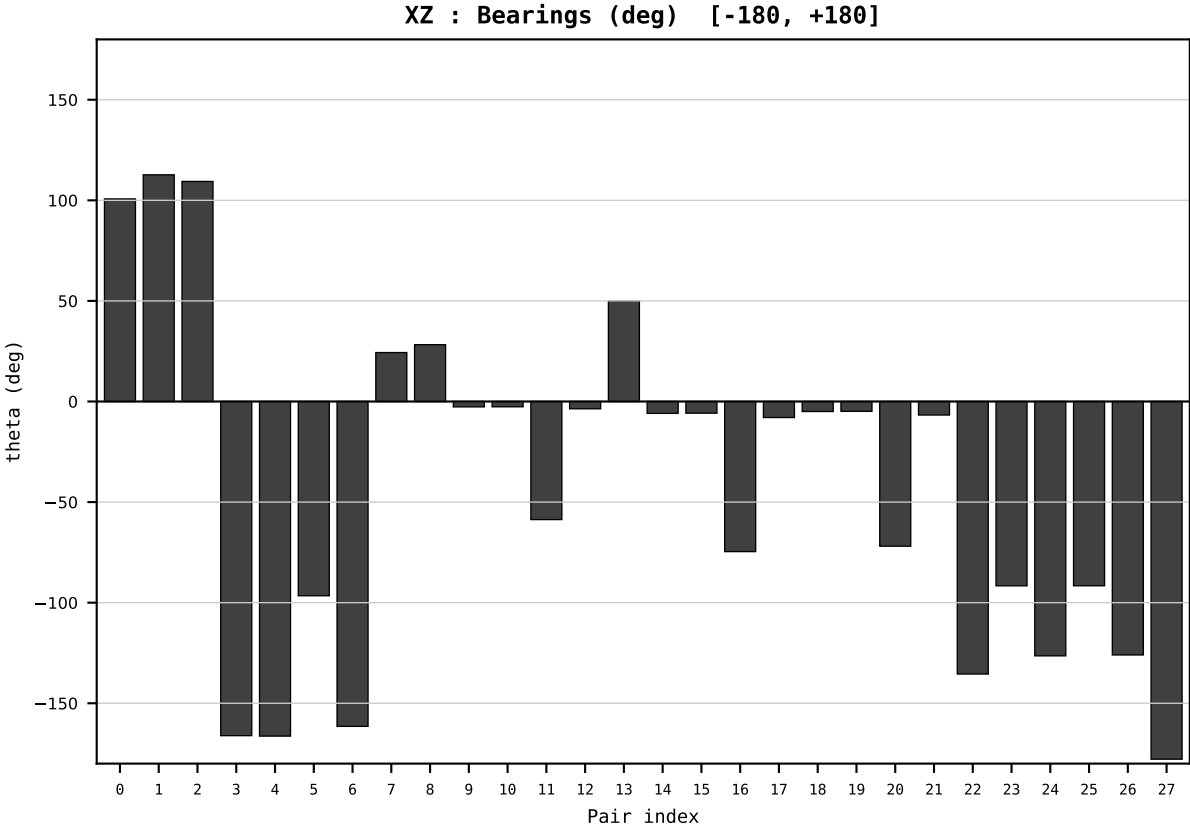
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2008
t : 8 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.503376
Ring = Hs-4
E_metric= 50.9375
kappa_HS= 9552.00
omega = 2.9028 deg
d_A = 0.363314
Helm = Nuclear
Helm d = -0.204435
DR = 9.165 HLR
DR ratio= 9552.0x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=9 | Year 2009 | D=8 N=26 pairs=28

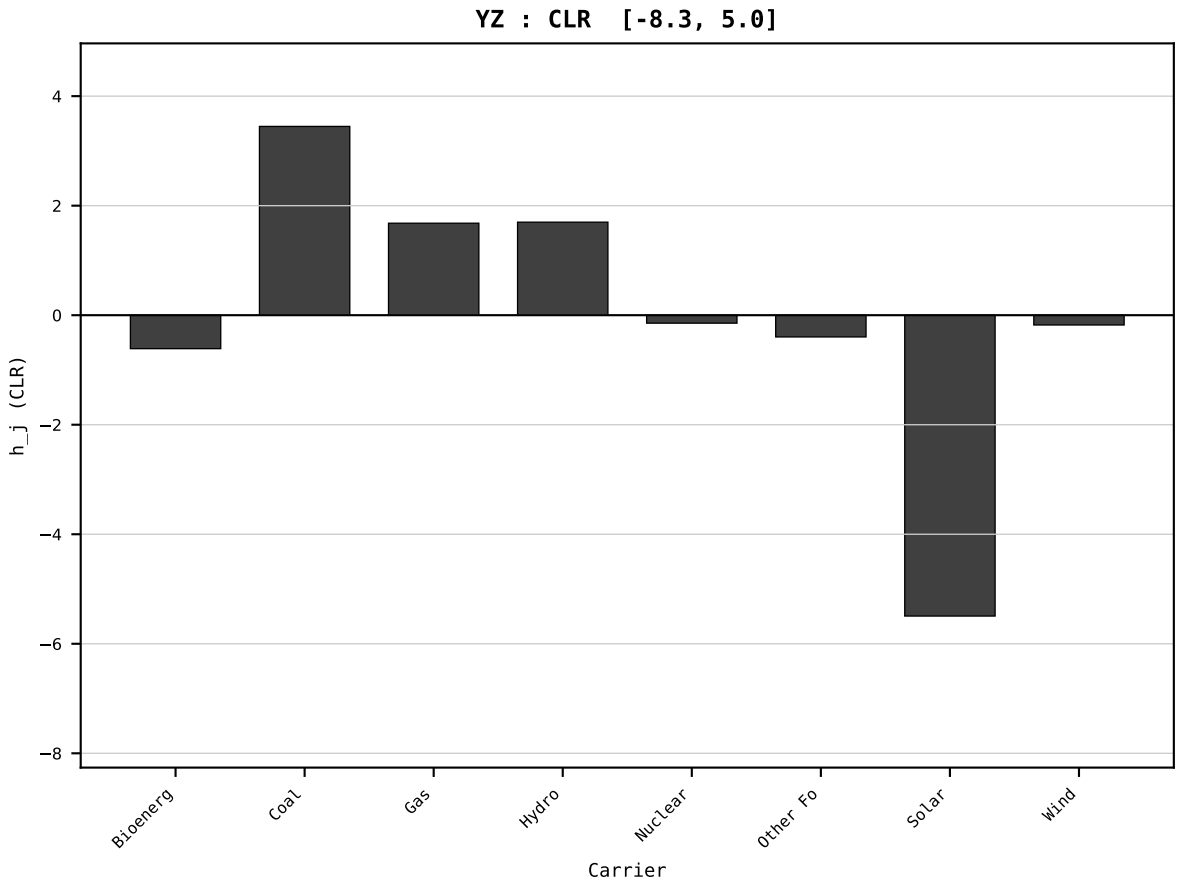
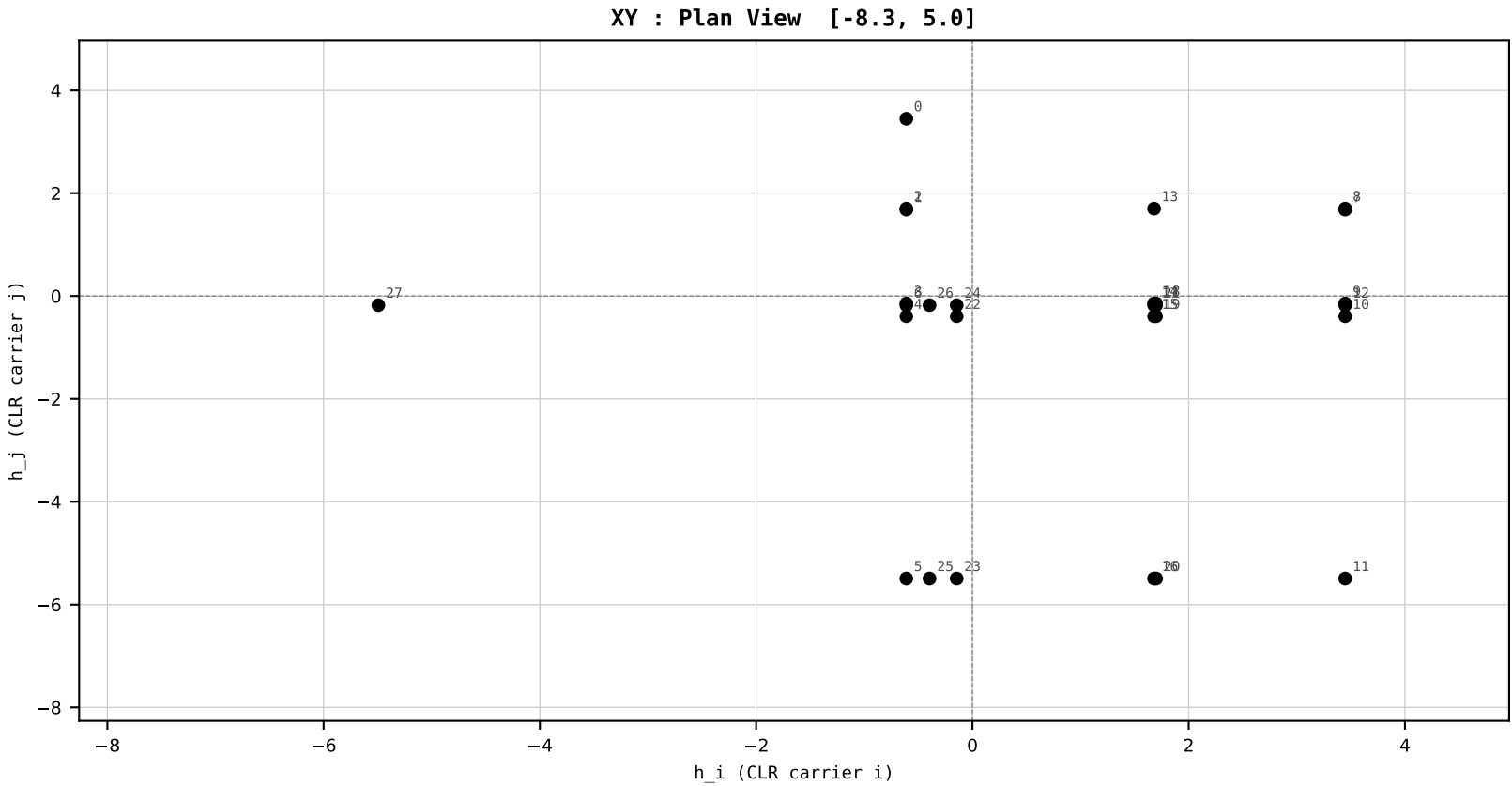
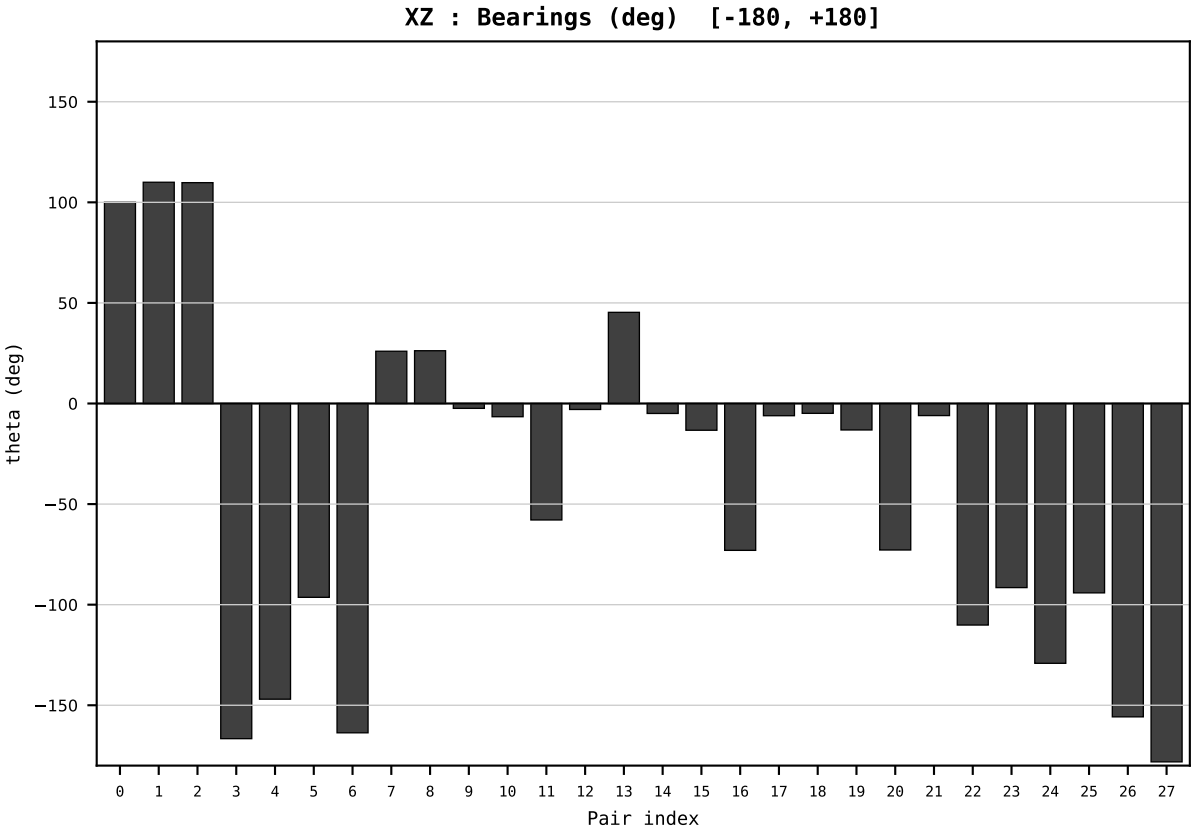
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2009
t : 9 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.505927
Ring = Hs-4
E_metric= 48.3562
kappa_HS= 7638.88
omega = 2.6745 deg
d_A = 0.376406
Helm = Other Fossil
Helm d = -0.237572
DR = 8.941 HLR
DR ratio= 7638.9x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=10 | Year 2010 | D=8 N=26 pairs=28

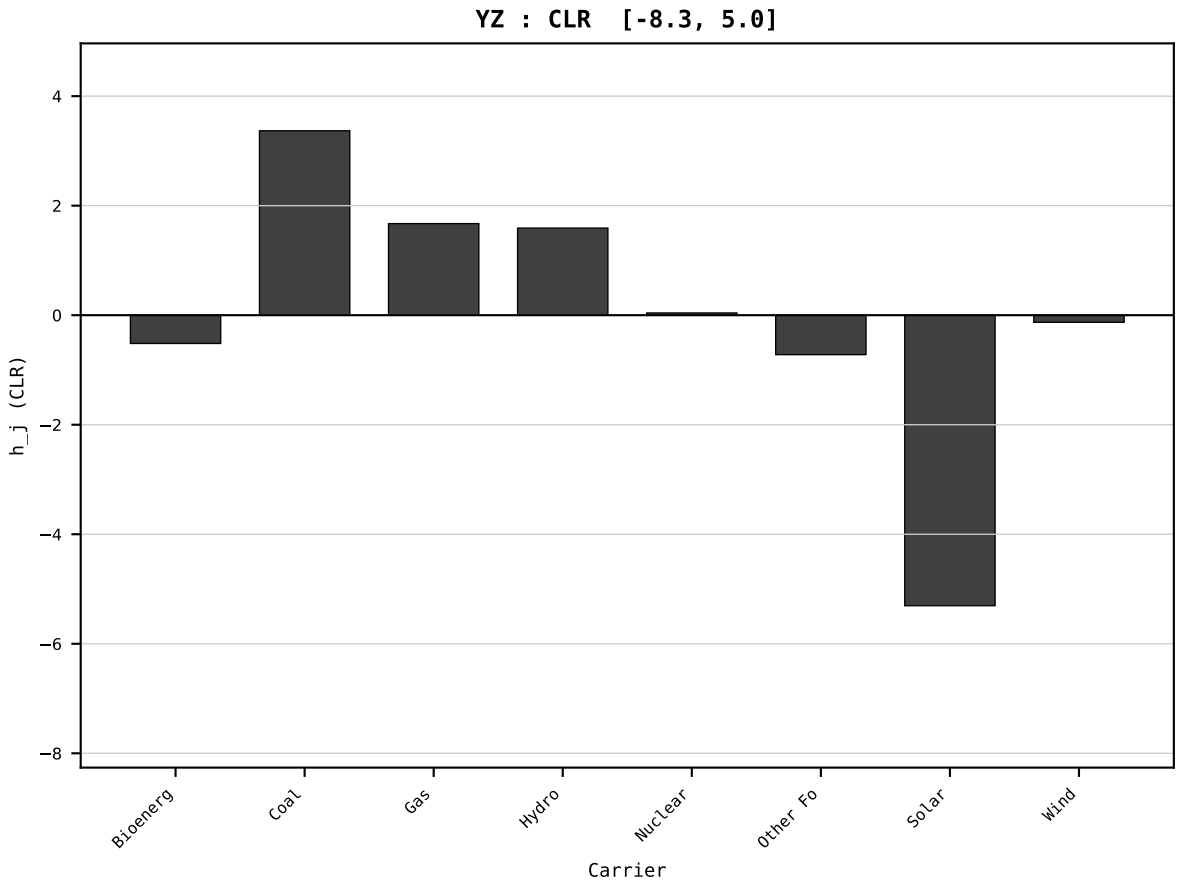
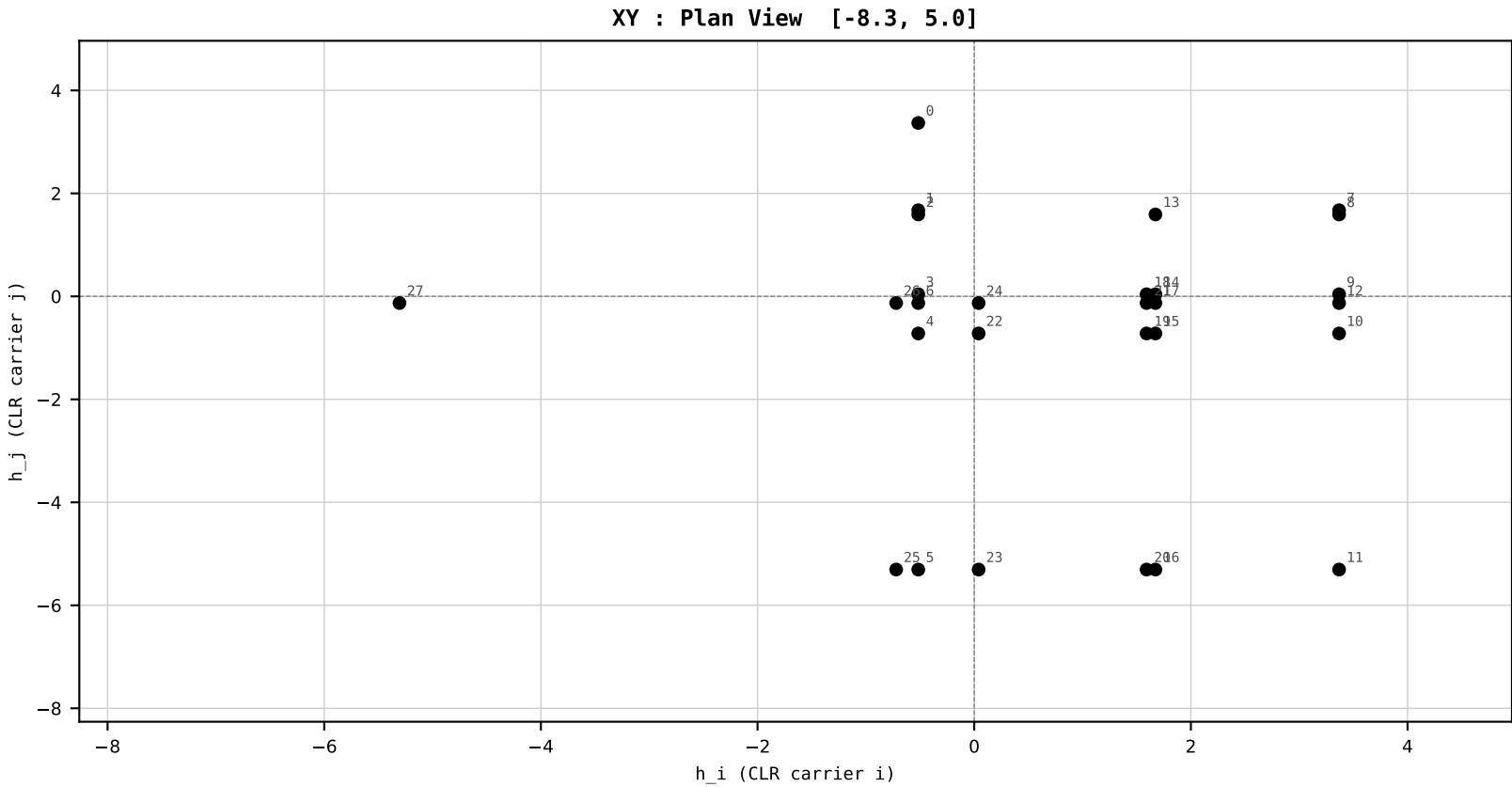
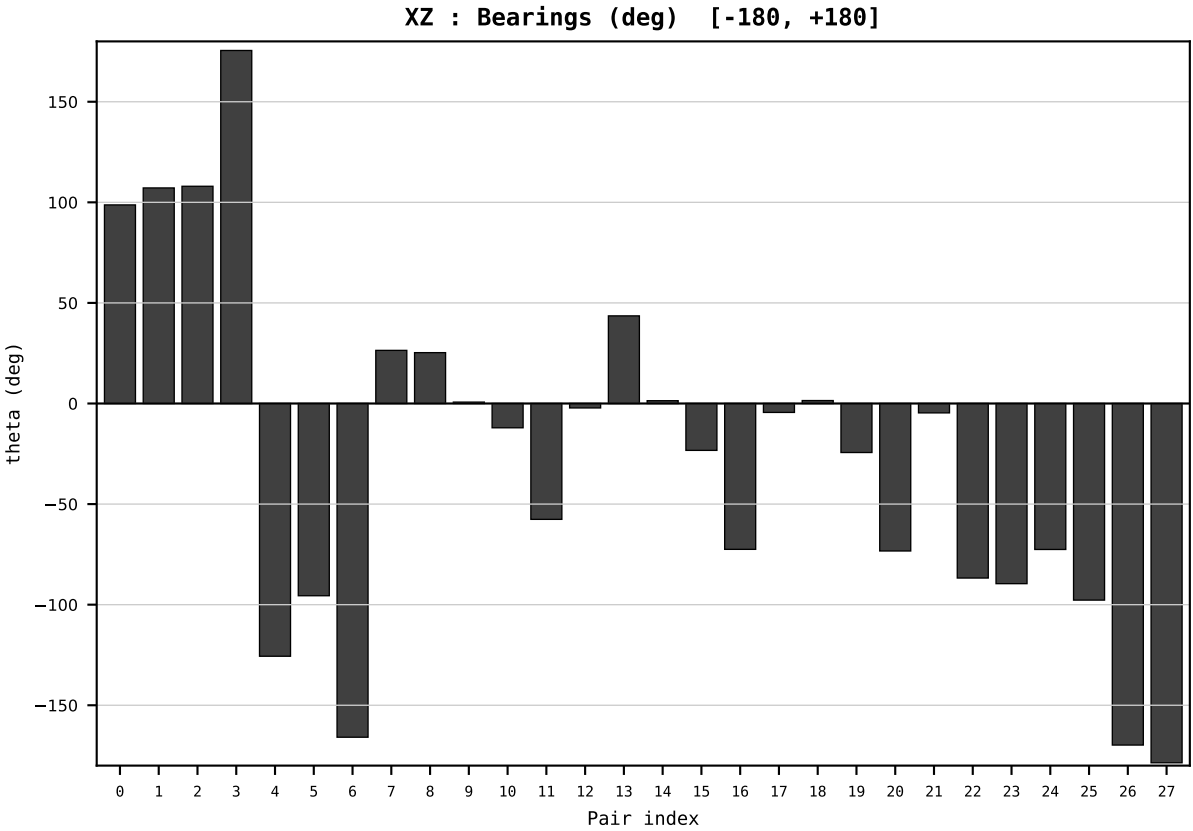
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2010
t : 10 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.493325
Ring = Hs-3
E_metric= 45.6238
kappa_HS= 5845.18
omega = 3.3901 deg
d_A = 0.451796
Helm = Other Fossil
Helm d = -0.323553
DR = 8.673 HLR
DR ratio= 5845.2x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=11 | Year 2011 | D=8 N=26 pairs=28

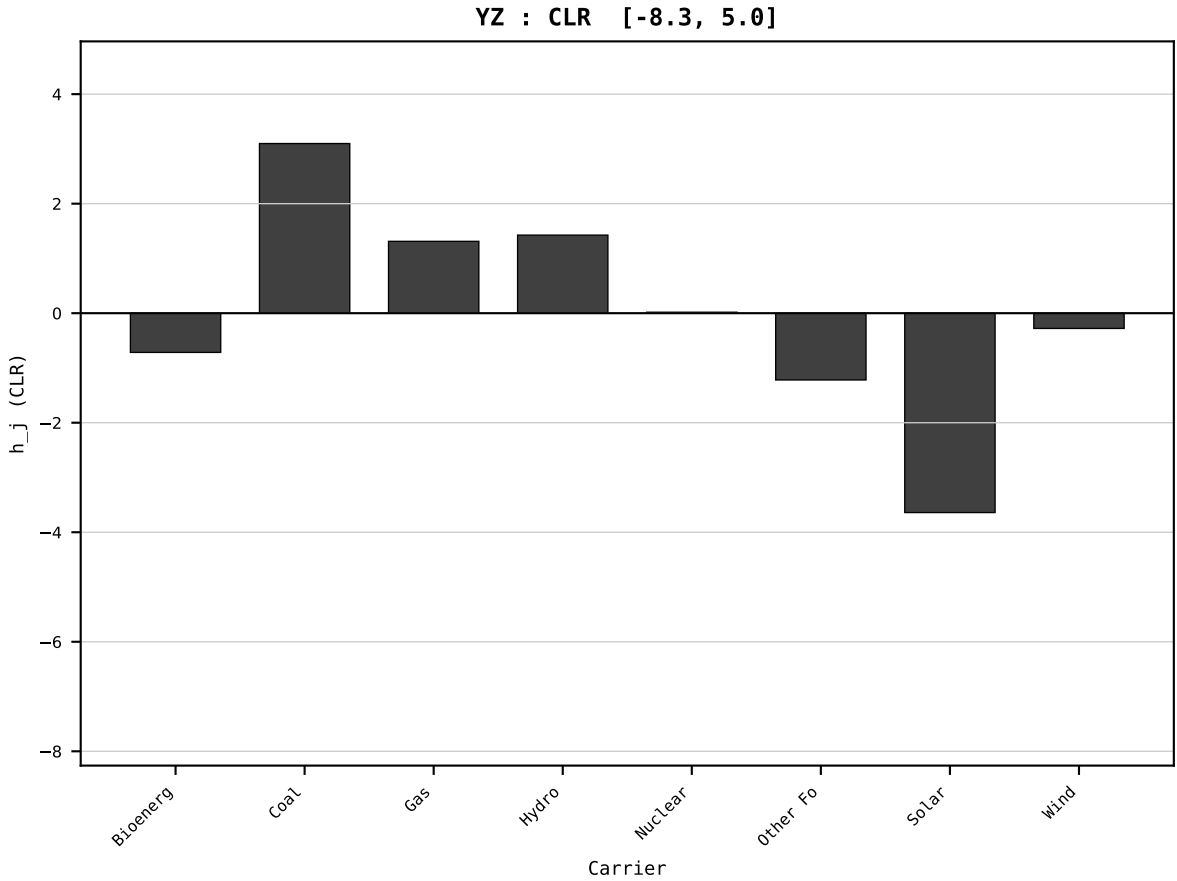
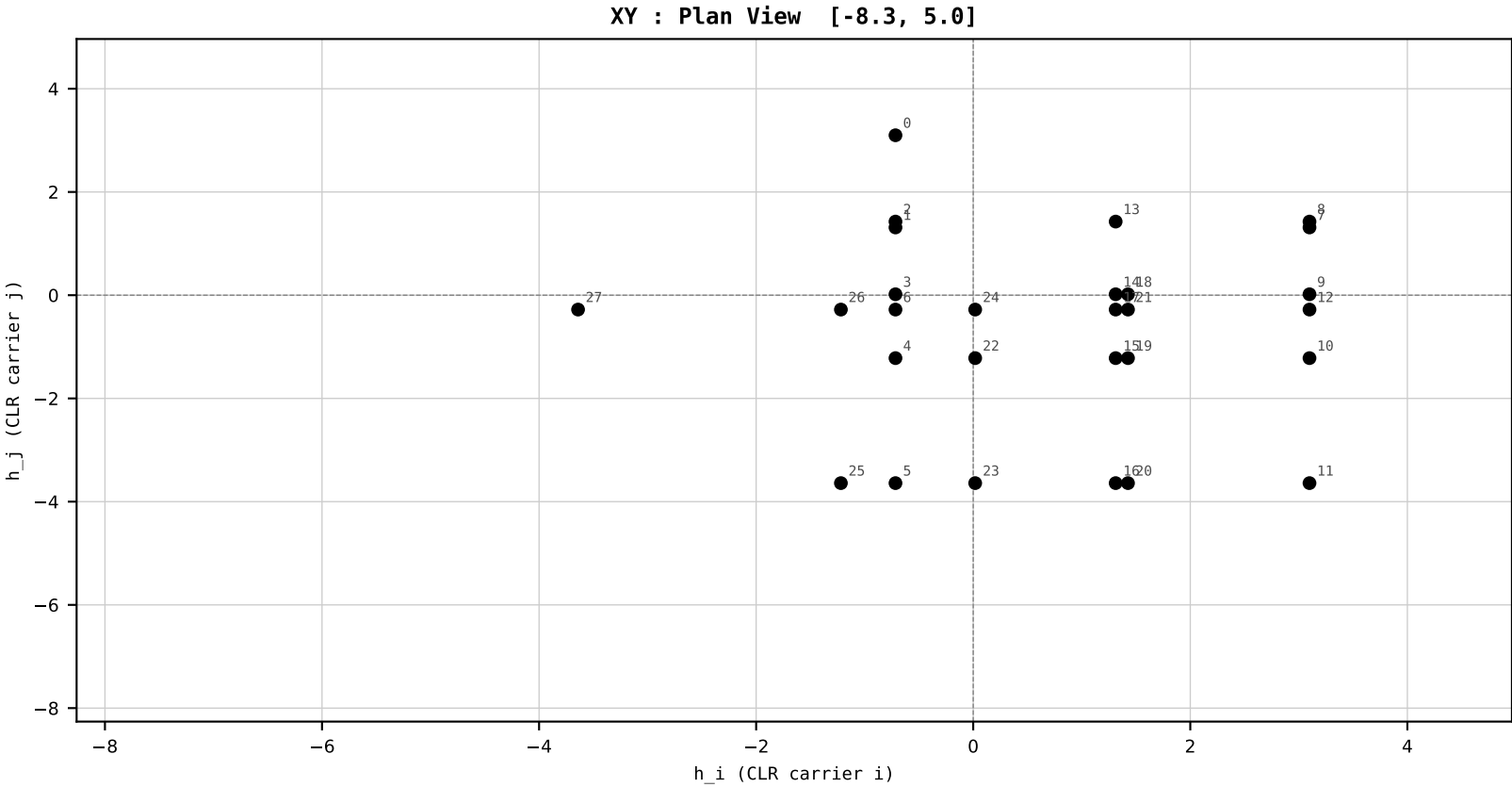
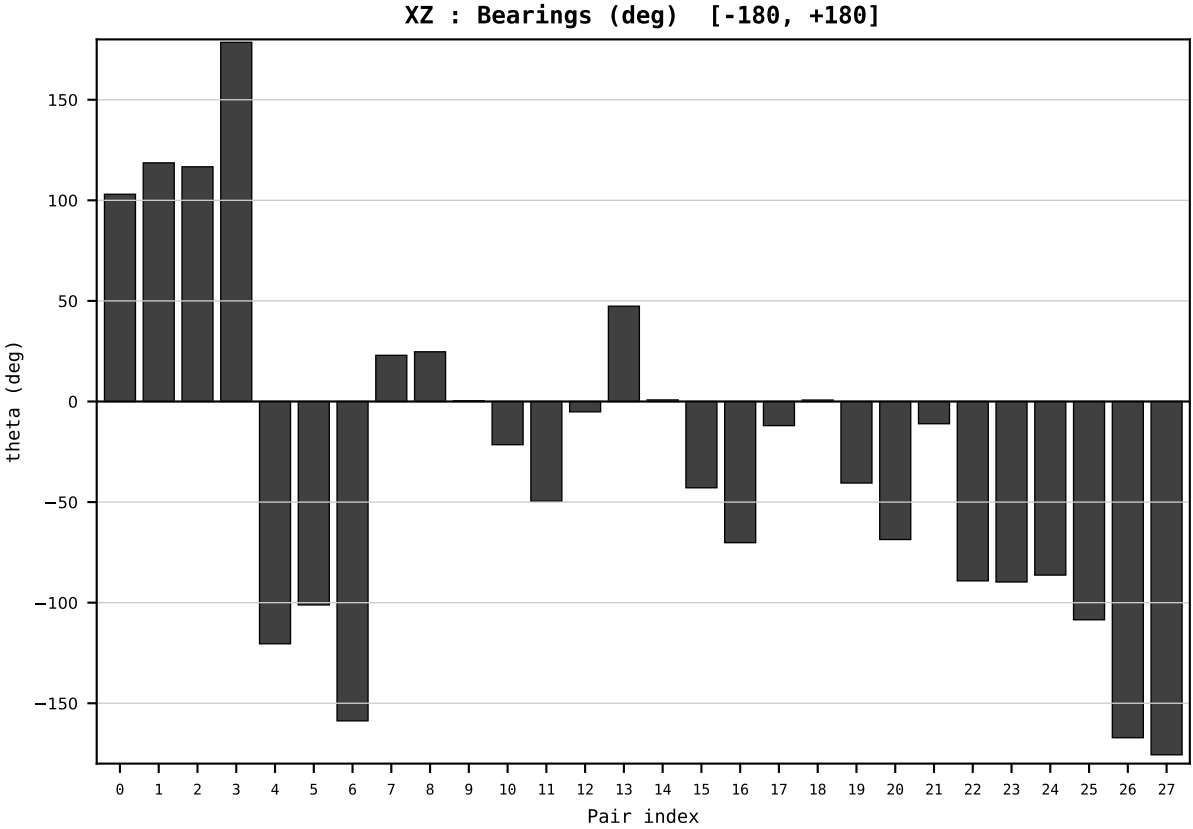
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2011
t : 11 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.480839
Ring = Hs-3
E_metric= 28.6888
kappa_HS= 844.88
omega = 11.1115 deg
d_A = 1.819832
Helm = Solar
Helm d = +1.664605
DR = 6.739 HLR
DR ratio= 844.9x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=12 | Year 2012 | D=8 N=26 pairs=28

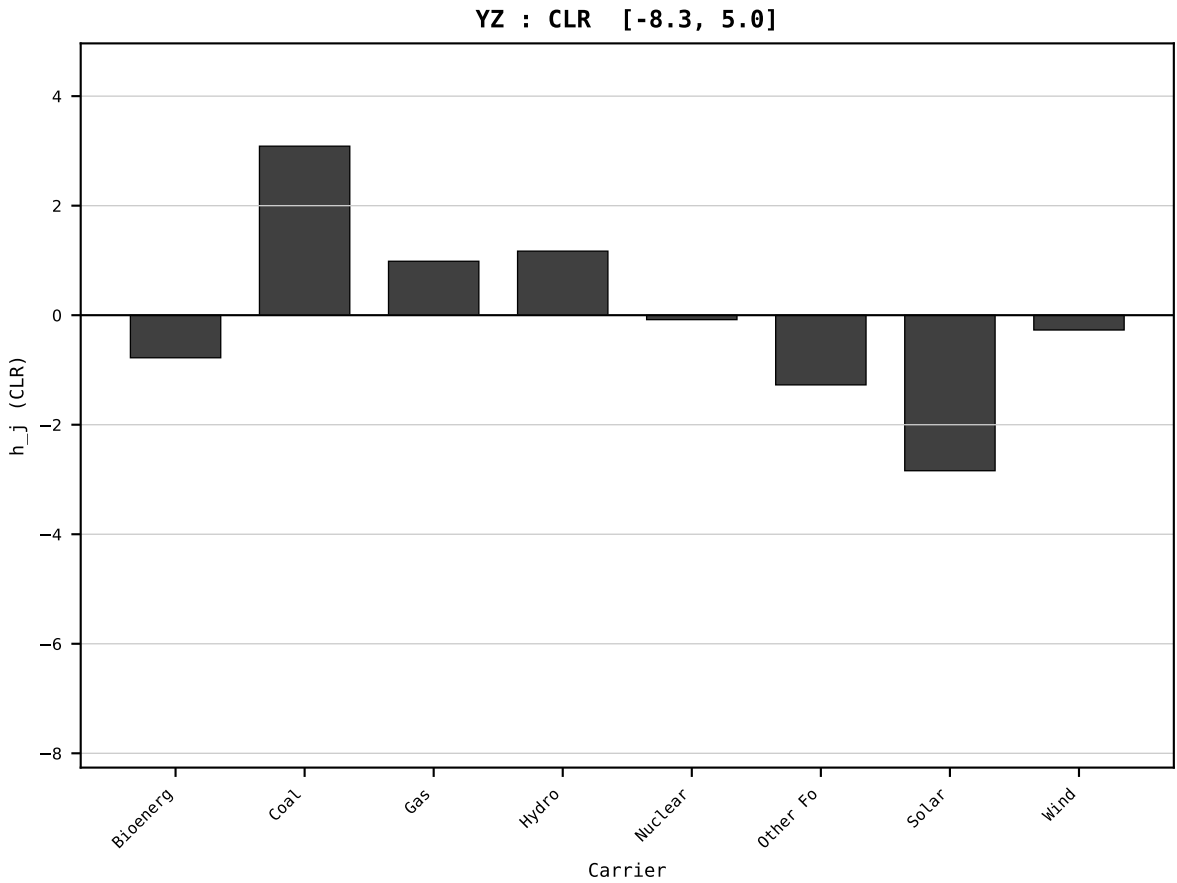
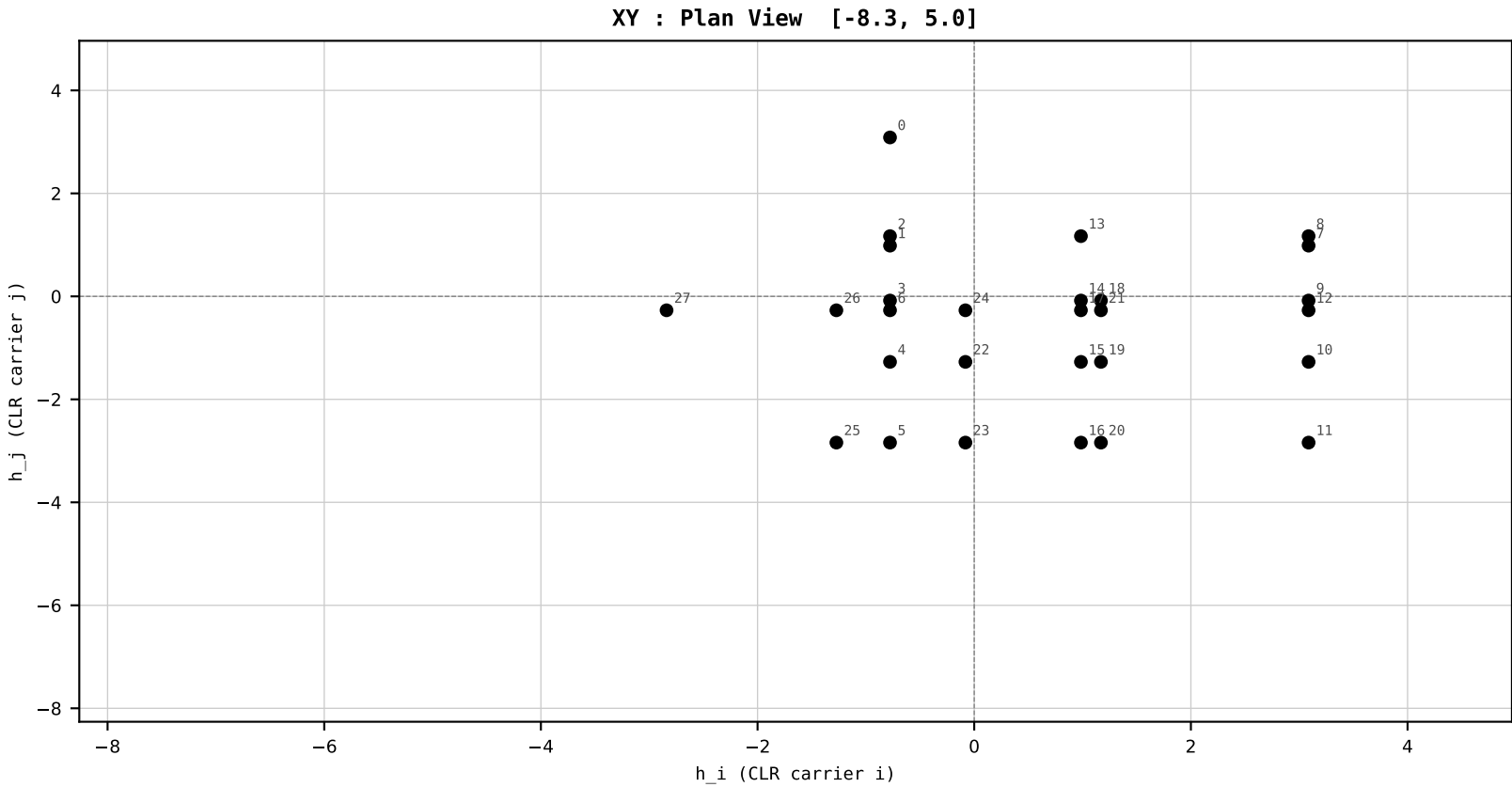
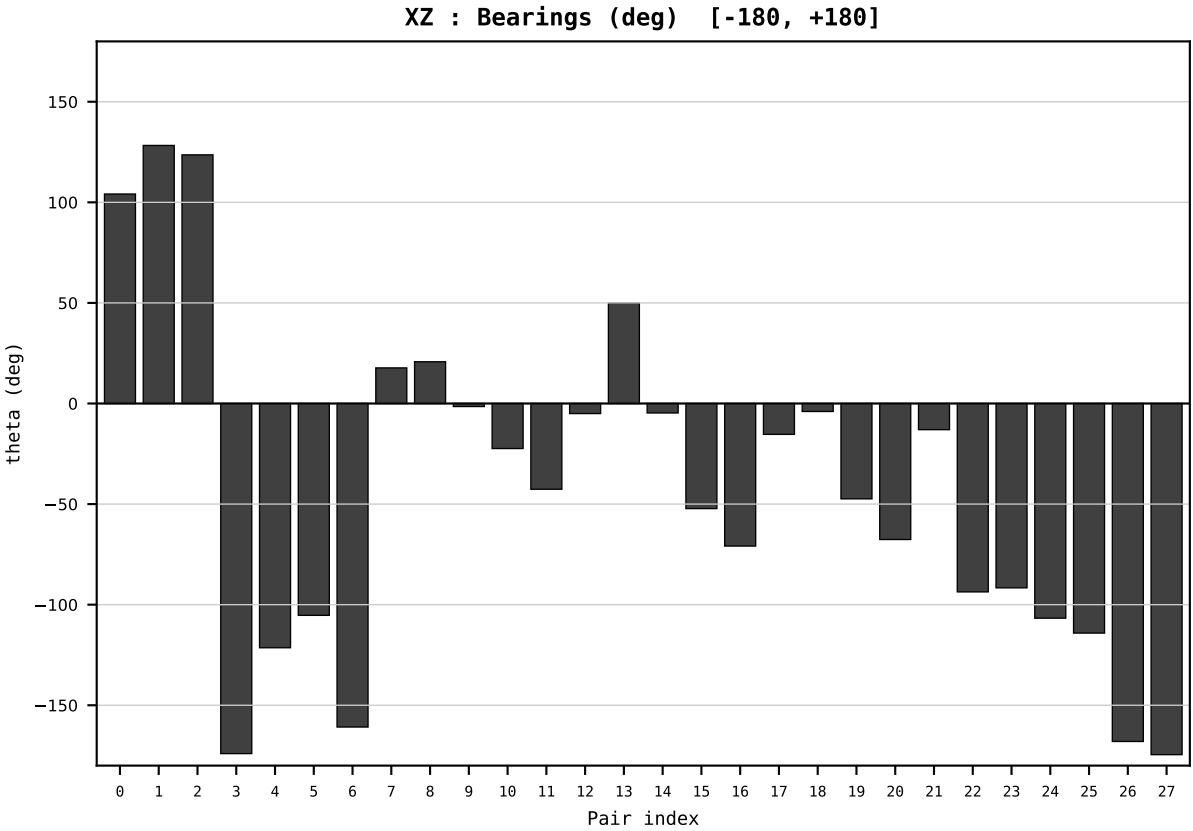
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2012
t : 12 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.516572
Ring = Hs-4
E_metric= 22.2333
kappa_HS= 374.82
omega = 7.3940 deg
d_A = 0.911514
Helm = Solar
Helm d = +0.800856
DR = 5.926 HLR
DR ratio= 374.8x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=13 | Year 2013 | D=8 N=26 pairs=28

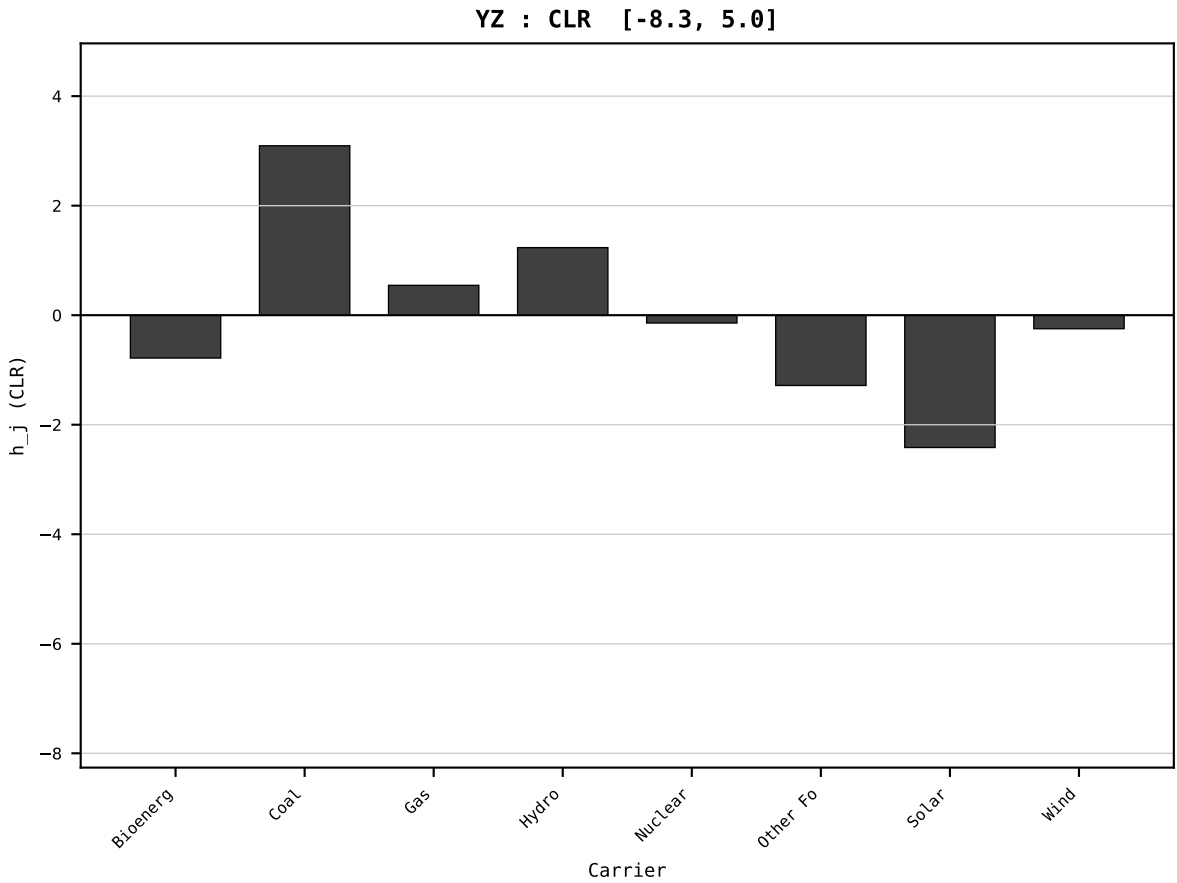
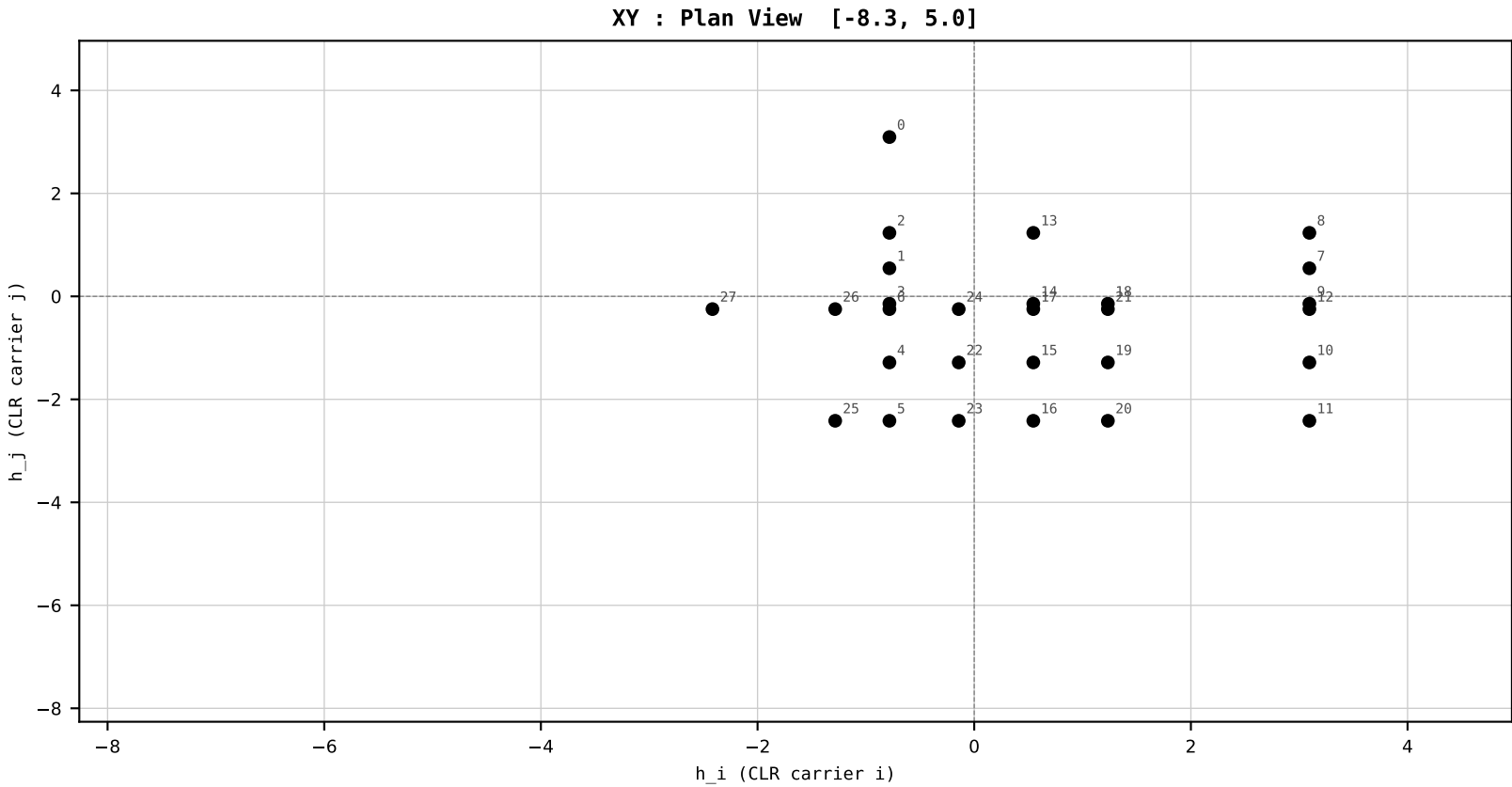
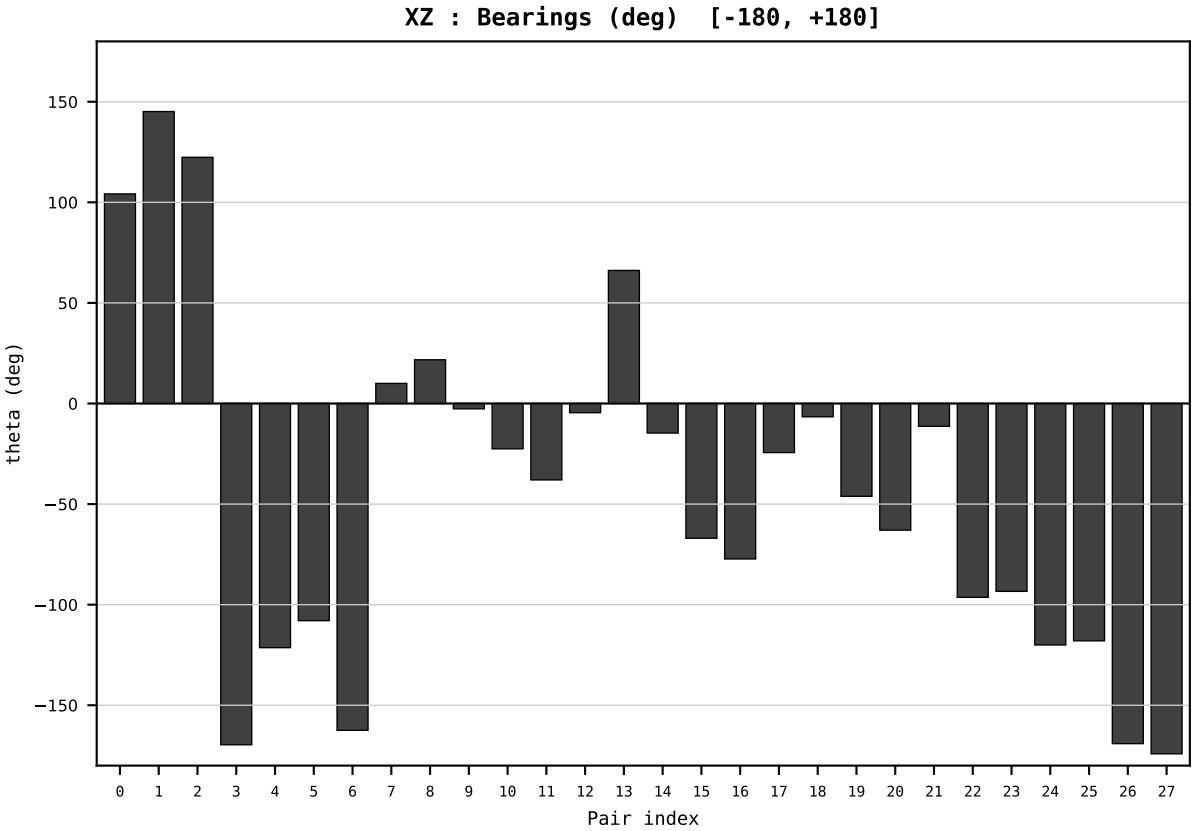
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2013
t : 13 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.538501
Ring = Hs-4
E_metric= 19.5721
kappa_HS= 247.22
omega = 6.8358 deg
d_A = 0.617545
Helm = Gas
Helm d = -0.439615
DR = 5.510 HLR
DR ratio= 247.2x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=14 | Year 2014 | D=8 N=26 pairs=28

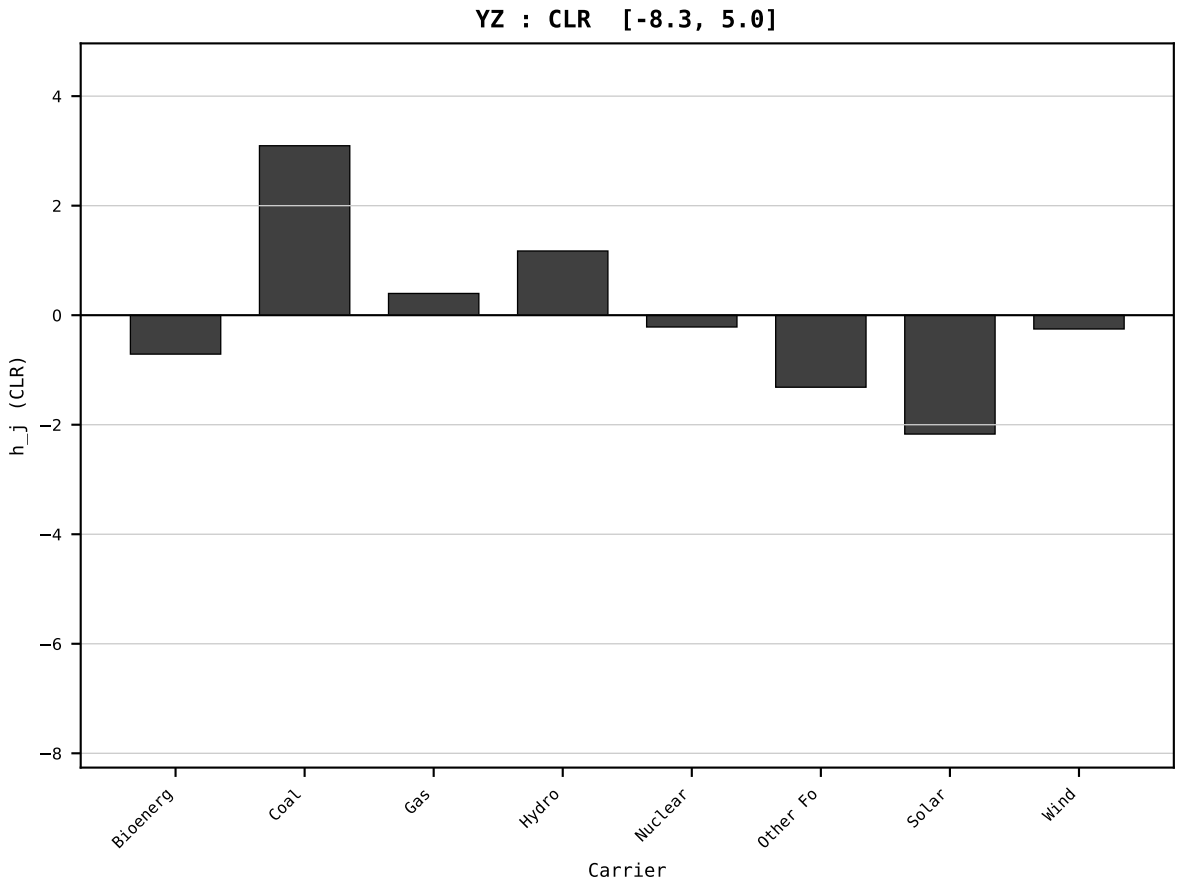
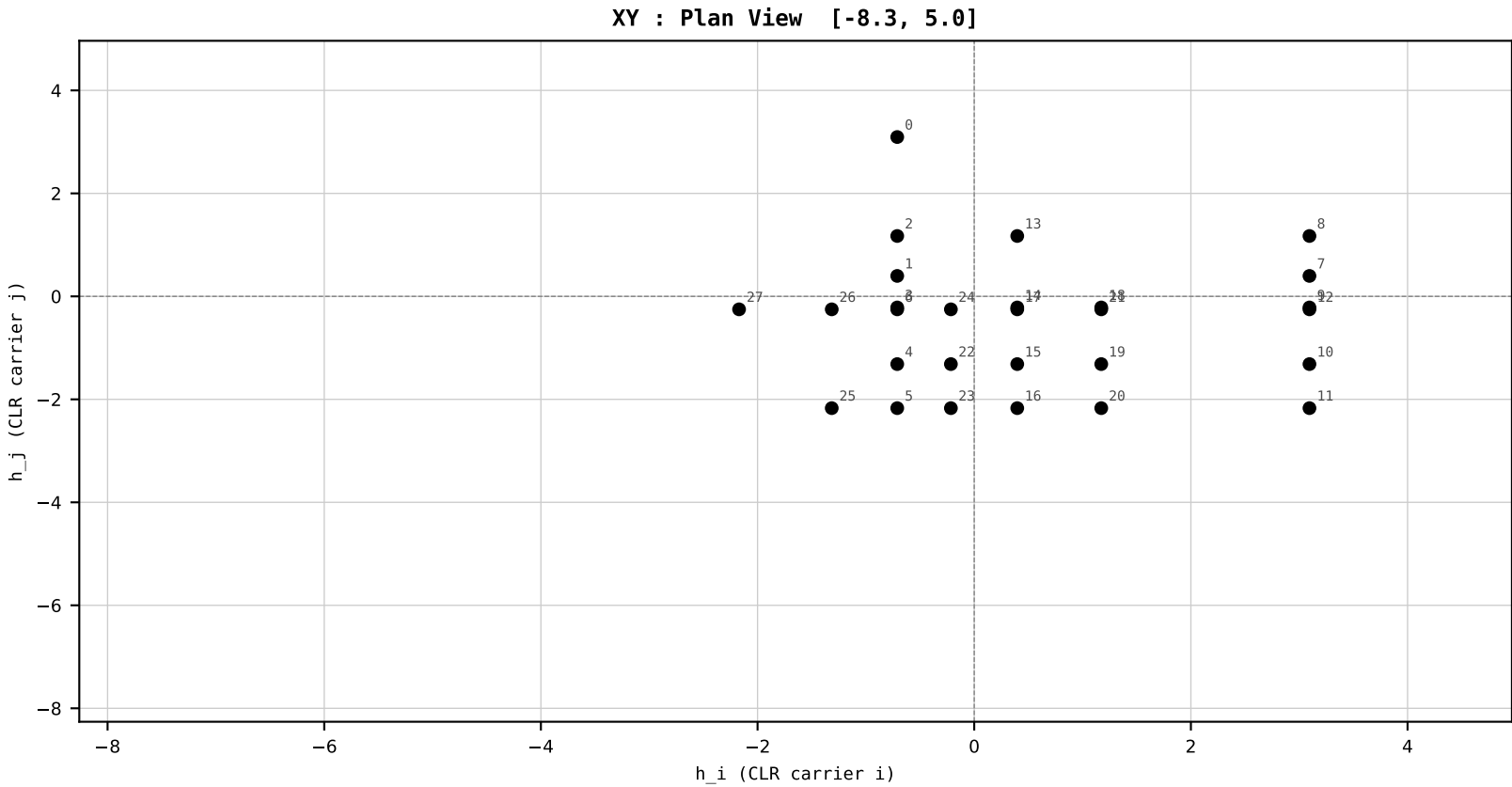
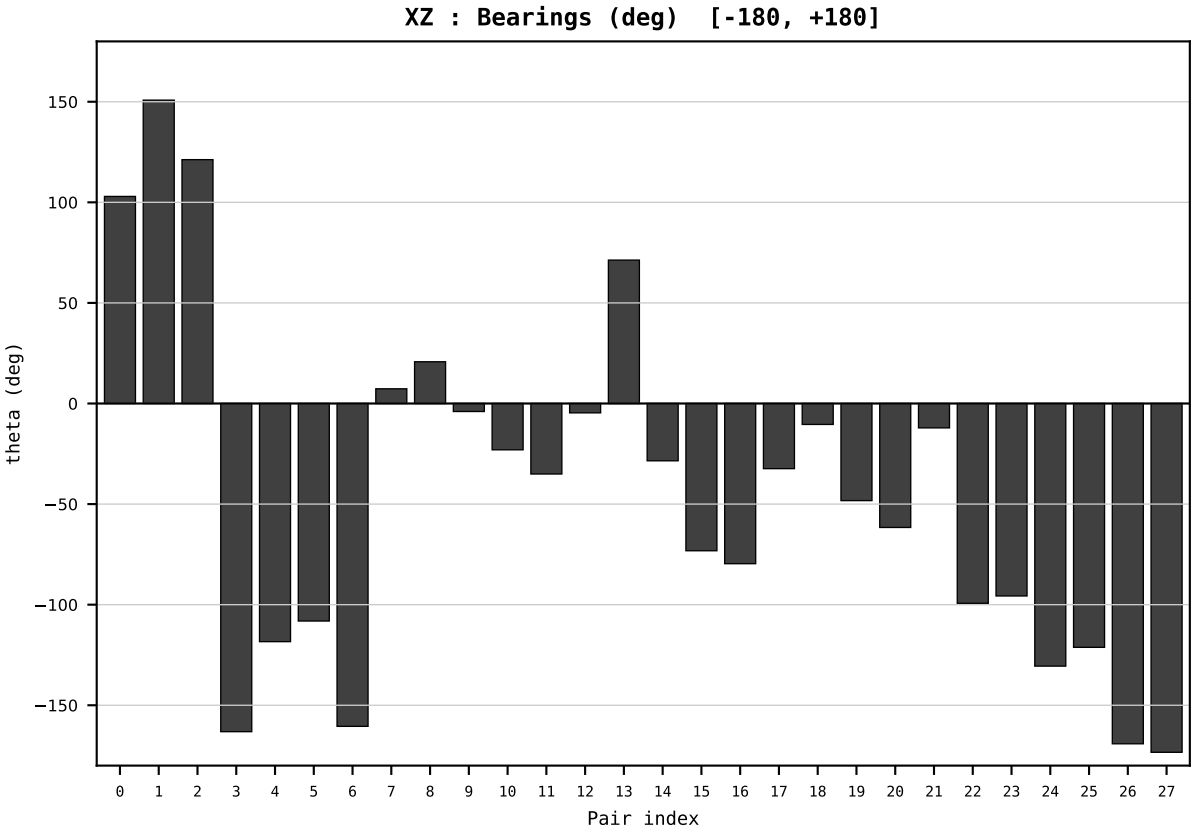
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2014
t : 14 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.549920
Ring = Hs-4
E_metric= 18.1649
kappa_HS= 193.47
omega = 3.5245 deg
d_A = 0.312363
Helm = Solar
Helm d = +0.245715
DR = 5.265 HLR
DR ratio= 193.5x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=15 | Year 2015 | D=8 N=26 pairs=28

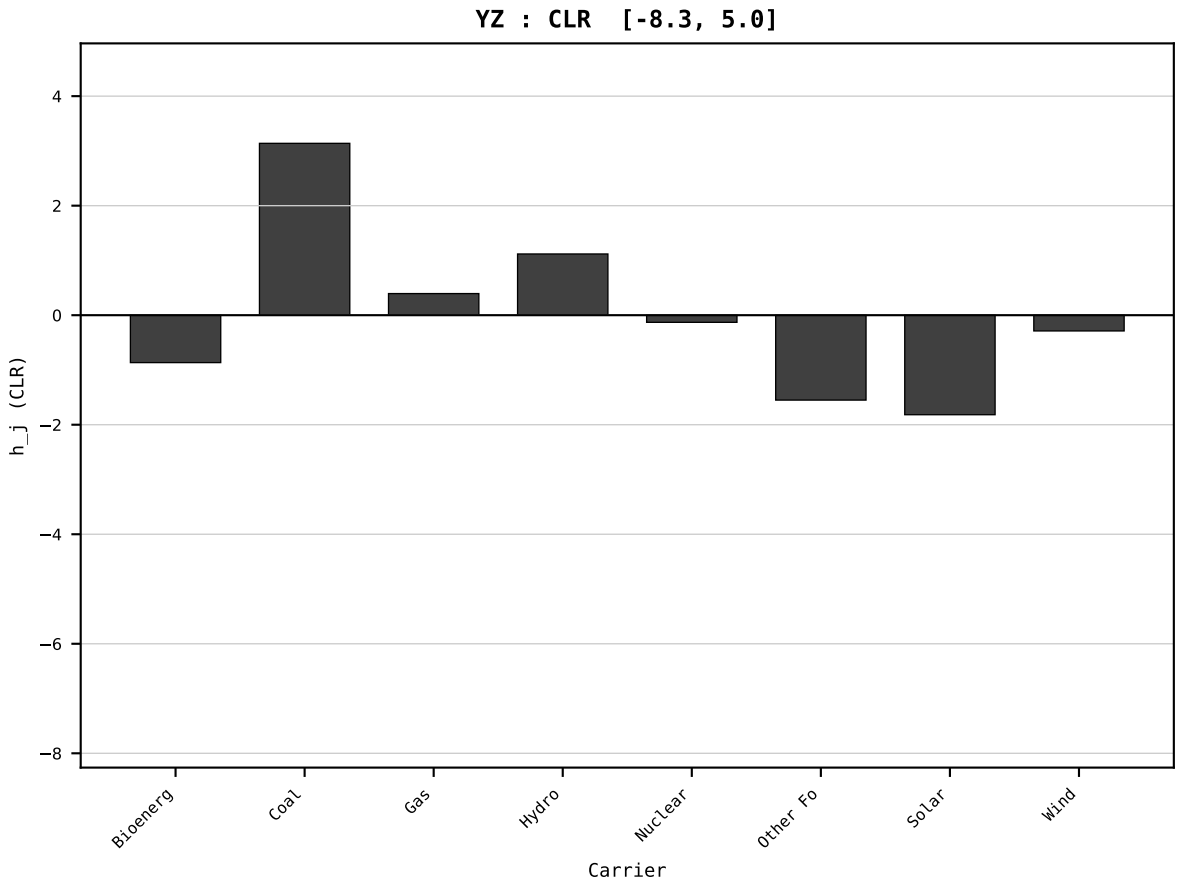
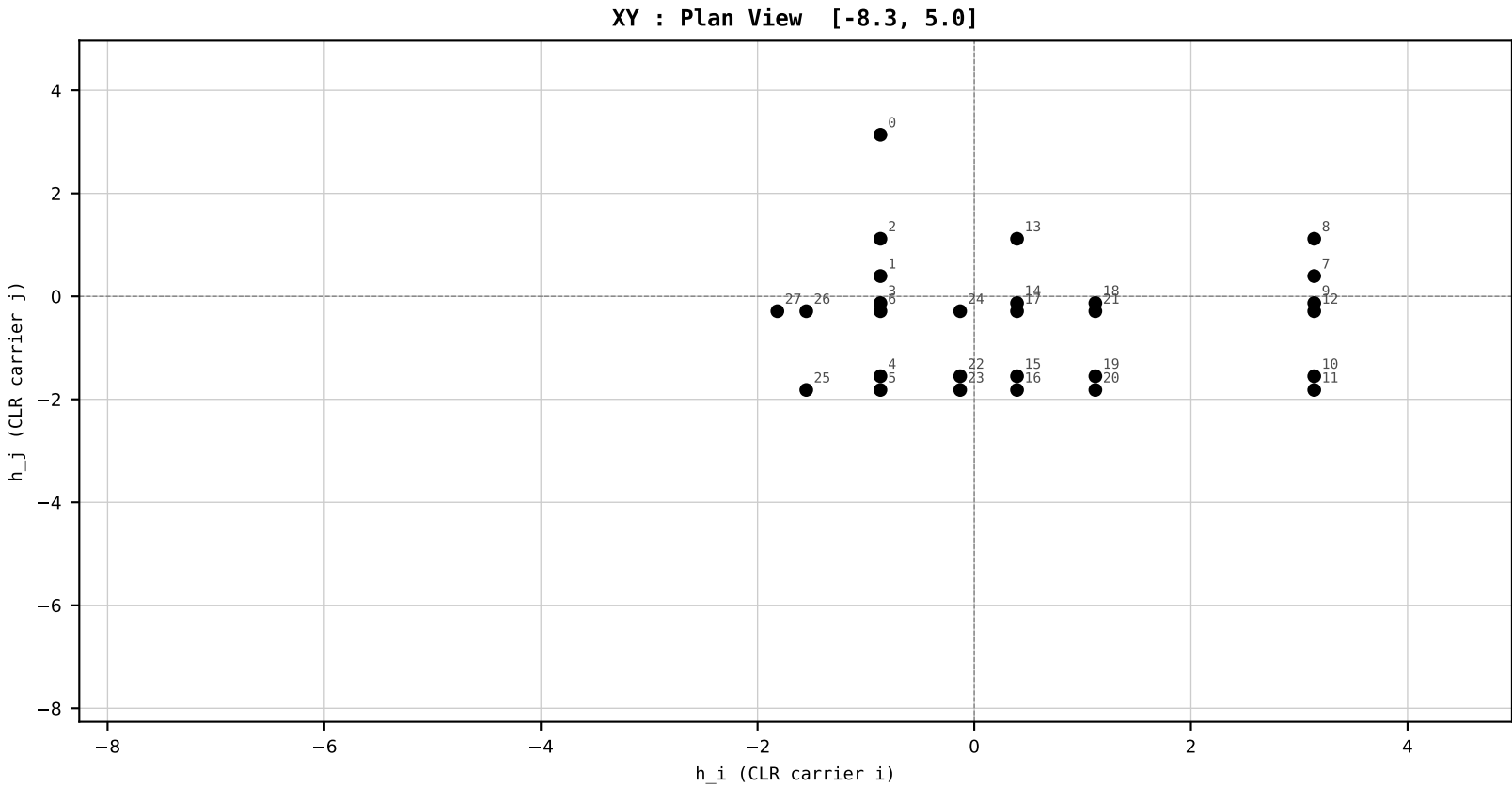
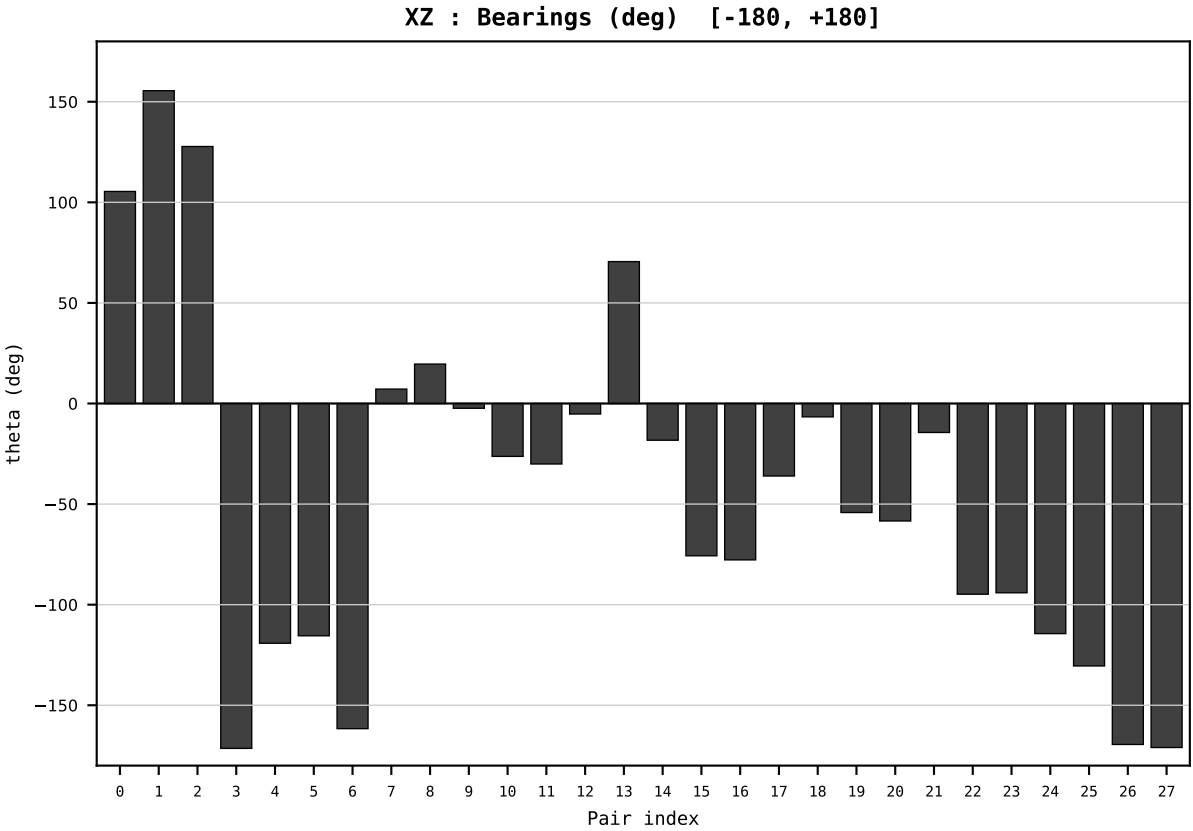
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2015
t : 15 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.566019
Ring = Hs-4
E_metric= 17.8098
kappa_HS= 141.97
omega = 6.2793 deg
d_A = 0.466443
Helm = Solar
Helm d = +0.353359
DR = 4.956 HLR
DR ratio= 142.0x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=16 | Year 2016 | D=8 N=26 pairs=28

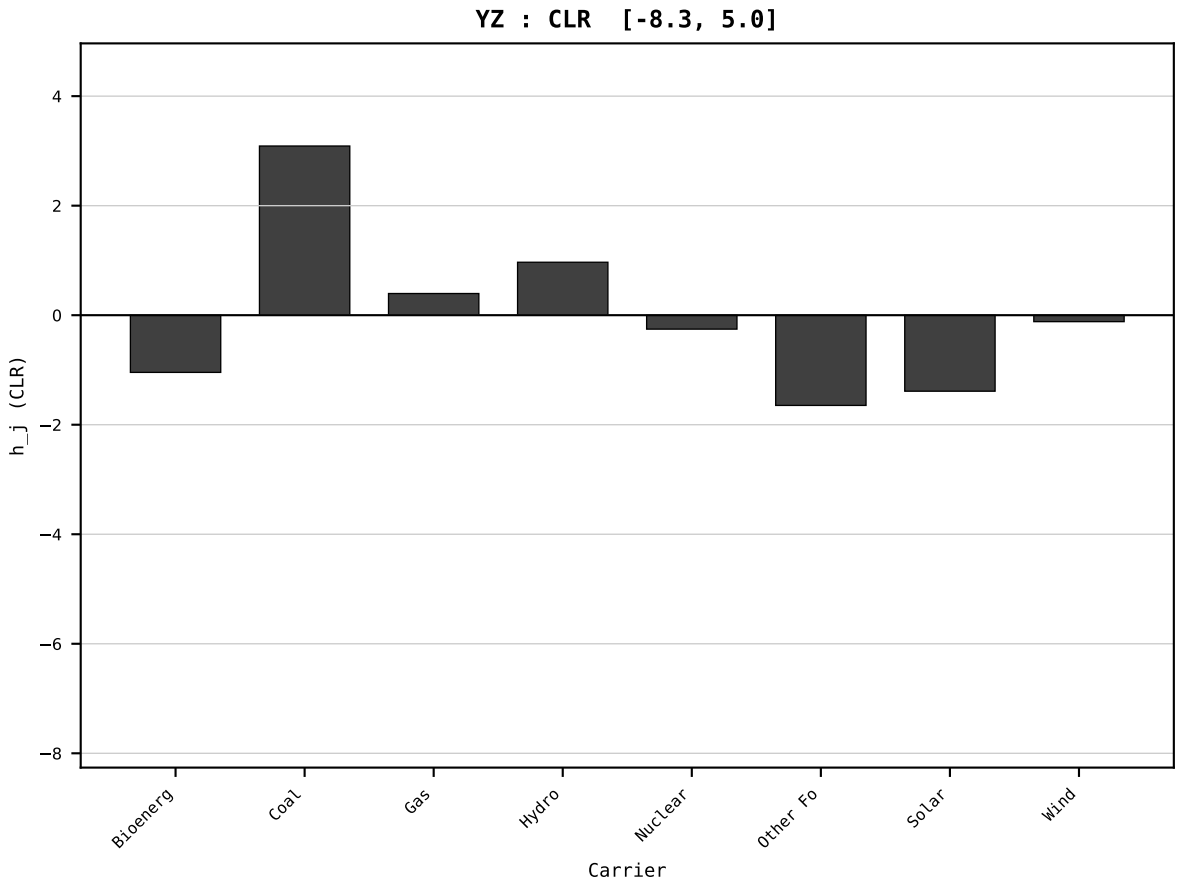
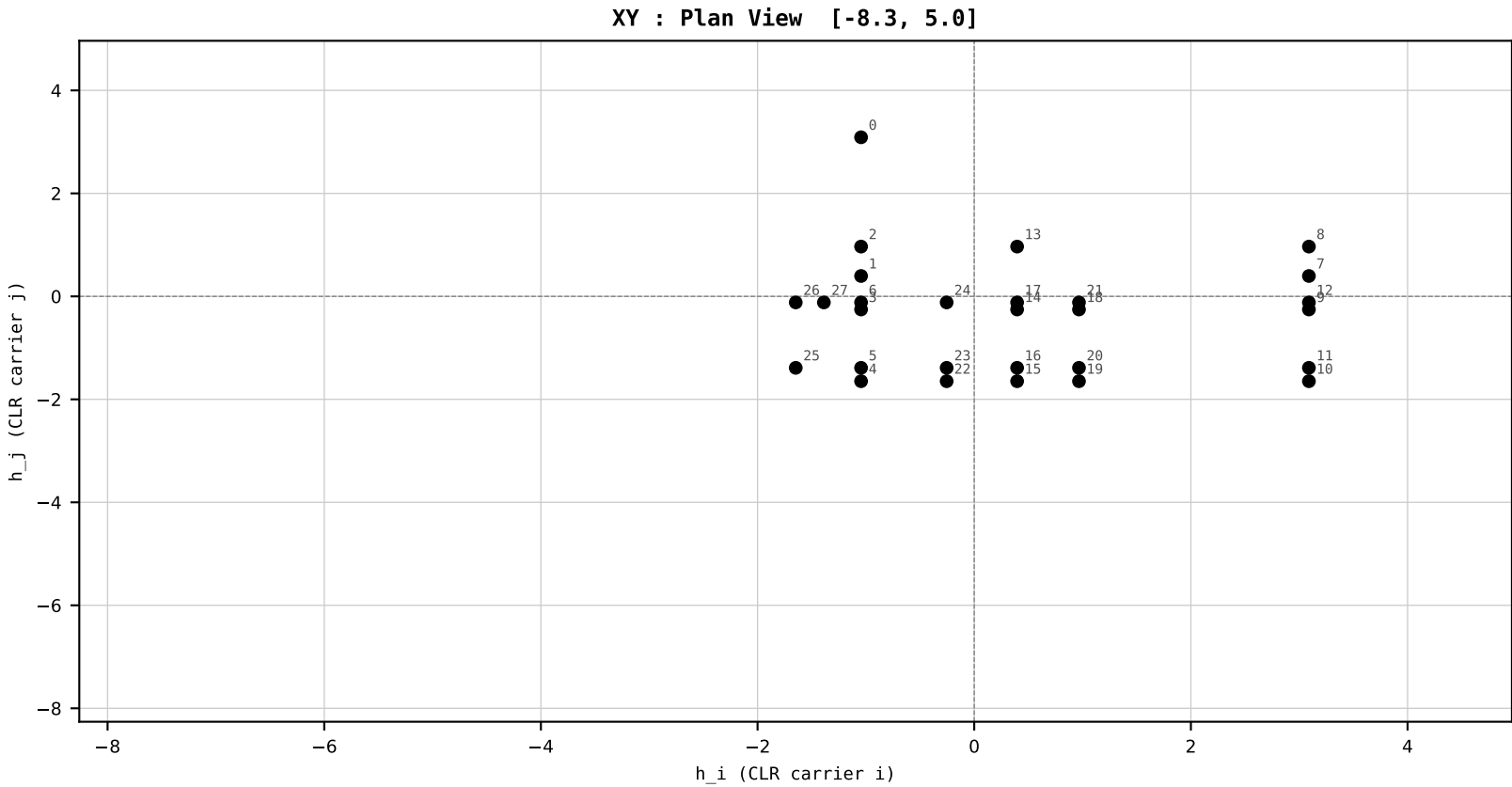
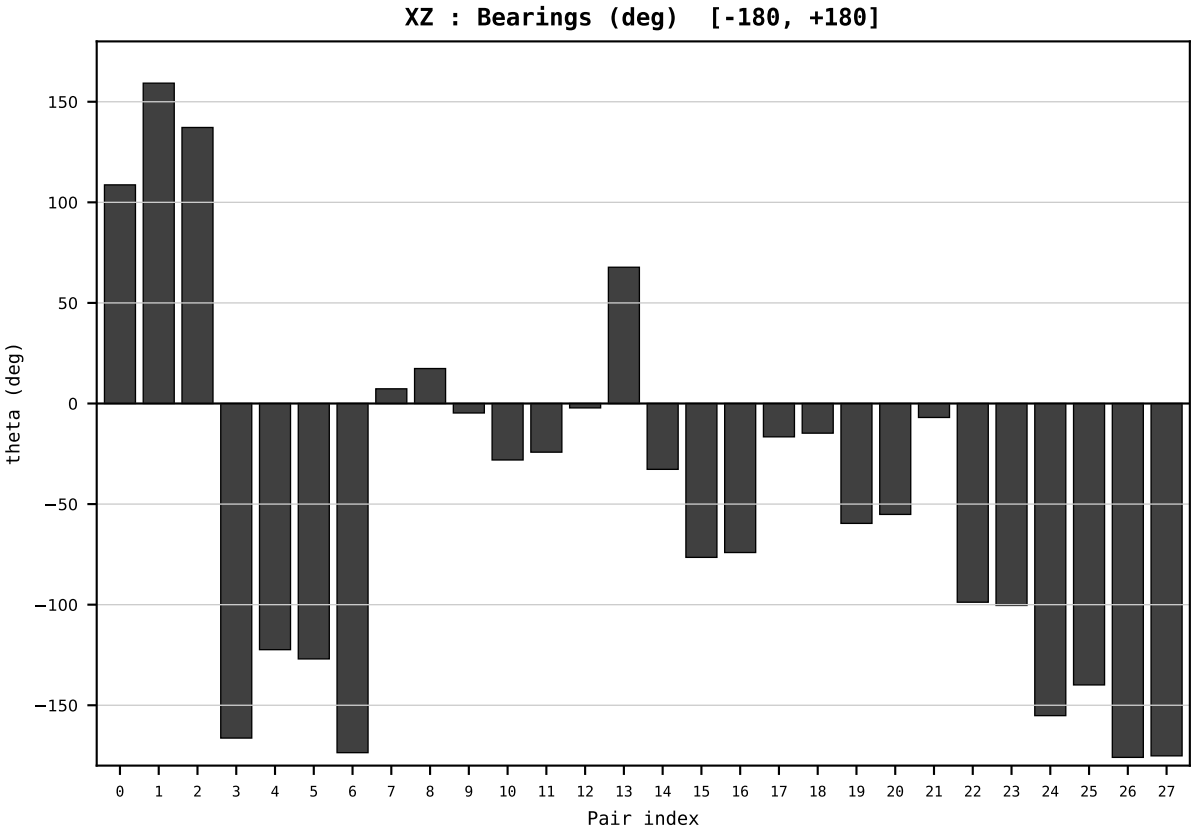
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2016
t : 16 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.562151
Ring = Hs-4
E_metric= 16.4465
kappa_HS= 114.09
omega = 7.1743 deg
d_A = 0.543254
Helm = Solar
Helm d = +0.429368
DR = 4.737 HLR
DR ratio= 114.1x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=17 | Year 2017 | D=8 N=26 pairs=28

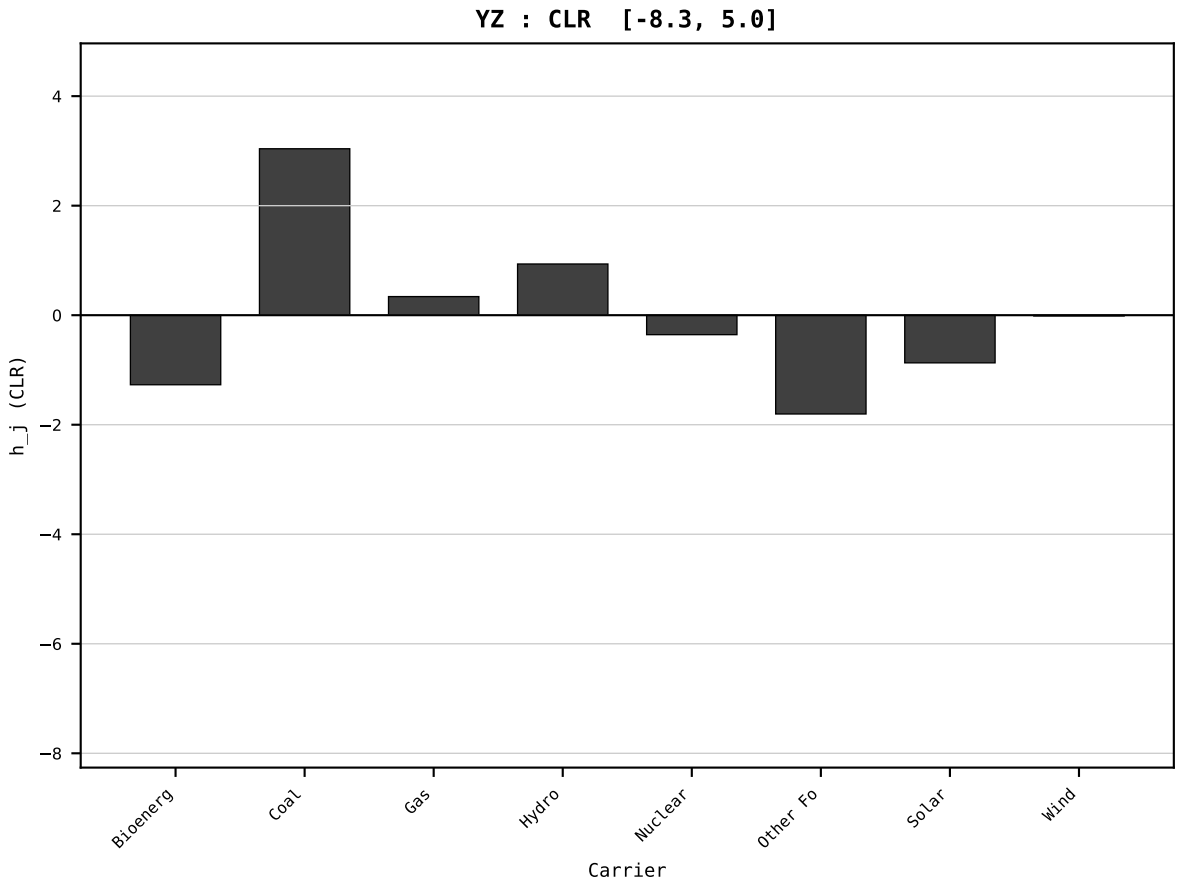
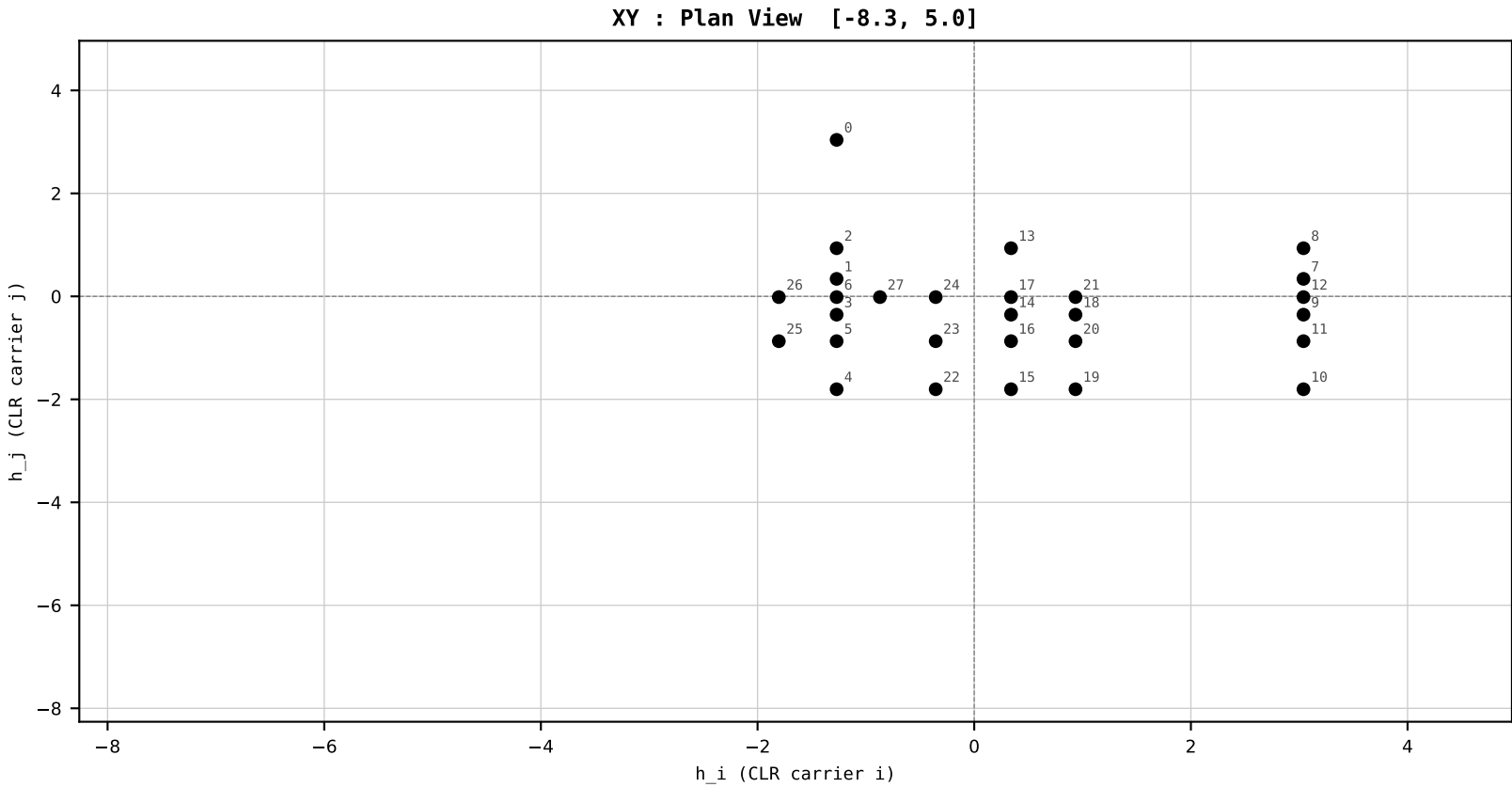
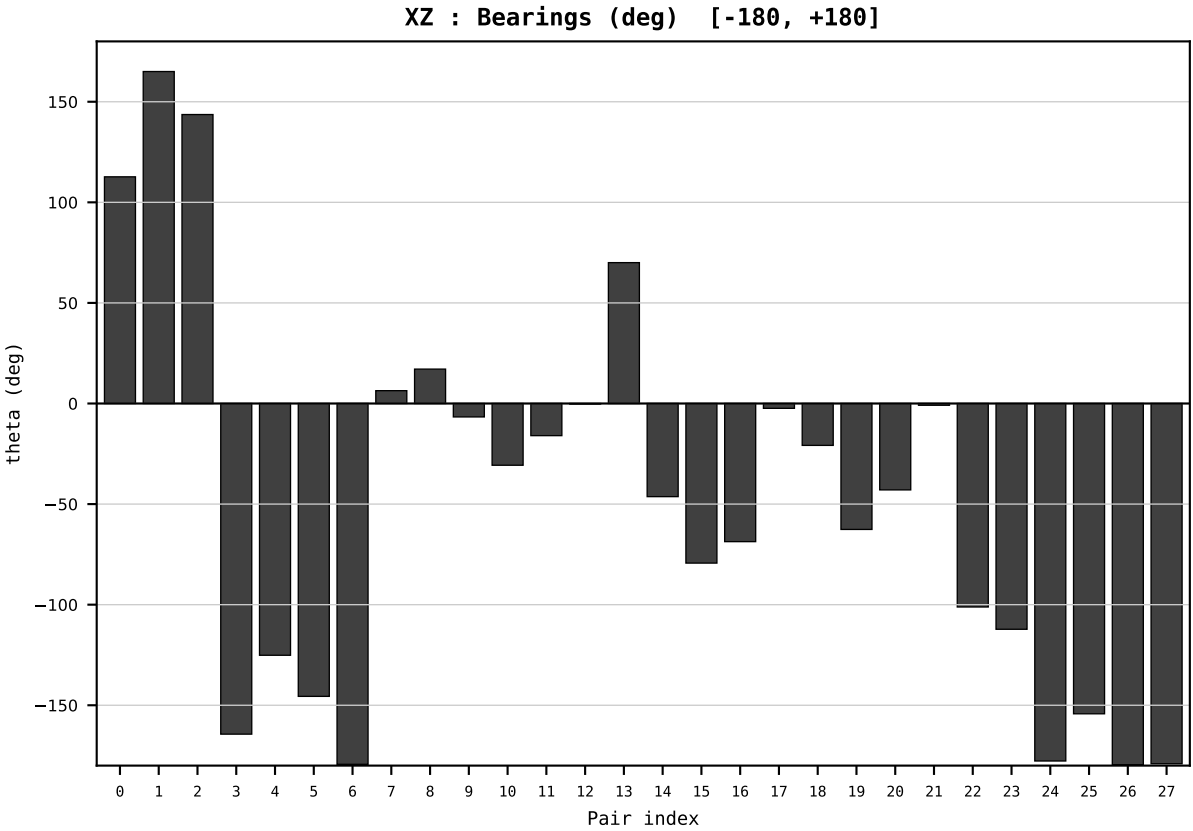
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2017
t : 17 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.550408
Ring = Hs-4
E_metric= 15.9784
kappa_HS= 126.88
omega = 8.6362 deg
d_A = 0.609086
Helm = Solar
Helm d = +0.517691
DR = 4.843 HLR
DR ratio= 126.9x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=18 | Year 2018 | D=8 N=26 pairs=28

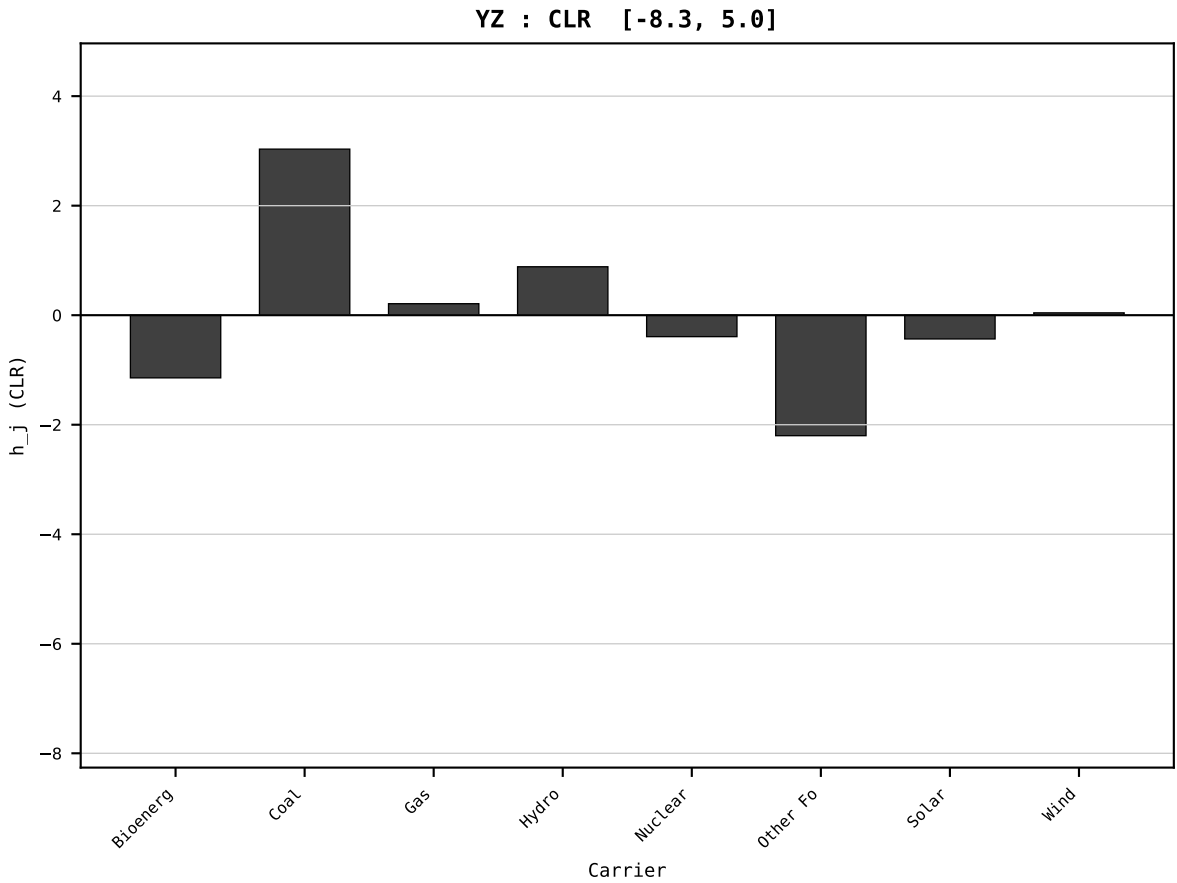
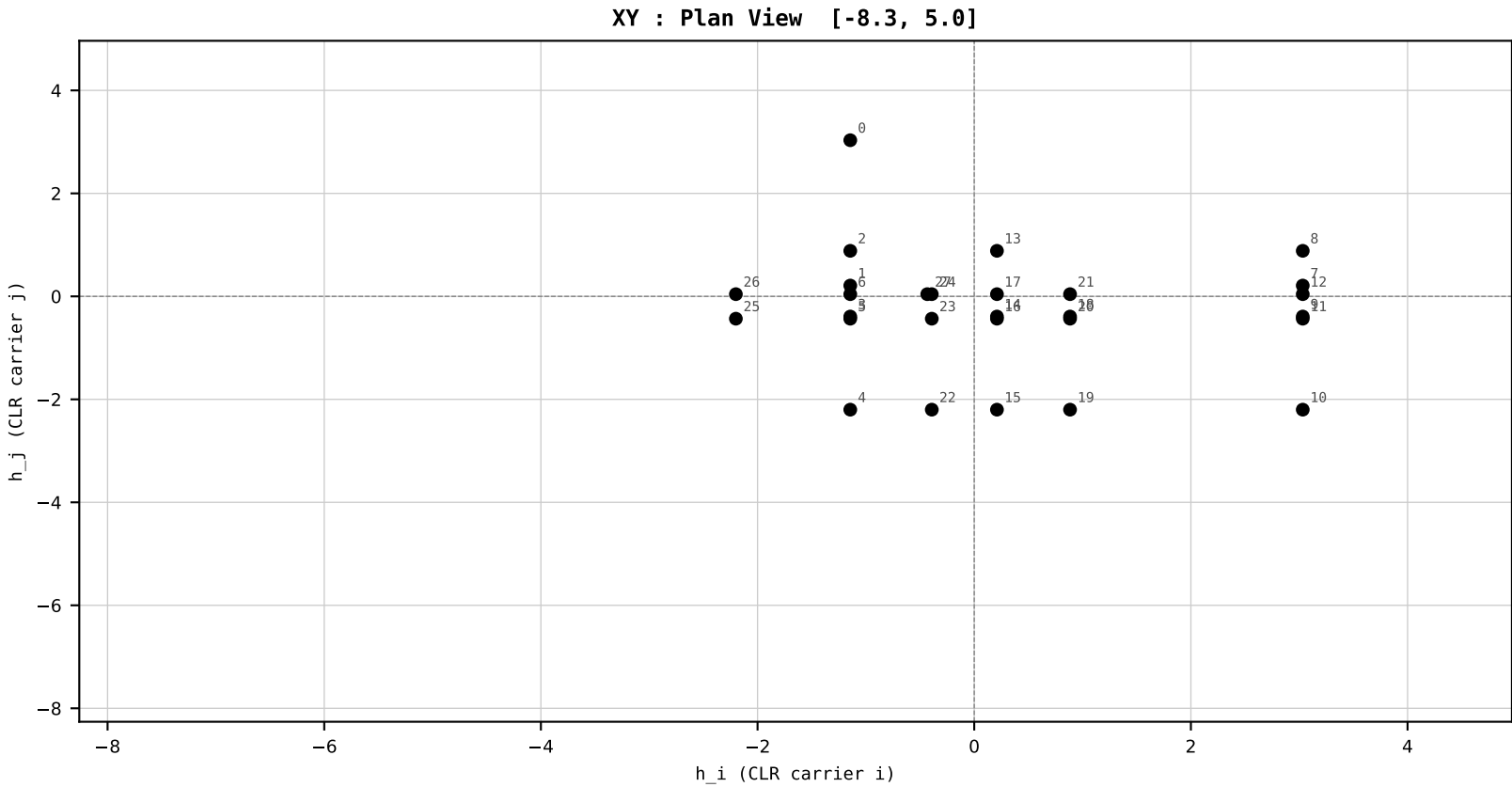
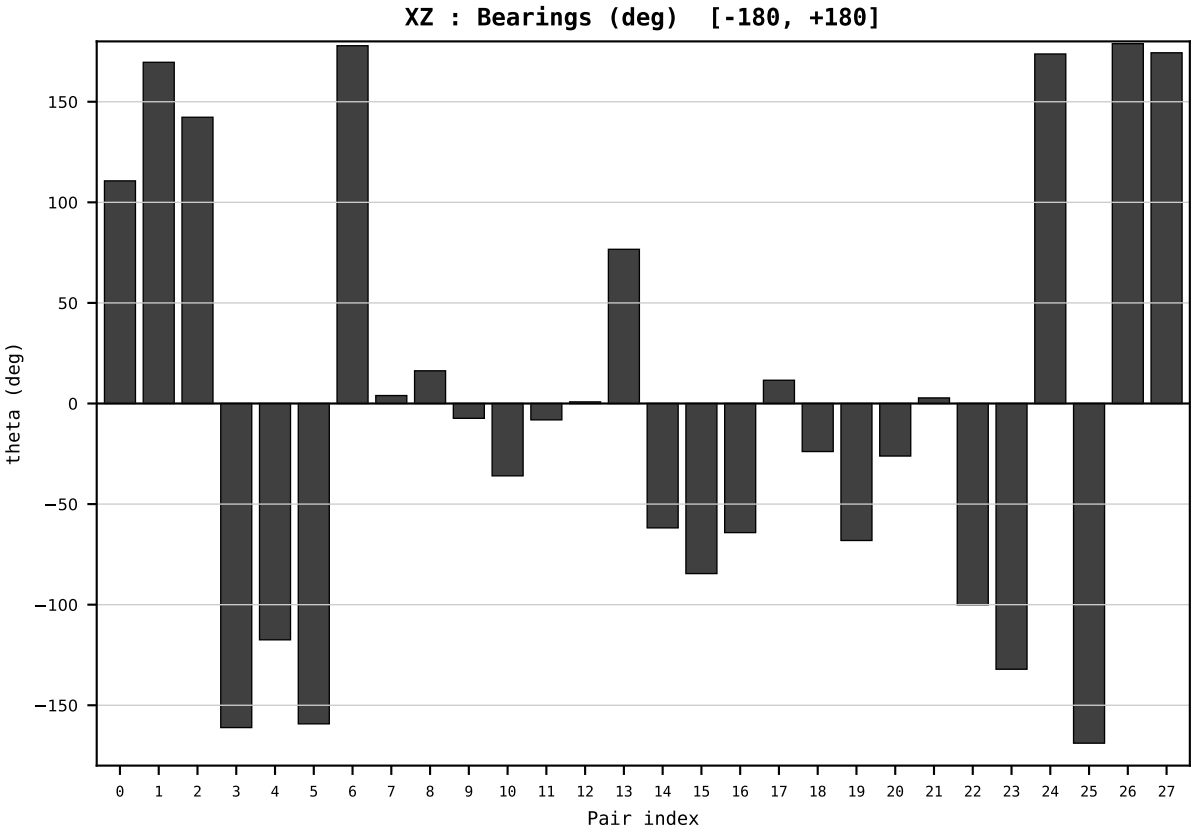
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2018
t : 18 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.546713
Ring = Hs-4
E_metric= 16.5164
kappa_HS= 187.32
omega = 8.8095 deg
d_A = 0.622688
Helm = Solar
Helm d = +0.436690
DR = 5.233 HLR
DR ratio= 187.3x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



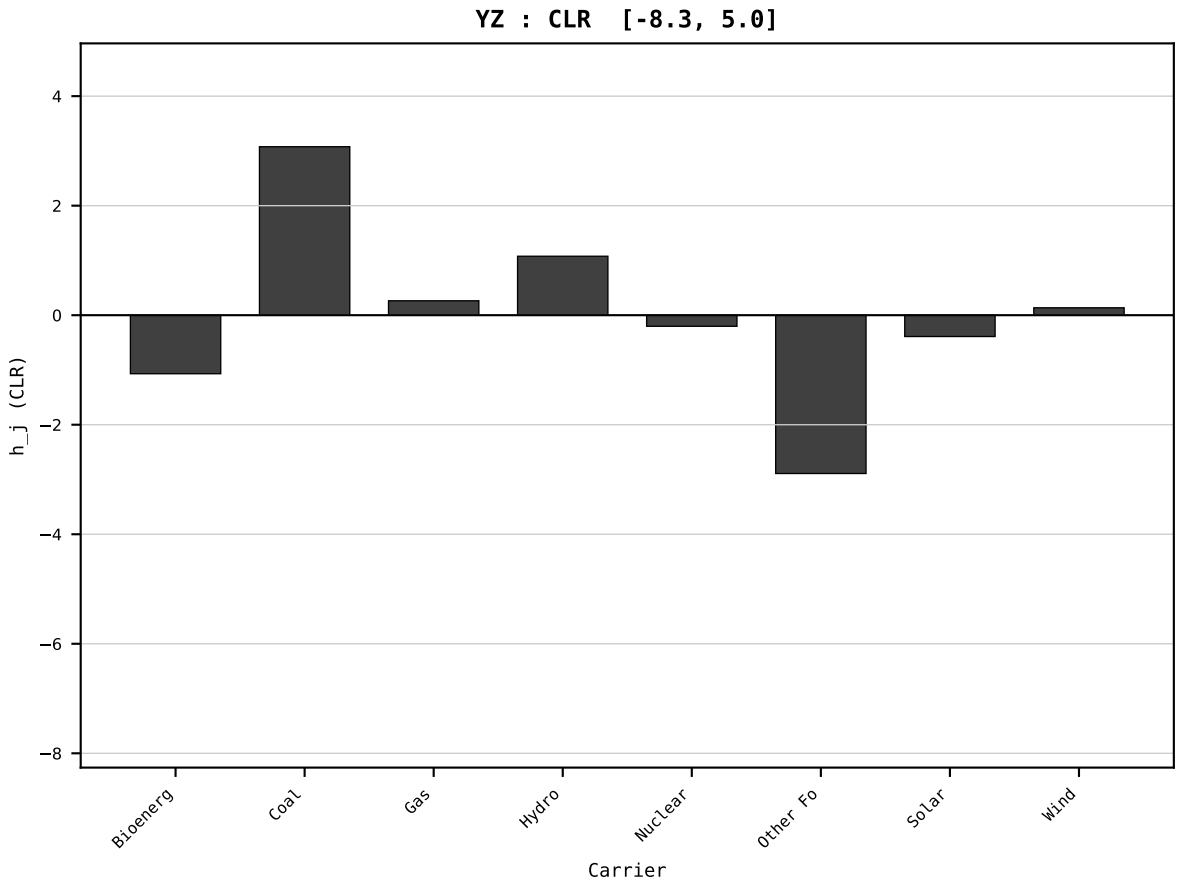
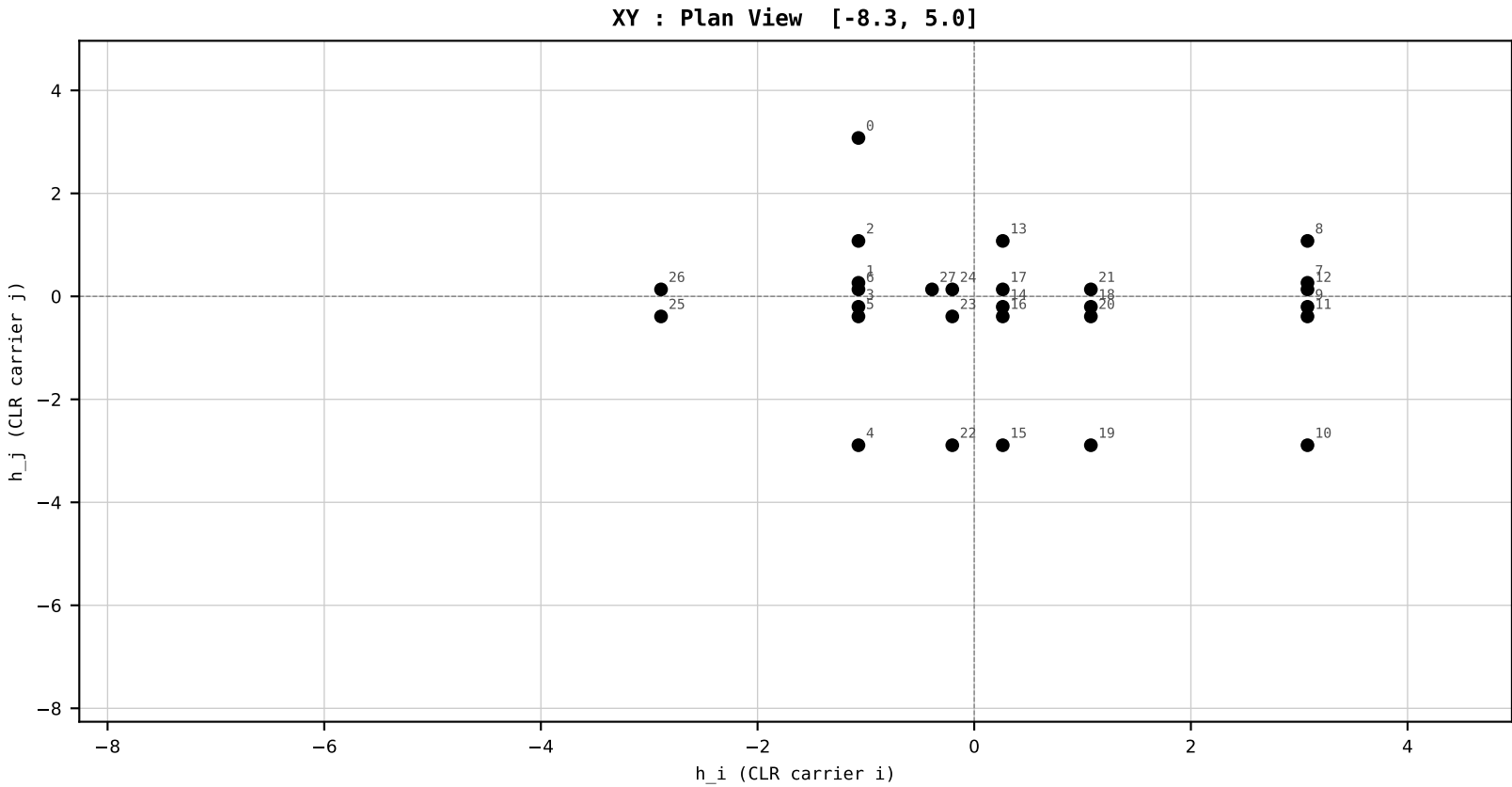
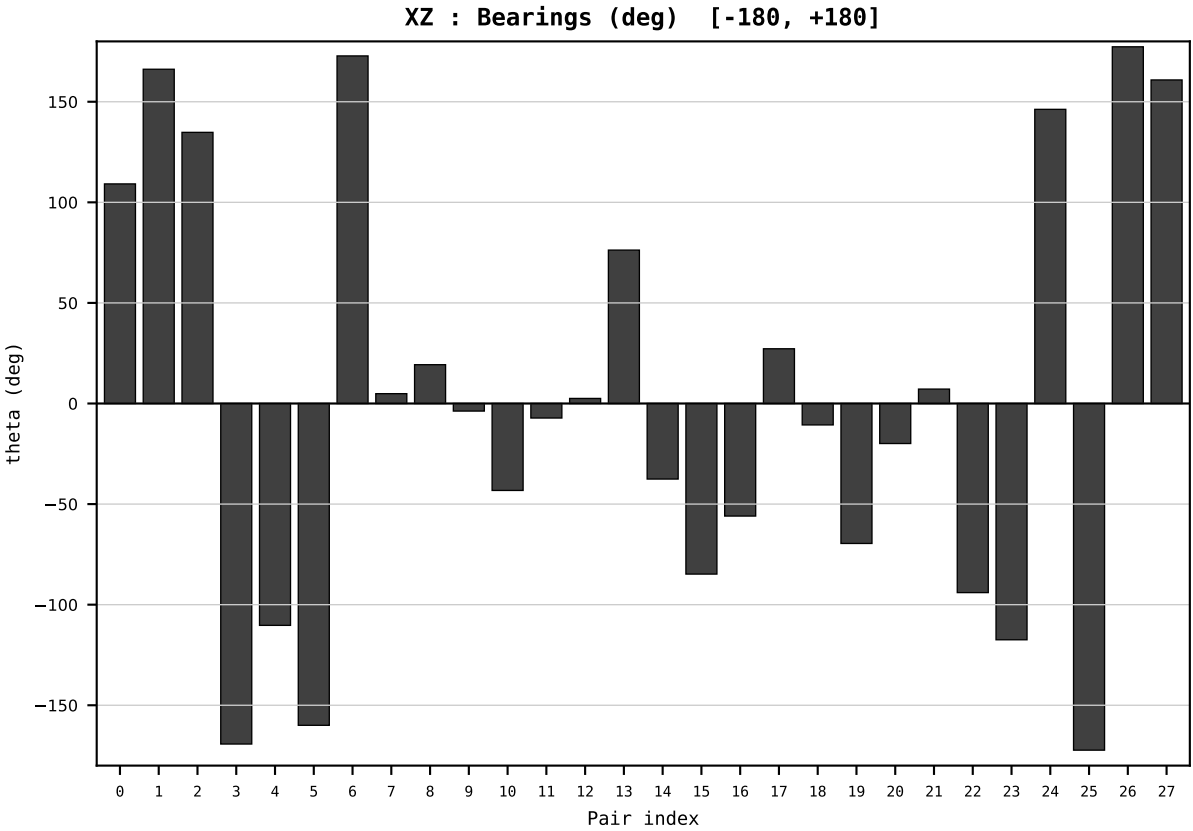
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2019
t : 19 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.534043
Ring = Hs-4
E_metric= 20.4039
kappa_HS= 390.48
omega = 8.0922 deg
d_A = 0.755525
Helm = Other Fossil
Helm d = -0.690711
DR = 5.967 HLR
DR ratio= 390.5x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=20 | Year 2020 | D=8 N=26 pairs=28

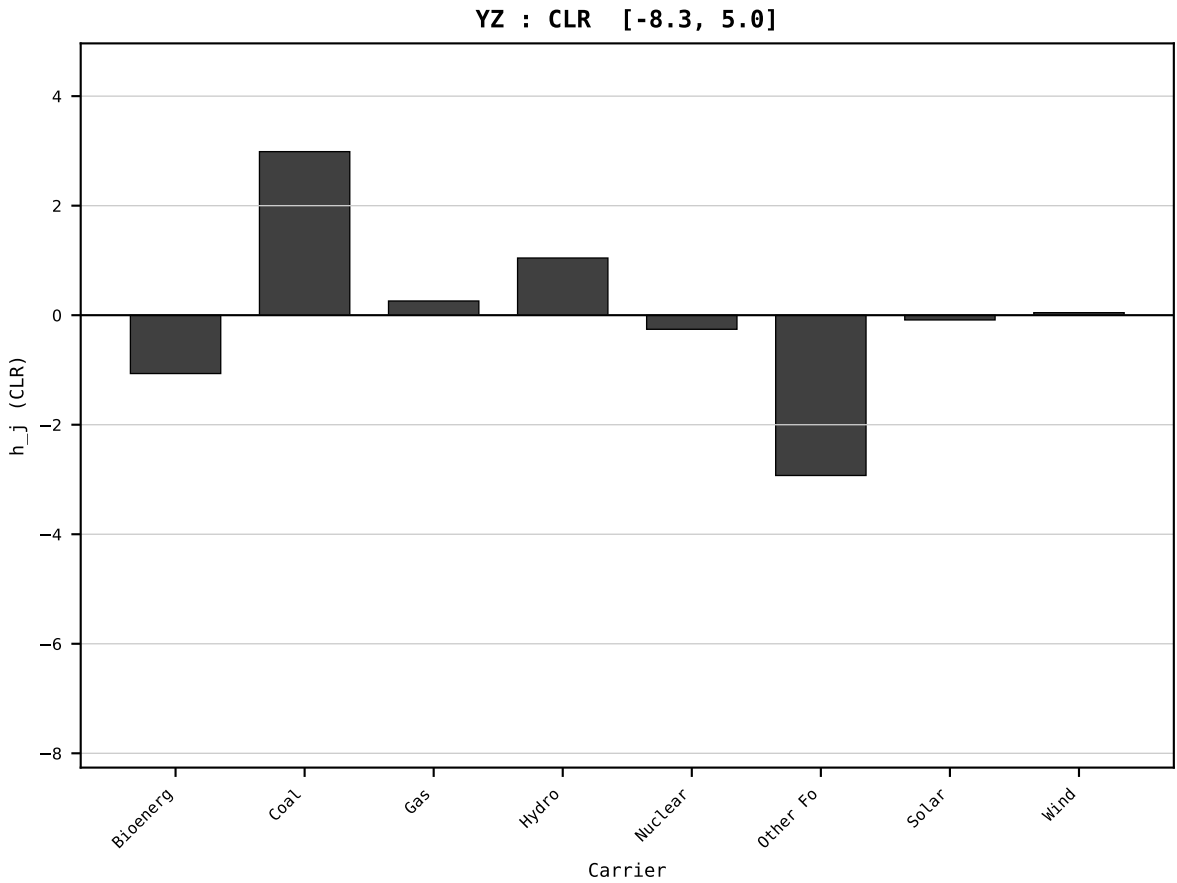
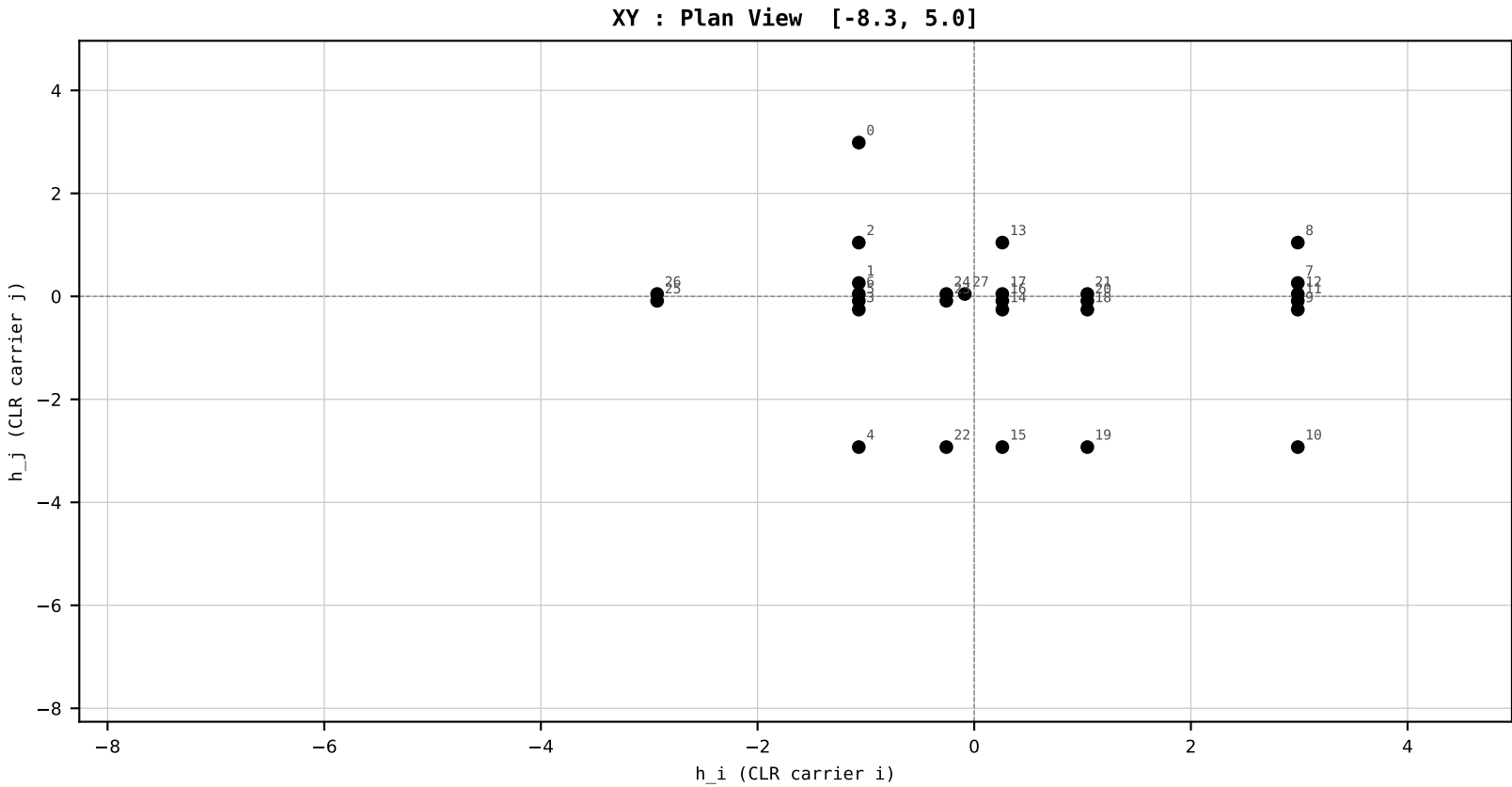
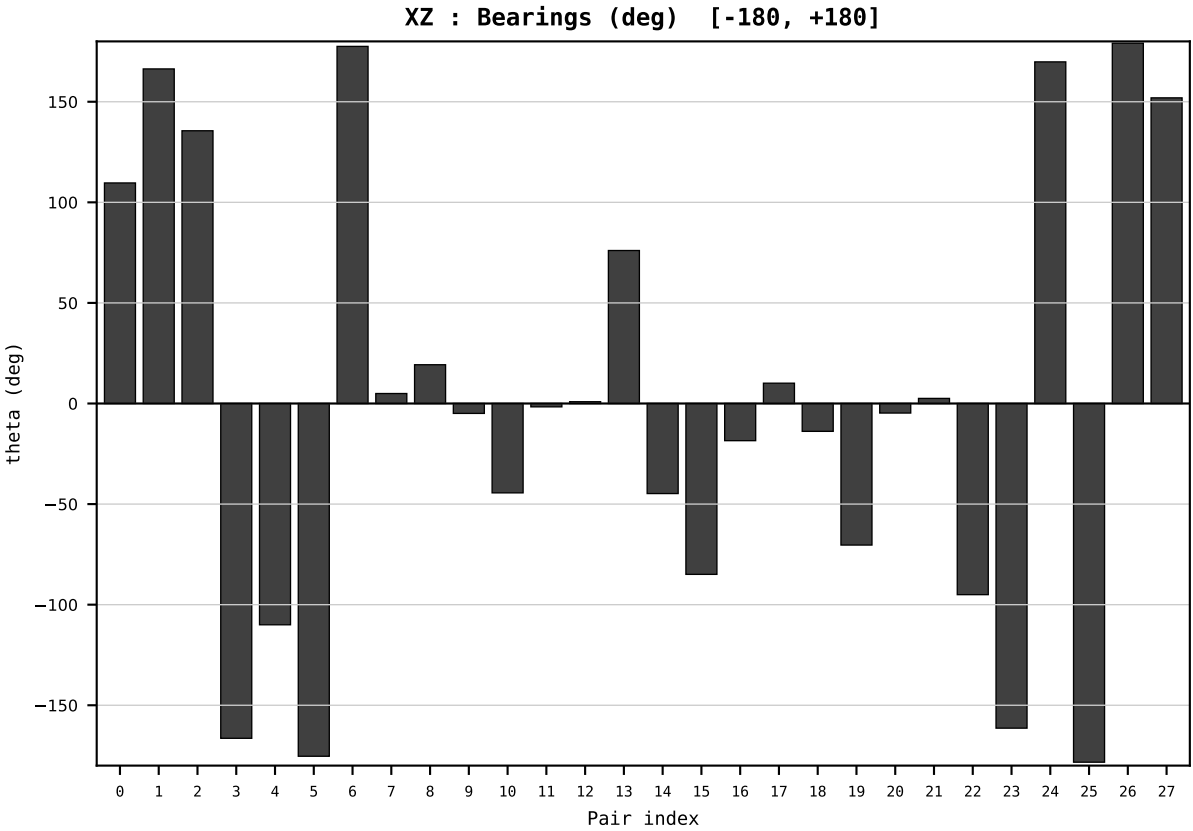
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2020
t : 20 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.509401
Ring = Hs-4
E_metric= 19.8580
kappa_HS= 370.16
omega = 4.2261 deg
d_A = 0.336395
Helm = Solar
Helm d = +0.302967
DR = 5.914 HLR
DR ratio= 370.2x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=21 | Year 2021 | D=8 N=26 pairs=28

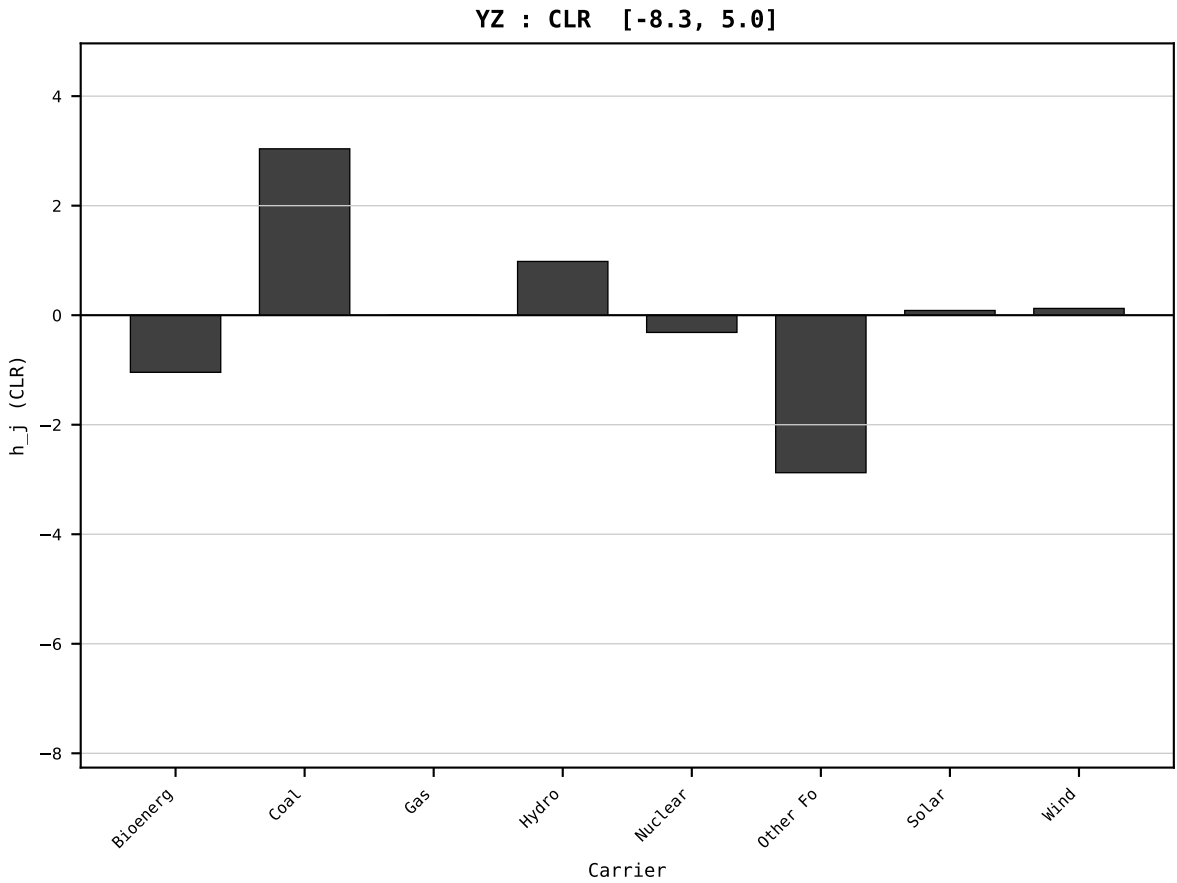
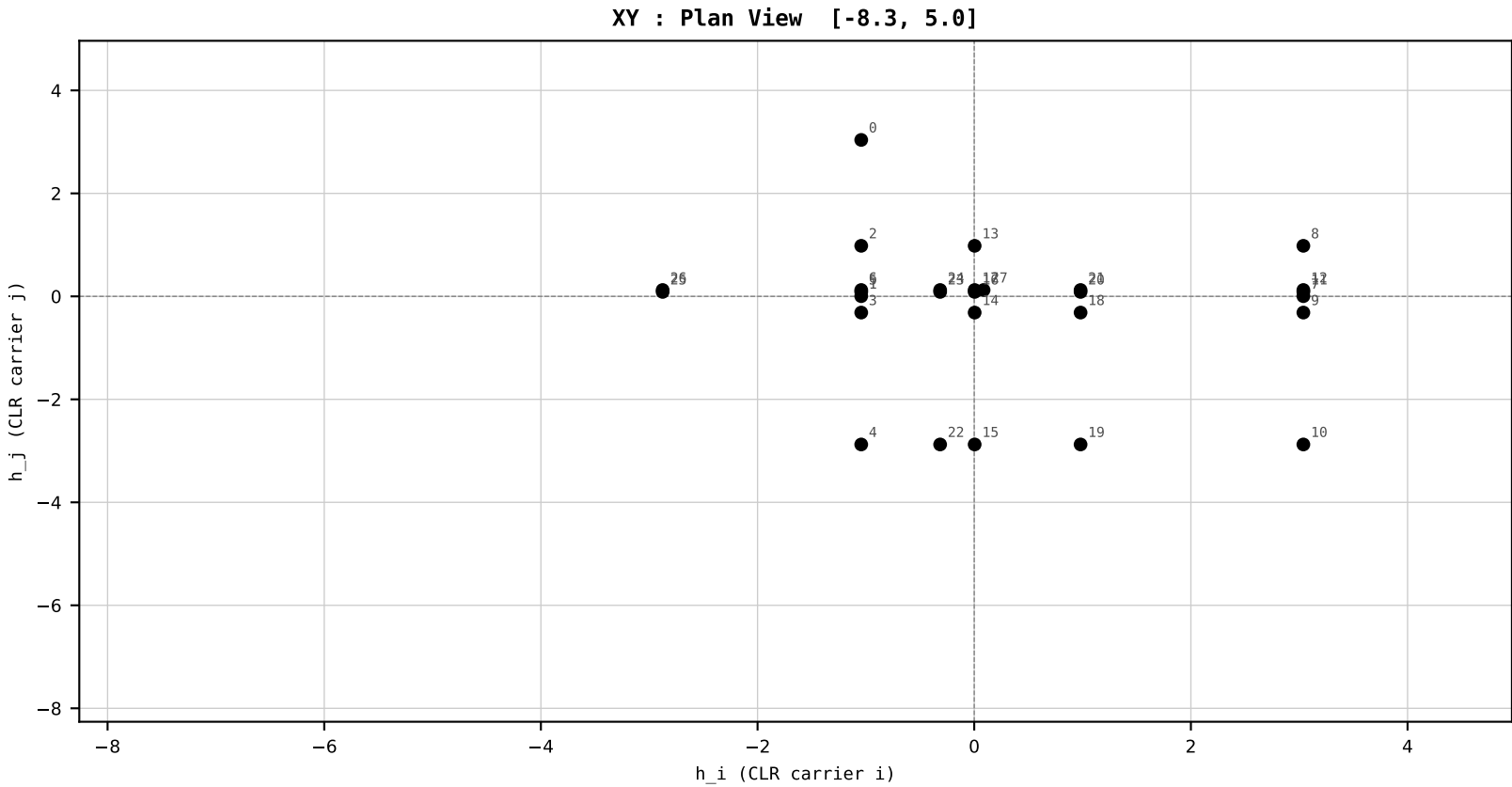
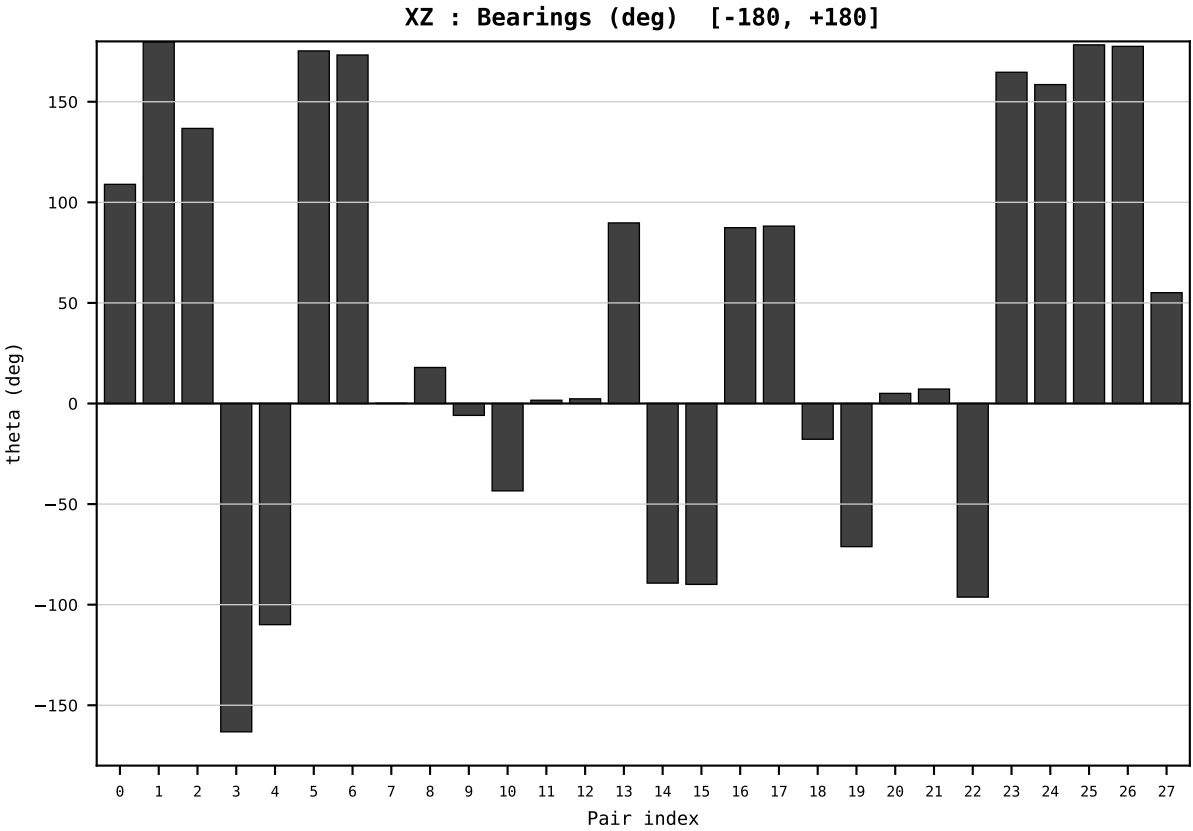
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2021
t : 21 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.528191
Ring = Hs-4
E_metric= 19.6773
kappa_HS= 370.37
omega = 4.3477 deg
d_A = 0.337902
Helm = Gas
Helm d = -0.255484
DR = 5.915 HLR
DR ratio= 370.4x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=22 | Year 2022 | D=8 N=26 pairs=28

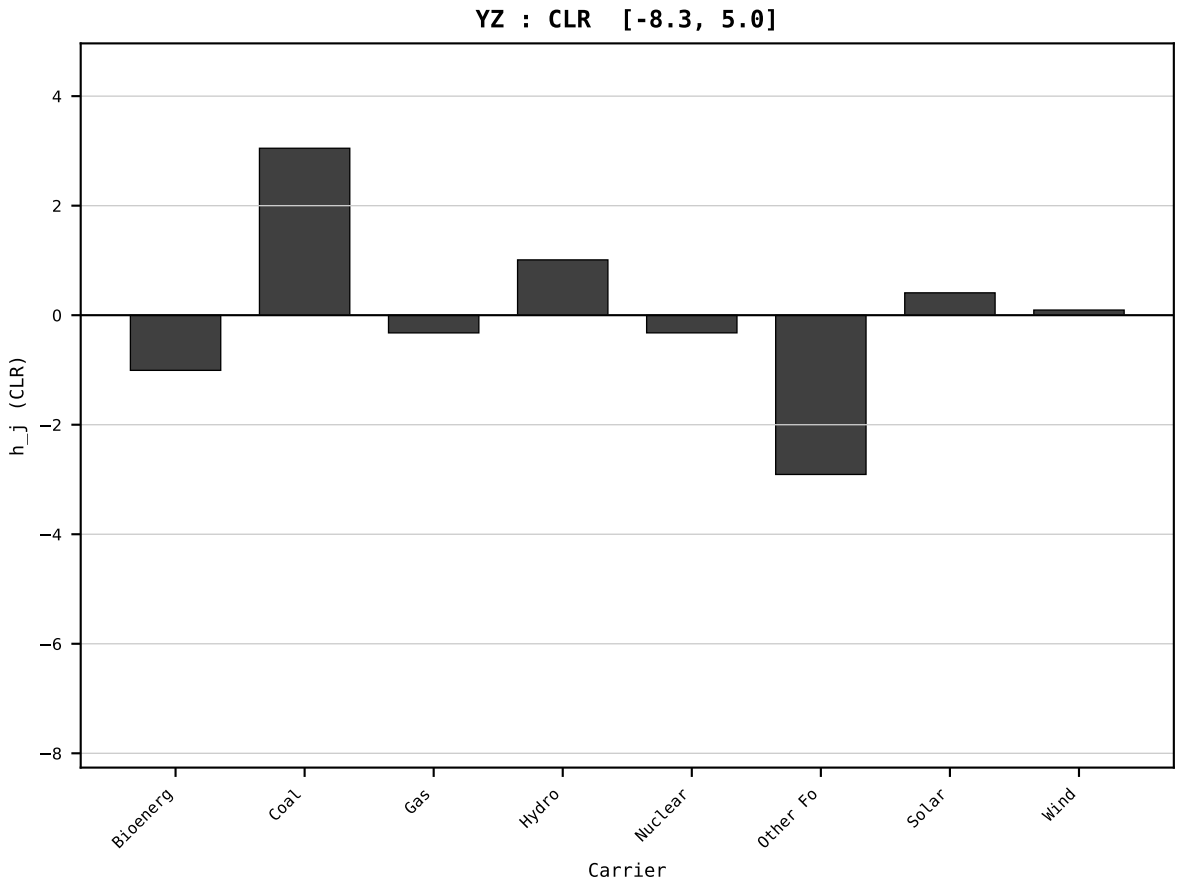
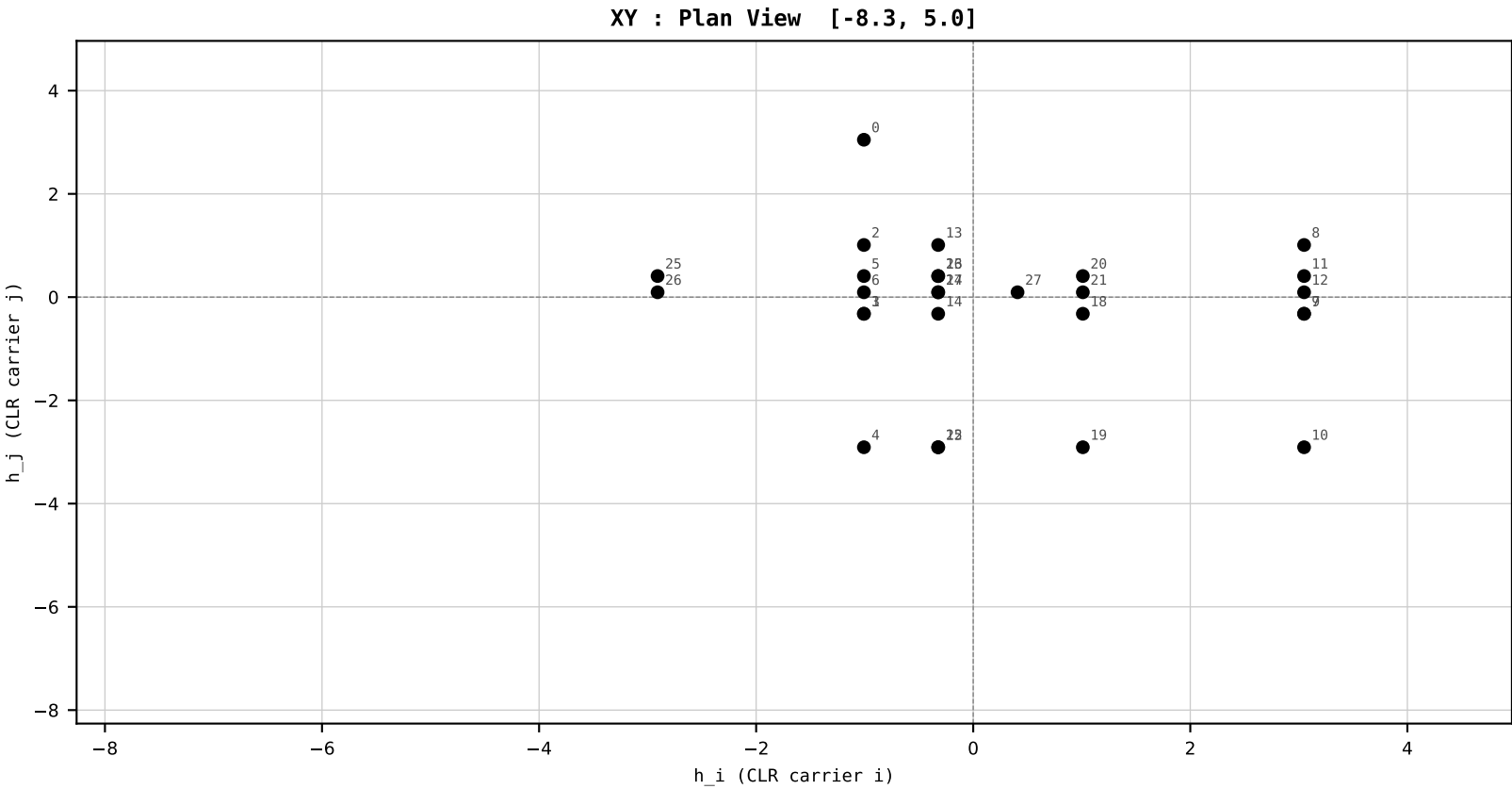
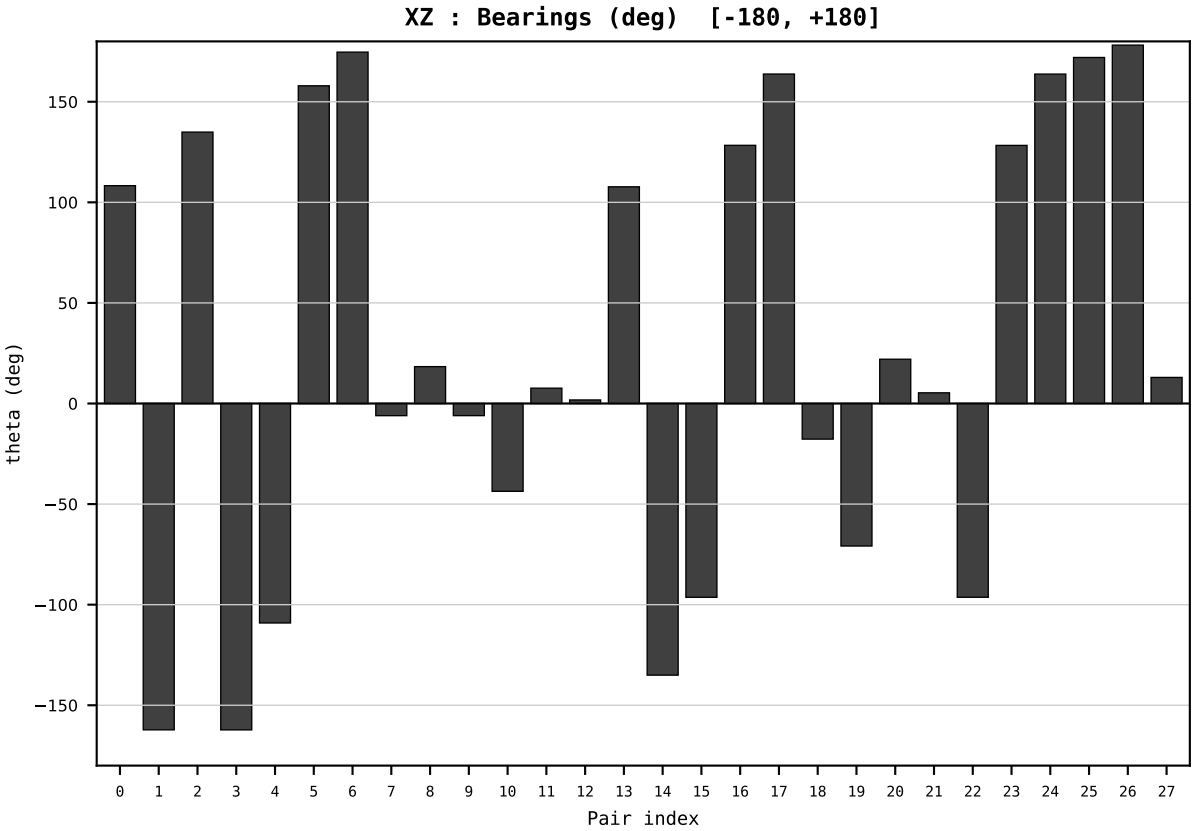
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2022
t : 22 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.527016
Ring = Hs-4
E_metric= 20.1694
kappa_HS= 386.33
omega = 5.9073 deg
d_A = 0.463267
Helm = Gas
Helm d = -0.326832
DR = 5.957 HLR
DR ratio= 386.3x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=23 | Year 2023 | D=8 N=26 pairs=28

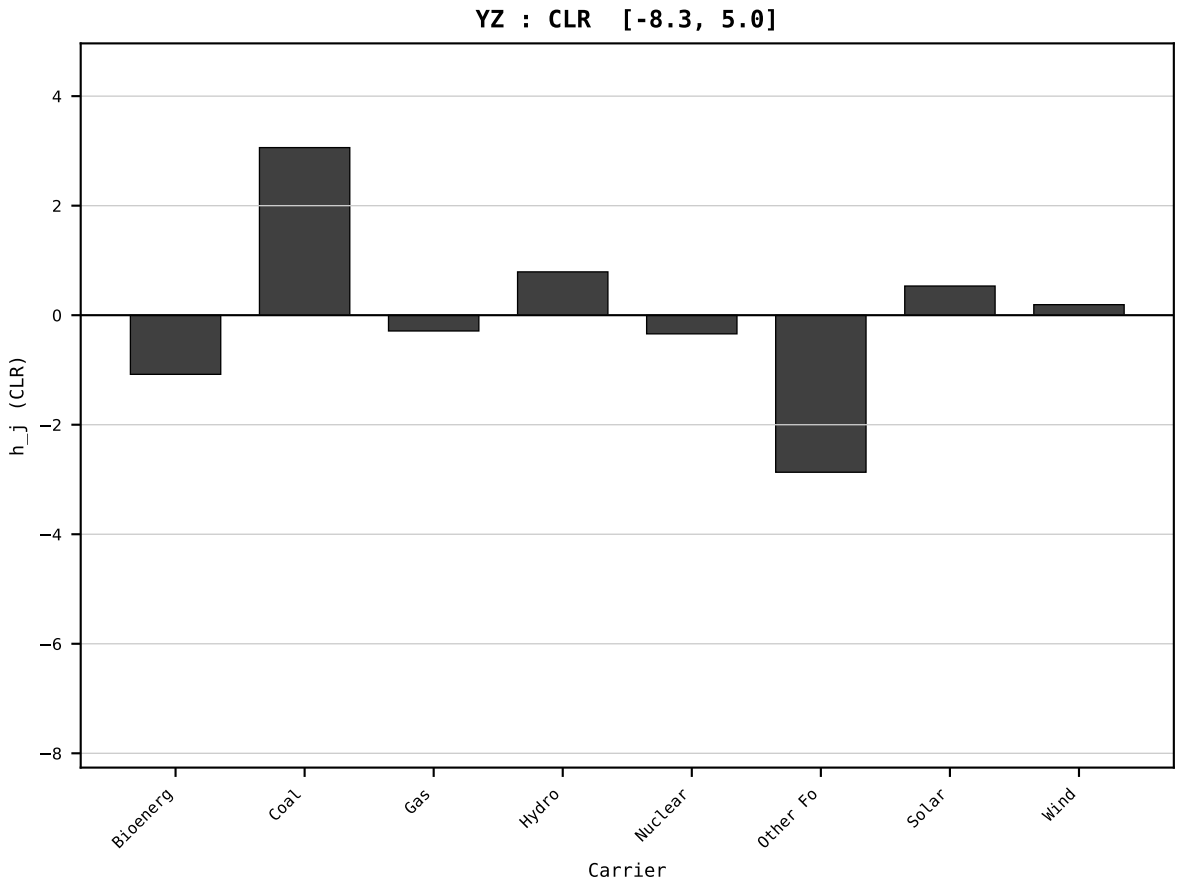
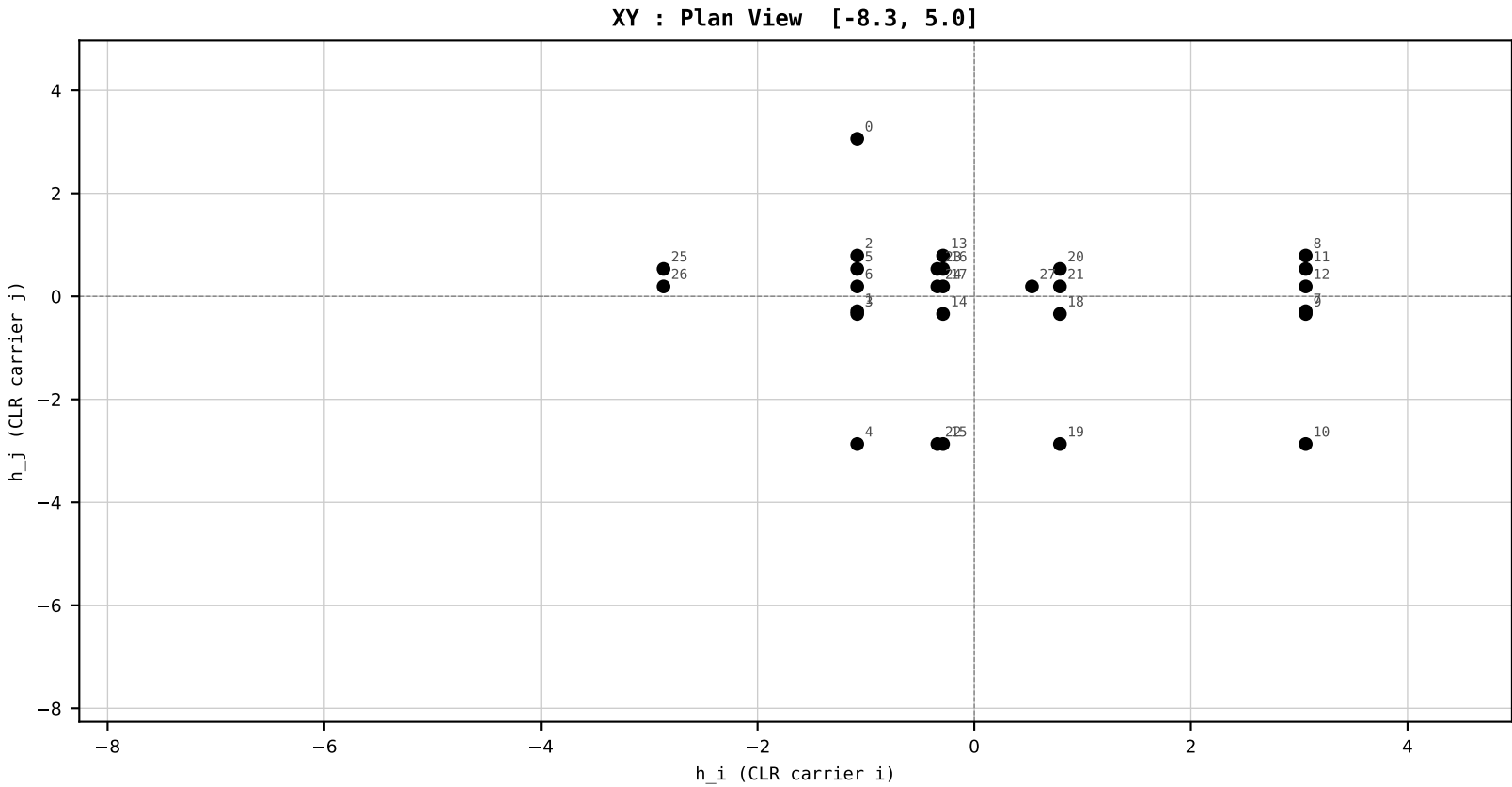
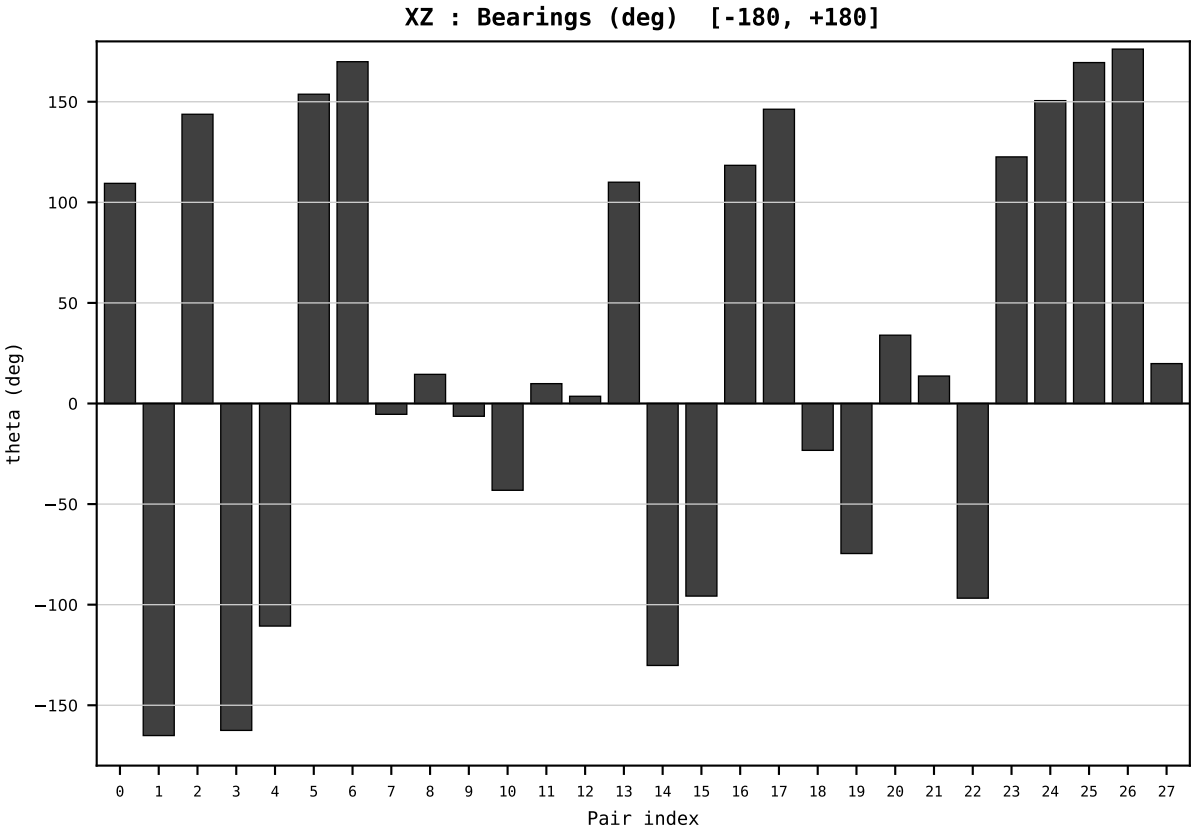
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2023
t : 23 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.533376
Ring = Hs-4
E_metric= 19.9012
kappa_HS= 375.44
omega = 3.6403 deg
d_A = 0.285909
Helm = Hydro
Helm d = -0.219097
DR = 5.928 HLR
DR ratio= 375.4x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=24 | Year 2024 | D=8 N=26 pairs=28

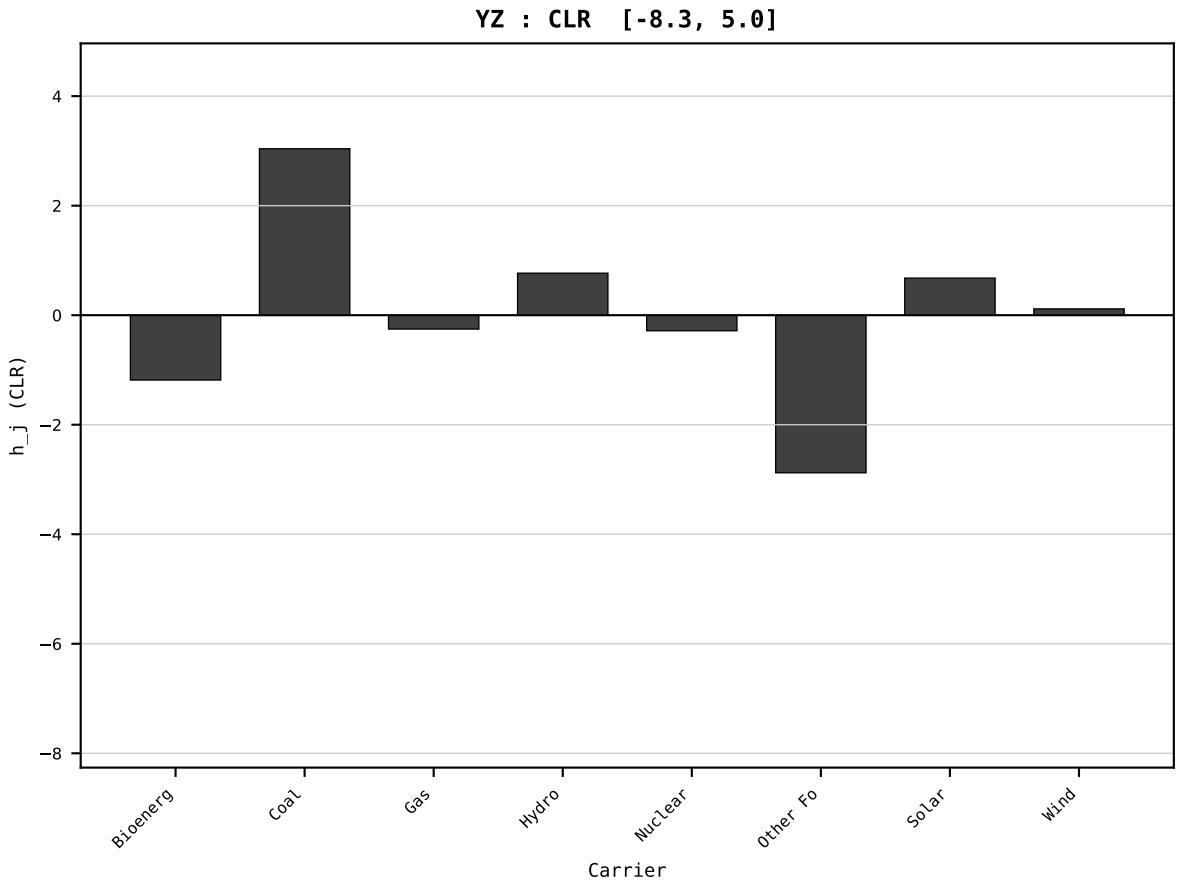
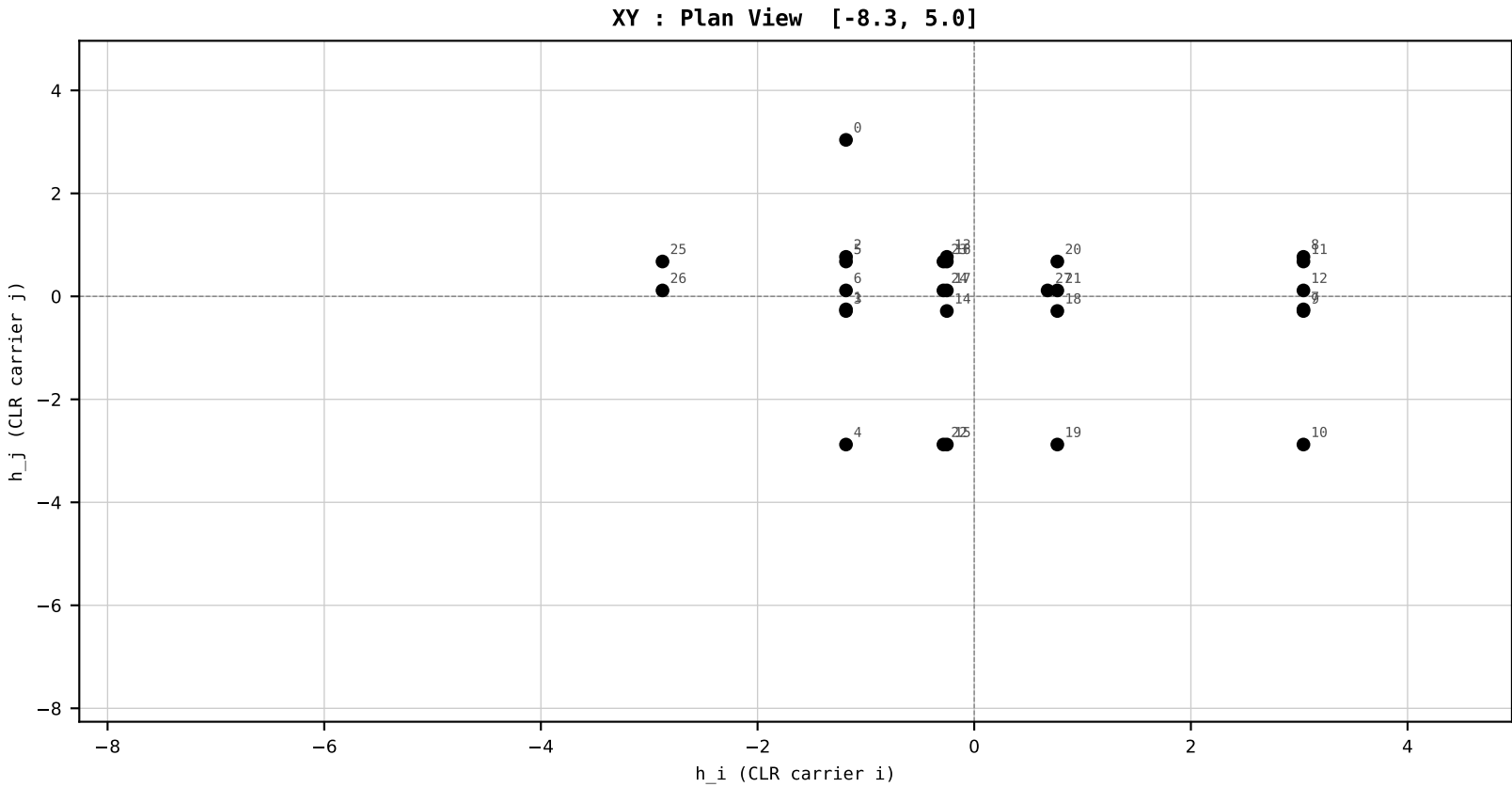
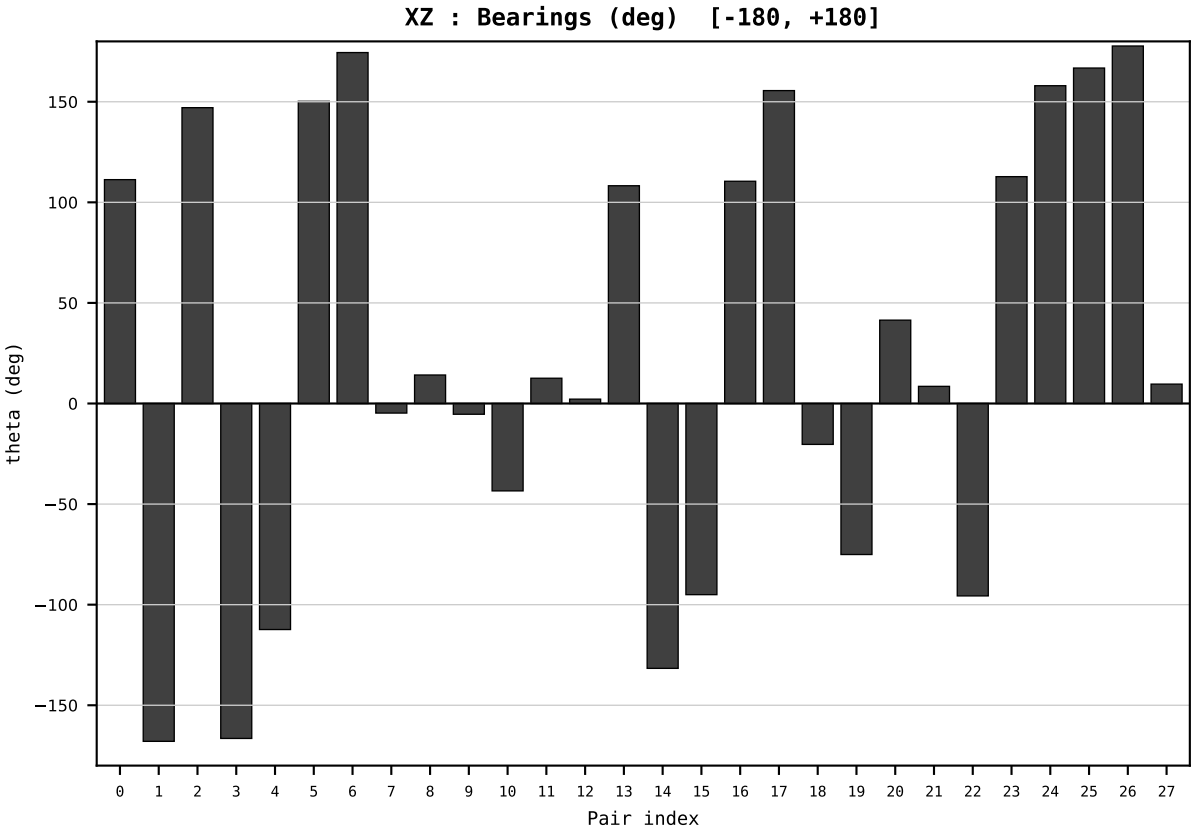
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2024
t : 24 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.523895
Ring = Hs-4
E_metric= 20.1222
kappa_HS= 371.12
omega = 2.6415 deg
d_A = 0.207695
Helm = Solar
Helm d = +0.144532
DR = 5.917 HLR
DR ratio= 371.1x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=25 | Year 2025 | D=8 N=26 pairs=28

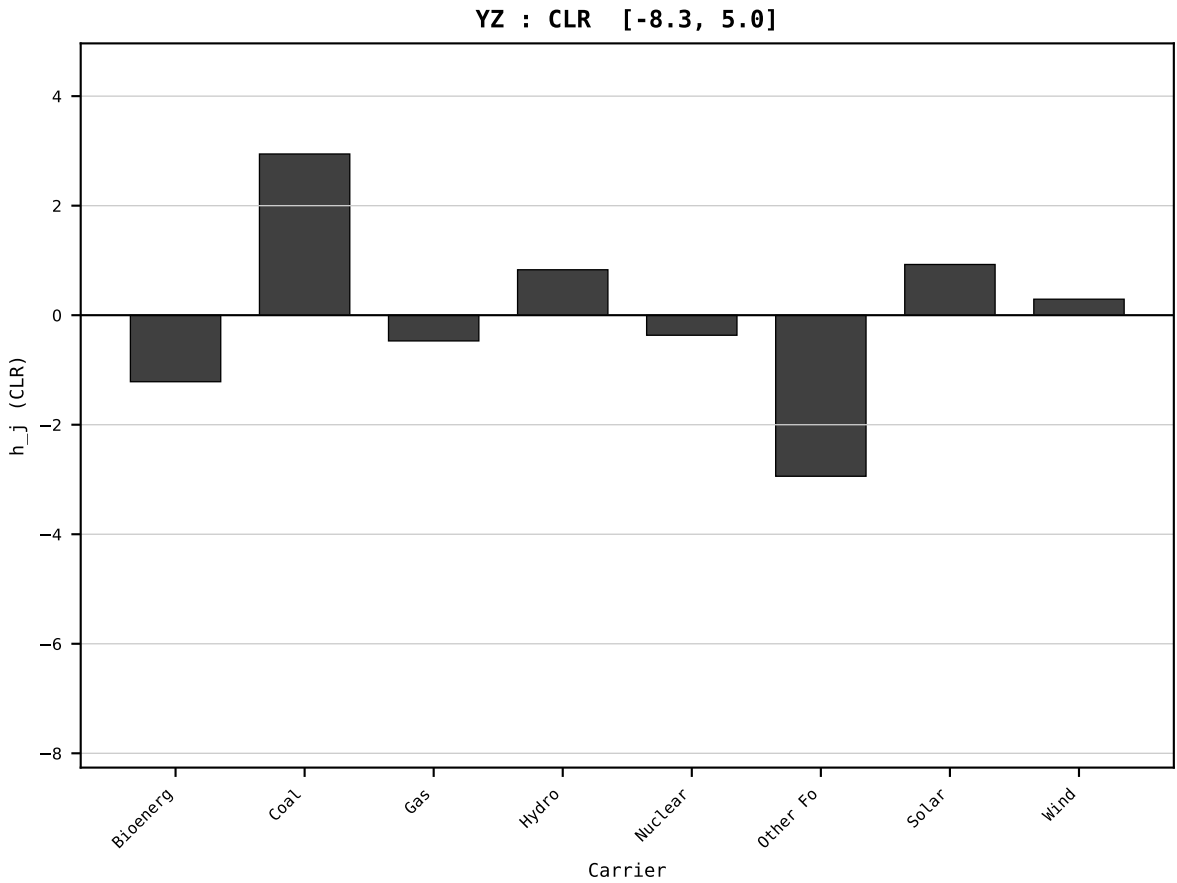
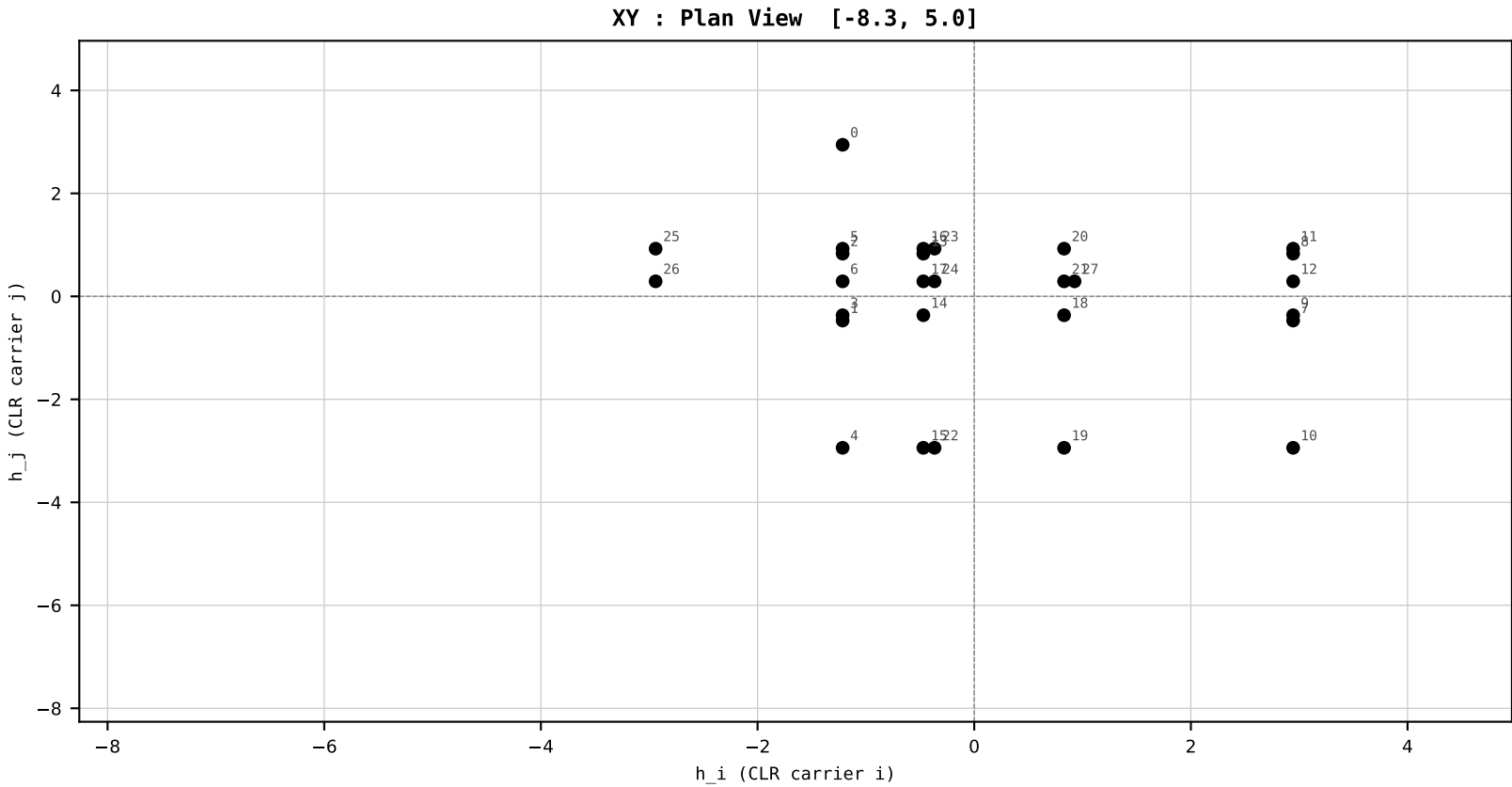
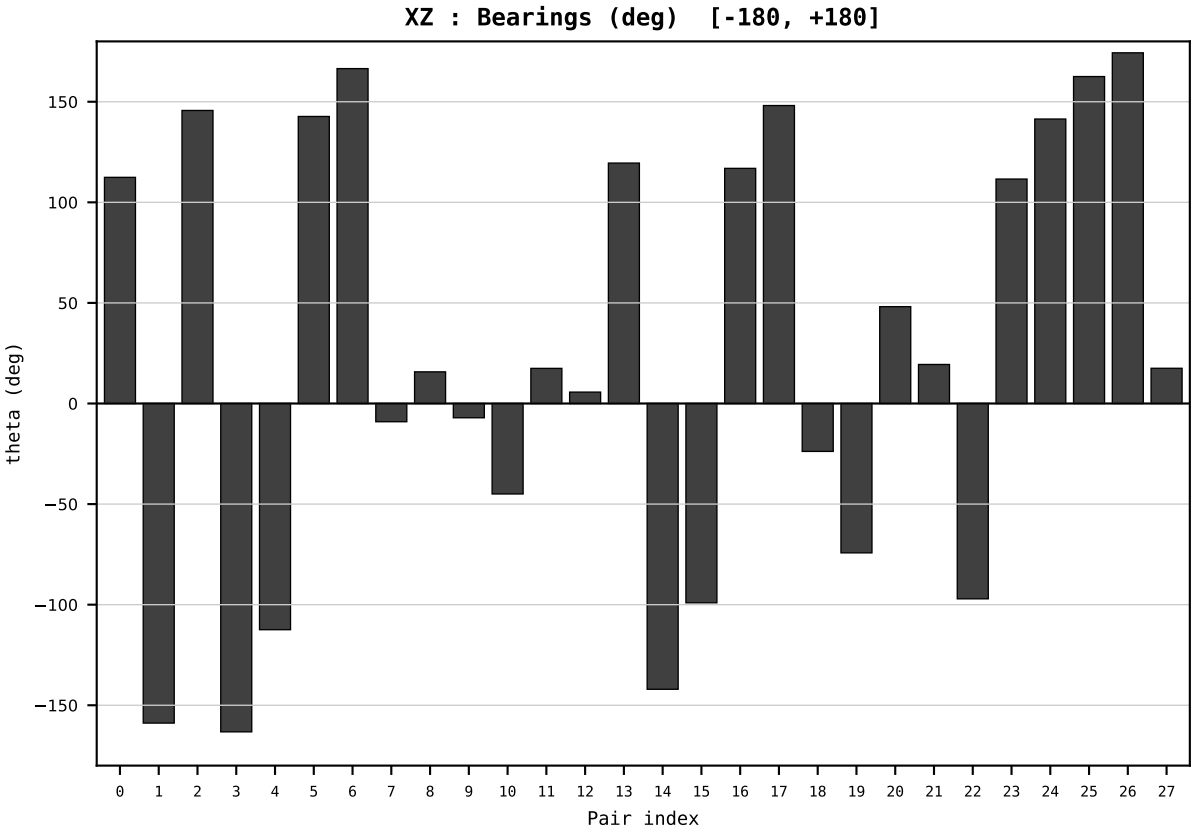
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

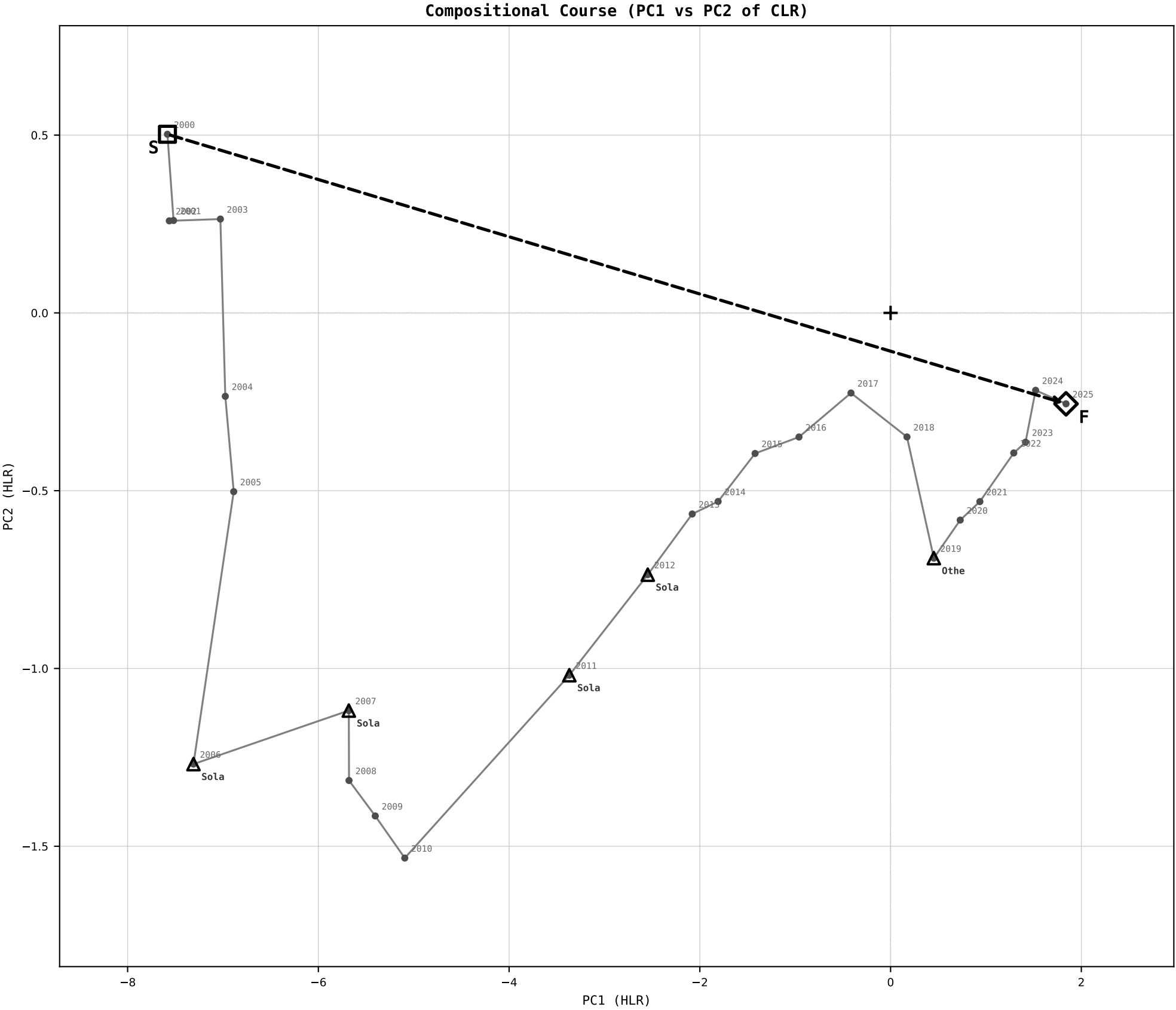
Dataset : ember_IND_India_generation_TWh.csv
Year : 2025
t : 25 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.484928
Ring = Hs-3
E_metric= 20.7772
kappa_HS= 359.55
omega = 5.0650 deg
d_A = 0.406114
Helm = Solar
Helm d = +0.248797
DR = 5.885 HLR
DR ratio= 359.5x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind





COURSE METRICS

Scale Provenance:
Y_j → x_j=Y_j/T → h_j=CLR
→ PC projection
Unit: HLR (Higgins Log-Ratio)

Net distance = 9.4603 HLR
Path length = 14.6188 HLR
Directness = 0.6471

DR start = 10.572 HLR
ratio = 39023.0x
DR final = 5.885 HLR
ratio = 359.5x

E_{metric} start = 75.8817
E_{metric} final = 20.7772

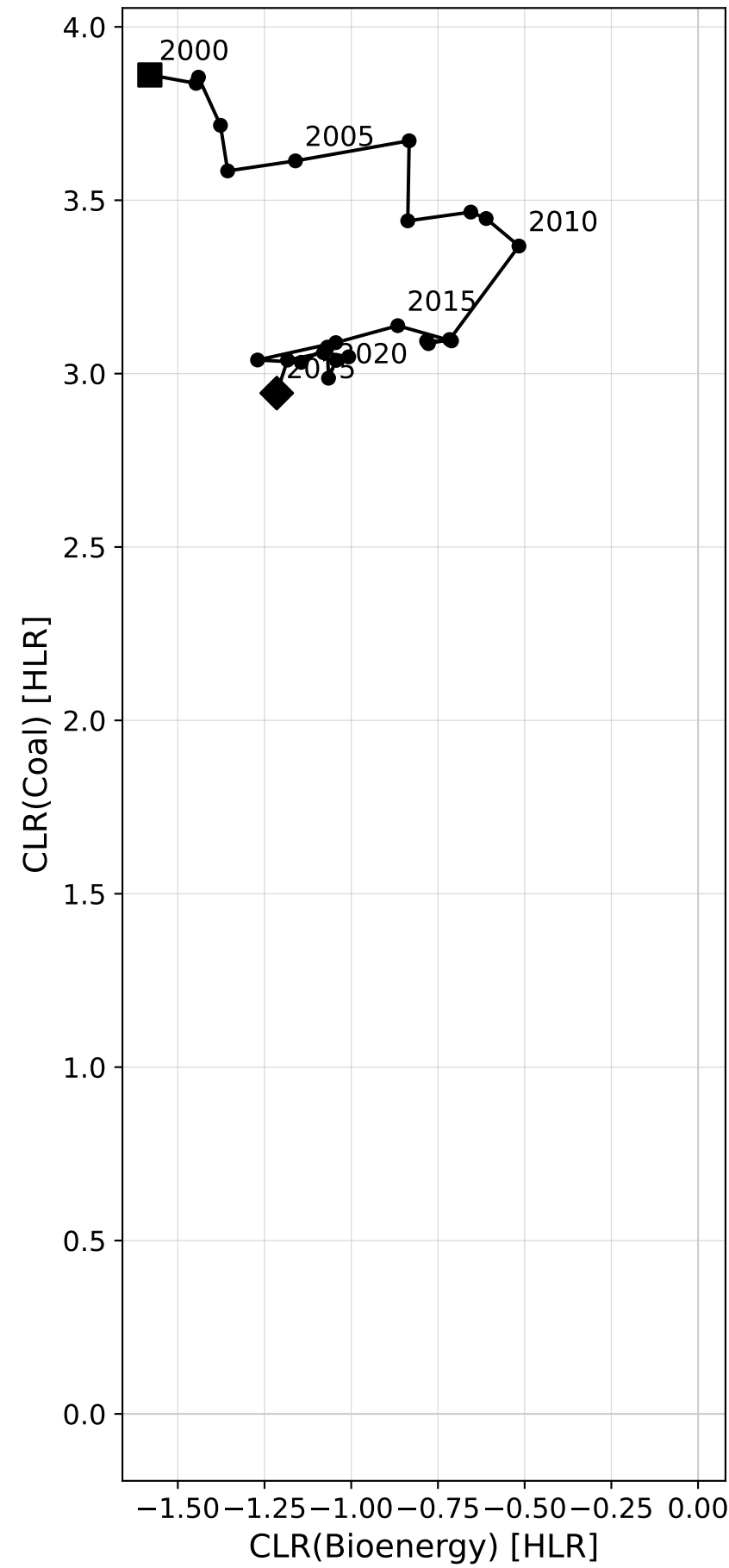
PC1 loadings:
Solar +0.841
Other Fo -0.425
Gas -0.241

PC2 loadings:
Wind -0.574
Bioenerg -0.487
Other Fo +0.458

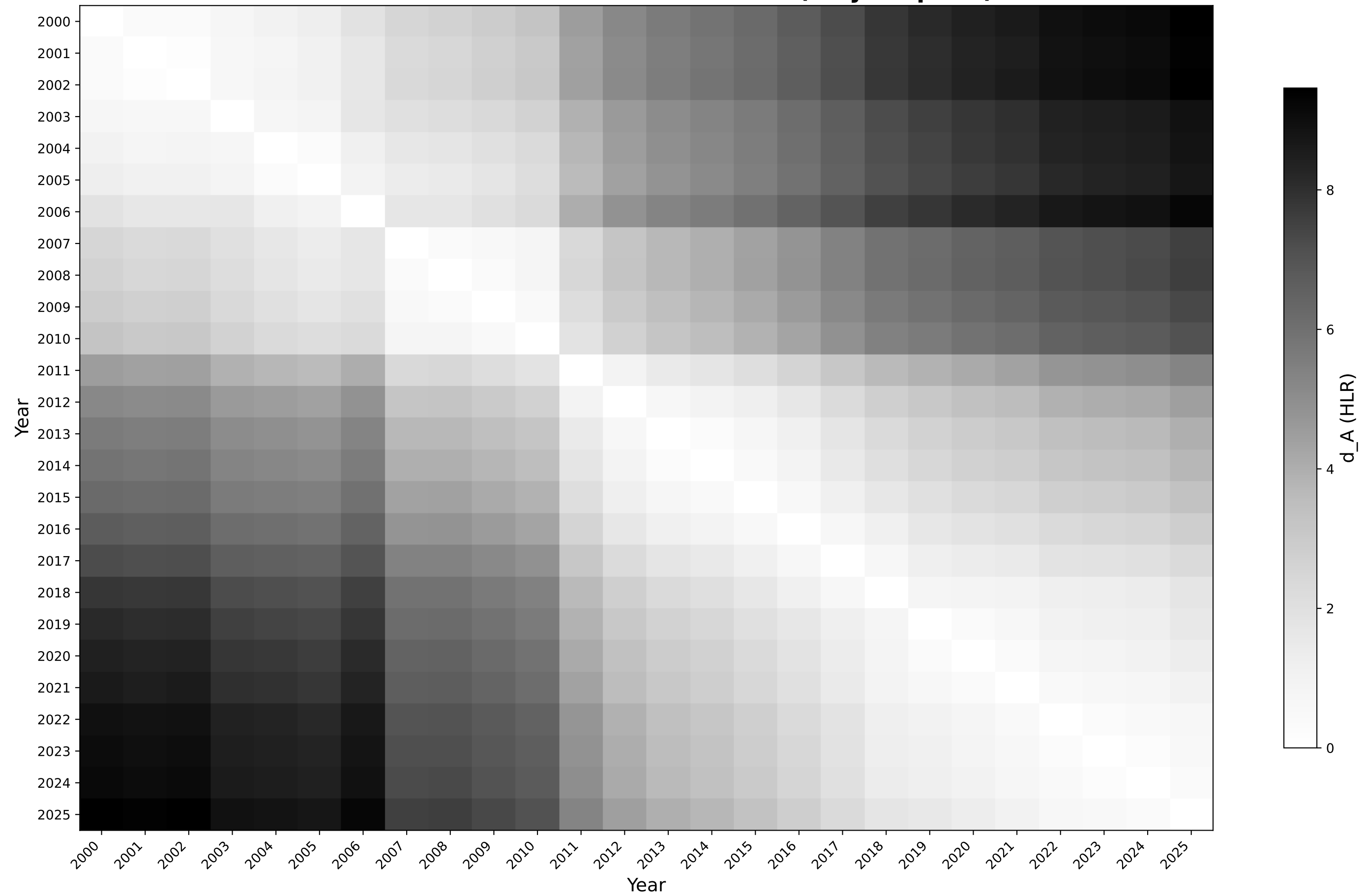
Top displacement events:
2011 Solar d=1.820
2007 Solar d=1.692
2006 Solar d=0.943
2012 Solar d=0.912
2019 Other Fo d=0.756

The instrument reads.
The expert decides.
The loop stays open.

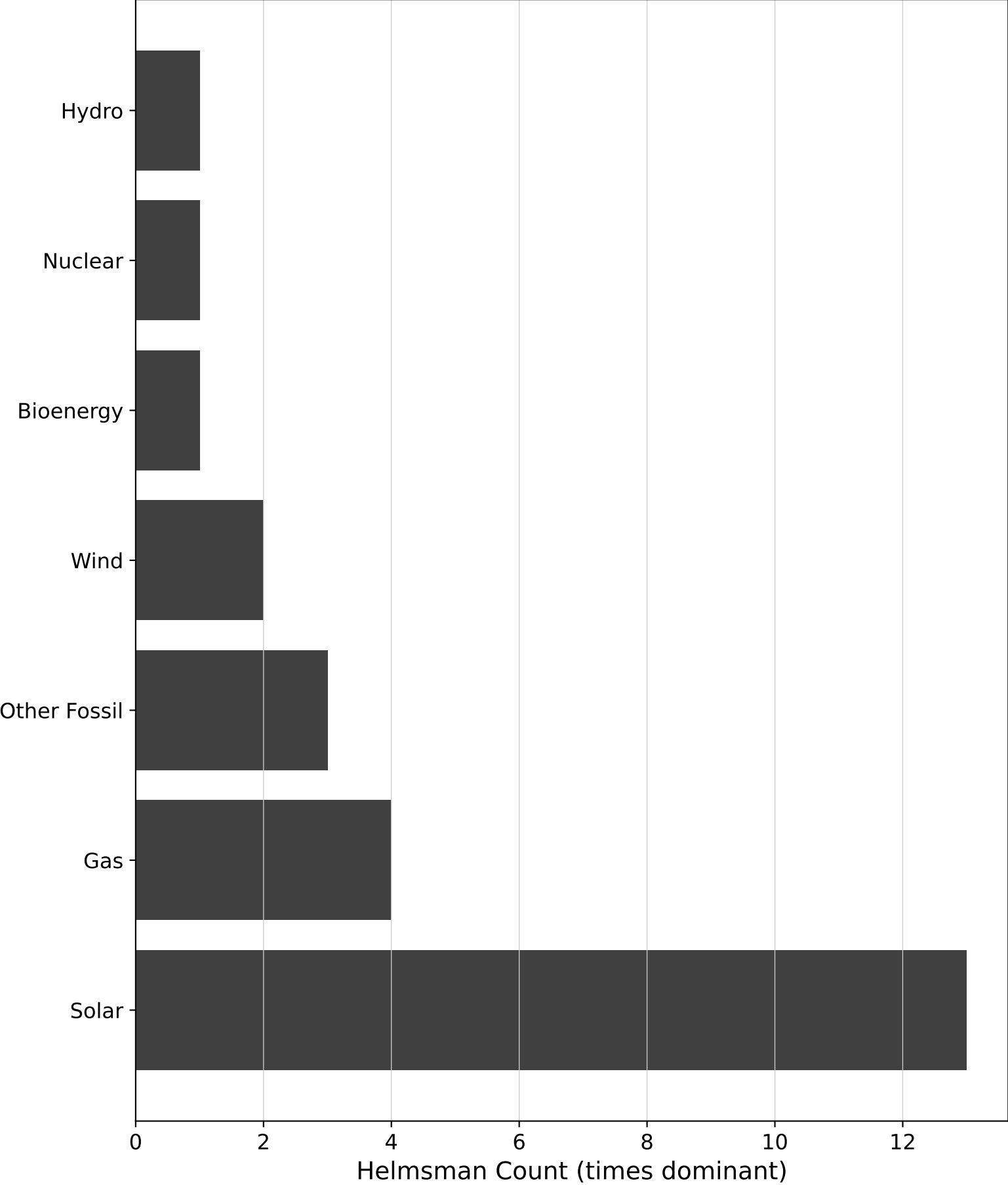
SYSTEM COURSE PLOT — XY Plan View (CLR Space)



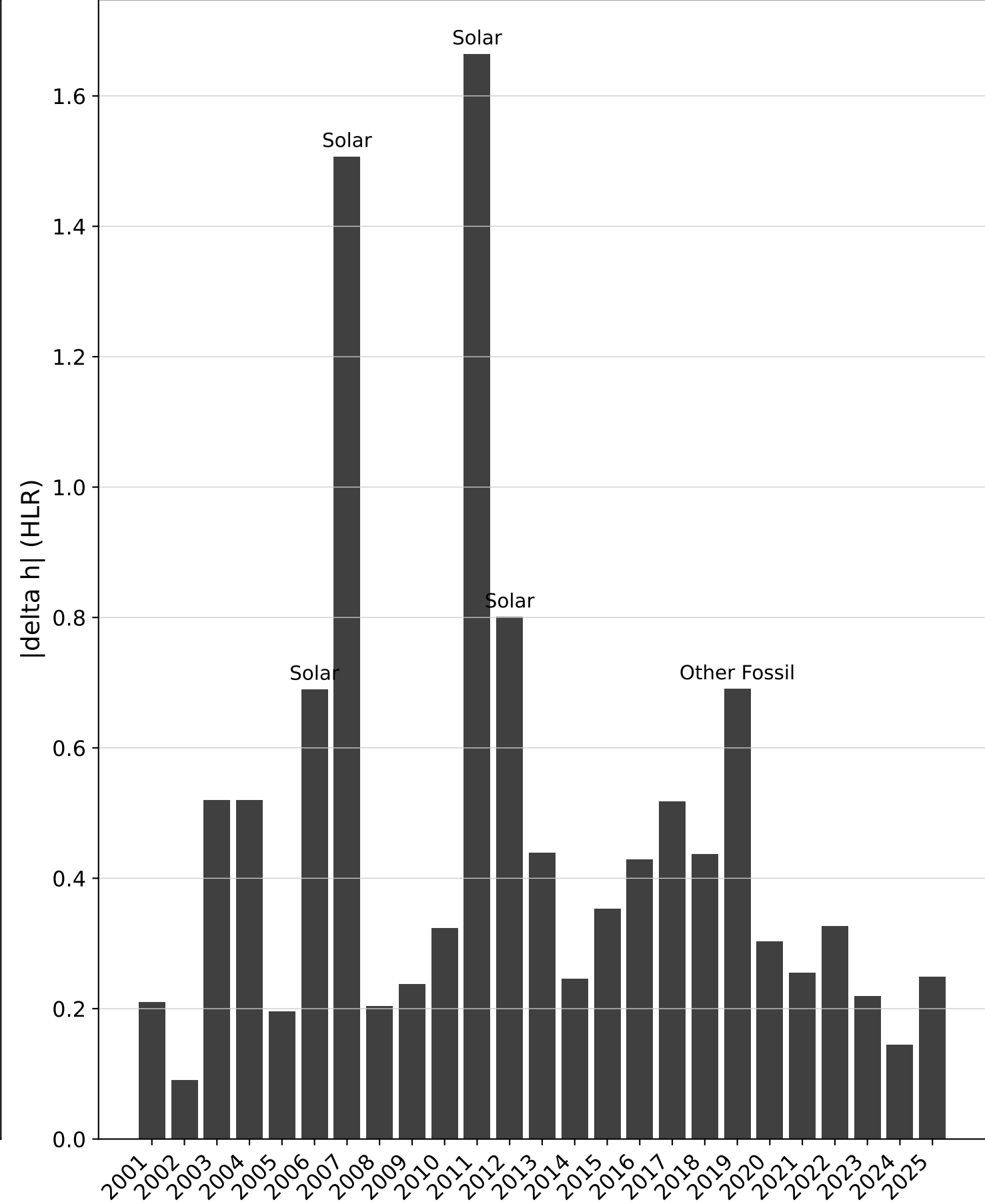
STAGE 2 — Aitchison Distance Matrix (all year pairs)



HELMSMAN FREQUENCY



HELMSMAN DISPLACEMENT PER YEAR



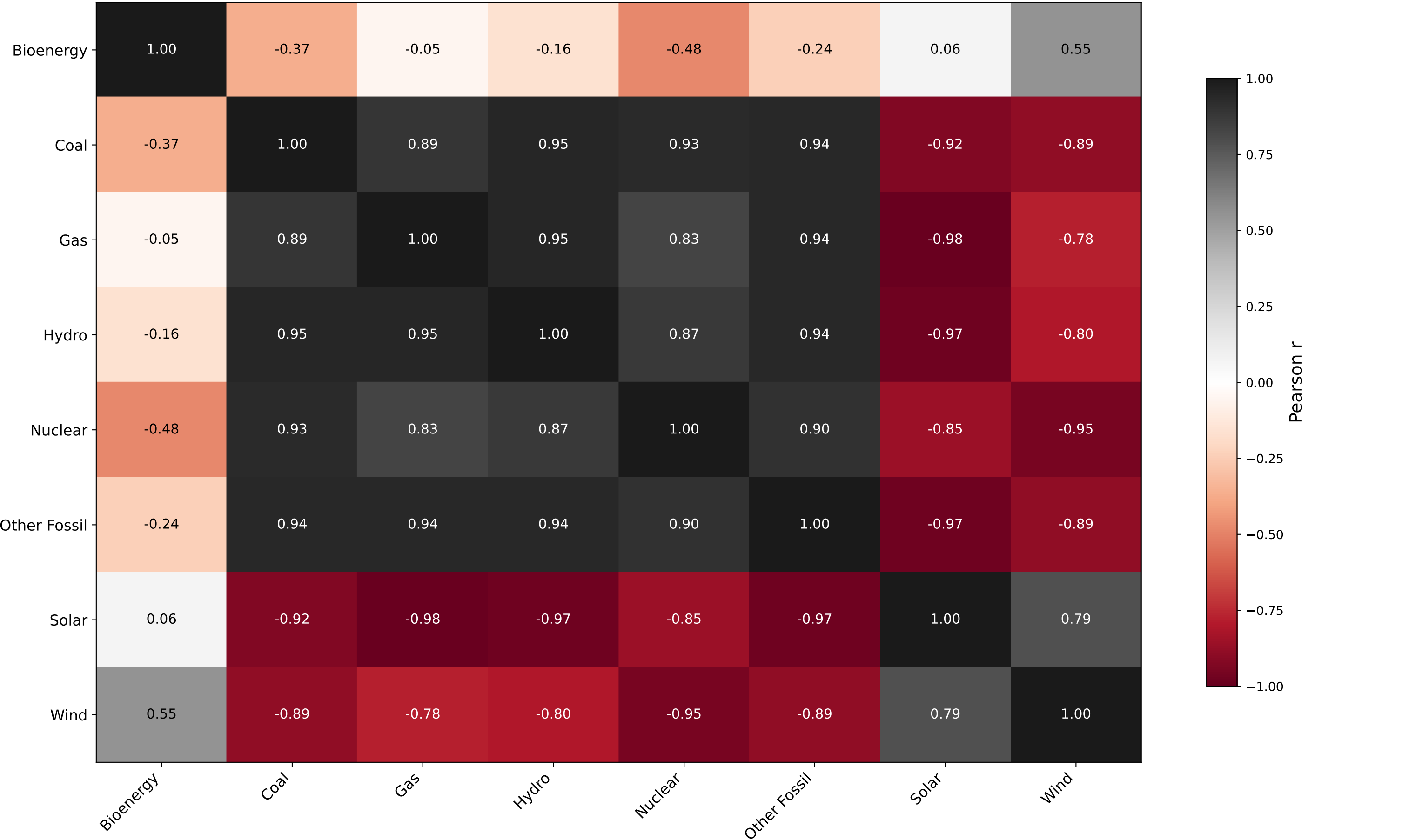
STAGE 2 — Pairwise Divergence Ranking (Top 15)

Rank	Year A	Year B	d_A (HLR)	Angle (deg)	Dominant Carrier
1	2000	2025	9.4603	84.8	Solar
2	2002	2025	9.4248	84.9	Solar
3	2001	2025	9.3775	84.6	Solar
4	2006	2025	9.2075	83.0	Solar
5	2000	2024	9.1358	80.8	Solar
6	2002	2024	9.0976	80.8	Solar
7	2001	2024	9.0528	80.6	Solar
8	2000	2023	9.0441	79.6	Solar
9	2002	2023	9.0023	79.6	Solar
10	2001	2023	8.9569	79.4	Solar
11	2000	2022	8.9232	77.9	Solar
12	2006	2024	8.8998	79.1	Solar
13	2003	2025	8.8960	84.2	Solar
14	2002	2022	8.8848	77.9	Solar
15	2001	2022	8.8372	77.6	Solar

STAGE 3 — Triadic Metric Triangle Area Ranking (Top 10)

Rank	Year A	Year B	Year C	Area (HLR^2)	Imbalance
1	2000	2010	2025	8.8156	2.92
2	2000	2006	2025	8.5653	5.08
3	2000	2010	2024	8.4875	2.82
4	2000	2009	2025	8.3089	3.24
5	2000	2006	2024	8.2823	4.91
6	2000	2010	2023	8.2104	2.80
7	2000	2006	2023	8.1635	4.86
8	2000	2010	2022	8.0866	2.76
9	2000	2006	2022	8.0472	4.79
10	2000	2009	2024	8.0278	3.13

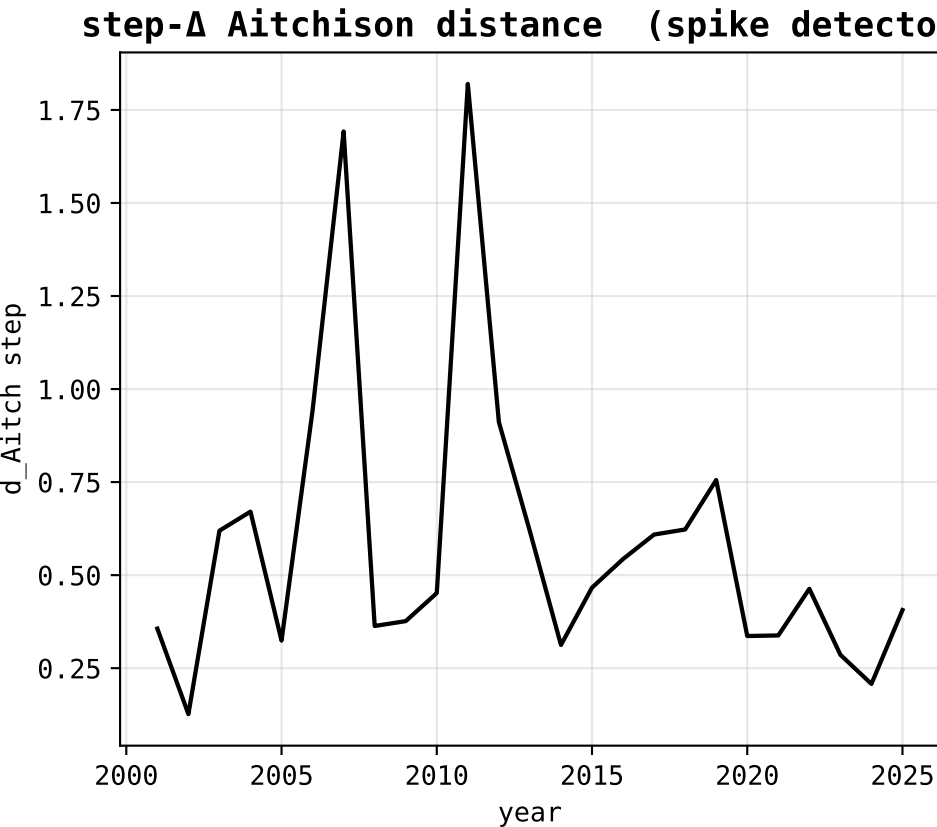
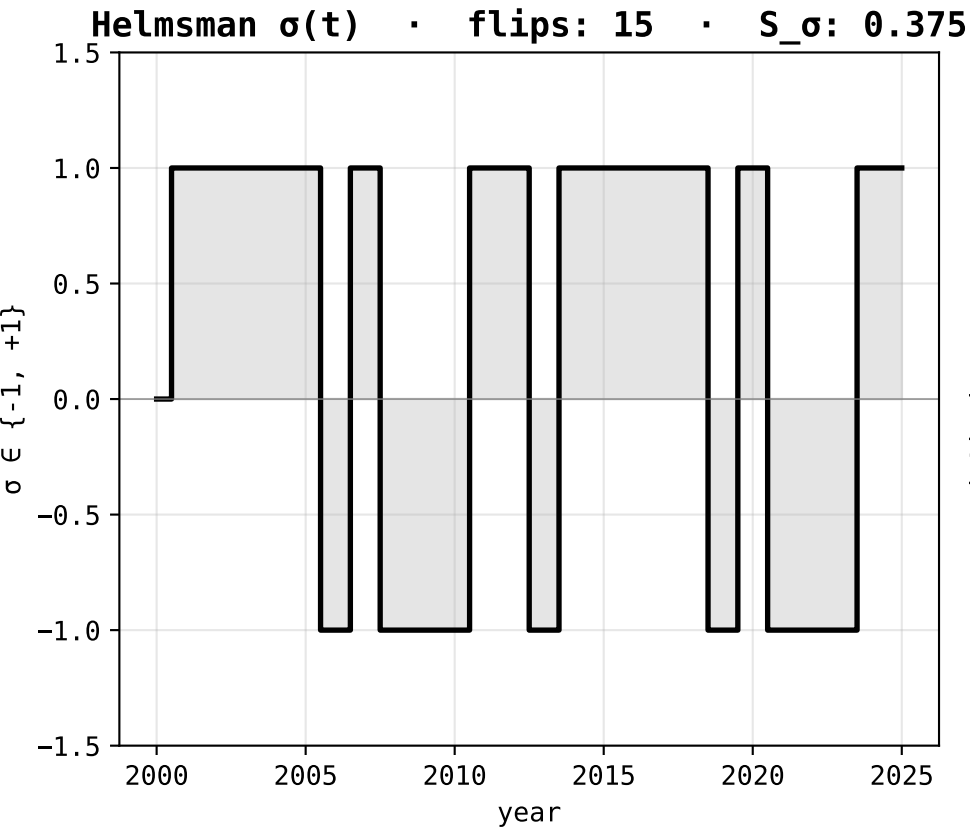
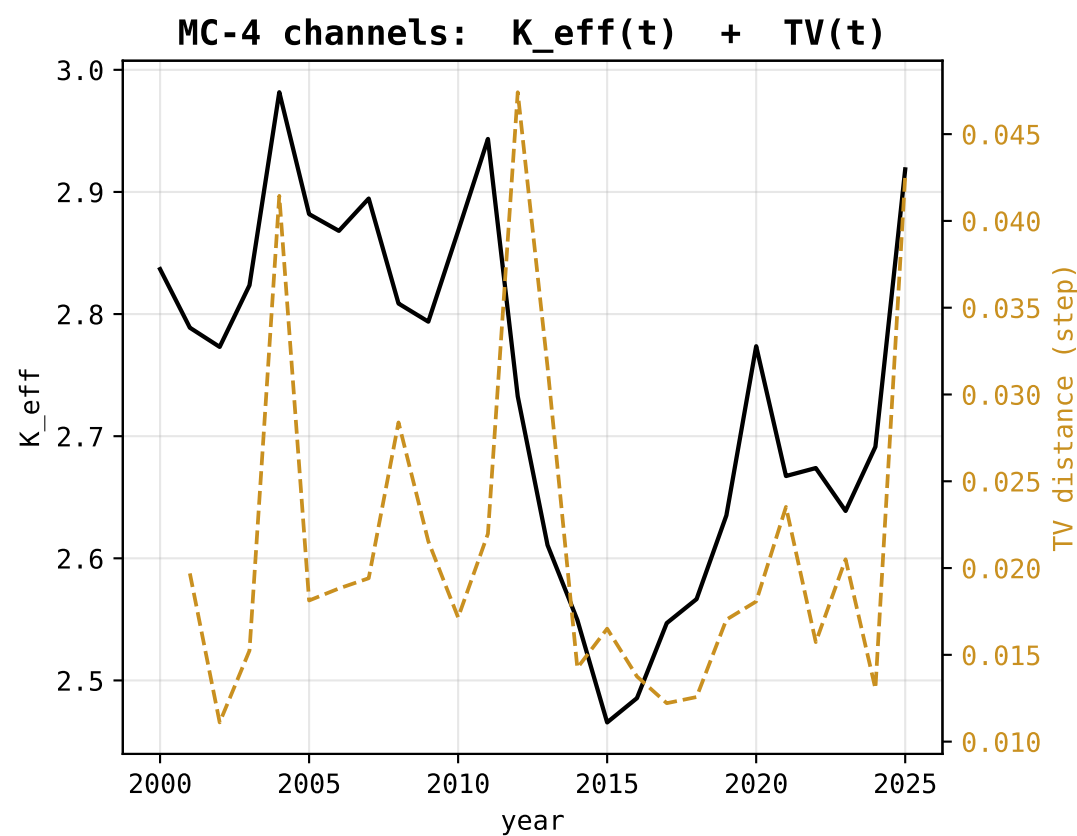
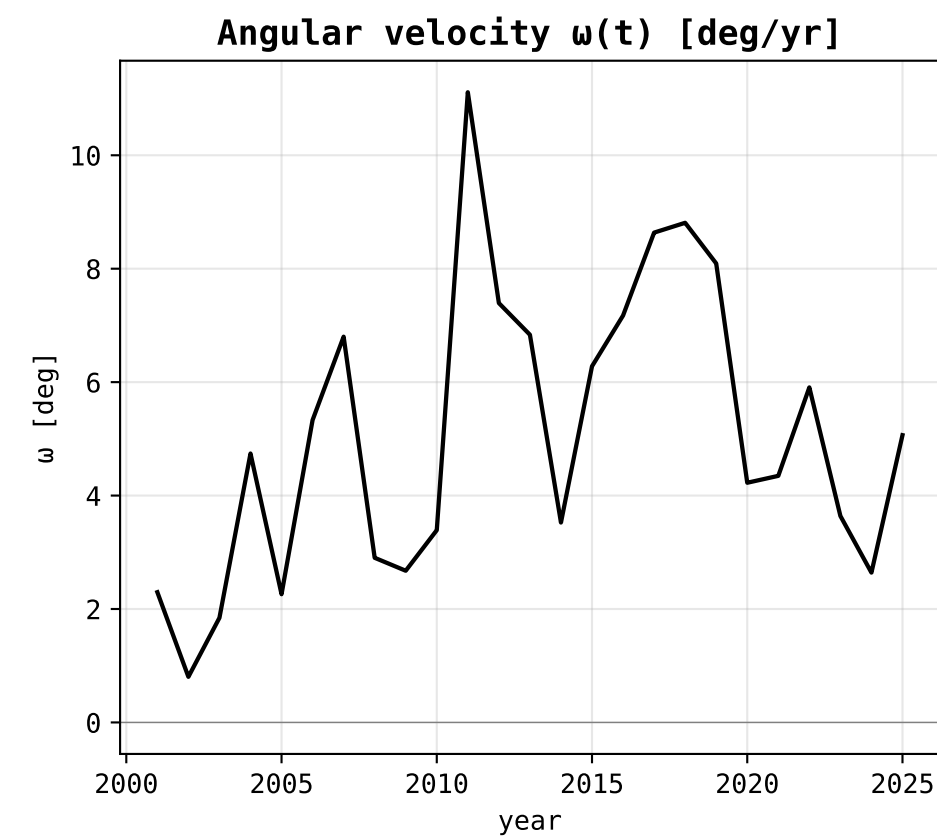
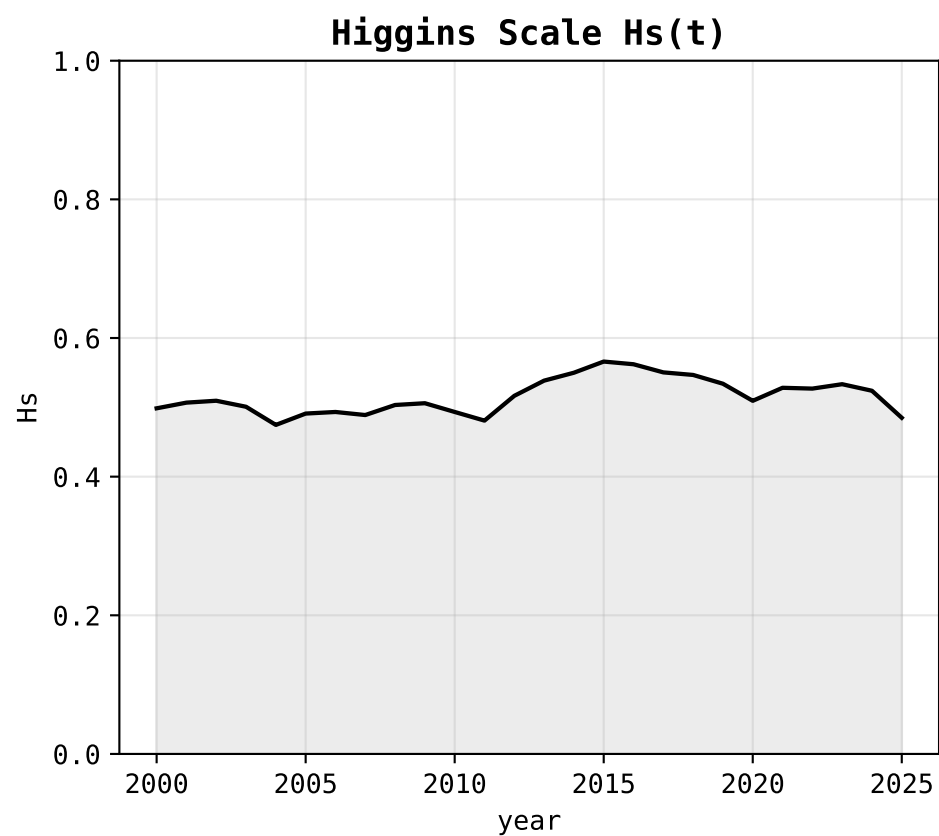
STAGE 3 — Carrier CLR Trajectory Correlation



NAVIGATION SUMMARY — All Stage 1 Metrics by Year

Year	Hs	Ring	E_metric	omega(deg)	d_A	Helmsman	kappa_max
2000	0.4986	Hs-3	75.9	0.00	0.0000	---	57139
2001	0.5068	Hs-4	74.5	2.30	0.3563	Wind	58609
2002	0.5095	Hs-4	75.1	0.80	0.1264	Gas	60964
2003	0.5008	Hs-4	65.8	1.84	0.6194	Solar	31906
2004	0.4746	Hs-3	65.6	4.74	0.6706	Wind	34916
2005	0.4910	Hs-3	64.6	2.26	0.3241	Bioenergy	35226
2006	0.4933	Hs-3	73.6	5.33	0.9426	Solar	74443
2007	0.4889	Hs-3	51.3	6.80	1.6922	Solar	13271
2008	0.5034	Hs-4	50.9	2.90	0.3633	Nuclear	13807
2009	0.5059	Hs-4	48.4	2.67	0.3764	Other Fossil	10983
2010	0.4933	Hs-3	45.6	3.39	0.4518	Other Fossil	8513
2011	0.4808	Hs-3	28.7	11.11	1.8198	Solar	1244
2012	0.5166	Hs-4	22.2	7.39	0.9115	Solar	518
2013	0.5385	Hs-4	19.6	6.84	0.6175	Gas	333
2014	0.5499	Hs-4	18.2	3.52	0.3124	Solar	256
2015	0.5660	Hs-4	17.8	6.28	0.4664	Solar	185
2016	0.5622	Hs-4	16.4	7.17	0.5433	Solar	148
2017	0.5504	Hs-4	16.0	8.64	0.6091	Solar	166
2018	0.5467	Hs-4	16.5	8.81	0.6227	Solar	246
2019	0.5340	Hs-4	20.4	8.09	0.7555	Other Fossil	521
2020	0.5094	Hs-4	19.9	4.23	0.3364	Solar	506
2021	0.5282	Hs-4	19.7	4.35	0.3379	Gas	495
2022	0.5270	Hs-4	20.2	5.91	0.4633	Gas	519
2023	0.5334	Hs-4	19.9	3.64	0.2859	Hydro	498
2024	0.5239	Hs-4	20.1	2.64	0.2077	Solar	498
2025	0.4849	Hs-3	20.8	5.07	0.4061	Solar	508

CNQ v2.0.0 – Quaternion View – IND India · D=8 · N=26



CNQ DIAGNOSTICS · IND (D = 8)

CHSH joint-coherence:

S value: 0.0800

classical bound: 2.0

Tsirelson bound: 2.8284271247461903

verdict: independent

Twin-quaternion factor:

enabled: True

partition axis: 6

max |residual|: 1.7617e+00

Attractor fit:

amplitude A: 0.616

contraction λ : -0.0187

period: 1

fitted: False

Helmsman family:

total flips: 15

stability S_σ : 0.375

§ JPN Japan

EMBER electricity-generation series · 2000-2025 · N = 26 · D = 8

INPUT

source: ember_JPN*_generation_TWh.csv
N records: 26
D carriers: 8
Bioenergy
Coal
Gas
Hydro
Nuclear
Other Fossil
Solar
Wind
source_sha256: e7643dca648ecadc96682877...

HEADLINE METRICS

Helmsman flips: ?
Helmsman stability S_σ: {'global': 0.29166666666666663, 'rolling': [0.5714285714285714, 0.42857142857142855]}
IR class: ?
Attractor amplitude: ?

CHSH S: ?
Tsirelson bound: $2\sqrt{2} \approx 2.828$

Bearing pairs (D(D-1)/2): 28

CNT v3.1.0 content_sha: ?...
CNQ v2.0.0 content_sha: ?...

Following pages: Stage 1 (XY/XZ/YZ × N years + course) → Stage 2/3 (7 pages) → CNQ (2 pages).

SECTION PLATE t=0 | Year 2000 | D=8 N=26 pairs=28

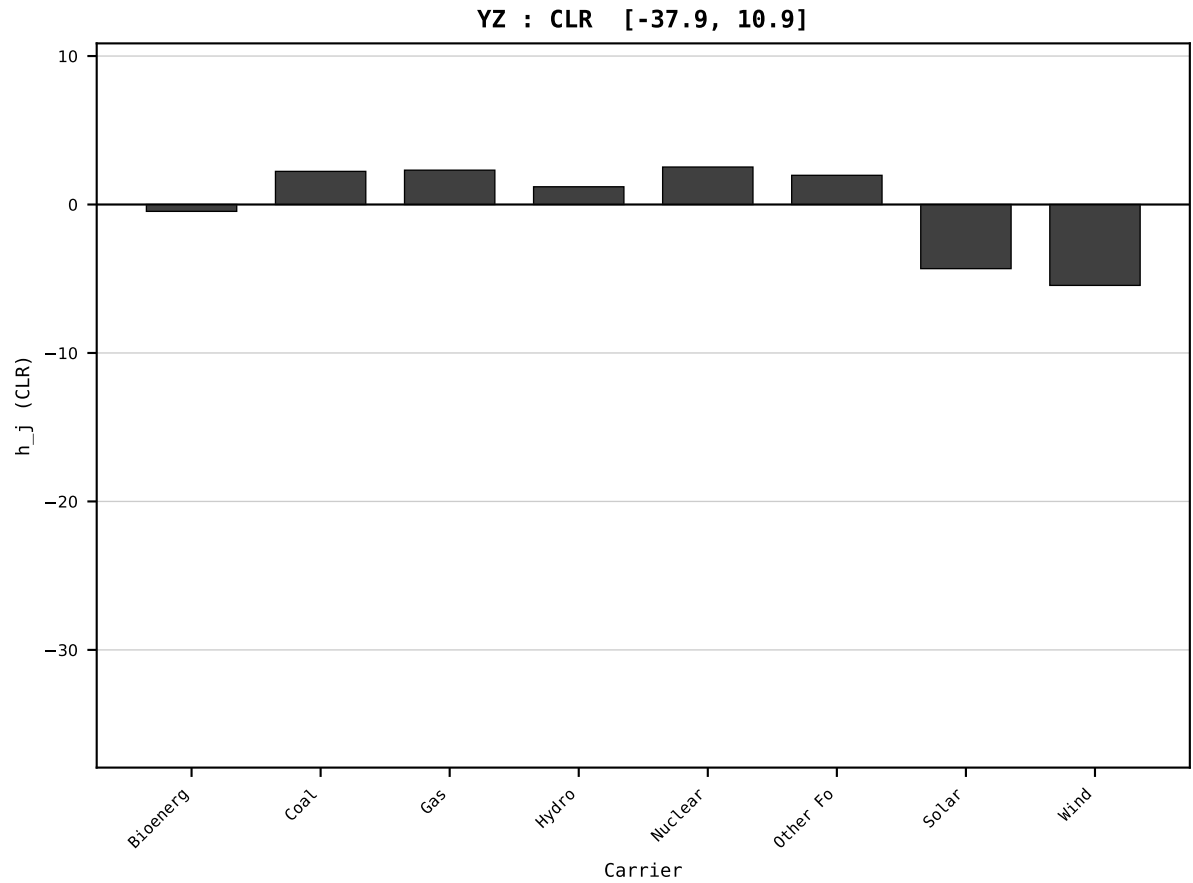
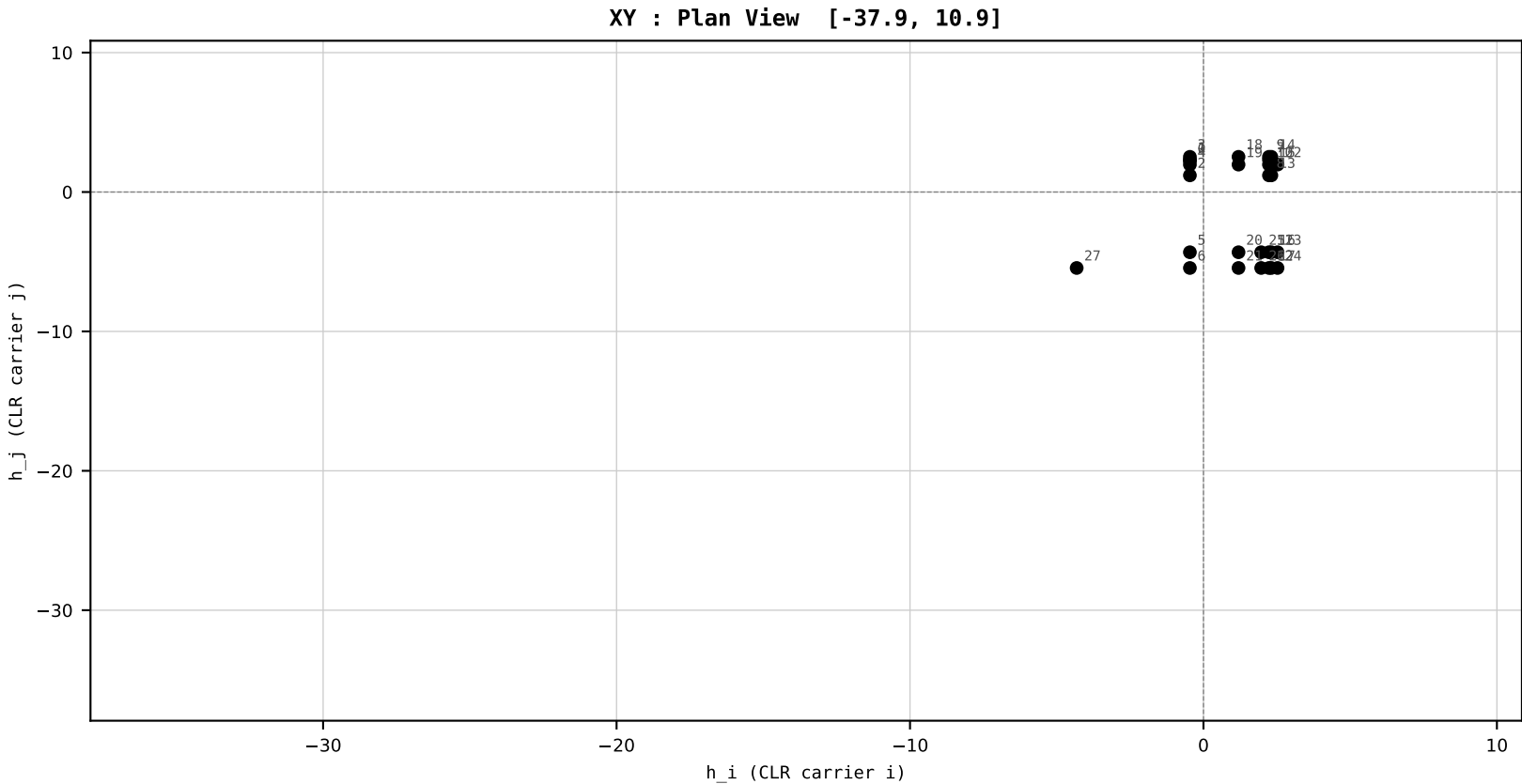
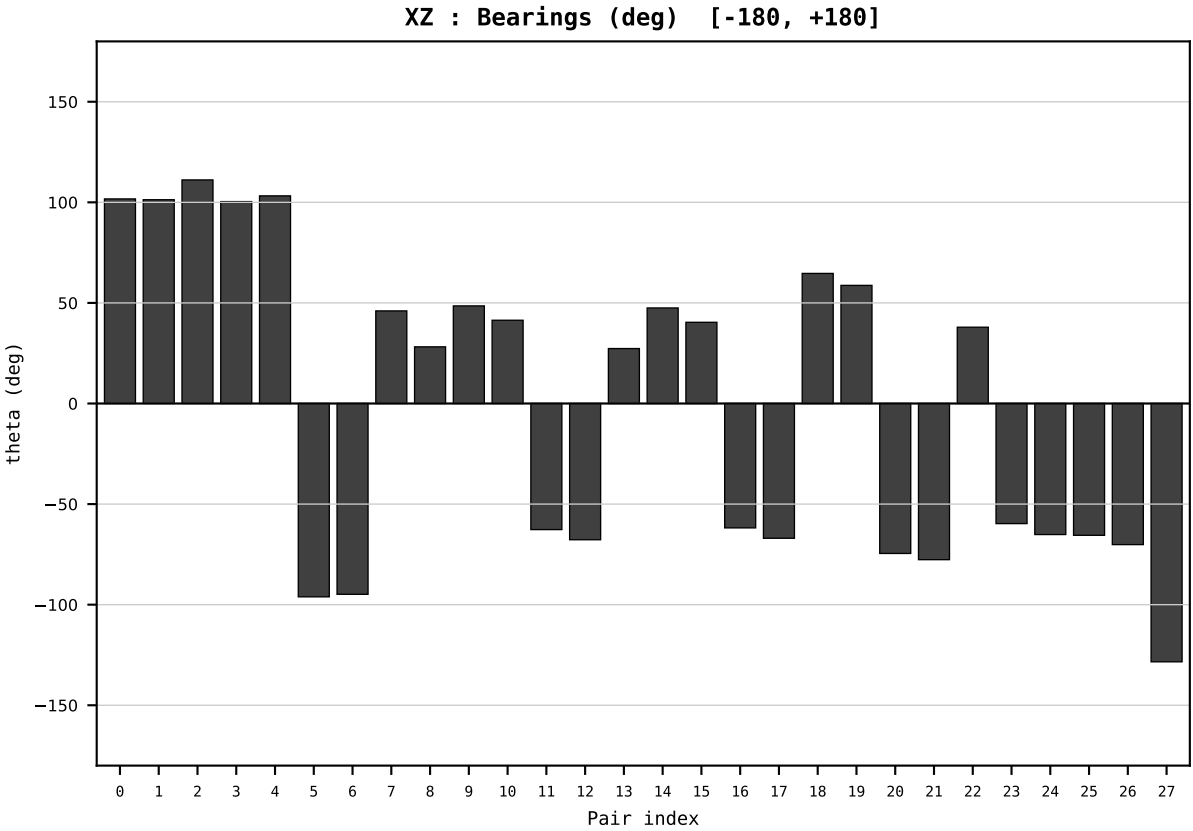
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2000
t : 0 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.234683
Ring = Hs-2
E_metric= 70.5673
kappa_HS= 2901.09
omega = 0.0000 deg
d_A = 0.000000
Helm = ---
Helm d = ---
DR = 7.973 HLR
DR ratio= 2901.1x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=1 | Year 2001 | D=8 N=26 pairs=28

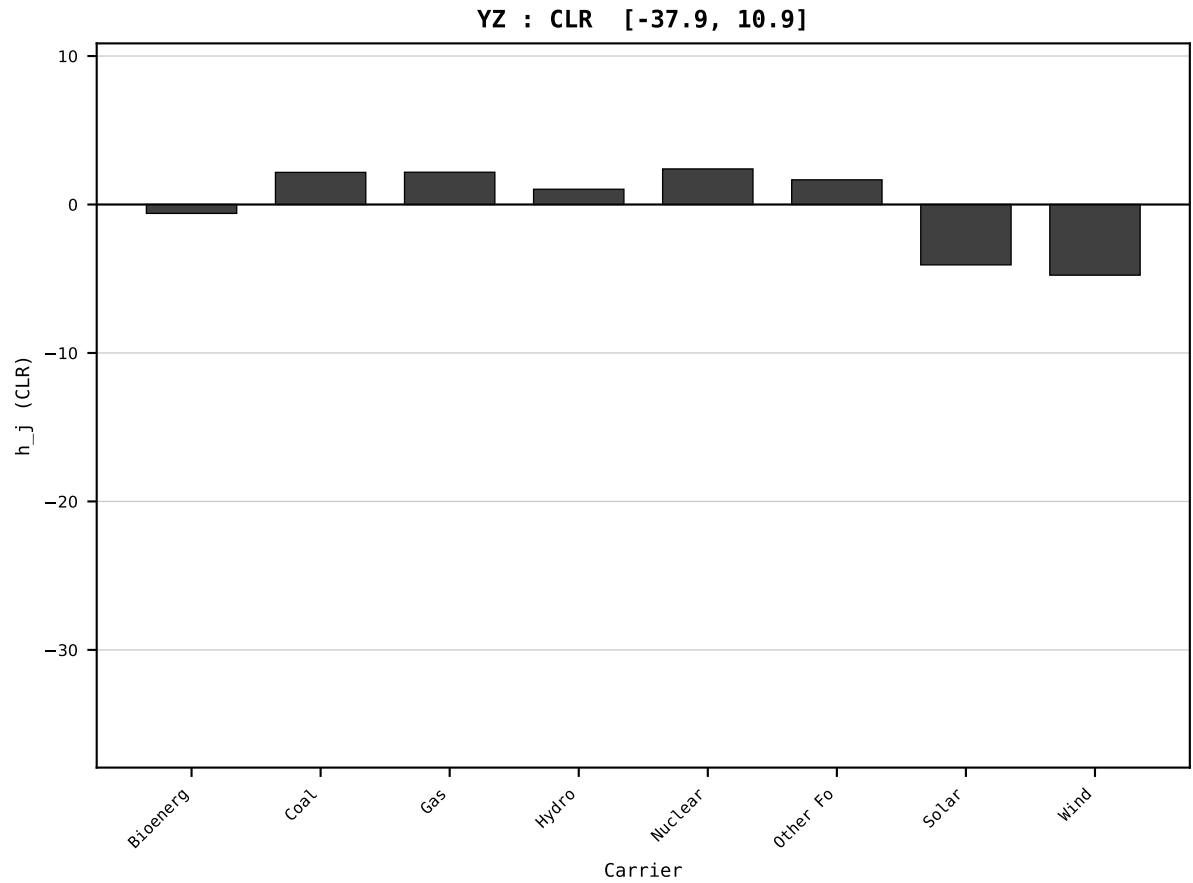
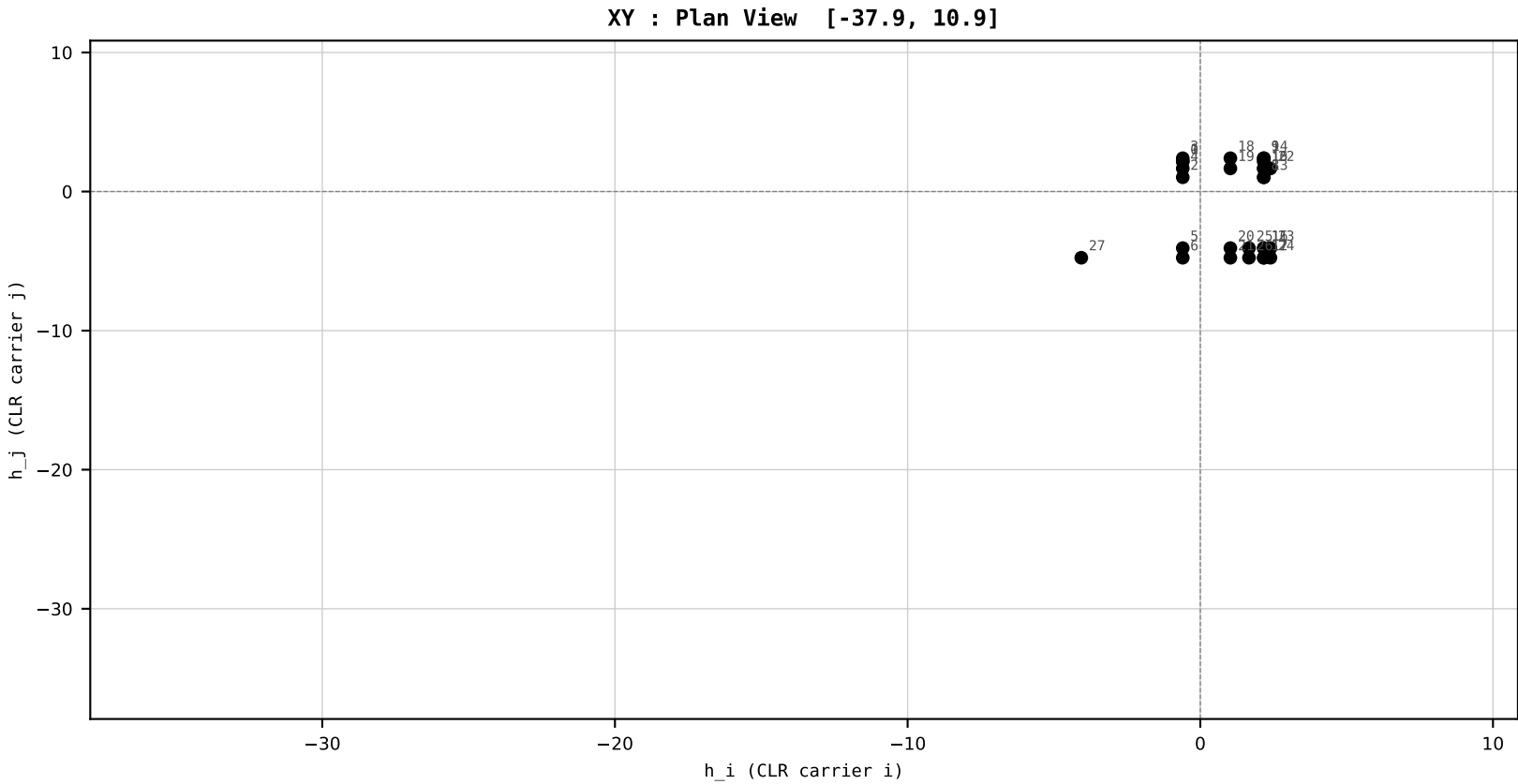
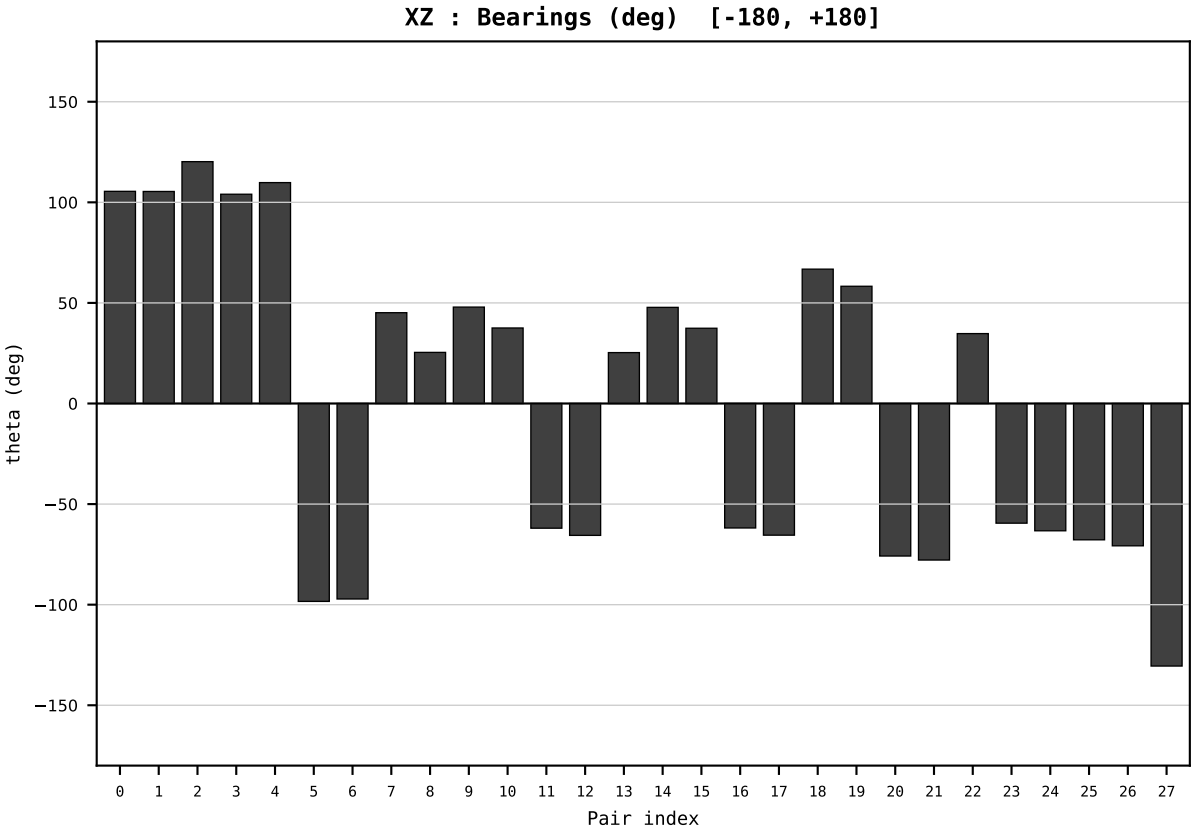
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2001
t : 1 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.239561
Ring = Hs-2
E_metric= 58.5208
kappa_HS= 1282.16
omega = 2.8189 deg
d_A = 0.847846
Helm = Wind
Helm d = +0.689167
DR = 7.156 HLR
DR ratio= 1282.2x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=2 | Year 2002 | D=8 N=26 pairs=28

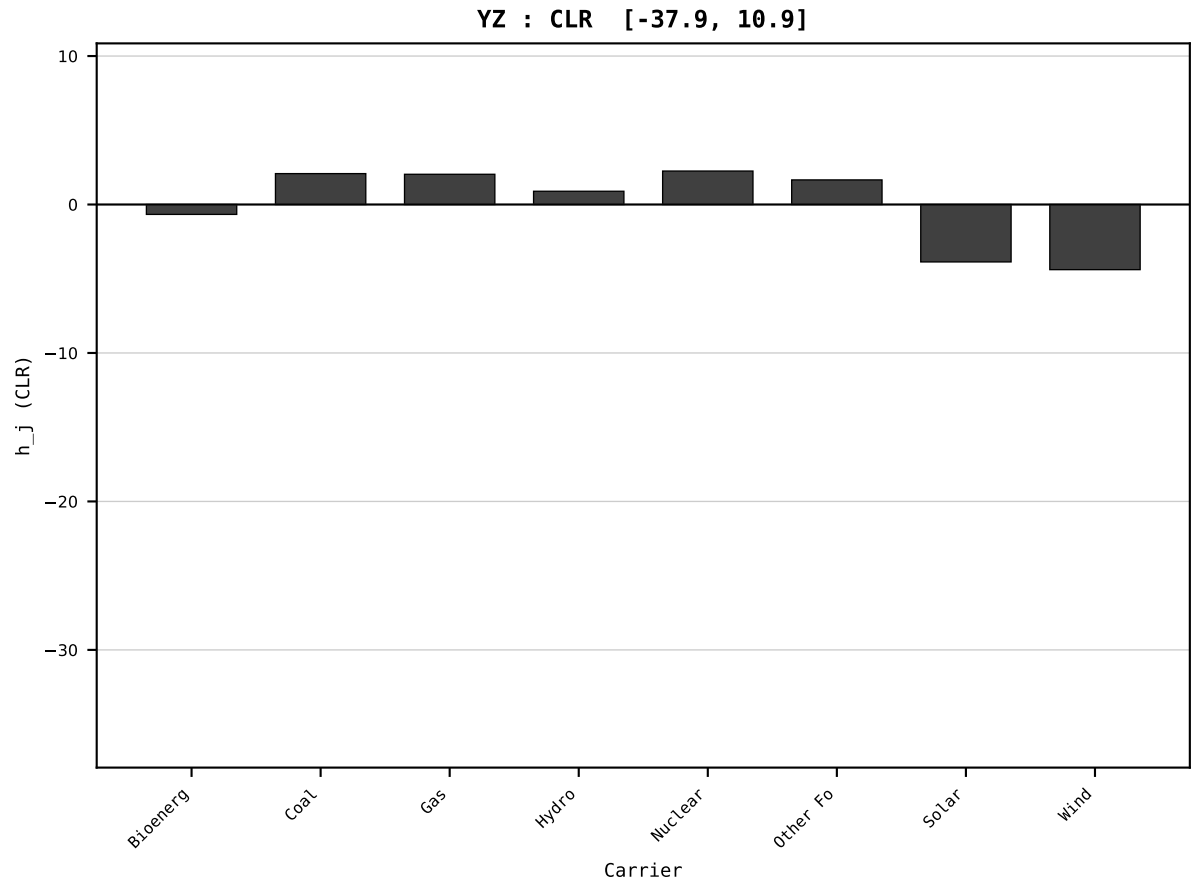
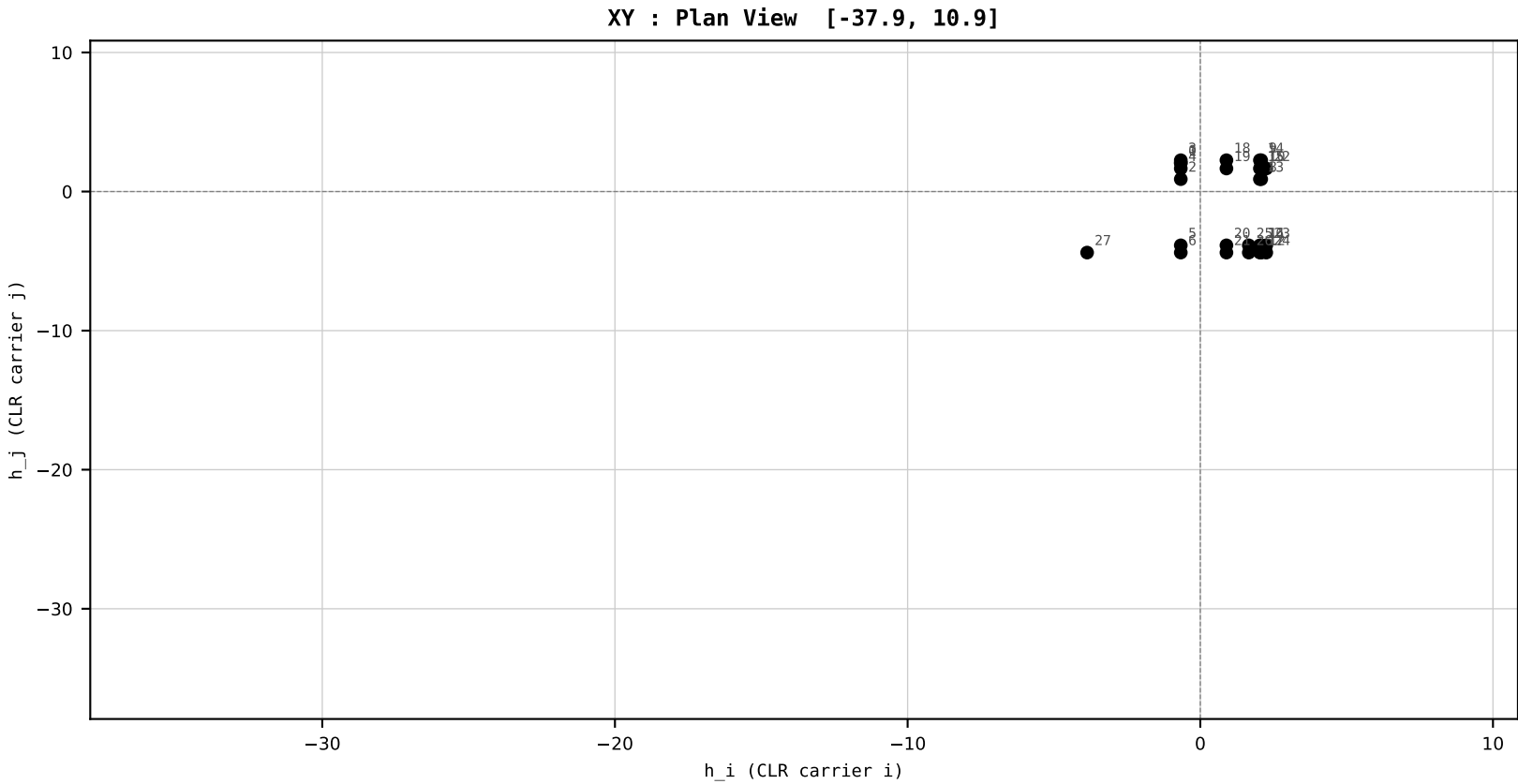
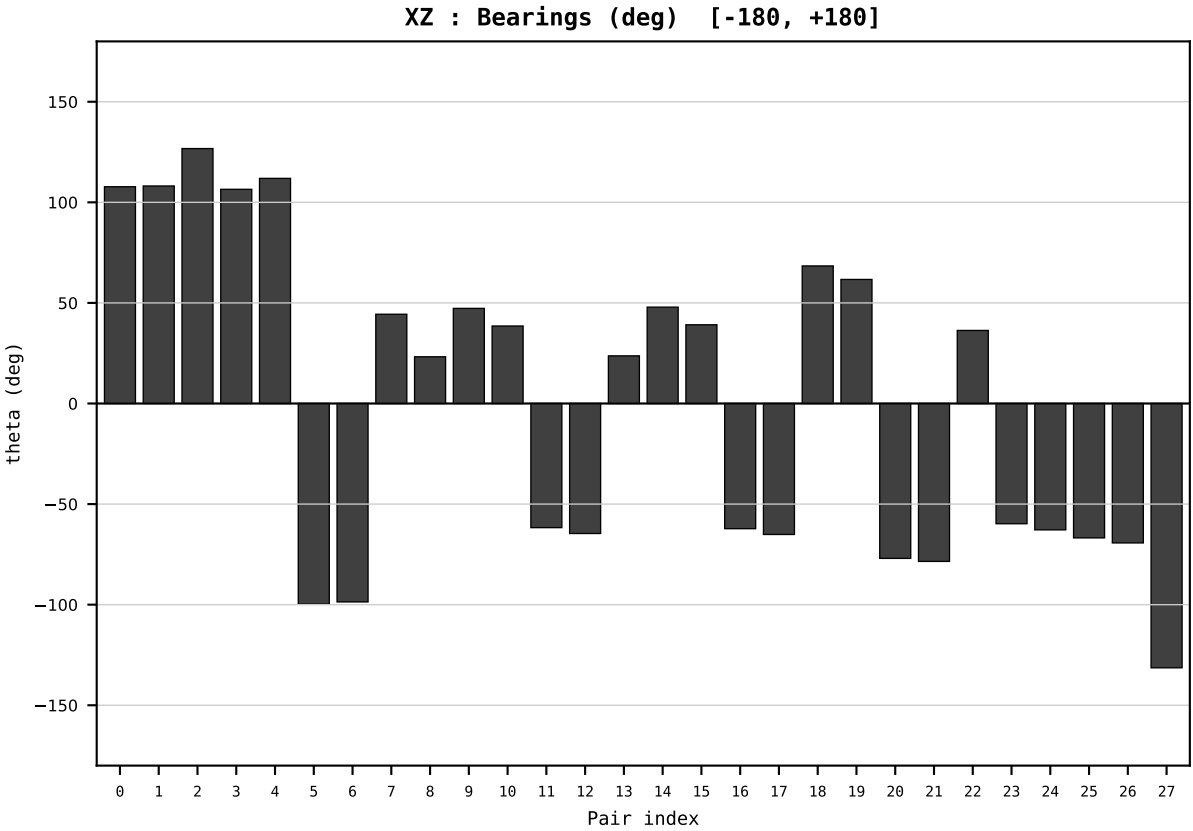
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2002
t : 2 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.234830
Ring = Hs-2
E_metric= 51.7554
kappa_HS= 766.49
omega = 1.5248 deg
d_A = 0.496689
Helm = Wind
Helm d = +0.371738
DR = 6.642 HLR
DR ratio= 766.5x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=3 | Year 2003 | D=8 N=26 pairs=28

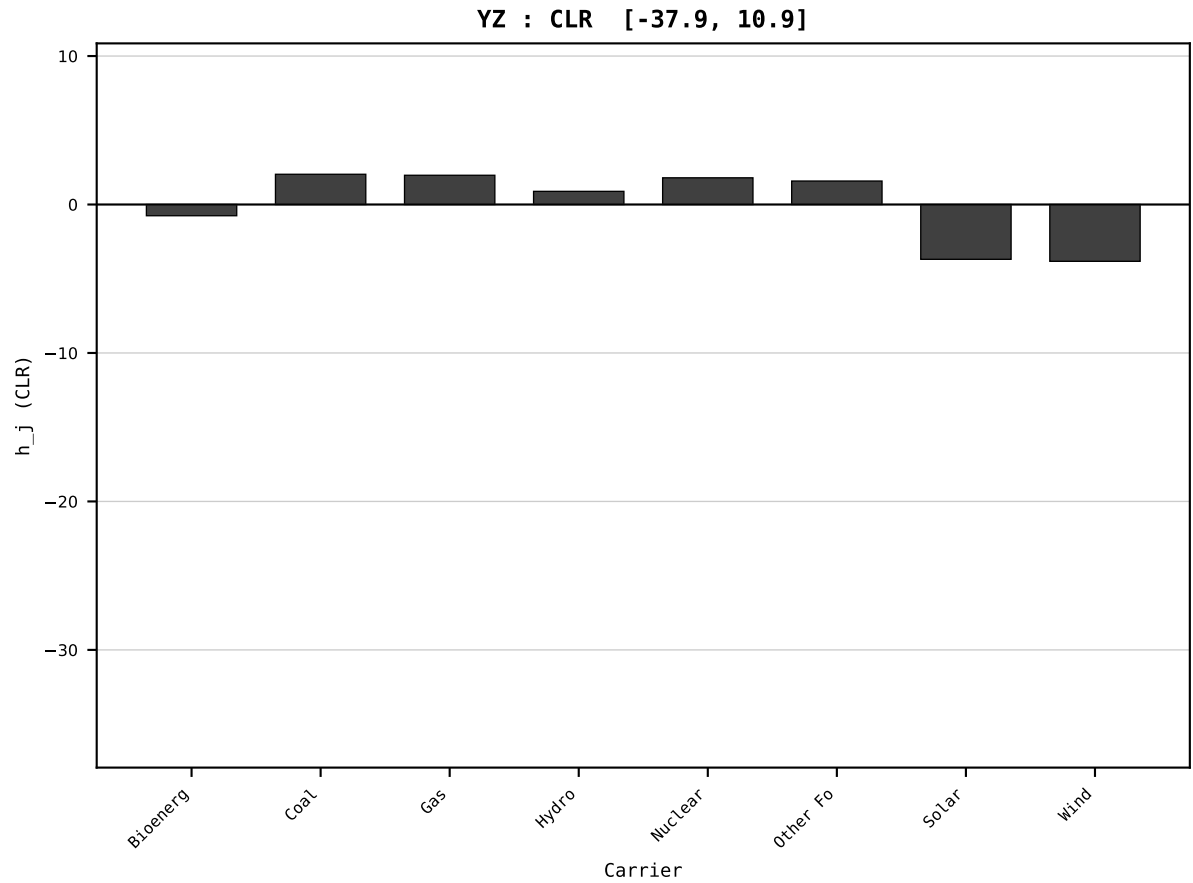
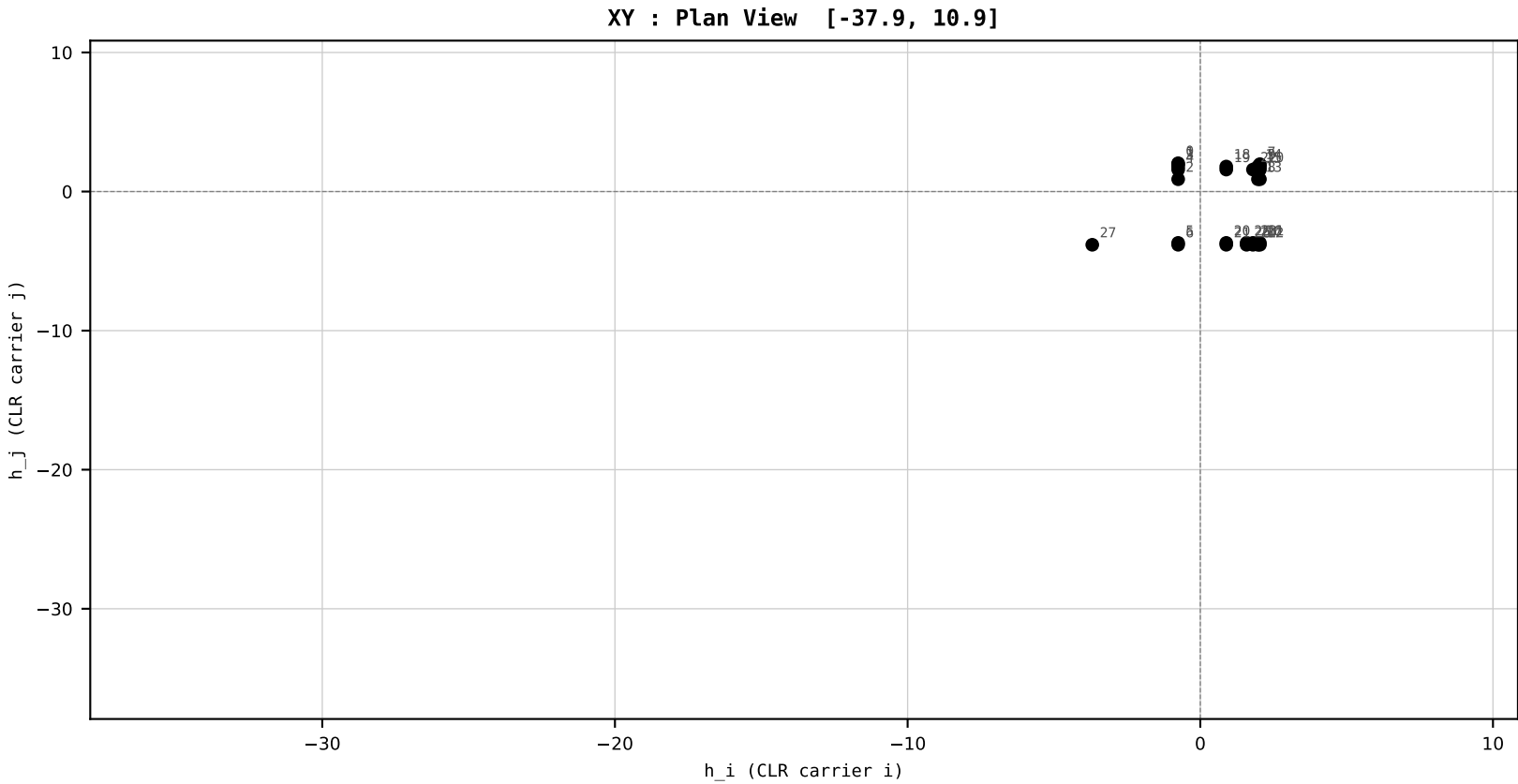
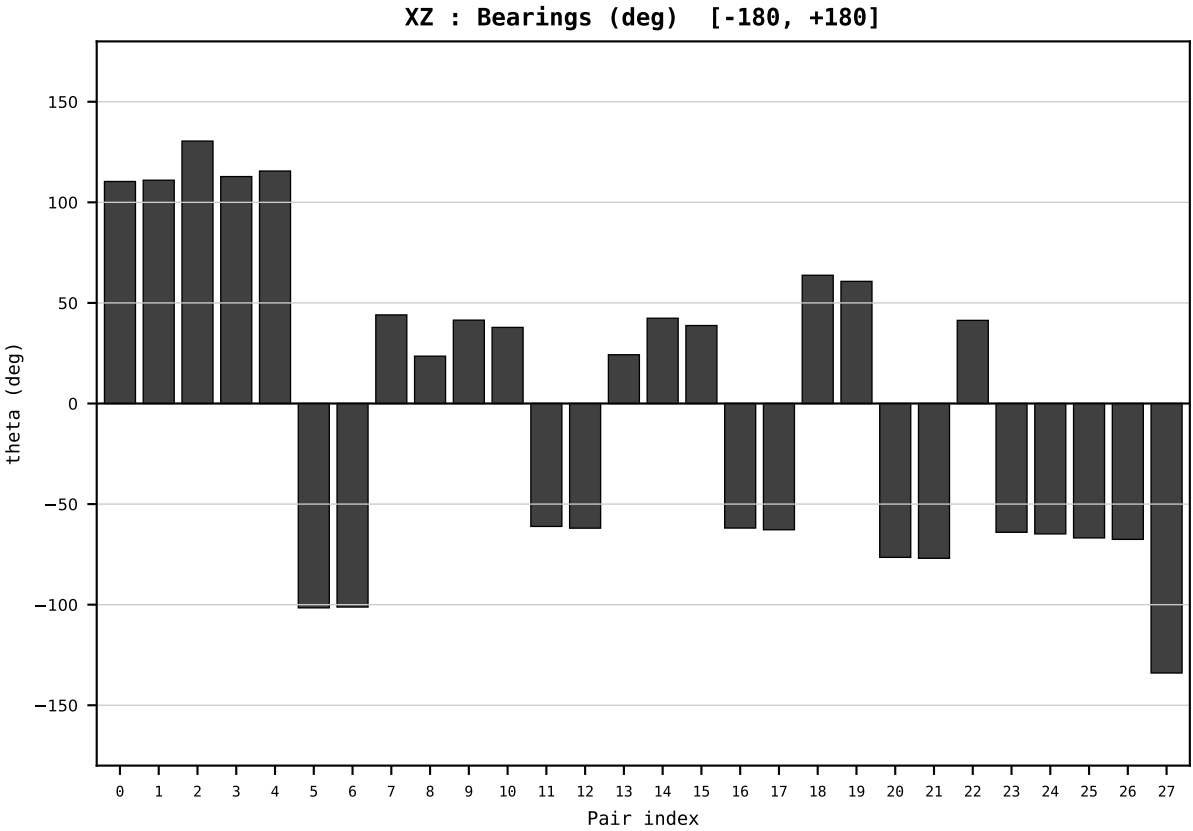
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2003
t : 3 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.222849
Ring = Hs-2
E_metric= 43.3901
kappa_HS= 351.88
omega = 3.7738 deg
d_A = 0.757601
Helm = Wind
Helm d = +0.561548
DR = 5.863 HLR
DR ratio= 351.9x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=4 | Year 2004 | D=8 N=26 pairs=28

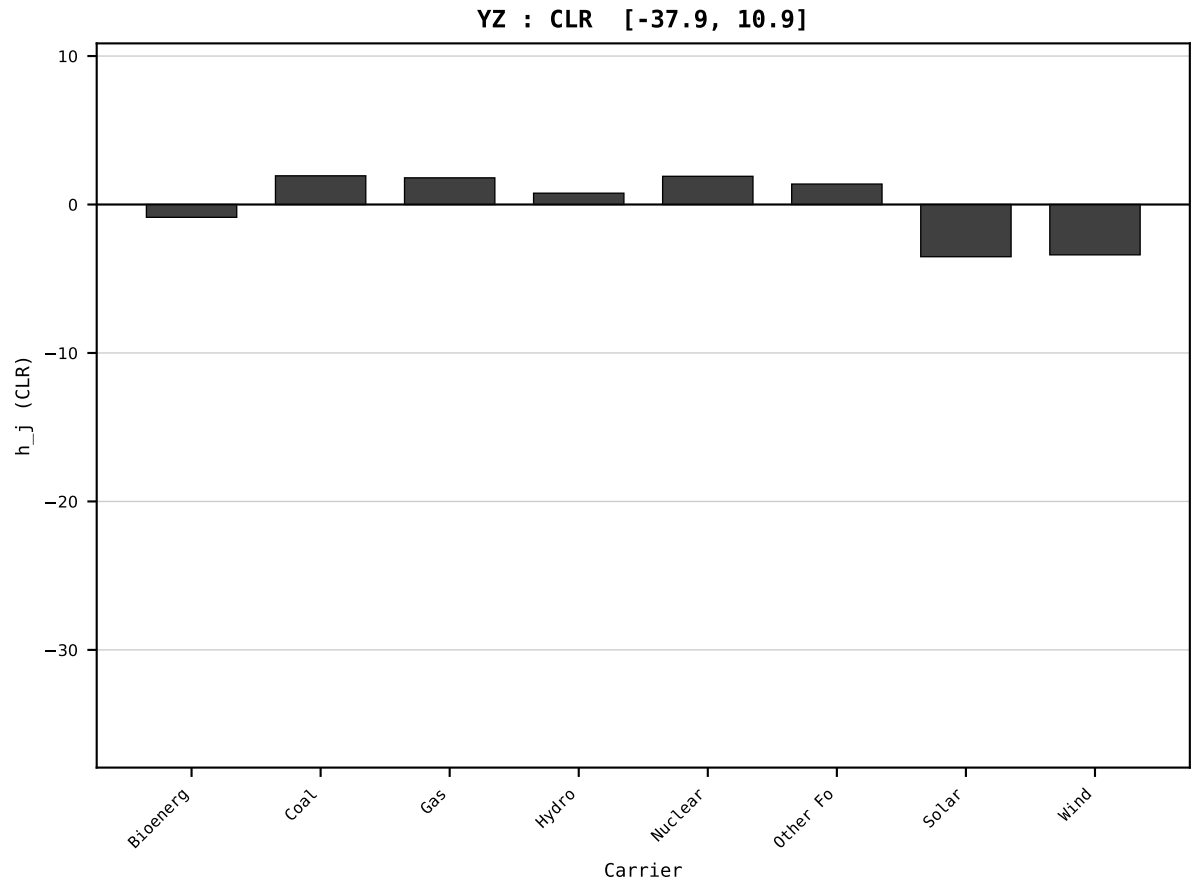
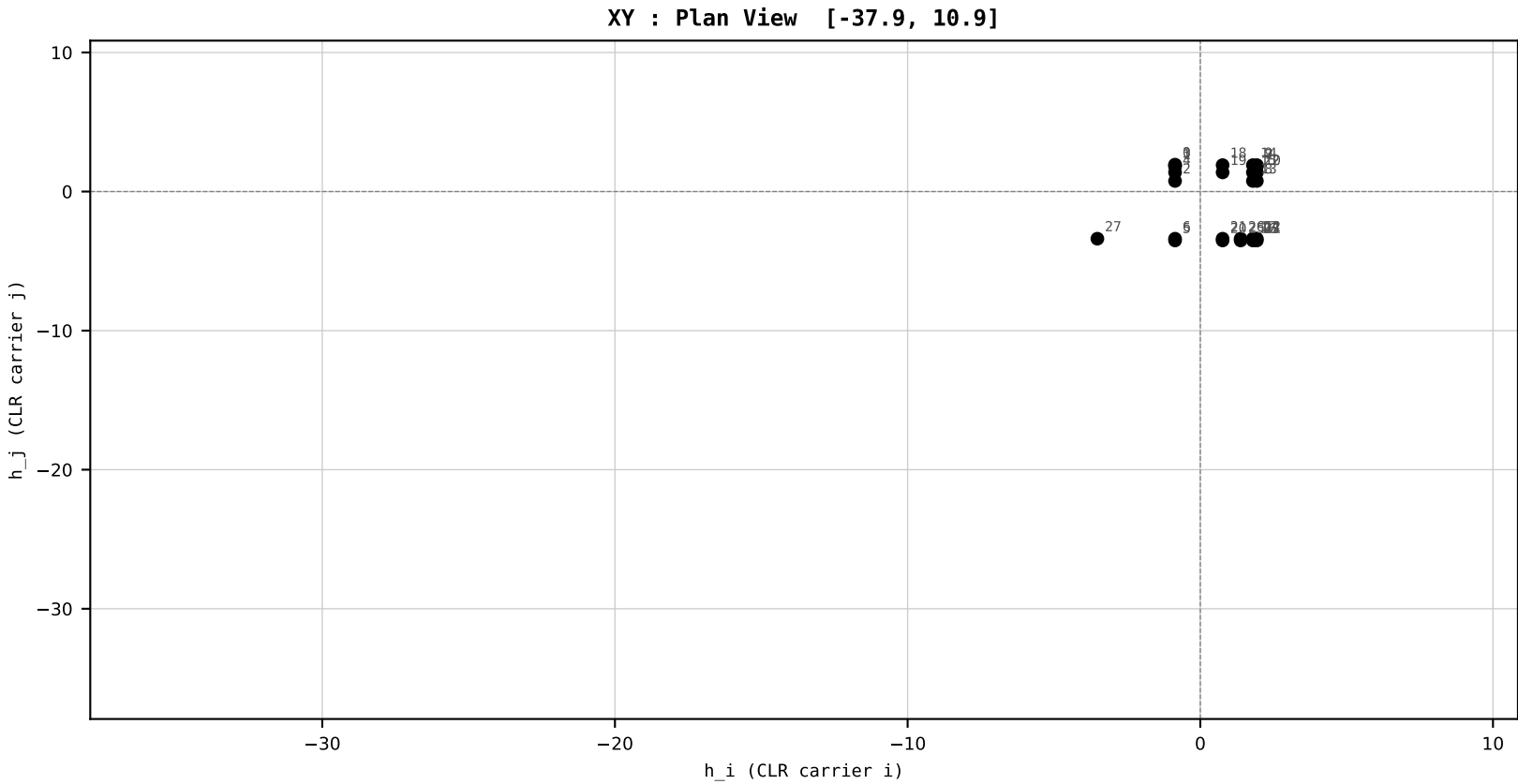
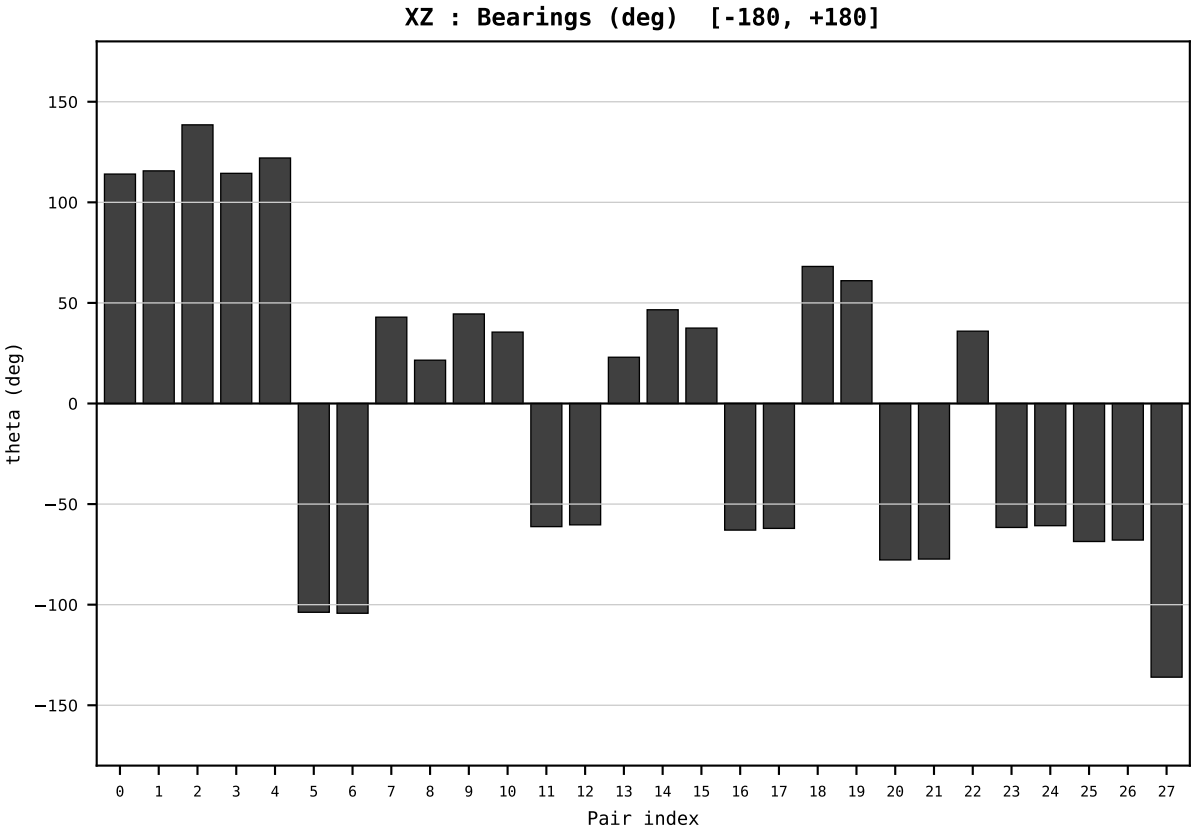
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2004
t : 4 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.225250
Ring = Hs-2
E_metric= 37.6631
kappa_HS= 232.59
omega = 3.3324 deg
d_A = 0.582491
Helm = Wind
Helm d = +0.435321
DR = 5.449 HLR
DR ratio= 232.6x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=5 | Year 2005 | D=8 N=26 pairs=28

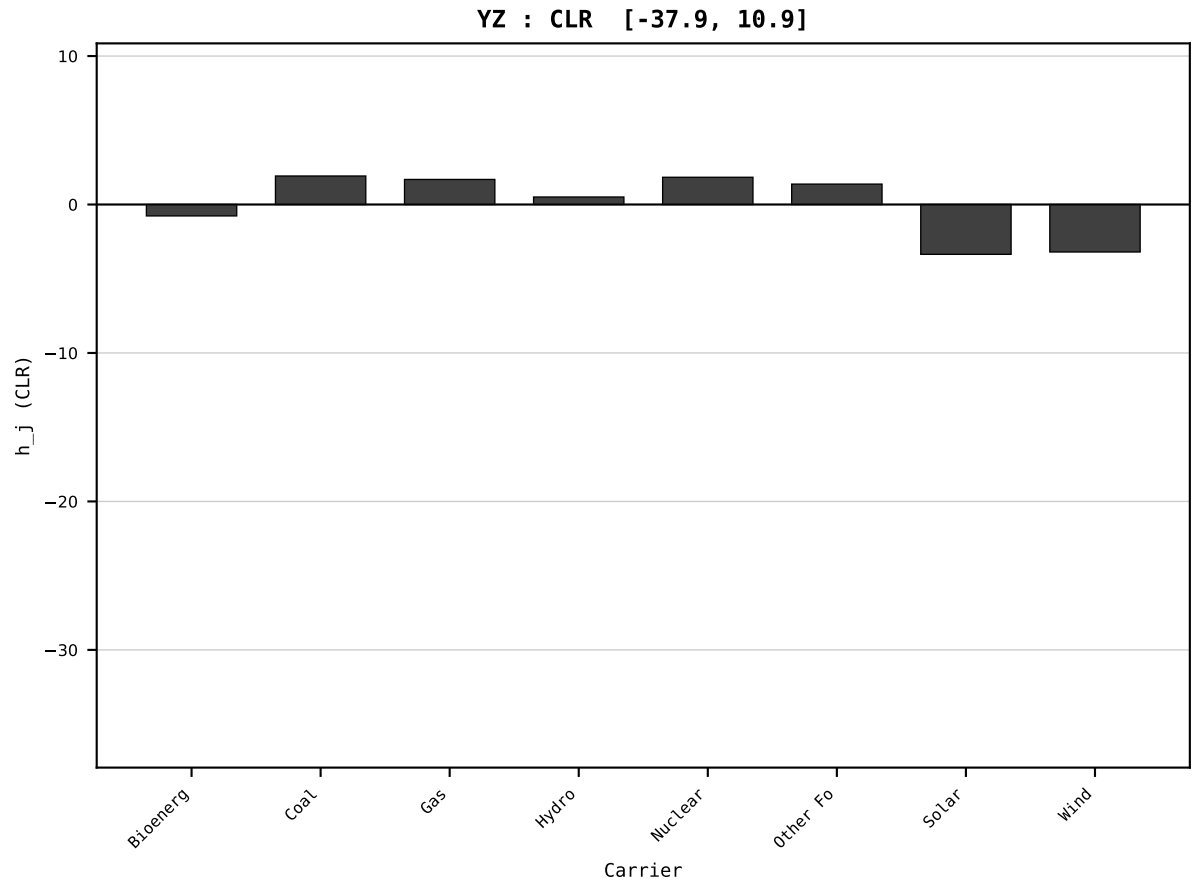
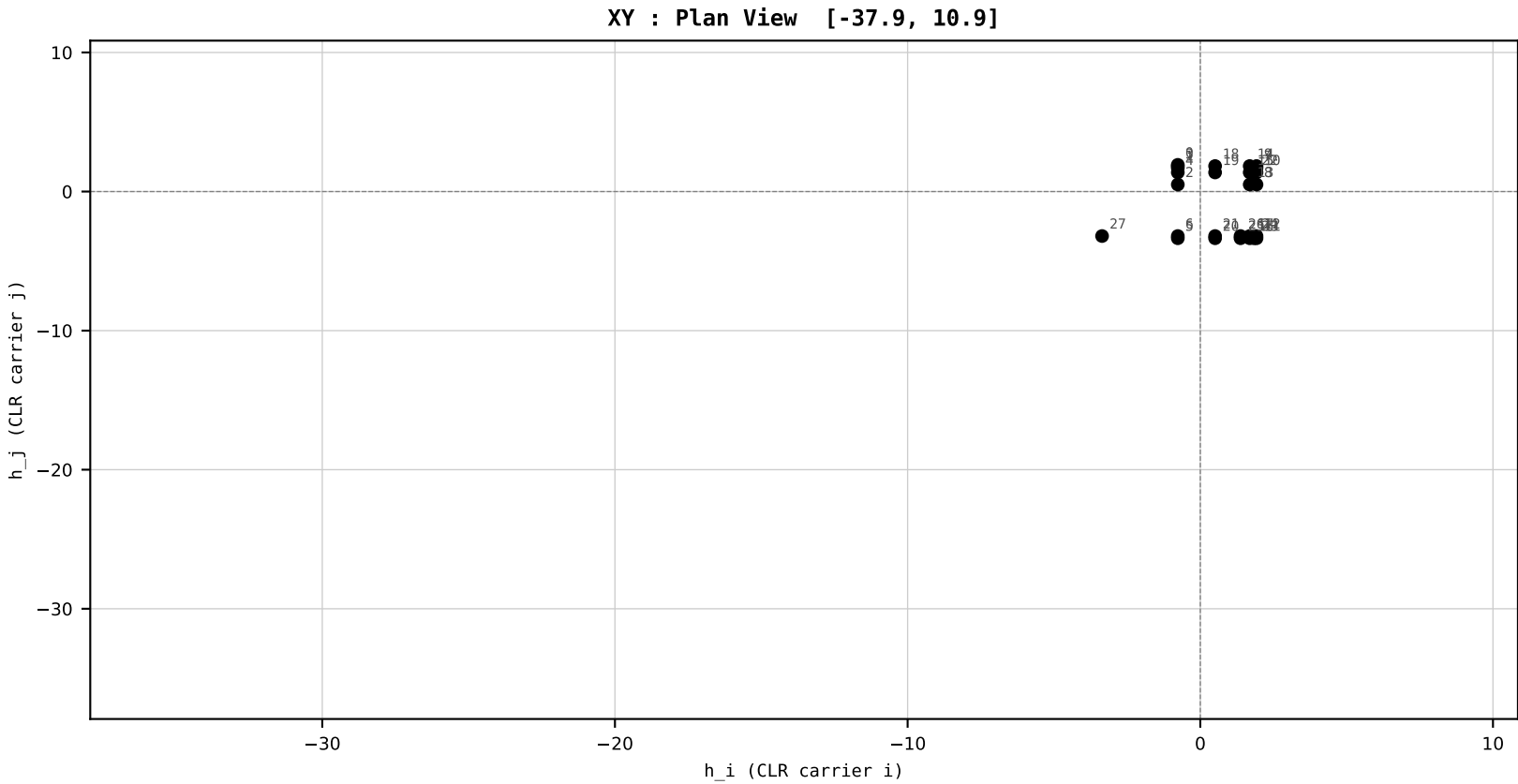
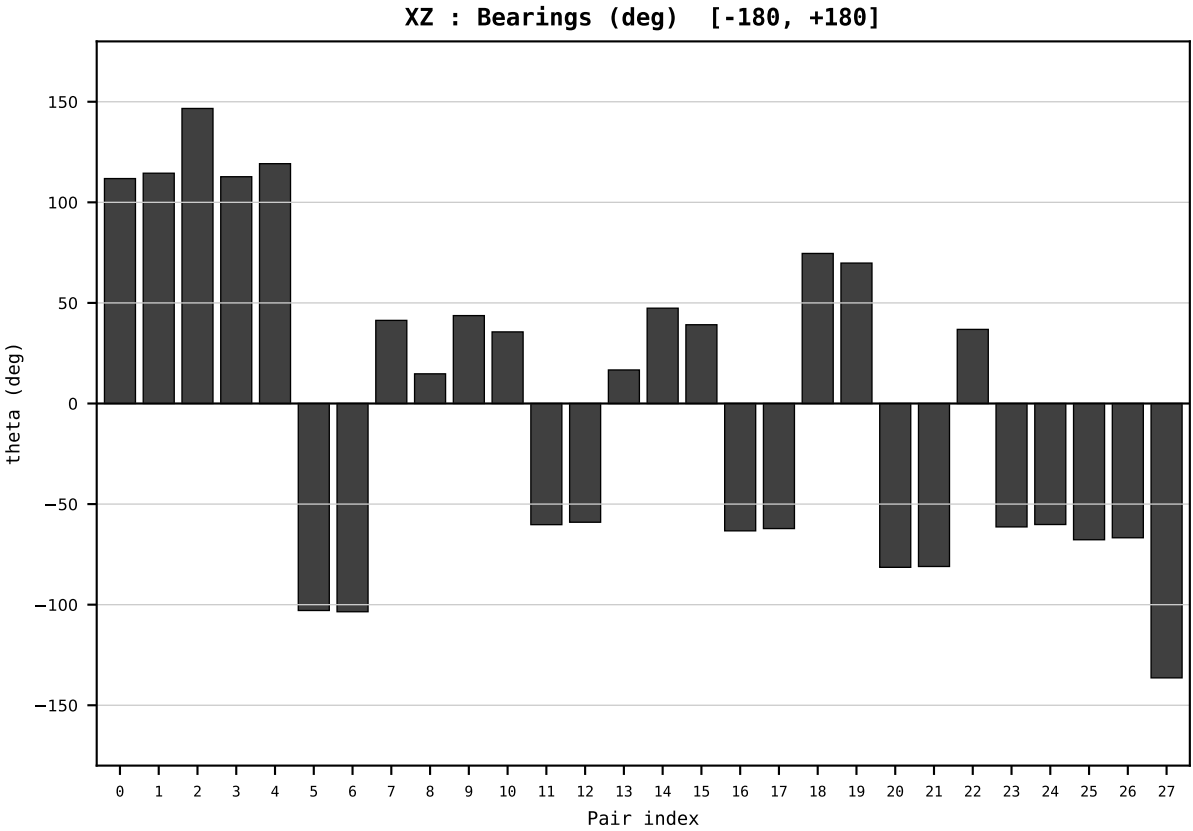
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2005
t : 5 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.227556
Ring = Hs-2
E_metric= 34.1389
kappa_HS= 195.83
omega = 2.4787 deg
d_A = 0.391966
Helm = Hydro
Helm d = -0.256970
DR = 5.277 HLR
DR ratio= 195.8x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=6 | Year 2006 | D=8 N=26 pairs=28

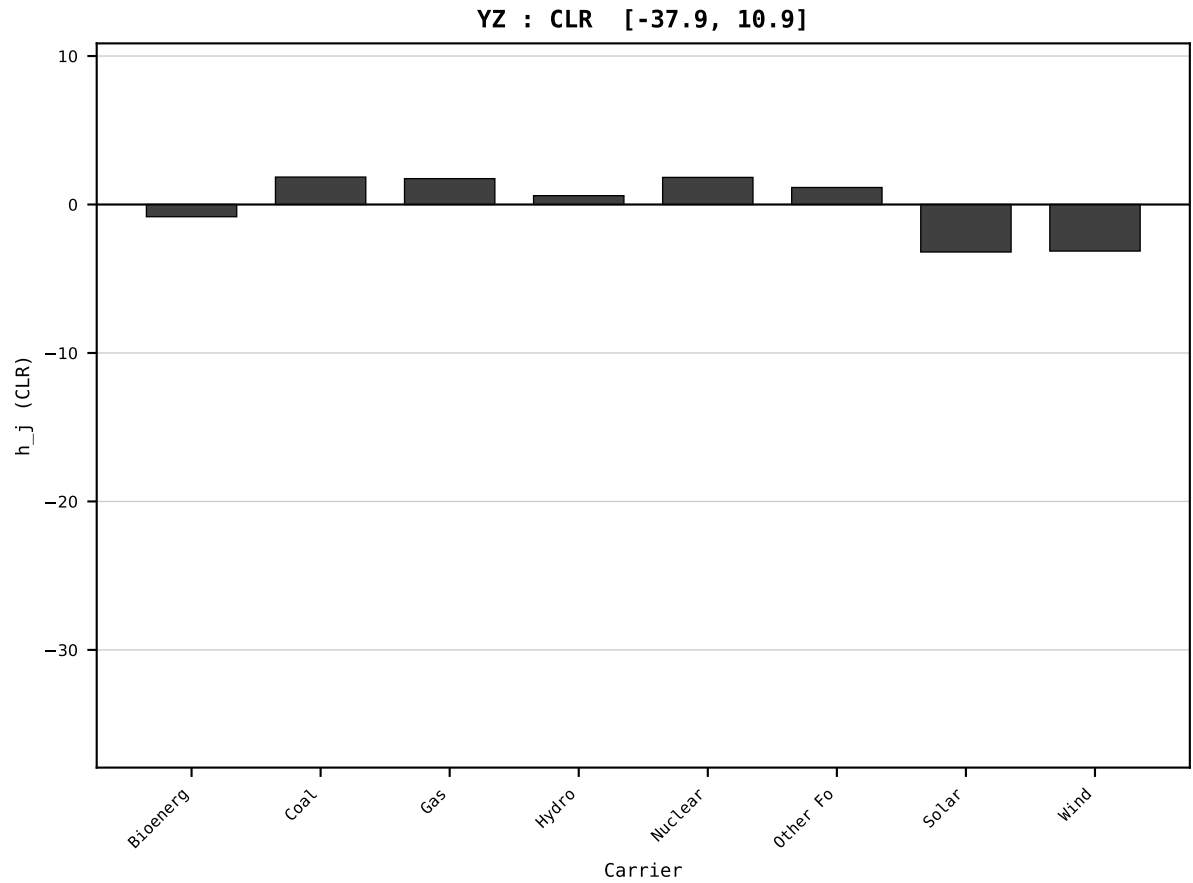
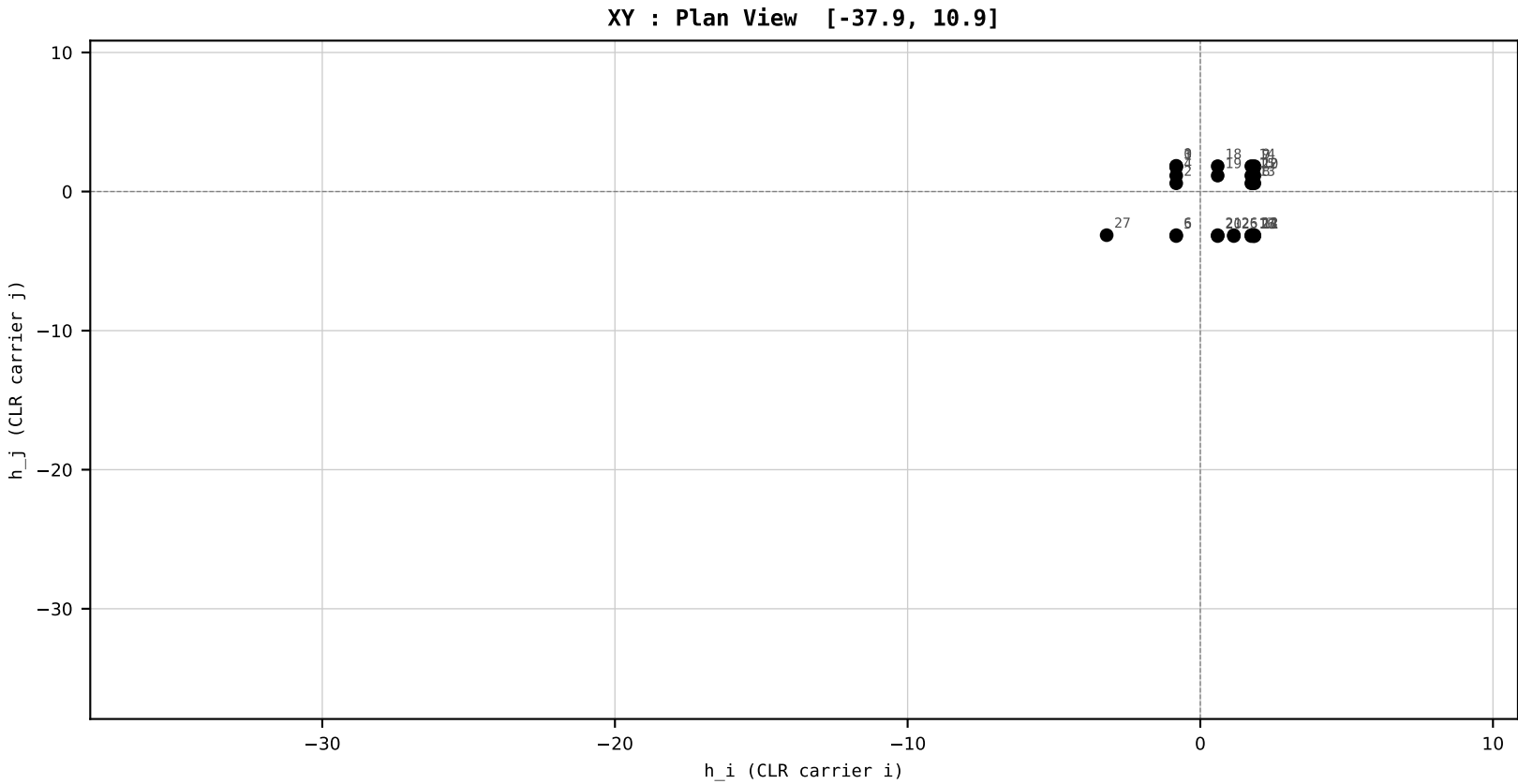
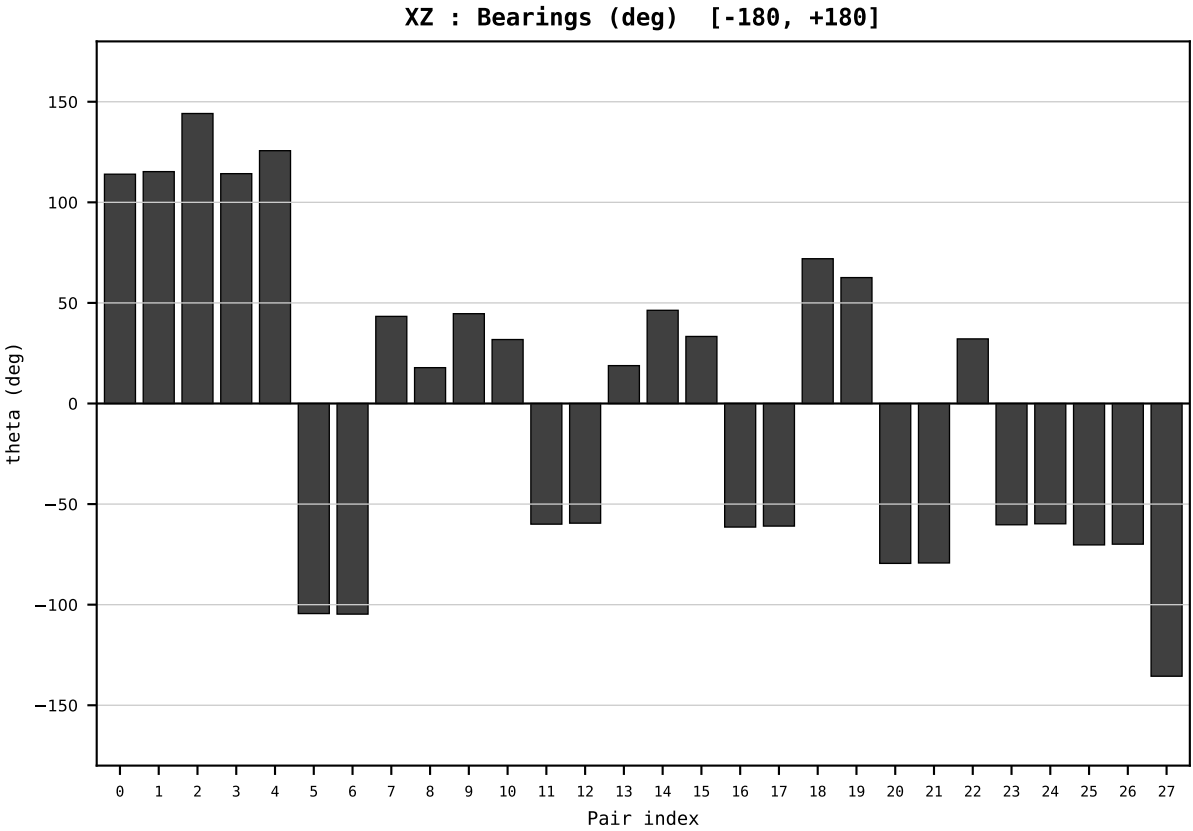
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2006
t : 6 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.227477
Ring = Hs-2
E_metric= 32.2009
kappa_HS= 155.67
omega = 2.6642 deg
d_A = 0.316211
Helm = Other Fossil
Helm d = -0.228493
DR = 5.048 HLR
DR ratio= 155.7x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=7 | Year 2007 | D=8 N=26 pairs=28

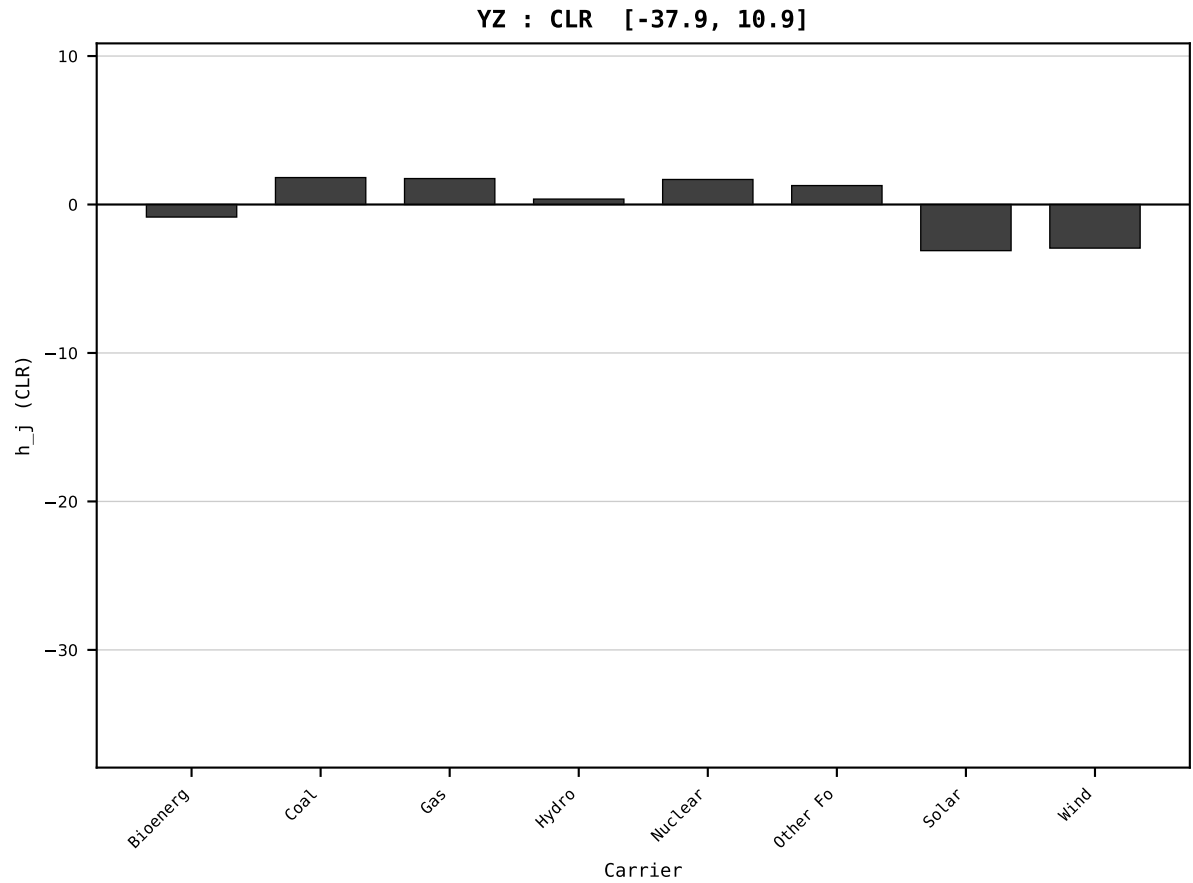
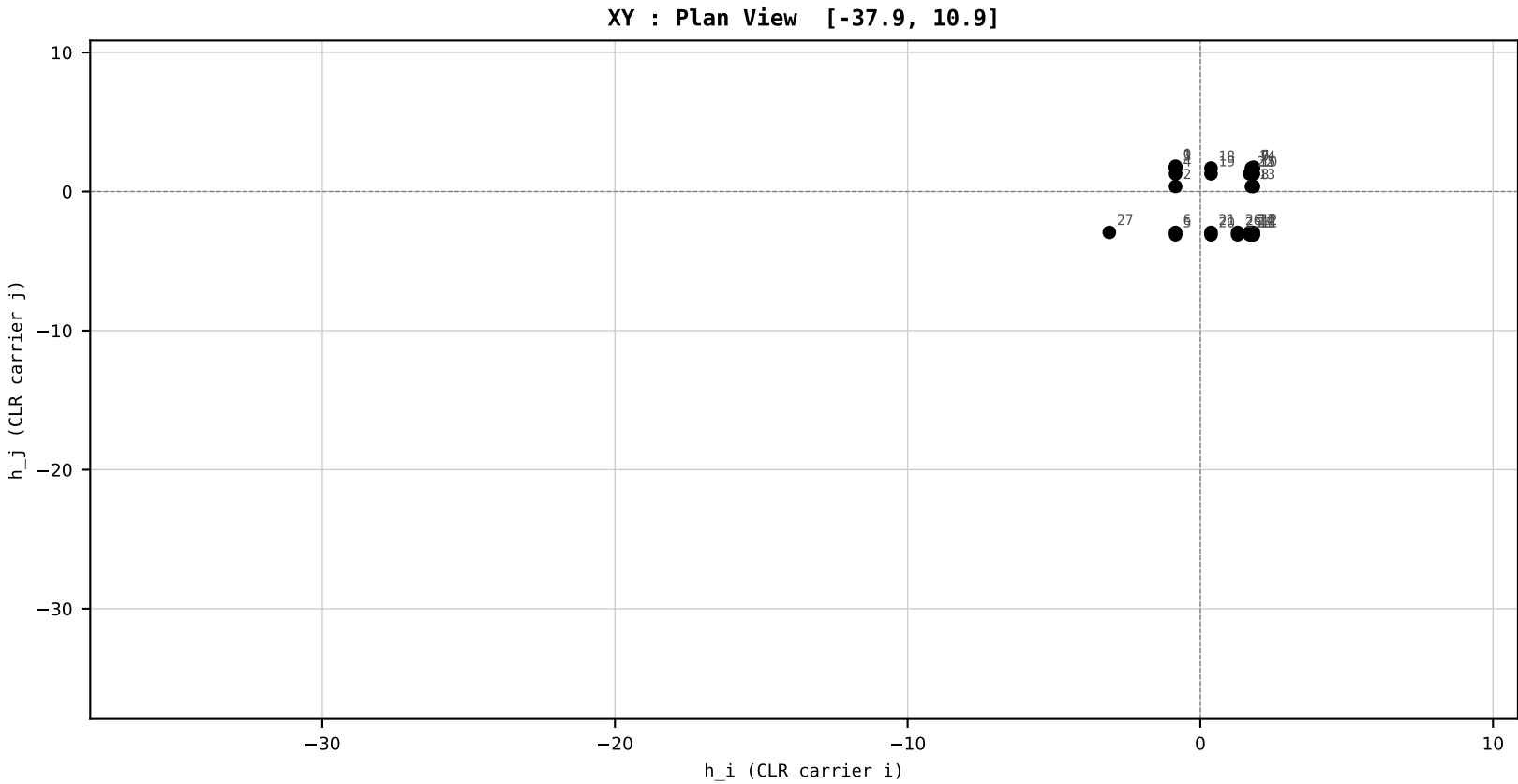
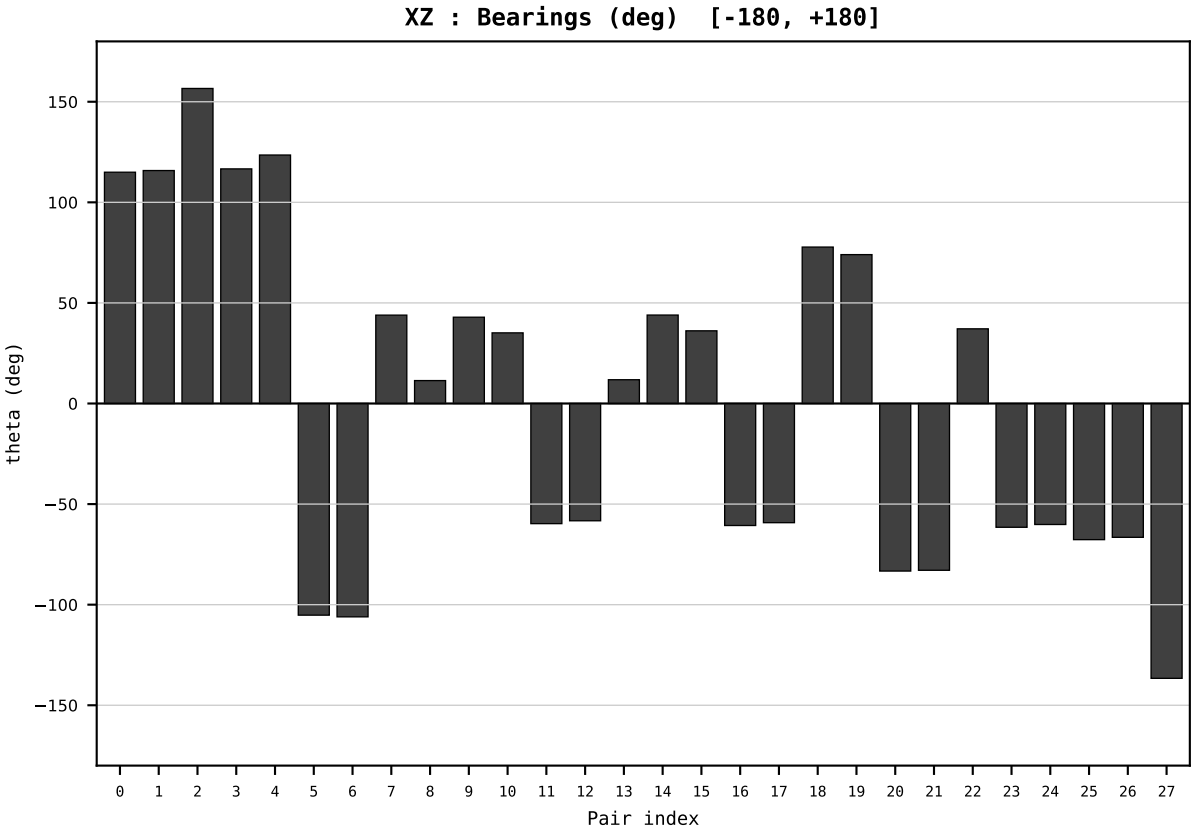
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2007
t : 7 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.227024
Ring = Hs-2
E_metric= 29.9546
kappa_HS= 137.36
omega = 3.2122 deg
d_A = 0.371740
Helm = Hydro
Helm d = -0.228639
DR = 4.923 HLR
DR ratio= 137.4x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=8 | Year 2008 | D=8 N=26 pairs=28

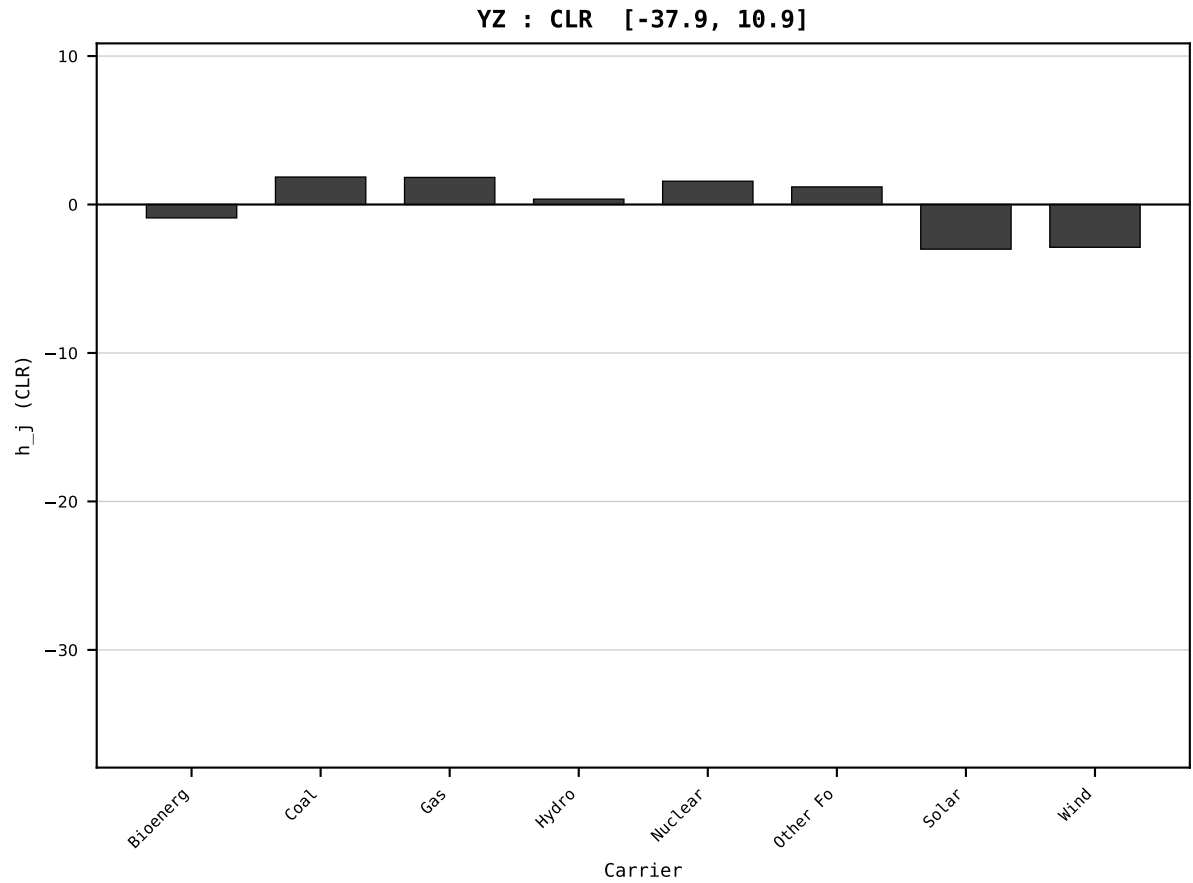
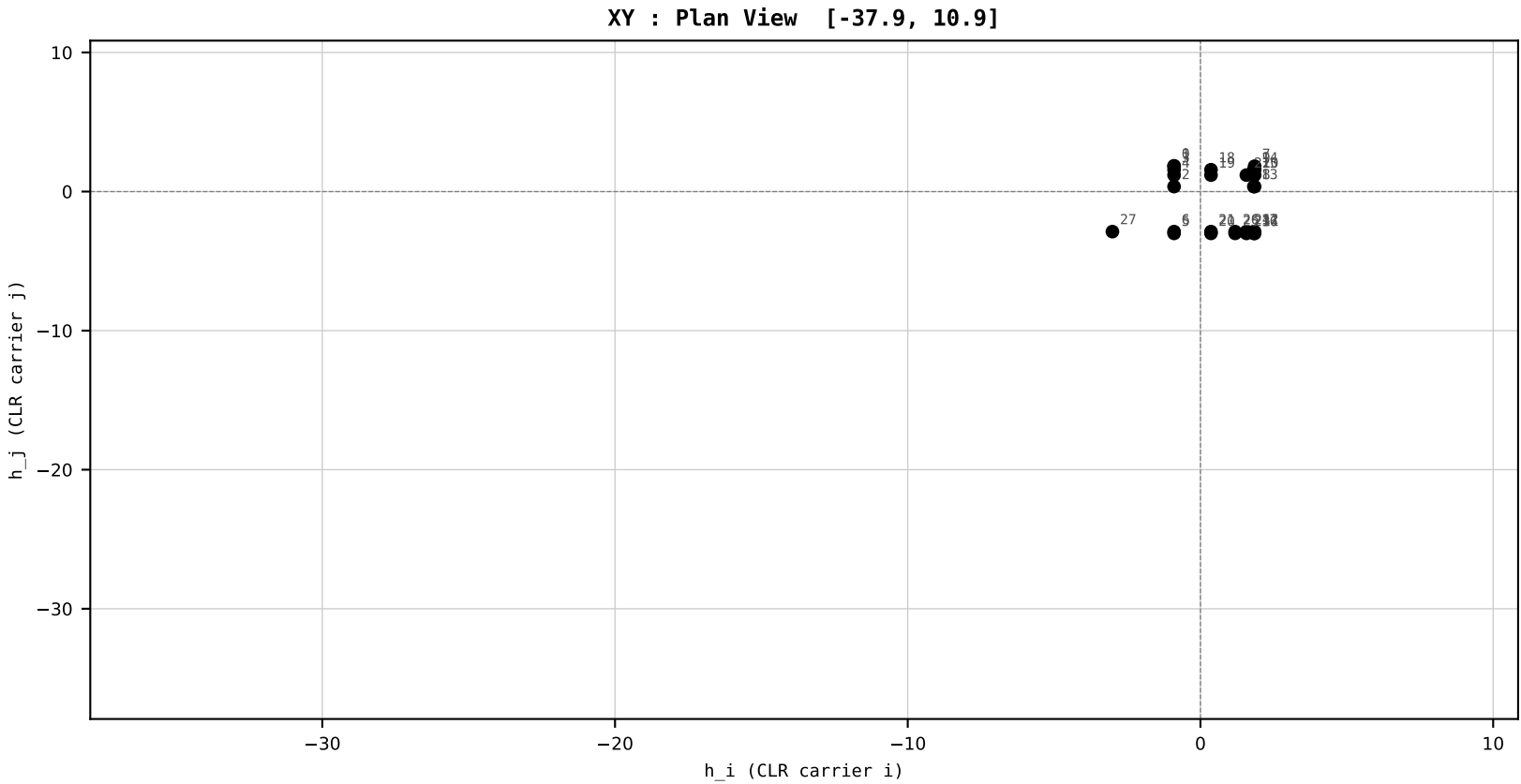
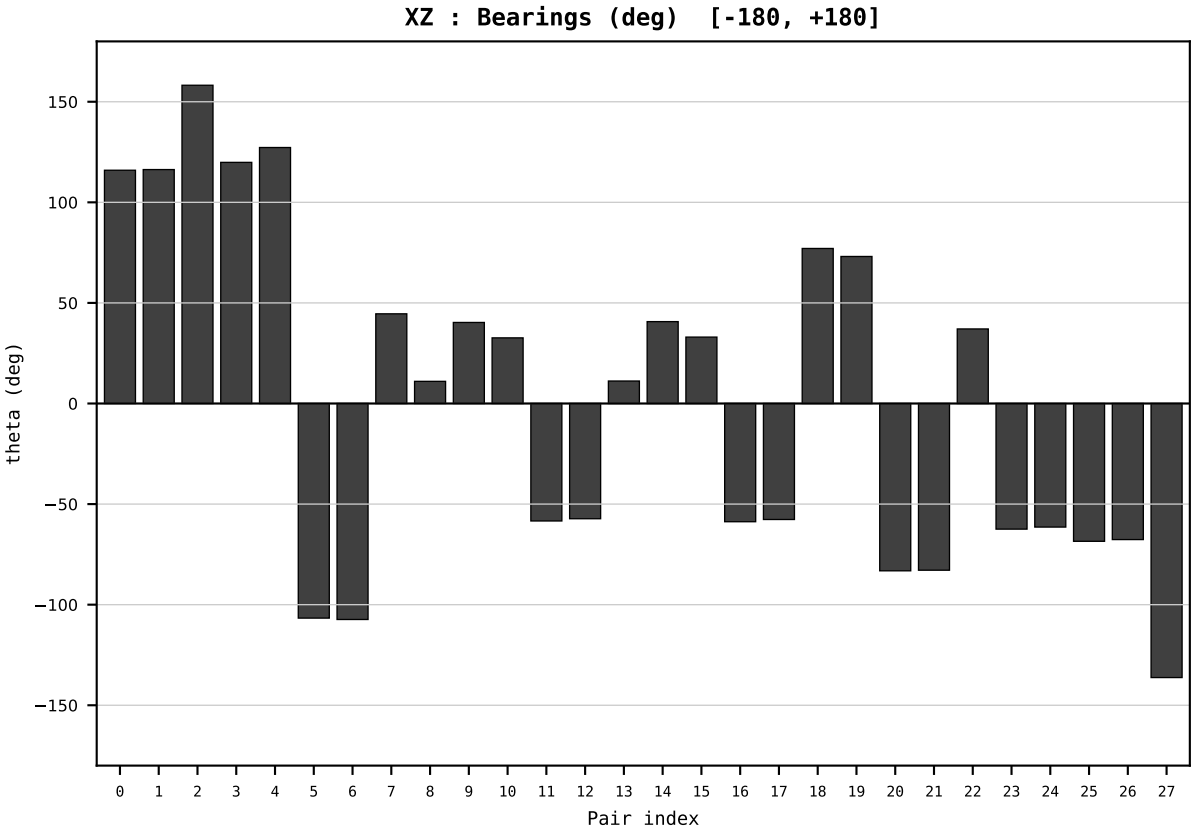
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2008
t : 8 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.232256
Ring = Hs-2
E_metric= 28.9064
kappa_HS= 128.84
omega = 2.0079 deg
d_A = 0.213233
Helm = Nuclear
Helm d = -0.116881
DR = 4.859 HLR
DR ratio= 128.8x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



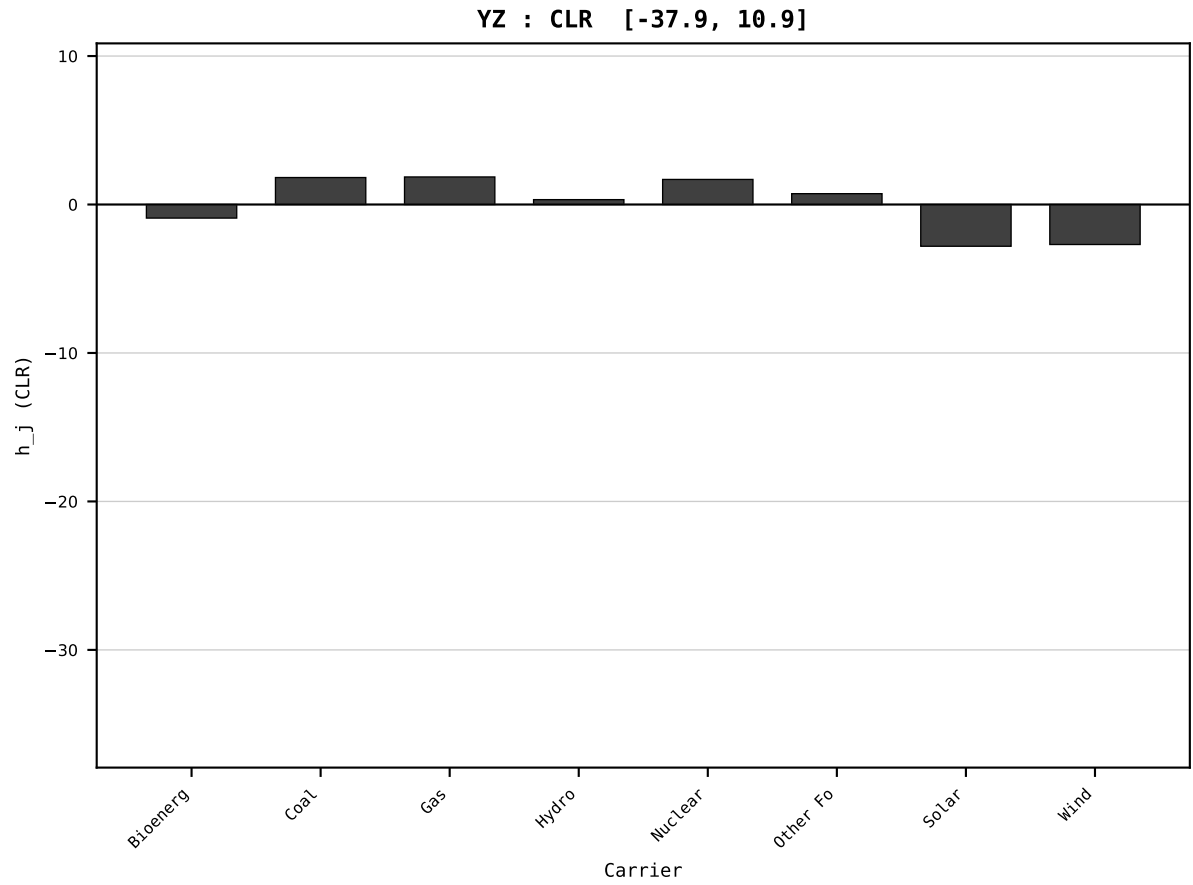
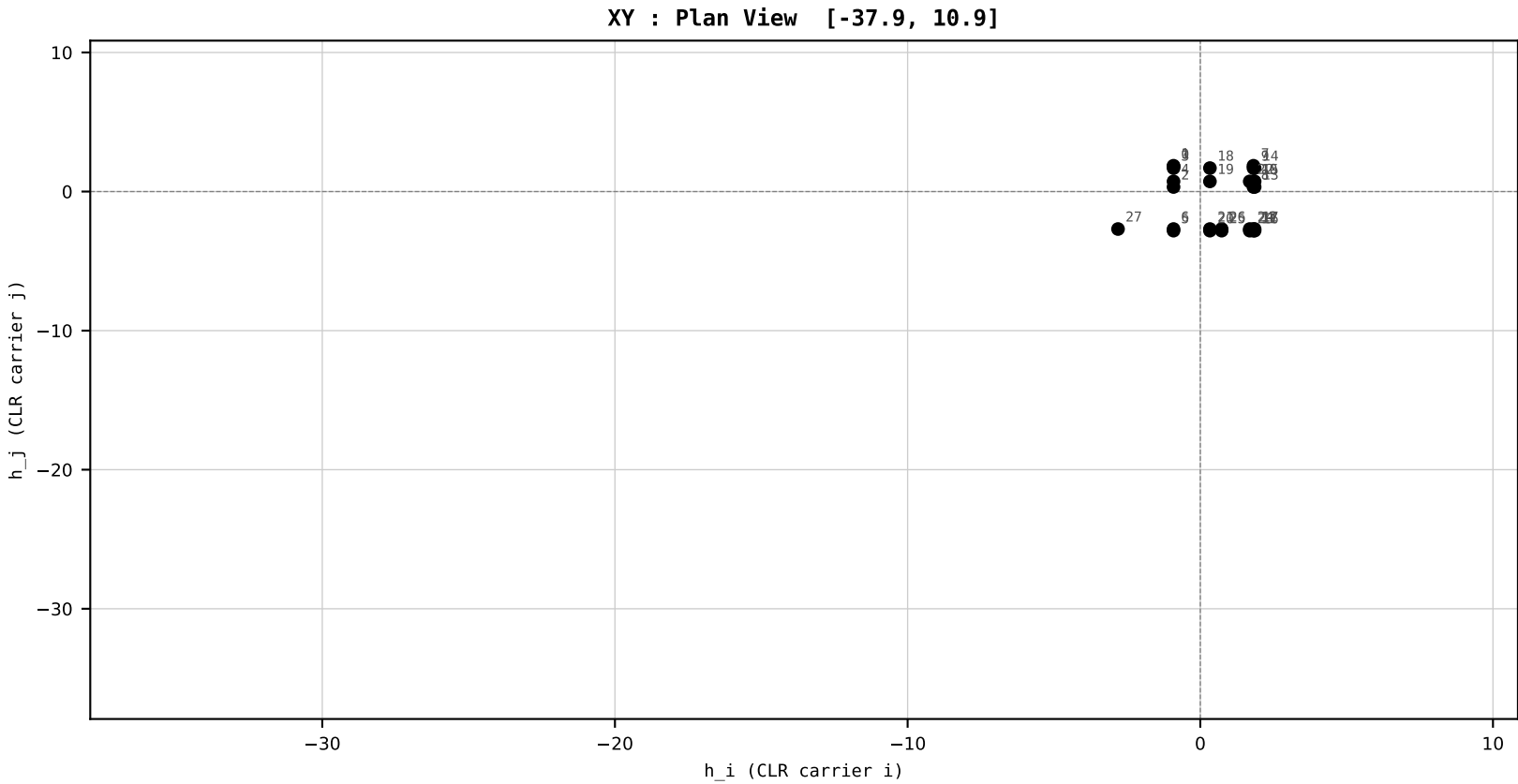
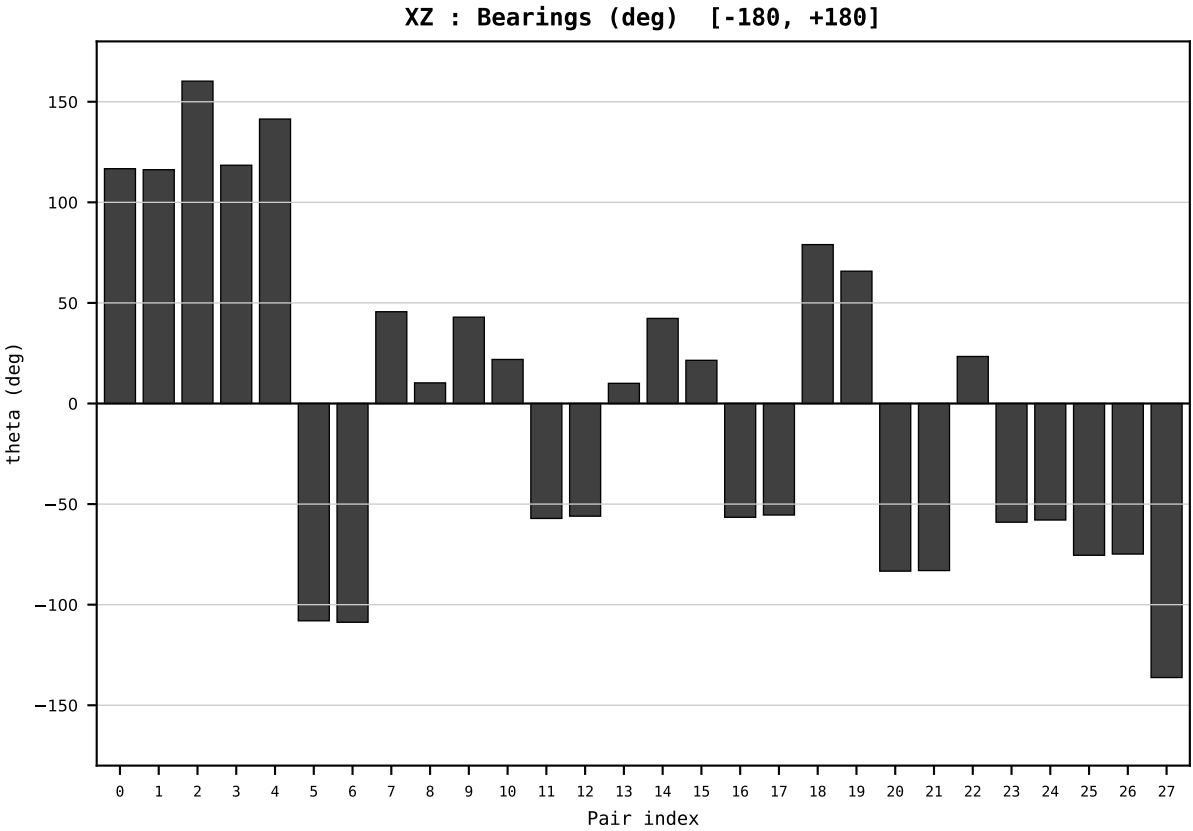
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2009
t : 9 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.247095
Ring = Hs-2
E_metric= 26.2382
kappa_HS= 106.47
omega = 5.2774 deg
d_A = 0.545961
Helm = Other Fossil
Helm d = -0.454982
DR = 4.668 HLR
DR ratio= 106.5x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=10 | Year 2010 | D=8 N=26 pairs=28

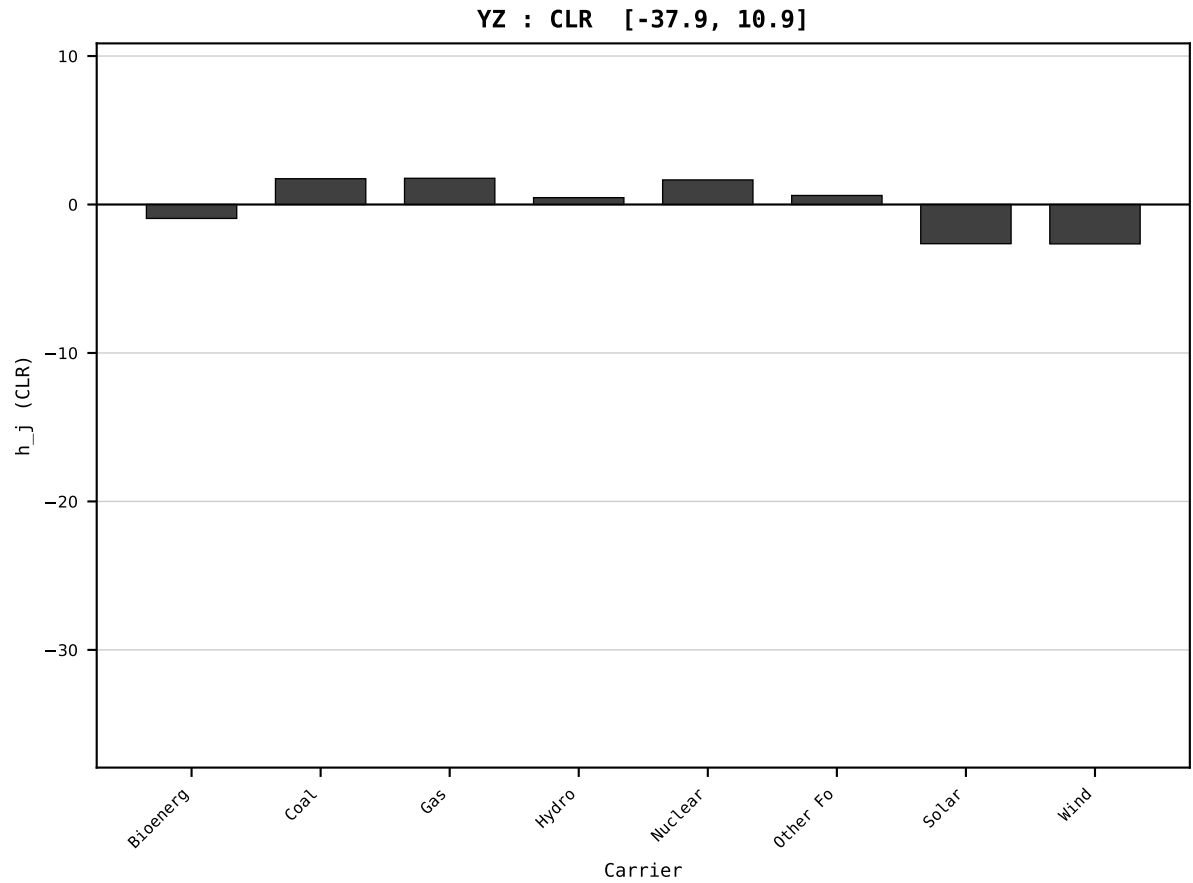
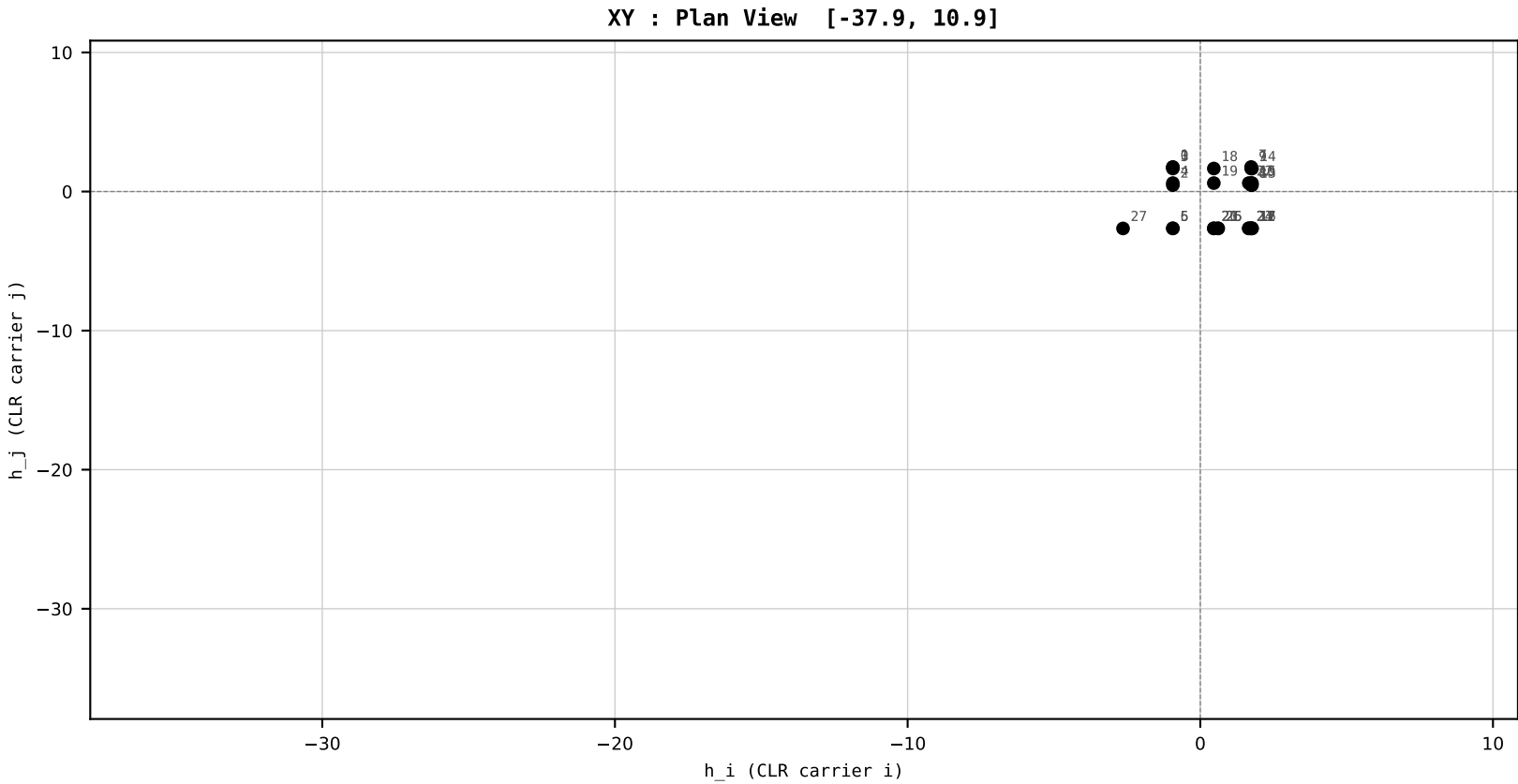
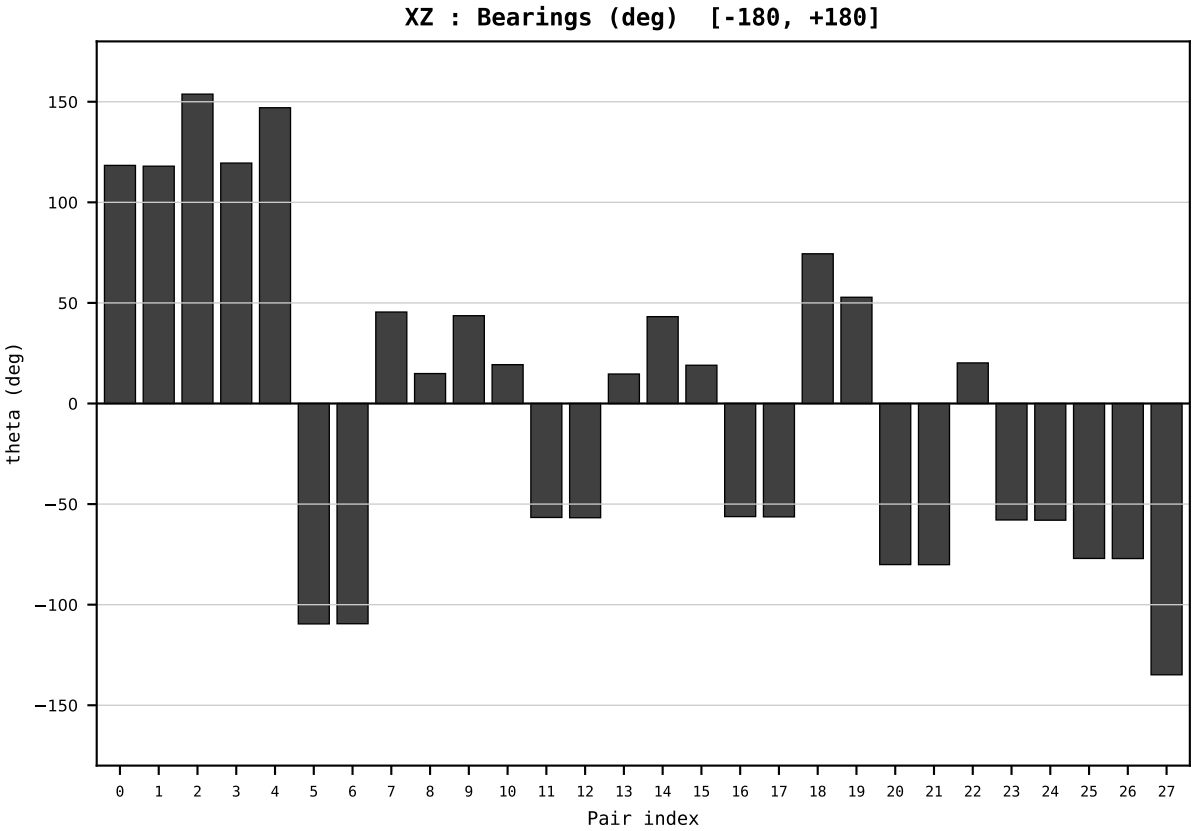
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2010
t : 10 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.237884
Ring = Hs-2
E_metric= 24.3502
kappa_HS= 82.97
omega = 2.4032 deg
d_A = 0.282318
Helm = Solar
Helm d = +0.171611
DR = 4.419 HLR
DR ratio= 83.0x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=11 | Year 2011 | D=8 N=26 pairs=28

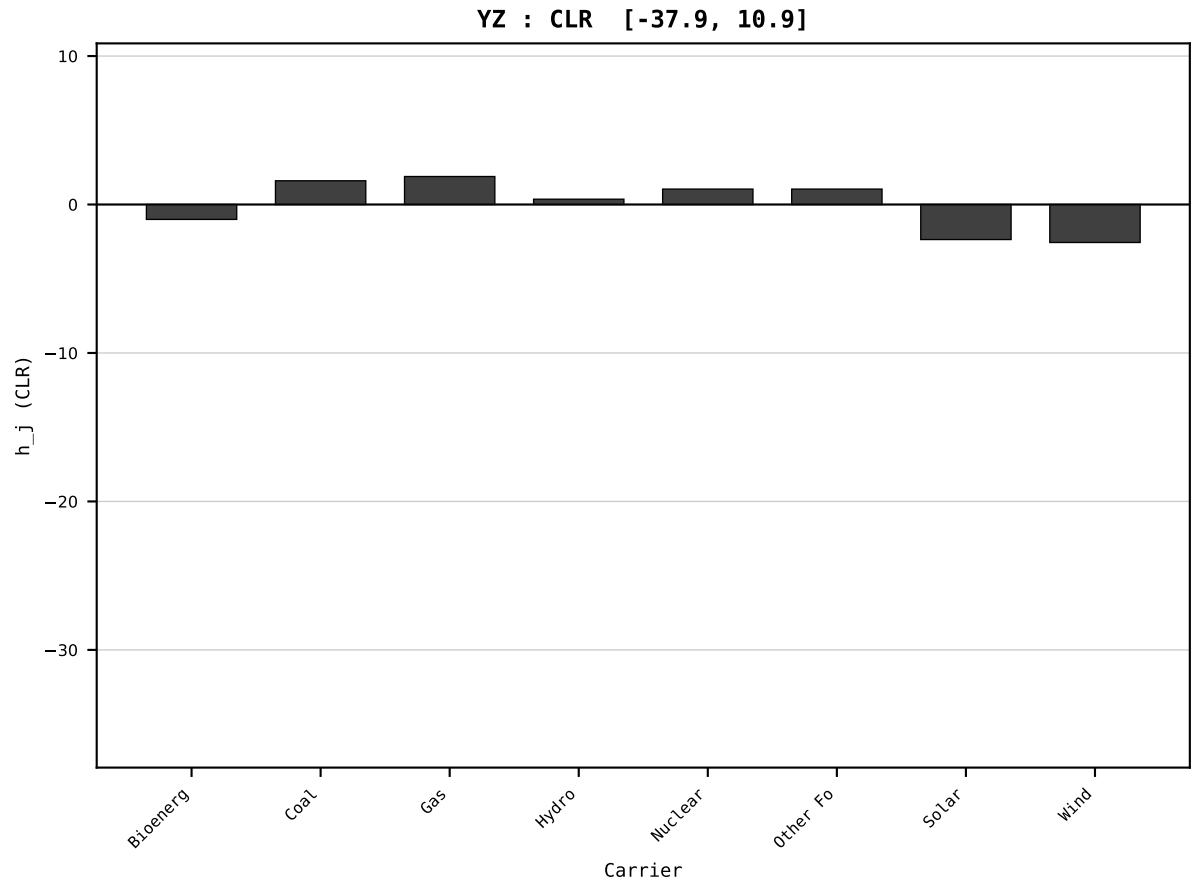
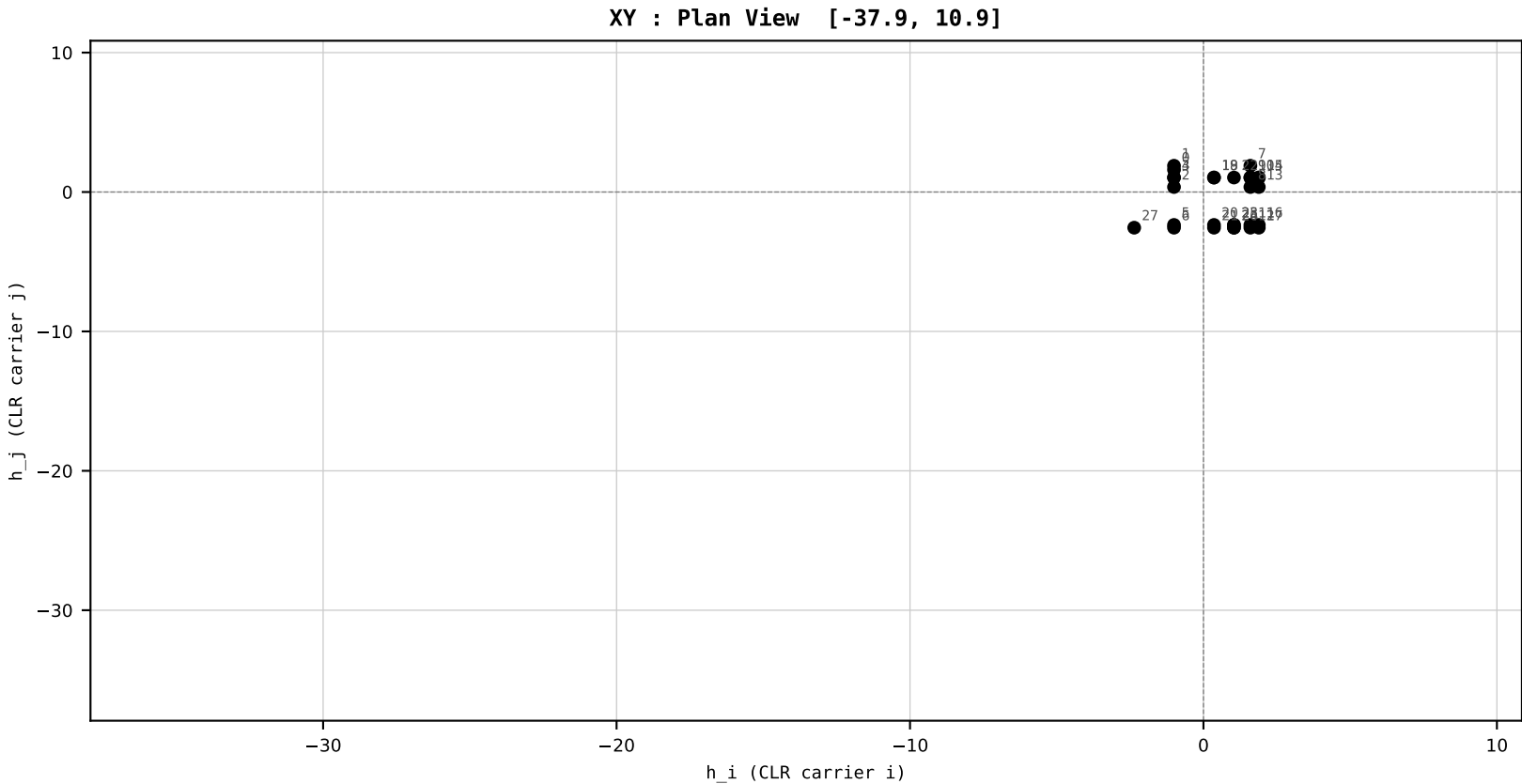
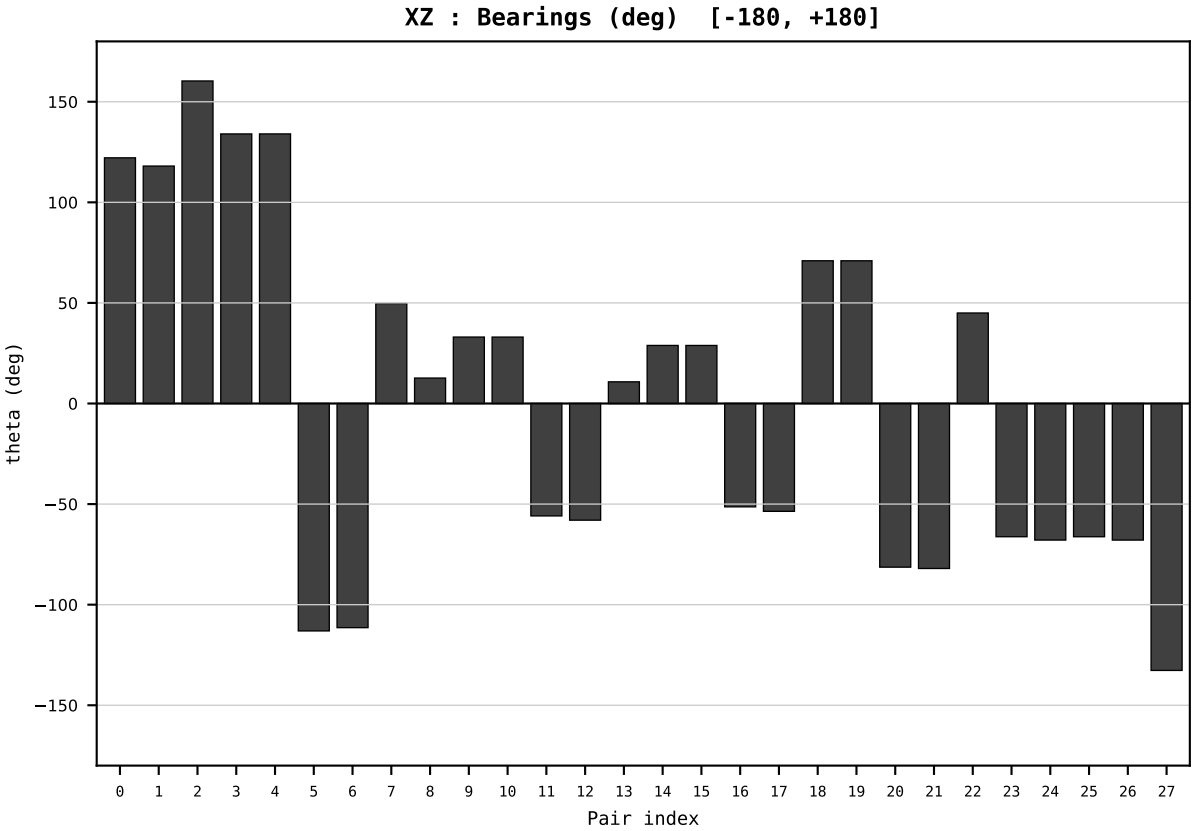
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2011
t : 11 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.230931
Ring = Hs-2
E_metric= 21.5271
kappa_HS= 85.16
omega = 9.4013 deg
d_A = 0.837843
Helm = Nuclear
Helm d = -0.617218
DR = 4.445 HLR
DR ratio= 85.2x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



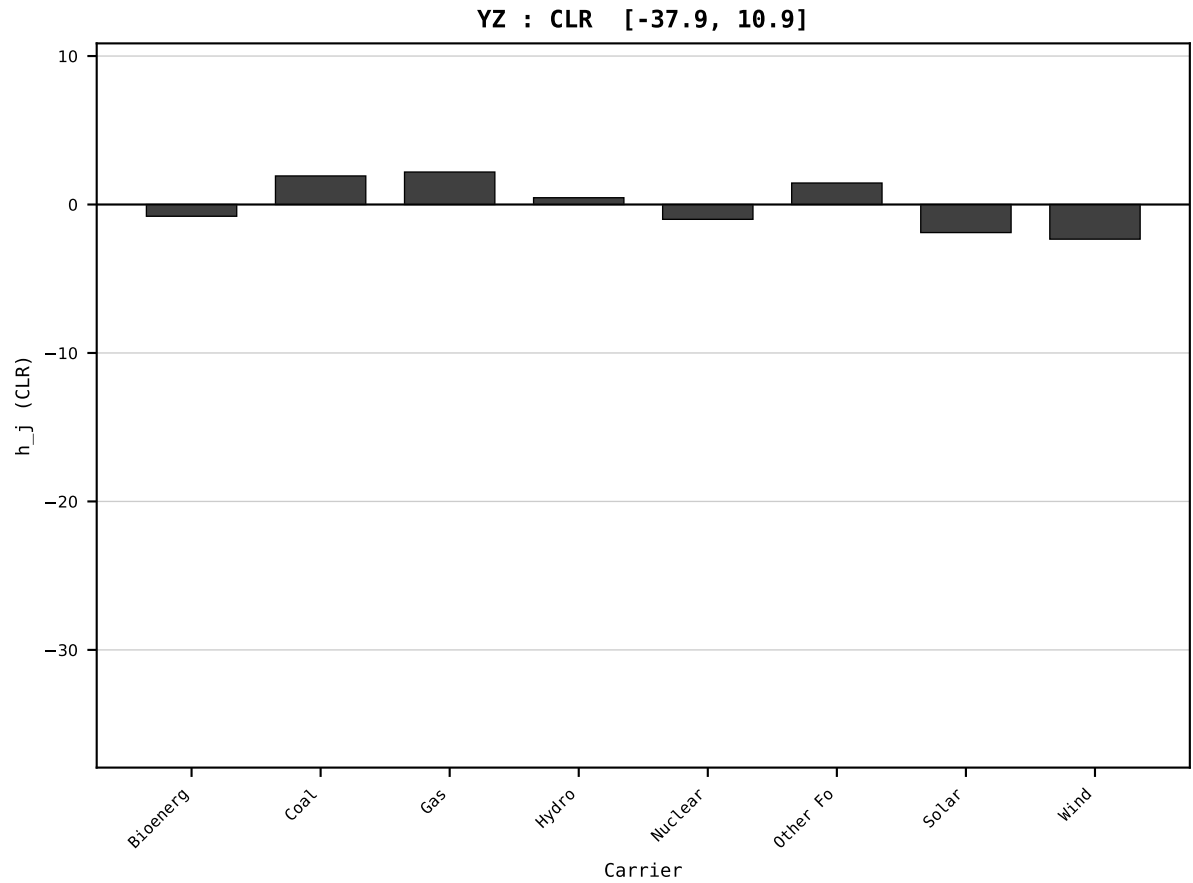
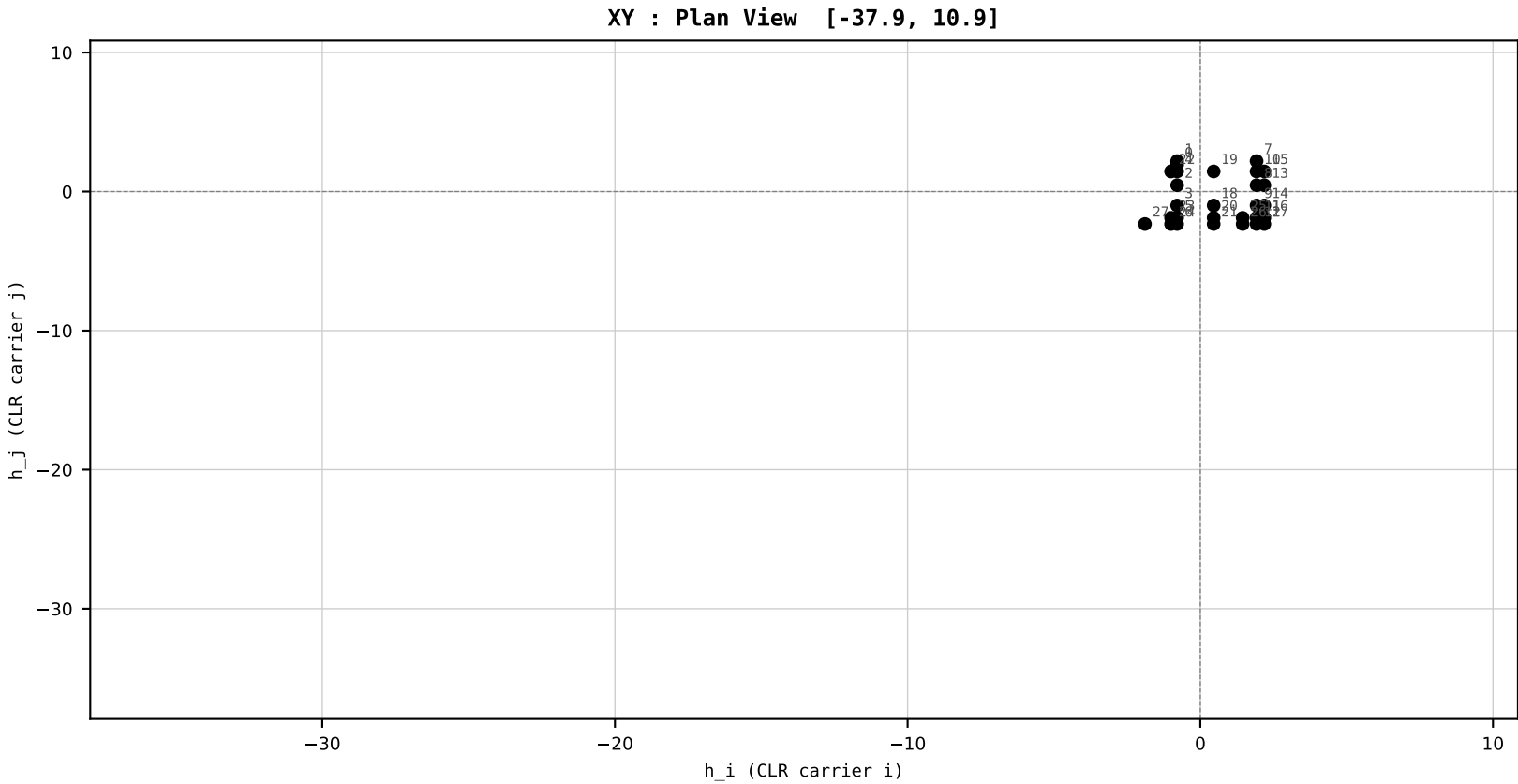
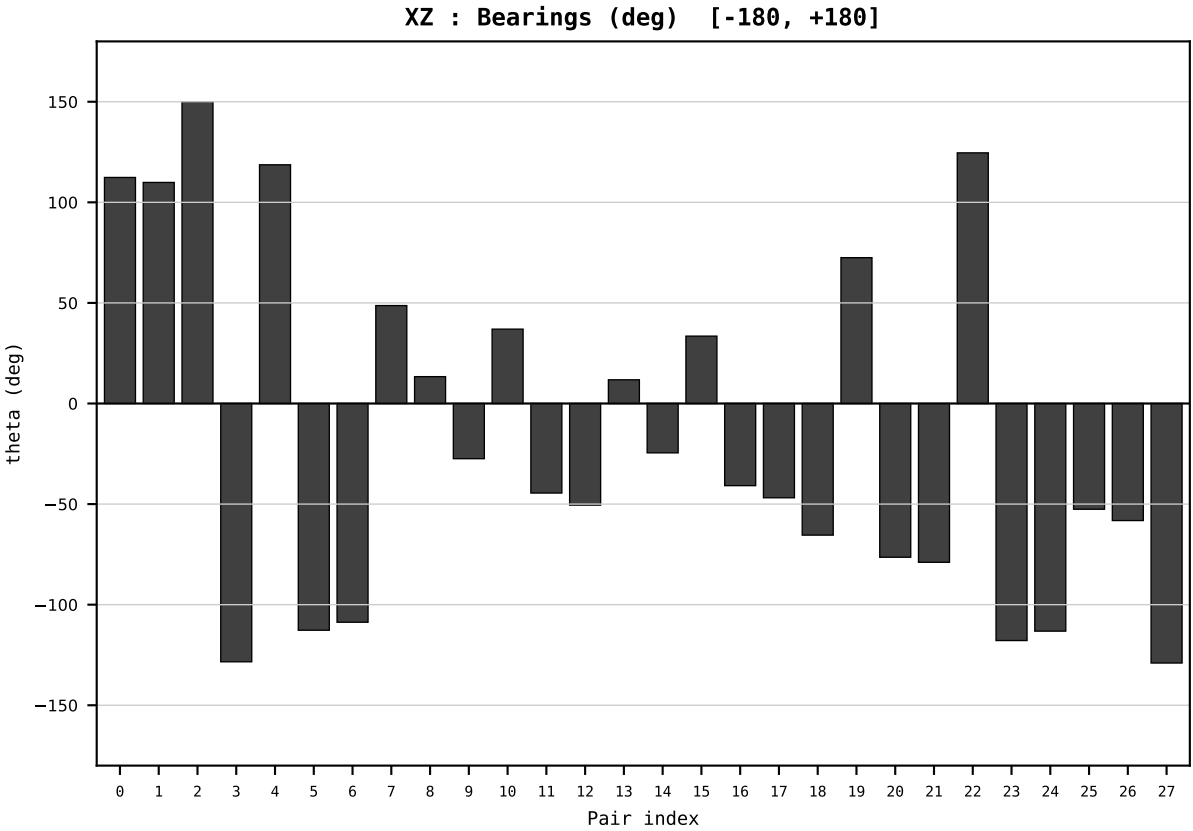
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2012
t : 12 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.312291
Ring = Hs-3
E_metric= 21.4426
kappa_HS= 92.07
omega = 27.4642 deg
d_A = 2.200635
Helm = Nuclear
Helm d = -2.038262
DR = 4.523 HLR
DR ratio= 92.1x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



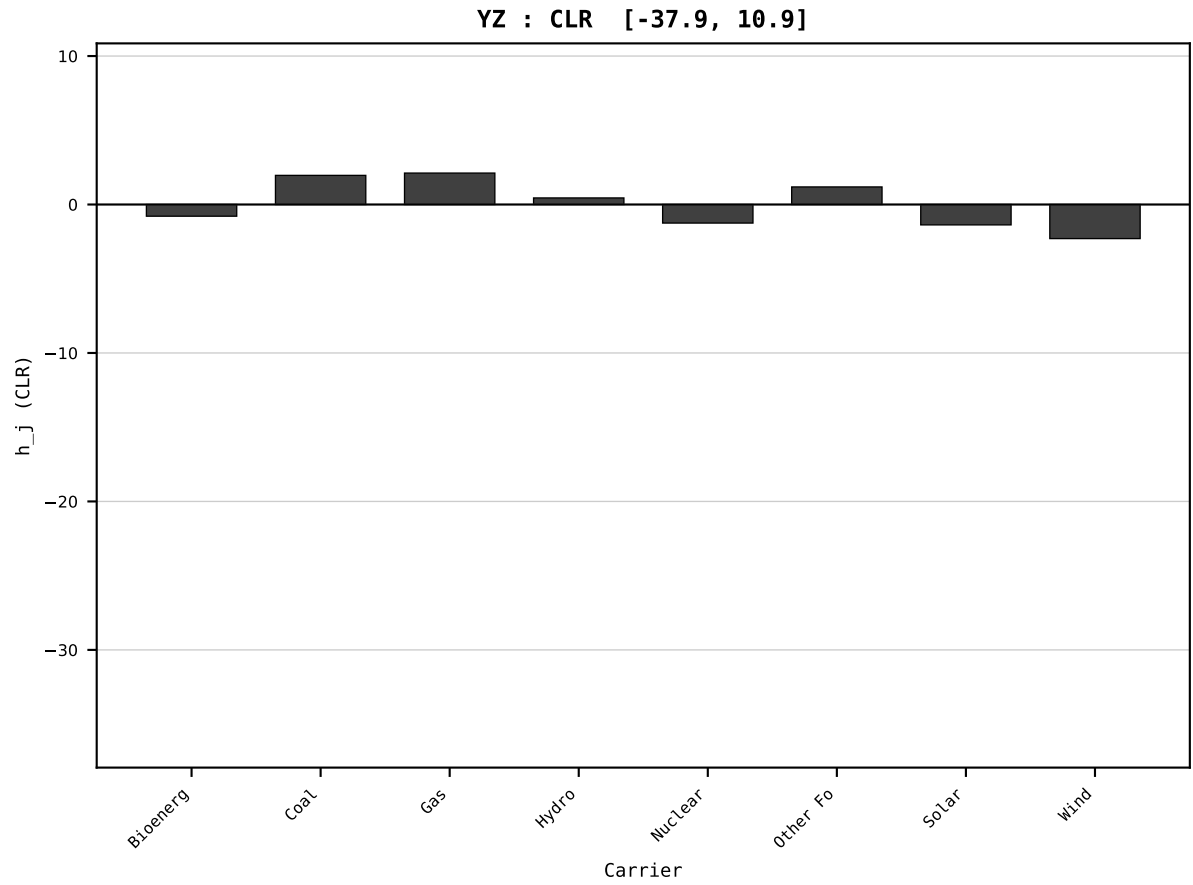
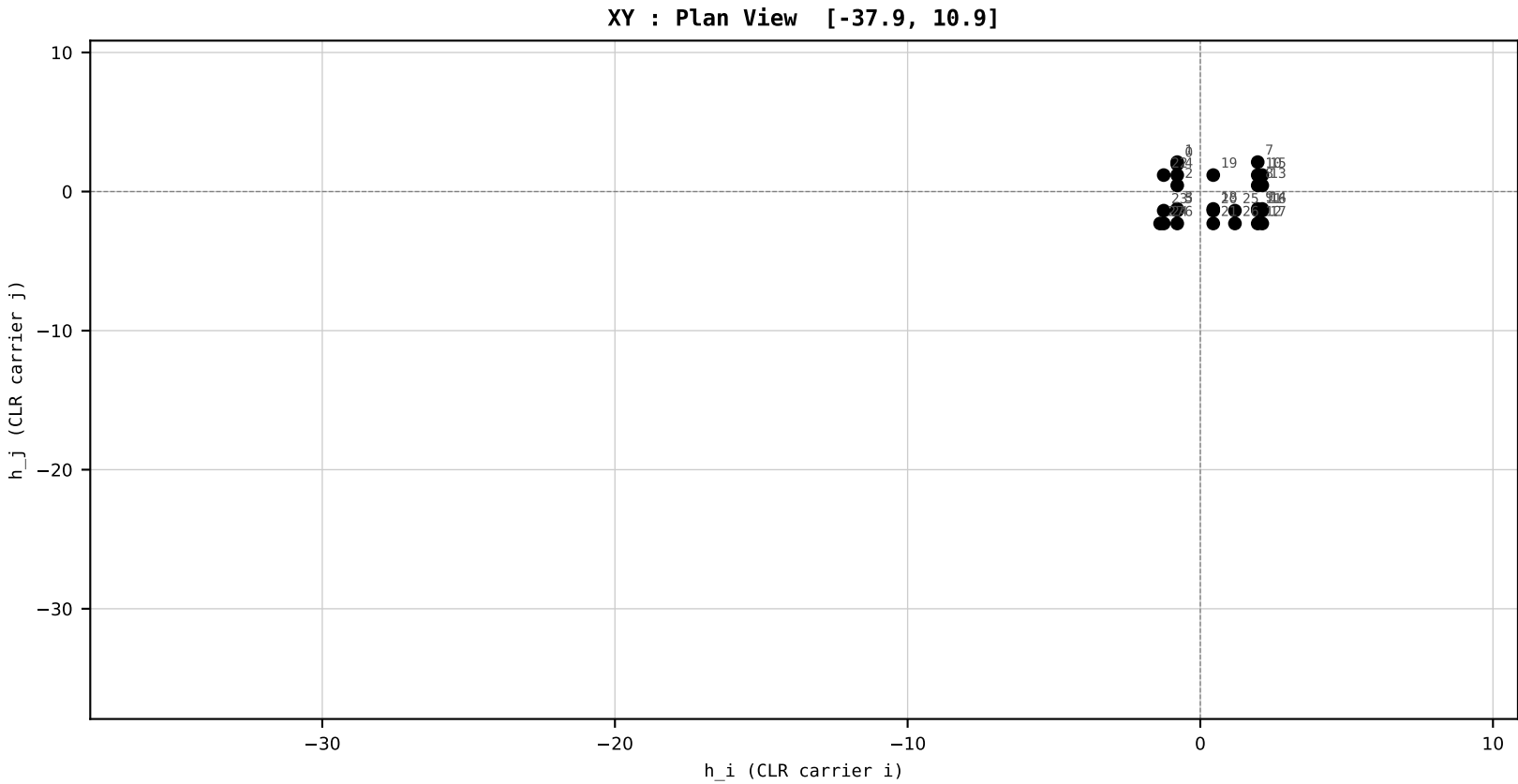
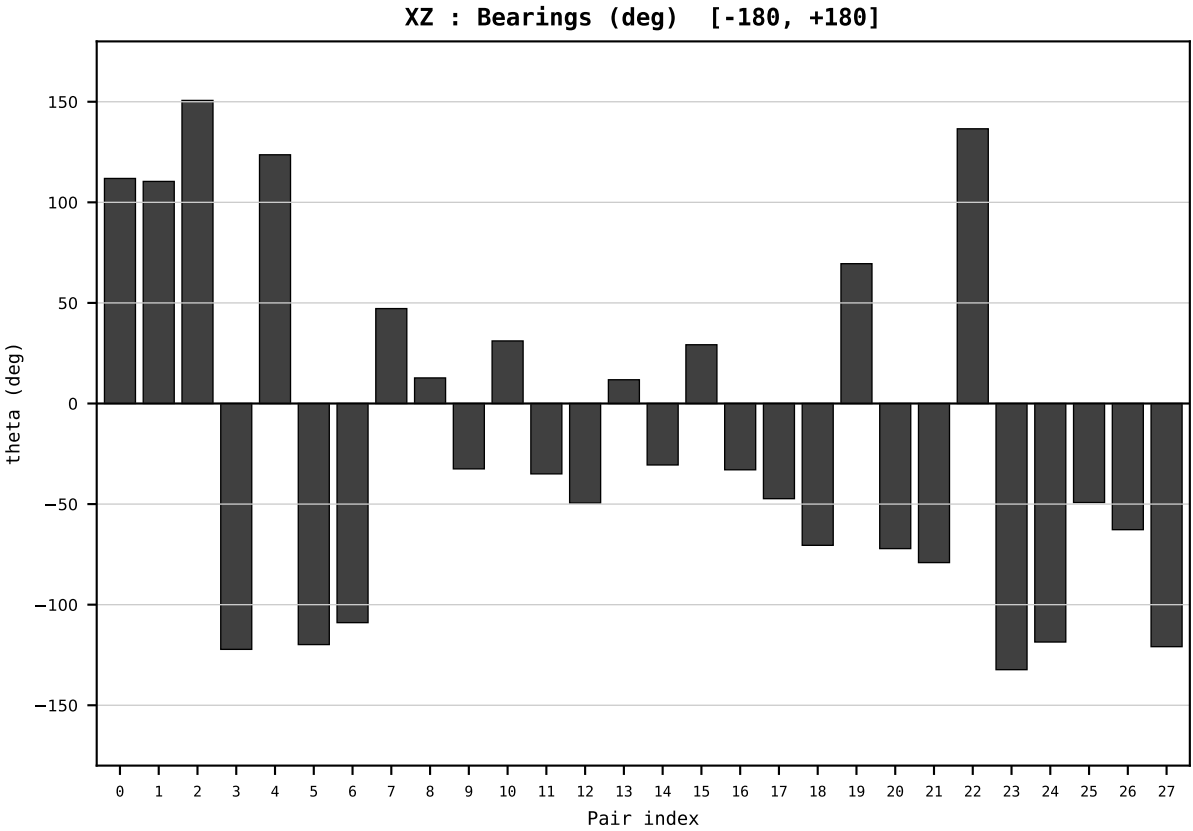
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2013
t : 13 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.312583
Ring = Hs-3
E_metric= 19.2716
kappa_HS= 82.57
omega = 7.5396 deg
d_A = 0.639857
Helm = Solar
Helm d = +0.518125
DR = 4.414 HLR
DR ratio= 82.6x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=14 | Year 2014 | D=8 N=26 pairs=28

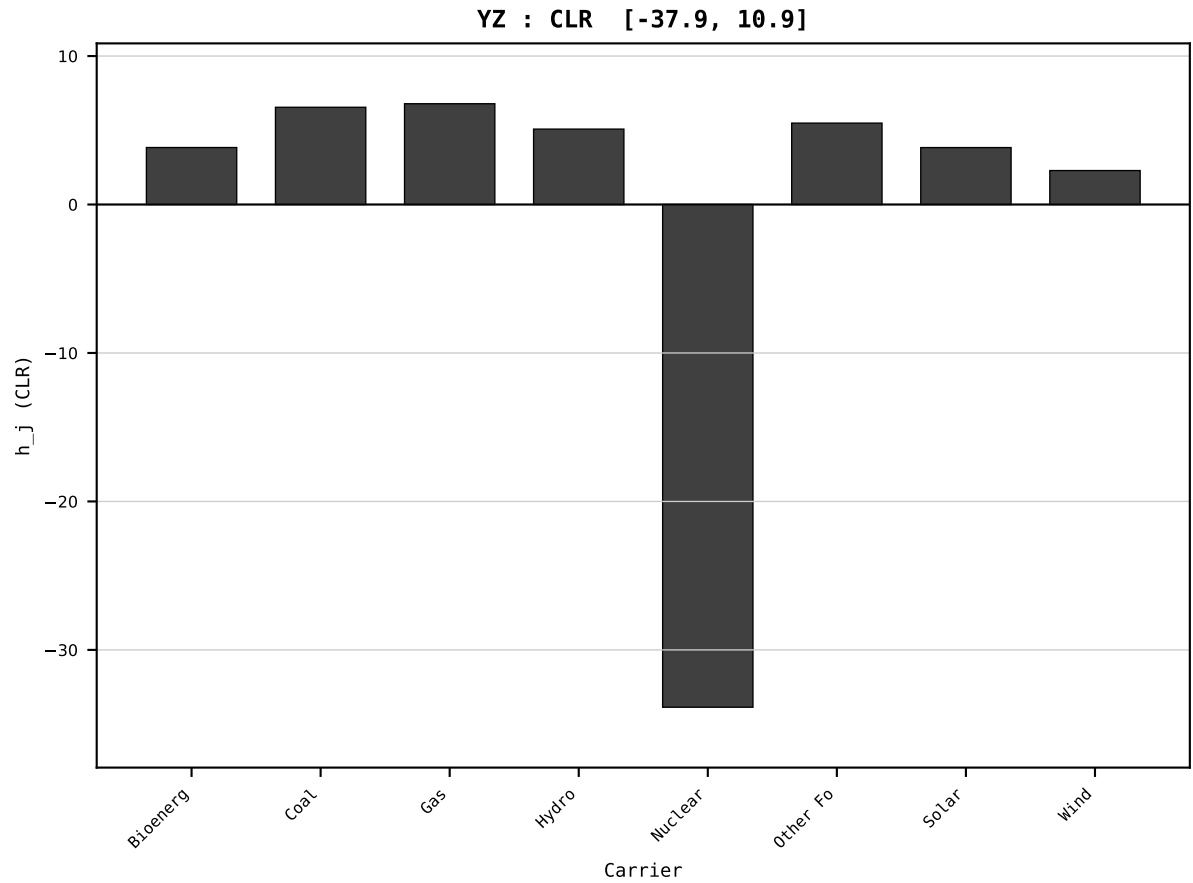
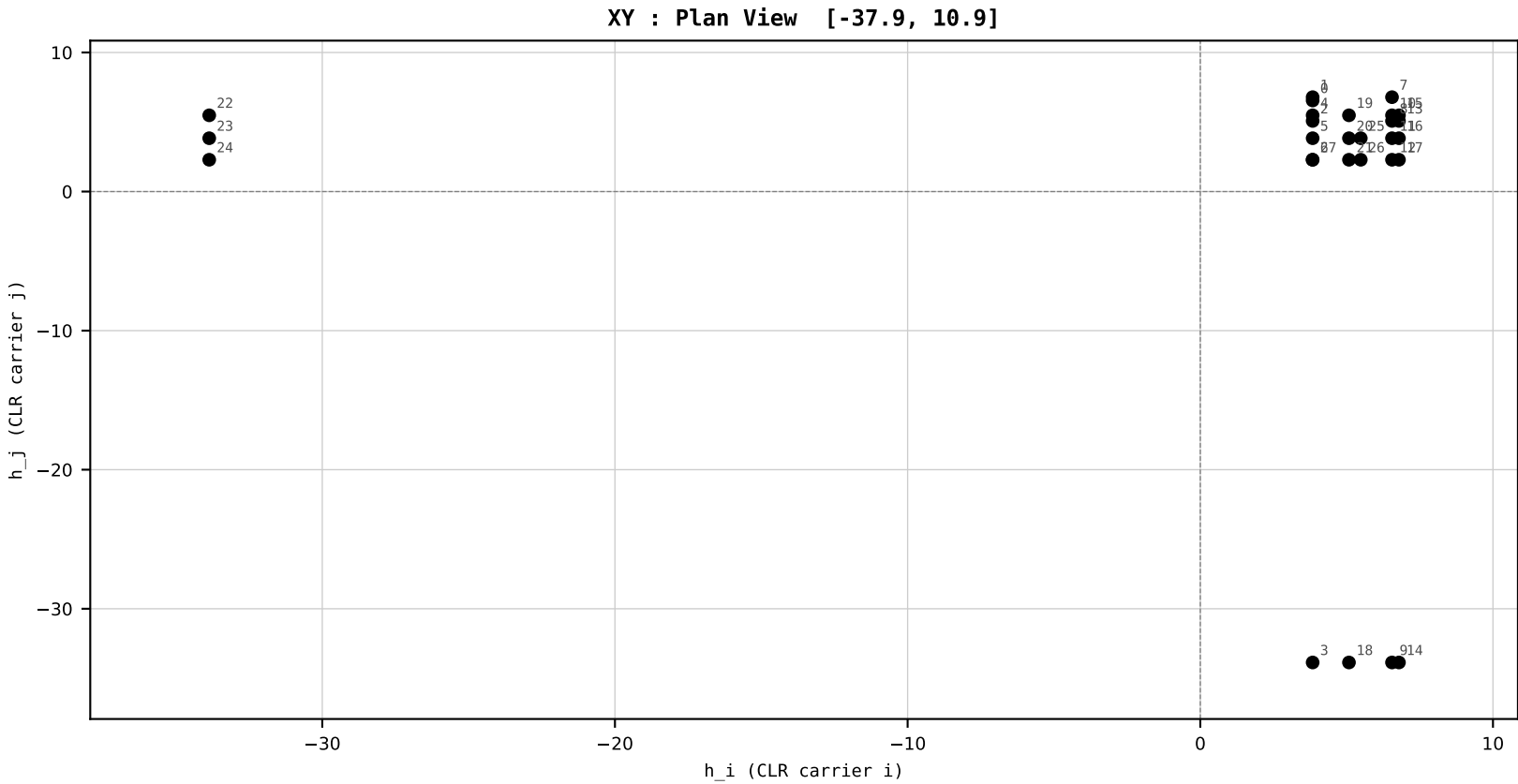
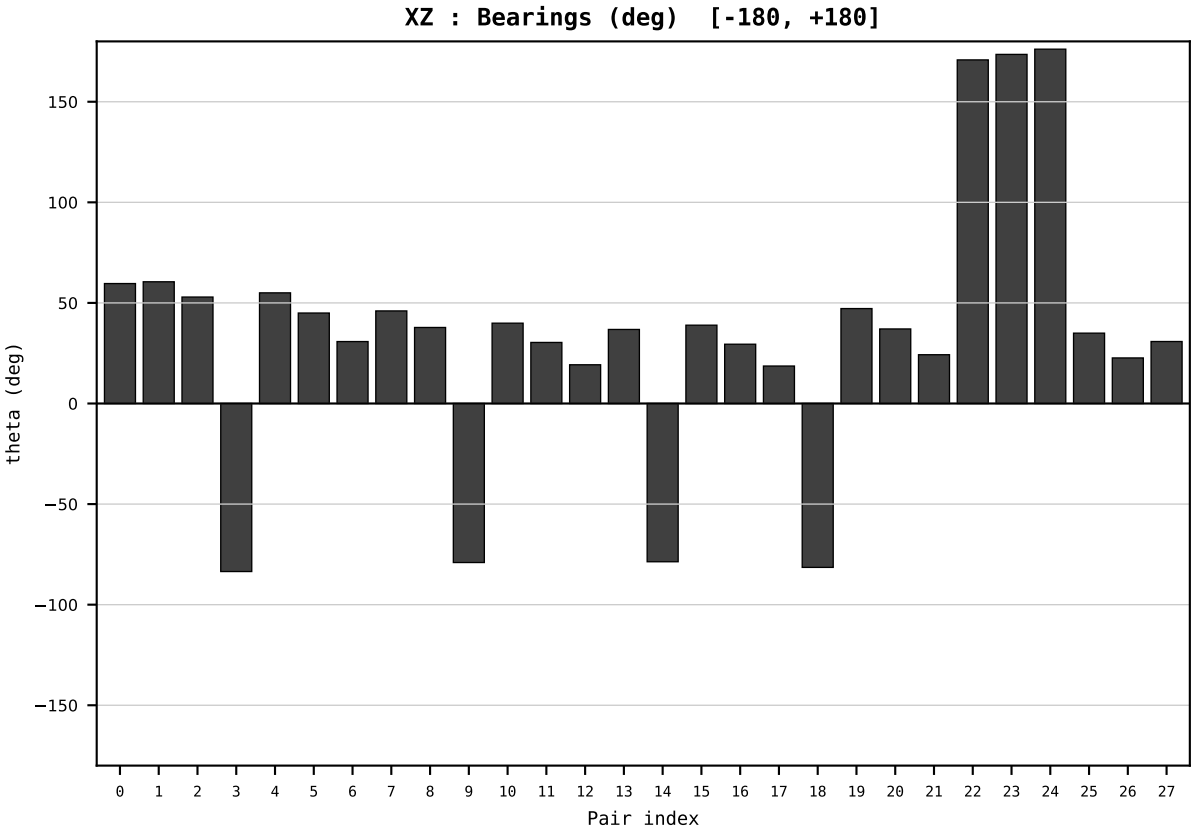
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2014
t : 14 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.340926
Ring = Hs-3
E_metric= 896.7320
kappa_HS= 45087999999999936.00
omega = 66.1070 deg
d_A = 34.870573
Helm = Nuclear
Helm d = -32.612482
DR = 40.650 HLR
DR ratio= 450879998367363840.0x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=15 | Year 2015 | D=8 N=26 pairs=28

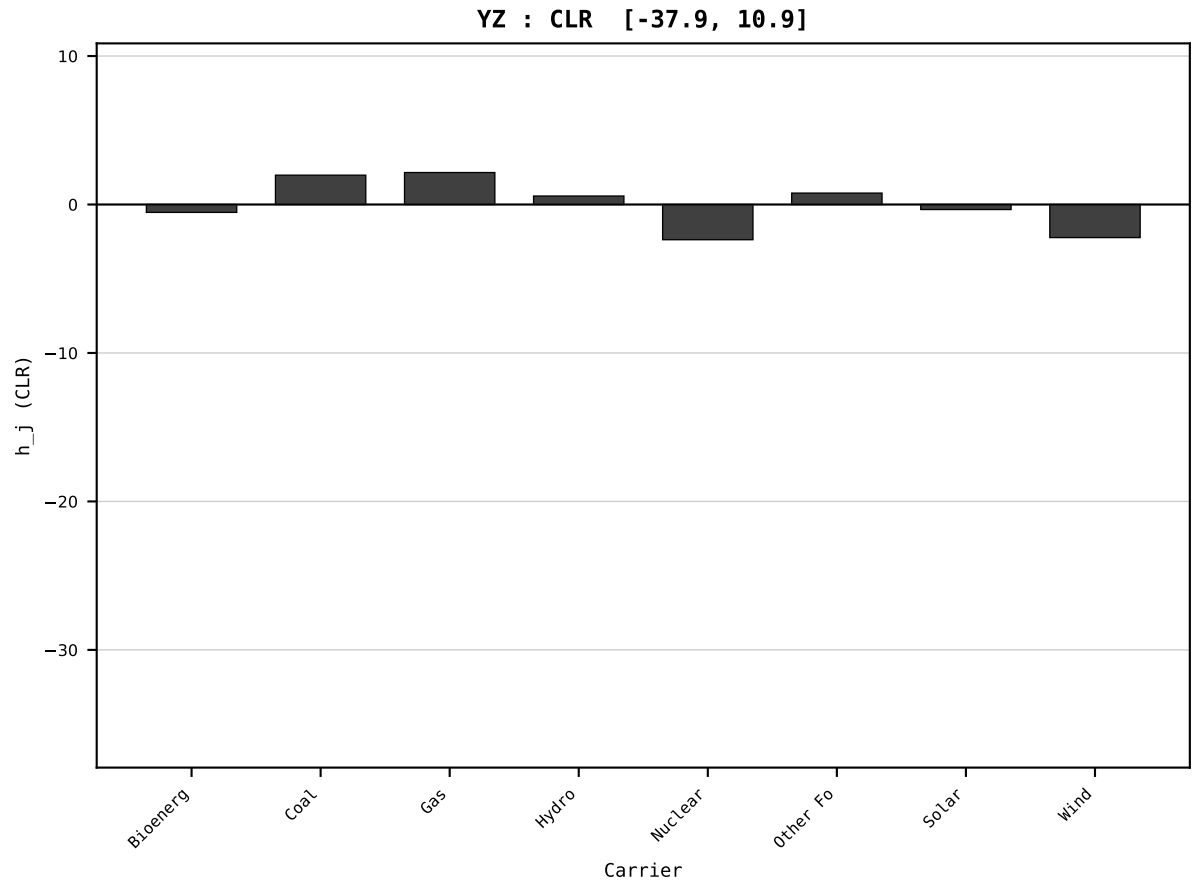
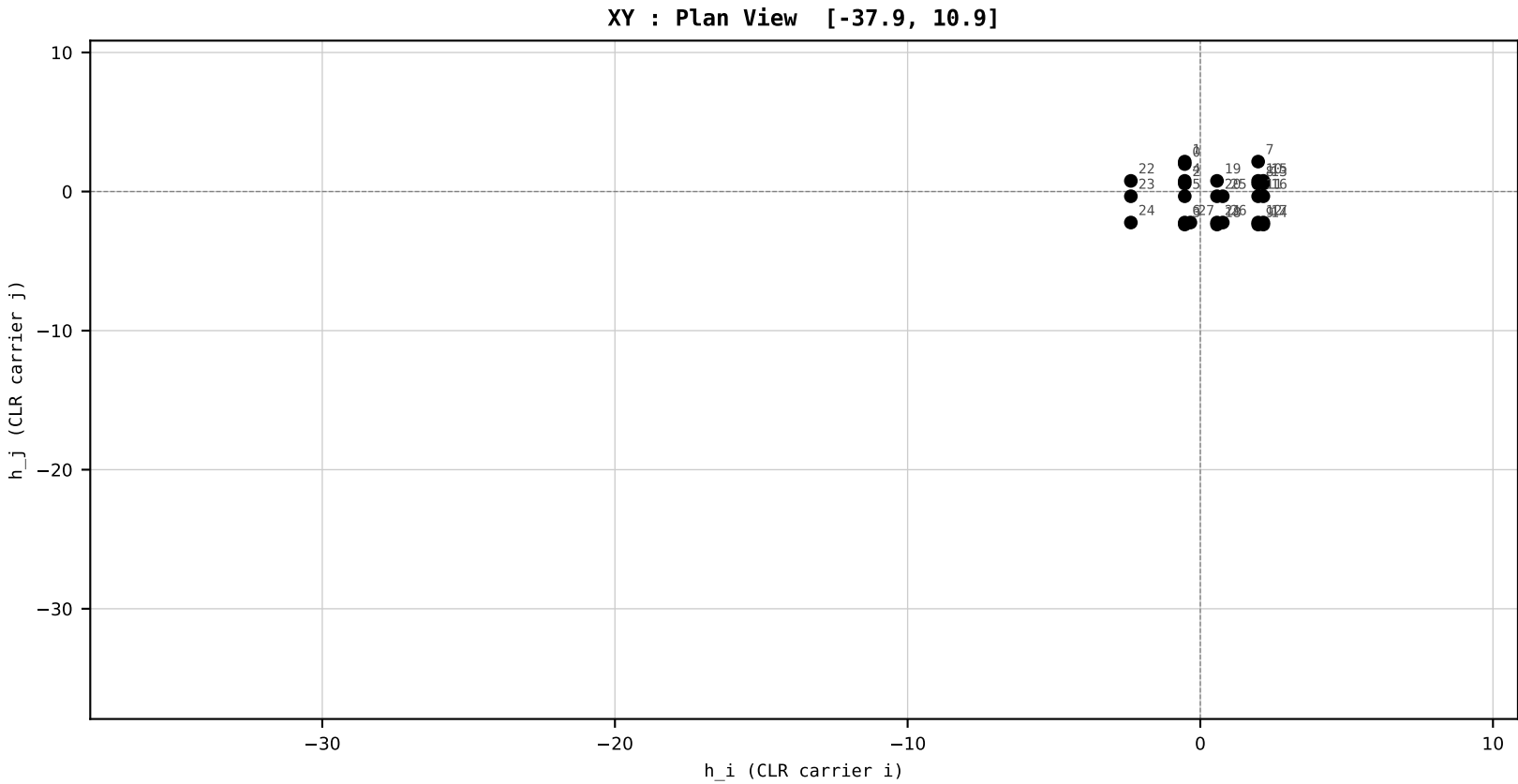
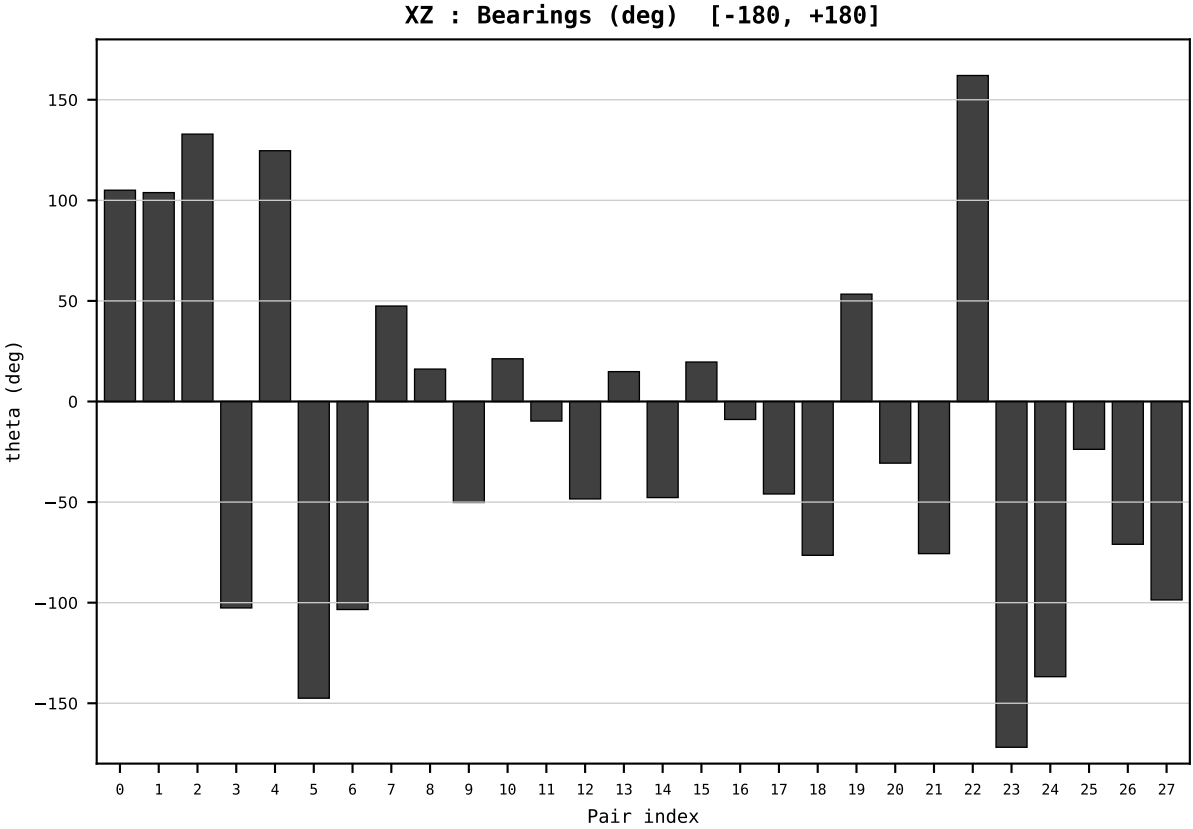
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2015
t : 15 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.309676
Ring = Hs-3
E_metric= 20.4479
kappa_HS= 92.37
omega = 49.6746 deg
d_A = 33.667556
Helm = Nuclear
Helm d = +31.490431
DR = 4.526 HLR
DR ratio= 92.4x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



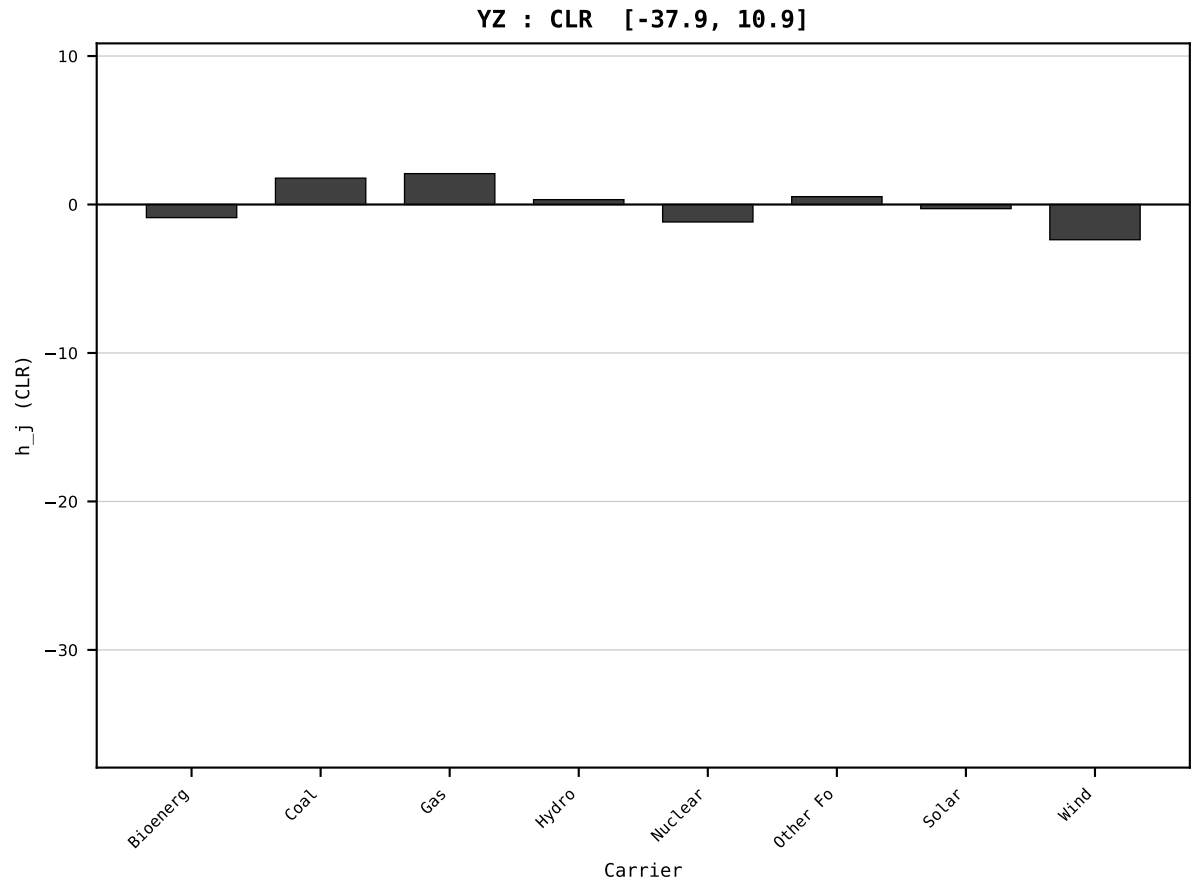
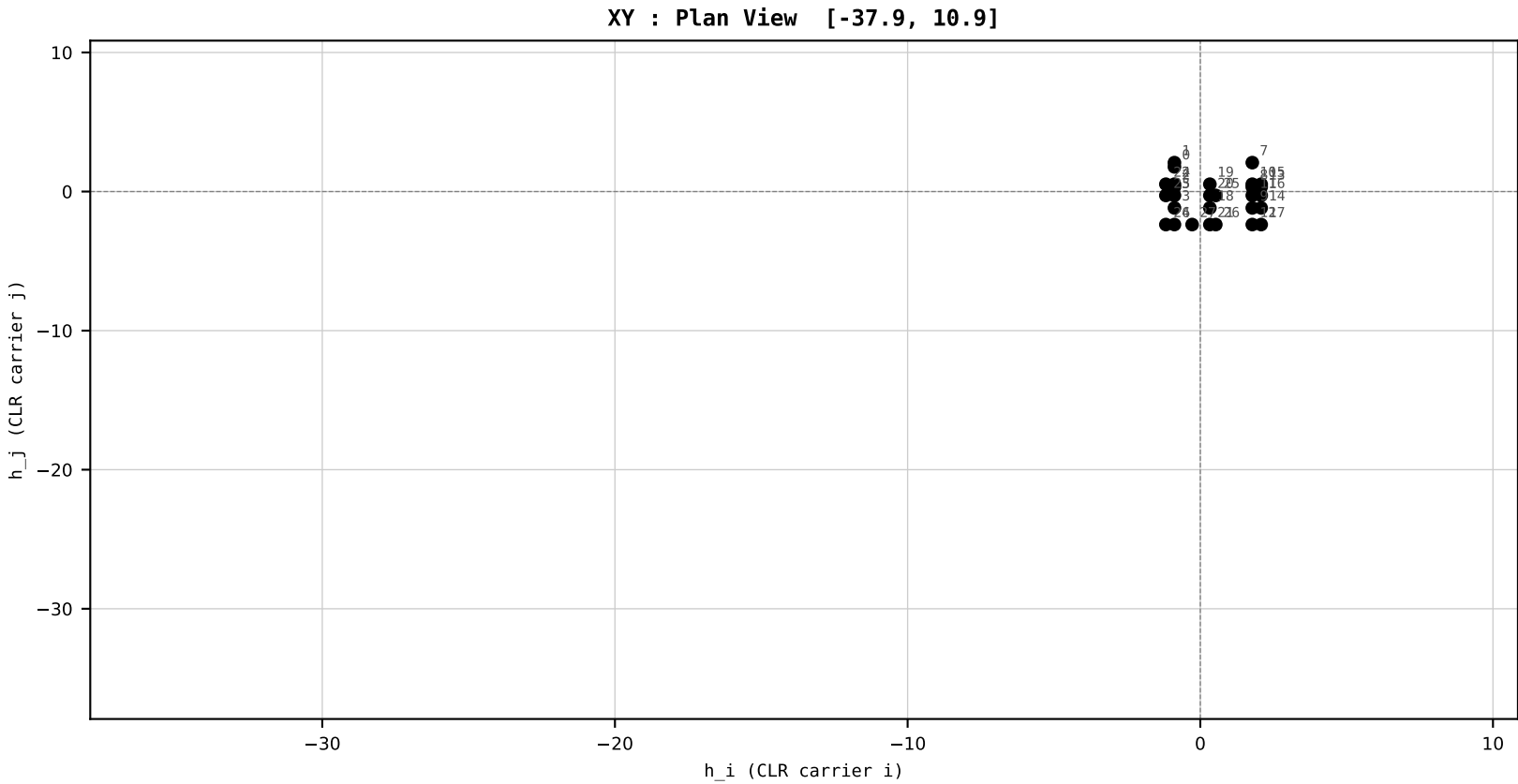
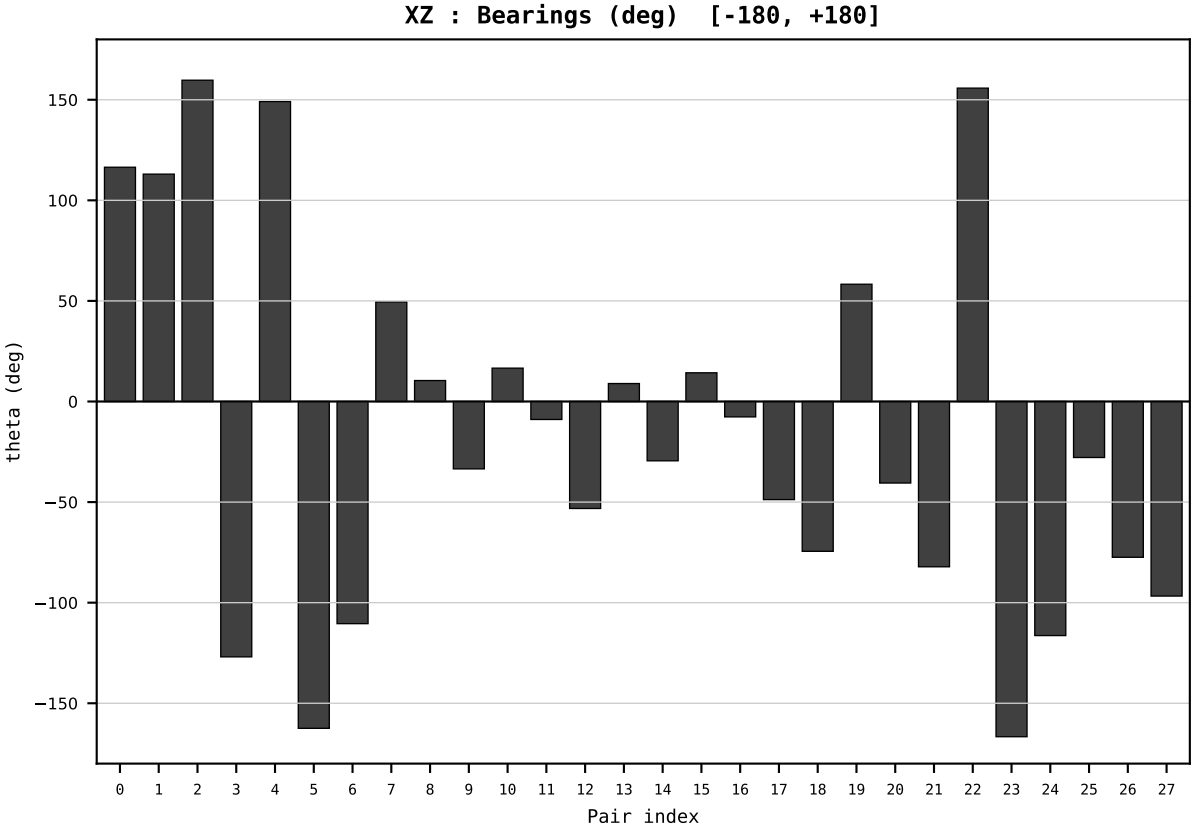
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2016
t : 16 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.303033
Ring = Hs-3
E_metric= 15.7404
kappa_HS= 85.85
omega = 16.2591 deg
d_A = 1.320048
Helm = Nuclear
Helm d = +1.196283
DR = 4.453 HLR
DR ratio= 85.8x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



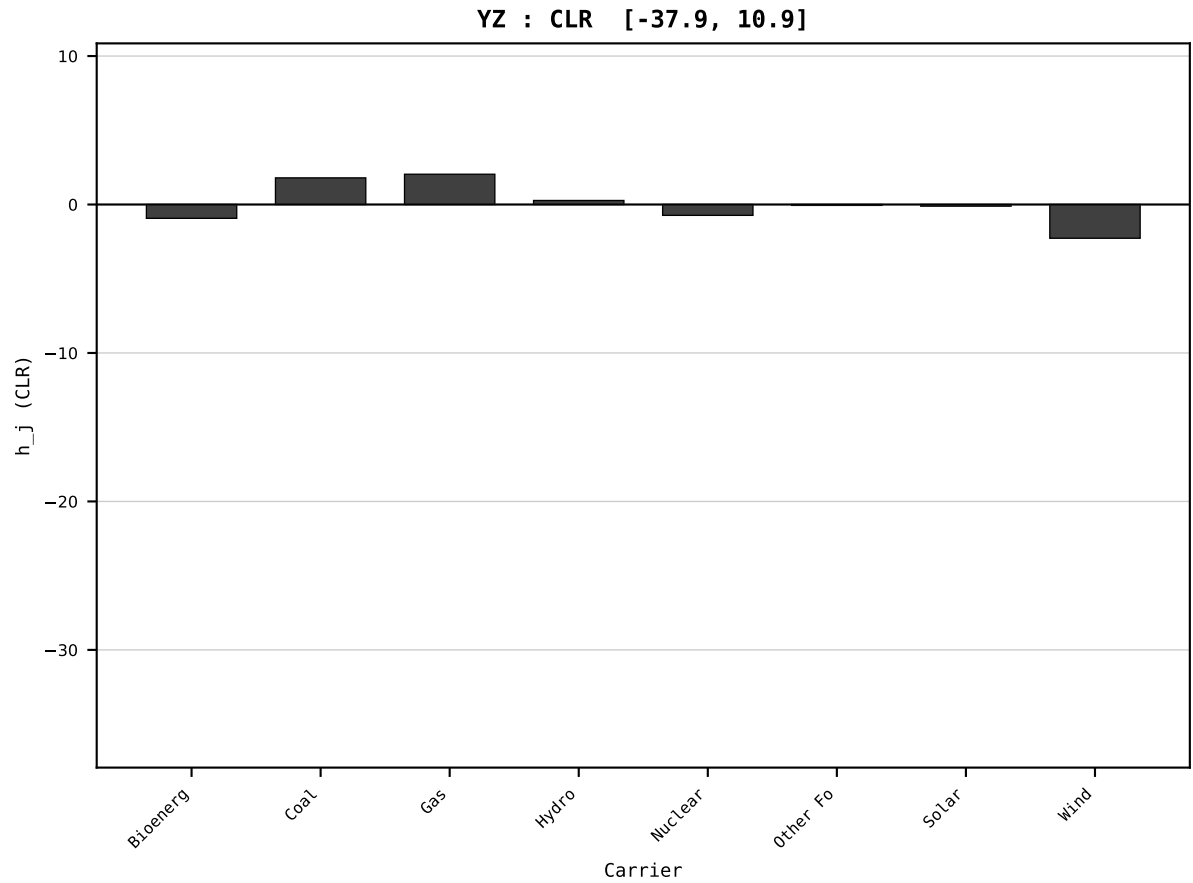
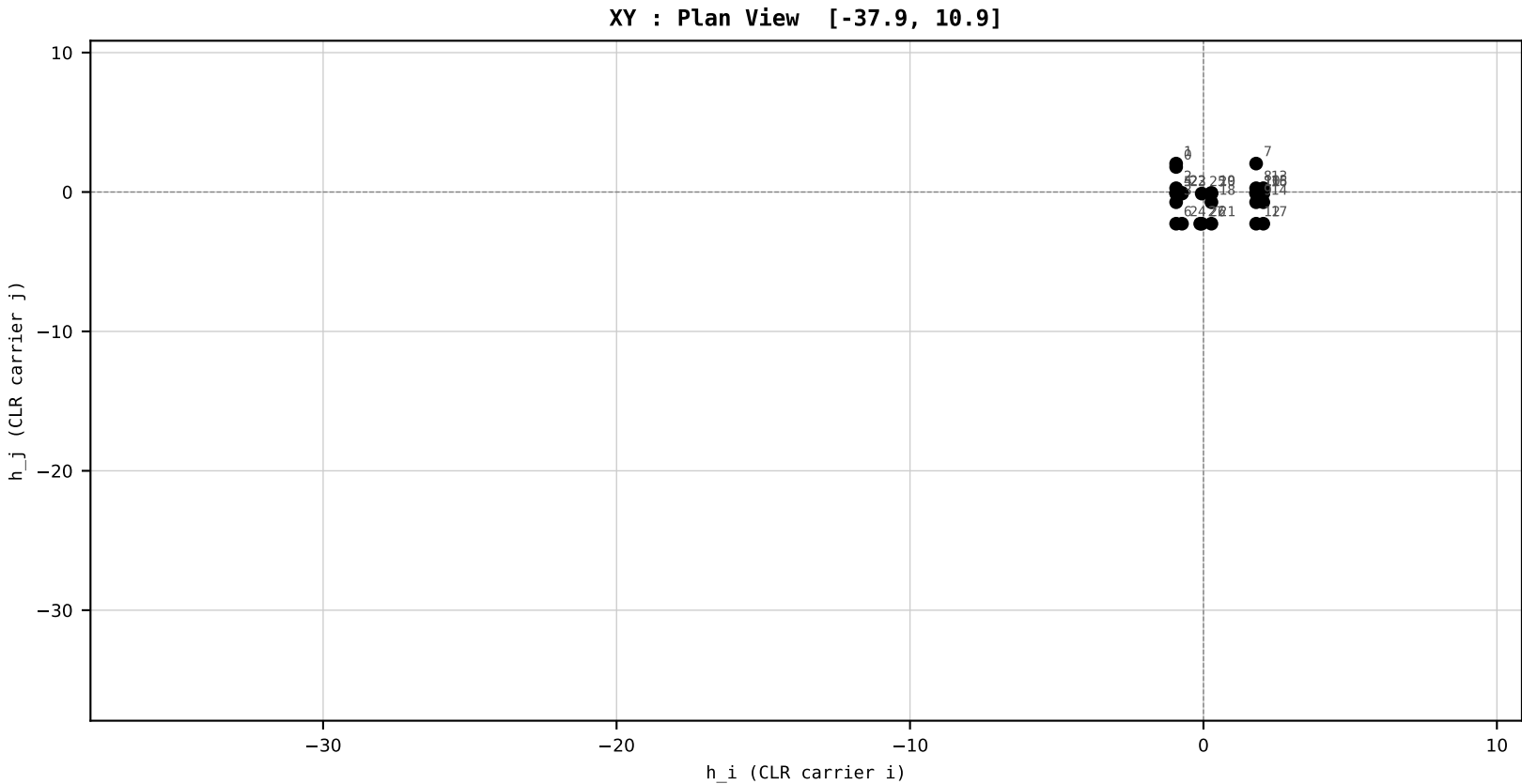
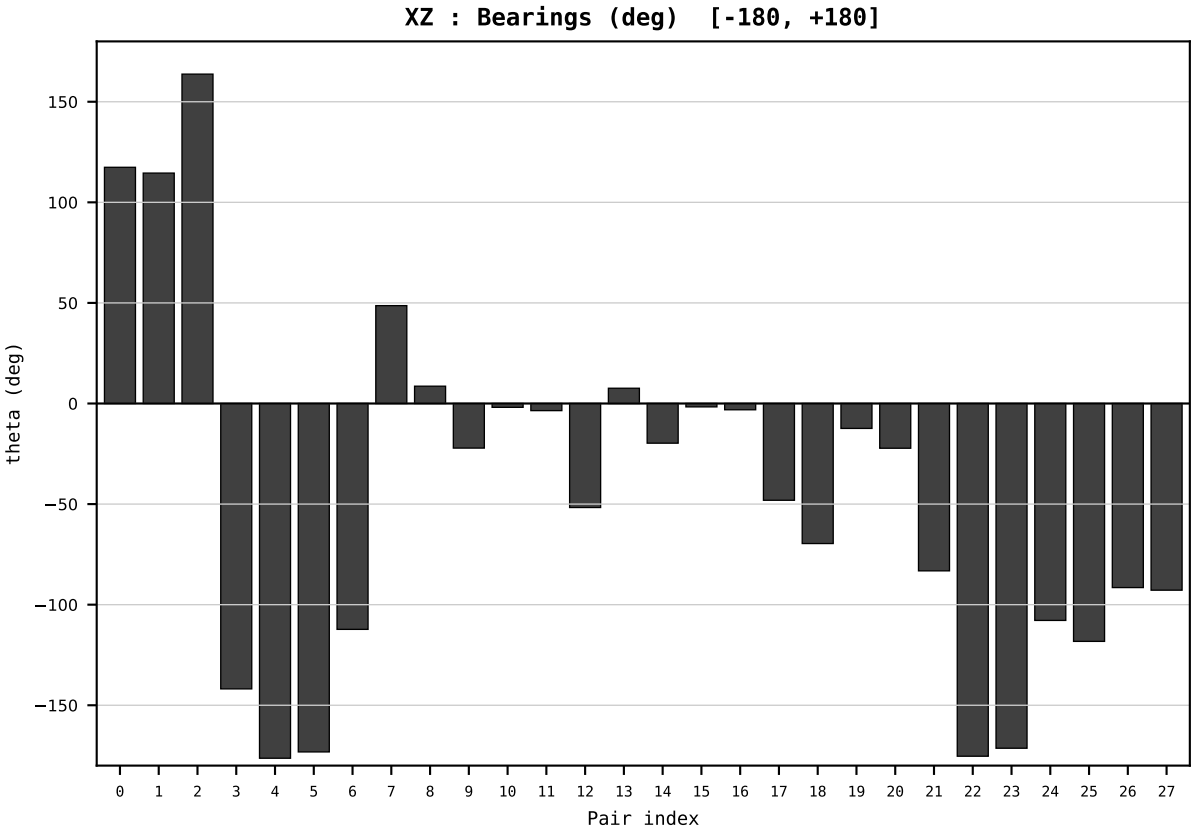
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2017
t : 17 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.308283
Ring = Hs-3
E_metric= 14.0385
kappa_HS= 74.62
omega = 10.9468 deg
d_A = 0.767882
Helm = Other Fossil
Helm d = -0.588595
DR = 4.312 HLR
DR ratio= 74.6x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=18 | Year 2018 | D=8 N=26 pairs=28

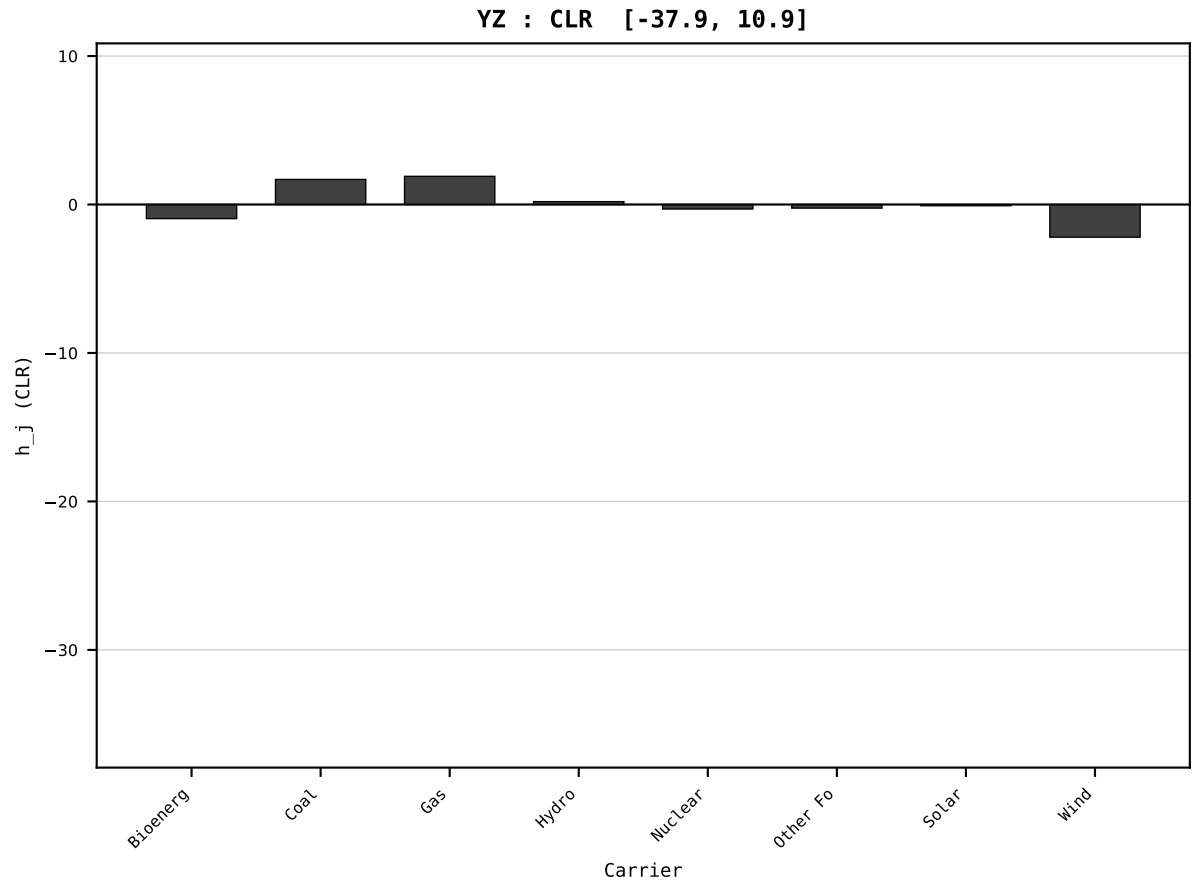
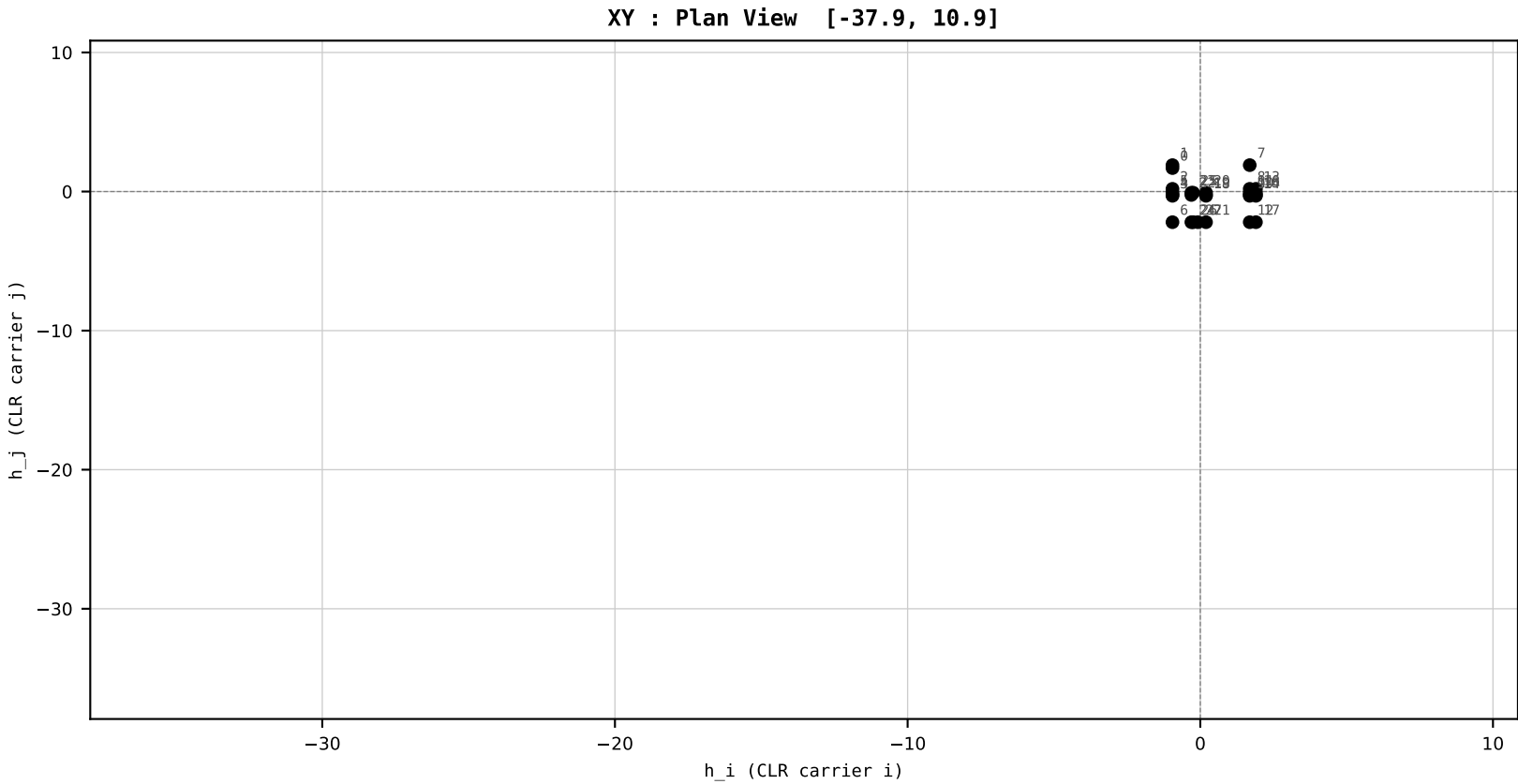
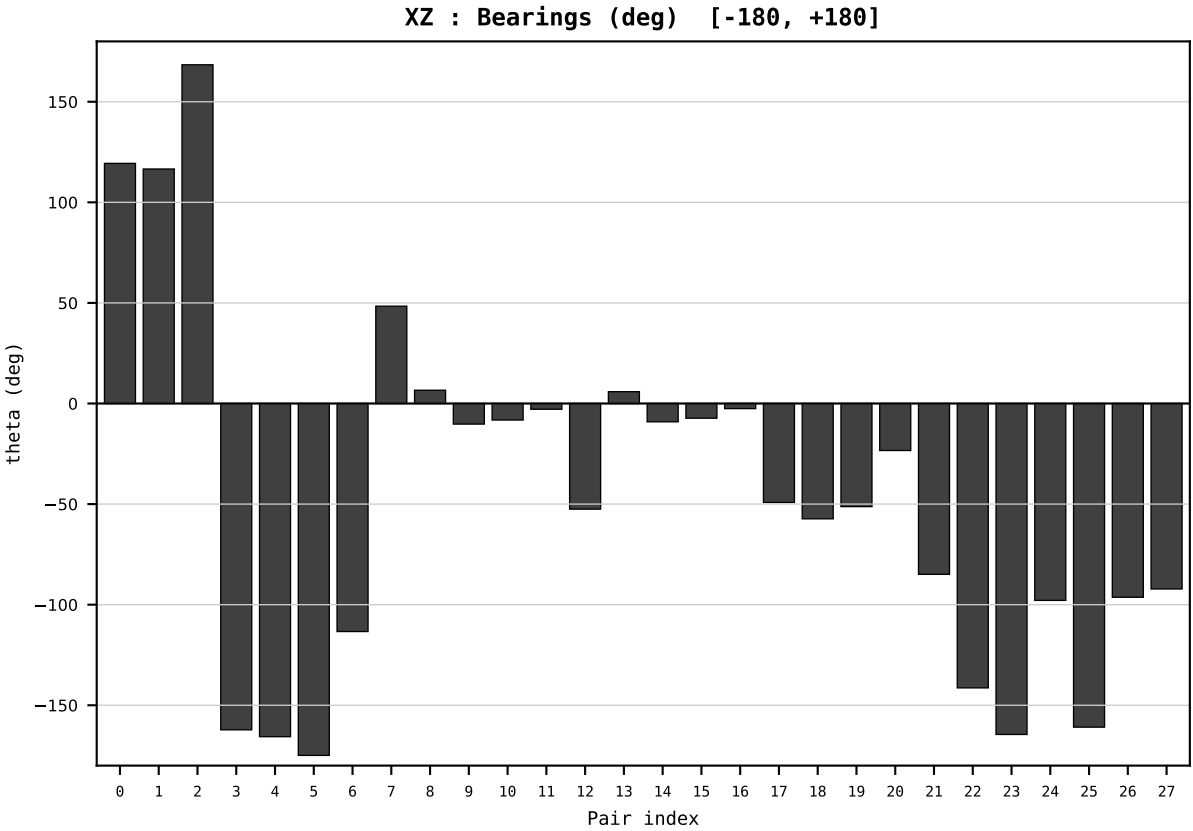
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2018
t : 18 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.279970
Ring = Hs-2
E_metric= 12.4125
kappa_HS= 60.46
omega = 7.1966 deg
d_A = 0.507945
Helm = Nuclear
Helm d = +0.426213
DR = 4.102 HLR
DR ratio= 60.5x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=19 | Year 2019 | D=8 N=26 pairs=28

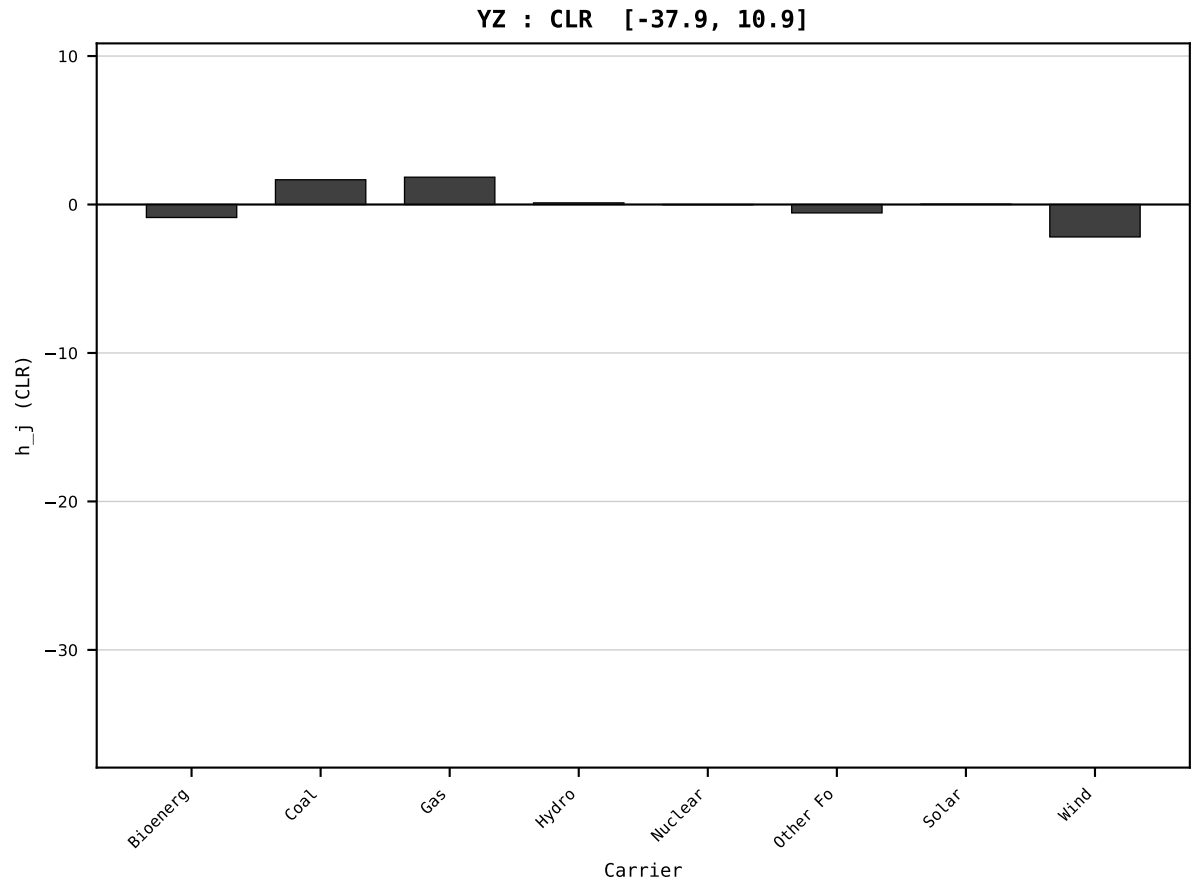
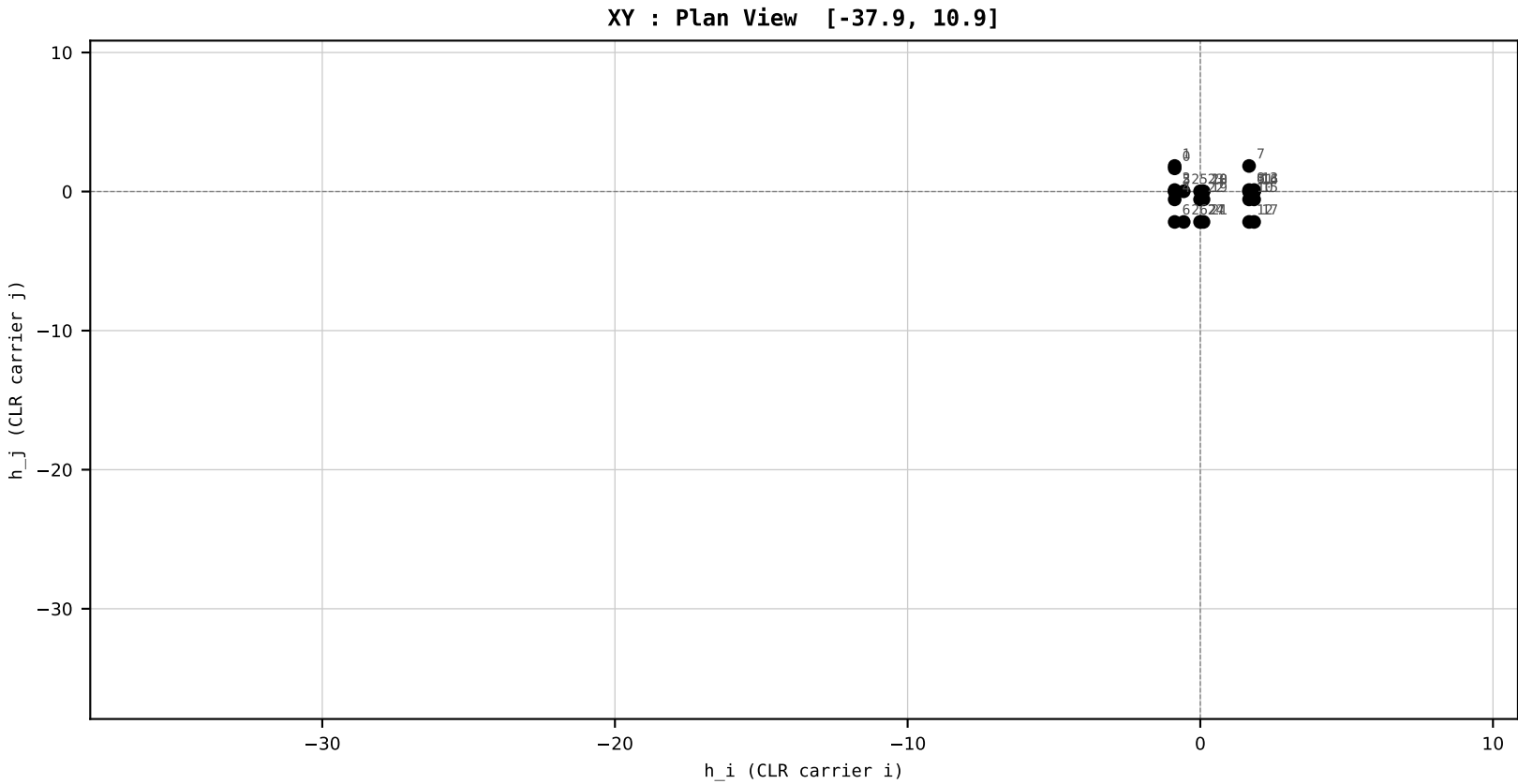
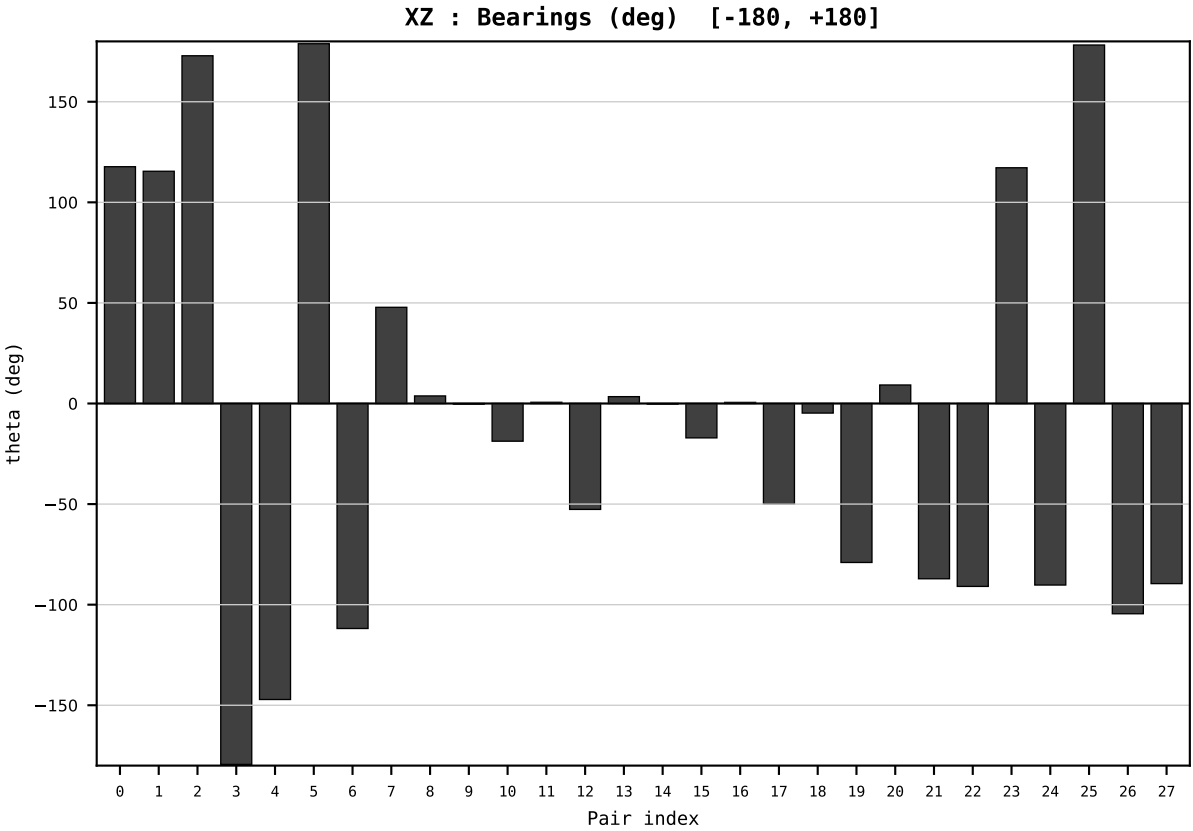
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2019
t : 19 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.268630
Ring = Hs-2
E_metric= 12.0301
kappa_HS= 55.89
omega = 7.6236 deg
d_A = 0.467986
Helm = Other Fossil
Helm d = -0.321852
DR = 4.023 HLR
DR ratio= 55.9x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=20 | Year 2020 | D=8 N=26 pairs=28

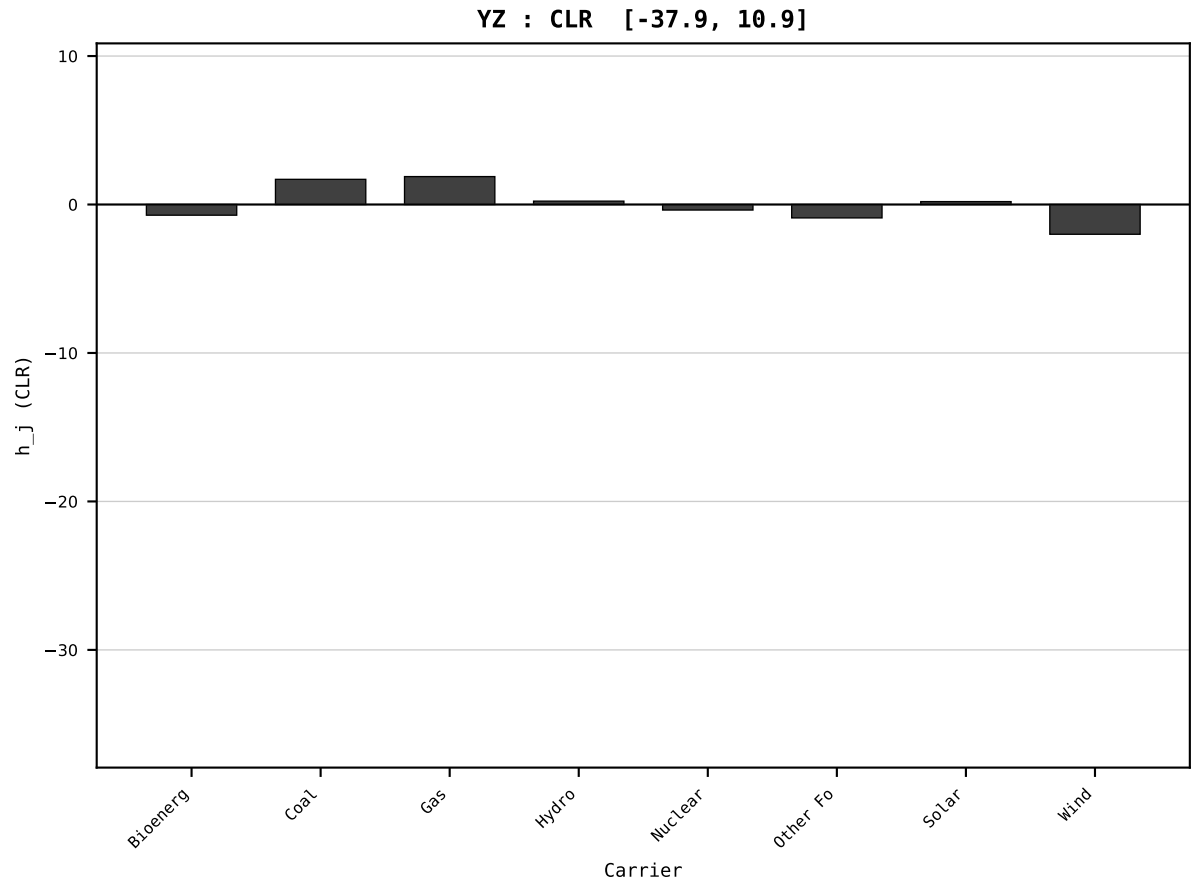
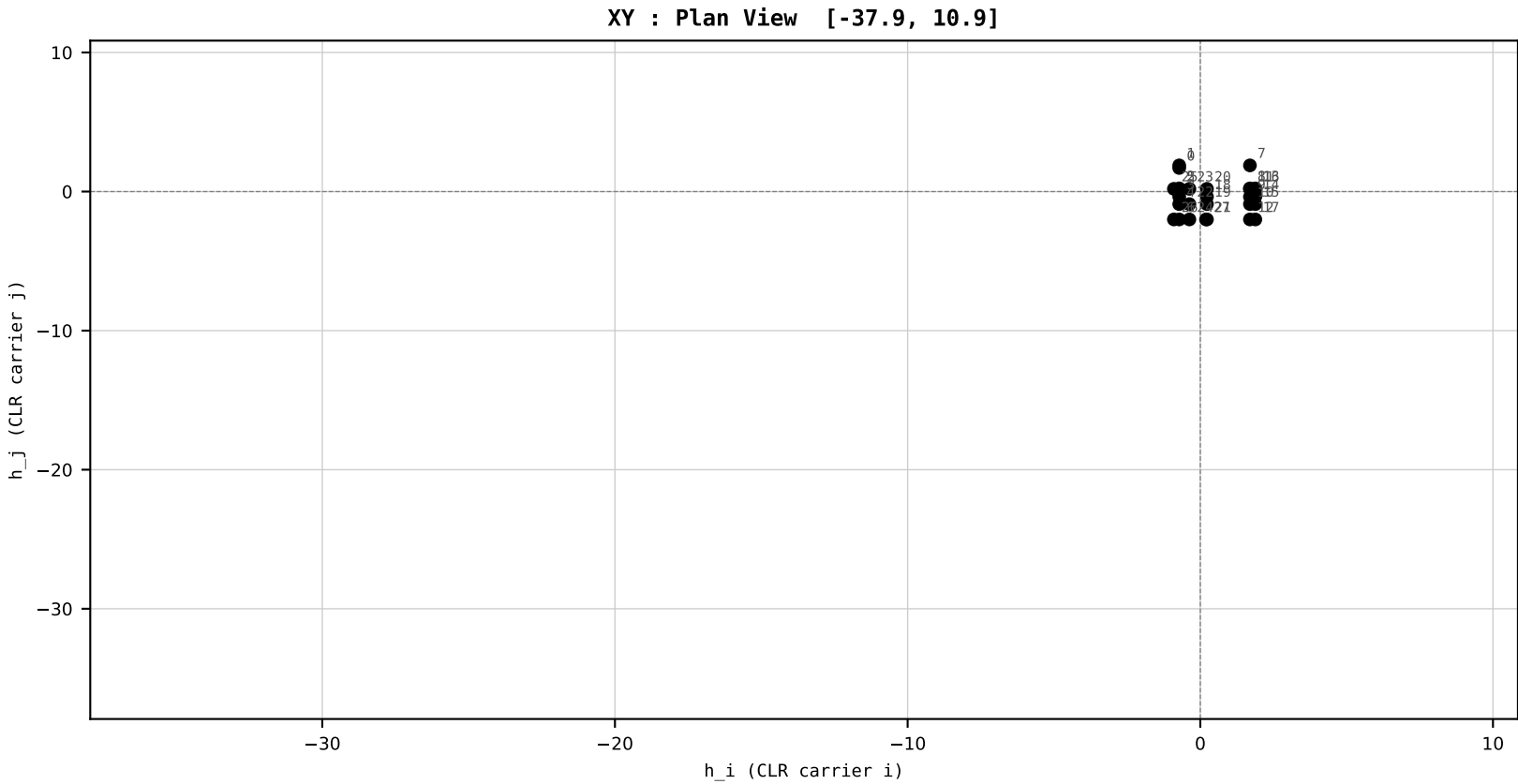
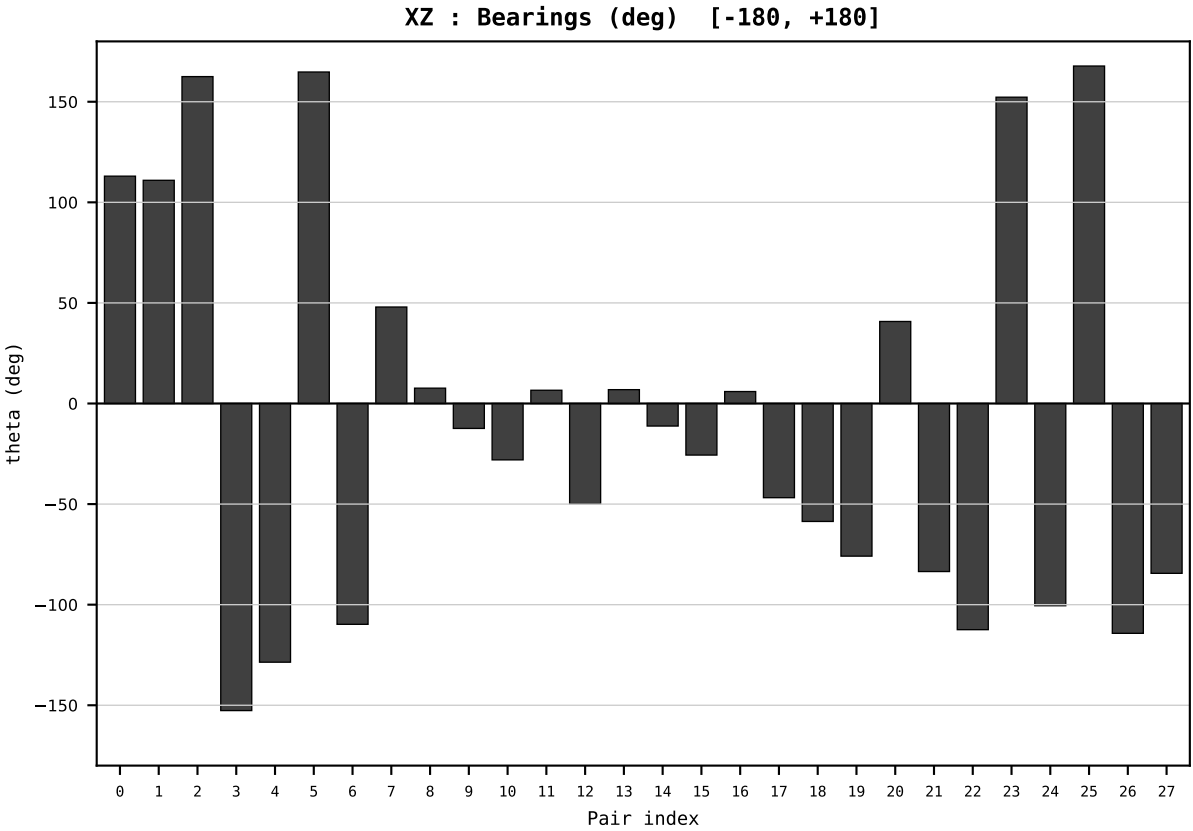
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2020
t : 20 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.280146
Ring = Hs-2
E_metric= 11.9940
kappa_HS= 48.68
omega = 9.7926 deg
d_A = 0.591660
Helm = Nuclear
Helm d = -0.363464
DR = 3.885 HLR
DR ratio= 48.7x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=21 | Year 2021 | D=8 N=26 pairs=28

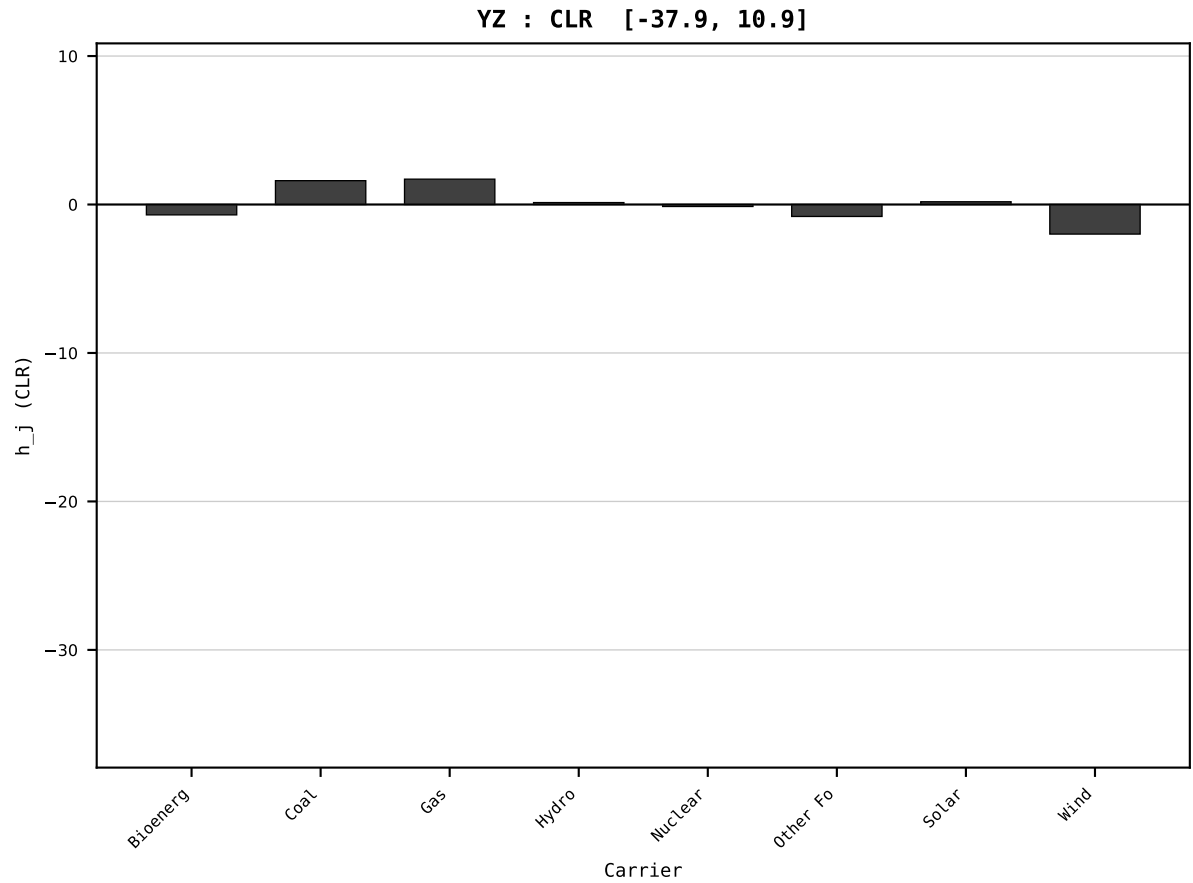
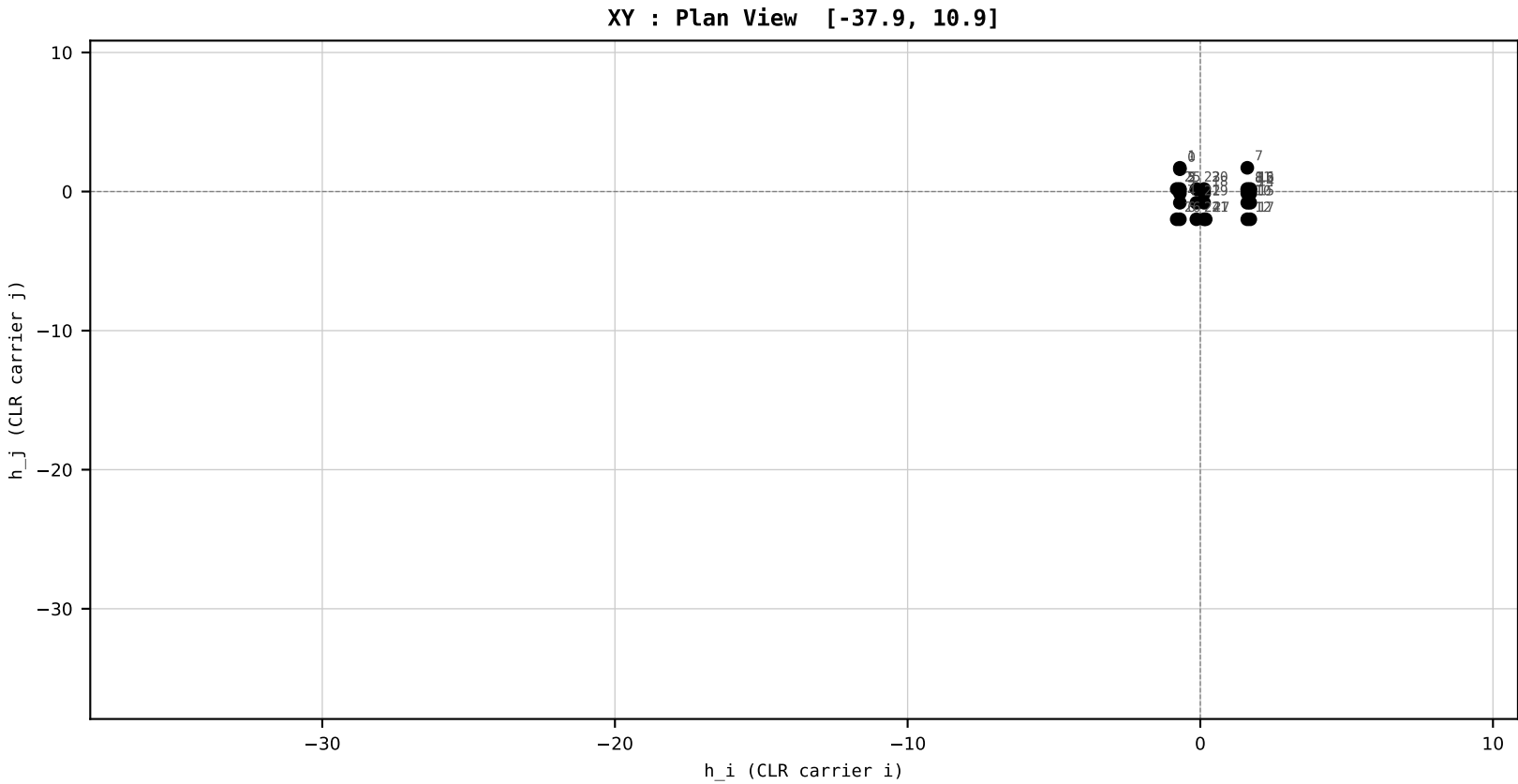
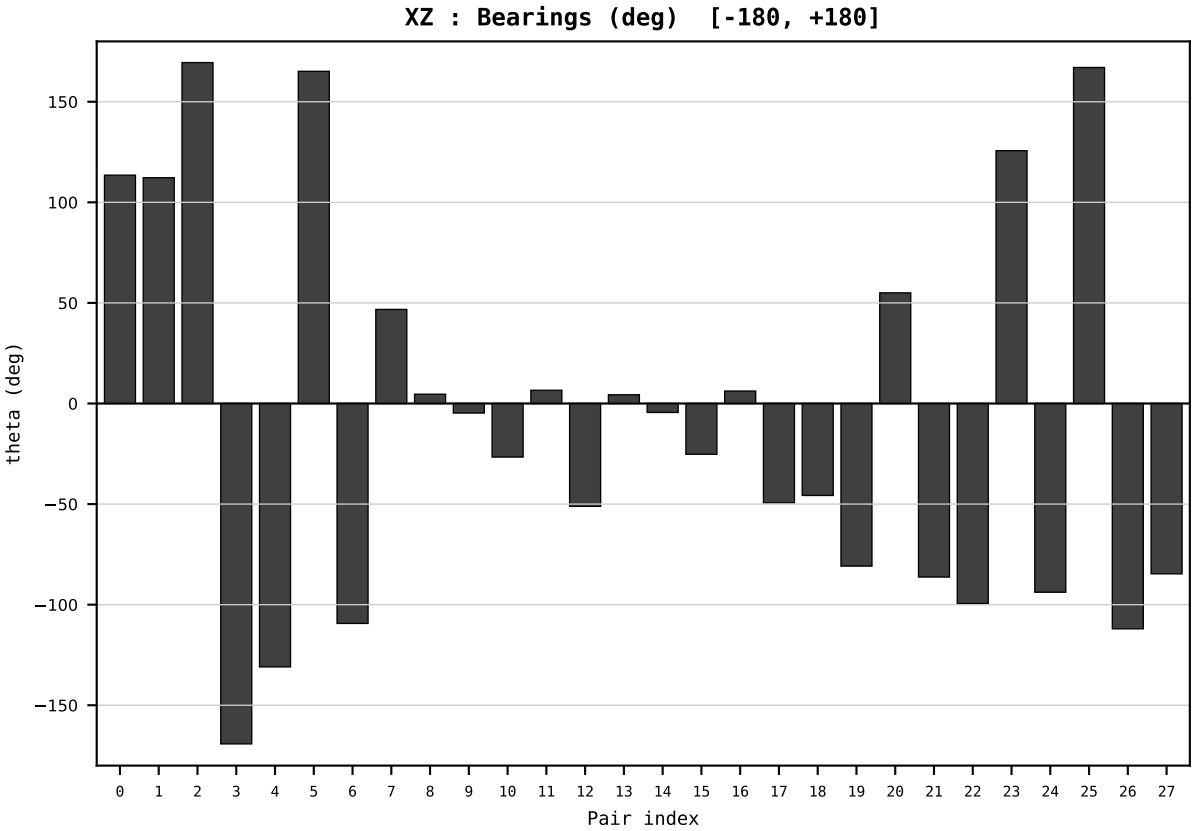
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2021
t : 21 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.246834
Ring = Hs-2
E_metric= 10.6662
kappa_HS= 40.44
omega = 4.7105 deg
d_A = 0.339626
Helm = Nuclear
Helm d = +0.239503
DR = 3.700 HLR
DR ratio= 40.4x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



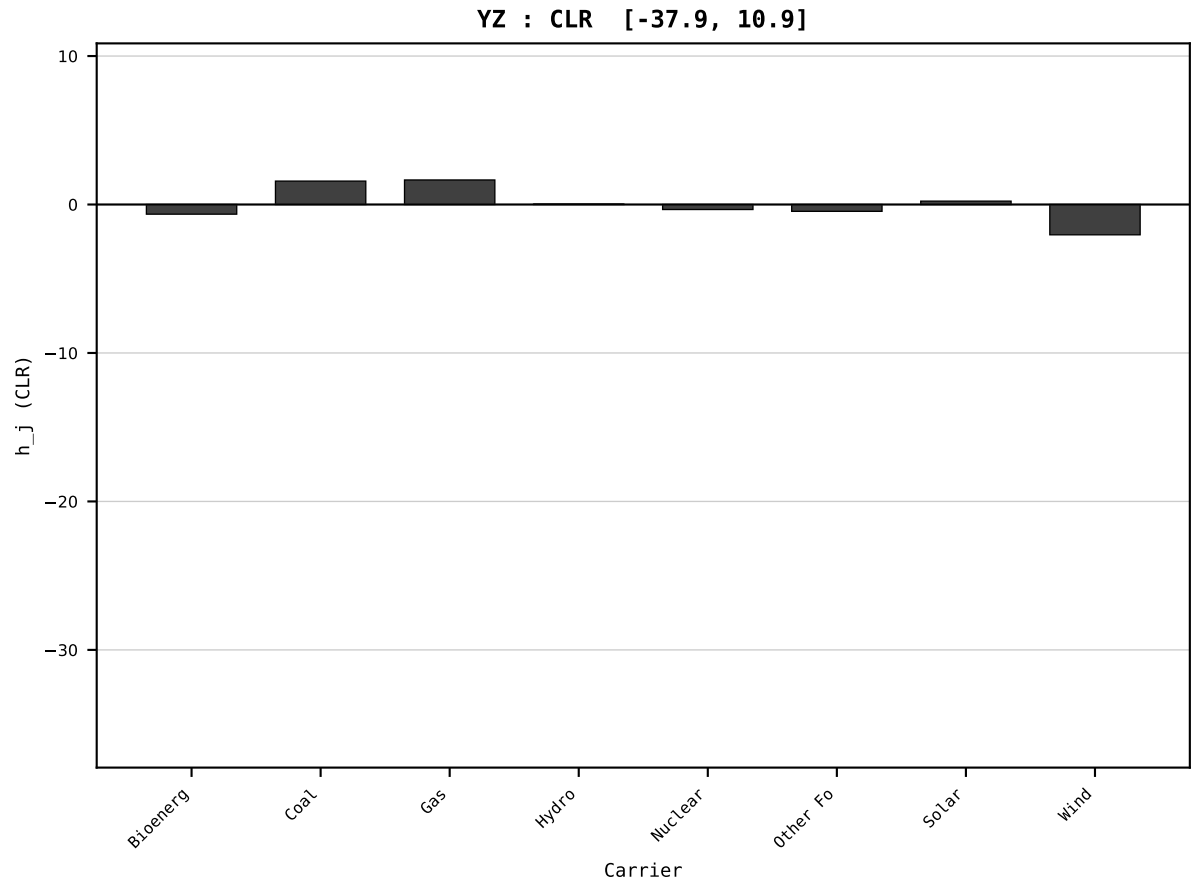
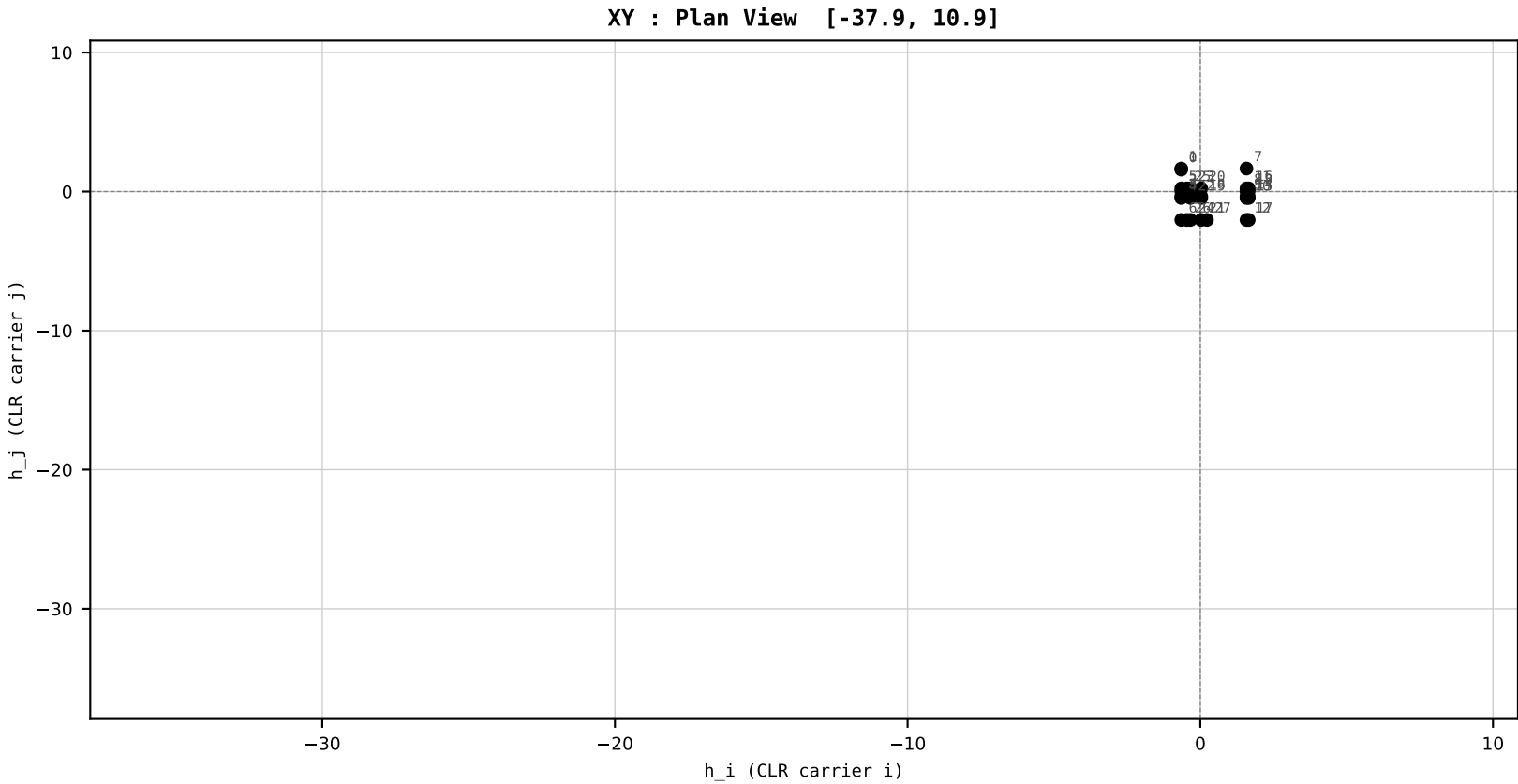
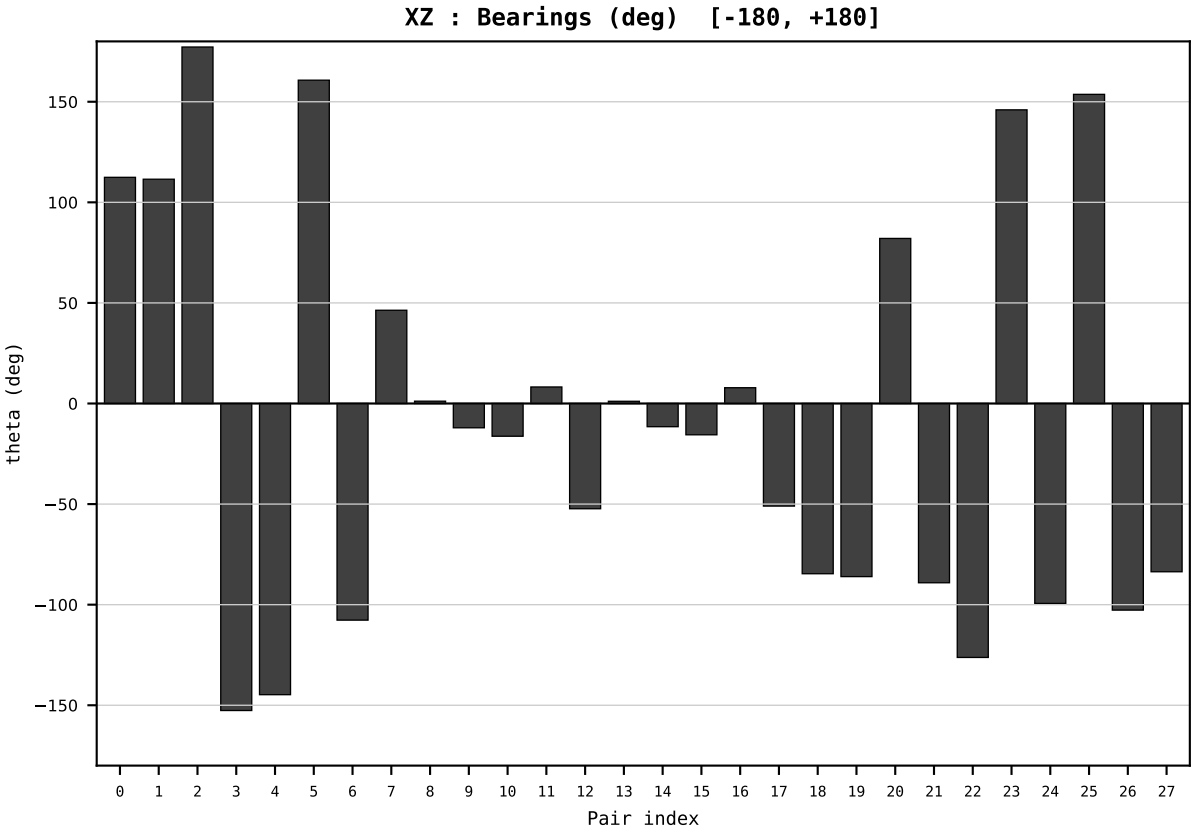
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2022
t : 22 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.236919
Ring = Hs-2
E_metric= 10.1872
kappa_HS= 40.22
omega = 7.4261 deg
d_A = 0.424692
Helm = Other Fossil
Helm d = +0.344772
DR = 3.694 HLR
DR ratio= 40.2x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



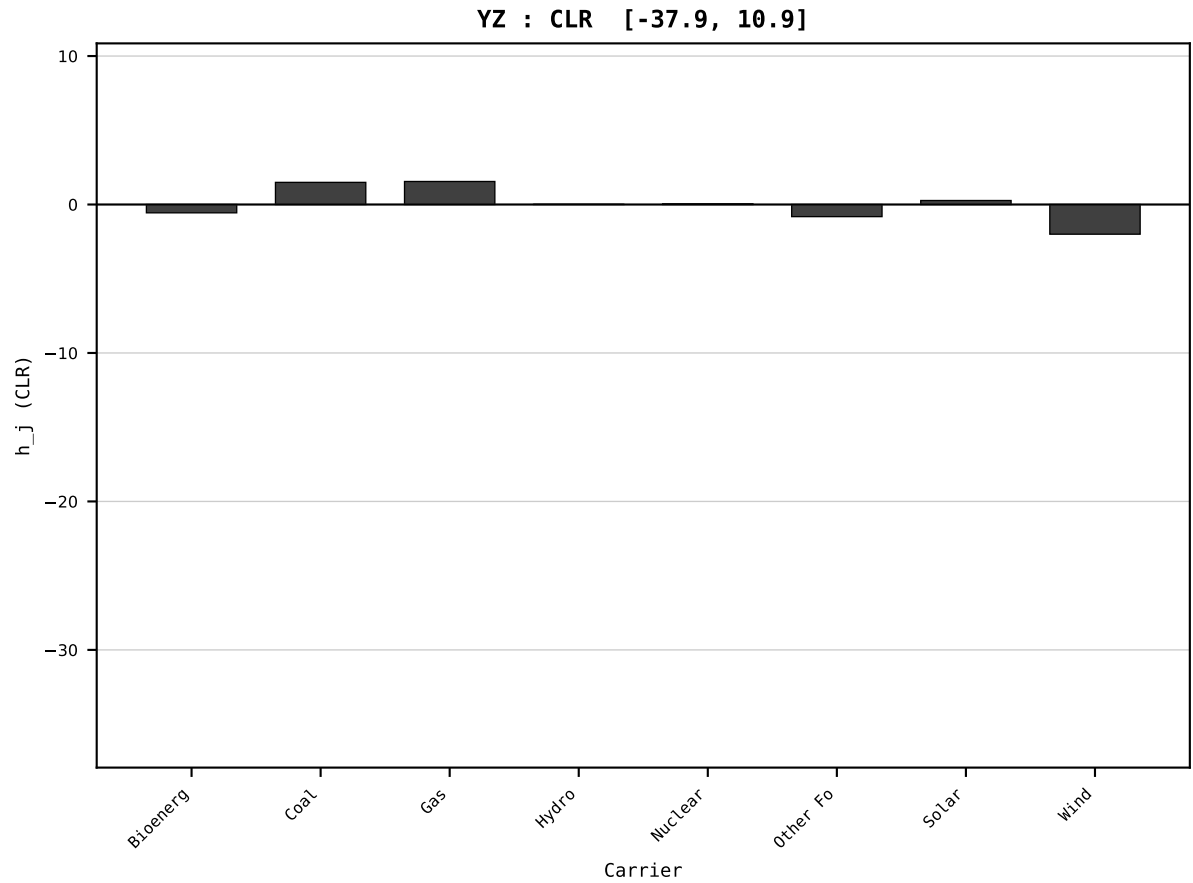
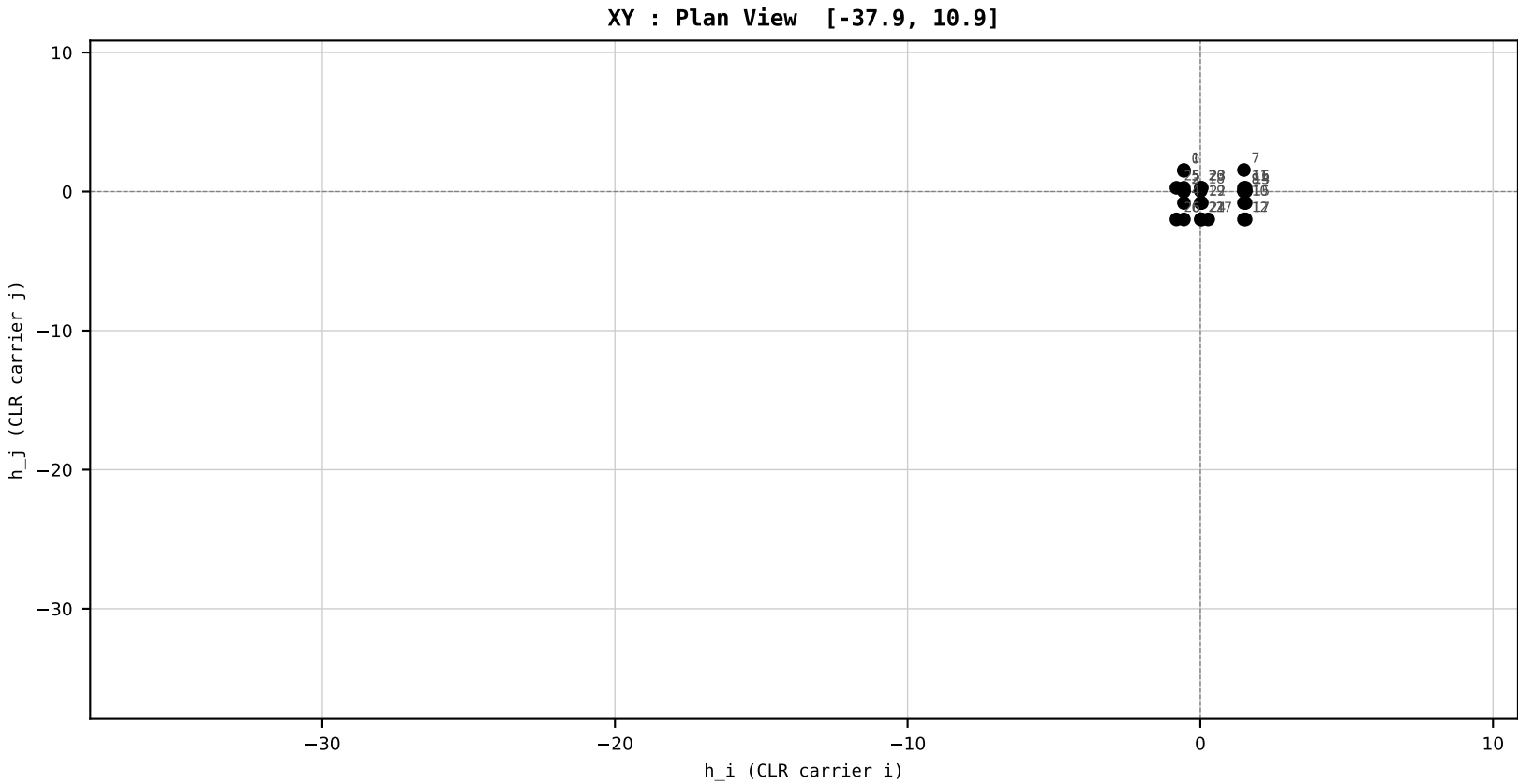
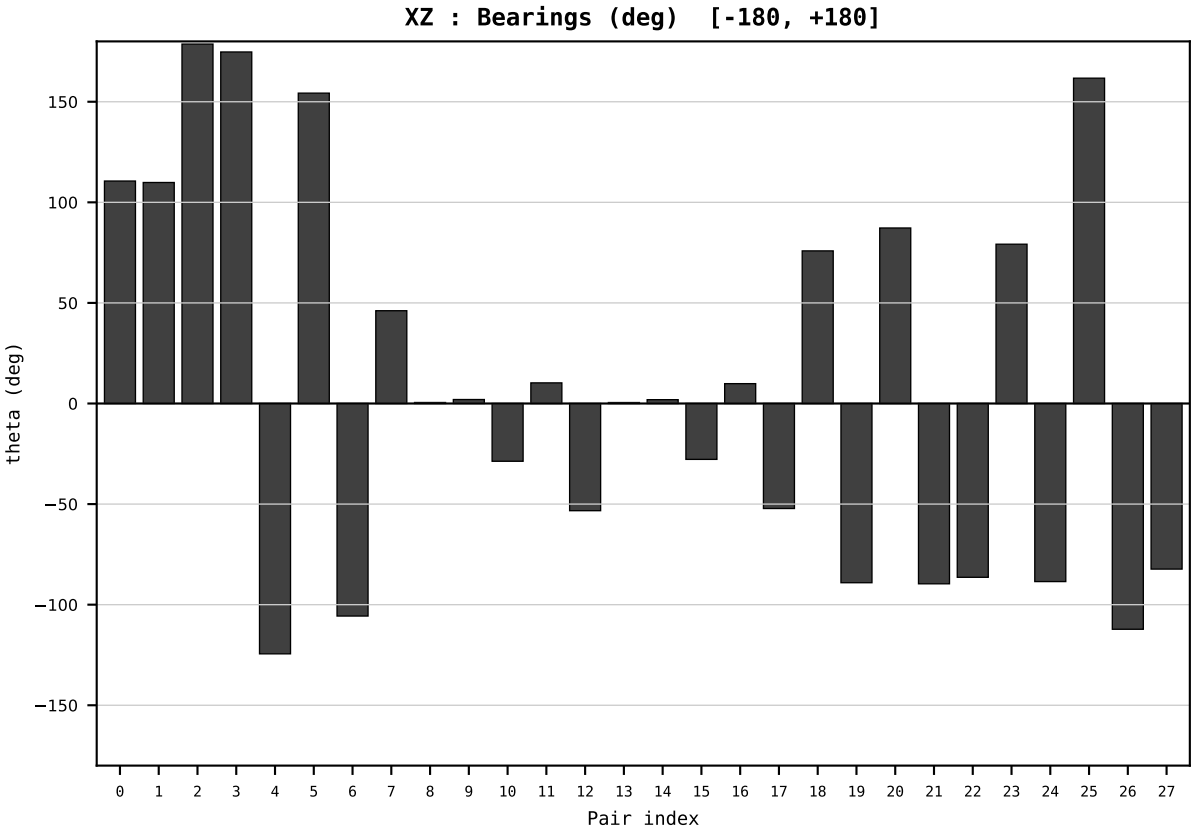
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2023
t : 23 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.213214
Ring = Hs-2
E_metric= 9.6860
kappa_HS= 34.84
omega = 10.0007 deg
d_A = 0.555145
Helm = Nuclear
Helm d = +0.388708
DR = 3.551 HLR
DR ratio= 34.8x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=24 | Year 2024 | D=8 N=26 pairs=28

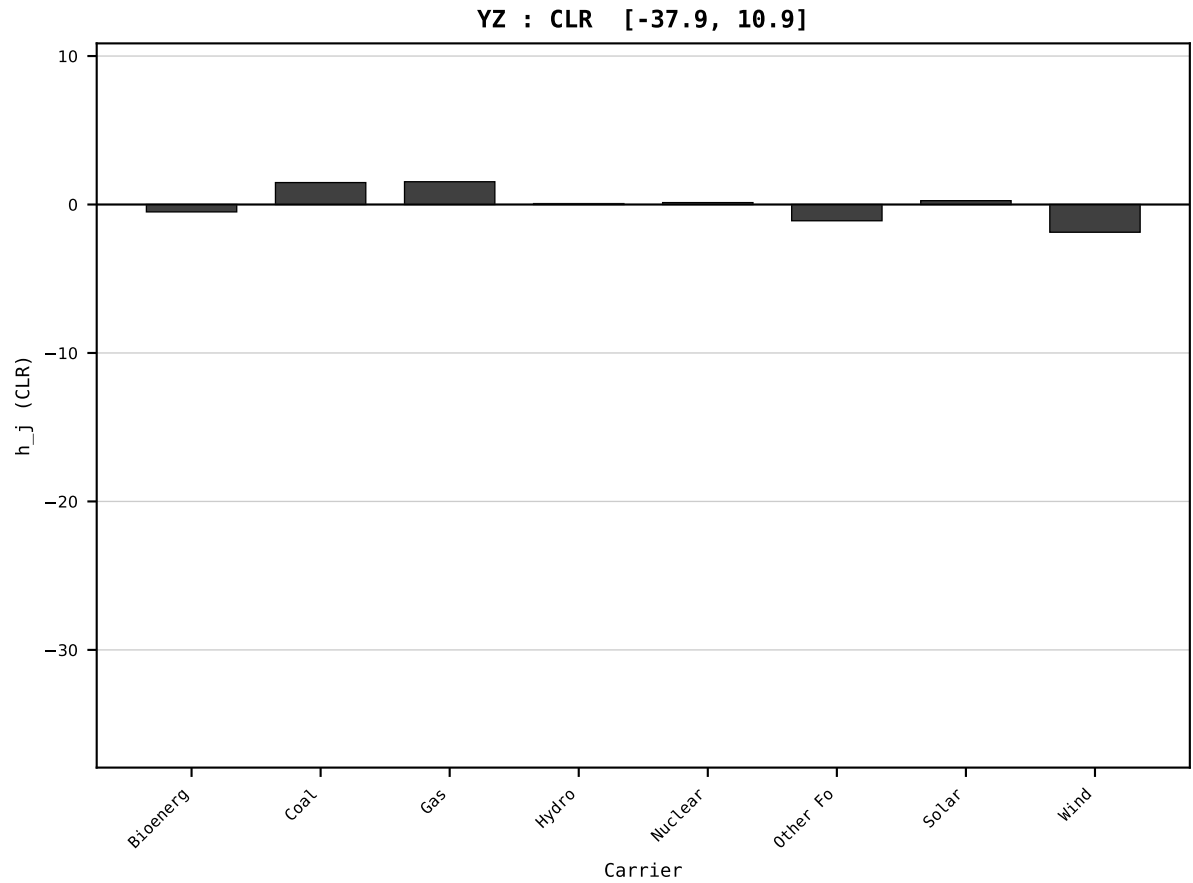
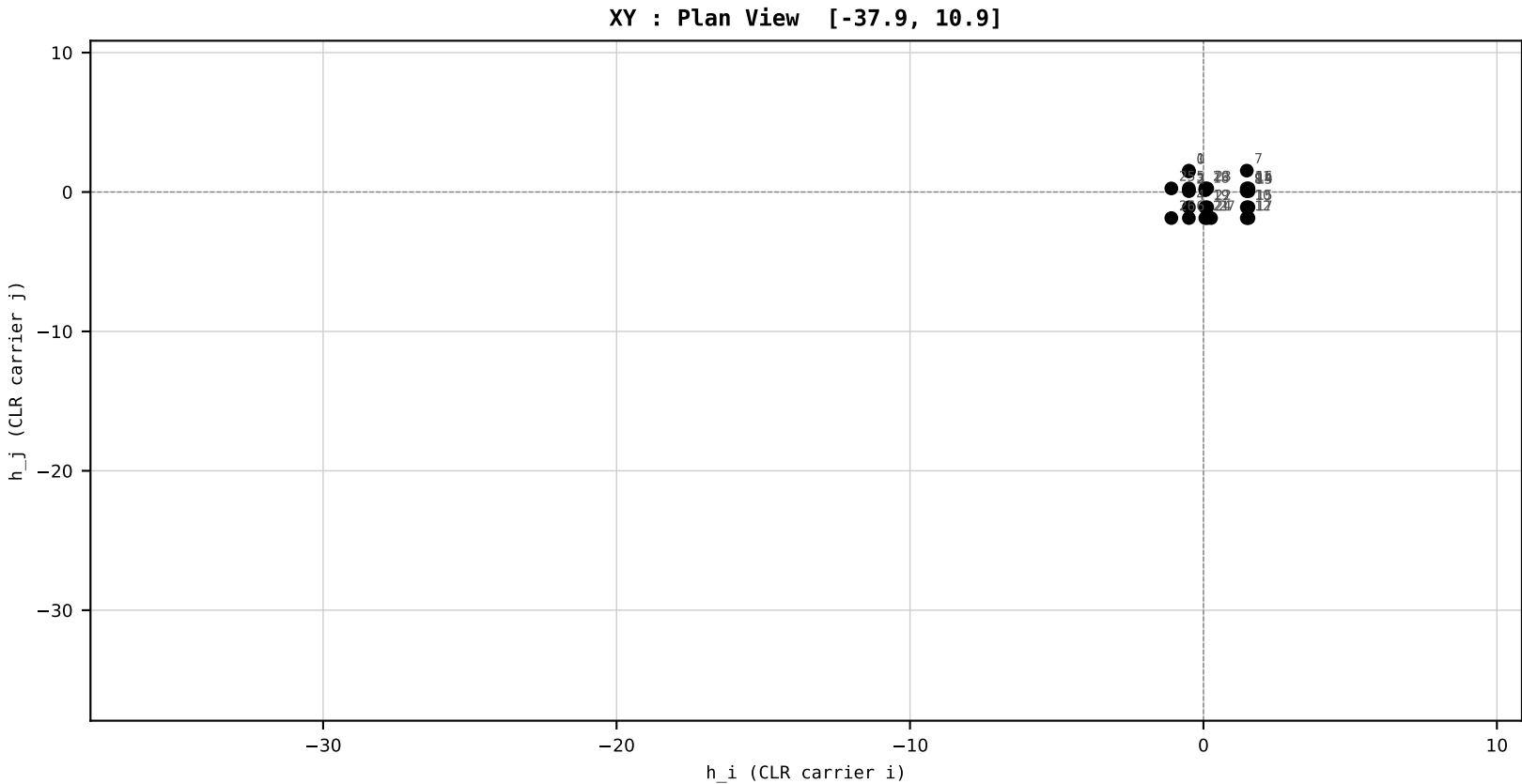
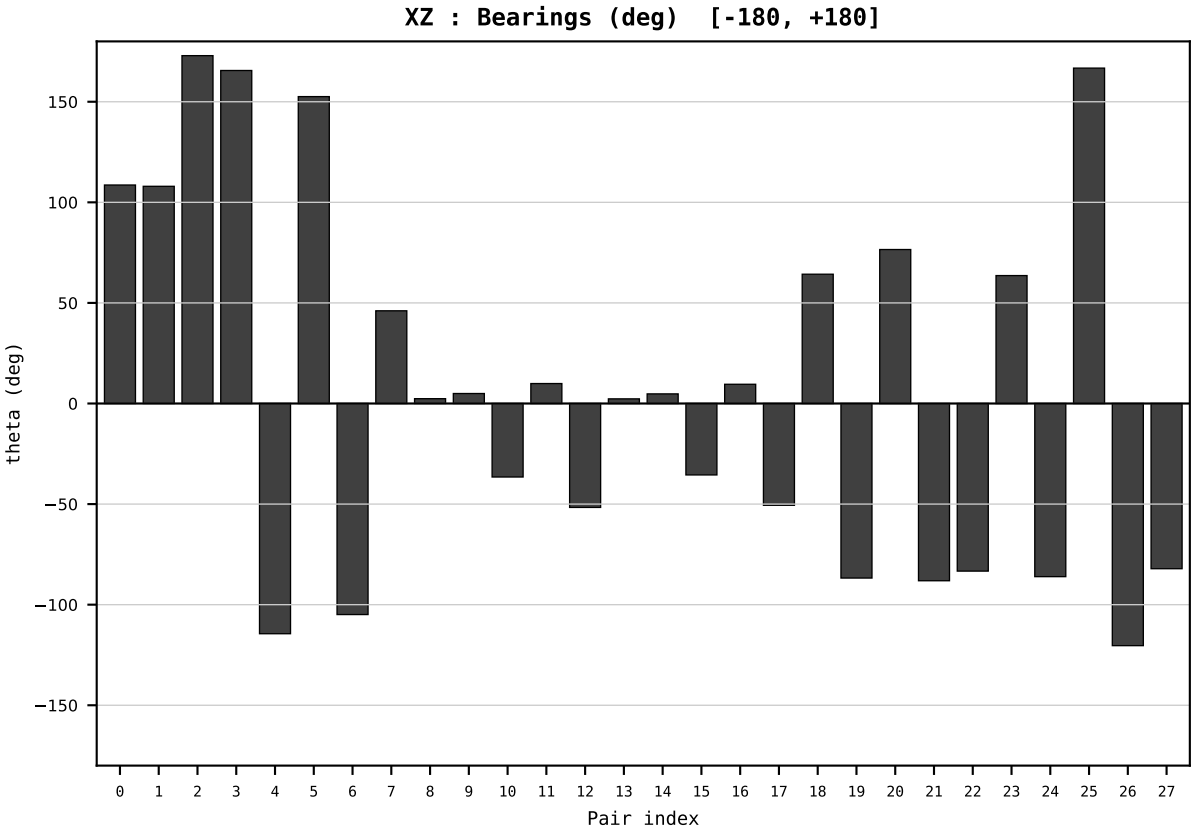
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2024
t : 24 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.209872
Ring = Hs-2
E_metric= 9.5540
kappa_HS= 30.00
omega = 6.0418 deg
d_A = 0.327601
Helm = Other Fossil
Helm d = -0.277449
DR = 3.401 HLR
DR ratio= 30.0x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=25 | Year 2025 | D=8 N=26 pairs=28

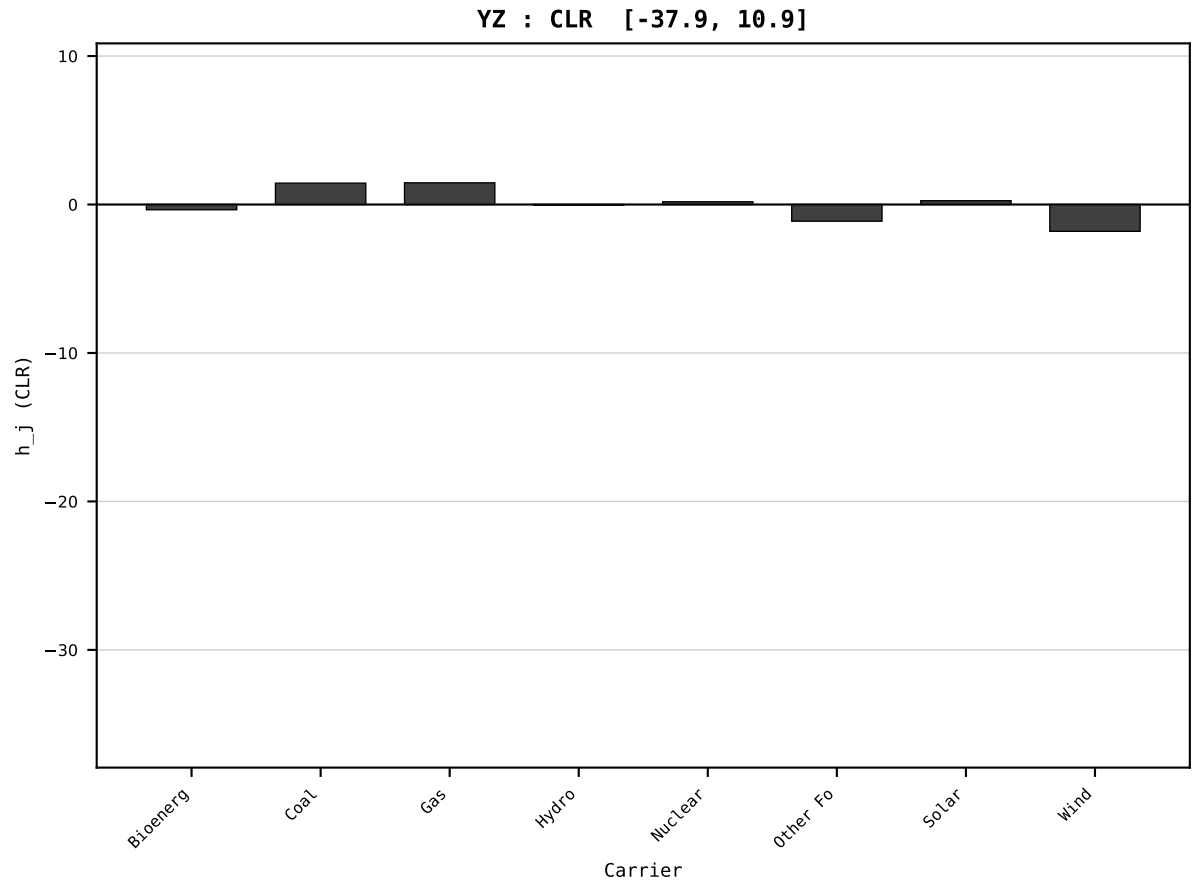
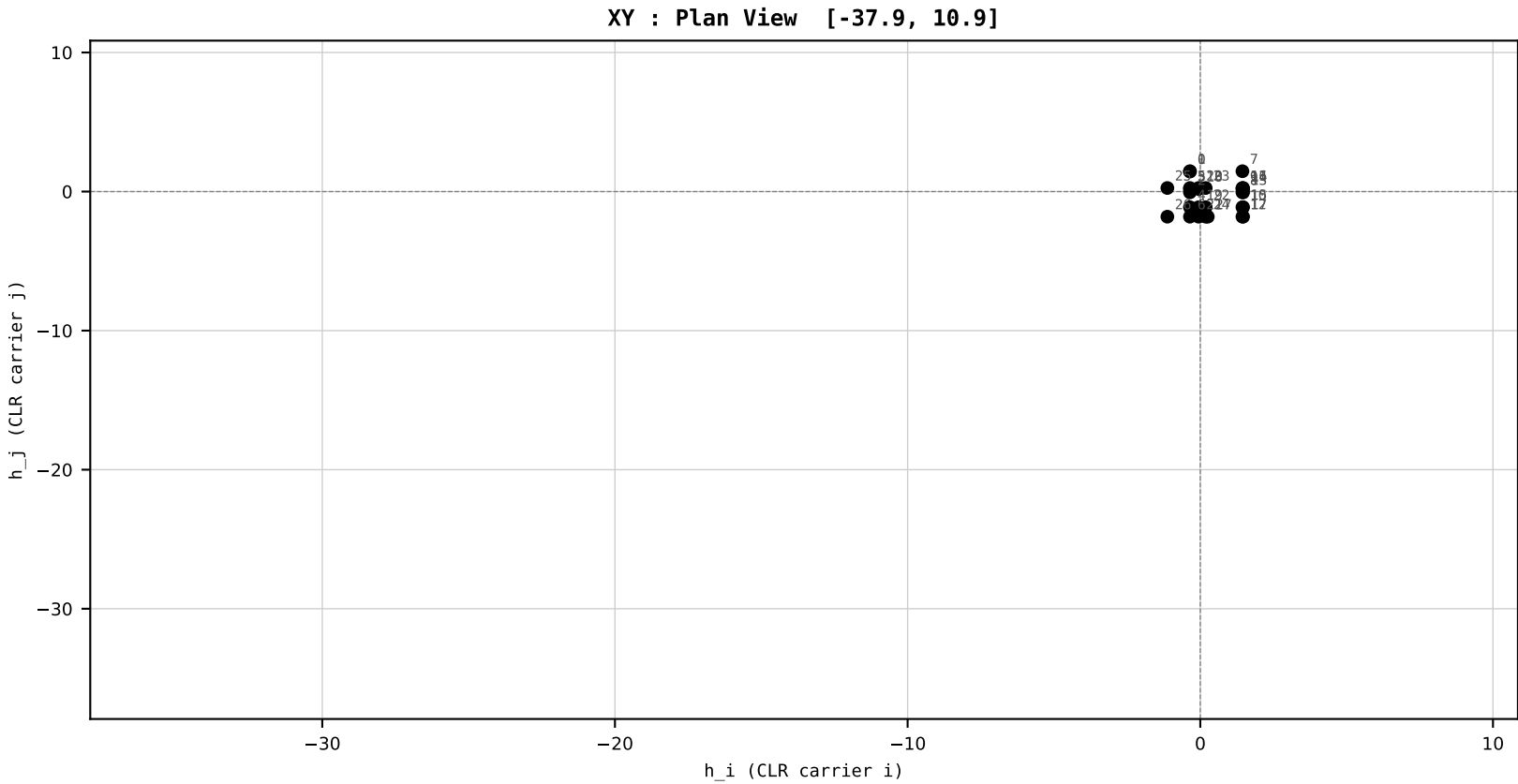
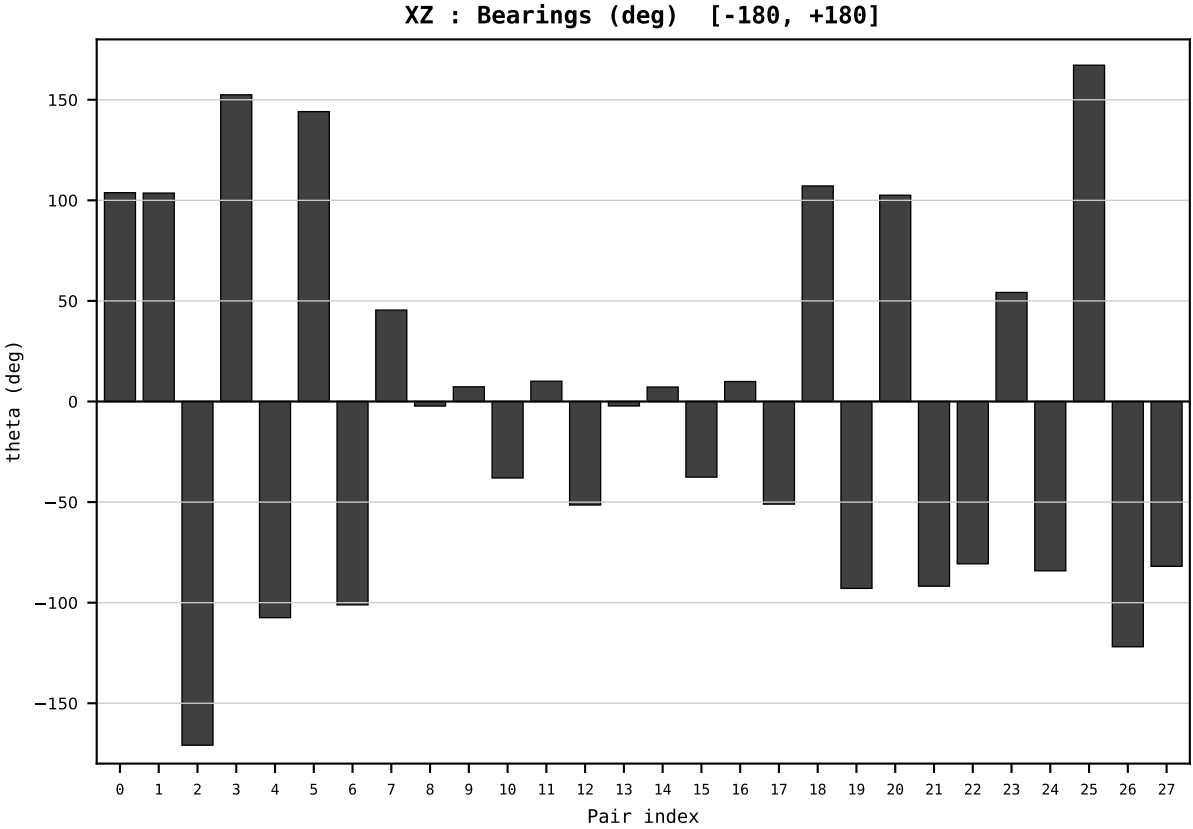
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

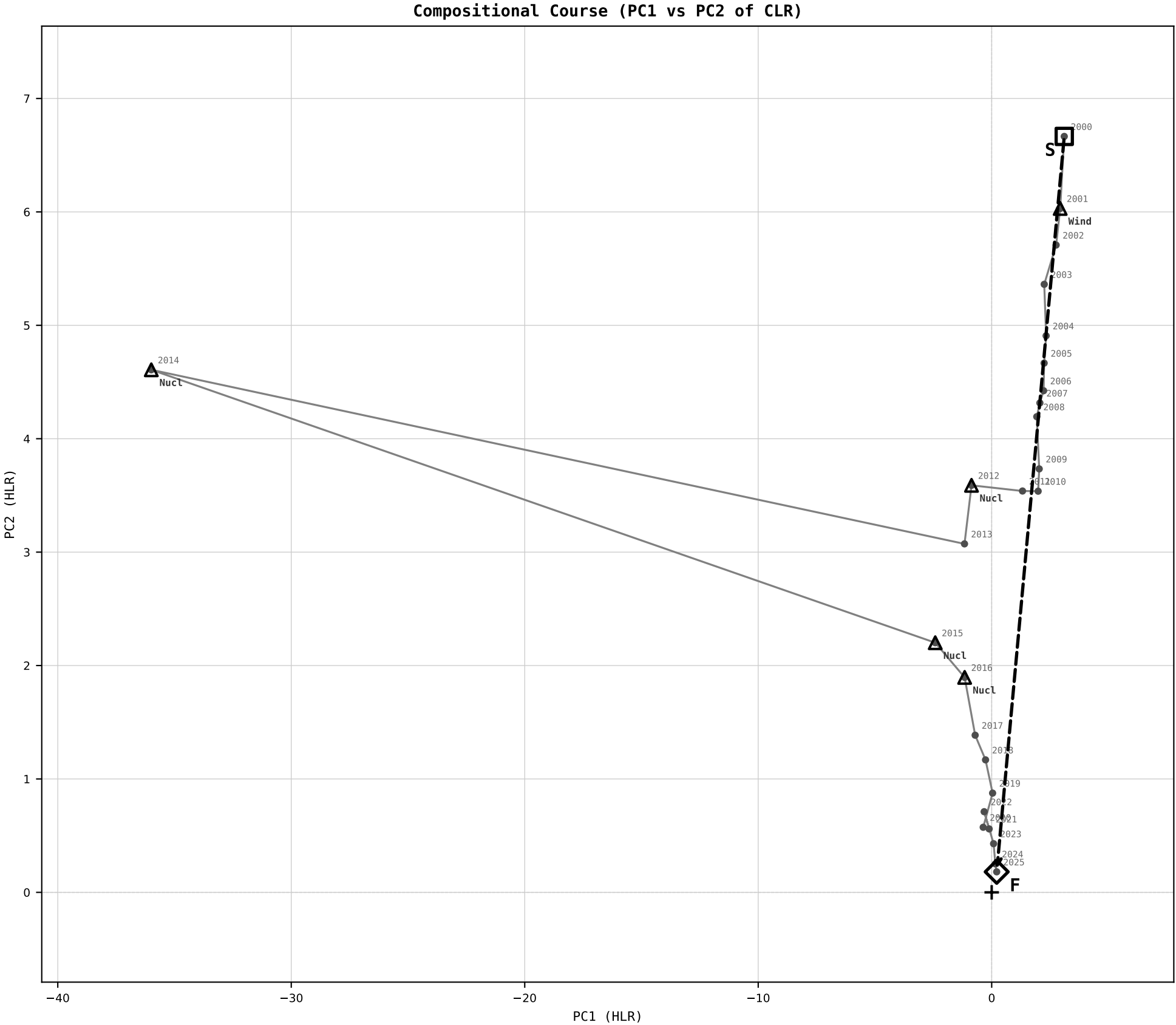
Dataset : ember_JPN_Japan_generation_TWh.csv
Year : 2025
t : 25 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.197825
Ring = Hs-2
E_metric= 8.9676
kappa_HS= 26.27
omega = 3.7598 deg
d_A = 0.221651
Helm = Bioenergy
Helm d = +0.143563
DR = 3.268 HLR
DR ratio= 26.3x

XZ [-180,180]
CLR[-37.9,10.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind





COURSE METRICS

Scale Provenance:
Y_j → x_j=Y_j/T → h_j=CLR
→ PC projection
Unit: HLR (Higgins Log-Ratio)

Net distance = 7.2233 HLR
Path length = 82.5468 HLR
Directness = 0.0875

DR start = 7.973 HLR
ratio = 2901.1x
DR final = 3.268 HLR
ratio = 26.3x

E_{metric} start = 70.5673
E_{metric} final = 8.9676

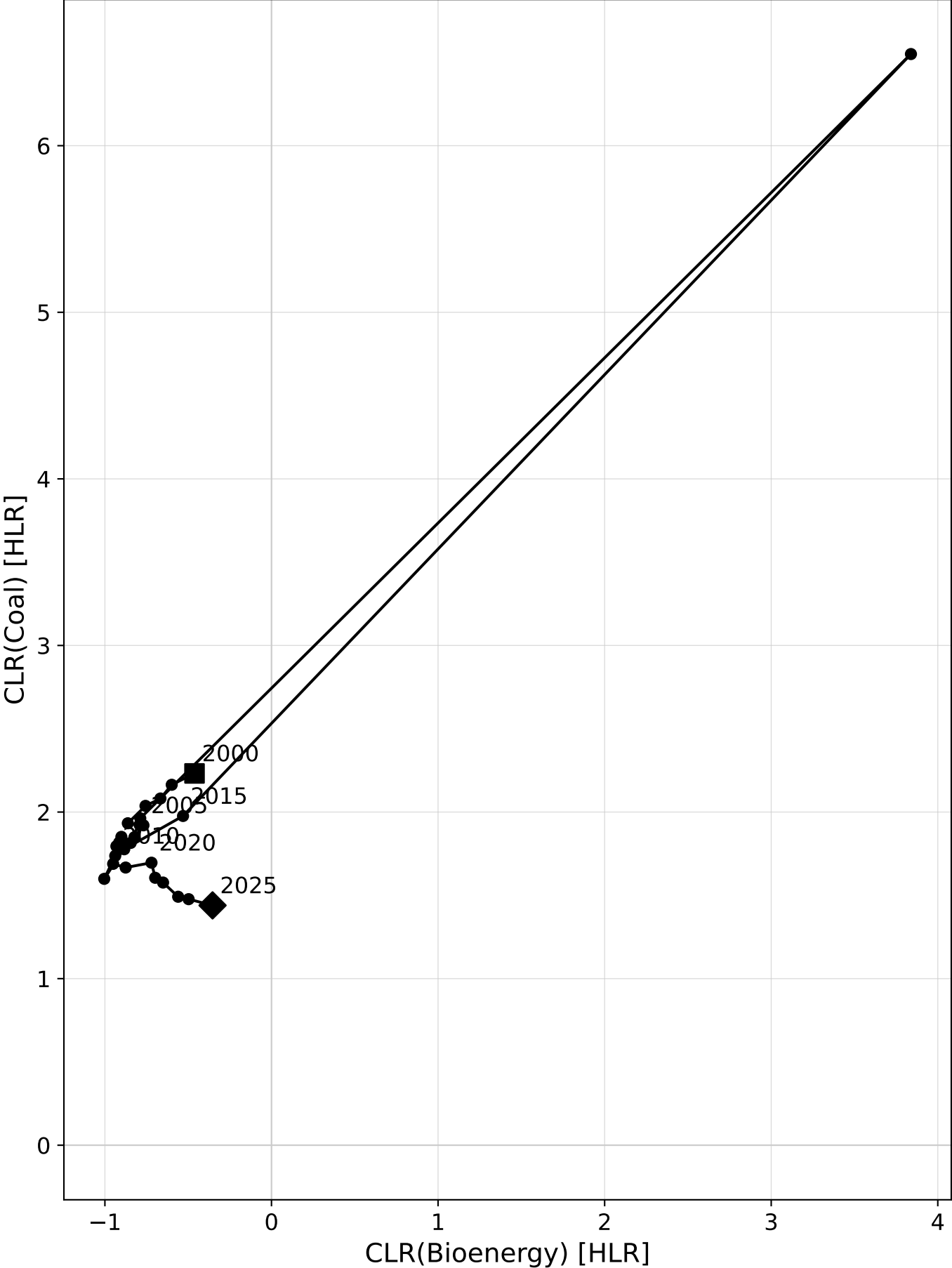
PC1 loadings:
Nuclear +0.934
Solar -0.184
Wind -0.152

PC2 loadings:
Solar -0.716
Other Fo +0.535
Wind -0.326

Top displacement events:
2014 Nuclear d=34.871
2015 Nuclear d=33.668
2012 Nuclear d=2.201
2016 Nuclear d=1.320
2001 Wind d=0.848

The instrument reads.
The expert decides.
The loop stays open.

SYSTEM COURSE PLOT — XY Plan View (CLR Space)

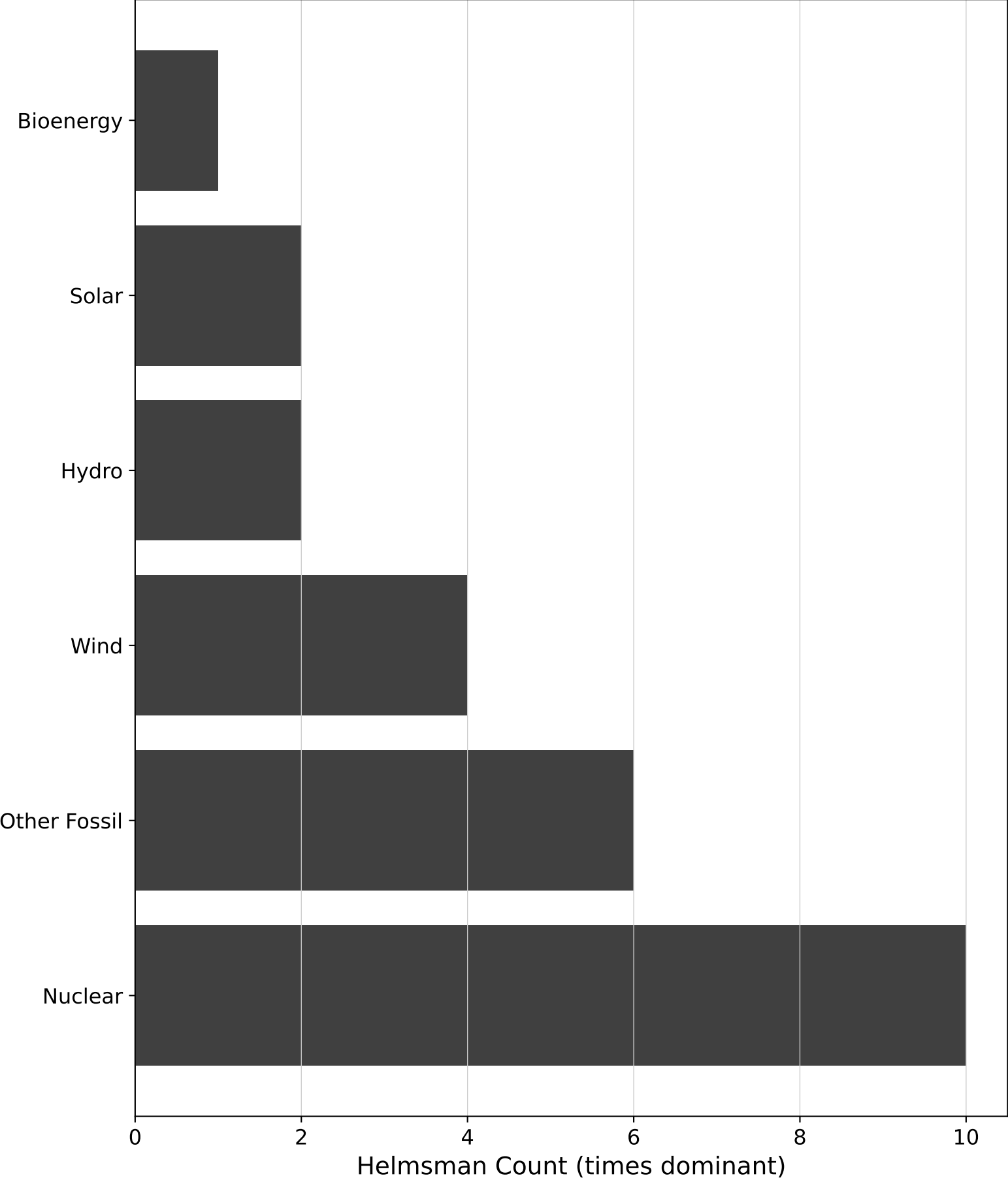


STAGE 2 – Aitchison Distance Matrix (all year pairs)

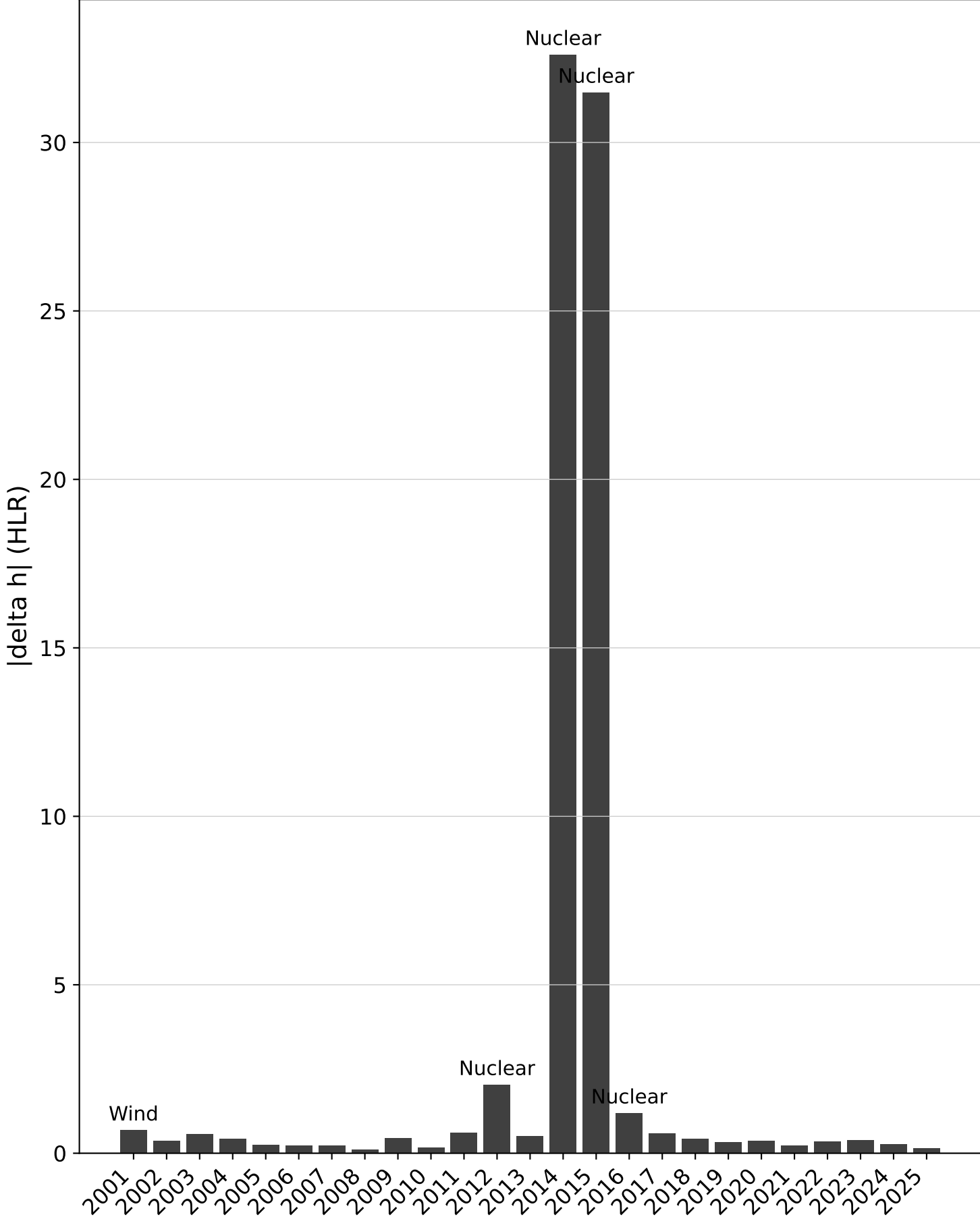


d_A (HLR)

HELMSMAN FREQUENCY



HELMSMAN DISPLACEMENT PER YEAR



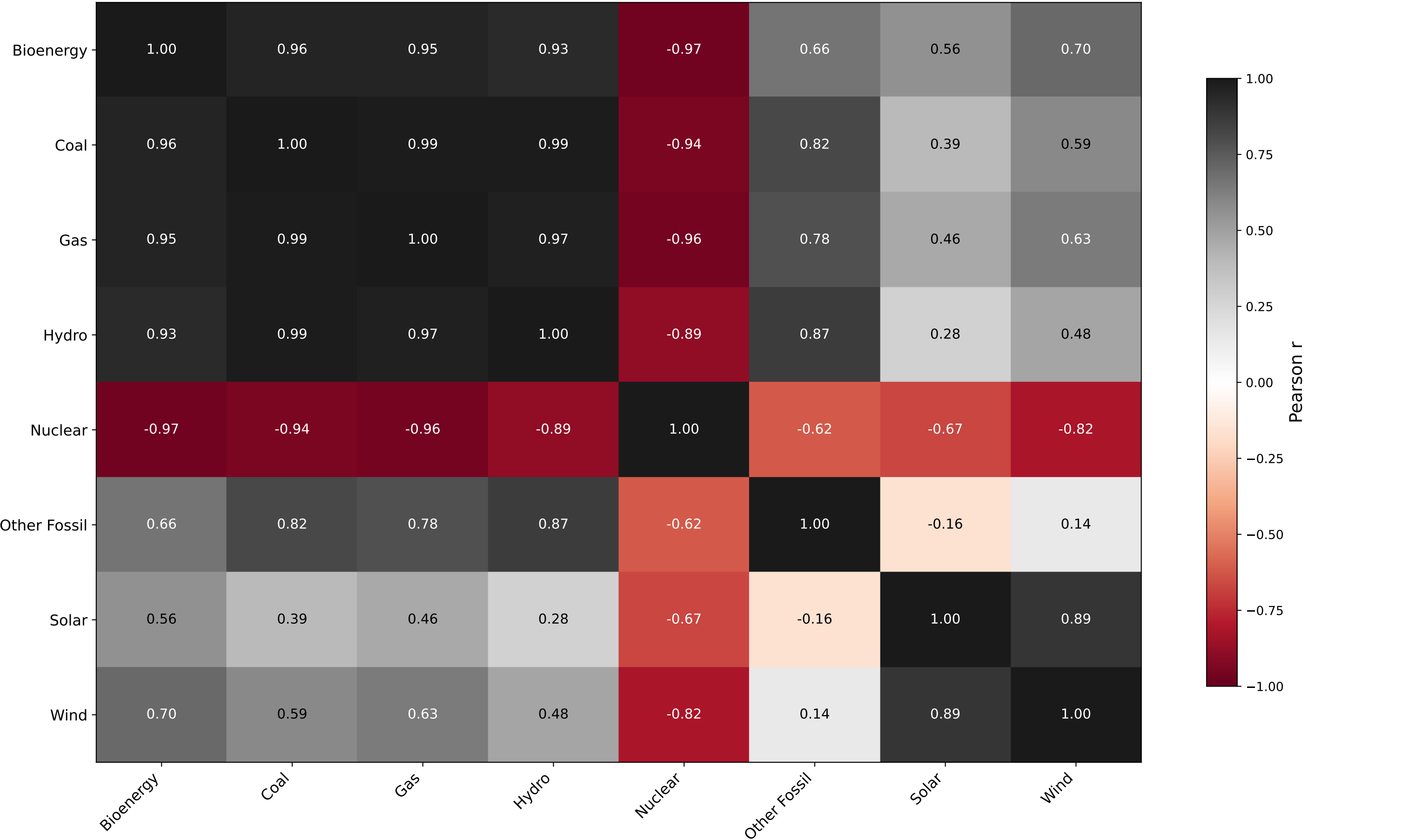
STAGE 2 — Pairwise Divergence Ranking (Top 15)

Rank	Year A	Year B	d_A (HLR)	Angle (deg)	Dominant Carrier
1	2000	2014	39.1787	103.1	Nuclear
2	2001	2014	38.9623	103.8	Nuclear
3	2002	2014	38.7724	103.8	Nuclear
4	2004	2014	38.3284	103.6	Nuclear
5	2003	2014	38.2521	101.3	Nuclear
6	2005	2014	38.2438	103.9	Nuclear
7	2006	2014	38.2199	104.3	Nuclear
8	2007	2014	38.0642	103.5	Nuclear
9	2009	2014	38.0444	104.7	Nuclear
10	2010	2014	38.0031	105.1	Nuclear
11	2008	2014	37.9324	102.4	Nuclear
12	2011	2014	37.3369	97.9	Nuclear
13	2014	2025	36.4813	88.9	Nuclear
14	2014	2024	36.4145	87.5	Nuclear
15	2014	2023	36.3177	85.7	Nuclear

STAGE 3 — Triadic Metric Triangle Area Ranking (Top 10)

Rank	Year A	Year B	Year C	Area (HLR^2)	Imbalance
1	2000	2014	2025	126.1654	5.42
2	2000	2014	2024	124.3973	5.47
3	2000	2014	2023	120.5923	5.58
4	2000	2014	2021	117.8599	5.60
5	2000	2014	2020	117.1553	5.52
6	2000	2014	2022	114.8469	5.62
7	2001	2014	2025	113.0317	5.99
8	2000	2014	2019	111.8742	5.89
9	2001	2014	2024	111.3093	6.04
10	2001	2014	2023	107.6440	6.17

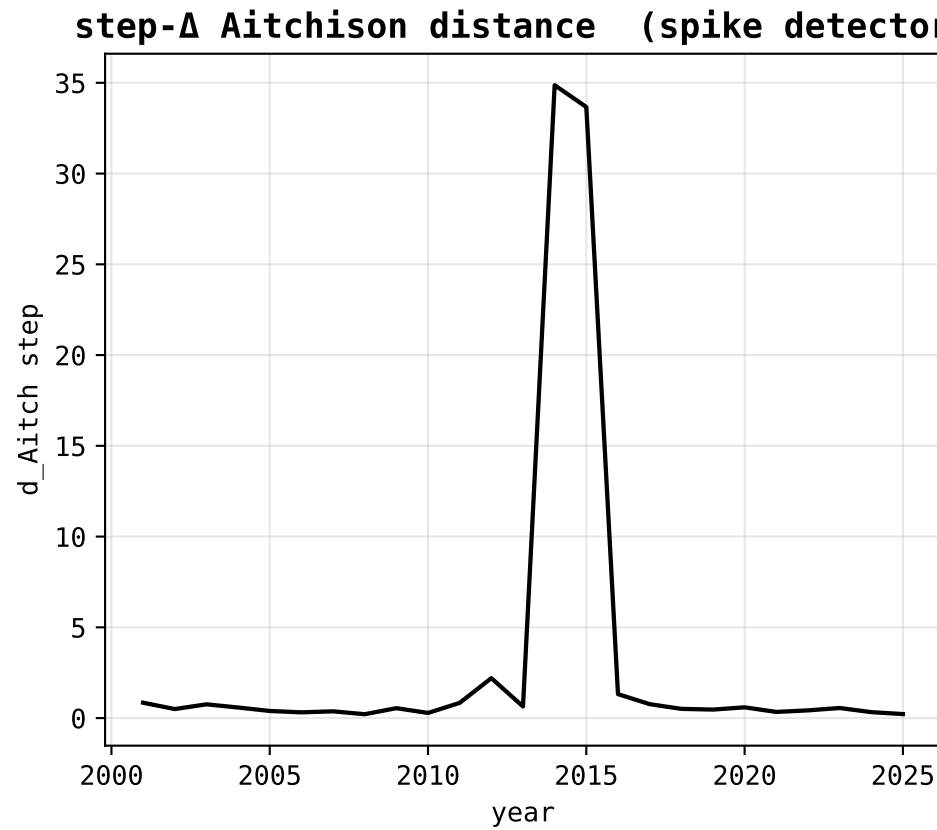
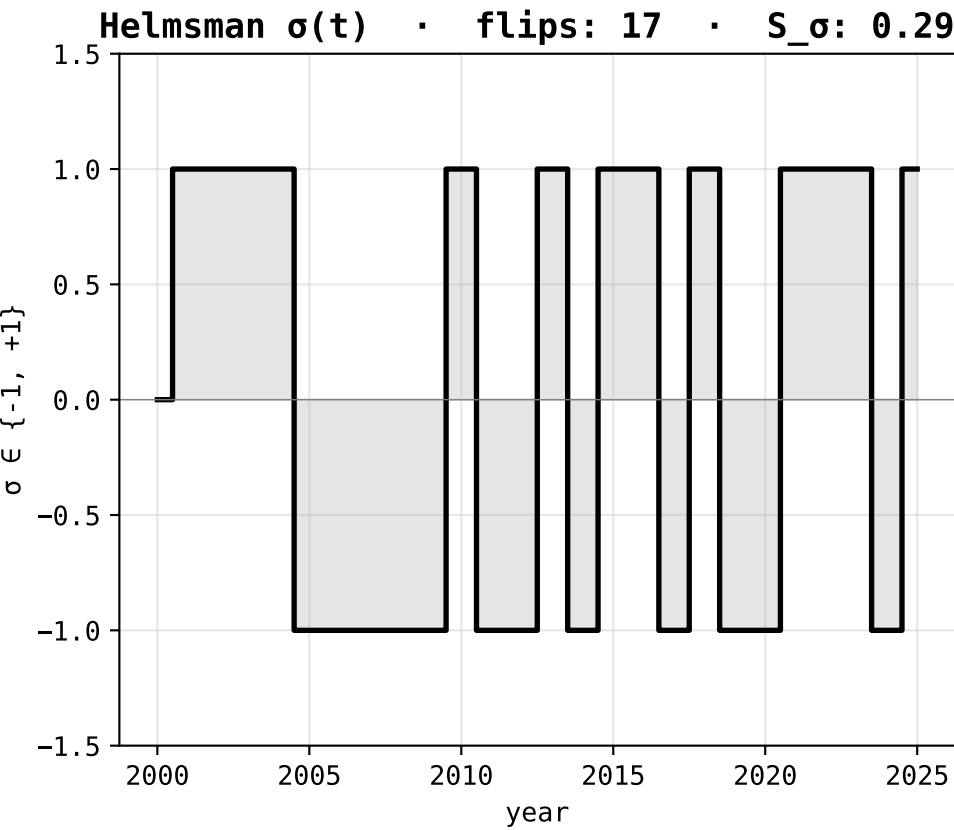
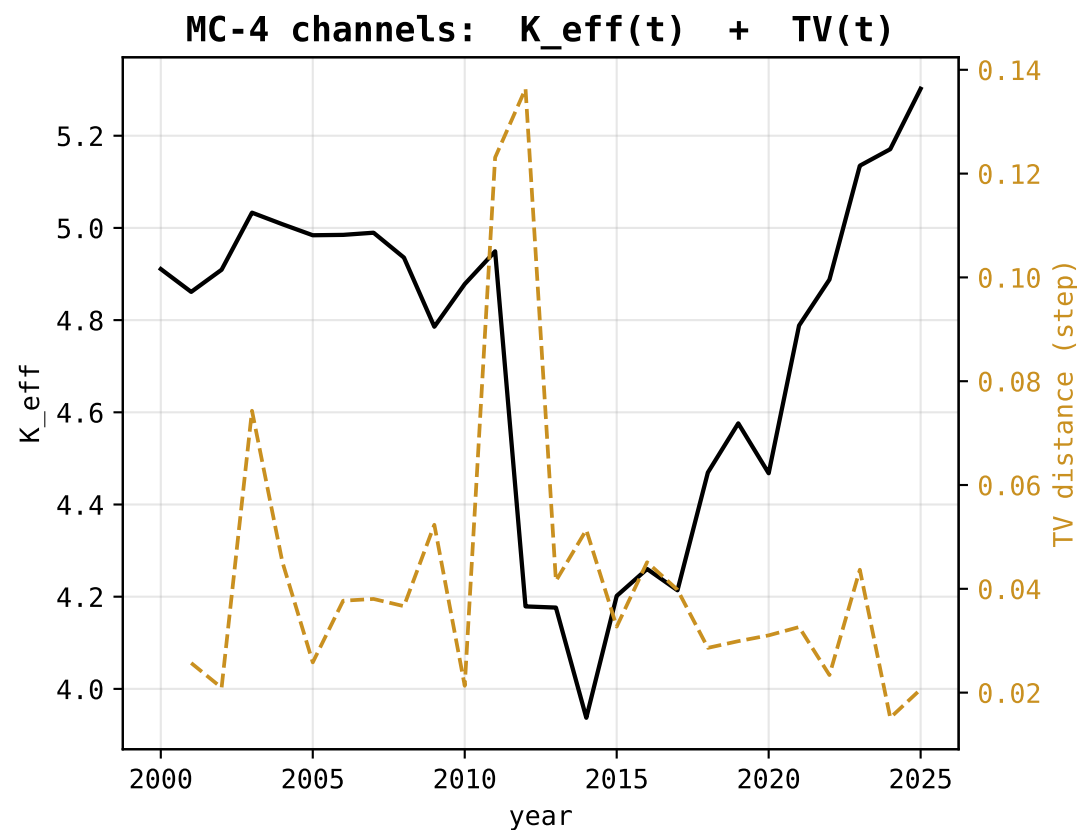
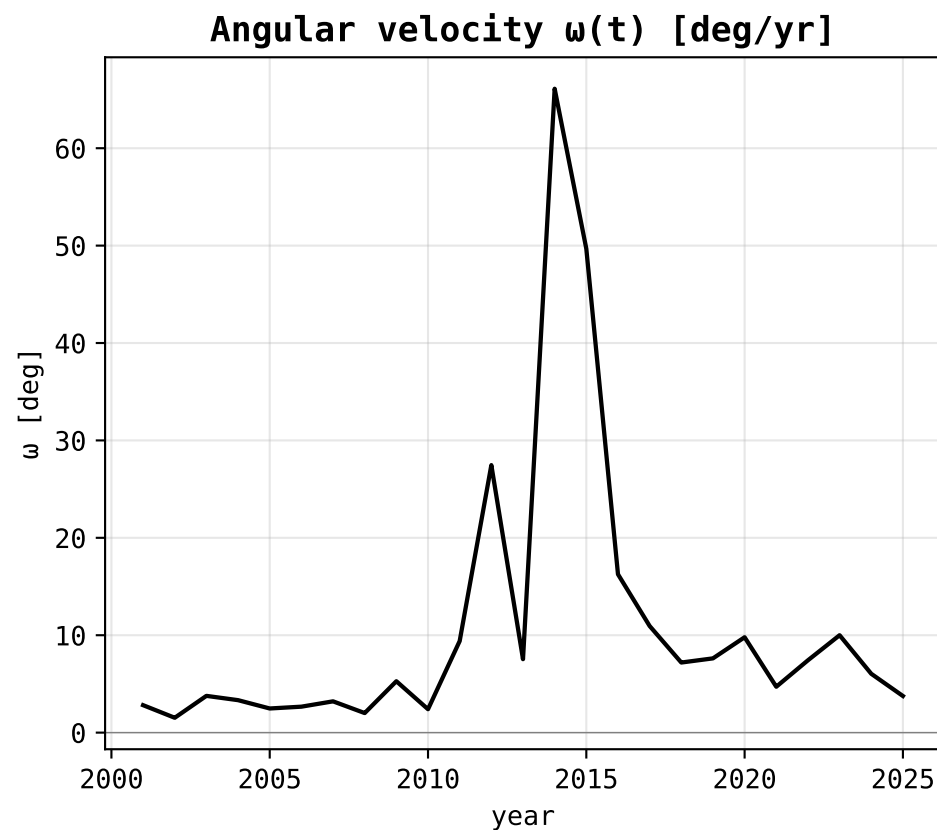
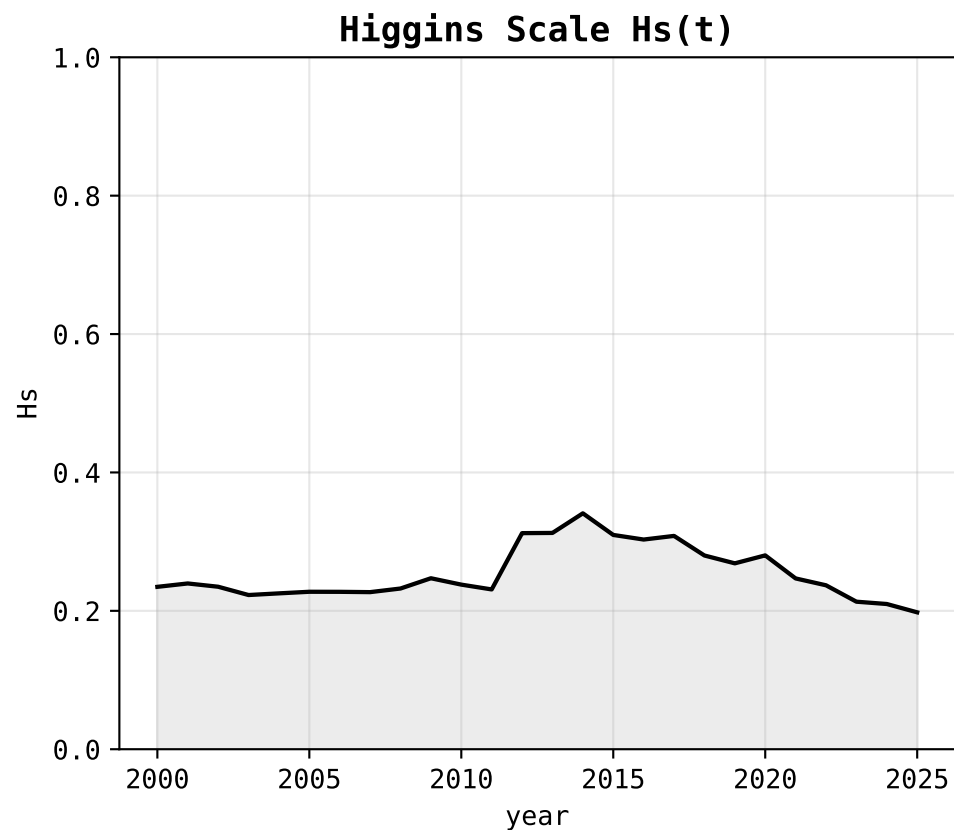
STAGE 3 — Carrier CLR Trajectory Correlation



NAVIGATION SUMMARY — All Stage 1 Metrics by Year

Year	Hs	Ring	E_metric	omega(deg)	d_A	Helmsman	kappa_max
2000	0.2347	Hs-2	70.6	0.00	0.0000	---	9997
2001	0.2396	Hs-2	58.5	2.82	0.8478	Wind	4333
2002	0.2348	Hs-2	51.8	1.52	0.4967	Wind	2690
2003	0.2228	Hs-2	43.4	3.77	0.7576	Wind	1316
2004	0.2253	Hs-2	37.7	3.33	0.5825	Wind	883
2005	0.2276	Hs-2	34.1	2.48	0.3920	Hydro	707
2006	0.2275	Hs-2	32.2	2.66	0.3162	Other Fossil	582
2007	0.2270	Hs-2	30.0	3.21	0.3717	Hydro	511
2008	0.2323	Hs-2	28.9	2.01	0.2132	Nuclear	457
2009	0.2471	Hs-2	26.2	5.28	0.5460	Other Fossil	365
2010	0.2379	Hs-2	24.4	2.40	0.2823	Solar	294
2011	0.2309	Hs-2	21.5	9.40	0.8378	Nuclear	248
2012	0.3123	Hs-3	21.4	27.46	2.2006	Nuclear	234
2013	0.3126	Hs-3	19.3	7.54	0.6399	Solar	212
2014	0.3409	Hs-3	896.7	66.11	34.8706	Nuclear	1000000000000000
2015	0.3097	Hs-3	20.4	49.67	33.6676	Nuclear	228
2016	0.3030	Hs-3	15.7	16.26	1.3200	Nuclear	199
2017	0.3083	Hs-3	14.0	10.95	0.7679	Other Fossil	173
2018	0.2800	Hs-2	12.4	7.20	0.5079	Nuclear	147
2019	0.2686	Hs-2	12.0	7.62	0.4680	Other Fossil	140
2020	0.2801	Hs-2	12.0	9.79	0.5917	Nuclear	120
2021	0.2468	Hs-2	10.7	4.71	0.3396	Nuclear	108
2022	0.2369	Hs-2	10.2	7.43	0.4247	Other Fossil	110
2023	0.2132	Hs-2	9.7	10.00	0.5551	Nuclear	101
2024	0.2099	Hs-2	9.6	6.04	0.3276	Other Fossil	88
2025	0.1978	Hs-2	9.0	3.76	0.2217	Bioenergy	80

CNQ v2.0.0 – Quaternion View – JPN Japan · D=8 · N=26



CNQ DIAGNOSTICS · JPN (D = 8)

CHSH joint-coherence:

S value:0.4000

classical bound:2.0

Tsirelson bound:2.8284271247461903

verdict:independent

Twin-quaternion factor:

enabled:True

partition axis:6

max |residual|:5.8250e+00

Attractor fit:

amplitude A:7.012

contraction λ :-0.0189

period:1

fitted:False

Helmsman family:

total flips:17

stability S_σ :0.292

§ USA United States

EMBER electricity-generation series · 2000-2025 · N = 25 · D = 9

INPUT

source: ember_USA*_generation_TWh.csv
N records: 25
D carriers: 9
Bioenergy
Coal
Gas
Hydro
Nuclear
Other Fossil
Other Renewables
Solar
Wind
source_sha256: ecdb50b25cb5807b38d4767a...

HEADLINE METRICS

Helmsman flips: ?
Helmsman stability S_σ : {'global': 0.6521739130434783, 'rolling': [0.2857142857142857, 0.2857142857142857]}
IR class: ?
Attractor amplitude: ?

CHSH S: ?
Tsirelson bound: $2\sqrt{2} \approx 2.828$

Bearing pairs $(D(D-1)/2)$: 36

CNT v3.1.0 content_sha: ?...
CNQ v2.0.0 content_sha: ?...

Following pages: Stage 1 (XY/XZ/YZ × N years + course) → Stage 2/3 (7 pages) → CNQ (2 pages).

SECTION PLATE t=0 | Year 2001 | D=9 N=25 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

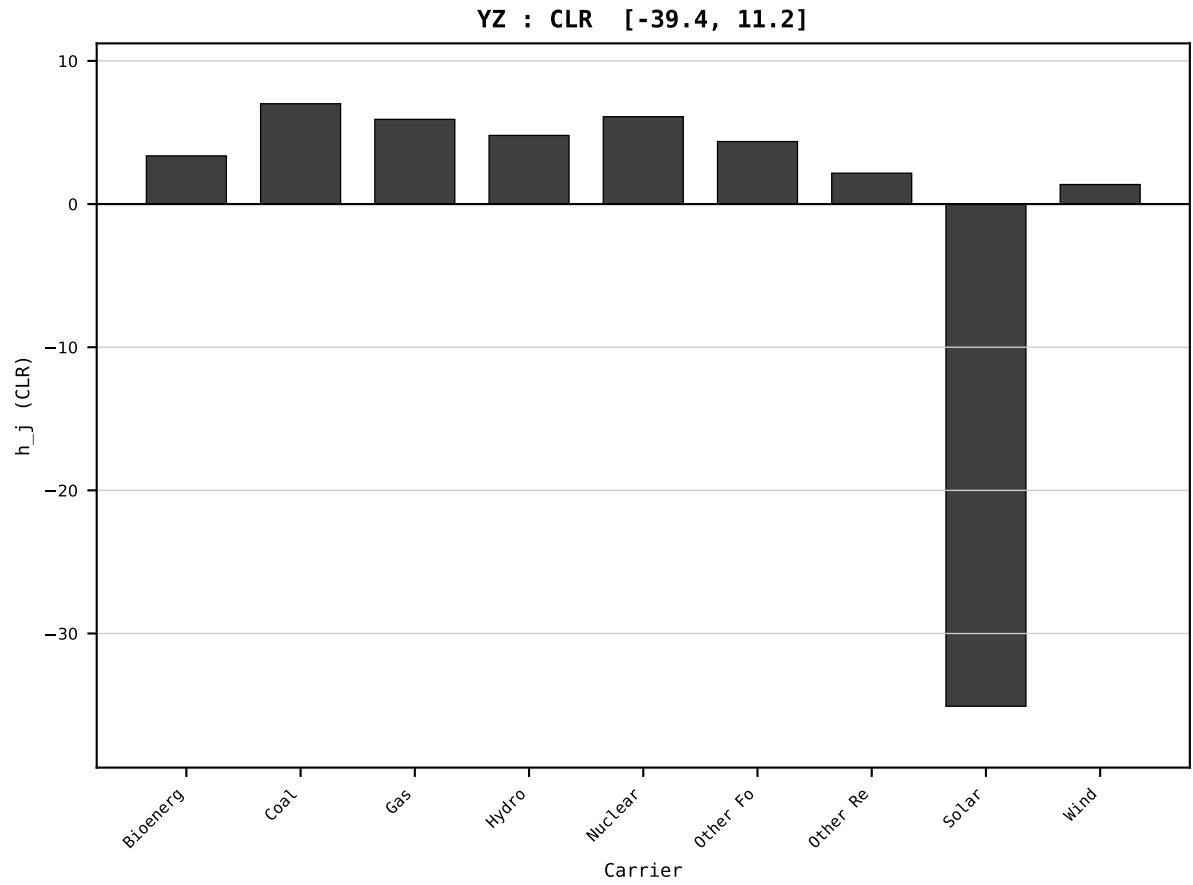
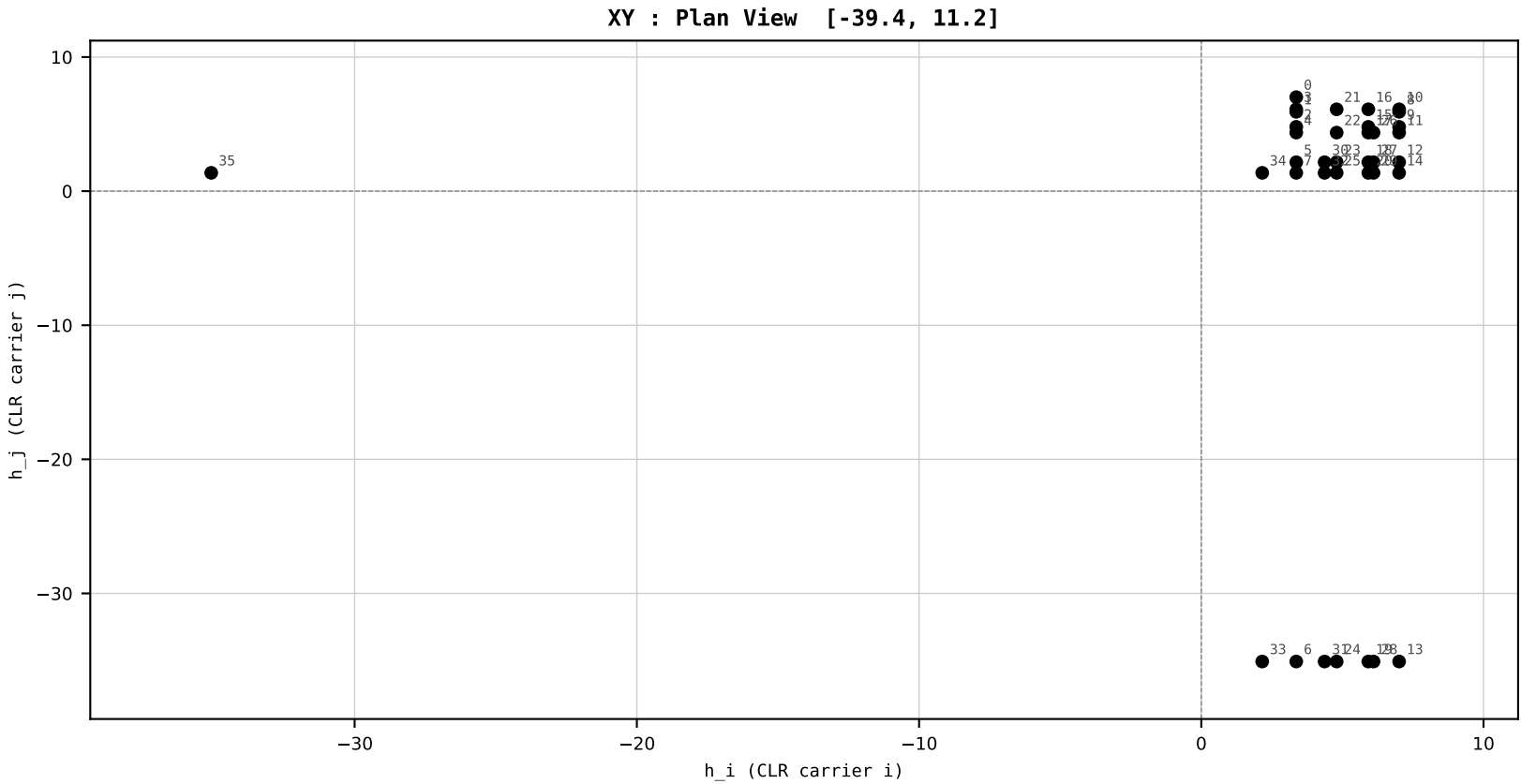
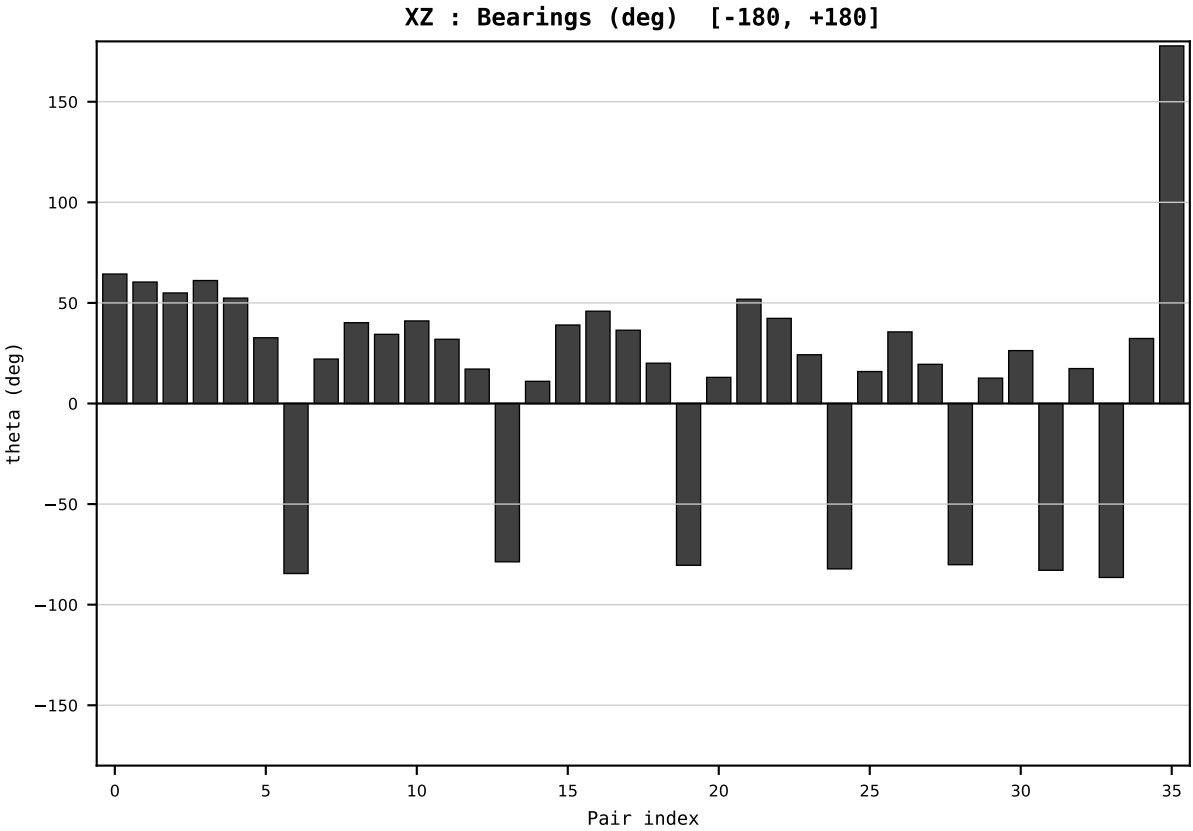
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2001
t : 0 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.388371
Ring = Hs-3
E_metric= 895.1337
kappa_HS= 190397000000000000.00
omega = 0.0000 deg
d_A = 0.000000
Helm = ---
Helm d = ---
DR = 42.090 HLR
DR ratio= 1903969992697630976.0x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=1 | Year 2002 | D=9 N=25 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

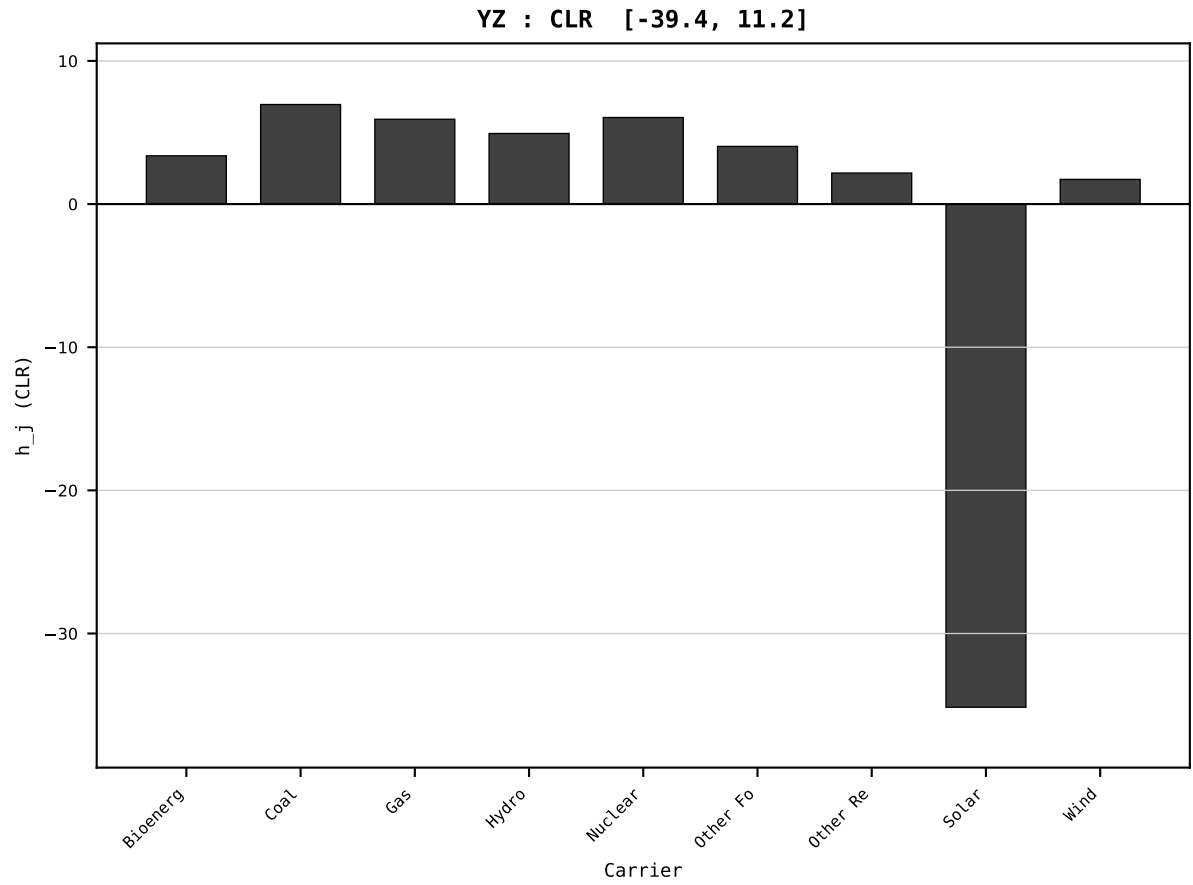
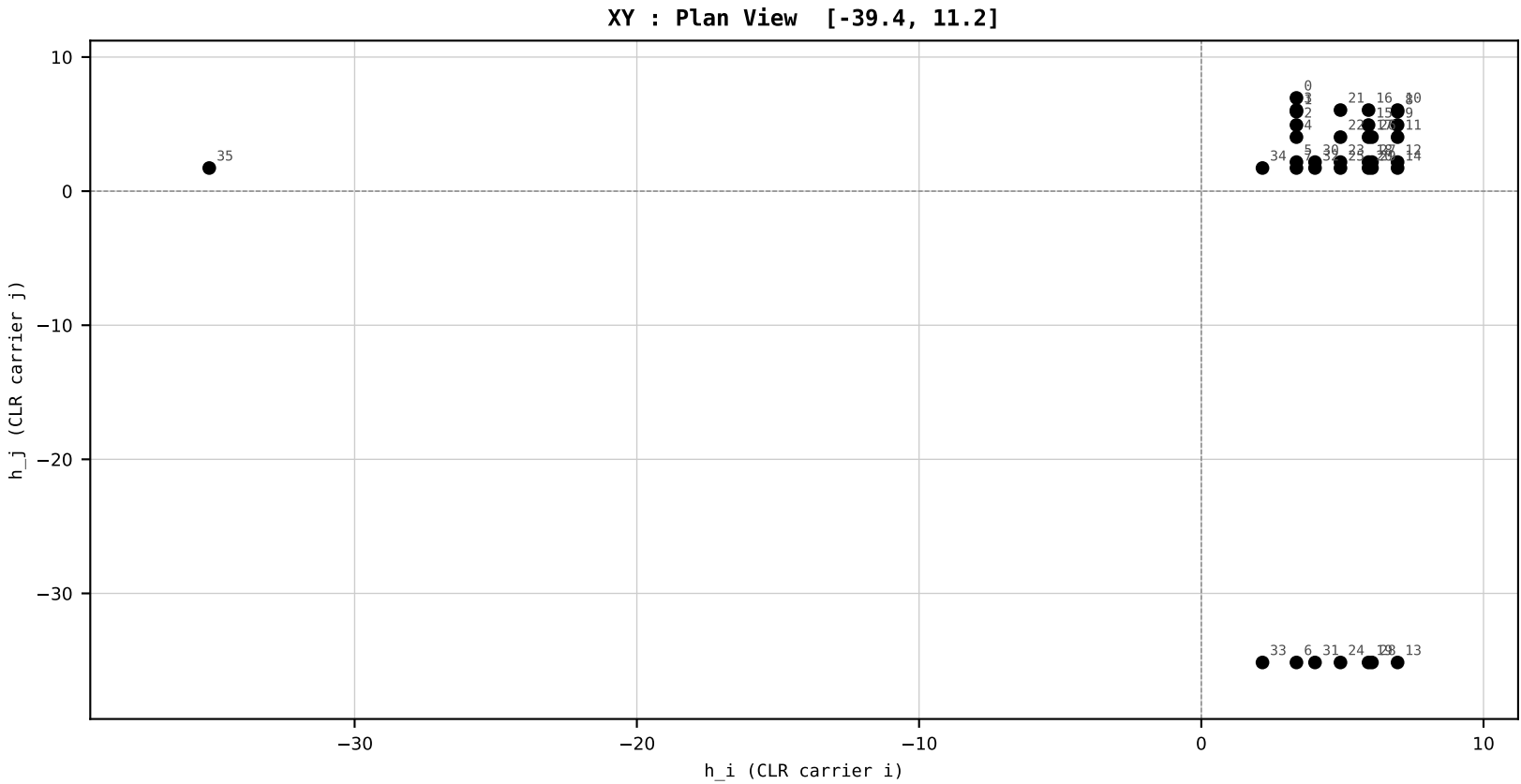
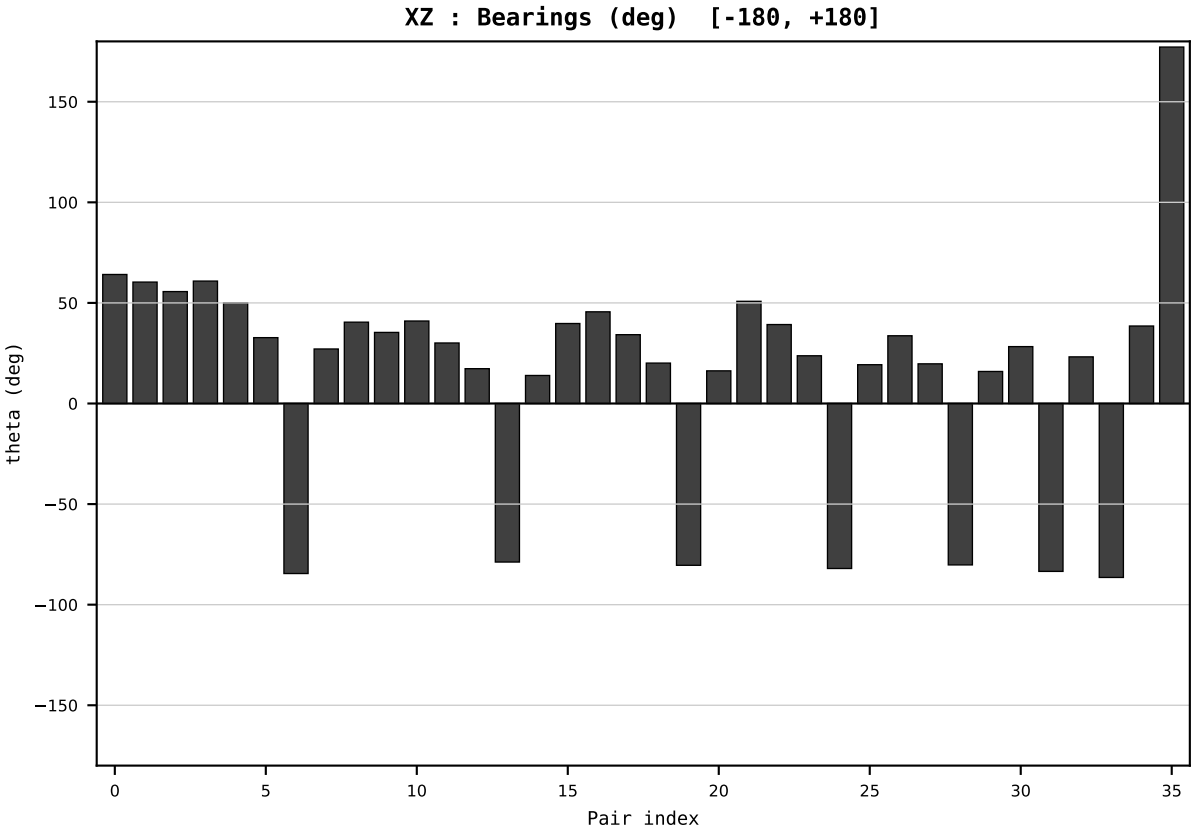
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2002
t : 1 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.383719
Ring = Hs-3
E_metric= 895.5128
kappa_HS= 193315000000000000.00
omega = 0.7941 deg
d_A = 0.522906
Helm = Wind
Helm d = +0.359790
DR = 42.106 HLR
DR ratio= 1933149998657515008.0x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=2 | Year 2003 | D=9 N=25 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

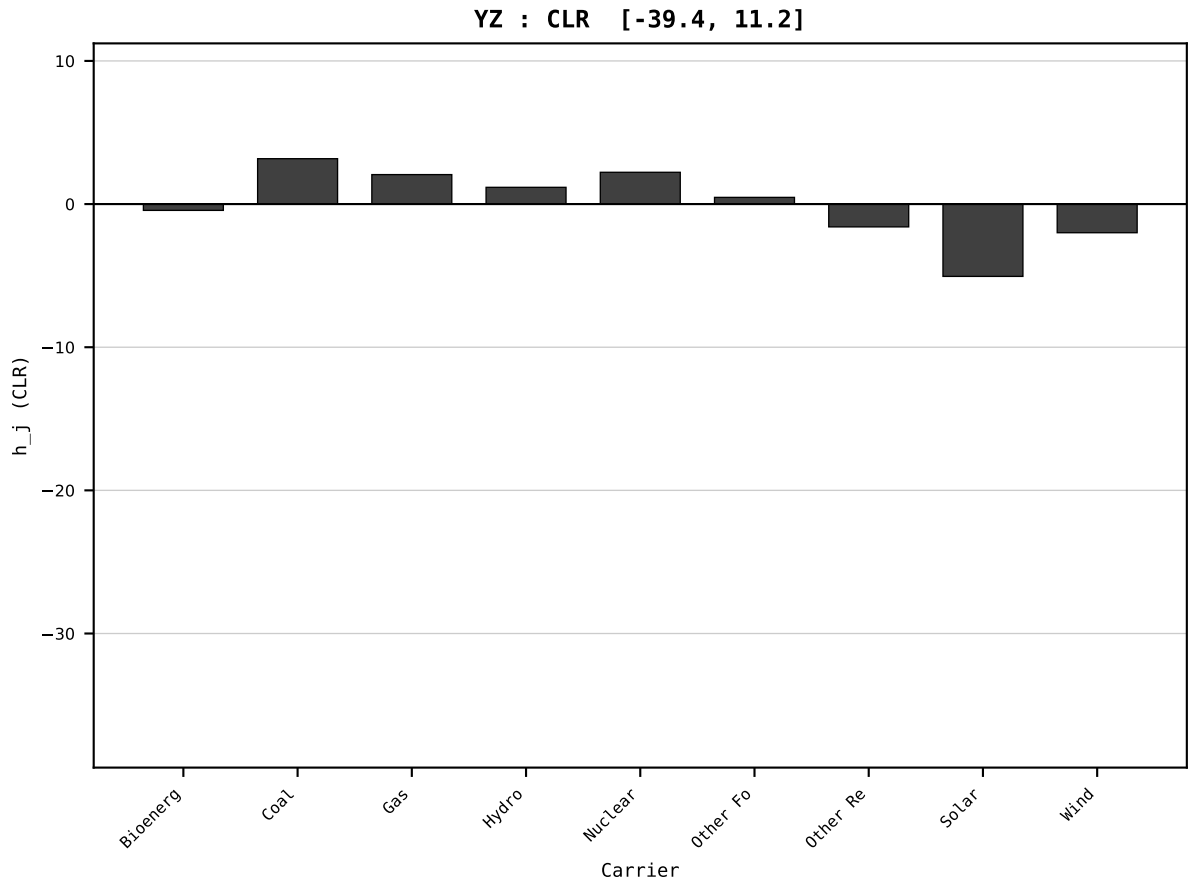
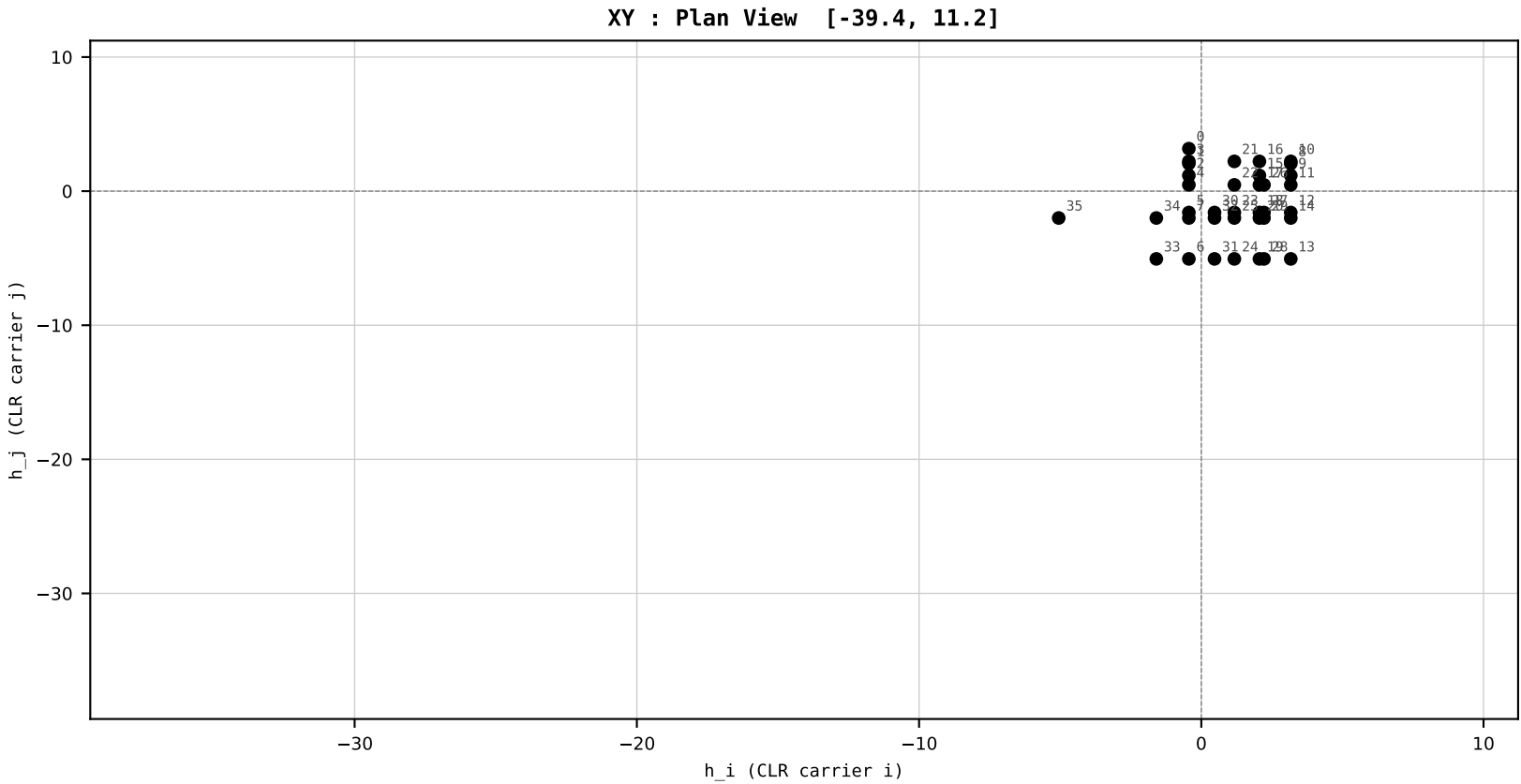
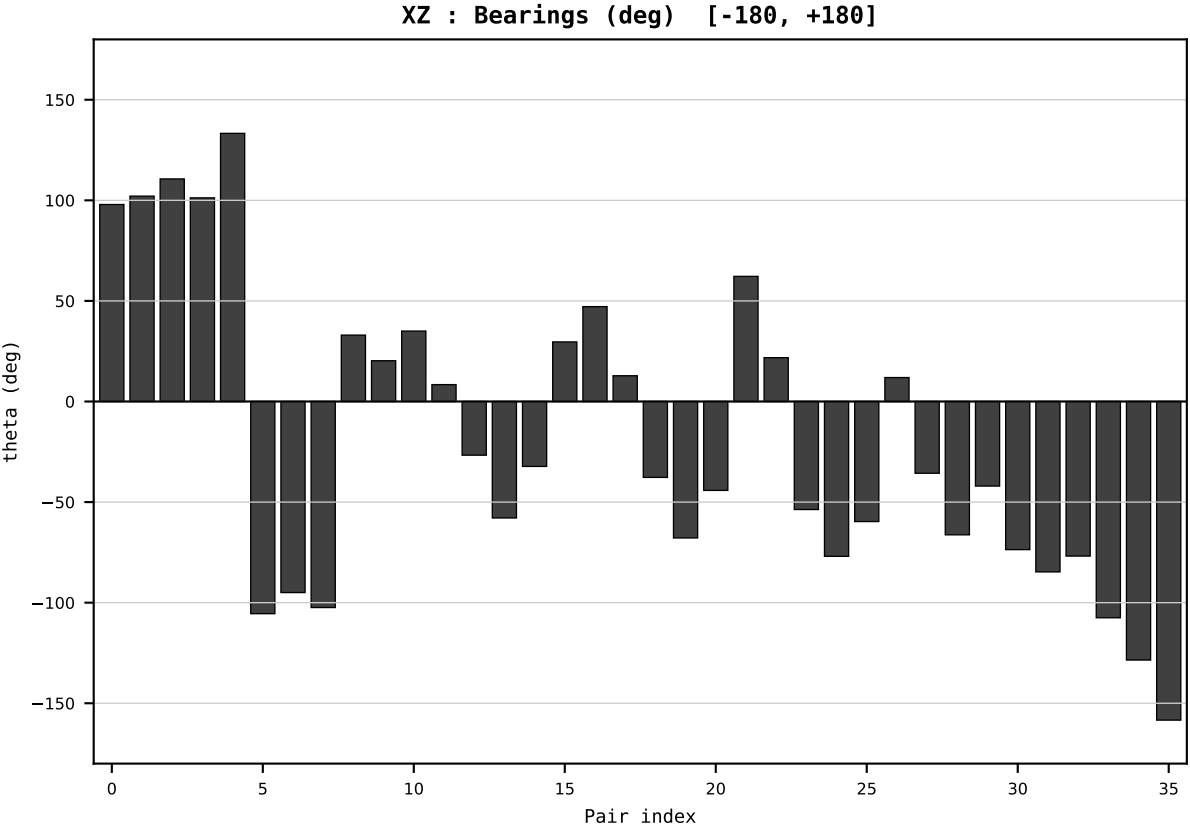
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2003
t : 2 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.379089
Ring = Hs-3
E_metric= 53.0775
kappa_HS= 3724.00
omega = 34.9884 deg
d_A = 31.925473
Helm = Solar
Helm d = +30.098741
DR = 8.223 HLR
DR ratio= 3724.0x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=3 | Year 2004 | D=9 N=25 pairs=36

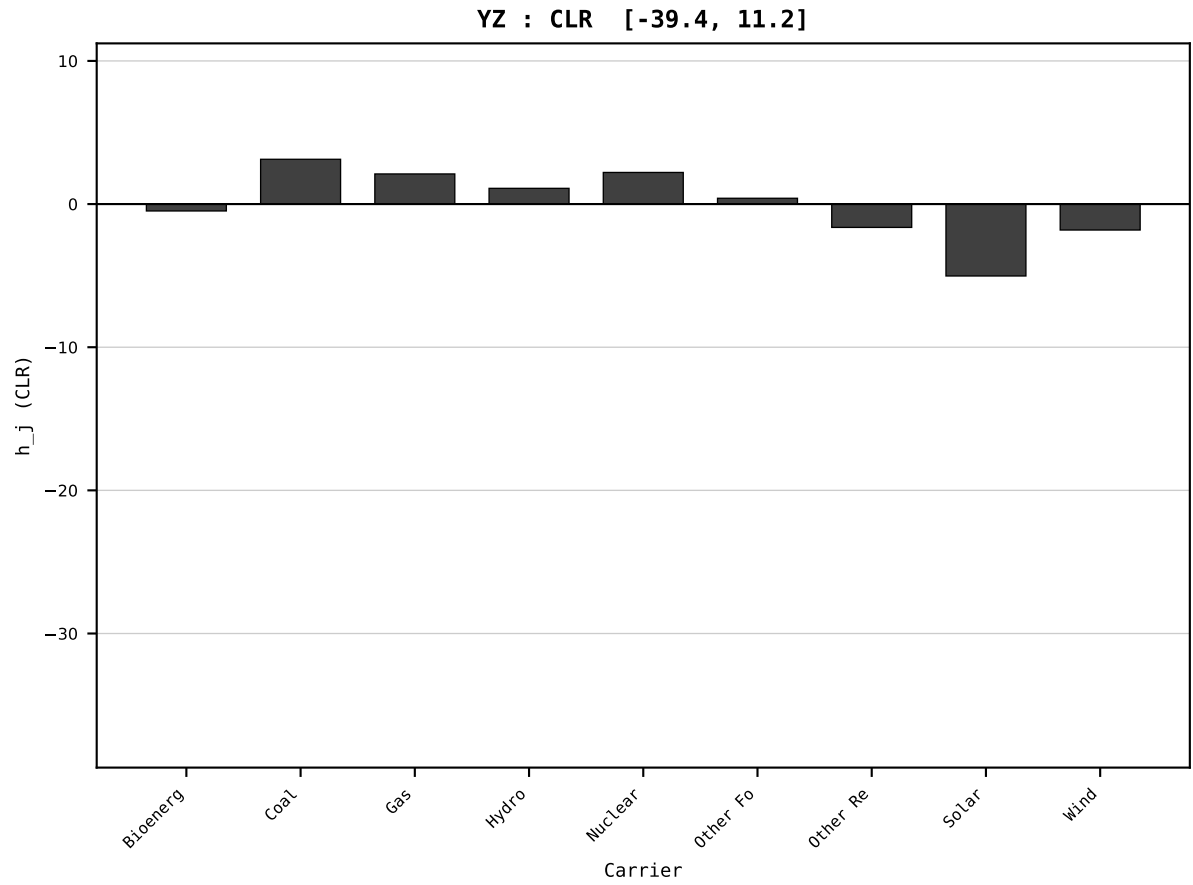
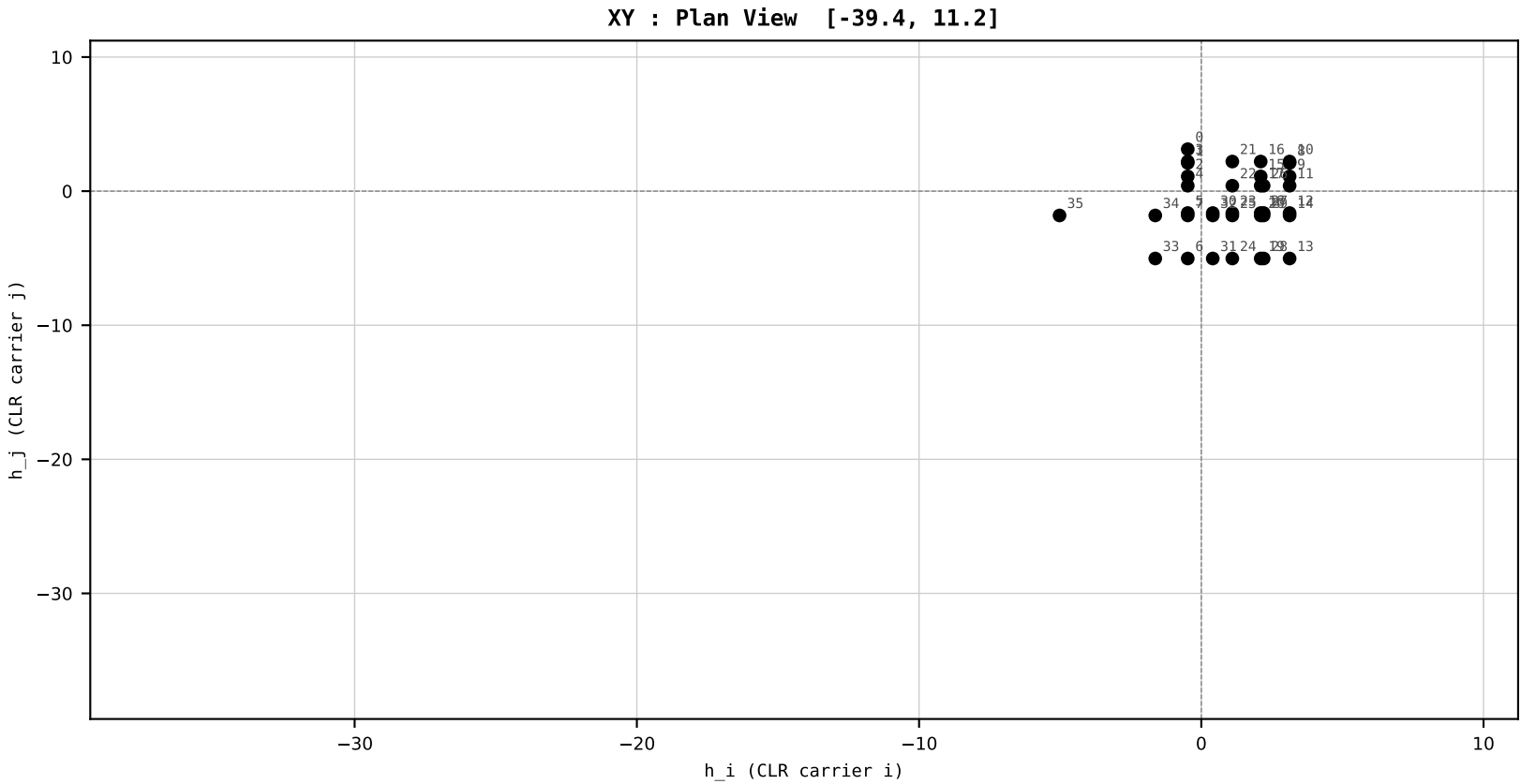
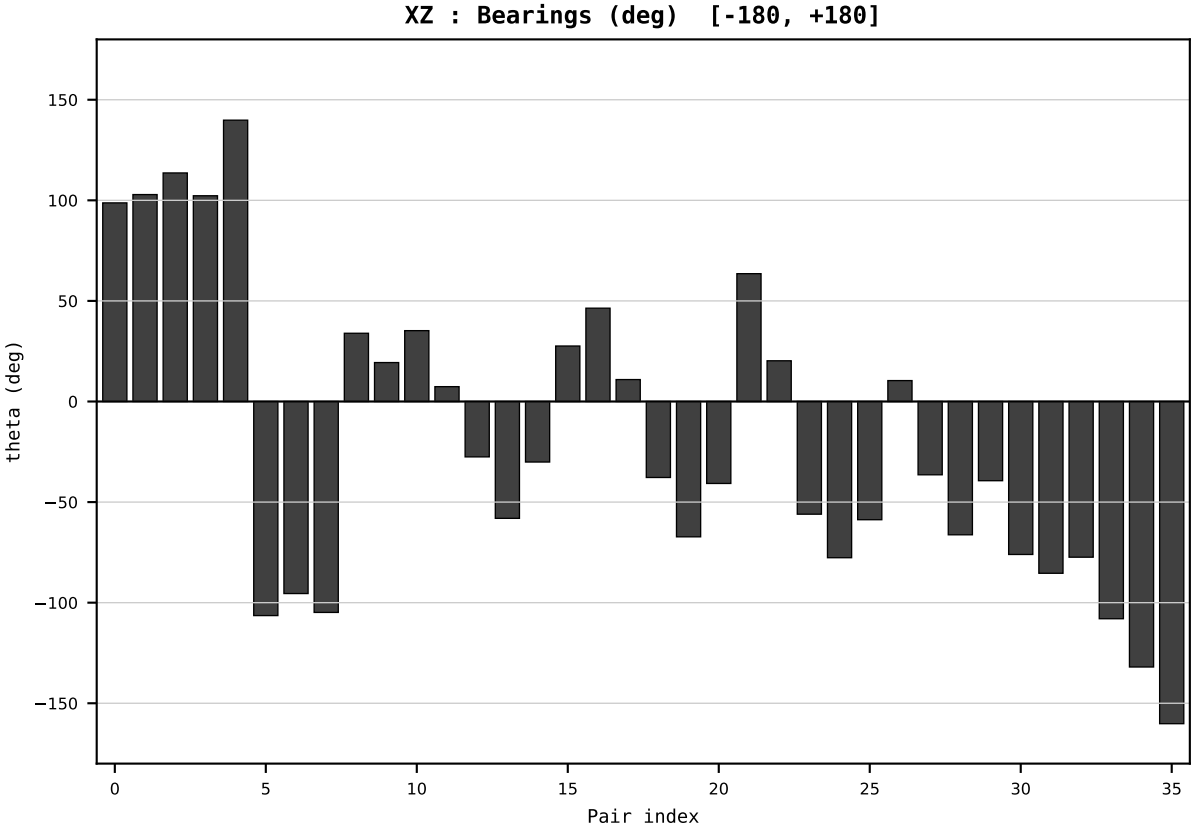
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2004
t : 3 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.376148
Ring = Hs-3
E_metric= 51.8804
kappa_HS= 3470.70
omega = 1.6887 deg
d_A = 0.228934
Helm = Wind
Helm d = +0.188988
DR = 8.152 HLR
DR ratio= 3470.7x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=4 | Year 2005 | D=9 N=25 pairs=36

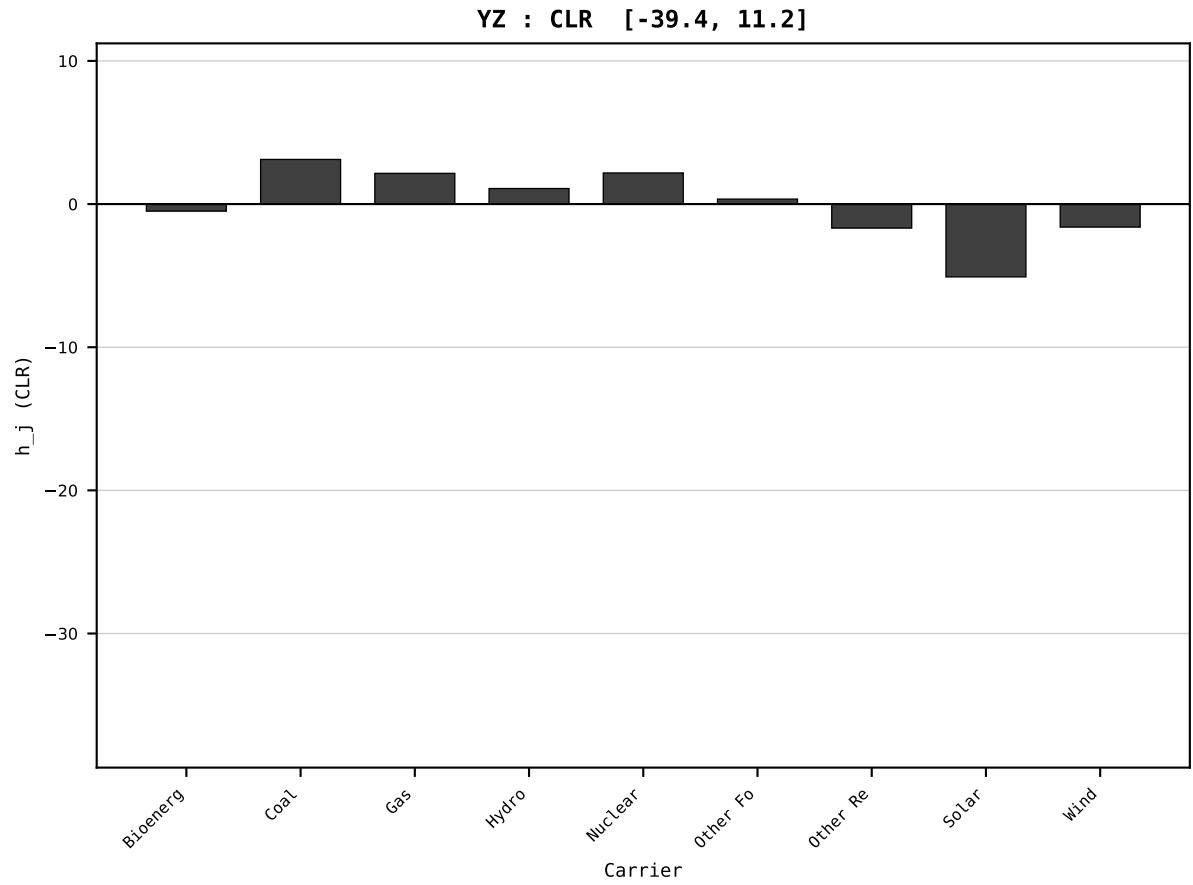
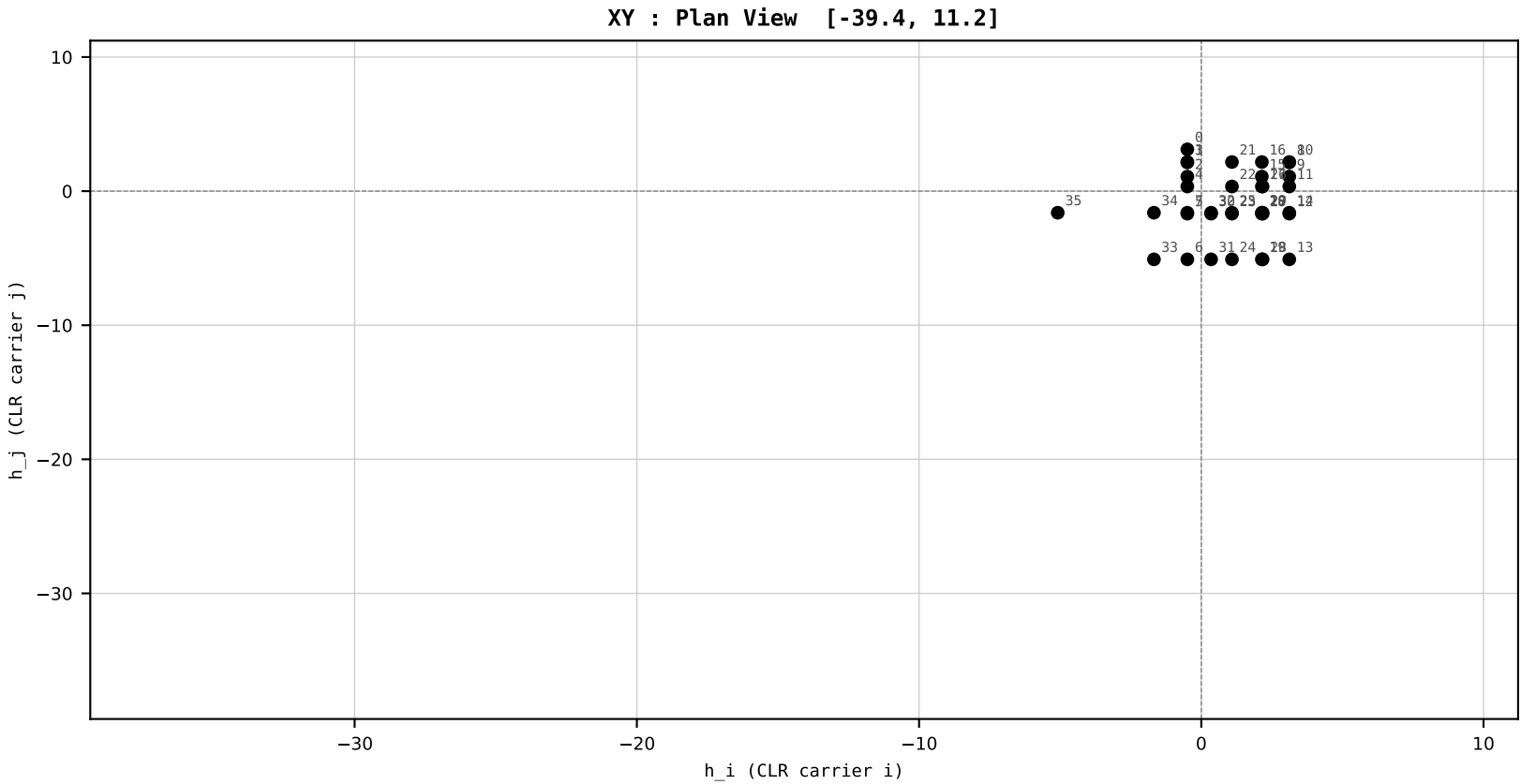
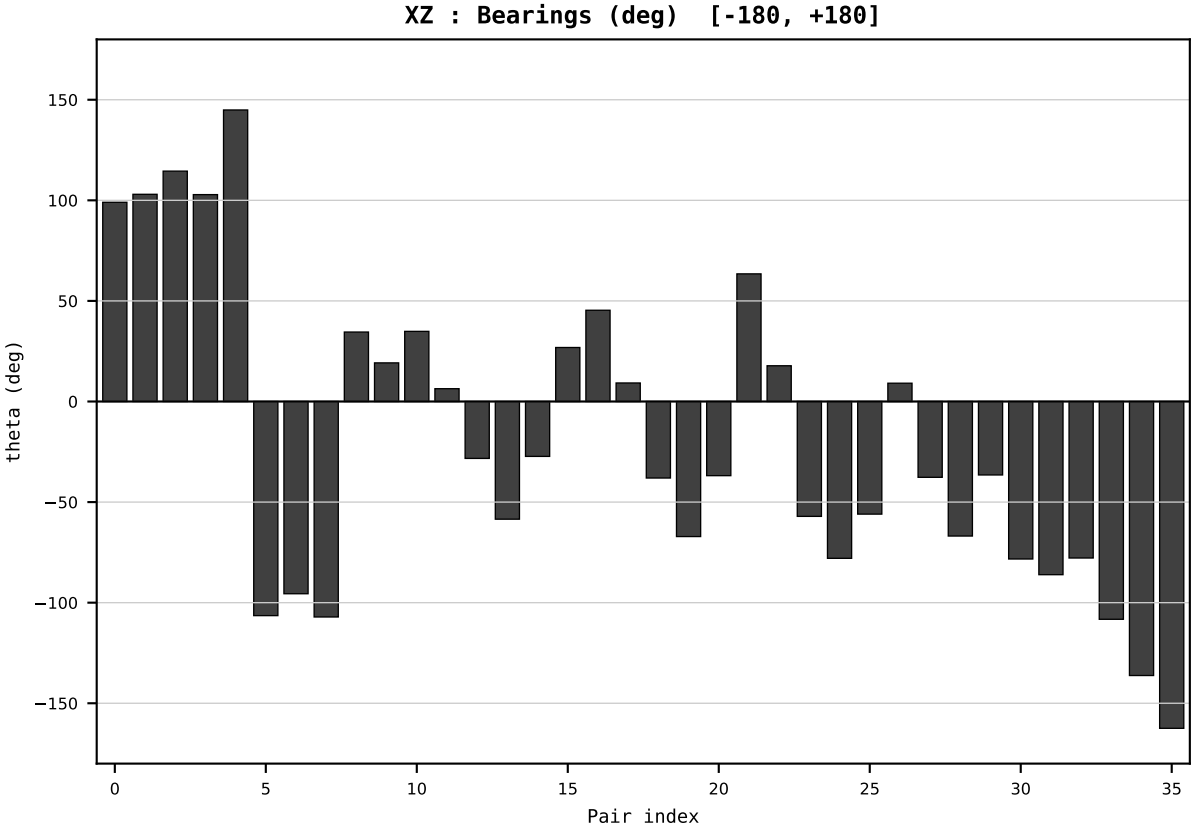
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2005
t : 4 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.375606
Ring = Hs-3
E_metric= 51.8713
kappa_HS= 3659.76
omega = 1.8643 deg
d_A = 0.234341
Helm = Wind
Helm d = +0.204228
DR = 8.205 HLR
DR ratio= 3659.8x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=5 | Year 2006 | D=9 N=25 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

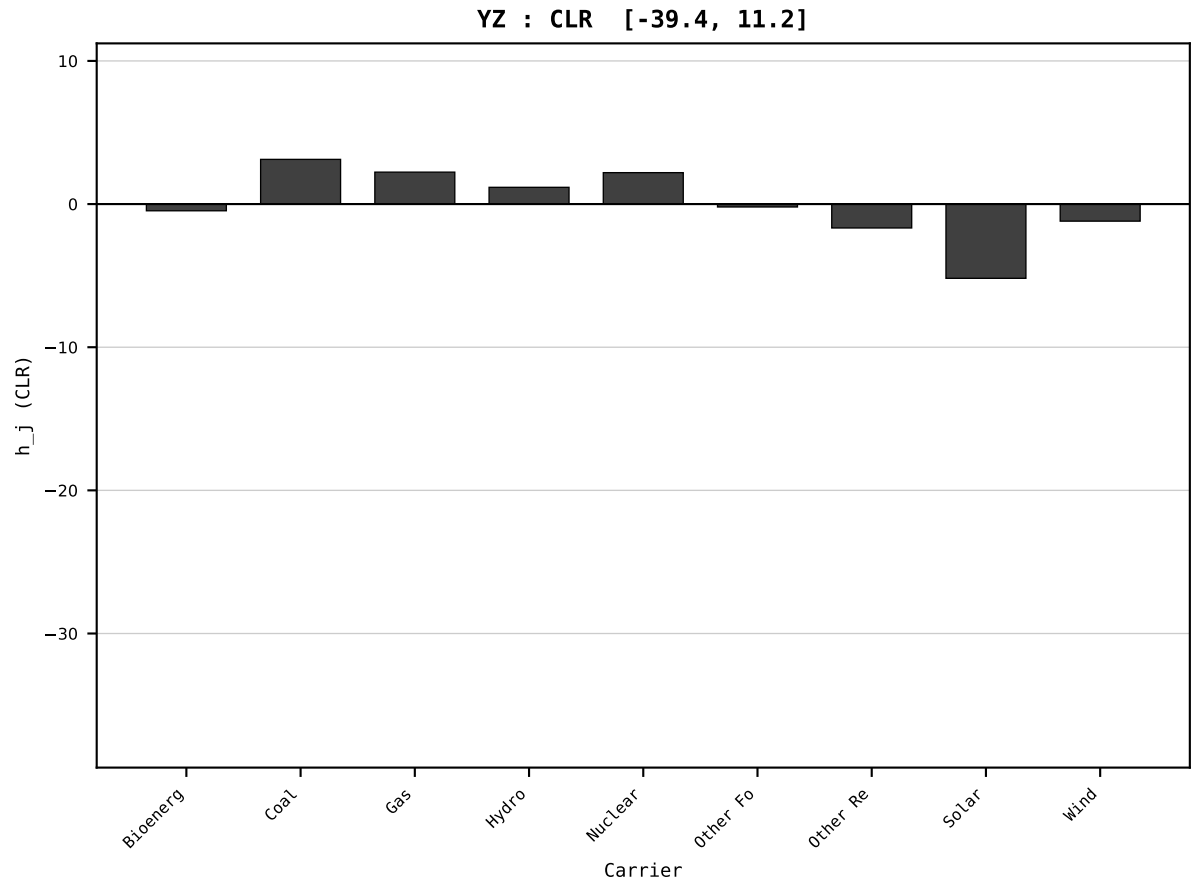
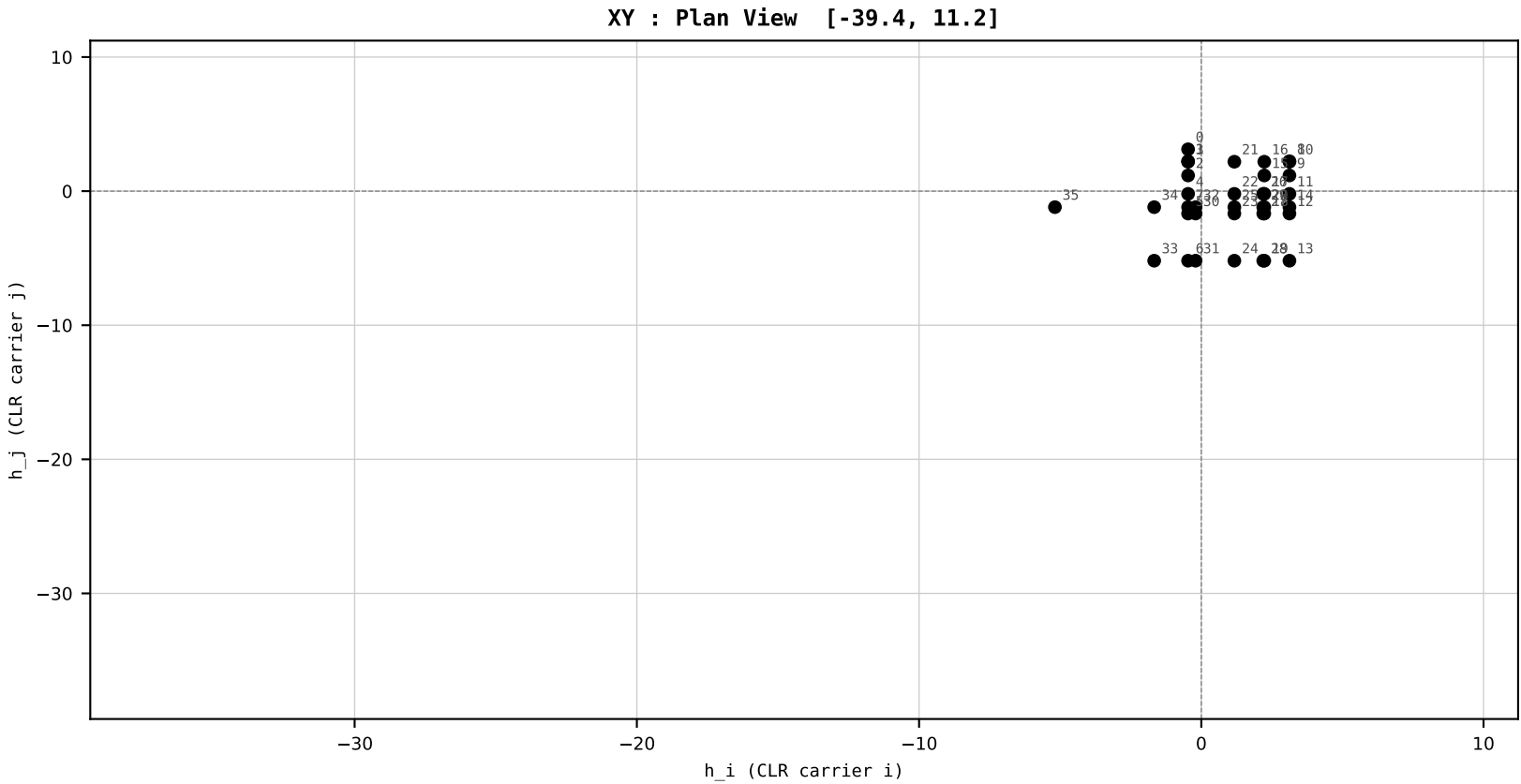
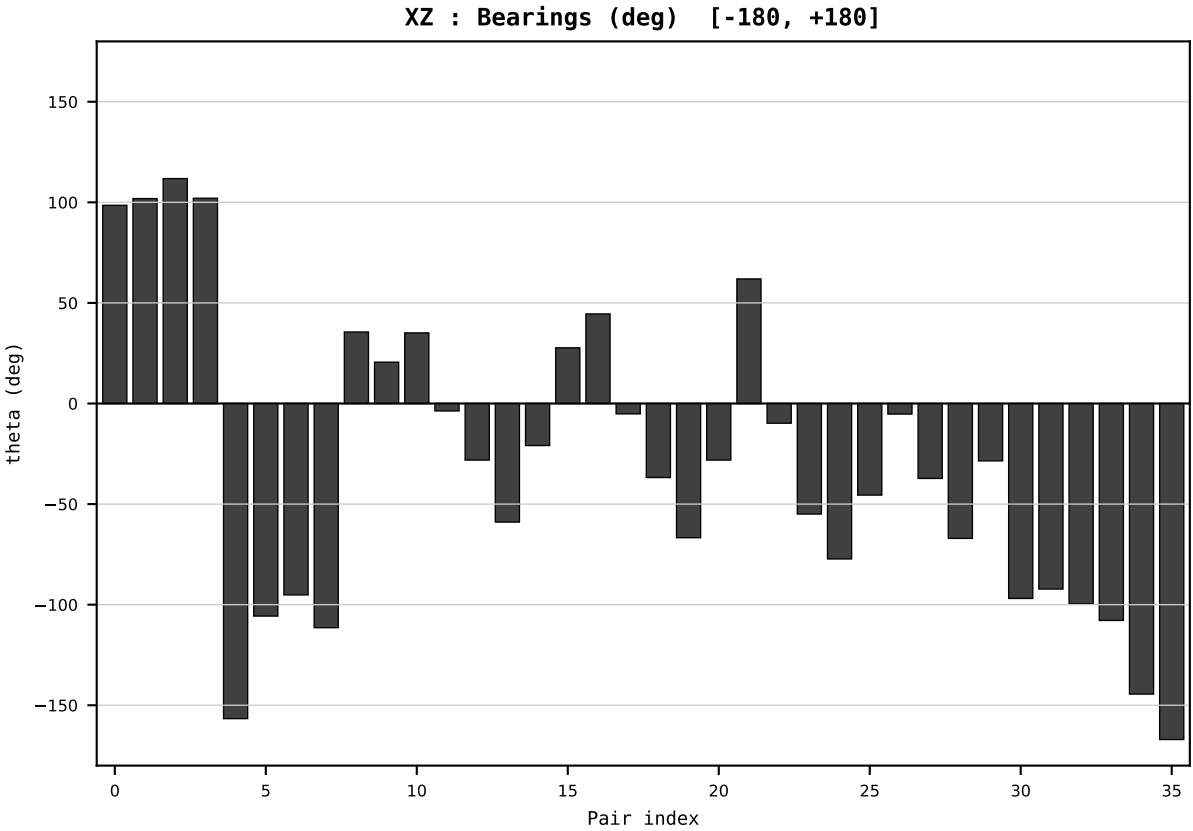
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2006
t : 5 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.379864
Ring = Hs-3
E_metric= 52.2936
kappa_HS= 4062.29
omega = 5.6182 deg
d_A = 0.707976
Helm = Other Fossil
Helm d = -0.550093
DR = 8.310 HLR
DR ratio= 4062.3x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=6 | Year 2007 | D=9 N=25 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

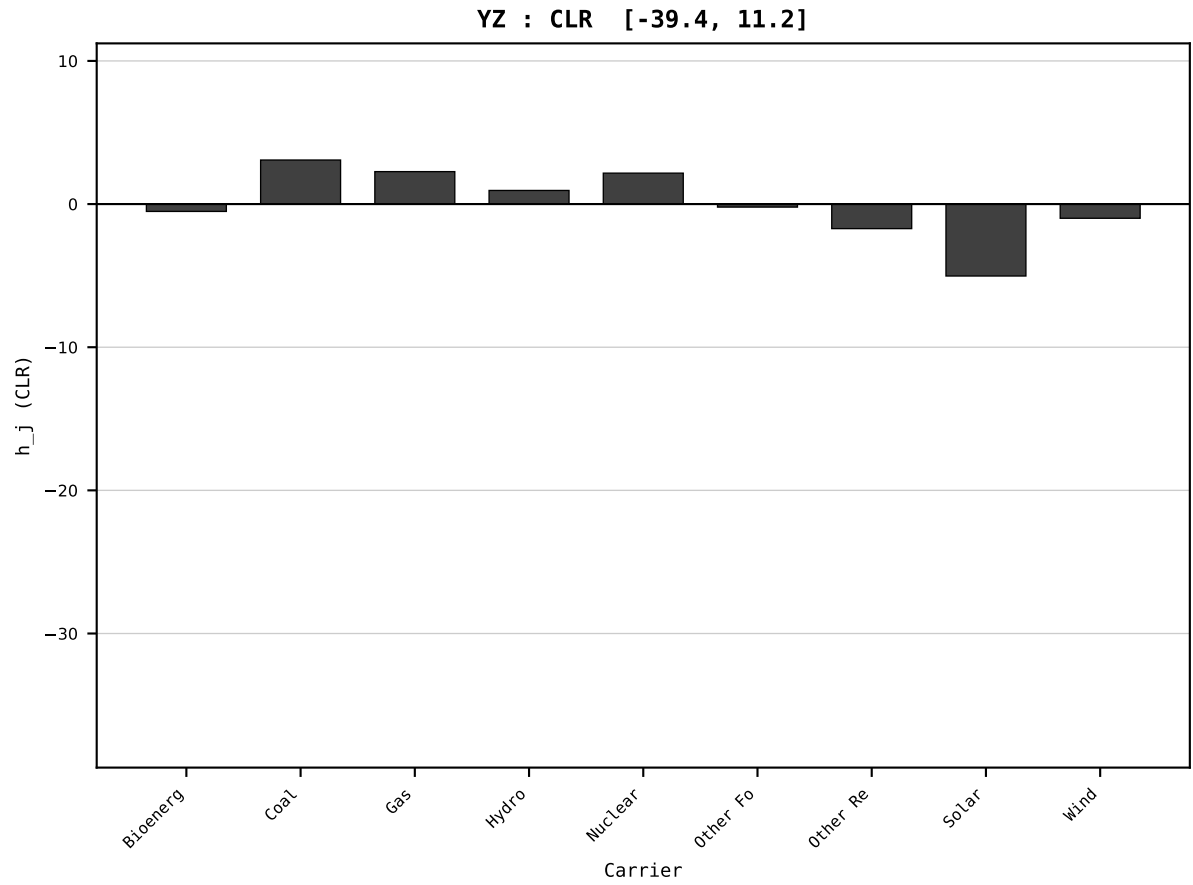
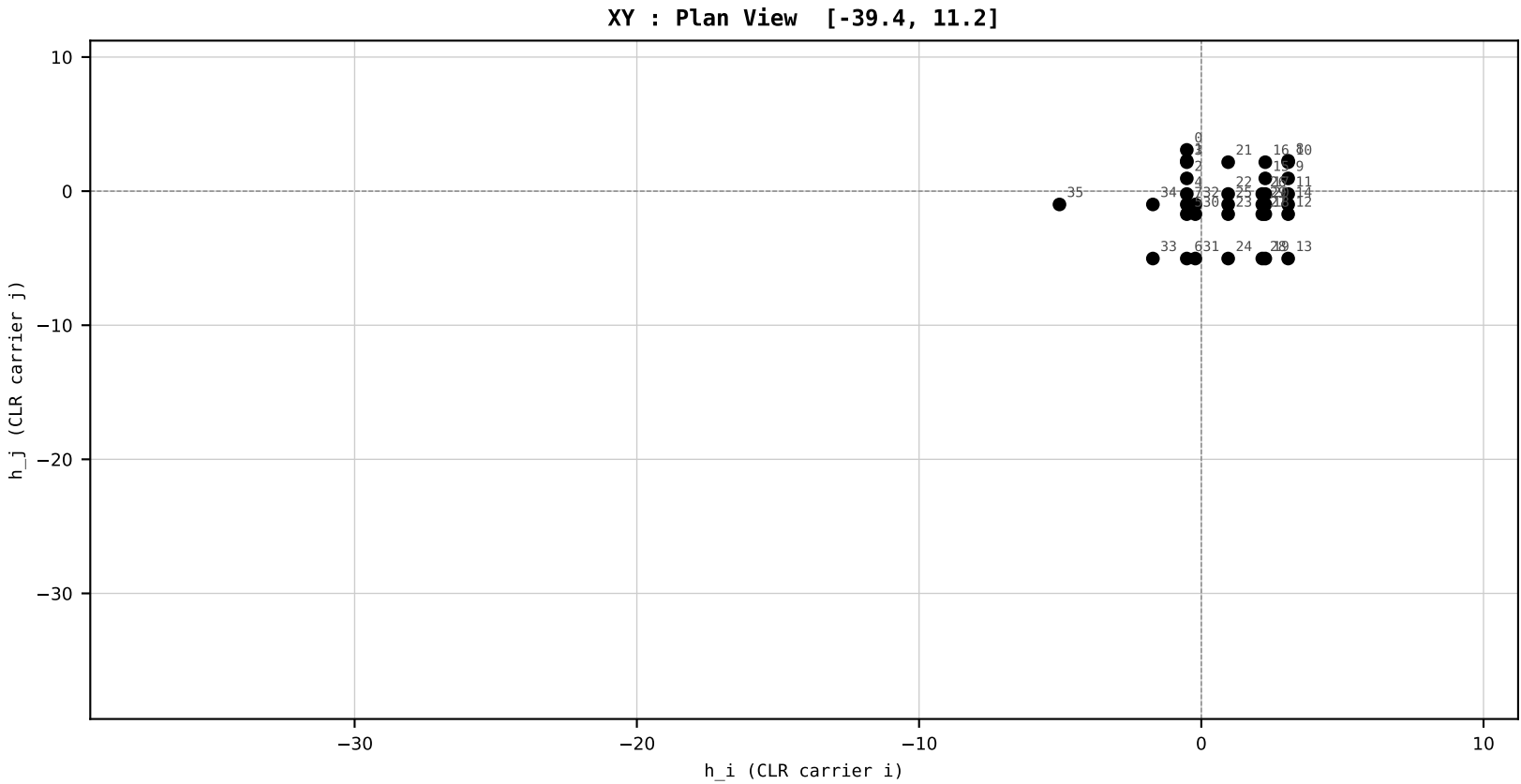
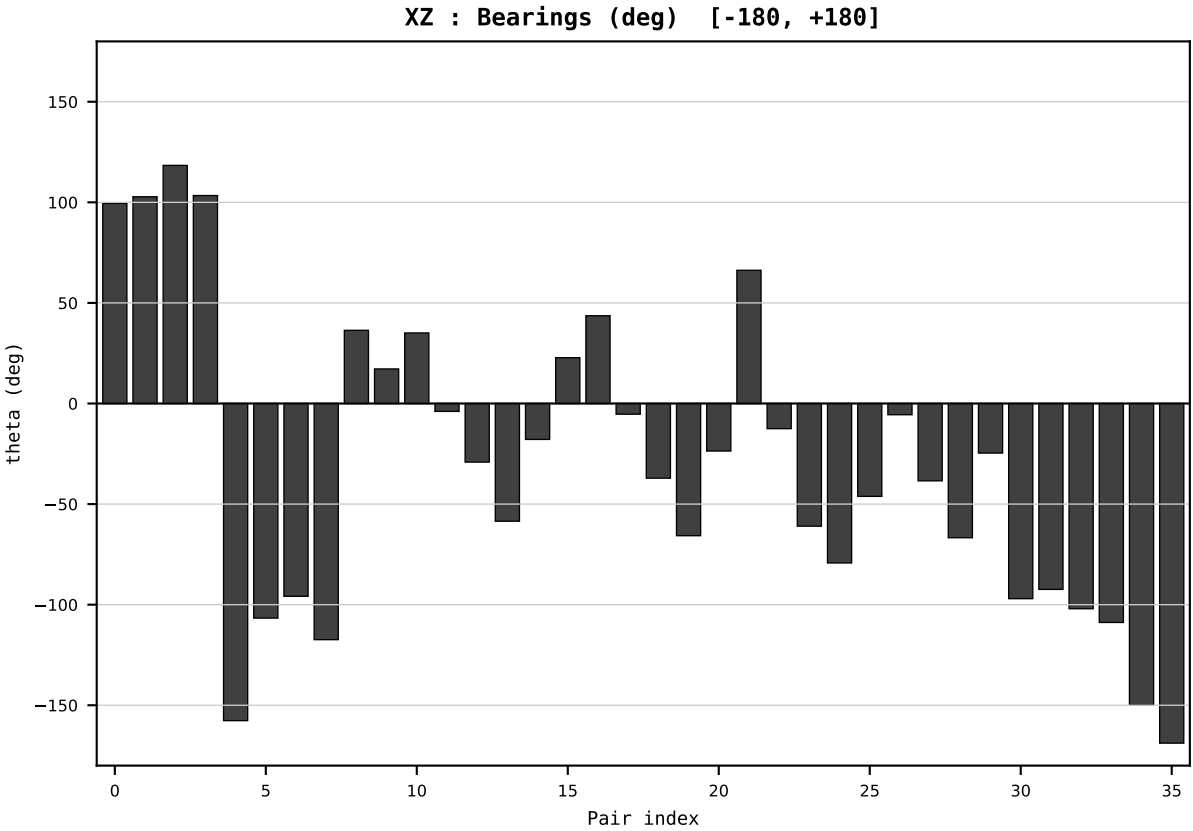
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2007
t : 6 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.381244
Ring = Hs-3
E_metric= 49.6813
kappa_HS= 3305.62
omega = 2.4037 deg
d_A = 0.350949
Helm = Hydro
Helm d = -0.219022
DR = 8.103 HLR
DR ratio= 3305.6x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=7 | Year 2008 | D=9 N=25 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

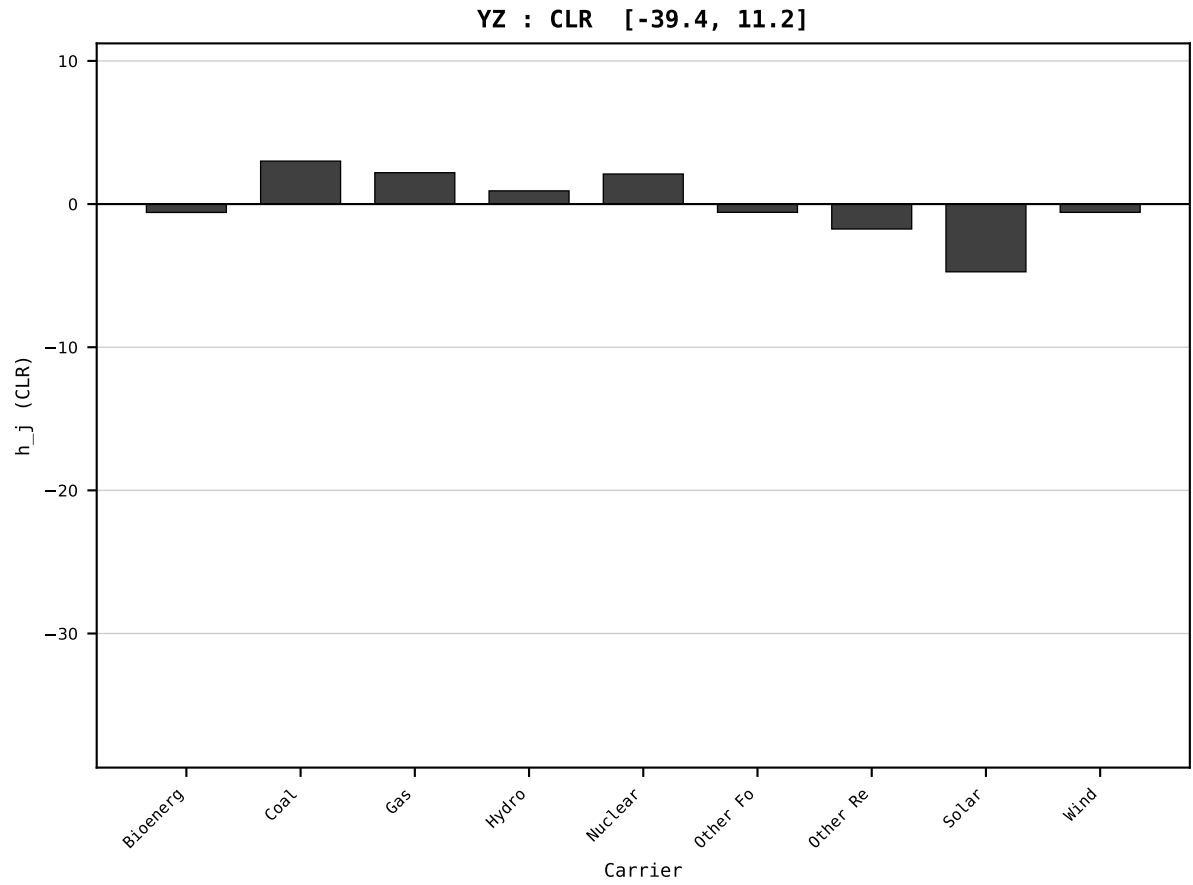
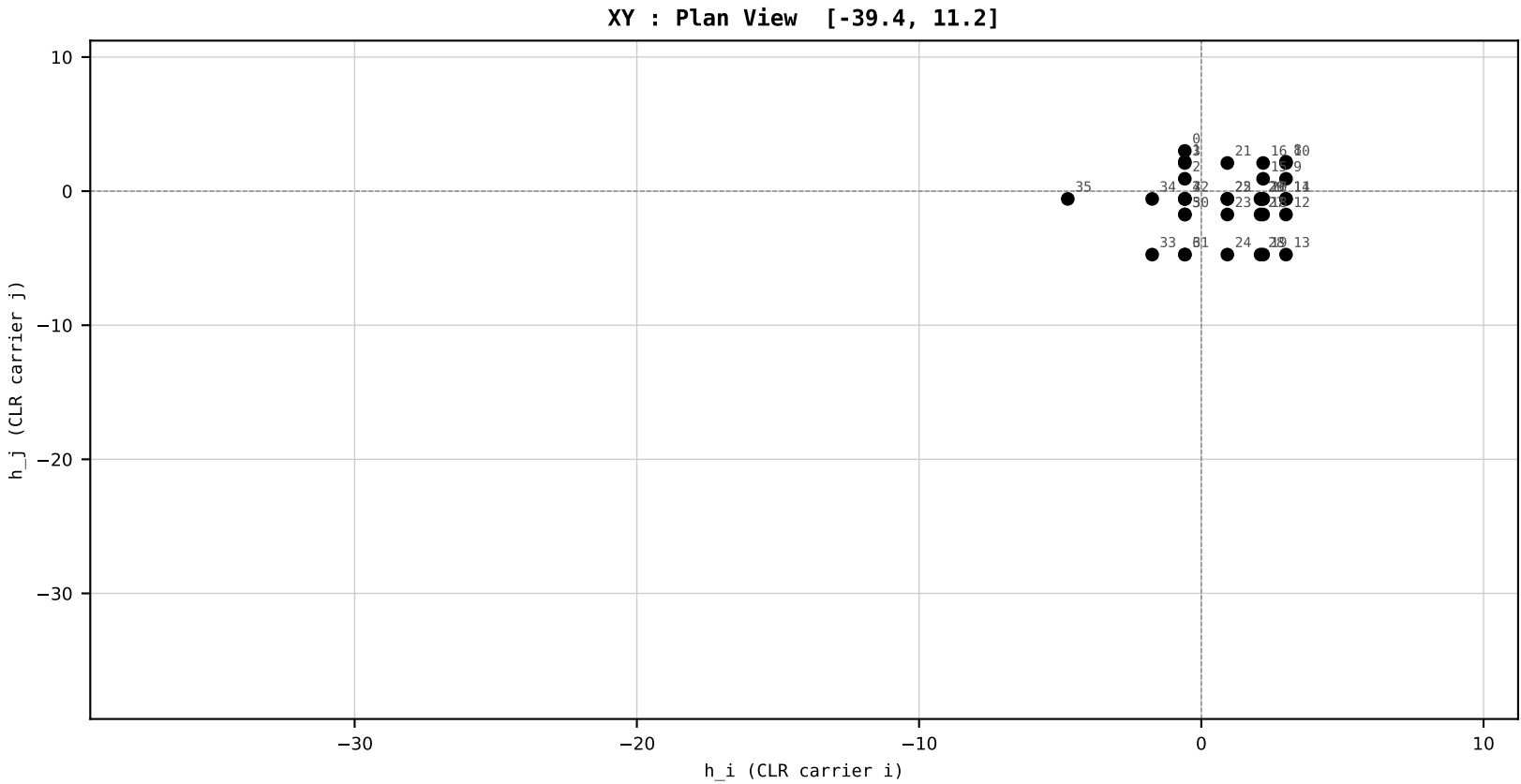
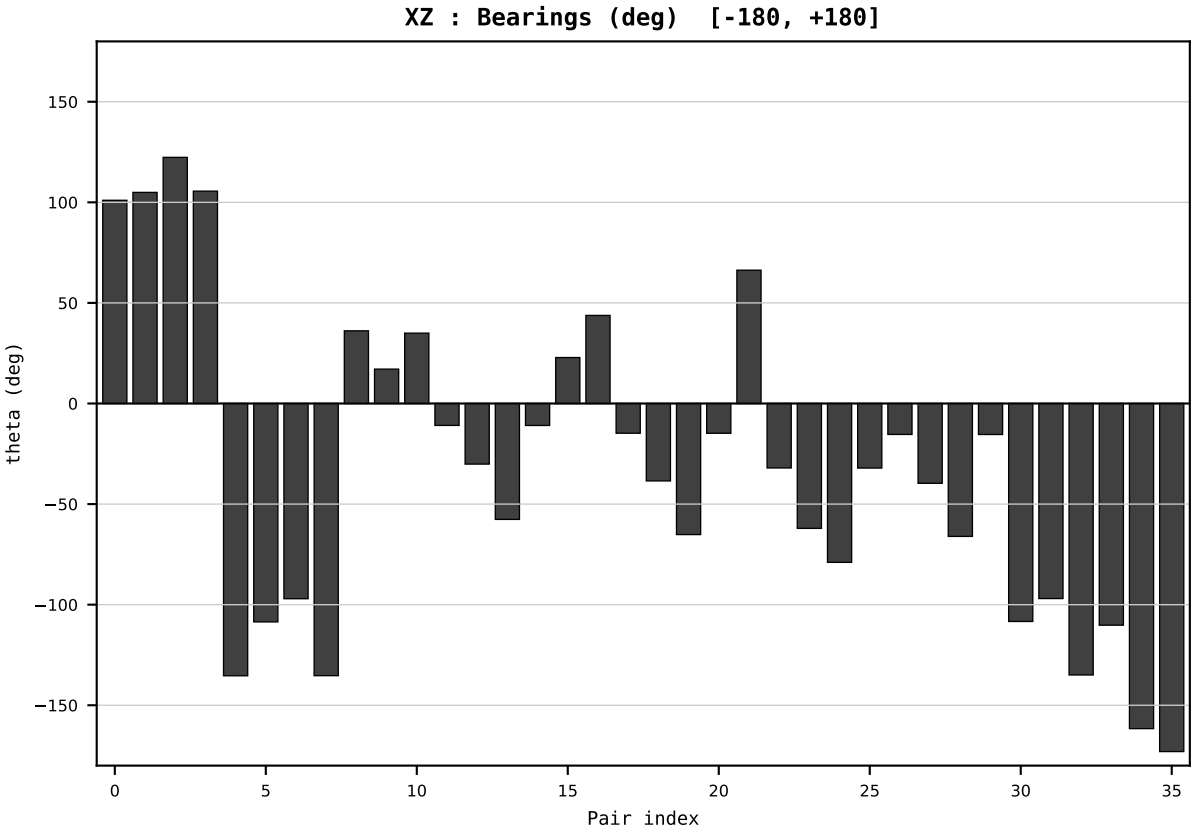
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2008
t : 7 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.376532
Ring = Hs-3
E_metric= 45.5003
kappa_HS= 2282.54
omega = 4.7104 deg
d_A = 0.642683
Helm = Wind
Helm d = +0.412825
DR = 7.733 HLR
DR ratio= 2282.5x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=8 | Year 2009 | D=9 N=25 pairs=36

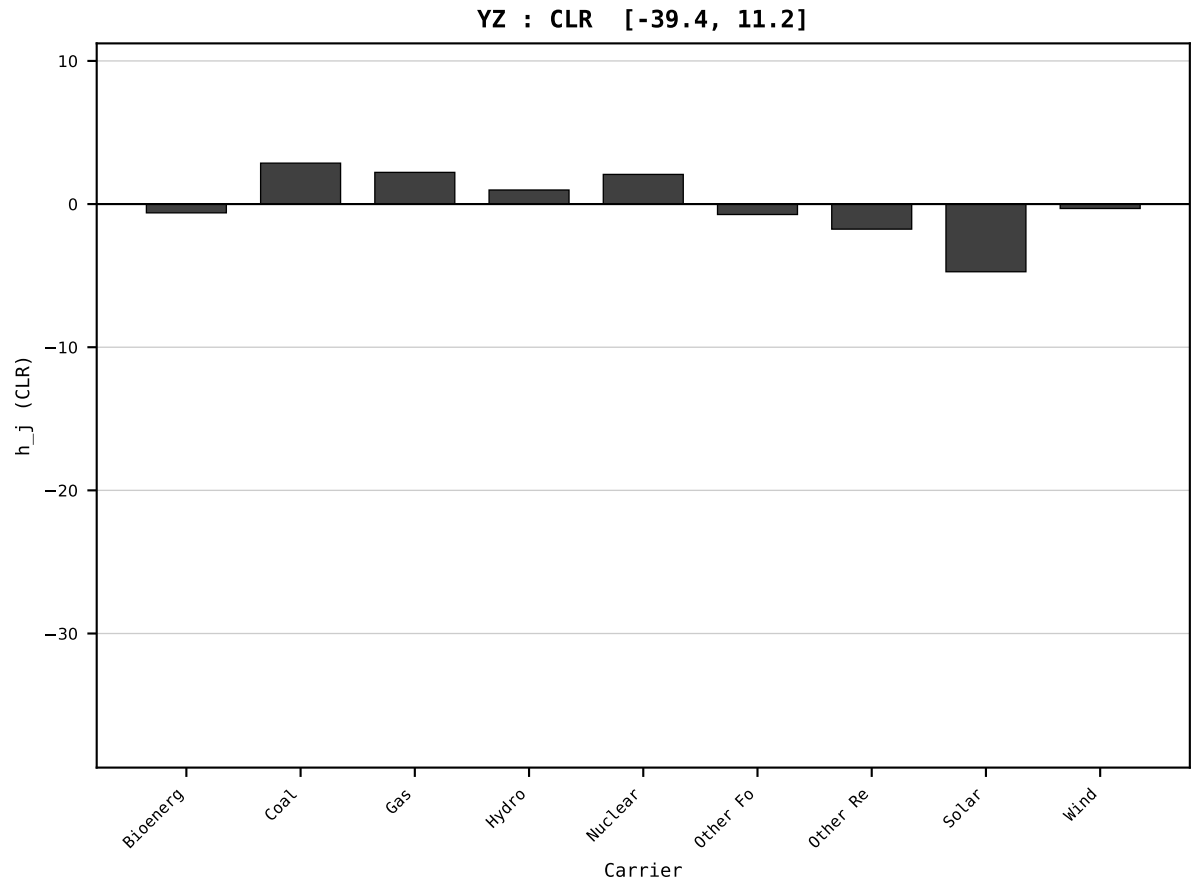
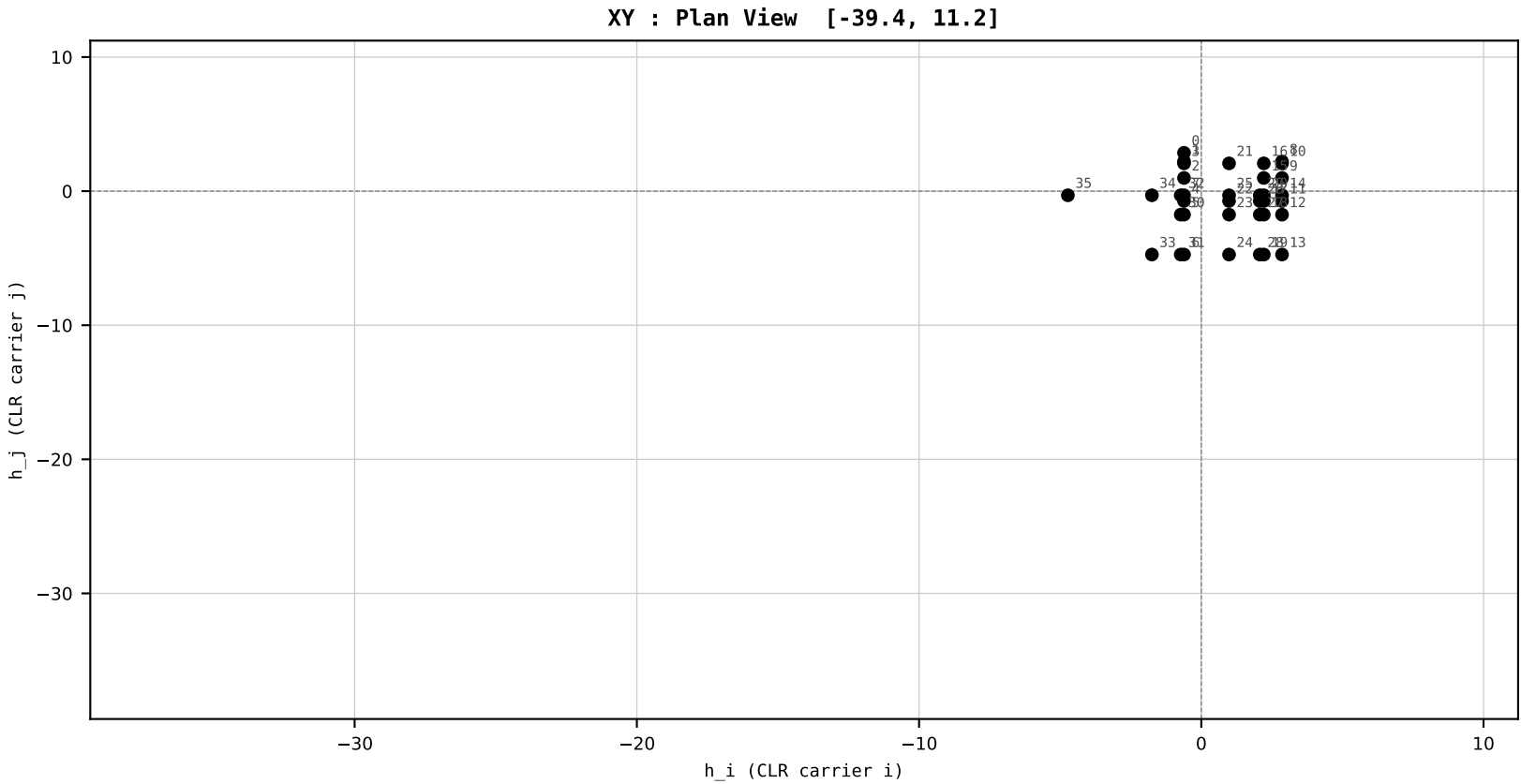
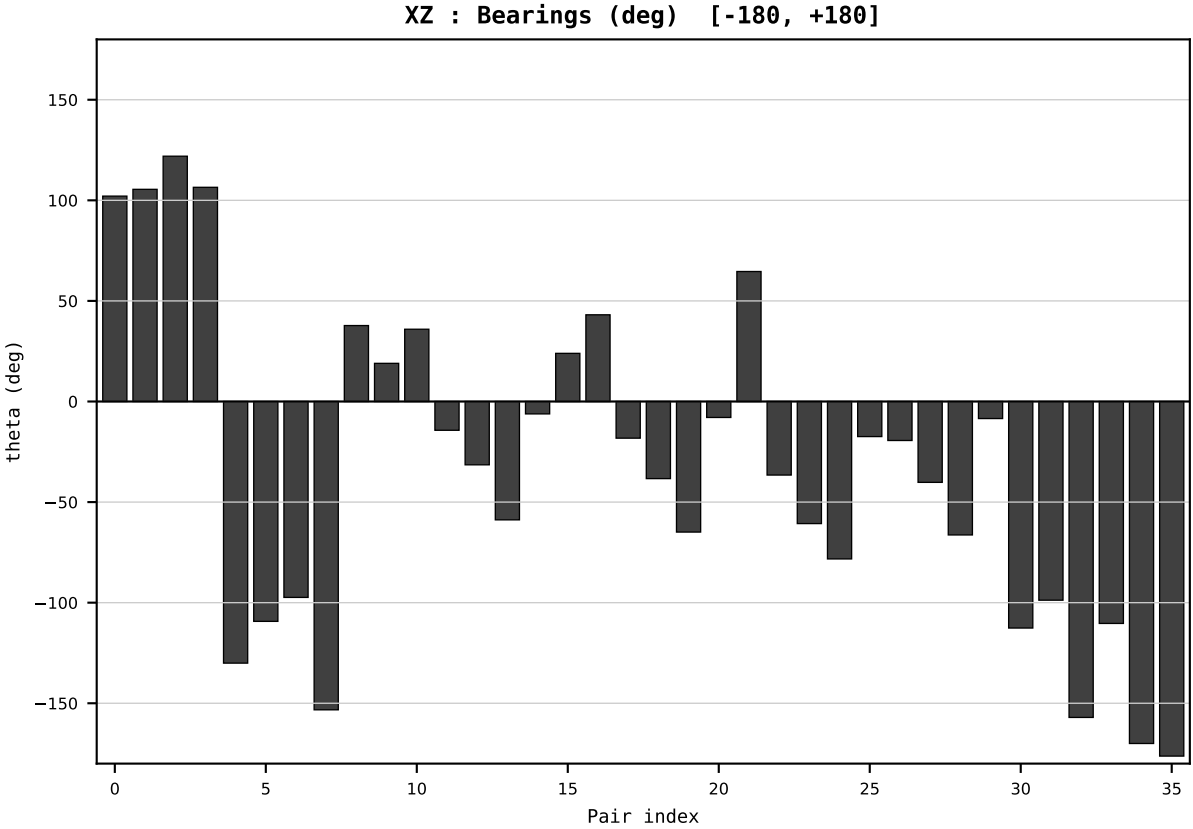
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2009
t : 8 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.353477
Ring = Hs-3
E_metric= 44.7578
kappa_HS= 1972.93
omega = 2.9378 deg
d_A = 0.348811
Helm = Wind
Helm d = +0.270064
DR = 7.587 HLR
DR ratio= 1972.9x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=9 | Year 2010 | D=9 N=25 pairs=36

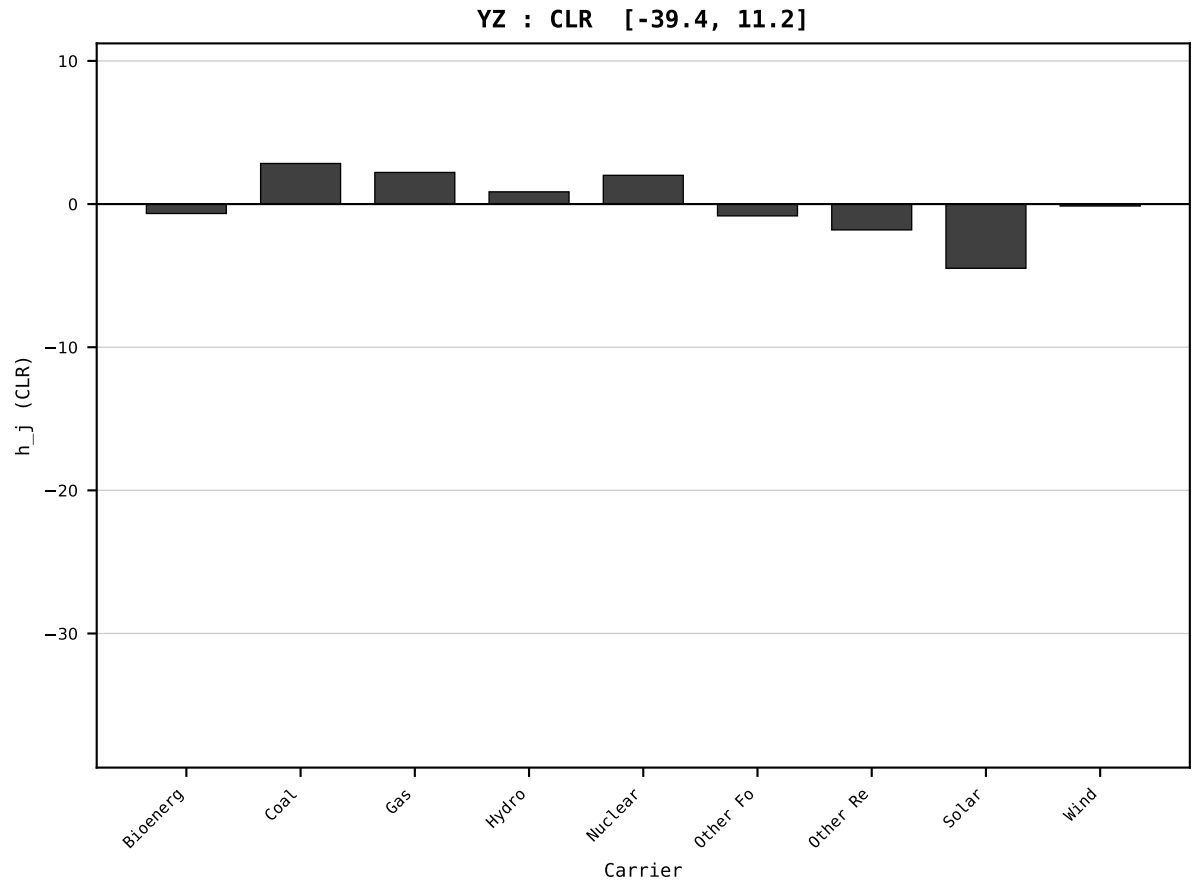
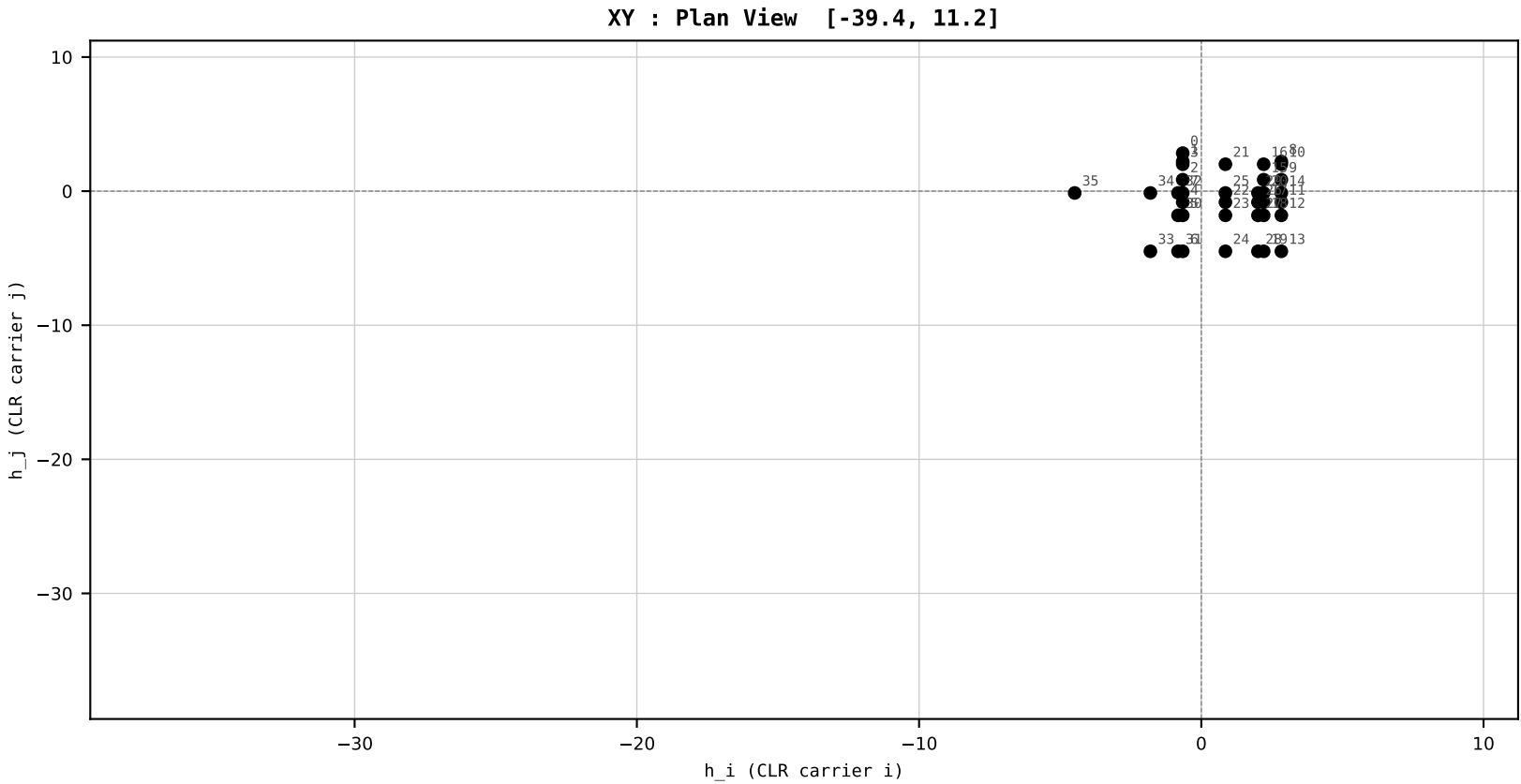
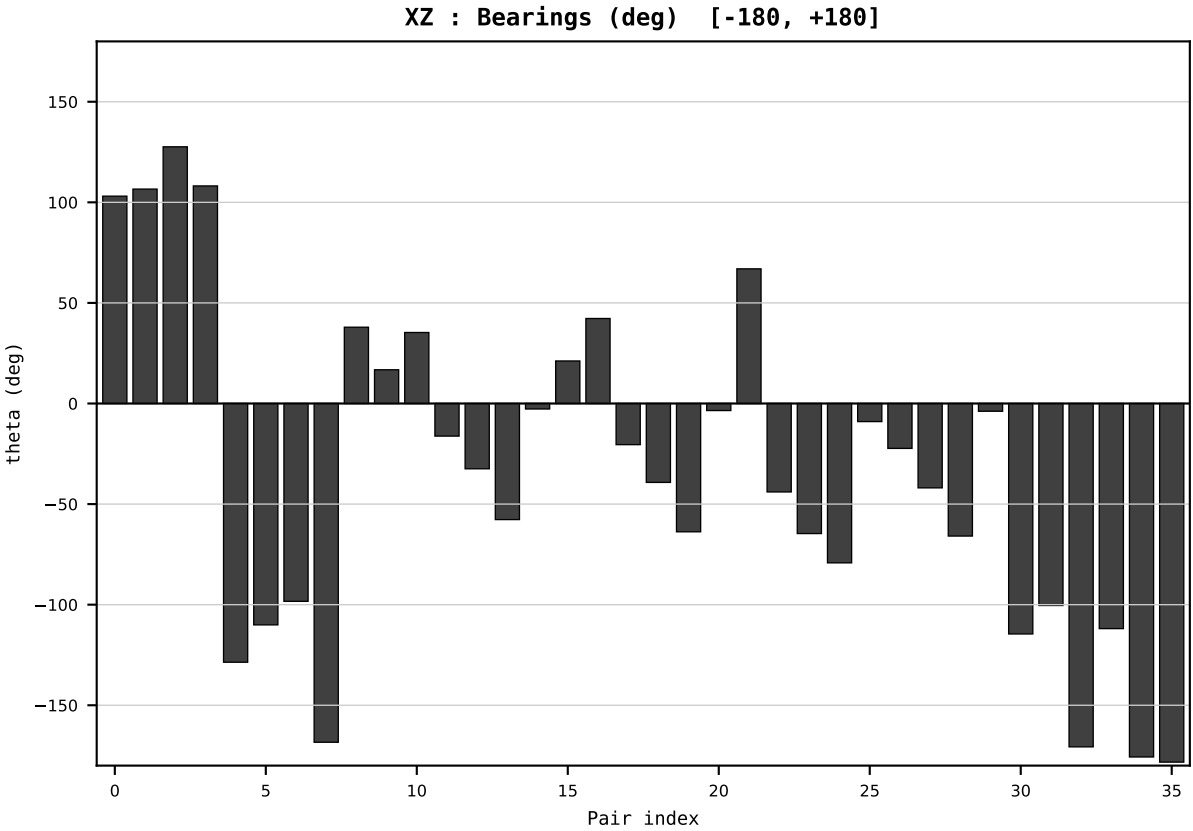
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2010
t : 9 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.355235
Ring = Hs-3
E_metric= 42.2037
kappa_HS= 1514.16
omega = 2.5467 deg
d_A = 0.351238
Helm = Solar
Helm d = +0.241100
DR = 7.323 HLR
DR ratio= 1514.2x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=10 | Year 2011 | D=9 N=25 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

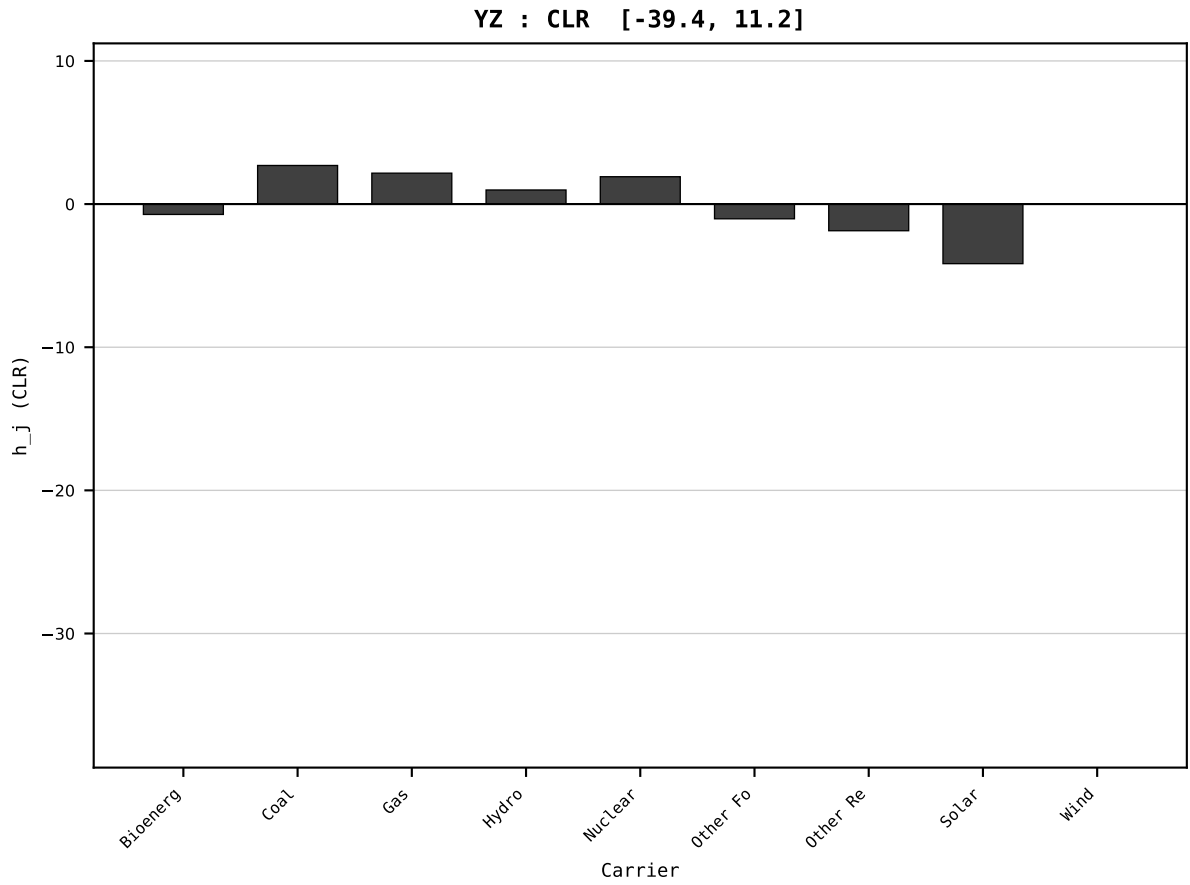
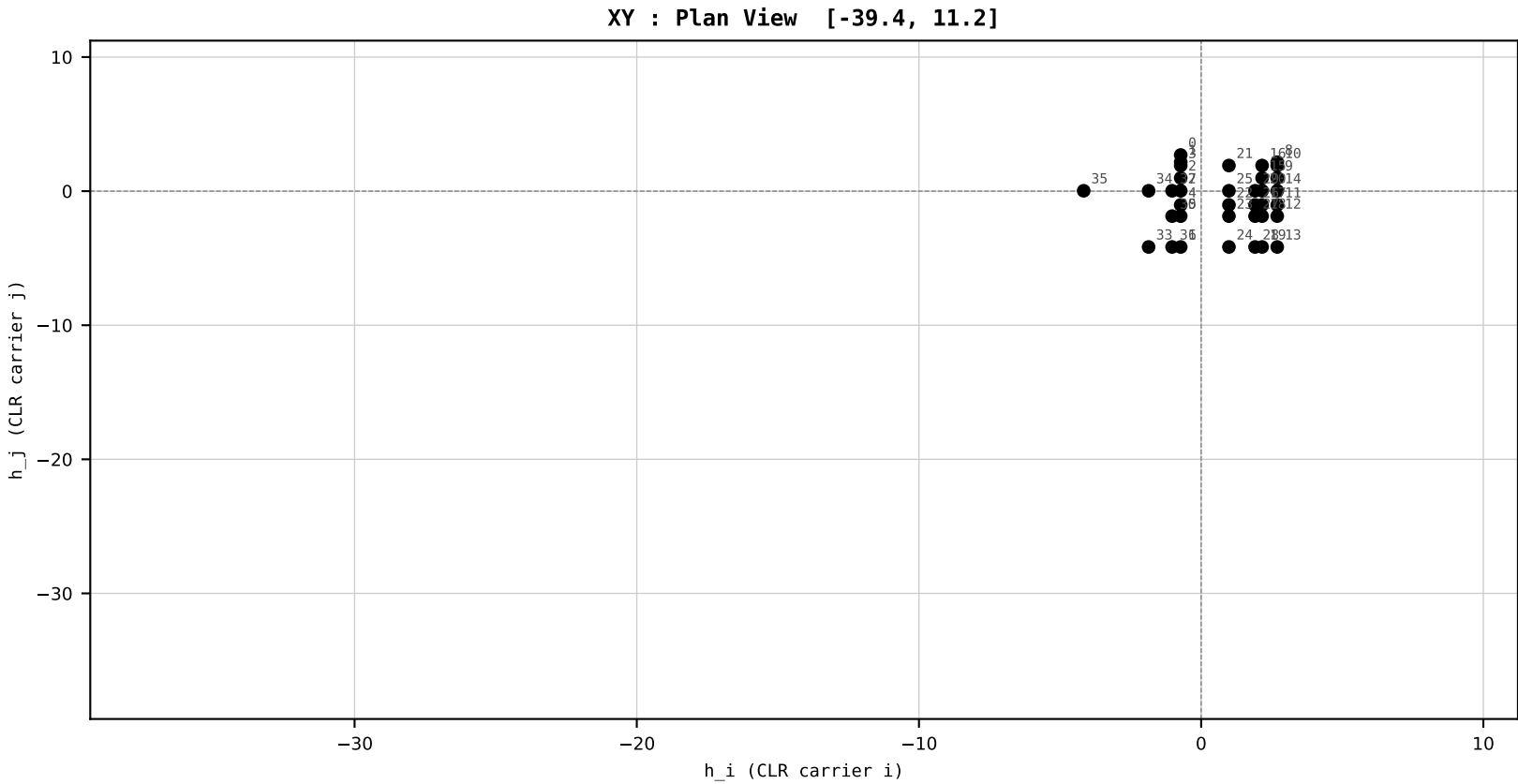
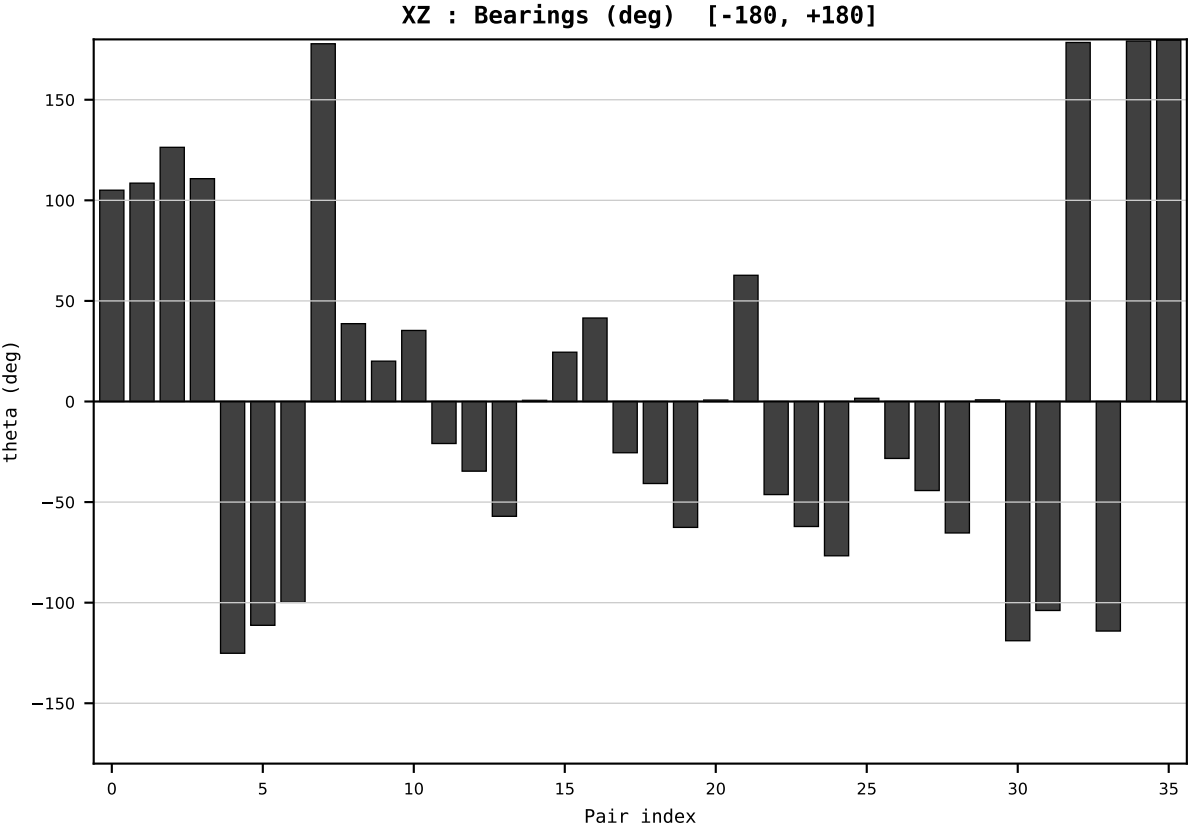
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2011
t : 10 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.334969
Ring = Hs-3
E_metric= 38.9355
kappa_HS= 952.43
omega = 3.6391 deg
d_A = 0.478871
Helm = Solar
Helm d = +0.323844
DR = 6.859 HLR
DR ratio= 952.4x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=11 | Year 2012 | D=9 N=25 pairs=36

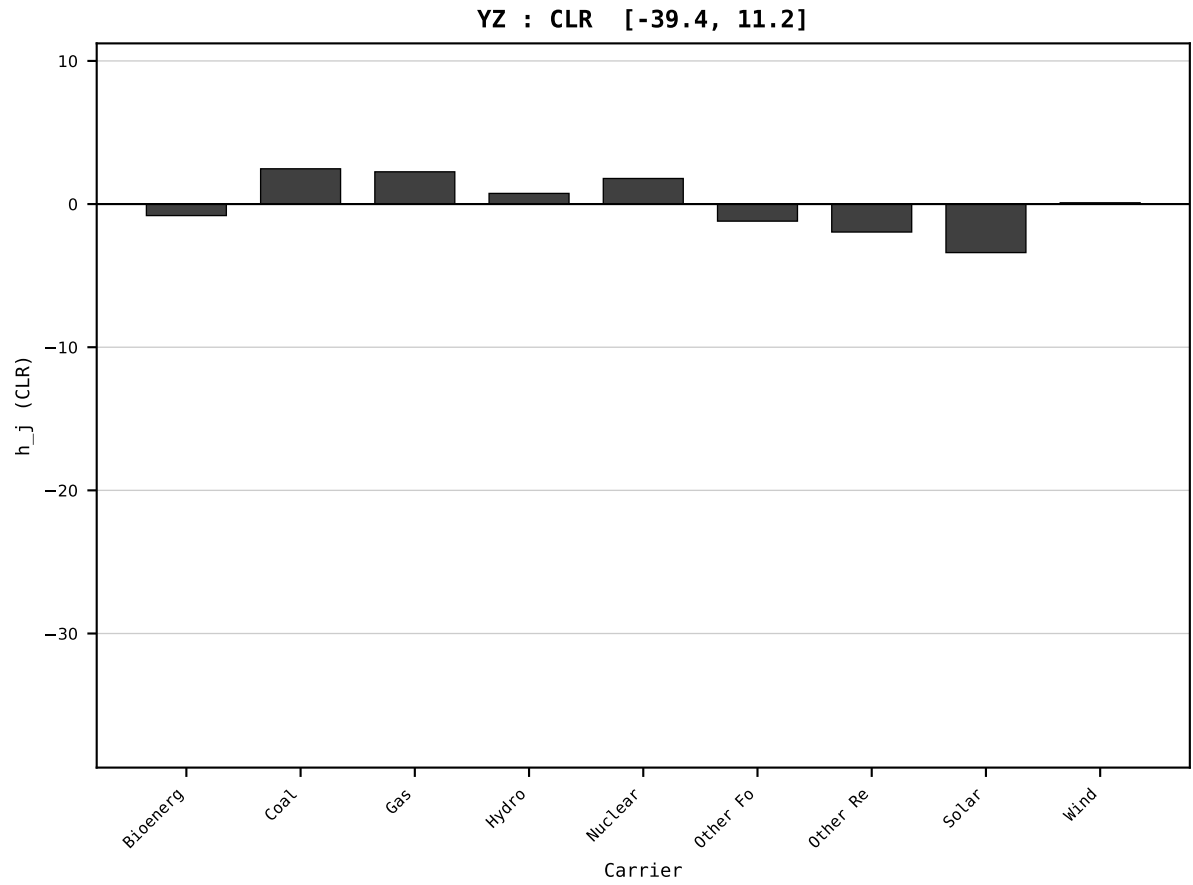
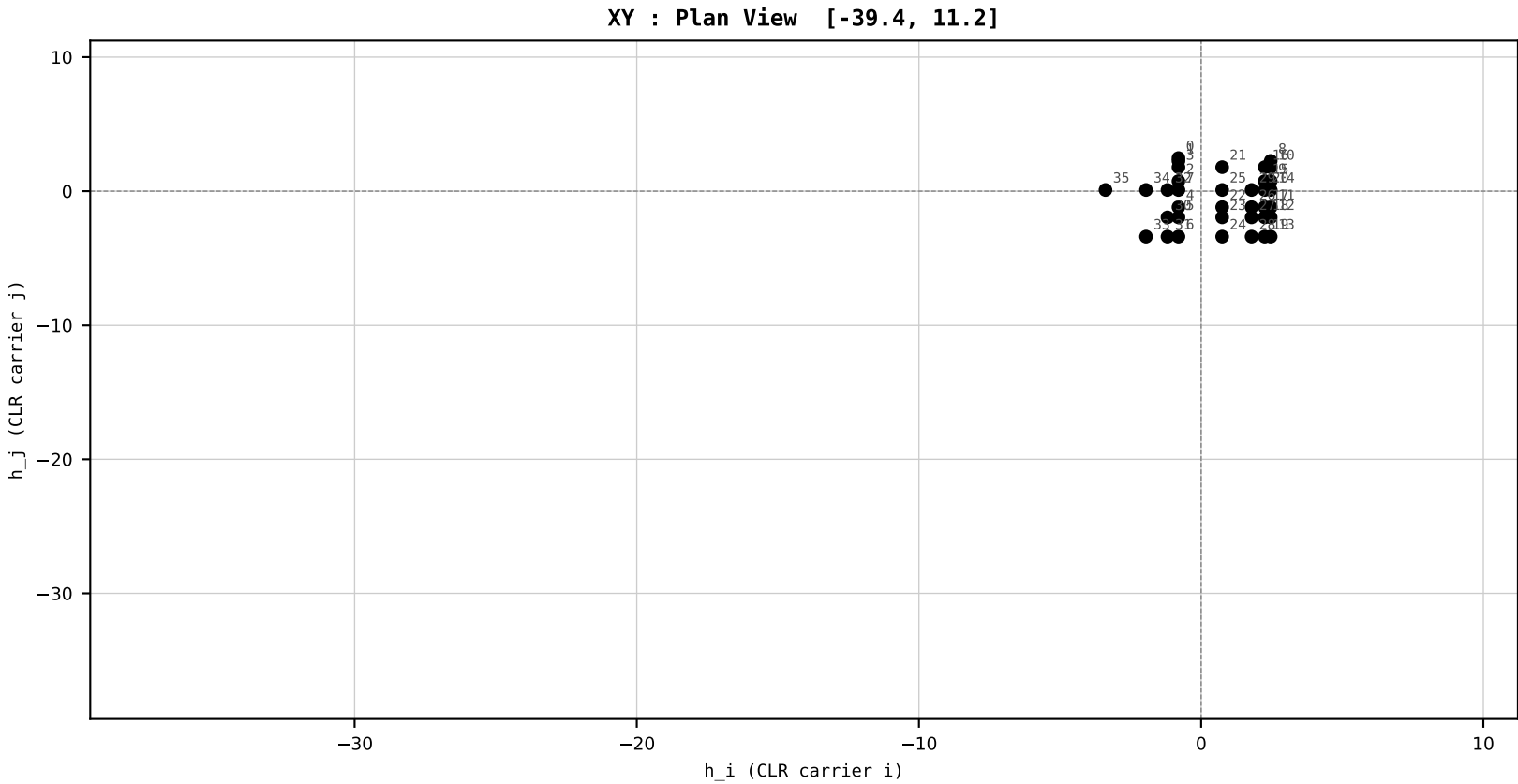
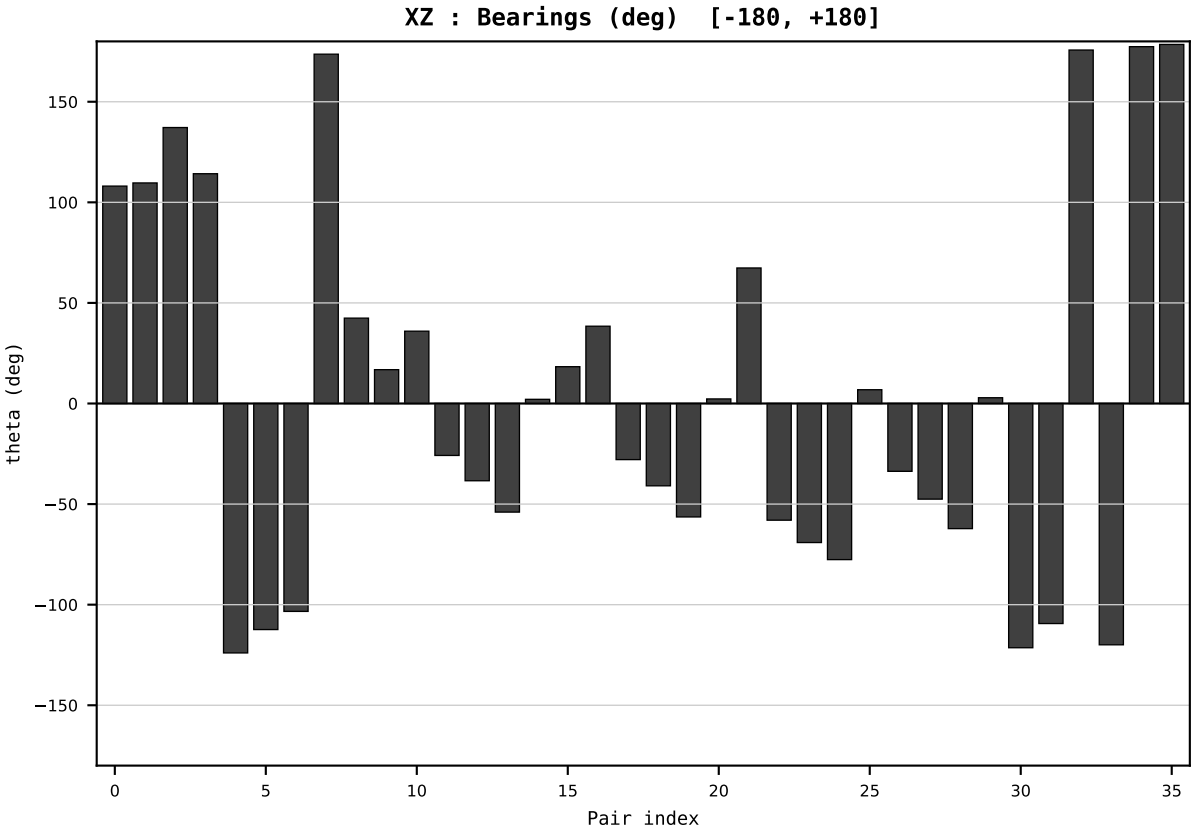
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2012
t : 11 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.325964
Ring = Hs-3
E_metric= 32.2877
kappa_HS= 348.86
omega = 6.5734 deg
d_A = 0.881530
Helm = Solar
Helm d = +0.772628
DR = 5.855 HLR
DR ratio= 348.9x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

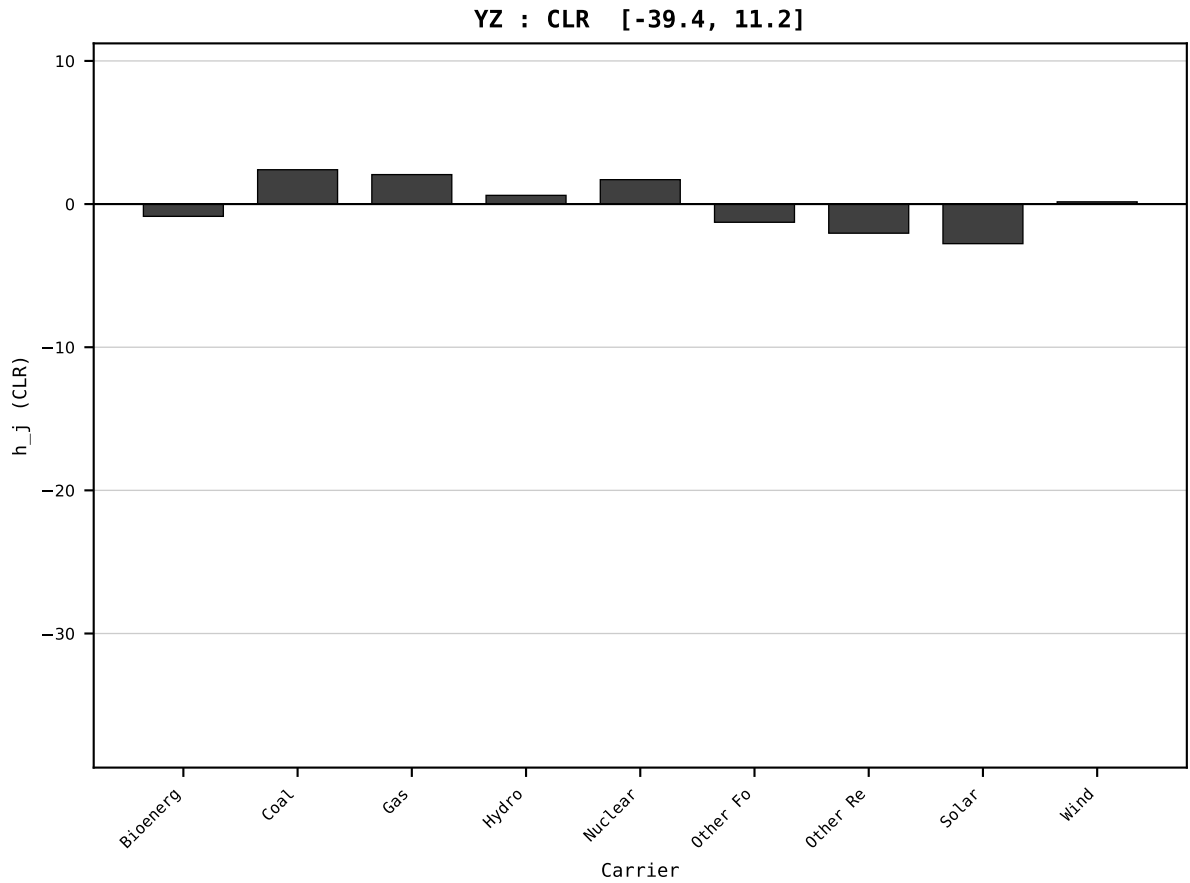
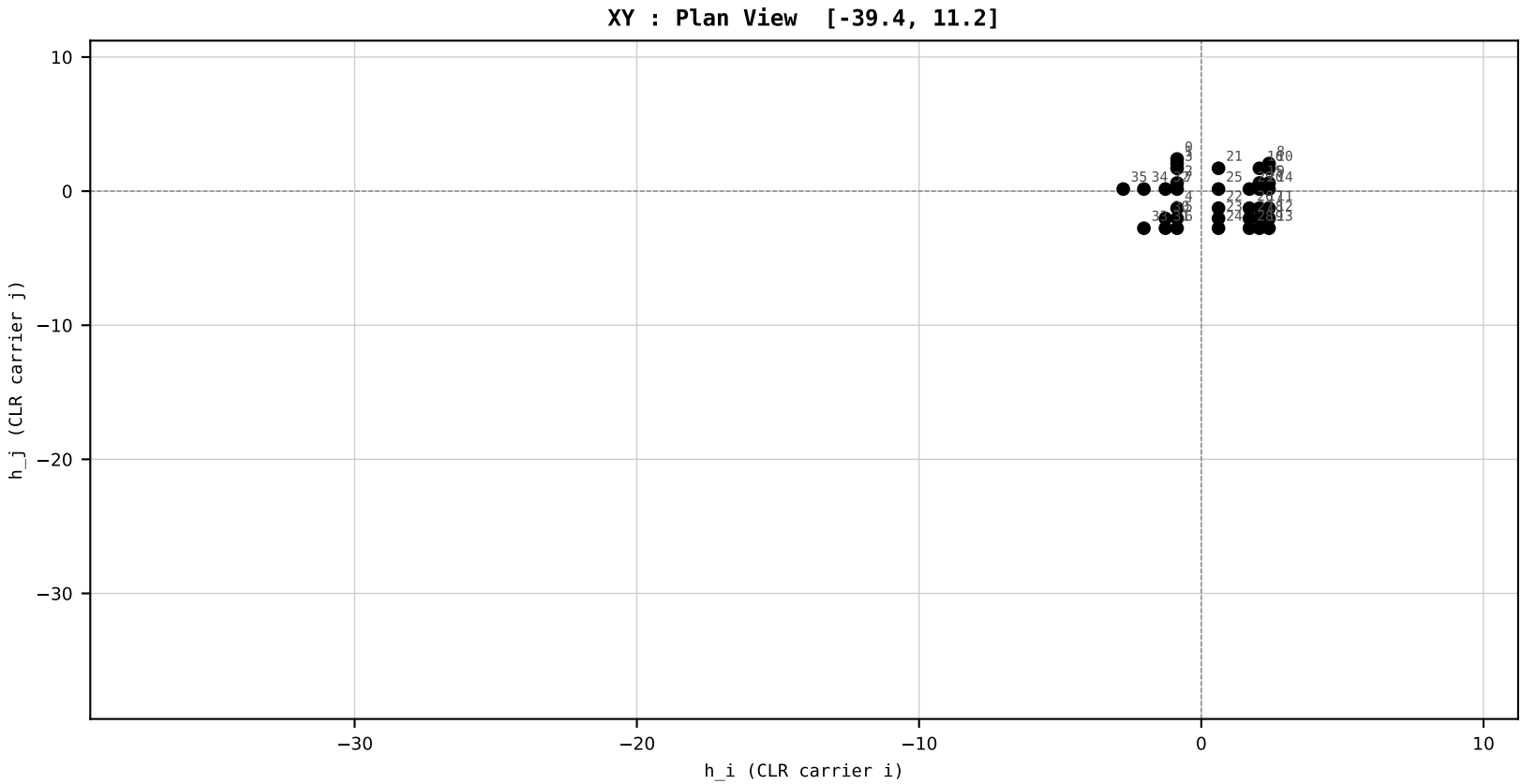
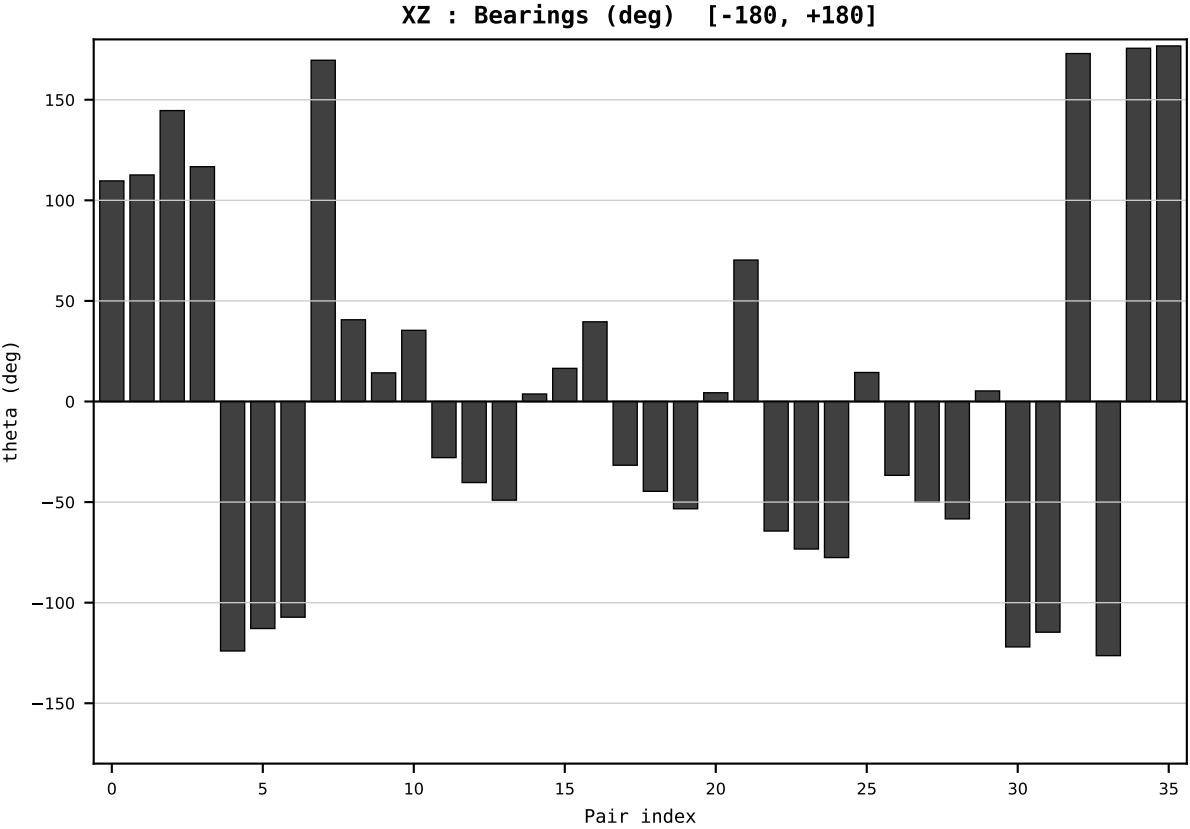
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2013
t : 12 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.318102
Ring = Hs-3
E_metric= 27.4388
kappa_HS= 175.10
omega = 5.5612 deg
d_A = 0.690898
Helm = Solar
Helm d = +0.624351
DR = 5.165 HLR
DR ratio= 175.1x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=13 | Year 2014 | D=9 N=25 pairs=36

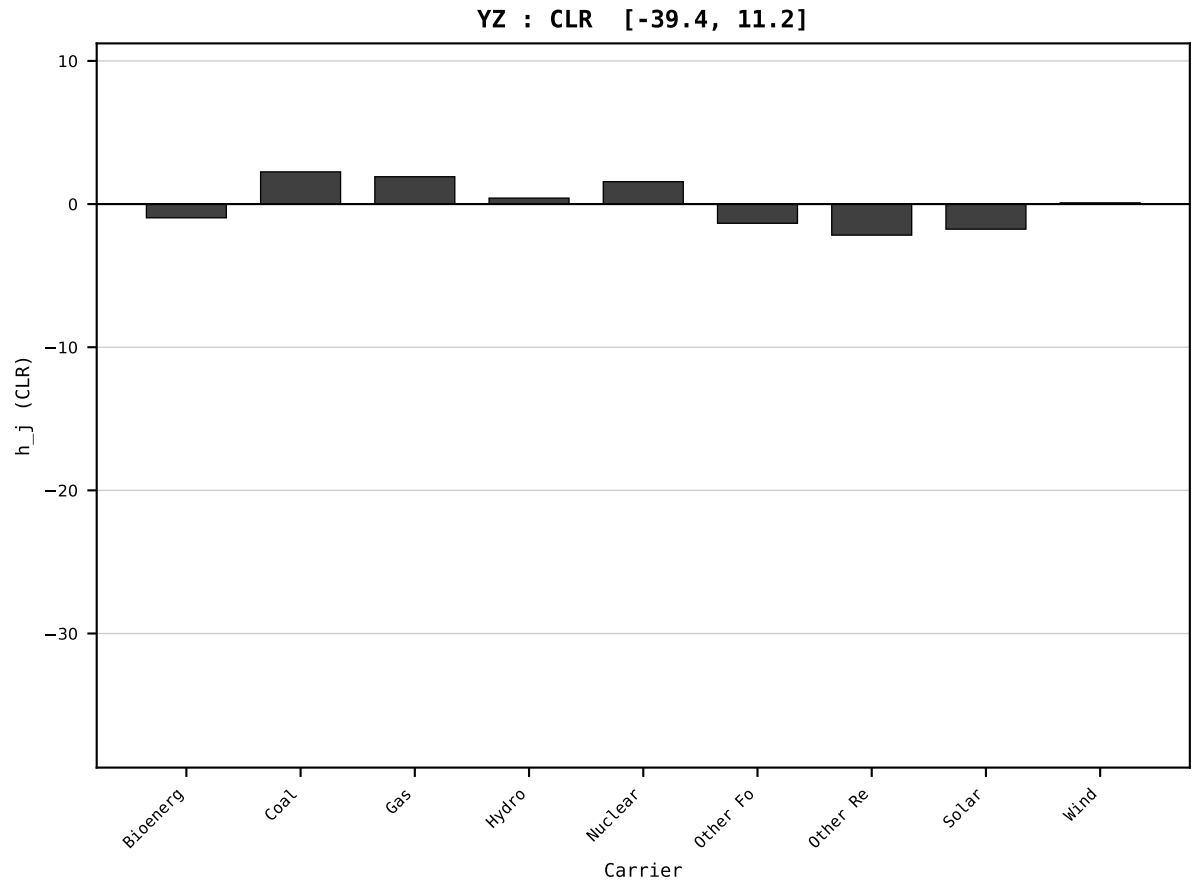
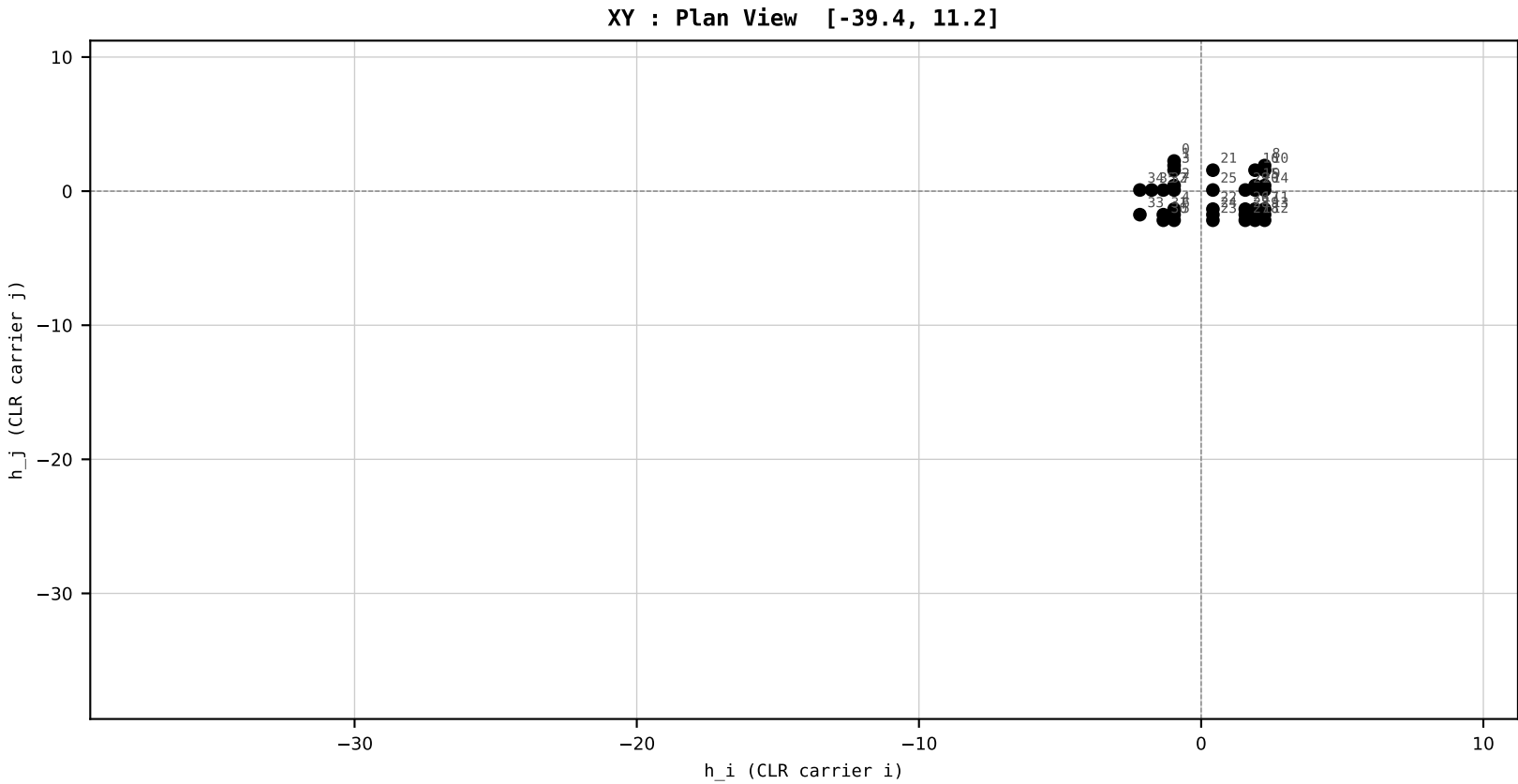
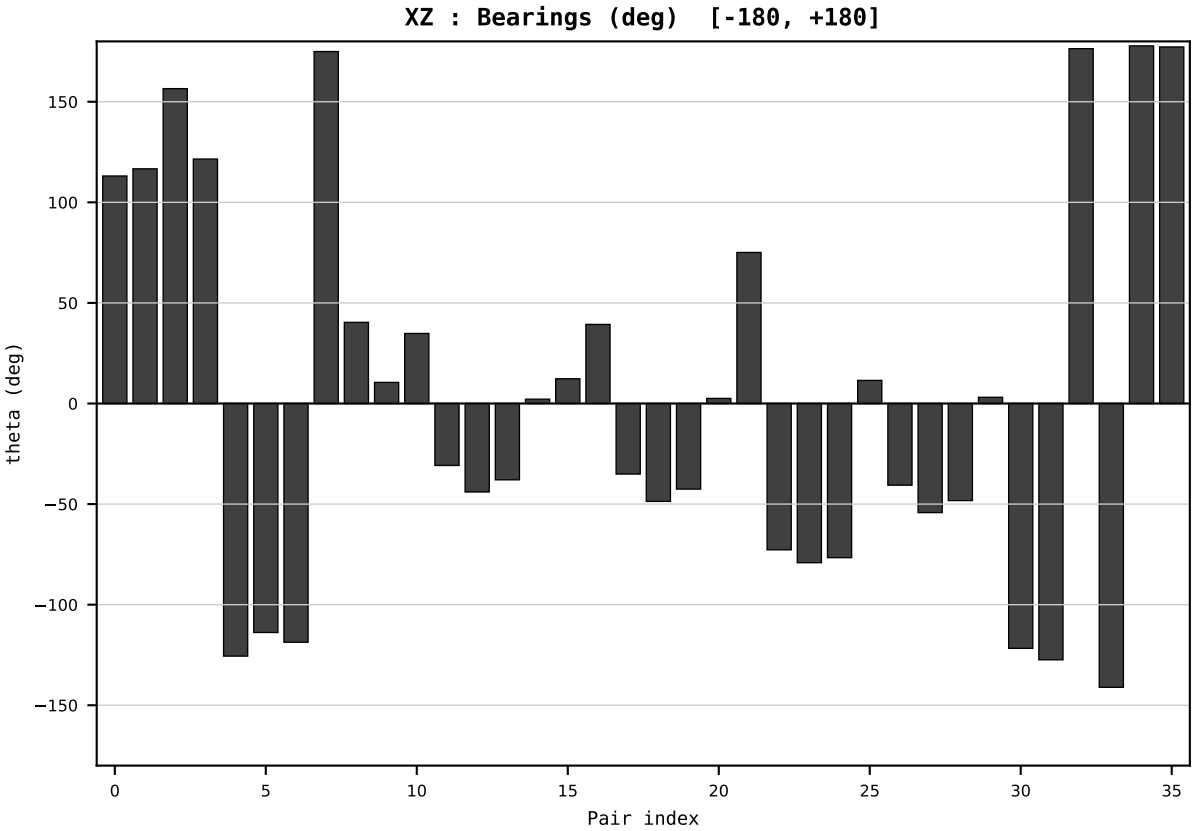
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2014
t : 13 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.306116
Ring = Hs-3
E_metric= 21.8268
kappa_HS= 83.03
omega = 10.6716 deg
d_A = 1.080374
Helm = Solar
Helm d = +1.013049
DR = 4.419 HLR
DR ratio= 83.0x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=14 | Year 2015 | D=9 N=25 pairs=36

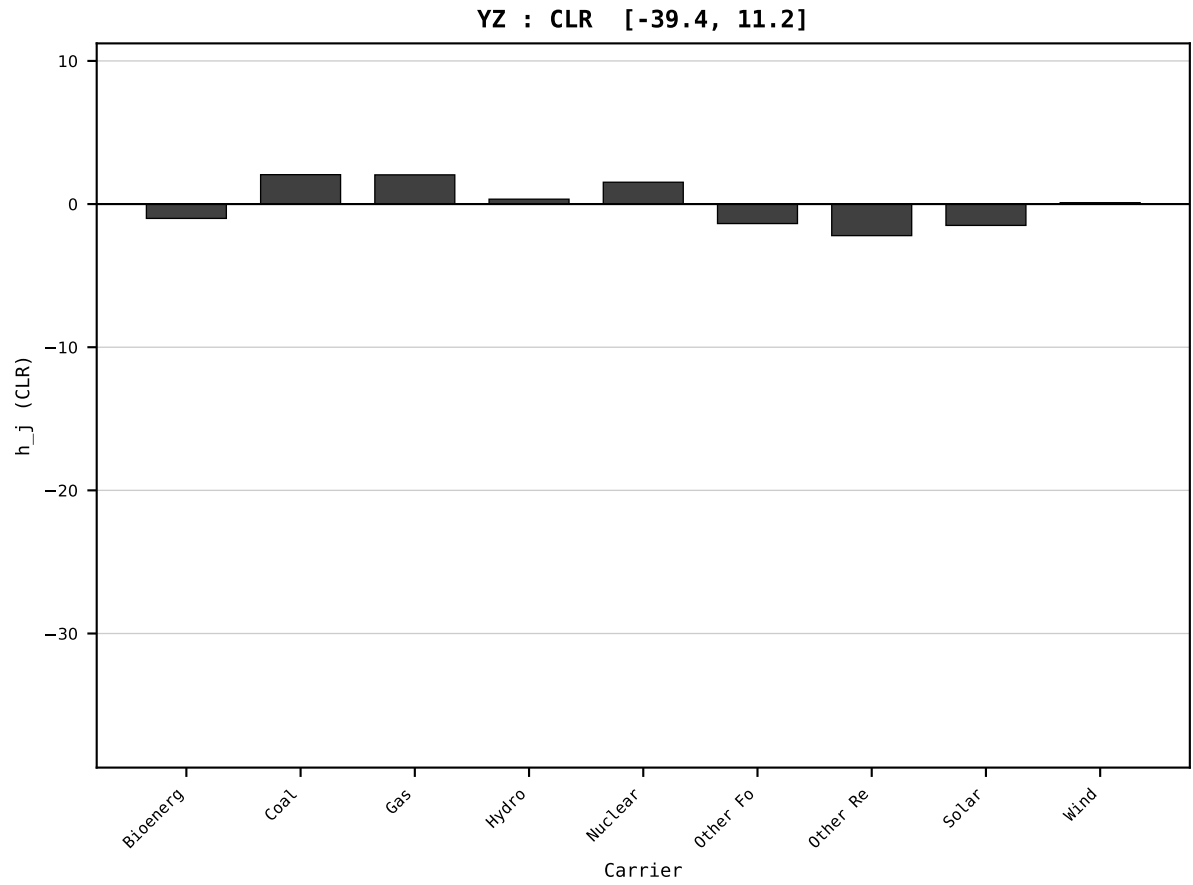
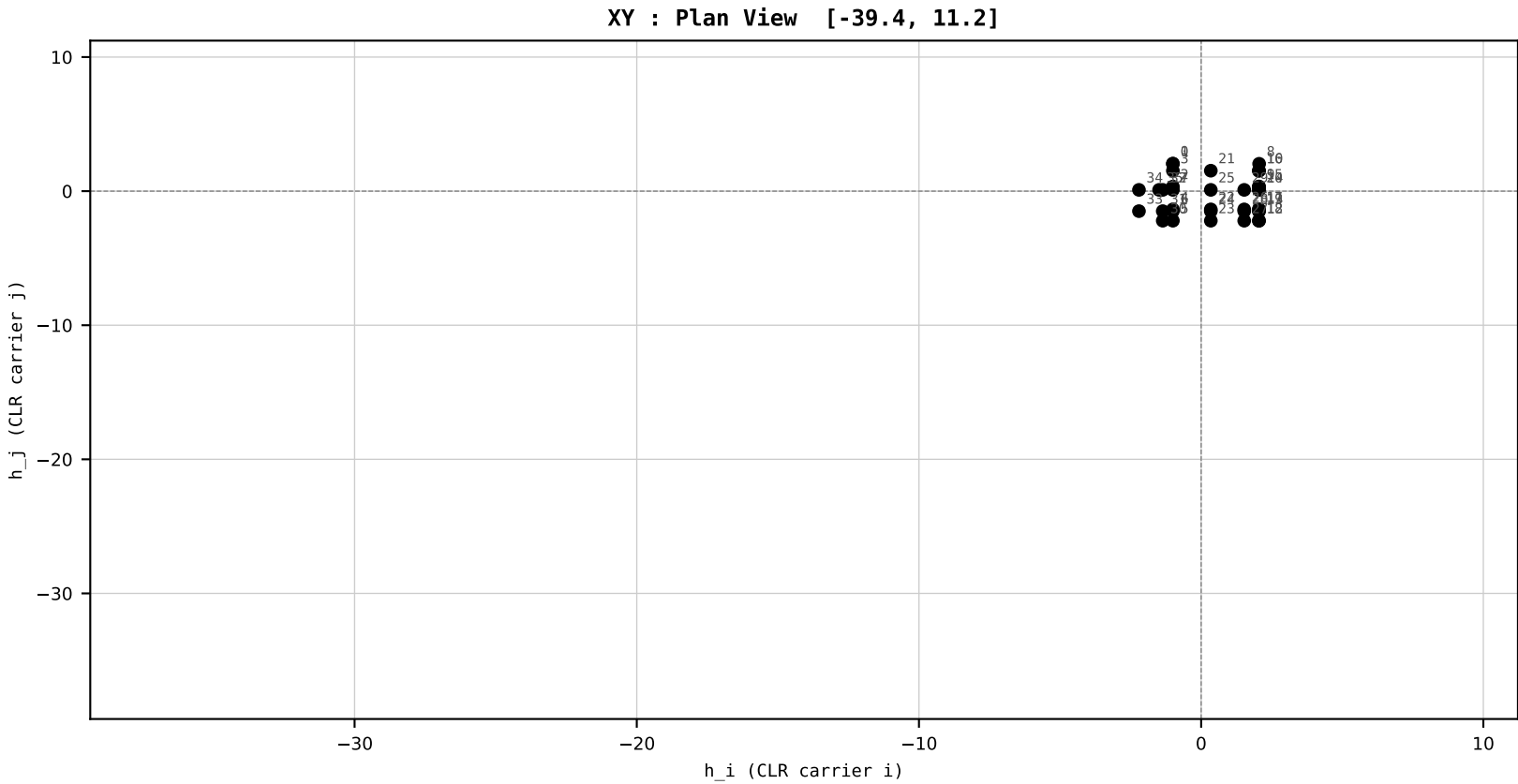
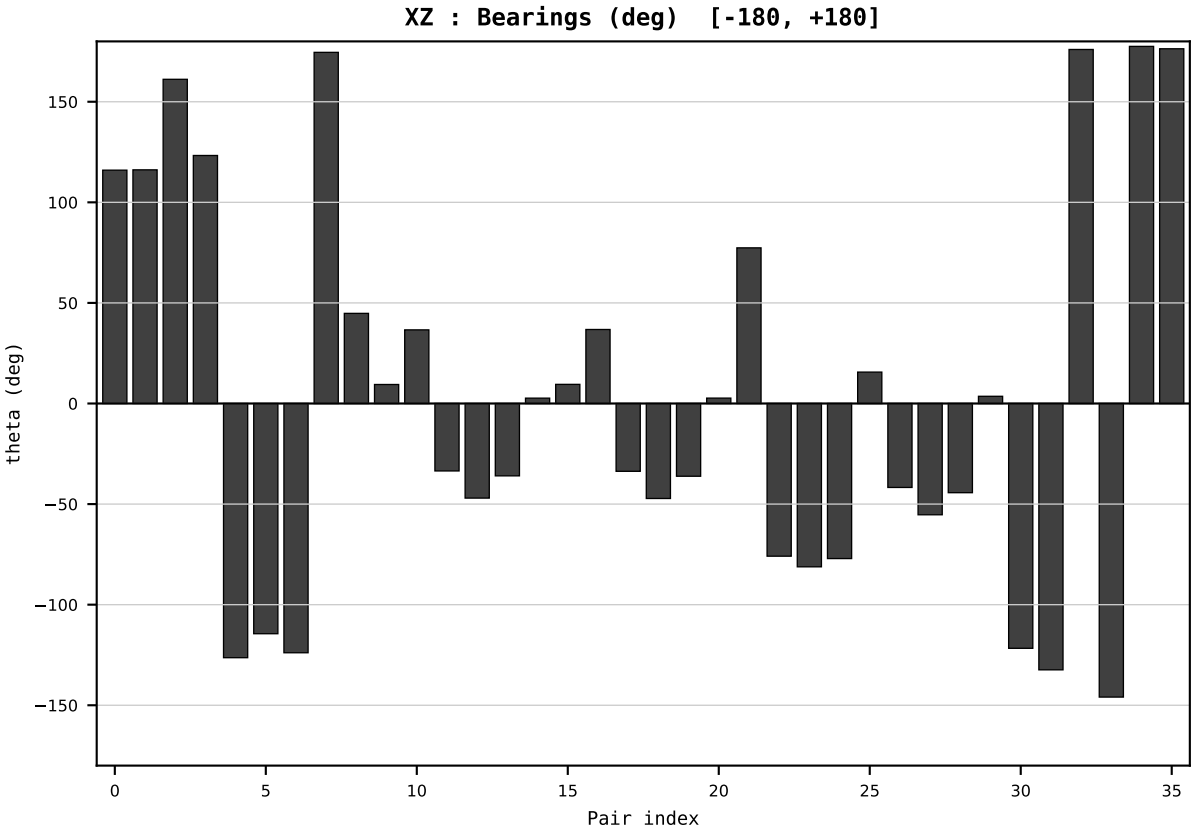
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2015
t : 14 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.296660
Ring = Hs-2
E_metric= 20.7750
kappa_HS= 70.73
omega = 4.3215 deg
d_A = 0.366154
Helm = Solar
Helm d = +0.262026
DR = 4.259 HLR
DR ratio= 70.7x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=15 | Year 2016 | D=9 N=25 pairs=36

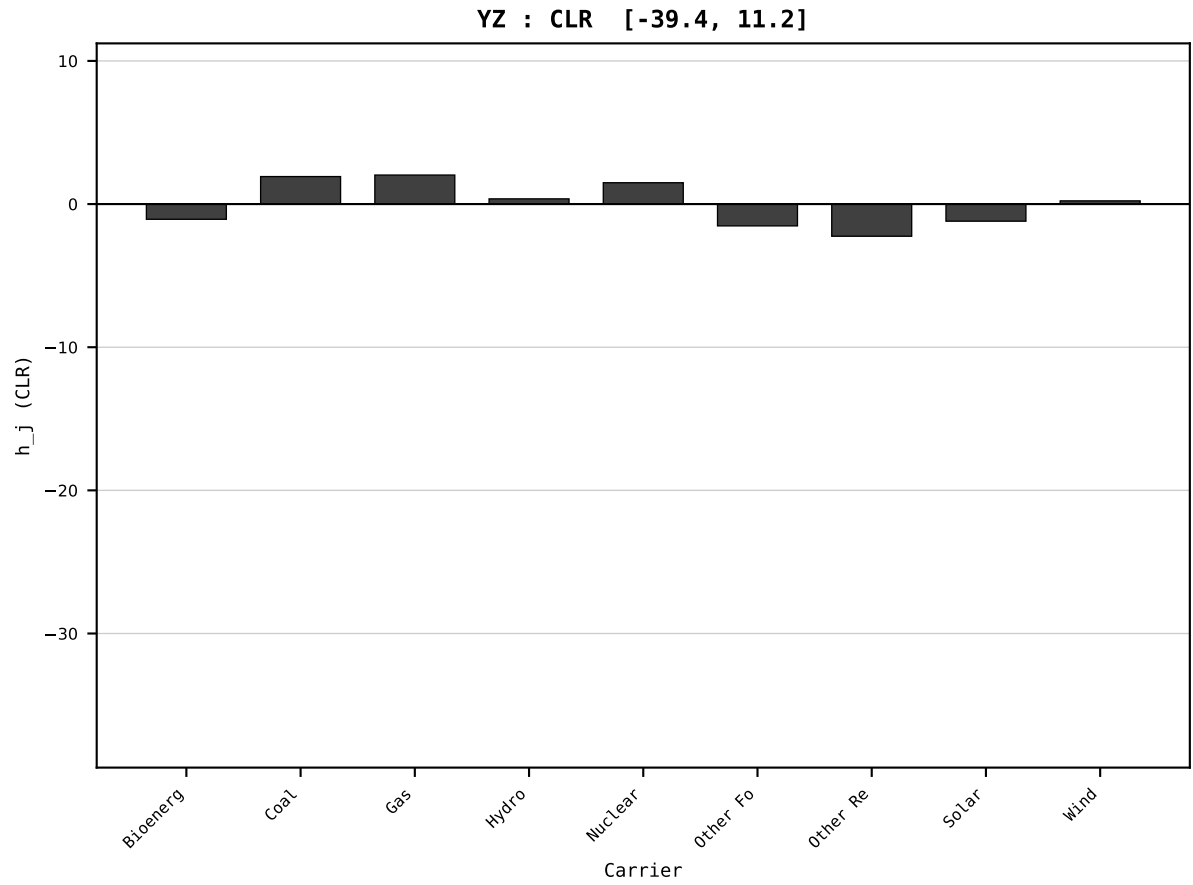
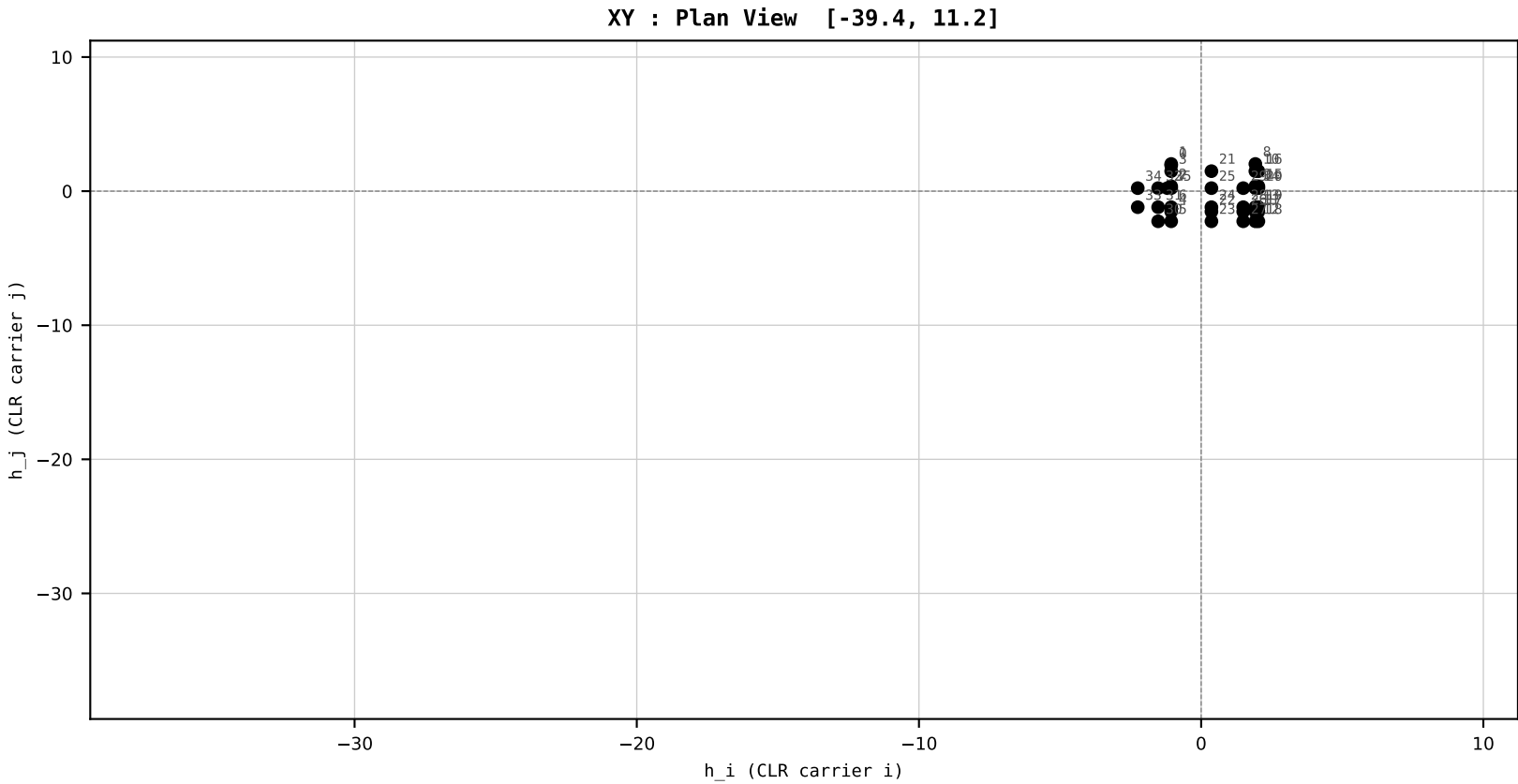
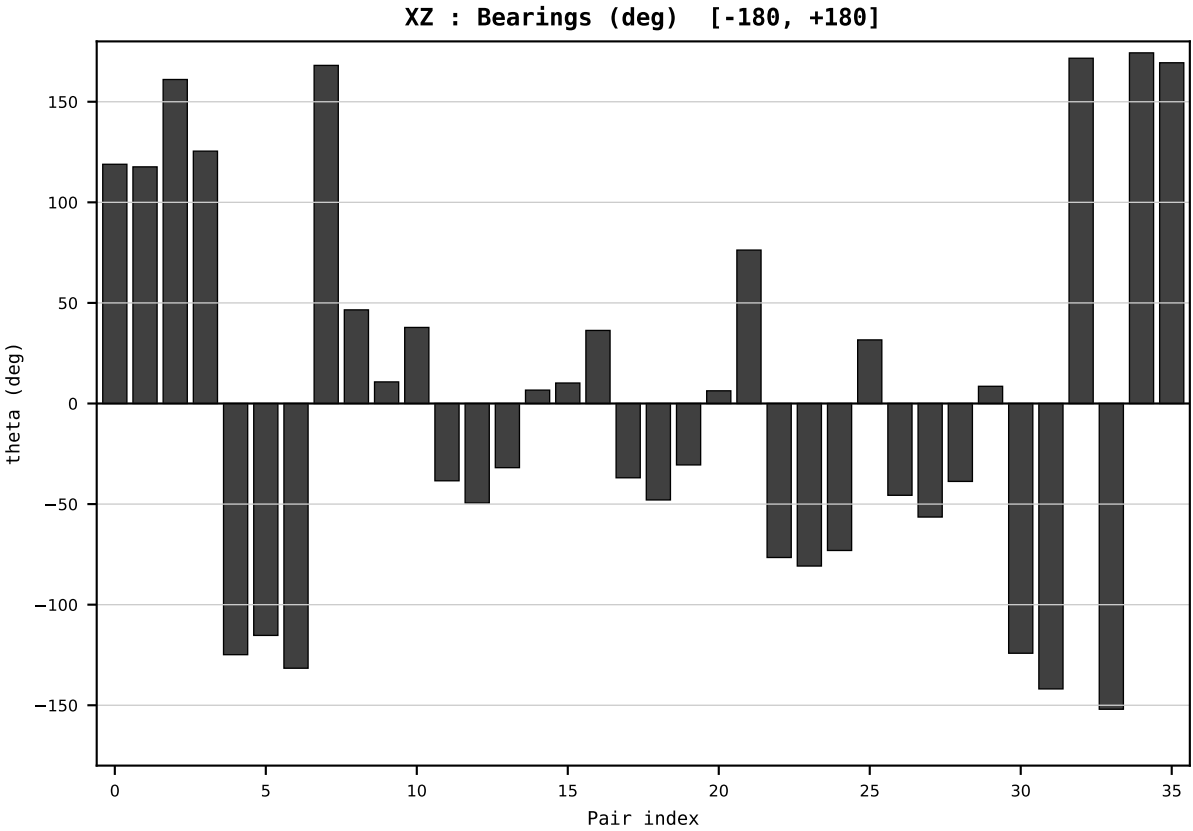
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2016
t : 15 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.282403
Ring = Hs-2
E_metric= 20.1405
kappa_HS= 71.90
omega = 4.9089 deg
d_A = 0.393670
Helm = Solar
Helm d = +0.294744
DR = 4.275 HLR
DR ratio= 71.9x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=16 | Year 2017 | D=9 N=25 pairs=36

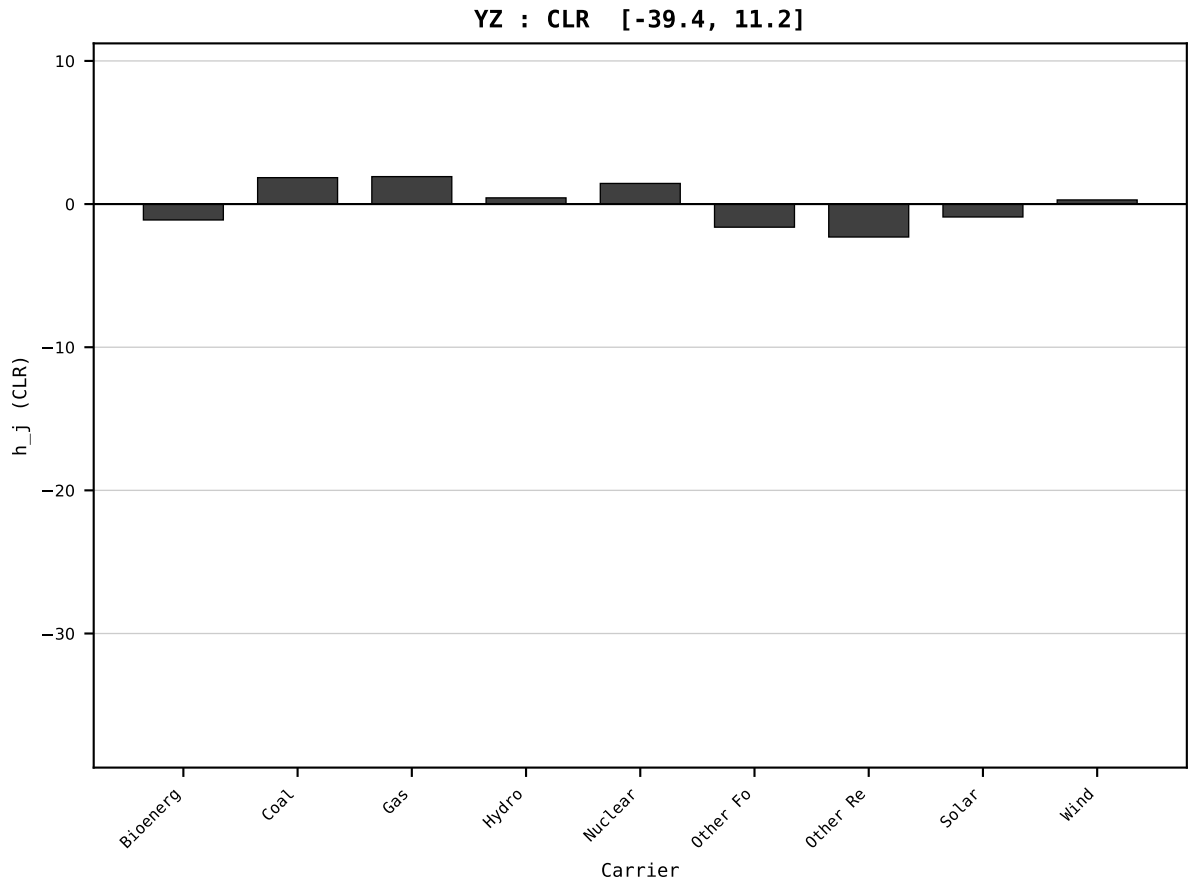
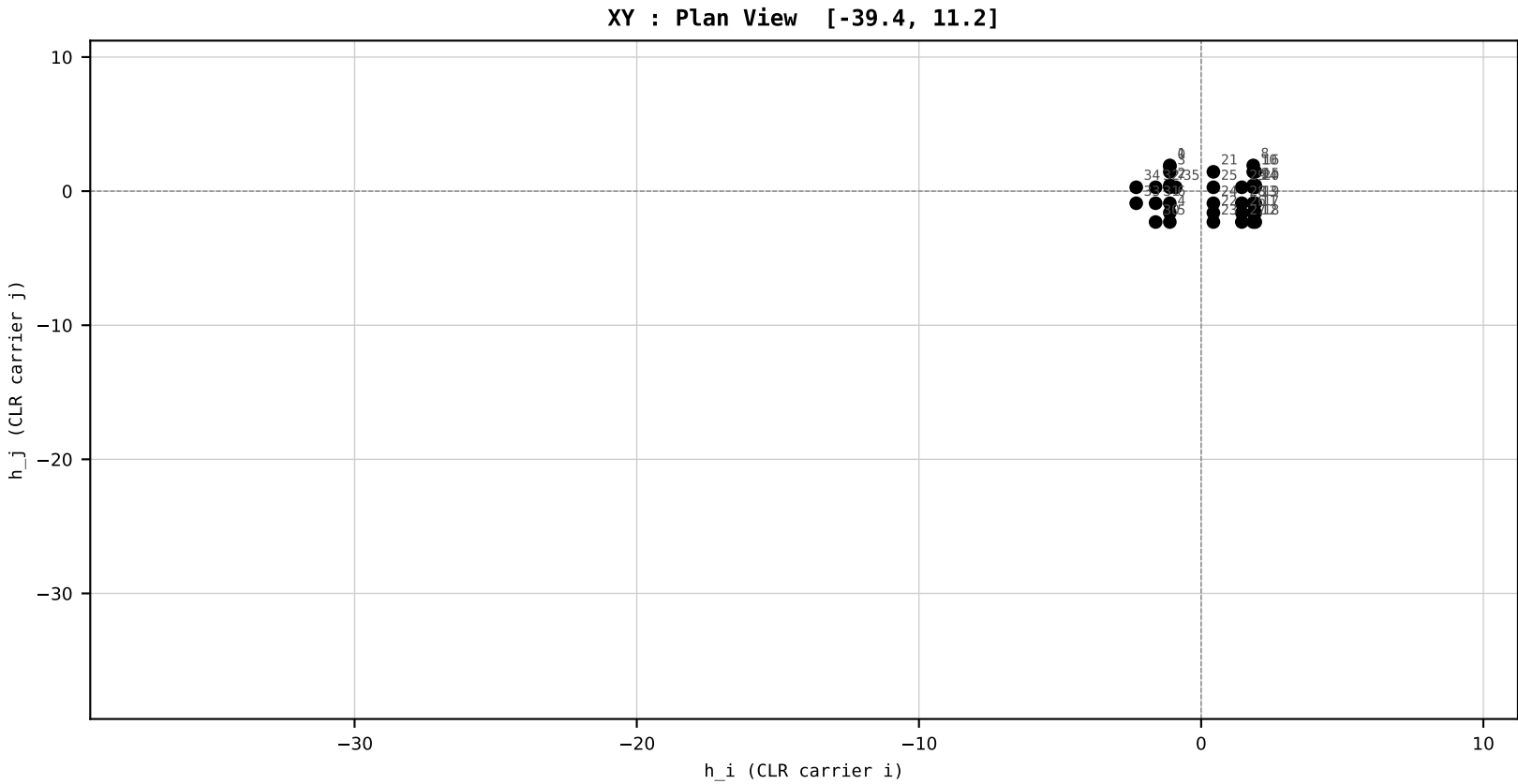
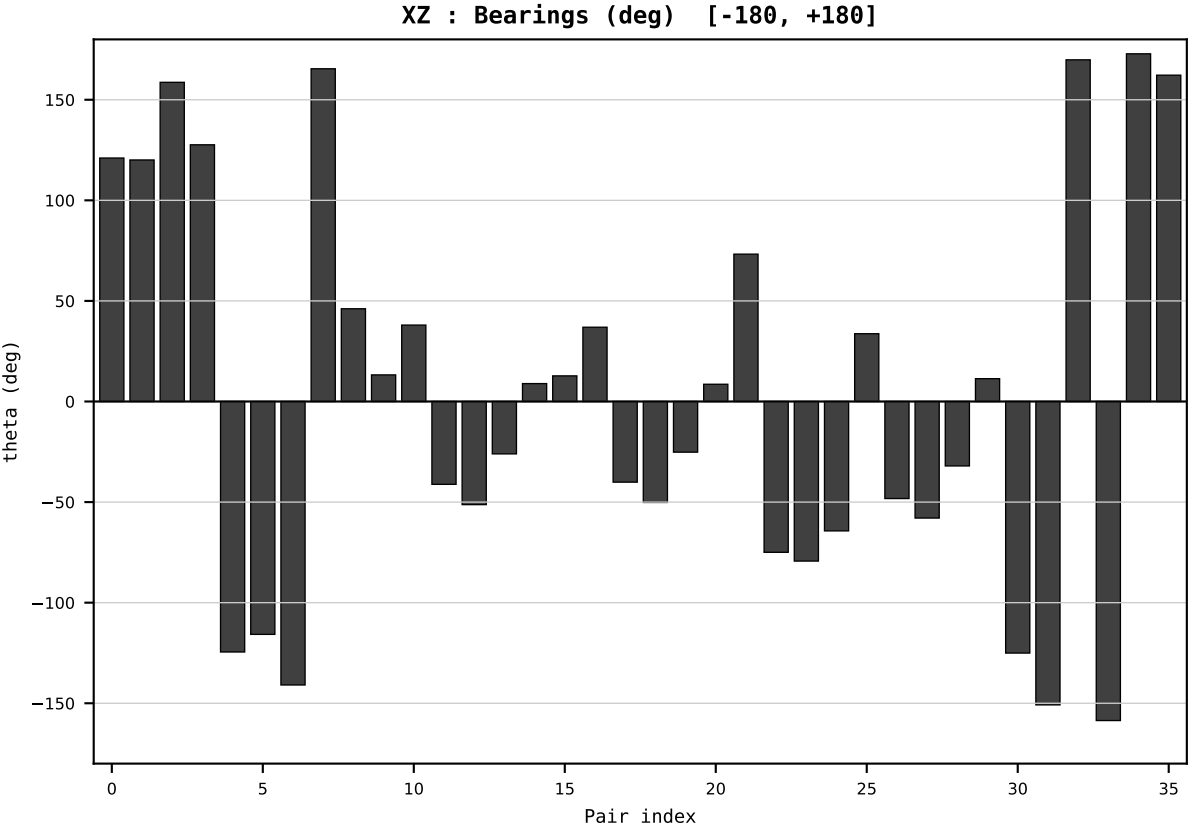
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2017
t : 16 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.263028
Ring = Hs-2
E_metric= 19.3839
kappa_HS= 67.98
omega = 4.5042 deg
d_A = 0.359563
Helm = Solar
Helm d = +0.294082
DR = 4.219 HLR
DR ratio= 68.0x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

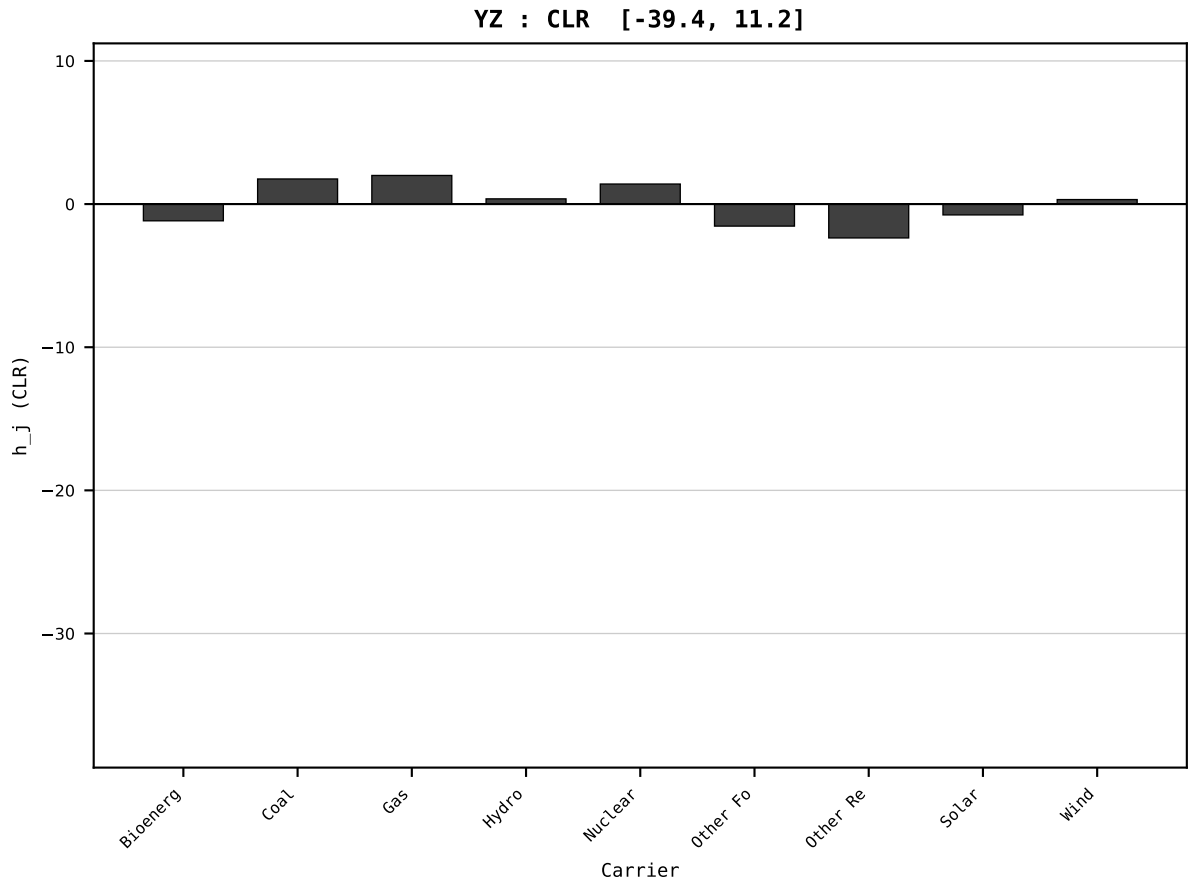
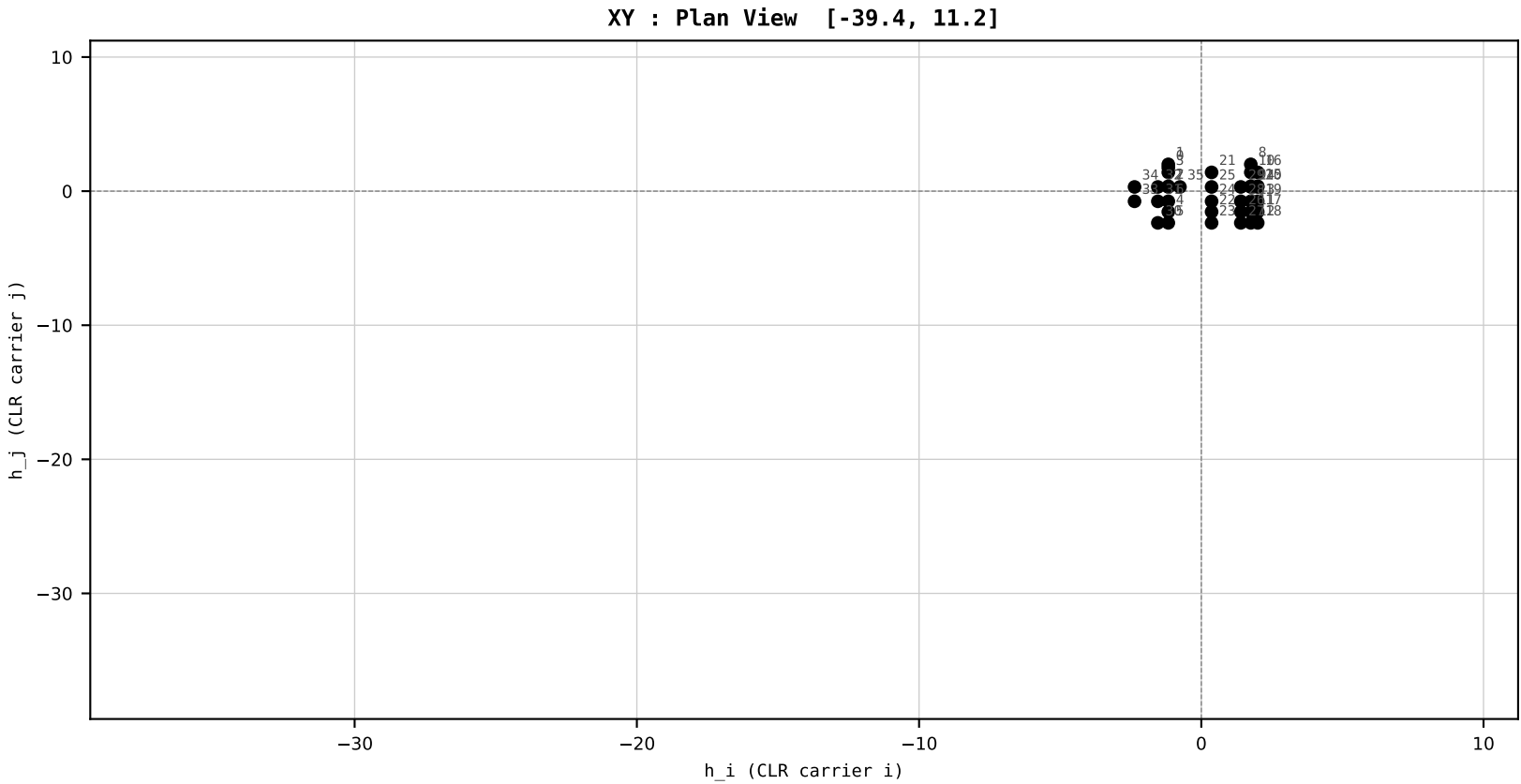
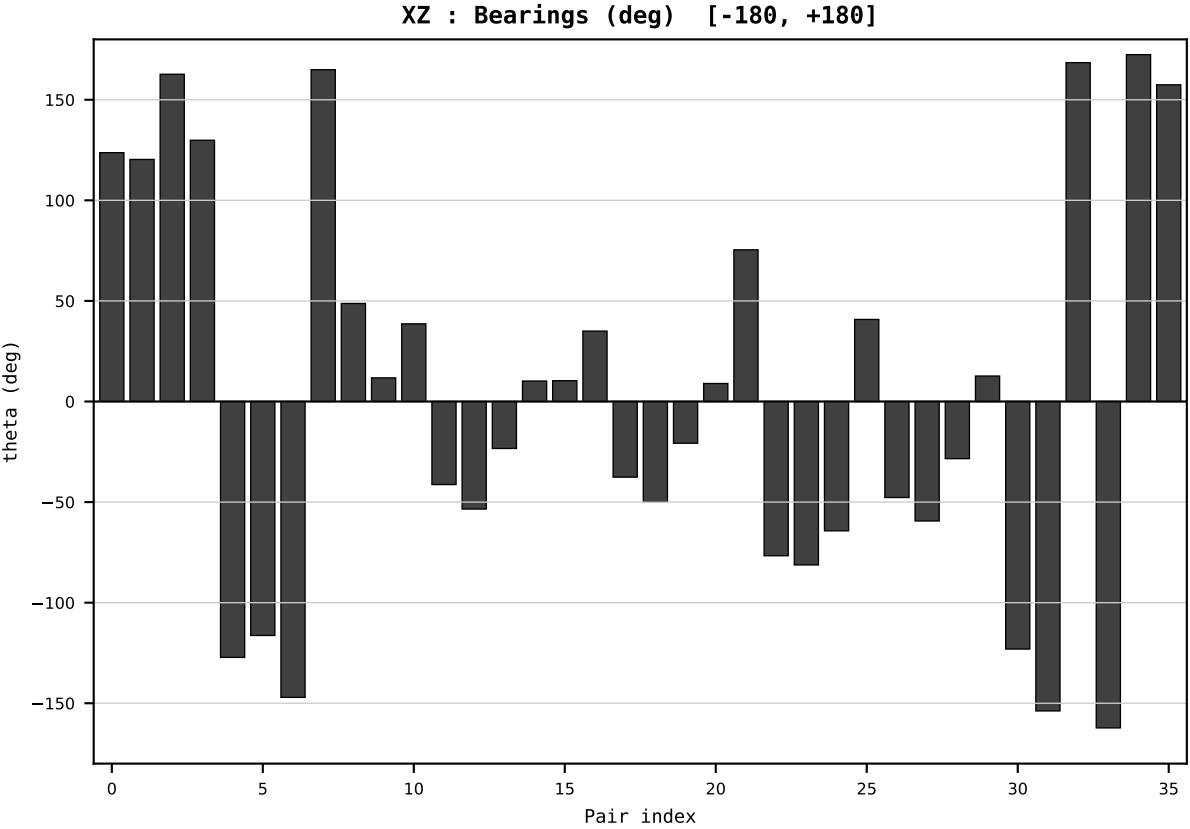
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2018
t : 17 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.264078
Ring = Hs-2
E_metric= 19.1683
kappa_HS= 78.61
omega = 3.0814 deg
d_A = 0.237367
Helm = Solar
Helm d = +0.144753
DR = 4.364 HLR
DR ratio= 78.6x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=18 | Year 2019 | D=9 N=25 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

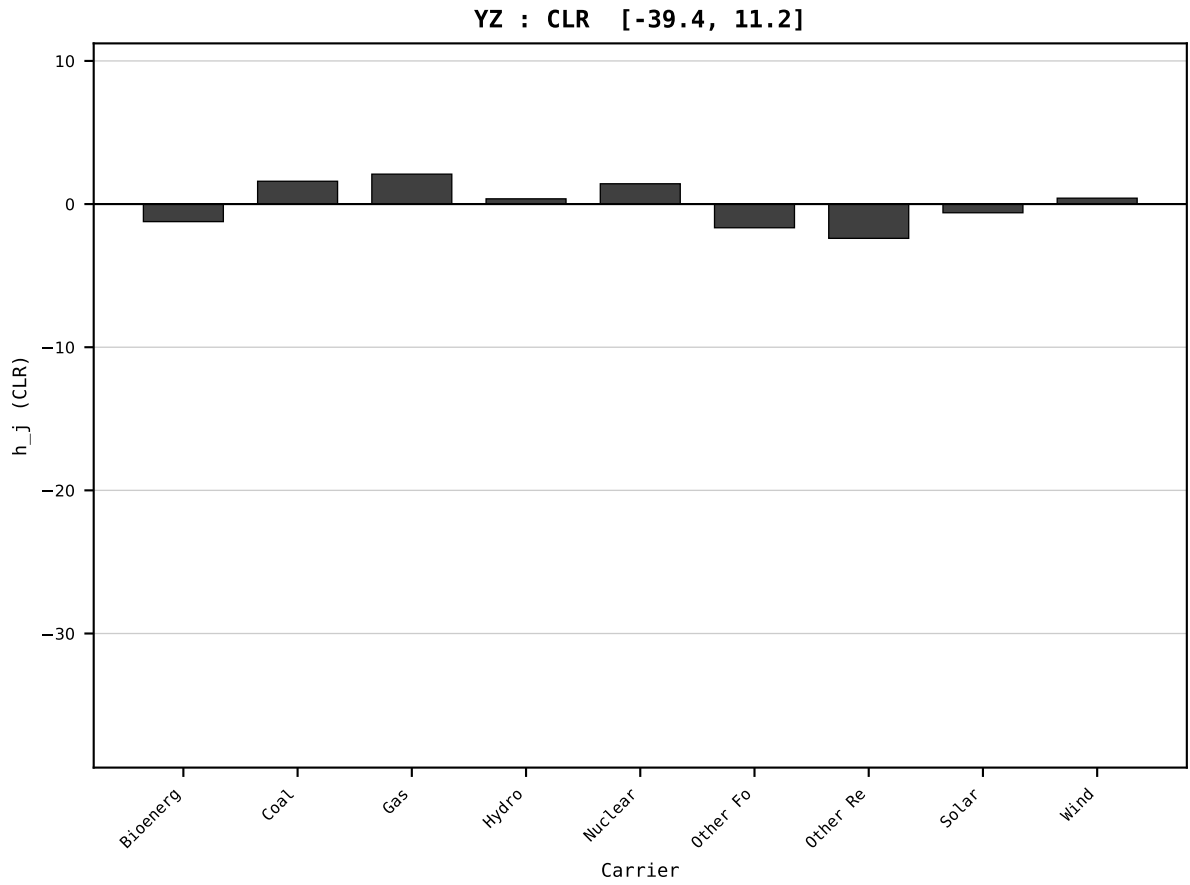
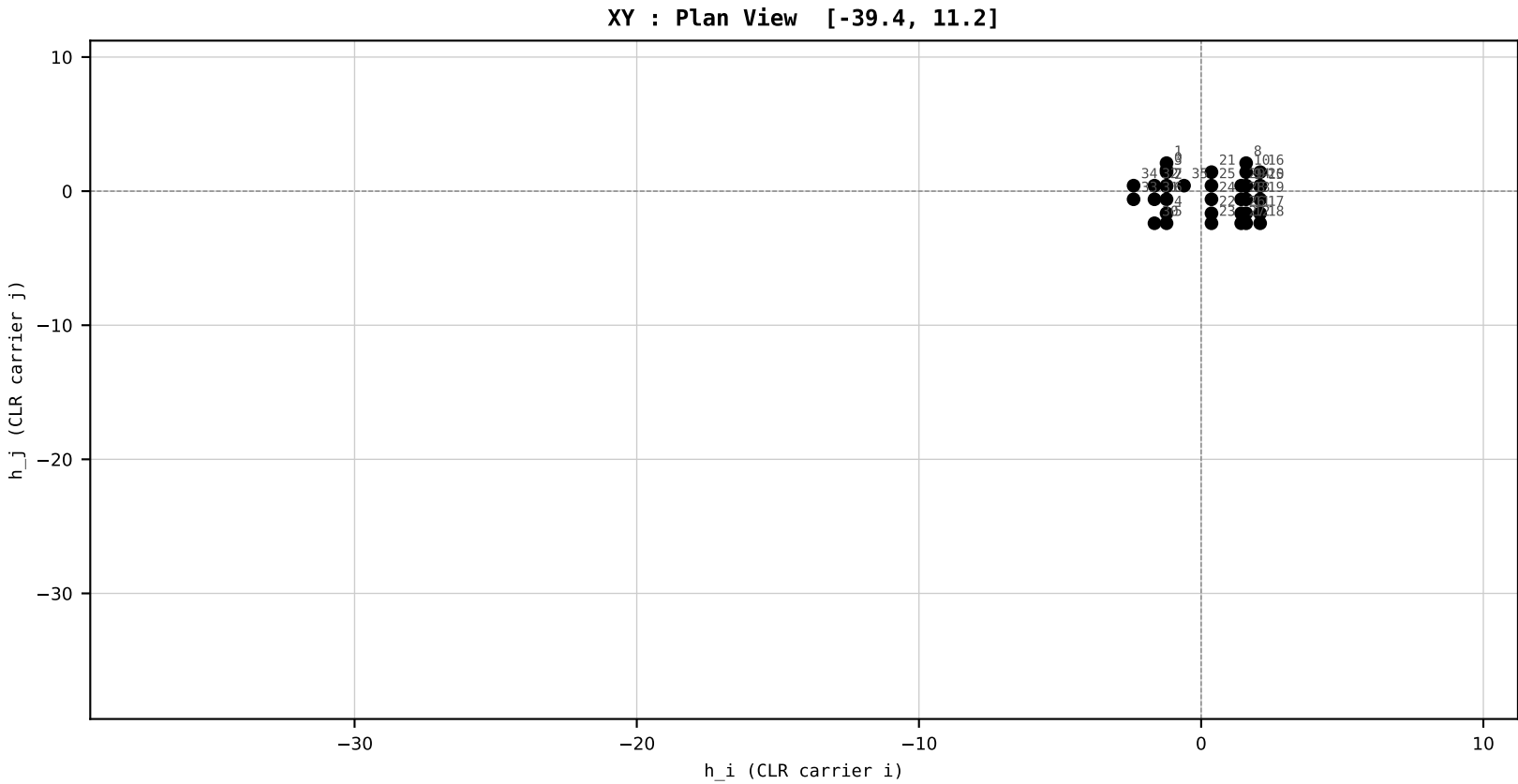
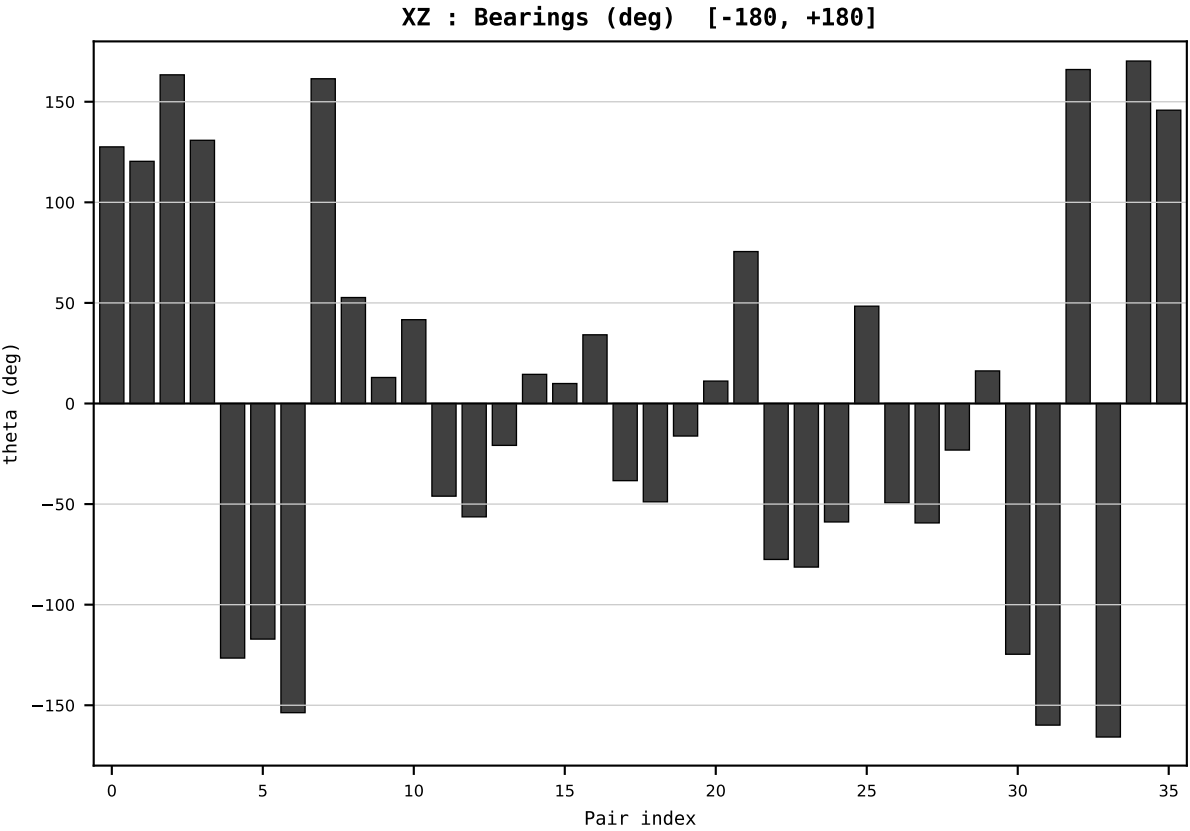
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2019
t : 18 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.264929
Ring = Hs-2
E_metric= 19.5686
kappa_HS= 88.69
omega = 3.7204 deg
d_A = 0.289310
Helm = Coal
Helm d = -0.159435
DR = 4.485 HLR
DR ratio= 88.7x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=19 | Year 2020 | D=9 N=25 pairs=36

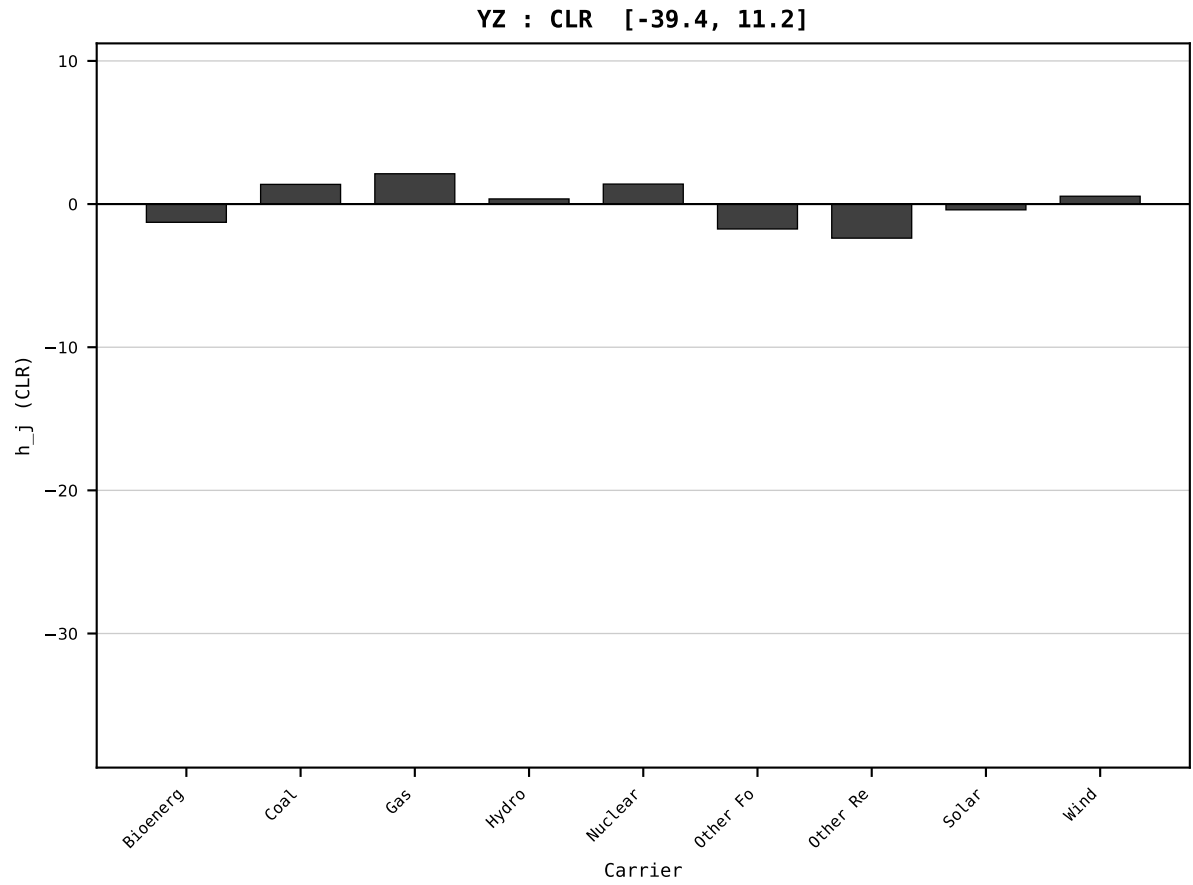
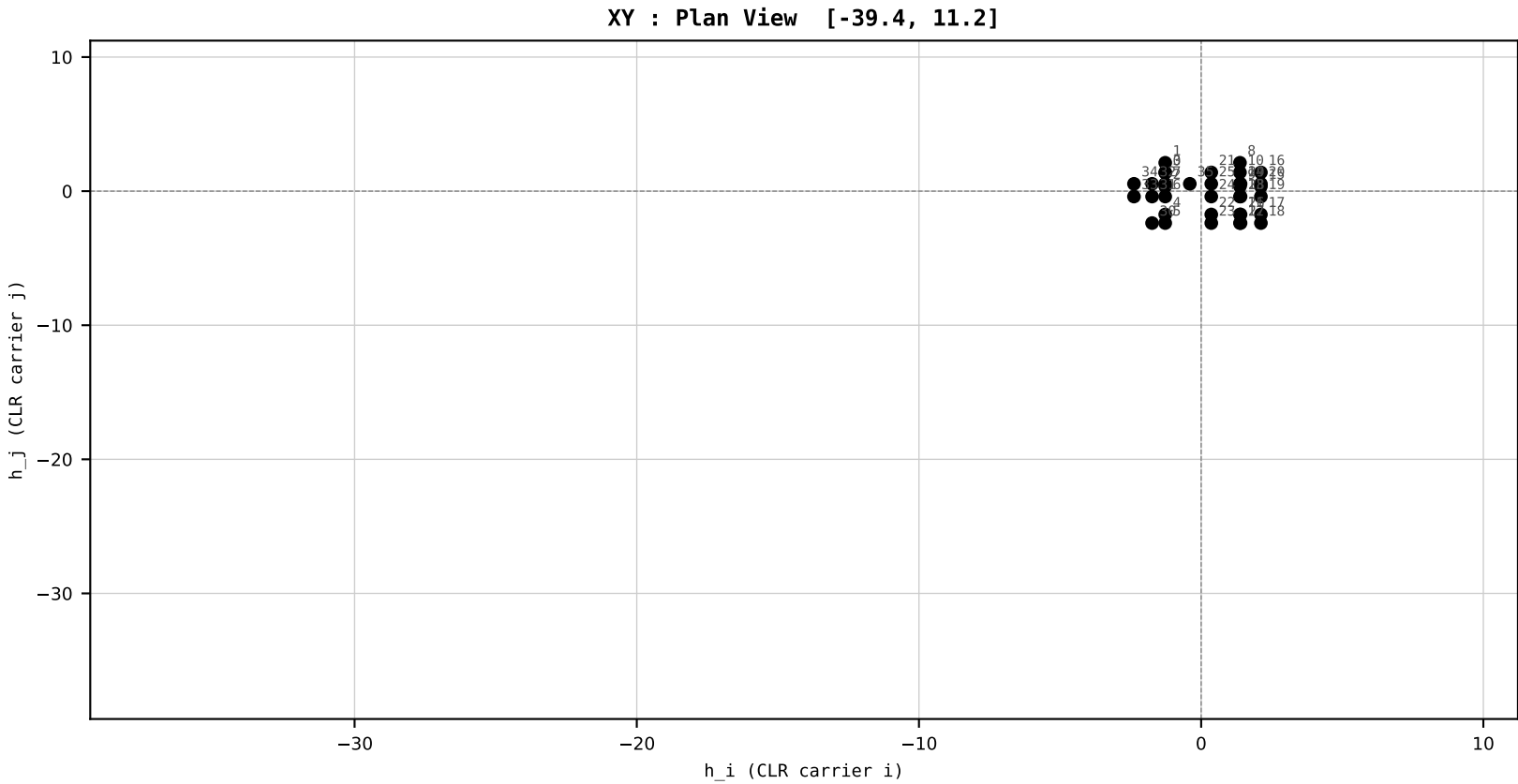
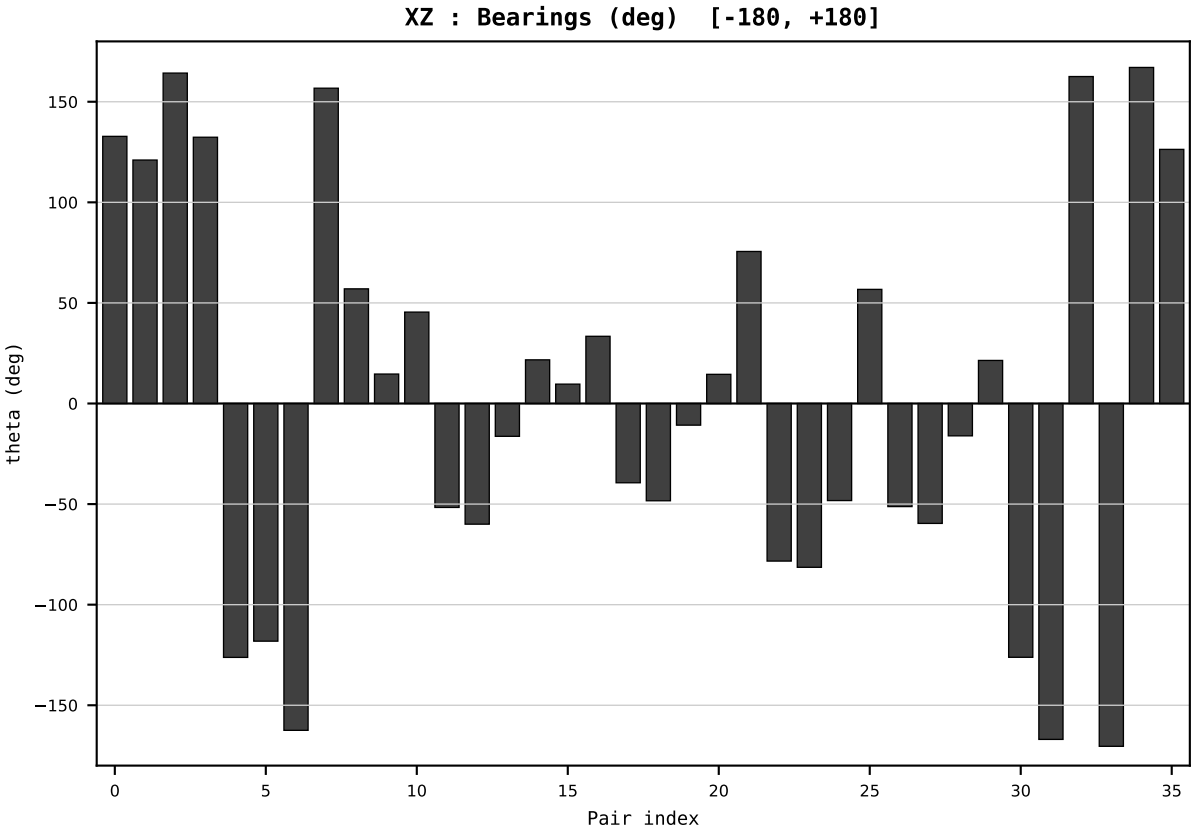
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2020
t : 19 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.259091
Ring = Hs-2
E_metric= 19.2298
kappa_HS= 89.83
omega = 4.4528 deg
d_A = 0.344356
Helm = Coal
Helm d = -0.218685
DR = 4.498 HLR
DR ratio= 89.8x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=20 | Year 2021 | D=9 N=25 pairs=36

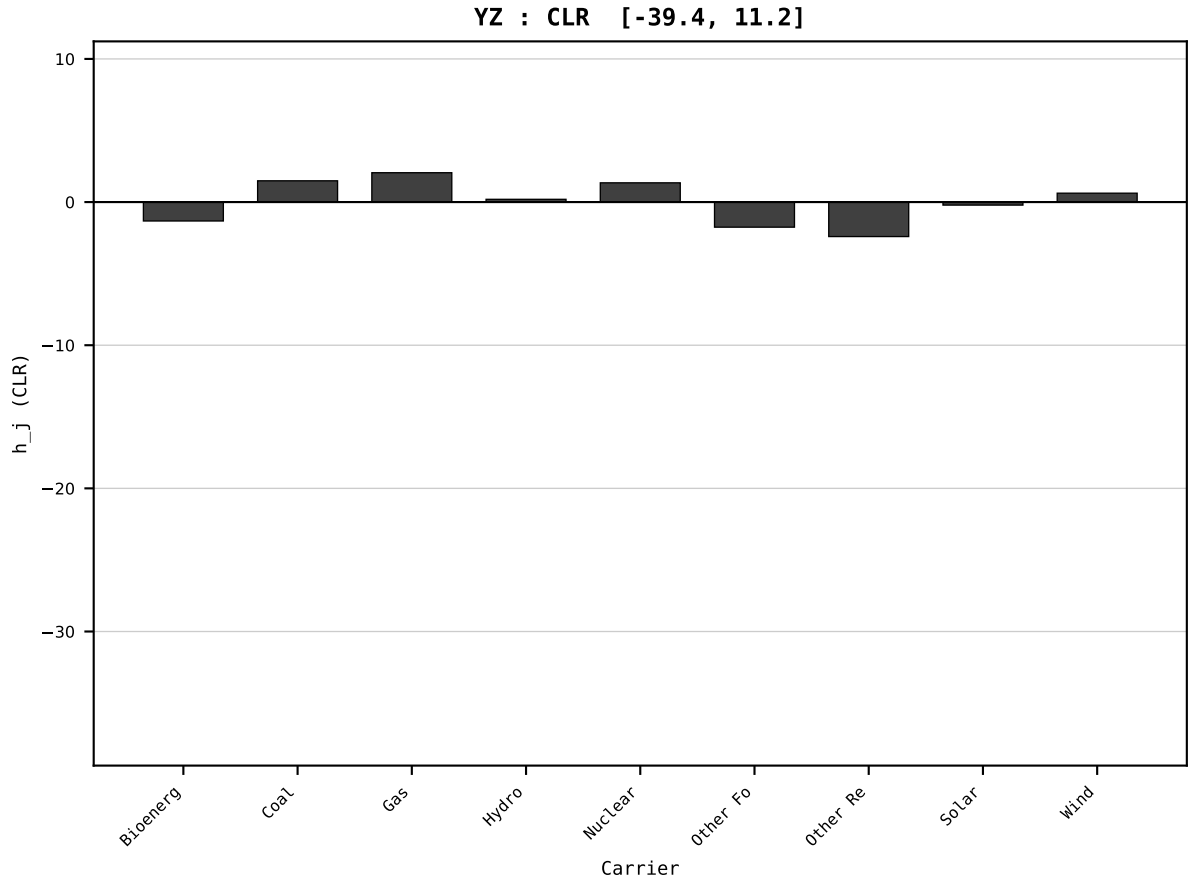
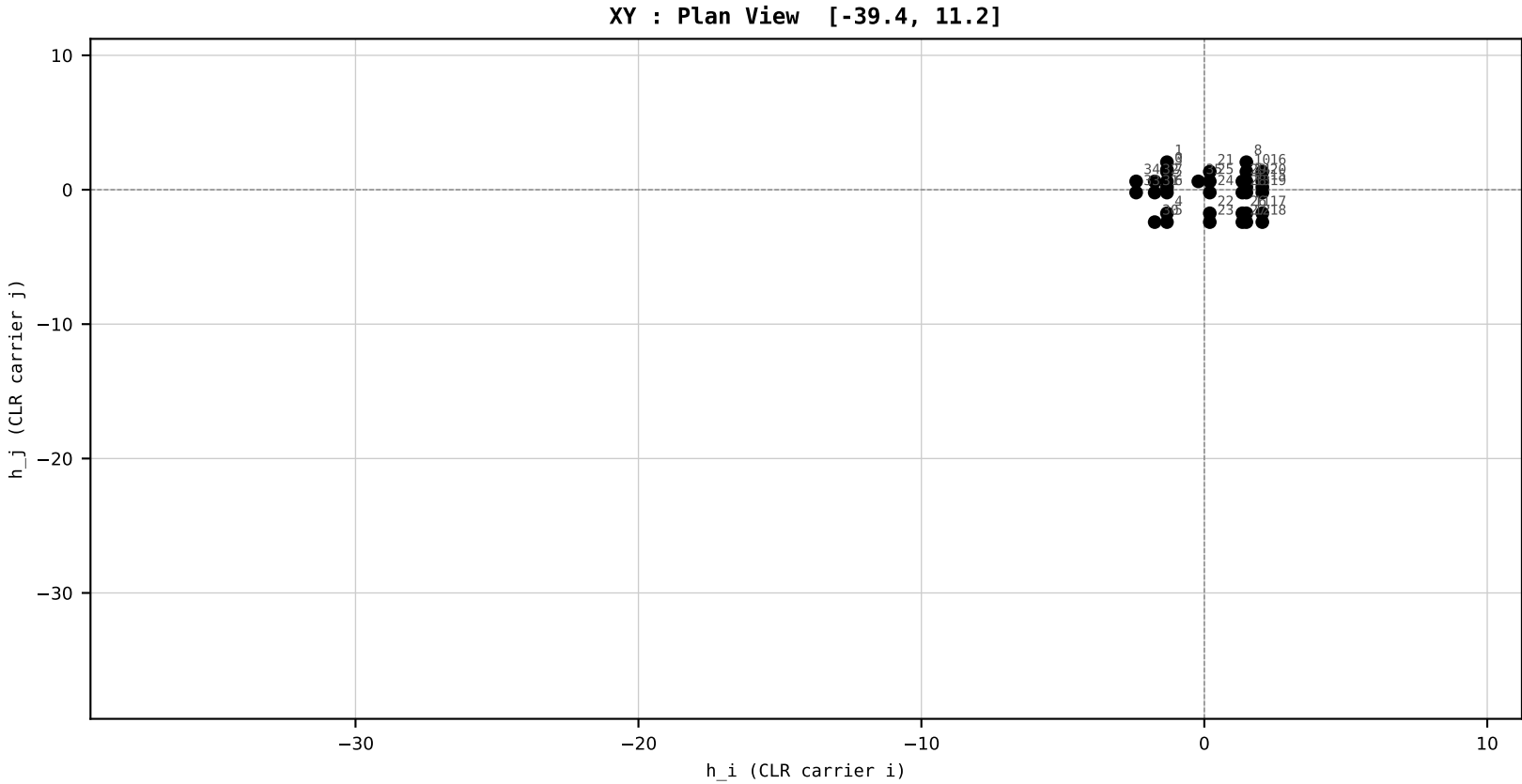
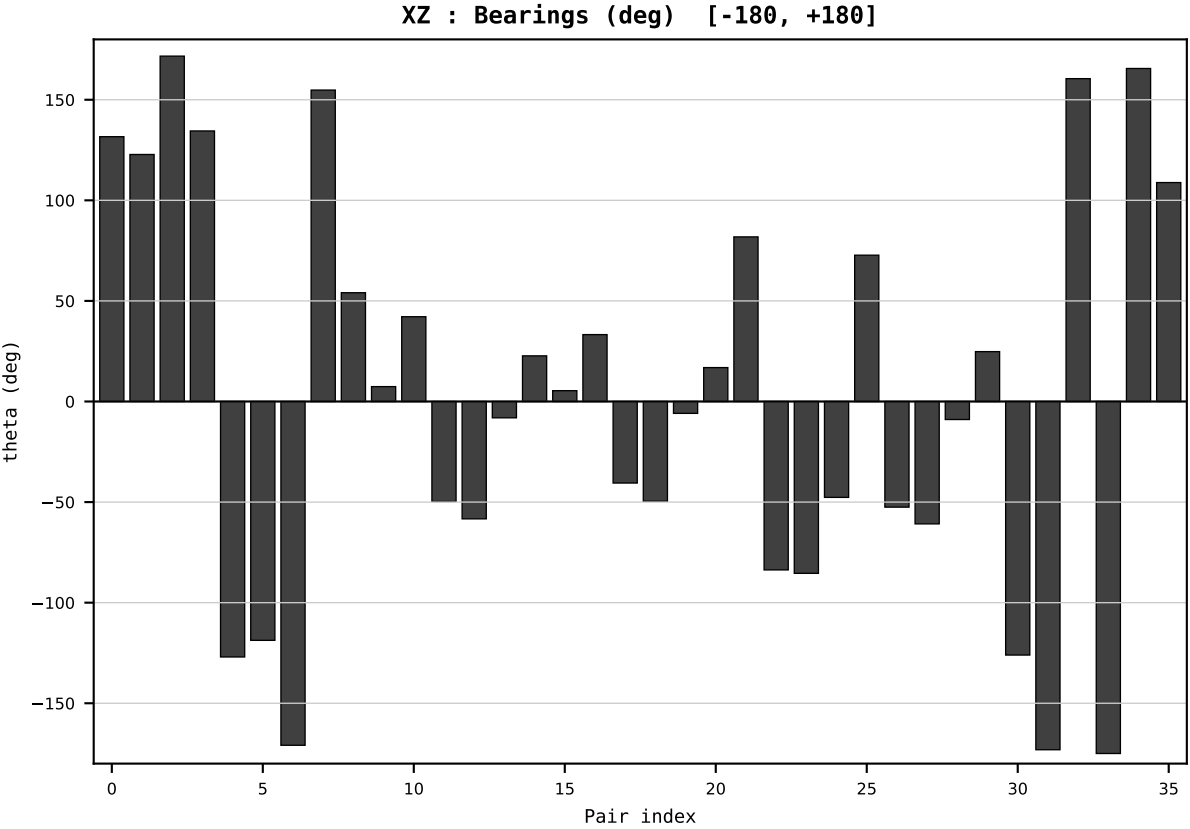
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2021
t : 20 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.250253
Ring = Hs-2
E_metric= 19.3124
kappa_HS= 86.48
omega = 3.9579 deg
d_A = 0.303328
Helm = Solar
Helm d = +0.190528
DR = 4.460 HLR
DR ratio= 86.5x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=21 | Year 2022 | D=9 N=25 pairs=36

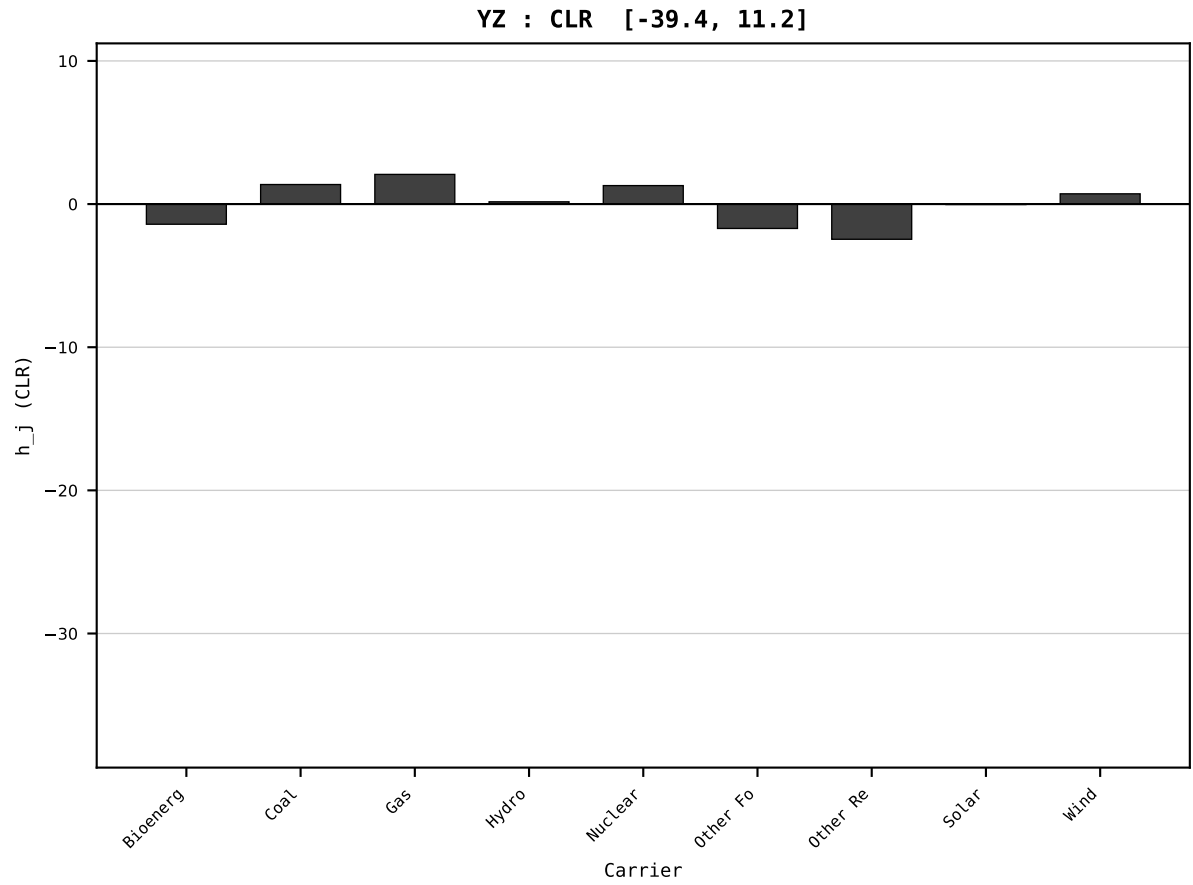
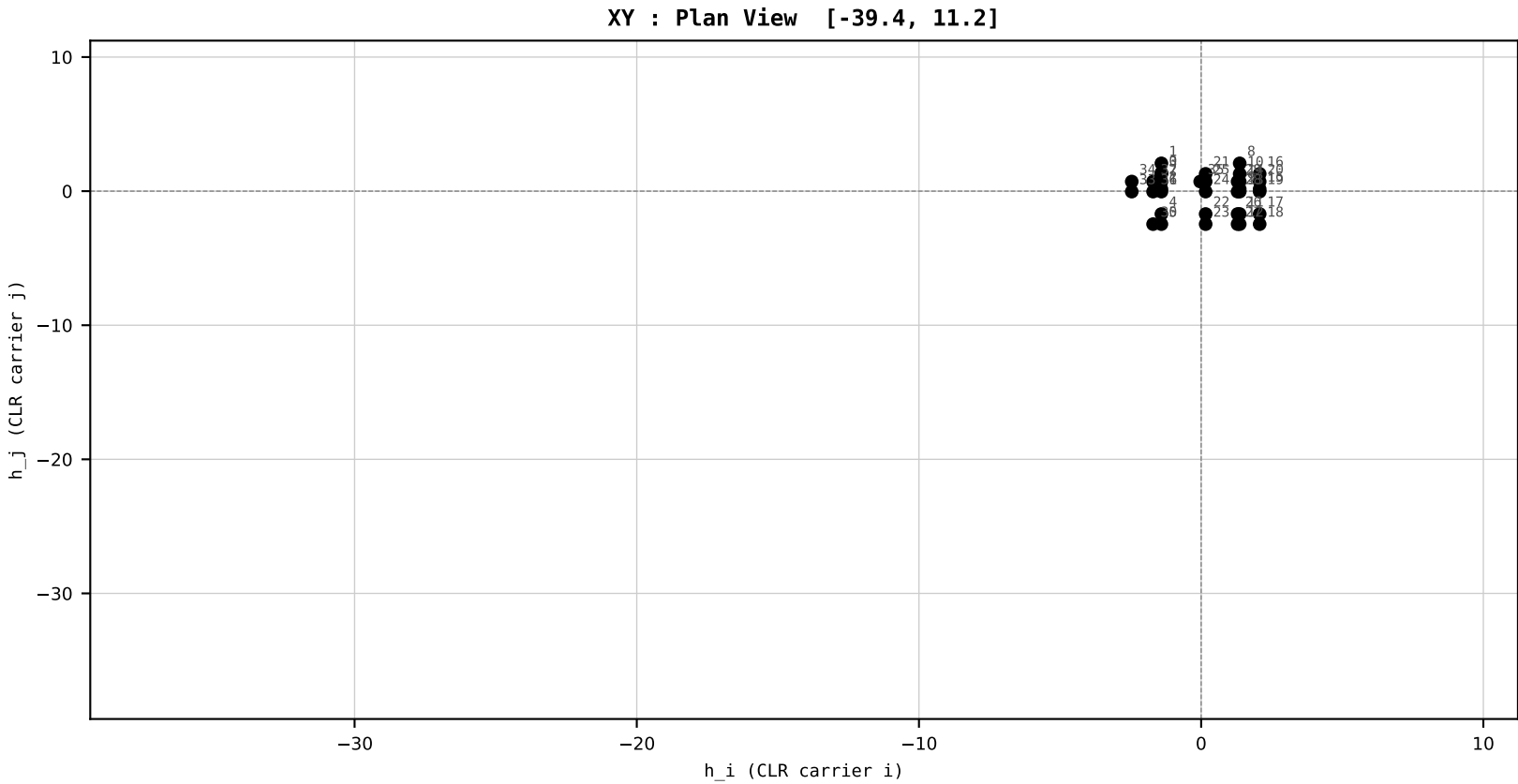
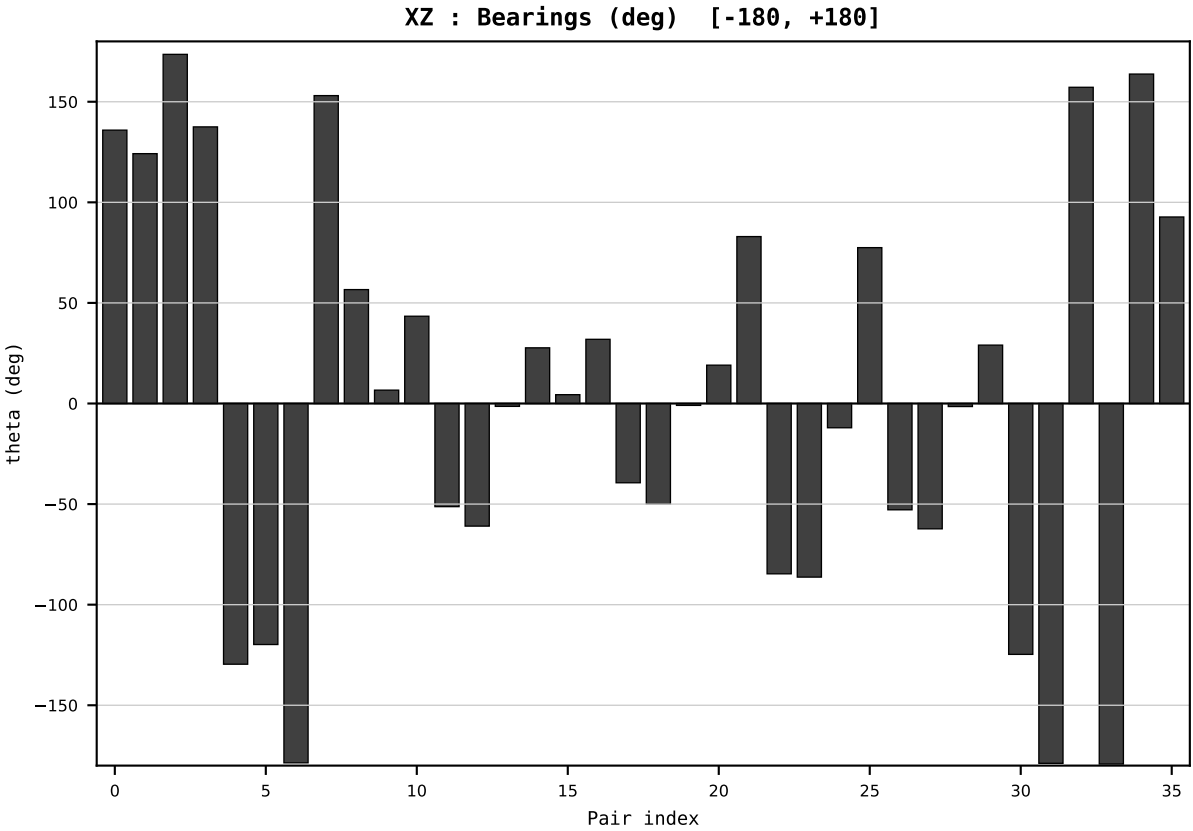
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2022
t : 21 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.246637
Ring = Hs-2
E_metric= 19.3055
kappa_HS= 93.00
omega = 3.5094 deg
d_A = 0.269105
Helm = Solar
Helm d = +0.177922
DR = 4.533 HLR
DR ratio= 93.0x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=22 | Year 2023 | D=9 N=25 pairs=36

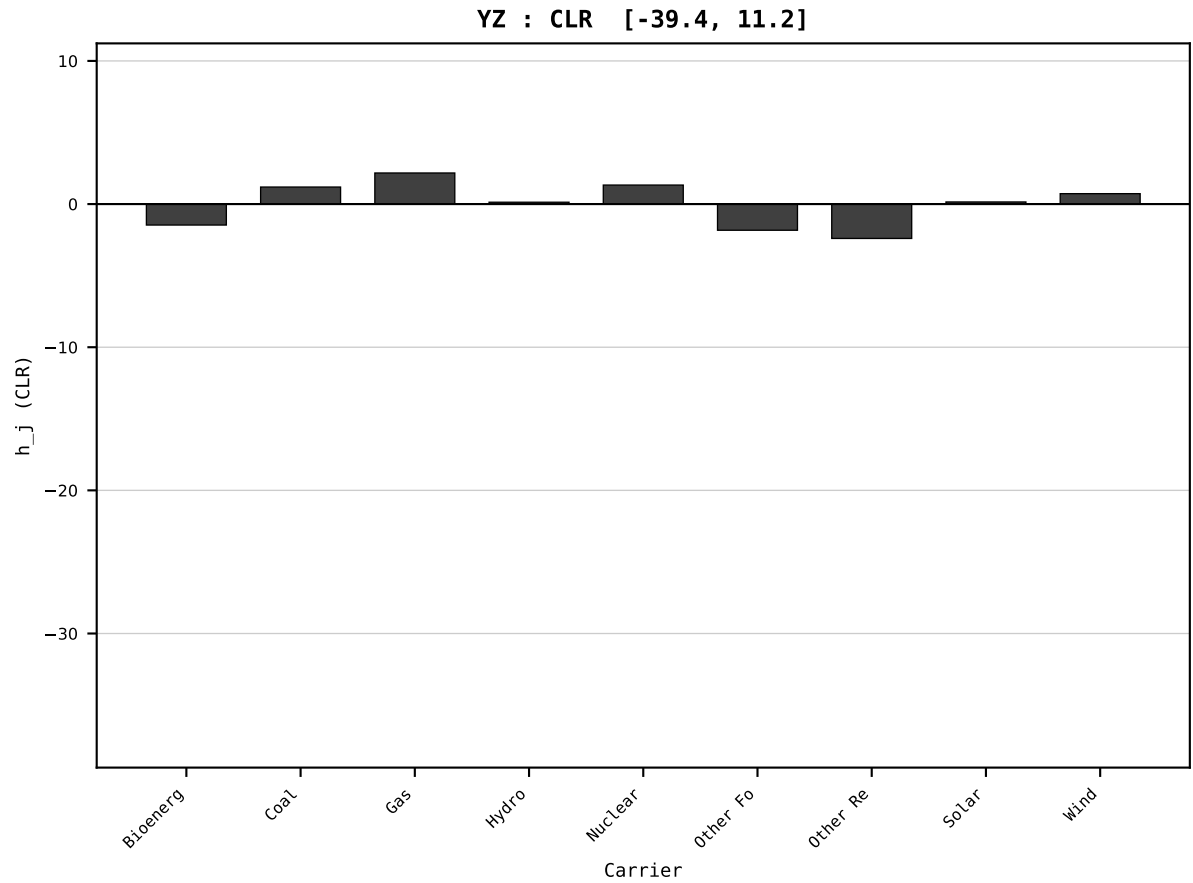
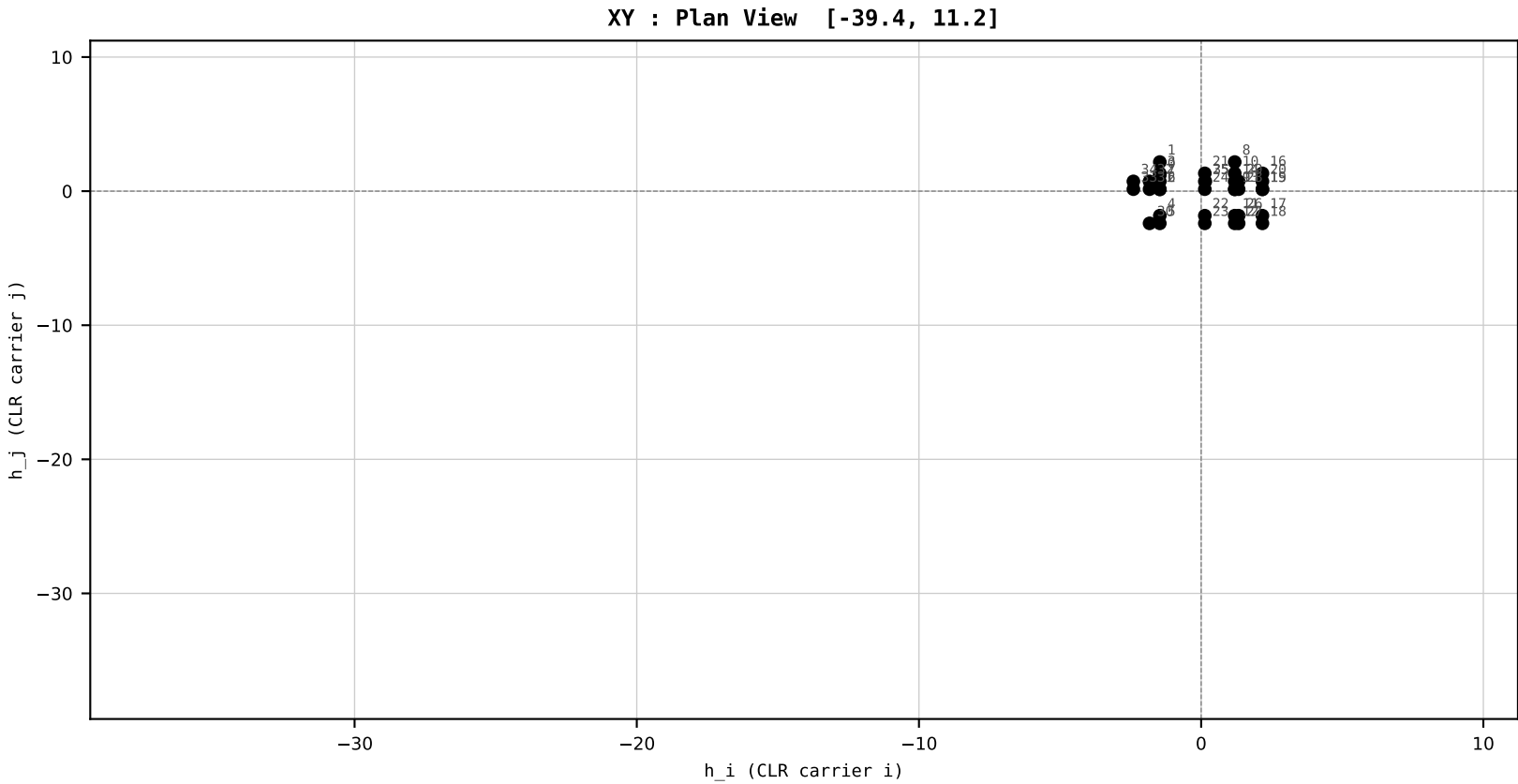
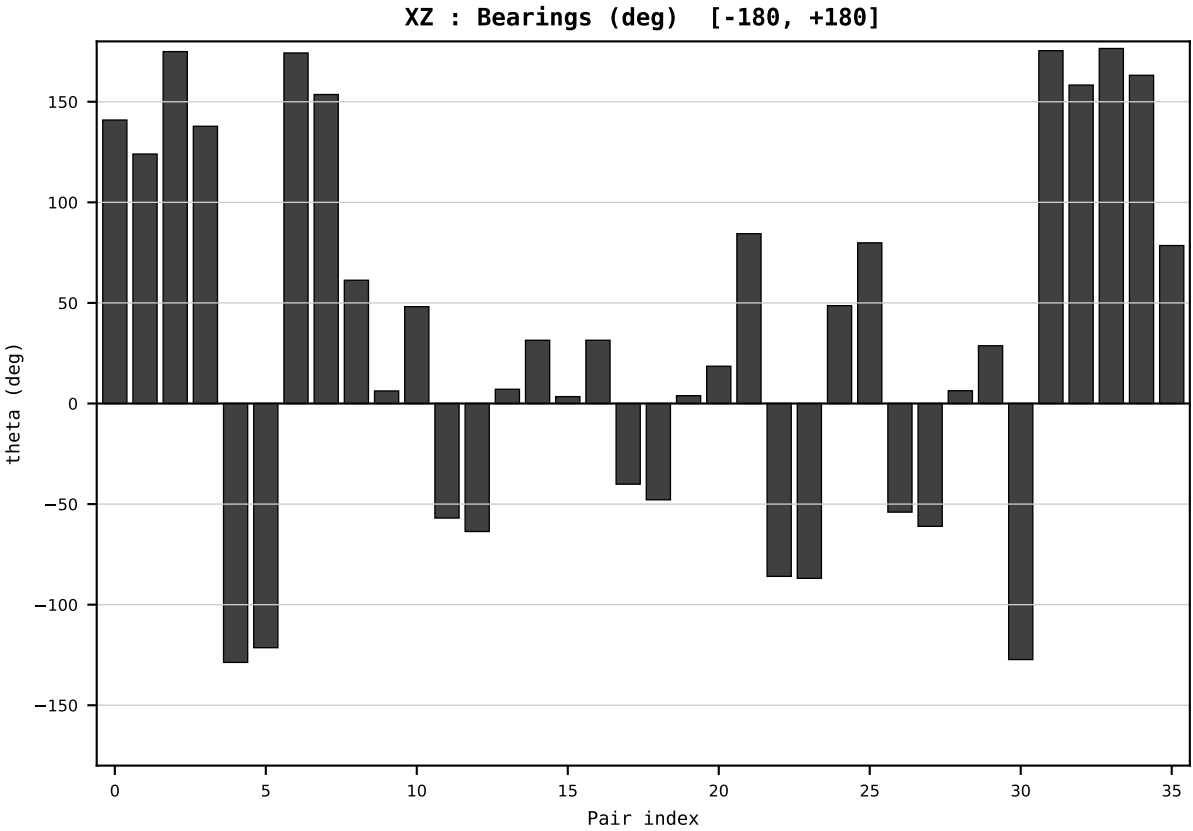
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2023
t : 22 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.258277
Ring = Hs-2
E_metric= 19.7208
kappa_HS= 96.83
omega = 4.0175 deg
d_A = 0.313215
Helm = Solar
Helm d = +0.181686
DR = 4.573 HLR
DR ratio= 96.8x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



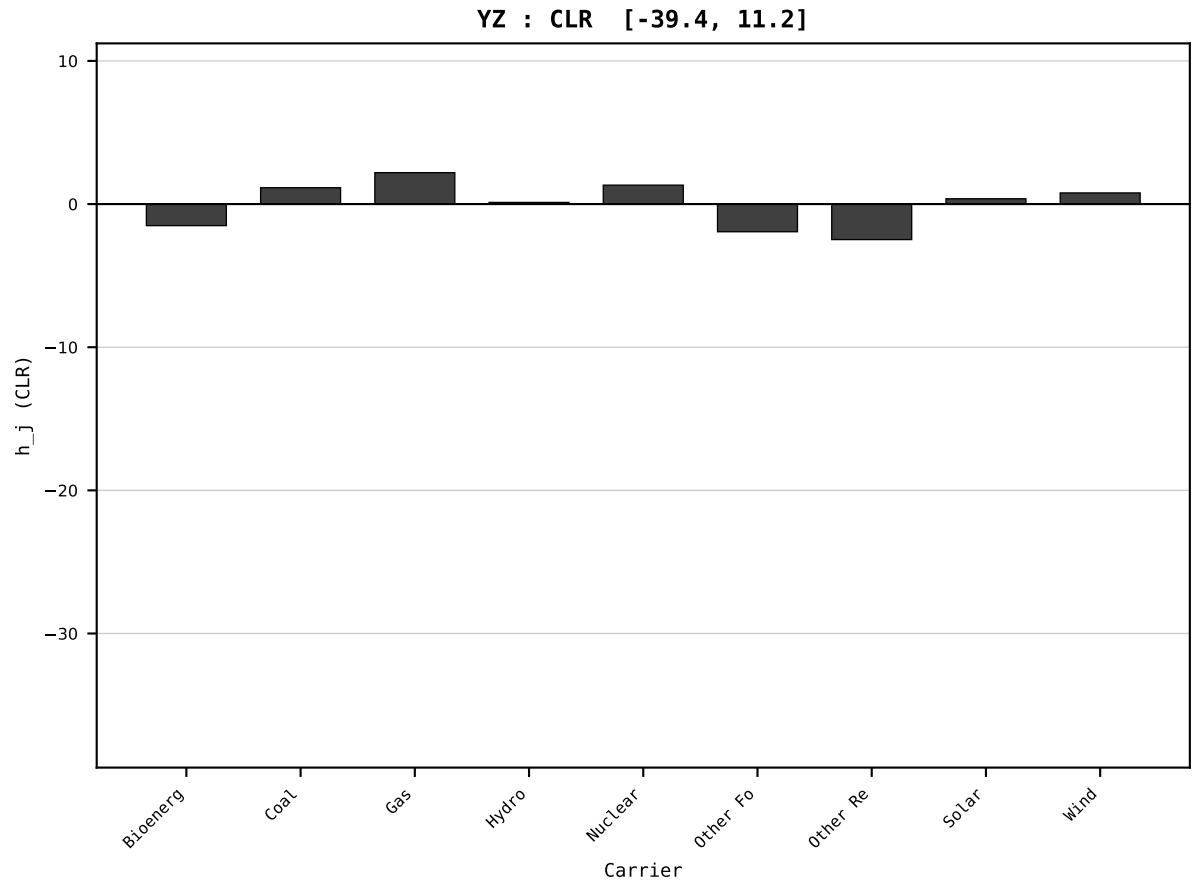
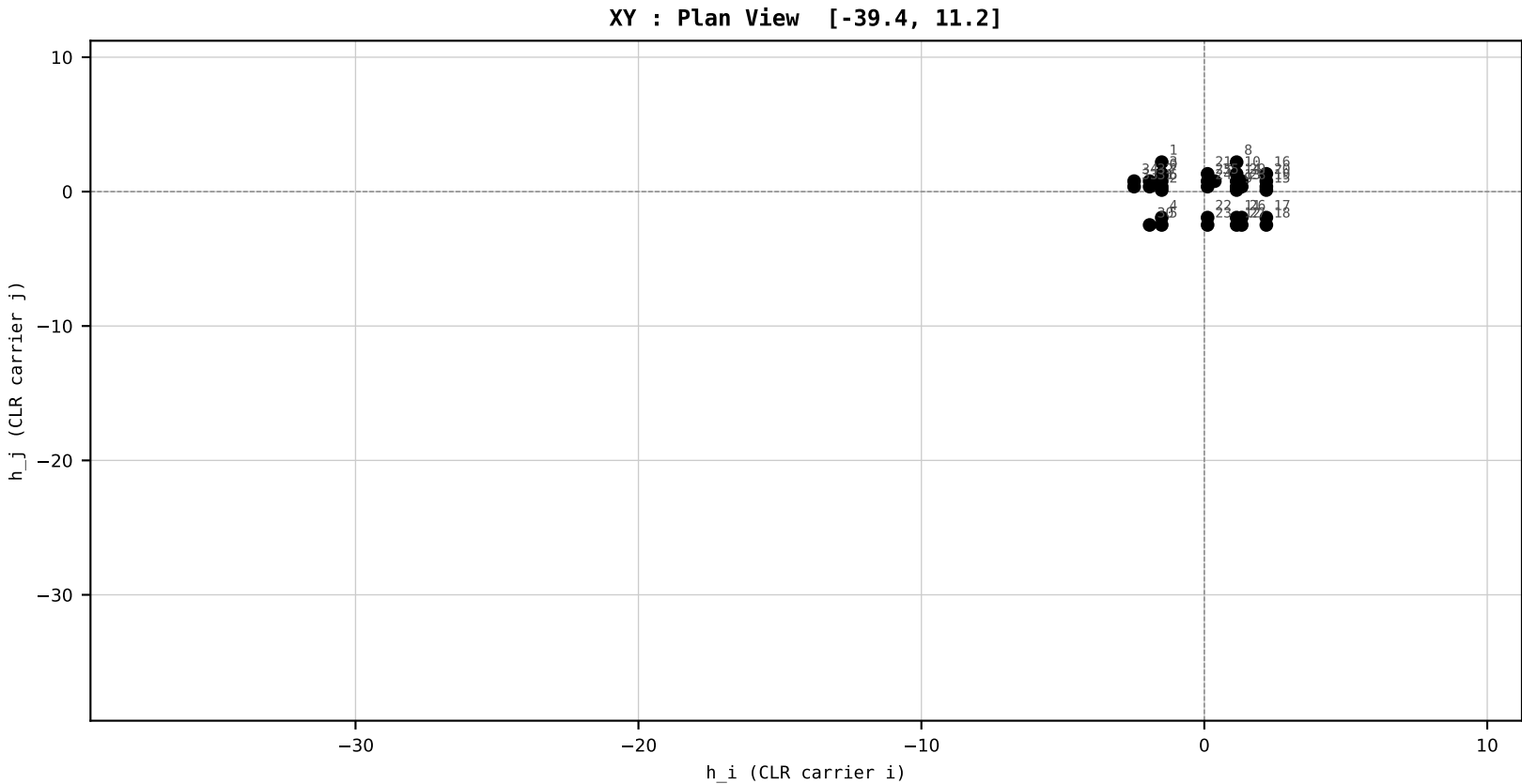
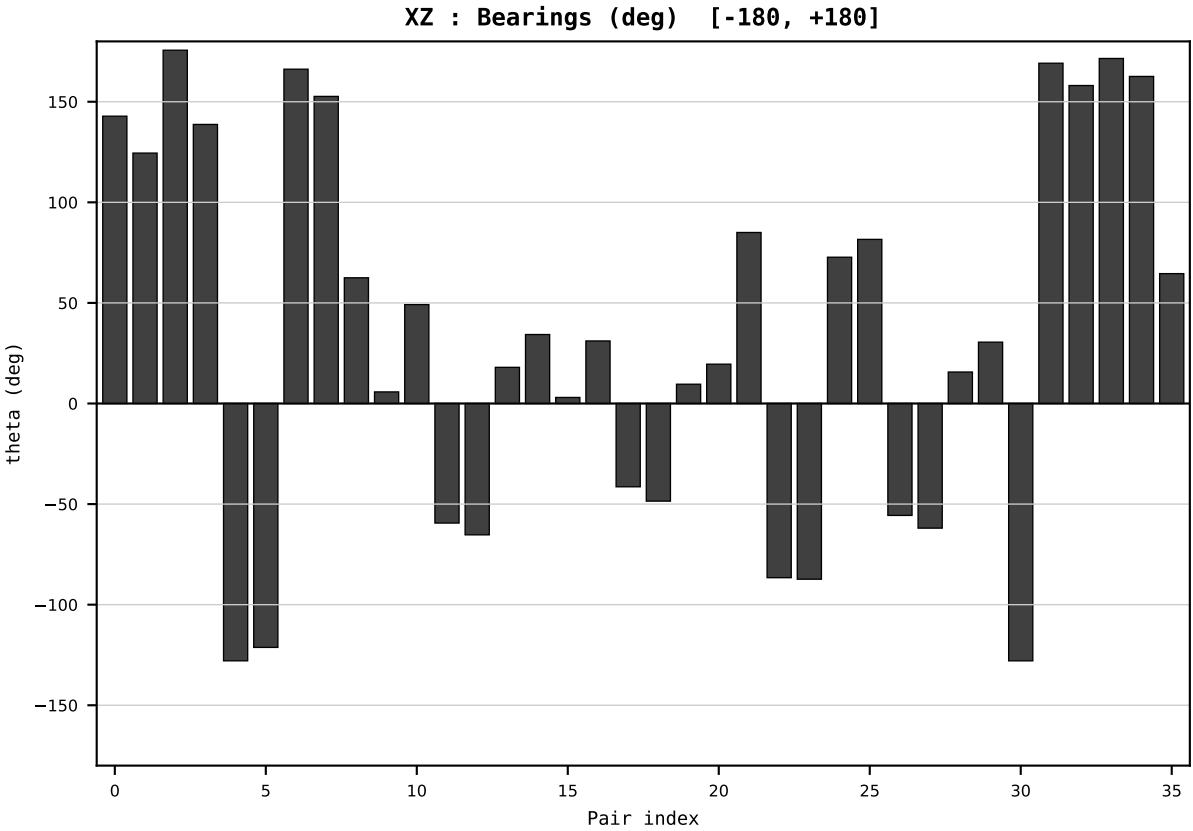
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2024
t : 23 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.256233
Ring = Hs-2
E_metric= 20.7697
kappa_HS= 107.02
omega = 3.1307 deg
d_A = 0.272026
Helm = Solar
Helm d = +0.222573
DR = 4.673 HLR
DR ratio= 107.0x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



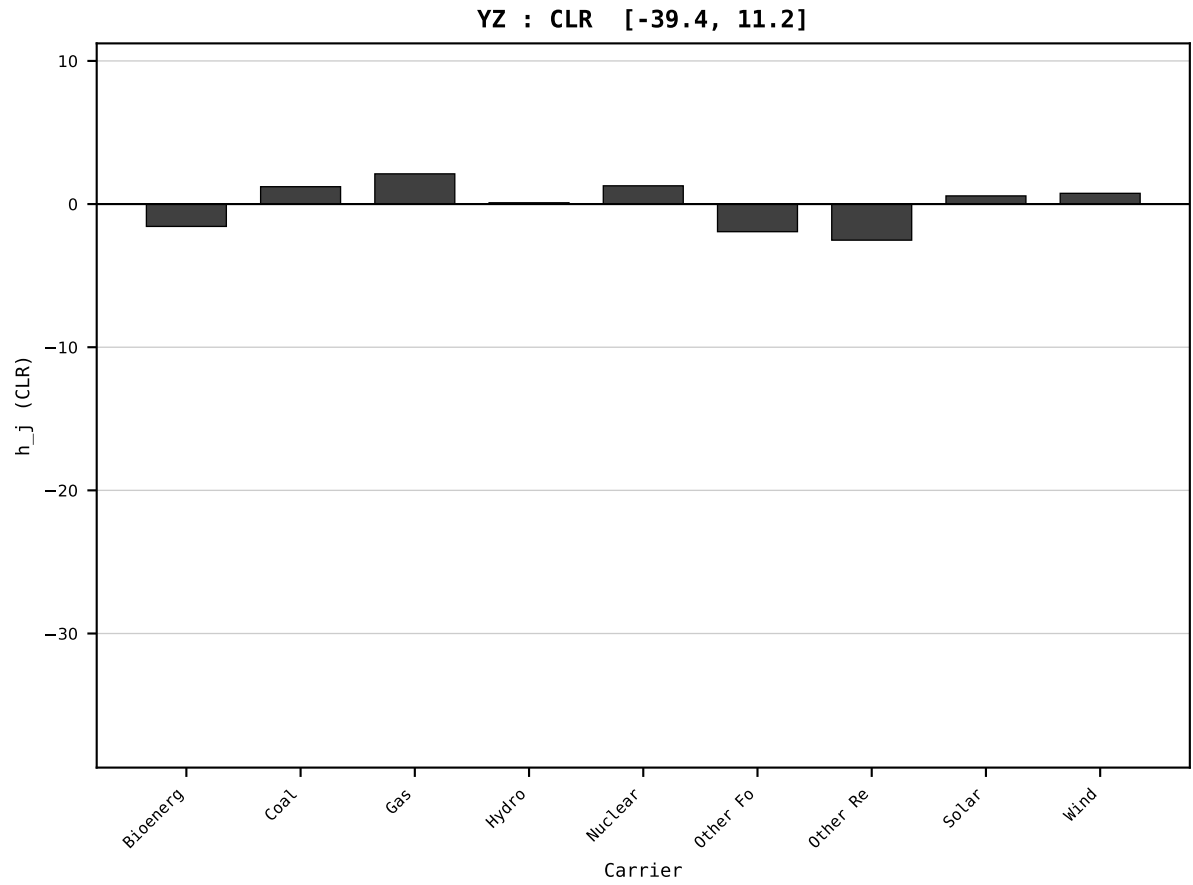
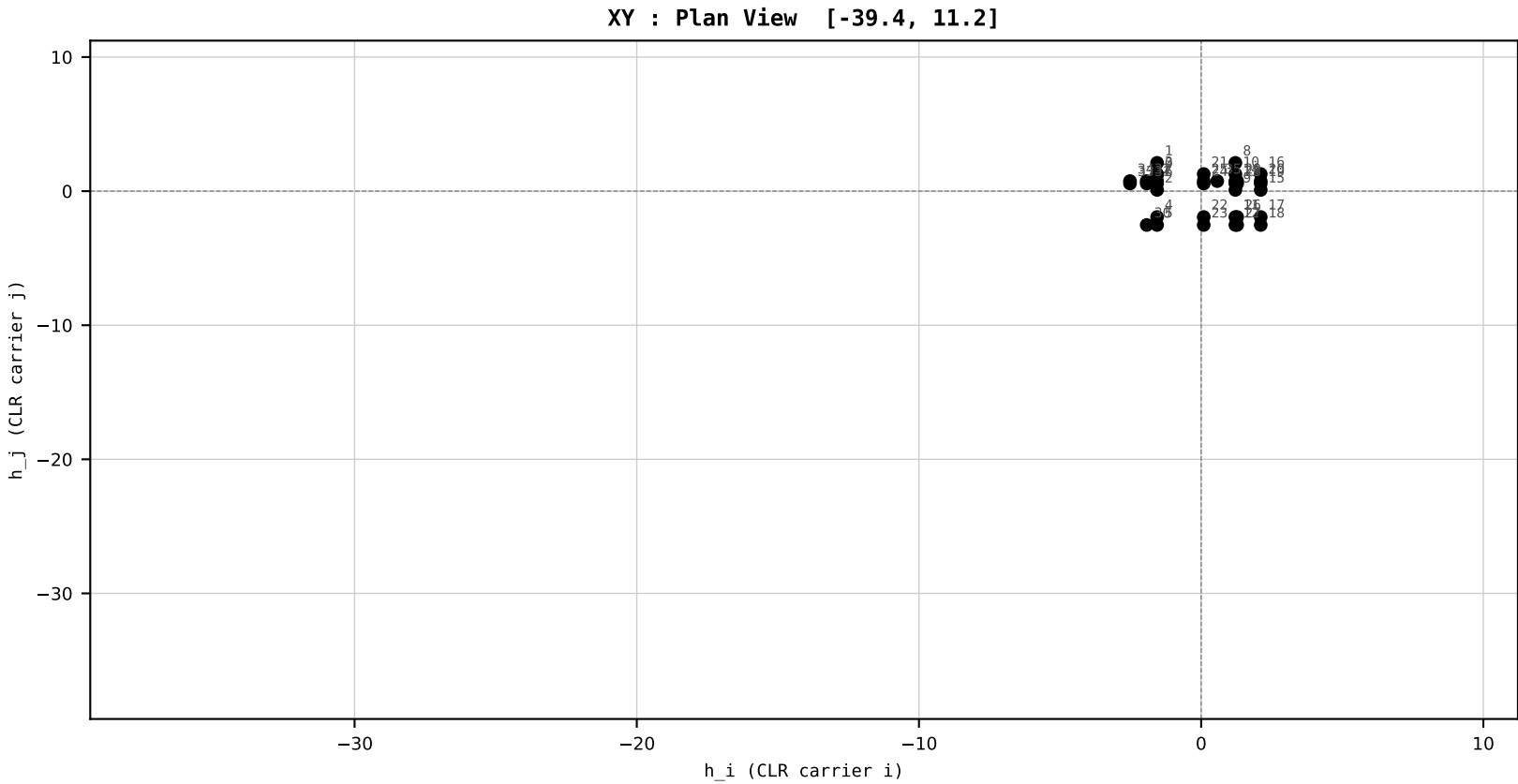
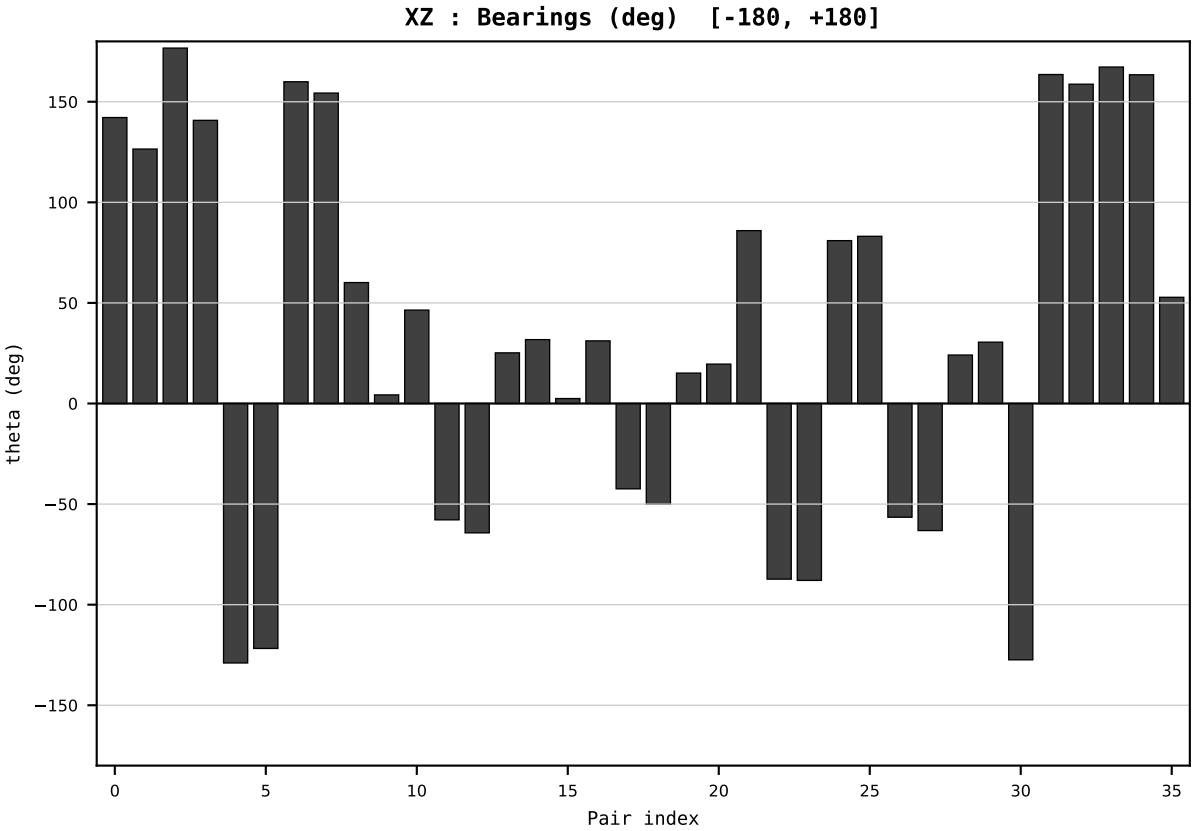
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

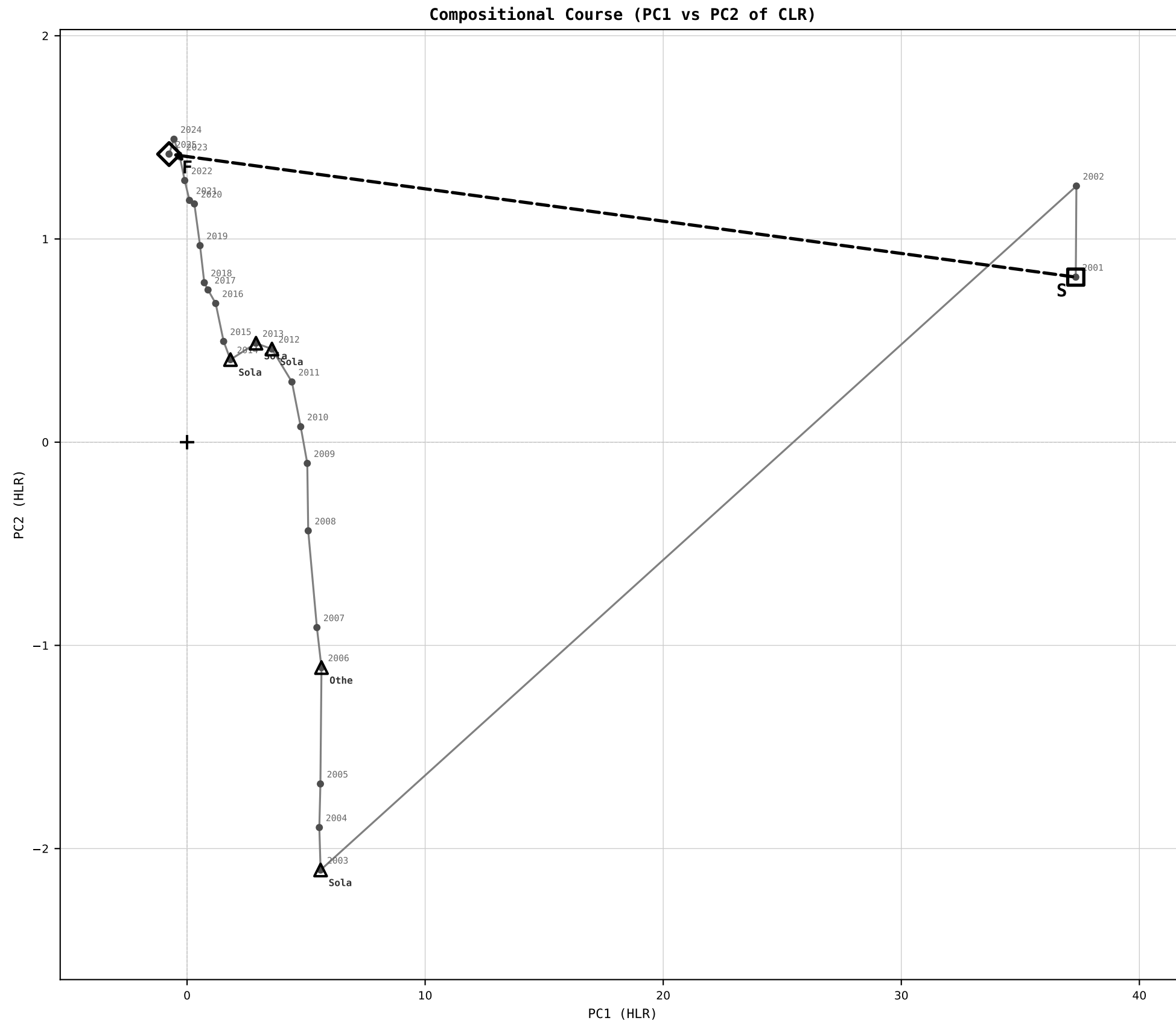
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2025
t : 24 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.240009
Ring = Hs-2
E_metric= 20.9704
kappa_HS= 102.78
omega = 3.0581 deg
d_A = 0.244792
Helm = Solar
Helm d = +0.200013
DR = 4.633 HLR
DR ratio= 102.8x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind





COURSE METRICS

Scale Provenance:

$$Y_j \rightarrow x_j = Y_j / T \rightarrow h_j = \text{CLR}$$

→ PC projection

Unit: HLR (Higgins Log-Ratio)

Net distance = 38.0885 HLR

Path length = 41.8379 HLR

Directness = 0.9104

DR start	= 42.090 HLR
----------	--------------

```
ratio = 1903969992697630976.0x
```

DR final	= 4.633 HLR
----------	-------------

$$\text{ratio} = 102.8x$$

```
E_metric start = 1411.9935
```

```
E_metric final = 20.9704
```

PC1 loadings:

Solar -0.938

Other Fo +0.159

Coal	+0.146
------	--------

PC2 loadings:

```

Wind      +0.831

```

Other Fo -0.385

Coal	-0.324
------	--------

Top displacement events:

2003 Solar d=31.925

2014 Solar d=1.080

2014 Solar $d=1.000$
2012 Solar $d=0.882$

2012 Total $d=0.882$
2006 Other Fo $d=0.708$

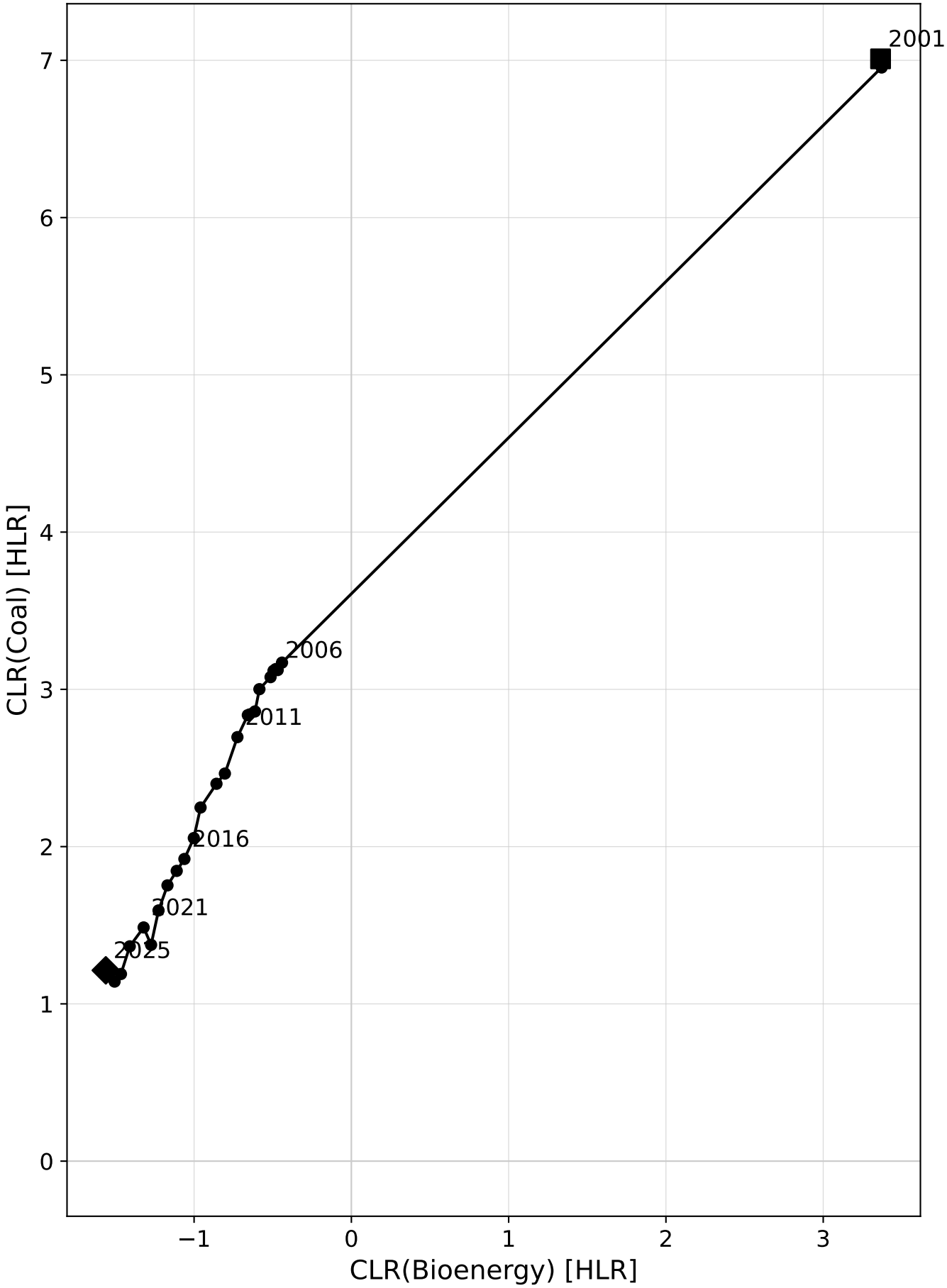
2013 Solar $d=0.691$

The instrument reads.

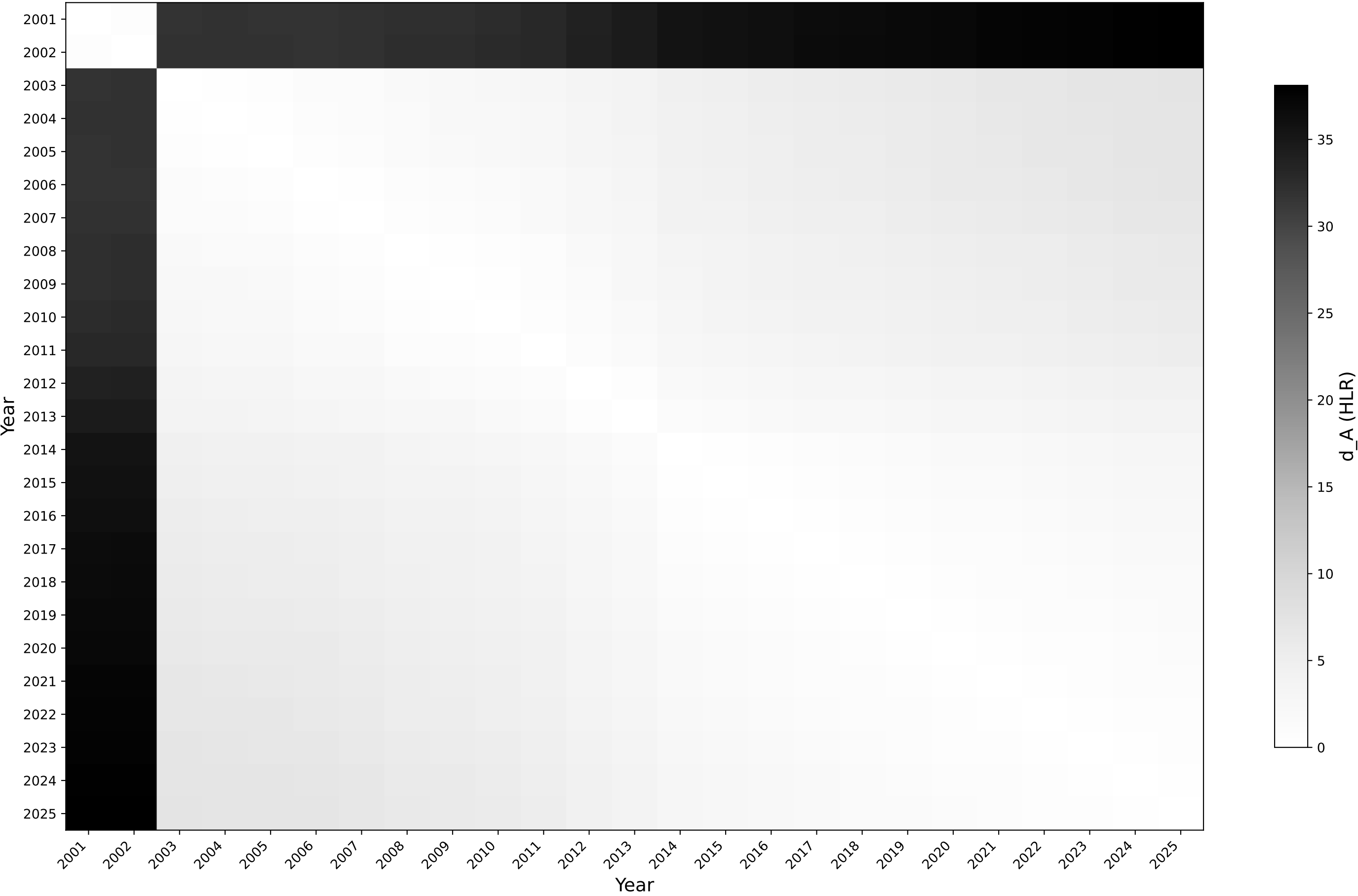
The expert decides.

The expert decides:
The loop stays open.

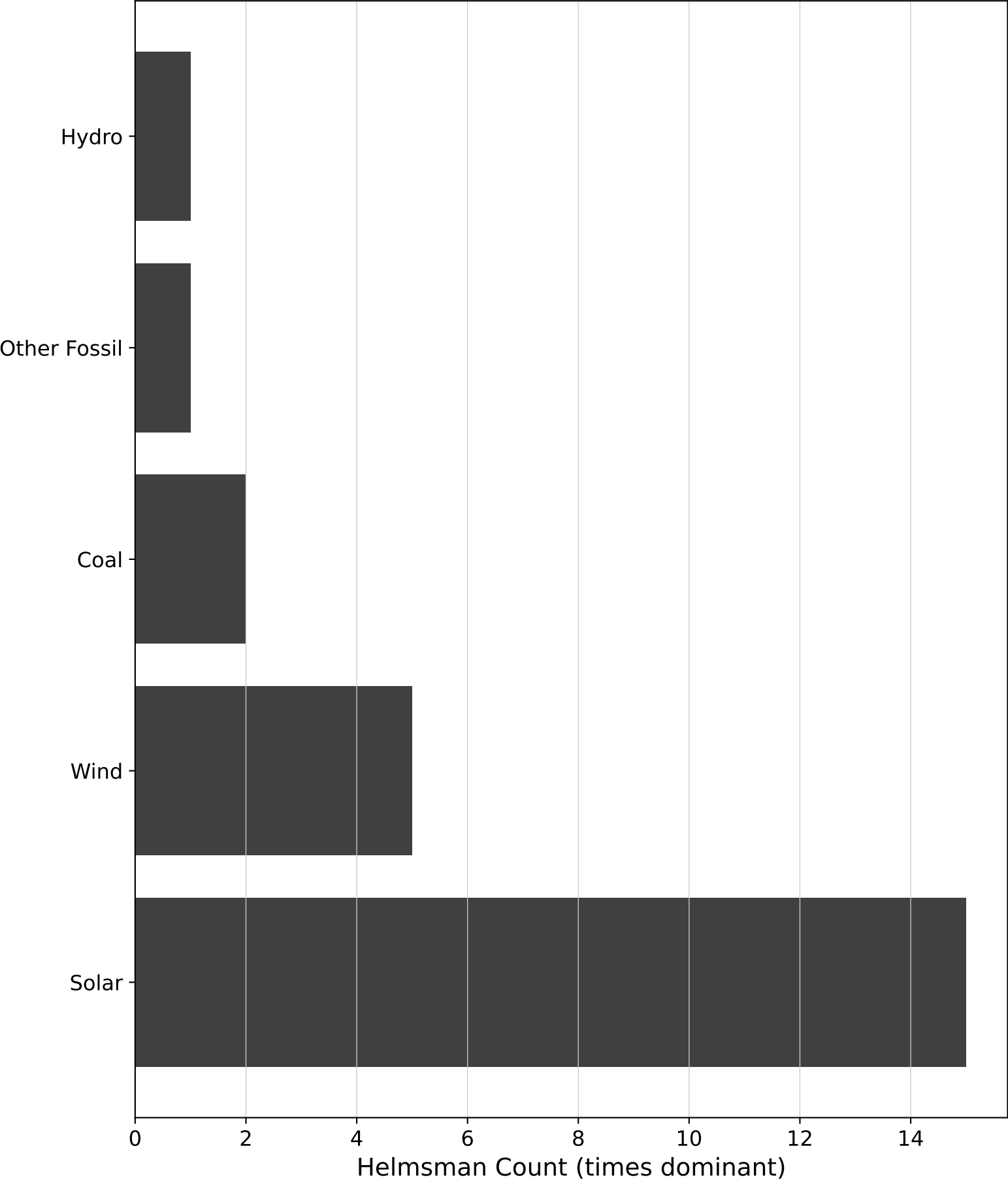
SYSTEM COURSE PLOT — XY Plan View (CLR Space)



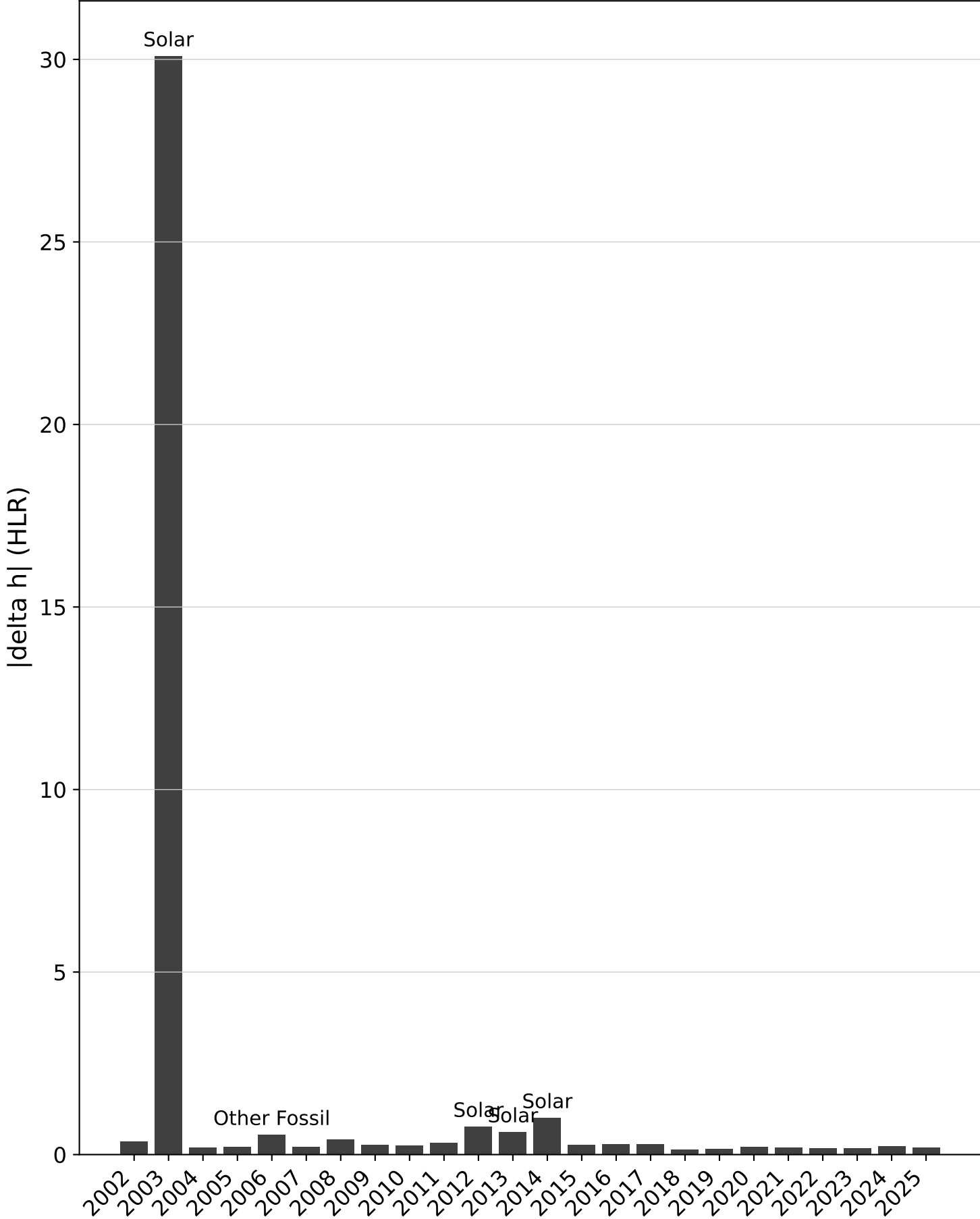
STAGE 2 — Aitchison Distance Matrix (all year pairs)



HELMSMAN FREQUENCY



HELMSMAN DISPLACEMENT PER YEAR



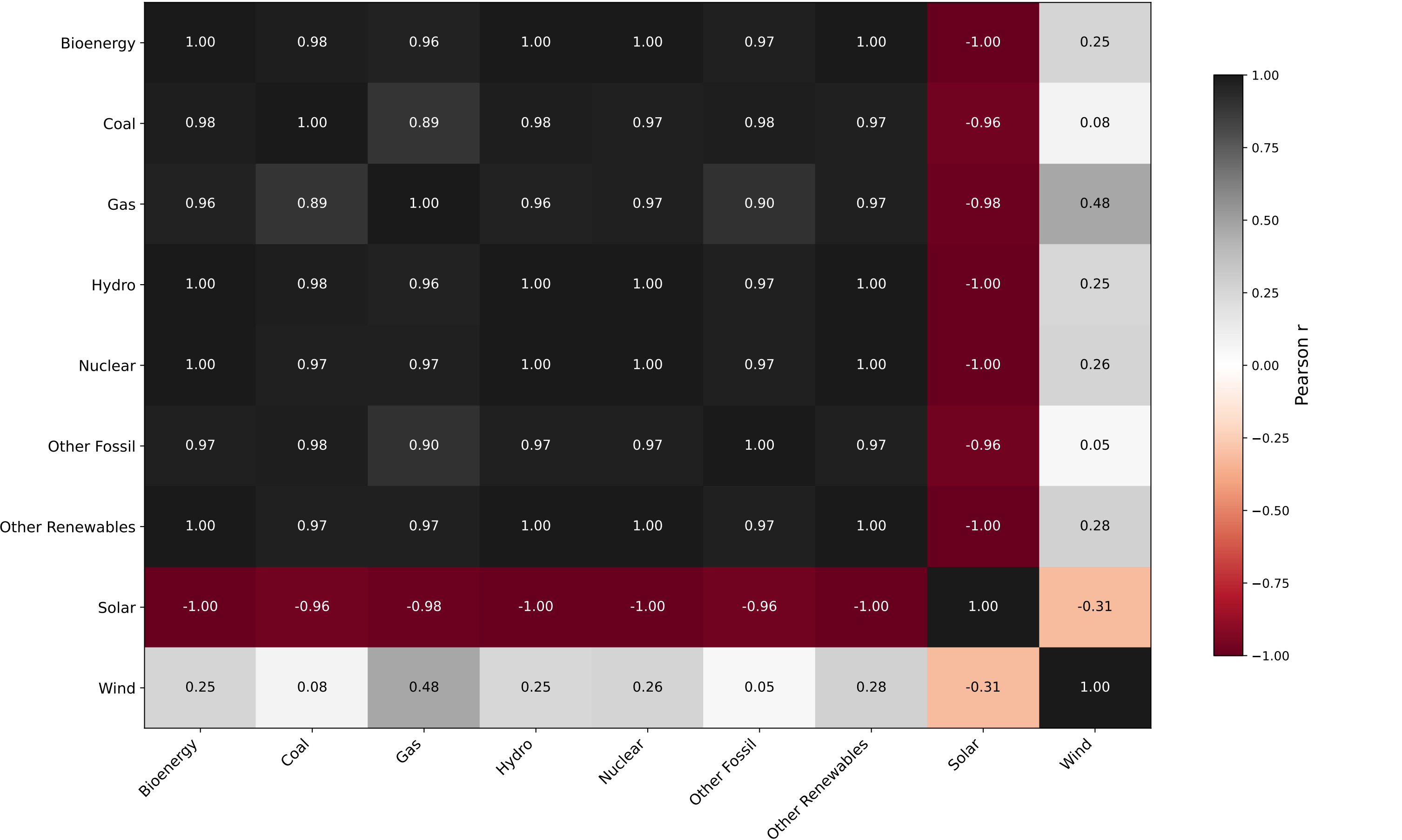
STAGE 2 — Pairwise Divergence Ranking (Top 15)

Rank	Year A	Year B	d_A (HLR)	Angle (deg)	Dominant Carrier
1	2002	2025	38.1117	92.7	Solar
2	2001	2025	38.0885	93.0	Solar
3	2002	2024	37.9059	90.1	Solar
4	2001	2024	37.8834	90.4	Solar
5	2002	2023	37.6593	87.1	Solar
6	2001	2023	37.6357	87.4	Solar
7	2002	2022	37.4542	84.5	Solar
8	2001	2022	37.4295	84.7	Solar
9	2002	2021	37.2539	81.9	Solar
10	2001	2021	37.2288	82.1	Solar
11	2002	2020	37.0480	79.2	Solar
12	2001	2020	37.0230	79.4	Solar
13	2002	2019	36.8108	76.1	Solar
14	2001	2019	36.7835	76.3	Solar
15	2002	2018	36.6338	73.7	Solar

STAGE 3 — Triadic Metric Triangle Area Ranking (Top 10)

Rank	Year A	Year B	Year C	Area (HLR^2)	Imbalance
1	2002	2003	2024	67.5061	5.31
2	2002	2003	2025	66.7205	5.24
3	2001	2003	2024	66.0803	5.31
4	2002	2003	2023	65.6965	5.48
5	2001	2003	2025	65.2449	5.23
6	2001	2003	2023	64.3308	5.47
7	2002	2003	2022	63.5557	5.64
8	2002	2004	2024	63.5441	5.43
9	2002	2004	2025	62.7167	5.35
10	2001	2003	2022	62.2329	5.64

STAGE 3 — Carrier CLR Trajectory Correlation

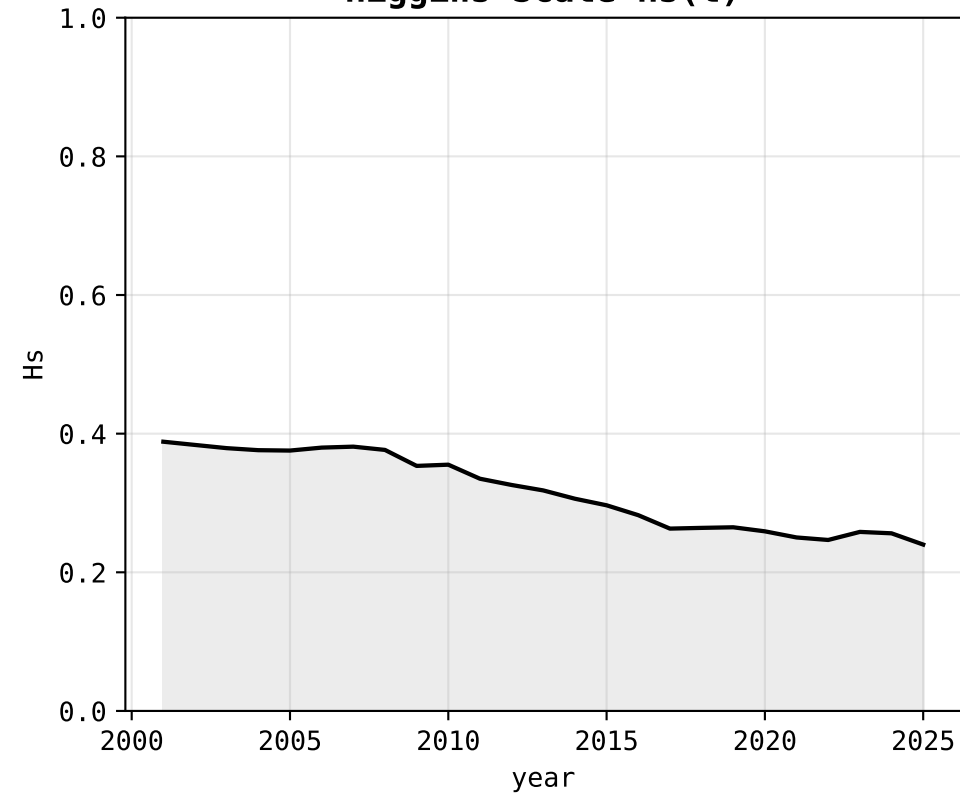


NAVIGATION SUMMARY — All Stage 1 Metrics by Year

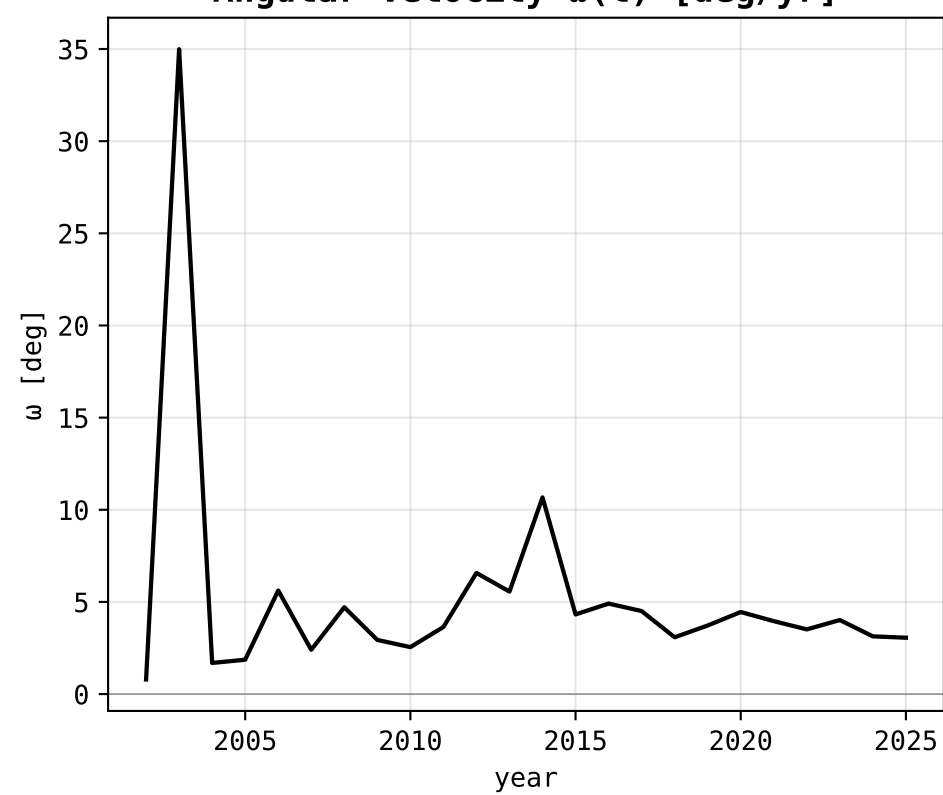
Year	Hs	Ring	E_metric	omega(deg)	d_A	Helmsman	kappa_max
2001	0.3884	Hs-3	895.1	0.00	0.0000	---	1000000000000000
2002	0.3837	Hs-3	895.5	0.79	0.5229	Wind	1000000000000000
2003	0.3791	Hs-3	53.1	34.99	31.9255	Solar	7300
2004	0.3761	Hs-3	51.9	1.69	0.2289	Wind	6933
2005	0.3756	Hs-3	51.9	1.86	0.2343	Wind	7336
2006	0.3799	Hs-3	52.3	5.62	0.7080	Other Fossil	8259
2007	0.3812	Hs-3	49.7	2.40	0.3509	Hydro	6791
2008	0.3765	Hs-3	45.5	4.71	0.6427	Wind	4721
2009	0.3535	Hs-3	44.8	2.94	0.3488	Wind	4427
2010	0.3552	Hs-3	42.2	2.55	0.3512	Solar	3372
2011	0.3350	Hs-3	38.9	3.64	0.4789	Solar	2247
2012	0.3260	Hs-3	32.3	6.57	0.8815	Solar	931
2013	0.3181	Hs-3	27.4	5.56	0.6909	Solar	449
2014	0.3061	Hs-3	21.8	10.67	1.0804	Solar	215
2015	0.2967	Hs-2	20.8	4.32	0.3662	Solar	214
2016	0.2824	Hs-2	20.1	4.91	0.3937	Solar	213
2017	0.2630	Hs-2	19.4	4.50	0.3596	Solar	212
2018	0.2641	Hs-2	19.2	3.08	0.2374	Solar	225
2019	0.2649	Hs-2	19.6	3.72	0.2893	Coal	233
2020	0.2591	Hs-2	19.2	4.45	0.3444	Coal	224
2021	0.2503	Hs-2	19.3	3.96	0.3033	Solar	227
2022	0.2466	Hs-2	19.3	3.51	0.2691	Solar	236
2023	0.2583	Hs-2	19.7	4.02	0.3132	Solar	228
2024	0.2562	Hs-2	20.8	3.13	0.2720	Solar	251
2025	0.2400	Hs-2	21.0	3.06	0.2448	Solar	257

CNQ v2.0.0 – Quaternion View – USA United States · D=9 · N=25

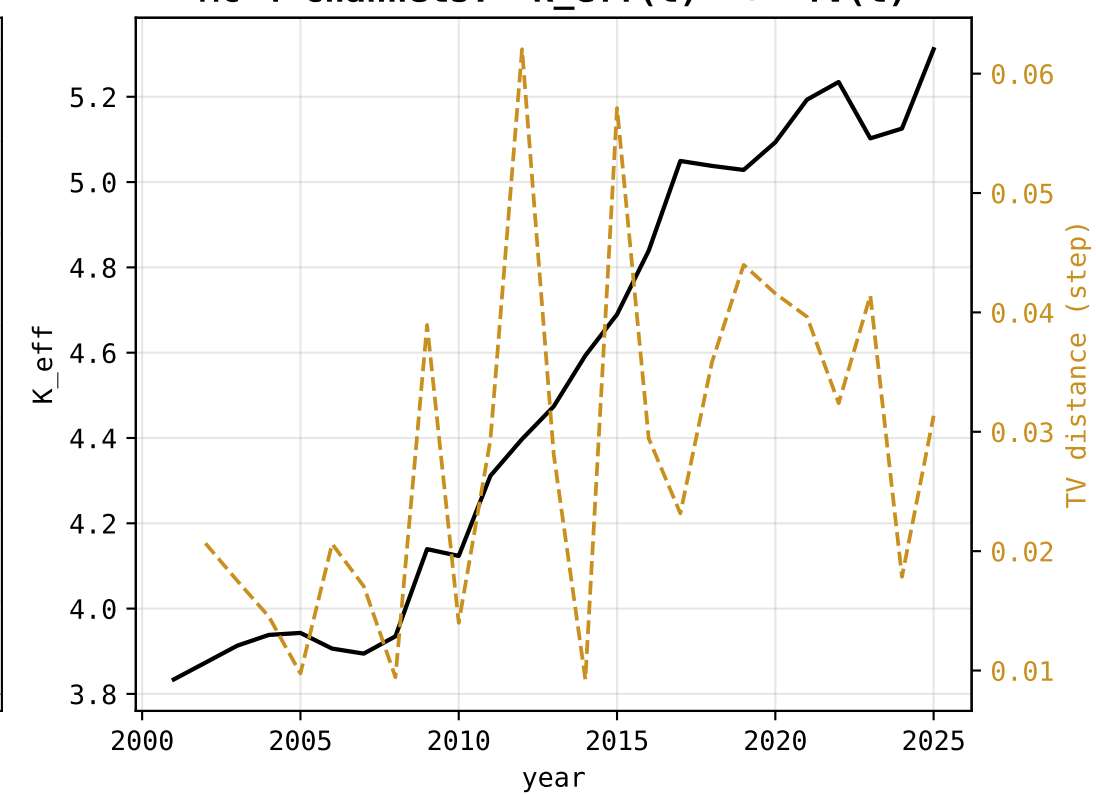
Higgins Scale $H_s(t)$



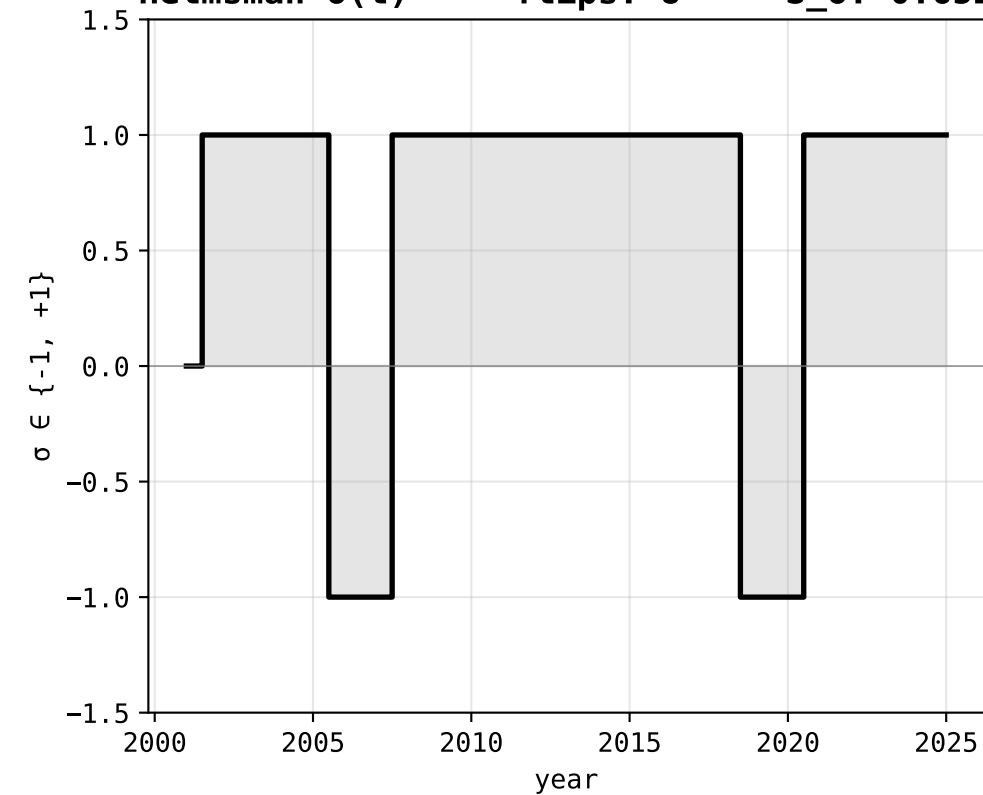
Angular velocity $\omega(t)$ [deg/yr]



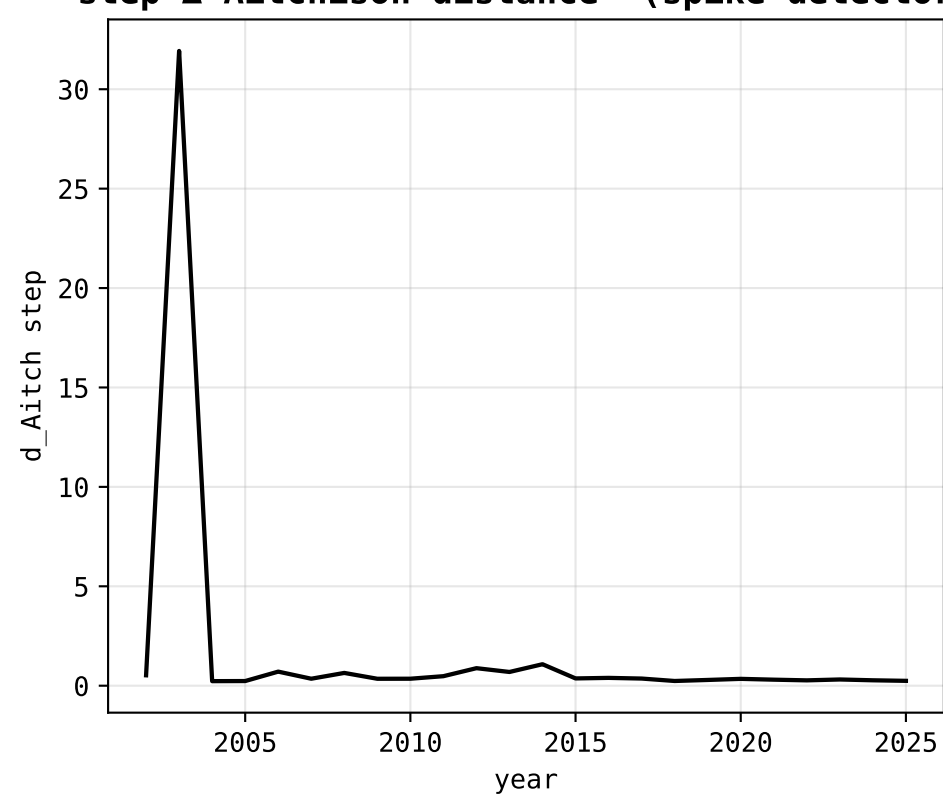
MC-4 channels: $K_{eff}(t)$ + $TV(t)$



Helmsman $\sigma(t)$ · flips: 8 · S_σ : 0.652



step- Δ Aitchison distance (spike detector)



CNQ DIAGNOSTICS · USA (D = 9)

CHSH joint-coherence:
not computed at D = 9
(CHSH requires D=8 twin factoring;
D = 9 does not factor as 4×4)

Twin-quaternion factor:
not computed at D = 9
(twin requires D=8 = 4+4 partition)

Attractor fit:
amplitude A: 9.688
contraction λ : 0.0843
period: 1
fitted: False

Helmsman family:
total flips: 8
stability S_σ : 0.652

§ WLD World

EMBER electricity-generation series · 2000-2025 · N = 26 · D = 9

INPUT

source: ember_WLD*_generation_TWh.csv
N records: 26
D carriers: 9
Bioenergy
Coal
Gas
Hydro
Nuclear
Other Fossil
Other Renewables
Solar
Wind
source_sha256: eff4a4d3ffe97ec12432bdda...

HEADLINE METRICS

Helmsman flips: ?
Helmsman stability S_σ: {'global': 0.8333333333333334, 'rolling': [0.4285714285714286, 0.4285714285714286]}
IR class: ?
Attractor amplitude: ?

CHSH S: ?
Tsirelson bound: $2\sqrt{2} \approx 2.828$

Bearing pairs (D(D-1)/2): 36

CNT v3.1.0 content_sha: ?...
CNQ v2.0.0 content_sha: ?...

Following pages: Stage 1 (XY/XZ/YZ × N years + course) → Stage 2/3 (7 pages) → CNQ (2 pages).

SECTION PLATE t=0 | Year 2000 | D=9 N=26 pairs=36

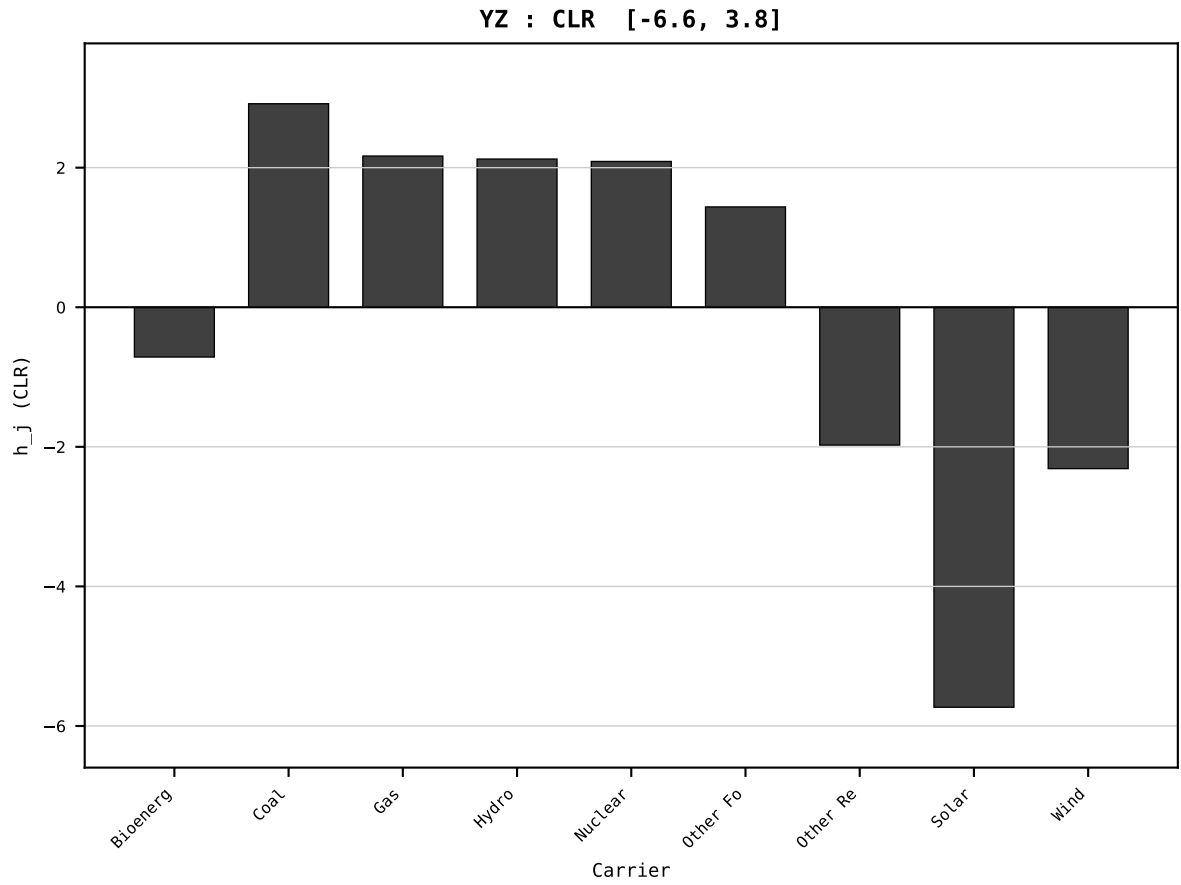
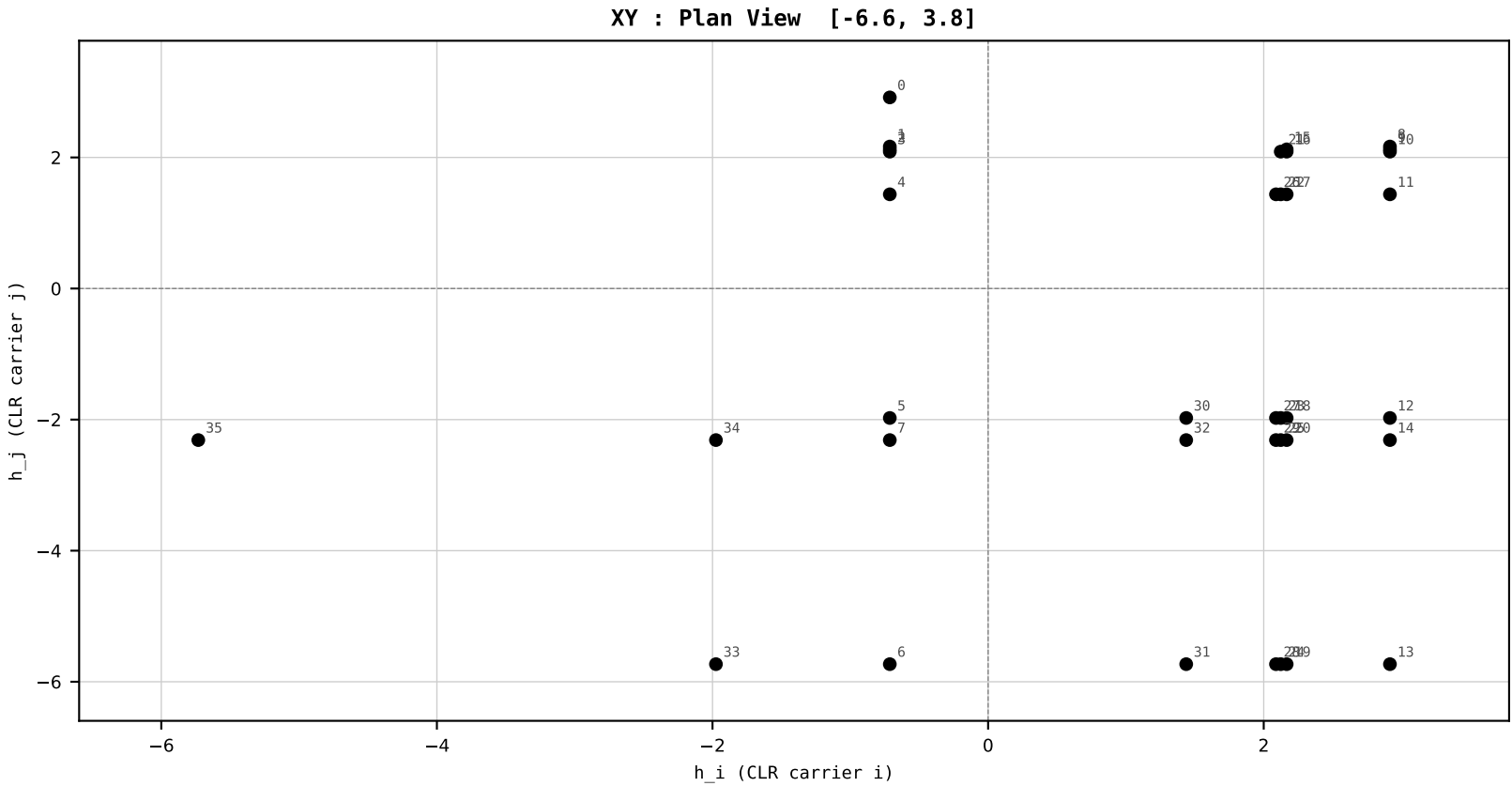
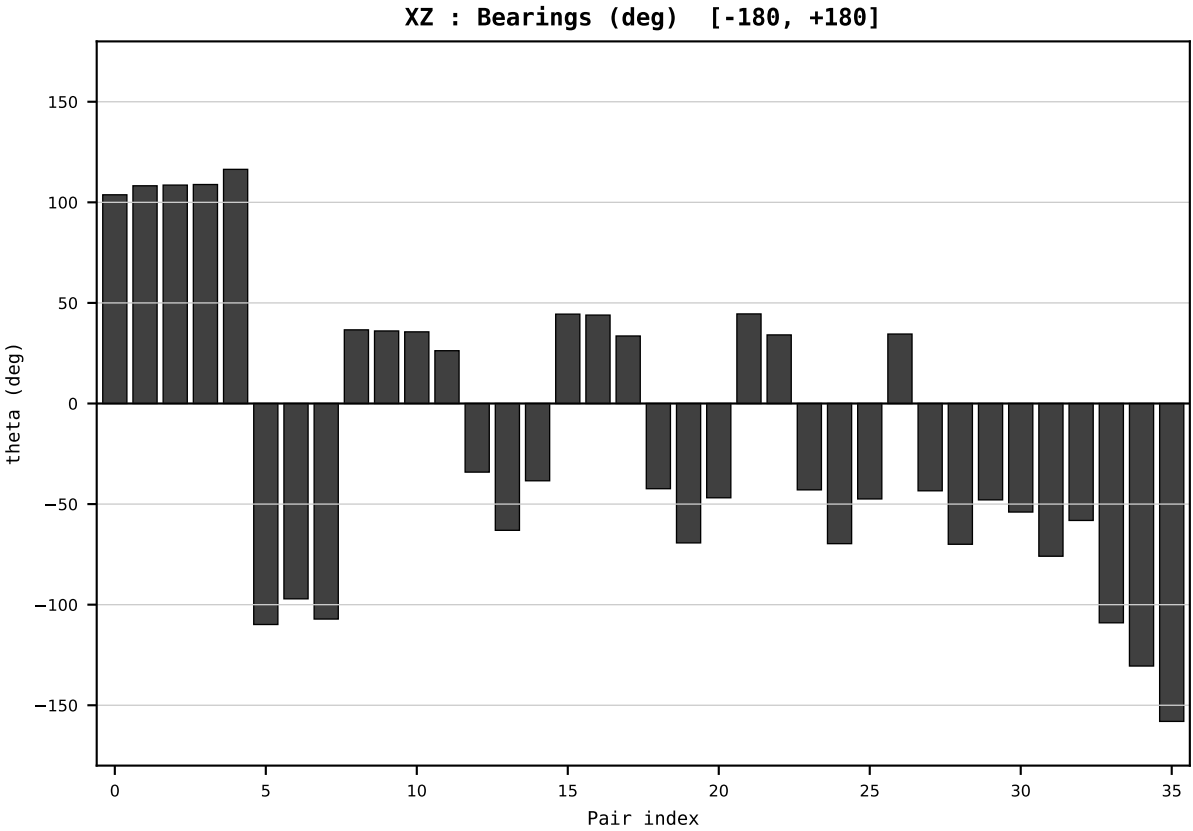
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2000
t : 0 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.287446
Ring = Hs-2
E_metric= 66.7408
kappa_HS= 5694.91
omega = 0.0000 deg
d_A = 0.000000
Helm = ---
Helm d = ---
DR = 8.647 HLR
DR ratio= 5694.9x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=1 | Year 2001 | D=9 N=26 pairs=36

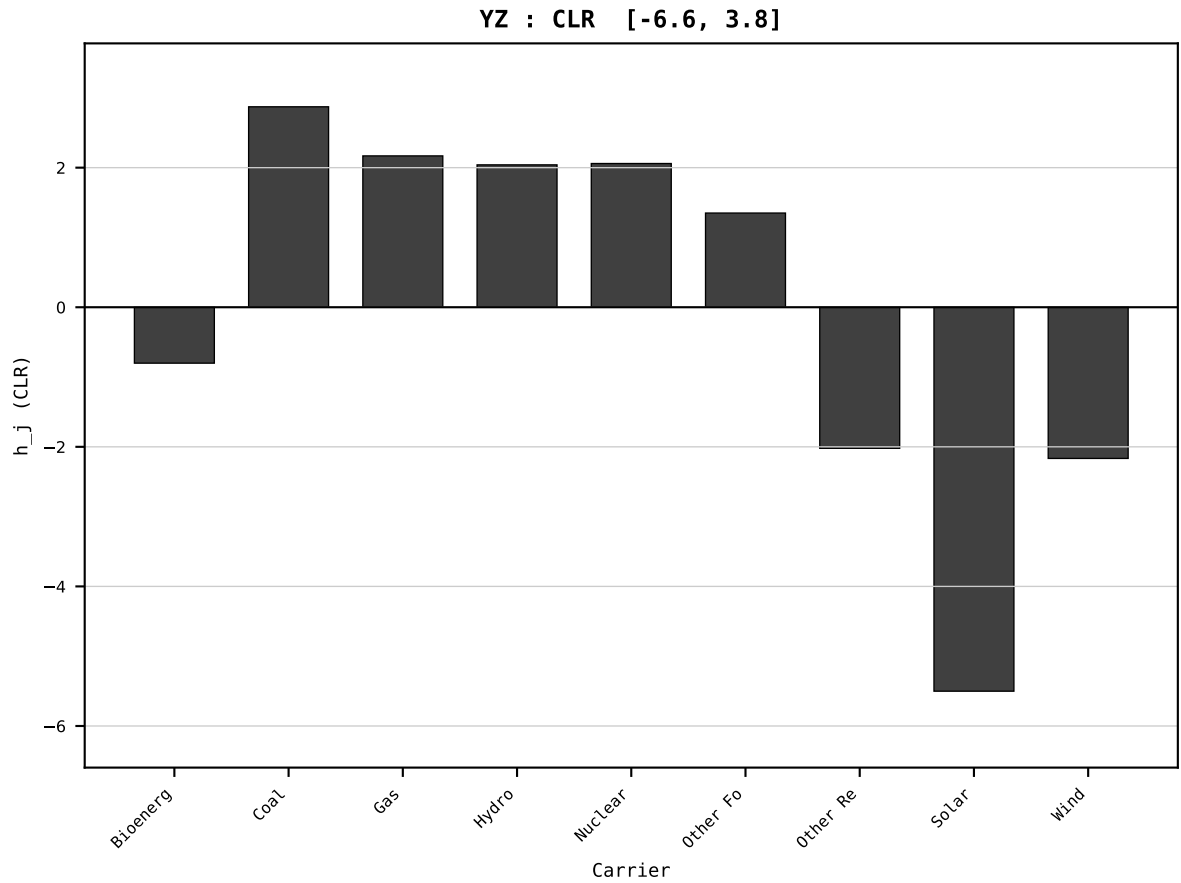
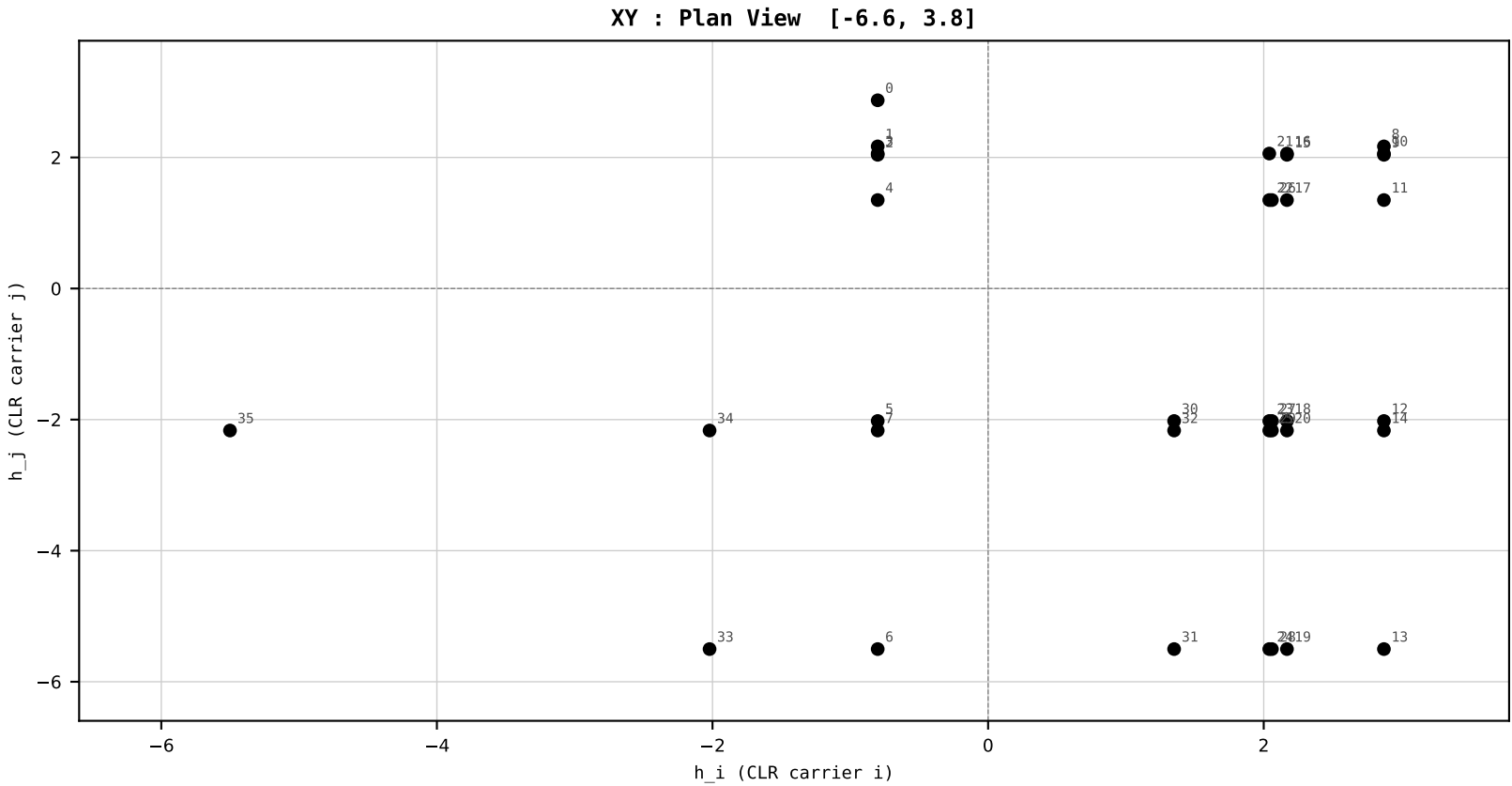
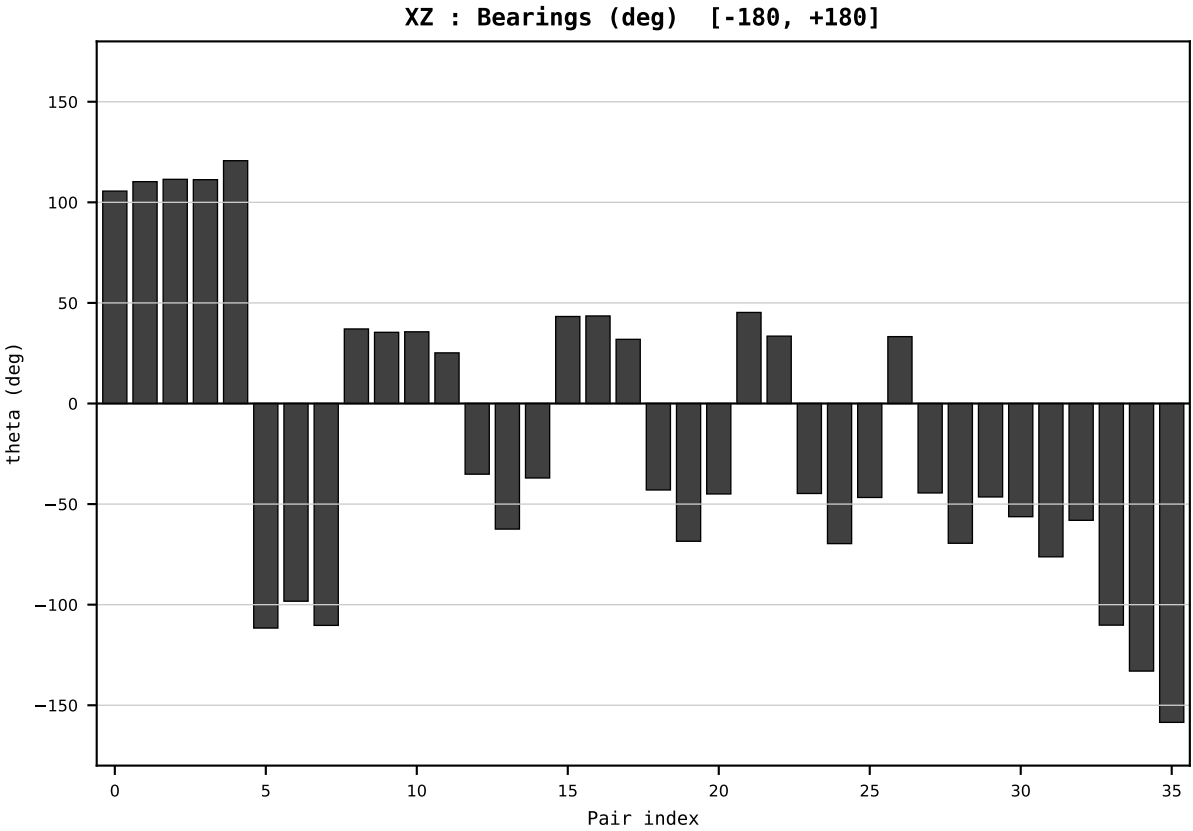
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2001
t : 1 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.288557
Ring = Hs-2
E_metric= 62.8650
kappa_HS= 4331.38
omega = 1.4848 deg
d_A = 0.318532
Helm = Solar
Helm d = +0.229979
DR = 8.374 HLR
DR ratio= 4331.4x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=2 | Year 2002 | D=9 N=26 pairs=36

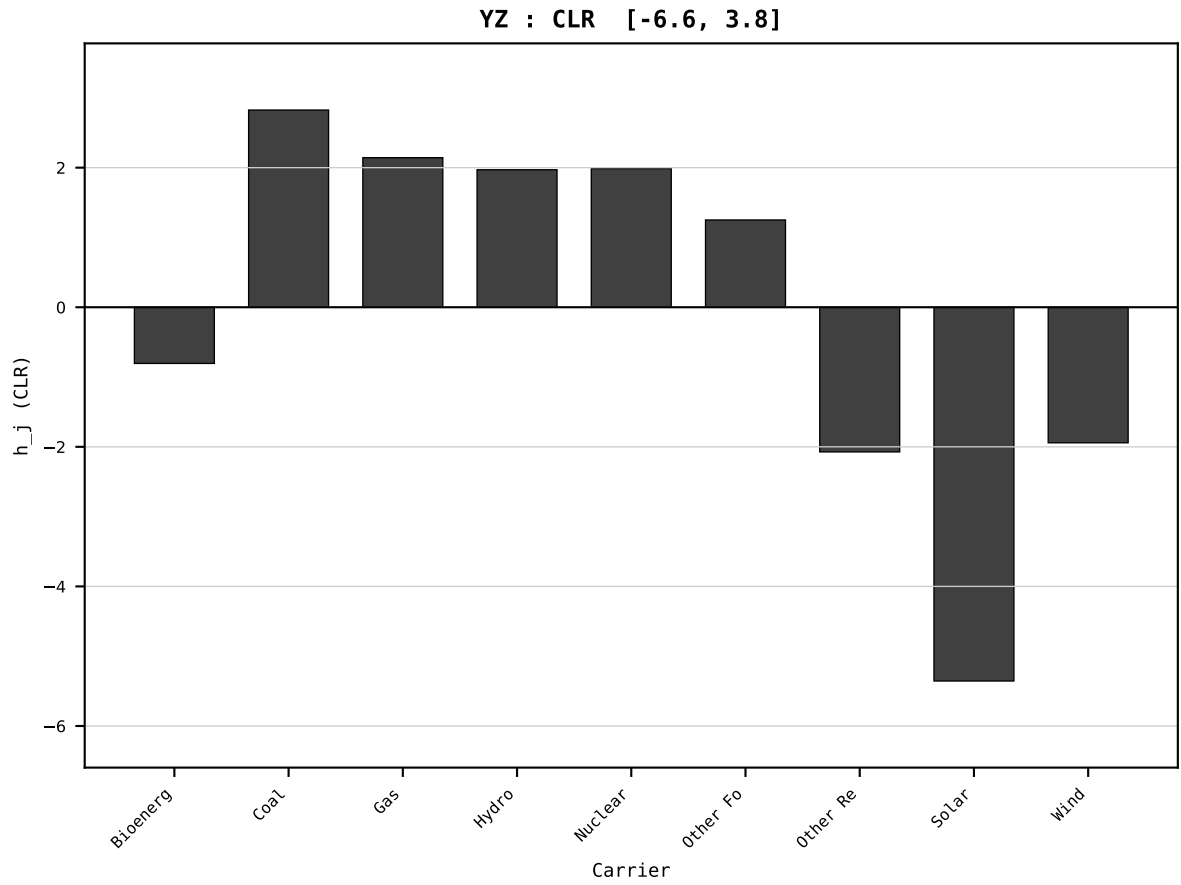
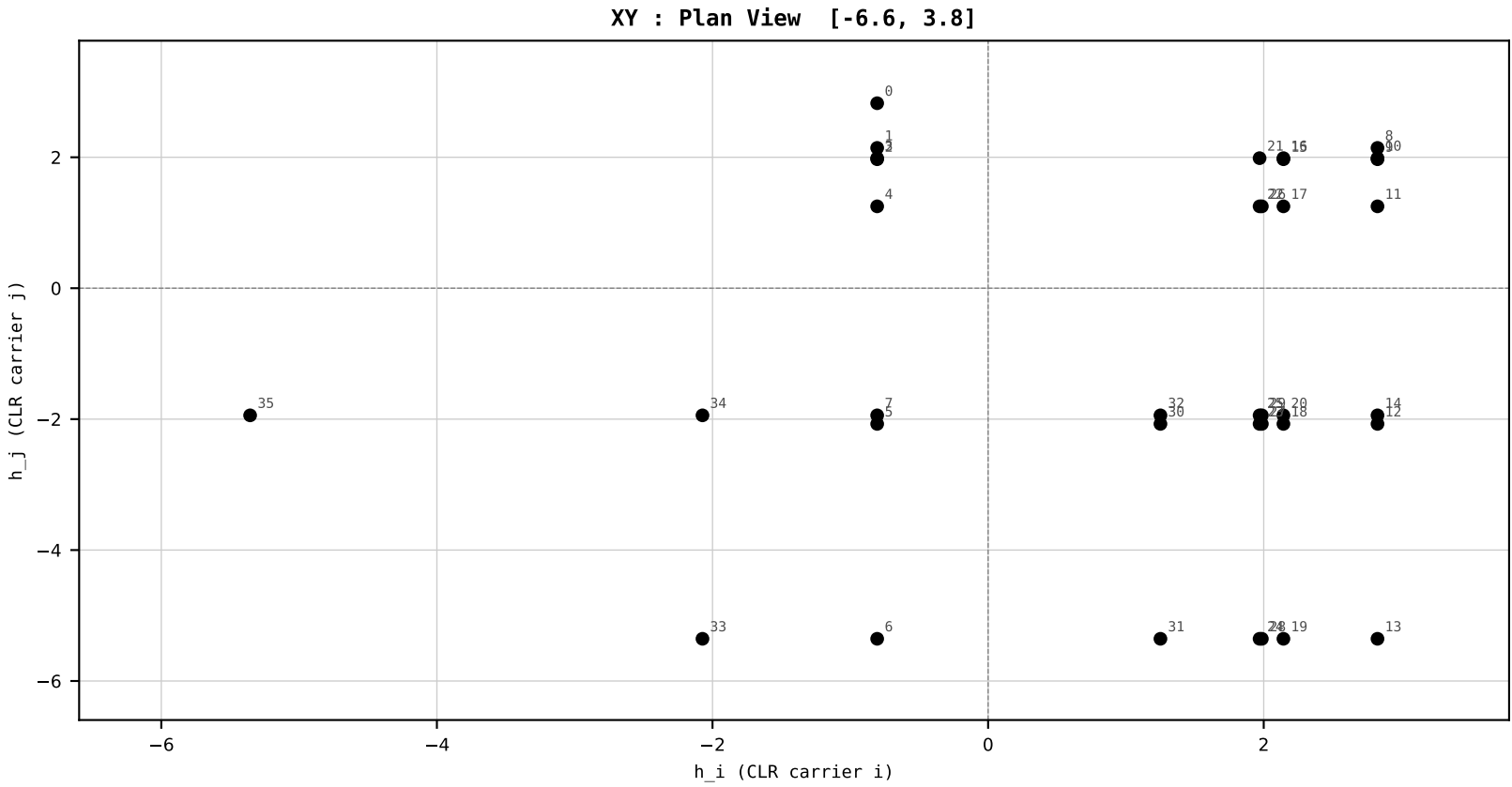
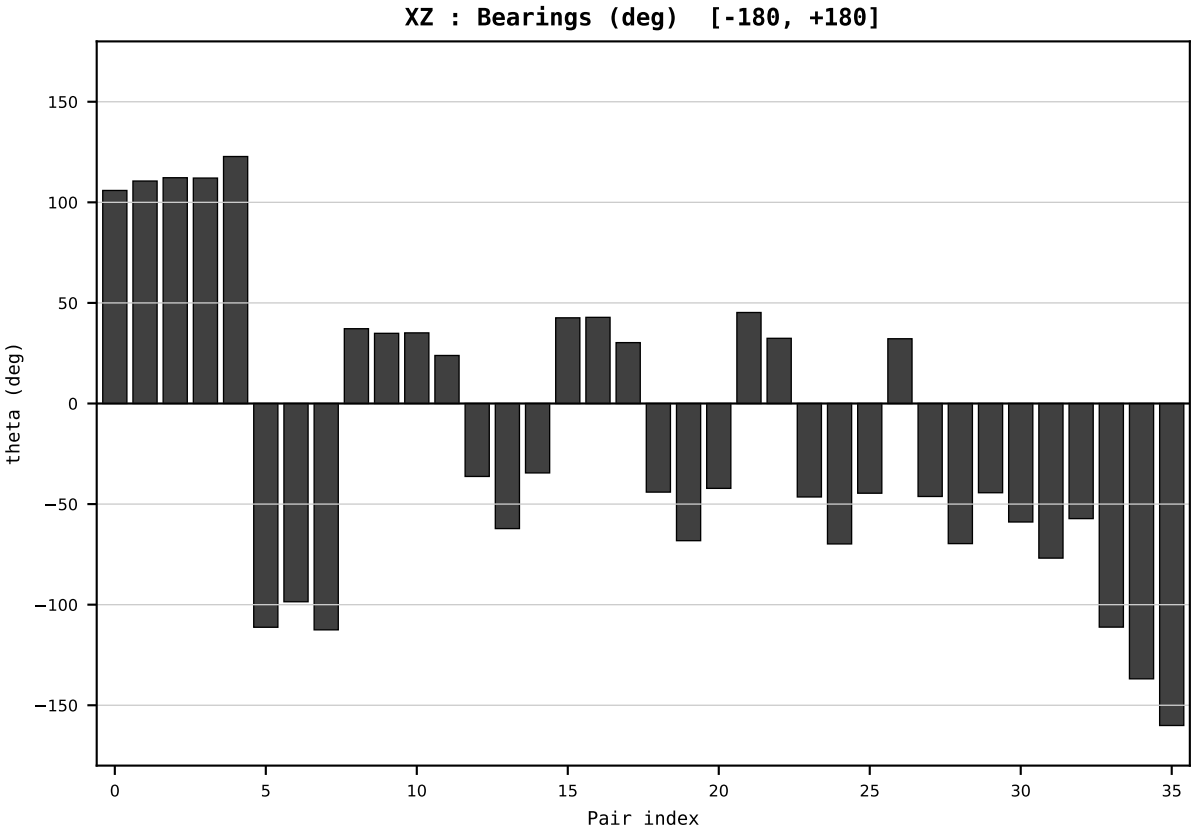
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2002
t : 2 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.288986
Ring = Hs-2
E_metric= 59.3639
kappa_HS= 3572.27
omega = 1.5895 deg
d_A = 0.311719
Helm = Wind
Helm d = +0.224097
DR = 8.181 HLR
DR ratio= 3572.3x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=3 | Year 2003 | D=9 N=26 pairs=36

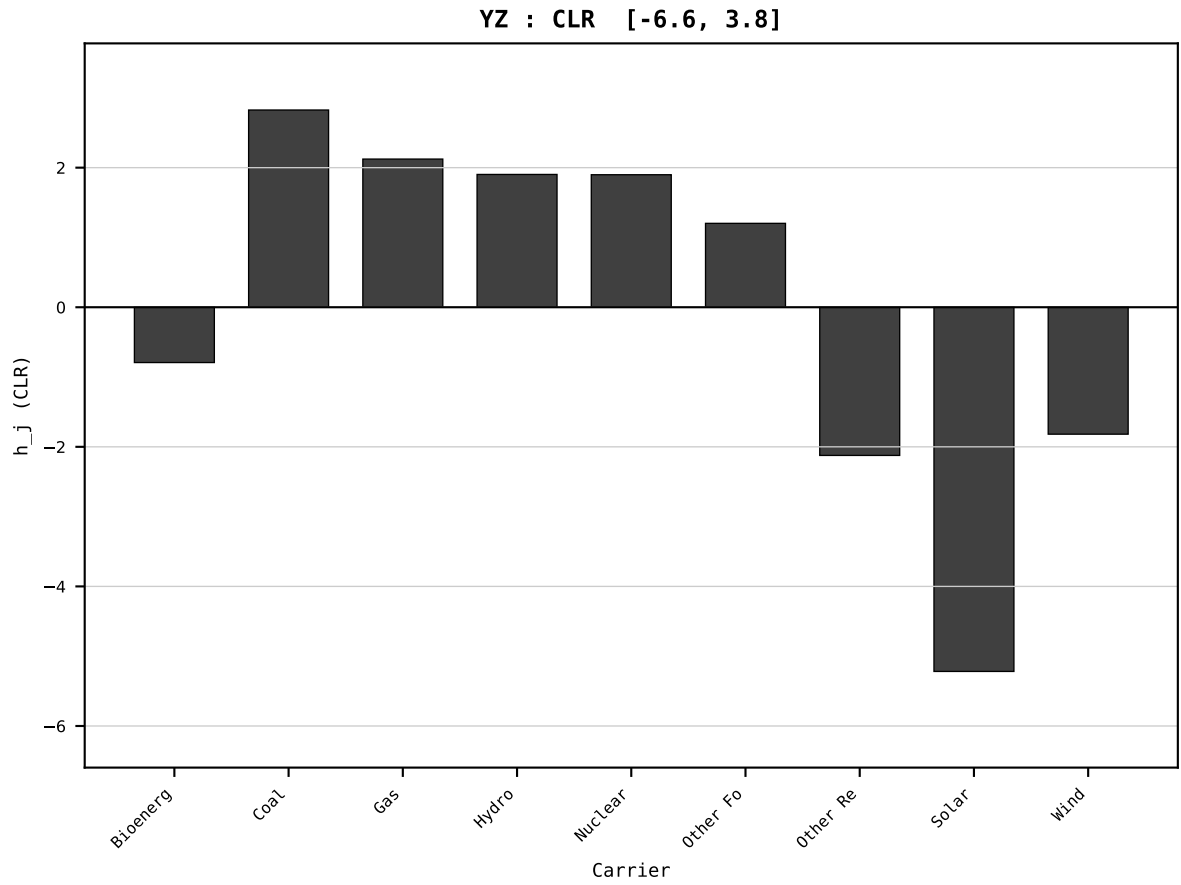
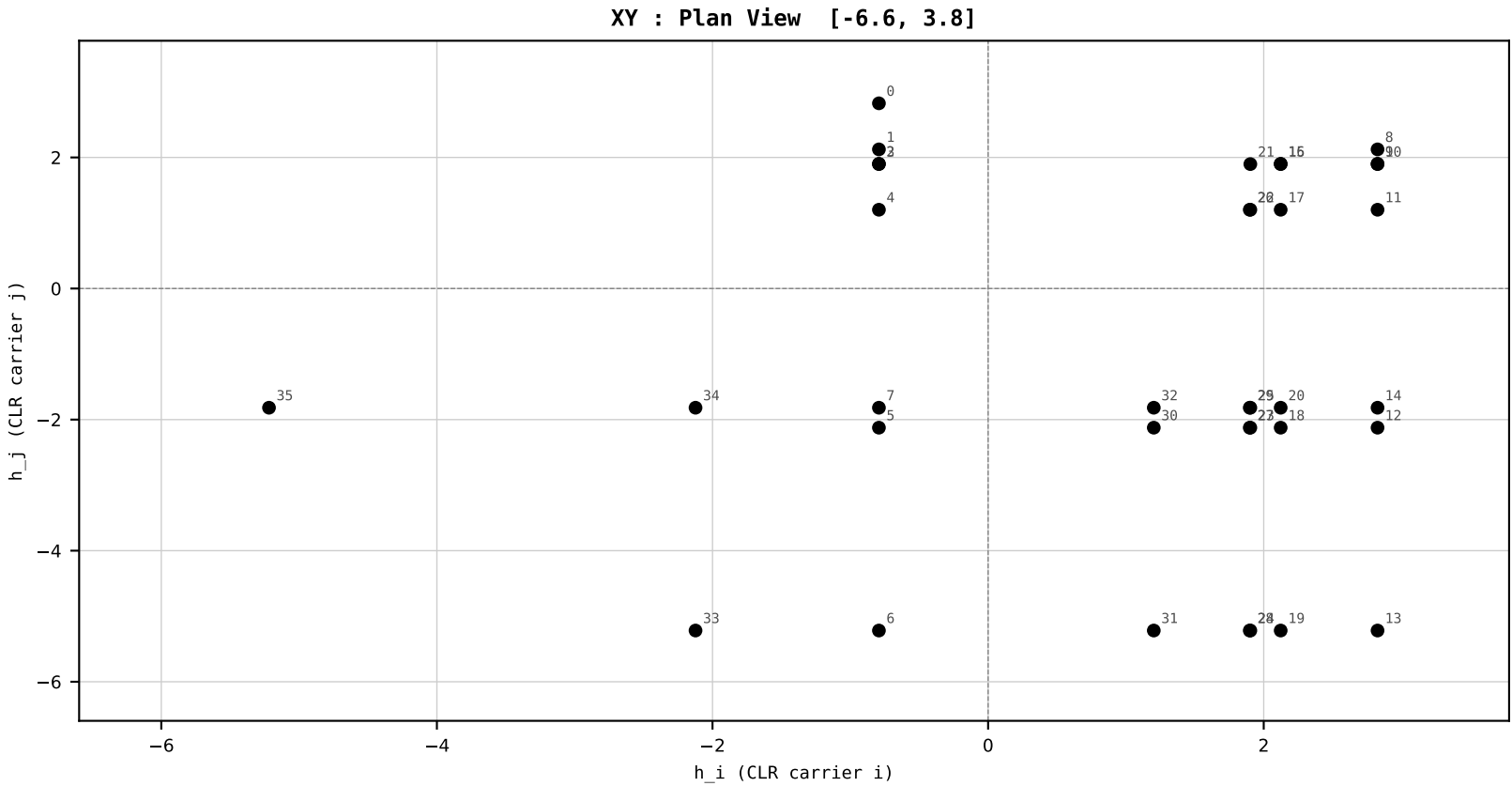
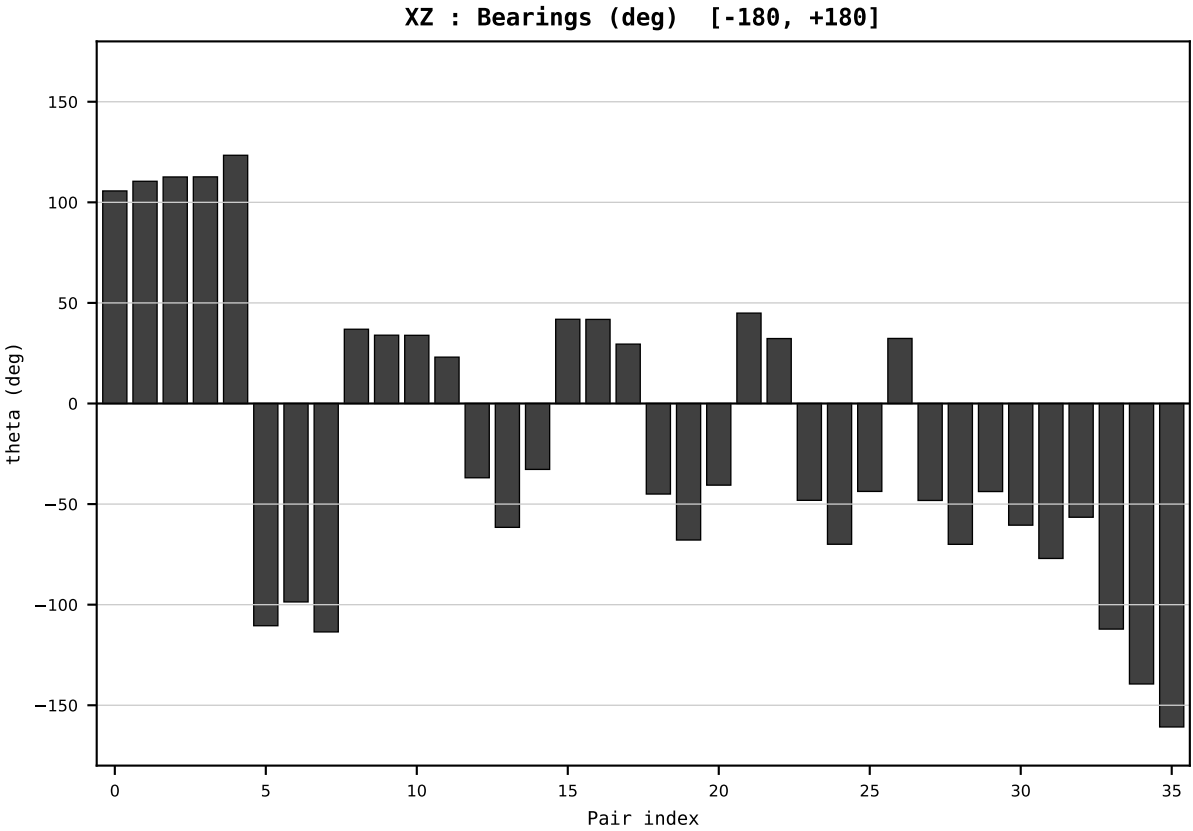
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2003
t : 3 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.293186
Ring = Hs-2
E_metric= 56.8394
kappa_HS= 3116.09
omega = 1.1711 deg
d_A = 0.227352
Helm = Solar
Helm d = +0.137038
DR = 8.044 HLR
DR ratio= 3116.1x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=4 | Year 2004 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

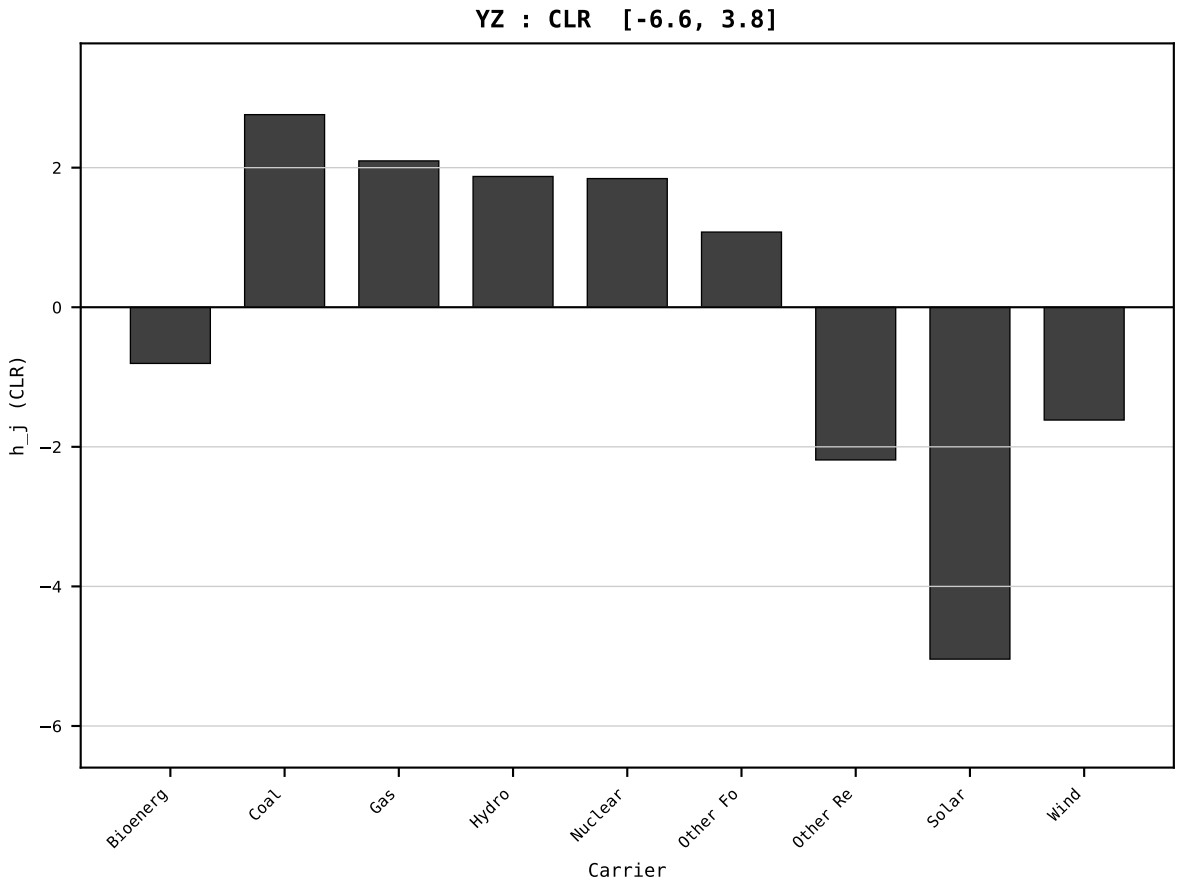
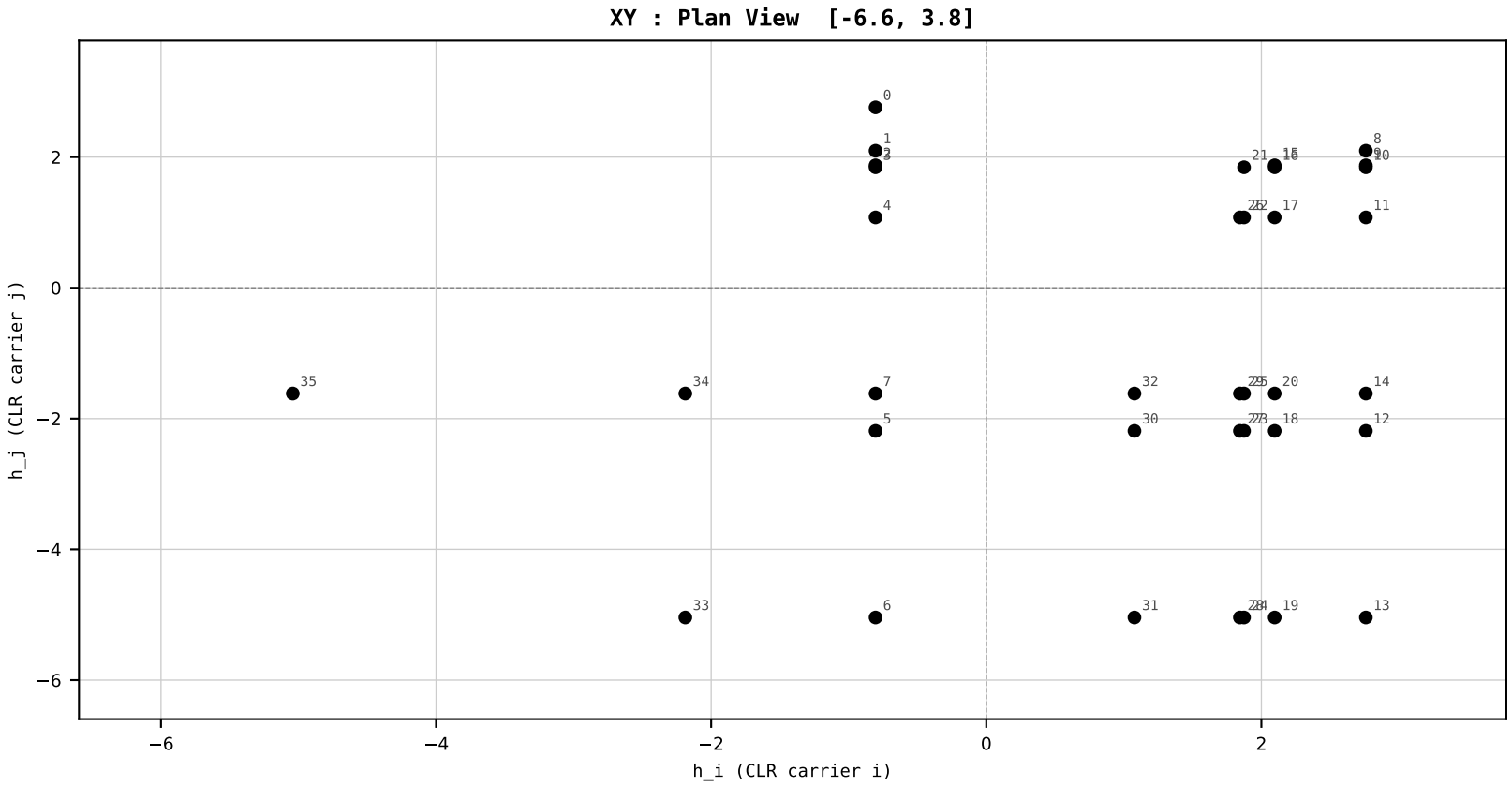
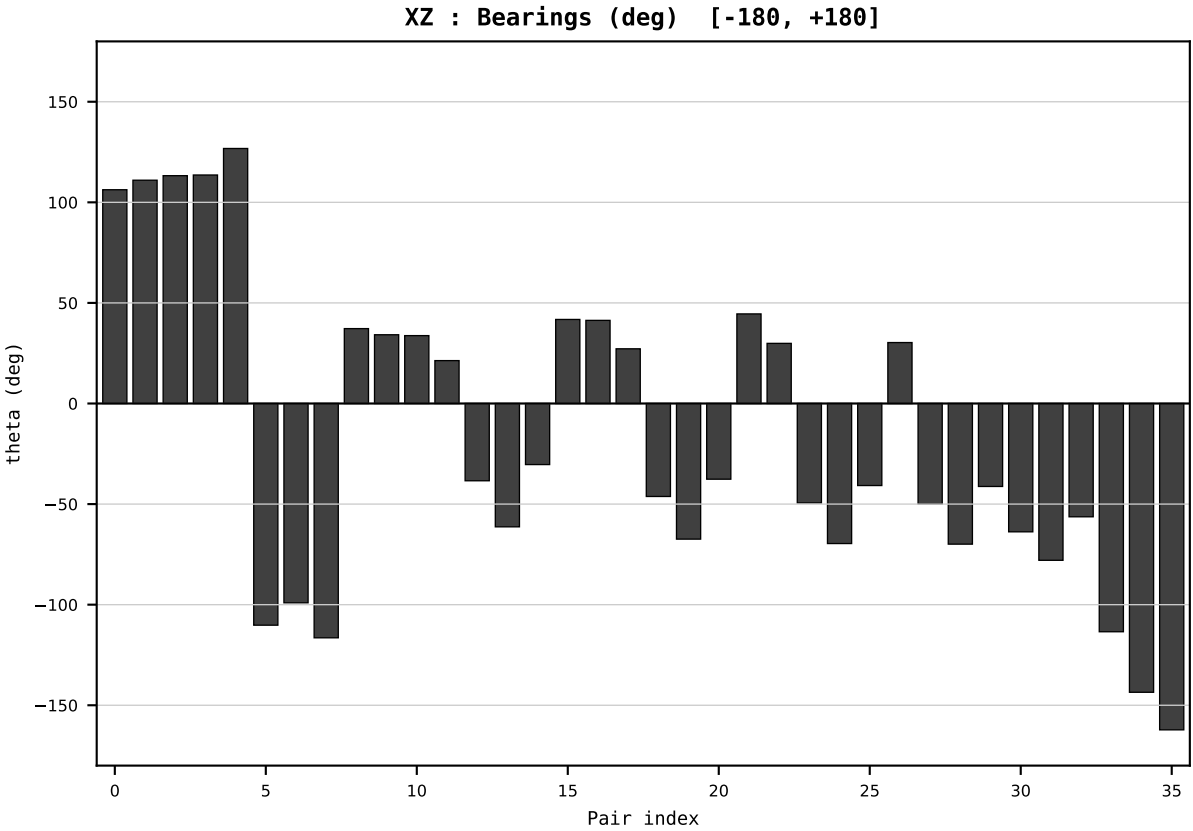
Dataset : ember_WLD_World_generation_TWh.csv
Year : 2004
t : 4 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.291066
Ring = Hs-2
E_metric= 53.5533
kappa_HS= 2445.07
omega = 1.7599 deg
d_A = 0.317755
Helm = Wind
Helm d = +0.202466
DR = 7.802 HLR
DR ratio= 2445.1x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=5 | Year 2005 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

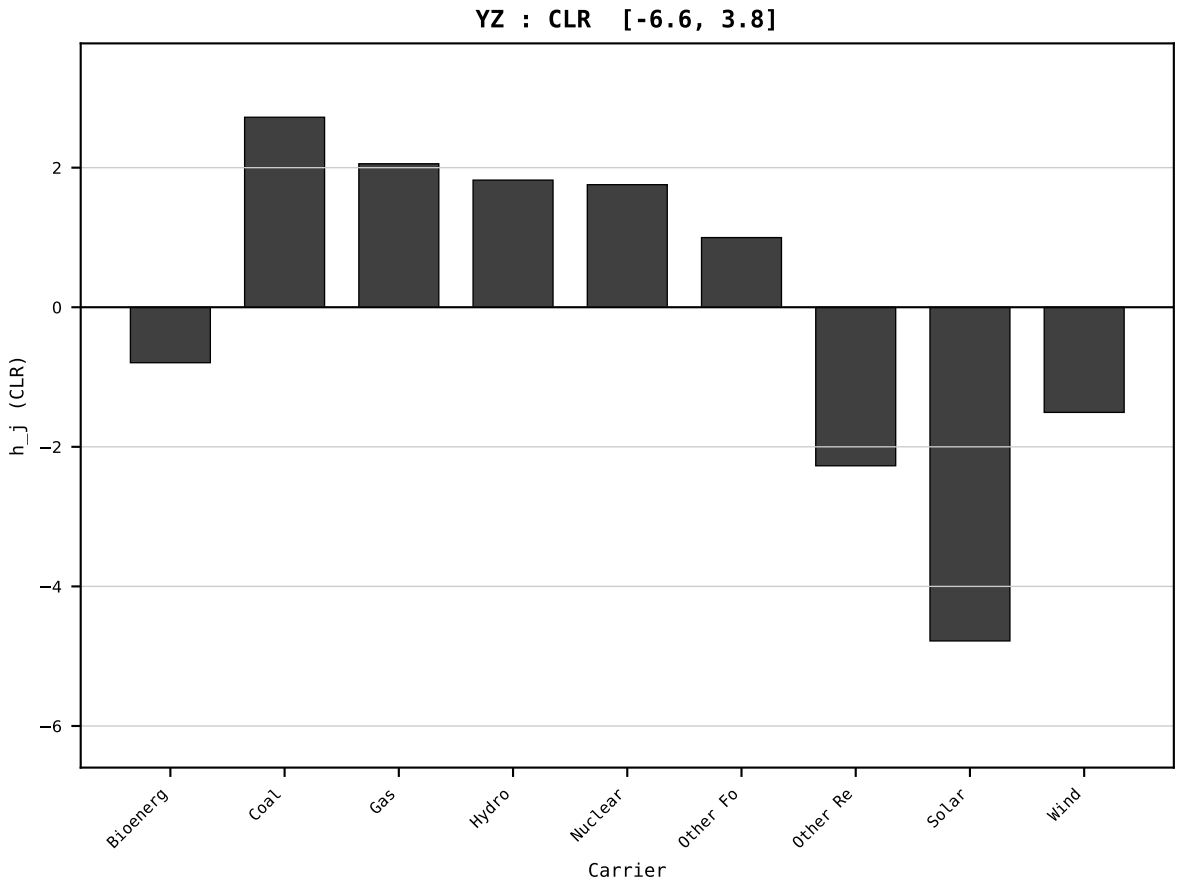
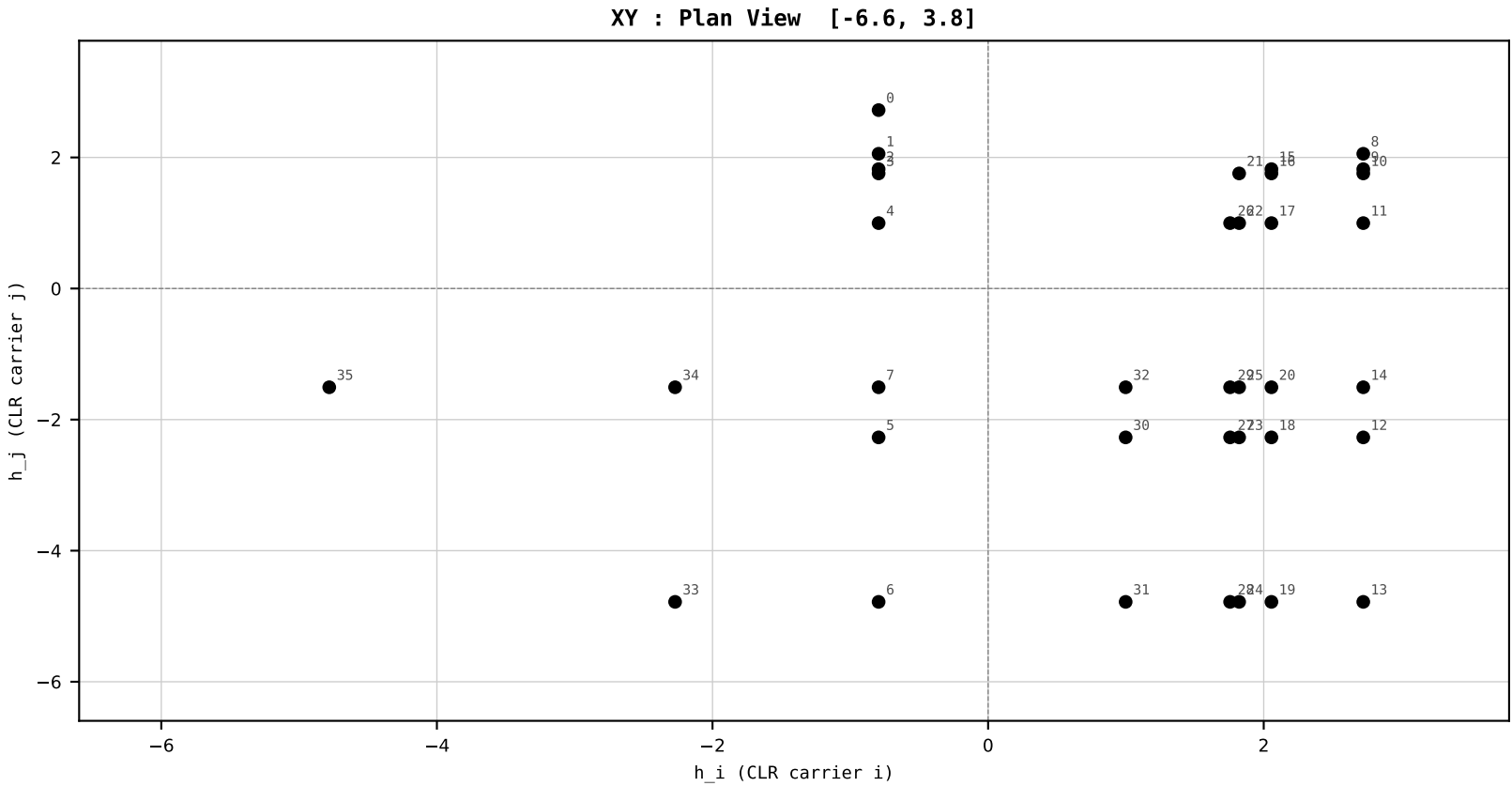
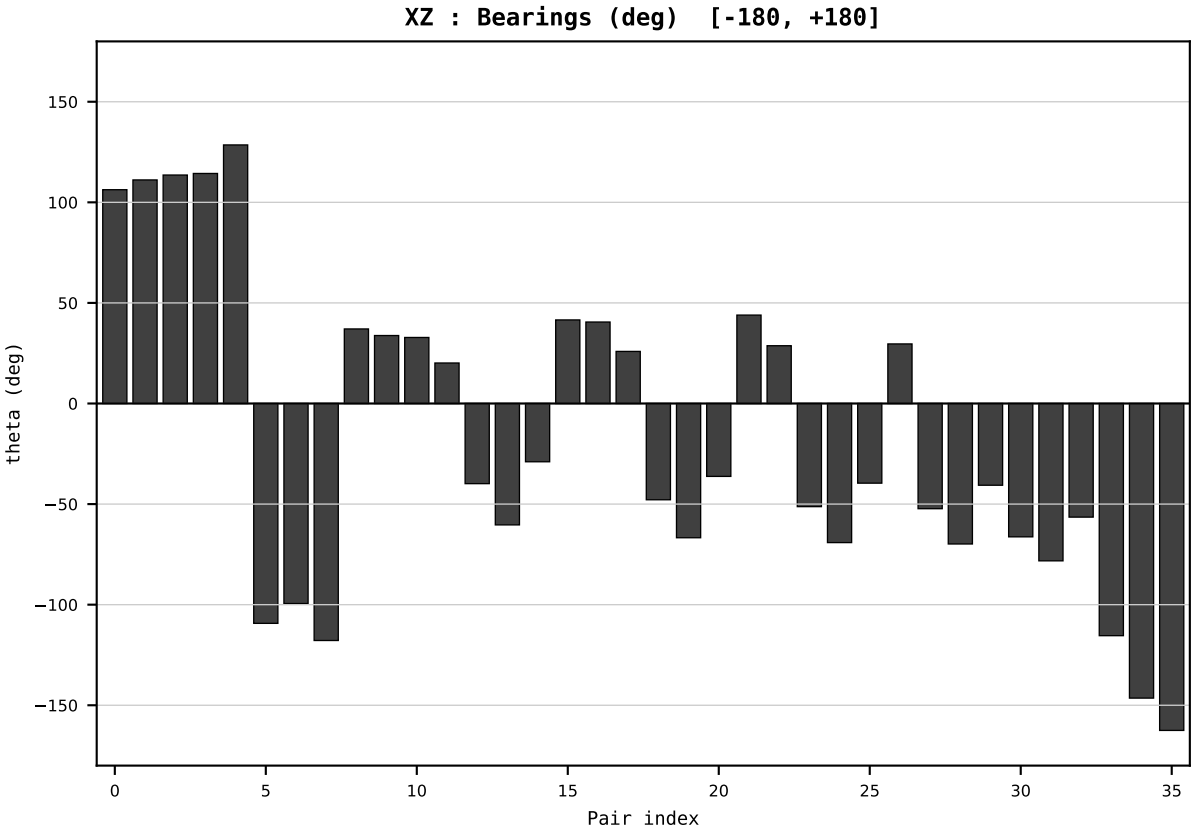
Dataset : ember_WLD_World_generation_TWh.csv
Year : 2005
t : 5 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.292003
Ring = Hs-2
E_metric= 49.9572
kappa_HS= 1814.48
omega = 1.6789 deg
d_A = 0.326946
Helm = Solar
Helm d = +0.260914
DR = 7.504 HLR
DR ratio= 1814.5x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=6 | Year 2006 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

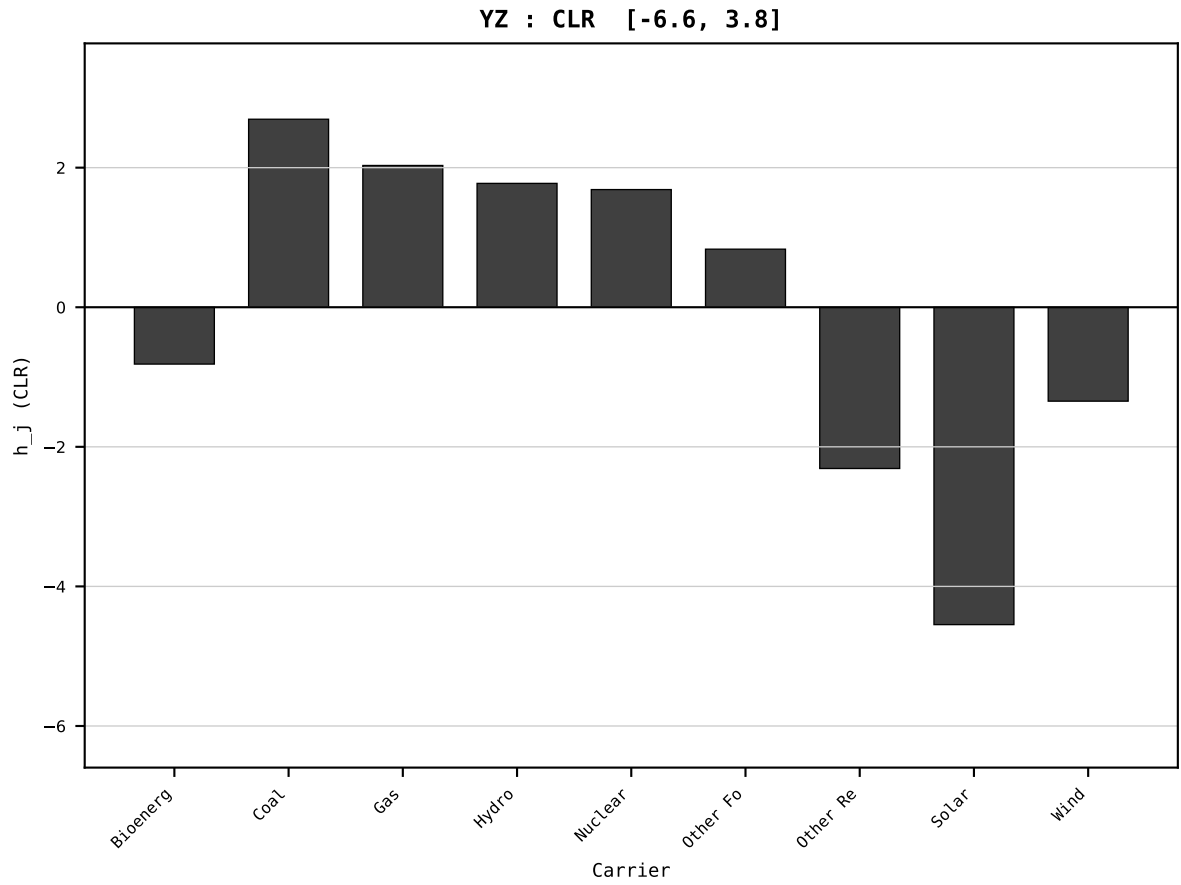
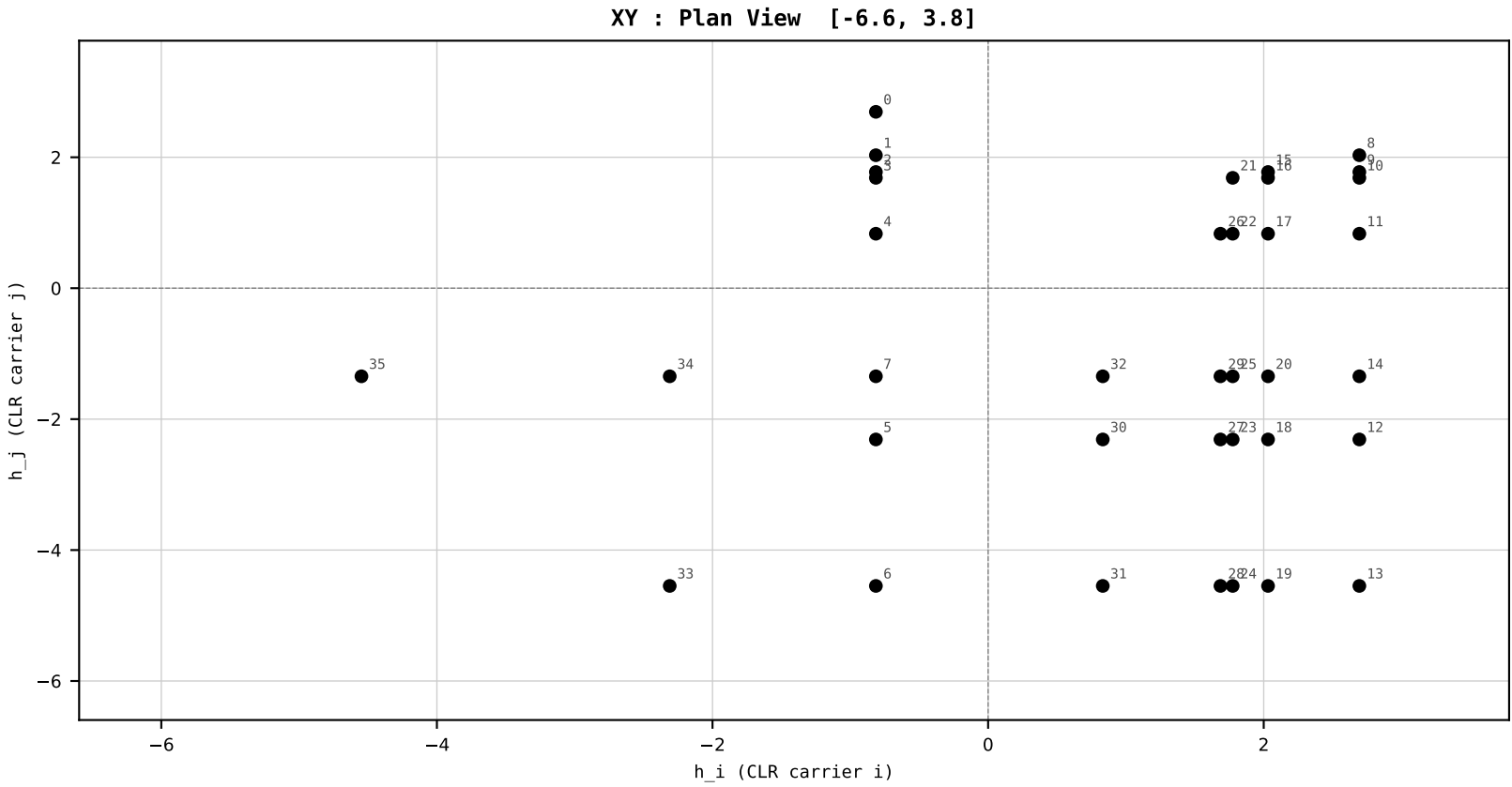
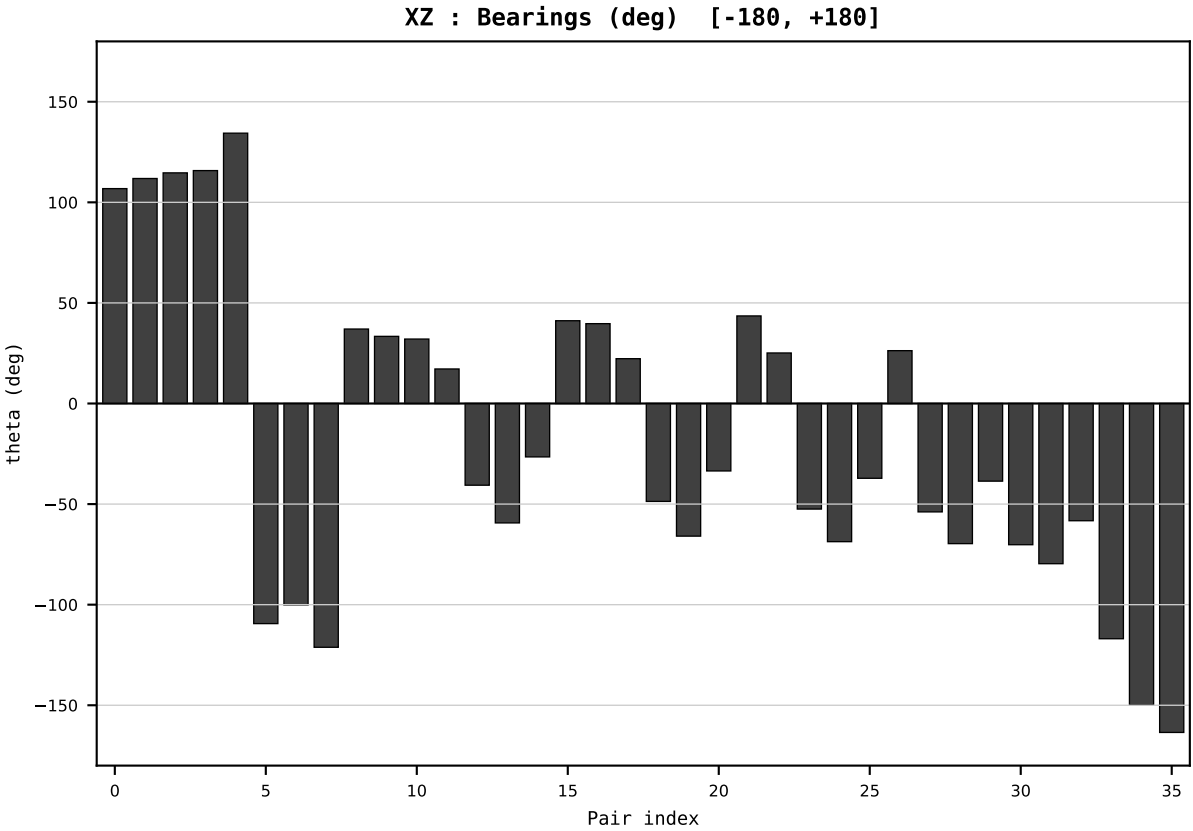
Dataset : ember_WLD_World_generation_TWh.csv
Year : 2006
t : 6 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.295981
Ring = Hs-2
E_metric= 46.5605
kappa_HS= 1396.12
omega = 1.9988 deg
d_A = 0.344208
Helm = Solar
Helm d = +0.233681
DR = 7.241 HLR
DR ratio= 1396.1x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=7 | Year 2007 | D=9 N=26 pairs=36

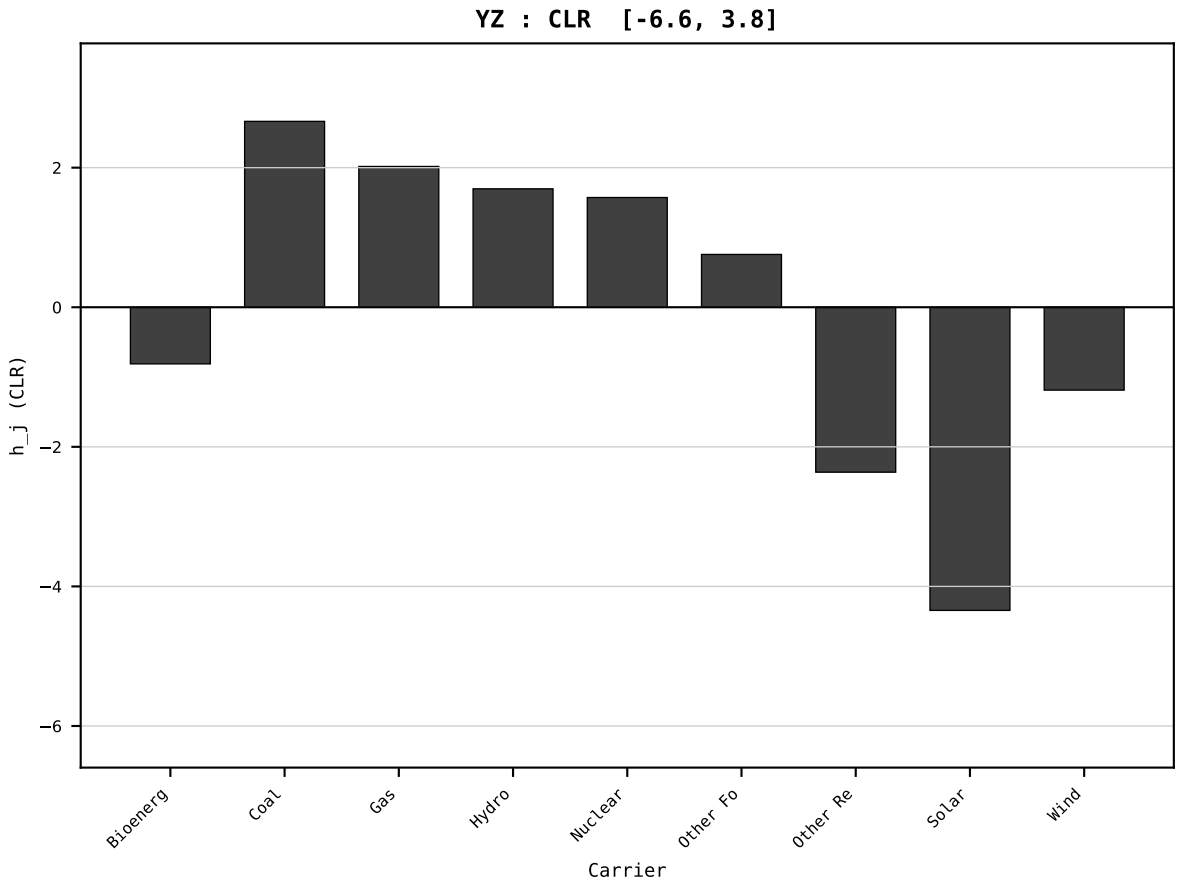
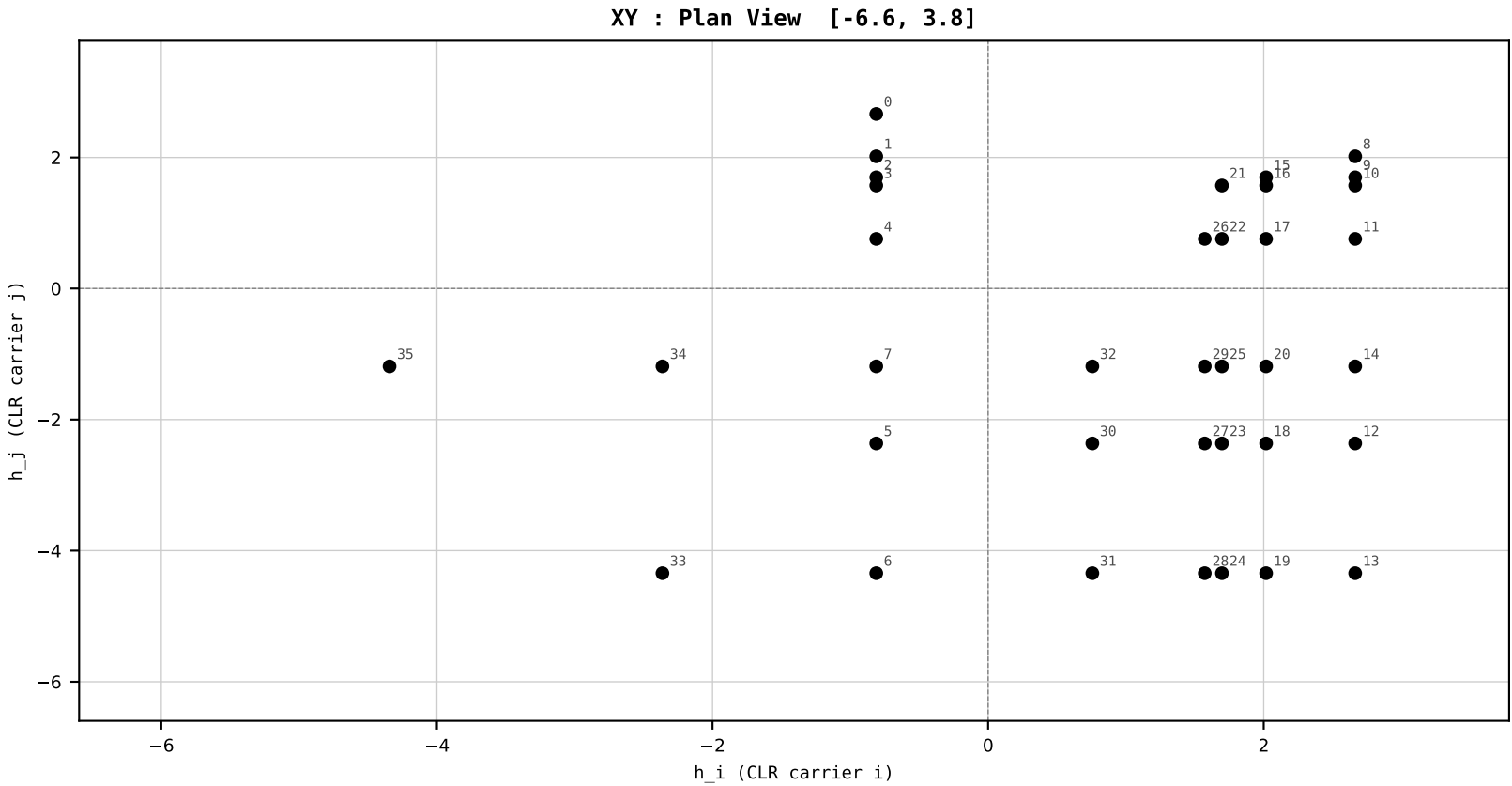
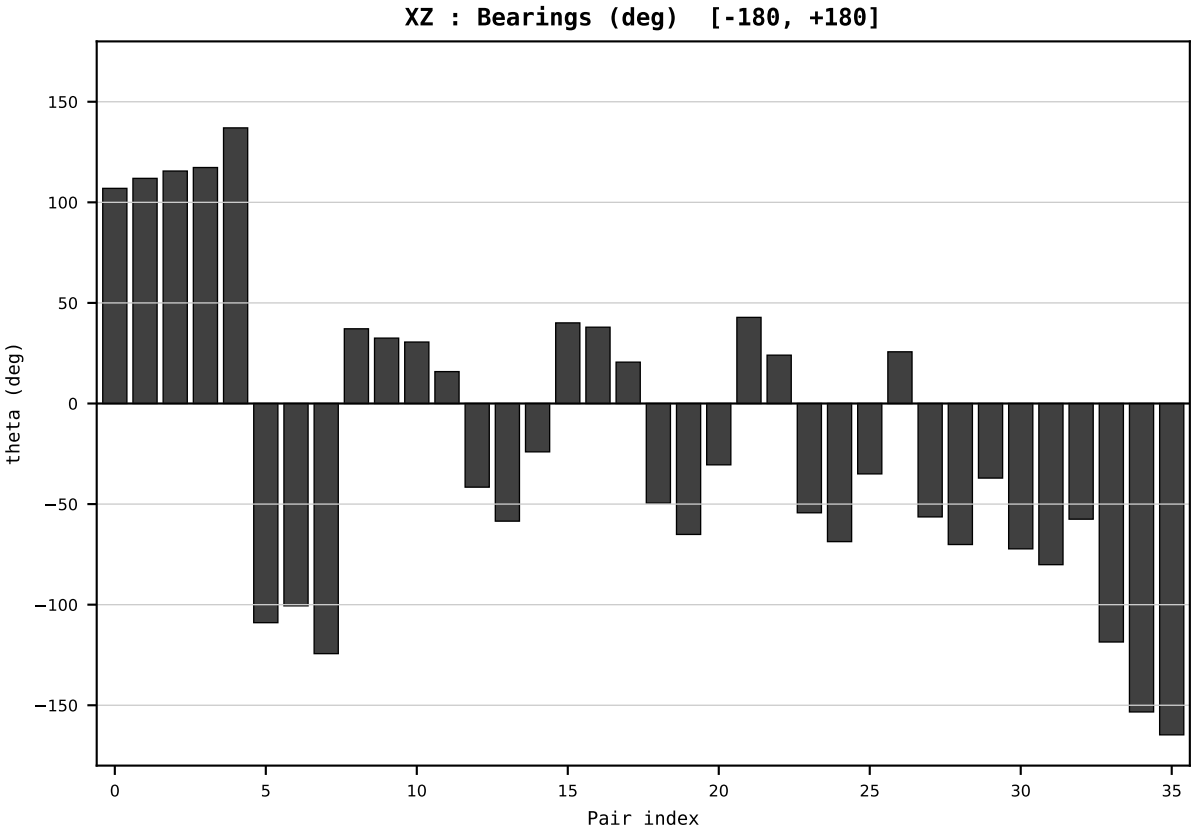
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2007
t : 7 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.297889
Ring = Hs-2
E_metric= 43.6095
kappa_HS= 1104.50
omega = 1.8487 deg
d_A = 0.308559
Helm = Solar
Helm d = +0.203824
DR = 7.007 HLR
DR ratio= 1104.5x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=8 | Year 2008 | D=9 N=26 pairs=36

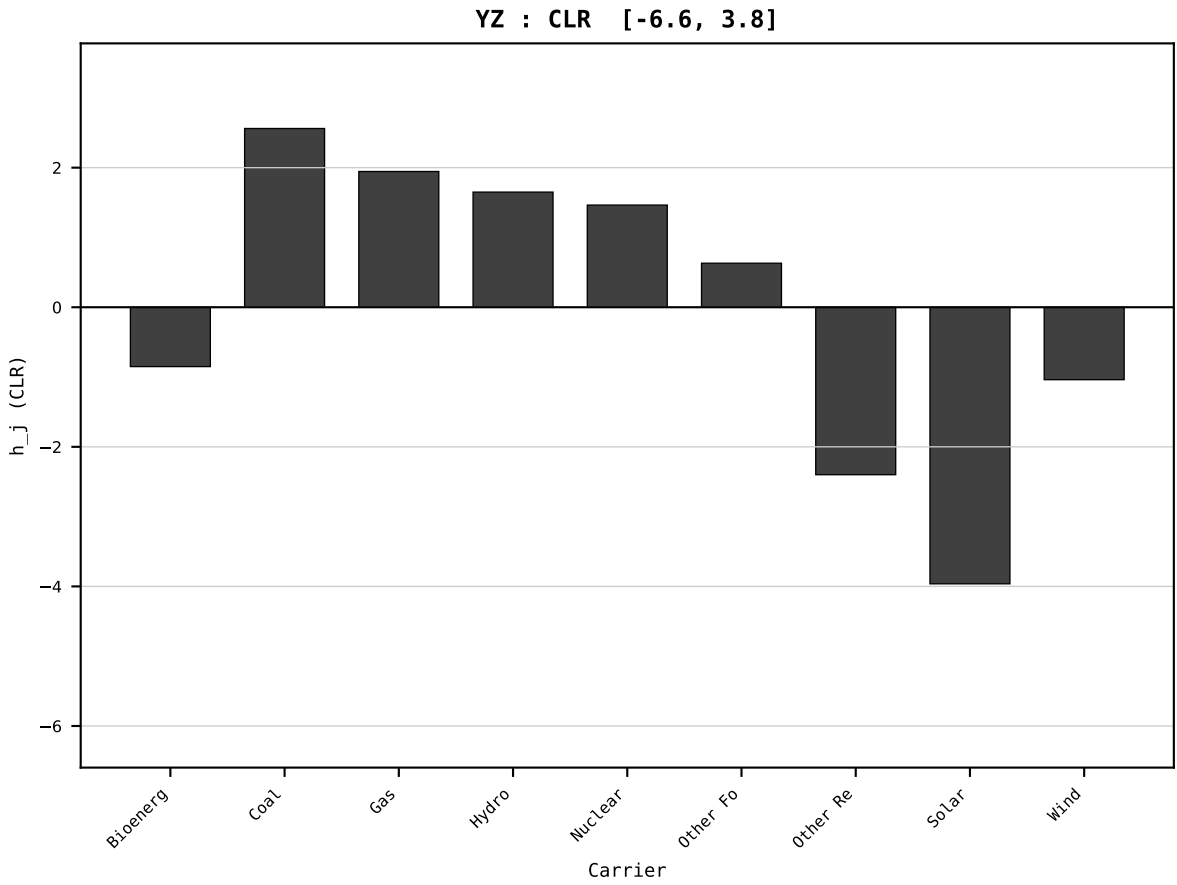
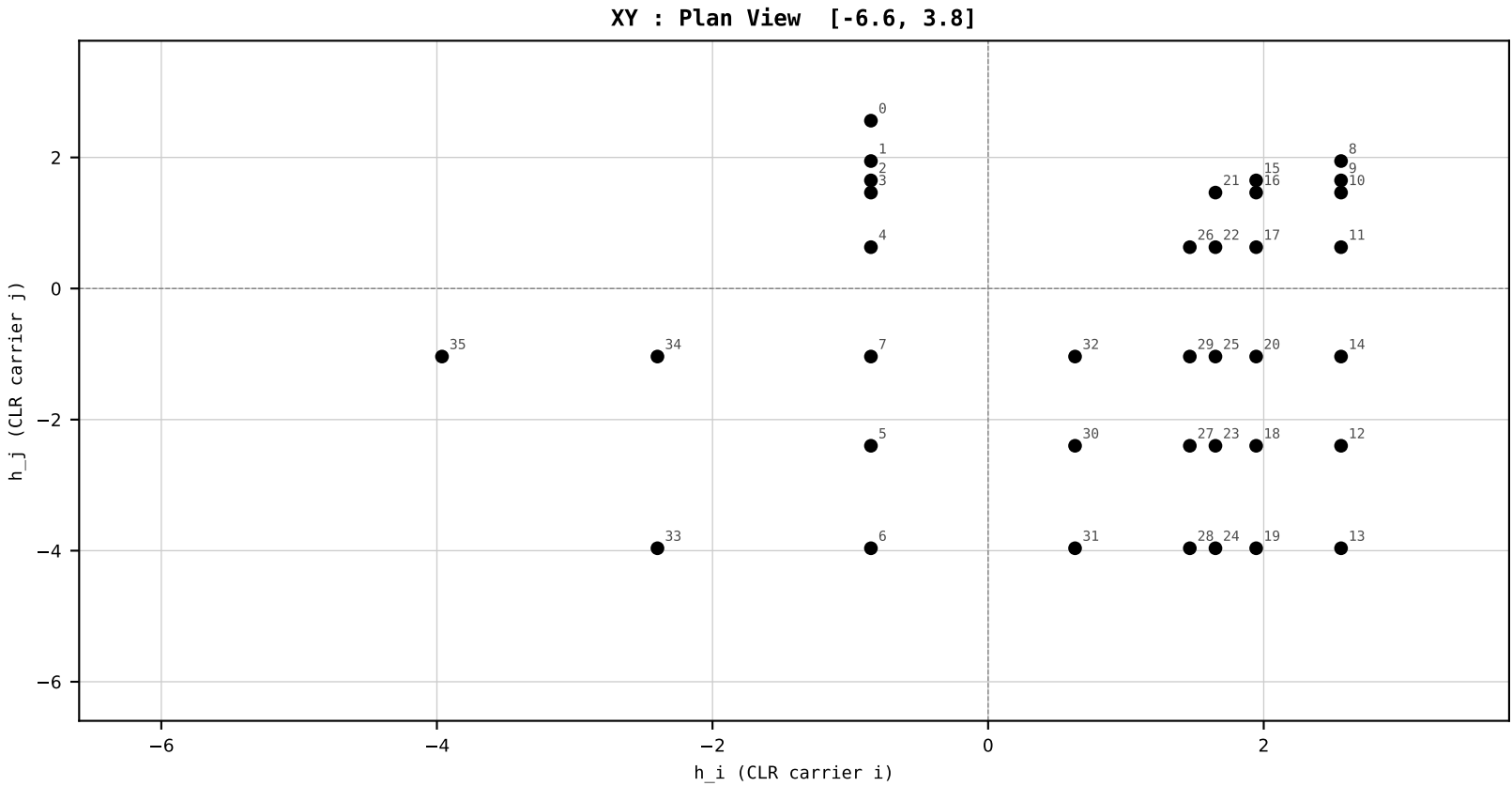
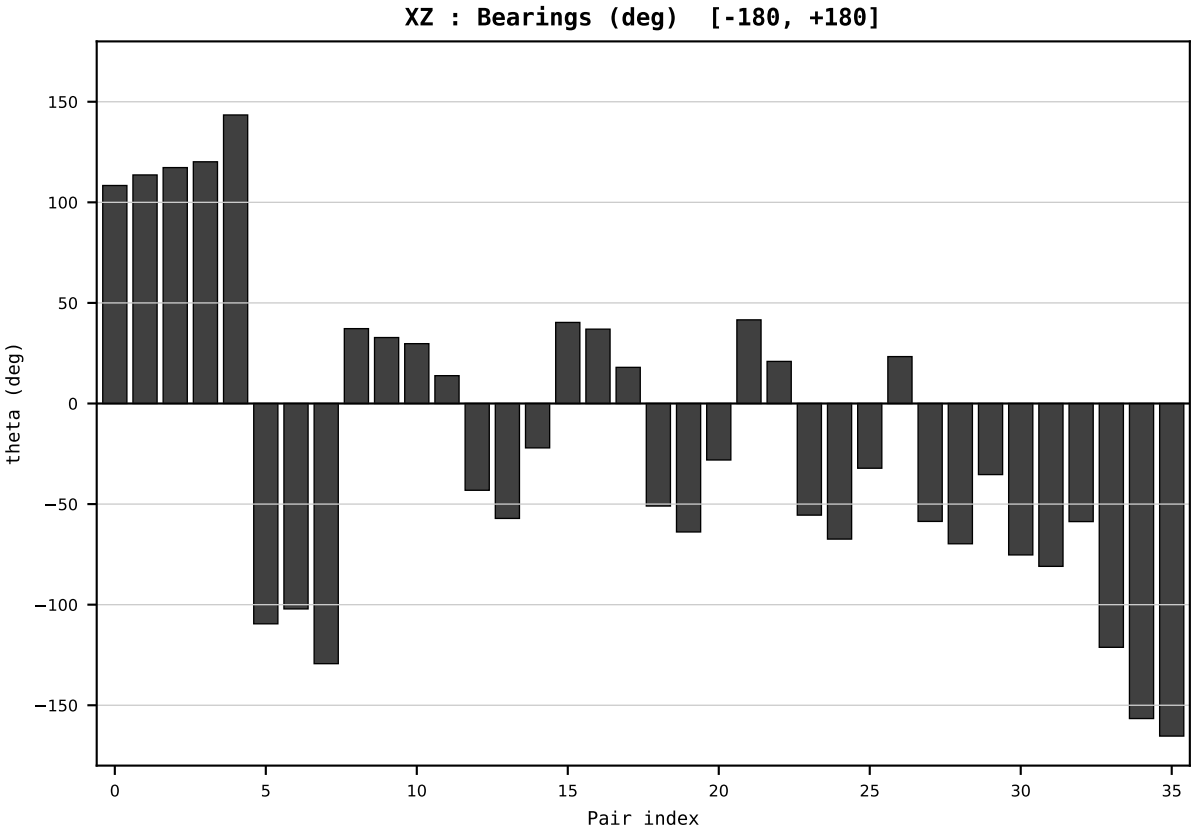
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2008
t : 8 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.292239
Ring = Hs-2
E_metric= 38.8679
kappa_HS= 681.37
omega = 2.5134 deg
d_A = 0.464351
Helm = Solar
Helm d = +0.380700
DR = 6.524 HLR
DR ratio= 681.4x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



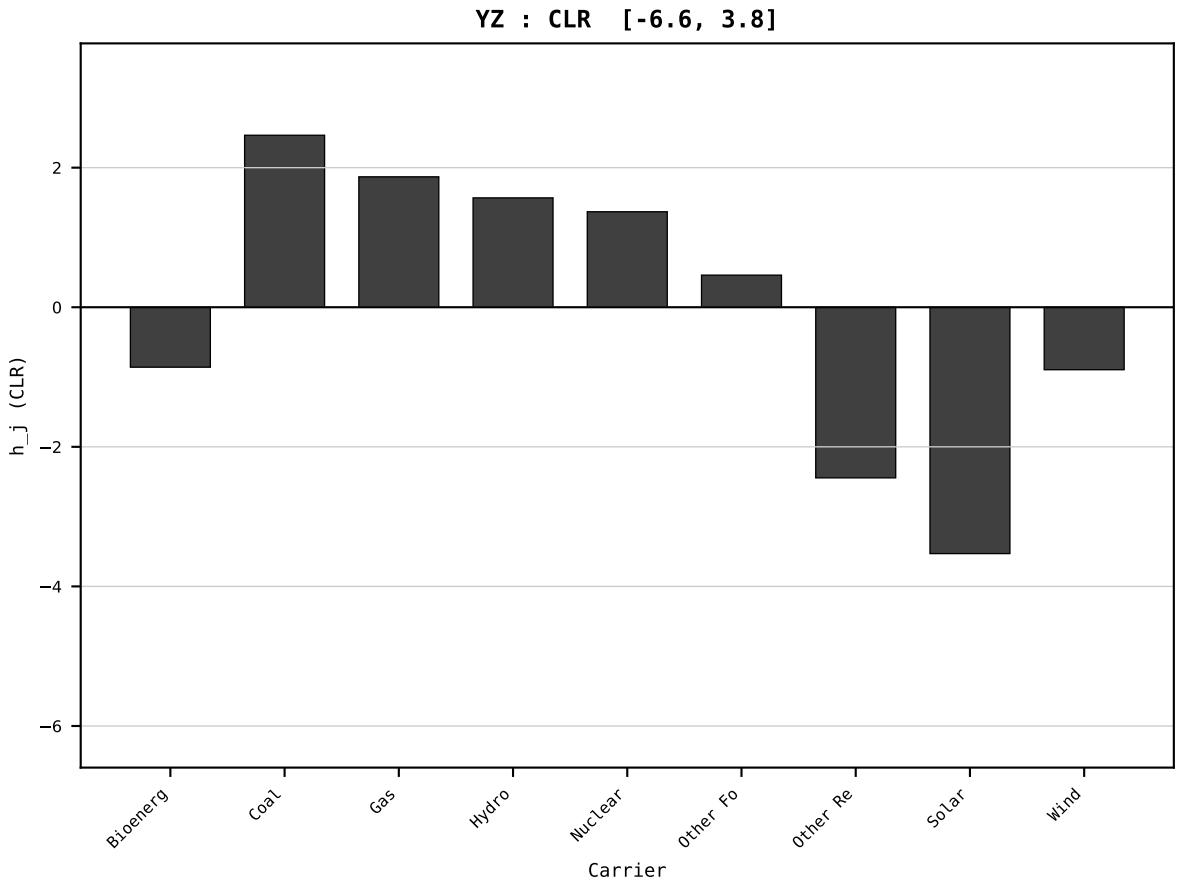
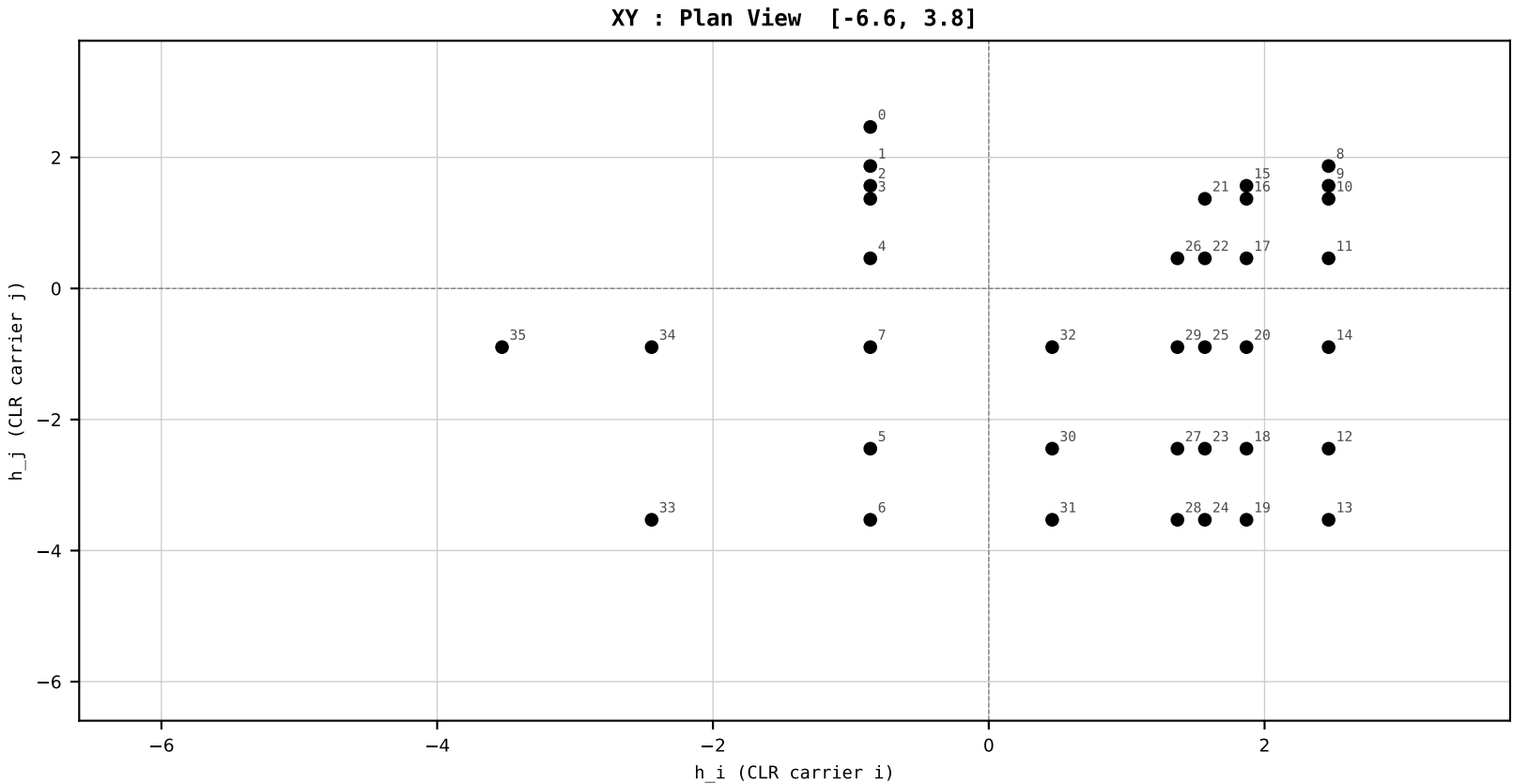
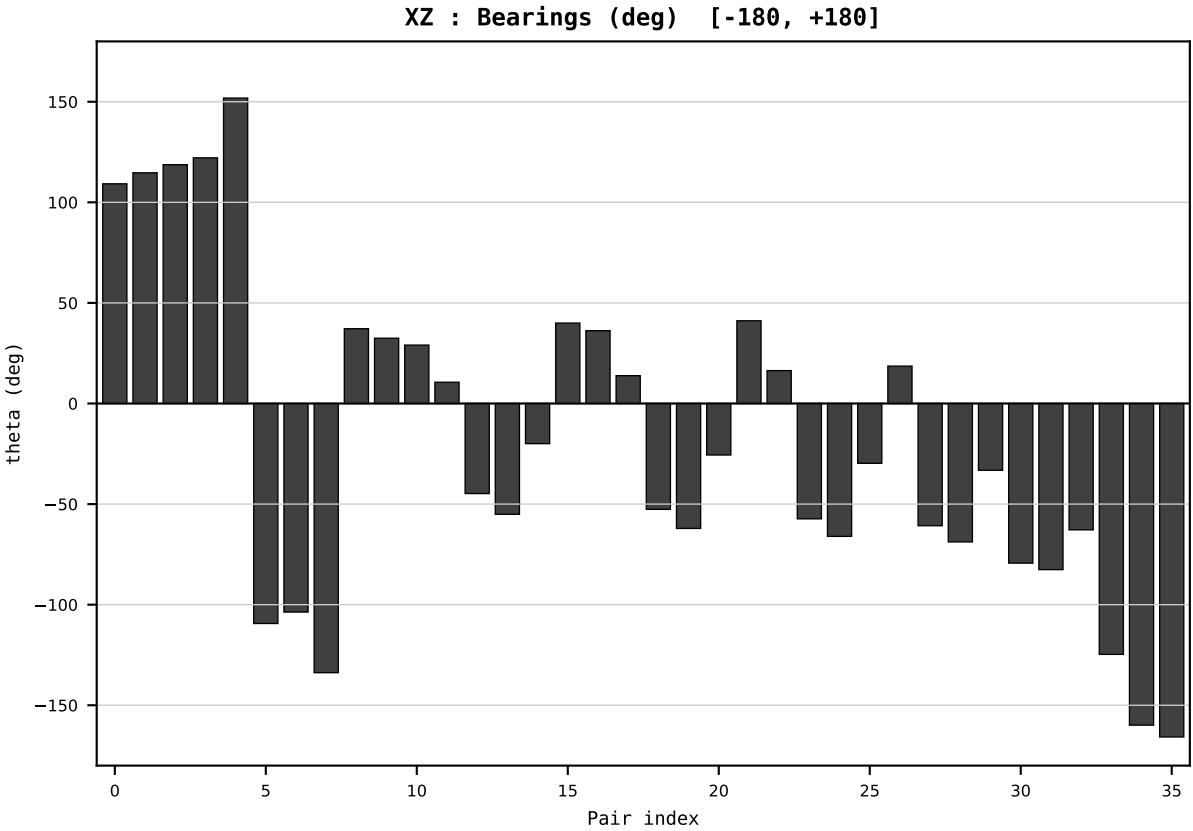
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2009
t : 9 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.287865
Ring = Hs-2
E_metric= 34.0820
kappa_HS= 401.15
omega = 3.1960 deg
d_A = 0.519977
Helm = Solar
Helm d = +0.433127
DR = 5.994 HLR
DR ratio= 401.1x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=10 | Year 2010 | D=9 N=26 pairs=36

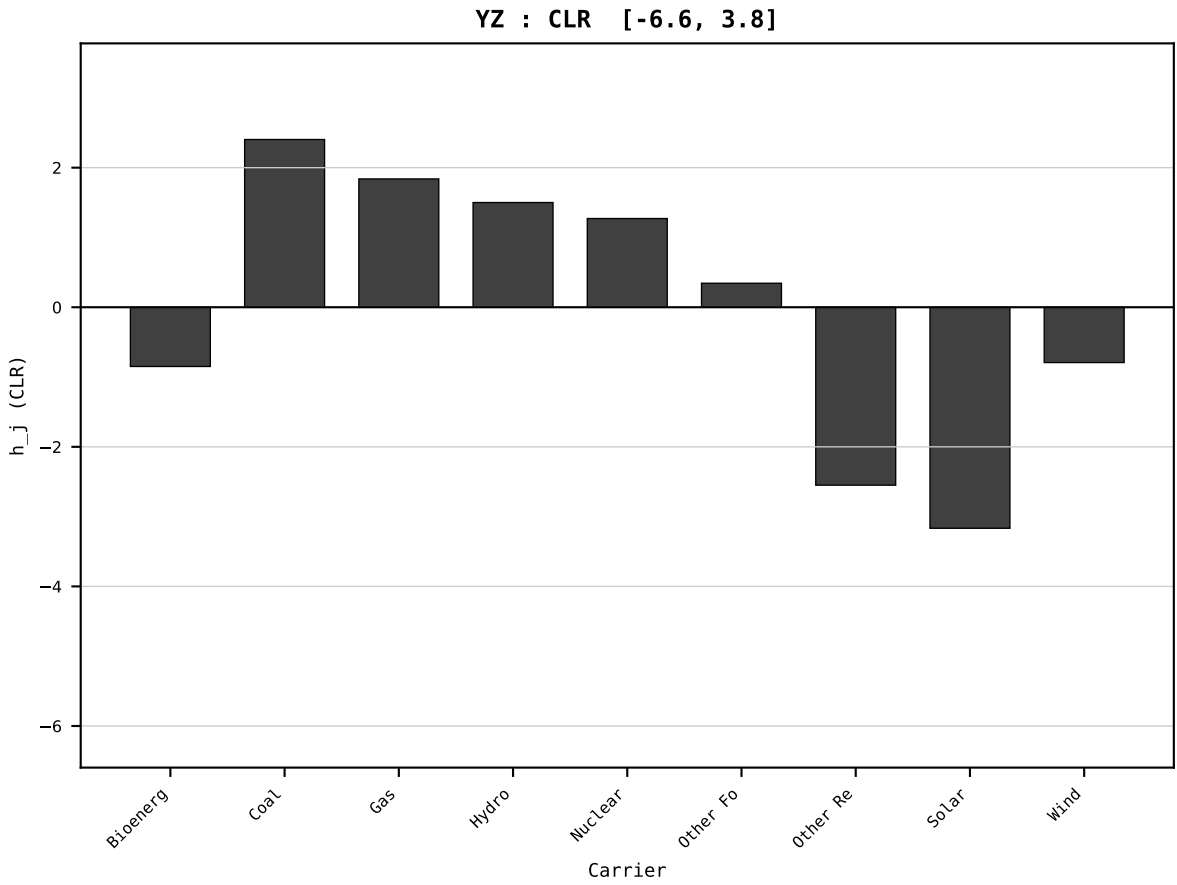
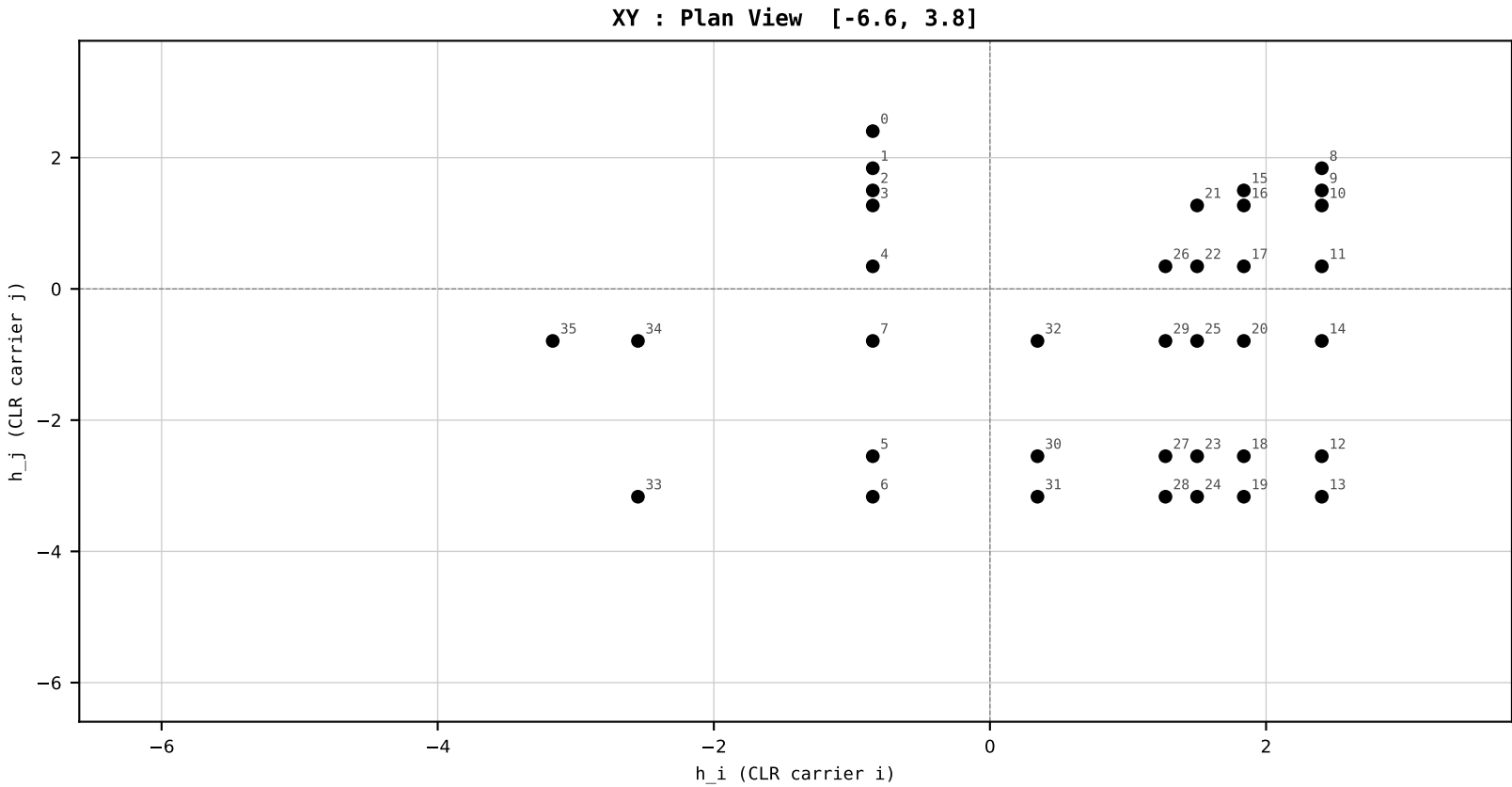
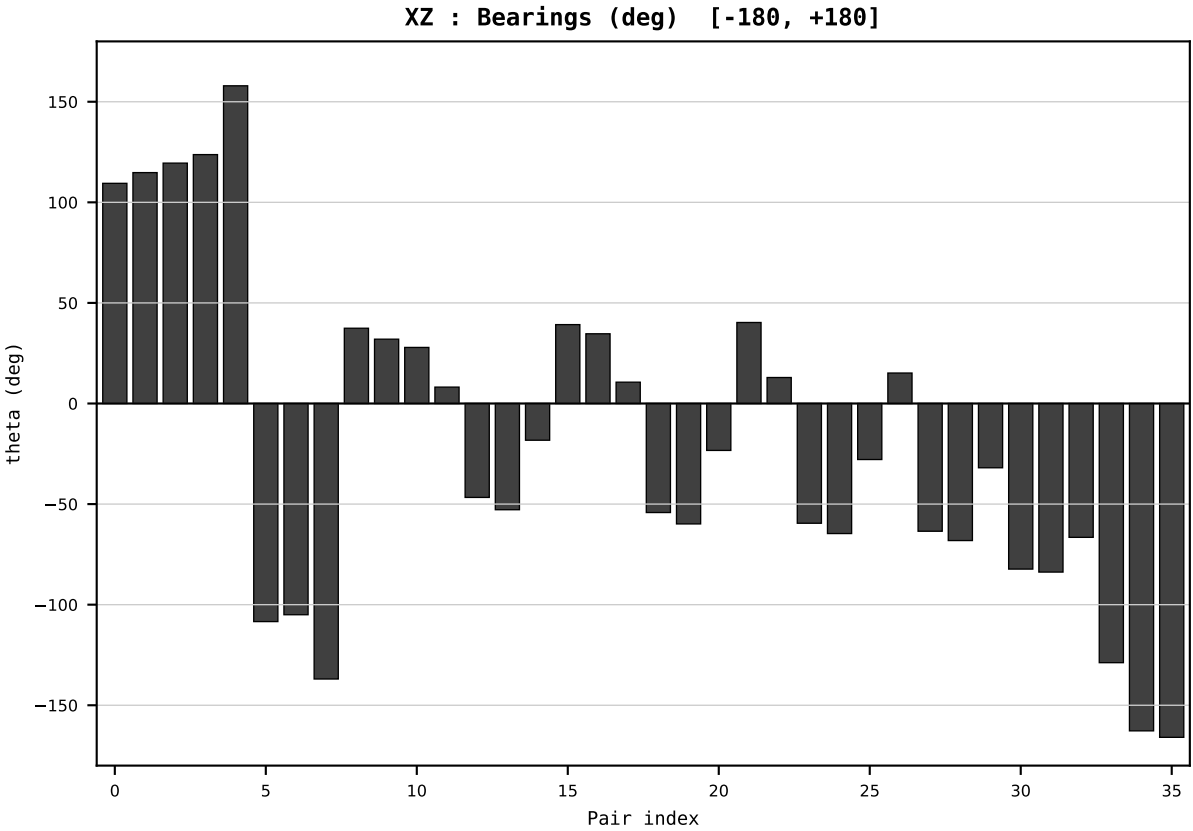
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2010
t : 10 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.285810
Ring = Hs-2
E_metric= 31.0237
kappa_HS= 262.59
omega = 3.3815 deg
d_A = 0.430229
Helm = Solar
Helm d = +0.362834
DR = 5.571 HLR
DR ratio= 262.6x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=11 | Year 2011 | D=9 N=26 pairs=36

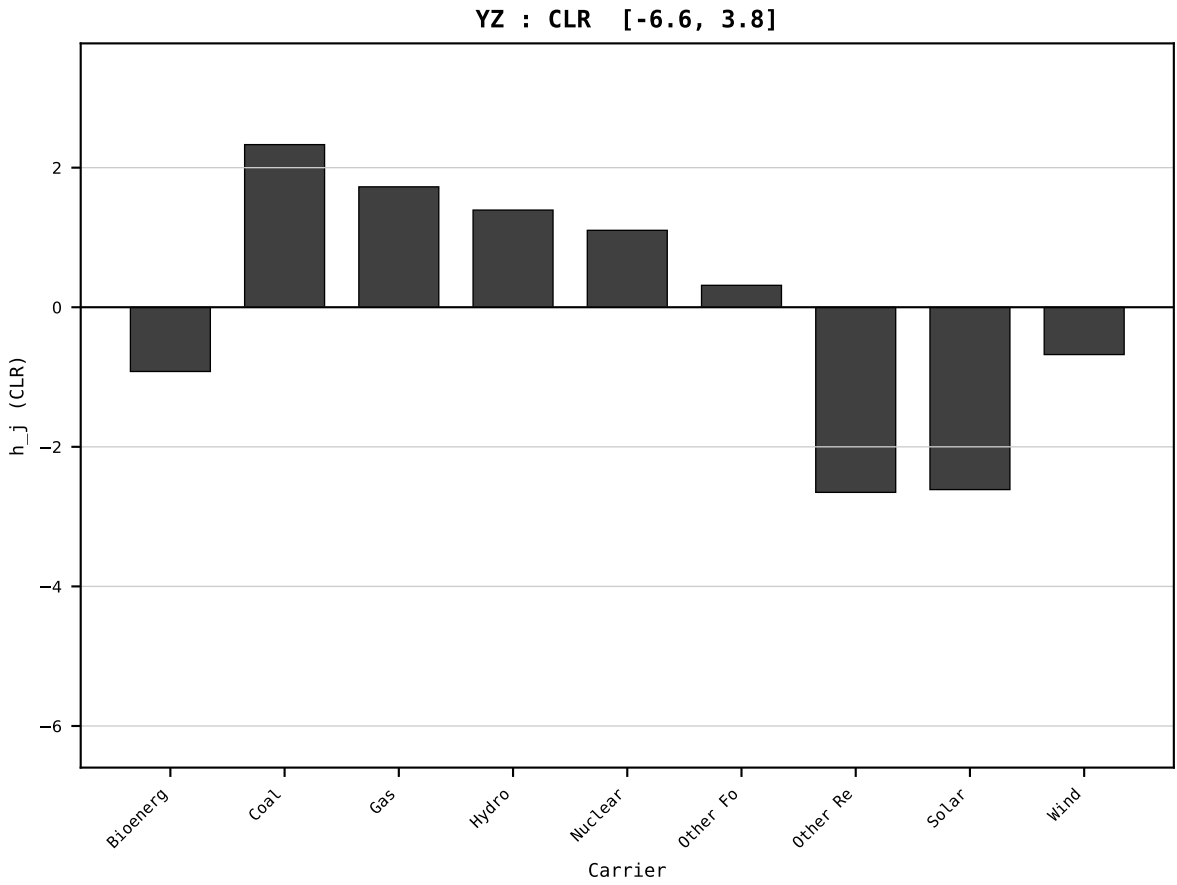
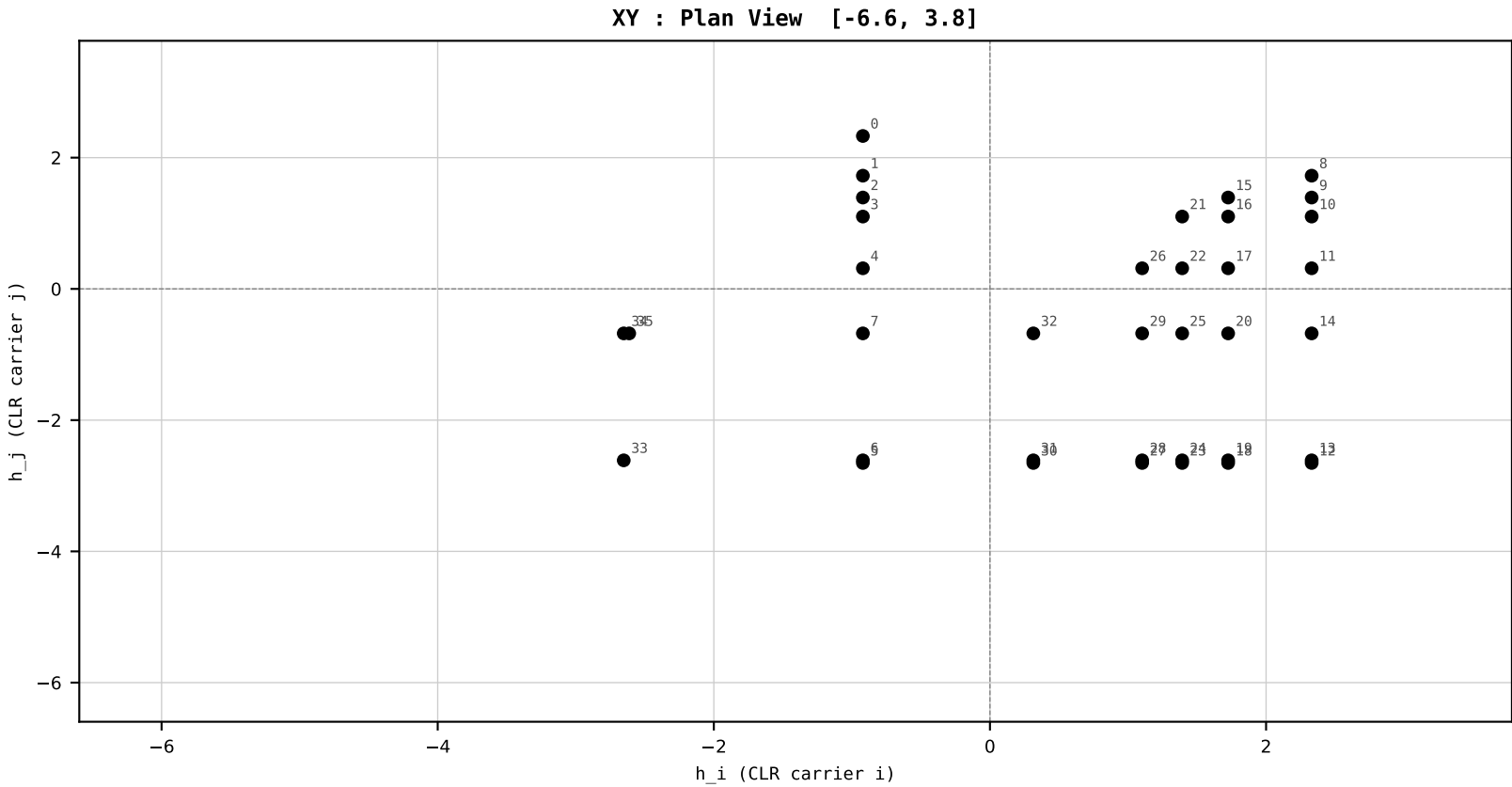
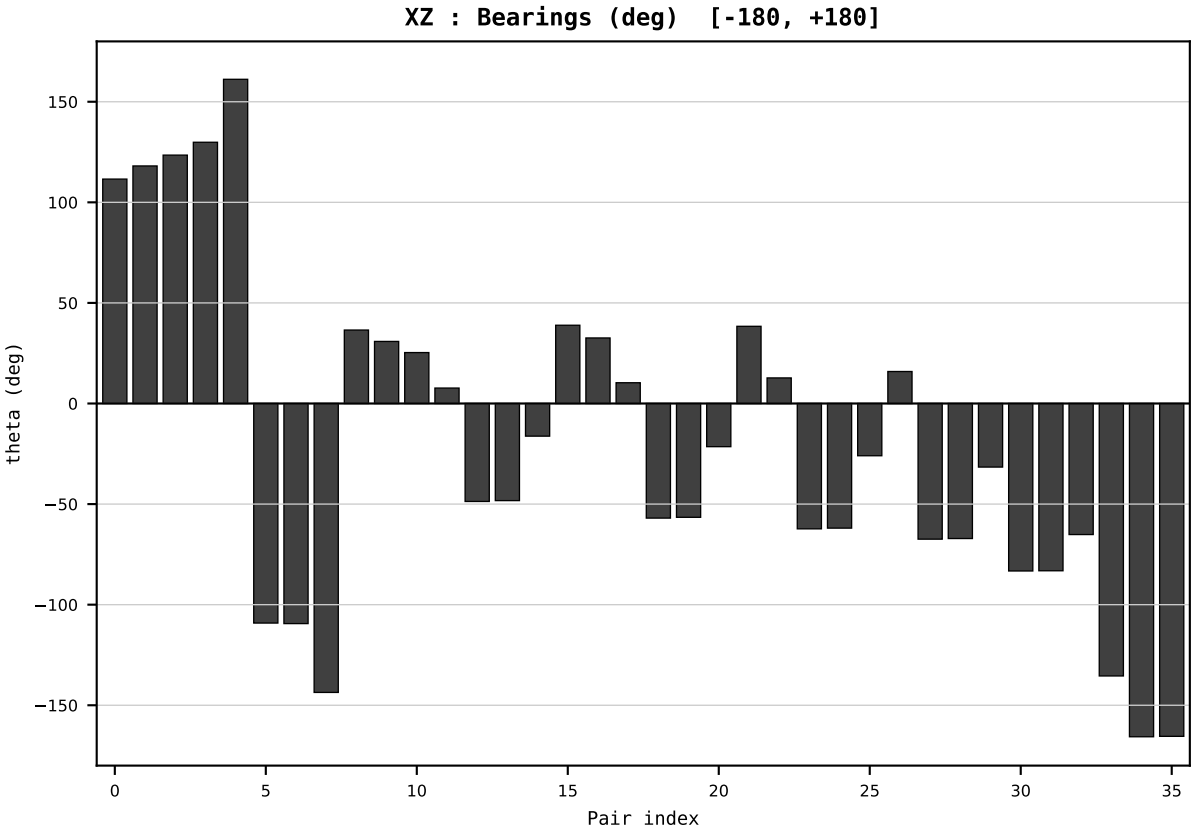
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2011
t : 11 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.281092
Ring = Hs-2
E_metric= 26.8292
kappa_HS= 145.89
omega = 5.2622 deg
d_A = 0.628840
Helm = Solar
Helm d = +0.554151
DR = 4.983 HLR
DR ratio= 145.9x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



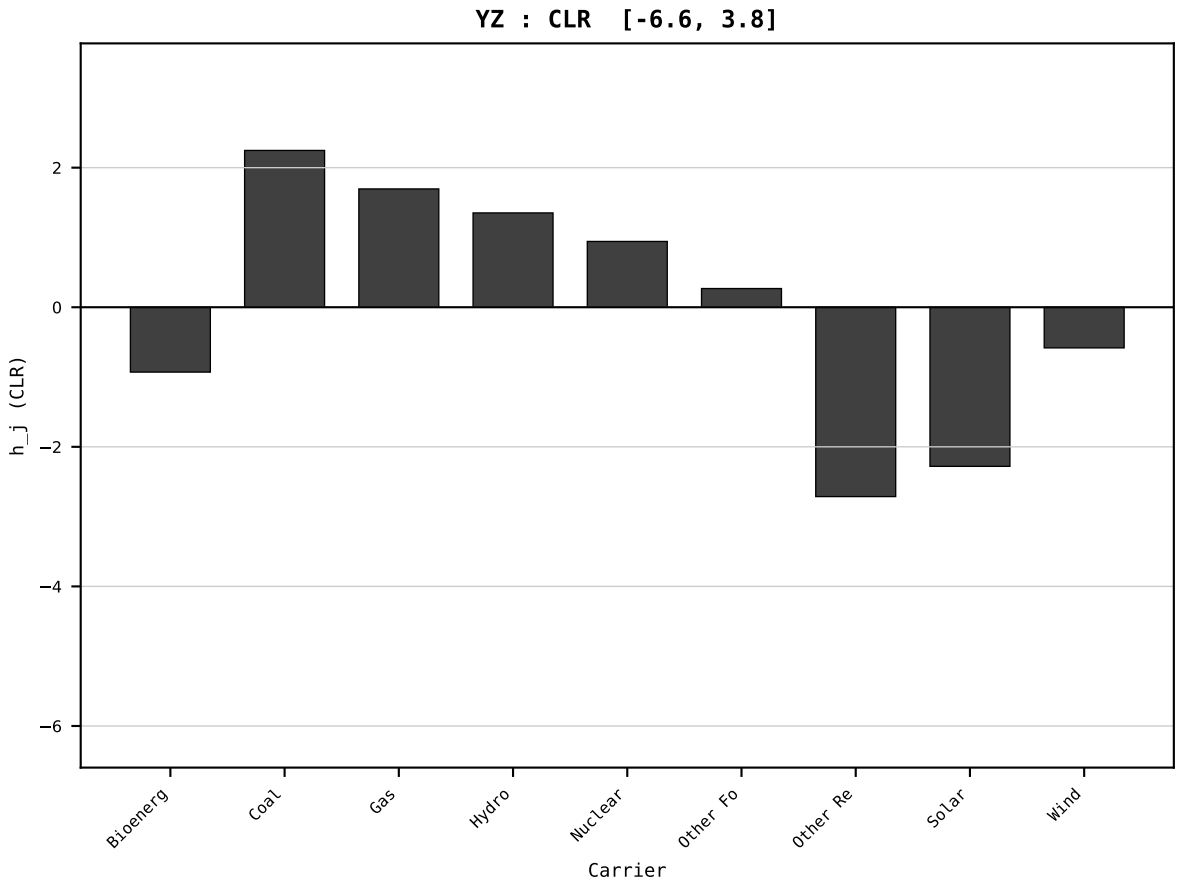
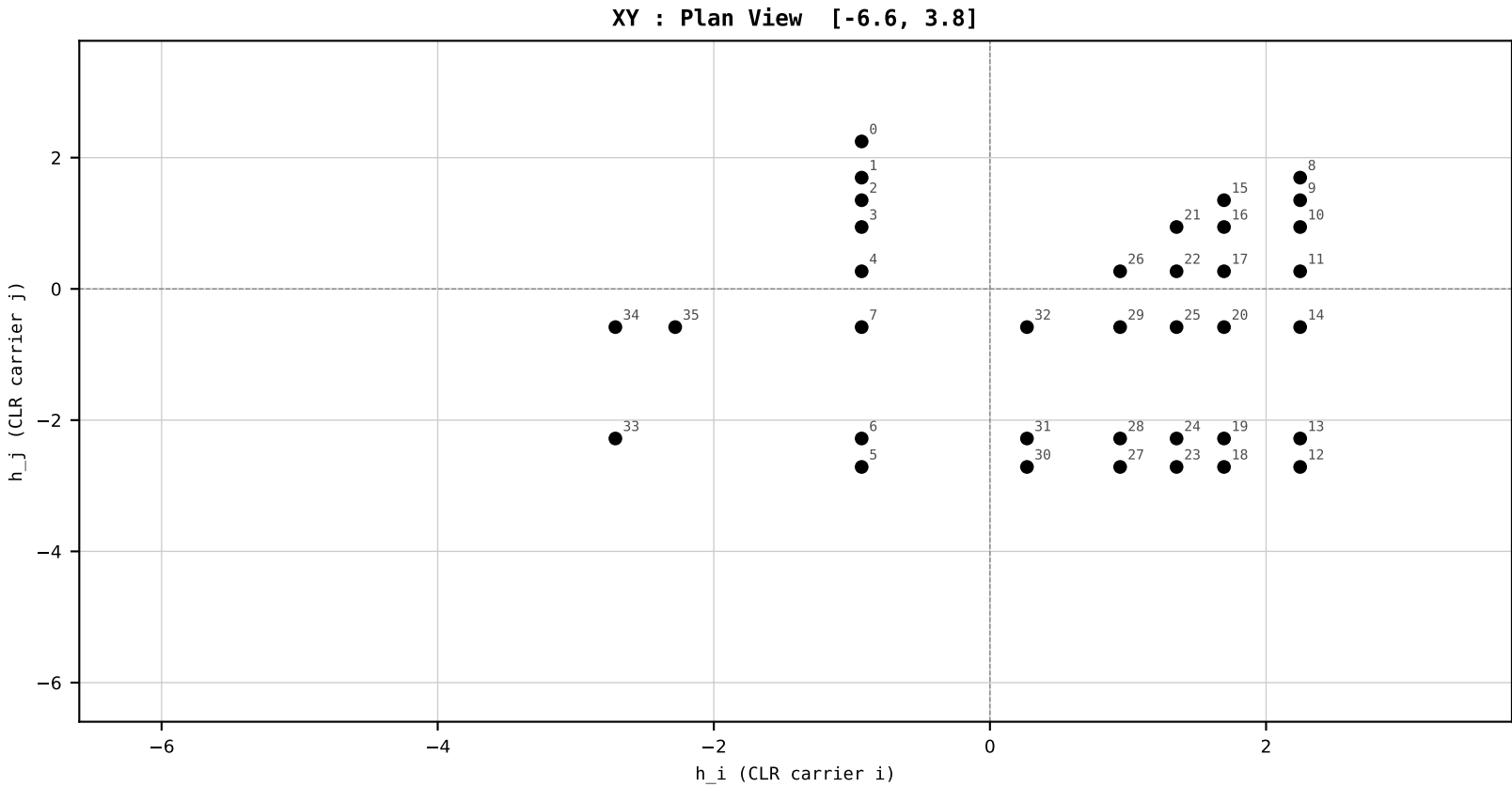
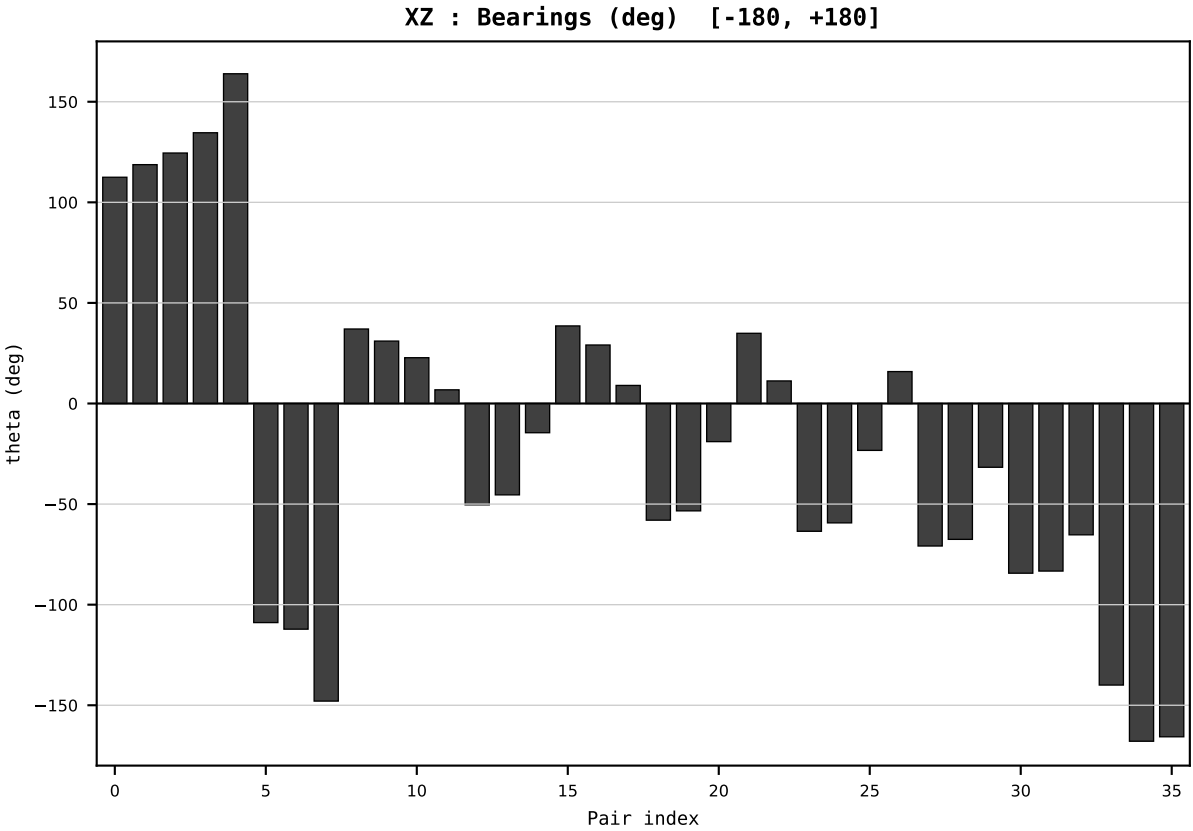
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2012
t : 12 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.273763
Ring = Hs-2
E_metric= 24.4711
kappa_HS= 142.59
omega = 3.6979 deg
d_A = 0.401150
Helm = Solar
Helm d = +0.333042
DR = 4.960 HLR
DR ratio= 142.6x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



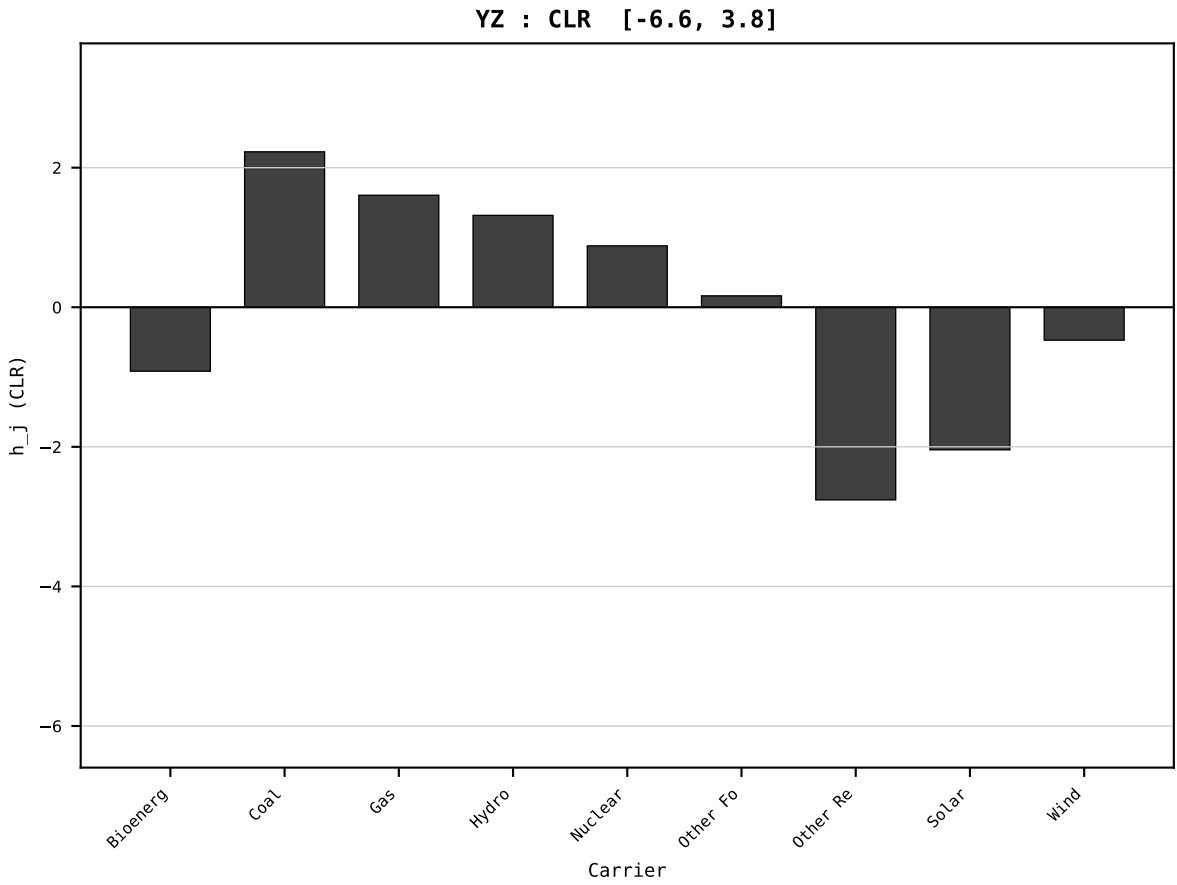
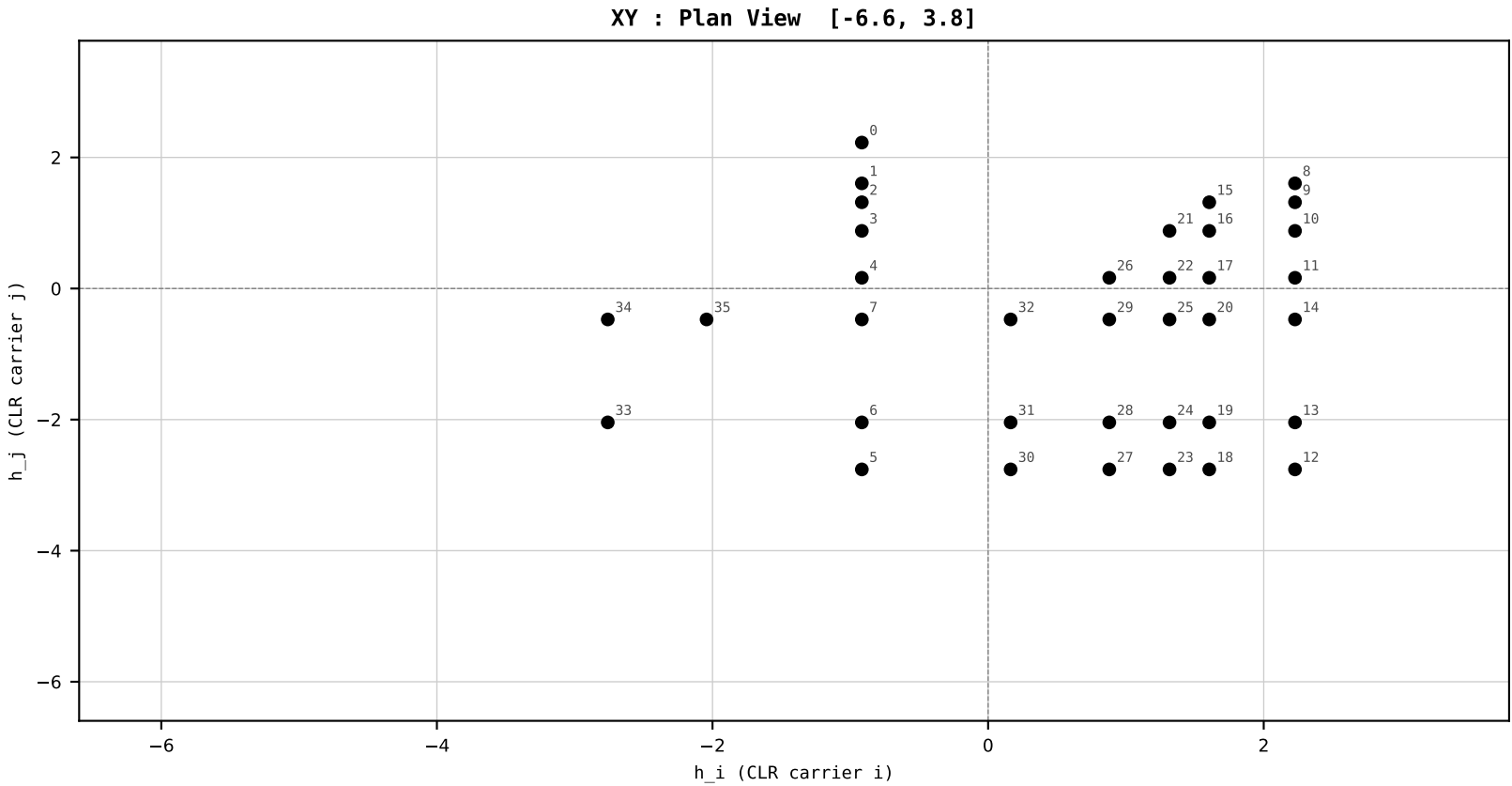
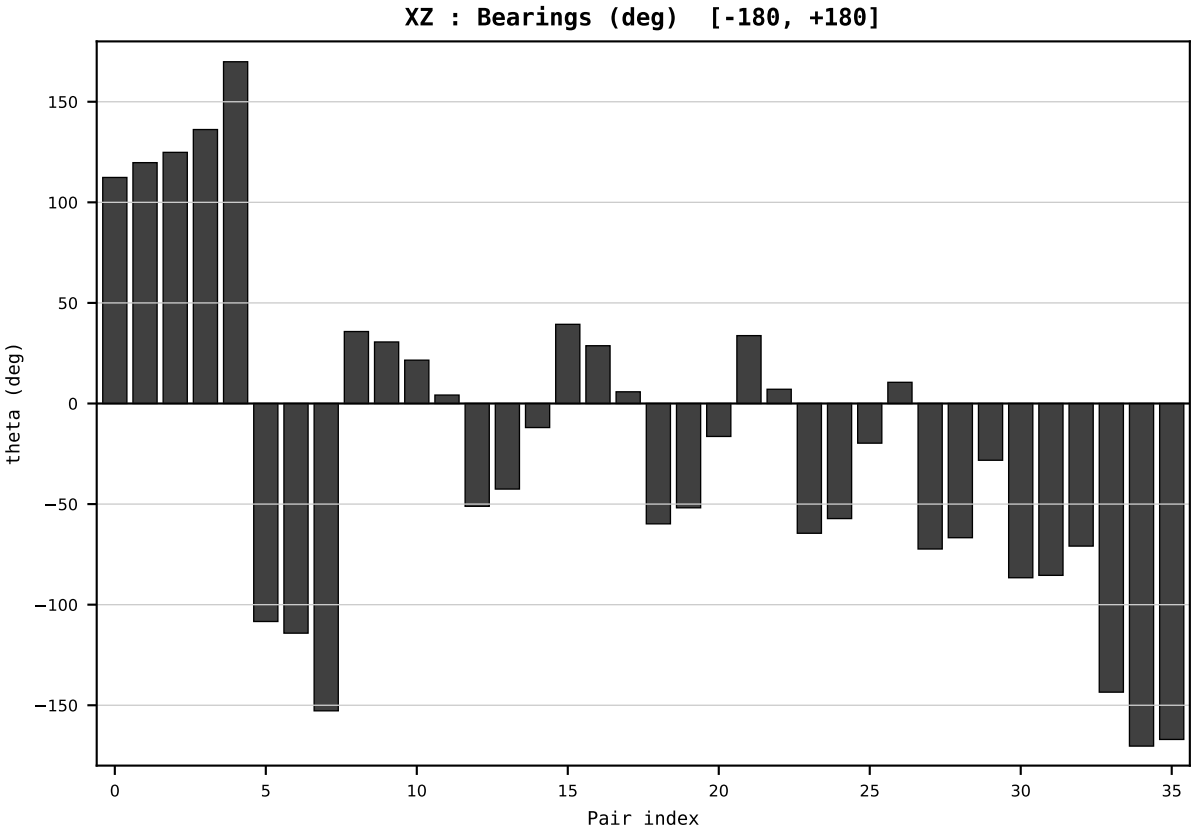
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2013
t : 13 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.271626
Ring = Hs-2
E_metric= 22.9224
kappa_HS= 146.52
omega = 3.1180 deg
d_A = 0.308917
Helm = Solar
Helm d = +0.236796
DR = 4.987 HLR
DR ratio= 146.5x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=14 | Year 2014 | D=9 N=26 pairs=36

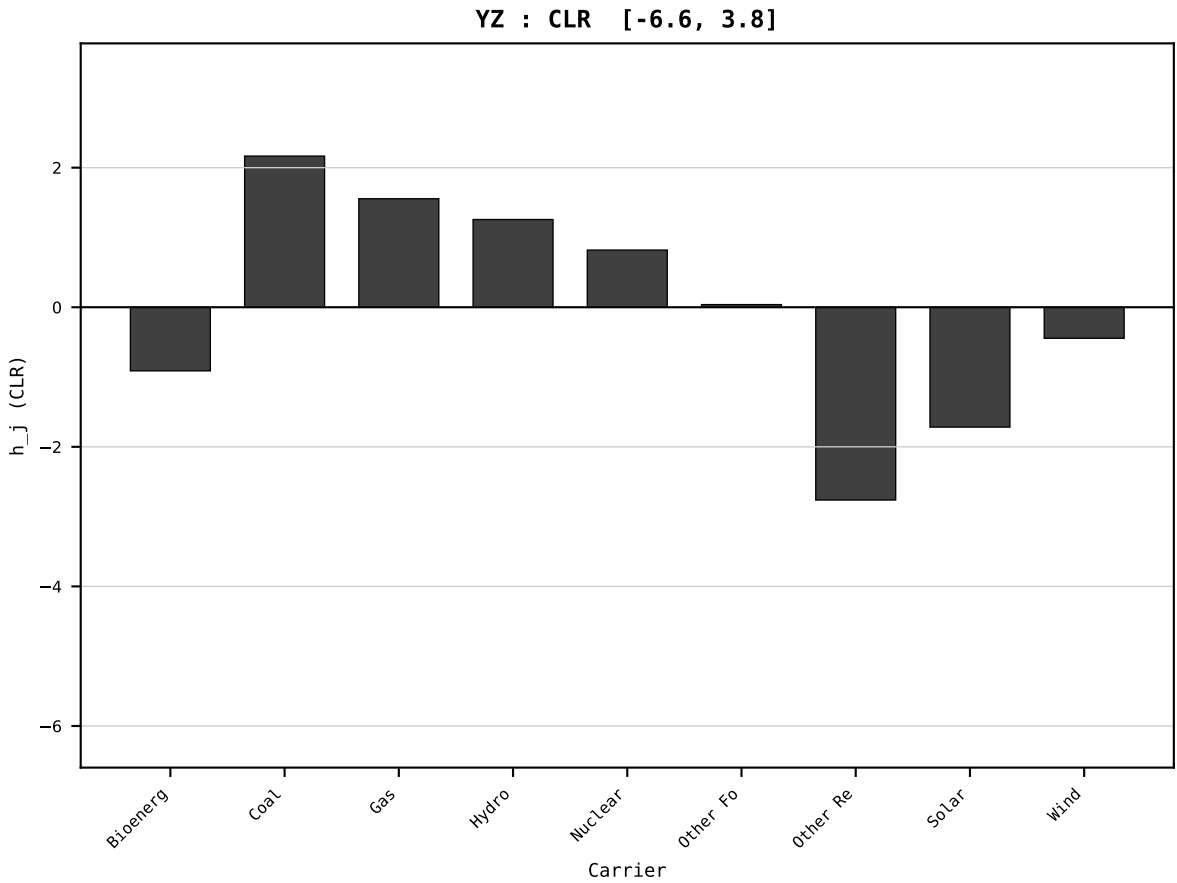
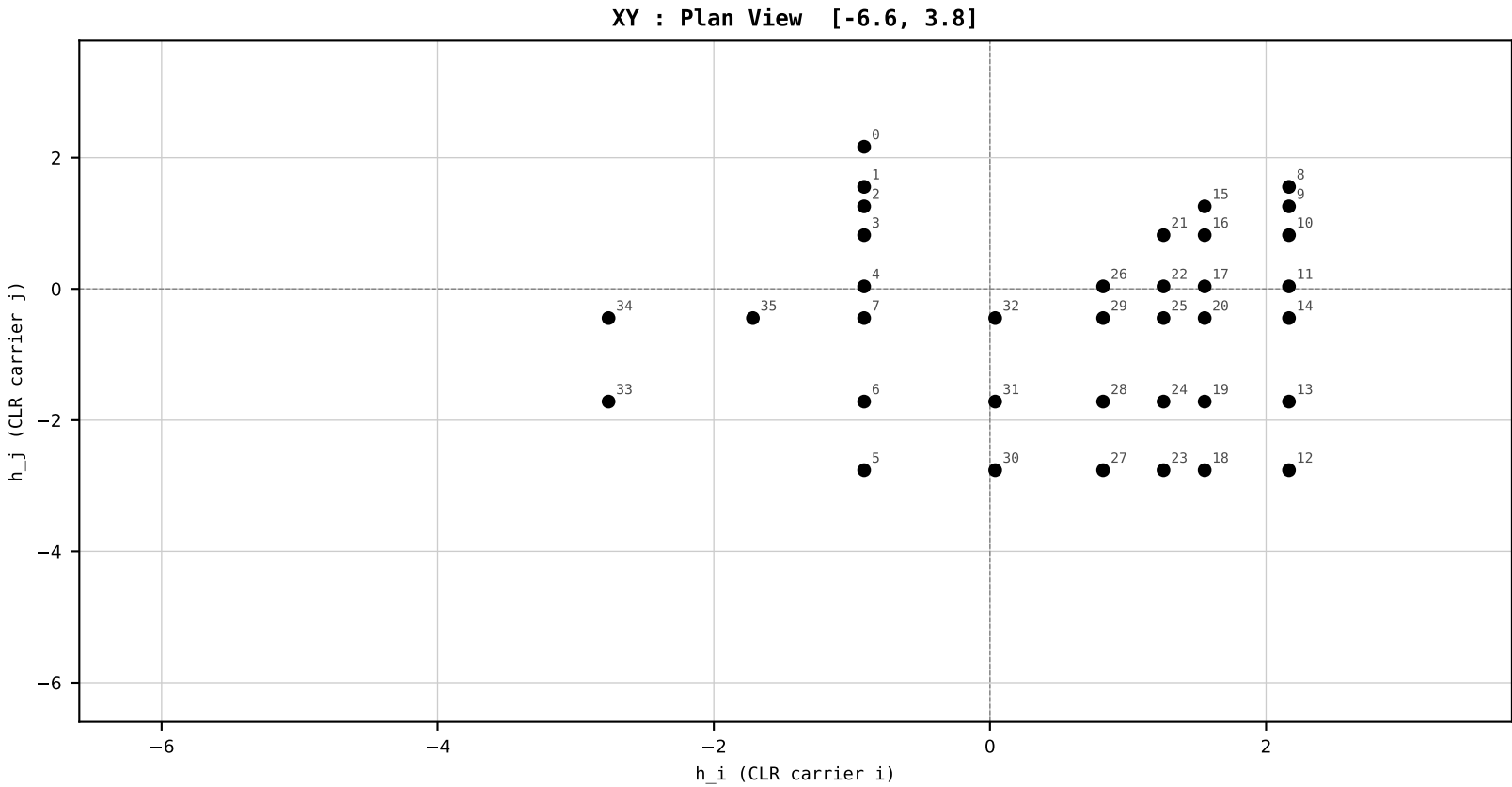
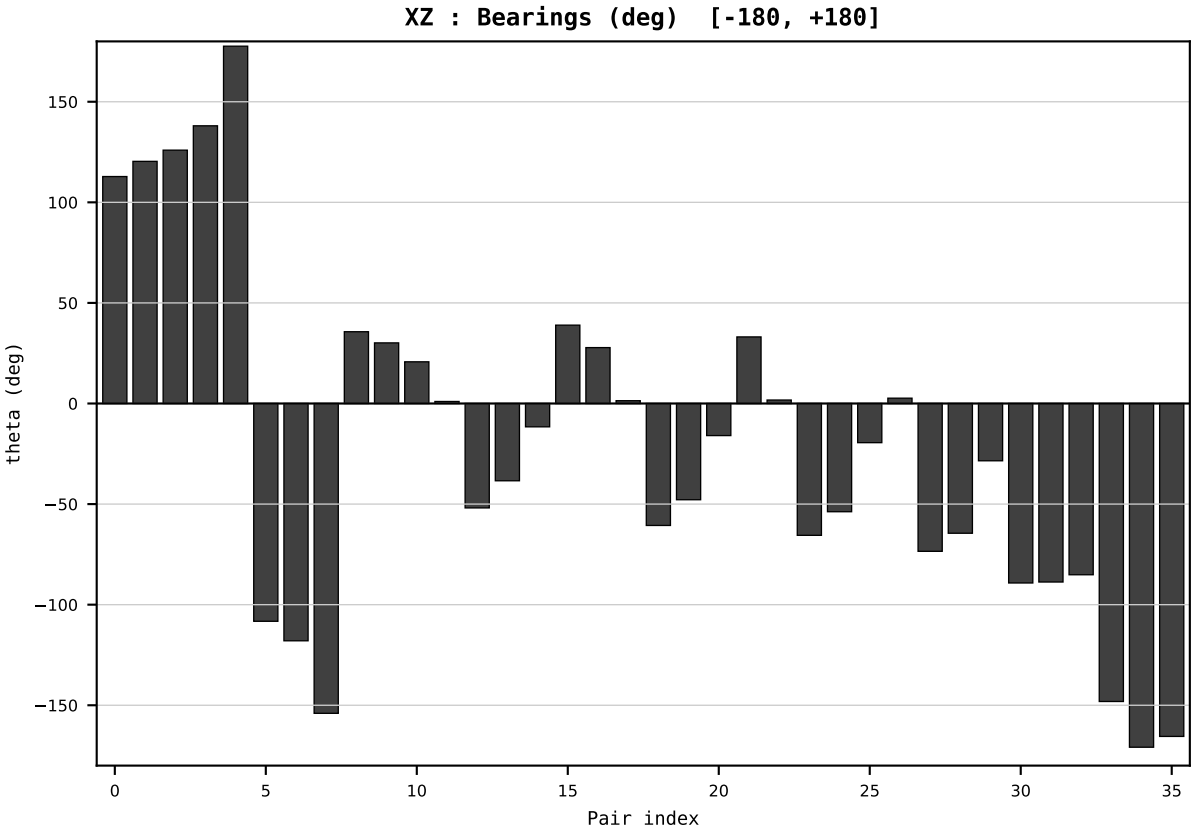
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2014
t : 14 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.265704
Ring = Hs-2
E_metric= 20.9690
kappa_HS= 138.13
omega = 3.7243 deg
d_A = 0.368906
Helm = Solar
Helm d = +0.325856
DR = 4.928 HLR
DR ratio= 138.1x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



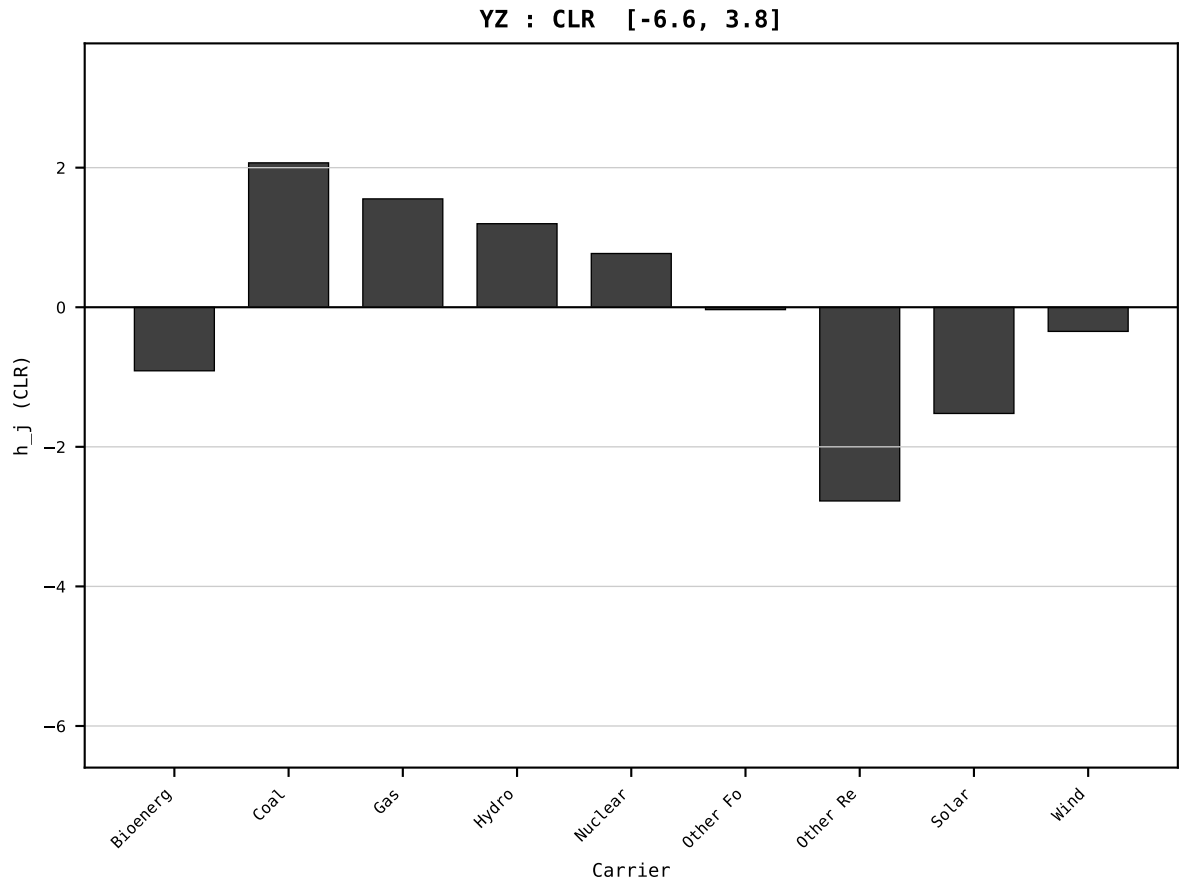
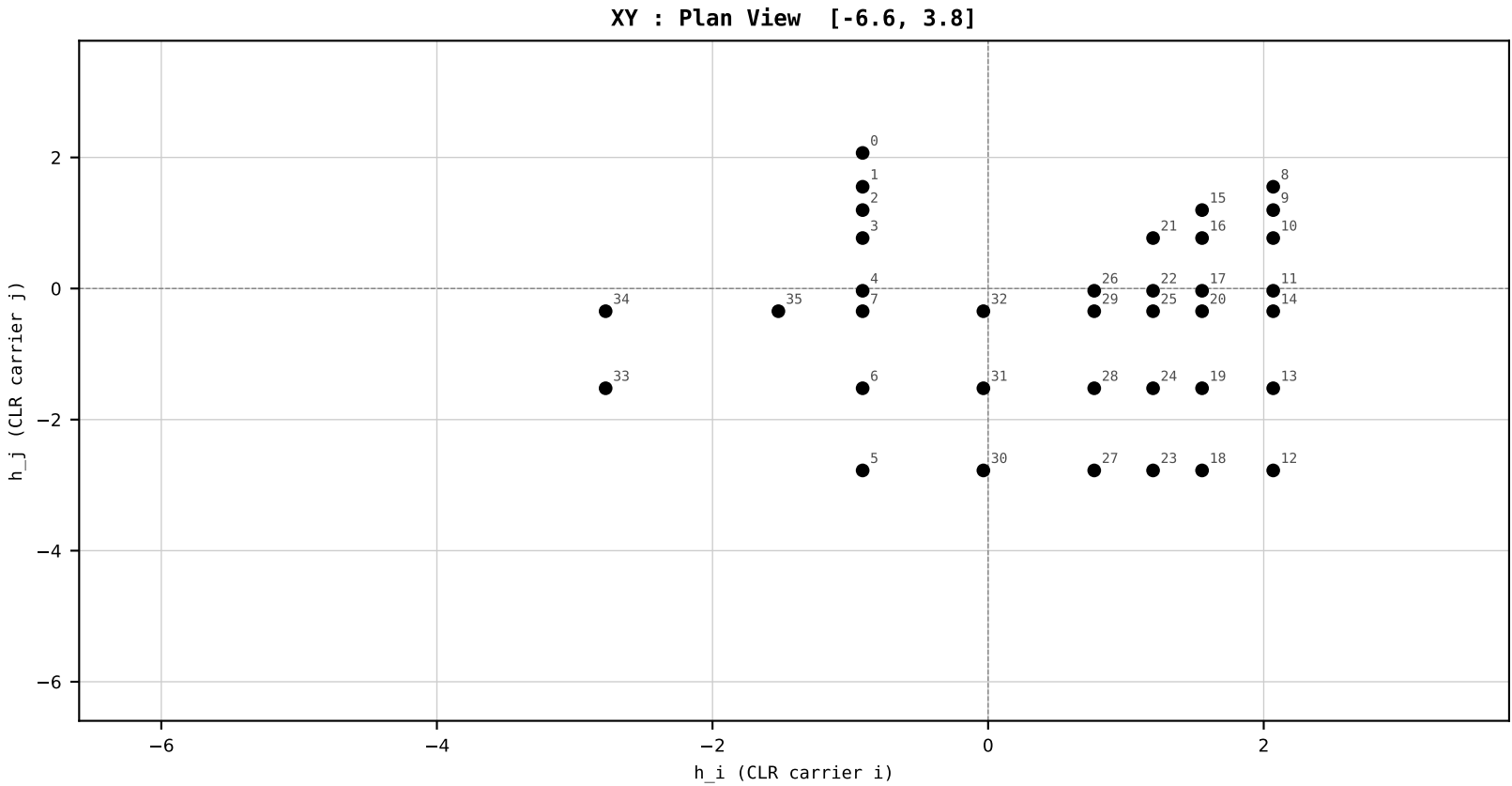
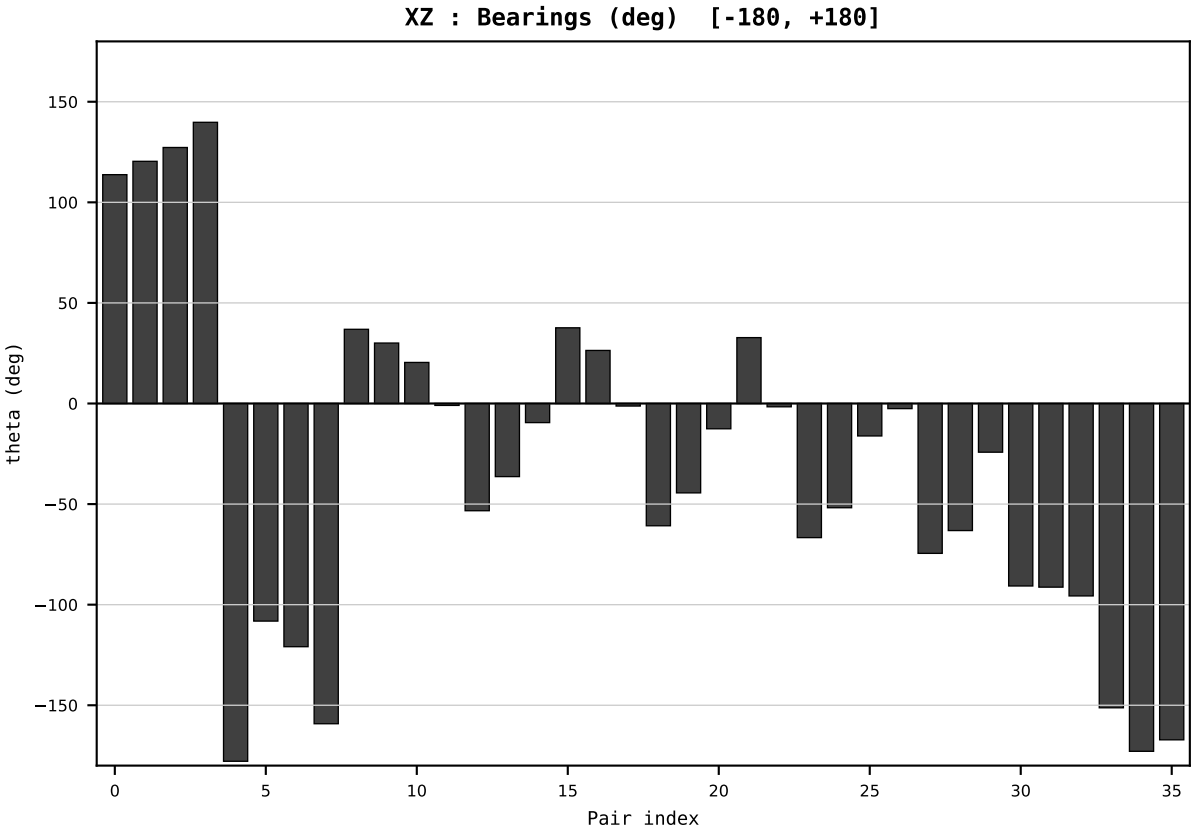
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2015
t : 15 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.252906
Ring = Hs-2
E_metric= 19.6868
kappa_HS= 127.03
omega = 2.7958 deg
d_A = 0.261905
Helm = Solar
Helm d = +0.195252
DR = 4.844 HLR
DR ratio= 127.0x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



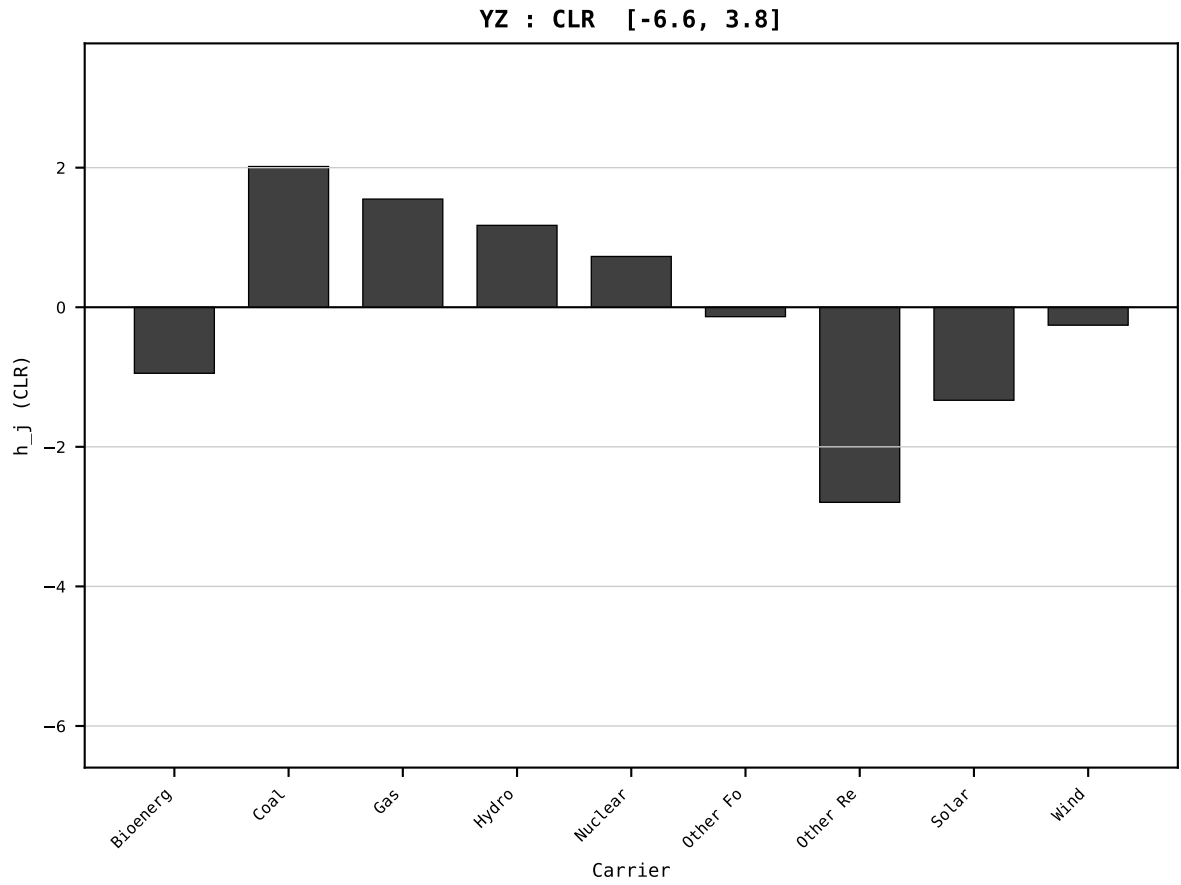
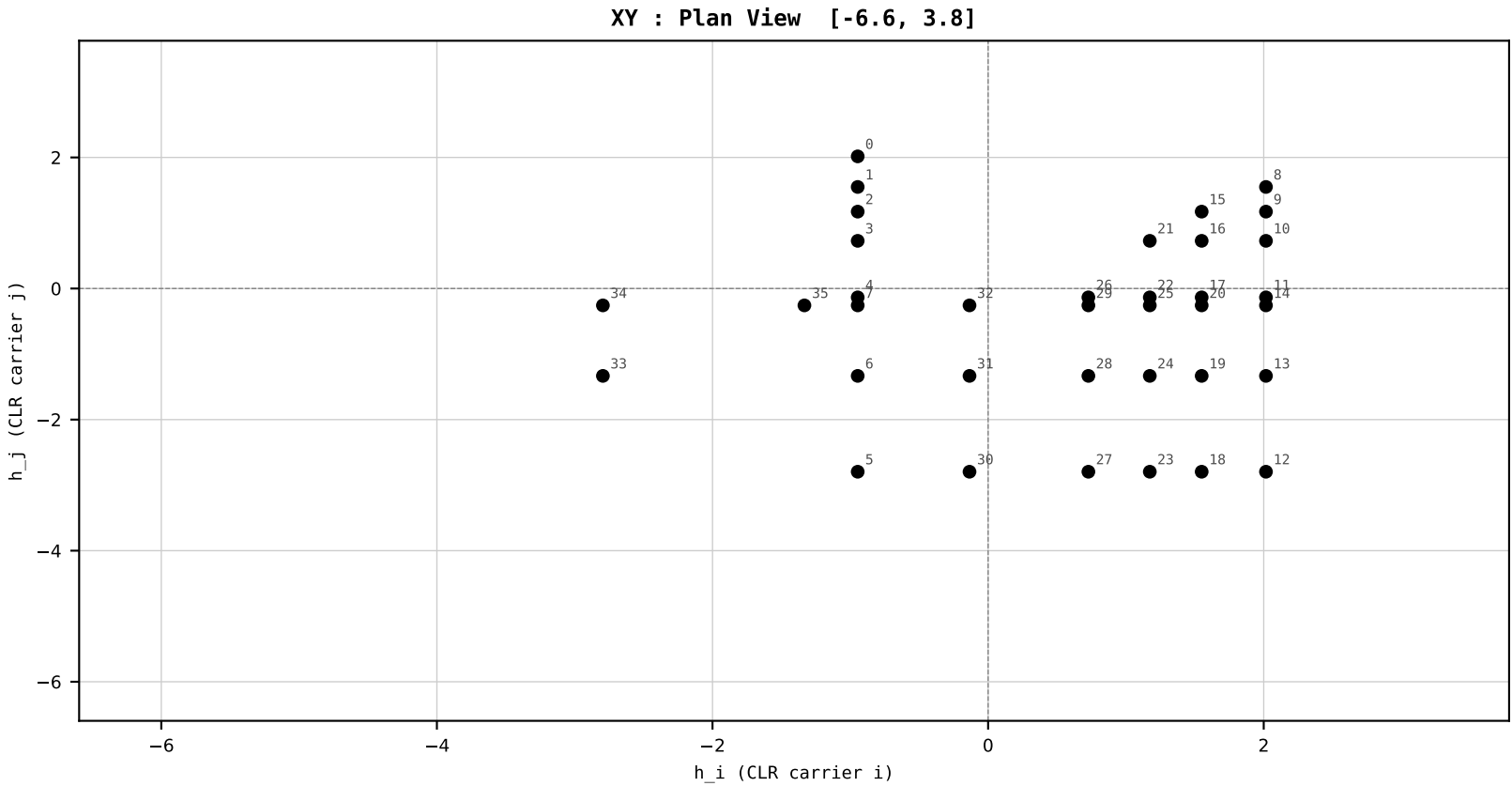
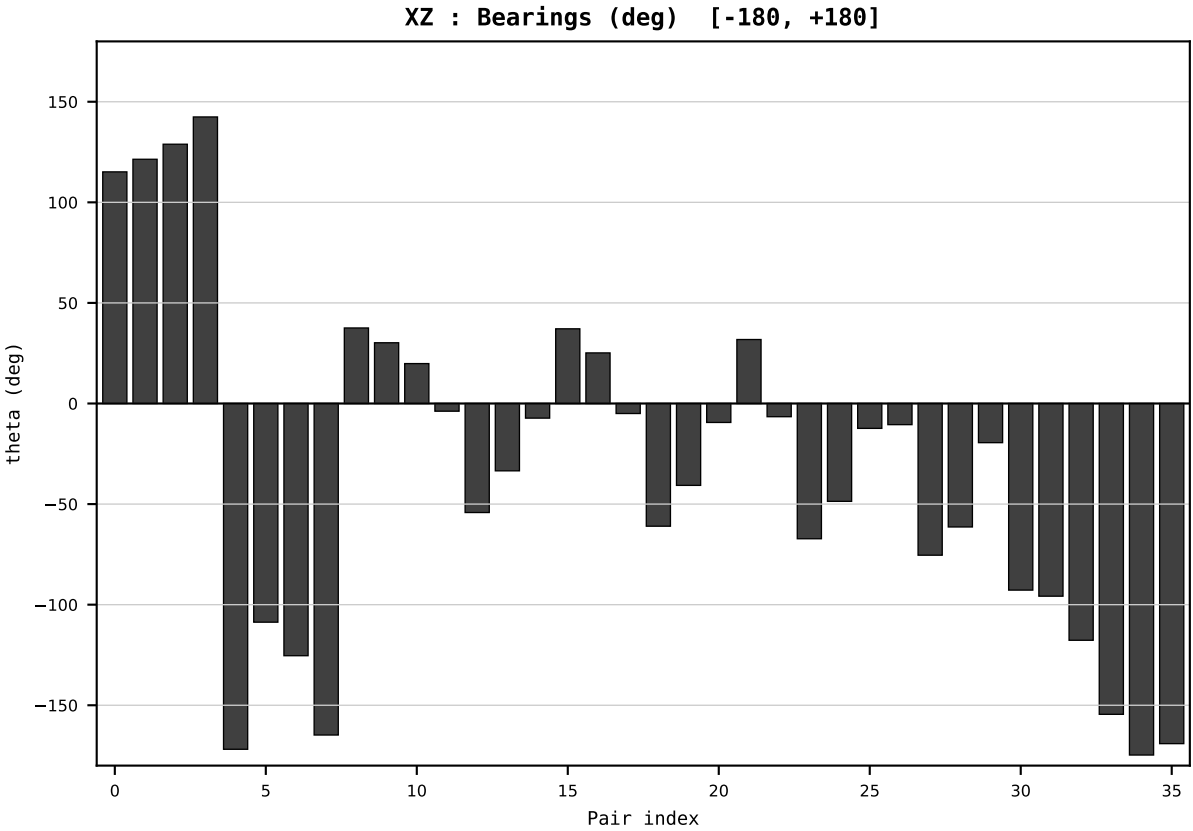
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2016
t : 16 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.246638
Ring = Hs-2
E_metric= 18.9446
kappa_HS= 122.95
omega = 3.0133 deg
d_A = 0.246037
Helm = Solar
Helm d = +0.188869
DR = 4.812 HLR
DR ratio= 122.9x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=17 | Year 2017 | D=9 N=26 pairs=36

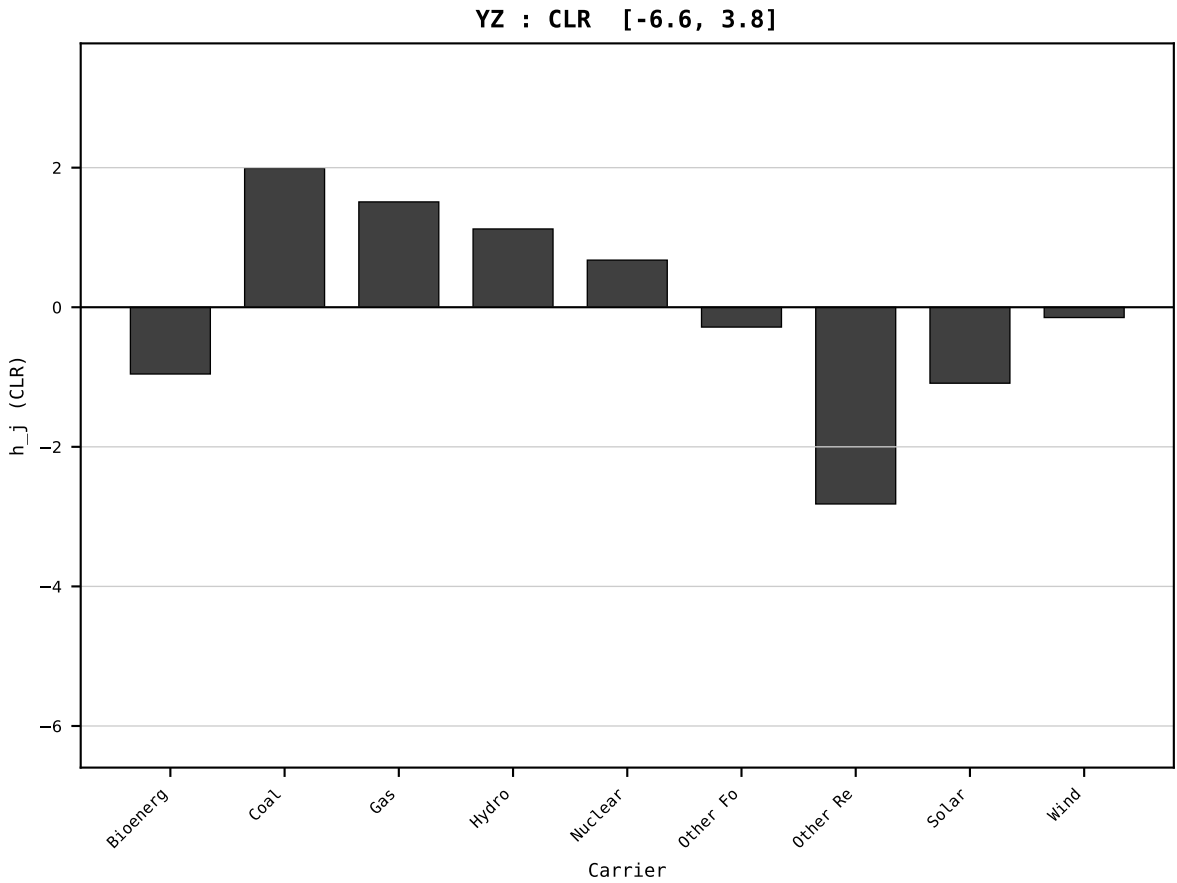
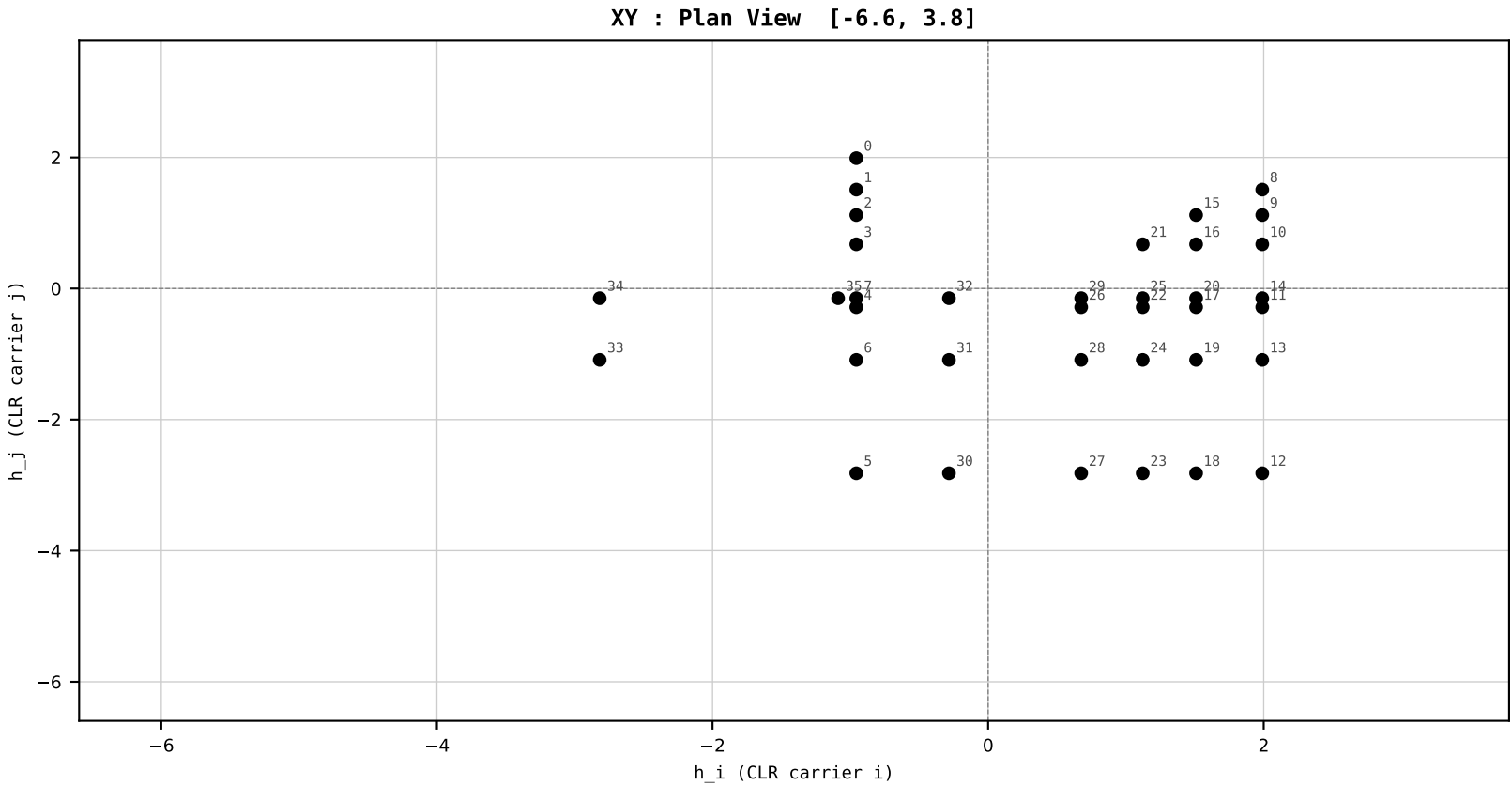
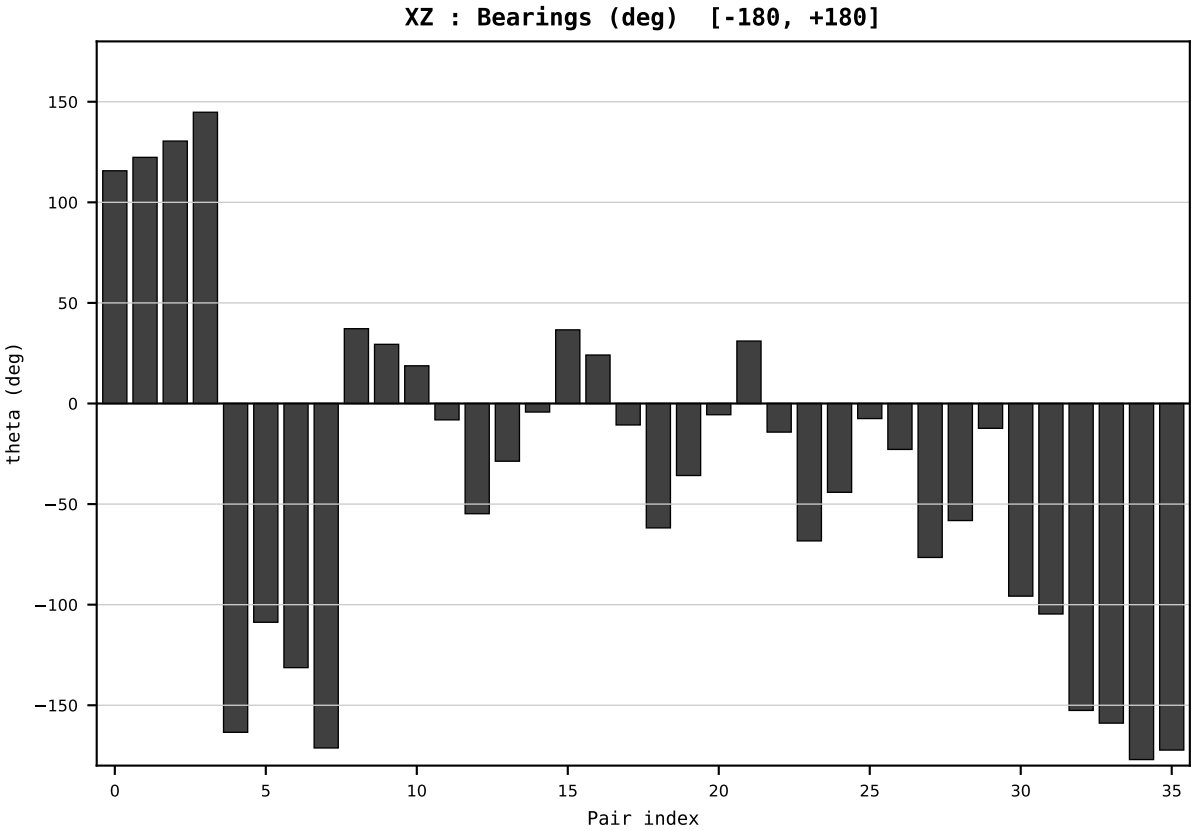
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2017
t : 17 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.241826
Ring = Hs-2
E_metric= 18.0970
kappa_HS= 122.49
omega = 4.0572 deg
d_A = 0.320167
Helm = Solar
Helm d = +0.244504
DR = 4.808 HLR
DR ratio= 122.5x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



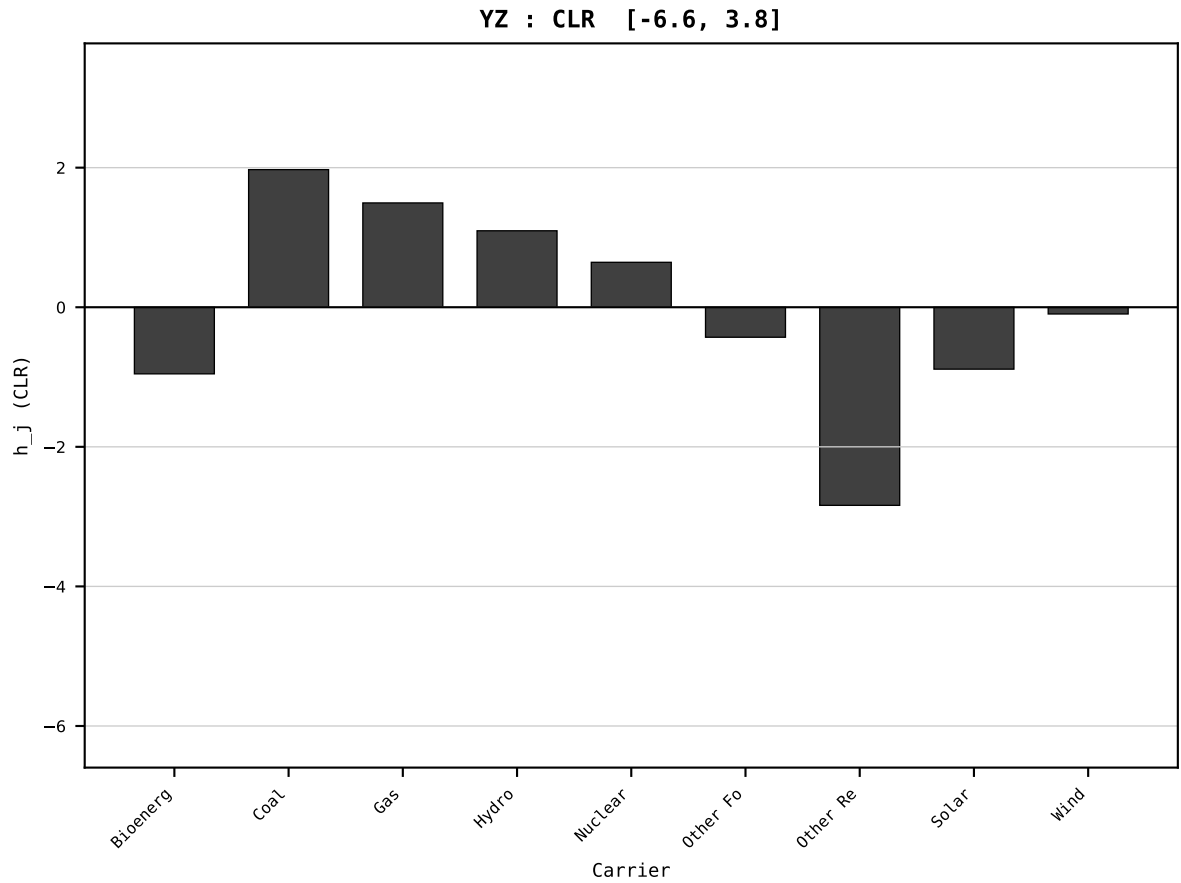
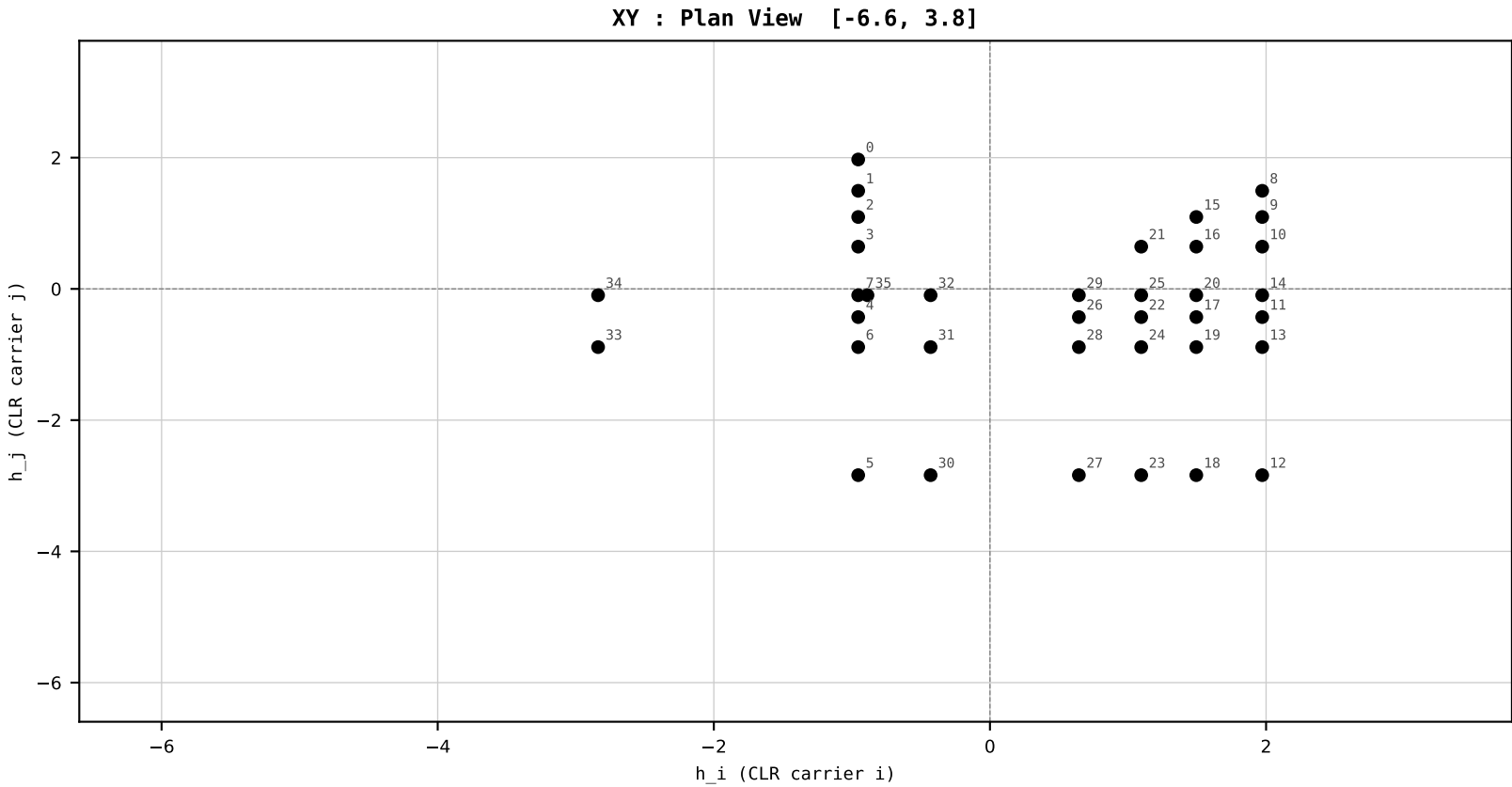
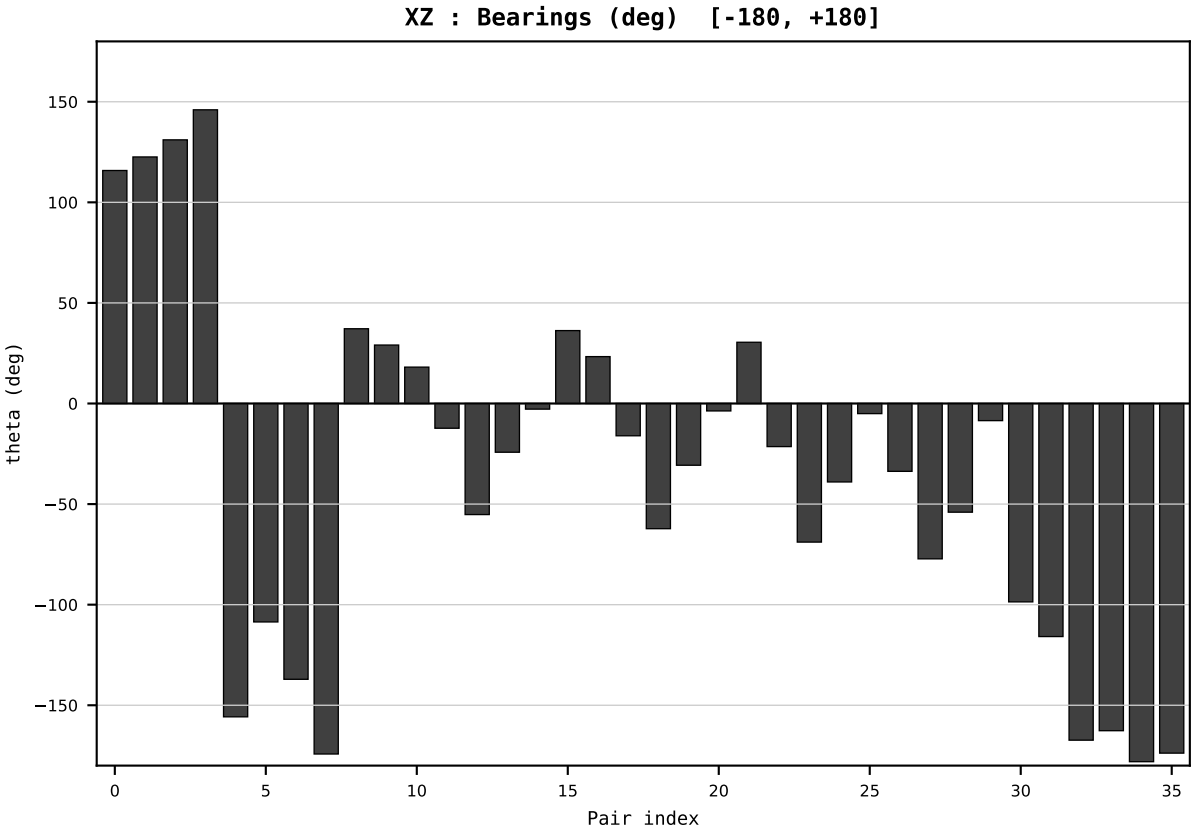
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2018
t : 18 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.239063
Ring = Hs-2
E_metric= 17.6853
kappa_HS= 122.77
omega = 3.4435 deg
d_A = 0.258782
Helm = Solar
Helm d = +0.201485
DR = 4.810 HLR
DR ratio= 122.8x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



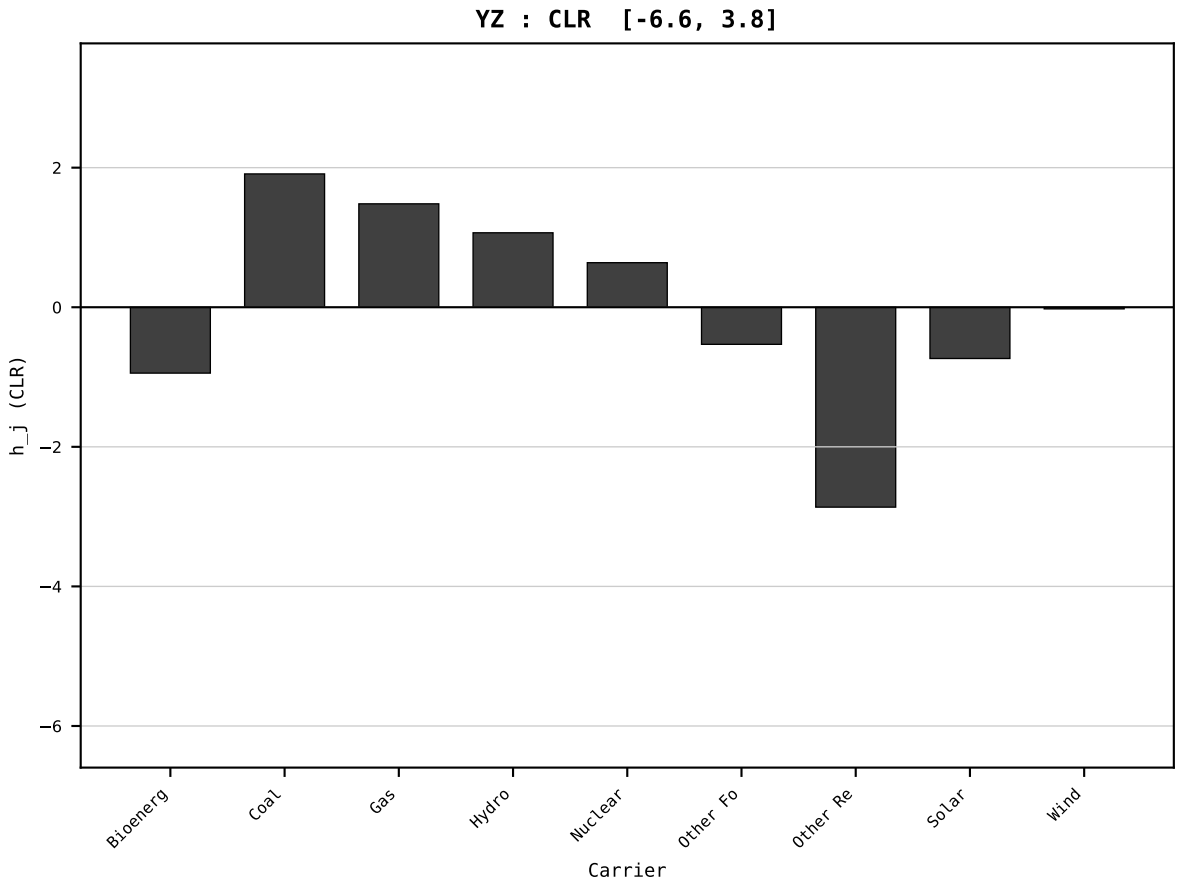
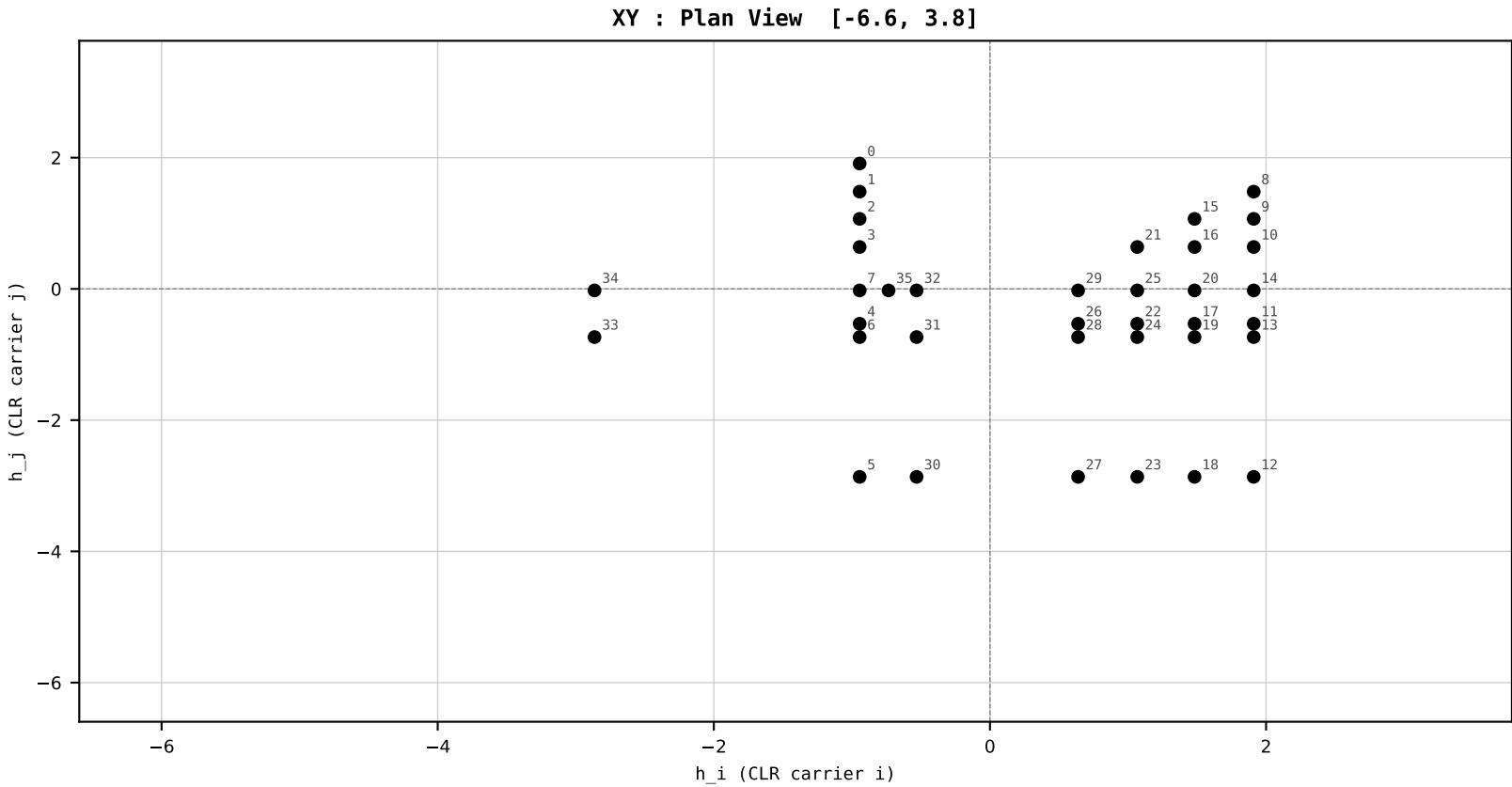
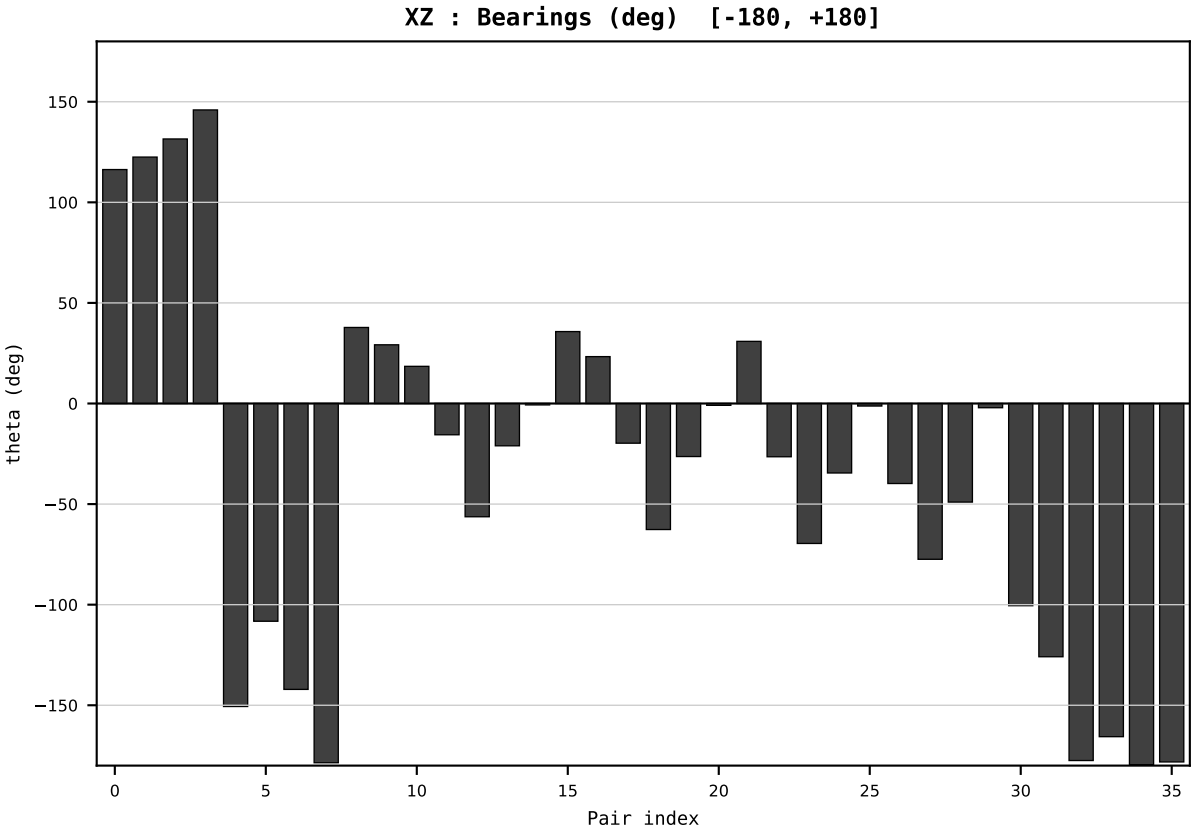
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2019
t : 19 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.228939
Ring = Hs-2
E_metric= 17.3079
kappa_HS= 118.46
omega = 2.8245 deg
d_A = 0.211059
Helm = Solar
Helm d = +0.152684
DR = 4.775 HLR
DR ratio= 118.5x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=20 | Year 2020 | D=9 N=26 pairs=36

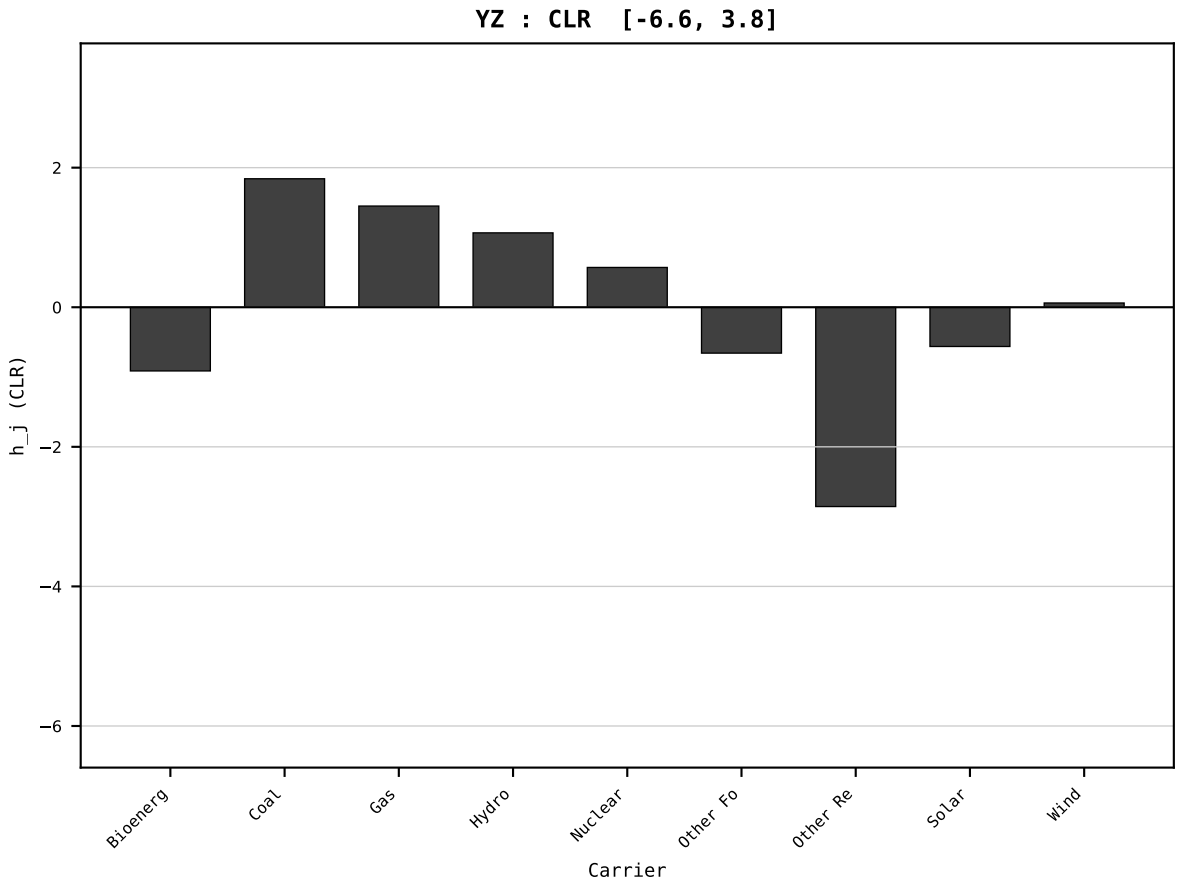
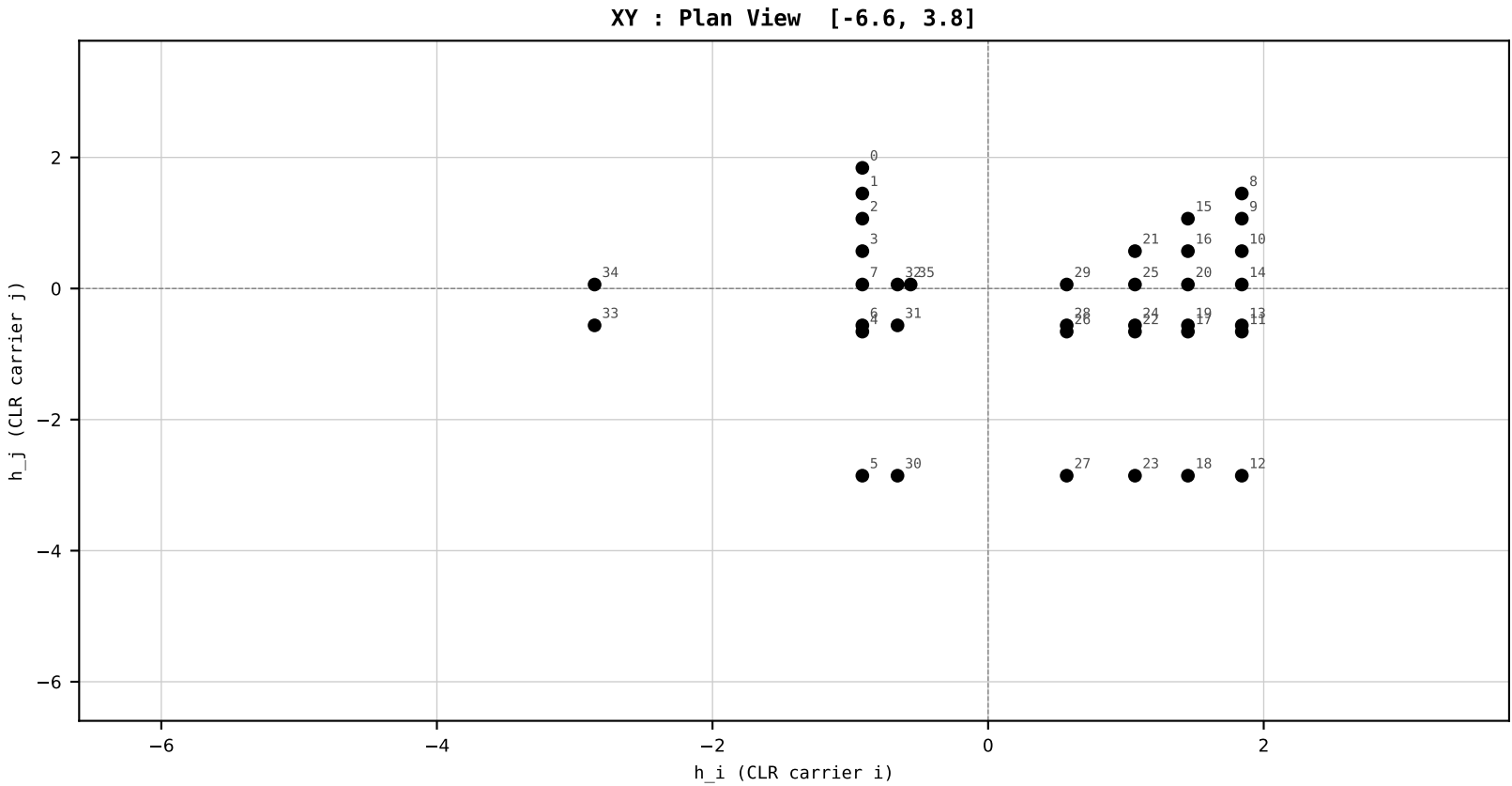
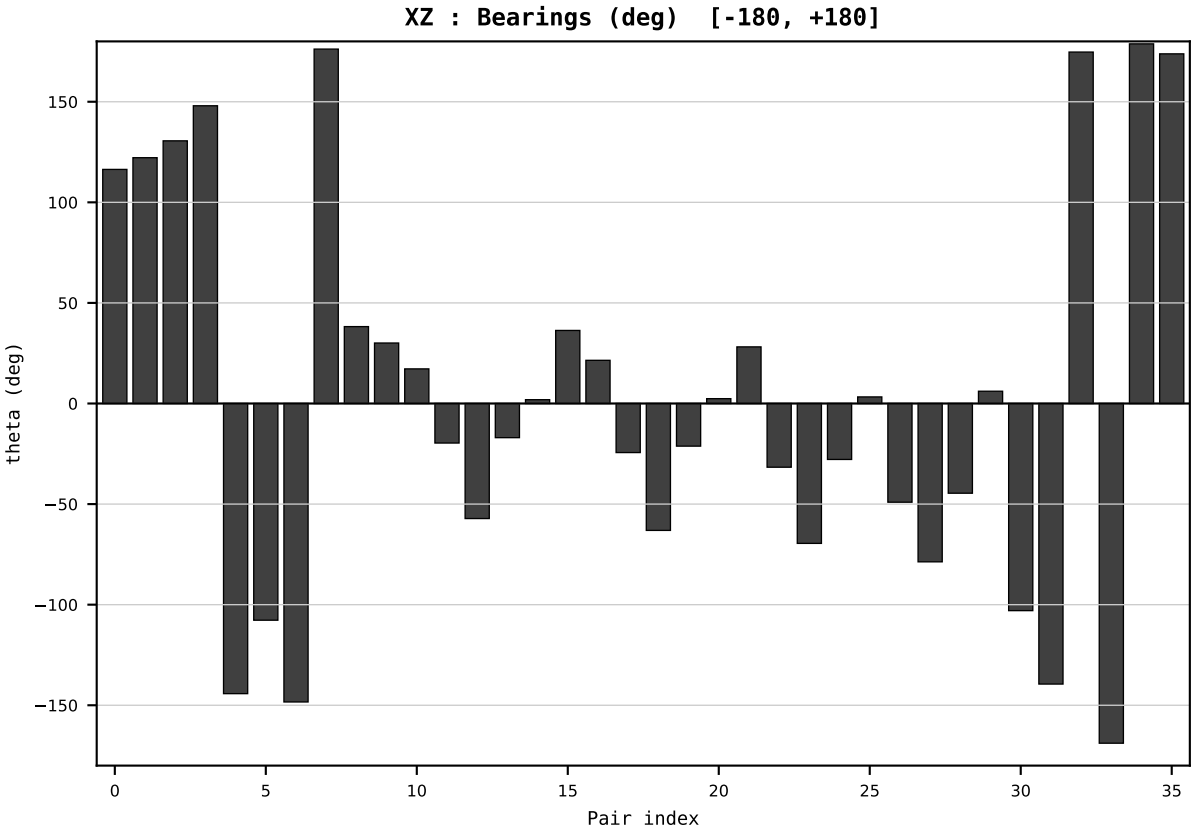
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2020
t : 20 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.217796
Ring = Hs-2
E_metric= 16.6924
kappa_HS= 109.58
omega = 3.3646 deg
d_A = 0.253314
Helm = Solar
Helm d = +0.172148
DR = 4.697 HLR
DR ratio= 109.6x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=21 | Year 2021 | D=9 N=26 pairs=36

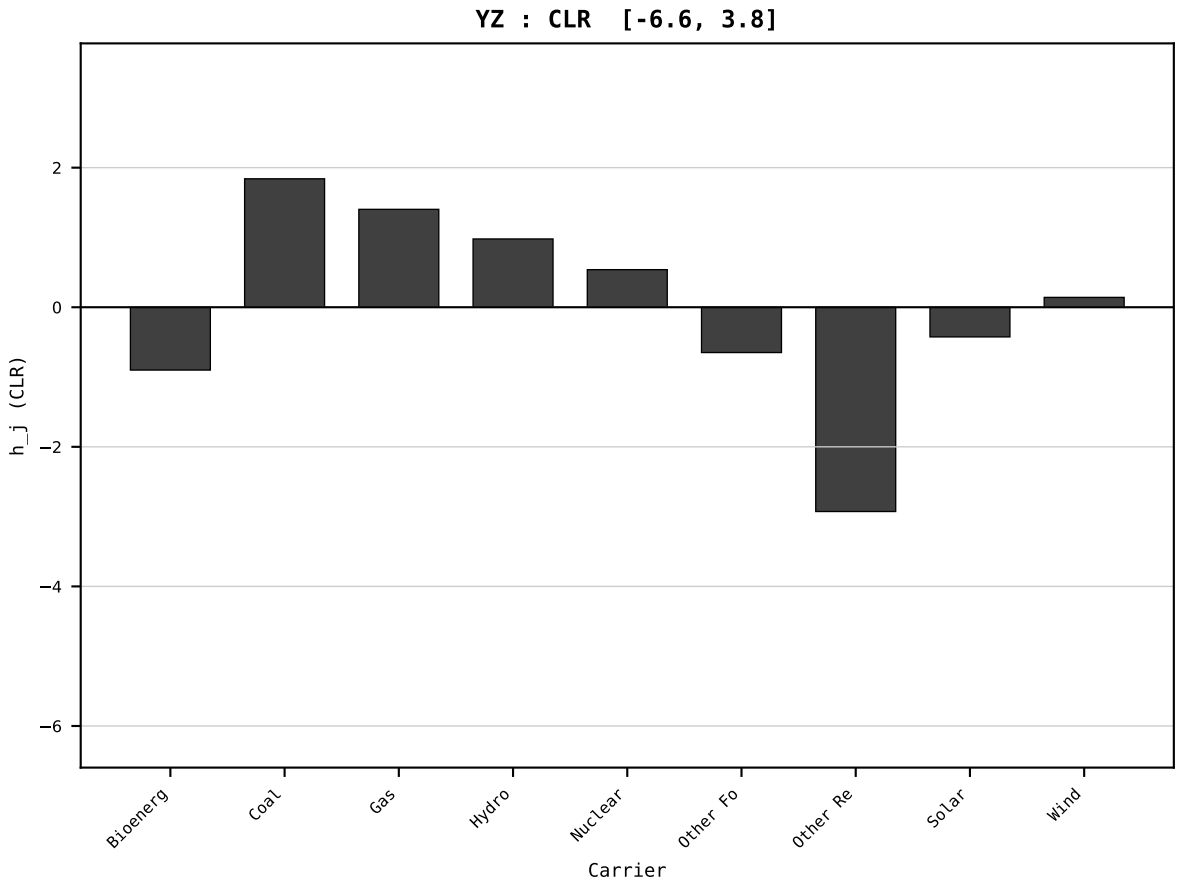
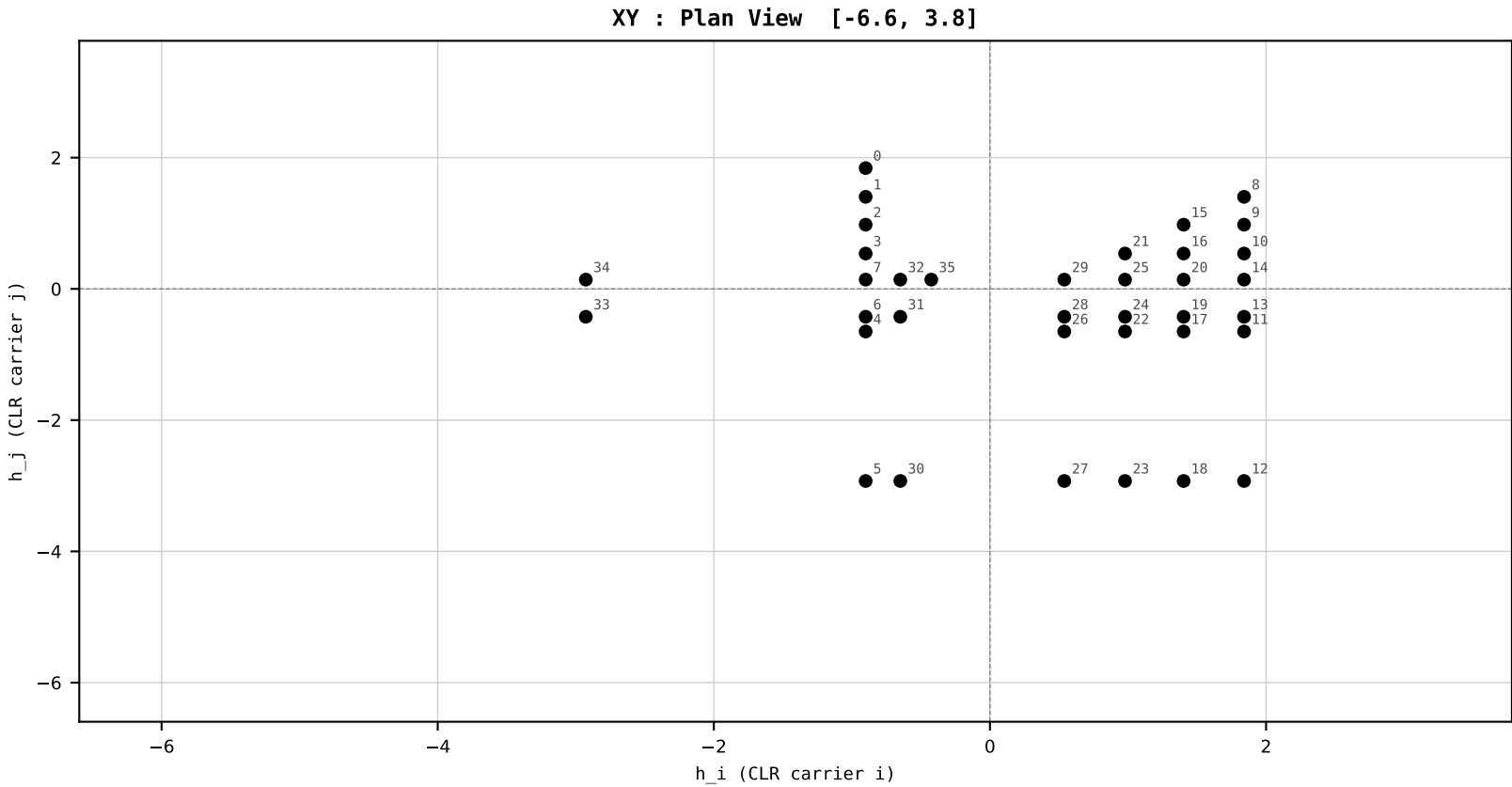
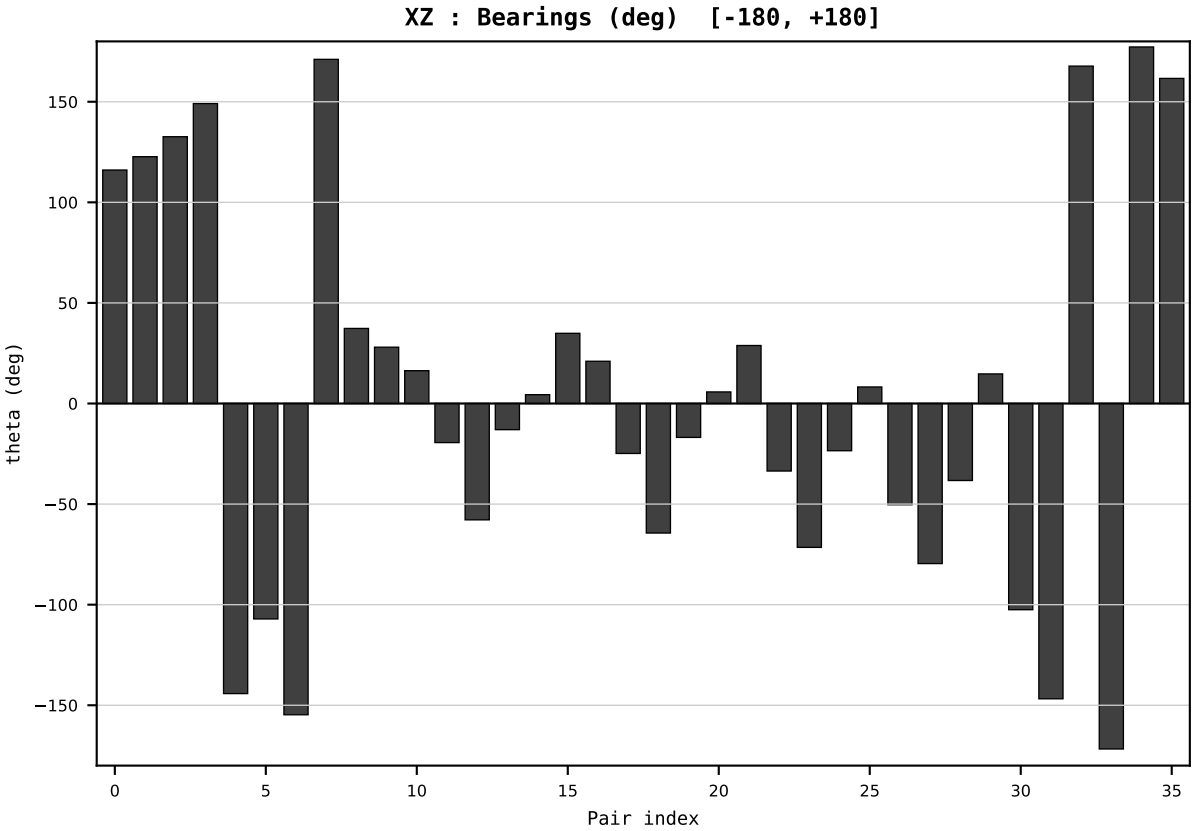
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2021
t : 21 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.211988
Ring = Hs-2
E_metric= 16.6038
kappa_HS= 117.65
omega = 2.8560 deg
d_A = 0.203654
Helm = Solar
Helm d = +0.137219
DR = 4.768 HLR
DR ratio= 117.6x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



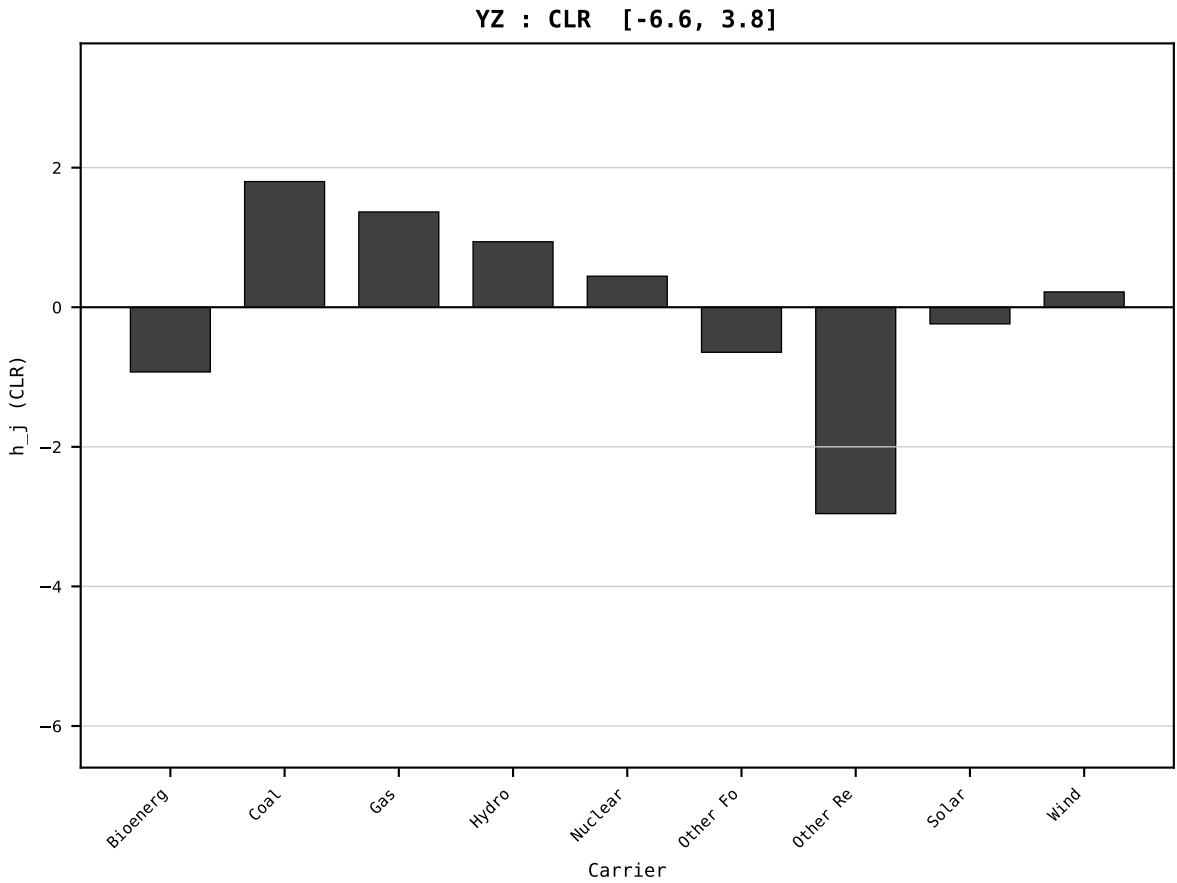
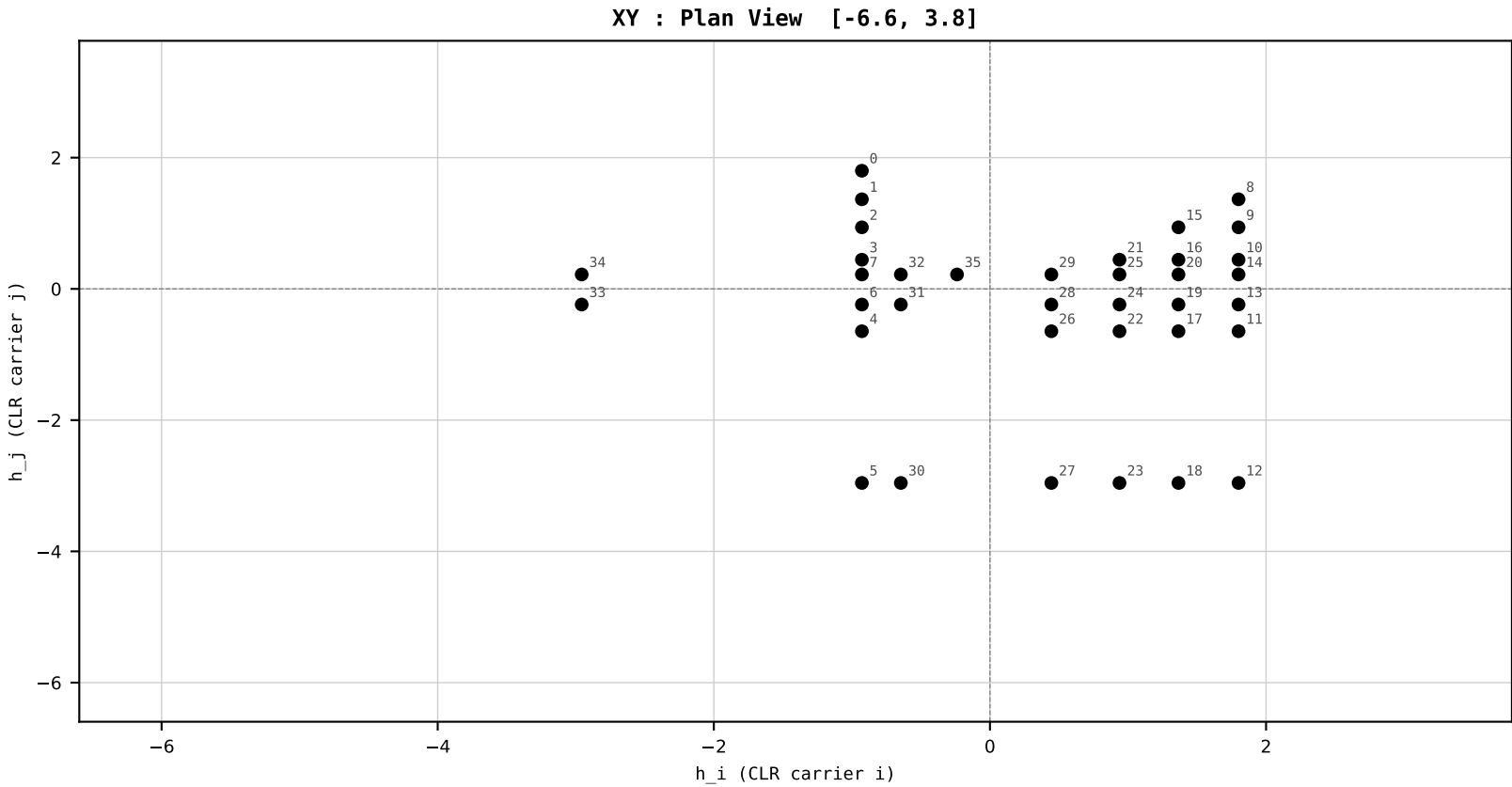
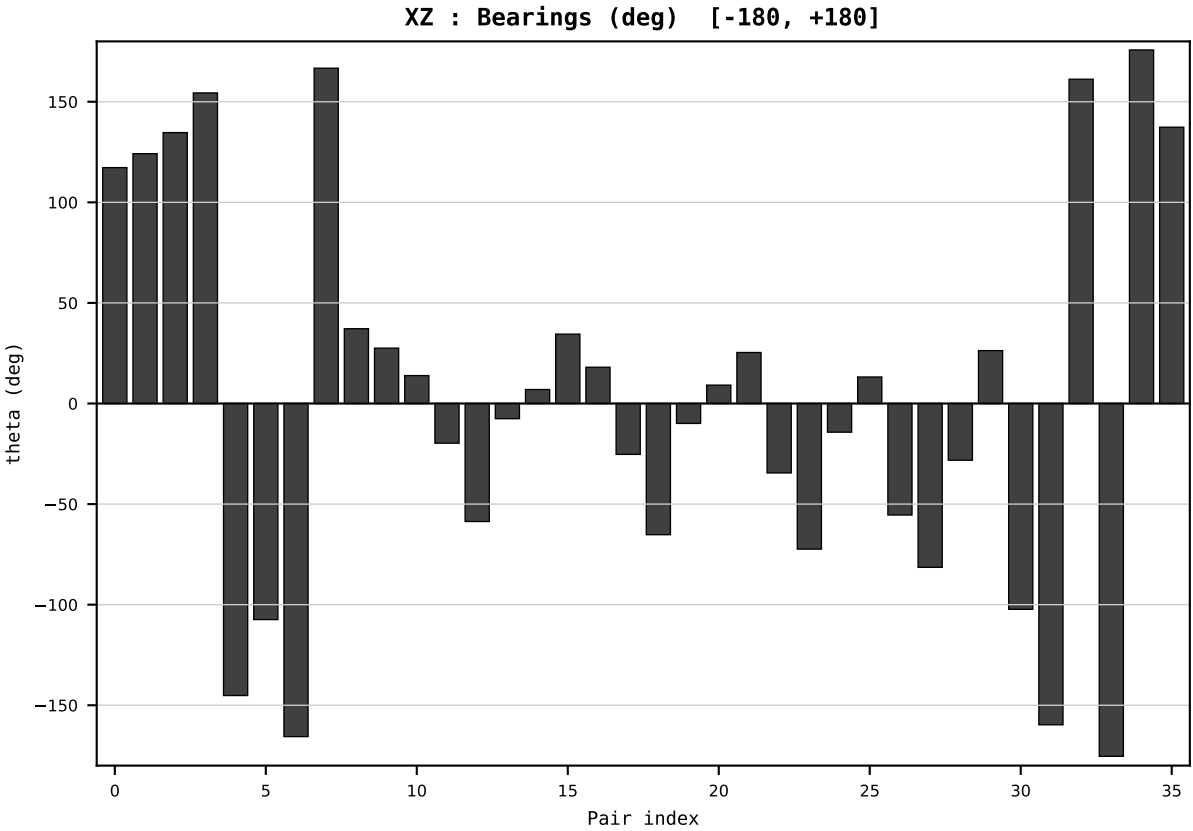
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2022
t : 22 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.202475
Ring = Hs-2
E_metric= 16.3100
kappa_HS= 116.45
omega = 3.3044 deg
d_A = 0.236707
Helm = Solar
Helm d = +0.186591
DR = 4.757 HLR
DR ratio= 116.4x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



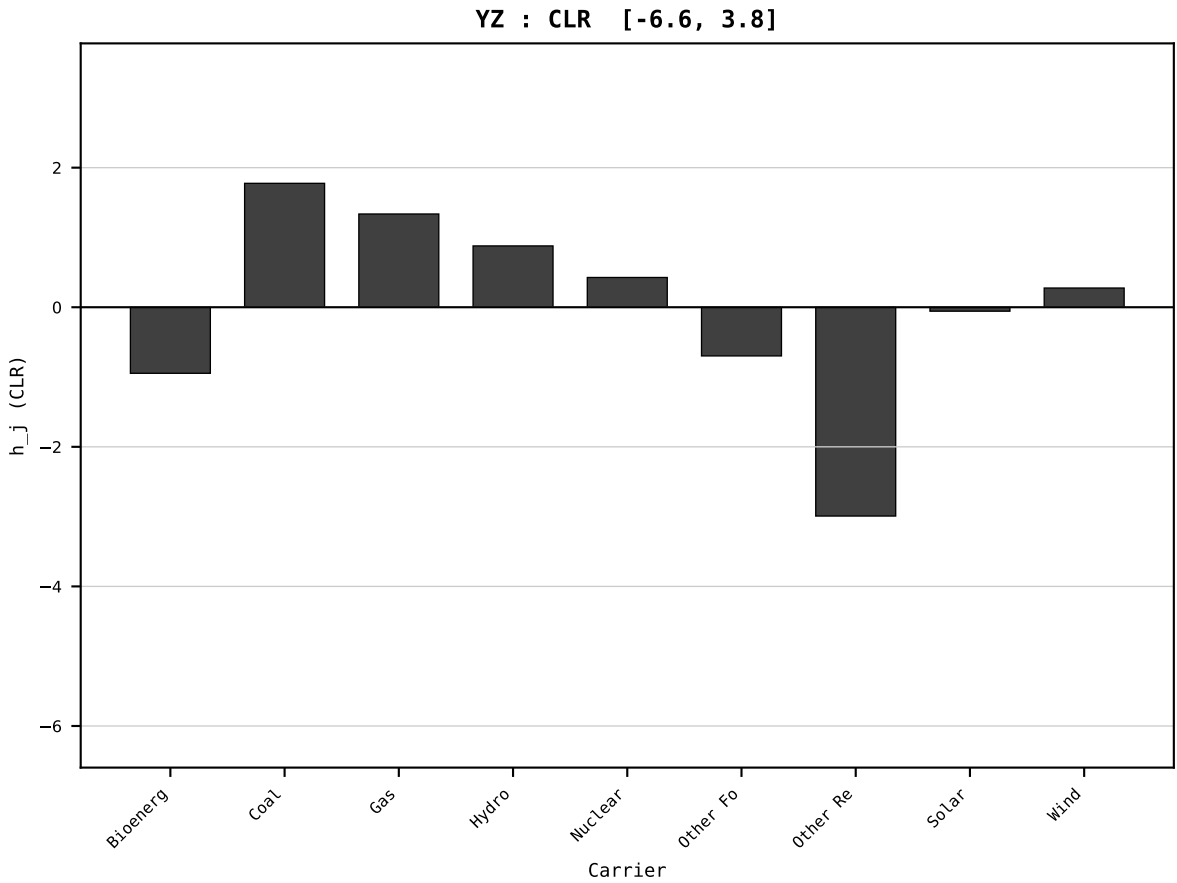
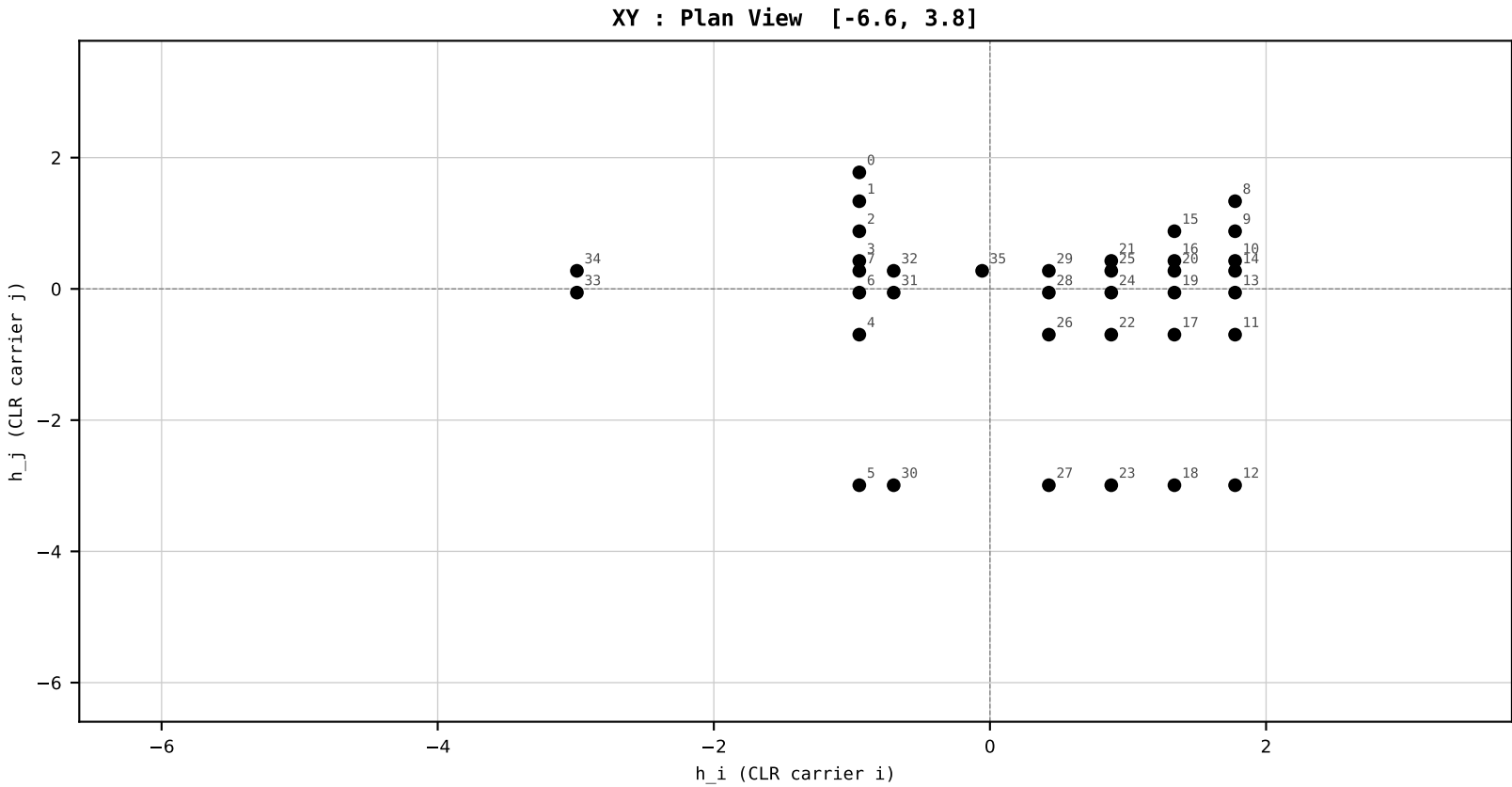
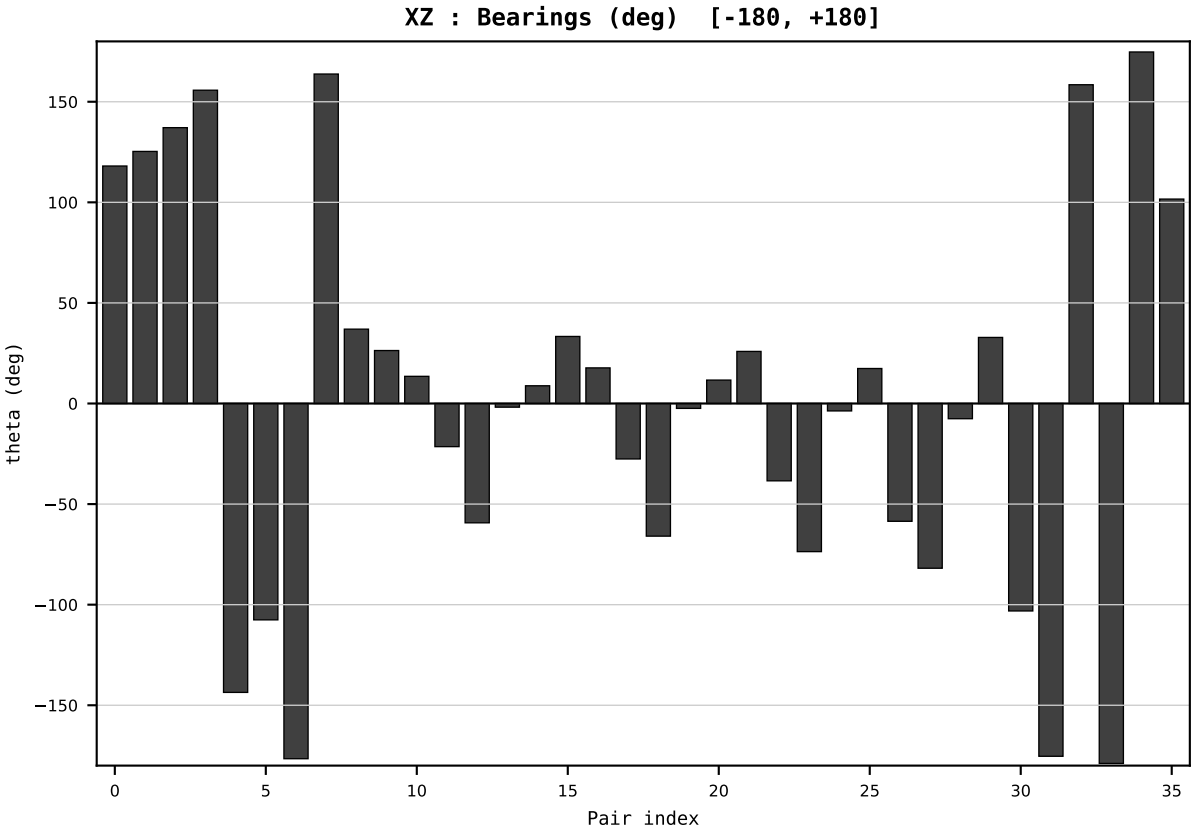
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2023
t : 23 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.195016
Ring = Hs-2
E_metric= 16.3032
kappa_HS= 117.60
omega = 3.0360 deg
d_A = 0.213954
Helm = Solar
Helm d = +0.181807
DR = 4.767 HLR
DR ratio= 117.6x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=24 | Year 2024 | D=9 N=26 pairs=36

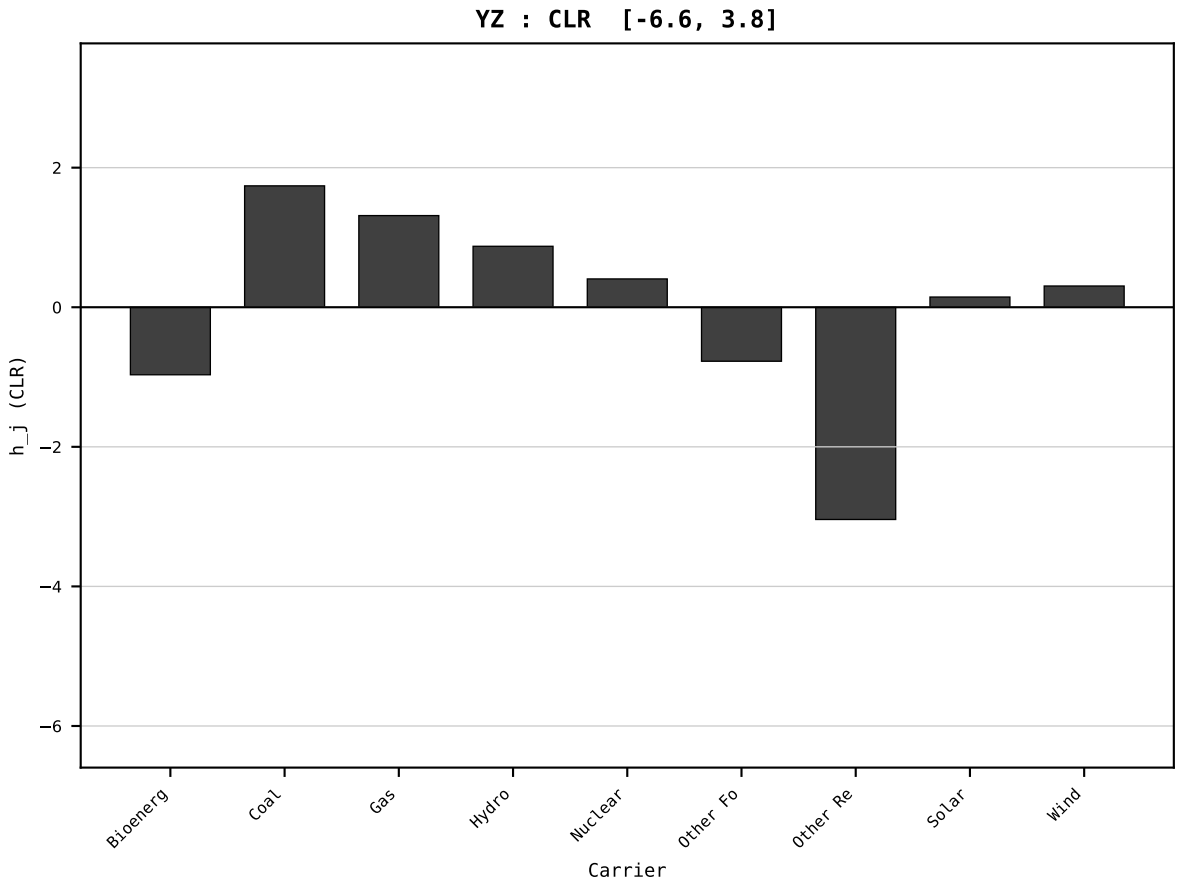
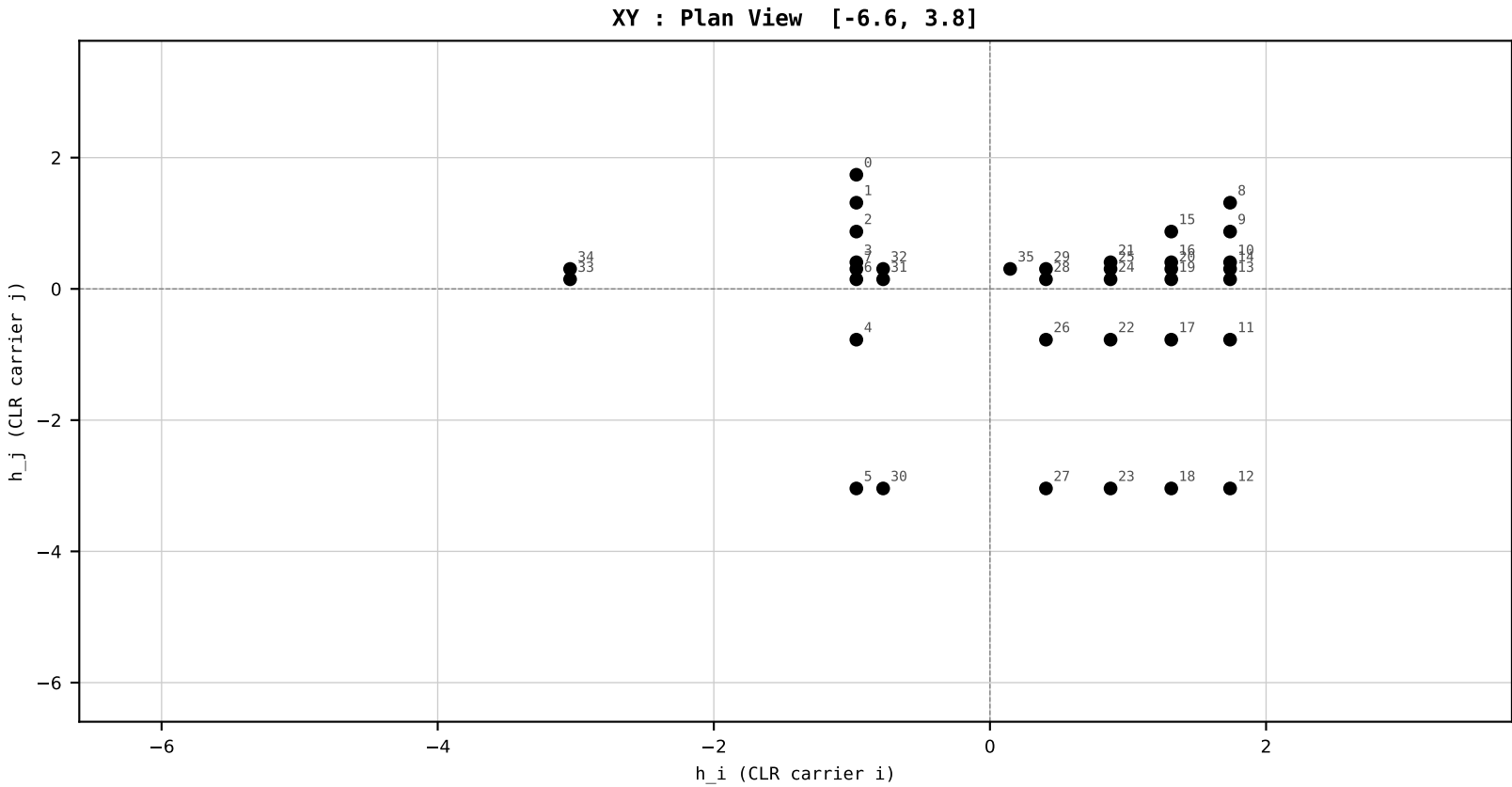
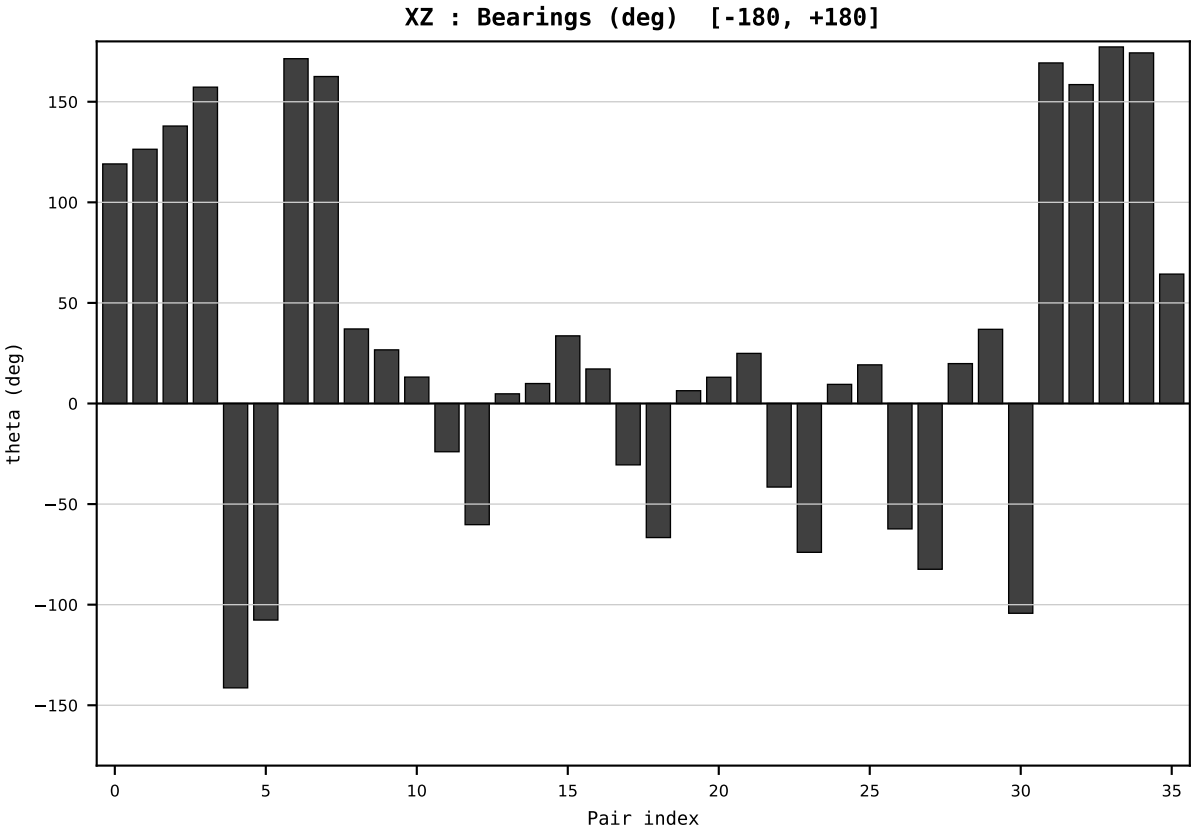
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_WLD_World_generation_TWh.csv
Year : 2024
t : 24 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.186834
Ring = Hs-2
E_metric= 16.5752
kappa_HS= 119.16
omega = 3.2199 deg
d_A = 0.230278
Helm = Solar
Helm d = +0.202924
DR = 4.781 HLR
DR ratio= 119.2x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=25 | Year 2025 | D=9 N=26 pairs=36

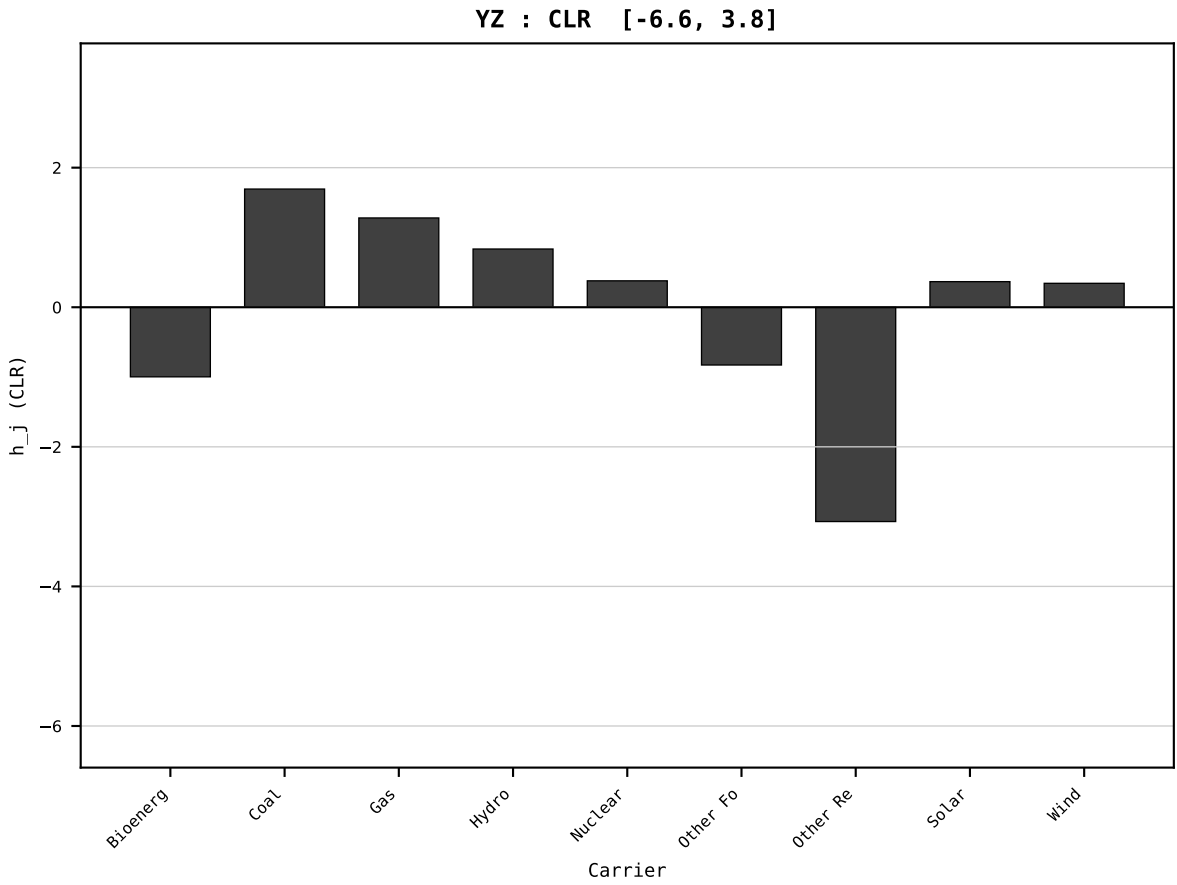
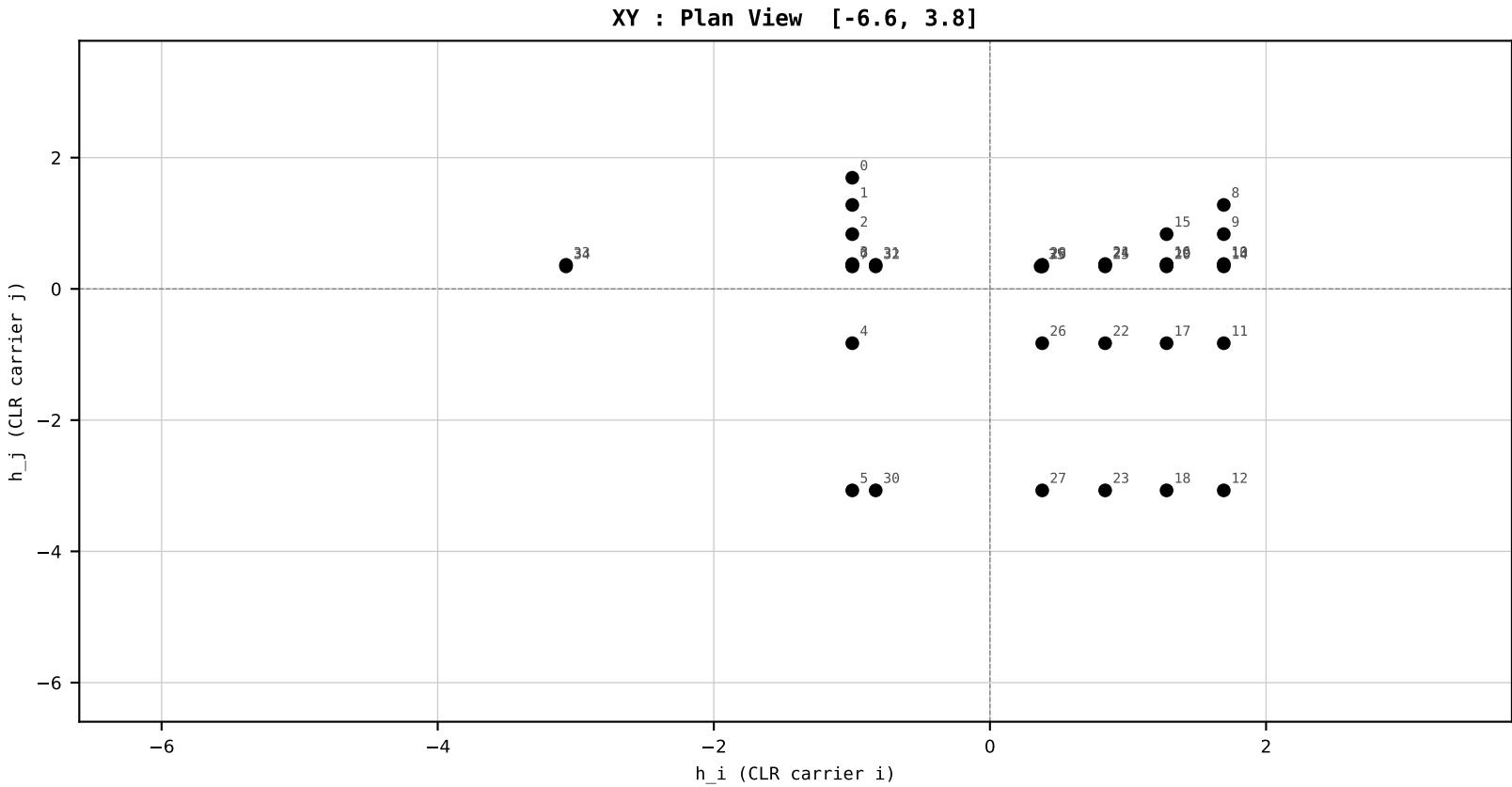
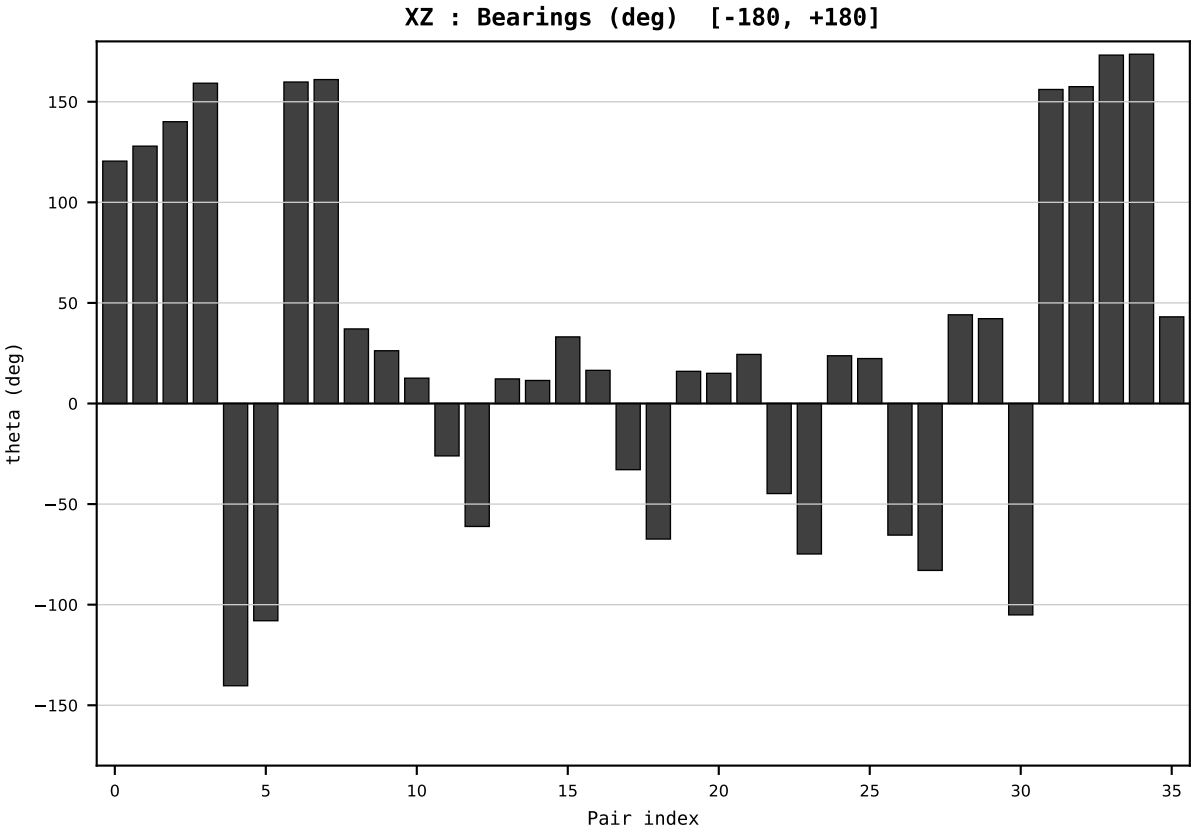
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

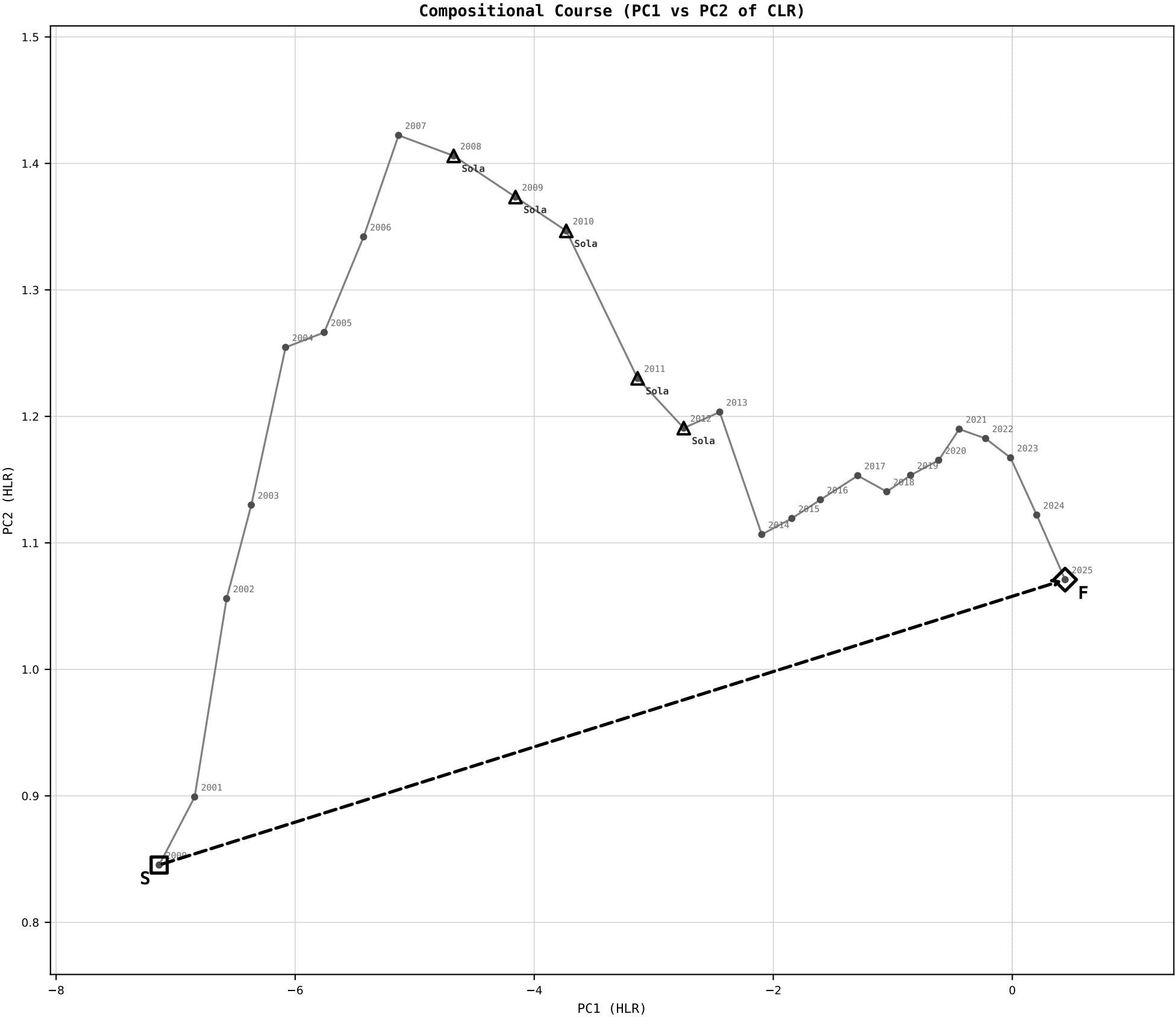
Dataset : ember_WLD_World_generation_TWh.csv
Year : 2025
t : 25 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.177094
Ring = Hs-2
E_metric= 16.7040
kappa_HS= 117.22
omega = 3.4412 deg
d_A = 0.245463
Helm = Solar
Helm d = +0.220443
DR = 4.764 HLR
DR ratio= 117.2x

XZ [-180,180]
CLR[-6.6,3.8]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind





COURSE METRICS

Scale Provenance:
 $Y_j \rightarrow x_j = Y_j / T \rightarrow h_j = \text{CLR}$
 $\rightarrow$ PC projection
Unit: HLR (Higgins Log-Ratio)

Net distance = 7.5841 HLR
Path length = 7.9588 HLR
Directness = 0.9529

DR start = 8.647 HLR
ratio = 5694.9x
DR final = 4.764 HLR
ratio = 117.2x

E_metric start = 66.7408
E_metric final = 16.7040

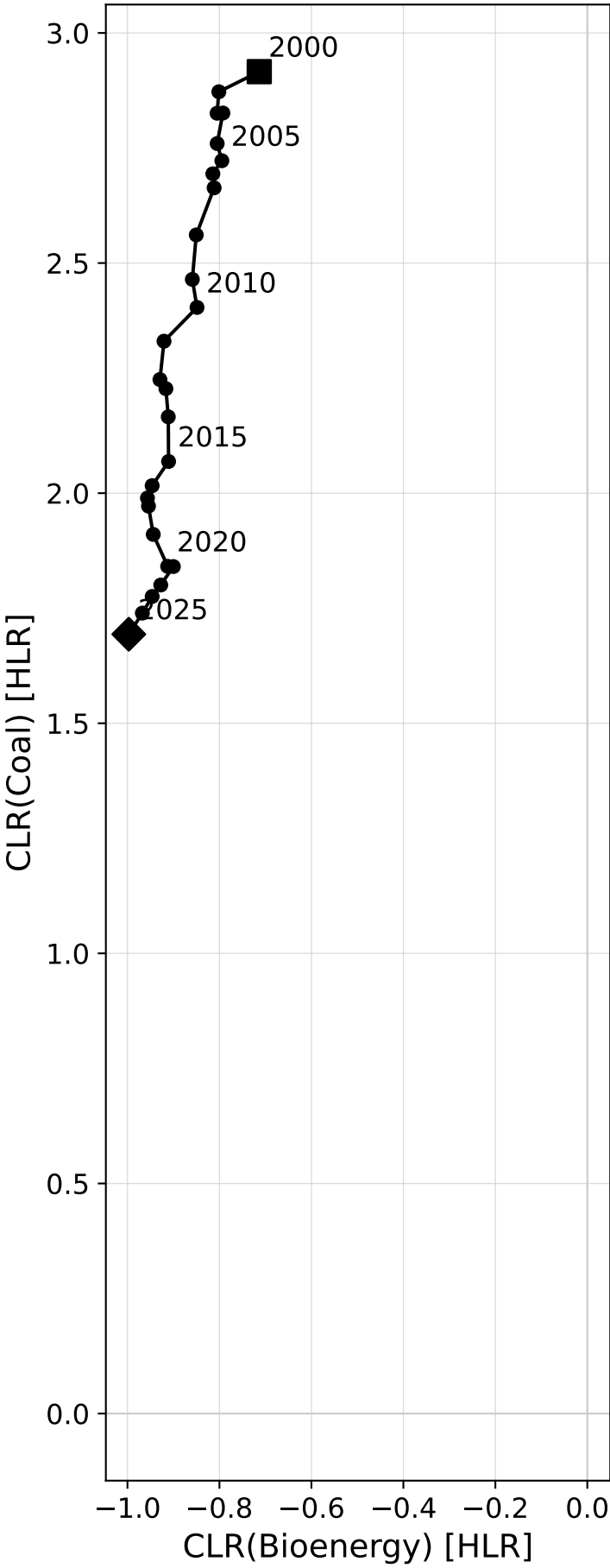
PC1 loadings:
Solar +0.816
Wind +0.324
Other Fo -0.296

PC2 loadings:
Wind +0.827
Solar -0.443
Other Fo -0.193

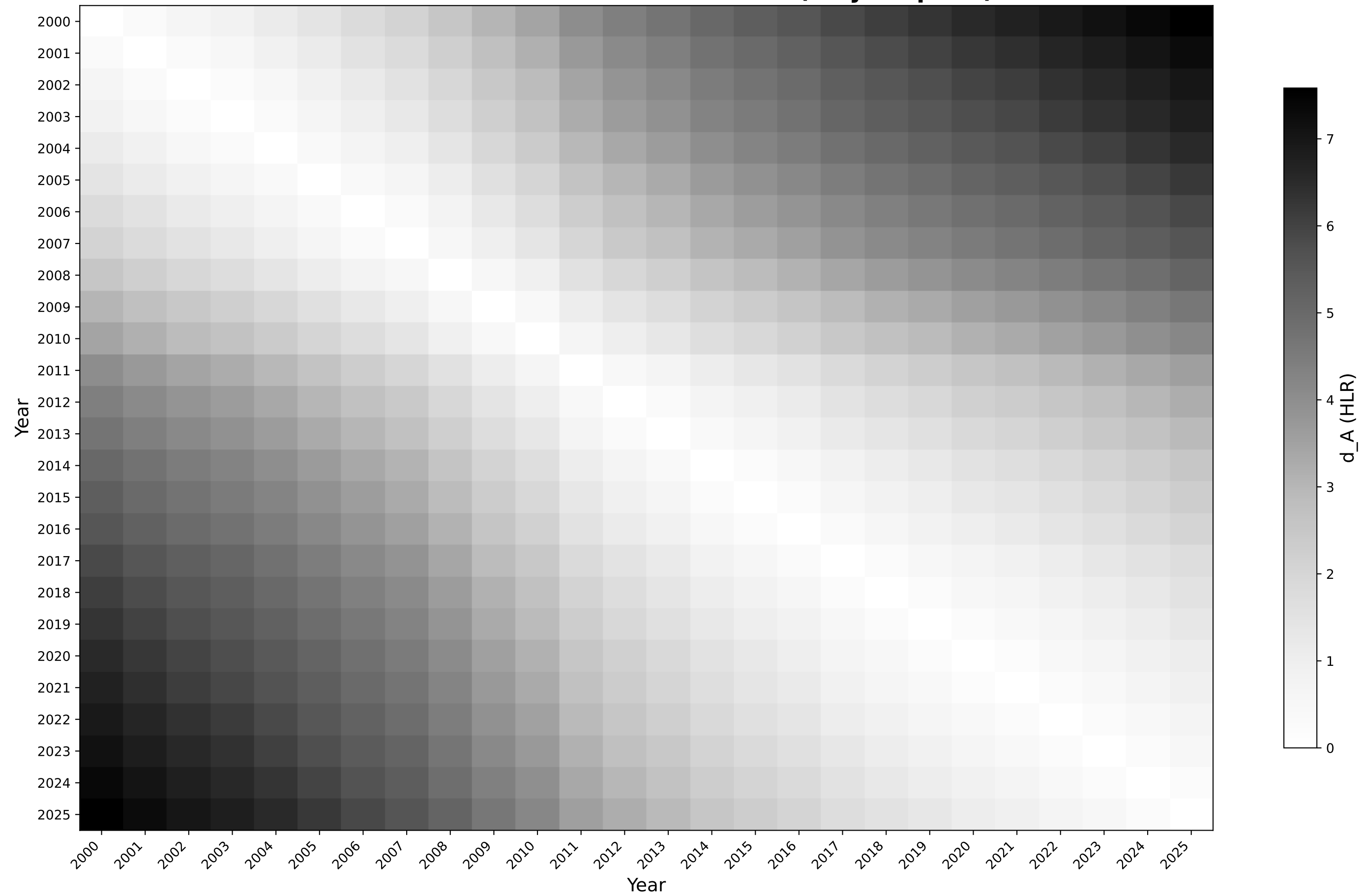
Top displacement events:
2011 Solar d=0.629
2009 Solar d=0.520
2008 Solar d=0.464
2010 Solar d=0.430
2012 Solar d=0.401

The instrument reads.
The expert decides.
The loop stays open.

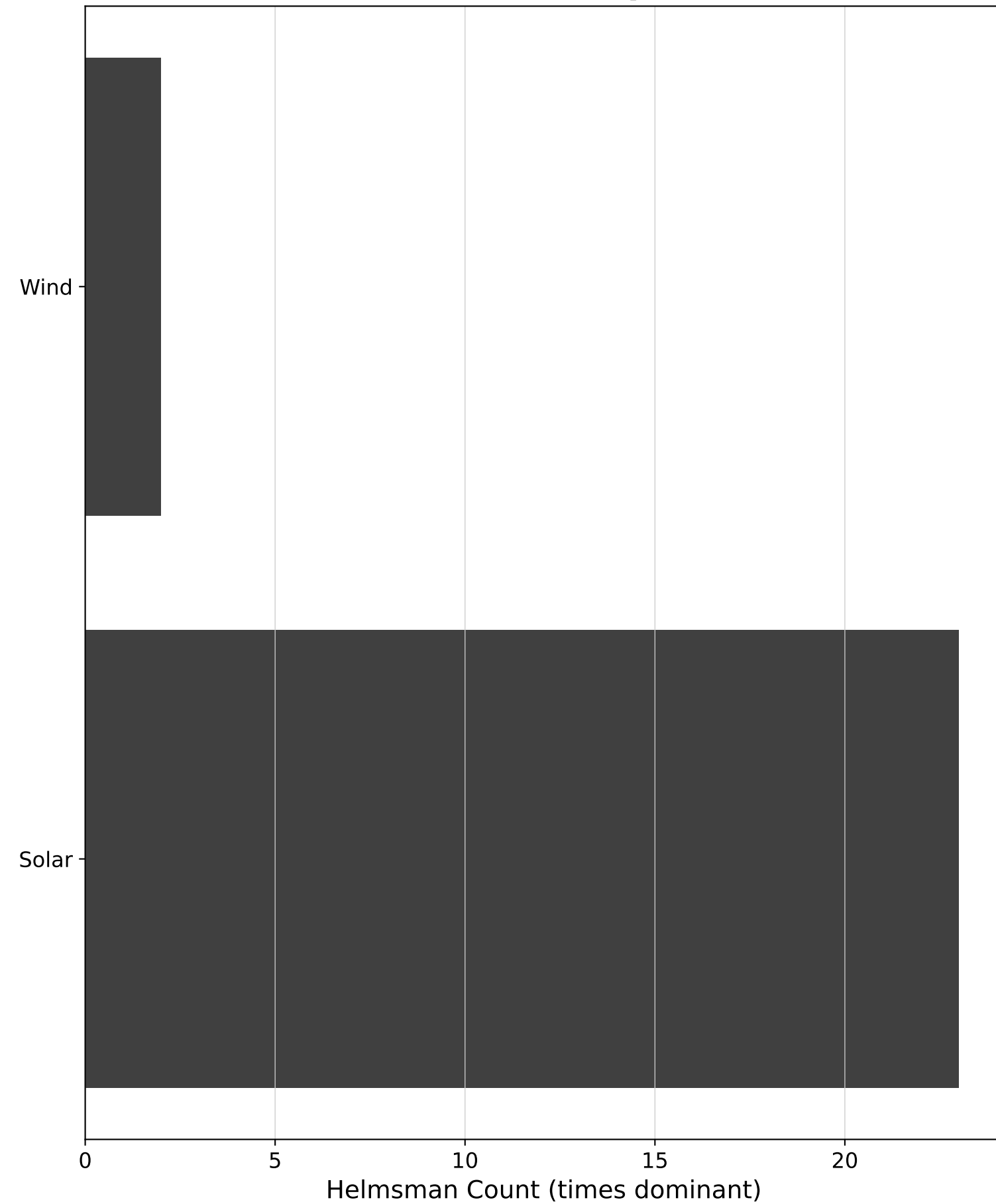
SYSTEM COURSE PLOT — XY Plan View (CLR Space)



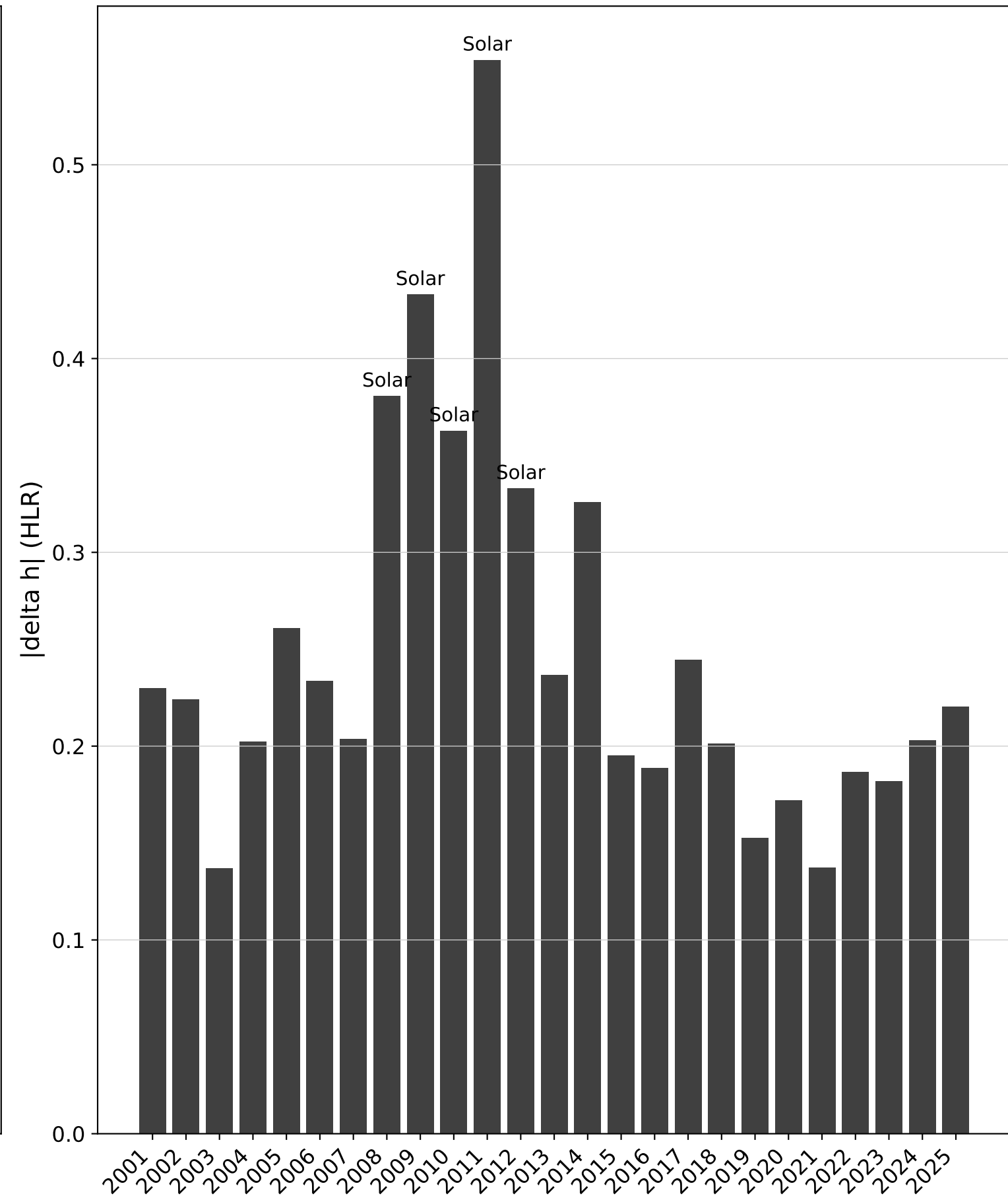
STAGE 2 — Aitchison Distance Matrix (all year pairs)



HELMSMAN FREQUENCY



HELMSMAN DISPLACEMENT PER YEAR



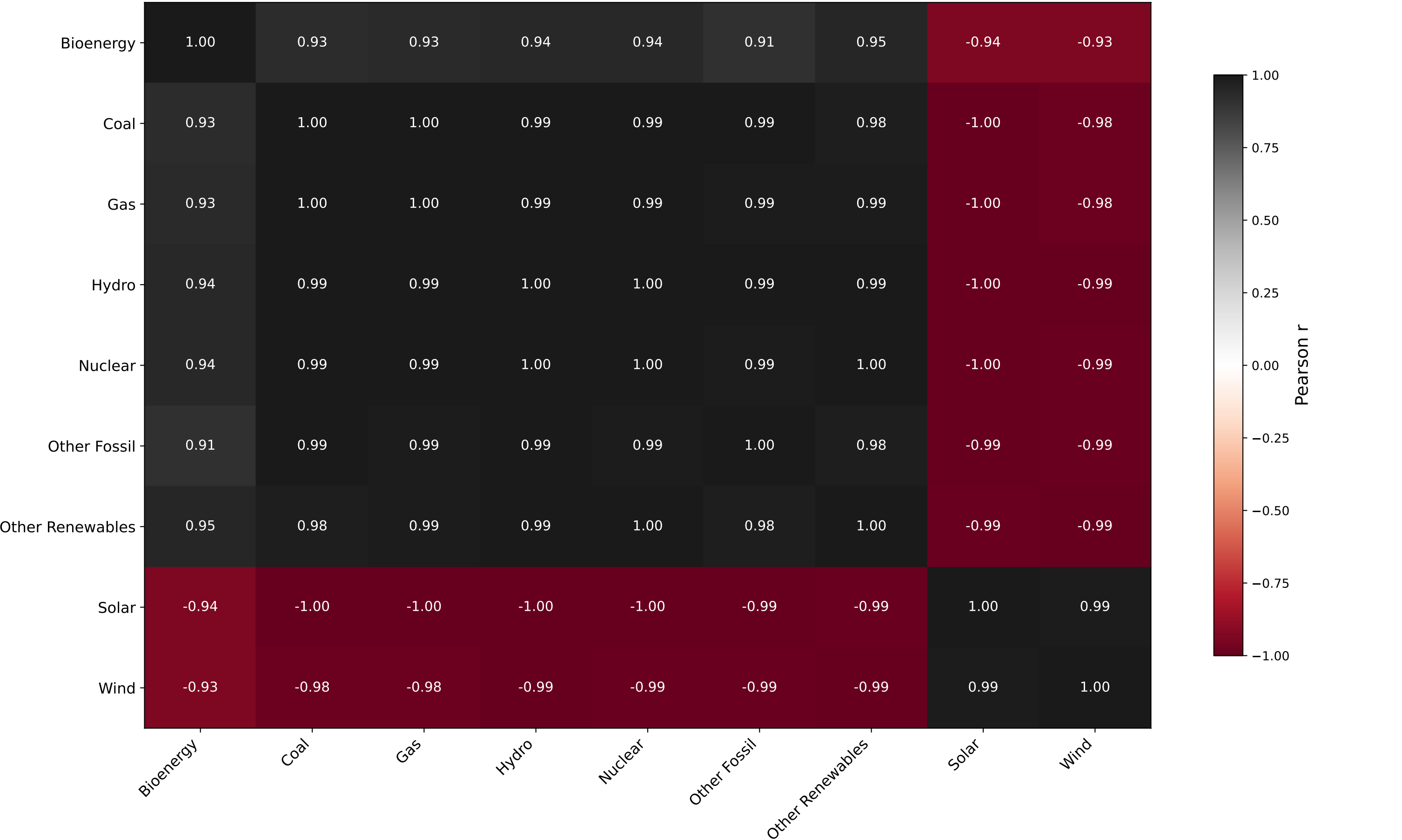
STAGE 2 — Pairwise Divergence Ranking (Top 15)

Rank	Year A	Year B	d_A (HLR)	Angle (deg)	Dominant Carrier
1	2000	2025	7.5841	67.2	Solar
2	2000	2024	7.3480	63.8	Solar
3	2001	2025	7.2851	65.9	Solar
4	2000	2023	7.1308	60.8	Solar
5	2001	2024	7.0491	62.6	Solar
6	2002	2025	7.0156	64.8	Solar
7	2000	2022	6.9229	57.8	Solar
8	2001	2023	6.8318	59.5	Solar
9	2003	2025	6.8097	63.8	Solar
10	2002	2024	6.7784	61.4	Solar
11	2000	2021	6.7029	54.8	Solar
12	2001	2022	6.6242	56.6	Solar
13	2003	2024	6.5719	60.5	Solar
14	2002	2023	6.5598	58.4	Solar
15	2000	2020	6.5308	52.3	Solar

STAGE 3 — Triadic Metric Triangle Area Ranking (Top 10)

Rank	Year A	Year B	Year C	Area (HLR^2)	Imbalance
1	2000	2007	2025	1.9693	3.64
2	2000	2008	2025	1.8595	3.00
3	2000	2007	2024	1.8505	3.52
4	2001	2007	2025	1.7748	4.08
5	2000	2007	2023	1.7414	3.42
6	2000	2008	2024	1.7317	2.91
7	2000	2006	2025	1.6982	4.26
8	2000	2009	2025	1.6787	2.50
9	2001	2008	2025	1.6760	3.27
10	2000	2007	2022	1.6689	3.32

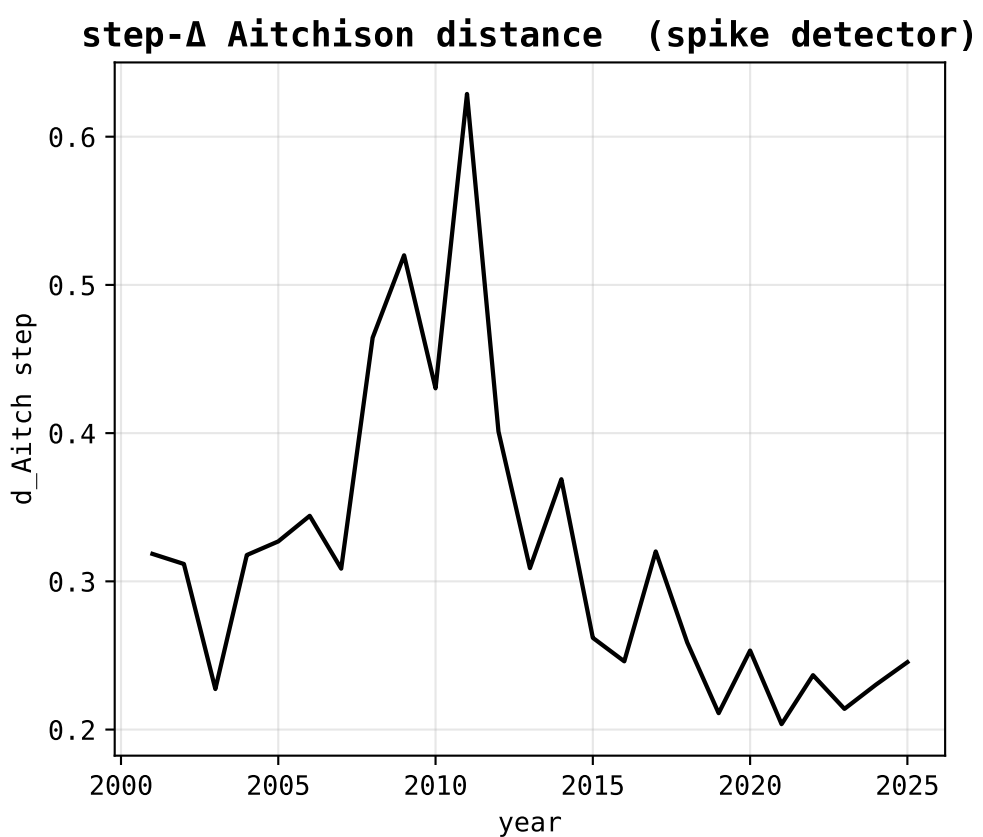
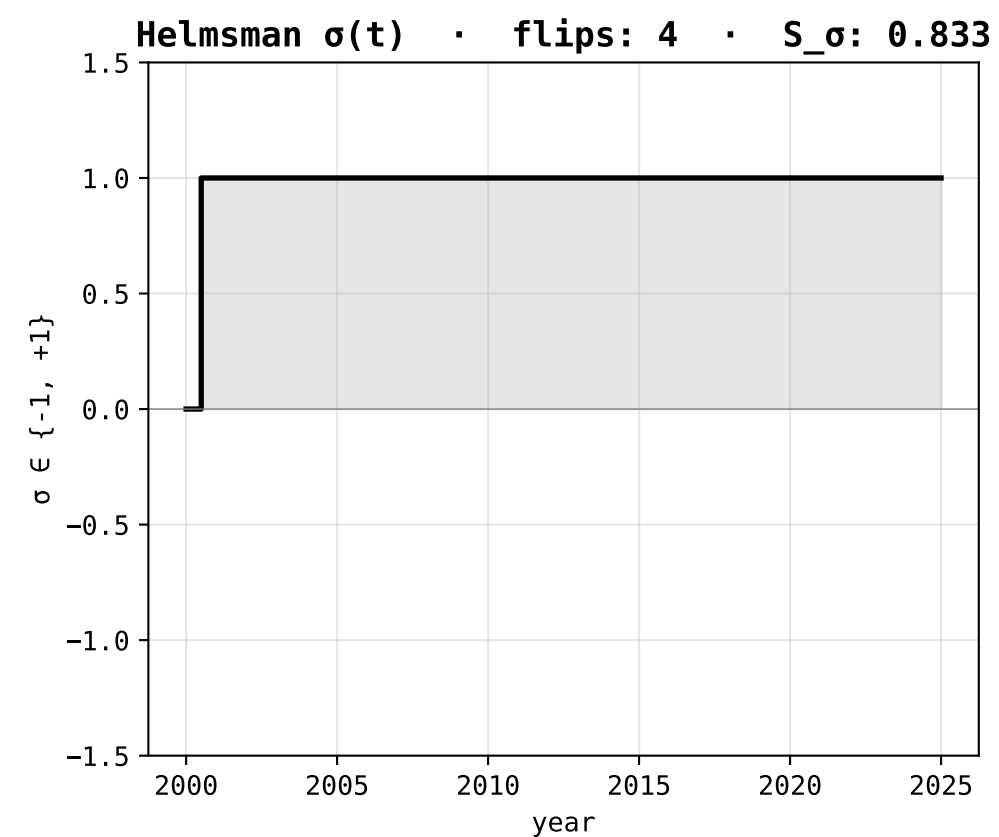
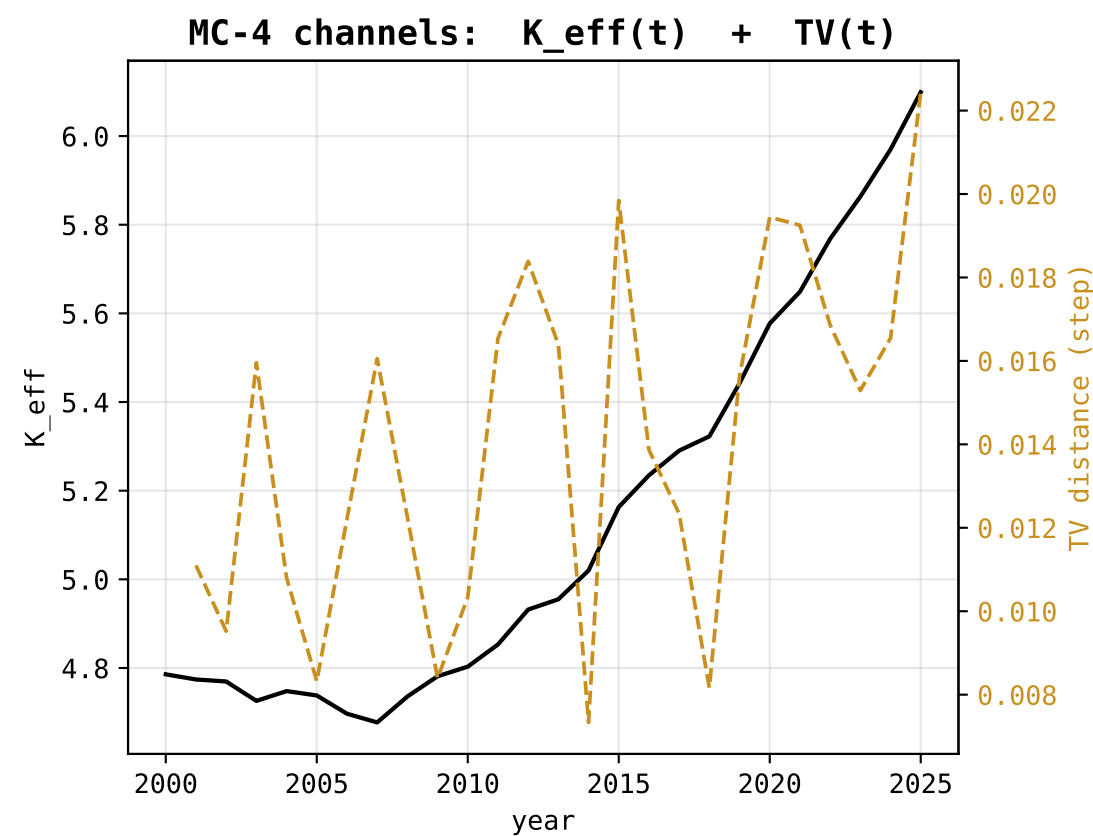
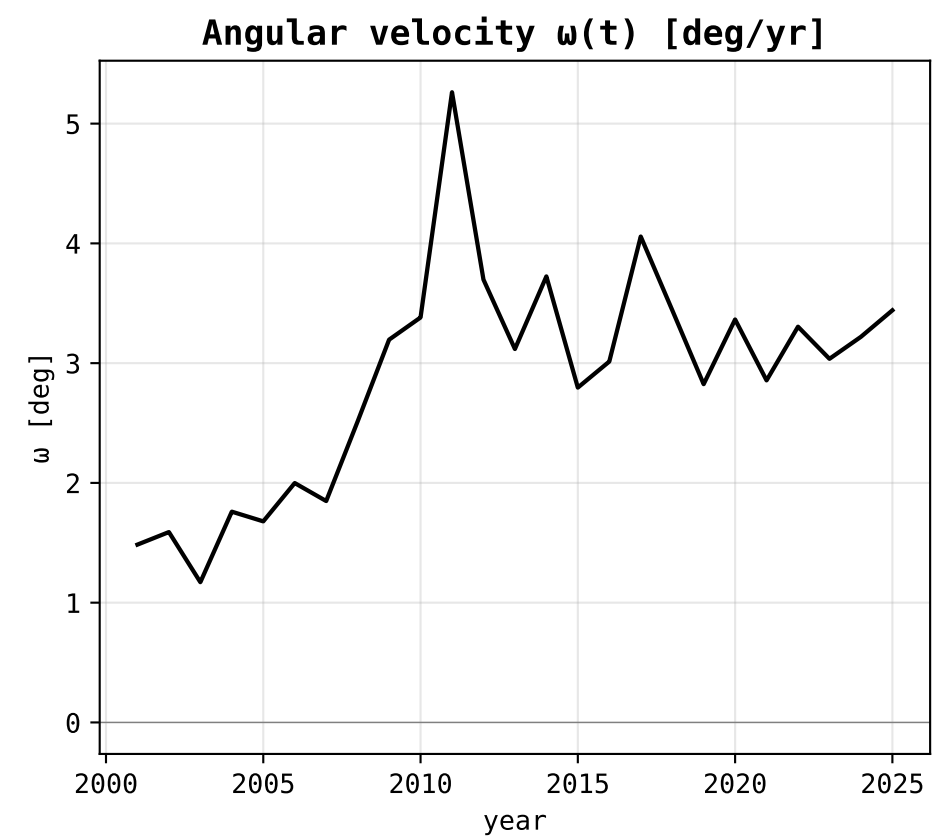
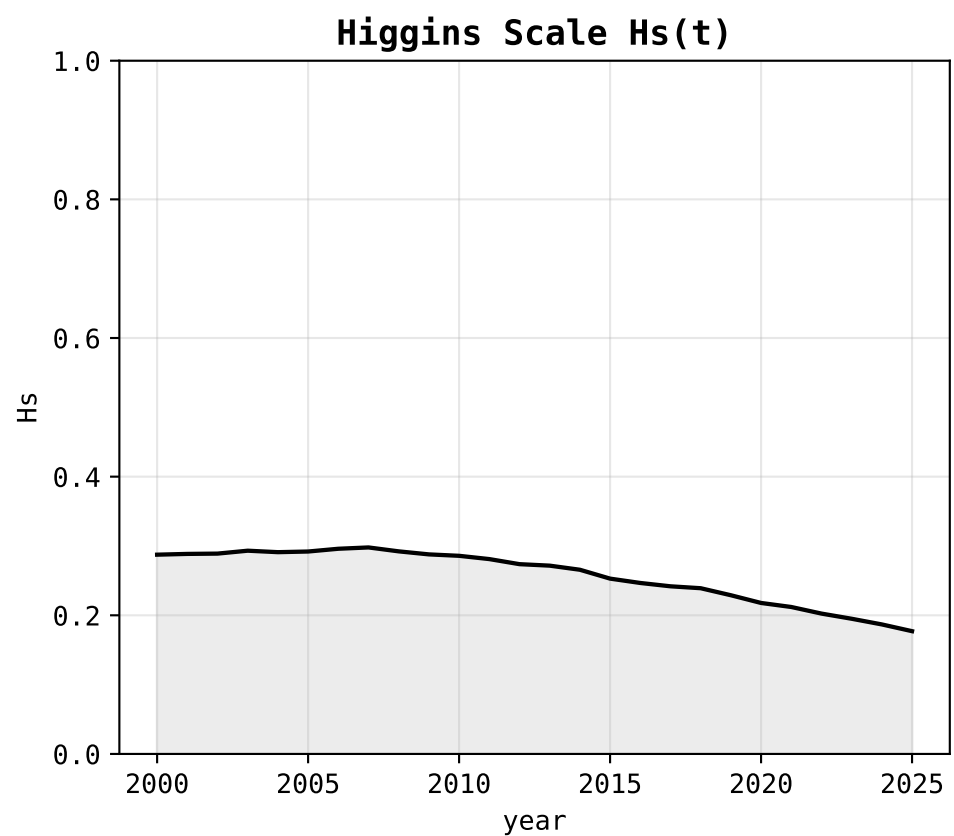
STAGE 3 — Carrier CLR Trajectory Correlation



NAVIGATION SUMMARY — All Stage 1 Metrics by Year

Year	Hs	Ring	E_metric	omega(deg)	d_A	Helmsman	kappa_max
2000	0.2874	Hs-2	66.7	0.00	0.0000	---	14979
2001	0.2886	Hs-2	62.9	1.48	0.3185	Solar	11398
2002	0.2890	Hs-2	59.4	1.59	0.3117	Wind	9331
2003	0.2932	Hs-2	56.8	1.17	0.2274	Solar	7881
2004	0.2911	Hs-2	53.6	1.76	0.3178	Wind	6264
2005	0.2920	Hs-2	50.0	1.68	0.3269	Solar	4591
2006	0.2960	Hs-2	46.6	2.00	0.3442	Solar	3476
2007	0.2979	Hs-2	43.6	1.85	0.3086	Solar	2704
2008	0.2922	Hs-2	38.9	2.51	0.4644	Solar	1696
2009	0.2879	Hs-2	34.1	3.20	0.5200	Solar	1006
2010	0.2858	Hs-2	31.0	3.38	0.4302	Solar	660
2011	0.2811	Hs-2	26.8	5.26	0.6288	Solar	360
2012	0.2738	Hs-2	24.5	3.70	0.4012	Solar	358
2013	0.2716	Hs-2	22.9	3.12	0.3089	Solar	360
2014	0.2657	Hs-2	21.0	3.72	0.3689	Solar	342
2015	0.2529	Hs-2	19.7	2.80	0.2619	Solar	328
2016	0.2466	Hs-2	18.9	3.01	0.2460	Solar	325
2017	0.2418	Hs-2	18.1	4.06	0.3202	Solar	323
2018	0.2391	Hs-2	17.7	3.44	0.2588	Solar	324
2019	0.2289	Hs-2	17.3	2.82	0.2111	Solar	323
2020	0.2178	Hs-2	16.7	3.36	0.2533	Solar	310
2021	0.2120	Hs-2	16.6	2.86	0.2037	Solar	327
2022	0.2025	Hs-2	16.3	3.30	0.2367	Solar	329
2023	0.1950	Hs-2	16.3	3.04	0.2140	Solar	336
2024	0.1868	Hs-2	16.6	3.22	0.2303	Solar	350
2025	0.1771	Hs-2	16.7	3.44	0.2455	Solar	356

CNQ v2.0.0 – Quaternion View – WLD World · D=9 · N=26



CNQ DIAGNOSTICS · WLD (D = 9)

CHSH joint-coherence:
not computed at D = 9
(CHSH requires D=8 twin factoring;
D = 9 does not factor as 4x4)

Twin-quaternion factor:
not computed at D = 9
(twin requires D=8 = 4+4 partition)

Attractor fit:
amplitude A: 0.135
contraction λ : -0.0042
period: 1
fitted: False

Helmsman family:
total flips: 4
stability S_σ : 0.833